"Game Theory" by Maschler, Solan and Zamir

Solution Manual

Chapter 1: The game of chess

1.1 No. The statement in the exercise holds for games without chance moves, but also for games with chance moves, like backgammon. The statement in Theorem 1.4, on the other hand, does not hold in backgammon; in this game, if a player is extremely lucky, he can win regardless of the strategy of the other player. In particular, the two statements cannot be equivalent.

1.2 Tic-Tac-Toe, Four in a row, Nim.

1.3 (a) Denote by N the number of game positions in chess. If after N^2 turns neither White wins nor Black wins, then there is at least one game position that is repeated twice so the play ends with draw. Hence, the conditions of Theorem 1.5 are satisfied and (a) holds.

(b) Assume that one of the players has a winning strategy in finite chess. Then he can guarantee by following this strategy that finite chess ends with his winning- before it reaches a repeated game position. Therefore, a strategy in which he follows his winning strategy as long as no game position is repeated, otherwise he chooses an arbitrary legal action is a winning strategy in chess. Indeed, by following this strategy in chess the game ends before it reaches a repeated game position as if it is finite chess, thus it leads to his winning.

(c) Assume that each player has a strategy guaranteeing at least a draw in finite chess. We show that each player has a strategy guaranteeing at least a draw in chess. We show it for White, the proof for Black is similar and thus it is omitted.
Denote by σ_W a strategy of White in finite chess that guarantees at least a draw. Consider the following strategy $\widehat{\sigma}_W$ for White in chess: implement the strategy σ_W until either the play of chess terminates or a game position repeats itself (at which point, the play of finite chess terminates). If the play of chess arrives to a game position x that has previously appeared, implement the strategy σ_W restricted to the subgame beginning at x until the play arrives at a game position y that has previously appeared, and so on.
We prove that the strategy $\widehat{\sigma}_W$ guarantees at least a draw in chess. Assume by contradiction that $\widehat{\sigma}_W$ does not guarantee at least a

draw. Let $\widehat{\sigma}_B$ be a strategy of Black that wins the game against White playing $\widehat{\sigma}_W$ with minimal number of moves. Since σ_W guarantees at least a draw in finite chess, when the player follow $(\widehat{\sigma}_W, \widehat{\sigma}_B)$, White faces at least one game position, say x, that appears at least twice before the winning of Black. According to $\widehat{\sigma}_W$, White selects the same move in x. Therefore, from the first repetition of x onward, Black could follow the moves he takes after the second repetition of x and win the game using less moves, which contradicts $\widehat{\sigma}_B$ wins with minimal number of moves.

Chapter 2: Utility theory

2.1 (a) Let $\succ$ be a strict preference.
$\succ$ is anti symmetric: if $x \succ y$ then $y \not\succsim x$, therefore $y \not\succ x$.
In addition, $\succ$ is transitive: if $x \succ y$ and $y \succ z$ then $x \succsim y$ and $y \succsim z$. Hence, by the transitivity of $\succsim$, one has $x \succsim z$. It is left to prove that $z \not\succsim x$. Assume to the contrary that $z \succsim x$, so by the transitivity of $\succsim$, one has $z \succsim y$, a contradiction. In conclusion, $z \not\succsim x$ and therefore $x \succ z$.

(b) Let $\approx$ be an indifference relation.
Then $\approx$ is symmetric: if $x \approx y$ then $x \succsim y$ and $y \succsim x$, therefore $y \approx x$.
In addition, $\approx$ is transitive: if $x \approx y$ and $y \approx z$ then $x \succsim y$ and $y \succsim x$, likewise $y \succsim z$ and $z \succsim y$. Hence, by the transitivity of $\succsim$, one has $x \succsim z$ and $z \succsim x$, so $x \approx z$.

2.2 Let O be a set of outcomes. Let $\succsim$ be a complete, reflexive, and transitive relation over O. Assume that u is a utility function representing $\succsim$. We prove that for every monotonically increasing function $v : \mathbf{R} \to \mathbf{R}$, the composition $v \circ u$ is also a utility function representing $\succsim$. Indeed, let $x, y \in O$. Then, since u represents $\succsim$, $x \succsim y$ if and only if $u(x) \geq u(y)$. But the last inequality holds if and only if $v(u(x)) \geq v\Big(u(y)\Big)$, since v is a monotonically increasing function. In particular, $v \circ u$ is also a utility function representing $\succsim$, as needed.

2.3 Let $O = \mathbf{Q}$, the set of the rational numbers. Define a relation over O as follows. For every $p, q \in O$, $p \succsim q$ if and only if $p \geq q$. We show that there is no utility function representing $\succsim$ that includes only integer values. Assume to the contrary that there is such a utility function. Then $0, 1 \in O$ and $u(0) < u(1)$ two integer values. But, there is an increasing countable series of rational numbers in $(0, 1)$, $p^1 < p^2 < ....$

Hence, $u(p^1) < u^(p^2) < ...$ an increasing countable series of integers in $\left(u\left(0\right), u\left(1\right)\right)$, a contradiction.

2.4 Assume that a transitive preference relation $\succsim_i$ satisfies the axioms of continuity and monotonicity. Let $A \succsim_i B \succsim_i C$ and $A \succ_i C$. We prove that there is a unique number $\theta_i \in [0,1]$ satisfying $B \approx \left[\theta_i\left(A\right), \left(1-\theta_i\right)\left(C\right)\right]$. By the continuity axiom, there is at least one such θ_i. We show that θ_i is unique. Let $\theta_i' \in [0,1]$ such that $B \approx \left[\theta_i'\left(A\right), \left(1-\theta_i'\right)\left(C\right)\right]$. Then, by transitivity, $\left[\theta_i\left(A\right), \left(1-\theta_i\right)\left(C\right)\right] \approx \left[\theta_i'\left(A\right), \left(1-\theta_i'\right)\left(C\right)\right]$. Hence, by the monotonicity axiom, one has $\theta_i' = \theta_i$.

2.5 We show that the axioms of VNM are independent. For each axiom we present a preference relation that does not satisfy that axiom, but satisfies all the other axioms.
Monotonicity: Let $O = \{x, y\}$ be a set of outcomes. Consider a preference relation that is determined by the total probabilities over the outcomes x, y as follows: $x \succ \left[p\left(x\right), \left(1-p\right)\left(y\right)\right] \succ y$ for every $p \in (0,1)$. Furthermore, for every $p, q \in (0,1)$ one has $\left[p\left(x\right), \left(1-p\right)\left(y\right)\right] \succsim \left[q\left(x\right), \left(1-q\right)\left(y\right)\right]$ if and only if $\max\{p, 1-p\} \geq \max\{q, 1-q\}$. The preference relation satisfies all the VNM axioms, but the monotonicity. Indeed, $x \succ y$ but $\left[0.25\left(x\right), 0.75\left(y\right)\right] \succ \left[0.5\left(x\right), 0.5\left(y\right)\right]$.
Continuity: Consider a set of outcomes $O = \{x, y, z\}$. Define a lexicographic preference over the set of lotteries over O as follows.

$$\left[p_1\left(x\right), p_2\left(y\right), p_3\left(z\right)\right] \succsim \left[q_1\left(x\right), q_2\left(y\right), q_3\left(z\right)\right] \Longleftrightarrow$$

$$\{p_1 > q_1\} \ \ or \ \ \{p_1 = q_1 \ \ and \ \ p_2 \geq q_2\}.$$

Finally, assume the preference relation is determined by the total probabilities over the outcomes x, y, z. This preference relation satisfies all the VNM axioms, but the continuity. Indeed, $x \succ y \succ z$, however, for every $0 < \theta \leq 1$, one has $\left[\theta\left(x\right), \left(1-\theta\right)\left(z\right)\right] \succ y$. In addition, for $\theta = 0$, $y \succ \left[\theta\left(x\right), \left(1-\theta\right)\left(z\right)\right]$.
The axiom of simplification of compound lotteries: Consider a set of outcomes $O = \{x, y\}$. Define a preference relation over the compound lotteries over O as follows. A player is indifferent between all simple lotteries (including outcomes). He also indifferent between all compound lotteries. However, he prefers every simple lottery over a compound lottery. Clearly, this preference relation satisfies all the VNM axioms, but the axiom of simplification of compound lotteries.

Independence: Let $O = \{x, y\}$ be a set of outcomes. Consider a preference relation that satisfies the axiom of simplification of compound lotteries such that, for every $p, q \in [0,1]$ one has

$$\left[p\left(x\right), \left(1-p\right)\left(y\right)\right] \succsim \left[q\left(x\right), \left(1-q\right)\left(y\right)\right] \Leftrightarrow \max\{p, 1-p\} \geq \max\{q, 1-q\}.$$

This preference relation satisfies all the VNM axioms, but the independence. Indeed, $\left[1\left(x\right)\right] \approx \left[1\left(y\right)\right]$ but $\left[0.5\left(x\right), 0.5\left(\left[1\left(x\right)\right]\right)\right] \succ \left[0.5\left(x\right), 0.5\left(\left[1\left(y\right)\right]\right)\right]$.

2.6 Let u be a linear utility function representing a preference relation $\succsim_i$ of player i. We prove that $\succsim_i$ satisfies the VNM axioms. First, since the relation $\geq$ over $\mathbf{R}$ is complete and transitive then $\succsim_i$ is complete and transitive as well.
Continuity: Let $A \succsim_i B \succsim_i C$ then $u\left(A\right) \geq u\left(B\right) \geq u\left(C\right)$. If $u\left(A\right) = u\left(C\right)$ then for each $\theta \in [0,1]$ one has $u\left(B\right) = \theta u\left(A\right) + \left(1-\theta\right) u\left(C\right)$. Otherwise, $u\left(A\right) > u\left(C\right)$, so $u\left(B\right) = \theta u\left(A\right) + \left(1-\theta\right) u\left(C\right)$ for $\theta = \frac{u(B)-u(C)}{u(A)-u(C)}$. By the linearity of u,

$$u\left(\left[\theta\left(A\right), \left(1-\theta\right)\left(C\right)\right]\right) = \theta u\left(A\right) + \left(1-\theta\right) u\left(C\right) = u\left(B\right).$$

So, $B \approx \left[\theta\left(A\right), \left(1-\theta\right)\left(C\right)\right]$.
Monotonicity: Let $\alpha, \beta \in [0,1]$, and $A \succ B$. By the linearity of u,

$$u\left(\left[\alpha\left(A\right), \left(1-\alpha\right)\left(B\right)\right]\right) = \alpha u\left(A\right) + \left(1-\alpha\right) u\left(B\right),$$

as well as,

$$u\left(\left[\beta\left(A\right), \left(1-\beta\right)\left(B\right)\right]\right) = \beta u\left(A\right) + \left(1-\beta\right) u\left(B\right).$$

By $A \succ B$ it follows that $u\left(A\right) > u\left(B\right)$. Hence,

$$\begin{gathered}\alpha \geq \beta \iff \\ \alpha u\left(A\right) + \left(1-\alpha\right) u\left(B\right) \geq \beta u\left(A\right) + \left(1-\beta\right) \iff \\ u\left(\left[\alpha\left(A\right), \left(1-\alpha\right)\left(B\right)\right]\right) \geq u\left(\left[\beta\left(A\right), \left(1-\beta\right)\left(B\right)\right]\right) \iff \\ \left[\alpha\left(A\right), \left(1-\alpha\right)\left(B\right)\right] \succsim \left[\beta\left(A\right), \left(1-\beta\right)\left(B\right)\right].\end{gathered}$$

The axiom of simplification of compound lotteries: For $j = 1, ..., J$, let $L_j = \left[p_1^j\left(A_1\right), ..., p_k^j\left(A_k\right)\right]$ be a simple lottery. Let $L = \left[q_1\left(L_1\right), ..., q_J\left(A_J\right)\right]$

be a compound lottery. Denote, $\widehat{L} = \left(r_1 \left(A_1\right), ..., r_K \left(A_K\right)\right)$, where $r_k = \sum_{j=1}^{J} q_j p_k^j$, for every $k = 1, ..., K$. Then'

$$
\begin{aligned}
u\left(L\right) &= q_1 u\left(L_1\right) + ... + q_J u\left(L_J\right) \\
&= q_1 \left(p_1^1 u\left(A_1\right) + ... + p_K^1\left(A_K\right)\right) + ... + q_J \left(p_1^J u\left(A_1\right) + ... + p_K^J\left(A_K\right)\right) \\
&= r_1 u\left(A_1\right) + ... + r_J u\left(A_J\right) \\
&= u\left(\widehat{L}\right) \Rightarrow \\
L &\approx \widehat{L}
\end{aligned}
$$

Independence: Let $\widehat{L} = \left[q_1\left(L_1\right), ..., q_K\left(L_K\right)\right]$ be a compound lottery, and let M be a simple lottery. Assume $L_j \approx_i M$ then $u\left(L_j\right) = u\left(M\right)$. Thus,

$$
\begin{aligned}
u\left(\widehat{L}\right) &= q_1 u\left(L_1\right) + ... + q_{j-1} u\left(L_{j-1}\right) + q_j u\left(L_j\right) + q_{j+1} u\left(L_{j+1}\right) + ... + q_J u\left(L_J\right) \\
&= q_1 u\left(L_1\right) + ... + q_{j-1} u\left(L_{j-1}\right) + q_j u\left(M\right) + q_{j+1} u\left(L_{j+1}\right) + ... + q_J u\left(L_J\right) \Rightarrow \\
\widehat{L} &\approx \left[q_1\left(L_1\right), ..., q_{j-1}\left(L_{j-1}\right), q_j\left(M\right), q_{j+1}\left(L_{j+1}\right), ..., q_J\left(L_J\right)\right].
\end{aligned}
$$

2.7 (a) Let $u\left(1000\right) = 1$ and $u\left(0\right) = 0$.

$$
\begin{aligned}
\left[1\left(500\right)\right] &\approx \left[\tfrac{2}{3}\left(1000\right), \tfrac{1}{3}\left(0\right)\right] \Rightarrow u\left(500\right) = \tfrac{2}{3} u\left(1000\right) + \tfrac{1}{3} u\left(0\right) = \tfrac{2}{3}, \\
\left[1\left(100\right)\right] &\approx \left[\tfrac{3}{8}\left(500\right), \tfrac{5}{8}\left(0\right)\right] \Rightarrow u\left(100\right) = \tfrac{3}{8} u\left(500\right) + \tfrac{5}{8} u\left(0\right) = \tfrac{1}{4}.
\end{aligned}
$$

(b) L_1 will be preferred by this person, indeed

$$
\begin{aligned}
u\left(L_1\right) &= \tfrac{3}{10} u\left(1000\right) + \tfrac{1}{10} u\left(500\right) + \tfrac{1}{2} u\left(100\right) + \tfrac{1}{10} u\left(0\right) = 0.492 > \\
u\left(L_2\right) &= \tfrac{2}{10} u\left(1000\right) + \tfrac{3}{10} u\left(500\right) + \tfrac{2}{10} u\left(100\right) + \tfrac{3}{10} u\left(0\right) = 0.45
\end{aligned}
$$

(c) One can observe that

$$
u\left(100\right) < u\left(L_1\right) < u\left(500\right) \quad \textit{and} \quad u\left(100\right) < u\left(400\right) < u\left(500\right),
$$

but it is not possible to ascertain whether $u\left(L_1\right) < u\left(400\right)$, $u\left(L_1\right) = u\left(400\right)$ or $u\left(L_1\right) > u\left(400\right)$ and therefore it is not possible to ascertain which will be preferred.

(d) One can observe that $u\left(100\right) < u\left(L_1\right) < u\left(500\right) < u\left(600\right)$, so he will prefer receiving \$600 with probability 1.

2.8 The preferences between the lotteries do not change by the suggested change, since it only define a positive affine transformation over the original utility function.

2.9 A person satisfies the VNM axioms and he prefers $A \succ_i D$. Set $u(A) = 1$ and $u(D) = 0$.

$$\begin{aligned}
&C \approx \left[\tfrac{3}{5}(A), \tfrac{2}{5}(D)\right] \Rightarrow u(C) = \tfrac{3}{5}u(A) + \tfrac{2}{5}u(D) = \tfrac{3}{5} \\
&B \approx \left[\tfrac{3}{4}(A), \tfrac{1}{4}(C)\right] \Rightarrow u(B) = \tfrac{3}{4}u(A) + \tfrac{1}{4}u(C) = \tfrac{9}{10} \\
&u(L_1) = \tfrac{2}{5}u(A) + \tfrac{1}{5}u(B) + \tfrac{1}{5}u(C) + \tfrac{1}{5}u(D) = \tfrac{35}{50} < \\
&u(L_2) = \tfrac{2}{5}u(B) + \tfrac{3}{5}u(C) = \tfrac{36}{50}.
\end{aligned}$$

L_2 will be preferred by this person.

2.10 Assume the preference in 2.9 is $D \succ_i A$ rather than $A \succ_i D$, then L_1 will be preferred. Indeed, set $u(A) = -1$ and $u(D) = 0$. Therfore,

$$\begin{aligned}
&C \approx \left[\tfrac{3}{5}(A), \tfrac{2}{5}(D)\right] \Rightarrow u(C) = \tfrac{3}{5}u(A) + \tfrac{2}{5}u(D) = -\tfrac{3}{5} \\
&B \approx \left[\tfrac{3}{4}(A), \tfrac{1}{4}(C)\right] \Rightarrow u(B) = \tfrac{3}{4}u(A) + \tfrac{1}{4}u(C) = -\tfrac{9}{10} \\
&u(L_1) = \tfrac{2}{5}u(A) + \tfrac{1}{5}u(B) + \tfrac{1}{5}u(C) + \tfrac{1}{5}u(D) = -\tfrac{35}{50} > \\
&u(L_2) = \tfrac{2}{5}u(B) + \tfrac{3}{5}u(C) = -\tfrac{36}{50}.
\end{aligned}$$

2.11 Assume that u is a linear utility function. Let $L = \left[q_1(L_1), q_2(L_2), ..., q_J(L_J)\right]$ be a compound lottery where $L_j = \left[p_1^j(A_1), p_2^j(A_2), ..., p_K^j(A_K)\right]$ for every $j = 1, 2, ..., J$. We prove that $u(L) = \sum_{j=1}^{J} q_j u\left(L_j\right)$.
By the axiom of simplification of compound lotteries,

$$L \approx \left[\sum_{j=1}^{J} p_1^j q_j(A_1), \sum_{j=1}^{J} p_2^j q_j(A_2), ..., \sum_{j=1}^{J} p_K^j q_j(A_K)\right] \Rightarrow$$

$$\begin{aligned}
u(L) &= u\left(\left[\sum_{j=1}^{J} p_1^j q_j(A_1), \sum_{j=1}^{J} p_2^j q_j(A_2), ..., \sum_{j=1}^{J} p_K^j q_j(A_K)\right]\right) \\
&= \sum_{k=1}^{K}\left(\sum_{j=1}^{J} p_K^j q_j\right) u(A_K) = \sum_{j=1}^{J} q_j \sum_{k=1}^{K} p_K^j u(A_K) \\
&= \sum_{j=1}^{J} q_j u\left(L_j\right).
\end{aligned}$$

2.12 Assume, $\succsim$ satisfies the VNM axioms. Let u be a linear utility function representing $\succsim$. We prove that

$$\left[\alpha\left(L_1\right),\left(1-\alpha\right)\left(L_3\right)\right] \succ \left[\alpha\left(L_2\right),\left(1-\alpha\right)\left(L_3\right)\right]$$

if and only if

$$\left[\alpha\left(L_1\right),\left(1-\alpha\right)\left(L_4\right)\right] \succ \left[\alpha\left(L_2\right),\left(1-\alpha\right)\left(L_4\right)\right],$$

for every four lotteries L_1, L_2, L_3, L_4 and $\alpha \in [0,1]$.

$$\begin{aligned}
\left[\alpha\left(L_1\right),\left(1-\alpha\right)\left(L_3\right)\right] \succ \left[\alpha\left(L_2\right),\left(1-\alpha\right)\left(L_3\right)\right] &\overset{\text{by } 2.11}{\Longleftrightarrow} \\
\alpha u\left(L_1\right)+\left(1-\alpha\right)u\left(L_3\right) > \alpha u\left(L_2\right)+\left(1-\alpha\right)u\left(L_3\right) &\Leftrightarrow \\
\alpha u\left(L_1\right)+\left(1-\alpha\right)u\left(L_4\right) > \alpha u\left(L_2\right)+\left(1-\alpha\right)u\left(L_4\right) &\Leftrightarrow \\
\left[\alpha\left(L_1\right),\left(1-\alpha\right)\left(L_4\right)\right] \succ \left[\alpha\left(L_2\right),\left(1-\alpha\right)\left(L_4\right)\right]&.
\end{aligned}$$

2.13 Suppose a person satisfies VNM axioms. Let u be a linear utility function representing $\succsim$. Assume also that his preferences with respect to lotteries L_1, L_2, L_3, L_4 are $L_1 \succ L_2$ and $L_3 \succ L_4$. We prove that for every $\alpha \in [0,1]$ one has $\left[\alpha(L_1),(1-\alpha)(L_3)\right] \succ \left[\alpha(L_2),(1-\alpha)(L_4)\right]$. For $\alpha = 0$ or $\alpha = 1$, it obviously holds. Let $\alpha \in (0,1)$, then

$$\begin{aligned}
L_1 \succ L_2 \Rightarrow u(L_1) > u(L_2) \quad &\Rightarrow \ \alpha u(L_1) > \alpha u(L_2), \text{ in addition,} \\
L_3 \succ L_4 \Rightarrow u(L_3) > u(L_4) \quad &\Rightarrow \ (1-\alpha)u(L_3) > (1-\alpha)u(L_4).
\end{aligned}$$

Hence,

$$\alpha u\left(L_1\right)+\left(1-\alpha\right)u\left(L_3\right) > \alpha u\left(L_2\right)+\left(1-\alpha\right)u\left(L_4\right).$$

So, by 2.11,

$$\left[\alpha(L_1),(1-\alpha)(L_3)\right] \succ \left[\alpha(L_2),(1-\alpha)(L_4)\right].$$

2.14 Suppose a person satisfies VNM axioms. Let u be a linear utility function representing $\succsim$. Assume also that his preference with respect to lotteries L_1, L_2 is $L_1 \succ L_2$. We prove that for every $0 < \alpha \leq 1$ one has

$\left[\alpha(L_1), (1-\alpha)(L_2)\right] \succ L_2.$

$$\begin{aligned}
& L_1 \succ L_2 && \Rightarrow \\
& u(L_1) > u(L_2) && \Rightarrow \\
& \alpha u(L_1) > \alpha u(L_2) && \Rightarrow \\
& \alpha u(L_1) + (1-\alpha) u(L_2) > \alpha u(L_2) + (1-\alpha) u(L_2) && \Rightarrow \\
& \alpha u(L_1) + (1-\alpha) u(L_2) > u(L_2) && \Rightarrow \\
& \left[\alpha(L_1), (1-\alpha)(L_2)\right] \succ L_2.
\end{aligned}$$

2.15 Suppose the tennis player satisfies VNM axioms.

(a) Let u be a linear utility function representing $\succsim$. Set $u(win) = 1$ and $u(lose) = 0$.

(b) Denote by $L_{s.great}$ ($L_{s.medium}$) the lottery that takes place when the player strikes the ball with great force (medium force, respectively) at the second attempt. Then

$$\begin{aligned}
L_{s.great} &= \left[0.65\left(\left[0.75(win), 0.25(lose)\right]\right), 0.35(lose)\right] \\
L_{s.medium} &= \left[0.9\left(\left[0.5(win), 0.5(lose)\right]\right), 0.1(lose)\right].
\end{aligned}$$

(c)

$$\begin{aligned}
L_{f.great,s.great} &= \left[0.65\left(\left[0.75(win), 0.25(lose)\right]\right), 0.35\left(L_{s.great}\right)\right] \\
L_{f.great,s.medium} &= \left[0.65\left(\left[0.75(win), 0.25(lose)\right]\right), 0.35\left(L_{s.medium}\right)\right] \\
L_{f.medium,s.great} &= \left[0.9\left(\left[0.5(win), 0.5(lose)\right]\right), 0.1\left(L_{s.great}\right)\right] \\
L_{f.medium,s.medium} &= \left[0.9\left(\left[0.5(win), 0.5(lose)\right]\right), 0.1\left(L_{s.medium}\right)\right]
\end{aligned}$$

(d)

$$\begin{aligned}
u\left(L_{s.great}\right) &= 0.65 \cdot 0.75u\,(win) + 0.65 \cdot 0.25u\,(lose) + 0.35u\,(lose) \\
&= 0.4875 \\
u\,(L_{s.medium}) &= 0.9 \cdot 0.5u\,(win) + 0.9 \cdot 0.5u\,(lose) + 0.1u\,(lose) \\
&= 0.45. \\
u\left(L_{f.great,s.great}\right) &= 0.65 \cdot 0.75u\,(win) + 0.65 \cdot 0.25u\,(lose) + 0.35u\left(L_{s.great}\right) \\
&= 0.6581 \\
u\left(L_{f.medium,s.great}\right) &= 0.9 \cdot 0.5u\,(win) + 0.9 \cdot 0.5u\,(lose) + 0.1u\left(L_{s.grat}\right) \\
&= 0.4988. \\
u\left(L_{f.great,s.medium}\right) &= 0.65 \cdot 0.75u\,(win) + 0.65 \cdot 0.25u\,(lose) + 0.35u\,(L_{s.medium}) \\
&= 0.645 \\
u\left(L_{f.medium,s.medium}\right) &= 0.9 \cdot 0.5u\,(win) + 0.9 \cdot 0.5u\,(lose) + 0.1u\,(L_{s.medium}) \\
&= 0.495.
\end{aligned}$$

The preferred lottery is strike the ball with great force in both attempts.

2.16 (a) Let p be the probability distribution over the black and the white chocolate balls in the container. Let n the number of chocolate balls in it. Then the set of outcomes is

$$O = \{p\} \bigcup \{w | w \in \{0, 1, ..., n\}\}$$

where w is the number of white chocolate balls in the container. The preference relation of Ron satisfies: (i) For every $w > w'$ one has $w \succ w'$ since Ron's excitement climbing higher the greater the number of white chocolate balls; and (ii) $p \succ w$ for every $w \in \{0, 1, .., n\}$.

(b) Ron's preferences relation does not satisfy the VNM axioms. He violates the monotonicity axiom according to which receiving a container with only white chocolate balls should be preferred to any lottery over the number of white chocolate balls as opposed to Ron's preferences ($p \succ w$).

2.17 There exists a preference relation over compound lotteries that satisfies VNM axioms and extends the lexicographic preference relation. There are only a finite number of pairs (x, y) such that $x, y = 0, 1, \ldots, 1000$. Indeed, one can define a utility function over the set of outcomes that maintain the lexicographic preference relation of the farmer (e.g. $u(0,0) = 0, u(0,1) = 1, ..., u(0,1000) = 1000, u(1,0) = 1001, ...$). Now, one may define a preference relation over compound lotteries as follows $L \succsim L'$ if and only if $Eu(L) \geq Eu(L')$.

2.18 Let $O = \{(x,y) \,|\, 0 \leq x, y \leq 1000\}$. We next prove that there is no preference relation over compound lotteries over O that satisfies VNM axioms and extends the lexicographic preference relation over O. Assume to the contrary that there is such a preference relation.

(a) Let $(x,y) \in [0,1000]^2$. In particular, $(1000,1000) \succsim (x,y) \succsim (0,0)$. So, by the continuity, there is at least one $\theta_{(x,y)}$, such that $(x,y) \approx \left[\theta_{(x,y)}(1000,1000), \left(1-\theta_{(x,y)}\right)(0,0)\right]$. But, by the monotonicity, $\theta_{(x,y)}$ is unique.

(b) We next prove that the function $(x,y) \mapsto \theta_{(x,y)}$ is injective. Let $(x,y) \neq (x',y')$ then necessarily either $(x,y) \succ (x',y')$ or $(x,y) \prec (x',y')$. Assume w.l.o.g. that $(x,y) \succ (x',y')$ then by transitivity

$$\left[\theta_{(x,y)}(1000,1000), \left(1-\theta_{(x,y)}\right)(0,0)\right] \succ$$
$$\left[\theta_{(x',y')}(1000,1000), \left(1-\theta_{(x',y')}\right)(0,0)\right].$$

Thus, by monotonicity, $\theta_{(x,y)} > \theta_{(x',y')}$.

(c) Set $A_x = \left\{\theta_{(x,y)} \,|\, y \in [0,1000]\right\}$. Then, by (b), $\theta_{(x,0)}, \theta_{(x,1000)} \in A_x$ and $\theta_{(x,0)} < \theta_{(x,1000)}$, so A_x contains at least two elements. In addition, if $\theta \in A_x$ and $x \neq x'$, then, by (b), $\theta \notin A_{x'}$, and the sets $\{A_x : x \in [0,1000]\}$ are pairwise disjoint.

(d) Let $x_1 < x_2$, $\theta_1 \in A_{x_1}$ and $\theta_2 \in A_{x_2}$. Then, there are y_1 and y_2 such that $\theta_{(x_1,y_1)} = \theta_1$ and $\theta_{(x_2,y_2)} = \theta_2$. However, since $x_1 < x_2$, one has $(x_1,y_1) \prec (x_2,y_2)$ and thus, by (b), $\theta_1 = \theta_{(x_1,y_1)} < \theta_{(x_2,y_2)} = \theta_2$.

(e) From (c) it follows that, for every $x \in [0,1000]$, there are at least two elements in A_x, say $\theta_x < \theta_{x'}$. By (d), $\{[\theta_x, \theta_{x'}] : x \in [0,1000]\}$ is an infinite set of closed disjoint segments contained in $[0,1000]$. But there can be only countable disjoint segments, a contradiction.

(f) Therefore, there is no utility function over $[0,1000]^2$ that represents the lexicographic preference relation.

(g) The monotonicity and continuity axioms are not satisfied by the preference relation.

2.19 Assume that v is a positive affine transformation of u. Then there are $a > 0$ and $b \in \mathbf{R}$ such that

$$v(\cdot) = a \cdot u(\cdot) + b.$$

In particular,

$$u(\cdot) = \frac{1}{a} v(\cdot) - \frac{b}{a}.$$

Thus, u is a positive affine transformation of v.

2.20 Assume that v is a positive affine transformation of u, and that w is a positive affine transformation of v. Then there are $a, b > 0$ and $c, d \in \mathbf{R}$ such that

$$v(\cdot) = a \cdot u(\cdot) + c$$

and

$$w(\cdot) = b \cdot v(\cdot) + d.$$

In particular,

$$w(\cdot) = ba \cdot u(\cdot) + (cb + d).$$

Thus, w is a positive affine transformation of u.

2.21 Suppose a person's preferences, which satisfies VNM axioms, are representable by two linear utility functions u and v. We prove that v is a positive affine transformation of u. If this person is indifference between each two outcomes, so both utility functions are necessarily constant, and we are done. Otherwise, let A and B be two outcomes such that $A \prec B$. Hence, $u(A) < u(B)$ and $v(A) < v(B)$. Set $a = \frac{v(B) - v(A)}{u(B) - u(A)} > 0$ and $b = v(A) - a \cdot u(A)$. We next prove that $v(\cdot) = a \cdot u(\cdot) + b$.

Case 1:

$$a \cdot u(A) + b = \frac{v(B) - v(A)}{u(B) - u(A)} u(A) + v(A) - \frac{v(B) - v(A)}{u(B) - u(A)} u(A) = v(A).$$

Case 2:

$$a \cdot u(B) + b = \frac{v(B) - v(A)}{u(B) - u(A)} u(B) + v(A) - \frac{v(B) - v(A)}{u(B) - u(A)} u(A) = v(B)$$

Case 3: $A \preceq C \preceq B$. By the continuity axiom, there is $\theta \in [0, 1]$ such that $C \approx \big[\theta(B), (1-\theta)(A)\big]$ Therefore

$$\begin{aligned} v(C) &= \theta \cdot v(B) + (1-\theta) v(A) \\ &= \theta(a \cdot u(B) + b) + (1-\theta)(a \cdot u(A) + b) \\ &= a(\theta \cdot u(B) + (1-\theta) u(A)) + b \\ &= a \cdot u(C) + b \end{aligned}$$

(the proof in case that $C \preceq A \prec B$ and $A \prec B \preceq C$ is similar to Case 3).

2.22 Let O be an infinite set of outcomes. Suppose that a player has a complete reflexive and transitive preference relation $\succsim$ over the set of all compound lotteries over a finite number of simple lotteries over a finite set of outcomes in O. Assume that the preference relation $\succsim$ satisfies VNM axioms, and also satisfies the property that O contains a most-preferred outcome A_K, and least preferred outcome A_1.

(a) We prove that there exists a linear utility function that represents the player preference's relation. First, set $u(A_K) = 1$ and $u(A_1) = 0$. For every $B \in O$ an outcome, one has $A_1 \preceq B \preceq A_K$. By the continuity axiom, there is $\theta_B \in [0,1]$ such that $B \approx \big[\theta_B(A_K), (1-\theta_B)(A_1)\big]$. By 2.4, θ_B is unique. Set $u(B) = \theta_B$.
For every simple lottery over a finite set of outcomes in O. Define,

$$u(\big[p_1(B_1), p_2(B_2), ..., p_n(B_n)\big]) = \sum_{i=1}^{n} p_i u(B_i) = \sum_{i=1}^{n} p_i \theta_{B_i}.$$

u is a linearly utility function, we show that u represents the preference relation. Let

$$L = \big[p_1(B_1), p_2(B_2), ..., p_n(B_n)\big]$$

and

$$L' = \big[q_1(B_1), q_2(B_2), ..., q_n(B_n)\big]$$

be two lotteries (note that one can assume without loss of generality that both lotteries are over the same set of outcome). Then, $L \succsim L'$ if and only if, by the independence axiom,

$$\Big[p_1(\big[\theta_{B_1}(A_K), (1-\theta_{B_1})(A_1)\big]), ..., p_n(\big[\theta_{B_n}(A_K), (1-\theta_{B_n})(A_1)\big])\Big] \succsim \\ \Big[q_1(\big[\theta_{B_1}(A_K), (1-\theta_{B_1})(A_1)\big]), ..., q_n(\big[\theta_{B_n}(A_K), (1-\theta_{B_n})(A_1)\big])\Big]$$

if and only if, by The axiom of simplification of compound lotteries,

$$\left[\left(\sum_{i=1}^{n} p_i\theta_{B_i}\right)(A_K), \left(1-\sum_{i=1}^{n} p_i\theta_{B_i}\right)(A_1)\right] \succsim \\ \left[\left(\sum_{i=1}^{n} q_i\theta_{B_i}\right)(A_K), \left(1-\sum_{i=1}^{n} q_i\theta_{B_i}\right)(A_1)\right]$$

if and only if, by the monotonicty axiom,

$$\sum_{i=1}^{n} p_i \theta_{B_i} \geq \sum_{i=1}^{n} q_i \theta_{B_i}$$

, if and only if $u(L) \geq u(L')$, as needed.

(b) To prove that if u and v are two linear utility functions that represents the player preference's relation, then v is a positive affine transformation of u, see Exercise 2.21.

(c) We prove that there exists a unique linear utility function (up to a positive affine transformation) representing the player preference's relation. By (a) it follows that there is a linear utility function u representing the player preference's relation. By (b), any other linear utility function representing the player preference's relation must be a positive affine transformation of u. Thus u is unique up to a positive affine transformation .

2.23 We prove that a player is a risk averse if and only if for each $p \in [0,1]$ and every pair of outcomes $x, y \in \mathbf{R}$, $u_i(\left[p(x),(1-p)(y)\right]) \leq u_i(\left[1(px+(1-p)y)\right])$. If a player is a risk averse then the inequality for each $p \in [0,1]$ and every pair of outcomes $x, y \in \mathbf{R}$ follows directly by the definition of risk averse. We prove the opposite direction. Assume that for each $p \in [0,1]$ and every pair of outcomes $x, y \in \mathbf{R}$, $u_i(\left[p(x),(1-p)(y)\right]) \leq u_i(\left[1(px+(1-p)y)\right])$. We prove that the player is a risk averse, that is for every lottery $L = \left[p_1(x_1), p_2(x_2), ..., p_n(x_n)\right]$, one has

$$u_i\left(\left[p_1(x_1), p_2(x_2), ..., p_n(x_n)\right]\right) \leq u_i\left(\left[1(\sum_{k=1}^{n} p_k x_k)\right]\right).$$

We prove that the last inequality holds for every lottery by induction over n. For $n = 2$, it holds by the above assumption. Assume it holds for every $n < N$. We prove for $n = N$. Let

$$L = \left[p_1(x_1), p_2(x_2), ..., p_N(x_N)\right]$$

be a lottery. Then,

$$u_i\left(\left[p_1\left(x_1\right), p_2\left(x_2\right), \ldots, p_N\left(x_N\right)\right]\right)$$

$$= u_i\left(\left(\sum_{i=1}^{N-1} p_i\right)\left(\left[\frac{p_1}{\sum_{i=1}^{N-1} p_i}\left(x_1\right), \ldots, \frac{p_{N-1}}{\sum_{i=1}^{N-1} p_i}\left(x_{N-1}\right)\right]\right), p_N\left(x_N\right)\right) \tag{1}$$

$$= \left(\sum_{i=1}^{N-1} p_i\right) u_i\left(\left[\frac{p_1}{\sum_{i=1}^{N-1} p_i}\left(x_1\right), \ldots, \frac{p_{N-1}}{\sum_{i=1}^{N-1} p_i}\left(x_{N-1}\right)\right]\right) + p_N u_i\left(x_N\right) \tag{2}$$

$$\leq \left(\sum_{i=1}^{N-1} p_i\right) u_i\left(\left[1\left(\frac{p_1}{\sum_{i=1}^{N-1} p_i}x_1 + \cdots + \frac{p_(N-1)}{\sum_{i=1}^{N-1} p_i}x_{N-1}\right)\right]\right) + p_N u_i\left(x_N\right) \tag{3}$$

$$\leq u_i\left(\left[1\left(\left(\sum_{i=1}^{N-1} p_i\right)\left(\frac{p_1}{\sum_{i=1}^{N-1} p_i}x_1 + \cdots + \frac{p_{N-1}}{\sum_{i=1}^{N-1} p_i}x_{N-1}\right) + p_N x_N\right)\right]\right) \tag{4}$$

$$= u_i\left(\left[1\left(\sum_{k=1}^{n} p_k x_k\right)\right]\right),$$

where (1) follows by the axiom of simplification of compound lotteries, (2) follows by the linearity of the utility function, and (3) and (4) by the induction hypothesis. The proof for risk neutral/risk seeking is similar; however the inequalities in (3) and (4) should be replaced by equalities/reverse inequalities.

2.24 (a) The function $u(x) = 2x + 5$ is an increasing linear function therefore it represents a risk neutral player.

(b) The function $u(x) = -7x + 5$ is decreasing in x which contradicts the assumption that a player prefers receiving more. In particular the function cannot represent a preference relation.

(c) The function $u(x) = 7x - 5$ is an increasing linear function therefore it represents a risk neutral player.

(d) The function $u(x) = x^2$ is decreasing in $x < 0$ which contradicts the assumption that a player prefers receiving more. In particular the function cannot represent a preference relation.

(e) The function $u(x) = x^3$ is concave for $x < 0$ and convex for $x > 0$ (since $u''(x) = 6x$) hence a player is not risk neutral/risk averse/risk seeking.

(f) The function $u(x) = e^x$ is convex since $u''(x) = e^x > 0$, so it represents a risk seeking player.

(g) The function $u(x) = \ln x$ is defined only for $x > 0$. In the definition range, the function is concave since $u''(x) = -\frac{1}{x^2} < 0$, so it represents a risk averse player.

(h) The function $u(x) = x$ for $x \geq 0$ and $u(x) = 6x$ for $x < 0$ represents a risk averse player. Indeed, consider a lottery over two outcome x, y with probabilities $p, 1-p$, respectively. If either $x, y \geq 0$ or $x, y \leq 0$ then the player is indifference between receiving the lottery or its expected value (since the utility function over these ranges are linear). On the other hand, if $x > 0$ and $y < 0$ then (since $x < 6x$ and $6y < y$)

$$\begin{aligned}
u\left(\left[p(x), (1-p)(y)\right]\right) &= px + (1-p)6y \\
&< \min\left\{p6x + (1-p)6y, px + (1-p)y\right\} \\
&= \min\left\{6\left[px + (1-p)y\right], \left[px + (1-p)y\right]\right\} \\
&\leq u\left(px + (1-p)y\right).
\end{aligned}$$

(i) The function $u(x) = 6x$ for $x \geq 0$ and $u(x) = x$ for $x < 0$ represents a risk seeking player. Indeed, consider a lottery over two outcome x, y with probabilities $p, 1-p$, respectively. If either $x, y \geq 0$ or $x, y \leq 0$ then the player is indifference between receiving the lottery or its expected value (since the utility function over these ranges are linear). On the other hand, if $x > 0$ and $y < 0$ then

$$\begin{aligned}
u\left(\left[p(x), (1-p)(y)\right]\right) &= px + (1-p)6y \\
&> \max\left\{p6x + (1-p)6y, px + (1-p)y\right\} \\
&= \max\left\{6\left[px + (1-p)y\right], \left[px + (1-p)y\right]\right\} \\
&\geq u\left(px + (1-p)y\right).
\end{aligned}$$

(j) The function $u(x) = x^{3/2}$ for $x \geq 0$ and x for $x < 0$ represents a risk seeking player. Indeed, for $x > 0$ one has $u''(x) = 3/4x^{-1/2} > 0$, and for every $x < 0$ one has $u''(x) = 0$.

(k) The function $u(x) = x\ln(2+x)$ for $x \geq 0$ and $u(x) = x$ for $x < 0$ represents a player that is represents a risk averse player.

Indeed, for $x > 0$ one has $u''(x) < 0$, and for every $x < 0$ one has $u''(x) = 0$.

2.25 Let $U : \mathbf{R} \to \mathbf{R}$ be a concave function, let X be a random variable with a finite expected value, and let Y be a random variable that is independent of X and has an expected value 0. Define $Z = X + Y$. We prove that $E\left[U(X)\right] \geq E\left[U(Z)\right]$. Let $x \in \mathbf{R}$. Then

$$\begin{aligned}
&E\left[U(Z)|X = x\right] \\
&= E\left[U(X+Y)|X = x\right] \\
&= E\left[U(x+Y)|X = x\right] \\
&= E\left[U(x+Y)\right] && (5)\\
&\leq U\left(E[x+Y]\right) && (6)\\
&= U\left(x + E[Y]\right) \\
&= U\left(x\right) && (7)
\end{aligned}$$

where (5) holds since X and Y are independent, (6) holds since U is concave, and (7) holds by the assumption that $E\left[Y\right] = 0$. Hence, by the law of total expectation, one has

$$\begin{aligned}
&E\left[U\left(Z\right)\right] \\
&= E\Big[E\left[U\left(Z\right)|X\right]\Big] \\
&\leq E\left[U\left(X\right)\right].
\end{aligned}$$

2.26 Let $U : \mathbf{R} \to \mathbf{R}$ be a concave function, let X be a random variable with a normal distribution, expected value μ and a standard deviation σ. Let $\lambda > 1$ and let Y be a random variable with a normal distribution, expected value μ and a standard deviation $\lambda\sigma$.

(a) For every $c > 0$ one has $\mu + c < \mu + \lambda c$ and $\mu - c > \mu - \lambda c$. Therefore, since U is concave, one has

$$U\left(\mu + c\right) + U\left(\mu - c\right) \geq U\left(\mu + \lambda c\right) + U\left(\mu - \lambda c\right).$$

(b)

$$\begin{aligned}
&\int_{-\infty}^{\infty} u(x)\frac{1}{\sqrt{2\pi}\sigma}e^{\frac{-(x-\mu)^2}{2\sigma^2}}dx \\
&= \int_{-\infty}^{\mu} u(x)\frac{1}{\sqrt{2\pi}\sigma}e^{\frac{-(x-\mu)^2}{2\sigma^2}}dx + \int_{\mu}^{\infty} u(x)\frac{1}{\sqrt{2\pi}\sigma}e^{\frac{-(x-\mu)^2}{2\sigma^2}}dx \\
&= \int_{0}^{\infty} u(\mu+t)\frac{1}{\sqrt{2\pi}\sigma}e^{\frac{-t^2}{2\sigma^2}}dt + \int_{0}^{\infty} u(\mu-t)\frac{1}{\sqrt{2\pi}\sigma}e^{\frac{-t^2}{2\sigma^2}}dt \\
&= \int_{0}^{\infty} \left\{u(\mu+t)+u(\mu-t)\right\}\frac{1}{\sqrt{2\pi}\sigma}e^{\frac{-t^2}{2\sigma^2}}dt \\
&= \int_{0}^{\infty} \left\{u\left(\mu+\frac{y}{\lambda}\right)+u\left(\mu-\frac{y}{\lambda}\right)\right\}\frac{1}{\sqrt{2\pi}\lambda\sigma}e^{\frac{-y^2}{2(\lambda\sigma)^2}}dy \\
&\geq \int_{0}^{\infty} \left\{u\left(\mu+\lambda\frac{y}{\lambda}\right)+u\left(\mu-\lambda\frac{y}{\lambda}\right)\right\}\frac{1}{\sqrt{2\pi}\lambda\sigma}e^{\frac{-y^2}{2(\lambda\sigma)^2}}dy \\
&\geq \int_{0}^{\infty} \left\{u(\mu+y)+u(\mu-y)\right\}\frac{1}{\sqrt{2\pi}\lambda\sigma}e^{\frac{-y^2}{2(\lambda\sigma)^2}}dy \\
&= \int_{-\infty}^{\infty} u(y)\frac{1}{\sqrt{2\pi}\lambda\sigma}e^{\frac{-(y-\mu)^2}{2(\lambda\sigma)^2}}dy
\end{aligned}$$

(c)

$$E\left[U(X)\right] = \int_{-\infty}^{\infty} u(x)\frac{1}{\sqrt{2\pi}\sigma}e^{\frac{-(x-\mu)^2}{2\sigma^2}}dx$$

$$\geq \int_{-\infty}^{\infty} u(y)\frac{1}{\sqrt{2\pi}\lambda\sigma}e^{\frac{-(y-\mu)^2}{2(\lambda\sigma)^2}}dy = E\left[U(Y)\right]$$

2.27 Let $U(x) = 1 - e^{-x}$.

(a) The player is risk averse. Indeed, $U''(x) = -e^{-x} < 0$.

(b) For every $a \in (0,1)$ and $p \in (0,1)$, denote by $X_{a,p}$ a random variable such that $P(X_{a,p} = 1 - a) = \frac{1-p}{2}$, $P(X_{a,p} = 1) = p$, and $P(X_{a,p} = 1 + a) = \frac{1-p}{2}$. Then $E\left[X_{a,p}\right] = 1$, and $Var(X_{a,p}) = (1-p)a^2$.

(c) Let $c^2 = (1-p)a^2$. Then,

$$\begin{aligned}
E\left[U(X_{a,p})\right] &= \frac{1-p}{2}U(1-a) + pU(1) + \frac{1-p}{2}U(1+a) \\
&= \frac{1-p}{2}(1-e^{-1+a}) + p(1-e^{-1}) + \frac{1-p}{2}(1-e^{-1-a}) \\
&= 1 - \frac{e^{-1}}{2}((1-p)(e^{-a}+e^{a}+2)-2) \\
&= 1 - \frac{e^{-1}}{2}((1-p)\frac{a^2}{a^2}(e^{-a}+e^{a}+2)-2) \\
&= 1 - \frac{e^{-1}}{2}(\frac{c^2}{a^2}(e^{-a}+e^{a}+2)-2).
\end{aligned}$$

(d) Let $a_1 = \frac{1}{4}$, $a_2 = \frac{1}{2}$,$p_1 = \frac{1}{2}$, $p_2 = \frac{7}{8}$ then $E\left[X_{a_1,p_1}\right] = E\left[X_{a_2,p_2}\right] = 1$ and $Var\left[X_{a_1,p_1}\right] = Var\left[X_{a_2,p_2}\right] = 1/32$. However,

$$E\left[U(X_{a_1,p_1})\right] < E\left[U(X_{a_2,p_2})\right].$$

(e) Let $a_1 = \frac{1}{4}$, $a_2 = \frac{1}{2}$,$p_1 = \frac{5}{8}$, $p_2 = \frac{7}{8}$ then $E\left[X_{a_1,p_1}\right] = E\left[X_{a_2,p_2}\right] = 1$ and $Var\left[X_{a_1,p_1}\right] < Var\left[X_{a_2,p_2}\right]$. However,

$$E\left[U(X_{a_1,p_1})\right] < E\left[U(X_{a_2,p_2})\right].$$

2.28 Let U_i be a monotonically increasing, strictly concave and twice continuously differentiable function over **R**.

(a) Assume the player has $\$x$, and is required to participate in a lottery in which he stands to gain or lose a small amount $\$h$, with equal probabilities. Denote by Y the amount of money the player will have after the lottery is conducted. Then, the expected value of Y is $E[Y] = \frac{1}{2}(x+h) + \frac{1}{2}(x-h) = x$, and the variance of Y is $Var(Y) = h^2$.

(b) The utility of the lottery is

$$u_i(Y) = \frac{1}{2}U_i(x+h) + \frac{1}{2}U_i(x-h).$$

Thus, the utility loss due to the fact that he is required to participate in the lottery is

$$\begin{aligned}
\Delta u_h = U_i(x) - u_i(Y) &= U_i(x) - \left(\frac{1}{2}U_i(x+h) + \frac{1}{2}U_i(x-h)\right) \\
&= -\frac{(U_i(x+h)-U_i(x)) - (U_i(x)-U_i(x-h))}{2}.
\end{aligned}$$

(c)

$$\begin{aligned}\lim_{h\to 0}\frac{\Delta u_h}{h^2} &= \lim_{h\to 0} -\frac{(U_i(x+h)-U_i(x))-(U_i(x)-U_i(x-h))}{2h^2}\\ &= \lim_{h\to 0} -\frac{\frac{U_i(x+h)-U_i(x)}{h}-\frac{U_i(x)-U_i(x-h)}{h}}{2h}\\ &= -\frac{U_{i''}(x)}{2}.\end{aligned}$$

(d) Denote by $y_{x,h}$ the amount of money that satisfies $U_i(y_{x,h}) = u_i(Y)$, and by Δ_{x_h} the difference $\Delta_{x_h} = x - y_{x,h}$. Then $\Delta_{x_h} \geq 0$. Indeed, since U_i is a monotonically increasing and strictly concave function then it necessarily represents a risk averse player. Therefore, $u_i(Y) \leq U_i(E[Y])$. Hence,

$$U_i(y_{x,h}) = u_i(Y) \leq U_i(E[Y]) = U_i(x).$$

Finally, from the fact that U_i is monotonically increasing, it follows that $y_{x,h} \leq x$, so $\Delta_{x_h} = x - y_{x,h} \geq 0$.

(e)

$$\begin{aligned}\lim_{h\to 0}\frac{\Delta_{x_h}}{Var(Y)} &= \lim_{h\to 0}\frac{\Delta_{x_h}}{h^2}\cdot\frac{\Delta_{u_h}}{\Delta_{u_h}}\\ &= \lim_{h\to 0}\left(\frac{\Delta_{u_h}}{h^2}\right)/\left(\frac{\Delta_{u_h}}{\Delta_{x_h}}\right)\\ &= \lim_{h\to 0} -\frac{U_i''(x)}{2U_i'(x)}\\ &= \frac{1}{2}r_{U_i}(x).\end{aligned}$$

(f) (a) Let $U_i(x) = x^\alpha$ for $0 < \alpha < 1$, then

$$r_{U_i}(x) = -\frac{\alpha(\alpha-1)x^{\alpha-2}}{\alpha x^{\alpha-1}} = \frac{1-\alpha}{x}.$$

(b) Let $U_i(x) = 1 - e^{-\alpha x}$ for $\alpha > 0$, then $r_{U_i}(x) = -\frac{-\alpha^2 e^{-\alpha x}}{\alpha e^{-\alpha x}} = \alpha$.

(g) The function $U_i(x) = x^\alpha$ for $0 < \alpha < 1$ exhibits decreasing absolute risk aversion.
The function $U_i(x) = 1 - e^{-\alpha x}$ for $\alpha > 0$ exhibits constant absolute risk aversion.

2.29 Apparently, the preferences expressed by the Second World War pilots violated the monotonicity axiom, since $Life \succ Death$ but they preferred $[1/2(Life), 1/2(Death)] \succ [3/4(Life), 1/4(Death)]$. However, it seems more likely that they violated the axiom of simplification of compound lotteries, so they were not indifference between the lottery

$$[3/4(Life), 1/4(Death)]$$

and the lottery

$$[3/4([1(Life)]), 1/4([1(Death)])].$$

Meaning, it seems that if the pilots were faced the lottery

$$[3/4(Life), 1/4(Death)]$$

they would have preferred it over the lottery

$$[1/2(Life), 1/2(Death)],$$

but they preferred the lottery

$$[1/2(Life), 1/2(Death)]$$

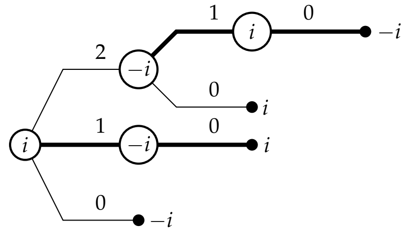
over the lottery

$$[3/4([1(Life)]), 1/4([1(Death)])].$$

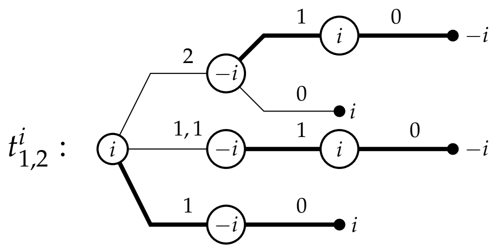
Chapter 3: Extensive-form games

3.1 Let Player I be the player that takes the first move in the game. In the game tree depicted below we represent the action taken by the current player as the piles which remain after the player removes matches. For instance, if at the first move Player I removed 2 matches from the pile contains 3 piles then the action is $1, 1, 2$. The outcome in the leaves represent the winning strategy. The bold lines represent the best reply of each player. To simplify the game tree, we first consider two subgames. Denote by t_3^i a similar game that starts with only one pile which contains 3 matches. Denote by i the opening player and by $-i$ his opponent. As one can see in the appropriate game tree, the opening player has a winning strategy.

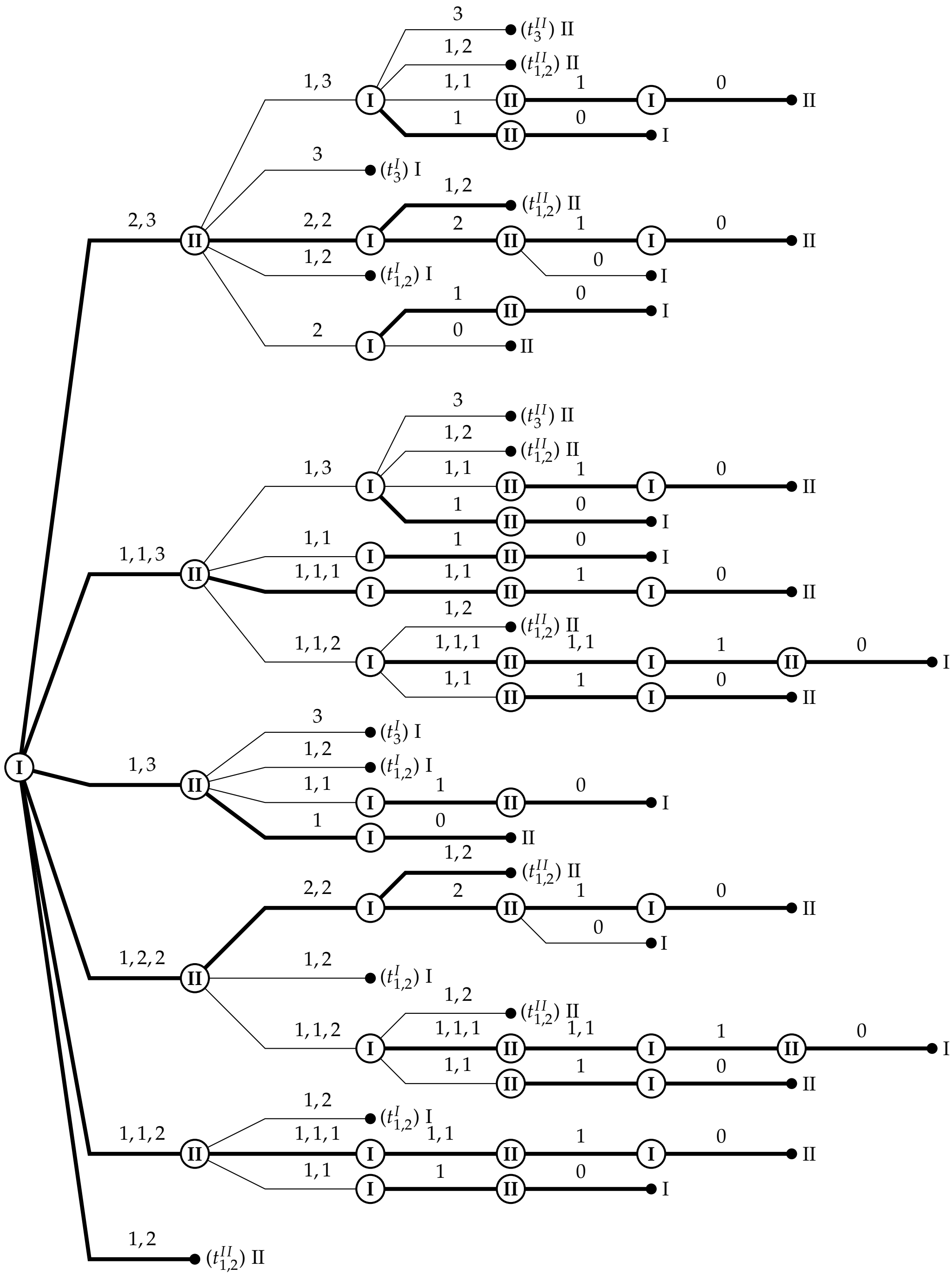
t_3^i :



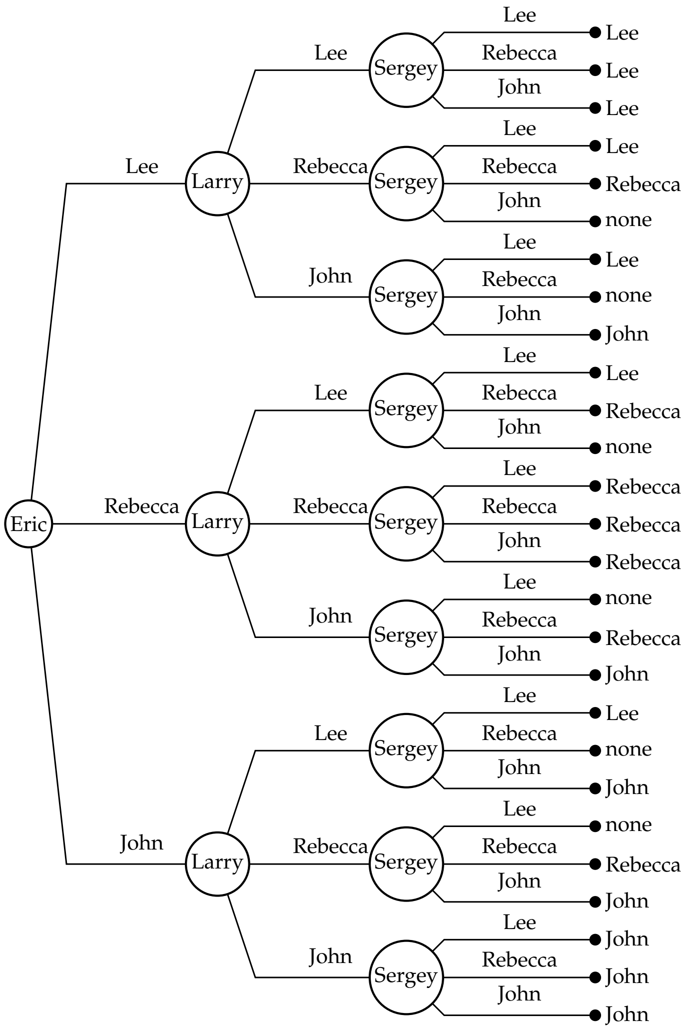
Let $t^i_{1,2}$ be a subgame that starts with 2 piles, one pile contains 2 matches and the other pile containse only one match. Denote by i the opening player and by $-i$ his opponent. In this game tree the opening player has a winning strategy.



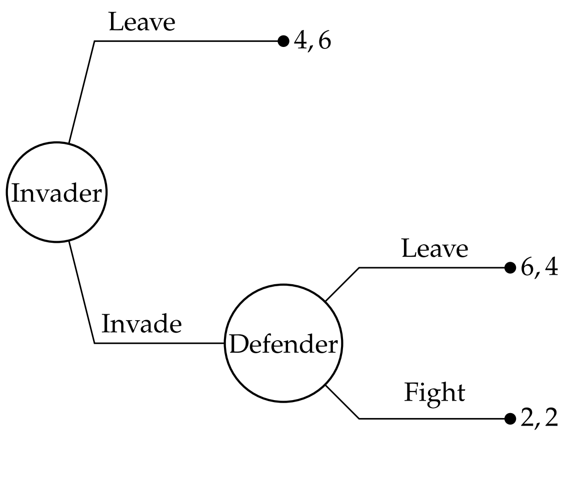
The game tree below represents the extensive form game in which Player II has a winning strategy appears as bold lines.



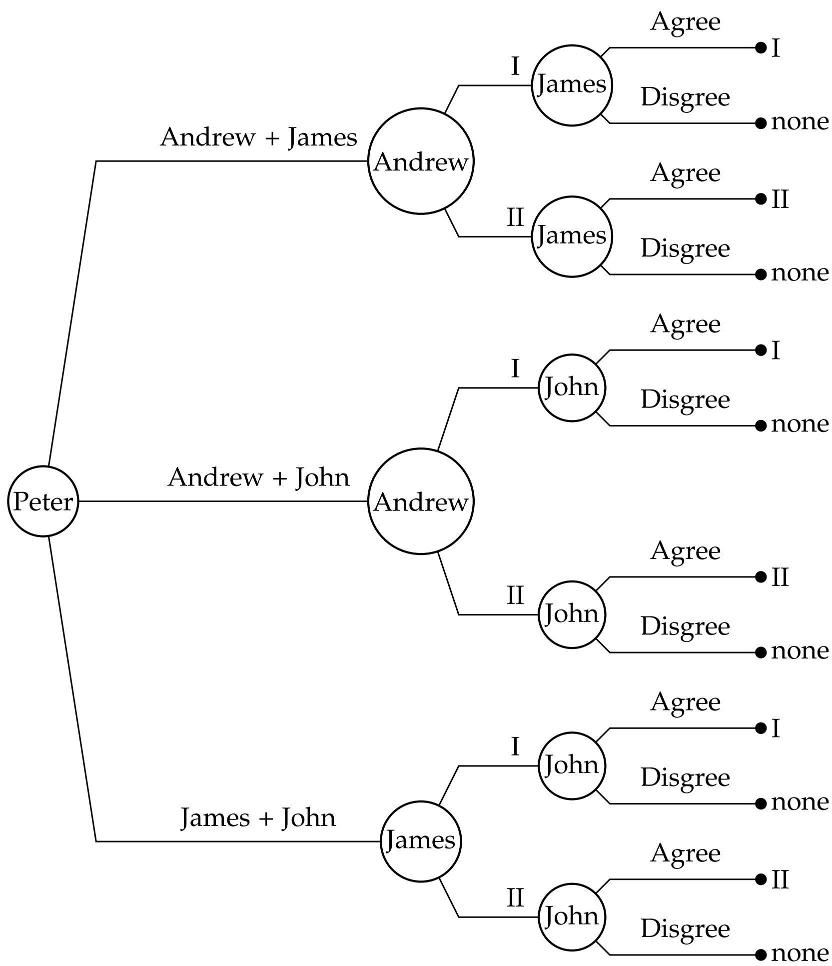
3.2 Candidate choice as a game tree:



3.3 **Does aggressiveness pay off?** We present the payoffs in the game as a pair of coordinates (i, j) where i is the expected number of descendants the invader will leave for posterity and j is the expected number of descendants the defender will leave for posterity.


Leave
4, 6
Invader
Leave
6, 4
Invade
Defender
Fight
2, 2

3·4


Peter
Andrew + James
Andrew + John
James + John
Andrew
Andrew
James
I
II
James
John
Agree
Disgree
I
II
none

3.5 (a) **Game A:** Player I has 6 strategies. Player II has 2 strategies.
Game B: Player I has 2 strategies. Player II has 6 strategies.
Game C: Player I has 12 strategies. Player II has 12 strategies. Player III has 4 strategies.

(b) **Game A:** The set of strategies of Player I is

$$\{T_1T_2, T_1M_2, T_1B_2, B_1T_2, B_1M_2, B_1B_2\}.$$

The set of strategies of Player II is $\{t, b\}$.
Game B: The set of strategies of Player I is $\{T, B\}$. The set of strategies of Player II is $\{t_1t_2, m_1t_2, b_1t_2, t_1b_2, m_1b_2, b_1b_2\}$.
Game C: The set of strategies of Player I is

$$\{T_1T_2T_3, M_1T_2T_3, B_1T_2T_3, T_1B_2T_3, M_1B_2T_3, B_1B_2T_3,$$
$$T_1T_2B_3, M_1T_2B_3, B_1T_2B_3, T_1B_2B_3, M_1B_2B_3, B_1B_2B_3\}.$$

The set of strategies of Player II is

$$\{t_1t_2t_3, m_1t_2t_3, b_1t_2t_3, t_1b_2t_3, m_1b_2t_3, b_1b_2t_3,$$
$$t_1t_2b_3, m_1t_2b_3, b_1t_2b_3, t_1b_2b_3, m_1b_2b_3, b_1b_2b_3\}.$$

The set of strategies of Player III is $\{\tau_1\tau_2, \tau_1\beta_2, \beta_1\tau_2, \beta_1\beta_2\}$.

(c) In Games A, B and C there are 5, 5 and 11 different plays respectively.

3.6 Denote by V_I the set of vertices controlled by Player I, then the number of strategies of Player I is $\prod_{x \in V_I} m_x$.

3.7 **The proof of von Neumann's Theorem:** The proof is similar to the proof of Theorem 1.5. The proof proceeds by induction on n_x, the number of vertices in the subgame $\Gamma(x)$.
Suppose x is a vertex such that $n_x = 1$, that is x is a terminal vertex. In particular, either Player I won or Player II won, or the game has ended in a draw. In either case, the empty strategy is a strategy guaranteeing either a win for Player I/a win for Player II/ a draw.
Next, suppose that x is a vertex such that $n_x > 1$. Assume by induction that at all vertices y satisfying $n_y < n_x$, one and only one of the alternatives (a), (b), or (c) is true in the subgame $\Gamma(y)$. Suppose, without loss of generality, that Player I has the first move in $\Gamma(x)$. Any vertex y that can be reached from x satisfies $n_y < n_x$, and so the inductive assumption is true in the corresponding subgame $\Gamma(y)$. Denote by $C(x)$

the collection of vertices that can be reached from x in one of Player I's moves.
If there is a vertex $y_0 \in C(x)$ such that Player I has a winning strategy in $\Gamma(y_0)$, then alternative (a) is true in $\Gamma(x)$: the winning strategy for Player I in $\Gamma(x)$ is to choose as his first move the move leading to vertex y_0, and to follow the winning strategy in $\Gamma(y_0)$ at all subsequent moves.
If Player II has a winning strategy in $\Gamma(y)$ for every vertex $y \in C(x)$, then alternative (b) is true in $\Gamma(x)$: Player II can win by ascertaining what the vertex y is after Player I's first move, and following his winning strategy in $\Gamma(y)$ at all subsequent moves.
Otherwise: there is at least one vertex $y_0 \in C(x)$ in which both players can guarantee a draw and in all other vertices in $C(x)$ Player II can guarantee at least a draw. Therefore, both players have a strategy guaranteeing at least a draw in $\Gamma(x)$: Player I can select y_0 and then follows the strategy guaranteeing him a draw in $\Gamma(y_0)$ at all subsequent moves. And Player II can guarantee at least a draw by ascertaining what the vertex y is after Player I's first move, and following his winning strategy/the strategy that guaranteeing a draw in $\Gamma(y)$ at all subsequent moves.

3.8 A player's strategy prescribes his selected action at each vertex in the game tree.Therefore we need to calculate how many vertices Player I has and how many actions he has in each one of them.

- A vertex whose depth is 0: the root. There are 9 actions.
- Vertices whose depth is 2: There are $8 \cdot 9$ vertices each one has 7 actions.
- Vertices whose depth is 4: There are $8 \cdot 9 \cdot 7 \cdot 6$ vertices each one has 5 actions.
- Vertices whose depth is 6: There are $8 \cdot 9 \cdot 7 \cdot 6 \cdot 5 \cdot 4$ vertices each one has 3 actions.
- Vertices whose depth is 8: There are $8 \cdot 9 \cdot 7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2$ vertices each one has 1 action.

The number of strategies of Player I is

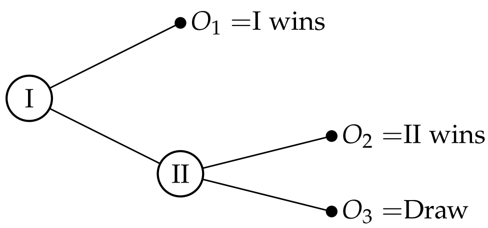
$$9^1 \cdot 7^{8\cdot 9} \cdot 5^{8\cdot 9\cdot 7\cdot 6} \cdot 3^{8\cdot 9\cdot 7\cdot 6\cdot 5\cdot 4}.$$

3.9 (a) The reduced strategies of Player I are T_1T_2, T_1M_2, T_1B_2 and B_1. The reduced strategies of Player II are $t_1t_3, t_1b_3, b_1t_2t_3, b_1t_2b_3, b_1b_2t_3$ and $b_1b_2b_3$. The reduced strategies of Player III are $\tau_1\tau_2, \tau_1\beta_2, \beta_1\tau_2$ and $\beta_1\beta_2$.

(b) The outcome of the game will obtain if the three players make use of the given reduced strategies is $2, 8, 3$.

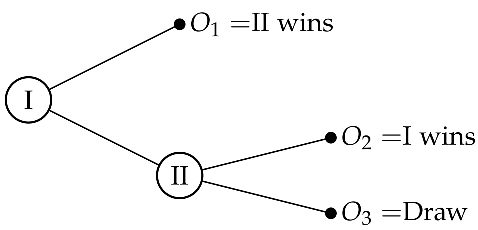
(c) Player I can increase his payoff from 2 to 3 by playing either T_1T_2 or T_1M_2 or T_1B_2. Player II cannot increase his payoff by unilaterally making use of a different strategy since 8 is th maximal payoff he can gain given Player I chooses B_1. Player III cannot increase his payoff by unilaterally making use of a different strategy since the play does not pass through vertices controlled by him.

3.10 (a) In the following game tree Player I can guarantee victory for himself by selecting O_1.

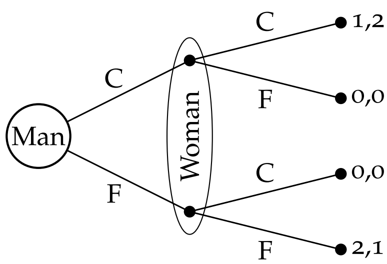


(b) There is no choice of O_1, O_2, and O_3 such that Player II can guarantee victory for himself. Indeed, Player I can guarantee at least a draw by choosing O_1 in case $O_1 \neq$II wins, otherwise by selecting an action yields to the vertex controlled by Player II.

(c) In the following game tree both players can guarantee for themselves at least a draw. Player I can guarantee at least a draw for himself by selecting an action yields to the vertex controlled by Player II, and Player II can guarantee it for himself by selecting O_3.



3.11 **The Battle of the Sexes** The payoffs appear as pairs x, y, where x is the payoff to the man and y is the payoff to the woman.



3.12 **The Centipede game** Both players is better off staying in the game as long as possible since the payoff increases along the game. However, each player prefers to stop the game just before his opponent does it to avoid losing $\$1,000$. Thus, by backward induction Player I, is better off stopping the game in his first move.

3.13 The following strategy is a winning strategy for Player I: in the opening move place a quarter in the center of the round table. From that point on, play symmetrically to Player II's actions to the center of the table. That is, if we represent the table as a unit circle centered at the origin, and Player II selects a point (i, j) to place on a center of a quarter, then Player I chooses the point $(-i, -j)$ as a response. Note that after the first move of Player I, the uncovered area is symmetric. Hence, if Player II can place an additional quarter on the table at his turn, then the symmetric place is also uncovered so can be used by Player I.

3.14 (a) The game includes finite number of matches from which at least one must be removed at each move, and thus the game is finite. Furthermore,the game has perfect information. In conclusion, the game satisfies the conditions of von Neumann's Theorem and therefore one of the three possibilities of the theorem must hold. Since the game cannot end in a draw, one of the players must have a winning strategy.

(b) Assume that the game is in a position that is not a winning position, i.e., there is a column in which the number of 1s in odd. Then it is possible to get to a winning position in one move as follows: choose the leftmost column with odd number of 1s. Select a pile in which there is 1 in the chosen column. Write an updated number of matches in the chosen pile, expressed in base 2: in every column that contains even number of 1 (except for in the chosen pile) write 0, otherwise write 1. The updated digits to the left of the chosen column remain unchanged, the 1 in the chosen column replaced by 0 thus the updated number is smaller than the original one. Remove matches from the chosen pile so the number of matches in the pile expressed in base 2 equals the updated number. By the choice of the updated number the new position is a winning position.

(c) Assume that the game at a winning position. In every legal action the number of matches in one pile decreases. Therefore, in the base 2 representation there must be 1 that is changed to 0 and thus in the appropriate column the number of 1s became an odd number which is a non-winning position.

(d) At the end of the game, each pile contains no match which can be expressed by zero. This is a winning position since 0 is even.

(e) If the game starts at a non-winning position then the opening player has a winning strategy (follow the move presented in Part (b)), otherwise the other player has a winning strategy. By following the winning strategy, each turn of Player II starts at a wining position, so by Part (c), every legal action he takes leads to a non-winning position-which is necessarily not the end of the game (by Part (d)).

3.15 (a) The game satisfies the conditions of von Neumann's Theorem and therefore one of the three possibilities of the theorem must hold. Since the game cannot end in a draw, one of the players must have a winning strategy.

(b) If the game starts at a non-winning position and there is at least one pile with more than one match then the opening player has a winning strategy, otherwise the other player has a winning strategy.

As long as there are more than one pile with more than one match, the wining strategy is identical to that presented in Exercise 3.14 (Part (e)). As soon as the game reaches a position in which there is only one pile with more than one match (which must be in Player I's turn since it is a non-winning position), Player I should remove matches from the pile contains more than one match and live only odd number of piles-each one contains only one match.

3.16 (a) In a game of Chomp played on a $2 \times \infty$ board, Player II has a wining strategy: As long as Player I captures squares of the form $(i, 2)$ for $i \geq 1$, Player II should capture square $(i+1, 1)$. If in some turn Player I captures square $(i, 1)$ then a game played on a $2 \times i$ board is obtained. Therefore, according to Theorem 3.16, the opening player in the game, namely, Player II, has a winning strategy. Otherwise, after finitely many turns, Player I captures square $(1,2)$ so Player II responses with $(2,1)$ and wins the game.

(b) Consider a game of Chomp played on an $m \times \infty$ board, where m is any finite integer. Then, if $m = 1$ Player I has a winning strategy according to which he captures square $(2,1)$ leaving only one square in a board and wins. If $m = 2$ then as proved in Part (a) Player II has a winning strategy. For $m > 2$ Player I has a winning strategy. In the open move capture square $(3,1)$ so a game on a $2 \times \infty$ board is obtained. In this game Player I is a second player whose has a winning strategy (by Part (a)).

(c) In a game of Chomp played on an $\infty \times \infty$ board Player I has two winning strategies. The first one is to capture square $(2,2)$ thus leaving only the squares in row $j = 1$ and column $i = 1$. From that point on, play symmetrically to Player II's actions. The second strategy is to capture square $(3,1)$ so a game on a $2 \times \infty$ board is obtained. From that point on, follow the winning strategy of the second player in the obtained game.

3.17 The conclusion of von Neumann's Theorem does not hold for the Matching Pennies game where we interpret the payoff $(1,-1)$ as victory for Player I and the payoff $(-1,1)$ as victory for Player II since it is a game with imperfect information. Indeed, if Player I plays H then he looses against t and if he plays T he looses against h, thus Player I has no winning strategy. Similarly, Player II has no winning strategy. Finally, since cannot end in a draw, no player has a strategy guaranteeing a draw.

3.18 **The proof of von Neumann's Theorem in games in extensive form with perfect information and without chance moves, in which the game tree has a countable number of vertices, but the depth of every vertex is bounded:** The proof proceeds by induction on n_x, the depth of the subgame $\Gamma(x)$.
Suppose x is a vertex such that $n_x = 0$, that is x is a terminal vertex. In particular, either Player I won or Player II won, or the game has ended in a draw. In either case, the $\emptyset$ is a strategy guaranteeing either a win for Player I/a win for Player II/ a draw.
Next, suppose that x is a vertex such that $n_x > 0$. Assume by induction that at all vertices y satisfying $n_y < n_x$, one and only one of the alternatives (a), (b), or (c) is true in the subgame $\Gamma(y)$. Suppose, without loss of generality, that Player I has the first move in $\Gamma(x)$. Any vertex

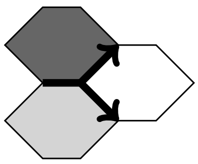
y that can be reached from x satisfies $n_y < n_x$, and so the inductive assumption is true in the corresponding subgame $\Gamma(y)$. Denote by $C(x)$ the collection of vertices that can be reached from x in one of Player I's moves.

If there is a vertex $y_0 \in C(x)$ such that Player I has a winning strategy in $\Gamma(y_0)$, then alternative (a) is true in $\Gamma(x)$: the winning strategy for Player I in $\Gamma(x)$ is to choose as his first move the move leading to vertex y_0, and to follow the winning strategy in $\Gamma(y_0)$ at all subsequent moves.

If Player II has a winning strategy in $\Gamma(y)$ for every vertex $y \in C(x)$, then alternative (b) is true in $\Gamma(x)$: Player II can win by ascertaining what the vertex y is after Player I's first move, and following his winning strategy in $\Gamma(y)$ at all subsequent moves.

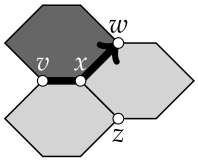
Otherwise: there is at least one vertex $y_0 \in C(x)$ in which both players can guarantee a draw and in all other vertices in $C(x)$ Player II can guarantee at least a draw. Therefore, both players have a strategy guaranteeing at least a draw in $\Gamma(x)$: Player I can select y_0 and then follows the strategy guaranteeing him a draw in $\Gamma(y_0)$ at all subsequent moves. And Player II can guarantee at least a draw by ascertaining what the vertex y is after Player I's first move, and following his winning strategy/the strategy that guaranteeing a draw in $\Gamma(y)$ at all subsequent moves.

3.19 (a) Assume to the contrary that there is a line that can be continued in two different manners as depicted below.



Then by the upper arrow the right hexagon should be light whereas by the bottom arrow should be dark, a contradiction.

(b) Assume by contradiction that the broken line may return to a vertex through which it previously passed and denote by x the first vertex that satisfies it. Below we depicted vertex x and the first time a broken line which passed through x.



The broken line cannot return to x through either v or w since it will contradict the fact that x is the first vertex that was visited twice. On the other hand, it cannot visit x trough z since the leg of the broken line must separate a game piece of one color from a game piece of the other color. That is x cannot be visited again, a contradiction.

(c) By Part (a), the broken line cannot end in an internal vertex. By Part (b), no vertex is visited twice. Thus, since there are only finitely many vertices, the broken line must end either in one of the sides of the rhombus or in some corner (that necessarily differs from W, by Part (b)). However, the hexagons in each side of the rhombus are from the same color, thus it cannot ends in any side which only leaves the corners.

(d) Note that above the broken line there are connected dark hexagons and below it there are connected light hexagons. Therefore, if the broken line ends at corner S, then the broken line is connected to both sides controlled by Dark, WN and SE, so there is a continuous path of dark hexagons that connects both sides of Dark and he wins. Similarly, if the broken line ends at corner N, then the broken line is connected to both sides controlled by Light, NE and SW, so there is a continuous path of light hexagons that connects both sides of Light and he wins.

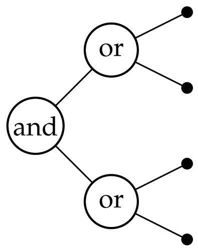
(e) Along the upper side of the broken line there are only dark hexagons and on the bottom side there are only light hexagons (as in vertex W) whereas in E the colors are inverted, hence E cannot be the end of the broken line.

(f) By Part (c) the game ends with a broken line that connect W to one of the other corner of the board. By Parts (d) and (e), the broken line necessarily ends either at S which case Dark wins or at N which case Light wins. In particular, a draw is impossible.

(g) Assume without loss of generality that Dark won the game. In particular, there is a continuous path of dark hexagons that connects the two sides of Dark and divides the board into two disconnected parts, one of them contains NE and the other contains WS. Hence there cannot be a a continuous light path that connect both sides of Light.

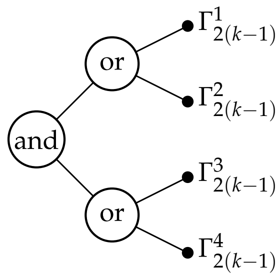
(h) Based on von Neumann's Theorem and previous claims, one (and only one) of the players has a winning strategy. Call the player with the opening move Player I, and the other player, Player II. Suppose by a contradiction that Player II has a winning strategy. We will prove then that Player I also has a winning strategy, contradicting von Neumann's Theorem. The winning strategy for Player I is as follows: in the opening move, place a game piece on any hexagon on the board. Call that game piece the "special piece." In subsequent moves, play as if (i) you are Player II (and use his winning strategy), (ii) the special piece has not been placed, and (iii) your opponent is Player I. If the strategy requires placing a game piece where the special game piece has already been placed, put a piece on any empty hexagon, and from there on, call that game piece the “special piece.” Since Player I plays the winning strategy of Player II thus he also has a winning strategy, a contradiction.

3.20 (a) Consider a game played on a tree of depth two.



Player I has a winning strategy in the game: choose a leaf of the top “or” sub-tree. Assign it the value 1. Then choose a leaf of the bottom “or” sub-tree that has not previously been selected. Assign it the value 1. Following this strategy Player I guarantees that the value of each “or” vertex is 1 and thus the value of the root is 1.

(b) Denote by Γ_{2k} a game played on a tree of depth $2k$, where k is any positive integer. We prove by induction over k that Player I has a winning strategy in Γ_{2k}. For $k = 1$, it holds by Part (a). Assume that Player I has a winning strategy in every game $\Gamma_{2k'}$ for every integer $k' < k$. Then the following strategy is a winning strategy of Player I in a game played on a tree of depth $2k$:



(1) Start by following a winning strategy in $\Gamma^1_{2(k-1)}$; (2) As long as Player II chooses leaves in either $\Gamma^1_{2(k-1)}$ or $\Gamma^2_{2(k-1)}$ continue following the winning strategy in $\Gamma^1_{2(k-1)}$; (3) At the first time Player II chooses a leaf in $\Gamma^s_{2(k-1)}$ for $s \in \{3, 4\}$, play a winning strategy

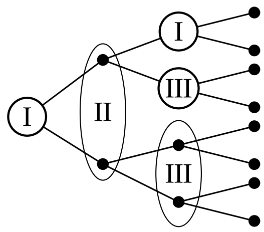
in $\Gamma^{7-s}_{2(k-1)}$. From that stage on, whenever Player II chooses a leaf in either $\Gamma^{1}_{2(k-1)}$ or $\Gamma^{2}_{2(k-1)}$ continue following the winning strategy in $\Gamma^{1}_{2(k-1)}$, whereas if Player II chooses a leaf in either $\Gamma^{3}_{2(k-1)}$ or $\Gamma^{4}_{2(k-1)}$ continue following the winning strategy in $\Gamma^{7-s}_{2(k-1)}$; (4) After all the leaves in $\Gamma^{1}_{2(k-1)}$ and $\Gamma^{7-s}_{2(k-1)}$ have been assigned choose an arbitrary leaf.
Player I always follows his wining strategies at least one step in front of Player II in both $\Gamma^{1}_{2(k-1)}$ and $\Gamma^{7-s}_{2(k-1)}$ and thus the values in both "or" vertices in depth 2 are 1, which leads that the root has value 1.

3.21 (a) In different information sets there must be different actions.
(b) The set of actions in two vertices in the same information set must be identical.
(c) The game must has a root.
(d) The set of actions in two vertices in the same information set must be identical. In particular, the number of actions in both sets must be the same.
(e) The sum of probabilities of all chance moves derived from the same vertex should be 1.
(f) The game must be on a tree, i.e., for every vertex in the game there should be only one path that connects it to the root.
(g) The sum of probabilities of all chance moves derived from the same vertex should be 1.
(h) Each information set of vertices at which a chance move is implemented includes only one vertex.

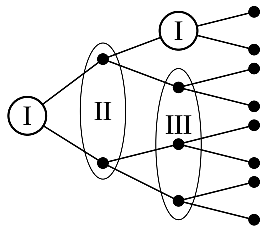
3.22 **Game A:** Player I has forgotten what he played.
Game B: Player I has forgotten whether he previously played or not.
Game C: Player I has forgotten what he previously played as well as what Player II played at the beginning of game.

3.23 In both information sets U^2_{II} and U^3_{II} Player II knows the action taken by Player I.

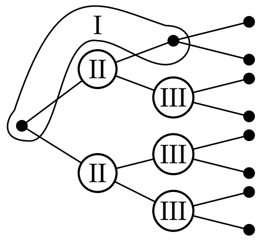
3.24 (a)



(b)



(c)



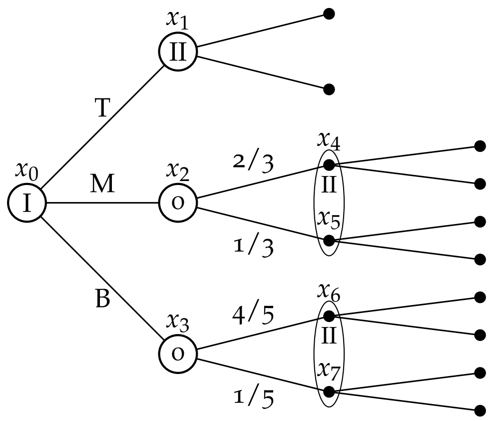
3.25 **Game A:** The game has no nontrivial subgames. In the information set of Player II, he does not know what Player I selected.

Game B: The game has no nontrivial subgames. In the information set of Player II, he does not remember what moves he and Player I made in the past. However, he knows that he chose t_1 if Player I selected T, otherwise he chose b_2. In the information set of Player III, he does not know what moves Player I made in the past and thus he does not know what Player II selected b_1 or t_2.

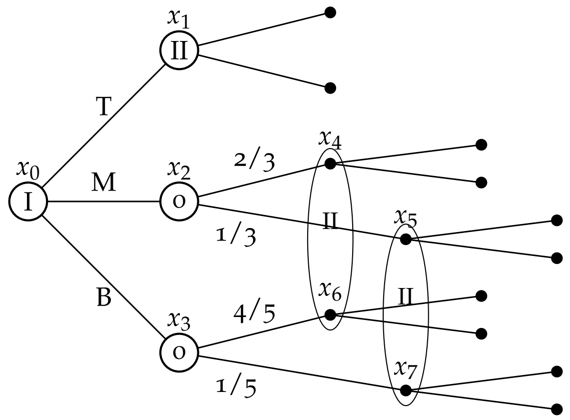
Game C: The game has no nontrivial subgames. In the information set of Player I, he does not remember what moves he made in the past and thus he does not know what Player II selected t_1 or b_2.

Game D: There is only one nontrivial subgames, $\Gamma(x_3)$. In the information set of Player II, he does not know what moves Player I made in the past.

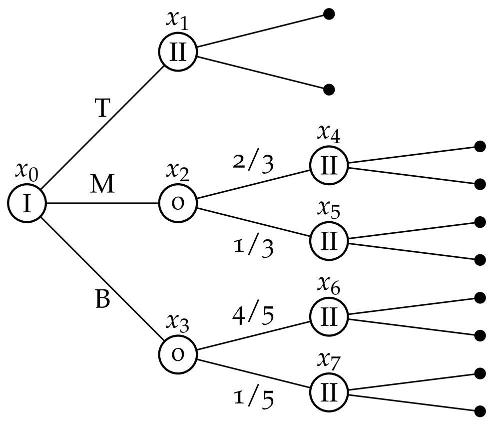
3.26 (a)



(b)



(c)



3.27 (a) In the information sets that includes two vertices Player I does not know the result of the chance move, and even does not remember whether he previously played or not. On the other hand, he knows what action he might select.

(b) Player I has 4 information sets, at each one of them, 2 actions are available, thus he has $2^4 = 16$ strategies.

(c) Player I should select B_3 and T_4 which strictly dominate T_3 and B_4 respectively. That leaves us with 4 optional strategies: $T_1T_2B_3T_4$, $T_1B_2B_3T_4$, $B_1T_2B_3T_4$ and $B_1B_2B_3T_4$. The expected payoffs of Player I under each strategy is as follows:

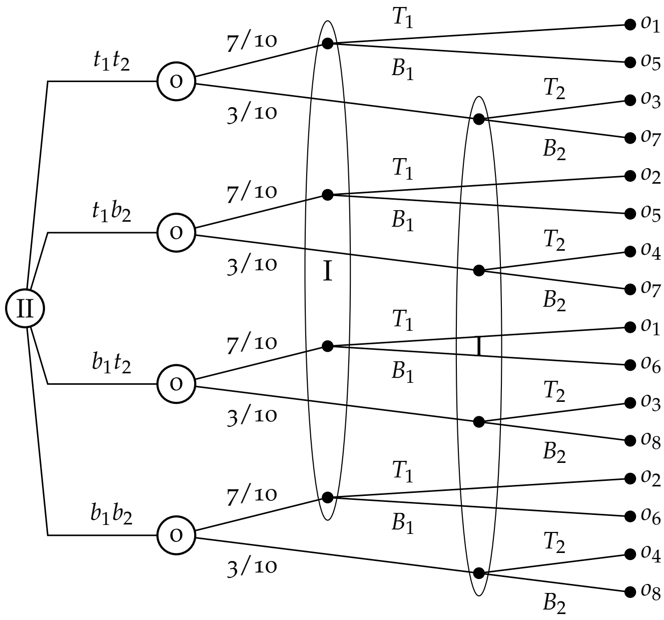
$$
\begin{aligned}
U(T_1T_2B_3T_4) &= \tfrac{3}{5} \cdot 20 + \tfrac{2}{5} \cdot 40 = 28 \\
U(T_1B_2B_3T_4) &= \tfrac{3}{5} \cdot 20 + \tfrac{2}{5} \cdot 25 = 22 \\
U(B_1T_2B_3T_4) &= \tfrac{3}{5} \cdot 25 + \tfrac{2}{5} \cdot 15 = 21 \\
U(B_1B_2B_3T_4) &= \tfrac{3}{5} \cdot 15 + \tfrac{2}{5} \cdot 25 = 19.
\end{aligned}
$$

Thus, Player I is better off selecting $T_1T_2B_3T_4$.

3.28 In Part(a), (b) and (d), two actions are available at each one of the 3 information sets of Player II,thus he has $2^3 = 8$ strategies. In Part (c), Player II has only 2 information set, to each one of them 2 actions are available so he has $2^2 = 4$ strategies.

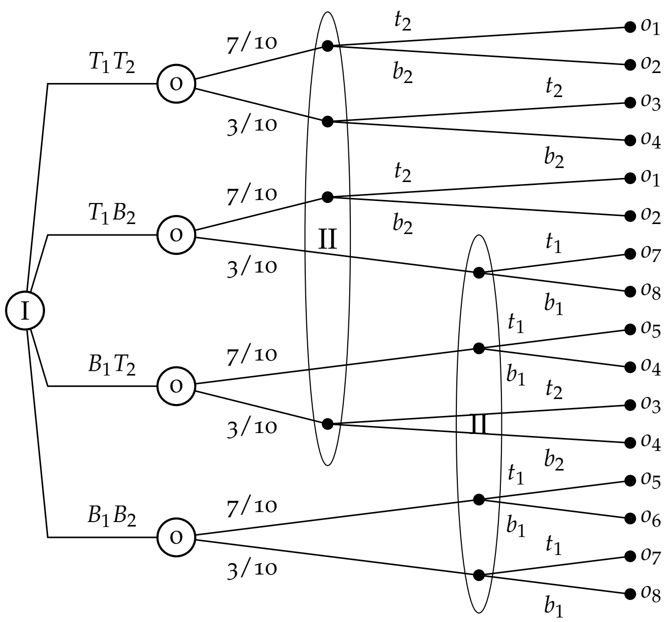
3.29 (a) Player II does not know the result of the chance move, at each of his information set. On the other hand he knows whether Player I selected the top action or whether he selected the bottom action.

(b)



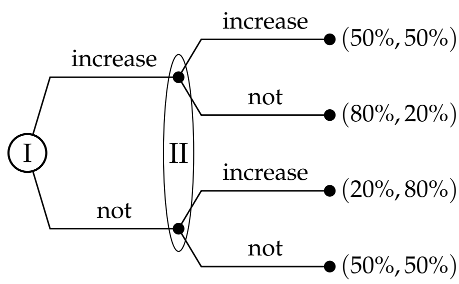
II
0
I
t_1t_2
t_1b_2
b_1t_2
b_1b_2
7/10
3/10
T_1
B_1
T_2
B_2
o_1
o_2
o_3
o_4
o_5
o_6
o_7
o_8

(c)

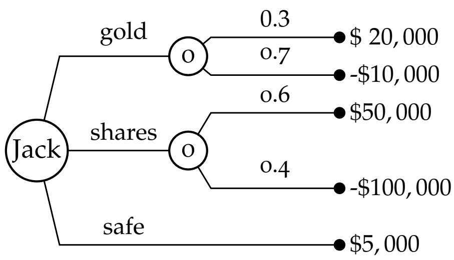


I
0
II
T_1T_2
T_1B_2
B_1T_2
B_1B_2
7/10
3/10
t_1
b_1
t_2
b_2
o_1
o_2
o_3
o_4
o_5
o_6
o_7
o_8

3.30



3.31



3.32 (a) Player I has two strategies $\{T, B\}$.

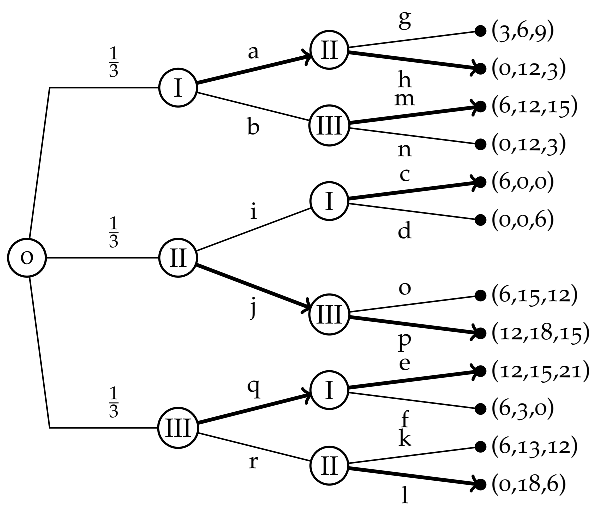
(b) Player II has 18 strategies as presented below.

$$\{t_1t_2t_3, t_1t_2b_3, t_1m_1t_3, t_1m_2b_3, t_1b_2t_3, t_1b_2b_3,\\ m_1t_2t_3, m_1t_2b_3, m_1m_1t_3, m_1m_2b_3, m_1b_2t_3, m_1b_2b_3,\\ b_1t_2t_3, b_1t_2b_3, b_1m_1t_3, b_1m_2b_3, b_1b_2t_3, b_1b_2b_3\}\,.$$

(c) The expected payoff is $(9, 5)$.

(d) The expected payoff is $\frac{2}{3}(0, 6) + \frac{1}{3}(9, 3) = (3, 5)$.

3.33 (a)

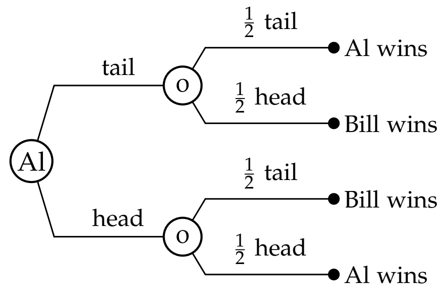


(b) The expected payoff is

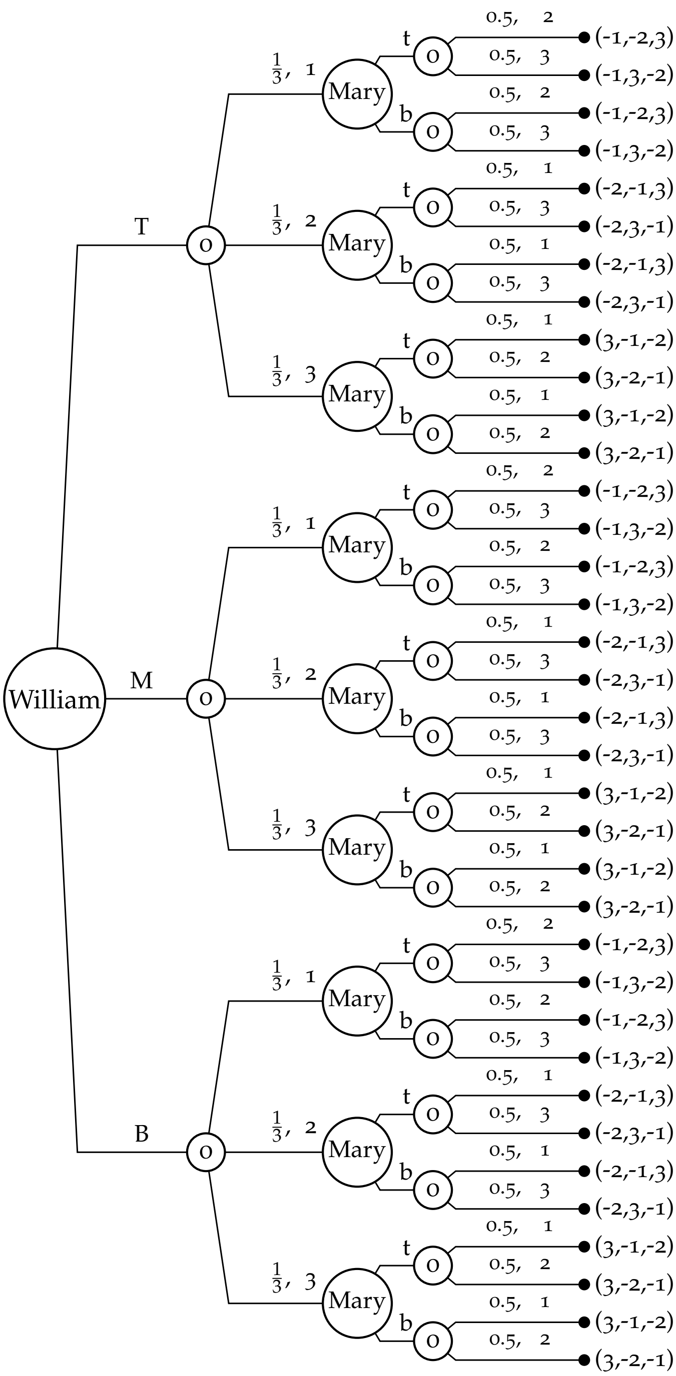
$$\tfrac{1}{3}(0, 12, 3) + \tfrac{1}{3}(12, 18, 15) + \tfrac{1}{3}(12, 15, 21) = (8, 15, 13).$$

(c) Player I is better off selecting the strategy (b, c, e) which dominates the other strategies (each action strictly dominates in the relevant subgame).

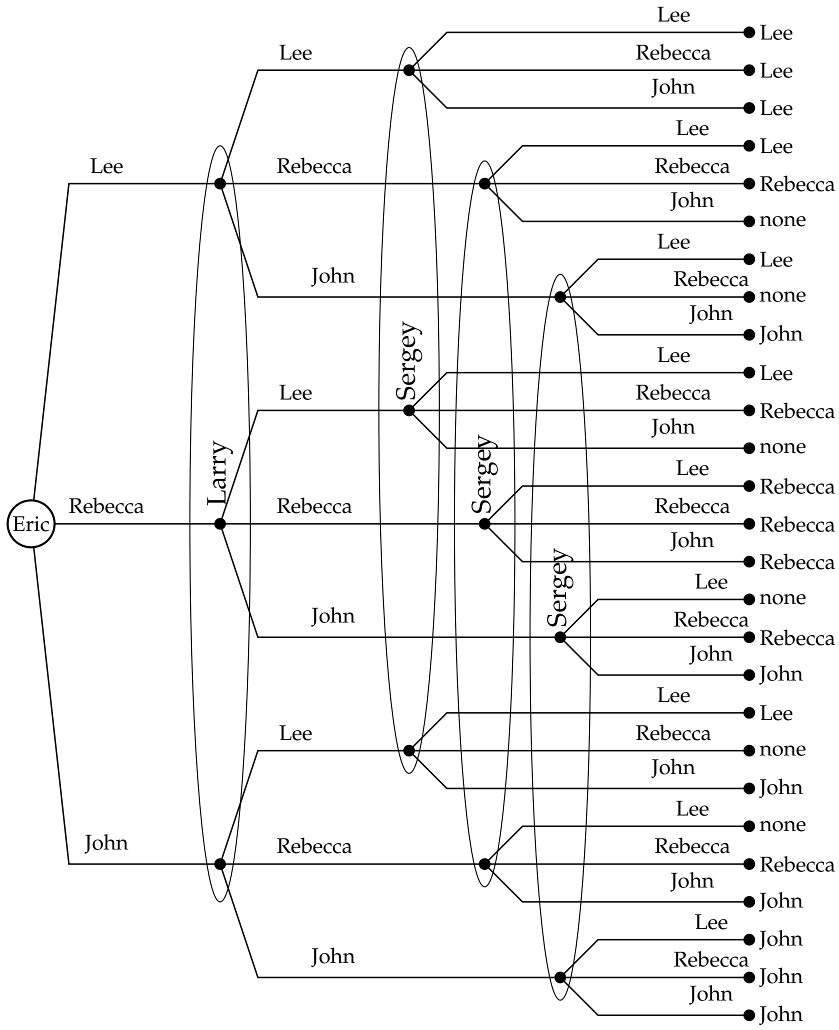
3.34



3.35 The outcome of the game is a triple (x, y, z) representing the payoff to each player, with x denoting the payoff to William, y the payoff to Mary and z the payoff to Anne.


William
T
M
B
0
1/3, 1
1/3, 2
1/3, 3
Mary
t
b
0.5, 2
0.5, 3
0.5, 1
(-1,-2,3)
(-1,3,-2)
(-1,-2,3)
(-1,3,-2)
(-2,-1,3)
(-2,3,-1)
(-2,-1,3)
(-2,3,-1)
(3,-1,-2)
(3,-2,-1)
(3,-1,-2)
(3,-2,-1)
(-1,-2,3)
(-1,3,-2)
(-1,-2,3)
(-1,3,-2)
(-2,-1,3)
(-2,3,-1)
(-2,-1,3)
(-2,3,-1)
(3,-1,-2)
(3,-2,-1)
(3,-1,-2)
(3,-2,-1)
(-1,-2,3)
(-1,3,-2)
(-1,-2,3)
(-1,3,-2)
(-2,-1,3)
(-2,3,-1)
(-2,-1,3)
(-2,3,-1)
(3,-1,-2)
(3,-2,-1)
(3,-1,-2)
(3,-2,-1)

3.36 (a)



Eric
Lee
Rebecca
John
Larry
Sergey
Sergey
Sergey
Lee
Lee
Rebecca
Lee
John
Lee
Lee
Lee
Rebecca
Rebecca
John
none
Lee
Lee
Rebecca
none
John
John
Lee
Lee
Rebecca
Rebecca
John
none
Lee
Rebecca
Rebecca
Rebecca
John
Rebecca
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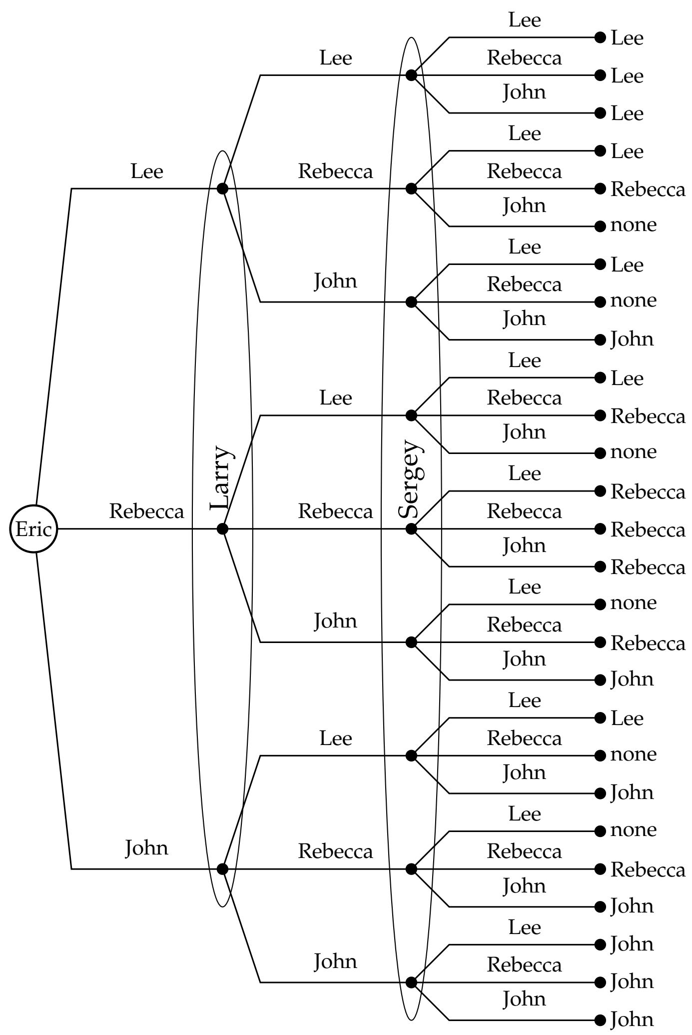
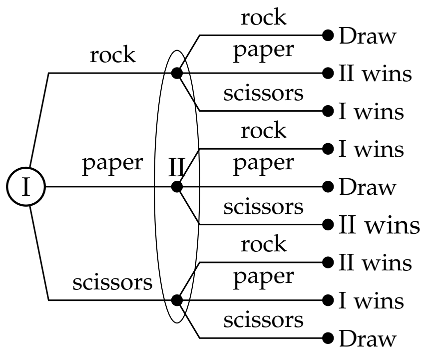
John
Lee
Lee
Rebecca
none
John
John
Lee
none
Rebecca
Rebecca
John
John
Lee
John
Rebecca
John
John
John

(b)



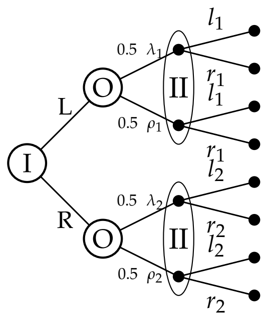
(c) The game tree presented in Part (a) represents also the current variant of the candidate game.

3.37 **Rock, Paper, Scissors**



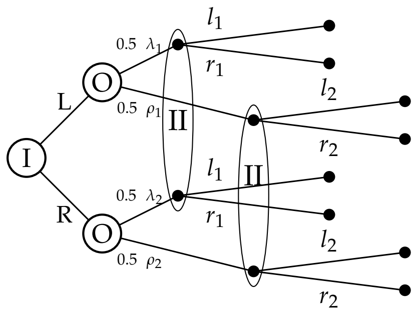
rock
paper
scissors
I
II
rock
paper
scissors
Draw
II wins
I wins
rock
paper
scissors
I wins
Draw
II wins
rock
paper
scissors
II wins
I wins
Draw

3.38 (a)



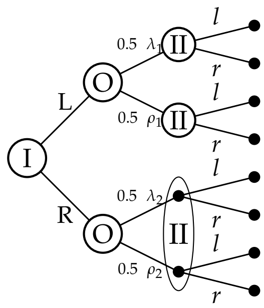
I
L
R
O
O
0.5 λ_1
0.5 ρ_1
0.5 λ_2
0.5 ρ_2
II
II
l_1
r_1
l_1
r_1
l_2
r_2
l_2
r_2

(b)

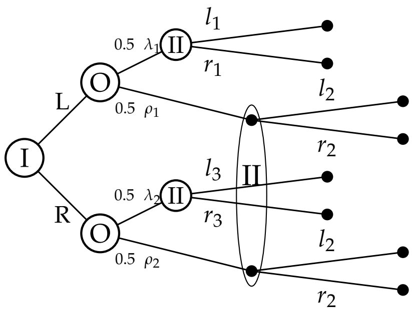


I
L
R
O
O
0.5 λ_1
0.5 ρ_1
0.5 λ_2
0.5 ρ_2
II
II
l_1
r_1
l_2
r_2
l_1
r_1
l_2
r_2

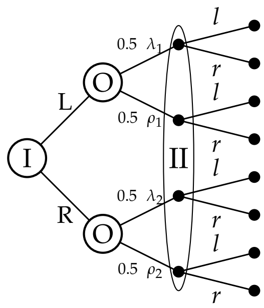
(c)



(d)



(e)



3.39 (a) In each information set, Player I does not remember if he previously made a move or not. However, he knows which move he may took.

(b) Action R_2 is strictly dominated by L_2 and thus can be eliminated. Thus, both actions R_1 and L_1 leads to the same expected payoff in the bottom information set. However, if Player I selects L_1 then his best response in the left information set is R_3 which leads to the expected payoff of $\frac{31}{3}$; whereas, if he selects R_1 then his best response in the left information set is L_3 which leads to the expected payoff of $\frac{35}{3}$. In conclusion, the recommendation to Player I is (R_1, L_2, L_3).

Chapter 4: Strategic-form games

4.1 The strategic-form game is as follows:

		Henry: Give me	Henry: Give the other participant
William	Give me	(\$1000,\$1000)	(\$5000,\$0)
	Give the other partticipant	(\$0,\$5000)	(\$4000,\$4000)

The strategy "Give me \$1000" strictly dominates the other strategy for both players. Hence, if both players are rational, they follow this strategy (which is also an equilibrium).

4.2 Game A:

		PlayerII t	b
PlayerI	T	O_1	O_1
	B	O_2	O_3

Game B:

		PlayerII cf	cg	df	dg	ef	eg
PlayerI	a	A	A	B	B	C	C
	b	D	E	D	E	D	E

Game C:

		Player 2 t_1t_2	t_1b_2	b_1t_2	b_1b_2
Player 1	T_1T_2	A	A	B	B
	T_1M_2	A	A	C	C
	T_1B_2	A	A	D	D
	B_1	F	E	F	E

Game D:

		Player II t_1t_2	t_1b_2	b_1t_2	b_1b_2
Player I	T_1T_2	A	A	B	B
	T_1B_2	A	A	C	C
	B_1	D	E	D	E

Game E:

	t_1 t_2 t_3	t_1 t_2 b_3	t_1 b_2 t_3	t_1 b_2 b_3	t_1 m_2 t_3	t_1 m_2 b_3	b_1 t_2 t_3	b_1 t_2 b_3	b_1 b_2 t_3
T	11,5	10,8	3,5	2,8	9,3	8,6	11,5	10,8	3,5
B	9,5	9,5	9,5	9,5	9,5	9,5	7,7	7,7	7,7

	b_1 b_2 b_3	b_1 m_2 t_3	b_1 m_2 b_3	m_1 t_2 t_3	m_1 t_2 b_3	m_1 b_2 t_3	m_1 b_2 b_3	m_1 m_2 t_3	m_1 m_2 m_1
T	2,8	9,3	8,6	11,5	10,8	3,5	2,8	(9,3)	8,6
B	7,7	7,7	7,7	5,9	5,9	5,9	5,9	5,9	5,9

Game F:

Player II

PlayerI	
T_1T_2	(6,10)
T_1B_2	(8,8)
B_1T_2	(7,10.5)
B_1B_2	(9,8.5)

Game G:

Player II

PlayerI	T	B
T	(5,4)	(9,4)
B	(2,8)	(8,3)

Game H:

Player II

PlayerI	t_1t_2	t_1b_2	b_1t_2	b_1b_2
T_1	(40,22))	(36,26)	(16,4)	(12,8)
B_1	(44,24)	(48,20)	(8,12)	(12,8)

Game I:

Player II

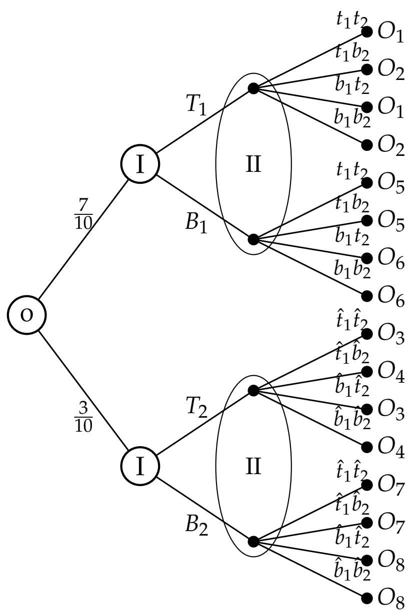
PlayerI	t_1t_2	t_1b_2	b_1t_2	b_1b_2
T_1T_2	$0.7O_1 + 0.3O_3$	$0.7O_2 + 0.3O_4$	$0.7O_1 + 0.3O_3$	$0.7O_2 + 0.3O_4$
T_1B_2	$0.7O_1 + 0.3O_7$	$0.7O_2 + 0.3O_7$	$0.7O_1 + 0.3O_8$	$0.7O_2 + 0.3O_8$
B_1T_2	$0.7O_5 + 0.3O_3$	$0.7O_5 + 0.3O_4$	$0.7O_6 + 0.3O_3$	$0.7O_6 + 0.3O_4$
B_1B_2	$0.7O_5 + 0.3O_7$	$0.7O_5 + 0.3O_7$	$0.7O_6 + 0.3O_8$	$0.7O_6 + 0.3O_8$

Game J:

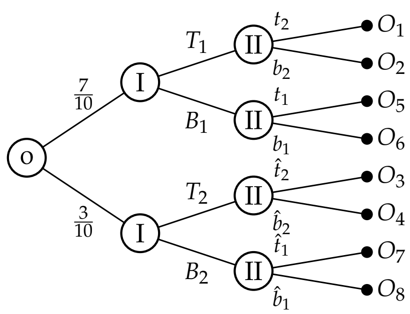
		Player II
Player I	T_1T_2	$[\frac{1}{3}(W), \frac{2}{3}(L)]$
	T_1B_2	$[\frac{1}{3}(W), \frac{2}{3}(L)]$
	B_1T_2	$[\frac{2}{3}(W), \frac{1}{3}(L)]$
	B_1B_2	$[\frac{2}{3}(W), \frac{1}{3}(L)]$

4.3 (a) Player II knows whether Player I chose B or T in each one of his information sets, but he does not know the result of the chance move.

(b) The extensive-form game:



(c) The extensive-form game:



(d) The strategic-form game for the original game is:

		Player 2			
		t_1t_2	t_1b_2	b_1t_2	b_1b_2
	T_1T_2	$0.7O_1 + 0.3O_3$	$0.7O_2 + 0.3O_4$	$0.7O_1 + 0.3O_3$	$0.7O_2 + 0.3O_4$
Player 1	T_1B_2	$0.7O_1 + 0.3O_7$	$0.7O_2 + 0.3O_7$	$0.7O_1 + 0.3O_8$	$0.7O_2 + 0.3O_8$
	B_1T_2	$0.7O_5 + 0.3O_3$	$0.7O_5 + 0.3O_4$	$0.7O_6 + 0.3O_3$	$0.7O_6 + 0.3O_4$
	B_1B_2	$0.7O_5 + 0.3O_7$	$0.7O_5 + 0.3O_7$	$0.7O_6 + 0.3O_8$	$0.7O_6 + 0.3O_8$

The strategic-form game for the games in (b) and (c) are identical:

	$t_1t_2\widehat{t}_1\widehat{t}_2$	$t_1t_2\widehat{t}_1\widehat{b}_2$	$t_1t_2\widehat{b}_1\widehat{t}_2$	$t_1t_2\widehat{b}_1\widehat{b}_2$
T_1T_2	$0.7O_1 + 0.3O_3$	$0.7O_1 + 0.3O_4$	$0.7O_1 + 0.3O_3$	$0.7O_1 + 0.3O_4$
T_1B_2	$0.7O_1 + 0.3O_7$	$0.7O_1 + 0.3O_7$	$0.7O_1 + 0.3O_8$	$0.7O_1 + 0.3O_8$
B_1T_2	$0.7O_5 + 0.3O_3$	$0.7O_5 + 0.3O_4$	$0.7O_5 + 0.3O_3$	$0.7O_5 + 0.3O_4$
B_1B_2	$0.7O_5 + 0.3O_7$	$0.7O_5 + 0.3O_7$	$0.7O_5 + 0.3O_8$	$0.7O_5 + 0.3O_8$

continued:

$t_1b_2\widehat{t}_1\widehat{t}_2$	$t_1b_2\widehat{t}_1\widehat{b}_2$	$t_1b_2\widehat{b}_1\widehat{t}_2$	$t_1b_2\widehat{b}_1\widehat{b}_2$
$0.7O_2 + 0.3O_3$	$0.7O_2 + 0.3O_4$	$0.7O_2 + 0.3O_3$	$0.7O_2 + 0.3O_4$
$0.7O_2 + 0.3O_7$	$0.7O_2 + 0.3O_7$	$0.7O_2 + 0.3O_8$	$0.7O_2 + 0.3O_8$
$0.7O_5 + 0.3O_3$	$0.7O_5 + 0.3O_4$	$0.7O_5 + 0.3O_3$	$0.7O_5 + 0.3O_4$
$0.7O_5 + 0.3O_7$	$0.7O_5 + 0.3O_7$	$0.7O_5 + 0.3O_8$	$0.7O_5 + 0.3O_8$

continued

$b_1t_2\widehat{t}_1\widehat{t}_2$	$b_1t_2\widehat{t}_1\widehat{b}_2$	$b_1t_2\widehat{b}_1\widehat{t}_2$	$b_1t_2\widehat{b}_1\widehat{b}_2$
$0.7O_1 + 0.3O_3$	$0.7O_1 + 0.3O_4$	$0.7O_1 + 0.3O_3$	$0.7O_1 + 0.3O_4$
$0.7O_1 + 0.3O_7$	$0.7O_1 + 0.3O_7$	$0.7O_1 + 0.3O_8$	$0.7O_1 + 0.3O_8$
$0.7O_6 + 0.3O_3$	$0.7O_6 + 0.3O_4$	$0.7O_6 + 0.3O_3$	$0.7O_6 + 0.3O_4$
$0.7O_6 + 0.3O_7$	$0.7O_6 + 0.3O_7$	$0.7O_6 + 0.3O_8$	$0.7O_6 + 0.3O_8$

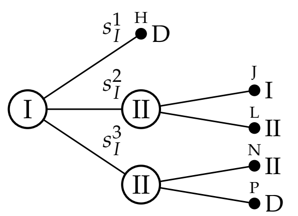
continued:

$b_1b_2\widehat{t}_1\widehat{t}_2$	$b_1b_2\widehat{t}_1\widehat{b}_2$	$b_1b_2\widehat{b}_1\widehat{t}_2$	$b_1b_2\widehat{b}_1\widehat{b}_2$
$0.7O_2 + 0.3O_3$	$0.7O_2 + 0.3O_4$	$0.7O_2 + 0.3O_3$	$0.7O_2 + 0.3O_4$
$0.7O_2 + 0.3O_7$	$0.7O_2 + 0.3O_7$	$0.7O_2 + 0.3O_8$	$0.7O_2 + 0.3O_8$
$0.7O_6 + 0.3O_3$	$0.7O_6 + 0.3O_4$	$0.7O_6 + 0.3O_3$	$0.7O_6 + 0.3O_4$
$0.7O_6 + 0.3O_7$	$0.7O_6 + 0.3O_7$	$0.7O_6 + 0.3O_8$	$0.7O_6 + 0.3O_8$

4.4 The game of Hex is a finite perfect information two-player game. By Exercise 3.19, the player with the opening move has a winning strategy, so it (weekly) dominates all other strategies. Therefore, the process whereby the two players eliminate (weakly) dominated strategies, ends with one strategy for each player. In particular, the normal-form game remains of with one cell with the result of winning of the player with the opening move.

4.5 There does not exists a two-player game in extensive form with perfect information, and possible outcomes I (Player I wins), II (Player II wins), and D (a draw), whose strategic-form description is as presented in the exercise. This follows the fact that this strategic-form does not include neither a winning strategy for one of the Players, nor a strategy for each player guaranteeing at least a draw while such an extensive form game must have (Von Neumann [1928]).

4.6 There does not exists a two-player game in extensive form with perfect information, and possible outcomes I (Player I wins), II (Player II wins), and D (a draw), whose strategic-form description is as presented in the exercise. Assume to the contrary that there is such a game. If player II is the first to play then each act he takes correspond a strategic subgame with three different row, hence Player I must have at least 9 strategies, a contradiction. Thus, Player I must be the first to play. Furthermore, his first decision must divide the strategic game into three rows (either by Von Neumann theorem [1928] or by the same arguments presented above for Player II). In conclusion, the only extensive form game that might correspond the strategic-form description presented in the exercise is as follows,



However, the strategic-form description given in the exercise does not correspond this extensive form game, a contradiction.

4.7 In Game C, the process of iterated elimination of strictly dominated strategies yields a single strategy vector, (γ, c) ,when it is completed. This vector is indeed the only Nash equilibrium of the game.

4.8 Game A: Player II should play R since it strictly dominates R. Thus Player I should play T since it strictly dominates H in the remaining game.
Game B: The player are recommended to play (T, R) since it the only equilibrium in the game furthermore it leads to the highest results in the game for both players.
Game C: The game has no equilibrium. Both players can follow their maxmin strategies. The maxmin strategies of Player I are δ and γ, which guaranties a payoff of at least 2. The maxmin strategy of Player

II is C which guaranties a payoff of at least 5.
Game D: The game has no equilibrium. Both players can follow their maxmin strategies. The maxmin strategy of Player I is α, which guaranties a payoff of at least -1. The maxmin strategies of Player II are C and D which guaranties a payoff of at least 1.

4.9 (a) Assume the process of iterative elimination of strictly dominated strategies results in a unique strategy vector s^*. By Corollary 4.37, s^* is the unique Nash equilibrium of the game. We next prove that this equilibrium is strict. Let i be a player and let $s_i \in S_i \setminus \{s_i^*\}$. Then, since s_i was eliminated then there is $s_i' \in S_i$ such that $u_i\left(s_{-i}^*, s_i\right) < u_i\left(s_{-i}^*, s_i'\right)$. In addition s^* is Nash equilibrium, so $u_i\left(s_{-i}^*, s_i'\right) \leq u_i\left(s^*\right)$. In conclusion, $u_i\left(s_{-i}^*, s_i\right) < u_i\left(s^*\right)$ and s^* is a strict Nash equilibrium.

(b) Assume $s^* = \left(s_i^*\right)_{i=1}^n$ is a strict Nash equilibrium. We prove that none of the strategies s_i^* can be eliminated by iterative elimination of dominated strategies (under either strict or weak domination). Assume to the contrary that it does not hold. Let s_i^* be the first strategy that was eliminated by iterative elimination of dominated strategies. In particular, there is a strategy $s_i \in S_i \setminus \{s_i^*\}$ such that $u_i\left(s_{-i}^*, s_i\right) \geq u_i\left(s^*\right)$. But this contradicts the assumption that s^* is a strict Nash equilibrium.

4.10 Let $G = \left(N, (S_i)_{i \in N}, (u_i)_{i \in N}\right)$ be a game in strategic form. Denote by $S^k = \left(S_i^k\right)_{i \in N}$ the set of the remaining strategies after the k-th elimination process of strictly dominated strategies has been completed, for $k \in \{1, 2\}$.
We show that $S^2 \subseteq S^1$.
For every natural number $n \geq 0$, denote by $S^{k,n} = \left(S_i^{k,n}\right)_{i \in N}$ the set of the remaining strategies after the n-th elimination according to the k-th elimination process of strictly dominated elimination.
Assume that s_i is strictly dominated by $\widehat{s}_i \in S_i^{k,n}$ against $S_{-i}^{k,n}$. We show that for every $m \geq n$, there must be $s_i' \in S_i^{k,m}$ such that s_i' strictly dominates s_i against $S_{-i}^{k,m}$. Indeed, for $m = n$, $\widehat{s}_i \in S_i^{k,n}$ dominates s_i against $S_{-i}^{k,n}$. Assume that there is $s_i' \in S_i^{k,m}$ such that s_i' strictly dominates s_i against $S_{-i}^{k,m}$ for $m \geq n$. Then, either $s_i' \in S^{k,m+1}$ so the claim holds since $(S_j^{k,n})_n$ is monotonic non-increasing by inclusion for every $j \in N$; or s_i' was the $m+1$-eliminated strategy so there is a

strategy $s_i'' \in S_i^{k,m+1}$ that strictly dominates s_i' against $S_{-i}^{k,m+1}$. Hence s_i'' strictly dominates s_i against $S_{-i}^{k,m+1}$.

We next prove that $S_i^2 \subseteq S_i^{1,n}$ for every i and every n, by induction over n, from which it follows that $S^2 \subseteq S^1$. For $n = 0$, $S_i^2 \subseteq S_i = S_i^{1,0}$ for every i. Assume that $S_i^2 \subseteq S_i^{1,n}$ for every i and every $n < M$. We prove that it also holds for M. Let s_i be the M-th eliminated strategy following the the first process, hence there is a strategy $s' \in S_i^{1,M-1}$ that strictly dominates s_i against $S_{-i}^{1,M-1}$. By the induction hypothesis, s_i' strictly dominates s_i against S_{-i}^2. Thus either $s_i' \in S_i^2$ (and it strictly dominates s_i against S_{-i}^2) or s_i' was eliminated during the second process and thus, by the above claim, there is a strategy $s_i'' \in S_i^2$ that strictly dominates s_i' against S_{-i}^2 so it also strictly dominates s_i against S_{-i}^2. In particular, $s_i \notin S_i^2$ so $S_i^2 \subseteq S_i^{1,M-1} \setminus \{s_i\} = S_i^{1,M}$ and $S_{-i}^2 \subseteq S_{-i}^{1,M-1} = S_{-i}^{1,M}$, as needed. In conclusion, $S^2 \subseteq S^1$. Furthermore, by changing the roles of S^1 and S^2 one has $S^1 \subseteq S^2$ so $S^1 = S^2$.

4.11 Game A: (β, b)(there are several orders of elimination that leads to the same result, e.g., the following order of elimination: a, c, α,d).
Game B: (α, a) (order of elimination: b, β).
Game C: (β, a) (there are several orders of elimination that leads to the same result, e.g., b, c, d, α, γ).
Game D: There are two rational strategy vectors (δ, b) (e.g. eliminate $a, c, \beta, d, \gamma, \delta$) and (δ, c) (e.g. eliminate $a, \beta, d, \gamma, \delta, b$).

4.12 The next game has one equilibrium (T, L).

	L	M	R
T	2,2	0,2	2,0
B	0,0	2,1	0,2

However, elimination of L, that is dominated by M, yields the following game which has no equilibria.

	M	R
T	0,2	2,0
B	2,1	0,2

4.13 We prove that a strictly dominated strategy cannot be an element of a game's equilibrium . Let $s_i \in S_i$ be a strategy of Player i that is strictly dominated by $s_i' \in S_i$. Let $s = (s_i, s_{-i})$ be a strategy vector in which player i using the strictly dominated strategy. Then,

$u_i(s_i', s_i) > u_i(s_i, s_{-i})$ so Player i can deviate and increase his payoff. Thus s cannot be an equilibrium.

4.14 (a) The strategy β_i^* of buyer i, in which he bids his private value for the item, does not weakly dominate all his other strategies. The payoff of buyer i under β_i^* is zero no netter how the other buyers bid. Whereas, by bidding less than the value, the buyer may gain more than zero if his bid wins.

(b) Every strategy b_i according to which buyer i bids $0 < b_i < v_i$ is weakly dominates β_i^*. As we already mentioned, $u_i(\beta_i^*, b_{-i}) = 0$, for every b_{-i}, while the payoff of buyer i following b_i is

$$u_i\left(b_i, b_{-i}\right) = \begin{cases} 0 & \text{if } b_i < \max_{j \neq i} b_j \\ \frac{v_i - b_i}{\#\{j \in N | bj = bi\}} & \text{if } b_i < \max_{j \neq i} b_j \geq u_i(\beta, b_{-i}). \\ v_i - b_i & \text{if } b_i > \max_{j \neq i} b_j \end{cases}$$

Note that in both (a) and (b) the answer depends only in the bids of the other buyers. In case that all buyers bid their value and the value is not known to the other players then the answer to (a) and (b) will not change. As well as, if the value of all buyers is at most the value of buyer i the answer will not change. However, if the values are common knowledge, and there is at least one buyer with a value higher than the value of i (and he bid his value), then the strategy β_i is weakly dominate all other strategies since i can win only with a bid of $b_i > v_i$ which case his payoff is $v_i - b_i < 0$.

4.15 The strategy vector s^* is a Nash equilibrium according to Definition 4.17, if and only if, for each $i \in N$ and and every strategy $s_i \in S_i$ one has

$$u_i\left(s^*\right) \geq u_i\left(s_i, s^*_{-i}\right) \tag{8}$$

But Equation (8) holds if and only if, for each $i \in N$,

$$u_i\left(s_i^*, s^*_{-i}\right) = u_i\left(s^*\right) = \max_{s_i \in S_i} u_i\left(s_i, s^*_{-i}\right).$$

Meaning, s_i^* is a best reply to s^*_{-i} for every player $i \in N$, so it a Nash equilibrium according to Definition 4.19.

4.16 Game A:

	a	b	c	d
γ	$\underline{7}, 3$	$\underline{6},3$	5,5	$4,\underline{7}$
β	4,2	$5,\underline{8}$	$\underline{8},6$	$5,\underline{8}$
α	6,1	3,8	2,4	$\underline{6},\underline{9}$

The equilibrium in the game is (α, d).

Game B:

	a	b	c	d
δ	5,2	3,1	2,2	$4,\underline{5}$
γ	$0,\underline{3}$	2,2	0,1	$-1,\underline{3}$
β	$\underline{8},\underline{4}$	$\underline{7},0$	$\underline{6},-1$	$\underline{5},2$
α	$0,\underline{5}$	1,-2	2,2	3,4

The equilibrium in the game is (β, a).

Game C:

	a	b	c	d
ϵ	0,0	$-1,\underline{1}$	$\underline{1},\underline{1}$	0,-1
δ	$\underline{1},-1$	$\underline{1},0$	$0,\underline{1}$	0,0
γ	$0,\underline{1}$	-1,-1	$\underline{1},0$	$\underline{1},-1$
β	$-1,\underline{1}$	0,-1	$-1,\underline{1}$	0,0
α	$\underline{1},\underline{1}$	0,0	-1,-1	0,0

The equilibria in the game are (α, a) and (ϵ, c).

4.17 The best response for each player in the game is:

	a	b
A	$0,\underline{0},\underline{5}$	$0,\underline{0},\underline{0}$
B	$\underline{2},\underline{0},\underline{0}$	$0,\underline{0},0$

α

	a	b
A	$\underline{1},\underline{2},3$	$0,0,\underline{0}$
B	$0,0,\underline{0}$	$\underline{1},\underline{2},3$

β

	a	b
A	$\underline{0},\underline{0},0$	$\underline{0},\underline{0},\underline{0}$
B	$\underline{0},\underline{5},\underline{0}$	$\underline{0},0,\underline{4}$

γ

Thus, the equilibria in the game are (B, a, α), (B, a, γ) and (A, b, γ).

4.18 The next game has no equilibrium.

	L	M	R
T	$3,10,\underline{8}$	$\underline{8},\underline{14},6$	$\underline{4},12,\underline{7}$
C	$\underline{4},\underline{7},2$	5,5,2	$2,2,\underline{8}$
B	3,-5,0	0,3,4	$-3,\underline{5},\underline{0}$

P

	L	M	R
T	$4,\underline{9},3$	$7,8,\underline{10}$	$\underline{5},7,-1$
C	$3,4,\underline{5}$	$17,3,\underline{12}$	$3,\underline{5},2$
B	$\underline{9},7,\underline{2}$	$\underline{20},0,\underline{13}$	$0,\underline{15},\underline{0}$

Q

4.19 We Prove that in the Centipede game, each strategy vector according to which Player I chooses S at the first move, and Player II chooses S at the second move (if he can) is an equilibrium in the game. Indeed, if Player I deviates to C, then Player II chooses S and he loses; and if Player II deviates, it does not effect the play so Player II does not gain by the deviation. Note that all other strategy vectors are not equilibria. If Player II plays S only after playing C for k times ($k > 0$), then Player I is better off playing S after playing C for k times, as well. However, the best reply of Player II to Player I's strategy is playing S after playing C for $k-1$ times and thus such a strategy vector cannot be an equilibrium.

4.20 Let G be a two-player symmetric game. Assume (s_1, s_2) is an equilibrium in G. Then,

$$u_1(s_1, s_2) = \max_{s \in S_1} u_1(s, s_2)$$

and

$$u_2(s_1, s_2) = \max_{s \in S_2} u_2(s_1, s)$$

However, by symmetry of the game

$$u_2(s_2, s_1) = u_1(s_1, s_2) = \max_{s \in S_1} u_1(s, s_2) = \max_{s \in S_2} u_2(s_2, s)$$

and

$$u_1(s_2, s_1) = u_2(s_1, s_2) = \max_{s \in S_2} u_2(s_1, s) = \max_{s \in S_1} u_1(s, s_1)$$

Thus (s_2, s_1) is also an equilibrium in G.

4.21 Game A:

	t_1t_2	t_1b_2	b_1t_2	b_1b_2
T	$2,\underline{5}$	$2,\underline{5}$	$\underline{5},2$	$5,2$
B	$\underline{4},\underline{8}$	$\underline{8},4$	$4,\underline{8}$	$\underline{8},4$

The equilibrium in the game is (B, t_1t_2).

Game B:

	t	b
T	$9,\underline{7}$	$4,\underline{7}$
B	$\underline{13},\underline{16}$	$\underline{13},\underline{16}$

The equilibria in the game are (B, t) and (B, b).

Game C:

	t	b
T_1T_2	$(\underline{5},\underline{10})$	$(\underline{7},8)$
T_1B_2	$(\underline{5},\underline{11})$	$(5,\underline{11})$
B_1T_2	$(1,\underline{6})$	$(3,2)$
B_1B_2	$(1,\underline{7})$	$(1,5)$

The equilibria in the game are (T_1T_2, t) and $T_1B_2, t)$.

Game D:

	t_1t_2	t_1b_2	b_1t_2	b_1b_2
T	$1,2,\underline{3}$	$\underline{1},2,\underline{3}$	$\underline{6},\underline{3},0$	$\underline{6},\underline{3},0$
B	$\underline{5},\underline{8},2$	$0,3,\underline{6}$	$5,\underline{8},2$	$0,3,\underline{6}$

τ

	t_1t_2	t_1b_2	b_1t_2	b_1b_2
T	$\underline{1},2,\underline{3}$	$\underline{1},2,\underline{3}$	$\underline{8},2,\underline{5}$	$\underline{8},2,\underline{5}$
B	$0,\underline{9},\underline{20}$	$0,3,\underline{6}$	$0,\underline{9},\underline{20}$	$0,3,\underline{6}$

β

The game has no equilibrium.

Game E:

	t_1t_2	t_1b_2	b_1t_2	b_1b_2
T	2,4,5	2,4,5	$\underline{3,8,2}$	$\underline{3,8,2}$
M	2,7,3	1,1,1	2,7,3	1,1,1
B	2,4,8	2,4,8	2,4,8	2,4,8

$\tau_1\tau_2$

	t_1t_2	t_1b_2	b_1t_2	b_1b_2
T	2,4,5	2,4,5	3,8,2	$\underline{3,8,2}$
M	4,0,5	1,1,1	4,0,5	1,1,1
B	2,4,8	2,4,8	2,4,8	2,4,8

$\tau_1\beta_2$

	t_1t_2	t_1b_2	b_1t_2	b_1b_2
T	2,4,5	2,4,5	3,8,2	3,8,2
M	2,7,3	1,1,1	2,7,3	1,1,1
B	27,9,3	27,9,3	27,9,3	27,9,3

$\beta_1\tau_2$

	t_1t_2	t_1b_2	b_1t_2	b_1b_2
T	2,4,5	2,4,5	3,8,2	3,8,2
M	4,0,5	1,1,1	4,0,5	1,1,1
B	27,9,3	27,9,3	27,9,3	27,9,3

$\beta_1\beta_2$

The equilibria in the game are $(T, b_1t_2, \tau_1\tau_1)$, $(T, b_1b_2, \tau_1\tau_1)$ and $(T, b_1b_2, \tau_1\beta_1)$.

Game F:

	$t_1t_2t_3$	$t_1t_2b_3$	$t_1m_2t_3$	$t_1m_2b_3$	$t_1b_2t_3$	$t_1b_2b_3$
T	11,5	$\underline{10,8}$	9,3	8,6	3,5	2,8
B	9,5	9,5	9,5	9,5	9,5	9,5

	$m_1t_2t_3$	$m_1t_2b_3$	$m_1m_2t_3$	$m_1m_2b_3$	$m_1b_2t_3$	$m_1b_2b_3$
	11,5	$\underline{10,8}$	9,3	8,6	3,5	2,8
	7,7	7,7	7,7	7,7	7,7	7,7

	$b_1t_2t_3$	$b_1t_2b_3$	$b_1m_2t_3$	$b_1m_2b_3$	$b_1b_2t_3$	$b_1b_2b_3$
	11,5	10,8	9,3	8,6	3,5	2,8
	5,9	5,9	5,9	5,9	$\underline{5,9}$	$\underline{5,9}$

The equilibria in the game are $(T, t_1 t_2 b_3)$, $(T, m_1 t_2 b_3)$, $(B, b_1 b_2 t_3)$ and $(B, b_1 b_2 b_3)$.

4.22 Let X and Y be two compact sets in $\mathbf{R}^m$, and let $f : X \times Y \to \mathbf{R}$ be a continuous function. We prove that the function $x \mapsto \min_{y \in Y} f(x, y)$ is also a continuous function.
Note that since f is a continuous function over compact sets it must attain a minimum. By the continuity of f, for every $\epsilon > 0$ there is $\delta > 0$ such that for every $(x_1, y_1), (x_2, y_2) \in X \times Y$ where $d\left((x_1, y_1), (x_2, y_2)\right) < \delta$ one has $\left|f(x_1, y_1), f(x_2, y_2)\right| < \epsilon$.
Let $x_1, x_2 \in X$ such that $d(x_1, x_2) < \delta$. Let $y_1, y_2 \in Y$ such that $f(x_1, y_1) = \min_{y \in Y} f(x_1, y)$ and $f(x_2, y_2) = \min_{y \in Y} f(x_2, y)$. Therefore, since $d\left((x_1, y_1), (x_2, y_1)\right) < \delta$,

$$\min_{y \in Y} f(x_2, y) \leq f(x_2, y_1) \leq f(x_1, y_1) + \epsilon = \min_{y \in Y} f(x_1, y) + \epsilon.$$

Similarly,

$$\min_{y \in Y} f(x_1, y) \leq f(x_1, y_2) \leq f(x_2, y_2) + \epsilon = \min_{y \in Y} f(x_2, y) + \epsilon.$$

Hence, $\min_{y \in Y} f(x, y)$ is continuous.

4.23 **Game A:** No strategy can be eliminated. The game has no value. The maxmin strategy of Player I is γ. The minmax strategy of Player II is b.
Game B: The strategies that are eliminated in a process of iterative elimination of dominated strategies are a, α, c, γ. The game has no value. The maxmin strategy of Player I is δ. The minmax strategies of Player II is b, d.
Game C: No strategy can be eliminated. The value of the game is 5. The maxmin strategy of Player I is γ. The minmax strategy of Player II is c.

4.24 We prove that in duopoly competition (Example 4.23) the pair of strategies (q_1^*, q_2^*) defined by $q_1^* = \frac{2-2c_1+c_2}{3}$, $q_2^* = \frac{2-2c_2+c_1}{3}$ is an equilibrium. We need to show that q_1^* is the best response of Player I to q_2^* and, similarly, q_2^* is the best response of Player II to q_1^*.

$$u_1(q_1, q_2^*) = q_1\left(2 - c_1 - q_1 - q_2^*\right) = -q_1^2 + (2 - c_1 - q_2^*) q_1.$$

Thus, $u_1(q_1, q_2^*)$ is a quadratic function in q_1, where the coefficient of

q_1^2 is -1. Thus the maximum is obtain in

$$q_1 = \frac{(2 - c_1 - q_2^*)}{2} = \frac{(2 - c_1 - \frac{2-2c_2+c_1}{3})}{2} = q_1 * .$$

In particular, q_1^* is the best response of Player I to q_2^*. The proof that q_2^* is the best response of Player II to q_1^*, is similar, so it was omitted.

4.25 **Proof. (Theorem 4.26)** Assume player i has a (weakly) dominant strategy s_i^*. Let $s_i \in S_i$ be a strategy of player i. Let $s'_{-i} \in S_{-i}$ be a strategy of $-i$ according to which $u_i(s_i^*, s'_{-i}) = \min_{s_{-i} \in S_{-i}} u_i(s_i^*, s_{-i})$. Then,

$$\min_{s_{-i} \in S_{-i}} u_i(s_i^*, s_{-i}) = u_i(s_i^*, s'_{-i}) \geq u_i(s_i, s'_{-i}) \geq \min_{s_{-i} \in S_{-i}} u_i(s_i, s_{-i}). \qquad (9)$$

Therefore, s_i^* is a maxmin strategy of player i.
In addition, for every $s_{-i} \in S_{-i}$ one has

$$u_i(s_i, s_{-i}) \leq u_i(s_i^*, s_{-i}) \qquad (10)$$

so s_i^* is the best reply of player i to every strategy vector of the other players. ■

4.26 **Proof. (Theorem 4.28)** Let G be a game in which every player i has a strategy s_i^* that strictly dominates all of his other strategies. By Exercise 4.25 and Definition 4.19, the strategy vector $(s_1^*, \ldots, s_n^*)$ is an equilibrium point of the game. Furthermore, since the dominance is strict, the weak inequality in Equation (10) should be changed to strong inequality and thus this equilibrium is unique.
In addition, by the first part of Exercise 4.25 and by changing the weak inequality in Equation (9) to strong inequality, the strategy vector $(s_1^*, \ldots, s_n^*)$ is a unique vector of maxmin strategies. ■

4.27 Let $G = (N, (S_i)_{i \in N}, (u_i)_{i \in N})$ be a game in strategic form, and let $\widehat{s}_i \in S_i$ be an arbitrary strategy of player i in this game. Let $\widehat{G}$ be the game derived from G by the elimination of strategy $\widehat{s}_i$. Hence, $\widehat{S}_j = S_j$ and $\widehat{S}_{-j} \subset S_{-j}$, for every player $j \neq i$.
Then,

$$\min_{s_{-j} \in S_{-j}} u_i(s_j, s_{-j}) \leq \min_{s_{-j} \in \widehat{S}_{-j}} u_i(s_j, s_{-j}) \quad \Rightarrow$$
$$\max_{s_j \in S_j} \min_{s_{-j} \in S_{-j}} u_i(s_j, s_{-j}) \leq \max_{s_j \in \widehat{S}_j} \min_{s_{-j} \in \widehat{S}_{-j}} u_i(s_j, s_{-j}).$$

That is, the maxmin value of player j,$j \neq i$, in the game $\widehat{G}$ is greater than or equal to his maxmin value in G.

The maxmin value of player i in game $\widehat{G}$ is not necessarily less than his maxmin value in G. In the following counterexample Player I is a row player and Player II is a column player, and $\widehat{G}$ is obtain by eliminating strategy B, however the maxmin of Player I in both game is 1.

G:

	a	b
A	1,0	1,0
B	2,1	0,1

$\widehat{G}$:

	a	b
A	1,0	1,0

4.28 In the example in previous exercise (A, a) is an equilibrium in $\widehat{G}$ but it is not an equilibrium in G.

4.29 Let $G = \big(N, (S_i)_{i \in N}, (u_i)_{i \in N}\big)$ be a game in strategic form. Let $\widehat{G}$ be the game derived from G by iterative elimination of K dominated strategies. Denote by G_k the game after the k-th elimination, in particular $\widehat{G} = G_K$ and $G = G_0$. We prove that every equilibriu, in $\widehat{G}$ is an equilibrium in G. By Theorem 4.32, every equilibrium in $\widehat{G} = G_K$ is an equilibrium in G_{K-1}. Assume that every equilibrium in $\widehat{G}$ is an equilibrium in G_k. Then, by theorem 4.32, every equilibrium in G_k is an equilibrium in G_{k-1}. Thus, and by the backward induction hypothesis, every equilibrium in $\widehat{G}$ is an equilibrium in G_{k-1}. In conclusion, by backward induction for $k = 0$ one has every equilibrium in $\widehat{G}$ is an equilibrium in $G_0 = G$, as needed.

4.30 In the next game (A, c) is the only equilibrium. However, A is dominated by C, and c is dominated by a.

	a	b	c
A	0,1	0,1	1,1
B	0,2	2,0	1,0
C	2,0	1,1	1,0

4.31 (a) The optimal strategy of Player I is T. The optimal strategy of Player II are $t_1 t_2$.

(b)

	t_1t_2	t_1b_2	b_1t_2	b_1b_2
T	12	12	15	15
B	8	10	8	10

(c)

	t_1t_2	t_1b_2	b_1t_2	b_1b_2	Min
T	12	12	15	15	12
B	8	10	8	10	8
Max	12	12	15	15	

The optimal strategy of Player I is T. The optimal strategies of Player II are t_1t_2, t_1b_2.

(d) In (a) we present the strategy t_1t_2 that is optimal in every subgame in particular it is optimal in the entire game. Whereas in (b) we present the strategies that are optimal in the entire game which include t_1t_2 and t_1b_2.

4.32 (a) $B = -A^T$ (where A^T is the transpose matrix of A).

(b)

		Player I	
		T	B
Player II	L	-3	2
	M	5	-8
	R	-7	-4

4.33 The value of the two-player zero-sum game given by the matrix A is 0. Then, it is not necessarily true that the value of the two-player zero-sum game given by the matrix $-A$ is also 0. In the counterexample which is given below the value of the two-player zero-sum game given by the matrix A is 0 while the value of the two-player zero-sum game given by the matrix $-A$ is -1.

	L	R	min
T	**0**	1	0
B	0	2	0
max	0	2	

A

	L	R	min
T	0	-1	-1
B	0	-2	-2
max	0	-1	

-A

4.34 Let A and B be two finite sets, and let $u : A \times B \to \mathbf{R}$ be an arbitrary function. We prove that $\max_{a\in A} \min_{b\in B} u(a,b) \leq \min_{b\in B} \max_{a\in A} u(a,b)$. Indeed,

$$\begin{aligned}
\min_{b\in B} u(a,b) &\leq u(a,b') && \forall b' \in B \Rightarrow \\
\max_{a\in A}\min_{b\in B} u(a,b) &\leq \max_{a\in A} u(a,b') && \forall b' \in B \Rightarrow \\
\max_{a\in A}\min_{b\in B} u(a,b) &\leq \min_{b'\in B}\max_{a\in A} u(a,b')
\end{aligned}$$

4.35 Game A:

	a	b	min
A	2	2	**2**
B	1	3	1
max	**2**	3	

The value is 2. The optimal strategy for Player I is A and for Player II is a.

Game B:

	a	b	c	min
A	1	2	3	1
B	4	3	0	0
max	4	3	3	

The game has no value.

Game C:

	a	b	c	d	min
A	3.5	3	4	12	3
B	7	5	6	13	**5**
C	4	2	3	0	0
max	7	**5**	6	13	

The value is 5. The optimal strategy for Player I is B and for Player II is b.

Game D:

	a	b	min
A	3	0	0
B	2	2	**2**
C	0	3	0
max	**3**	**3**	

The game has no value.

4.36 **Proof.** **(Theorem 4.48)** Let G be a two-player zero-sum game. Then $\left(s_I^*, s_{II}^*\right)$ is a saddle point if and only if

$$u\left(s_I^*, s_{II}^*\right) \geq u\left(s_I, s_{II}^*\right), \qquad \forall s_I \in S_I, \tag{11}$$

$$u\left(s_I^*, s_{II}^*\right) \leq u\left(s_I^*, s_{II}\right), \qquad \forall s_{II} \in S_{II}. \tag{12}$$

But Inequality (11) holds if and only if

$$u\left(s_I^*, s_{II}^*\right) \geq \max_{s_I \in S_I} u\left(s_I, s_{II}^*\right) \geq \min_{s_{II} \in S_{II}} \max_{s_I \in S_I} u\left(s_I, s_{II}\right) = \overline{v}.$$

In addition, Inequality (12) holds if and only if

$$u\left(s_I^*, s_{II}^*\right) \leq \min_{s_{II} \in S_{II}} u\left(s_I^*, s_{II}\right) \leq \max_{s_I \in S_I} \min_{s_{II} \in S_{II}} u\left(s_I, s_{II}\right) = \underline{v}.$$

But thus

$$u\left(s_I^*, s_{II}^*\right) \leq \underline{v} \leq \overline{v} = u\left(s_I^*, s_{II}^*\right) \Rightarrow v = \underline{v} = \overline{v}.$$

In particular, all the weak inequalities are actually equalities and therefore, s_I^* is an optimal strategy for Player I and s_{II}^* is an optimal strategy for Player II. ■

4.37 Let A and B be two finite-dimensional matrices with positive payoffs. Then the game

$$T = \begin{array}{|c|c|} \hline A & 0 \\ \hline 0 & B \\ \hline \end{array}$$

has no value. Indeed, $T_{ij} \geq 0$ (from the matrix positivity) and there is at least one zero at each row. Hence,

$$\underline{v} = \max_i \min_j T_{ij} = \max_i 0 = 0.$$

On the other hand, since A and B are positive, there is at least one positive value at each column. So, by the finite dimensions of the matrices, the maximum value over each column is positive. In particular,

$$\overline{v} = \min_j \max_i T_{ij} > 0 = \underline{v},$$

from which it follows that the game has no value.

4.38 (a) The equilibria obtained by backward induction in game A are $(af, c), (af, d)(bf, c)$ and (bf, d).
The equilibrium obtained by backward induction in game B is (b, dfg).

(b) Game A in strategic form:

	c	d
ae	3	2
af	3	3
be	3	3
bf	3	3

Game B in strategic form:

	egc	egd	ehc	ehd	fgc	fgd	fhc	fhd
a	1	1	8	8	-7	-7	0	0
b	5	4	5	4	5	4	5	4

(c) There are other Nash equilibria in addition to those found by a backward induction. In game A-(be, c) and (be, d), and in game B-(b, egd) and (b, ehd).

4.39 **Proof of Theorem 4.49 (Kuhn):** We will prove the claim by backward induction. If the game tree is comprised of a single vertex, then the unique strategy vector is $(\emptyset, \ldots, \emptyset)$ (so there are no available deviations), and it is therefore the unique Nash equilibrium. Assume by induction that the claim is true for each perfect information game in extensive form containing fewer than K vertices, and consider a game G with K vertices. Let x be a vertex all of whose children are leaves. Since the game has perfect information and perfect recall, the player choosing an action at vertex x (say player i) knows that the play of the game has arrived at that vertex (and not at a larger information set containing x) and he therefore chooses the leaf l giving him the maximal payoff. Consider the game tree obtaining by erasing the leaves following x and thus turning x into a leaf with a payoff equal to the payoff of l. The resulting game tree has fewer vertices than the original tree, so we can apply the induction hypothesis to it, i.e. there is Nash equilibrium s^{-x} in the resulting game. Let s^* be a strategy vector according to which the players follow s^{-x} in all vertices but x, whereas in x player i who control x chooses l. Since player i chooses l in x according to s^*, the resulting game and the original game are the same from the point of view of each player $j \neq i$, so he can not gain by deviating. Assume to the contrary that player i can gain by deviating to s_i. Then, by the maximality of l, player i can gain by deviating to s_i^l according to which at x he chooses l. But then he can gain by deviating to s_i^l in the resulting game, which contradicts the assumption that S^{-x} is Nash equilibrium in the resulting game.

4.40 As shown in the game tree there are two Nash equilibria that are obtained by backward induction: $(B_1B_2T_3, t_1b_2t_3b_4, \beta)$ and $(B_1B_2B_3, t_1b_2t_3b_4, \beta)$. $(B_1T_2B_3, t_1b_2t_3b_4, \beta)$ is an additional Nash equilibrium of this game.

4.41 In a two-player zero-sum game on the unit square where Player I's strategy set is $X = [0, 1]$ and Player II's strategy set is $Y = [0, 1]$.

(a)

$$\min_{y\in[0,1]} u(x,y) = \min_{y\in[0,1]} y(1-5x)+1+4x = \begin{cases} 1+4x & \forall x \in [0,0.2] \\ 2-x & \forall x \in [0.2,1]. \end{cases}$$
$$\Rightarrow \underline{v} = \max_{x\in[0,1]} \min_{y\in[0,1]} u(x,y) = 1.8.$$

$$\max_{x\in[0,1]} u(x,y) = \max_{x\in[0,1]} x(4-5y)+1+y = \begin{cases} 5-4y & \forall y \in [0,0.8] \\ 1+y & \forall y \in [0.8,1]. \end{cases}$$
$$\Rightarrow \overline{v} = \min_{y\in[0,1]} \max_{x\in[0,1]} u(x,y) = 1.8.$$

So, the value is $v = 1.8$. The optimal strategy for Player I is $x = 0.2$. The optimal strategy for Player II is $y = 0.8$.

(b)

$$\min_{y\in[0,1]} u(x,y) = \min_{y\in[0,1]} y(2-4x)+4 = \begin{cases} 4 & \forall x \in [0,0.5] \\ 6-4x & \forall x \in [0.5,1]. \end{cases}$$
$$\Rightarrow \underline{v} = \max_{x\in[0,1]} \min_{y\in[0,1]} u(x,y) = 4.$$

$$\max_{x\in[0,1]} u(x,y) = \max_{x\in[0,1]} -4yx+4+2y = 4+2y \qquad \forall y \in [0,1]$$
$$\Rightarrow \overline{v} = \min_{y\in[0,1]} \max_{x\in[0,1]} u(x,y) = 4.$$

So, the value is $v = 4$. The optimal strategies for Player I are $x \in [0,0.5]$. The optimal strategy for Player II is $y = 0$.

4.42 Consider a two-player non-zero-sum game on the unit square in which Player I's strategy set is $X = [0,1]$, Player II's strategy set is $Y = [0,1]$. Assume that (x^*,y^*) is a unique equilibrium, where $x^*, y^* \in (0,1)$. We prove that the equilibrium payoff to each player equals his maxmin value. We show that the claim holds for Player I, the proof that it also holds for Player II is identical.
Let $u(x,y) = axy+bx+cy+d$ be the payoff function of Player I for every $x,y \in [0,1]$.
Since (x^*,y^*) is an equilibrium, for every $x \in [0,1]$

$$u(x,y^*) \le u(x^*,y^*) \Rightarrow$$
$$axy^* + bx + cy^* + d \le ax^*y^* + bx^* + cy^* + d \Rightarrow$$
$$x(ay^*+b) \le x^*(ay^*+b).$$

However, since $x^* \in (0,1)$, $ay^* + b = 0$ and $u(x, y^*) = u(x^*, y^*)$ for every $x \in [0,1]$. Furthermore, since the equilibrium is unique, one of the following holds: either (1) $a > 0$, $b < 0$ and $a + b > 0$; or (2) $a < 0$, $b > 0$ and $a + b < 0$.
Let $x \in [0,1]$, then

$$\begin{aligned}\min_{y\in[0,1]} u(x,y) &= \min_{y\in[0,1]} (ax+c)y + bx + d \\ &= \begin{cases} bx+d & ax+c \geq 0 \\ (ax+c)+bx+d & ax+c \leq 0 \end{cases} \\ &= \begin{cases} bx+d & ax+c \geq 0 \\ (a+b)x+c+d & ax+c \leq 0 \end{cases}.\end{aligned}$$

Therefore, if (1) holds then $bx + c$ is monotonic decreasing, and $(a + b)x + c + d$ is monotonic increasing from which it follows that the maximum is obtained at x' satisfying $ax' + c = 0$. Similarly, if (2) holds then $bx + c$ is monotonic increasing, and $(a+b)x + c + d$ is monotonic decreasing from which it follows that the maximum is obtained at x' satisfying $ax' + c = 0$. Either case, $u(x', y)$ is independent of y so

$$\max_{x\in[0,1]} \min_{y\in[0,1]} u(x,y) = u(x', y^*) = u(x^*, y^*),$$

as claimed.

4.43 The strategic form game described in the exercise is given by $N = \{1, 2, \ldots, 50\}$ a set of players; $S_i = \{0, 1, \ldots, 100\}$ a strategy set for player i for every $i \in N$; and

$$u_i(x_1, \ldots, x_{50}) = \begin{cases} \frac{1000}{\#\left|\arg\min_{j\in N} |x_j - \frac{2}{3}x|\right|} & \text{if } i \in \arg\min_{j\in N} |x_j - \frac{2}{3}x| \\ 0 & \text{otherwise .} \end{cases}$$

where x is the average of all the numbers x_j written by the players.

There are two Nash equilibria in the game. The first is the strategy vector in which all players write 1 which case $\frac{2}{3}x = \frac{2}{3}$ so each player gets $\frac{1000}{50}$ whereas by deviating he gets zero. The second is strategy vector according to which all players write zero which case $\frac{2}{3}x = 0$ and again each player gets $\frac{1000}{50}$ whereas by deviating he gets zero.

We show that each other strategy vector in is not a Nash equilibrium. Let $v = (x, x, \cdots, x)$ be a strategy vector such that each player writes the same number $x > 1$. The payoff for every player under v is

$u_i(x, x, \cdots, x) = 1000/50$ whereas by deviating to $x - 1$ a player can gain 1000.
We next consider the strategy vector $(x_1, x_2, \cdots, x_{50})$ such that each player writes one of two different numbers a or b (each number is written at least once). Denote by N_a and N_b the sets of players that write a and b, respectively. We first handle the case where $|b - \frac{2}{3}x| = |a - \frac{2}{3}x|$. In particular, each player receives $\frac{1000}{50}$. However, at least one of the sets N_a or N_b includes more than one player, say N_a. Then each player $i \in N_a$ can guarantee that he still wins by deviates to b, and the number of players that win decreases so his payoff increases.
Otherwise, w.l.o.g. $|b - \frac{2}{3}x| < |a - \frac{2}{3}x|$. Then, each player $i \in N_a$ receives zero, and he gets positive payoff by deviating to b.
Finally, let $v = (x_1, x_2, \cdots, x_{50})$ be a strategy vector that includes at least three different numbers. Then, there must be at least one player whose payoff under v is zero. If that player deviates then $\frac{2}{3}$ of the average will changed by at most $\frac{2}{3} \cdot \frac{100}{50} = 1.33$ hence he can deviates to a number under which he wins the game.

4.44 (a) The next game has only one equilibrium in which all players bet.

	Don't bet	Bet.
Don't bet	$\frac{1}{3}, \frac{1}{3}, \frac{1}{3}$	$\frac{1}{6}, \frac{2}{3}, \frac{1}{6}$
Bet.	$\frac{2}{3}, \frac{1}{6}, \frac{1}{6}$	$\frac{11}{24}, \frac{11}{24}, \frac{1}{12}$

Don't bet

	Don't bet	Bet.
Don't bet	$\frac{1}{6}, \frac{1}{6}, \frac{2}{3}$	$\frac{1}{12}, \frac{11}{24}, \frac{11}{24}$
Bet.	$\frac{11}{24}, \frac{1}{12}, \frac{11}{24}$	$\frac{1}{3}, \frac{1}{3}, \frac{1}{3}$

Bet

(b) The following game also has only one equilibrium in which all players bet.

	Don't bet	Bet.
Don't bet	$\frac{1}{3}, \frac{1}{3}, \frac{1}{3}$	$\frac{1}{6}, \frac{2}{3}, \frac{1}{6}$
Bet.	$\frac{2}{3}, \frac{1}{6}, \frac{1}{6}$	$\frac{5}{12}, \frac{5}{12}, \frac{1}{6}$

Don't bet

	Don't bet	Bet.
Don't bet	$\frac{1}{6}, \frac{1}{6}, \frac{2}{3}$	$\frac{1}{6}, \frac{5}{12}, \frac{5}{12}$
Bet.	$\frac{5}{12}, \frac{1}{6}, \frac{5}{12}$	$\frac{1}{3}, \frac{1}{3}, \frac{1}{3}$

Bet

4.45 (a) The Partnership Game is given by $S_i = [0, \infty)$ a strategy set for player i for every $i \in \{1, 2\}$; and $u_i(e_1, e_2) = \frac{e_1^{\alpha_1} e_2^{\alpha_2}}{2} - ce_i$ a payoff function for player i .

(b) First note that if one of the players does not put an effort in the business then the opponent best response is to put no effort as well, so $(0,0)$ is an equilibrium. Assume next that player $-i$ chooses to put an effort of $e_{-i} > 0$ in the business. To find the best response e_i of player i we search for the maximum of $u_i(e_i, e_{-i})$ using derivatives

$$\frac{\partial u_i\left(e_i, e_{-i}\right)}{\partial e_i} = \alpha_i e^{\alpha_i - 1} e^{\alpha_{-i}} - c = 0 \Rightarrow \frac{\alpha_i}{e_i} = \frac{c}{e_i^{\alpha_i} e_{-i}^{\alpha_{-i}}}.$$

and

$$\frac{\partial^2 u_i\left(e_i, e_{-i}\right)}{\partial e_i^2} = \left(\alpha_i - 1\right) \alpha_i e^{\alpha_i - 2} e^{\alpha_{-i}} < 0.$$

Then one can deduce that the Nash equilibrium set of the game contains every strategy vector (e_1, e_2) such that $\alpha_1 e_2 = \alpha_2 e_1$.

4.46 (a) The set of players is $N = \{1, 2, \cdots, 60\}$. For every player $i \in N$ the strategy set is given by $S_i = \{\text{Mountain View}, \text{Cupertino}\}$. For every player i and strategy vector s denote by M_s the number of cars that chooses to take the road via Mountain View, then the amount of travel time of player i is given by

$$u_i\left(s_1, s_2, \cdots, s_{60}\right) = \begin{cases} 52 + 1.1\left(M_s\right) & \text{if } s_i = \text{ Mountain View} \\ 52 + 1.1\left(60 - M_s\right) & \text{if } s_i = \text{ Cupertino.} \end{cases}$$

(b) Every strategy vector s such that $M_s = 30$ is an equilibrium. At these equilibria, the time the trip takes at an early morning hour is $52 + 1.1 \cdot 30 = 85$, whereas, if some car deviates, then its travel time increases to $52 + 1.1 \cdot 31 = 86.1$.
Note that, for every strategy s for which $M_s < 30$, any car that travel via Cupertino is better off deviates so its travel time decreases from $52 + 1.1\left(60 - n_s\right) > 85$ to $52 + 1.1\left(n_s + 1\right) \leq 85$, thus s is not an equilibrium. Similarly, every strategy s for which $M_s > 30$ is not an equilibrium.

(c) Let s be a strategy in the new game. Denote by MD_s the number of cars that take the road via Mountain View directly to San Jose (without using the new road) and by MC_s the number of cars that take the new road via Mountain and Cupertino. Then, in an equilibrium, the amount of time that takes via each possible way must be the same otherwise there is a car that is better off deviating. Thus

$$\begin{aligned}
&(1 + MD_s + MC_s) + (51 + 0.1MD_s) = \\
&(1 + MD_s + MC_s) + (10 + 0.1MC_s) + (1 + 60 - MD_s) = \\
&\big(51 + 0.1(60 - MD_S - MC_s)\big) + (1 + 60 - MD_s) \\
&\Rightarrow MD_s = 20, MC_s = 20.
\end{aligned}$$

That is, 20 cars take each way.

(d) The construction of the additional road doesn't improve the travel time, moreover, it increases the travel time for all travellers from 85 to 94.

4.47 (a) Before the new road is constructed, the travel time of each truck between its headquarters and Chicago is $20 + 2\sqrt{10}$.

(b) The strategic-form game corresponds to the situation after the construction of the new road is given as follows. There are two players: Player I- Davis' manager and Player II- Roland's manager. Each player $i \in \{1, 2\}$ chooses how many truck to send on the road connecting his company with Chicago, i.e the strategy set $S_i = \{0, 1, \cdots, 10\}$. The the total travel time to Chicago of all the trucks in the fleet of player i is

$$\begin{aligned}
u_i(x_i, x_{-i}) = x_i &\left(20 + 2\sqrt{x_i + 10 - x_{-i}}\right) \\
&+ (10 - x_i)\left(20.2 + 2\sqrt{10 - x_i + x_{-i}}\right).
\end{aligned}$$

(c) The strategy vector in which both Davis and Roland send their entire fleets directly to Chicago, ignoring the new road, is not a Nash equilibrium, whereas

$$u_1(10, 10) = 263.2 > 260.8 = u_1(9, 10).$$

(d) The strategy vector in which both Davis and Roland send six drivers directly to Chicago and four via the new road is an equilibrium. Indeed, if a player decides to deviate by sending m drivers from the six via the new road instead of directly, then the time increases by

$$0.2m + 2\left(\sqrt{10 + m} - \sqrt{10 - m}\right) > 0.$$

On the other hand, if a player decides to deviate by sending m drivers from the four directly to Chicago, then the time increases by

$$-0.2m + 2\left(\sqrt{10 + m} - \sqrt{10 - m}\right) > 0.$$

The total travel time of the trucks of each company in this equilibrium is

$$6\left(20+2\sqrt{10}\right)+4\left(20.2+2\sqrt{10}\right).$$

In particular, the construction of a new road increases the total travel time.

(e) The game has another equilibrium according to which Davis and Roland send five drivers directly to Chicago and five via the new road.

		Roland										
		0	1	2	3	4	5	6	7	8	9	10
Davis	0	265,265	268,262	271,261	274,260	277,261	280,262	282,265	285,269	287,274	289,281	291,289
	1	262,268	265,265	268,263	270,262	272,262	274,263	276,265	278,269	280,273	281,279	282,287
	2	261,271	263,268	265,265	267,263	268,263	270,264	271,265	273,268	274,272	275,278	275,285
	3	260,274	262,270	263,267	265,265	266,264	267,264	268,265	269,268	269,271	269,276	270,283
	4	261,277	262,272	263,268	264,266	264,264	265,264	265,265	266,267	266,270	266,274	265,280
	5	262,280	263,274	264,270	264,267	264,265	264,264	264,265	264,266	264,269	263,272	262,278
	6	269,282	265,276	265,271	265,268	265,265	265,264	264,264	263,265	263,267	261,270	260,275
	7	269,288	269,278	269,273	268,269	267,266	266,264	265,263	264,264	263,266	261,268	259,272
	8	274,287	273,280	273,274	271,269	270,266	269,264	267,263	266,263	264,264	262,266	260,269
	9	281,289	279,281	278,275	276,269	274,266	272,263	270,261	268,261	266,261	263,263	261,266
	10	289,291	287,282	285,275	283,270	280,265	278,262	275,260	272,259	269,260	266,261	263,263

4.48 (a) The game is given by a set of players $N = \{1, 2, ..., a\}$; a set of action $S_i = \{\text{northern road, southern road}\}$ of player i for every $i \in N$; and a payoff function of player $i \in N$ defined by

$$u_i(s) = \begin{cases} 1 & s_i = \text{northern road} \\ \frac{N(s)}{1000} & s_i = \text{southern road,} \end{cases}$$

where $N(s)$ is the the number of vehicles using southern road according to s.

(b) The Nash equilibria of this game are all $s \in S$ such that $N(s) = \min\{1000, a\}$ or $N(s) = \min\{999, a\}$. Indeed, if $N(s) > 1000$ then the travel time on the northern road is shorter than the travel time on the southern road and thus a driver on the southern road is better of deviating. If $N(s) < 999$ and there is a driver on the northern road, then he is better of deviating to the northern. The average time is $\min\{\frac{a}{1000}, 1\}$.

(c) Suppose that the police can order each driver to use a specific road. Denote by N the number of cars that will use the northern road, then the average travel time from Liverpool to Manchester is

$$AT(N) = \frac{1}{a}\left(N + \frac{(a-N)^2}{1000}\right).$$

By differentiating the function $T(N)$ and equating its directional derivatives to 0, we deduce that the optimal N that minimizing $T(N)$ is $N = \min\{a - 500, 0\}$. That is, $N = \min\{a - 500, 0\}$ cars will use the northern road, and $\min\{500, a\}$ cars that will use the southern road. The average time of the trip from Liverpool to Manchester in this case is either $\frac{a}{1000}$ if $a \leq 500$ or $1 - \frac{250}{a}$ if $a > 500$.

(d) The price of anarchy in this game as a function of a is $1 - \frac{250}{a} - 1 = -\frac{250}{a}$ if $a > 1000$; $1 - \frac{250}{a} - \frac{a}{1000}$ if $500 < a \leq 1000$; and $\frac{a}{1000} - \frac{a}{1000} = 0$ if $a < 500$.

4.49 (a) There are two players in the game, Player I- Pete's Coffee and Player II- Caribou Coffee. The set of strategies of player i is $S_i =$

$[0,1]$, for every $i \in \{1,2\}$. The payoff function is

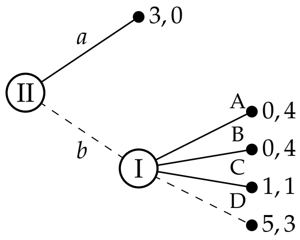
$$u_i(s_1, s_2) = \begin{cases} \frac{1}{2}(s_{-i} - s_i) + s_i & \text{if } s_i < s_{-i} \\ \frac{1}{2} & \text{if } s_i = s_{-i} \\ \frac{1}{2}(s_i - s_{-i}) + (1 - s_i) & \text{if } s_i > s_{-i}. \end{cases}$$

(b) For every strategy vector (s_1, s_2) such that $s_1 \neq s_2$ each player can gain by decreasing the gap between their strategies, so it cannot be an equilibrium.
For every strategy vector (s_1, s_2) such that $s_1 = s_2 \neq \frac{1}{2}$ each player can gain by deviating to $\frac{1}{2}$, so it is not an equilibrium.
Finally, under the strategy vector $(\frac{1}{2}, \frac{1}{2})$ both players receive $\frac{1}{2}$ and if either one of them deviates he receives less than $\frac{1}{2}$, so it is an equilibrium.

(c) We prove that if three chains were to compete for a location in Cambridge, the resulting game would have no equilibrium. For every strategy vector (s_1, s_2, s_3) such that $s_1 = s_2 = s_3$, the players receive $\frac{1}{3}$ so each player can gain by deviating to any location differs than s_1 which is close to $\frac{1}{2}$.
Otherwise, either the rightmost chain or the leftmost chain is located alone so it can gain by decreasing the gap to the chain that is close to it, thus it is not an equilibrium as well.

4.50 (a) Game A cannot be represented as an extensive-form game with perfect information. Assume to the contrary that it can be represented. Then, the extensive-form game has one vertex controlled by Player II (with two actions a and b). This vertex must be the root of the tree since there is no strategy s_1 of Player I in which $\big(u_1(s_1, a), u_2(s_1, a)\big) = \big(u_1(s_1, b), u_2(s_1, b)\big)$. Then, under each action chosen by Player II there are two different payoff vectors that correspond to two actions taken by Player I. If Player II chooses a then Player I chooses between $(1,1)$ and $(3,0)$. Otherwise, if Player II chooses b then Player I chooses between $(5,3)$ and $(0,4)$. In particular there must be a strategy of Player I in which the payoff vector is either (3.0) (in case Player II choses a) or $(0,4)$ (in case Player II chooses b), a contradiction.

(b) The following extensive-form game corresponds to Game B.



4.51 Let Γ be a game in extensive form. The agent-form game derived from Γ is a strategic-form game in which a player in the game is defined by a couple (i, U_i) for each player i and each information set U_i of player i. The set of strategies of player (i, U_i) is $A(U_i)$. The strategy vector $\sigma = (\sigma_i)_{i \in N}$ in Γ corresponds to the strategy vector $(\sigma_i(U_i))_{\{i,U_i\}}$ in the agent-form game. The payoff function of player (i, Ui) in the agent-form game is the payoff function of player i in the game Γ.
Assume that $\sigma = (\sigma_i)_{i \in N}$ is a Nash equilibrium in the game Γ and assume to the contrary that the corresponding strategy vector $(\sigma_i(U_i))_{\{i,U_i\}}$ is not a Nash equilibrium in the agent-form game derived from Γ. Thus there is a player (i, U_i) that can gain by deviating to $\sigma_i'(U_i)$. Therefore, player i can gain by deviating in Γ to a strategy according to which he follows $\sigma_i'(U_i)$ in the information set U_i, otherwise he follows σ_i, a contradiction.

Chapter 5: Mixed strategies

5.1 Let S be a finite set and $\Delta(S) = \left\{p : S \to [0,1] \mid \sum_{s \in S} p(s) = 1\right\}$ be the simplex over S.
We first prove that $\Delta(S)$ is convex. Let $p, q \in \Delta(S)$ and $\alpha \in [0,1]$. Then $\alpha p(s) + (1-\alpha)q(s) \geq 0$ since $p(s), q(s) \geq 0$ for every $s \in S$ and $\alpha \geq 0$. In addition,

$$\sum_{s \in S} \left(\alpha p\left(s\right) + \left(1-\alpha\right) q\left(s\right)\right) = \alpha \sum_{s \in S} p(s) + (1-\alpha) \sum_{s \in S} q(s) = \alpha + (1-\alpha) = 1.$$

Hence, $\alpha p + (1-\alpha)q \in \Delta(S)$ and $\Delta(S)$ is convex.
We next prove that $\Delta(S)$ is compact. The simplex is bonded by the unit cube. Moreover, let $p_1, p_2, ... \in \Delta(S)$ be a series such that $\lim_{n\to\infty} p_n = p$. For every $s \in S$ and every $n \in \mathbf{N}$, $p_n(s) \geq 0$, hence $p(s) = \lim_{n\to\infty} p_n(s) \geq 0$. Finally,

$$\sum_{s \in S} p(s) = \sum_{s \in S} \lim_{n\to\infty} p_n(s) = \lim_{n\to\infty} \sum_{s \in S} p_n(s) = \lim_{n\to\infty} 1 = 1,$$

where the second equality follows the fact that the limit of a sum over non-negative real numbers is the sum of the limits. Thus, $p \in \Delta(S)$

and $\Delta(S)$ is closed. In conclusion, $\Delta(S)$ is closed and bounded so it is compact.

5.2 Let $A \subseteq \mathbf{R}^n$ and $B \subseteq \mathbf{R}^m$ be two compact sets. We prove that the product set $A \times B \subset R^{n+m}$ is a compact set as well.
A and B are bounded in $\mathbf{R}^n$ and $\mathbf{R}^m$, respectively. In particular, there are two balls where the origin is their center, one is in R^n and of radius R_A and the second one is in R^m and of radius R_B. The first ball contains A and the second one contains B. Hence, $A \times B$ is contained in a ball in $\mathbf{R}^{n+m}$ of which its center is the origin and of radius $\max\{R_A, R_B\}$ so $A \times B$ is bounded.
Next, let $(a_1, b_1), (a_2, b_2), ... \in A \times B$ be a series such that $(a, b) = \lim_{n\to\infty}(a_n, b_n)$. Then, by the closeness of A and B, $a = \lim_{n\to\infty} a_n \in A$ and $b = lim_{n\to\infty} b_n \in B$. In particular $(a, b) \in A \times B$, so $A \times B$ is closed in addition to bounded. That is, $A \times B$ is compact.

5.3 Let $A \subseteq R^n$ and $B \subseteq R^m$ be two convex sets. We prove that the product set $A \times B \subseteq R^{n+m}$ is a convex set.
Let $(a_1, b_1), (a_2, b_2) \in A \times B$, and $\alpha \in [0, 1]$. Then,

$$\alpha(a_1, b_1) + (1-\alpha)(a_2, b_2) = \left(\alpha a_1 + (1-\alpha)a_2, \alpha b_1 + (1-\alpha)b_2\right).$$

Since A is convex, $\alpha a_1 + (1-\alpha)a_2 \in A$. Similarly, by the convexity of B, $\alpha b_1 + (1-\alpha)b_2 \in B$. Thus, $\alpha(a_1, b_1) + (1-\alpha)(a_2, b_2) \in A \times B$, and $A \times B$ is a convex set.

5.4 Let $f : \Sigma \to \mathbf{R}$ be a multilinear function. That is

$$f(\sigma_1, \cdots, \sigma_n) = \sum_{s \in S} a(s_1, \cdots, s_n)\sigma(s_1) \cdot \sigma(s_2) \cdots \sigma(s_n).$$

We show that f is a continuous function. First, for every $i \in N$ and $s \in S$, the function $g_i^s : \Sigma \to \mathbf{R}$ where $g_i^s(\sigma_1, \cdots, \sigma_n) = \sigma_i(s_i)$ is a projection function which is a continuous function. In addition, the function $g_0^s(\sigma_1, \cdots, \sigma_n) = a(s_1, \cdots, s_n)$ is a constant function and thus it is a continuous function. Therefore, the function $g^s(\sigma_1, \cdots, \sigma_n) = a(s_1, \cdots, s_n)\sigma(s_1) \cdot \sigma(s_2) \cdots \sigma(s_n)$ is a continuous function as a product of continuous functions (since $g^s = \Pi_{i=0}^n g_i^s$). Finally, f as a finite sum of continuous functions ($f = \sum_{s\in S} g^s$) is also continuous.

5.5 We prove that for every player i the set of extreme points of player i's collection of mixed strategies is his set of pure strategies.

First, we show that the set of extreme points of player i's collection of mixed strategies is contained in his set of pure strategies. Let $(x_1, x_2, \ldots, x_M) \in \Sigma_i$ such that for some $m \in \{1, \ldots, M\}$ one has $0 < x_m < 1$. Then $(x_1, x_2, \ldots, x_M)$ is a linear combination of two other points in Σ as follows

$$\begin{aligned}
&(x_1, \ldots, x_{m-1}, x_m, x_{m+1}, \ldots, x_M) \\
&= x_m \,(0, \ldots, 0, 1, 0, \ldots, 0) \\
&+ (1 - x_m) \left(\frac{x_1}{1-x_m}, \ldots, \frac{x_{m-1}}{1-x_m}, 0, \frac{x_{m+1}}{1-x_m}, \ldots, \frac{x_M}{1-x_m} \right).
\end{aligned}$$

Hence $(x_1, x_2, \ldots, x_M)$ is not an extreme point of player i's collection of mixed strategies.
It is left to show that every pure strategy is an extreme point. We prove that s_1 is an extreme point, the prove for all other pure strategies is similar, and thus it is omitted. Assume to the contrary that s_1 is not an extreme point, so it is a a linear combination of two other points in Σ,

$$(1, 0, \ldots, 0) = \alpha(x_1, \cdots, x_M) + (1-\alpha)(y_1, \cdots, y_M)$$

where $0 < \alpha < 1$ and $(x_1, \cdots, x_M) \neq (y_1, \cdots, y_M)$. However, for every $i > 1$, since $\alpha x_i + (1-\alpha) y_i = 0$ and $\alpha \in (0,1)$ then $x_i = y_i = 0$ from which it follows that $x_1 = y_1 = 1$. Therefore, $(x_1, \cdots, x_M) = (y_1, \cdots, y_M)$, a contradiction.

5.6 Consider a two-player zero-sum game over the unit square with bilinear payoff function. Namely, the set of strategies of each player is $[0,1]$, and the payoff function is

$$u(x, y) = ax + by + cxy + d \quad \forall x, y \in [0,1],$$

where $a, b, c, d \in \mathbf{R}$.
Then it is actually the extension to mixed strategies of the following zero-sum two-player game

		Player II	
		y	1-y
Player I	x	$a+b+c+d$	$a+d$
	1-x	$b+d$	d

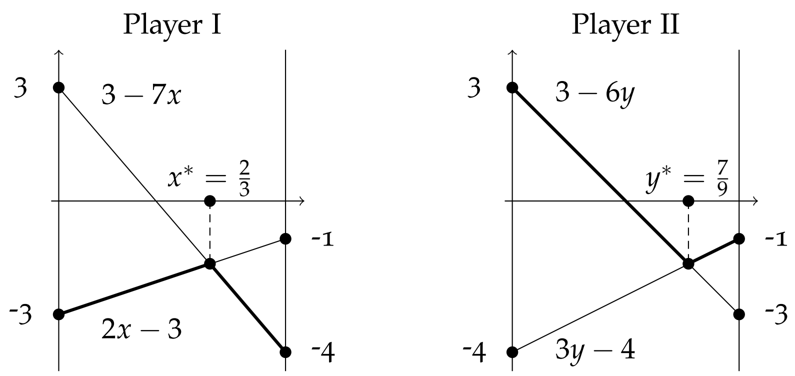
5.7 Let σ_{-i} be a vector of mixed strategies of the all the players accept for player i. The best reply of player i is the mixed strategy of player i that maximizes player i's payoff function. By Exercise 5.1, the set of mixed strategies of player i is compact and convex. By Exercise 5.4, the payoff function of player i is continuous. In conclusion the function attains its maximum at some extreme point. By Exercise 5.5, the set of extreme points of player i's collection of mixed strategies is his set of pure strategies. Hence, one can conclude that player i has a best reply to σ_{-i} that is a pure strategy.

5.8 In all the games in this exercise the set of mixed strategies of Player I is $\Sigma_1 = \Big\{ \big[x(T),(1-x)(B)\big] \,|x \in [0,1]\Big\}$ and the set of mixed strategies of Player II is $\Sigma_2 = \big\{[y(L),(1-y)(R)]|y \in [0,1]\big\}$. We next present the value of each game.

Game A:

		Player II		
		y	1-y	
Player I	x	-1	-4	3y-4
	1-x	-3	3	3-6y
		2x-3	3-7x	

We present in the following graphs the payoff of each player with regards to the mixed strategy he chooses. The thick line plots in the left-hand graph representing the minimum payoff that Player I can receive if he plays x. Similarly, the thick line plots in the right-hand graph representing the maximum payoff that Player II can pay if he plays y.

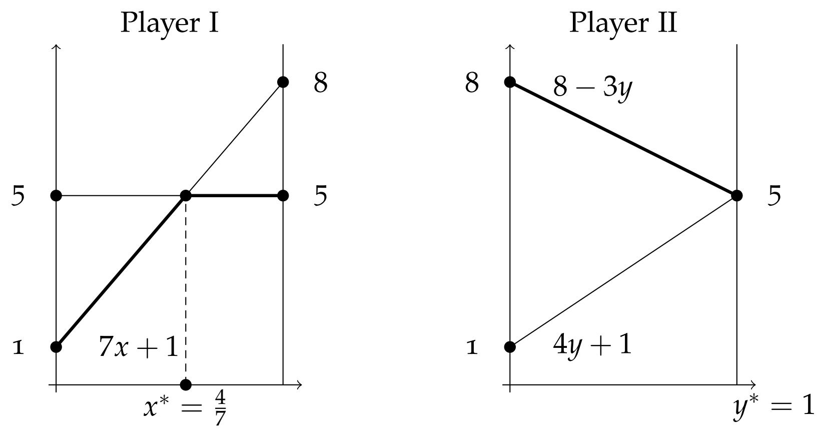


Therefore, the value of Game A is $v = -\frac{5}{3}$. The optimal strategy of Player I is $[\frac{2}{3}(T), \frac{1}{3}(B)]$. The optimal strategy of Player II is $[\frac{7}{9}(L), \frac{2}{9}(R)]$.

Game B:

		Player II		
		y	1-y	
Player I	x	5	8	-3y+8
	1-x	5	1	4y+1
		5	7x+1	

We present it in the following graphs:



The value of Game B is $v = 5$. The set of optimal strategies of Player I is $\left\{[x^*(t),(1-x^{)}(B)\|x^* \in [\frac{4}{7},1]\right\}$. The optimal strategy of Player II is L.

Game C:

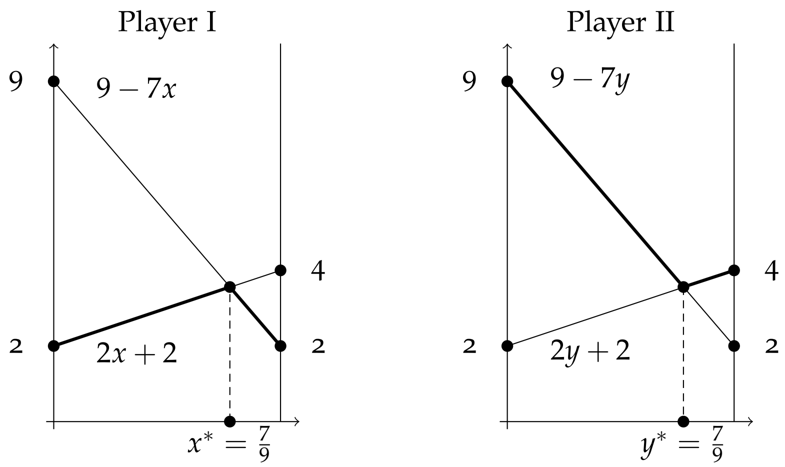
		Player II	
		L	R
Player I	T	5	4
	B	2	3

Strategy B of player I is strictly dominated by T and thus can be eliminated. Therefore, strategy L is strictly dominated by R so it also can be eliminated. In particular, the value of Game C is $v = 4$. The optimal strategy of Player I is T. The optimal strategy of Player II is R.

Game D:

		Player II		
		y	1-y	
Player I	x	4	2	2y+2
	1-x	2	9	-7y+9
		2x+2	-7x+9	

We present it in the following graphs:



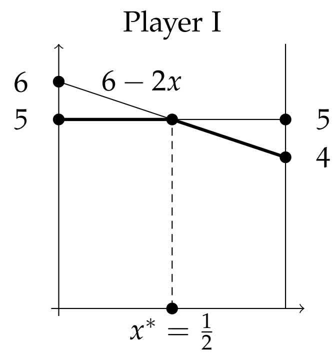
The value of Game D is $v = \frac{32}{9}$. The optimal strategy of Player I is $[\frac{7}{9}(T), \frac{2}{9}(B)]$. The optimal strategy of Player II is $[\frac{7}{9}(L), \frac{2}{9}(R)]$.

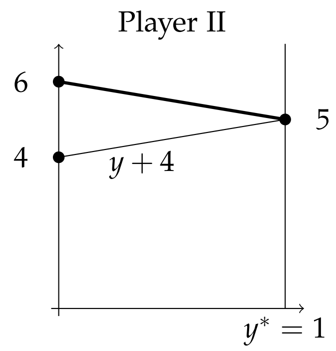
Game E:

		Player II		
		y	1-y	
Player I	x	5	4	y+4
	1-x	5	6	-y+6
		5	-2x+6	

We present it in the following graphs:

$6 - y$

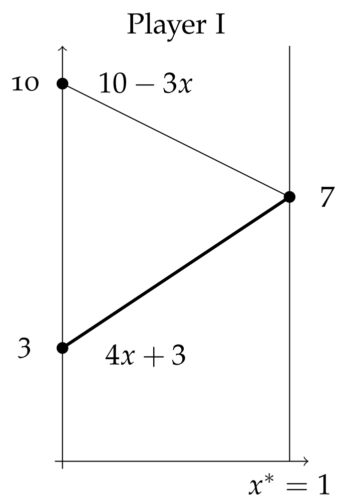


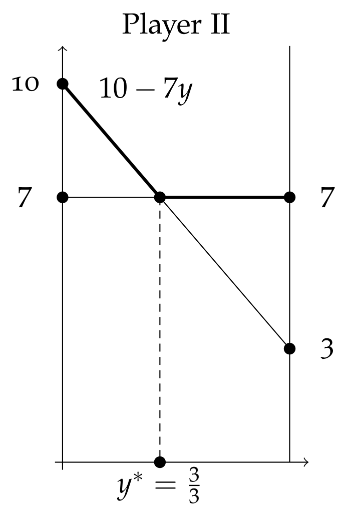


The value of Game E is $v = 5$. The set of optimal strategies of Player I is $\left\{[x^*(T), (1-x^*)(B) \| x^* \in [0, \frac{1}{2}]\right\}$. The optimal strategy of Player II is $y^* = 1$.

Game F:

		Player II		
		y	1-y	
Player I	x	7	7	7
	1-x	3	10	-7y+10
		4x+3	-3x+10	





The value of Game F is $v = 7$. The optimal strategy of Player I is T ($x^* = 1$). The set of optimal strategies of Player II is $\left\{ [y^*(L), (1-y^*)(R)] \, | y^* \in \left[\frac{3}{7}, 1\right] \right\}$.

5.9 **Game A:**

		Player II		
		y	1-y	
Player I	x_1	2	6	6-4y
	x_2	5	5	5
	x_3	7	4	4+3y
		$2x_1 + 5x_2 + 7x_3$	$6x_1 + 5x_2 + 4x_3$	

The value of the game is $v = 5$. The set of optimal strategies of Player II is $\left\{ [y^*(L), (1-y^*)(B) \| y^* \in [\frac{1}{4}, \frac{1}{3}] \right\}$. The optimal strategy of Player I is M (indeed, if $x_1 \leq x_3$ and $x_1 > 0$ then, by playing L, Player II can guarantees that the payoff is less than 5. Similarly, if $x_1 < x_3$, Player II can guarantees that the payoff is less than 5, by playing R).

Game B:
First note that Strategy b can be eliminated since it is strictly dominated by the mix strategy $[\frac{1}{2}(a), \frac{1}{2}(d)]$. Similarly, Strategy c can be eliminated since it is strictly dominated by the mix strategy $[0.27(a), 0.73(d)]$. Therefore, the following game is obtained.

		Player II		
		y	1-y	
Player I	x	15	-8	23y-8
	1-x	-3	8	8-11y
		18x-3	8-16x	

The value of Game B is $v = \frac{48}{17}$. The optimal strategy of Player I is $[\frac{11}{34}(d), \frac{23}{34}(a)]$. The optimal strategy of Player II is $[\frac{8}{17}(L), \frac{9}{17}(R)]$.

Game C:
Strategy a can be eliminated since it is strictly dominated by c. Strategy b can be eliminated since it is strictly dominated by the mix strategy $[\frac{1}{2}(c), \frac{1}{2}(d)]$. Therefore, the next game is obtained.

		Player II		
		y	1-y	
Player I	x	4	0	4y
	1-x	-5	6	6-11y
		9x-5	6-6x	

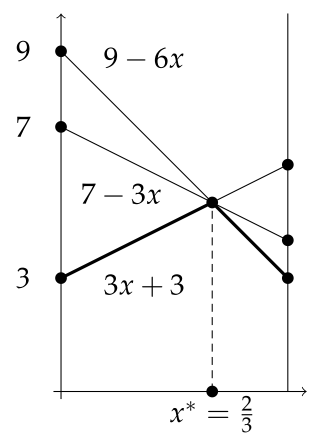
The value of Game C is $v = \frac{8}{5}$. The optimal strategy of Player I is $[\frac{11}{15}(T), \frac{4}{15}(B)]$. The optimal strategy of Player II is $[\frac{2}{5}(c), \frac{3}{5}(d)]$.

Game D:

		Player II			
		y_1	y_2	$1 - y_1 - y_2$	
Player I	x	6	4	3	$3 + (3y_1 + y_2)$
	1-x	3	7	9	$9 - 2(3y_1 + y_2)$
		3+3x	7-3x	9-6x	

Following the next graph the optimal strategy of Player I is $[\frac{2}{3}(T), \frac{1}{3}(B)]$. Hence the value is $v = 5$.

Player I



It is left to find the optimal strategy of Player II.

$$\max\left\{3 + \left(3y_1 + y_2\right), 9 - \left(3y_1 + y_2\right)\right\} \Leftrightarrow 3y_1 + y_2 = 2 \Leftrightarrow y_2 = 2 - 3y_1.$$

In addition, we require that $0 \leq y_2 = 2 - 3y_1 \leq 1$, so $\frac{1}{3} \leq y_1 \leq \frac{2}{3}$. Thus the optimal strategy of Player II is

$$\left\{\left[y_1(L), y_2(M), (1 - y_1 - y_2)(R)\right] \in \Sigma_2 | \frac{1}{3} \leq y_1 \leq \frac{2}{3}, y_2 = 2 - 3y_1\right\}.$$

5.10 We use the result in Exercise 5.6.

(a)

	L	R
T	7	-4
B	4	-2

(b)

	L	R
T	-5	4
B	-7	0

(c)

	L	R
T	4,18	1,19
B	5,4	5,12

(d)

	L	R
T	-2,11	-8,19
B	-2,4	-5,12

5.11

Case A

	L	M	R
x	4	6	7
$1-x$	7	3	5
	$7-3x$	$3+3x$	$5+2x$

$v = 4$

$x^* = \frac{1}{3}$

Case B

	L	R
x	4	3
$1-x$	1	2
	$1+3x$	$2+x$

$v = 3$

$x^* = 1$

Case C

	L	R
x	2	1
$1-x$	3	4
	$3-x$	$4-3x$

$v = 3$

$x^* = 0$

Case D

	L	R
x	4	2
$1-x$	1	2
	$1+3x$	2

$v = 2$

$x^* \in \left[\frac{1}{3}, 1\right]$

Case E

	L	M	R
x	4	1	2
$1-x$	0	5	2
	$4x$	$5-4x$	2

$v = 2$

$x^* \in \left[\frac{1}{2}, \frac{3}{4}\right]$

Case F

	L	M	R
x	4	2	1
$1-x$	1	3	4
	$1+3x$	$3-x$	$4-3x$

$v = \frac{5}{2}$

$x^* = \frac{1}{2}$

5.12 Let $A = \left(a_{i,j}\right)_{i,j}$ be a (finite) antisymmetric square matrix, i.e. $a_{i,j} = -a_{j,i}$ for all i and j. Assume that the matrix is a payoff matrix of a two-player zero-sum game. Then, since the matrix is square, the sets of strategies of both player are equal, $\Sigma_1 = \Sigma_2$. Furthermore,

$$\begin{aligned}
v &= \underline{v} \\
&= \max_{x\in\Sigma_1} \min_{y\in\Sigma_2} \sum_{i,j} x_i y_j a_{i,j} \\
&= \max_{x\in\Sigma_1} \min_{y\in\Sigma_2} \sum_{i,j} x_i y_j \left(-a_{j,i}\right) \\
&= \max_{x\in\Sigma_1} \left(- \max_{y\in\Sigma_2} \sum_{i,j} x_i y_j a_{j,i} \right) \\
&= - \min_{x\in\Sigma_1} \max_{y\in\Sigma_2} \sum_{i,j} x_i y_j a_{j,i} \\
&= - \min_{x\in\Sigma_2} \max_{y\in\Sigma_1} \sum_{i,j} x_i y_j a_{j,i} \\
&= -\overline{v} \\
&= -v.
\end{aligned}$$

Hence, $v = -v = 0$. In addition, x^* is an optimal strategy of Player I, if and only if

$$\begin{aligned}
&\forall y \in \Sigma_2, \quad \sum_{i,j} x_i^* y_j a_{i,j} \geq 0 \Leftrightarrow \\
&\forall y \in \Sigma_2, \quad -\sum_{i,j} x_i^* y_j a_{j,i} \geq 0 \Leftrightarrow \\
&\forall y \in \Sigma_1, \quad \sum_{i,j} x_i^* y_j a_{j,i} \leq 0.
\end{aligned}$$

However, the last inequality holds for all $y \in \Sigma_1$ if and only if x^* is an optimal strategy of Player II. Thus, Player I's set of optimal strategies is identical to that of Player II.

5.13 Let $G = (V, E)$ be a directed graph, where V is a set of vertices, and E is a set of edges. A directed edge from vertex x to vertex y is denoted by (x, y). Suppose that the graph is complete, i.e., for every pair of edges $x, y \in V$, either $(x, y) \in E$ or $(y, x) \in E$, but not both. In particular, $(x, x) \in E$ for all $x \in E$.

(a) Define a two-player zero-sum game in which the set of pure strategies of the two players is V, and the payoff function is defined as follows:

$$u(x,y) = \begin{cases} 0 & \text{if } \ x = y \\ 1 & \text{if } \ x \neq y, (x,y) \in E \\ -1 & \text{if } \ x \neq y, (x,y) \notin E \end{cases}$$

Then the payoff matrix is square. In addition,

$$\begin{array}{lll} u(x,y)=0 & \Leftrightarrow x=y & \Leftrightarrow u(y,x)=0 \\ u(x,y)=1 & \Leftrightarrow x\neq y,(x,y)\in E & \Leftrightarrow u(y,x)=-1 \\ u(x,y)=-1 & \Leftrightarrow x\neq y,(x,y)\notin E & \Leftrightarrow u(y,x)=1. \end{array}$$

That is, the payoff matrix of this game is an anti-symmetric matrix. Therefore, one can deduce, by Exercise 5.12, that the value of this game in mixed strategies is 0.

(b) Let q be an optimal strategy of Player II. Then, for every $x \in V$ one has,

$$\begin{aligned} 0 &\geq \sum_{\{y\in V\}} q(y)u(x,y) \\ &= \sum_{\{y\in V\setminus\{x\}:(x,y)\in E\}} q(y)u(x,y) + \sum_{\{y\in V\setminus\{x\}:(x,y)\notin E\}} q(y)u(x,y) \\ &= \sum_{\{y\in V\setminus\{x\}:(x,y)\in E\}} q(y) - \sum_{\{y\in V\setminus\{x\}:(x,y)\notin E\}} q(y) \\ &= \sum_{\{y\in V:(x,y)\in E\}} q(y) - \sum_{\{y\in V:(x,y)\notin E\}} q(y) \\ &= \sum_{\{y\in V:(x,y)\in E\}} q(y) - \sum_{\{y\in V:(y,x)\in E\}} q(y). \end{aligned}$$

Hence,

$$\sum_{\{y\in V:(x,y)\in E\}} q(y) \leq \sum_{\{y\in V:(y,x)\in E\}} q(y).$$

Therefore,

$$\begin{aligned} 1 &= \sum_{\{y\in V\}} q(y) \\ &= \sum_{\{y\in V:(x,y)\in E\}} q(y) + \sum_{\{y\in V:(y,x)\in E\}} q(y) \\ &\leq \sum_{\{y\in V:(y,x)\in E\}} q(y) + \sum_{\{y\in V:(y,x)\in E\}} q(y) \\ &= 2\sum_{\{y\in V:(y,x)\in E\}} q(y). \end{aligned}$$

In conclusion,

$$\sum_{\{y\in V:(y,x)\in E\}} q(y) \geq \frac{1}{2}.$$

5.14 Let WD be a set of weakly dominated mixed strategies of player i. We prove that WD is a convex set.
Let $a \in (0,1)$ and let $\sigma_i^1, \sigma_i^2 \in WD$ be two weakly dominated mixed strategies of player i. In particular, there are two mixed strategies $\widehat{\sigma}_i^1, \widehat{\sigma}_i^2$ such that for every $\sigma_{-i} \in \Sigma_{-i}$ and every $k \in \{1,2\}$ one has,

$$U(\sigma_i^k, \sigma_{-i}) \leq U(\widehat{\sigma}_i^k, \sigma_{-i}),$$

and for every $k \in \{1,2\}$ there is a strategy $\widehat{\sigma}_{-i}^k$ such that

$$U(\sigma_i^k, \widehat{\sigma}_{-i}) < U(\widehat{\sigma}_i^k, \widehat{\sigma}_{-i}).$$

We next show that the strategy $[a(\sigma_i^1), (1-a)(\sigma_i^2)]$ is weakly dominated by $[a(\widehat{\sigma}_i^1), (1-a)(\widehat{\sigma}_i^2)]$. Indeed,

$$\begin{aligned} U([a(\sigma_i^1), (1-a)(\sigma_i^2)], \sigma_{-i}) &= aU(\sigma_i^1, \sigma_{-i}) + (1-a)U(\sigma_i^2, \sigma_{-i}) \\ &\leq aU(\widehat{\sigma}_i^1, \sigma_{-i}) + (1-a)U(\widehat{\sigma}_i^2, \sigma_{-i}) \\ &= U([a(\widehat{\sigma}_i^1), (1-a)(\widehat{\sigma}_i^2)], \sigma_{-i}). \end{aligned}$$

In particular, the weak inequality become a strong one, if either $\sigma_{-i} = \widehat{\sigma}_{-i}^1$ or $\sigma_{-i} = \widehat{\sigma}_{-i}^2$.

5.15 (a) The claim is incorrect. Consider the following two player game.

		Player II L	Player II R
Player I	T	1,0	5,1
Player I	M	4,0	4,0
Player I	B	5,1	1,0

The strategy $[0.5(T), 0.5(B)]$ is strictly dominated by M. However, both strategy profile, (T,R) and (B,L), are equilibria.

(b) The claim is correct. Let $\sigma^* = (\sigma_i^*)_{\{i \in I\}}$ be an equilibrium. Assume that the mixed strategy $\widehat{\sigma}_i$ is strictly dominated by σ_i. Assume to the contrary that for every $s_i \in S_i$ such that $\widehat{\sigma}_i(s_i) > 0$ one has $\sigma_i^*(s_i) \neq 0$.
Then, by Theorem 5.18, for every $s_i \in S_i$ such that $\widehat{\sigma}_i(s_i) > 0$,

$$U_i(\sigma^*) = U_i\left(s_i, \sigma_{-i}^*\right).$$

Thus

$$U_i(\sigma^*) = U_i\left(\widehat{\sigma}_i, \sigma_{-i}^*\right) < U_i\left(\sigma_i, \sigma_{-i}^*\right),$$

which contradicts the fact that σ^* is an equilibrium.

5.16 Suppose player i has a pure strategy s_i that is chosen with positive probability in each of his maxmin strategies. We prove that s_i is not weakly dominated by any other strategy (pure or mixed).
Assume to the contrary that s_i is weakly dominated.

We first prove that there is a strategy σ_i that weakly dominates s_i such that $\sigma(s_i) = 0$. Indeed, if $\widehat{\sigma}_i$ is weakly dominates s_i. Then,

$$\forall \sigma_{-i} \in \Sigma_{-i}, \qquad U_i(s_i, \sigma_{-i}) \leq U_i(\widehat{\sigma}_i, \sigma_{-i}), \text{ and} \tag{13}$$

$$\exists \widehat{\sigma}_{-i} \in \Sigma_{-i}; \qquad U_i(s_i, \widehat{\sigma}_{-i}) < U_i(\widehat{\sigma}_i, \widehat{\sigma}_{-i}). \tag{14}$$

Then, by Inequality (14), $\widehat{\sigma}_i(s_i) < 1$. Thus the strategy σ_i where for every $s \in S$

$$\sigma_i(s) = \begin{cases} 0 & \text{if } s = s_i \\ \frac{\widehat{\sigma}_i(s)}{1-\widehat{\sigma}_i(s_i)} & \text{if } s \neq s_i \end{cases}$$

is weakly dominates s_i, since for every σ_{-i} once has

$$\begin{aligned} U_i(\sigma_i, \sigma_{-i}) &= \sum_{s \in S_i} \sigma_i(s) U_i(s, \sigma_{-i}) \\ &= \frac{1}{1-\widehat{\sigma}_i(s_i)} \sum_{s \neq s_i} \widehat{\sigma}_i(s) U_i(s, \sigma_{-i}) \\ &= \frac{1}{1-\widehat{\sigma}_i(s_i)} \sum_{s \in S_i} \widehat{\sigma}_i(s) U_i(s, \sigma_{-i}) - \frac{1}{1-\widehat{\sigma}_i(s_i)} \widehat{\sigma}_i(s_i) U_i(s, \sigma_{-i}) \\ &= \frac{1}{1-\widehat{\sigma}_i(s_i)} U_i(\widehat{\sigma}_i, \sigma_{-i}) - \frac{1}{1-\widehat{\sigma}_i(s_i)} \widehat{\sigma}_i(s_i) U_i(s, \sigma_{-i}) \\ &\geq \frac{1}{1-\widehat{\sigma}_i(s_i)} U_i(s_i, \sigma_{-i}) - \frac{1}{1-\widehat{\sigma}_i(s_i)} \widehat{\sigma}_i(s_i) U_i(s, \sigma_{-i}) \\ &= U_i(s_i, \sigma_{-i}). \end{aligned}$$

We next use σ_i to construct a maxmin strategy for player i in which s_i is not played which contradicts the assumption. Let σ_i^* be a maxmin strategy of player i. Then the strategy

$$\bar{\sigma}_i(s) = \begin{cases} 0 & \text{if } s = s_i \\ \sigma_i^*(s) + \sigma_i^*(s_i)\sigma_i(s) & \text{if } s \neq s_i \end{cases}$$

is a maxmin strategy. Indeed,

$$\begin{aligned}
U_i(\bar{\sigma}_i, \sigma_{-i}) &= \sum_{s \in S_i} \bar{\sigma}_i(s) U_i(s, \sigma_{-i}) \\
&= \sum_{s \neq s_i} (\sigma_i^*(s) + \sigma_i^*(s_i)\sigma_i(s)) U_i(s, \sigma_{-i}) \\
&= \sum_{s \neq s_i} \sigma_i^*(s) U_i(s, \sigma_{-i}) + \sigma_i^*(s_i) \sum_{s \neq s_i} \sigma_i(s)) U_i(s, \sigma_{-i}) \\
&\geq \sum_{s \neq s_i} \sigma_i^*(s) U_i(s, \sigma_{-i}) + \sigma_i^*(s_i) U_i(s_i, \sigma_{-i}) \\
&= \sum_{s \in S_i} \sigma_i^*(s) U_i(s, \sigma_{-i}) \\
&= U_i(\sigma_i^*, \sigma_{-i}) \\
&= \max_{\sigma_i' \in \Sigma_i} \min_{\sigma_i \in \Sigma_{-i}} U_i(\sigma_i', \sigma_{-i}).
\end{aligned}$$

Thus $\bar{\sigma}_i$ is a maxmin strategy in which $\bar{\sigma}_i(s_i) = 0$, which leads a contradiction.

5.17 Suppose player i has a pure strategy s_i that is chosen with positive probability in one of his maxmin strategies. Then s_i is not necessarily chosen with positive probability in each of player i's maxmin strategies. Indeed, consider the following two-player zero-sum game:

		Player II	
		L	R
	T	2	0
Player I	M	1	1
	B	0	2

The value in the above game is 1. Both strategies, M and $[0.5(T), 0.5(B)]$, are maxmin strategies of Player I. However, M is chosen with probability 1 in the first strategy, and with probability 0 in the second one.

5.18 Suppose player i has a pure strategy s_i that is not weakly dominated by any of his other pure strategies. Then s_i is not necessarily chosen with positive probability in one of player i's maxmin strategies. As shown in the next game, M is not weakly dominated by any of Player I other pure strategies, but it is not chosen with positive probability in one of player I's maxmin strategies.

		Player II	
		L	R
	T	2	0
Player I	M	0.5	0.5
	B	0	2

5.19 Let σ^* be an equilibrium in a strategic-form game and let s_i and $\widehat{s}_i$ be two pure strategies of Player i.

(a) Assume that $U_i(s_i, \sigma^*_{-i}) < U_i(\sigma^*)$. Assume by contradiction that $\sigma^*_i(s_i) > 0$. By Theorem 5.18, for every $\widehat{s}_i \in S_i$ satisfying $\sigma^*_i(\widehat{s}_i) > 0$ one has $U_i(s_i, \sigma^*_{-i}) = U_i(\widehat{s}_i, \sigma^*_{-i})$. Therefore,

$$\begin{aligned} U_i(\sigma^*) &= \sum_{\widehat{s}_i \in S_i} \sigma^*_i(\widehat{s}_i) U_i(\widehat{s}_i, \sigma^*_{-i}) \\ &= \sum_{\substack{\widehat{s}_i \in S_i, \\ \sigma^*_i(\widehat{s}_i) > 0}} \sigma^*_i(\widehat{s}_i) U_i(\widehat{s}_i, \sigma^*_{-i}) \\ &= \sum_{\substack{\widehat{s}_i \in S_i, \\ \sigma^*_i(\widehat{s}_i) > 0}} \sigma^*_i(\widehat{s}_i) U_i(s_i, \sigma^*_{-i}) \\ &= U_i(s_i, \sigma^*_{-i}), \end{aligned}$$

a contradiction.

(b) Assume that $\sigma^*_i(s_i) > 0$. By Part (a) and since σ^* is an equilibrium,

$$U_i(s_i, \sigma^*_{-i}) = U_i(\sigma^*) \geq U_i(\widehat{s}_i, \sigma^*_{-i}).$$

(c) The proof is presented in Theorem 5.18.

(d) Suppose that s_i is strictly dominated by $\widehat{s}_i$. Then, for every $s_{-i} \in S_{-i}$,

$$u_i(s_i, s_{-i}) < u_i(\widehat{s}_i, s_{-i}).$$

Hence,

$$\begin{aligned} U_i(s_i, \sigma^*_{-i}) &= \sum_{s_{-i} \in S_{-i}} \sigma^*_{-i}(s_{-i}) u_i(s_i, s_{-i}) < \sum_{s_{-i} \in S_{-i}} \sigma^*_{-i}(s_{-i}) u_i(\widehat{s}_i, s_{-i}) \\ &= U_i(\widehat{s}_i, \sigma^*_{-i}), \end{aligned}$$

so by Part (b), $\sigma^*_i(s_i) = 0$.

5.20 Consider the following two-player game where Julie is the row player and Sam is the column player. Assume that $a_{ii} = \sum_{j \neq i} a_{ij}$ for every $i = 1, ..., n$.

	1	2	$\cdots$	i	$\cdots$	j	$\cdots$	n
1	$-a_{11}$	a_{12}	$\cdots$	a_{1i}	$\cdots$	a_{1j}	$\cdots$	a_{1n}
2	a_{21}	$-a_{22}$		a_{2i}		a_{2j}		a_{2n}
$\vdots$	$\vdots$		$\ddots$					
i	a_{i1}	a_{i2}		$-a_{ii}$		a_{ij}		a_{in}
$\vdots$	$\vdots$				$\ddots$			
j	a_{j1}	a_{j2}		a_{ji}		$-a_{jj}$		a_{jn}
$\vdots$	$\vdots$						$\ddots$	
n	a_{n1}	a_{n2}		a_{ni}		a_{nj}		$-a_{nn}$

Then the column player, Sam, can guarantee that the payoff is zero by following the strategy σ_2 in which every pure strategy of his is chosen with equal probability of $\frac{1}{n}$. Indeed, for every strategy σ_1 of Julie (the row player) one has,

$$\begin{aligned} U(\sigma_1, \sigma_2) &= \sum_{i=1}^{n}\sum_{j=1}^{n} \sigma_1(i)\sigma_2(j)U(i,j) \\ &= \sum_{i=1}^{n} \frac{\sigma_1(i)}{n}\left(-a_i i + \sum_{j\neq i} a_{ij}\right) \\ &= \sum_{i=1}^{n} \frac{\sigma_1(i)}{n} 0 = 0. \end{aligned}$$

In particular, the value of the game is $v \leq 0$. Assume to the contrary that $v < 0$ so there is a strategy $\widehat{\sigma}_2$ of Sam in which the payoff in the game is negative no matter how Julie plays. That is, σ_2 is strictly dominated by $\widehat{\sigma}_2$. Therefore, by Exercise 5.15(b), for each equilibrium in the game Sam has a pure strategy j that is chosen with probability zero. The best response of Julie to such a strategy is to play $i = j$ which case the equilibrium payoff is at least zero. However, that contradict the assumption that the value of the game and thus the equilibrium payoff is negative.

5.21 (a) The mixed strategy $\sigma_1^* = \left[\frac{2}{5}[T], \frac{3}{5}[M], 0[B]\right]$ of Player I that guarantees him the same payoff of (2.4) against any pure strategy of Player II.

(b) The mixed strategy $\sigma_2^* = \left[\frac{22}{25}[L], \frac{2}{25}[C], \frac{1}{25}[R]\right]$ of Player II that guarantees him the same payoff of (2.4) against any pure strategy of Player I.

(c) Note that, by (a), for every strategy $\sigma_2 \in \Sigma_2$ one has,

$$U_1(\sigma_1^*, \sigma_2) = \sum_{s_2 \in S_2} \sigma_2(s_2) U_1(\sigma_1^*, s_2)$$
$$= \sum_{s_2 \in S_2} \sigma_2(s_2) 2.4 = 2.4,$$
$$\Rightarrow v \geq 2.4.$$

Likewise, by (b), for every strategy $\sigma_1 \in \Sigma_1$ one has,

$$U_2(\sigma_1, \sigma_2^*) = \sum_{s_1 \in S_1} \sigma_1(s_1) U_2(s_1, \sigma_2^*)$$
$$= \sum_{s_1 \in S_1} \sigma_1(s_1) 2.4 = 2.4,$$
$$\Rightarrow v \leq 2.4.$$

Thus $v = 2.4$ and the strategies σ_1^* and σ_2^* are the optimal strategies of Player I and Player II, respectively.

(d) Suppose a two-player zero-sum game is represented by an $n \times m$ matrix. Suppose each player i has an equalizing strategy σ_i^* guaranteeing him the same payoff of c_i against any pure strategy his opponent may play. We prove that σ_i^* is an optimal strategy of player i.
For every strategy $\sigma_{-i} \in \Sigma_{-i}$ of the opponent of player i one has,

$$U_i(\sigma_{-i}, \sigma_i^*) = \sum_{s_{-i} \in S_{-i}} \sigma_{-i}(s_{-i}) U_i(s_{-i}, \sigma_i^*)$$
$$= \sum_{s_{-i} \in S_{-i}} \sigma_{-i}(s_{-i}) c_i = c_i.$$

In particular, if both players follow their equalizing strategy the payoff is c_1 as well c_2, so $c_1 = c_2$.
Therefore, Player I can guarantee that the payoff equals c_1 by following σ_1^*. Similarly, Player II can guarantee that the payoff equals c_1 by following σ_2^*. Therefore the value is $v = c_1$ and any equalizing strategy is an optimal strategy.

(e) Consider the next two-player zero-sum game

	L	R
T	3	4
B	2	1

Only Player I has an equalizing strategy $[0.5(T), 0.5(B)]$, but this strategy is not optimal. Notice that this is not a contradiction to (d), since the assumption in item (d) is that both players has equalizing strategies and not just one.

5.22 In the game presented in the exercise, no player has an optimal pure strategy. Thus there is no equilibrium in pure strategies. Then the inequalities the numbers $a,\ b,\ c,\ d$ must satisfies are either

$$\min\{a,d\} > \max\{b,c\}$$

or

$$\min\{b,c\} > \max\{a,d\}.$$

Under the above assumption, the optimal strategies of both players must be in mixed strategies, $[x(T),(1-x)(B)]$ for Player I and $[y(L),(1-y)(R)]$ for Player II. Furthermore, the payoff under each pure strategy of each player must be identical. That is,

$$\begin{aligned} ax + c(1-x) &= bx + d(1-x) \\ ay + b(1-y) &= cy + d(1-y). \end{aligned}$$

$$\begin{aligned} \Rightarrow x &= \frac{d-c}{a-b+d-c}, \quad y = \frac{d-b}{a-c+d-b} \\ v &= \frac{da-bc}{a-b+d-c}. \end{aligned}$$

5.23 We prove that in any n-person game, at Nash equilibrium, each player's payoff is greater than or equal to his maxmin value.
Let σ be Nash equilibrium. Let $\underline{v}_i$ be the maxmin value of player i and let τ_i be his maxmin strategy. Then

$$U_i(\sigma) \geq U_i(\tau_i, \sigma_{-i}) \geq \underline{v}_i$$

where the left inequality holds since σ is Nash equilibrium, and the right inequality holds since τ_i is the maxmin strategy of player i.

5.24 Let $G = \left(N, (S_i)_{i\in N}, (u_i)_{i\in N}\right)$ be a two-player zero-sum game in which $N = \{I, II\}$. For each pair of mixed strategies $\sigma_I = \left[p_I^1\left(s_I^1\right), \ldots, p_I^{m_I}\left(s_I^{m_I}\right)\right]$ and $\widehat{\sigma}_I = \left[\widehat{p}_I^1\left(s_I^1\right), \ldots, \widehat{p}_I^{m_I}\left(s_I^{m_I}\right)\right]$, and each real number in the unit line interval $\alpha \in [0,1]$, define a vector $q_I = \left(q_I^j\right)_{j=1}^{m_I}$ as follows:

$$q_I^j = \alpha p_I^j + (1-\alpha)\,\widehat{p}_I^j.$$

(a) $q_I^j \geq 0$ is non-negative for every j, since $0 \leq \alpha \leq 1$ and $p_I^j, \widehat{p}_I^j \geq 0$. Moreover,

$$\begin{aligned}\sum_{j=1}^{m_I} q_I^j &= \sum_{j=1}^{m_I} \left(\alpha p_I^j + (1-\alpha)\, \widehat{p}_I^j\right) \\ &= \alpha \sum_{j=1}^{m_I} p_I^j + (1-\alpha) \sum_{j=1}^{m_I} \widehat{p}_I^j \\ &= \alpha + (1-\alpha) = 1.\end{aligned}$$

Hence, $q_I = \left(q_I^j\right)_{j=1}^{m_I}$ is a probability distribution.

(b) Define a mixed strategy τ_I of Player I as follows:

$$\tau_I = \left[q_I^1 \left(s_I^1\right), \ldots, q_I^{m_I} \left(s_I^{m_I}\right)\right].$$

For every mixed strategy σ_{II} of Player II,

$$\begin{aligned}U(\tau_I, \sigma_{II}) &= \sum_{j=1}^{m_I} q_I^j\, U\left(s_I^j, \sigma_{II}\right) \\ &= \alpha \sum_{j=1}^{m_I} p_I^j\, U\left(s_I^j, \sigma_{II}\right) + (1-\alpha) \sum_{j=1}^{m_I} \widehat{p}_I^j\, U\left(s_I^j, \sigma_{II}\right) \\ &= \alpha\, U(\sigma_I, \sigma_{II}) + (1-\alpha)\, U(\widehat{\sigma}_I, \sigma_{II}).\end{aligned}$$

(c) Assume that σ_I and $\widehat{\sigma}_I$ two strategies of Player I such that for every strategy σ_{II} of Player II,

$$U(\sigma_I, \sigma_{II}) \geq v \text{ and } U(\widehat{\sigma}_I, \sigma_{II}) \geq v.$$

Therefore, by (b),

$$U(\tau_I, \sigma_{II}) = \alpha\, U(\sigma_I, \sigma_{II}) + (1-\alpha)\, U(\widehat{\sigma}_I, \sigma_{II}) \geq \alpha v + (1-\alpha) v = v.$$

(d) Let σ_I and $\widehat{\sigma}_I$ be two optimal strategies of Player I. Thus, both strategies guarantees the maxmin payoff. Thus, by (c), τ_I also guarantees the maxmin payoff, so it is an optimal strategy.

(e) By (d), it can be deduced that for every σ_I and $\widehat{\sigma}_I$ two optimal strategies of Player I and every $\alpha \in [0,1]$ then $[\alpha(\sigma_I), (1-\alpha)(\widehat{\sigma}_I)]$ is also an optimal strategy of Player I. In particular the set of optimal strategies of Player I is a convex set in $\Delta(S_I)$.

5.25 Consider a two-player zero-sum game. Let $\left(\sigma_I^k\right)_{k\in N}$ be a sequence of optimal strategies of Player I, and for each $k \in N$ denote $\sigma_I^k = \left[p_I^{k,1}(s_I^1),\dots,p_I^{k,m_I}(s_I^{m_I})\right]$. Suppose that for each $j = 1,2,\dots,m_I$, the limit $p_I^{*,j} = \lim_{k\to\infty} p_I^{k,j}$ exists.

(a) Since $p_I^{k,j} \geq 0$ for every k and every j, $p_I^{*,j} = \lim_{k\to\infty} p_I^{k,j} \geq 0$. Moreover,

$$\sum_{k=1}^{m_I} p_I^{*,j} = \sum_{k=1}^{m_I} \lim_{k\to\infty} p_I^{k,j} = \lim_{k\to\infty} \sum_{k=1}^{m_I} p_I^{k,j} = \lim_{k\to\infty} 1 = 1,$$

so the vector $(p_I^{*,j})_{j=1}^{m_I}$ is a probability distribution over S_I.

(b) Define a mixed strategy as follows

$$\sigma_I^* = \left[p_I^{*,1}(s_I^1),\dots,p_I^{*,m_I}(s_I^{m_I})\right].$$

Then for every mixed strategy of Player II σ_{II},

$$\begin{aligned}
U(\sigma_I^*,\sigma_{II}) &= \sum_{j=1}^{m_I} p_I^{*,j}(s_I^j)U(s_I^j,\sigma_{II}) \\
&= \sum_{j=1}^{m_I} \lim_{k\to\infty} p_I^{k,j}(s_I^j)U(s_I^j,\sigma_{II}) \\
&= \lim_{k\to\infty} \sum_{j=1}^{m_I} p_I^{k,j}(s_I^j)U(s_I^j,\sigma_{II}) \\
&= \lim_{k\to\infty} U(\sigma_I^k,\sigma_{II}) \\
&\geq \lim_{k\to\infty} \underline{v}_I = \underline{v}_I.
\end{aligned}$$

In particular, σ_I^* is an optimal strategy of Player I.

(c) By the previous sections it can be deduced that that Player I's set of optimal strategies is closed. In addition, it is bounded by the unit cube in $\mathbf{R}^{m_I}$. Hence the set of optimal strategy of Player I is a compact subset of the set of mixed strategies $\Delta(S_I)$.

5.26 (a) **Game A:**
The sets of mixed strategy of the players are

$$\Delta(S_1) = \left\{[x(T),(1-x)[B]|x\in[0,1]\right\}$$
$$\Delta(S_2) = \left\{[y(L),(1-y)[R]|y\in[0,1]\right\}.$$

For every vector of mixed strategies $(\sigma_1, \sigma_2) \in \Delta(S_1) \times \Delta(S_2)$, the payoff functions are

$$\begin{aligned} U_1(\sigma_1, \sigma_2) &= xy + 4x(1-y) + 2(1-x)y + 3(1-x)(1-y) \\ &= -2xy + x - y + 3 \\ U_2(\sigma_1, \sigma_2) &= xy + 0x(1-y) + 10(1-x)y + 5(1-x)(1-y) \\ &= -4xy - 5x + 5y + 5. \end{aligned}$$

Game B:
The sets of mixed strategy of the players are the same as in Game A. For every vector of mixed strategies $(\sigma_1, \sigma_2) \in \Delta(S_1) \times \Delta(S_2)$, the payoff functions are

$$\begin{aligned} U_1(\sigma_1, \sigma_2) &= xy + 2x(1-y) + 0(1-x)y + (1-x)(1-y) \\ &= x - y + 1 \\ U_2(\sigma_1, \sigma_2) &= 2xy + 2x(1-y) + 3(1-x)y + 1(1-x)(1-y) \\ &= -2xy + x + 2y + 1. \end{aligned}$$

Game C:
The sets of mixed strategy of the players are

$$\begin{aligned} &\Delta(S_1) = \left\{[x(T), (1-x)[B] | x \in [0,1]\right\} \\ &\Delta(S_2) = \left\{[y(L), z(M), (1-y-z)[R] | y, z \in [0,1], y+z \leq 1\right\}. \end{aligned}$$

For every vector of mixed strategies $(\sigma_1, \sigma_2) \in \Delta(S_1) \times \Delta(S_2)$, the payoff functions are

$$\begin{aligned} &U_1(\sigma_1, \sigma_2) = xy + 2x(1-y-z) + (1-x)z - (1-x)(1-y-z) \\ &U_2(\sigma_1, \sigma_2) = xy + 2xz + 3(1-x)(1-y-z). \end{aligned}$$

(b) **Game A:**
Strategies R and T are eliminated during a process of elimination of strictly dominated strategies. Thus, (L, B) is the only equilibrium, in the game.
Game B:
Strategy B is eliminated during a process of elimination of strictly dominated strategies, which case Player II is indifferent between L and R. In particular, the set of equilibria is

$$\left\{ \left(T, \left[y(L), (1-y)(R)\right]\right) \mid y \in [0,1] \right\}.$$

Game C: Note that there is no equilibrium in which either Player I or Player II plays a pure strategy. Thus, in every equilibrium

Player I plays both pure strategies with positive probability. Therefore, Player II plays L with zero probability in every equilibrium (since M dominates L) whereas he plays both M and R with positive probability. Therefore, to find the equilibrium one needs to guarantee that Player I is indifferent between T and B, and Player II is indifferent between M and R:

$$2(1-y) = y-(1-y)$$
$$2x = 3(1-x).$$

Hence the equilibrium is

$$\Big(\,[0.6(T), 0.4(B)]\,,\ [0.75(M), 0.25(R)]\,\Big)$$

5.27 The game has no equilibrium in which one of the players plays pure. In particular, in an equilibrium Player I plays $[x(T),(1-x)(B)]$ where $x \in (0,1)$.
If $x > \frac{3}{4}$ then the best response of II is the pure action A which cannot define an equilibrium.
If $x = \frac{3}{4}$ then the best response of II is the pure actions A, B. Therefore, there is an equilibrium $([\frac{3}{4}(T), \frac{1}{4}(B)][\frac{1}{2}(A), \frac{1}{2}(B)])$.
If $\frac{1}{2} < x < \frac{3}{4}$ then the best response of II is the pure action B which cannot define an equilibrium.
If $x = \frac{1}{2}$ then the best response of II is the pure actions B, C. Therefore, there is an equilibrium $([\frac{1}{2}(T), \frac{1}{2}(B)][\frac{1}{2}(B), \frac{1}{2}(C)])$.
If $\frac{1}{4} < x < \frac{1}{2}$ then the best response of II is the pure action C which cannot define an equilibrium.
If $x = \frac{1}{4}$ then the best response of II is the pure actions C, D. Therefore, there is an equilibrium $([\frac{1}{4}(T), \frac{3}{4}(B)][\frac{1}{2}(C), \frac{1}{2}(D)])$.
If $x < \frac{1}{4}$ then the best response of II is the pure action D which cannot define an equilibrium.

5.28 The game has no equilibrium in which Player II plays pure. Let $[y(L),(1-y)(R)]$ be the strategy taken by Player II in an equilibrium where $y \in (0,1)$.
The best response of Player I to Player II's strategy is M if $y \in (0, \frac{1}{4})$; M and B if $y = \frac{1}{4}$; B if $y \in (\frac{1}{4}, \frac{3}{4})$; T and B if $y = \frac{3}{4}$; and T if $y \in (\frac{3}{4}, 1)$.
However, if Player I plays $[x(T),(1-x)(B)]$ (or $[x(M),(1-x)(B)]$) where $x > 0$ then Player II is better off playing L (or R, respectively) rather than the mixed action and thus it cannot be an equilibrium.
In conclusion the set of equilibria is $(B, [y(L),(1-y)(R)])$ where $y \in [\frac{1}{4}, \frac{3}{4}]$.

5.29 **Game A:**
Strategy B strictly dominates T so B is the optimal strategy of Player I. Similarly, strategy R strictly dominates L so L is the optimal strategy of Player II. Therefore, (B, R) is the only equilibrium in the game.
Game B:
The game has two pure equilibria: (T, L) which yields a payoff of $(9, 5)$ and (B, R) which yields a payoff of $(15, 6)$. To find the equilibrium in mixed strategies we solve the following equations:

$$5x + 4(1 - x) = 4x + 6(1 - x) \qquad \Rightarrow x = \frac{2}{3}$$
$$9y + 10(1 - y) = 8y + 15(1 - y) \qquad \Rightarrow y = \frac{5}{6}.$$

Therefore, the equilibrium in mixed strategies is

$$\left(\left[\frac{2}{3}(T), \frac{1}{3}(B)\right], \left[\frac{5}{6}(L), \frac{1}{6}(R)\right]\right)$$

which yields a payoff of $(9.16, 4.67)$.
The optimal strategy of Player I is T. The optimal strategy of Player II is $[\frac{2}{3}(L), \frac{1}{3}(R)]$ (which is obtained by solving $5y + 4(1 - y) = 4y + 6(1 - y)$).
The equilibrium (B, R) is the best action for both players since each one of them achieves his maximal payoff in the game.
Game C:
There is no equilibrium in the game in which at least one of the player plays a pure strategy. The equilibrium in mixed strategies is

$$\left(\left[\frac{1}{2}(T), \frac{1}{2}(B)\right], \left[\frac{7}{18}(L), \frac{11}{18}(R)\right]\right)$$

which yields a payoff of $(11.11, 11.5)$.
The optimal strategy of Player I is $[\frac{8}{18}(T), \frac{10}{18}(B)]$. The optimal strategy of Player II is $[\frac{7}{16}(L), \frac{9}{3}(16)]$.
The payoff of each player in the equilibrium equals his maxmin payoff given the player follows his optimal strategy. Thus it is better for each player to follow his optimal strategy.
Game D:
Strategy T strictly dominates B so T is the optimal strategy of Player I. Similarly, strategy L strictly dominates R so R is the optimal strategy of Player II. Therefore, (T, L) is the only equilibrium in the game.
Game E:
The equilibria in the game are (T, L), (B, R) and $\left(\left[\frac{3}{4}(T), \frac{1}{4}(B)\right], \left[\frac{1}{2}(L), \frac{1}{2}(R)\right]\right)$.

The optimal strategy of Player I is T, and the optimal strategy of Player II is L.
The equilibrium (B, R) is the best action for both players since each one of them achieves his maximal payoff in the game.
Game F:
The equilibria in the game are (B, L), (T, R) and $\left(\left[\frac{2}{3}(T), \frac{1}{3}(B)\right], \left[\frac{1}{3}(L), \frac{2}{3}(R)\right]\right)$.
The optimal strategy of Player I is $\left[\frac{2}{3}(T), \frac{1}{3}(B)\right]$, and the optimal strategy of Player II is L.
Player I is better off following his optimal strategy, thus Player II should follow the mixed equilibrium.
Game G:
The set of equilibria is

$$\left\{([x(T), (1-x)(B)],\ R) \mid x \in [0,1]\right\}.$$

Every strtegy of Player I in the game is an optimal strategy. The optimal strategy of Player II is R.

5.30 The game has no equilibrium in which one of the players play pure strategy. Therefore, we find an equilibrium $([z(T), (1-z)(B)], [y(L), (1-y)(R)])$. By the indifferent principle,

$$\begin{aligned}(3+x)y &= 2y + (1-y)\\ 1-z &= z,\end{aligned}$$

which yields to a unique equilibrium $([\frac{1}{2}(T), \frac{1}{2}(B)], [\frac{1}{2+x}(L), \frac{1+x}{2+x}(R)])$. The equilibrium payoff is $\frac{3+x}{2+x} = 1 + \frac{1}{2+x}$ which is monotonic decreasing in x.

5.31 (a) We first prove that there is no equilibrium in which one of the players chooses one of his pure strategy with probability zero. Assume Player I chooses T with probability zero. Therefore, we may eliminate T and the players faces the following game.

	L	C	R
M	6,7	0,0	7,6
B	7,6	6,7	0,0

However, the only equilibrium in this subgame is (B, L) (by a strictly dominated elimination process). But (B, L) is not an equilibrium in the original game, since Player I is better off deviating to T. In particular there is no equilibrium, in which Player I chooses T with probability 0. By the symmetry of the game, one can deduce that each pure strategy in the game is played with positive probability. Finally, the equilibrium in which both players play completely mixed strategies, is the solution of the next linear equation system:

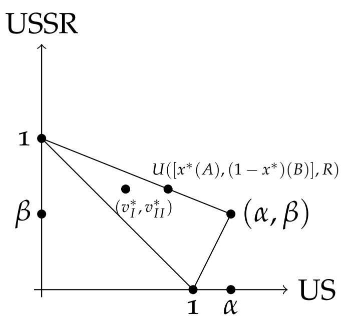
$$7y_2 + 6(1 - y_1 - y_2) = 6y_1 + 7(1 - y_1 - y_2) = 7y_1 + 6y_2$$
$$7x_2 + 6(1 - x_1 - x_2) = 6x_1 + 7(1 - x_1 - x_2) = 7x_1 + 6x_2$$

which is $x_1 = x_2 = x_3 = y_1 = y_2 = y_3 = \frac{1}{3}$.

(b) If Player I deviates to T, then Player II best reply is R that leads to the payoffs of $(6,7)$ whereas the equilibrium payoffs are $(4\frac{1}{3}, 4\frac{1}{3})$. Although the payoff under the proposed deviation is higher than the equilibrium payoff, Player II will not play that strategy since he knows that if he does so, then both players start a sequence of deviations which may end at $(0,0)$.

5.32 (a) The value of α is greater than 1 from which it follows that US prefers that the PTBT is not violated. On the other hand, the value of β is in $(0,1)$ and thus USSR prefers that the PTBT is violated but without it being revealed by the US.

(b)



(c) The pure strategies of the US are (1) Send an inspection team on date A; (2) Send an inspection team on date B; and (3) Do not send a team.

The pure strategies of the USSR are (1) Conduct underground nuclear test on date A; (2) Do not conduct a nuclear test on date A. Conduct a nuclear test on date B; (3) Do not conduct a nuclear test on date A. Conduct a nuclear test on date B, only if the US sent an inspection team on date A; (4) Do not conduct a nuclear test on date A. Conduct a nuclear test on date B, only if the US did not send an inspection team on date A; and (5) Do not conduct a nuclear tests on both dates A and B.

(d)

	L	R
A	$(1,0)$	$(0,1)$
B	$(0,1)$	(α,β)

(e) The pure strategy (3) of the US is weakly dominated by B. Therefore, all of the other pure strategies of the USSR ((2), (4) and (5) in Part (c)) are weakly dominated by R.

(f) The game in Part (d) has no equilibrium in which one of the players play pure. By the indifference principle, the equilibrium $([x^*(A),(1-x^*)(B)],[y^*(L),(1-y^*)(R)])$ must satisfies

$$\begin{aligned} y^* &= \alpha(1-y^*) \qquad & 1-x^* &= x^*+\beta(1-x^*) \Rightarrow \\ y^* &= \tfrac{\alpha}{1+\alpha} & x^* &= \tfrac{1-\beta}{2-\beta}. \end{aligned}$$

In particular, the equilibrium payoffs of the US and the USSR are $v_I^* = \frac{\alpha}{1+\alpha}$ and $v_{II}^* = \frac{1}{2-\beta}$.

(g) The payoff $U([x^*(A),(1-x^*)(B)],R) = (\frac{\alpha}{2-\beta},\frac{1}{2-\beta})$. The line segment connecting $(0,1)$ with (α,β) is $U_{II} = \frac{\beta-1}{\alpha}U_I + 1$. In particular, since

$$\tfrac{\beta-1}{\alpha}\cdot\tfrac{\alpha}{2-\beta}+1 = \tfrac{1}{2-\beta},$$

the payoff $U([x^*(A),(1-x^*)(B)],R)$ is on the line segment connecting $(0,1)$ with (α,β).

(h) Consider the following possible strategy of the US: play $[(x^*+\epsilon)(A),(1-x^*-\epsilon)(B)]$, where $\epsilon>0$ is small, and commit to playing this mixed strategy. Then,

$$\begin{aligned} U_{II}([(x^*+\epsilon)(A),(1-x^*-\epsilon)(B)],L) &= 1-x^*-\epsilon = \tfrac{1}{2-\beta}-\epsilon \\ U_{II}([(x^*-\epsilon)(A),(1-x^*-\epsilon)(B)],R) &= x^*+\epsilon+\beta(1-x^*-\epsilon) \\ &= \tfrac{1}{2-\beta}+(1-\beta)\epsilon. \end{aligned}$$

Hence, the best reply of the USSR to this mixed strategy is to play strategy R which leads to an higher payoff than the equilibrium payoff. The payoff to the US from the strategy vector

$$U_{II}([(x^*+\epsilon)(A),(1-x^*-\epsilon)(B)],R) = \alpha(\tfrac{1}{2-\beta}-\epsilon).$$

If $\epsilon < \frac{\alpha+\beta-1}{(2-\beta)(1+\alpha)} = \frac{1}{2-\beta}-\frac{1}{1+\alpha}$ then the payoff to the US from the strategy vector is higher than the equilibrium payoff (note that since $\alpha>1$ and $0<\beta<1$ then $\frac{\alpha+\beta-1}{(2-\beta)(1+\alpha)}$).

(i) The USSR can guarantee itself a payoff of v_{II}^*, regardless of the mixed strategy used by the US, when it plays its maxmin strategy $[\frac{1-\beta}{2-\beta}(L),\frac{1}{2-\beta}(R)]$.

(j) The USSR can threaten that it plays its maxmin strategy in case that the US does not commit to the strategy presented in Part (h). By Part (i), this threat is credible. In addition, the US is better committing, or else it loses.

5.33 (a) Let Country A be the row player and Country B be the column player. Country A has has two pure actions: Develop nuclear weapons (D); and Do not develop nuclear weapons (N). Country B has 4 pure actions: Do not bomb (a); Bomb regardless the spy reports (b); Bomp only if the spy report is "Country A develops" (c); and Bomp only if the spy report is "Country A does not develop" (d). The strategic-form game corresponds to this situation is given by the following matrix.

Country B

Country A		a	b	c	d
	D	$1,0$	$0,\frac{3}{4}$	$1-\alpha,\frac{3}{4}\alpha$	$\alpha,\frac{3}{4}-\frac{3}{4}\alpha$
	N	$\frac{3}{4},1$	$\frac{1}{2},\frac{1}{2}$	$\frac{1}{2}+\frac{1}{4}\alpha,\frac{1}{2}+\frac{1}{2}\alpha$	$\frac{3}{4}-\frac{1}{4}\alpha,1-\frac{1}{2}\alpha$

(b) If the quality of Country B's spy ring is $\alpha = \frac{1}{2}$ then it is equivalent to tossing a coin. If the quality of Country B's spy ring is $\alpha = 1$ then it is equivalent to perfect information.

(c) We first consider the case of $\alpha = 1$. If Country A chooses D with some positive probability, then Country B chooses either b or c, which case Country A is better of deviating. Hence, in an equilibrium Country A chooses N. Therefore, the best reply of Country B is either a or c. To prevent a deviation of Country A, the set of equilibria is $\{(N, [y(a),(1-y)(c)]) : y \leq \frac{3}{4}\}$.

Assume next that $\frac{1}{2} \leq \alpha < 1$. There is no equilibrium in which one of the players plays pure. In particular, Country A plays a completely mixed strategy $\sigma_I = [x(D),(1-x)(N)]$. Hence, the expected payoffs of Country B are

$$U_{II}(\sigma_I, a) = 1 - x,$$
$$U_{II}(\sigma_I, b) = \frac{1}{2} + \frac{1}{4}x,$$
$$U_{II}(\sigma_I, c) = \frac{1}{2}(1+\alpha) + (\frac{1}{4}\alpha - \frac{1}{2})x,$$
$$U_{II}(\sigma_I, d) = 1 - \frac{1}{2}\alpha - (\frac{1}{4}\alpha + \frac{1}{4})x.$$

If $\alpha = \frac{1}{2}$ then for $x = \frac{2}{5}$ Country B is indifferent between its pure strategies. Therefore, using the indifference principle, the set of equilibria is

$$\{([\tfrac{2}{5}(D), \tfrac{3}{5}(N)], [y_1(a), y_2(b), y_3(c), (1-y_1-y_2-y_3)(d)]) : y_1 - y_2 = \tfrac{1}{3}\}.$$

If $\frac{1}{2} < \alpha < 1$ then d is strictly dominated by c and can be eliminated. Furthermore, if $x > \frac{2}{5}$ then Country B is better off playing b and c which case Country A deviates to N, a contradiction. Hence, in an equilibrium, $x \leq \frac{2}{5}$ so $U_{II}(\sigma_I, b) < U_{II}(\sigma_I, c)$ and thus Country B only plays a and c with some positive probability. Using the indifference principle, the equilibrium is

$$\{([\tfrac{2-2\alpha}{2+\alpha}(D), \tfrac{3\alpha}{2+\alpha}(N)], [\tfrac{5\alpha-2}{5\alpha-1}(a), \tfrac{1}{5\alpha-1}(c)])\}.$$

(d) The set of equilibrium payoffs is:

$$(U_I^*(\alpha), U_{II}^*(\alpha)) \begin{cases} (\frac{2}{3}, \frac{3}{5}) & \alpha = \frac{1}{2}, \\ (\frac{4\alpha-1}{5\alpha-1}, \frac{3\alpha}{2+\alpha}) & \frac{1}{2} < \alpha < 1, \\ (\frac{3}{4}, 1) & \alpha = 1. \end{cases}$$

In particular, at $\alpha = 1$, the maximal equilibrium payoffs of both countries are obtained.

(e) Assuming both countries play their equilibrium strategy, then the probability that Country A will manage to develop nuclear weapons without being bombed is zero if $\alpha = 1$; is

$$\tfrac{2-2\alpha}{2+\alpha}\left(\tfrac{5\alpha-2}{5\alpha-1} + (1-\alpha)\cdot\tfrac{1}{5\alpha-1}\right) = \tfrac{(2-2\alpha)(4\alpha-1)}{(2+\alpha)(5\alpha-1)}$$

if $\frac{1}{2} < \alpha < 1$; and is

$$\tfrac{2}{5}\left(y_1 + \tfrac{y_3}{2} + \tfrac{1-y_1-y_2-y_3}{2}\right) = \tfrac{2}{5}\cdot\tfrac{1+y_1-y_2}{2} = \tfrac{4}{15}$$

if $\alpha = \frac{1}{2}$.

5.34 For every $\sigma_I \in \Sigma_I$ and $\sigma_{II} \in \Sigma_{II}$,

$$\begin{aligned} U_I(\sigma_I, \sigma_{II}) &= \sum_{s_{II} \in S_{II}} \sigma_{II}(s_{II}) U_I(\sigma_I, s_{II}) \\ &\geq \sum_{s_{II} \in S_{II}} \sigma_{II}(s_{II}) \min_{s'_{II} \in S_{II}} U_I(\sigma_I, s'_{II}) \\ &= \min_{s'_{II} \in S_{II}} U_I(\sigma_I, s'_{II}) \end{aligned}$$

Therefore,

$$\min_{\sigma_{II} \in \Sigma_{II}} U_I(\sigma_I, \sigma_{II}) \geq \min_{s'_{II} \in S_{II}} U_I(\sigma_I, s'_{II}).$$

However, since $s_{II} \in \Sigma_{II}$ then

$$\begin{aligned}\min_{\sigma_{II}\in\Sigma_{II}} U_I(\sigma_I,\sigma_{II}) &\leq \min_{s'_{II}\in S_{II}} U_I(\sigma_I,s'_{II}) \Rightarrow \\ \min_{\sigma_{II}\in\Sigma_{II}} U_I(\sigma_I,\sigma_{II}) &= \min_{s'_{II}\in S_{II}} U_I(\sigma_I,s'_{II}) \Rightarrow \\ \max_{\sigma_I\in\Sigma_I}\min_{\sigma_{II}\in\Sigma_{II}} U_I(\sigma_I,\sigma_{II}) &= \max_{\sigma_I\in\Sigma_I}\min_{s'_{II}\in S_{II}} U_I(\sigma_I,s'_{II}).\end{aligned}$$

5.35 (a) Let σ_i be a best reply of player i to σ_{-i}. Thus,

$$\begin{aligned}\max_{\substack{s_i\in S_i,\\ \sigma_i(s_i)>0}} U_i(s_i,\sigma_{-i}) \leq U_i(\sigma_i,\sigma_{-i}) \\ &= \sum_{\substack{s_i\in S_i,\\ \sigma_i(s_i)>0}} \sigma_i(s_i)U_i(s_i,\sigma_{-i}) \\ &\leq \max_{\substack{s_i\in S_i,\\ \sigma_i(s_i)>0}} U_i(s_i,\sigma_{-i}).\end{aligned}$$

In particular,

$$U_i(\sigma_i,\sigma_{-i}) = \sum_{\substack{s_i\in S_i,\\ \sigma_i(s_i)>0}} \sigma_i(s_i)U_i(s_i,\sigma_{-i}) = \max_{\substack{s_i\in S_i,\\ \sigma_i(s_i)>0}} U_i(s_i,\sigma_{-i}).$$

Thus, by a property of an expectation, $U_i(\sigma_i,\sigma_{-i}) = U_i(s_i,\sigma_{-i})$ for every pure strategy in the support of σ_i.

(b) Let $\widehat{\sigma}_i$ be a mixed strategy of player i whose support is contained in the support of σ_i.

$$\begin{aligned}U_i(\widehat{\sigma}_i,\sigma_{-i}) &= \sum_{\substack{s_i\in S_i,\\ \widehat{\sigma}_i(s_i)>0}} \widehat{\sigma}_i(s_i)U_i(s_i,\sigma_{-i}) = \sum_{\substack{s_i\in S_i,\\ \widehat{\sigma}_i(s_i)>0}} \widehat{\sigma}_i(s_i)U_i(\sigma_i,\sigma_{-i}) \\ &= U_i(\sigma_i,\sigma_{-i})\end{aligned}$$

where the second equality follows (a).

(c) By (a), every mixed strategy in player i's set of best replies to σ_{-i} is a convex combination of pure strategies that gives player i a maximal payoff against σ_{-i} .
By (b), every convex combination of pure strategies that gives player i a maximal payoff against σ_{-i} is in player i's set of best replies to σ_{-i}.
Hence, player i's set of best replies to σ_{-i} is the convex hull of the pure strategies that give him a maximal payoff against σ_{-i} .

5.36 Let G be a symmetric game. We prove that there exists a symmetric equilibrium $\sigma = (\sigma_i)_{i \in N}$ in mixed strategies satisfying $\sigma_i = \sigma_j$ for each $i, j \in N$.
Denote by $\Sigma^* = \left\{(\sigma_1, \sigma_1, ..., \sigma_1) | \sigma_1 \in \Sigma_1\right\}$ the set of all symmetric mixed profiles in the game. The set Σ^* is compact and convex.
Consider the function f defined in Equation (5.81) in the proof of Nash's Theorem. By Claim 5.28, $f : \Sigma \to \Sigma$, and by Claim 5.29, f is a continuous function. We next restrict f to the set of symmetric profiles and show that $f : \Sigma^* \to \Sigma^*$. Indeed,by the symmetric of the game, for every $\sigma_1 \in \sigma_1$,

$$\begin{aligned} g_i^{s_1}(\sigma_1, ..., \sigma_1) &= \max \left\{ 0, U_i(\sigma_1, ..., \sigma_1, \underbrace{s_1}_{\substack{\text{the } i\text{'th} \\ \text{coordinate}}}, \sigma_1, ..., \sigma_1) \right\} \\ &= \max \left\{ 0, U_j(\sigma_1, ..., \sigma_1, \underbrace{s_1}_{\substack{\text{the } j\text{'th} \\ \text{coordinate}}}, \sigma_1, ..., \sigma_1) \right\} \\ &= g_j^{s_1}(\sigma_1, ..., \sigma_1). \end{aligned}$$

Hence,

$$\begin{aligned} f_i^{s_1}(\sigma_1, ...\sigma_1) &= \frac{\sigma_1(s_1) + g_i^{s_1}(\sigma_1, ..., \sigma_1)}{1 + \sum_{s_1' \in S_1} g_i^{s_1'}(\sigma_1, ..., \sigma_1)} \\ &= \frac{\sigma_1(s_1) + g_j^{s_1}(\sigma_1, ..., \sigma_1)}{1 + \sum_{s_1' \in S_1} g_j^{s_1'}(\sigma_1, ..., \sigma_1)} \\ &= f_j^{s_1}(\sigma_1, ...\sigma_1). \end{aligned}$$

$f : \Sigma^* \to \Sigma^*$, is a continuous function from a convex and compact set to itself, so by Brouwer's Fixed Point Theorem, f has a fixed point $(\sigma_1, ..., \sigma_1)$. However, by Claim 5.31, $(\sigma_1, ..., \sigma_1)$ is a symmetric equilibrium in the game.

5.37 (a) The game is given by $N = \{1, 2, ..., 10\}$, a set of players;

$$S_i = \{\text{Confess(C), Don't confess(D)}\},$$

a set of pure action of each player $i \in N$; and a payoff function of each player $i \in N$

$$u_i(s_1, ..., s_{10}) = \begin{cases} 0 & \text{if } s_1 = ... = s_{10} = D \\ 9 & \text{if } s_i = C \\ 10 & \text{Otherwise} \end{cases} . \tag{15}$$

(b) There are ten pure equilibria according to which only one player confesses and all other players do not confess. These equilibria can be achieved in case there is a hierarchy between the players and there is one prominent player who is expected to volunteer. For example, if there is a hierarchy among the criminals, it is expected that the junior criminal will volunteer. It is clear to him that no one will volunteer in his place and that everyone expects him to do so.

(c) Consider a symmetric strategy vector σ in mixed strategies according to which $\sigma_i = [p(D), (1-p)(C)]$ for every player i, where $p \in (0,1)$. Then σ is an equilibrium if and only if

$$u_i(C, \sigma_{-i}) = u_i(D, \sigma_{-i}) \iff$$
$$9 = p^9 \cdot 0 + (1-p^9) \cdot 10 \iff p = 10^{-1/9}.$$

In particular, the probability that at this equilibrium no one volunteers to confess is $10^{-10/9}$.

(d) Assume next that the number of suspects is not 10, but n. There is a symmetric equilibrium σ in mixed strategies according to which $\sigma_i = [10^{-1/n-1}(D), (1-10^{-1/n-1})(C)]$ for every player i. The probability that at this equilibrium no one volunteers to confess is $10^{-n/n-1}$ which goes to 0.1 as n as n goes to infinity. One may conclude from this analysis that as larger the groups as smaller the probability of volunteering since one may think that even if he will not volunteer someone else probably will do it. Note that the limit probability that no one volunteers depends on the ratio between the cost and the gain of volunteering.

5.38 (a) The set of strategies of player i is the power set of $\{1, 2, ..., K\}$,

$$S_i = \left\{ s_i \,|\, s_i \subseteq \{1, 2, ..., K\} \right\}.$$

For every vector of pure strategies $(s_1, s_2, ..., s_n)$ let $D(s_1, s_2, ..., s_n)$ be the set of of all the numbers that have been purchased by only one of the players, i.e., $D(s_1, s_2, ..., s_n) = \bigcup_{i=1}^{n} \left(s_i \setminus (\bigcup_{j \neq i} s_j) \right)$. The payoff function of player i is given by

$$u_i(s_1, ..., s_n) = \begin{cases} M - |s_i| & \text{if } \min_{k \in D(s_1, s_2, ..., s_n)} k \in s_i \\ -|s_i| & \text{otherwise} \end{cases}.$$

(b) The game is symmetric so, by Exercise 5.36, the game necessarily has a symmetric equilibrium. However, since every symmetric pure strategy vector is not an equilibrium. The symmetric equilibrium is an equilibrium in mixed strategies.

(c) For $p_1 \in (0,1)$, consider the following mixed strategy $\sigma_i(p_1)$ of player i: with probability p_1 purchase only the number 1, and with probability $1 - p_1$ do not purchase any number. The strategy vector $\sigma = (\sigma_i)_{i \in N}$ is an equilibrium if the following conditions hold.

$$u_i(\{1\}, \sigma_{-i}) = u_i(\emptyset, \sigma_{-i}) \tag{16}$$

$$u_i(\{1\}, \sigma_{-i}) \geq u_i(\{k\}, \sigma_{-i}) \qquad \forall k \geq 2 \tag{17}$$

$$u_i(\{1\}, \sigma_{-i}) \geq u_i(\{1, k\}, \sigma_{-i}) \qquad \forall k \geq 2 \tag{18}$$

By Inequality (17), $n \leq 2$. Otherwise, the probability of winning with $k \geq 2$ is higher than the probability of winning with 1 which contradicts Inequality (17). By Equation (16),

$$u_i(\{1\}, \sigma_{-i}) = 0 = (1 - p_1) \cdot M - 1 \Rightarrow M = \frac{1}{1 - p_1}.$$

Finally, by Inequality (18), since under $(\{1, 2\}, \sigma_{-i})$ player i wins, one can derive that

$$0 \geq M - 2 = \frac{1}{1 - p_1} - 2 \Rightarrow p_1 \leq \tfrac{1}{2}.$$

If (16)-(18) hold then $u_i(\sigma) \geq u_i(s_i, \sigma_{-i})$ for every $s_i \in S_i$.

(d) Let σ be an equilibrium in which there is positive probability that player i will not purchase any number. Thus $\emptyset$ is in the support of σ_i and by Exercise 5.35, the expected payoff of player i satisfies $u_i(\sigma) = u_i(\emptyset, \sigma_{-i}) = 0$.

(e) Assume that $M < n$, meaning that the number of participants is greater than the value of the prize. We show that at equilibrium, there is at least one player that with positive probability does not purchase a number. Assume by a contrary that with probability 1 every player purchases at least one number. Thus the expected number of natural numbers purchased by all the players together is greater than the value of the prize M, and hence there is a player whose expected payoff is negative, which case that player can gain by avoiding purchasing any number, a contradiction. In particular, at every symmetric equilibrium, every player select to not purchase any number with positive probability, hence, by Part (d), the expected payoff of every player is 0.

5.39 We prove that the set of equilibria in mixed strategies is a compact set. First, the set of equilibria in mixed strategies is bounded since it a subset of the bounded product space $\Delta(S1) \times \Delta(S2) \times ... \times \Delta(Sn)$.

We next prove that it is also closed and thus it is compact. Let $(\sigma(n))_{n\in\mathbf{N}}$ be a sequence of equilibria in mixed strategies that convergences to a limit σ. By Exercises 5.1 and 5.2, σ is a mixed strategy vector. Furthermore, for every player i and every mixed strategy y_i of player i,

$$U_i(\sigma(n)) \geq U_i(y_i, \sigma(n)_{-i}) \quad \forall n \in \mathbf{N}.$$

However, U_i is a continuous function and thus, by passing to the limit, it follows that

$$U_i(\sigma) \geq U_i(y_i, \sigma_{-i}) \quad \forall n \in \mathbf{N},$$

so σ is an equilibrium, as needed.
Note that the set of equilibria is not necessarily convex. Consider the game given by a game matrix.

	L	R
T	1,1	0,0
B	0,0	2,2

Whereas the strategy vectors (T, L) and (B, R) are equilibria, the strategy vector $\left([\frac{1}{2}(T), \frac{1}{2}(B)], [\frac{1}{2}(L), \frac{1}{2}(R)]\right)$ is not a equilibrium.

5.40 Let $M_{n,m}$ be the space of matrices of order $n \times m$ representing two-player zero-sum games in which Player I has n pure strategies and Player II has m pure strategies. We prove that the function that associates with every matrix $A = \left(a_{ij}\right) \in M_{n,m}$ the value in mixed strategies of the game that it represents is continuous in $\left(a_{ij}\right)$.
Let $\left(A^k\right)_{k\in\mathbf{N}}$ be a sequence of matrix of order $n \times m$. For every $k \in \mathbf{N}$, let v^k be the value of the game matrix A^k in mixed strategies, and let x^k and y^k the corresponding optimal strategies of Player I and Player II respectively.
Since the set of mixed strategy vectors is compact, there is subsequence $(x^{k_l}, y^{k_l})_{l\in\mathbf{N}}$ converging to a limit (x, y). By the optimality of x^{k_l} and y^{k_l} one has

$$\sum_{i=1}^{n} x_i^{k_l} a_{ij}^{k_l} \geq v^{k_l} \qquad \forall j \in \{1, 2, ..., m\} \text{ and } l \in \mathbf{N}$$

$$\sum_{j=1}^{m} y_j^{k_l} a_{ij}^{k_l} \leq v^{k_l} \qquad \forall i \in \{1, 2, ..., n\} \text{ and } l \in \mathbf{N}.$$

Passing to the limit as $l \to \infty$ yields to,

$$\sum_{i=1}^{n} x_i a_{ij} \geq \limsup_{l \to \infty} v^{k_l} \qquad \forall j \in \{1, 2, ..., m\}$$
$$\sum_{j=1}^{m} y_j a_{ij} \leq \liminf_{l \to \infty} v^{k_l} \qquad \forall i \in \{1, 2, ..., n\}.$$

In particular, $\limsup_{l \to \infty} v^{k_l} \leq \liminf_{l \to \infty} v^{k_l}$. Since $\limsup_{l \to \infty} v^{k_l} \geq \liminf_{l \to \infty} v^{k_l}$ is always attained, $\lim_{l \to \infty} v^{k_l} = \limsup_{l \to \infty} v^{k_l} = \liminf_{l \to \infty} v^{k_l}$. Note that since we showed that for every subsequence of optimal strategies the limit defines the optimal strategies in A, and since the value is unique, $\lim_{k \in \mathbf{N}} v^k = v$, as needed.

5.41 Let $A = \left(a_{ij}\right)$ and $B = \left(b_{ij}\right)$ be two $n \times m$ matrices representing two-player zerosum games in strategic form. Let $v(A)$ and $v(B)$ be the values of the game matrices A and B respectively. Assume without loss of generality that $v(A) \geq v(B)$. Denote by x^A (x^B) the optimal strategy of Player I and by y^A (y^B) the optimal strategy of Player II in the game matrix A (B). Then,

$$v(A) = \sum_{i=1}^{n} \sum_{j=1}^{m} x_i^A y_j^A a_{ij} \leq \sum_{i=1}^{n} \sum_{j=1}^{m} x_i^A y_j^B a_{ij} \tag{19}$$
$$v(B) = \sum_{i=1}^{n} \sum_{j=1}^{m} x_i^B y_j^B b_{ij} \geq \sum_{i=1}^{n} \sum_{j=1}^{m} x_i^A y_j^B b_{ij}, \tag{20}$$

where Inequality (19) holds since the strategy y^B is not necessarily an optimal strategy of Player II in A. Similarly, Inequality (20) holds since the strategy x^A is not necessarily an optimal strategy of Player I in B. Hence it can be deduced that

$$\begin{aligned}\left|v(A) - v(B)\right| = v(A) - v(b) &= \sum_{i=1}^{n} \sum_{j=1}^{m} x_i^A y_j^A a_{ij} - \sum_{i=1}^{n} \sum_{j=1}^{m} x_i^B y_j^B b_{ij} \\ &\leq \sum_{i=1}^{n} \sum_{j=1}^{m} x_i^A y_j^B a_{ij} - \sum_{i=1}^{n} \sum_{j=1}^{m} x_i^A y_j^B b_{ij} = \sum_{i=1}^{n} \sum_{j=1}^{m} x_i^A y_j^B \left(a_{ij} - b_{ij}\right) \\ &\leq \sum_{i=1}^{n} \sum_{j=1}^{m} x_i^A y_j^B \left|a_{ij} - b_{ij}\right| \leq \max_{i=1}^{n} \max_{j=1}^{m} \left|a_{ij} - b_{ij}\right|.\end{aligned}$$

5.42 The values of both A and B presented below is 1, whereas the value of the matrix $0.5A + 0.5B$ is only 0.5.

	L	R
T	1	1
B	0	0

A

	L	R
T	0	0
B	1	1

B

	L	R
T	0.5	0.5
B	0.5	0.5

$0.5A + 0.5B$

5.43 Let Γ and $\widehat{\Gamma}$ be two strategic-form games given by the payoff matrices G and $\widehat{G}$ respectively.

	L	R
T	1,1	4,0
B	0,0	5,5

G

	L	R
T	1,1	6,0
B	0,0	5,5

$\widehat{G}$

Both games have the same set of pure strategies. The maximal difference between the payoff functions of the two games is $c = 2$. By eliminating strictly dominated strategies, there is a unique equilibrium (T, L) of $\widehat{\Gamma}$ corresponding the payoff $(1, 1)$ whereas $(5, 5)$ is an equilibrium payoff of Γ so the set of equilibria of Γ is not close to the set of equilibria of $\widehat{\Gamma}$.
Note that such a phenomenon cannot exist in two-player zero-sum games since every equilibrium payoff of the game equals its value which, by Exercise 5.41, is bounded by c.

5.44 When the payoffs of a strategic form game increase, the equilibrium payoffs do not necessarily increase. The two-player non-zero games given by the following matrices constitute a counter example.

	L	R
T	5,5	2,2
B	6,1	3,3

	L	R
T	4,4	1,1
B	0,0	0,0

Using the process of iterated elimination of strictly dominated strategies, one can verify that each game has only one equilibrium. The right game equilibrium payoff is $(4, 4)$ whereas the left game equilibrium payoff is $(3, 3)$ although the payoffs in the left game are higher than the payoffs in the right game.

5.45 Let $\Gamma = (N, (S_1, S_2), (u_1, u_2))$ be a two-player strategic-form game. Consider the two-player zero-sum $\widehat{\Gamma}_i = (N, (S_1, S_2), u_i)$ obtained by considering only the payoffs of player i in Γ. By von Neumann's Minmax Theorem the game $\widehat{\Gamma}_i$ has a value v^i in mixed strategies. In particular, both the minmax value and maxmin value of player i in $\widehat{\Gamma}_i$ in mixed strategies are identical. However, the minmax (maxmin) value of player i in Γ in mixed strategies equals his minmax (maxmin) value in $\widehat{\Gamma}_i$ in mixed strategies. In conclusion, the minmax value in mixed strategies of a player equals his maxmin value in mixed strategies.

5.46 (a) We prove that the payoff of Player I at the equilibrium equals his maxmin value in mixed strategies, the proof for Player II is similar and thus it was omitted.
Assume without loss of generality that $a \geq c$. By the assumption that the game has a unique equilibrium, given by a completely mixed strategy, no pure strategy vector is an equilibrium from which it can be deduced that

$$a \geq c \Rightarrow f > b \Rightarrow g > e \Rightarrow d > h \Rightarrow a > c.$$

Furthermore, since the maxmin strategies of both players are also completely mixed, neither $c \geq g$ nor $e \geq a$ (otherwise the optimal strategies of Player I is pure). Denote by

$$\sigma^* := \big([x^*(T), (1-x^*)(B)], [y^*(L), (1-y^*)(R)]\big)$$

the equilibrium of the game. Since σ^* is a completely mixed equilibrium

$$\begin{aligned} &U_I(T, \sigma^*_{II}) = U_I(B, \sigma^*_{II}) = U_I(\sigma^*) \Rightarrow \\ &\max_{\sigma_I \in \Sigma_I} U_I(\sigma_I, \sigma^*_{II}) = U_I(\sigma^*) \end{aligned} \tag{21}$$

In addition, since $g > c$, for every mixed strategy

$$\sigma_{II} = [y(L), (1-y)(R)]$$

of Player II such that $y < y^*$,

$$\max_{\sigma_I \in \Sigma_I} U_I(\sigma_I, \sigma_{II}) \geq U_I(B, \sigma_{II}) > U_I(B, \sigma^*_{II}) = U_I(\sigma^*). \tag{22}$$

Finally, since $a > e$, for every mixed strategy

$$\sigma_{II} = [y(L), (1-y)(R)]$$

of Player II such that $y > y^*$,

$$\max_{\sigma_I \in \Sigma_I} U_I(\sigma_I, \sigma_{II}) \geq U_I(T, \sigma_{II}) > U_I(T, \sigma^*_{II}) = U_I(\sigma^*). \tag{23}$$

By Equality (21) and Inequalities (22) and (23), one can conclude that

$$\min_{\sigma_{II} \in \Sigma_{II}} \max_{\sigma_I \in \Sigma_I} U_I(\sigma_I, \sigma_{II}) = U_I(\sigma^*).$$

By Exercise 5.45 it follows that the payoff of Player I at σ^* equals his maxmin value in mixed strategies.

(b) The minmax startegy of Player I is $[\frac{g-c}{a+g-c-e}(T), \frac{a-e}{a+g-c-e}(B)]$ which can be obtained by solving the equation

$$ax + c(1-x) = ex + g(1-x).$$

The minmax strategy of Player II is $[\frac{h-f}{b+h-d-f}(L), \frac{b-d}{b+h-d-f}(R)]$ which can be obtained by solving the equation

$$by + f(1-y) = dy + h(1-y).$$

The completely mixed equilibrium is found by the indifference principle.

$$\begin{aligned} ay + e(1-y) &= cy + g(1-y) \\ bx + d(1-x) &= fx + h(1-x). \end{aligned}$$

Hence the equilibrium is

$$([\tfrac{g-e}{a+g-c-e}(T), \tfrac{a-c}{a+g-c-e}(B)], [\tfrac{h-d}{b+h-d-f}(L), \tfrac{b-f}{b+h-d-f}(R)]).$$

5.47 We prove that the only equilibrium in the given three-player game is (T,L,W).
We first prove that in every equilibrium of the game neither Player I plays the pure strategy B nor Player II plays the pure strategy R nor Player III plays the pure strategy E. By the the symmetry between the players, it is enough to prove that there is no equilibrium in which Player I plays the pure strategy B. Assume by contradiction that there is an equilibrium in which Player I plays B. Then the best response of Player II is L. Thus, Player III prefers to play E, so Player I is better off deviating to T, a contradiction.
We next show that if in an equilibrium one of the players plays pure, then all of them necessarily play pure (which leads to the equilibrium (T, L, W)). Assume for instance that Player I plays T. If in addition Player II plays R with positive probability, then Player III's best response is W which case Player I is better off deviating to B. Thus in an equilibrium Player II must play L. If Player III plays E with positive probability, then Player II is better off deviating to R, a contradiction. Thus, it derives the equilibrium (T, L, W).
We finally show that there is no equilibrium with completely mixed strategy. Assume by contradiction that there is such an equilibrium. By the indifference principle the following equations hold.

$$\left.\begin{aligned} &\text{Player I} &: zy + (1-z)(3y + 1 - y) = z \\ &\text{Player II} &: xz + (1-x)(3z + 1 - z) = x \\ &\text{Player III} &: yx + (1-y)(3x + 1 - x) = y \end{aligned}\right\} \iff x = y = z = 1,$$

a contradiction.

5.48 (a) Assume that Player 3 plays the mixed action $[z(a), (1-z)(b)]$ then the payoff of Player 1 as a function of his pure action is

$$U_1(a, [\tfrac{1}{2}(a), \tfrac{1}{2}(b)], [z(a), (1-z)(b)]) = \tfrac{1}{2}z + \tfrac{1}{2}(1-z) = \tfrac{1}{2}$$
$$U_1(b, [\tfrac{1}{2}(a), \tfrac{1}{2}(b)], [z(a), (1-z)(b)]) = \tfrac{1}{2}(1-z) + \tfrac{1}{2}z = \tfrac{1}{2},$$

from which it follows that Player 1 is indifferent between his two pure strategies, regardless of the mixed strategy played by Player 3.

(b) In the three-player game in which the payoff function of each player coincides with u_1 there are continuum of equilibria in mixed strategies. For instance, $([\frac{1}{2}(a), \frac{1}{2}(b)], [\frac{1}{2}(a), \frac{1}{2}(b)], [z(a), (1-z)(b)])$ for $z \in [0,1]$. Indeed, as a result of item (a), Each player is indifferent between his two pure strategies since there is another player playing the mixed action $[\frac{1}{2}(a), \frac{1}{2}(b)]$.

(c) In the following three-player game, where Player 1 chooses a row, Player 2 chooses a column, and Player 3 chooses a matrix, each player has two pure strategies, no payoff appears twice in the payoff matrix, and there are continuum of equilibria in mixed strategies. For instance, $([\frac{1}{2}(T), \frac{1}{2}(B)], [\frac{1}{2}(L), \frac{1}{2}(R)], [z(W), (1-z)(E)])$ for $z \in [0,1]$.

W	L	R
T	1,5,9	-1,-6,-9
B	-2,-5,-10	2,6,10

E	L	R
T	3,7,11	-3,-8,-11
B	-4,-7,-12	4,8,12

5.49 **The proof of Theorem 5.56:**

(a) Let (x_1, y_1) and (x_2, y_2) be two equilibria of a two-player strategic-form game with payoffs (a, c) and (b, d) satisfying $\text{supp}(x_1) = \text{supp}(x_2)$ and $\text{supp}(y_1) = \text{supp}(y_2)$. Let $0 \leq \alpha, \beta \leq 1$. Then, by Exercise 5.35, $\alpha x_1 + (1-\alpha)x_2$ is a best response to y_1 as well as to y_2. Therefore, for every $x \in \Sigma_1$ one holds

$$\begin{aligned}
&u_1(\alpha x_1 + (1-\alpha)x_2, \beta y_1 + (1-\beta)y_2) \\
&= \beta u_1(\alpha x_1 + (1-\alpha)x_2, y_1) + (1-\beta)u_1(\alpha x_1 + (1-\alpha)x_2, y_2) \\
&= \beta u_1(x_1, y_1) + (1-\beta)u_1(x_2, y_2) \\
&\geq \beta u_1(x, y_1) + (1-\beta)u_1(x, y_2) \\
&= u_1(x, \beta y_1 + (1-\beta)y_2).
\end{aligned}$$

In particular, $\alpha x_1 + (1-\alpha)x_2$ is a best response to $\beta y_1 + (1-\beta)y_2$. The proof that $\beta y_1 + (1-\beta)y_2$ is a best response to $\alpha x_1 + (1-\alpha)x_2$

is similar.
In conclusion, $(\alpha x_1 + (1-\alpha)x_2, \beta y_1 + (1-\beta)y_2)$ is a Nash equilibrium with the same support, and with payoff $(\alpha a + (1-\alpha)c, \beta b + (1-\beta)d)$.

(b) Let S'_I be a subset of Player I's pure strategies, and S'_{II} be a subset of Player II's pure strategies. Denote by $E(S'_I, s'_{II})$ the set of Nash equilibria payoffs yielded by any strategy vectors (x, y) satisfying $\text{supp}(x) = S'_I$ and $\text{supp}(y) = S'_{II}$. We show that $E(S'_I, S'_{II})$ is a rectangle.
Set

$$\begin{aligned} a &:= \max \left\{a' \mid \exists b' \text{ such that } (a', b') \in E(S'_I, S'_{II})\right\} \\ c &:= \min \ \left\{a' \mid \exists b' \text{ such that } (a', b') \in E(S'_I, S'_{II})\right\} \\ b &:= \max \left\{b' \mid (a, b') \in E(S'_I, S'_{II})\right\} \\ d &:= \min \ \left\{b' \mid (c, b') \in E(S'_I, S'_{II})\right\} \end{aligned}$$

By Part (a) and the definition of $E(S'_I, S'_{II})$

$$[a, c] \times [b, d] \subseteq E(S'_I, S'_{II}).$$

On the other hand, for every $(f, g) \in E(S'_I, S'_{II})$ one has

$$c \leq f \leq a.$$

In addition, by Part (a), $(a, g), (c, g) \in E(S'_I, S'_{II})$ so $d \leq g \leq b$. Thus $E(S'_I, S'_{II}) \subseteq [a, c] \times [b, d]$ which derives that $[a, c] \times [b, d] = E(S'_I, S'_{II})$.

(c) By Part (b), for every couple of supports (S'_I, S'_{II}), the set of Nash equilibria payoffs yielded by any strategy corresponding this couple is a rectangle of the form $[a, b] \times [c, d]$. However, since the number of pure strategies is finite and so the number of possible supports is finite as well, the set of equilibrium payoffs of every two-player game in strategic form is a union of a finite number of rectangles of the form $[a, b] \times [c, d]$.

(d) We prove that the set of equilibrium payoffs in the given game is A.
Denote by R_i the pure strategy of Player I in which he selects the i-th row from top to bottom, and by C_j the pure strategy of Player II in which he selects the j-th column from left to right.
If Player I plays any strategy that is not of the form

$$[\alpha(R_{2k-1}), (1-\alpha)(R_{2k})]$$

then Player II is better off selecting a strategy of the form

$$[\beta(C_{2K-1}), (1-\beta)(C_{2K})]$$

thus Player I deviates to $[\alpha(R_{2K-1}), (1-\alpha)(R_{2K})]$. Hence, in every equilibrium of the game Player I plays a strategy of the form $[\alpha(R_{2k-1}), (1-\alpha)(R_{2k})]$. By similar arguments, in every equilibrium of the game Player II plays a strategy of the form $[\beta(C_{2k-1}), (1-\beta)(C_{2k})]$.

For every $k \in \{1, 2..., K\}$ and every $\alpha \in [0,1]$, the best response of Player II to $[\alpha(L_{2k-1}), (1-\alpha)(L_{2k})]$ in which no player has an incentive to deviate is $[\beta(C_{2k-1}), (1-\beta)(C_{2k})]$ for any $\beta \in [0,1]$, and vice versa. In conclusion, the set of equilibrium payoffs in the game is A which corresponding the set of equilibrium

$$([\alpha(L_{2k-1}), (1-\alpha)(L_{2k})], [\beta(C_{2k-1}), (1-\beta)(C_{2k})])$$

for every $k \in \{1, ..., K\}$ and every $\alpha, \beta \in [0,1]$.

5.50 We start by proving that in every equilibrium of the game Player III Plays W with probability 1. Assume by contradiction that Player III plays E with some positive probability. Then Player I's best response is B which dominates T, and Player II's best response is L which dominates R, which case Player III is better off Playing W with probability 1, a contradiction.

Assume next that Player III plays W with probability 1. In this case, Player I is indifferent between T and B so his best response is $[x(T), (1-x)(B)]$, for every $x \in [0,1]$. In addition, Player II is indifferent between L and R so his best response is $[y(L), (1-y)(R)]$, for every $y \in [0,1]$. However, to prevent Player III deviating to E the following inequality must hold

$$3xy + (1-xy) \geq 4xy$$

that is, $\frac{1}{2} \geq xy$. Therefore, one can conclude that the set of equilibria in the game is

$$\left\{ \left([x(T), (1-x)(B)], [y(L), (1-y)(R)], W \right) : 0 \leq x, y \leq 1, xy \leq \tfrac{1}{2} \right\}$$

which corresponds the set of equilibrium payoffs

$$\left\{ (y, 1-x, 1+2xy) : 1 \leq x, y \leq 1, xy \leq \tfrac{1}{2} \right\}.$$

This set of equilibrium payoffs is not a union of polytopes in $\mathbf{R}^3$.

5.51 We first find all equilibria in which at least one of the players plays pure. Assume for instance that Player III plays the pure strategy W. If Player I plays B with some positive probability then Player II responses with R which case Player I prefers to deviate to the pure strategy T.

If Player II plays R with some positive probability, then Player I responses with T which case Player III deviates to E. Therefore, there is only one equilibrium in which Player III plays W which is (T, L, W). In a similar way one can verify that there is another equilibrium (B, R, E). By the symmetry between the players these are the only equilibria in which at least one of the players plays pure.
Consider next the completely mixed strategy vector

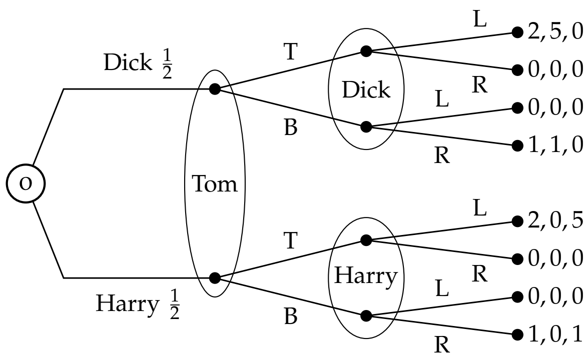
$$\left([x(T), (1-x)(B)], [y(L), (1-y)(R)], [z(W), (1-z)(E)]\right).$$

This vector is an equilibrium if and only if the following inequalities hold

$$\begin{aligned}(1-y)z &= y(1-z)\\(1-x)z &= x(1-z)\\(1-x)y &= x(1-y)\end{aligned}$$

if and only if $x = y = z \in [0, 1]$.

5.52 (a) The extensive form of the game is depicted below.



(b) The strategic form of the game is as follows where Tom is the row player, Dick is the column player and Harry selects a matrix.

L

	L	R
T	2, 2.5, 2.5	1, 0, 2.5
B	0, 0, 0	0.5, 0.5, 0

R

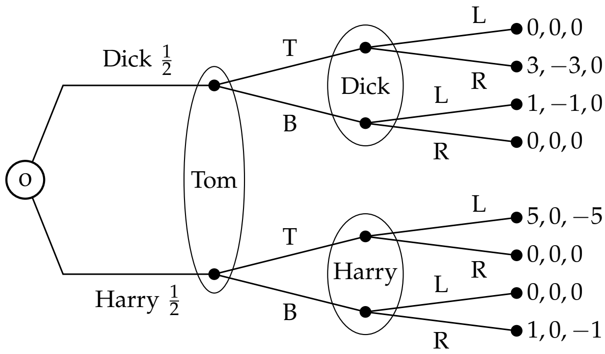
	L	R
T	1, 2.5, 0	0, 0, 0
B	0.5, 0, 0.5	1, 0.5, 0.5

(c) There are 2 pure equilibria of the game which are (T, L, L) and (B, R, R).

(d) An additional equilibrium in mixed strategies is

$$\left([\tfrac{1}{6}(T), \tfrac{5}{6}(B)], [\tfrac{1}{3}(L), \tfrac{2}{3}(R)], [\tfrac{1}{3}(L), \tfrac{2}{3}(R)]\right).$$

5.53 (a) The extensive form of the game is depicted below.



(b) The strategic form of the game is as follows where Tom is the row player, Dick is the column player and Harry selects a matrix.

L

	L	R
T	$2.5, 0, -2.5$	$4, -1.5, -2.5$
B	0.5,-0.5,0	$0, 0, 0$

R

	L	R
T	$0, 0, 0$	$1.5, -1.5, 0$
B	$1, -0.5, -0.5$	$0.5, 0, -0.5$

(c) We will show that the game has only one equilibrium

$$\left([\tfrac{1}{4}(T), \tfrac{3}{4}(B)], [\tfrac{1}{2}(L), \tfrac{1}{2}(R)], [1(R)]\right).$$

Consider Tom's strategy $[x(T), (1-x)(B)]$ where $x \in [0, 1]$. Dick is indifferent between L and R if and only if $-0.5(1-x) = -1.5x$ if and only if $x = \frac{1}{4}$. In particular if $x > \frac{1}{4}$ he prefers L, otherwise, if $x < \frac{1}{4}$ he prefers R.

Harry is indifferent between L and R if and only if $-0.5(1-x) = -2.5x$ if and only if $x = \frac{1}{6}$. If $x > \frac{1}{6}$ he prefers R, whereas, if $x < \frac{1}{6}$ he prefers L.

Therefore, there is no equilibrium in which Tom plays T with probability $x < \frac{1}{4}$, otherwise, Dick selects R, which case Tom selects T, a contradiction.

In addition, there is no equilibrium in which $x > \frac{1}{4}$, otherwise, Dicks plays L and Harry plays R which case Tom deviates to B, a contradiction.

Thus, in an equilibrium, Tom plays T with probability $x = \frac{1}{4}$, so Harry plays R. However, to guarantee that Tom will play a mixed strategy, Dick must follow the strategy $[y(L), (1-y)(R)]$ such that $1.5(1-y) = y + 0.5(1-y)$ if and only if $y = \frac{1}{2}$, as claimed above.

5.54 Consider a two-player game on the unit square, i.e., $S_1 = S_2 = [0,1]$. Assume that the payoff functions of the two players are bilinear (not necessarily zero sum)

$$u_1(x,y) = (a_1 y + b_1)x + c_1 y + d_1$$
$$u_2(x,y) = (a_2 x + c_2)y + b_2 x + d_2$$

We prove that the game has an equilibrium in pure strategies.

The best response of Player I is either $x = 1$ if and only $a_1 y + b_1 \geq 0$; or $x = 0$ if and only $a_1 y + b_1 \leq 0$; or $x \in [0,1]$ if and only $a_1 y + b_1 = 0$.

The best response of Player II is either $y = 1$ if and only $a_2 x + c_2 \geq 0$; or $y = 0$ if and only $a_2 x + c_2 \leq 0$; or $y \in [0,1]$ if and only $a_2 x + c_2 = 0$.

Thus, if $a_1 + b_1 \geq 0$ and $a_2 + c_2 \geq 0$ then $(1,1)$ is an equilibrium.
If $b_1 \geq 0$ and $a_2 + c_2 \leq 0$ then $(1,0)$ is an equilibrium.
If $a_1 + b_1 \leq 0$ and $c_2 \geq 0$ then $(0,1)$ is an equilibrium.
If $b_1 \leq 0$ and $c_2 \leq 0$ then $(0,0)$ is an equilibrium.
If non of the above holds then either $c_2 > 0$, $a_1 + b_1 > 0$, $a_2 + c_2 < 0$ and $b_1 < 0$ or $b_1 > 0$, $a_2 + c_2 > 0$, $a_1 + b_1 < 0$ and $c_2 < 0$. However, both cases lead to $0 < -\frac{b1}{a_1}, -\frac{c_2}{a_2} < 1$ so $(-\frac{c_2}{a_2}, -\frac{b_1}{a_1})$ is an equilibrium.

5.55 Let Γ be a two-player zero-sum game in which Player I's set of pure strategies S_I, is finite, Player II's set of pure strategies $S_{II} = \{1,2,3,...\}$ is a countable set, and the payoff function u is bounded. Let Γ^n be a two-player zero-sum game in which Player I's set of pure strategies is S_I, Player II's set of pure strategies is $S_{II}^n = \{1,2,...,n\}$, and the payoff functions are identical to those of Γ. Let v^n be the value of the game Γ^n, and let $\sigma_I^n \in \Delta(S_I)$ and $\sigma_{II}^n \in \Delta(S_{II}^n)$ be the optimal strategies of the two players in this game, respectively.

(a) For every $n \in \mathbf{N}$, $S_I^n = S_I^{n+1}$ and $S_{II}^n \subseteq S_{II}^{n+1}$. Therefore, by following σ_{II}^n in Γ^{n+1} Player II can guarantee that the payoff is at most v^n so $v^n \geq v^{n+1}$. In particular, $(v^n)_{n\in\mathbf{N}}$ is a sequence of nonincreasing real numbers that are also bounded. Hence, it can be deduced that $v := \lim_{n\to\infty} v^n$ exists.

(b) Let σ_I be an accumulation point of the sequence $(\sigma_I^n)_{n\in\mathbf{N}}$; i.e., there exists a subsequence $(\sigma_I^{n_k})_{k\in\mathbf{N}}$ converging to σ_I.
Let $\sigma_{II} \in \Delta(S_{II})$ be a strategy of Player II. Set $\widehat{\sigma}_{II}^{n_k}(i) := \frac{\sigma_{II}(i)}{\sum_{m=1}^{n_k} \sigma_{II}(m)}$, so $\widehat{\sigma}_{II}^{n_k} \in \Delta(S_{II}^{n_k})$ and $\lim_{k\to\infty} \widehat{\sigma}_{II}^{n_k} = \sigma_{II}$. Then, by the continuity of u one has

$$v = \lim_{k\to\infty} v^{n_k} \leq \lim_{k\to\infty} u(\sigma_I^{n_k}, \widehat{\sigma}_{II}^{n_k}) = u(\lim_{k\to\infty} \sigma_I^{n_k}, \lim_{k\to\infty} \widehat{\sigma}_{II}^{n_k}) = u(\sigma_I, \sigma_{II}),$$

$$\Rightarrow v \leq \inf_{\sigma_{II} \in \Delta(S_{II})} u(\sigma_I, \sigma_{II}).$$

(c) σ_{II}^n is an optimal strategy of Player II in the game Γ^n which has the same payoff function as Γ, thus

$$v^n \geq \sup_{\sigma_I \in \Delta(S_I)} u(\sigma_I, \sigma_{II}^n).$$

(d) By Parts (b) and (c) it follows that

$$\begin{aligned} v &\leq \inf_{\sigma_{II} \in \Delta(S_{II})} u(\sigma_I, \sigma_{II}) \leq \sup_{\sigma_I \in \Delta(S_I)} \inf_{\sigma_{II} \in \Delta(S_{II})} u(\sigma_I, \sigma_{II}) \\ &\leq \inf_{\sigma_{II} \in \Delta(S_{II})} \sup_{\sigma_I \in \Delta(S_I)} u(\sigma_I, \sigma_{II}) \leq \sup_{\sigma_I \in \Delta(S_I)} u(\sigma_I, \sigma_{II}^n) \leq v^n \xrightarrow[n \to \infty]{} v \end{aligned}$$

Hence

$$v = \sup_{\sigma_I \in \Delta(S_I)} \inf_{\sigma_{II} \in \Delta(S_{II})} u(\sigma_I, \sigma_{II}) = \inf_{\sigma_{II} \in \Delta(S_{II})} \sup_{\sigma_I \in \Delta(S_I)} u(\sigma_I, \sigma_{II}).$$

(e) Assume that the payoff function is $u(s, n) = \frac{1}{n}$ for every strategy $s \in S_I$ of Player I and every $n \in S_{II}$. Then, $\sigma_{II}^n = n$ is an optimal strategy of Player II in Γ^n. However, the sequence $(\sigma_{II}^n)_{n \in \mathbf{N}}$ has no accumulation point.

(f) Part (d) does not necessarily hold when S_I is also countably infinite. Consider for instance a two player zero-sum game where $S_I = S_{II} = \mathbf{N}$ and for every $n, m \in \mathbf{N}$

$$u(n, m) = \begin{cases} 1 & n > m \\ 0 & n = m \\ -1 & n < m. \end{cases}$$

Then, for every $\epsilon > 0$ and every $\sigma_I \in \Delta(S_I)$ there is $N \in \mathbf{N}$ such that $\sum_{k=N+1}^{\infty} \sigma_I(k) < \epsilon$. Hence $u(\sigma_I, N+1) < -1 + 2\epsilon$. Hence

$$\sup_{\sigma_I \in \Delta(S_I)} \inf_{\sigma_{II} \in \Delta(S_{II})} u(\sigma_I, \sigma_{II}) = -1,$$

whereas, by similar arguments,

$$\inf_{\sigma_{II} \in \Delta(S_{II})} \sup_{\sigma_I \in \Delta(S_I)} u(\sigma_I, \sigma_{II}) = 1.$$

5.56 (a) Let $(s_1, s_2, ...)$ be a pure strategy vector. If $\sum_{j=1}^{\infty} s_j < \infty$ then there is at least one player i such that $s_i = 0$ and his payoff is 0. Hence by deviating to 1 he can gain at least 1, so $(s_1, s_2, ...)$ is not an equilibrium. On the other hand, if $\sum_{j=1}^{\infty} s_j = \infty$ then there is at least one player i such that $s_i = 1$ and his payoff is -1. However, by deviating to 0 he can gain at least 1, hence it is not an equilibrium.

(b) Let $(\sigma_1, \sigma_2, ...)$ be a mixed strategy vector such that $\sigma_j = [x_j(1), (1 - x_j)(0)]$ for every $j \in \mathbf{N}$. Then by Kolmogorov's 0-1 Law, since the strategies of the players are independent, the probability of the event according to which infinitely many players selects 1 is either 1 or 0. If the probability of the event is 1 then there is at least one player that plays 1 who can gain by deviating to 0. Likewise, if the probability of the event is 0 then there is at least one player that plays 0 who can gain by deviating to 1.

5.57 (a) Let $X \subseteq \mathbf{R}^n$ be a nonempty convex, and compact set, and let $f : X \to \mathbf{R}^n$ be a continuous function. We prove that there is a point $x^* \in X$ such that $\langle x - x^*, f(x^*) \rangle \leq 0$ for every $x \in X$, where $\langle x, y \rangle := \sum_{i=1}^{n} x_i y_i$ is the scalar product in $\mathbf{R}^n$.
Define, $g : X \to X$ by $g(x) := \pi_X(x + f(x))$ for every $x \in X$, where π_X is the projection on X. g is continuous as a composition of two continuous functions. Following Brouwer's Fixed Point Theorem, there is a point $x^* \in X$ such that $g(x^*) = x^*$. Hence, for every $x \in X$ and every $\alpha \in (0, 1]$, $\alpha x + (1 - \alpha)x^* \in X$ so

$$\begin{aligned}
\|f(x^*)\|_2^2 &= \|x^* + f(x^*) - x^*\|_2^2 \\
&\leq \|x^* + f(x^*) - (\alpha x + (1 - \alpha)x^*)\|_2^2 \\
&= \|\alpha(x^* - x) + f(x^*)\|_2^2 \\
&= \|f(x^*)\|_2^2 + 2\alpha\langle x^* - x, f(x^*)\rangle + \alpha^2\|x^* - x\|_2^2,
\end{aligned}$$

from which it follows that for every $\alpha \in (0, 1]$ and every $x \in X$ one has

$$-\alpha\frac{\|x^* - x\|_2^2}{2} \leq \langle x^* - x, f(x^*)\rangle \Rightarrow 0 = \lim_{\alpha \to 0} -\alpha\frac{\|x^* - x\|_2^2}{2} \leq \langle x^* - x, f(x^*)\rangle.$$

(b) Let $G = (N, (S_i)_{i \in N}, (u_i)_{i \in N})$ be a strategic-form game and let $\Gamma = (N, (\Sigma_i)_{i \in N}, (U_i)_{i \in N})$ be its mixed extension. Let $f : \Sigma \to \mathbf{R}^{\sum_{i \in I} |S_i|}$ be the function defined by

$$f_{i,s_i}(\sigma) := U_i(s_i, \sigma_{-i}), \quad \forall i \in N, s_i \in S_i.$$

Set $f(\sigma) = (f_{i,s_i}(\sigma))_{i \in N, s_i \in S_i}$. $f : \Sigma \to \mathbf{R}^{|N|}$ is a continuous function over a compact and convex set and thus, by Part (a), there is $\sigma^* \in \Sigma$ satisfying $\langle \sigma, f(\sigma^*) \rangle \leq \langle \sigma^*, f(\sigma^*) \rangle$ for every $\sigma \in \Sigma$. We prove that σ^* is Nash equilibrium. For every player i and every strategy $\sigma_i \in \Sigma_i$, set $\sigma = (\sigma_i, \sigma^*_{-i})$.

$$\begin{aligned}
U_i(\sigma_i, \sigma^*_{-i}) &= \sum_{s_i \in S_i} \sigma_i(s_i) U_i(s_i, \sigma^*_{-i}) \\
&= \sum_{j \in N} \sum_{s_j \in S_j} \sigma_j(s_j) U_j(s_j, \sigma^*_{-j}) - \sum_{j \neq i} \sum_{s_j \in S_j} \sigma_j(s_j) U_j(s_j, \sigma^*_{-j}) \\
&= \langle \sigma, f(\sigma^*) \rangle - \sum_{j \neq i} \sum_{s_j \in S_j} \sigma_j(s_j) U_j(s_j, \sigma^*_{-j}) \\
&\leq \langle \sigma^*, f(\sigma^*) \rangle - \sum_{j \neq i} \sum_{s_j \in S_j} \sigma^*_j(s_j) U_j(s_j, \sigma^*_{-j}) \\
&= U_i(\sigma^*),
\end{aligned}$$

as claimed.

5.58 Let $\Gamma = (N, (S_i)_{i \in N}, (u_i)_{i \in N})$ be a two-player constant-sum game; i.e., there is $c \in \mathbf{R}$ such that for every $s_1 \in S_1$ and every $s_2 \in S_2$ one has $u_1(s_1, s_2) + u_2(s_1, s_2) = c$ which leads to $u_2(s_1, s_2) = c - u_1(s_1, s_2)$. Therefore, the zero-sum game Γ' with the same sets of pure strategies and a payoff function u_1 is strategically equivalent to Γ. Indeed, the payoff function in Γ, u_1 and u_2 are positive affine transformations of the payoff function in Γ', u_1 and $-u_1$, respectively.

5.59 Let (σ_I, σ_{II}) is the solution of the system of linear equations (5.70)–(5.79). By (5.74)-(5.79), σ_I and σ_{II} are appropriate mixed strategies of Player I and Player II, respectively. By (5.76) and (5.77) Y_I and Y_{II} are the support of σ_I and σ_{II} respectively. By (5.70) and (5.72), every pure strategy in the support of σ_I is a best response to σ_{II}, hence σ_I is also a best response to σ_{II}. Similarly, by (5.71) and (5.72) σ_{II} is also a best response to σ_I. In conclusion, (σ_I, σ_{II}) is an equilibrium.

5.60 (a) Denote by u_1 the utility function of Player I. Assume without loss of generality that $u_1(1500) = 1$ and $u_1(300) = 0$. Then,

$$\begin{aligned}
u_1(600) &= \tfrac{1}{4} u_1(300) + \tfrac{3}{4} u_1(1500) = \tfrac{3}{4} \\
u_1(800) &= \tfrac{1}{2} u_1(600) + \tfrac{1}{2} u_1(1500) = \tfrac{7}{8}.
\end{aligned}$$

Denote by u_2 the utility function of Player II. Assume without loss of generality that $u_1(-1500) = -1$ and $u_1(-300) = 0$. Then,

$$u_1(-800) = \tfrac{1}{8}u_1(-300) + \tfrac{7}{8}u_1(-1500) = -\tfrac{7}{8}$$
$$u_1(-600) = \tfrac{1}{7}u_1(-300) + \tfrac{6}{7}u_1(-800) = -\tfrac{3}{4}.$$

(b) The payoff functions satisfies $u_2(-1500) = -u_1(1500)$, $u_2(-800) = -u_1(800)$, $u_2(-600) = -u_1(600)$ and $u_2(-300) = -u_1(300)$ thus it is indeed a zero-sum game.

(c) The zero-sum game is depicted in the following matrix.

		Player II		
		L	M	R
Player I	T	0	7/8	1
	B	1	3/4	0

The value of the game is $\frac{1}{2}$. The optimal strategy of Player I is $[\frac{1}{2}(T), \frac{1}{2}(B)]$. The optimal strategy of Player II is $[\frac{1}{2}(L), \frac{1}{2}(R)]$.

5.61 For every game of the following games that are strategically equivalent to two-player zero-sum games, denote by u_1 and u_2 the payoff functions in the original game and by v_1 and v_2 the payoff functions in its equivalent zero-sum game.
Game A: The game is equivalent to a zero-sum game if there are $a > 0$ and $b \in \mathbf{R}$ such that $v_1 = u_1$ and $-v_1 = v_2 = a \cdot u_2 + b$. Namely,

$$\begin{aligned} -11 &= 2a + b \\ -5 &= 4a + b \\ 7 &= 8a + b \\ -17 &= b. \end{aligned}$$

The above system of linear equations has a solution which is $a = 3$ and $b = -17$. Hence, the game A is indeed equivalent to a zero sum game where $v_1 = u_1$ and $v_2 = 3u_2 - 17$.
Game B: $v_1 = u_1$ and $v_2 = u_2 - 9$.
Game D: $v_1 = u_1$ and $v_2 = 3u_2 - 17$.

Note that in Game C both players gain if Player II plays R instead of C given Player I plays B, thus it cannot be equivalent to a zero-sum game. In Game F, given that Player II plays L and Player I deviates from T to B then the payoffs of Player II increases while the payoff of Player I does not change so the game cannot be equivalent to a zero-sum game. Game E is not equivalent to a zero-sum game since the appropriate system of linear equations has no solution.

5.62 (a) The value of the game is $v = 10$. The optimal strategy of Player I is $[\frac{3}{4}(T), \frac{1}{4}(B)]$. The optimal strategy of Player II is $[\frac{1}{2}(L), \frac{1}{2}(R)]$.

(b) The game is equivalent to the zero-sum game in Part (a) and thus it has only one equilibrium according to which both players follow their optimal strategies,

$$\left([\tfrac{3}{4}(T), \tfrac{1}{4}(B)], [\tfrac{1}{2}(L), \tfrac{1}{2}(R)]\right).$$

The equilibrium payoff of player II will become $-10 + 18 = 8$.

(c) The game in (c) is also equivalent to the zero-sum game in Part (a) and thus it has the same equilibrium as in Part (b). The equilibrium payoff of Player I will be changed by the given transformation and become 17.

5.63 Let $G = (N, (S_i)_{i \in N}, (u_i)_{i \in N})$ and $\widehat{G} = (N, (S_i)_{i \in N}, (v_i)_{i \in N})$ be two strategically equivalent strategic-form games, i.e., for each player $i \in N$ there exist $\alpha_i > 0$ and $\beta_i \in \mathbf{R}$ such that $v_i(s) = \alpha_i u_i(s) + \beta_i$ for every $s \in S$. Let σ be an equilibrium in mixed strategies of G. Thus, for every player $i \in N$ and every mixed $\sigma_i' \in \Sigma_i$ of player i

$$\begin{aligned}
U_i(\sigma) &\geq U_i(\sigma_i', \sigma_{-i}) && \Longleftrightarrow \\
\sum_{s \in S} \sigma(s) u_i(s) &\geq \sum_{s \in S} \sigma_i'(s_i) \sigma_{-i}(s_{-i}) u_i(s) && \Longleftrightarrow \\
\sum_{s \in S} \sigma(s) \alpha_i u_i(s) + b_i &\geq \sum_{s \in S} \sigma_i'(s_i) \sigma_{-i}(s_{-i}) \alpha_i u_i(s) + b_i && \Longleftrightarrow \\
\sum_{s \in S} \sigma(s) \left(\alpha_i u_i(s) + b_i\right) &\geq \sum_{s \in S} \sigma_i'(s_i) \sigma_{-i}(s_{-i}) \left(\alpha_i u_i(s) + b_i\right) && \Longleftrightarrow \quad (24) \\
\sum_{s \in S} \sigma(s) v_i(s) &\geq \sum_{s \in S} \sigma_i'(s_i) \sigma_{-i}(s_{-i}) v_i(s) && \Longleftrightarrow \\
V_i(\sigma) &\geq V_i(\sigma_i', \sigma_{-i}) && \quad (25)
\end{aligned}$$

where Inequality (24) holds since σ and (σ_i', σ_{-i}) define distributions over S. In particular, by (25), σ is an equilibrium in mixed strategies of $\widehat{G}$.

5.64 (a) The game has no equilibrium in which at least one of the player plays pure: if Player I plays T then Player II's best response is R which case Player I prefers to deviate to B. Likewise, if Player I plays B then Player II's best response is L which case Player I prefers to deviate to T.

The strategy vector $\left([\frac{1}{2}(T), \frac{1}{2}(B)], [\frac{1}{2}(L), \frac{1}{2}(R)]\right)$ is an equilibrium in mixed strategy since Player I is indifferent between T and B

$(\frac{1}{2} \cdot 1 + \frac{1}{2} \cdot (-1) = 0)$ as well as Player II is indifferent between L and R $(\frac{1}{2} \cdot 0 + \frac{1}{2} \cdot 1 = \frac{1}{2} \cdot 1 + \frac{1}{2} \cdot 0)$.

(b) Note that in the zero-sum game the strategy R strictly dominates L, hence the optimal strategy of Player I is B which guarantees him 0. In particular, the value of the game is 0.

(c) If Player II plays $[\frac{1}{2}(L), \frac{1}{2}(R)]$ then Player I is indifferent between the strategies $[\frac{1}{2}(T), \frac{1}{2}(B)]$ and B since his expected payoffs under both strategies are equal.

5.65 (a) Let $\sigma_I = [x(T), (1-x)(B)]$ and $\sigma_{II} = [y(L), (1-y)(R)]$. Then

$$U(\sigma_I, \sigma_{II}) = x(1-y) - (1-x)y = x - y.$$

Therefore,

$$\max_{x \in [0,1]} \min_{\{y \in [0,1] : xy \leq \frac{1}{2}\}} U(\sigma_I, \sigma_{II}) = 1 - \tfrac{1}{2} = \tfrac{1}{2}.$$

$$\min_{y \in [0,1]} \max_{\{x \in [0,1] : xy \leq \frac{1}{2}\}} U(\sigma_I, \sigma_{II}) = \tfrac{1}{2} - 1 = -\tfrac{1}{2}.$$

(b) The strategy B is strictly dominated by T so in an equilibrium Player I will ascribe greater probability to T. Likewise, the strategy R is strictly dominated by L so in an equilibrium Player II will ascribe greater probability to R. Thus, every strategy vector $\big([x(T), (1-x)(B)], [y(L), (1-y)(R)]\big)$ such that $xy = \frac{1}{2}$ is an equilibrium.

(c)

$$\begin{aligned}
&\max_{\sigma_I \in \Sigma_I} \min_{\{\sigma_{II} \in \Sigma_{II} : C(\sigma_I, \sigma_{II}) \leq \gamma\}} U(\sigma_I, \sigma_{II}) \geq \max_{\sigma_I \in \Sigma_I} \min_{\sigma_{II} \in \Sigma_{II}} U(\sigma_I, \sigma_{II}) \\
&= \min_{\sigma_{II} \in \Sigma_{II}} \max_{\sigma_I \in \Sigma_I} U(\sigma_I, \sigma_{II}) \geq \min_{\sigma_{II} \in \Sigma_{II}} \max_{\{\sigma_I \in \Sigma_I : C(\sigma_I, \sigma_{II}) \leq \gamma\}} U(\sigma_I, \sigma_{II}),
\end{aligned}$$

where the first inequality holds since the minimization is taken over a larger set and the second inequality holds since the maximization is taken over a smaller set.

The result does not contradict Theorem 5.40 since the sets over which the minimizations and the maximizations are taken (in the right-hand side and left-hand side) are not identical.

(d) By the compactness of Σ_i and Σ_{-i} and since C is continuous,

$$\sup_{\sigma_{-i} \Sigma_{-i}} \inf_{\{\sigma_i \in \Sigma_i : C(\sigma_i, \sigma_{-i}) \leq \gamma\}} C(\sigma_i, \sigma_{-i}) = \max_{\sigma_{-i} \Sigma_{-i}} \min_{\{\sigma_i \in \Sigma_i : C(\sigma_i, \sigma_{-i}) \leq \gamma\}} C(\sigma_i, \sigma_{-i}).$$

Let $\sigma^*_{-i} \in \Sigma_{-i}$ be a strategy vector such that

$$\max_{\sigma_{-i}\Sigma_{-i}} \min_{\{\sigma_i \in \Sigma_i : C(\sigma_i,\sigma_{-i}) \leq \gamma\}} C(\sigma_i, \sigma_{-i}) = \min_{\{\sigma_i \in \Sigma_i : C(\sigma_i,\sigma^*_{-i}) \leq \gamma\}} C(\sigma_i, \sigma^*_{-i}).$$

By the Slater condition, there is $\sigma^*_i \in \Sigma_i$ such that $C(\sigma^*_i, \sigma^*_{-i}) < \gamma$. Hence,

$$\min_{\{\sigma_i \in \Sigma_i : C(\sigma_i,\sigma^*_{-i}) \leq \gamma\}} C(\sigma_i, \sigma^*_{-i}) \leq C(\sigma^*_i, \sigma^*_{-i}) < \gamma,$$

as needed.

(e) Let σ be a strategy vector satisfying $C(\sigma) \leq \gamma$. Let i be a player, and let $(\sigma^k_{-i})_{k=1}^{\infty}$ be a sequence of strategy vectors converging to σ_{-i}. By Part (d), there exists a sequence $(\widehat{\sigma}^k_i)_{k=1}^{\infty}$ such that for every $k \in \mathbf{N}$, $C(\widehat{\sigma}^k_i, \sigma^k_{-i}) < \gamma$. Denote by $\sigma^k_i = \alpha^k \sigma_i + (1-\alpha^k)\widehat{\sigma}^k_i$ such that

$$\alpha^k = \begin{cases} 1 & \text{if } C(\sigma_i, \sigma^k_{-i}) \leq \gamma \\ \frac{\gamma - C(\widehat{\sigma}^k_i, \sigma^k_{-i})}{C(\sigma_i,\sigma^k_{-i}) - C(\widehat{\sigma}^k_i, \sigma^k_{-i})} & \text{if } C(\sigma_i, \sigma^k_{-i}) > \gamma. \end{cases}$$

For every $k \in \mathbf{N}$, if $C(\sigma_i, \sigma^k_{-i}) \leq \gamma$ then $\sigma^k_i = \sigma_i$ so $C(\sigma^k_i, \sigma^k_{-i}) \leq \gamma$. If $C(\sigma_i, \sigma^k_{-i}) > \gamma$, then

$$\begin{aligned} C(\sigma^k_i, \sigma^k_{-i}) =& \frac{\gamma - C(\widehat{\sigma}^k_i, \sigma^k_{-i})}{C(\sigma_i,\sigma^k_{-i}) - C(\widehat{\sigma}^k_i, \sigma^k_{-i})} C(\sigma_i, \sigma^k_{-i}) \\ &+ \frac{C(\sigma_i,\sigma^k_{-i}) - \gamma}{C(\sigma_i,\sigma^k_{-i}) - C(\widehat{\sigma}^k_i, \sigma^k_{-i})} C(\sigma^k_i, \sigma^k_{-i}) = \gamma. \end{aligned}$$

In addition, since $(\sigma^k_{-i})_{k=1}^{\infty}$ converges to σ_{-i} and by the continuity of C, $\lim_{k\to\infty} C(\sigma_i, \sigma^k_{-i}) = C(\sigma) \leq \gamma$. Thus $(\alpha^k)_{k=1}^{\infty}$ converges to 1 and $(\sigma^k_i)_{k=1}^{\infty}$ converges to σ_i as needed.

(f) Let $F : \Sigma \to \Sigma$ be a function defined as follows

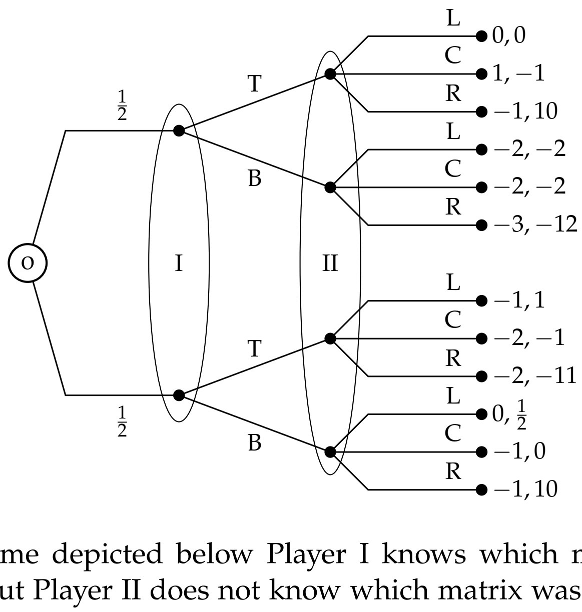
$$F(\sigma) = \left\{ \widehat{\sigma} \in \Sigma \ : \ \begin{array}{c} \forall i \in N, C(\widehat{\sigma}_i, \sigma_{-i}) \leq \gamma \text{ and} \\ \forall i \in N \forall \sigma'_i \in \Sigma_i, \text{ s.t. } C(\sigma'_i, \sigma_{-i}) \leq \gamma \\ \Rightarrow U_i(\widehat{\sigma}_i, \sigma_{-i}) \geq U_i(\sigma'_i, \sigma_{-i}) \end{array} \right\}$$

F assigns the best response of each player among all permissible strategies of that player. Notice that since C is multilinear and by the Slater condition, the set $\left\{\widehat{\sigma}_i \in \Sigma_i \ : \ C(\widehat{\sigma}_i, \sigma_{-i}) \leq \gamma\right\}$ is non-empty and compact, for every player i and every σ_{-i}. Therefore F is not empty. In addition, F is convex and compact. Therefore, by Kakutani's Fixed Point Theorem, there is a strategy vector σ^* such that $\sigma^* \in F(\sigma^*)$. In particular, (1) $C(\sigma^*) = C(\sigma_1^*, \sigma_{-1}^*) \leq \gamma$, namely, σ^* is permissible; and (2) for every player $i \in N$ and every strategy $\sigma_i' \in \Sigma_i$ such that $C(\sigma_i', \sigma_{-i}^*) \leq \gamma$ one has $U_i(\widehat{\sigma}_i, \sigma_{-i}^*) \geq U_i(\sigma_i', \sigma_{-i}^*)$. That is, σ^* is an equilibrium of the strategic-form game with constraints.

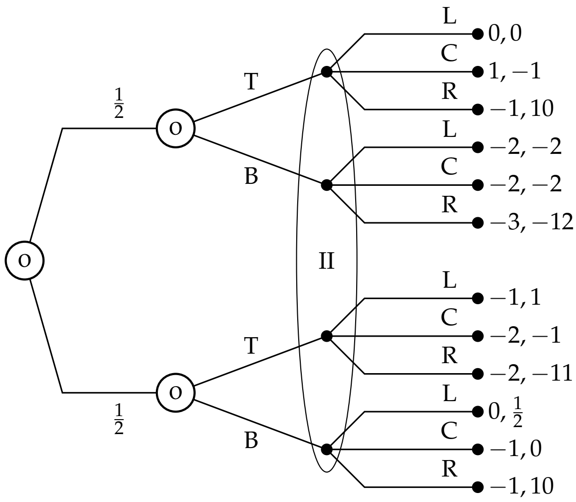
5.66 Let Γ be a two-player zero-sum game in extensive form and let Γ' be the game derived from Γ by splitting several information sets of Player I. By Theorem 5.44, the maxmin value of Player I in mixed strategies in the game Γ' is greater than or equal to his maxmin value in mixed strategies in Γ. However, in a zero-sum game the maxmin value of Player I in mixed strategies equals the value of the game in mixed strategies. Therefore, the value of the game Γ' in mixed strategies is greater than or equal to the value of Γ in mixed strategies.

5.67 The unique equilibrium payoff of Game A is $(5, 5)$ which is obtained by the equilibrium $([\frac{1}{2}(l), \frac{1}{2}(r)], [\frac{1}{2}(L), \frac{1}{2}(R)])$. The unique equilibrium payoff of Game B is $(20, 1)$ which is obtained by the equilibrium $(l_1 r_2, M)$. By adding an information to Player I his equilibrium payoff increases.

5.68 (a) In the game depicted below no player knows which matrix was chosen, so the only equilibrium is (T, L) which corresponds to the equilibrium payoff $(-\frac{1}{2}, \frac{1}{2})$.



(b) In the game depicted below Player I knows which matrix was chosen, but Player II does not know which matrix was chosen.



In this game, if the left matrix is chosen then Player I plays T since it strictly dominates B, whereas if the right matrix is chosen he plays B that strictly dominates T. Therefore the best response of Player II is R. Hence, the only equilibrium payoff is $(-1, 10)$. The equilibrium payoff of Player I decreases by adding him information whereas the equilibrium payoff of Player II increases.

5.69 See Exercise 5.45

5.70 (a) The dual program, in the variables $x = (x_i)_{i=1}^n$ is:

$$\begin{aligned} \text{Compute: } & Z_D := \max b \\ \text{subject to: } & xA \geq \overrightarrow{b} \\ & \sum_{i=1}^{n} x_i = 1 \\ & x \geq \overrightarrow{0} \end{aligned}$$

where b is a real number, and $\overrightarrow{b}$ is an m-dimensional vector, all of whose coordinates equal b.

(b) The set $Y = \left\{ y \in \mathbf{R}^m \ : \ y \geq \overrightarrow{0}, \sum_{j=1}^m y_j = 1 \right\}$ is a non-empty, compact and convex set and thus the set $\left\{ Ay^T \ : \ y \in Y \right\}$ is non-empty compact and convex as well. Hence, it can be derived that the set $\left\{ c \in \mathbf{R} : \exists y \in Y \text{ s.t. } Ay^T \leq \overrightarrow{c} \right\}$ is convex, closed and bounded from below so it has a minimum which is Z_P. Therefore, Z_P is finite.

(c) Let y be an optimal solution to the primal program, therefore, $y \in Y$ defines a mixed strategy for Player II. Furthermore, since $Ay^T \leq \overrightarrow{Z_P}$, for every pure strategy of Player I (which is equivalent to a row) the payoff is at most Z_P, i.e., y guarantees Player II an expected payoff of at most Z_P.

(d) Following a similar arguments as in Part (b), the optimal solution to the dual program satisfies $\sum_{i=1}^{n} x_i = 1$ and $x \geq \overrightarrow{0}$ so it defines a mixed strategy for Player I. Furthermore, by the inequality $xA \geq \overrightarrow{Z_D}$ the solution guarantees an expected payoff of at least Z_D.

(e) By Part (b) Z_p is finite so the Duality Theorem is applicable here. Furthermore, by Part (b), the minmax value of the game is at most Z_P, i.e, $\overline{v} \leq Z_P$. By Part (c) the maxmin value of game is at least Z_D, i.e., $\underline{v} \geq Z_D$. Therefore, by Theorem 5.40,

$$Z_D \leq \underline{v} \leq \overline{v} \leq Z_P. \tag{26}$$

However, the Duality Theorem implies that $Z_P = Z_D$, therefore the inequalities of (26) obtain as equalities so Z_P is the value of the game.

5.71 (a) Let G be a game, and let G' be a game obtained from G by adding information to one of the players, say player i. Let j be a player differs from i. The set Σ_{-j} of mixed strategy vectors of $-j$ in G is contained in the set Σ'_{-j} of mixed strategy vectors of $-j$ in G', hence $\min_{\sigma'_{-j} \in \Sigma'_{-j}} U_j(\sigma_j, \sigma'_{-j}) \leq \min_{\sigma_{-j} \in \Sigma_{-i}} U_j(\sigma_i, \sigma_{-j})$. Therefore,

$$\max_{\sigma_j \in \Sigma_j} \min_{\sigma'_{-j} \in \Sigma'_{-j}} U_j(\sigma_j, \sigma'_{-j}) \leq \max_{\sigma_j \in \Sigma_j} \min_{\sigma_{-j} \in \Sigma_{-j}} U_j(\sigma_j, \sigma_{-j}).$$

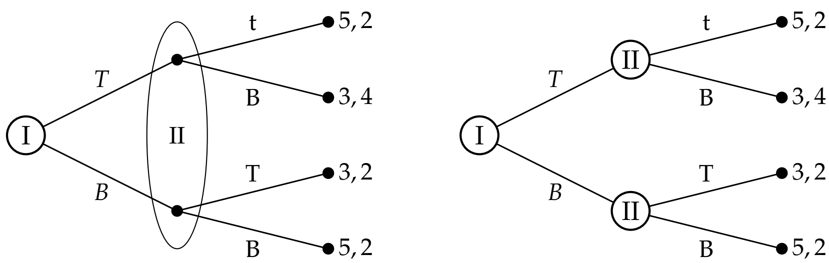
Namely, the maxmin value of the other players does not increase.

(b) Let G, G', Σ_{-j} and Σ'_{-j} be as defined in Part (a). Let $\sigma^*_{-j} \in \Sigma_{-j}$ such that $\min_{\sigma_{-j} \in \Sigma_{-j}} \max_{\sigma_j \in \Sigma_j} U_j(\sigma_j, \sigma_{-j}) = \max_{\sigma_j \in \Sigma_j} U_j(\sigma_j, \sigma^*_{-j})$. Since $\sigma^*_{-j} \in \Sigma'_{-j}$ it follows that

$$\min_{\sigma_{-j} \in \Sigma_{-j}} \max_{\sigma_j \in \Sigma_j} U_j(\sigma_j, \sigma_{-j}) = \max_{\sigma_j \in \Sigma_j} U_j(\sigma_j, \sigma^*_{-j}) \geq \min_{\sigma'_{-j} \in \Sigma'_{-j}} \max_{\sigma_j \in \Sigma_j} U_j(\sigma_j, \sigma'_{-j}).$$

Hence, the minmax value of the other players does not increase.

(c) As the following example shows adding information to one of the players may have no effect on his maxmin value. Indeed, in both games Player II's maxmin value is 2 even though Player II has more information in the right-hand side game.



(d) The example presented in Part (c) shows that adding information to one of the players may decrease the maxmin value of the other players. In the example, adding information to Player II decreases the maxmin value of Player I from 4 to 3.

5.72 Notice that if Player I plays R with positive probability then Player II is better off plays l_1 rather than r_1 which case Player I prefers to play R with probability 0. Hence, there is no equilibrium in which Player I plays R with positive probability. Similarly, there is no equilibrium in which Player I plays L with positive probability. In conclusion, in every equilibrium Player I plays M. To prevent a deviation of Player I, Player II must play l_1 and r_2 with total probability of at most $\frac{1}{10}$. Hence, the set of equilibria is

$$\left\{ \left(M, [x(l_1), (1-x)(r_1)], [y(l_2), (1-y)(r_2)]\right) \,:\, x+y \in [0, 0.1] \right\}.$$

The equilibrium payoff which corresponding the above equilibria is $(1, 1)$.

5.73 **Proof of Lemma 5.57:** Let s_i be a strictly dominated pure strategy. That is, there exists a mixed strategy $\sigma_i \in \Delta(S_i)$ such that $U_i(\sigma_i; s_{-i}) > Ui(s_i; s_{-i})$ for every $s_{-i} \in S_{-i}$. In particular, for every $\sigma_{-i} \in \times_{j \neq i} \Delta(S_j)$,

$$U_i(\sigma_i; \sigma_{-i}) > Ui(s_i; \sigma_{-i}). \tag{27}$$

Assume by contradiction that there is a collection of nonempty sets of pure strategies $(D_j)_{j \neq i}$ such that $s_i \in br_i(D_{-i})$. Hence, there is $\sigma_{-i} \in \times_{j \neq i} \Delta(D_j)$,

$$Ui(s_i; \sigma_{-i}) \geq U_i(s_i'; \sigma_{-i}) \quad \forall s_i' \in S_i,$$

from which it follows that $Ui(s_i; \sigma_{-i}) \geq U_i(\sigma_i; \sigma_{-i})$, a contradiction to Equation (27).

5.74 **Proof of Lemma 5.58:** We show that every pure rationalizable strategy profile survives the process of iterated elimination of strictly dominated strategies.

Let E^k be the set of strategy profiles after the k-th elimination in the process of iterated elimination of strictly dominated strategies. We prove by induction over k that $R_i \subseteq E_i^k$ for every $i \in N$ and every k. For $k = 0$, $R_i \subseteq S_i = E_i^0$ for every player $i \in N$. Assume that it holds for k. Then, by the induction hypothesis and the definition of R_i, $br_i(R_{-i}) = R_i \subseteq E_i^k$ for every $i \in N$. In particular, for every player $i \in N$ and every strategy $s_i \in R_i$ there is $\sigma_{-i} \in \times_{j \neq i}\Delta(R_j) \subseteq \times_{j \neq i}\Delta(E_j^k)$ such that $s_i \in br_i(\sigma_{-i})$, and thus there is no strategy in $\Delta(E_i^k)$ that strictly dominates s_i against E_{-i}^k so $s_i \in E_i^{k+1}$. In conclusion, $R_i \subseteq E^{k+1}$ for every player $i \in N$, as needed. That is, every rationalizable pure strategy profile survives the process of iterated elimination of strictly dominated strategies.

5.75 **Game A:** $R_I = \{T, B\}, \quad R_{II} = \{L, M\}$.
Indeed, $br_I(L) = \{T\}, br_I(M) = \{B\}, br_{II}(T) = \{L\}$ and $br_{II}(B) = \{M\}$.
Game B: $R_I = \{T, B\}, \quad R_{II} = \{L, M\}$.
Indeed, $br_I(L) = \{T, B\}, br_I(M) = \{B\}, br_{II}(T) = \{L, M\}$ and $br_{II}(B) = \{M\}$.
Game C: $R_I = \{T, B\}, \quad R_{II} = \{L, C, R\}$.
M is strictly dominated by T and thus it is eliminated.
Game D: $R_I = \{T\}, \quad R_{II} = \{L\}$.
Indeed, by the process of iterated elimination of strictly dominated strategies: R is eliminated since it is strictly dominated by $[\frac{7}{10}(L), \frac{3}{10}(C)]$; then M and B are strictly dominated by T and thus are eliminated; finally, C is eliminated.

5.76 **Proof of Corollary 5.53:** Let $\sigma^* = (\sigma_i^*)_{i \in N}$ be a mixed strategy Nash equilibrium. Denote by $D_i = \mathbf{supp}(\sigma_i^*)$ for every player $i \in N$. By Exercise 5.35, for every $s_i \in D_i$ one has $s_i \in br_i(\sigma_{-i}^*) \subseteq br_i(D_{-i})$ and thus $D_i \subseteq br_i(D_{-i})$. Therefore, following Theorem 5.51, $D_i \subseteq R_i$. In conclusion, if $s = (s_i)_{i \in N}$ is a pure strategy profile that is played under σ^* with positive probability then s is a rationalizable pure strategy profile.

5.77 (a) Denote by E_i the set of strategies of player i that survive the process of iterated elimination of strictly dominated strategies. By Lemma 5.58, $R_i \subseteq E_i$ for every player $i \in N$. Therefore, if a strategic-form game is solvable by dominance, then $1 \leq |R_i| \leq |E_i| = 1$ and thus the game has a unique rationalizable strategy profile.

(b) Let Γ be a strategic-form game which is solvable by dominance. By Nash's Theorem (Theorem 5.10), the game has an equilibrium. By Corollary 5.53, the support of every Nash equilibrium σ^* is contained in $(R_i)_{i\in N}$. That is, if $\sigma_i^*(s_i) > 0$ then $s_i \in R_i$, for every $i \in N$ and every $s_i \in S_i$. However, since Γ is solvable by dominance, then $|R_i| = 1$ (item (a)) and thus for every player $i \in N$, there is a unique $s_i \in R_i$ such that $\sigma_i^*(s_i) = 1$. In particular, σ^* is the unique rationalizable strategy profile (which survive the process of iterated elimination of strictly dominated strategies) and it is the only Nash equilibrium in mixed strategies of the game.

5.78 Consider the following three players game where Player 1 chooses the row, Player 2 chooses the column, and Player 3 chooses the matrix:

W	L	R
T	8,1,1	0,1,1
C	3,1,1	3,1,1
B	4,1,1	4,1,1

E	L	R
T	4,1,1	4,1,1
C	3,1,1	3,1,1
B	0,1,1	8,1,1

Since the payoff functions of both Player 2 and Player 3 are constant, $R_2 = \{L, R\}$ and $R_3 = \{W, E\}$. To find R_1, let σ_{-1} be the strategy profile according to which Player 2 plays $[x(L), (1-x)(R)]$ and Player 3 plays $[y(W), (1-y)(E)]$. Then

$$\begin{aligned}
U_1(T, \sigma_{-1}) &= 8xy + 4(1-y),\\
U_1(C, \sigma_{-1}) &= 3,\\
U_1(B, \sigma_{-1}) &= 4y + 8(1-x)(1-y).
\end{aligned}$$

Therefore, if $x < \frac{1}{2}$ then $br_1(\sigma_{-1}) = \{B\}$; if $x > \frac{1}{2}$ then $br_1(\sigma_{-1}) = \{T\}$; and if $x = \frac{1}{2}$ then $br_1(\sigma_{-1}) = \{T, B\}$. In conclusion, $R_1 = \{T, B\}$ and C is never-best-reply strategy. Note that also C is never-best-reply strategy, it is not strictly dominated. Indeed, for every mixed strategy $\sigma_1 = [z(T), (1-z)(B)]$, if $z \leq \frac{1}{2}$ then $u_1(C, R, W) > U_1(\sigma_1, R, W)$; and if $z \geq \frac{1}{2}$ then $u_1(C, L, E) > U_1(\sigma_1, L, E)$.

5.79 (a) We prove the analog of Theorem 5.51: For every player $i \in N$ there exists a nonempty set of pure strategies $T_i \subseteq S_i$ such that $T_i = br_i^*(T_{-i})$, where $T_{-i} = (T_j)_{j\neq i}$. Moreover, every collection of nonempty sets $(D_i)_{i\in N}$ that satisfies $D_i \subseteq br_i^*(D_{-i})$ for each $i \in N$ also satisfies $D_i \subseteq T_i$ for each $i \in N$.
Define

$$D_i^0 := S_i, \quad \forall i \in N,$$

and iteratively define, for every $k \geq 0$,

$$D_i^{k+1} := br_i^*(D_{-i}^k), \quad \forall i \in N,$$

where $D^k_{-i} := (D^k_j)_{j \neq i}$. We prove by induction that $(D^k_i)_{k \geq 0}$ is a weakly decreasing sequence of sets, that is, $D^{k+1}_i \subseteq D^k_i$ for every $k \geq 0$ and every $i \in N$. Indeed, for $k = 0$ we have by definition

$$D^1_i \subseteq S_i = D^0_i, \quad \forall i \in N.$$

Assume now by induction that $D^{k+1}_j \subseteq D^k_j$ for every $j \in N$. Then the set $\Delta(\times_{j \neq i} D^{k+1}_j)$ is a subset of $\Delta(\times_{j \neq i}(D^k_j)$. By Equation (5.186), the set $br^*_i(D^{k+1}_{-i})$ is a union of fewer sets than the set $br^*_i(D^k_{-i})$, and hence $D^{k+1}_i \subseteq D^k_i$.

Since the sequence $(D^k_i)_{k \geq 0}$ is weakly decreasing for every $i \in N$, and since the sets $(S_i)_{i \in N}$ are finite, it follows that there exists $k^* \in \mathbf{N}$ such that $D^{k^*+1}_i = D^{k^*}_i$ for every $i \in N$. Set $R_i := D^{k^*}_i$ for every $i \in N$ and $R_{-i} = (R_j)_{j \neq i}$. Then $R_i = br^*_i(R_{-i})$, as we wanted to show.

Let now $(D_i)_{i \in N}$ be a collection of sets that satisfy $D_i \subseteq br^*_i(D_{-i})$ for every $i \in N$, and fix $i \in N$. To prove that $D_i \subseteq R_i$, we will prove by induction that $D_i \subseteq D^k_i$ for every $k \geq 0$. For $k = 0$ we have $D_i \subseteq S_i = D^0_i$. Assume by induction that for some $k \geq 0$ we have $D_j \subseteq D^k_j$ for every $j \in N$. As in the first part of the proof, this implies that $br^*_i(D_{-i}) \subseteq br^*_i(D^k_{-i})$. Hence, we obtain

$$D_i \subseteq br^*_i(D_{-i}) \subseteq br^*_i(D^k_{-i}) = D^{k+1}_i,$$

as claimed.

(b) We prove the analog of Theorem 5.59: Let $\Gamma = (N; (S_i)_{i \in N}; (u_i)_{i \in N})$ be a strategic-form game where the sets of pure strategies of the players are finite. Let $i \in N$ be a player and let $s_i \notin br^*_i(S_{-i})$. We prove that the strategy s_i is strictly dominated by a mixed strategy.

Let $\widehat{s}_i \in S_i$ be a never-best-reply pure strategy of player i. Then for every probability distribution $\sigma_{-i} \in \Delta(S_{-i})$ there is a pure strategy $b(\sigma_{-i}) \in S_i$ such that

$$U_i(b(\sigma_{-i}), \sigma_{-i}) > U_i(\widehat{s}_i, \sigma_{-i}). \tag{28}$$

Consider an auxiliary two-player zero-sum strategic-form game $\widehat{\Gamma}$ in which player i is the row player and all other players together are the column player, the sets of pure strategies are S_i and S_{-i}, and the payoff function $\widehat{U}$ is given by

$$\widehat{u}(s_i, s_{-i}) := u_i(s_i, s_{-i}) - u_i(\widehat{s}_i, s_{-i}).$$

Let $\widehat{\sigma}_i$ and $\widehat{\sigma}_{-i}$ be optimal strategies of the two players in the game $\widehat{\Gamma}$. Then for every pure strategy s_{-i} of the column player,

$$\begin{aligned} U_i(\widehat{\sigma}_i, s_{-i}) - U_i(\widehat{s}_i, s_{-i}) &= \widehat{U}(\widehat{\sigma}_i, s_{-i}) & (29)\\ &\geq \widehat{U}(\widehat{\sigma}_i, \widehat{\sigma}_{-i}) & (30)\\ &\geq \widehat{U}(b(\widehat{\sigma}_i), \widehat{\sigma}_{-i}) & (31)\\ &= U_i(b(\widehat{\sigma}_i), \widehat{\sigma}_{-i}) - U_i(\widehat{s}_i, \widehat{\sigma}_{-i}) & (32)\\ &> 0, & (33)\end{aligned}$$

where Eqs. (29) and (32) hold by the definition of $\widehat{u}$, Eq. (30) holds since $\widehat{\sigma}_{-i}$ is an optimal strategy of Player 2, Eq. (31) holds since $\widehat{\sigma}_i$ is an optimal strategy of the row player, and Eq. (33) holds by Eq. (28). We deduce that $U_i(\widehat{\sigma}_i, s_{-i}) > U_i(\widehat{s}_i, s_{-i})$. Since this holds for every $s_{-i} \in S_{-i}$, the pure strategy $\widehat{s}_{-i}$ is strictly dominated by the mixed strategy $\widehat{\sigma}_i$.

(c) We prove the analog of Corollary 5.60: For any number of players, the set $\times_{i \in N} T_i$ is the set E of pure strategy profiles that survive iterated elimination of strictly dominated strategies.

We first show that $T_i \subseteq E_i$ for every $i \in N$. Using Part (a), T_{-i} is a collection non empty set of pure strategies, for every $i \in N$. Let s_i be a strictly dominated pure strategy. That is, there exists a mixed strategy $\sigma_i \in \Delta(S_i)$ such that $U_i(\sigma_i; s_{-i}) > Ui(s_i; s_{-i})$ for every $s_{-i} \in S_{-i}$. In particular, for every $\sigma_{-i} \in \Delta(S_{-i})$,

$$U_i(\sigma_i; \sigma_{-i}) > Ui(s_i; \sigma_{-i}). \quad (34)$$

Assume by contradiction that $s_i \in br_i(T_{-i})$. Hence, there is $\sigma_{-i} \in \Delta(T_{-i})$,

$$Ui(s_i; \sigma_{-i}) \geq U_i(s'_i; \sigma_{-i}) \;\; \forall s'_i \in S_i,$$

from which it follows that $Ui(s_i; \sigma_{-i}) \geq U_i(\sigma_i; \sigma_{-i})$, a contradiction to Equation (34). Therefore, $T_i \subseteq E_i$ for every $i \in N$.

On the other hand by item (b), for every $i \in N$ one has $E_i \subseteq T_i$ and therefore $T_i = E_i$, as claimed.

5.80 **Game A:** The equilibria of the game are $(Dove, Hawk)$, $(Hawk, Dove)$ and $([\frac{1}{2}(Dove), \frac{1}{2}(Hawk)], [\frac{1}{2}(Dove), \frac{1}{2}(Hawk)])$. Both pure equilibria are not symmetric hence cannot define an evolutionarily stable strategy. By Theorem 5.65, the mixed equilibrium defines an evolutionarily stable strategy $[\frac{1}{2}(Dove), \frac{1}{2}(Hawk)]$. Indeed, for every strategy $[x(Dove), (1-x)(Hawk)]$, one has

$$u_1(x, \tfrac{1}{2}) = 5 = U_1(\tfrac{1}{2}, \tfrac{1}{2}).$$

and

$$u_1(x,x) = -2x^2 - 3x + 7 < 7.5 - 5x = u_1(\tfrac{1}{2}, x),$$

so the condition in Equation (5.141) holds.

Game B: The unique equilibrium of the game is $(Hawk,,Hawk)$. The equilibrium is strict hence it defines an evolutionarily stable strategy *Hawk*.

Game C: There are two strict pure equilibria of the game which are $(Hawk,,Hawk)$ and $(Dove, Dove)$ so both strategies *Hawk* and *Dove* are evolutionarily stable strategies. In addition, the strategy vector $([\frac{7}{8}(Dove), \frac{1}{8}(Hawk)], [\frac{7}{8}(Dove), \frac{1}{8}(Hawk)])$ is a completely mixed equilibrium. Therefore, every mixed strategy is a best response to $[\frac{7}{8}(Dove), \frac{1}{8}(Hawk)]$, so the condition in Equation (5.140) does not satisfy. In addition, for every $x = 1$, $u_1(1,1) = 2 > \frac{15}{8} = u_1(\frac{7}{8}, 1)$ so the condition in Equation (5.141) does not hold, thus it does not defined evolutionarily stable strategy.

Game D: Every strategy vector in the game is an equilibria, hence, by Theorem 5.65, no strategy is evolutionarily stable.

Game E: The equilibria of the game are $(Dove, Hawk)$, $(Hawk, Dove)$ and $([\frac{1}{7}(Dove), \frac{6}{7}(Hawk)], [\frac{1}{7}(Dove), \frac{6}{7}(Hawk)])$. Both pure equilibria are not symmetric hence cannot define an evolutionarily stable strategy. By Theorem 5.65, the mixed equilibrium defines an evolutionarily stable strategy $[\frac{1}{7}(Dove), \frac{6}{7}(Hawk)]$. Indeed, for every strategy $[x(Dove), (1-x)(Hawk)]$, one has

$$u_1(x, \tfrac{1}{7}) = \tfrac{50}{7} = U_1(\tfrac{1}{7}, \tfrac{1}{7}).$$

and

$$u_1(x,x) = -7x^2 + 2x + 7 < \tfrac{50}{7} = u_1(\tfrac{1}{2}, x),$$

so the condition in Equation (5.141) holds.

Game F: There are two pure equilibrium: (1) $(Hawk, Hawk)$ which is a strict equilibrium and thus *Hawk* is evolutionarily stable; (2) $(Dove, Dove)$ which does not define a evolutionarily stable strategy by Theorem 5.65.

5.81 The given symmetric two-player game in which each player has two pure strategies and all payoffs are nonnegative has an ESS if and only if either $a \neq c$ or $b \neq d$.

If either $a > c$ or $a = c$ but $b < d$ then (T, T) is an equilibrium satisfying either the condition in Equation (5.140) or the condition in Equation (5.141) (respectively) so T is an ESS.

Similarly, if either $b > d$ or $b = d$ but $a < c$ then (B, B) is an equilibrium satisfying either the condition in Equation (5.140) or the condition in Equation (5.141) (respectively) so B is an ESS.

If $a < c$ and $b < d$ then $x^* = ([\frac{d-b}{c-a+d-b}(T), \frac{c-a}{c-a+d-b}(B)][\frac{d-b}{c-a+d-b}(T), \frac{c-a}{c-a+d-b}(B)])$ is a completely mixed equilibrium (as shown in Exercise 5.22) and thus it does not satisfy the condition in Equation (5.140). We show that x^* satisfies the condition in Equation (5.141). Let $x = [p(T), (1-p)(B)]$ where $x \neq x^*$. If $p < \frac{d-b}{c-a+d-b}$ then $U_I(T,x) > U_I(B,x)$, indeed

$$\begin{aligned} U_I(T,x) - U_I(B,x) &= (ax + (1-x)d) - (cx + (1-x)b) \\ &= (d-b) - (c-a+d-b)p \\ &> (d-b) - (c-a+d-b)\tfrac{d-b}{c-a+d-b} = 0. \end{aligned}$$

Therefore,

$$\begin{aligned} U_I(x,x) &= pU_I(T,x) + (1-p)U_I(B,x) \\ &< \tfrac{d-b}{c-a+d-b}U_I(T,x) + \tfrac{c-a}{c-a+d-b}U_I(B,x) \\ &= U_I(x^*,x). \end{aligned}$$

Similarly, if $p > \frac{d-b}{c-a+d-b}$ then $U_I(T,x) < U_I(B,x)$ and thus

$$\begin{aligned} U_I(x,x) &= pU_I(T,x) + (1-p)U_I(B,x) \\ &< \tfrac{d-b}{c-a+d-b}U_I(T,x) + \tfrac{c-a}{c-a+d-b}U_I(B,x) \\ &= U_I(x^*,x). \end{aligned}$$

In conclusion, x^* is an ESS.
Finally, if $a = c$ and $b = d$ then for every strategy x, $U_I(T,x) = U_I(B,x)$ so neither the condition in Equation (5.140) nor the condition in Equation (5.141) can be satisfied, hence there is no ESS.

5.82 We prove that the unique Nash equilibrium of Rock, Paper, Scissors (Example 4.3) is

$$([\tfrac{1}{3}(Rock), \tfrac{1}{3}(Paper), \tfrac{1}{3}(Scissors)], [\tfrac{1}{3}(Rock), \tfrac{1}{3}(Paper), \tfrac{1}{3}(Scissors)]).$$

We first show that an equilibrium is completely mixed. Assume by contradiction that there is a an equilibrium which is not completely mixed. That is, one of the players, say Player I, plays one of its pure action, say Scissors, with probability 0. Hence, Player II plays Rock with probability 0 (since it is strictly dominated by Paper). Therefore, Player I also plays Paper with probability 0, so he necessarily plays Rock. The best response of Player II is then Paper which case Player I is better of deviating to Scissors, a contradiction. By the symmetry of the game (among the strategies as well as the players) every equilibrium must be completely mixed.

Let

$$([x_1(Rock), x_2(Paper), (1-x_1-x_2)(Scissors)],$$
$$[y_1(Rock), y_2(Paper), (1-y_1-y_2)(Scissors)])$$

be a completely mixed equilibrium. By the indifference principle:

$$\begin{aligned} -y_2+(1-y_1-y_2) = \quad & y_1-(1-y_1-y_2) = \quad & -y_1+y_2 \\ -x_2+(1-x_1-x_2) = \quad & x_1-(1-x_1-x_2) = \quad & -x_1+x_2 \end{aligned}$$

The solution of this system of equation is $x_1 = x_2 = y_1 = y_2 = \frac{1}{3}$ which leads to the given equilibrium.

5.83 (a) If $V-c > \alpha V$ and $\alpha V - c > 0$, then by eliminating Abandon which is strictly dominated by Care then (Care,Care) is a strict and unique equilibrium and thus Care is the only evolutionarily stable strategy.

(b) If $V-c < \alpha V$ and $\alpha V - c < 0$, then by eliminating Care which is strictly dominated by Abandon then (Abandon,Abandon) is a strict and unique equilibrium and thus Abandon is the only evolutionarily stable strategy.

(c) If $\alpha < \frac{1}{2}$ (in this case $(1-\alpha)V > \alpha V$), and investment in caring for offspring satisfies $(1-\alpha)V > c > \alpha V)$, then $V-c > V-(1-\alpha)V = \alpha V$ and $\alpha V - c < 0$. In particular, the condition in Equation (5.140) holds for both (Care,Care) and (Abandon, Abandon). Hence, both strategies, Care and Abandon, are evolutionarily stable strategies. If initially there are more caring species, then Care equilibrium emerges, whereas, if initially there are more abandoning species, then Abandon equilibrium emerges.

(d) If $\alpha > \frac{1}{2}$ (in this case $(1-\alpha)V < \alpha V$, and investment in caring for offspring satisfies $(1-\alpha)V < c < \alpha V)$, then by Exercise 5.81 the only evolutionarily stable equilibrium is the mixed strategy in which Care is chosen with probability $\frac{\alpha V - c}{(2\alpha-1)V}$.

5.84 Although a single male leopard can mate with all the female leopards on the savanna, every generation of leopards is composed of 50% males and 50% females. These proportions are evolutionarily stable. If there were more male leopards than female leopards, then each female leopard has more offspring than a male leopard which case it is preferred being a female. Therefore, the proportion of females will increase. Similarly, if there were more female leopards than male leopards, then each male leopard has more offspring than a female leopard which case it is preferred being a male. Therefore, the proportion of females will increase. Hence, the evolutionarily stable strategy is that at which the number of male leopards born equals the number of female leopards born.

5.85 If either $x < 0$ or $y < 0$ then by the process of eliminated strictly dominated strategies the game has a unique equilibrium (B, R) corresponding to the payoff $(1,1)$. If $x = 0$ and $y \geq 0$ (or $y = 0$ and $x \geq 0$) then there are two equilibria (B, R) corresponding to the payoff $(1,1)$ and (T, L) corresponding to the payoff (x, y). Finally, if both x and y are positive then there are three equilibria (B, R) corresponding to the payoff $(1,1)$, (T, L) corresponding to the payoff (x, y) and $([\frac{1}{1+y}(T), \frac{y}{1+y}(B)], [\frac{1}{1+x}(L), \frac{x}{1+x}(R)]$ corresponding to the payoff $(\frac{x}{1+x}, \frac{y}{1+y})$.

5.86 Let N be a finite set of players and for every player $i \in N$ and let S_i be a finite set of pure strategies of that player. Let $(u^k)_{k=1}^{\infty}$ be an infinite sequence of payoff functions. For every $k \in \mathbf{N}$ let σ^k be an equilibrium of the strategic form game $(N, (S_i)_{i \in N}, (u_i^k)_{i \in N})$. Suppose that for every player $i \in N$ and every $s \in \times_{i \in N} S_i$ we have:

- $\lim_{k \to \infty} u_i^k(s)$ exists and is equal to $u_i^*(s)$.
- $\lim_{k \to \infty} \sigma_i^k(s_i)$ exists and is equal to $\sigma_i^*(s_i)$.

Since $\mathbf{R}^N$ is a compact set then u_i^* is a payoff function for every player $i \in N$ and thus the game equilibrium of the game $(N, (S_i)_{i \in N}, (u_i^*)_{i \in N})$ is well defined. Similarly, by Exercise 5.1, the set of mixed strategies of player i is compact and thus σ_i^* is also a mixed strategy of player i in the game, for every $i \in N$. We next show that σ^* is an equilibrium by showing that $U_i^*(\sigma^*) \geq U_i(s_i, \sigma_{-i}^*) - \epsilon$ for every $i \in N$, every $s_i \in S_i$ and every $\epsilon > 0$.

Set $M = \max\{u_i(s) : i \in N, s \in S\}$ and let $\epsilon > 0$. Let K be sufficiently large such that for every $k \geq K$ one has

$$|u_i^k(s) - u_i^*(s)| < \frac{\epsilon}{4} \tag{35}$$

and

$$|\sigma^*(s) - \sigma^k(s)| < \frac{\epsilon}{4M} \tag{36}$$

for every $i \in N$ and every $s \in S$. Therefore, for every $i \in N$ and every

$s_i \in S_i$ one has

$$\begin{aligned}
U_i^*(\sigma^*) &= \sum_{s \in S} \sigma^*(s) u_i^*(s) \\
&\geq \sum_{s \in S} \sigma^k(s) u_i^*(s) - \tfrac{\epsilon}{4} && (37) \\
&\geq \sum_{s \in S} \sigma^k(s) u_i^k(s) - \tfrac{\epsilon}{2} && (38) \\
&\geq \sum_{s \in S} \sigma_{-i}^k(s_{-i}) u_i^k(s_i, s_{-i}) - \tfrac{\epsilon}{2} && (39) \\
&\geq \sum_{s \in S} \sigma_{-i}^k(s_{-i}) u_i^*(s_i, s_{-i}) - \tfrac{3\epsilon}{4} && (40) \\
&\geq \sum_{s \in S} \sigma_{-i}^*(s_{-i}) u_i^*(s_i, s_{-i}) - \epsilon && (41) \\
&= U_i^*(s_i, \sigma_{-i}^*) - \epsilon,
\end{aligned}$$

where Equations (37) and (41) follows by Equation (36), Equations (38) and (40) follows by Equation (35) and Equation (39) holds since σ^k is an equilibrium of $(N, (S_i)_{i \in N}, (u_i^k)_{i \in N})$. Since $U_i^*(\sigma^*) \geq U_i^*(s_i, \sigma_{-i}^*) - \epsilon$ for every $\epsilon > 0$ then $U_i^*(\sigma^*) \geq U_i^*(s_i, \sigma_{-i}^*)$, as needed.

5.87 Let $\eta : \mathcal{G}_i \to \widetilde{\mathcal{G}}_i(N, S) \times \overline{\mathcal{G}}_i(S_i)$ be the function defined by $\eta(u_i) := (\widetilde{u}_i, \overline{u}_i)$, where $\widetilde{u}_i$ is defined as in Equation (5.146) and $\overline{u}_i$ is defined as in Equation (5.145).

(a) We first show that the function η is one-to-one. Let $u_i, v_i \in \mathbf{R}^S$ satisfying that $\eta(u_i) = \eta(v_i)$. Then $\widetilde{u}_i = \widetilde{v}_i$ and $\overline{u}_i = \overline{v}_i$. Hence,

$$u_i(s) = \widetilde{u}_i(s) + \overline{u}_i(s_i) = \widetilde{v}_i(s) + \overline{v}_i(s_i) = v_i(s) \quad \forall s \in S,$$

and thus $u_i = v_i$.

We next show that the function η is onto. Let $z \in \widetilde{\mathcal{G}}_i(N, S)$ and $w \in \overline{\mathcal{G}}_i(S_i)$. Set

$$u_i(s) = z_i(s) + w_i(s_i) \in \mathbf{R} \quad \forall s \in S. \tag{42}$$

Then,

$$\overline{u}_i(s_i) = \overline{z}_i(s_i) + \overline{w}_i(s_i) = 0 + \overline{w}_i(s_i) = \overline{w}_i(s_i),$$

and

$$\widetilde{u}_i(s) = u_i(s) - \overline{u}_i(s_i) = (z_i(s) + w_i(s_i)) - w_i(s_i) = z_i(s).$$

That is, $\eta(u_i) = (z, w)$.

(b) The inverse function η^{-1} as defined in Equation (42) is a linear function of two variables and thus it is continuous.

5.88 We prove that the function φ defined in Equation (5.161) has the following property: if $(u^k, \sigma^k)_{k\in\mathbf{N}}$ is a sequence of points in E satisfying $\lim_{k\to\infty} \|u^k\|_\infty = \infty$, then $\lim_{k\to\infty} \|\varphi(u^k, \sigma^k)\|_\infty = \infty$. That is, we show that for every $M > 0$ there is K sufficiently large, such that for every $k > K$ one has $\|\varphi(u^k, \sigma^k)\|_\infty > M$.

Since $\lim_{k\to\infty} \|u^k\|_\infty = \infty$ there is K sufficiently large, such that for every $k > K$ one has $\|u^k\|_\infty > 3M$. For every $k > K$ let $i^k \in N$ and $s^k \in S$ such that $\|u^k\|_\infty = |u^k_{i^k}(s^k)|$. If $\|\widetilde{u}^k_{i^k}\|_\infty > M$ then $\|\varphi(u^k, \sigma^k)\|_\infty \geq \|\widetilde{u}^k_{i^k}\|_\infty > M$, as claimed. Otherwise,

$$M \geq \|\widetilde{u}^k_{i^k}\|_\infty \geq |\widetilde{u}^k_{i^k}(s)| = |u^k_{i^k}(s) - \overline{u}^k_{i^k}(s_{i^k})| \qquad \forall s \in S.$$

Therefore, for every $s_{-i^k} \in S_{-i^k}$

$$u^k_{i^k}(s^k_{i^k}, s_{-i^k}) \geq |\overline{u}^k_{i^k}(s^k_{i^k})| - M \geq |u^k_{i^k}(s^k_{i^k}, s^k_{-i^k})| - 2M > M$$

In particular,

$$\|\varphi(u^k, \sigma^k)\|_\infty \geq \|z_{i^k}(u^k, \sigma^k)\|_\infty \geq |z_{i^k, s^k_{i^k}}(u^k, \sigma^k)| \geq U^k_{i^k}(s^k_{i^k}, \sigma^k_{-i^k}) > M,$$

as needed.

Chapter 6: Behavior strategies and Kuhn's Theorem

6.1 **Game A:** Player I has perfect recall.
Game B: both players have perfect recall.
Game C: Player II has prefect recall. On the other hand, Player I has imperfect recall, he may forget how many moves he made in the past.
Game D: Player I has imperfect recall, he may forget the result of the chance move and what move he made in the past.
Game E: Player II has prefect recall. On the other hand, Player I has imperfect recall, he may forget things he knew at earlier stages of the games, such as the result of a chance move and whether or not he made a move at all in the past.
Game F: both players have perfect recall.
Game G: Player II has prefect recall. On the other hand, Player I has imperfect recall, he may forget what moves he made in the past.
Game H: both players have perfect recall.

6.2 (a) The set of pure strategies of Player I is $S_I = \{T, B\}$ so b_I is also a mixed strategy.

(b) Player II has 2^3 pure strategies. The mixed strategy equivalent to b_{II} is

$$\sigma_{II} = \Big[\tfrac{8}{45}(t_1t_2t_3), \tfrac{4}{45}(t_1t_2b_3), \tfrac{16}{135}(t_1b_2t_3), \tfrac{8}{135}(t_1b_2b_3),$$
$$\tfrac{2}{9}(b_1t_2t_3), \tfrac{1}{9}(b_1t_2b_3), \tfrac{4}{27}(b_1b_2t_3), \tfrac{2}{27}(b_1b_2b_3)\Big].$$

(c) Player II has 2^2 pure strategies. The mixed strategy equivalent to b_{II} is

$$\sigma_{II} = \Big[\tfrac{1}{9}(t_1t_2), \tfrac{3}{9}(t_1b_2), \tfrac{5}{36}(b_1t_2), \tfrac{15}{36}(b_1b_2)\Big].$$

6.3 **Game A:** Player I can guarantee himself a payoff of 80 by choosing T (this action is both a optimal mixed strategy as well as optimal behavior strategy since Player I has only one information set).
Game B: Player I can guarantee himself a payoff of 100. The optimal mixed strategies guaranteeing this payoff are $[x(T_1T_2), (1-x)(B_1B_2)]$ for every $x \in [0,1]$. The optimal behavior strategies are $([1(T_1)], [1(T_2)])$ and $([1(B_1)], [1(B_2)])$.
Game C: Player I has only one information set. Thus

$$\Sigma_I = \Big\{ \big[x(T), (1-x)(B)\big] \,|x \in [0,1] \Big\}$$

is his mixed strategy set as well as his behavior strategy set. For every strategy $\sigma_I = \big[x(T), (1-x)(B)\big]$,

$$u(\sigma_I) = 6x^2 + 8x(1-x) + 10x(1-x) + 2(1-x)^2.$$

The maximal payoff is obtained at $x = 0.7$. In particular, Player I can guarantee himself a payoff of 6.9 by selecting the strategy $[0.7(T), 0.3(B)]$.
Game D: The strategic form of this game is given in the following matrix

	t	b
T_1T_2	100	0
T_1B_2	0	0
B_1T_2	0	0
B_1B_2	0	100

The value of the game in mixed strategies is $v = 50$. An optimal mixed strategy guaranteeing this payoff to Player I is $\sigma_I = \Big[\tfrac{1}{2}(T_1T_2), \tfrac{1}{2}(B_1B_2)\Big]$. Player II's optimal (mixed/bahavior) strategy is $\sigma_{II} = \Big[\tfrac{1}{2}(t), \tfrac{1}{2}(b)\Big]$. We next check what Player I can guarantee himself in behavior strategies

$$b_I = \Big(\big[x(T_1), (1-x)(B_1)\big], \big[y(T_2), (1-y)(B_2)\big]\Big).$$

Then,

$$u(b_I, t) = 100xy \text{ and}$$
$$u(b_I, b) = 100(1-x)(1-y) = 100(xy + 1 - x - y).$$

Thus Player I can guarantee $\bar{v} = \min_{b_I \in \mathcal{B}_I} \max_{b_{II} \in \mathcal{B}_{II}} = 25$ by following the bahavior strategy $\left(\left[0.5(T_1), 0.5(B_1)\right], \left[0.5(T_2), 0.5(B_2)\right]\right)$.

Game E: The strategic form of this game is given in the following matrix

	t	b
$T_1T_2T_3$	0	0
$T_1T_2B_3$	0	100(1-p)
$T_1B_2T_3$	0	0
$T_1B_2B_3$	0	0
$B_1T_2T_3$	100p	0
$B_1T_2B_3$	0	100(1-p)
$B_1B_2T_3$	100p	100
$B_1B_2B_3$	0	100

The value of the game in mixed strategies is $v = 100p(1-p)$. For every $0 \leq x \leq p$, and every $0 \leq y \leq 1-p$ the mixed strategy $\sigma_I = \left[x(T_1T_2B_3), (p-x)(B_1T_2B_3), y(B_1T_2T_3), (1-p-y)(B_1B_2T_3)\right]$ is an optimal strategy guaranteeing the payoff of $100p(1-p)$ to Player I. The behavior strategy of Player I guaranteeing him this payoff is

$$b_I = ([1(B_1)], [1(T_2)], [(1-p)(T_3), p(B_3)]).$$

Player II's optimal (mixed/bahavior) strategy is $\sigma_{II} = \left[(1-p)(t), p(b)\right]$.

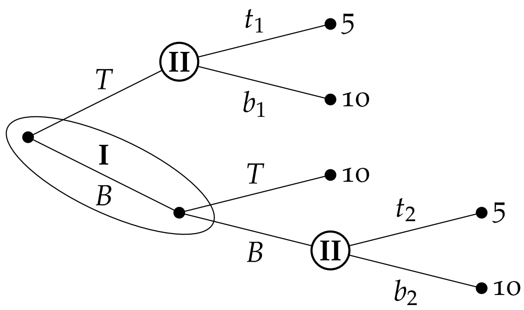
6.4 Assume player i in an extensive-form game has only one information set, i.e. $\mathcal{U}_i = \{u_i\}$.

(a) $\Sigma_i = \Delta(S_i) = \Delta\left(\times_{\bar{u}_i \in \mathcal{U}_i} A(\bar{u}_i)\right) = \Delta\left(A(u_i)\right)$.

(b) $\mathcal{B}_i = \times_{\bar{u}_i \in \mathcal{U}_i} \Delta\left(A(\bar{u}_i)\right) = \Delta\left(A(u_i)\right)$.

(c) From Parts (a) and (b) one can conclude that there is a natural identification between the set of mixed strategies of Player I and the set of his behavior strategies. Note that two strategies that are identified with each other are not necessarily equivalent when the game does not have perfect recall since in a mixed strategy the lottery is conducted once whereas in a behavior strategy a lottery is conducted at each vertex in the information set. Indeed, the mixed strategy $[\frac{1}{2}(T), \frac{1}{2}(B)]$ in Example 6.9 cannot yield the outcome O_2 whereas its equivalent behavior strategy may yield the outcome O_2 (with probability 0.25).

6.5 The following two-player zero-sum extensive-form game (which is a version of Example 6.9) has a value in mixed strategies and a value in behavior strategies, but these two values are not equal to each other.



Note that since Player I cannot play BT in mixed strategies, the value in mixed strategies is $v = 5$. Every mixed strategy $[x(T), (1-x)(B)]$ of Player I, is an optimal strategy of Player I. The optimal mixed strategy of Player II is $[1(t_1t_2)]$. On the other hand, the value in behavior strategies is $v = 6.25$. The optimal behavior strategy of Player I is $[0.5(T), 0.5(B)]$. The optimal behavior strategy of Player II is $([1(t_1)], [1(t_2)])$.

6.6 Let b_i be a behavior strategy of player i that is equivalent to the mixed strategy σ_i, i.e., for every vertex x in the game tree and every strategy vector σ_{-i} of the other players, $\rho(x; \sigma_i, \sigma_{-i}) = \rho(x; b_i, \sigma_{-i})$. Denote by V^L the set of leaves of the game tree. Then for every strategy vector σ_{-i} of the other players and every player $j \in N$,

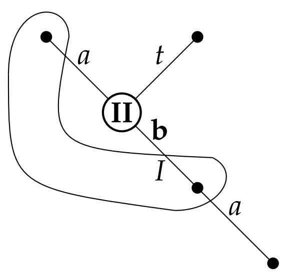
$$u_j(\sigma_i, \sigma_{-i}) = \sum_{x \in V^L} \rho(x; \sigma_i, \sigma_{-i})\, u_j(x) = \sum_{x \in V^L} \rho(x; b_i, \sigma_{-i})\, u_j(x) = u_j(b_i, \sigma_{-i}).$$

6.7 We next prove that in Example 6.4 for every mixed strategy $\sigma_{II} = [x(t), (1-x)(b)]$ of Player II the probability distribution induced by (σ_I, σ_{II}) over the leaves of the game tree is identical with the probability distribution induced by (b_I, σ_{II}) over the leaves of the game tree.

$$
\begin{array}{rrrr}
\rho(x_5, \sigma_I, \sigma_{II}) = & \frac{6}{10} \cdot (1-x) = & \frac{3}{5} \cdot (1-x) = & \rho(x_5, b_I, \sigma_{II}) \\
\rho(x_{10}, \sigma_I, \sigma_{II}) = & \frac{1}{10} \cdot x = & \frac{2}{5} \cdot x \cdot \frac{1}{4} = & \rho(x_{10}, b_I, \sigma_{II}) \\
\rho(x_{11}, \sigma_I, \sigma_{II}) = & \frac{3}{10} \cdot x = & \frac{2}{5} \cdot x \cdot \frac{3}{4} = & \rho(x_{11}, b_I, \sigma_{II}) \\
\rho(x_{12}, \sigma_I, \sigma_{II}) = & \frac{1}{10} \cdot (1-x) = & \frac{2}{5} \cdot (1-x) \cdot \frac{1}{4} = & \rho(x_{12}, b_I, \sigma_{II}) \\
\rho(x_{13}, \sigma_I, \sigma_{II}) = & \frac{3}{10} \cdot (1-x) = & \frac{2}{5} \cdot (1-x) \cdot \frac{3}{4} = & \rho(x_{13}, b_I, \sigma_{II}) \\
\rho(x_{14}, \sigma_I, \sigma_{II}) = & \frac{2}{10} \cdot x \cdot \frac{2}{3} = & \frac{3}{5} \cdot x \cdot \frac{2}{3} \cdot \frac{1}{3} = & \rho(x_{14}, b_I, \sigma_{II}) \\
\rho(x_{15}, \sigma_I, \sigma_{II}) = & \frac{4}{10} \cdot x \cdot \frac{2}{3} = & \frac{3}{5} \cdot x \cdot \frac{2}{3} \cdot \frac{2}{3} = & \rho(x_{15}, b_I, \sigma_{II}) \\
\rho(x_{16}, \sigma_I, \sigma_{II}) = & \frac{6}{10} \cdot x \cdot \frac{1}{3} = & \frac{3}{5} \cdot x \cdot \frac{1}{3} \cdot 1 = & \rho(x_{16}, b_I, \sigma_{II}) \\
\rho(x_{17}, \sigma_I, \sigma_{II}) = & 0 = & \frac{3}{5} \cdot x \cdot \frac{1}{3} \cdot 0 = & \rho(x_{17}, b_I, \sigma_{II}).
\end{array}
$$

6.8 Assume there exists a path from the root to vertex x that passes through x_1 and x_2, two vertices in the same information set U_i of player i. Assume also that the action leading to x is not the same action in each of these vertices, i.e., $a_i(x_1 \to x) \neq a_i(x_2 \to x)$. Then, every behavior strategy b_i in which every action $a_i \in \bigcup_{U_i \in \mathcal{U}_i} A(U_i)$ is chosen with positive probability, has no equivalent mixed strategy. Indeed, for every behavior strategy vector $\bar{\sigma}_{-i}$ of the players $N \setminus \{i\}$ according to which every player selects every action at each information set with positive probability satisfies $\rho(x, b_i, \bar{\sigma}_{-i}) > 0$. However, since each pure strategy of player i would require player i to choose the same action every time he gets U_i, he cannot choose both $a_i(x_1 \to x)$ and $a_i(x_2 \to x)$ at U_i. Thus, for every mixed strategy σ_i of player i and every mixed/behavior strategy σ_{-i} of the players $N \setminus \{i\}$, $\rho(x, \sigma_i, \sigma_{-i}) = 0$. In particular, $\rho(x, \sigma_i, \bar{\sigma}_{-i}) = 0$. Hence, every strategy b_i as defined before has no equivalent mixed strategy.

6.9 Consider the following game tree,



The game does not satisfy the condition that there are at least two possible actions at each vertex. In addition, every behavior strategy of player i has an equivalent mixed strategy also there is an information set of player i that intersects twice a path emanating from the root. In conclusion, Theorem 6.11 does not hold without the condition that there are at least two possible actions at each vertex.

6.10 Equation (6.37) in the proof of Theorem 6.11 does not necessarily hold when a game does not have perfect recall. If there is an information set U_i of player i that intersects a path from the root to a leaf more than once then $b_i(s_i(U_i); U_i)$ (which could be in $(0, 1)$) should appear more than one in the product of Equation (6.37).

6.11 If a player does not know whether or not he has previously made a move during the play of a game, then there is a path from the root to a leaf in the game tree that pass trough at least two vertices controlled by the player who cannot distinct between the two vertices, i.e., the vertices must be in the same information set. In particular, there is an information set of a player that intersects a path from the root to a leaf at least twice so the player has imperfect recall.

6.12 Assume that a player knows during the play of a game how many moves he has previously made, but later forgets this. Then there must be at least two paths to an information set of the player in which the number of moves he made in each path is different. Hence, there are two paths from the root that end in the same information set of player i and pass through different information sets of player i thus the player does not have perfect recall.

6.13 Assume that a player knows during the play of a game which action another player has chosen at a particular information set, but later forgets this. Then, there is an information set of the player in which he does not distinct between at least two actions of another player, whereas during at least two paths from the root that end in that information set he distinct between these actions made by another player. In particular these paths pass through different information sets of the player and the player has imperfect recall.

6.14 If a player does not know what action he chose at a previous information set in a game, then there are at least two paths from the root that end in the same information set in which the player chooses different actions. Hence the player has imperfect recall.

6.15 Assume that at a particular information set, say U_i^1 in a game player i knows which player made the move leading to that information set, but later, in the information set U_i^2, forgets this. Then there must be two paths from the root to U_i^2, one of them passes trough U_i^1 and the other one does not. Therefore, player i does not have perfect recall.

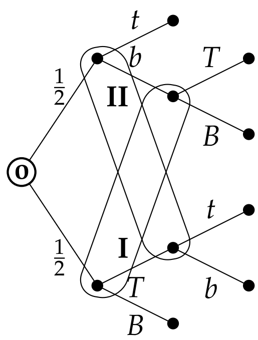
6.16 Let x_1 and x_2 be two vertices in the same information set of player i, and assume that player i has perfect recall in the game. In particular, if the decision vertices of player i on the path from the root to x_1 are $x_1^1, x_1^2, ..., x_1^{L_1} = x_1$ and his decision vertices on the path from the root to x_2 are $x_2^1, x_2^2, ..., x_2^{L_2} = x_2$, then $L_1 = L_2$, and for every $l \in \{1, 2, ..., L_1\}$ one has $U_i(x_1^l) = U_i(x_2^l)$ and $a_i(x_1^l \to x_1) = a_i(x_2^l \to x_2)$. Let $s_i \in S_i^*(x_1)$ be a pure strategy of player i under which at each information set $U_i(x_1^l)$, $1 \leq l \leq L_1$, he chooses the action $a(U_i(x_1^l) \to x_1)$. Thus, since a player has perfect recall and x_1 and x_2 are in the same information set of player i, at each information set $U_i(x_2^l)$, $1 \leq l \leq L_2$, player i chooses the action $a(U_i(x_1^l) \to x_1) = a(U_i(x_2^l) \to x_2)$ thus $s_i \in S_i^*(x_2)$. Hence $S_i^*(x_2) \subseteq S_i^*(x_1)$. By the same arguments $S_i^*(x_1) \subseteq S_i^*(x_2)$, so we can conclude that $S_i^*(x_2) = S_i^*(x_1)$.

6.17 (a) Let U be an information set of a player with perfect recall. Suppose by contradiction that that U precedes U. Then there exist a vertex $x \in U$ and a vertex $\widehat{x} \in U$ such that the path from the root to $\widehat{x}$ passes through x. However, the information set U of player i intersects a path from the root through x and $\widehat{x}$ to a leaf at least twice, so player i has an imperfect recall, a contradiction.

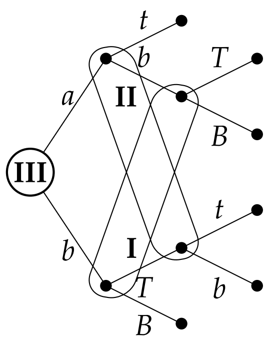
(b) Consider a two-player game without chance moves, where both players have perfect recall. Assume that the information set U precedes $\widehat{U}$. Then, if U and $\widehat{U}$ are information sets of the same player, say player i, by the perfect recall of that player, every two paths from the root that end in the same information set of player i pass through the same information sets of player i, and in the same order, so it must be that $\widehat{U}$ does not precede U.
Assume next that U and $\widehat{U}$ are information sets of player I and player II, respectively. Since U precedes $\widehat{U}$ there exist a vertex $x \in U$ and a vertex $\widehat{x} \in \widehat{U}$ such that the path from the root to $\widehat{x}$ passes through x. However, by the perfect recall of Player I, every path that starts in the root and ends in U passes through the same information sets of player I, and in the same order, and in every such information set the same action is chosen. Similarly, by the perfect recall of Player II, every path that starts in the root and ends in $\widehat{U}$ passes through the same information sets of player II, and in the same order, and in every such information set the same action is chosen. Therefore, since the game has no chance moves, the path from the root to x and then to $\widehat{x}$ is determined by the pure actions chosen by the two players. Therefore, we can

conclude that every path that passes through U and $\widehat{U}$ must visit $x \in U$ before it reaches $\widehat{U}$. In particular, since every information set intersects every path from the root to a leaf at most once, $\widehat{U}$ does not precede U.

(c) The following game tree is a two-player game with chance moves, where both players have perfect recall and there exist two information sets U and $\widehat{U}$ such that U precedes $\widehat{U}$, and $\widehat{U}$ precedes U.



(d) The following game tree is a three-player game , where all players have perfect recall and there exist two information sets U and $\widehat{U}$ such that U precedes $\widehat{U}$, and $\widehat{U}$ precedes U.



6.18 (a) $b_I = ([\frac{1}{2}(B_1), \frac{1}{2}(T_1)], [1(B_2), 0(T_2)])$.

(b) $b_I = ([\frac{4}{7}(B_1), \frac{3}{7}(T_1)], [\frac{3}{4}(B_2), \frac{1}{4}(T_2)], [0(B_3), \frac{2}{3}(M_3), \frac{1}{3}(T_3)])$,
$b_{II} = ([\frac{4}{7}(b_1), \frac{3}{7}(t_1)], [\frac{4}{7}(b_2), \frac{3}{7}(t_2)])$.

6.19 (a) Let i be a player with perfect recall in an extensive-form game and let σ_i be a mixed strategy of player i. Suppose that there is a strategy vector σ_{-i} of the other players such that $\rho(x; \sigma_i, \sigma_{-i}) > 0$ for each leaf x in the game tree. By the perfect recall of player i, there is a behavior strategy b_i of player i equivalent to σ_i (Theorem 6.15). Since $\rho(x;\sigma_i,\sigma_{-i}) > 0$ for each leaf x in the game tree, $\rho(x;\sigma_i,\sigma_{-i}) > 0$ for each vertex x in the game tree. For every information set $U_i \in \mathcal{U}_i$ of player i, every vertex $x \in U_i$ and every action $a \in A(x)$, denote by x^a the first vertex that is visited if player i selects a in x, then

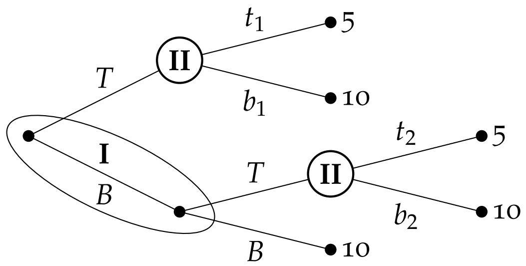
$$\rho(x^a;\sigma_i,\sigma_{-i}) = \rho(x^a;b_i,\sigma_{-i}) = b_i(a,U_i)\rho(x;b_i,\sigma_{-i}) = b_i(a,U_i)\rho(x;\sigma_i,\sigma_{-i})$$
$$\Rightarrow b_i(a,Ui) = \frac{\rho(x^a;\sigma_i,\sigma_{-i})}{\rho(x;\sigma_i,\sigma_{-i})}.$$

In particular, b_i is defined uniquely.

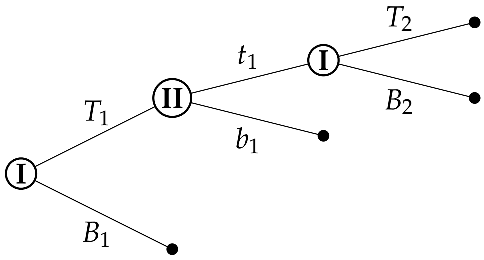
(b) Player II in the next game tree has perfect recall, but every mixed strategy $\sigma_{II} = [q_1(t_1t_2), q_2(t_1b_2), q_3(b_1t_2), q_4(b_1b_2)]$ has infinitely many equivalent behavior strategies of the form

$$b_{II} = ([(q_1+q_2)(t_1),(q_3+q_4)(b_1)],[p(t_2),(1-p)(b_2)])$$

for every $p \in [0,1]$.



6.20 Consider the following game tree, where at each information set each player chooses each action with probability 0.5. Then every mixed strategy $[0.25(T_1T_2), 0.25(T_1B_2), x(B_1T_2), (0.5-x)(B_1B_2)]$ of Player I is equivalent to the behavior strategy of Player I.



6.21 The optimal behavior strategies are $b_I = ([1(T_1), 0(B_1)], [\frac{2}{5}(T_2), \frac{3}{5}(B_2)], [\frac{1}{2}(T_3), \frac{1}{2}(B_3)])$ and $b_{II} = ([\frac{1}{4}(t_1), \frac{3}{4}(b_1)], [\frac{4}{5}(t_2), \frac{1}{5}(b_2)])$. There must be optimal behavior strategies since both players have perfect recall.

6.22 The value of the game in mixed strategies as well as in behavior strategies is $v = 2$. The optimal mixed strategy of Player I is $\sigma_I = [1(T_1B_2T_3)]$ which is equivalent to the behavior strategy $([1(T_1)], [1(B_2)], [1(T_3)])$. The optimal mixed/behavior strategy of Player II is $\sigma_{II} = [1(t)]$.

6.23 (a) In order to find the value of the game in mixed strategies we consider the game given by the following matrix

	t	b
T_1T_2	1	-2
T_1B_2	1	4
B_1T_2	2	4
B_1B_2	-1	-1

Therefore, the value of the game in mixed strategies is $v = 2$. The optimal mixed strategy of Player I is B_1T_2. The optimal mixed strategy of Player II is t .

(b) The security value in behavior strategies of both players is 2.

Note that if $u(B_1T_2, b) = -4$ then the security values in mixed strategies and in behavior strategies are not identical. Indeed, the value of the game in mixed strategies is $v = \frac{4}{3}$. The optimal mixed strategy of Player I is $\sigma_I = [\frac{2}{3}(T_1B_2), \frac{1}{3}(B_1T_2)]$. The optimal mixed strategy of Player II is $\sigma_{II} = [\frac{8}{9}(t), \frac{1}{9}(b)]$.

However, to find the security values in behavior strategies, assume that Player I plays

$$b_I = ([x(T_1), (1-x)(B_1)], [y(T_2), (1-y)(B_2)])$$

and Player II plays $b_{II} = [z(t), (1-z)(b)]$. Then the payoff is

$$\begin{aligned} U(b_I, b_{II}) &= -3xz - 3xy + 6yz + 5x - 3y - 1 \\ &= 3z(2y - x) - 3xy + 5x - 3y - 1. \end{aligned}$$

Hence, if $2y - x < 0$ then Player II can minimize the expected payoff by selecting $z = 1$ which case the maximal payoff Player I can guarantee is 1 (by $x = 1$ and $y < \frac{1}{2}$). If $2y - x \geq 0$ then Player II can minimize the expected payoff by selecting $z = 0$ which case the maximal payoff Player I can guarantee is 1 (by $x = 1$ and $y = \frac{1}{2}$). Therefore, the security value of Player I in behavior strategies is 1. On the other hand, Player II can guarantee a payoff of at most $\frac{4}{3}$ by following the behavior strategy $b_I = [\frac{8}{9}(t), \frac{1}{9}(b)]$.

6.24 Let (V, E, x^0) be a (finite or infinite) game tree such that $|C(x)| < \infty$ for each vertex x. Denote by H the set of maximal paths in the game tree. Let $p : V \to [0,1]$ be a function satisfying $p(x^0) = 1$ and $p(x) = \sum_{x' \in C(x)} p(x')$ for each vertex $x \in V$ that is not a leaf. We next prove that there exists a unique probability distribution $\widehat{p}$ over $(H, \mathcal{H})$ satisfying $\widehat{p}(H(x)) = p(x)$ for all $x \in V$.

Assume w.l.o.g. that each maximal path in the game is an infinite path (otherwise, if a path ends in a leaf, it can be continued by infinitely many vertices with only one action).

Let X_n be the set of vertices of depth n. Set $X^\infty = \times_{n=1}^{\infty} X_n$ and $X^N = \times_{n=1}^{N} X_n$ for every $N \in \mathbf{N}$. Note that p define a probability distribution over X_n for every $n \in \mathbf{N}$: since $p(x) \in [0,1]$ for vertex x it is enough to prove that $\sum_{x_n \in X_n} p(x_n) = 1$ for every $n \in \mathbf{N}$. We prove it by induction. For $n = 1$, $\sum_{x_1 \in X_1} p(x_1) = p(x^0) = 1$. Assume by induction that the claim is true for n. Then by the induction hypothesis it follow that

$$\sum_{x_{n+1} \in X_{n+1}} p(x_{n+1}) = \sum_{x_n \in X_n} \sum_{x_{n+1} \in C(x_n)} p(x_{n+1}) = \sum_{x_n \in X_n} p(x_n) = 1.$$

Let H^N be the set of every $(x_n)_{n=1}^N \in X^N$ such that $(x_n)_{n=0}^N$ is a path in the game tree that connects the root to x_N. Define $p^N : X^N \to [0,1]$ for every $(x_n)_{n=1}^N \in X^N$as follows:

$$p^N\left((x_n)_{n=1}^N\right) = \begin{cases} p(x_N) & \text{if } (x_n)_{n=0}^N \in H^N \\ 0 & \text{if } (x_n)_{n=0}^N \notin H^N \end{cases}$$

For every $x_N \in X_N$ there is only one path in the game tree from the root to x_N, thus there is only one $(x_n)_{n=1}^N \in X^N$ such that $p^N\left((x_n)_{n=1}^N\right) = p(x_N)$ thus it defines a probability distribution over X^N. In addition, it satisfies the condition

$$p^N(A) = p^{N+1}(A \times X_{N+1}), \quad \forall N \in \mathbf{N}, \ \forall A \in X^N$$

Therefore, by Theorem 6.21, there exists a unique probability distribution $\widehat{p}$ over $(X^\infty, \mathcal{Y})$ extending $(p^N)_{N \in \mathbf{N}}$, i.e.,

$$p^N(A) = \widehat{p}(A \times X_{N+1} \times X_{N+2} \times ...), \quad \forall n \in \mathbf{N}, \ \forall A \in X^N.$$

In particular, for every $n \in \mathbf{N}$,

$$\widehat{p}(H^N \times X^{N+1} \times X^{N+2} \times ...) = p^N(H^N) = 1$$

So
$$\widehat{p}(H) = \lim_{N\to\infty} \widehat{p}(H^N \times X^{N+1} \times X^{N+2} \times ...) = 1.$$
Thus, $\widehat{p}$ is a probability distribution over over $(H, \mathcal{H})$. Finally, for every $x \in V$, there is a unique path $(x_n)_{n=0}^N$ from the root to $x_N = x$. Hence
$$\begin{aligned}\widehat{p}(H(x)) &= \widehat{p}((x_n)_{n=1}^N \times X^{N+1} \times X^{N+1} \times ...)\\ &\quad - \widehat{p}((x_n)_{n=1}^N \times X^{N+1} \times X^{N+1} \times ... \setminus H)\\ &= p^N((x_n)_{n=1}^N) - 0 = p(x),\end{aligned}$$
as needed.

6.25 let σ be a mixed/behavior strategy vector in a (finite or infinite) extensive-form game. Then $\rho(\cdot,\sigma) : V \to [0,1]$. By Theorem 6.22 (Exercise 6.24), in order to prove that there exists a unique probability distribution μ_σ over $(H, \mathcal{H})$ satisfying $\mu_\sigma(H(x)) = \rho(x;\sigma)$ for each vertex x it is enough to prove that $\rho(x,\sigma) = \sum_{x'\in C(x)} \rho(x',\sigma)$ for every $x \in V$.

Let $x \in V$. Denote by i_x the player such that $x \in U_{i_x}$.
Assume first that σ is a behavior strategy vector. Then $\rho_{i_x}(x',\sigma) = b_{i_x}(a_{i_x}(x \to x'), U_{i_x})\rho_{i_x}(x,\sigma_{i_x})$, and for every player $i \neq i_x$, $\rho_i(x,\sigma_i) = \rho_i(x',\sigma_i)$. In particular,
$$\begin{aligned}\sum_{x'\in C(x)} \rho(x',\sigma) &= \sum_{x'\in C(x)} \left(\prod_{i\in N} \rho_i(x',\sigma_i)\right)\\ &= \sum_{x'\in C(x)} \left(\prod_{i\neq i_x} \rho_i(x,\sigma_i)\right) \left(b_{i_x}(a_{i_x}(x \to x'), U_{i_x}\right) \rho_{i_x}(x,\sigma_{i_x})\\ &= \left(\prod_{i\in N} \rho_i(x,\sigma_i)\right) \sum_{x'\in C(x)} b_{i_x}(a_{i_x}(x \to x'), U_{i_x}) = \rho(x,\sigma),\end{aligned}$$
as needed.
Assume next that σ is a mixed strategy vector and x is of depth T. Then, for every player $i \neq i_x$, $\rho_i(x,\sigma_i) = \rho_i(x',\sigma_i)$. In addition, since every pure strategy of player i_x in G^{T+1} in which player i_x selects the actions leading to x is a strategy of player i_x in G^{T+1} in which player i_x in selects the actions leading to some $x' \in C(x)$, $\rho_{i_x}(x,\sigma_{i_x}) =$

$\sum_{x' \in C(x)} \rho_{i_x}(x', \sigma_{i_x})$ thus

$$\begin{aligned}\sum_{x' \in C(x)} \rho(x', \sigma) &= \sum_{x' \in C(x)} \left(\prod_{i \in N} \rho_i(x', \sigma_i) \right) \\ &= \left(\prod_{i \neq i_x} \rho_i(x, \sigma_i) \right) \sum_{x' \in C(x)} \rho_{i_x}(x', \sigma_{i_x}) \\ &= \prod_{i \in N} \rho_i(x, \sigma_i) = \rho(x, \sigma),\end{aligned}$$

which conclude the proof.

6.26 We prove that the mixed strategy σ_i constructed in the proof of Theorem 6.26 is equivalent to the given behavior strategy b_i.
Let $\sigma_{-i} = (\sigma_j)_{j \neq i}$ be a mixed/behavior strategy vector of the other players. For each $T \in \mathbf{N}$, let σ_j^T be the strategy σ_j restricted to the game G^T. Denote $\sigma_{-i}^T = (\sigma_j^T) j \neq i$. Since the strategies σ_i^T and b_i^T are equivalent in the game G^T,

$$\rho(x; \sigma_i^T, \sigma_{-i}^T) = \rho(x; b_i^T, \sigma_{-i}^T)$$

for each vertex whose depth is less than or equal to T. By definition, it follows that for each vertex x

$$\rho(x; \sigma_i, \sigma_{-i}) = \rho(x; b_i, \sigma_{-i})$$

Theorem 6.25 implies that σ_i and b_i are equivalent strategies.

Chapter 7: Equilibrium refinements

7.1 (a) In a game with perfect information whose game tree has eight vertices there are eight subgames since each vertex define a subgame begining at this vertex.

(b) The number of subgames in a game whose game tree has eight vertices and one information set, which contains two vertices (with all other information sets containing only one vertex) is either six in case the players have perfect recall or seven if one of the two vertices in the information set is the root and the other is his child.

(c) The number of subgames in a game whose game tree has eight vertices, three of which are chance vertices is eight as long as each information set of each players contains only one vertex.

7.2 See Exercise 4.38.

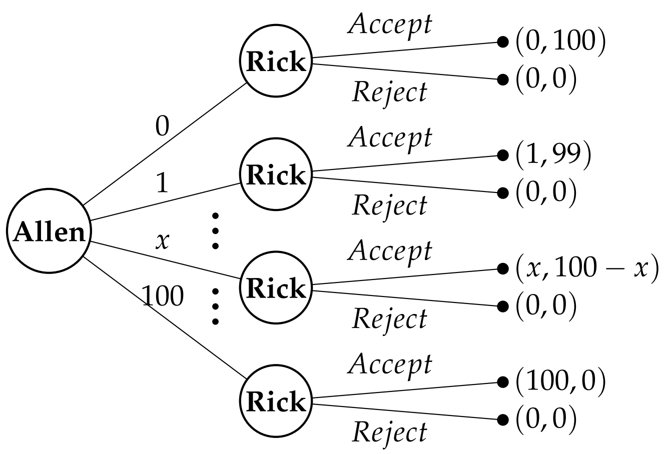
7.3 See Exercise 4.31.

7.4 **Game A:** Note that the game has only one subgame which is the original game so every equilibrium of the game is also a subgame perfect equilibrium. The only subgame perfect equilibrium of the game is $([1(T)], [1(t)])$.
Game B: $\big([1(T)], ([x(t_1), (1-x)(b_1)], [1(b_2)])\big)$ for every $x \in [0,1]$.
Game C: Player II is indifferent between t and b. Hence the set of subgame perfect equilibria is

$$([1(B)], [x(t), (1-x)(b)]) \;\; \forall x \geq \tfrac{1}{3}$$
$$([1(T)], [x(t), (1-x)(b)]) \;\; \forall x \leq \tfrac{1}{3}.$$

Game D: The only subgame perfect equilibrium is $([1(T), 1(b)]$.

7.5 (a) The extensive-form game is given by the following game tree.



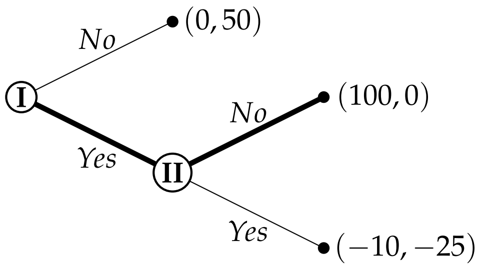
(b) The set of pure strategies of Allen is $S_A = \{0, 1, ..., 100\}$. The set of strategies of Rick is $S_R = \{Reject, Accept\}^{100}$.

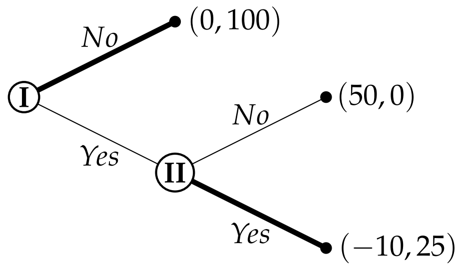
(c) For every $a \in \{0, 1, ..., 100\}$, every strategy vector according to which Allen chooses a and Rick Reject each offer $x > a$ and accept $x = a$ is an equilibrium corresponds the result $(a, 100-a)$. Indeed, Allen cannot gain by deviating since higher offer will be rejected so he gets zero. On the other hand, if Rick deviates then the result will be changed only in case of rejection of a which case both of them gets zero.

(d) In a subgame perfect equilibrium, Rick must accept each offer $a \in \{0, 1, ..., 99\}$, whereas if the offer is 100 he is indifferent between accepting and rejecting so he may play $[y(Accept)(1-y)(Reject)]$.

Thus if $y < 0.99$ Allen chooses 99, if $y > 0.99$ Allen chooses 100 for $y = 0.99$ Allen may play $[z(100), (1-z)(99)]$ for every $z \in [0, 1]$.

7.6 (a) The next game trees represent the the mechanism implemented by the king. In the left game tree Elizabeth (I) is the true mother of the child, and in the right game tree Mary (II) is the true mother of the child.

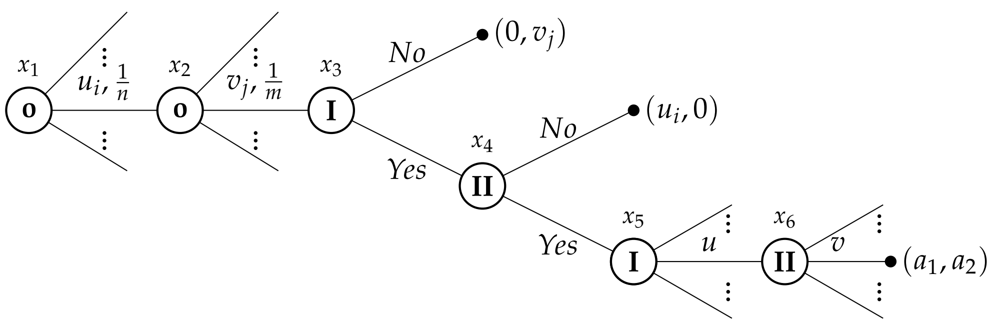




(b) The subgame perfect equilibrium of each game consist of the actions appear as bold lines in the appropriate game tree. As one can see in the left game tree in which Elizabeth (I) is the true mother of the child, she gets the child and neither woman pays anything at all. Similarly, in the right game tree in which Mary (II) is the true mother of the child, she gets the child and neither woman pays anything at all.

(c) Another equilibrium of each game, which is not the subgame perfect equilibrium is that the mother claim she is not the mother while the pretender claim she is the mother.

7.7 (a) In the next game tree Claudius is Player I and Marcus is Player II.



where the payoff vector (a_1, a_2) is given by

$$(a_1, a_2) = \begin{cases} (u_i - u + \epsilon/4, -\epsilon/4) & \text{if } u > v \\ (-\epsilon/4, v_i - v + \epsilon/4) & \text{if } u < v \\ (-\epsilon/4, -\epsilon/4) & \text{if } u = v. \end{cases}$$

(b) We next find the subgame perfect equilibrium of the game. Consider first the subgame starts at x_6. Assume first that $u \geq v_i$. In case Player II suggests $v \leq u$ his payoff is $-\epsilon/4$ whereas if he suggests a payoff $v > u \geq v_j$ then $v > u + \epsilon$ and his payoff satisfy

$$v_j - v + \tfrac{\epsilon}{4} \leq u - v + \tfrac{\epsilon}{4} < -\epsilon + \tfrac{\epsilon}{4} = -\tfrac{3\epsilon}{4}$$

so he is better off suggesting $v \leq u$. on the other hand, if $u < v_j$ then the best response of Player II is to suggest $v = \min_{v'_j > u} v'_j$. In particular $v \leq v_j$ and his payoff is $v_j - v + \epsilon/4 > -\epsilon/4$.

Hence, in the subgame starts at x_5, Player I knows he wins the horse if his suggestion $u > v_j$ and he loses the horse if $u < v_j$. Therefore, using the same arguments presented above, if $u_i > v_j$ his best response is $u = \min_{u'_i > v_j} u'_i$ and the payoff in this subgame is $(u_i - u + \varepsilon/4, -\varepsilon/4)$. If $u_i < v_j$ then to win the game he must suggest $u > v_j > u_i$ but then his payoff is less than $-\frac{3\varepsilon}{4}$ so his best response is $u \leq v_j$.

Therefore, one can conclude that in a subgame equilibrium of the subgame that starts at x_5, the player who values the horse the most wins the horse so his payoff is positive whereas his opponent's payoff is negative. Therefore, player II in x_4 as well as Player I in x_3 is better telling the truth, which leads to the result that the player who values the horse the most receiving the horse (in case both friends equally value the horse, Claudius receives the horse).

7.8 Theorem 7.9: Every (finite) extensive-form game with perfect information has a subgame perfect equilibrium in pure strategies.
Proof. We will prove the claim by backward induction. If the game tree is comprised of a single vertex, then the unique strategy vector is $(\emptyset, \ldots, \emptyset)$ (so there are no available deviations), and it is therefore the unique subgame perfect equilibrium. Assume by induction that the claim is true for each perfect information game in extensive form containing fewer than K vertices, and consider a game G with K vertices. Let x be a vertex all of whose children are leaves. Since the

game has perfect information and perfect recall, the player choosing an action at vertex x (say player i) knows that the play of the game has arrived at that vertex (and not at a larger information set containing x) and he therefore chooses the leaf l giving him the maximal payoff. Consider the game tree obtaining by erasing the leaves following x and thus turning x into a leaf with a payoff equal to the payoff of l. The resulting game tree has fewer vertices than the original tree, so we can apply the induction hypothesis to it, i.e. there is subgame perfect equilibrium s^{-x} in the resulting game. Let s^* be a strategy vector according to which the players follow s^{-x} in all vertices but x, whereas in x player i who controls x chooses l. Since player i chooses l in x according to s^*, the resulting game and the original game are the same from the point of view of each player $j \neq i$, so he can not gain by deviating in each subgame. Assume to the contrary that player i can gain by deviating to s_i. Then, by the maximality of l, player i can gain by deviating to s_i^l according to which at x he chooses l. But then he can gain by deviating to s_i^l in the resulting game, which contradicts the assumption that S^{-x} is Nash equilibrium in the resulting game. ■

7.9 We prove that in the n-times-repeated Prisoner's Dilemma game, the only subgame perfect equilibrium is the one where both players choose D in all stages of the game (after every history of previous actions), for every $n \in \mathbf{N}$ in particular for $n = 100$. We prove the claim by backward induction over n. If $n = 1$ then the game has only one subgame with only one equilibrium (which is necessarily subgame perfect equilibrium) in which both players play D. Assume the claim holds whenever the number of repetition is less than n. For every subgame which does not start at the root the number of repetition is less than n then by the induction hypothesis, from the second repetition onwards both players select D. Thus in the first repetition in the n-times-repeated game both players face the following game:

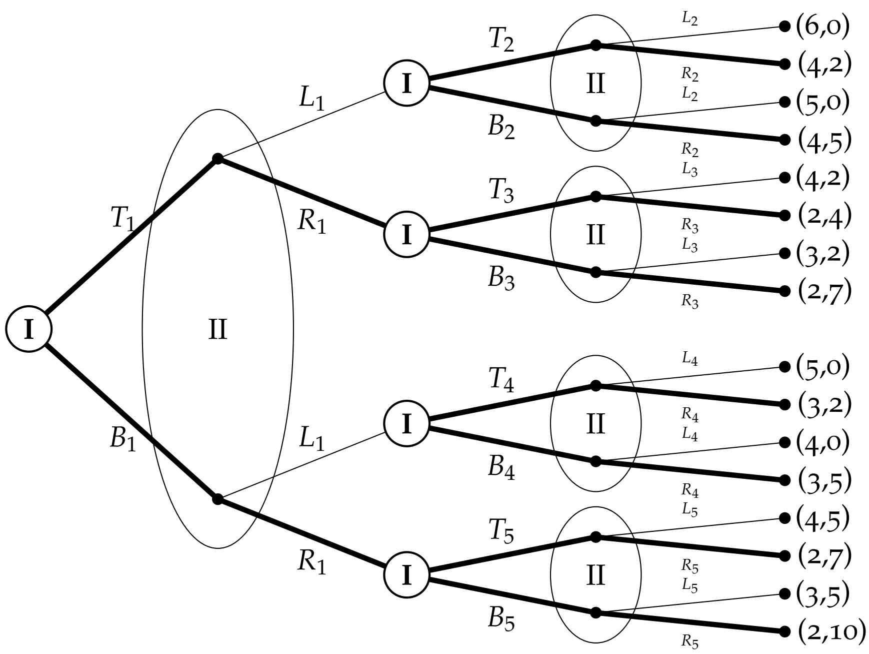
	D	C
D	$1+n, 1+n$	$4+n, 0+n$
C	$0+n, 4+n$	$3+n, 3+n$

However, by eliminating strictly dominated strategies this game has only one equilibrium in which both players play D. In conclusion, there is only one subgame perfect equilibrium in which both players always play D.

7.10 (a) The set of equilibria is

$$\left\{ ([x(T), (1-x)(B)], [1(R)]) \quad \middle| \quad x \in [0,1] \right\}.$$

(b) The two-stage extensive-form game is given by the following game tree.



(c) There are five subgames of the two-stage game, each one of them starts at a vertex controlled by Player I.

(d) Every subgame perfect equilibrium of the two-stage game consists of the actions appear as bold lines in the above game tree (b).

7.11 We prove that a strategy vector $\sigma^* = (\sigma_i^*)_{i \in N}$ in an extensive-form game with perfect information is a subgame perfect equilibrium if and only if for each player $i \in N$, every decision vertex x, and every strategy $\widehat{\sigma}_i$ of player i that is identical to σ_i^* at every one of his decision vertices except for x,

$$u_i(\sigma^* | x) \geq u_i((\widehat{\sigma}_i, \sigma_{-i}^*) | x). \tag{43}$$

If $\sigma^* = (\sigma_i^*)_{i \in N}$ in an extensive-form game with perfect information is a subgame perfect equilibrium then Inequality (43) holds for every vertex x, every player i, and every strategy $\widehat{\sigma}_i$, in particular it holds for a strategy $\widehat{\sigma}_i$ of player i that is identical to σ_i^* at every one of his decision vertices except for x.

We next prove the other direction. Assume that Inequality (43) holds for each player $i \in N$, every decision vertex x, and every strategy $\widehat{\sigma}_i$

of player i that is identical to σ_i^* at every one of his decision vertices except for x. For every player i and every strategy σ_i of player i denote by $\mathcal{X}_i(\sigma_i)$ the set of all vertices such that σ_i differs from σ_i^*. We show by induction over $|\mathcal{X}_i(\sigma_i)|$ that for every player i, every strategy σ_i of player i and every vertex x one has

$$u_i(\sigma^*|x) \geq u_i((\sigma_i, \sigma_{-i}^*)|x).$$

For $|\mathcal{X}_i(\sigma_i)| = 1$ the claim holds by the assumption. Assume that the claim holds whenever $|\mathcal{X}_i(\sigma_i)| < n$. Let σ_i be such that $|\mathcal{X}_i(\sigma_i)| = n$. From the vertices in $\mathcal{X}_i(\sigma_i)$, choose a "highest" vertex $\widehat{x}$, i.e., a vertex such that every path from the root to it does not pass through any other vertex in $\mathcal{X}_i(\sigma_i)$. Define a strategy $\widehat{\sigma}_i$ such that for every vertex x' which is a descendant of $\widehat{x}$, $\widehat{\sigma}_i(x') = \sigma_i(x')$ otherwise $\widehat{\sigma}_i(x') = \sigma_i^*(x')$. Then by the induction hypothesis, for every vertex x, $u_i(\sigma^*|x) \geq u_i((\widehat{\sigma}_i, \sigma_{-i}^*)|x)$. Hence,

$$\begin{aligned}
u_i((\sigma_i, \sigma_{-i}^*)|\widehat{x}) &= \sum_{x' \in C(\widehat{x})} \sigma_i(a_i(\widehat{x} \to x'))u_i((\sigma_i, \sigma_{-i}^*)|x') \\
&= \sum_{x' \in C(\widehat{x})} \sigma_i(a_i(\widehat{x} \to x'))u_i((\widehat{\sigma}_i, \sigma_{-i}^*)|x') \\
&\leq \sum_{x' \in C(\widehat{x})} \sigma_i(a_i(\widehat{x} \to x'))u_i(\sigma^*|x') \leq u_i(\sigma^*|\widehat{x}),
\end{aligned}$$

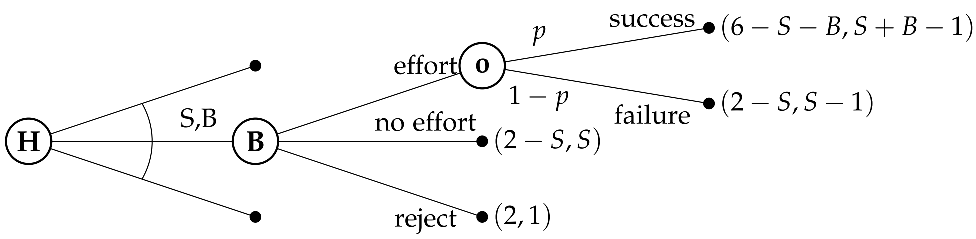
where the last inequality follows the induction hypothesis. Finally, let $\bar{\sigma}_i$ be a strategy such that for every vertex x' in the subgame $\Gamma(\widehat{x})$, $\bar{\sigma}_i(x') = \sigma_i^*(x')$ otherwise $\bar{\sigma}_i(x') = \sigma_i(x')$. Then, since the induction hypothesis holds for $\bar{\sigma}_i$, for every vertex x

$$\begin{aligned}
u_i((\sigma_i, \sigma_{-i}^*)|\widehat{x}) &\leq u_i((\bar{\sigma}_i, \sigma_{-i}^*)|\widehat{x}) \\
&\leq u_i((\sigma^*)|\widehat{x}),
\end{aligned}$$

as needed.

7.12 If σ is a subgame perfect equilibria in Γ then it defines a Nash equilibria, in every subgame, and thus it also defines a subgame in the agent-form game (Exercise 4.51). The other direction follows by Exercise 7.11.

7.13 (a) In the extensive-form game presented in the following game tree Hillary (H) is Player I and Bill (B) is Player II. The graph depicts only one subgame tree, after Hillary has chosen the salary and bonus. All the other possible subtrees are identical to the one shown here.



(b) We use a backward induction to find a subgame perfect equilibrium.

- If Bill takes the job offered by Hillary and invests effort on the project then the expected payoffs are

$$(4p + 2 - (S + pB), (S + pB) - 1).$$

- If $S + pB - 1 \geq \max\{S, 1\}$ (so $B \geq 1/p$ and $S + pB \geq 2$) then Bill chooses to take the job and to invest effort so the payoff vector is

$$(4p + 2 - (S + pB), (s + pB) - 1).$$

  In particular, the payoff of Hillary is $4p + 2 - (S + pB) \leq 4p$.
- If $S \geq \max\{1, S + pB - 1\}$ (so $B \leq 1/p$) then Bill chooses to take the job but to invest no effort so the payoff vector is $(2 - S, S)$.
- If $1 \geq \max\{S, S + pB - 1\}$ then Bill chooses to reject the job so the payoff vector is $(2, 1)$.
- Therefore, if $4p < 2$ Hillary is better off offering Bill a salary and bonus that he rejects ($S < 1$ and $B < (2 - S)/p$). If $4p > 2$ Hillary is better off offering Bill the minimal salary and bonus that still provide him an incentive to take the job and invest effort ($B \leq 1/p$ and $S + pB = 2$).

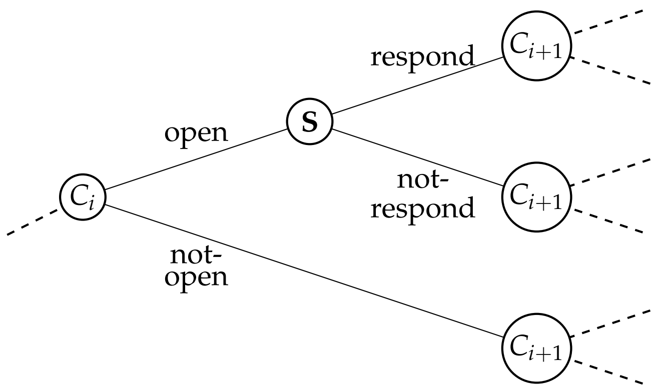
In conclusion, if $p \geq 0.5$, every subgame perfect equilibrium in the game is the strategy vector according to which Hillary suggests a salary S and a bonus B such that $B \leq 1/p$ and $S + pB = 2$ and Bill takes the Job and invests effort. If $p \leq 0.5$, every subgame perfect equilibrium in the game is the strategy vector according to which Hillary suggests a salary S and a bonus B such that $S < 1$ and $B < (2 - S)/p$ and Bill rejects the Job.

(c) Hillary need to persuade Bill during their job interview, that the probability of success is even higher than it really is which leads to a lower demands from Bill, and thus increases her expected payoff at equilibrium.

7.14 (a) The extensive-form game includes a set of players $N = \{S, C_1, C_2, ..., C_{10}\}$ where S is the chainstore and C_i is the i's competitor. The action set at every vertex x controlled by the chainstore is $A_S(x) = \{respond, not - respond\}$. The action set at every vertex x controlled by the i-th competitor is $A_{C_i}(x) = \{open, not - open\}$. The payoff of the chainstore is

$$5(10 - O) + 3(O - R) - R = 50 - 2O - 4R$$

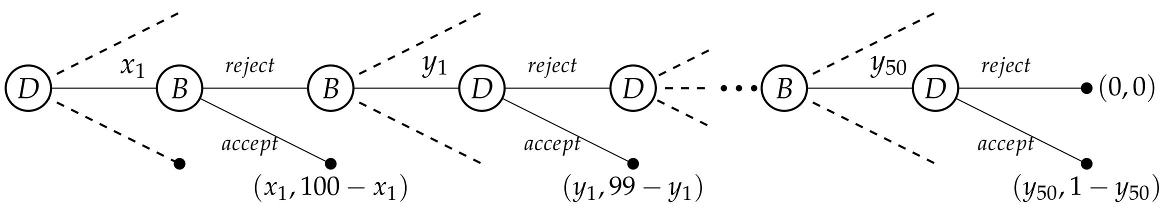
where O is the number of competitors that decide to "open" a rival electronics store and R is the the number of times the chainstore decide to "respond" ($R \leq O$).The payoff of the i's competitor is zero if he decides to not open, -1 if he decides to open and the chainstore respond and 3 if he decides to open and the chainstore does not respond. The game starts with a vertex of competitor i. We present part of the game tree, it continuous in the same manner until the last competitor (C_{10}).



(b) We show by backward induction that the only subgame is the one in which every competitor opens a rival electronics store in his city, and the the national chainstore does not respond by undercutting prices. We start with each vertex controlled by the national chainstore who should select a response to the opening a rival electronics store by the last competitor (C_{10}). The best action of the national chainstore is to not respond by undercutting prices (and gain additional 3 rather than lose 1). Thus the last competitor is better off opening a rival electronics store in his city (and gaining additional 3 rather than zero). Assume that from every vertex controlled by competitor i onwards the competitors open rival electronics stores in their city, and the the national chainstore does not respond by undercutting prices. Then by similarly arguments the best response of the national chainstore is to not respond by undercutting prices and the i's competitor prefers to open a rival electronics store.

(c) The strategy vector in which the national cainstore respond by undercutting prices whenever a competitor decides to open a rival electronics store in his city, and no competitor opens a store is Nash equilibrium. No competitor gains by deviating (since if he opens a store he loses 1 rather than zero) as well as the national chainstore who gains her maximal possible payoff in this situation). However, this is not a subgame perfect equilibrium since the threat of the national chainstore is not credible.

7.15 The extensive-form game is presented in the following game tree.



We use a backward induction to determine the subgame perfect equilibrium. In the last offer (round 100), Debby accepts any offer $(y_{50}, 1 - y_{50})$ (if $y_{50} = 0$ she is indifferent between accepting and rejecting the offer). Thus Barack offers $(0, 1)$. Therefore, Debby must offer Barack at least 1, so she offers $(1, 1)$. Thus in round 98 Barack must offer Debby at least 1, hence he is better offering $(1, 2)$ and so on. Hence in a subgame perfect equilibrium, for every odd k, at the k'th round, Barack offers $\left(\frac{100-k}{2}, 1 + \frac{100-k}{2}\right)$ and Debby accepts. For every even k, at the k'th round, Debby offers $\left(\frac{101-k}{2}, \frac{101-k}{2}\right)$ and Barack accepts. In particular, the game starts with Debby offering $(50, 50)$ which is accepted by Barack.

7.16 We prove that every extensive-form game with perfect recall has a subgame perfect equilibrium in behavior strategies by induction over the number of subgames. The existence of a subgame perfect equilibrium in mixed strategies will follow by Kuhn's Theorem. If the game has only one subgame which is the game itself then the existence of a subgame perfect equilibrium in behavior strategies follows Theorem 6.16 (since the game has perfect recall). Assume that every extensive-form game with perfect recall and less than n subgames has a subgame perfect equilibrium in behavior strategies. We prove that it also holds for extensive-form game Γ with perfect recall and n subgames. Let x be a vertex in Γ differs from the root such that the subgame $\Gamma(x)$ that starts at x has the most subgames (hence it is not contained in any subgame except for Γ). $\Gamma(x)$ has perfect recall and less than n subgames, hence, by the induction hypothesis, it has a subgame perfect equilibrium in behavior strategies denoted by σ^x which leads a payoff of $(u_i(\sigma^x|x))_{i\in N}$. Consider the game $\widehat{\Gamma}$ that is identical to Γ but if the game reaches x the game is over with the payoff of $(u_i(\sigma^x|x))_{i\in N}$. $\widehat{\Gamma}$ has perfect recall and less than n subgames therefore, by the induction hypothesis, it has subgame perfect equilibrium $\widehat{\sigma}$. Define next a strategy vector σ^* in Γ as follows. As long as the game does not reach x, all of the players follow $\widehat{\sigma}$ as soon as it reaches x, they follow σ^x. We show that σ^* is a subgame perfect equilibrium in Γ. First, every subgame that is not Γ itself either does not contain $\Gamma(x)$ or it is contained in $\Gamma(x)$ therefore σ^* is an equilibrium in that subgame by the way σ^* is defined. It is left to prove that it is also an equilibrium in Γ. For every player i and every behavior strategy σ_i one has

$$u_i(\sigma^*|x) = u_i(\sigma^x|x) \geq u_i(\sigma_i, \sigma^x_{-i}|x) = u_i(\sigma_i, \sigma^*_{-i}|x).$$

Hence, the payoff $u_i(\sigma_i, \sigma^*_{-i})$ in Γ is at most the payoff of player i under $(\sigma_i, \sigma^*_{-i})$ restricted to $\widehat{\Gamma}$ which is at most $u_i(\sigma^*)$, so $u_i(\sigma_i, \sigma^*_{-i}) \leq u_i(\sigma^*)$ as needed.

7.17 (a) The set of all subgames in the game (not including the trivial subgames) includes every subgame $\Gamma(x_{a,b})$ ($a \in [0,1]$ and $b \in \{T, B\}$) that starts at a vertex $x_{a,b}$ controlled by Player III.

(b) For every strategy $a < 0$ of Player I, the best response of Player III is $(b_1 b_2)$ and the best response of Player IV is $\beta_1\beta_2$. For every strategy $a > 0$ of Player I, the best response of Player III is $(t_1 t_2)$ and the best response of Player IV is $\tau_1\tau_2$.

(c) If $a = 0$, every strategy of Player III and Player IV is a best response of them.

(d) There does not exist a subgame perfect equilibrium such that Player I plays $a = 0$, Player II plays $\beta = \frac{1}{2}$, and if $a = 0$, Player III chooses t_1 with probability $\frac{1}{4}$, and chooses t_2 with probability $\frac{1}{8}$, while Player IV chooses τ_1 and τ_2 with probability 1. Indeed, under this strategy vector, the expected payoff of Player I is $-20(\frac{1}{2} \cdot \frac{7}{8} + \frac{1}{2} \cdot \frac{3}{4}) = -16.25$. Whereas, if for Player I deviates to $a \neq 0$ then Players III and IV follow the strategies presented in (b), and the payoff of Player I is now $-|a|^2 \geq -1$.

(e) The expected payoff of Player I in the case where $a \neq 0$ is given by $-a^2 + 2a(2\beta - 1)$.

(f) The upper bound of the possible payoffs Player I can receive, in the case where $a = 0$ is zero.

(g) Player I's best reply when $\beta < \frac{1}{2}$ is $a = 2\beta - 1 < 0$. The best replies of Players III and IV, given Player I's strategy are $(b_1 b_2)$ and $\beta_1\beta_2$, respectively. Player II's best reply to these strategies of Players I, III, and IV is $\beta = 1$. In particular, there is no subgame perfect equilibrium in the game in which $\beta < \frac{1}{2}$.

(h) Player I's best reply when $\beta > \frac{1}{2}$ is $a = 2\beta - 1 > 0$. The best replies of Players III and IV, given Player I's strategy are $(t_1 t_2)$ and $\tau_1\tau_2$, respectively. Player II's best reply to these strategies of Players I, III, and IV is $\beta = 0$. In particular, there is no subgame perfect equilibrium in the game in which $\beta > \frac{1}{2}$. Hence and by (g), one can conclude that every subgame perfect equilibrium must satisfies that $\beta = \frac{1}{2}$.

(i) Suppose that $\beta = \frac{1}{2}$. By (e), if Player I plays $a \neq 0$ then his payoff is $-a^2 < 0$. Therefore, he can guarantee himself a payoff as close to zero as he wants so for every $a \neq 0$ there is a worthwhile deviation. Moreover, if he plays $a = 0$ and his expected payoff is $v < 0$, then he is better off deviating to $min\{\frac{v}{2}, 1\}$. By (h), under σ Player II must play $\beta = \frac{1}{2}$, thus the expected payoff of Player I is zero, so Player I necessarily plays a = 0. In particular, one can deduce that under σ Players III and IV play $([x(t_1), (1 - x)(b_1)], [y(t_2), (1 - y)(b_2)])$ and $([x(\tau_1), (1 - x)(\beta_1)], [y(\tau_2), (1 - y)(\beta_2)])$ for $x, y \in \{0, 1\}$, respectively. The best response of Player II is playing a pure strategy $\beta = 1 - y$.

(j) There is no subgame perfect equilibrium of this game such that $\beta \neq \frac{1}{2}$ (by(g) and (h)) nor subgame perfect equilibrium in which $\beta = \frac{1}{2}$, so this game has no subgame perfect equilibrium.

(k) The game has Nash equilibrium: Player I plays $a = 0$, Player II plays $\beta = \frac{1}{2}$, Player III plays $([\frac{1}{2}(t_1), \frac{1}{2}(b_1)], [\frac{1}{2}(t_2), \frac{1}{2}(b_2)])$ and $\tau_1\tau_2$.

7.18 We prove that for each player i, and every vector of perturbations ϵ_i, the set of strategies $\Sigma_i(\epsilon_i)$ is compact (closed and bounded) and convex. The set Σ_i is bounded, so $\Sigma_i(\epsilon_i) \subseteq \Sigma_i$ is also bounded.
Let $\sigma_i^1, \sigma_i^2, ...$ be a sequence of a strategies in $\Sigma_i(\epsilon_i)$ such that $\sigma_i = \lim_{k\to\infty} \sigma_i^k$. Then, $\sigma_i \in \Sigma_i$ since Σ_i is a closed set. In addition, for every $k \in \mathbf{N}$ and every s_i one has $\sigma_i^k(s_i) \geq \epsilon_i(s_i)$ so

$$\sigma_i(s_i) = \lim_{k\to\infty} \sigma_i^k(s_i) \geq \epsilon_i(s_i)$$

and $\Sigma_i(\epsilon_i)$ is a closed set.
Finally, we prove that $\Sigma_i(\epsilon_i)$ is convex. Let $\sigma_i^1, \sigma_i^2 \in \Sigma_i(\epsilon_i)$ and $\alpha \in$

$[0,1]$. Then, by the convexity of Σ_i, $\alpha\sigma_i^1 + (1-\alpha)\sigma_i^2 \in \Sigma_i$. Moreover, for every s_i one has

$$\alpha\sigma_i^1(s_i) + (1-\alpha)\sigma_i^2(s_i) \geq \alpha\epsilon_i(s_i) + (1-\alpha)\epsilon_i(s_i) = \epsilon_i(s_i),$$

which leads to $\alpha\sigma_i^1 + (1-\alpha)\sigma_i^2 \in \Sigma_i(\epsilon_i)$ and $\Sigma_i(\epsilon_i)$ is convex.

7.19 Let $\sigma_i \in \Sigma_i$ be a mixed strategy. We prove that σ_i can be approximated by a completely mixed strategy. Let $0 < \delta < 1$. For every pure strategy $s_i \in S_i$ set

$$\sigma'_i(s_i) = \delta\frac{1}{|S_i|} + (1-\delta)\sigma_i(s_i).$$

Then σ'_i is a completely mixed strategy. Indeed, $\sigma'_i(s_i) \geq \frac{\delta}{|S_i|} > 0$, and

$$\sum_{s_i\in S_i}\sigma'_i(s_i) = \sum_{s_i\in S_i}\left(\delta\frac{1}{|S_i|} + (1-\delta)\sigma_i(s_i)\right) = |S_i|\frac{\delta}{|S_i|} + (1-\delta)\sum_{s_i\in S_i}\sigma_i(s_i) = 1.$$

Furthermore, for every $s_i \in S_i$, $|\sigma_i(s_i) - \sigma'_i(s_i)| = |-\frac{\delta}{|S_i|} + \delta\sigma_i(s_i)| \leq \delta$. Note that every completely mixed strategy that is δ-approximation of σ_i is also δ'-approximation of σ_i, for $\delta' > \delta$, thus the claim holds for every $\delta > 0$.

7.20 We prove that the set of perfect equilibria of a strategic-form game is a closed subset of $\Sigma = \times_{i\in N}\Sigma_i$.
Let $\sigma^1, \sigma^2, \ldots$ be a sequence of perfect equilibria of a strategic-form game that converges to a mixed strategy profile $\sigma \in \Sigma$. Then, for every $m \in \mathbf{N}$, there is a sequence of perturbation vectors $(\epsilon^{m,k})_{k\in N}$ satisfying $\lim_{k\to\infty} M(\epsilon^{m,k}) = 0$, and for each $k \in N$ there exists an equilibrium $\sigma^{m,k}$ of $\Gamma(\epsilon^{m,k})$ such that $\lim_{k\to\infty}\sigma^{m,k} = \sigma^m$. For every $m \in \mathbf{N}$, let $k_m \in \mathbf{N}$ such that $M(\epsilon^{m,k_m}) < \frac{1}{m}$ and

$$\left\|\sigma^{m,k_m} - \sigma^m\right\| < \|\sigma - \sigma^m\|.$$

Hence, we can deduce that $(\epsilon^{m,k_m})_{m\in N}$ is a sequence of perturbation vectors satisfying

$$0 \leq \lim_{m\to\infty} M(\epsilon^{m,k_m}) \leq \lim_{m\to\infty}\tfrac{1}{m} = 0.$$

Moreover, for each $m \in N$, σ^{m,k_m} is an equilibrium of $\Gamma(\epsilon^{m,k_m})$ such that

$$0 \leq \left\|\sigma - \sigma^{m,k_m}\right\| \leq \|\sigma - \sigma^m\| + \left\|\sigma^m - \sigma^{m,k_m}\right\| < 2\|\sigma - \sigma^m\| \xrightarrow[m\to\infty]{} 0.$$

In conclusion $\lim_{m\to\infty}\sigma^{m,k_m} = \sigma$ which implies that σ is a perfect equilibrium and thus the set of perfect equilibria of a strategic-form game is a closed set.

7.21 **Game A:** (T, L) is the only perfect equilibrium of the game. By Theorem 7.28, since action B is weakly dominated, it is chosen with probability zero in every perfect equilibrium. Thus Player I necessarily plays T in a perfect equilibrium which exist by Theorem 7.27, and Player II best replay is L.

Game B: (T, L) is the only perfect equilibrium of the game. Actions B and R are eliminated during an elimination process of strictly dominated strategies. Therefore, actions M and C are played with probability zero in a perfect equilibrium (by Theorem 7.28) which leads to a unique equilibrium (T, L) (Theorem 7.27).

7.22 (a) We next show that $\sigma = ([x_1(T), x_2(M), (1 - x_1 - x_2)(B)], L)$ is a Nash equilibrium of this game if and only if $\frac{1}{3} \leq x_1 \leq \frac{2}{3}$, $0 \leq x_2 \leq 2 - 3x_1$, and $x_1 + x_2 \leq 1$. Obviously, Player I cannot profit by deviating since his payoff is 1 under every mixed strategy given Player II plays L. Therefore σ is a Nash-equilibrium if and only if

$$u_{II}(\sigma_I, L) = 2 - x_2 \geq 3 - 3x_1 - x_2 = u_{II}(\sigma_I, C) \qquad (44)$$

$$u_{II}(\sigma_I, L) = 2 - x_2 \geq 3x_1 = u_{II}(\sigma_I, R) \qquad (45)$$

From Inequality (44) one can deduce that $x_1 \geq \frac{1}{3}$, and from Inequality (45) it follows that $2 - 3x_1 \geq x_2 \geq 0$ so $\frac{2}{3} \geq x_1$.

(b) The equilibria identified in Part (a) are all the Nash equilibria of the game.

First, there is no equilibrium, in which Player I plays pure:

- If he plays T then Player II prefers to play R and thus Player I is better off playing B.
- If he plays $M(/B)$ then Player II prefers to play C and thus Player I is better off playing T.

In addition, there is no equilibrium in which Player I plays exactly two actions with positive probabilities except of those identified in Part (a).

- If Player I plays action T and M ($/T$ and B) with positive probabilities then Player II Plays at most one action of actions R and C with positive probability. However, if Player II selects R (and maybe L) then Player I's best response is B. Otherwise, if Player II selects C (and maybe L) then Player I's best response is T.
- If Player I plays action M and B with positive probabilities then Player II plays C and thus Player I is better off selecting T.

Finally, there is no Nash equilibrium in which Player I plays a completely mixed strategy except for those identified in Part (a). In such an equilibrium the expected payoffs of Player I under all pure strategies are identical. Denote by $[y_1(L), y_2(C), y_3(R)]$ the strategy of Player II in such an equilibrium, then

$$y_1 + 3y_2 = y_1 + 2y_2 + 2y_3 = y_1 + 3y3$$
$$\Rightarrow y_2 = 2y_3 \text{ and } y_2 = y_3 \Rightarrow y_1 = y_2 = 0.$$

(c) Assume that $\sigma = ([x_1(T), x_2(M), (1 - x_1 - x_2)(B)], L)$ is a perfect equilibrium; i.e., there exists a sequence of perturbation vectors $(\epsilon^k)_{k\in N}$ satisfying $\lim_{k\to\infty} M(\epsilon^k) = 0$, and for each $k \in \mathbf{N}$ there exists an equilibrium σ^k of $\Gamma(\epsilon^k)$ such that $\lim_{k\to\infty} \sigma^k = \sigma$. By Corollary 7.26, σ is a Nash equilibrium and by Part (a) it satisfies $x_1 \geq \frac{1}{3}$. Hence, there is $K \in \mathbf{N}$ such that for every $k \geq K$ one has $\sigma_I^k(T) > \epsilon^k(T)$. If by contradiction, $\sigma_I^k(B) > \epsilon^k(B)$ then from some k onward, the expected payoffs of Player I in $\Gamma(\epsilon^k)$ under (T, σ_{II}^k) and (B, σ_{II}^k) are identical, thus $\sigma_{II}^k(C) = \sigma_{II}^k(R)$ which case Player I is better off Playing M so $\sigma_I^k(T) = \epsilon^k(T)$, a contradiction. Hence one may derive that

$$1 - x_1 - x_2 = \sigma_I(B) = \lim_{k\to\infty} \sigma_I^k(B) = \lim_{k\to\infty} \epsilon^k(B) = 0.$$

(d) We show that for every $x_1 \in (\frac{1}{3}, \frac{1}{2})$ the strategy vector $\sigma = ([x_1(T), (1 - x_1)(M)], L)$ is a perfect equilibrium. Let $x_1 \in (\frac{1}{3}, \frac{1}{2})$. For every $k \geq 4$ set

$$\epsilon_I^k(T) = \tfrac{2}{k}, \;\; \epsilon_I^k(M) = \tfrac{2}{k}, \;\; \epsilon_I^k(B) = \tfrac{2}{k}$$
$$\epsilon_{II}^k(L) = \tfrac{1}{k}, \;\; \epsilon_{II}^k(C) = \tfrac{2}{k}, \;\; \epsilon_{II}^k(R) = \tfrac{1}{k}.$$

We show that the strategy vector

$$\sigma^k = ([(x_1 - \tfrac{1}{k})(T), (1 - x_1 - \tfrac{1}{k})(M), \tfrac{2}{k}(B)], [(1 - \tfrac{3}{k})(L), \tfrac{2}{k}(C), \tfrac{1}{k}(R)])$$

is an equilibrium of $\Gamma(\epsilon^k)$. The best response of Player I to σ_{II}^k is T and B since

$$1 \cdot (1 - \tfrac{3}{k}) + 3 \cdot \tfrac{2}{k} = 1 \cdot (1 - \tfrac{3}{k}) + 2 \cdot \tfrac{2}{k} + 2 \cdot \tfrac{1}{k} > 1 \cdot (1 - \tfrac{3}{k}) + 3 \cdot \tfrac{1}{k}.$$

The best response of Player II to σ_I^k is L. Indeed, since $x_1 \in (\frac{1}{3}, \frac{1}{2})$, for large enough k one has

$$x_1 + 1 + \tfrac{1}{k} > 2 - 2x_1 + \tfrac{4}{k}, \;\; 3 + x_1 - \tfrac{3}{k}.$$

In conclusion, there exists a sequence of perturbation vectors $(\epsilon^k)_{k\in N}$ satisfying $\lim_{k\to\infty} M(\epsilon^k) = 0$, and for each $k \in \mathbf{N}$ there exists an equilibrium σ^k of $\Gamma(\epsilon^k)$ such that $\lim_{k\to\infty} \sigma^k = \sigma$ thus σ is a perfect equilibrium of the game.

(e) Using Exercise 7.20 and Part (d), one may conclude that the set of perfect equilibria includes every strategy vector $\sigma = ([x_1(T), (1 - x_1)(M)], L)$ for which $x_1 \in [\frac{1}{3}, \frac{1}{2}]$. We show that there are no other perfect equilibria. By Corollary 7.26, the set of perfect equilibria is a subset of Nash equilibria found in Part (b). By Part (c) $1 - x_1 - x_2 = 0$ so $1 - x_1 = x_2 \leq 2 - 3x_1$ so every perfect equilibrium is of the form $([x_1(T), x_2(M)], L)$ such that $\frac{1}{3} \leq x_1 \leq \frac{2}{3}$, $0 \leq x_2 \leq 2 - 3x_1$, and $x_1 + x_2 = 1$, which leads to the condition $\frac{1}{3} \leq x_1 \leq \frac{1}{2}$, as needed.

7.23 Let σ be a perfect equilibrium of the game; i.e., there exist a sequence of perturbation vectors $(\epsilon^k)_{k \in N}$ satisfying $\lim_{k\to\infty} M(\epsilon^k) = 0$, and for each $k \in \mathbf{N}$ there exists an equilibrium σ^k of $\Gamma(\epsilon^k)$ such that $\lim_{k\to\infty} \sigma^k = \sigma$. Assume that a pure strategy s'_i of player i is wealy dominated by a strategy $\bar{s}_i$. Then, the expected payoff of player i under $(\bar{s}_i, \sigma_{-i})$ is higher than his expected payoff under (s'_i, σ_{-i}) for every $k \in \mathbf{N}$, since σ_{-i} is completely mixed. Thus, the equilibrium σ^k of $\Gamma(\epsilon^k)$ must satisfy that $\sigma_i^k(s'_i) = \epsilon_i^k(s'_i)$ and therefore

$$\sigma_i(s'_i) = \lim_{k\to\infty} \sigma_i^k(s'_i) = \lim_{k\to\infty} \epsilon_i^k(s'_i) = 0.$$

7.24 The claim is incorrect. There are optimal strategies σ_1 and σ_2 of two players in a two-player zero-sum game, such that (σ_1, σ_2) is not a perfect equilibrium. Consider the following two-player zero-sum game.

Player II

Player I	L	C	R
T	0	1	1
B	0	-1	1

Then $\sigma_1 = [\frac{1}{2}(T), \frac{1}{2}(B)]$ is an optimal strategy of Player I and L is an optimal strategy of Player II, but B is a weakly dominated strategy and therefore, by Exercise 7.23 (σ_1, σ_2) is not a perfect equilibrium.

7.25 Let σ be a perfect equilibrium of the game; i.e., there exist a sequence of perturbation vectors $(\epsilon^k)_{k \in N}$ satisfying $\lim_{k\to\infty} M(\epsilon^k) = 0$, and for each $k \in \mathbf{N}$ there exists an equilibrium σ^k of $\Gamma(\epsilon^k)$ such that $\lim_{k\to\infty} \sigma^k = \sigma$. Assume that a pure strategy s'_i of player i is wealy dominated by a mixed strategy σ'_i (w.l.o.g $\sigma'_i(s'_i) = 0$). Assume by contradiction that $\sigma_i^k(s'_i) > \epsilon_i^k(s'_i)$. Then player i may deviate to the strategy $\widehat{\sigma}_i^k$ in $\Gamma(\epsilon^k)$ defined as follows

$$\widehat{\sigma}_i^k(s_i) = \begin{cases} \sigma_i^k(s_i) + \left(\sigma_i^k(s_i') - \epsilon_i^k(s_i')\right)\sigma_i'(s_i) & \text{if } s_i \neq s_i' \\ \epsilon_i^k(s_i') & \text{if } s_i = s_i' \end{cases}$$

Since s_i' is weakly dominated by σ_i' and σ_{-i}^k is completely mixed, the expected payoff of player i under $(\widehat{\sigma}_i^k, \sigma_{-i}^k)$ is higher than his expected payoff under (σ^k) which leads to a contradiction. In particular, for every $k \in \mathbf{N}$, $\sigma_i^k(s_i') = \epsilon_i^k(s_i')$ and therefore

$$\sigma_i(s_i') = \lim_{k\to\infty} \sigma_i^k(s_i') = \lim_{k\to\infty} \epsilon_i^k(s_i') = 0.$$

7.26 (a) The game has at least one perfect equilibrium. By Exercise 7.23, the weakly dominated action, M ,is played with probability zero in a perfect equilibrium, so Player II must play L. The only best response of Player I to L is T. Thus, (T, L) is the only perfect equilibrium in pure strategies of the following game.

(b) We show that the strategy vector (B, M) is a perfect equilibrium. For every $k \in \mathbf{N}$ set

$$\epsilon_I^k(T) = \tfrac{1}{2k},\ \ \epsilon_I^k(B) = \tfrac{1}{2k},\ \ \epsilon_I^k(I) = \tfrac{1}{2k}$$
$$\epsilon_{II}^k(L) = \tfrac{1}{2k},\ \ \epsilon_{II}^k(M) = \tfrac{1}{2k},\ \ \epsilon_{II}^k(R) = \tfrac{2}{k}.$$

The strategy vector

$$\sigma^k = ([(\tfrac{1}{2k})(T), (1-\tfrac{5}{2k})(B), \tfrac{4}{2k}(I)], [(\tfrac{1}{2k})(L), (1-\tfrac{5}{2k})(M), \tfrac{4}{2k}(R)])$$

is an equilibrium of $\Gamma(\epsilon^k)$. The best response of Player I to σ_{II}^k is B since

$$4 - \tfrac{6}{2k} > \tfrac{8}{2k}, \tfrac{2}{2k}.$$

The best response of Player II to σ_I^k is M:

$$4 - \tfrac{12}{2k} > 4 - \tfrac{14}{2k},\ \ \tfrac{8}{2k}.$$

In conclusion, we found a sequence of perturbation vectors $(\epsilon^k)_{k\in N}$ satisfying $\lim_{k\to\infty} M(\epsilon^k) = 0$, and for each $k \in \mathbf{N}$ there exists an equilibrium σ^k of $\Gamma(\epsilon^k)$ such that $\lim_{k\to\infty} \sigma^k = (B, M)$ thus (B, M) is a perfect equilibrium of the game.

7.27 (a) L strictly dominates R and W strictly dominates E.

(b) The set of Nash equilibria is $\{([x(T),(1-x)(B)],L,W) \mid x \in [0,1]\}$.

(c) Note that if Players II and III play R and E, respectively, with small enough probabilities, then Player I is better off playing T with as high probability as he can. Therefore, there is only one perfect equilibrium which is (T,L,W).

7.28 We prove that Definition 7.61 of perfect equilibrium is equivalent to Definition 7.25. Let σ be a perfect equilibrium according to 7.25. Then, there is a sequence of perturbation vectors $(\epsilon^k)_{k\in N}$ satisfying $\lim_{k\to\infty} M(\epsilon^k)=0$, and for each $k\in\mathbf{N}$ there exists an equilibrium σ^k of $\Gamma(\epsilon^k)$ such that $\lim_{k\to\infty}\sigma^k=\sigma$. $(\sigma^k)_{k\in\mathbf{N}}$ is a sequence of vectors of completely mixed strategies that converges to σ. In particular, there is $K\in\mathbf{N}$ such that $\sigma_i^k(s_i)>\epsilon_i^k(s_i)$ for every $k\geq K$, every player i and every pure strategy s_i of player i in the support of σ_i. Therefore, for every $k\geq K$, every pure strategy s_i in the support of σ_i is a best reply of player i to σ_{-i}^k and thus σ_i is a best reply of player i to σ_{-i}^k. One may conclude that σ is a perfect equilibrium according to Definition 7.61.
In the other direction, let σ be a perfect equilibrium according to Definition 7.61. Hence, there exists a sequence $(\sigma^k)_{k\in\mathbf{N}}$ of vectors of completely mixed strategies satisfying: (i) For each player $i\in\mathbf{N}$, $\sigma_i=\lim_{k\to\infty}\sigma_i^k$; and (ii) σ_i is a best reply to σ_i^k, for each $k\in\mathbf{N}$, and each player $i\in\mathbf{N}$. Then, for every $k\in\mathbf{N}$, every player i and every pure strategy s_i of player i set $\epsilon_i^k(s_i)=\frac{\sigma_i(s_i)}{2^k}$ if s_i is in the support of σ_i, otherwise set $\epsilon_i^k(s_i)=\sigma_i^k(s_i)$. For every player i and every pure strategy s_i in the support of σ_i, $\lim_{k\to\infty}\epsilon_i^k(s_i)=\lim_{k\to\infty}\frac{\sigma_i(s_i)}{2^k}=0$. For every player i and every s_i that is not in the support of σ_i, $\lim_{k\to\infty}\epsilon_i^k(s_i)=\lim_{k\to\infty}\sigma_i^k(s_i)=\sigma_i(s_i)=0$. Therefore, the sequence of perturbation vectors $(\epsilon^k)_{k\in N}$ satisfies $\lim_{k\to\infty}M(\epsilon^k)=0$. Furthermore, for every $k\in\mathbf{N}$, every player i and every pure strategy s_i of player i one has $\sigma_i^k(s_i)>\epsilon_i^k(s_i)$ if and only if s_i is in the support of σ_i if and only if s_i is a best reply of player i to σ_{-i}^k. In particular, σ^k is an equilibrium of $\Gamma(\epsilon^k)$. In conclusion, σ is a perfect equilibrium according to Definition 7.25.

7.29 In the given game, (B,L) is a Nash equilibrium since no player can change his payoff by deviation. On the other hand, (B,L) is not a perfect equilibrium since B is dominated by the mixed strategy $[\frac{1}{2}(T),\frac{1}{2}(C)]$, and thus, by Exercise 7.25, B is played with probability zero in every perfect equilibrium.

7.30 Set $\delta_i^k(a_i)=\frac{1}{3^k}$, for every $k\in\mathbf{N}$, every player $i\in\{1,2\}$ and every action a_i of player i. Then

$$\sigma^k = \left(\left[\tfrac{1}{3^k}(C), (1 - \tfrac{2}{3^k})(B), \tfrac{1}{3^k}(A)\right], \left[(1 - \tfrac{1}{3^k})(R), \tfrac{1}{3^k}(L)\right] \right)$$

is an equilibrium of $\Gamma(\delta^k)$ and $(B, R) = \lim_{k\to\infty} \sigma^k$. Hence (B, R) is an extensive-form perfect equilibrium. Note that if Player I plays A with probability less than 1, then Player II is better off Playing R, which case Player I's best Replay is B. In conclusion, the game has no other Nash equilibrium.

7.31 For every $k \in \mathbf{N}$, set $\delta_1^k(T) = \delta_1^k(B) = \frac{1}{3^{k+1}}$, $\delta_2^k(t) = \delta_2^k(b) = \frac{1}{3^k}$ and $\delta_3^k(\tau) = \delta_2^k(\beta) = \frac{1}{3^{k+2}}$. Let

$$\sigma^k = \left(\left[\left(1 - \tfrac{1}{3^{k+1}}\right)(T), \tfrac{1}{3^{k+1}}(B) \right], \left[\left(1 - \tfrac{1}{3^k}\right)(t), \tfrac{1}{3^k}(b) \right], \left[\tfrac{1}{3^{k+2}}(\tau), \left(1 - \tfrac{1}{3^{k+2}}\right)(\beta) \right] \right).$$

The payoff of each player in this game is at least $\frac{4}{9}$ (and converges to 1 as k goes to infinity) whereas the payoff of each player decreases by deviating. Thus σ^k is an equilibrium of $\Gamma(\delta^k)$ and it can be deduced that $(T, t, \beta) = \lim_{k\to\infty} \sigma^k$ is an extensive-form perfect equilibrium.
Assume by contradiction that (B, t, τ) is also an extensive-form perfect equilibrium. Thus there exists a sequence of perturbation vectors $\delta^k_{k\in\mathbf{N}}$ satisfying $\lim_{k\to\infty} M(\delta^k) = 0$, and for each $k \in \mathbf{N}$ there exists an equilibrium σ^k of $\Gamma(\delta^k)$, such that $\lim_{k\to\infty} \sigma^k = (B, t, \tau)$. In particular, in every $\Gamma(\delta^k)$ all vertices are reached with some positive probability. Moreover, from some $k \in \mathbf{N}$ onwards, Player III plays τ with probability that is closed to 1, which case Player II is better off selecting b over t and playing t with probability $\delta_2^k(t)$ that converges to zero as k goes to infinity. This contradicts the assumption that $\lim_{k\to\infty} \sigma_2^k(t) = 1$.

7.32 We prove that every extensive-form perfect equilibrium is a Nash equilibrium in behavior strategies.
Let σ be an extensive-form perfect equilibrium; i.e., there is a sequence of perturbation vectors $(\delta^k)_{k\in\mathbf{N}}$ such that $\lim_{k\to\infty} M(\delta^k) = 0$, and for every $k \in \mathbf{N}$ there is a profile σ^k which is an equilibrium of $\Gamma(\delta^k)$ such that $\sigma = \lim_{k\to\infty} \sigma^k$. We prove that σ is a Nash equilibrium of the original game Γ.
Let $\widehat{\sigma}_i$ be a behavior strategy of player i, for $i \in N$. For every information set U_i of player i and every action $a_i \in A(U_i)$ define

$$\widehat{\sigma}_i^k(a_i) = (1 - \sum_{a\in A(U_i)} \delta_i^k(a))\widehat{\sigma}_i(a_i) + \delta_i^k(a_i).$$

Since σ^k is an equilibrium in $\Gamma(\delta^k)$ and $\widehat{\sigma}_i^k$ is a strategy of player i in $\Gamma(\delta^k)$, one has

$$u_i(\sigma^k) \geq u_i(\widehat{\sigma}_i^k, \sigma_{-i}^k).$$

Finally, $\lim_{k\to\infty} \widehat{\sigma}_i^k = \widehat{\sigma}_i$ and u_i is continuous, therefore

$$u_i(\sigma) = \lim_{k\to\infty} u_i(\sigma^k) \geq \lim_{k\to\infty} u_i(\widehat{\sigma}_i^k, \sigma_{-i}^k) = u_i(\widehat{\sigma}_i, \sigma_{-i}),$$

so σ is Nash equilibrium.

7.33 Every finite extensive-form game can be presented as a strategic-form game where N is a finite set of players, $S_i = \times_{U_i \in \mathcal{U}_i} A(U_i)$ is a finite set of pure strategies of player i, for every player $i \in N$. By Theorem 7.27, this strategic-form game has at least one perfect equilibrium which defines a strategic-form perfect equilibrium in the given extensive-form game.

7.34 Let δ be a perturbation vector, let σ^* be a Nash equilibrium (in behavior strategies) in the game $\Gamma(\delta)$, and let $\Gamma(x)$ be a subgame of Γ. Then every action in every information set of each player is played with positive probability. Therefore, under σ^* every vertex x is reached with positive probability. Hence, by in Theorem 7.5, the strategy vector σ^*, restricted to the subgame $\Gamma(x)$, is a Nash equilibrium (in behavior strategies) of $\Gamma(x; \delta)$.

7.35 Let $(\delta^k)_{k\in\mathbf{N}}$ be a sequence of perturbation vectors satisfying $\lim_{k\to\infty} M(\delta^k) = 0$. Let $\sigma^i \in \mathcal{B}_i$ be a behavior strategy of player i. For every information set $U_i \in \mathcal{U}_i$ and every action $a_i \in A(U_i)$ set

$$\sigma_i^k(a_i) = \delta_i^k(a_i) + \left(1 - \sum_{a\in A(U_i)} \delta_i^k(a)\right)\sigma_i(a_i).$$

Then, since $\sum_{a\in A(U_i)} \delta_i^k(a) \leq 1$ and $\sigma_i(a_i) \geq 0$ one has $\sigma_i^k(a_i) \geq \delta_i^k(a_i)$. In addition,

$$\begin{aligned}\sum_{a_i\in A(U_i)} \sigma_i^k(a_i) &= \sum_{a_i\in A(U_i)} \left(\delta_i^k(a_i) + \left(1 - \sum_{a\in A(U_i)} \delta_i^k(a)\right)\sigma_i(a_i)\right) \\ &= \sum_{a_i\in A(U_i)} \delta_i^k(a_i) + \left(1 - \sum_{a\in A(U_i)} \delta_i^k(a)\right) \sum_{a_i\in A(U_i)} \sigma_i(a_i) = 1,\end{aligned}$$

which derives that $\sigma_i^k \in \mathcal{B}_i(\delta_i)$.
Finally, since $\lim_{k\to\infty} M(\delta^k) = 0$, for every information set $U_i \in \mathcal{U}_i$ and

every action $a_i \in A(U_i)$, $\lim_{k\to\infty} \sigma_i^k(a_i) = \sigma_i(a_i)$. Thus, $\lim_{k\to\infty} \sigma_i^k$ exists and equals σ_i.

7.36 Consider the behavior strategy vector (T_1B_2, b). To show that this strategy is an extensive-form perfect equilibrium, choose for instance $\delta_2^k(t) = \frac{1}{k^3}$ and $\delta_i^k(a_i) = \frac{1}{k}$ for every other action $a_i \neq t$ of each player i. Then,

$$\sigma^k = \left(\left([(1-\tfrac{1}{k})(T_1), \tfrac{1}{k}(B_1)], [\tfrac{1}{k}(T_2)(1-\tfrac{1}{k})(B_2)]\right), [\tfrac{1}{k^3}(t)(1-\tfrac{1}{k^3})(b)]\right)$$

is an equilibrium in the perturbed game $\Gamma(\delta^k)$ such that $\lim_{k\to\infty} \sigma^k = (T_1B_2, b)$.
On the other hand, (T_1B_2, b) is not a strategic-form perfect equilibrium since the strategy T_1B_2 is weakly dominated by the strategy B_1B_2.
Every strategy of the form

$$\left(\left([x(T_1), (1-x)(B_1)], [1(B_2)]\right), b\right),$$

for every $x \in [0,1]$ is a subgame perfect equilibrium.

7.37 (a) By Theorem 7.36, the game has an extensive-form perfect equilibrium. By Theorem 7.33 every extensive-form perfect equilibrium of the game is a subgame perfect equilibrium. However, since the game has only one subgame perfect equilibrium (T_1B_2, t), it must also be the only extensive-form perfect equilibrium.

(b) In the corresponding strategic-form game given by the matrix game, Player I is the row player and Player II is the column player.

	t	b
T_1T_2	1,1	1,1
T_1B_2	1,1	1,1
B_1T_2	0,2	0,3
B_1B_2	0,2	2,0

The game has a continuum of strategic-form perfect equilibria: $\sigma^y = ([y(T_1T_2), (1-y)(T_1B_2)], t)$ for $y \in [0,1]$. Indeed, let $\epsilon_i^k(s_i) = \frac{1}{k}$ for every player i and every strategy s_i of player i. Then the strategy vector $\sigma^{k,y}$ where

$$\sigma_1^{k,y}(s_1) = (1-\tfrac{4}{k})\sigma_1^y(s_1) + \tfrac{1}{k} \quad \forall s_1 \in S_1 \text{ and}$$
$$\sigma_2^k(s_2) = (1-\tfrac{2}{k})\sigma_2^y(s_2) + \tfrac{1}{k} \quad \forall s_2 \in S_2$$

is an equilibrium in $\Gamma(\epsilon^k)$ such that $\lim_{k\to\infty} \sigma^{k,y} = \sigma^y$.

(c) Following Parts (a) and (b), the game has strategic-form perfect equilibria that are not a subgame perfect equilibrium.

7.38 The strategy vector (B, R) is the unique Nash equilibrium. Indeed, assume Player II plays $[x(R), (1-x)(L)]$. If $x < 1$, A is strictly dominated by C. If $x = 1$, A is strictly dominated by B. Hence, in every Nash equilibrium Player I plays A with probability zero which case Player II's best reply is R. The Nash equilibria is also a unique subgame perfect equilibrium since the game has no nontrivial subgames. By Theorem 7.36 and 7.33, (B, R) is also a unique extensive-form perfect equilibrium. Finally, (B, R) is also a unique strategic-form perfect equilibrium. As presented in the game matrix depicted below, L is weakly dominated by R so in a strategic-form perfect equilibrium L is played with probability zero, which case B is the best reply of Player I.

	L	R
A	1,2	1,2
B	0,0	3,1
C	2,1	1,3

7.39 Let Γ be an extensive-form game in which every player has a single information set. Then, for every player $i \in N$, the set of pure actions of player i and the set of pure strategies of player i are identical. Assume that the behavior strategy vector σ is an extensive-form perfect equilibrium; i.e., there exists a sequence of perturbation vectors $(\delta^k)_{k \in \mathbf{N}}$ satisfying $\lim_{k\to\infty} M(\delta^k) = 0$, and for each $k \in \mathbf{N}$ there exists an equilibrium in behavior strategies σ^k of $\Gamma(\delta^k)$, such that $\lim_{k\to\infty} \sigma^k = \sigma$ is satisfied. However, the set of perturbation over the actions and over the pure strategies are the same and σ and σ^k are also mixed strategy vectors so σ is a strategic-form perfect equilibrium as well. The other direction is proved similarly, and the proof is omitted.

7.40 (a) The pair is not a consistent assessment. First, the distribution over the information set of Player I should not be determined by the strategy of that information set. In addition the set $\{x_4, x_5, x_6\}$ is not an information set of neither one of the players.

(b) The pair is not a consistent assessment since the set $\{x_4, x_5, x_6\}$ is not an information set of neither one of the players.

(c) The pair is not a consistent assessment since the set $\{x_4, x_6\}$ is not an information set of neither one of the players.

(d) The pair is not a consistent assessment since the distribution over the information set of Player II is wrong.

7.41 Let σ^* be a Nash equilibrium in behavior strategies. We show that the pair $(\sigma^*, \mu_{\sigma^*})$ is sequentially rational in every information set U satisfying $P_{\sigma^*}(U) > 0$.
Let σ_i be a behavior strategy of player i and let U_i' be an information set of of player i such that $P_{\sigma^*}(U_i') > 0$. Denote by $u^*_{i,U_i'}$ player i's expected payoff under σ^*, given that the play of the game does not pass through U_i'.
Define a behavior strategy $\widehat{\sigma}_i$ of player i as follows: follow σ_i^* at each information set U_i of player i such that every path from the root to a vertex in U_i does not pass through U_i' otherwise, follow σ_i. Then, under $(\widehat{\sigma}_i, \sigma^*_{-i})$ the play of the game will not pass through U_i' with probability $1 - P_{\sigma^*}(U_i')$ which case player i's expected payoff is $u^*_{i,U_i'}$.
From the above argument and since σ^* is a Nash equilibrium one has,

$$
\begin{aligned}
u_i(\sigma^*) &= (1 - P_{\sigma^*}(U_i')) \cdot u^*_{i,U_i'} + P_{\sigma^*}(U_i') u_i(\sigma^* | U_i', \mu_{\sigma^*}) \\
&\geq u_i(\widehat{\sigma}_i, \sigma^*_{-i}) = (1 - P_{\sigma^*}(U_i')) \cdot u^*_{i,U_i'} + P_{\sigma^*}(U_i') u_i(\sigma_i', \sigma^*_{-i} | U_i', \mu_{\sigma^*}).
\end{aligned}
$$

However, $P_{\sigma^*}(U_i') > 0$ so

$$
u_i(\sigma^* | U_i', \mu_{\sigma^*}) \geq u_i(\sigma_i', \sigma^*_{-i} | U_i', \mu_{\sigma^*}).
$$

which is what we wanted to prove.

7.42 Let Γ be a game with perfect information. That is, for every information set U in the game there is a vertex x such that $U = \{x\}$. Hence, this game has only one complete belief system denoted by $\widehat{\mu}$, and it must satisfies $\widehat{\mu}_U = [1(x)]$ for every information set $U = \{x\}$.
Therefore, a behavior strategy σ is a subgame perfect equilibrium if and only if σ is a Nash equilibrium in the subgame that start at vertex x for every vertex x if and only if, by Theorem 7.44 and Theorem 7.45, the pair $(\sigma, \widehat{\mu})$ is sequentially rational at each vertex in the game.

7.43 The consistent assessments of the extensive-form game are

$$
\Big([y(T), (1-y)(B)], [(v(t), (1-v)(b)], [\tfrac{2}{3}(x_2), \tfrac{1}{3}(x_3)], [\tfrac{2-2y}{3-2y}(x_5), \tfrac{y}{3-2y}(x_6), \tfrac{1-y}{3-2y}(x_7)]\Big),
$$

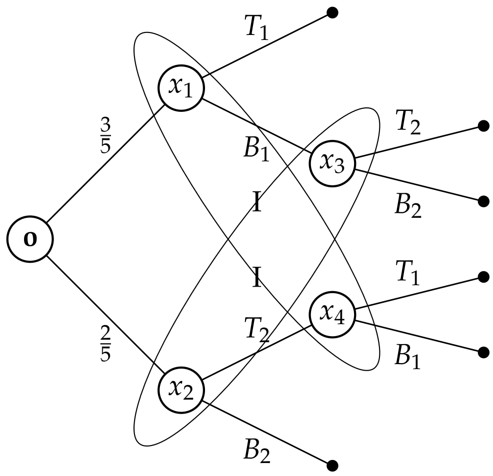
for every $y, v \in [0,1]$.

7.44 Denote by x_1 the root of the game and by x_2, x_3 and x_4 the children of x_1 given Player I select T, M and B, respectively. The consistent assessments of the extensive-form game are

$$\left([y_1(T), y_2(M), (1-y_1-y_2)(B)], [(v(t), (1-v)(b)], [1(x_1))], [w(x_2), 1-w(x_3)]\right),$$

for every $y_1, y_2, v, w \in [0,1]$ such that if $y_1 + y_2 > 0$ then $w = \frac{y_1}{y_1+y_2}$.

7.45 Denote the vertices of the information sets by x_1, x_2, x_3, x_4 as presented below.



The consistent assessments of the extensive-form game are

$$\left([(1-y)(T_1), y(B_1)], [(v(T_2), (1-v)(B_2)], [\tfrac{3}{3+2v}(x_1), \tfrac{2v}{3+2v}(x_4)], [\tfrac{2}{2+3y}(x_2), \tfrac{3y}{2+3y}(x_3)]\right),$$

for every $y, v \in [0,1]$.

7.46 Denote by x_1 the vertex controlled by Player I, by x_2 the vertex controlled by Player II, and by x_3 and x_4 the top and the bottom vertices controlled by Player III, respectively.
The consistent assessments are

$$\begin{aligned}&([y(T), (1-y)(B)], [v(t), (1-v)(b)], [w(\tau), (1-w)(\beta)],\\ &\quad [1(x_1)], [1(x_2)], [p(x_3), (1-p)(x_4)])\end{aligned}$$

for every $y, v, w, p \in [0,1]$ such that if $yv \neq 1$ then $p = \frac{y-yv}{1-yv}$.
We next find the sequentially rational assessments.

- If $p > \frac{2}{3}$ then player III is better off playing τ so Player I and II selects T and t, respectively.
- If $p < \frac{2}{3}$ Player III is better off Playing β so Player I and II prefer selecting T and b, respectively, which leads to a deviation of Player II to τ.

- If $p = \frac{2}{3}$ Player III is indifferent between choosing τ and β. However, to ensure that $p = \frac{2}{2}$ the payoff of Player I as well as for Player II should be 1, if and only if $w = \frac{3}{11}$.

In conclusion, the sequentially rational assessments are

$$\big([1(T)],[1(t)(b)],[1(\tau)],[1(x_1)],[1(x_2)],[p(x_3),(1-p)(x_4)]\big)$$

where $p > \frac{2}{3}$, and

$$\begin{aligned}([y(T),(1-y)(B)],[v(t),(1-v)(b)],[\tfrac{3}{11}(\tau),\tfrac{8}{11}(\beta)],\\ [1(x_1)],[1(x_2)],[\tfrac{2}{3}(x_3),\tfrac{1}{3}(x_4)])\end{aligned}$$

for every $y \geq \frac{2}{3}$ and $v = \frac{2-2y}{y}$.

7.47 Let x_1 be the root controlled by Player I, and let x_2, x_3 and x_4 be the children of x_1 given Player I selects A, B and C, respectively. For every set of believes over the information set of Player II, he is better off selecting R so Player I's best reply is B. Therefore, the sequential equilibria of the game are

$$\big([1(B)],[1(R)],[1(x_1)],[p(x_2),(1-p)(x_3)]\big)$$

for every $p \in [0,1]$.
To show that it is consistent define

$$\sigma^k = \Big([(1-\tfrac{1}{k})(B),\tfrac{p}{k}(A),\tfrac{1-p}{k}(C)],[(1-k)(R),k(L)],[1(x_1)]\Big).$$

If $p = 1$ define

$$\sigma^k = \Big([(1-\tfrac{1}{k}-\tfrac{1}{k^2})(B),\tfrac{1}{k}(A),\tfrac{1}{k^2}(A)],[(1-k)(R),k(L)],[1(x_1)]\Big).$$

The case where $p = 0$ is similar to $p = 1$.

7.48 Denote by x_1 and x_2 the top and the bottom vertices, respectively, controlled by Player I. Denote by x_3 and x_4 the top and the bottom vertices, respectively, controlled by Player II.

(a) We prove that at every Nash equilibrium Player I plays (T_1, B_2). Assume by contradiction that there is a Nash equilibrium in which Player I plays the strategy $([(1-y)(T_1), y(B_1)],[z(T_2),(1-z)(B_2)])$ where $y + z > 0$. Therefore the belief system of Player II is $[\frac{y}{y+z}(x_3), \frac{z}{y+z}(x_4)]$. Then,

- If $\frac{y}{y+z} < \frac{1}{3}$ ($z > 2y$), Player II selects c with probability 1, which case Player I plays $z = 0$ and thus $y = 0$, a contradiction.

- If $\frac{y}{y+z} = \frac{1}{3}$ $(z = 2y)$, Player II plays b with probability 0, which case Player I plays $z = 0$ and thus $y = 0$, a contradiction.
- If $\frac{1}{3} < \frac{y}{y+z} < \frac{2}{3}$, Player II selects b with probability 1, which case Player I plays $z = 0$ and $y = 0$, a contradiction.
- If $\frac{y}{y+z} = \frac{2}{3}$ $(y = 2z)$, Player II selects b with probability 0, which case Player I plays $y = 0$ and thus $z = 0$, a contradiction.
- If $\frac{y}{y+z} > \frac{2}{3}$ $(y > 2z)$, Player II selects b with probability 1, which case Player I plays $y = 0$ and thus $z = 0$, a contradiction.

In conclusion $y = z = 0$, as needed.

(b) The Nash equilibria of the game are:

$$\big([1(T_1)], [1(B_2)], [(1 - p_1 - p_2)(a), p_1(b), p_2(c)]\big)$$

for every $p_1, p_2 \leq \frac{1}{3}$.

(c) The set of sequential equilibria of the game are

$$\big([1(T_1)], [1(B_2)], [1(b)], [1(x_1)], [1(x_2)], [q(x_3), (1-q)(x_4)]\big) \quad \forall \tfrac{1}{3} \leq q \leq \tfrac{2}{3}$$
$$\Big([1(T_1)], [1(B_2)], [(1-p)(a), p(b)], [1(x_1)], [1(x_2)], [\tfrac{2}{3}(x_3), \tfrac{1}{3}(x_4)]\Big)$$
$$\Big([1(T_1)], [1(B_2)], [(1-p)(a), p(c)], [1(x_1)], [1(x_2)], [\tfrac{1}{3}(x_3), \tfrac{2}{3}(x_4)]\Big).$$

7.49 Denote by x_1 the vertex controlled by Player I, by x_2 the vertex controlled by Player II, and by x_3 and x_4 the top and the bottom vertices, respectively, controlled by Player III. Assume that Player III's belief is $[p(x_3), (1-p)(x_4)]$. Then

- If $p < \frac{1}{2}$ then player III prefers playing β and hence Player I selects T which case $p = 1$, a contradiction.
- If $p > \frac{1}{2}$ then player III prefers playing τ and hence Player II selects t and Player I is better off Playing B. In conclusion, it defines a sequential equilibrium given by

$$\big([1(B)], [1(b)], [1(\tau)], [1(x_1)], [1(x_2)], [p(x_3), (1-p)(x_4)\big) \;\; \forall p > \tfrac{1}{2}.$$

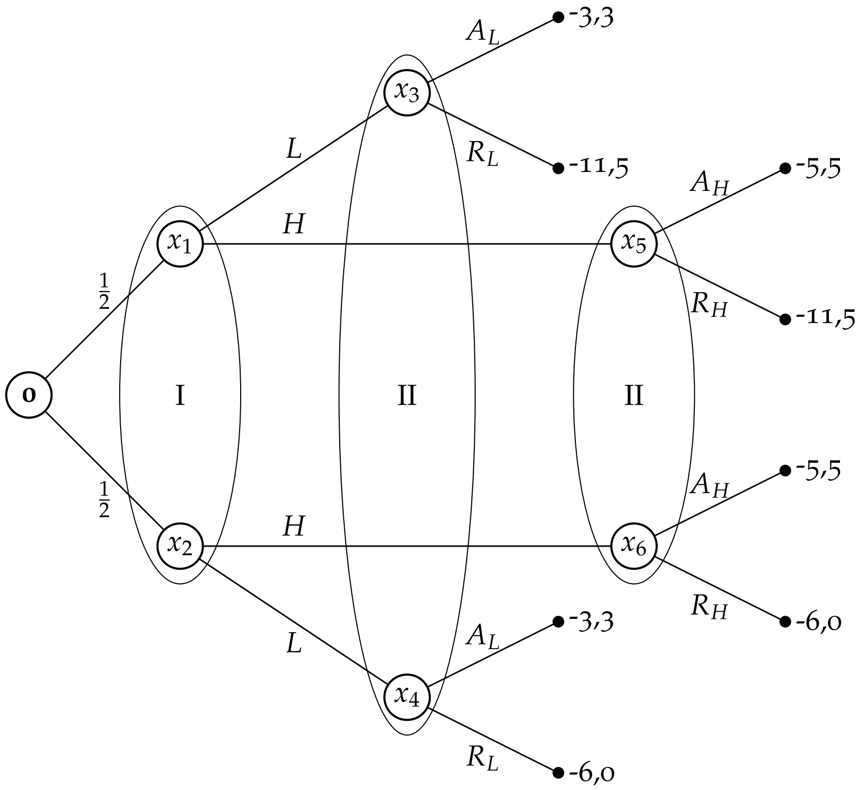
- If $p = \frac{1}{2}$ then Player III is indifferent between playing β andτ. If Player III plays β with probability $q > \frac{1}{3}$ then Player I best reply is T which case $p = 1$, a contradiction. If Player III plays β with probability $q \leq \frac{1}{3}$ then Player II best reply is b which case Player I plays B. In conclusion, it defines a sequential equilibrium given by

$$\Big([1(B)], [1(b)], [q(\tau), (1-q)(\beta)], [1(x_1)], [1(x_2)], [\tfrac{1}{2}(x_3), \tfrac{1}{2}(x_4)]\Big) \;\; \forall q \leq \tfrac{1}{3}.$$

7.50 In Game A, $\big([1(T)],[1(b)],[1(x_1)],[0.2(x_2),0.8(x_3)]\big)$ is a sequential equilibrium corresponding the payoff $(2,2)$ where x_1 is the vertex controlled by Player I and x_2 and x_3 are the top and the bottom vertices, respectively, controlled by Player II.
On the other hand, in Game B, in every is a sequential equilibrium Player I must selects M since M dominates R and thus Player II necessarily prefers t over b, which case Player I better off playing I leading a payoff of $(3,3)$.

7.51 In an extensive-form game with perfect recall, not every Nash equilibrium is a part of a sequential equilibrium, for instance in Game B in Exercise 7.50, the strategy vector $([1(T)],[1(R)],[1(b)])$ is a Nash equilibrium but not a sequential equilibrium.

7.52 (a) Let Player I be the contractor and Player II be the municipality. Let L and H be the actions of player I corresponding the alternatives "Low offer" and "High offer". Let A_L and R_L (A_H and R_H) be the actions of Player II given Player I's offer is low (high) corresponding the alternatives "Accept" and "Rejects", respectively. The extensive-form game represents this situation is depicted below where the payoffs are in \$100,000.



(b) The significance of assumption that a chance move determines with equal probability whether the contractor was negligent or not is that there is no evidence that makes one of the alternatives most likely.

(c) Denote by p the probability that the contractor was negligent. Assume that $p < \frac{3}{5}$ (as for $p = \frac{1}{2}$).
If the play corresponding to a Nash equilibrium reaches the information set $\{x_3, x_4\}$ with positive probability then Player II's best reply is A_L.
If the play corresponding to a Nash equilibrium reaches the information set $\{x_5, x_6\}$ with positive probability then Player II's best reply is A_H.

i. In case that Player I plays L with positive probability, Player II response with $[1(A_L)], [q(A_H), (1-q)(R_H)]$ which case Player I prefers playing L with probability 1. In conclusion, it defines Nash equilibria of the form

$$\big([1(L)], [1(A_L)], [q(A_H), (1-q)(R_H)]\big) \quad \forall q \in [0, 1].$$

ii. In case that Player I plays L with probability zero, Player II response with $[r(A_L), (1-r)(R_L)], [1(A_H)]$. To prevent Player

I's deviation, one needs to require that

$$-5 \geq -6 - 5p + (3+5p)r \iff r \leq \tfrac{1+5p}{3+5p}.$$

Therefore, it defines Nash equilibria of the form

$$\big([1(H)], [r(A_L), (1-r)(R_L)], [1(A_2)]\big) \quad \forall r \leq \tfrac{1+5p}{3+5p}.$$

(d) Since the belief over every information is distributed as the distribution over the chance moves, and by the sequentially rational condition, in a sequential equilibrium Player II plays A_L, A_H thus the sequential equilibria of the game are

$$\begin{gathered}([1(L)], [1(A_L)], [1(A_H)], [p(x_1), (1-p)(x_2)],\\ [p(x_3), (1-p)(x_4)], [p(x_5), (1-p)(x_6)]).\end{gathered}$$

(e) We next repeat Parts (c) and (d) for every $p \geq \frac{3}{5}$.
The case of $\frac{3}{5} < p < 1$:
If the play corresponding to a Nash equilibrium reaches the information set $\{x_3, x_4\}$ with positive probability then Player II's best reply is R_L.
If the play corresponding to a Nash equilibrium reaches the information set $\{x_5, x_6\}$ with positive probability then Player II's best reply is A_H.

i. In case that Player I plays L and H with positive probabilities, Player II responses with $[1(R_L)], [1(A_H)]$ which case Player I prefers playing H with probability 1, a contradiction.
ii. In case that Player I plays L with probability 1, Player II responses with $[1(R_L)], [q(A_H), (1-q)(R_H)]$. However, to prevent Player I's deviation, Player II must play $[1(A_H)]$. In conclusion, it defines a Nash equilibrium which is

$$\big([1(L)], [1(R_L)], [1(R_H)]\big).$$

Note that this equilibrium is not a part of a sequential equilibrium.
iii. In case Player I plays H with probability 1, Player II responses with $[r(A_L), (1-r)(R_L)], [1(A_H)]$. To prevent Player I's deviation, one needs to require that

$$-5 \geq -6 - 5p + (3+5p)r \iff r \leq \tfrac{1+5p}{3+5p}.$$

Therefore, it defines Nash equilibria of the form

$$\big([1(H)], [r(A_L), (1-r)(R_L)], [1(A_H)]\big) \quad \forall r \leq \tfrac{1+5p}{3+5p}.$$

In case that $r = 0$, the equilibrium is a part of the only sequential equilibrium in the game, given by

$$([1(H)], [1(R_L)], [1(A_H)],$$
$$[p(x_1), (1-p)(x_2)], [p(x_3), (1-p)(x_4)], [p(x_5), (1-p)(x_6)]).$$

The case of $p = 1$:
If the play corresponding to a Nash equilibrium reaches the information set $\{x_3, x_4\}$ with positive probability then Player II's best reply is R_L.
If the play corresponding to a Nash equilibrium reaches the information set $\{x_5, x_6\}$ with positive probability then Player II's is indifferent between A_H and R_H.

i. In case that Player I plays L with positive probability, Player II responses with $[1(R_L)], [q(A_H), (1-q)(R_H)]$. However, if $q > 0$ then Player I prefers playing H with probability 1 which leads to a contradiction. Thus it defines a Nash equilibrium

$$([y(L), (1-y)(H)], [1(R_L)], [1(R_H)])$$

which is a part of a sequential equilibrium

$$([y(L), (1-y)(H)], [1(R_L)], [1(R_H)], [1(x_1)], [1(x_3)], [1(x_5)]).$$

ii. In case that Player I plays H with probability 1, Player II responses with $[r(R_L), (1-r)(R_L)], [q(A_H), (1-q)(R_H)]$. However, to prevent Player I's deviation, one needs to require that

$$-11 + 6q \geq -11 + 8r \iff r \leq \tfrac{3}{4}q.$$

In conclusion, it defines a Nash equilibrium which is

$$\big([1(H)], [r(A_L), (1-r)(R_L)], [q(A_H), (1-q)(R_H)]\big) \quad \forall r \leq \tfrac{3}{4}q.$$

If $r = 0$, the equilibrium is a part of the following sequential equilibrium in the game

$$\big([1(H)], [1(R_L)], [q(A_H), (1-q)(R_H)], [1(x_3)], [1(x_5)]\big) \quad \forall q \in [0, 1].$$

The case of $p = \frac{3}{5}$:
If the play corresponding to a Nash equilibrium reaches the information set $\{x_3, x_4\}$ with positive probability then Player II is indifferent between A_L and R_L.
If the play corresponding to a Nash equilibrium reaches the information set $\{x_5, x_6\}$ with positive probability then Player II's best reply is A_H.

i. In case that Player I plays H with positive probability, Player II response with $[r(A_L), (1-r)(R_L)], [1(A_H)]$. However, to prevent Player I's deviation, one needs to require that

$$-5 \geq -6 - 5p + r(3+5p) \iff r \leq \tfrac{2}{3}q.$$

In particular, it defines two form of Nash equilibria which are a part of sequential equilibria

$$[y(L), (1-y)(H)], [\tfrac{2}{3}(A_L), \tfrac{1}{3}(R_L)], [1(A_H)] \quad \forall y \in [0,1]$$
$$[1(H)], [r(A_L), (1-r)(R_L)], [1(A_H)] \quad \forall r \leq \tfrac{2}{3}.$$

ii. In case that Player I plays H with probability 0, Player II responses with $[r(R_L), (1-r)(R_L)], [q(A_H), (1-q)(R_H)]$. However, to prevent Player I's deviation, one needs to require that

$$-9 + 4q \leq -9 + 6r \iff r \geq \tfrac{2}{3}q.$$

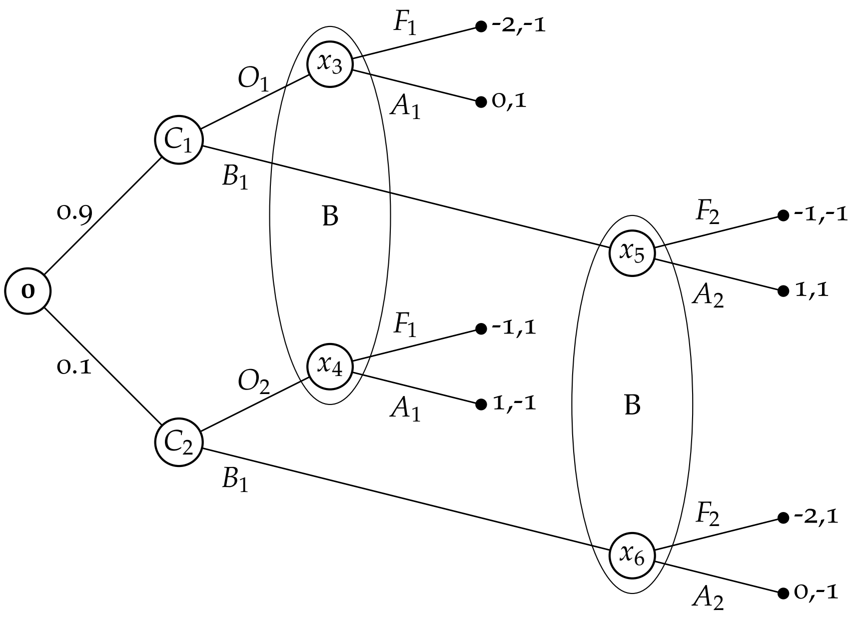
In conclusion, it defines a Nash equilibrium which is

$$\big([1(L)], [r(A_L), (1-r)(R_L)], [q(A_H), (1-q)(R_H)]\big) \quad \forall r \geq \tfrac{2}{3}q.$$

If $q = 1$, the equilibrium is a part of a sequential equilibrium in the game, given by

$$([1(L)], [r(A_L), (1-r)(R_L)], [1(A_H)],$$
$$[\tfrac{3}{5}(x_1), \tfrac{2}{5}(x_2)], [\tfrac{3}{5}(x_3), \tfrac{2}{5}(x_4)], [\tfrac{3}{5}(x_5), \tfrac{2}{5}(x_6)])) \quad \forall r \geq \tfrac{2}{3}.$$

7.53 (a) The game tree depicted below represents the Signaling game.



(b) We next find all the Nash equilibria of this game.

i. Assume, Brutus always fights. Then Caesar prefers to drink a beer in case he is brave, otherwise, he prefers an orange juice. Therefore, if Caesar will drink a beer, Brutus will know that Caesar is brave and avoid a fight, a contradiction.

ii. Assume, Brutus always avoids fighting. Then Caesar prefers to drink a beer in case he is brave otherwise he prefers an orange juice. Therefore, if Caesar will drink an orange juice, Brutus will know that Caesar is cowardly and fights, a contradiction.

iii. Assume, Brutus fights in case that Caesar drinks an orange juice and avoids fighting if he drinks a beer. Then Caesar prefers to drink a beer either way which leads Brutus to believe that it is more likely that Caesar is brave and it is better to avoid fighting with him. These strategies define an equilibrium.

iv. Assume, Brutus fights in case that Caesar drinks a beer and avoids fighting if he drinks an orange juice. Then Caesar prefers to drink a beer either way which leads Brutus to believe that it is more likely that Caesar is brave and it is better off avoiding fighting with him. These strategies define an equilibrium.

v. Assume that Brutus conducts a lottery over his action set in case that Caesar drinks a beer. Therefore, he necessarily indifferent between his actions. Thus $0.9q = 0.1r$, where q and r are the probabilities that Caesar drinks beer given he is brave and cowardly respectively. Therefore,

$$0.9(1-q) > 0.1(1-r).$$

In particular, if Caesar drinks an orange juice it seems more likely that he is brave over cowardly which case Brutus prefer to avoid fighting. Thus, if Caesar is indeed cowardly then he is better off drinking an orange juice;i.e., $r = 0$ so $q = 0$ as well. To prevent Caesar drinking an orange juice in case he is brave, Brutus must select fighting with probability at least $\frac{1}{2}$.

vi. Assume that Brutus conducts a lottery over his action set in case that Caesar drinks an orange juice. Therefore, he necessarily indifferent between his actions. Thus

$$0.9(1-q) = 0.1(1-r),$$

where q and r are the probabilities that Caesar drinks a beer given he is brave and cowardly respectively. Therefore,

$$0.9q > 0.1r.$$

In particular, if Caesar drinks a beer it seems more likely that he is brave over cowardly which case Brutus prefer to avoid fighting. Thus, if Caesar is indeed brave then he is better off drinking a beer; i.e, $q = 1$ so $s = 1$ as well. To prevent Caesar drinking a beer in case he is brave, Brutus must select fighting with probability at least $\frac{1}{2}$.

In conclusion there are two types of Nash equilibria:

$$([1(B_1)], [1(B_2)], [t(F_1), (1-t)(A_1)], [1(A_2)]) \quad \forall t \geq \tfrac{1}{2}$$
$$([1(O_1)], [1(O_2)], [1(A_1)], [t(F_2), (1-t)(A_2)]) \quad \forall t \geq \tfrac{1}{2}.$$

(c) Every Nash equilibrium of this game is a part of a sequential equilibrium. For instance, let

$$\sigma = ([1(B_1)], [1(B_2)], [t(F_1), (1-t)(A_1)], [1(A_2)]),$$

for $t \geq \frac{1}{2}$. Then One can define a sequence of completely mixed strategies

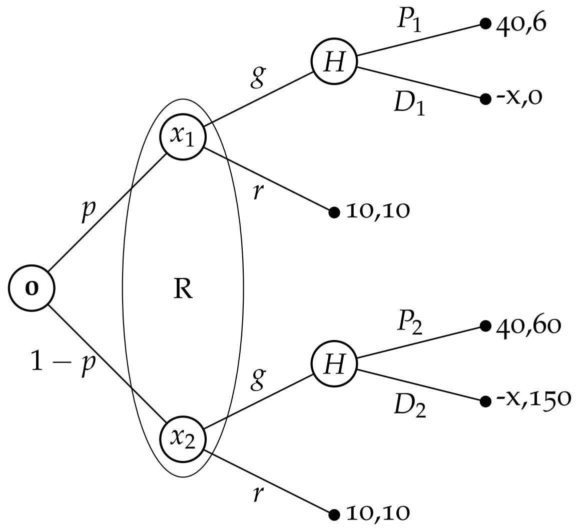
$$\sigma^k = ([\tfrac{1}{k}(O_1), (1-\tfrac{1}{k})(B_1)], [\tfrac{9}{k}(O_2), (1-\tfrac{9}{k})(B_2)],$$
$$[t(F_1), (1-t)(A_1)], [\tfrac{1}{k}(F_2), (1-\tfrac{1}{k})(A_2)]).$$

Then $\lim_{k\to\infty} \sigma^k = \sigma$ and the belief over $\{x_3, x_4\}$ is a uniform distribution. Hence, there are two types of sequential equilibrium.

$$([1(B_1)], [1(B_2)], [t(F_1), (1-t)(A_1)],$$
$$[1(A_2)], [0.5(x_3), 0.5(x_4)], [0.9(x_3), 0.1(x_4)]) \qquad \forall t \geq \tfrac{1}{2}$$

$$([1(O_1)], [1(O_2)], [1(A_1)], [t(F_2), (1-t)(A_2)],$$
$$[0.9(x_3), 0.1(x_4)], [0.5(x_5), 0.5(x_6)]) \qquad \forall t \geq \tfrac{1}{2}.$$

7.54 (a) The game tree depicted below represents the described situation.



(b) i. Assume $p = \frac{1}{3}$ and $x = 100$. The Nash equilibria in the game are

$$\left([1(r)], [t(P_1), (1-t)(D_1)], [q(P_2), (1-q)(D_2)]\right) \quad \forall t + 2q \leq \tfrac{33}{14}.$$

There is only one sequential equilibrium given by

$$\left([1(r)], [1(P_1)], [1(D_2)][\tfrac{1}{3}(x_1), \tfrac{2}{3}(x_2)]\right).$$

ii. Assume $p = 0.1$ and $x = 50$. The Nash equilibria in the game are

$$\left([1(r)], [t(P_1), (1-t)(D_1)], [q(P_2), (1-q)(D_2)]\right) \quad \forall 9t + 81q \leq 60.$$

There is only one sequential equilibrium in case that $t = 1, q = 0$ given by

$$\left([1(r)], [1(P_1)], [1(D_2)][0.1(x_1), 0.9(x_2)]\right).$$

iii. Assume $p = 0$ and $x = 75$. The set of Nash equilibria in the game which are also sequential equilibria is given by

$$\left([1(r)], [q(P_2), (1-q)(D_2)]\right) \qquad \forall q \leq \tfrac{85}{115}.$$

7.55 Let (σ, μ) be a consistent assessment in an extensive-form game Γ with perfect recall. We prove that the assessment (σ, μ) is a sequential equilibrium if and only if for each player $i \in N$, and every information set U_i

$$u_i(\sigma | U_i, \mu) \geq u_i(\widehat{\sigma}_i, \sigma_{-i} | U_i, \mu), \tag{46}$$

under every strategy $\widehat{\sigma}_i$ that differs from σ_i only at the information set U_i.

If (σ, μ) is a sequential equilibrium then Inequality (46) holds for every strategy $\widehat{\sigma}_i$ of player i, in particular for $\widehat{\sigma}_i$ that differs from σ_i only at the information set U_i.

We next prove the second direction. That is, if Inequality (46) holds for every strategy $\widehat{\sigma}_i$ of player i that differs from σ_i only at the information set U_i, then it also holds for every strategy σ_i'. That will be done by induction over the number of information sets of player i over which σ_i' differs from σ_i.

Assume that a strategy σ_i' of player i differs from σ_i at one information set of player i, name U_i'. Let U_i be an information set of player i. Denote by $P_{\sigma,\mu}(U_i'|U_i)$ the probability that, when the strategy vector σ is played from U_i associating with μ onward, a play of the game will pass through U_i'. Denote by $u^*_{\sigma,\mu}$ the expected payoff of player i from U_i onward, under the strategy vector σ given the play of the game will not pass through U_i'. Note that $P_{\sigma,\mu}(U_i'|U_i)$ and $u^*_{\sigma,\mu}$ are independent of the choice of player i in U_i', hence $P_{\sigma,\mu}(U_i'|U_i) = P_{\sigma_i',\sigma_{-i},\mu}(U_i'|U_i)$ and $u^*_{\sigma,\mu} = u^*_{\sigma_i',\sigma_{-i},\mu}$. Therefore,

$$\begin{aligned}
u_i(\sigma|U_i,\mu) &= P_{\sigma,\mu}(U_i'|U_i)u_i(\sigma|U_i',\mu) + (1 - P_{\sigma,\mu}(U_i'|U_i))u^*_{\sigma,\mu} \\
&= P_{\sigma_i',\sigma_{-i},\mu}(U_i'|U_i)u_i(\sigma|U_i',\mu) + (1 - P_{\sigma_i',\sigma_{-i},\mu}(U_i'|U_i))u^*_{\sigma_i',\sigma_{-i},\mu} \\
&\geq P_{\sigma_i',\sigma_{-i},\mu}(U_i'|U_i)u_i(\sigma_i',\sigma_{-i}|U_i',\mu) + (1 - P_{\sigma_i',\sigma_{-i},\mu}(U_i'|U_i))u^*_{\sigma_i',\sigma_{-i},\mu} \\
&= u_i(\widehat{\sigma}_i,\sigma_{-i}|U_i,\mu).
\end{aligned}$$

Assume that the condition holds for every strategy that differs from σ_i by at most n information sets of player i. Let σ_i' be a strategy that differs from σ_i at $n+1$ information sets of player i. Let U_i' be an information set of player i in which σ_i' differs from σ_i. Denote by $\widehat{\mathcal{U}}_i$ the set of all information sets $\widehat{U}_i$ of player i such that (1) there is a path from U_i' that passes trough $\widehat{U}_i$; (2) σ_i' differs from σ_i at $\widehat{U}_i$; and (3) there is no other information set of player i along the path from U_i' to $\widehat{U}_i$ that satisfies (1) and (2). Note that, every information set $\widehat{U}_i \in \widehat{\mathcal{U}}_i$ satisfies Inequality (46) by the induction hypothesis. For every vertex $x \in U_i'$ and every action $a \in A_i(x)$, let $P_{\sigma,\mu}(\widehat{U}_i|x,a)$ be the probability that, when player i selects a at x and then the strategy vector σ is played onward, a play of the game will pass through $\widehat{U}_i$. Let $u^{x,a}_{\sigma,\mu}$ be the expected payoff of player i from x onward, given Player i chooses a at x and then all of the players follow the strategy vector σ and the play of the game does not pass through any information sets in $\widehat{\mathcal{U}}_i$. Note that $P_{\sigma,\mu}(\widehat{U}_i|x,a)$ and $u^{x,a}_{\sigma,\mu}$ are independent of the choice of player i neither in U_i' nor in $\widehat{\mathcal{U}}_i$, hence $P_{\sigma,\mu}(\widehat{U}_i|x,a) = P_{\sigma_i',\sigma_{-i},\mu}(\widehat{U}_i|x,a)$ and

$u^{x,a}_{\sigma,\mu} = u^{x,a}_{\sigma'_i,\sigma_{-i},\mu}$. Therefore,

$$
\begin{aligned}
u_i(\sigma|U'_i,\mu) &= \sum_{x\in U'_i}\sum_{a\in A(x)} \mu(x)\sigma_i(a)\Bigg(\sum_{\widehat{U}_i\in\widehat{\mathcal{U}}_i} P_{\sigma,\mu}(\widehat{U}_i|x,a)u_i(\sigma|\widehat{U}_i,\mu) \\
&+ \Big(1-\sum_{\widehat{U}_i\in\widehat{\mathcal{U}}_i} P_{\sigma,\mu}(\widehat{U}_i|x,a)\Big)u^{x,a}_{\sigma,\mu}\Bigg) \\
&\geq \sum_{x\in U'_i}\sum_{a\in A(x)} \mu(x)\sigma'_i(a)\Bigg(\sum_{\widehat{U}_i\in\widehat{\mathcal{U}}_i} P_{\sigma,\mu}(\widehat{U}_i|x,a)u_i(\sigma|\widehat{U}_i,\mu) \\
&+ \Big(1-\sum_{\widehat{U}_i\in\widehat{\mathcal{U}}_i} P_{\sigma,\mu}(\widehat{U}_i|x,a)\Big)u^{x,a}_{\sigma,\mu}\Bigg) \\
&\geq \sum_{x\in U'_i}\sum_{a\in A(x)} \mu(x)\sigma'_i(a)\Bigg(\sum_{\widehat{U}_i\in\widehat{\mathcal{U}}_i} P_{\sigma'_i,\sigma_{-i},\mu}(\widehat{U}_i|x,a)u_i(\sigma'_i,\sigma_{-i}|\widehat{U}_i,\mu) \\
&+ \Big(1-\sum_{\widehat{U}_i\in\widehat{\mathcal{U}}_i} P_{\sigma'_i,\sigma_{-i},\mu}(\widehat{U}_i|x,a)\Big)u^{x,a}_{\sigma'_i,\sigma_{-i},\mu}\Bigg) \\
&= u_i(\widehat{\sigma}_i,\sigma_{-i}|U_i,\mu).
\end{aligned}
$$

where the first inequality follows by the condition, and the second inequality follows by the induction hypothesis. Finally, the proof for every other information set of player i is similar to the last arguments, and thus it was omitted.

Chapter 8: Correlated equilibria

8.1 (a)

$$S = \left\{xy(1,2) + (1-x)(1-y)(2,1)|x,y\in[0,1]\right\}.$$

(b)

$$S = \left\{x(1,2) + y(2,1)|x,y\in[0,1]\right\}.$$

8.2 Let σ^* be a Nash equilibrium of $G = \big(N,(S_i)_{i\in N},(u_i)_{i\in N}\big)$. We prove that the probability distribution p_{σ^*} is a correlated equilibrium.
Since σ^* is a Nash equilibrium then for every strategy $s'_i\in S_i$ one has

$$u_i(\sigma^*) \geq u_i(s'_i,\sigma^*_{-i}).$$

In addition, for every $s_i \in \text{supp}(\sigma_i^*)$ one has

$$u_i(\sigma^*) = u_i(s_i, \sigma_{-i}^*).$$

In conclusion, for every $s_i \in \text{supp}(\sigma_i^*)$ one has

$$\begin{aligned}\sum_{s_{-i}\in S_{-i}} \sigma^*(s_i, s_{-i}) u_i(s_i, s_{-i}) &= u_i(s_i, \sigma_{-i}^*) = u_i(\sigma^*) \\ &\geq u_i(s_i', \sigma_{-i}^*) = \sum_{s_{-i}\in S_{-i}} \sigma^*(s_i, s_{-i}) u_i(s_i', s_{-i}),\end{aligned}$$

Hence, σ^* is a correlated equilibrium.

8.3 By Theorem 8.7 (Exercise 8.2), the set W of all probability distributions p_σ over the set of action vectors S that are induced by Nash equilibria σ is a subset of correlated equilibria. By Theorem 8.9, the set of correlated equilibria is convex, hence any point in the convex hull of W is a correlated equilibrium.

8.4 Let σ^* be a correlated equilibrium. We prove that the payoff to each player i in σ^* is at least his maxmin value in mixed strategies

$$\underline{v}_i = \max_{\sigma_i \in \Sigma_i} \min_{\sigma_{-i}\in\Sigma_{-i}} U_i(\sigma_i, \sigma_{-i}).$$

Denote by $\widehat{\sigma}_i$ the optimal strategy of player i; i.e., $\underline{v}_i = \min_{\sigma_{-i}\in\Sigma_{-i}} U_i(\widehat{\sigma}_i, \sigma_{-i})$. Note that since U_i is a function that is linear in σ_{-i} , the minimum is attained at an extreme point of Σ_{-i}, namely, at a point $\widehat{s}_{-i} \in S_{-i}$, so

$$\underline{v}_i = U_i(\widehat{\sigma}_i, \widehat{s}_{-i}).$$

Since σ^* is a correlated equilibrium, then for every player i and every strategies $s_i, s_i' \in S_i$ one has

$$\sum_{s_{-i}\in S_{-i}} \sigma^*(s_i, s_{-i}) u_i(s_i, s_{-i}) \geq \sum_{s_{-i}\in S_{-i}} \sigma^*(s_i, s_{-i}) u_i(s_i', s_{-i}).$$

Therefore,

$$\begin{aligned}\sum_{s_{-i}\in S_{-i}} \sigma^*(s_i, s_{-i}) u_i(s_i, s_{-i}) &\geq \sum_{s_i'\in S_i} \widehat{\sigma}_i(s_i') \sum_{s_{-i}\in S_{-i}} \sigma^*(s_i, s_{-i}) u_i(s_i', s_{-i}) \\ &= \sum_{s_{-i}\in S_{-i}} \sigma^*(s_i, s_{-i}) \sum_{s_i'\in S_i} \widehat{\sigma}_i(s_i') u_i(s_i', s_{-i}) \\ &\geq \sum_{s_{-i}\in S_{-i}} \sigma^*(s_i, s_{-i}) U_i(\widehat{\sigma}_i, \widehat{s}_{-i}) = \underline{v}_i.\end{aligned}$$

The payoff to each player i under every recommendation is at least $\underline{v}_i$, so the payoff in the entire game is at least $\underline{v}_i$ as well.

8.5 Let $G = \big(N, (S_i)_{i\in N}, (u_i)_{i\in N}\big)$ be a strategic-form game. The set of correlated equilibria is the set of solution vectors for the following linear program (which is given by the constraints only).

Subject to

$$\sum_{s_{-i}\in S_{-i}} p_{s_i,s_{-i}} u_i(s_i, s_{-i}) \geq \sum_{s_{-i}\in S_{-i}} p_{s_i,s_{-i}} u_i(s_i', s_{-i}) \quad \forall i \in N, \forall s_i \in S_i, \forall s_i' \in S_i$$

$$\sum_{s\in S} p_s = 1$$

$$p_s \geq 0 \quad \forall s \in S$$

8.6 Let $G = \big(N, (S_i)_{i\in N}, (u_i)_{i\in N})\big)$ and $\widehat{G} = \big(N, (S_i)_{i\in N}, (\widehat{u}_i)_{i\in N})\big)$ be strategically equivalent games, i.e., for each player $i \in N$ there exist $\alpha_i > 0$ and $\beta_i \in \mathbf{R}$ such that

$$\widehat{u}_i(s) = \alpha_i u_i(s) + \beta_i, \quad \forall s \in S.$$

A probability distribution p over the set of action vectors S is a correlated equilibrium of G if and only if, for every player i and every strategies $s_i, s_i' \in S_i$ one has

$$\sum_{s_{-i}\in S_{-i}} \sigma^*(s_i, s_{-i}) u_i(s_i, s_{-i}) \geq \sum_{s_{-i}\in S_{-i}} \sigma^*(s_i, s_{-i}) u_i(s_i', s_{-i}) \iff$$
$$\sum_{s_{-i}\in S_{-i}} \sigma^*(s_i, s_{-i}) \left(\alpha_i u_i(s_i, s_{-i}) + \beta_i\right) \geq \sum_{s_{-i}\in S_{-i}} \sigma^*(s_i, s_{-i}) \left(\alpha_i u_i(s_i', s_{-i}) + \beta_i\right).$$

Thus p is also a correlated equilibrium of $\widehat{G}$. In conclusion, the sets of correlated equilibria of both games are identical and the set of correlated equilibrium payoffs of $\widehat{G}$ of each player i is the appropriate affine transformation of the set of correlated equilibrium payoffs of G of that player.

8.7 Let $G = \big(N, (S_i)_{i\in N}, (u_i)_{i\in N}\big)$ be a game in strategic form, and let $\widehat{G}$ be the game derived from G by a process of iterated elimination of strictly dominated strategies. Then the set of correlated equilibria of G and the set of correlated equilibria of $\widehat{G}$ are identical.
We show that by eliminating a strategy $\bar{s}_j$ which is strictly dominated by $\widehat{s}_j$ the set of correlated equilibrium remains unchanged. Let G' be the game obtained by eliminating $\bar{s}_j$ from G. Then, every correlated equilibrium σ^* of G satisfies that

$$\sum_{s_{-i}\in S_{-i}} \sigma^*(s_i, s_{-i}) u_i(s_i, s_{-i}) \geq \sum_{s_{-i}\in S_{-i}} \sigma^*(s_i, s_{-i}) u_i(s_i', s_{-i}),$$

for every player i and every strategies $s_i, s_i' \in S_i$. However, since

$$u_j(\bar{s}_j, s_{-j}) < u_j(\hat{s}_j, s_{-j}) \ \ \forall s_{-j} \in S_{-j}$$

one has $\sigma^*(\bar{s}_j) = 0$. Therefore, σ^* is an appropriate distribution over the strategies in G' so it is also a correlated equilibrium of G'.
If, on the other hand, σ^* is a correlated equilibrium of G', then

$$\begin{aligned}\sum_{s_{-j} \in S_{-j}} \sigma^*(s_j, s_{-j}) u_j(s_j, s_{-j}) &\geq \sum_{s_{-j} \in S_{-j}} \sigma^*(s_j, s_{-j}) u_j(\hat{s}_j, s_{-j}) \\ &\geq \sum_{s_{-j} \in S_{-j}} \sigma^*(s_j, s_{-j}) u_j(\bar{s}_j, s_{-j})\end{aligned}$$

so σ^* is a correlated equilibrium of G.

8.8 **Example 8.1:** Denote a probability distribution over the action vectors by $p = [\alpha(F,F), \beta(F,C), \gamma(C,F), \delta(C,C)]$. Then we need to find p that maximizes the sum of the players' payoffs $3\alpha + 3\delta$ subject to the following constrains

$$\left.\begin{array}{l} 2\alpha \geq \beta \\ \delta \geq 2\gamma \\ \alpha \geq 2\gamma \\ 2\delta \geq \beta \\ \alpha + \beta + \gamma + \delta = 1 \end{array}\right\}$$

Hence the correlated equilibria that maximizes the sum of the players' payoffs are $[\alpha(F,F), (1-\alpha(C,C)]$ for every $\alpha \in [0,1]$.
Example 8.3: Denote a probability distribution over the action vectors by $p = [\alpha(T,L), \beta(T,R), \gamma(B,L), \delta(B,R)]$. Then we need to find p that maximizes the sum of the players' payoffs $12\alpha + 9\beta + 9\gamma$ subject to the following constrains

$$\left.\begin{array}{l} 6\alpha + 2\beta \geq 7\alpha \\ 7\gamma \geq 6\gamma + 2\delta \\ 6\alpha + 2\gamma \geq 7\alpha \\ 7\beta \geq 6\beta + 2\delta \\ \alpha + \beta + \gamma + \delta = 1 \end{array}\right\} \iff \begin{cases} 2\beta \geq \alpha \\ \gamma \geq 2\delta \\ 2\gamma \geq \alpha \\ \beta \geq 2\delta \end{cases}$$

Hence the only correlated equilibrium that maximizes the sum of the players' payoffs is $[\frac{1}{2}(F,F), \frac{1}{4}(T,R), \frac{1}{4}(B,L)]$.

8.9 A correlated equilibrium whose expected payoff is $(\frac{40}{9}, \frac{36}{9})$ in the game of "Chicken" is $[0(T,L), \frac{172}{405}(T,R), \frac{208}{405}(B,L), \frac{25}{405}(B,R)]$.

8.10 The game has only one Nash equilibrium

$$([\tfrac{1}{3}(T), \tfrac{1}{3}(M), \tfrac{1}{3}(B)], [\tfrac{1}{3}(L), \tfrac{1}{3}(C), \tfrac{1}{3}(R)]).$$

In particular, the following correlated equilibrium of the game is not in the convex hull of the Nash equilibria

$$[\tfrac{1}{6}(T, C), \tfrac{1}{6}(T, R), \tfrac{1}{6}(M, L), \tfrac{1}{6}(M, R), \tfrac{1}{6}(B, L), \tfrac{1}{6}(B, C)].$$

8.11 The Nash equilibria of the game are (B, L), (T, R) and

$$([\tfrac{3}{4}(T), \tfrac{1}{4}(B)], [\tfrac{3}{4}(L), \tfrac{1}{4}(R)]).$$

The following correlated equilibrium of the game is not in the convex hull of the Nash equilibria

$$[\tfrac{1}{2}(T, R), \tfrac{1}{2}(B, L)].$$

8.12 (a) A behavior strategy of player i in the game $\Gamma^*_M(q)$ is a function $\sigma_i : M_i \to \Delta(S_i)$, thus the set of behavior strategy of player i is $(\Delta(S_i))^{M_i}$.

(b) Let σ be a vector of behavior strategies in $\Gamma^*_M(q)$. Denote by $\sigma_i(s_i|m_i)$ the probability that under σ_i player i selects s_i given he gets a signal m_i. Then σ induces a probability distribution over the set of action vectors $S = \times_{i \in N} S_i$ as follows:

$$p(s) = \sum_{m \in M} q(m) \prod_{i \in N} \sigma_i(s_i|m_i) = \sum_{m \in M} q(m)\sigma(s|m) \ \ \forall s \in S.$$

(c) Let σ be a Nash equilibrium of $\Gamma^*_M(q)$. We show that the probability distribution p on the set of pure strategy vectors S induced by σ is a correlated equilibrium.
Since σ is a Nash equilibrium, for every player i, every message $m_i \in M_i$ and every strategies $s_i, s'_i \in S_i$ one has

$$\sum_{m_{-i} \in M_{-i}} \sum_{s_{-i} \in S_{-i}} q(m_i, m_{-i})\sigma_i(s_i|m_i)\sigma_{-i}(s_{-i}|m_{-i})u_i(s_i, s_{-i}) \geq$$
$$\sum_{m_{-i} \in M_{-i}} \sum_{s_{-i} \in S_{-i}} q(m_i, m_{-i})\sigma_i(s_i|m_i)\sigma_{-i}(s_{-i}|m_{-i})u_i(s'_i, s_{-i}),$$

that is, whenever player i receives (with positive probability) a message m_i he is better off following a strategy s_i in the support of $\sigma_i(\cdot|m_i)$ (if either $q(m_i) = 0$ or $\sigma_i(s_i|m_i) = 0$ the above inequality holds trivially). Therefore, for every player i and every strategies $s_i, s_i' \in S_i$ one has

$$\sum_{m_i \in M_i} \sum_{m_{-i} \in M_{-i}} \sum_{s_{-i} \in S_{-i}} q(m_i, m_{-i}) \sigma_i(s_i|m_i) \sigma_{-i}(s_{-i}|m_{-i}) u_i(s_i, s_{-i}) \geq$$
$$\sum_{m_i \in M_i} \sum_{m_{-i} \in M_{-i}} \sum_{s_{-i} \in S_{-i}} q(m_i, m_{-i}) \sigma_i(s_i|m_i) \sigma_{-i}(s_{-i}|m_{-i}) u_i(s_i', s_{-i}) \iff$$
$$\sum_{s_{-i} \in S_{-i}} u_i(s_i, s_{-i}) \sum_{m \in M} q(m) \sigma_i(s_i|m_i) \sigma_{-i}(s_{-i}|m_{-i}) \geq$$
$$\sum_{s_{-i} \in S_{-i}} u_i(s_i', s_{-i}) \sum_{m \in M} q(m) \sigma_i(s_i|m_i) \sigma_{-i}(s_{-i}|m_{-i}) \iff$$
$$\sum_{s_{-i} \in S_{-i}} u_i(s_i, s_{-i}) p(s_i, s_{-i}) \geq \sum_{s_{-i} \in S_{-i}} u_i(s_i', s_{-i}) p(s_i, s_i).$$

In particular, the probability distribution p on the set of pure strategy vectors S induced by σ is a correlated equilibrium.

8.13 Denote a probability distribution over the action vectors by

$$p = [\alpha(T, L), \beta(T, R), \gamma(B, L), \delta(B, R)].$$

The distribution p is a correlated equilibrium if it satisfies the following constrains

$$\left.\begin{array}{l} \alpha + c\beta \geq (1+a)\beta \\ (1+a)\delta \geq \gamma + c\delta \\ \gamma \geq (1+d)\alpha + b\gamma \\ (1+d)\beta + b\delta \geq \delta \\ \alpha + \beta + \gamma + \delta = 1 \\ \alpha, \beta, \gamma, \delta \geq 0 \end{array}\right\} \iff \left\{\begin{array}{l} \alpha \geq (1+a-c)\beta \\ (1+a-c)\delta \geq \gamma \\ (1-b)\gamma \geq (1+d)\alpha \\ (1+d)\beta \geq (1-b)\delta \\ \alpha + \beta + \gamma + \delta = 1 \\ \alpha, \beta, \gamma, \delta \geq 0 \end{array}\right.$$

Hence the correlated equilibrium satisfies

$$\begin{aligned} &\alpha \geq (1+a-c)\beta \geq \frac{(1+a-c)(1-b)}{1+d}\delta \geq \frac{1-b}{1+d}\gamma \geq \alpha \Rightarrow \\ &\alpha = (1+a-c)\beta = \frac{(1+a-c)(1-b)}{1+d}\delta = \frac{1-b}{1+d}\gamma = \alpha. \end{aligned} \tag{47}$$

However, there is only one solution of the Equation 47 which satisfies $\alpha + \beta + \gamma + \delta = 1$. Furthermore, if $a, b, c, d \in (-\frac{1}{4}, \frac{1}{4})$ then the solution also satisfies $\alpha, \beta, \gamma, \delta \geq 0$ as needed. Note that the limit of the correlated equilibrium payoff as a, b, c, and d approach 0 is $(\frac{1}{2}, \frac{1}{2})$.

8.14 Let $\bar{s}_i$ be a strictly dominated action by $\widehat{s}_i$ of player i in a strategic form game G. Then for every correlated equilibrium p of G, $\bar{s}_i$ is chosen with zero probability, i.e., for every $s_{-i} \in S_{-i}$ one has $p(\bar{s}_i, s_{-i}) = 0$. This result follows Exercise 8.7. We prove it directly. Since p is a correlated equilibrium, it satisfies that

$$\sum_{s_{-i} \in S_{-i}} p(s_i, s_{-i}) u_i(s_i, s_{-i}) \geq \sum_{s_{-i} \in S_{-i}} p(s_i, s_{-i}) u_i(s_i', s_{-i})$$

for every strategies $s_i, s_i' \in S_i$ of player i. Thus, the above inequality also holds for $\bar{s}_i$ and $\widehat{s}_i$:

$$\sum_{s_{-i} \in S_{-i}} p(\bar{s}_i, s_{-i}) u_i(\bar{s}_i, s_{-i}) \geq \sum_{s_{-i} \in S_{-i}} p(\bar{s}_i, s_{-i}) u_i(\widehat{s}_i, s_{-i}).$$

However, since $\widehat{s}_i$ strictly dominates $\bar{s}_i$ one has

$$u_i(\bar{s}_i, s_{-i}) < u_i(\widehat{s}_i, s_{-i}) \;\; \forall s_{-i} \in S_{-i}$$

which leads to $p(\bar{s}_i, s_{-i}) = 0$ for every $s_{-i} \in S_{-i}$.

8.15 We prove that in a two-player zero-sum game G, every correlated equilibrium payoff to Player I is the value v of the game in mixed strategies. Let p be a correlated equilibrium in a zero-sum game, then

$$\sum_{s_2 \in S_2} p(s_1, s_2) u(s_1, s_2) \geq \sum_{s_2 \in S_2} p(s_1, s_2) u(s_1', s_2) \;\; \forall s_1, s_1' \in S_1 \tag{48}$$

$$\sum_{s_1 \in S_1} p(s_1, s_2) u(s_1, s_2) \leq \sum_{s_1 \in S_1} p(s_1, s_2) u(s_1, s_2') \;\; \forall s_2, s_2' \in S_2. \tag{49}$$

Denote by σ_1 and σ_2 the optimal strategies of Player I and Player II respectively. By Inequality (48) it follows that if $p(s_1) := \sum_{s_2 \in S_2} p(s_1, s_2) > 0$ then

$$\begin{aligned}
\sum_{s_2 \in S_2} p(s_1, s_2) u(s_1, s_2) &\geq \sum_{s_1' \in S_1} \sigma_1(s_1') \sum_{s_2 \in S_2} p(s_1, s_2) u(s_1', s_2) \\
&= p(s_1) \sum_{s_1' \in S_1} \sum_{s_2 \in S_2} \sigma_1(s_1') \frac{p(s_1, s_2)}{p(s_1} u(s_1', s_2) \\
&= p(s_1) v.
\end{aligned}$$

Thus the correlated equilibrium payoff to Player I is

$$\sum_{s_1 \in S_1} \sum_{s_2 \in S_2} p(s_1, s_2) u(s_1, s_2) \geq \sum_{s_1 \in S_1} p(s_1) v = v. \tag{50}$$

On the other hand, by Inequality (49) it follows that if $p(s_2) := \sum_{s_1 \in S_1} p(s_1, s_2) > 0$ then

$$\begin{aligned}\sum_{s_1 \in S_1} p(s_1, s_2) u(s_1, s_2) &\leq \sum_{s'_2 \in S_2} \sigma_2(s'_2) \sum_{s_1 \in S_1} p(s_1, s_2) u(s_1, s'_2) \\ &= p(s_2) \sum_{s'_2 \in S_2} \sum_{s_1 \in S_1} \sigma_2(s'_2) \frac{p(s_1, s_2)}{p(s_2} u(s_1, s'_2) \\ &= p(s_2) v.\end{aligned}$$

Thus the correlated equilibrium payoff to Player I is

$$\sum_{s_1 \in S_1} \sum_{s_2 \in S_2} p(s_1, s_2) u(s_1, s_2) \leq \sum_{s_2 \in S_2} p(s_2) v = v. \tag{51}$$

Thus, by Inequalities (50) and (51) the correlated equilibrium payoff to Player I is the value in mixed strategies.

8.16 We prove that if p is a correlated equilibrium of a two-player zero-sum game, then for every recommendation s_I that Player I receives with positive probability, the conditional probability

$$\big(p(s_{II} | s_I)\big)_{s_{II} \in S_{II}} = \left(\tfrac{p(s_I, s_{II})}{p(s_I)}\right)_{s_{II} \in S_{II}}$$

is an optimal strategy for Player II.
By Exercise 8.15, if $p(s_I) > 0$ then for every mixed strategy $\sigma_I \in \Sigma_I$ of Player I one has

$$v = \sum_{s_{II} \in S_{II}} \tfrac{p(s_I, s_{II})}{p(s_I)} u(s_I, s_{II}) \geq \sum_{s'_I \in S_I} \sum_{s_{II} \in S_{II}} \sigma_1(s'_I) \tfrac{p(s_I, s_{II})}{p(s_I)} u(s'_I, s_{II})$$

therefore, $\big(p(s_{II} | s_I)\big)_{s_{II} \in S_{II}}$ is an optimal strategy for Player II. Similarly, the conditional probability $\big(p(s_I | s_{II})\big)_{s_I \in S_I}$ is an optimal strategy for Player I. Hence it can be concluded that the marginal distribution of p over the set of actions of each of the players is an optimal strategy for that player.

8.17 The value of the game is $\frac{1}{2}$.
The optimal strategies of Player I are $\left[\frac{1}{2}(T), x(M), (\frac{1}{2} - x)(B)\right]$ for every $x \in \left[0, \frac{1}{2}\right]$.
The optimal strategies of Player II are $\left[\frac{1}{2}(L), y(C), (\frac{1}{2} - y)(R)\right]$ for every $y \in \left[0, \frac{1}{2}\right]$.
To find the set of correlated equilibria let p be a distribution over the set of action vectors as depicted below.

	L	C	R
T	a	b	c
M	d	e	f
B	g	h	i

The distribution p is a correlated equilibrium if it satisfies the following constrains.

$$\left.\begin{aligned} c &\geq a+b \\ d+e &\geq f \\ g+h &\geq i \end{aligned}\right\} \text{Player I}$$

$$\left.\begin{aligned} d+g &\leq a \\ e+h &\leq b \\ c &\leq f+i \end{aligned}\right\} \text{Player II}$$

Hence

$$\begin{aligned} &c \geq a+b \geq (d+g)+(e+h) \geq f+i \geq c \Rightarrow \\ &\quad c = a+b = (d+g)+(e+h) = f+i = c \end{aligned}$$

But since $a+b+c+d+e+f+g+h+i=1$, the correlated equilibria of the game are

$$\begin{aligned} &c = a+b = d+g+e+h = f+i = \tfrac{1}{4} \\ &a,b,d,e,f,g,h,i \geq 0. \end{aligned}$$

Not every correlated equilibrium lies in the convex hull of the product distributions that correspond to pairs of optimal strategies, for instance $a=c=e=i=\frac{1}{4}$ is a correlated equilibrium that does not lie in such a convex hull.

8.18 We prove that the set-valued function that assigns to every game its set of correlated equilibria is an upper semi-continuous mapping. Let $(G^k)_{k\in N}$ be a sequence of games $G^k = \left(N, (S_i)_{i\in N}, (u_i^k)_{i\in N}\right)$, all of which share the same set of players N and the same sets of actions $(S_i)_{i\in N}$. Further suppose that for each player i,

$$\lim_{k\to\infty} u_i^k(s) = u_i(s), \quad \forall s \in S. \tag{52}$$

Suppose that for each $k \in \mathbf{N}$ the probability distribution p^k is a correlated equilibrium of G^k, and

$$\lim_{k\to\infty} p^k(s) = p(s), \quad \forall s \in S. \tag{53}$$

Let $\delta > 0$. By Equations (52) and (53) there is a large enough $K \in \mathbf{N}$ such that for every $k \geq K$ one has

$$|u_i^k(s) - u_i(s)| < \frac{\delta}{4} \qquad \forall s \in S \tag{54}$$

$$|p^k(s) - p(s)| < \frac{\delta}{4|S| \max_{i,s} |u_i(s)|} \qquad \forall s \in S \tag{55}$$

Then, for every player i and every strategies $s_i, s_i' \in S_i$

$$\sum_{s_{-i} \in S_{-i}} p(s_i, s_{-i}) u_i(s_i, s_{-i}) \geq \sum_{s_{-i} \in S_{-i}} p^k(s_i, s_{-i}) u_i(s_i, s_{-i}) - \tfrac{\delta}{4} \tag{56}$$

$$\geq \sum_{s_{-i} \in S_{-i}} p^k(s_i, s_{-i}) u_i^k(s_i, s_{-i}) - \tfrac{\delta}{2} \tag{57}$$

$$\geq \sum_{s_{-i} \in S_{-i}} p^k(s_i, s_{-i}) u_i^k(s_i', s_{-i}) - \tfrac{\delta}{2} \tag{58}$$

$$\geq \sum_{s_{-i} \in S_{-i}} p^k(s_i, s_{-i}) u_i(s_i', s_{-i}) - \tfrac{3\delta}{4} \tag{59}$$

$$\geq \sum_{s_{-i} \in S_{-i}} p(s_i, s_{-i}) u_i(s_i', s_{-i}) - \delta \tag{60}$$

where Inequalities (56) and (60) follow by Inequality (55); Inequalities (57) and (59) obtain by Inequality (54); and Inequality (58) holds since p^k is a correlated equilibrium of G^k.
In conclusion, since

$$\sum_{s_{-i} \in S_{-i}} p(s_i, s_{-i}) u_i(s_i, s_{-i}) \geq \sum_{s_{-i} \in S_{-i}} p(s_i, s_{-i}) u_i(s_i', s_{-i}) - \delta$$

for every $\delta > 0$, one can deduce that for every player i and every strategies $s_i, s_i' \in S_i$,

$$\sum_{s_{-i} \in S_{-i}} p(s_i, s_{-i}) u_i(s_i, s_{-i}) \geq \sum_{s_{-i} \in S_{-i}} p(s_i, s_{-i}) u_i(s_i', s_{-i})$$

so p is a correlated equilibrium of the game $(N, (S_i)_{i \in N}, (u_i)_{i \in N})$.

8.19 (a) Not every game in strategic form has a strict correlated equilibrium. Consider, for instance, the following two-player game:

	L	R
T	1,1	0,0
B	1,1	0,0

R is strictly dominated by L so it will be chosen with probability zero in every correlated equilibrium (Exercise 8.14). Hence the correlated equilibria of the game are

$$[x(T, L), (1 - x)(B, L)]$$

for every $x \in [0,1]$. However, whenever Player I is recommended to play T he will not lose by playing B and vice versa.

(b) Denote a probability distribution over the action vectors by $p = [\alpha(T,L), \beta(T,R), \gamma(B,L), \delta(B,R)]$. p is a strict correlated equilibrium if it satisfies the next constrains.

$$\left.\begin{array}{ll} 4\alpha & > 5\alpha \\ 5\gamma & > 4\gamma + 3\delta \\ 2\alpha + \gamma & > 4\alpha \\ 4\beta & > 2\beta + \delta \end{array}\right\} \iff \begin{cases} \alpha & < \frac{\gamma}{2}, 3\beta \\ \delta & < \frac{\gamma}{3}, 2\beta. \end{cases}$$

Set $\gamma = 1 - \alpha - \beta - \delta$.

Assume first that $2\alpha \geq 3\delta$. Then

$$\left.\begin{array}{l} \beta > \frac{\alpha}{3} \\ 1 - \alpha - \beta - \delta > 2\alpha \end{array}\right\} \Rightarrow \frac{\alpha}{3} < \beta < 1 - 3\alpha - \delta \Rightarrow 2\alpha < 0.6 - 0.6\delta.$$

Using the assumption that $2\alpha \geq 3\delta$ one can derive the following set of strict correlated equilibria

$$\left\{ [\alpha(T,L), \beta(T,R), (1-\alpha-\beta-\delta)(B,L), \delta(B,R)] \,\middle|\, 2\alpha \geq 3\delta, \tfrac{1}{6} < \delta, 3\beta > \alpha \right\}.$$

Assume next that $2\alpha < 3\delta$. Then

$$\left.\begin{array}{l} \beta > \frac{\delta}{2} \\ 1 - \alpha - \beta - \delta > 3\delta \end{array}\right\} \Rightarrow \frac{\delta}{2} < \beta < 1 - \alpha - 4\delta \Rightarrow 3\delta < \frac{2}{3} - \frac{2}{3}\alpha.$$

Using the assumption that $2\alpha < 3\delta$ one can derive the following set of strict correlated equilibria

$$\left\{ [\alpha(T,L), \beta(T,R), (1-\alpha-\beta-\delta)(B,L), \delta(B,R)] \,\middle|\, 2\alpha < 3\delta, \alpha < \tfrac{1}{4},\ 2\beta > \delta \right\}.$$

8.20 (a) Harry's pure strategies in this game are V-choosing a vector, T and B. Harriet's pure strategies in this game are L and R.

(b) The strategic form game can be represented by the following matrix.

	L	R
V	2,1	2,1
T	0,0	1,3
B	4,2	0,0

The set of Nash equilibria of the game is

$$\left\{([1(V)],[y(L),(1-y)(R)])\,\middle|\,\forall y\in[\tfrac{1}{2},1]\right\}\bigcup(B,L).$$

(c) The set of correlated equilibria of the game is

$$\left\{[\alpha(V,L),\beta(V,R),(1-\alpha-\beta)(B,L)]\;|0\le\alpha\le\beta,\alpha+\beta\le 1\right\}.$$

8.21 Consider the following strategic form game.

	C_1	C_2	C_3	...	C_n
R_1	x_1, y_1	0,0	0,0	...	$0, y_1$
R_2	0,0	x_2, y_2	0,0	...	$0, y_2$
R_3	0,0	0,0	x_3, y_3	...	$0, y_3$
...	...	...	...	...	...
R_n	$x_1, 0$	$x_2, 0$	$x_3, 0$	...	x_n, y_n

(a) The Nash equilibria of the game are (R_k, C_k) for every $k\in\{1,2,...,n\}$. Indeed, if Player I used a mixed strategy then Player II best reply is C_n hence Player I is better off deviating to R_n. Similarly, there is no Nash equilibrium in which Player II plays a mixed strategy.

(b) Let represent the probability distribution over the action vector by the next matrix.

	C_1	C_2	C_3	...	C_n
R_1	$p_{1,1}$	$p_{1,2}$	$p_{1,3}$	...	$p_{1,n}$
R_2	$p_{2,1}$	$p_{2,2}$	$p_{2,3}$	...	$p_{2,n}$
R_3	$p_{3,1}$	$p_{3,2}$	$p_{3,3}$	...	$p_{3,n}$
...	...	...	...	...	...
R_n	$p_{n,1}$	$p_{n,2}$	$p_{n,3}$	...	$p_{n,n}$

Then for every $k\in\{1,2,...,n-1\}$ one has

$$p_{k,k}x_k\ge\sum_{m=1}^{n}p_{k,m}x_m$$

so $p_{k,m}=0$ for every $k\neq m$. Similarly, for every $k\in\{1,2,...,n-1\}$ one has

$$p_{k,k}y_k\ge\sum_{m=1}^{n}p_{m,k}y_m$$

so $p_{m,k}=0$ for every $k\neq m$. Therefore, the set of correlated equilibria is

$$\left\{[p_{1,1}(R_1,C_1),...,p_{n,n}(R_n,C_n)]\,\middle|\,\sum_{m=1}^{n}p_{m,m}=1\right\},$$

i.e., the convex hull of the set of Nash equilibria.

8.22 Let A and B be two sets in $\mathbf{R}^2$ satisfying: $A \subseteq B$; A is a union of a finite number of rectangles

$$A = \bigcup_{k=1}^{n} [a_i, b_i] \times [c_i, d_i] \quad \text{such that } a_i \leq b_i,\ c_i \leq d_i;$$

B is the convex hull of a finite number of points.
We next define a strategic form game G satisfying the property that its set of Nash equilibrium payoffs is A, and its set of correlated equilibrium payoffs is B.
Choose $m \in \mathbf{R}$ such that $m < a_i, b_i, c_i, d_i$ for every $i \in \{1, 2, ..., n\}$. Consider the following two-player strategic form game.

	C_1	C_2	C_3	C_4	...	C_{2n-1}	C_{2n}
R_1	a_1, c_1	b_1, c_1	m, m	m, m	...	m, c_1	m, c_1
R_2	a_1, d_1	b_1, d_1	m, m	m, m	...	m, d_1	m, d_1
R_3	m, m	m, m	a_2, c_2	b_2, c_2	...	m, c_2	m, c_2
R_4	m, m	m, m	a_2, d_2	b_2, d_2	...	m, d_2	m, d_2
...	...	...	...	...	...	...	...
R_{2n-1}	a_1, m	b_1, m	a_2, m	b_2, m	...	a_n, c_n	b_n, c_n
R_{2n}	a_1, m	b_1, m	a_2, m	b_2, m	...	a_n, d_n	b_n, d_n

Note that we can assume w.l.o.g. that $a_i, b_i, c_i, d_i > 0$ and $m = 0$, otherwise one can implement a positive affine transformation $v(x) = u(x) - m$ for all payoff in the game. By Theorem 5.35, the sets of Nash equilibria in both games are identical and the equilibrium payoffs change in accordance with the positive affine transformation that has been implemented. Furthermore, by Exercise 8.6 the same holds for the set of correlated equilibria.
Hence, by Exercise 5.49, A is the set of Nash equilibrium payoffs of the game.
Define a probability distribution over the action vectors.

	C_1	C_2	C_3	C_4	...	C_{2n-1}	C_{2n}
R_1	$p_{1,1}$	$p_{1,2}$	$p_{1,3}$	$p_{1,4}$	...	$p_{1,2n-1}$	$p_{1,2n}$
R_2	$p_{2,1}$	$p_{2,2}$	$p_{2,3}$	$p_{2,4}$	...	$p_{2,2n-1}$	$p_{2,2n}$
R_3	$p_{3,1}$	$p_{3,2}$	$p_{3,3}$	$p_{3,4}$	...	$p_{3,2n-1}$	$p_{3,2n}$
R_4	$p_{4,1}$	$p_{4,2}$	$p_{4,3}$	$p_{4,4}$	...	$p_{4,2n-1}$	$p_{4,2n}$
...	...	...	...	...	...	...	...
R_{2n-1}	$p_{2n-1,1}$	$p_{2n-1,2}$	$p_{2n-1,3}$	$p_{2n-1,4}$	...	$p_{2n-1,2n-1}$	$p_{2n-1,2n}$
R_{2n}	$p_{2n,1}$	$p_{2n,2}$	$p_{2n,3}$	$p_{2n,4}$	...	$p_{2n-1,2n}$	$p_{2n,2n}$

Using a similar arguments as in Exercise 8.21 (Part (b)), since R_{2n} weakly controls every strategy R_l for $l < 2n-1$, then if $u_I(R_l, C_k) = m$ then $p_{l,k} = 0$. Similarly, if $u_{II}(R_l, C_k) = m$ then $p_{l,k} = 0$. Hence the set of correlated equilibrium payoffs is the convex hull of A, i.e., B.

8.23 Let x, y, a, b be positive numbers. Consider the two-player strategic-form game with the following payoff matrix, in which Player I chooses a row, and Player II chooses a column.

	C_1	C_2	C_3	C_4	C_5
R_1	$x-1, y+1$	$0,0$	$0,0$	$x+1, y-1$	$0, y$
R_2	$x+1, y-1$	$x-1, y+1$	$0,0$	$0,0$	$0, y$
R_3	$0,0$	$x+1, y-1$	$x-1, y+1$	$0,0$	$0, y$
R_4	$0,0$	$0,0$	$x+1, y-1$	$x-1, y+1$	$0, y$
R_5	$x, 0$	$x, 0$	$x, 0$	$x, 0$	a,b

(a) The only Nash equilibrium of the game is (R_5, C_5). Assume by contradiction that there is another Nash equilibrium of the form

$$[([q_1(R_1), q_2(R_2), q_3(R_3), q_4(R_4), q_5(R_5)],$$
$$[p_1(C_1), p_2(C_2), p_3(C_3), p_4(C_4), p_5(C_5)])$$

where, for instance, $q_1 > 0$. Therefore, the expected payoff of Player I under R_1 is at least his expected payoff under R_5 so $p_4 \geq p_1$ and $p_4 > 0$. Hence, Player II plays C_4 with positive probability thus his expected payoff under C_4 is at least his expected payoff under C_5 so $q_4 \geq q_1$. Continue with similar arguments, one can deduce that

$$p_3 \geq p_4 \Rightarrow q_3 \geq q_4 \Rightarrow p_2 \geq p_3 \Rightarrow q_2 \geq q_3 \Rightarrow p_1 \geq p_2 \Rightarrow q_1 \geq q_2.$$

In particular, $q_1 = q_2 = q_3 = q_4 > 0$ and $p_1 = p_2 = p_3 = p_4 > 0$ however, Player I is better off deviating to R_5, a contradiction.

(b) Define the probability distribution over the action vectors as depicted below.

	C_1	C_2	C_3	C_4	C_5
R_1	$p_{1,1}$	$p_{1,2}$	$p_{1,3}$	$p_{1,4}$	$p_{1,5}$
R_2	$p_{2,1}$	$p_{2,2}$	$p_{2,3}$	$p_{2,4}$	$p_{2,5}$
R_3	$p_{3,1}$	$p_{3,2}$	$p_{3,3}$	$p_{3,4}$	$p_{3,5}$
R_4	$p_{4,1}$	$p_{4,2}$	$p_{4,3}$	$p_{4,4}$	$p_{4,5}$
R_5	$p_{5,1}$	$p_{5,2}$	$p_{5,3}$	$p_{5,4}$	$p_{5,5}$

Comparing the expected payoff of Player I by following the recommendation of R_1 over R_5 yields the necessary condition such that $p_{1,1} \leq p_{1,4}$. We implement similar considerations for all possible recommendations of both players and thus

$$p_{1,1} \overbrace{\leq}^{R_1} p_{1,4} \overbrace{\leq}^{C_4} p_{4,4} \overbrace{\leq}^{R_4} p_{4,3} \overbrace{\leq}^{C_3} p_{3,3} \overbrace{\leq}^{R_3} p_{3,2} \overbrace{\leq}^{C_2} p_{2,2} \overbrace{\leq}^{R_2} p_{2,1} \overbrace{\leq}^{C_1} p_{1,1}.$$

Therefore,

$$p_{1,1} = p_{1,4} = p_{4,4} = p_{4,3} = p_{3,3} = p_{3,2} = p_{2,2} = p_{2,1},$$

Finally, in a correlated equilibrium, for every $l, k \in \{1, 2, ..., 5\}$ such that either $u_I(R_l, C_k) = 0$ or $u_{II}(R_k, C_l) = 0$ one has $p_{l,k} = 0$. In conclusion, every correlated equilibrium is of the form

$$p_{5,5} \geq 0,\ p_{1,1} = p_{1,4} = p_{4,4} = p_{4,3} = p_{3,3} = p_{3,2} = p_{2,2} = p_{2,1} = \tfrac{1-p_{55}}{8}.$$

Chapter 9: Games with incomplete information and common priors

9.1 We prove that the knowledge operator K_i of each player i satisfies the following properties:

(a) For every $\omega \in Y$ one has $F_i(\omega)$ is a nonempty subsets of Y, thus

$$K_i Y = \{\omega \in Y : F_i(\omega) \subseteq Y\} = Y.$$

(b)

$$\begin{aligned} K_i A \cap K_i B &= \{\omega \in Y : F_i(\omega) \subseteq A\} \cap \{\omega \in Y : F_i(\omega) \subseteq B\} \\ &= \{\omega \in Y : F_i(\omega) \subseteq A \cap B\} = K_i(A \cap B). \end{aligned}$$

(c) Note that for every $\omega \in Y$ and every $\omega' \in F_i(\omega)$ one has $\omega' \in F_i(\omega) = F_i(\omega')$. Thus,

$$\begin{aligned} &F_i(\omega) \subseteq \{w' \in Y : F_i(\omega') \not\subseteq A\} \iff w \in \{w' \in Y : F_i(\omega') \not\subseteq A\} \\ &\iff F_i(\omega) \not\subseteq A. \end{aligned}$$

Therefore,

$$\begin{aligned} K_i((K_i A)^c) &= \{\omega \in Y : F_i(\omega) \subseteq (K_i A)^c\} \\ &= \Big\{\omega \in Y : F_i(\omega) \subseteq \{\omega' \in Y : F_i(\omega') \not\subseteq A\}\Big\} \\ &= \{\omega \in Y : F_i(\omega) \not\subseteq A\} = (K_i A)^c. \end{aligned}$$

9.2 (a) Let $\omega \in Y$ and $A \subseteq Y$ such that $w \in K(A)$ (by (i) there is such a set). Hence, by (iii), $\omega \in A$. In particular,

$$\omega \in \cap \left\{ A \subseteq Y : \omega \in K(A) \right\} = F(\omega).$$

(b) To prove that $F(\omega') = F(\omega)$ it is sufficient to prove that $\omega \in K(A)$ if and only if $\omega' \in K(A)$.

Let $A \subseteq Y$ and suppose that $\omega \in K(A)$. By (iv), $\omega \in K(A) = K(K(A))$. Therefore the set $K(A)$ is one of the sets in the intersection in the definition of $F(\omega)$, hence $F(\omega) \subseteq K(A)$. Since $\omega' \in F(\omega)$, it follows that $\omega' \in K(A)$.

Let now $A \subseteq Y$ and suppose that $\omega \notin K(A)$. By (v), $\omega \in (K(A))^c = K((K(A))^c)$. Therefore the set $(K(A))^c$ is one of the sets in the intersection in the definition of $F(\omega)$, hence $F(\omega) \subseteq (K(A))^c$. Since $\omega' \in F(\omega)$, it follows that $\omega' \in (K(A))^c$, and therefore $\omega' \notin K(A)$.

By (iii), for every A such that $\omega \in K(A)$ we have $\omega \in A$. The definition of $F(\omega)$ implies then that $\omega \in F(\omega)$. The two properties (a) $\omega \in F(\omega)$ and (b) $F(\omega') = F(\omega)$ whenever $\omega' \in F(\omega)$, imply that $\mathcal{F}$ is a partition of Y.

(c) Let $A \subseteq Y$ and let $\omega \in Y$. If $\omega \in K(A)$, then by the definition of $F(\omega)$, $F(\omega) \subseteq A$. On the other hand, if $F(\omega) \subseteq A$ then by Part (b) and (iii)

$$F(\omega) = K(F(\omega)) = K(F(\omega) \cap A) = K(F(\omega)) \cap K(A) = F(\omega) \cap K(A).$$

Hence, $F(\omega) \subseteq K(A)$ so, by Part (a), $\omega \in F(\omega) \subseteq K(A)$. In conclusion,

$$K(A) = \{\omega \in Y : \omega \in K(A)\} = \{\omega \in Y : F(\omega) \subseteq A\} = K'(A).$$

(d) All properties except for (iv) was used in order to prove Part (c).

9.3 We show that in Kripke's S5 system, the fourth property, is a consequence of the other four properties. Indeed, for Y it follows directly by (i). For every $A \subseteq Y$ it follows directly by (v),

$$K_i A = \left((K_i A)^c\right)^c = K_i \left(\left((K_i A)^c\right)^c\right) = K_i(K_i A).$$

9.4

$$
\begin{aligned}
K_1 A &= \{1\}, &(61)\\
K_2 A &= \emptyset, &(62)\\
K_2 K_1 A &= \emptyset, &(63)\\
K_1 K_2 A &= \emptyset, &(64)\\
K_1 B &= \{1,2,3,4,5\}, &(65)\\
K_2 B &= \{1,2,3,4,5,6\}, &(66)\\
K_2 K_1 B &= \{1,2,3,4\}, &(67)\\
K_1 K_2 B &= \{1,2,3,4,5\}, &(68)\\
K_1 K_2 K_1 B &= \{1,2,3\}, &(69)\\
K_2 K_1 K_2 B &= \{1,2,3,4\}. &(70)
\end{aligned}
$$

9.5 An Aumann model of incomplete information that describes the situation of Emily, Marc, and Thomas is $(N, Y, (\mathcal{F}_i)_{i \in N}, s)$ where

- $N = \{$(E)mily, (M)arc, (T)homas$\}$.
- $S = Y = \left\{ \begin{array}{c} (p_E, p_M, p_T), (p_E, p_M, n_T), (p_E, n_M, p_T), (p_E, n_M, n_T), \\ (n_E, p_M, p_T), (n_E, p_M, n_T), (n_E, n_M, p_T), (n_E, n_M, n_T) \end{array} \right\}.$

  where p_i is associated with the situation in which $i \in N$ is a poet and n_i is associated with the situation in which $i \in N$ is a novelist.
- $\mathcal{F}_E = \left\{ \begin{array}{c} \{(p_E, p_M, p_T)\}, \{(p_E, p_M, n_T)\}, \{(p_E, n_M, p_T)\}, \{(p_E, n_M, n_T)\}, \\ \{(n_E, p_M, p_T), (n_E, p_M, n_T), (n_E, n_M, p_T), (n_E, n_M, n_T)\} \end{array} \right\},$

  $\mathcal{F}_M = \left\{ \begin{array}{c} \{(p_E, p_M, p_T)\}, \{(p_E, p_M, n_T)\}, \{(n_E, p_M, n_T)\}, \{(n_E, p_M, p_T)\}, \\ \{(p_E, n_M, p_T), (p_E, n_M, n_T), (n_E, n_M, p_T), (n_E, n_M, n_T)\} \end{array} \right\},$

  $\mathcal{F}_T = \left\{ \begin{array}{c} \{(p_E, p_M, p_T)\}, \{(n_E, p_M, p_T)\}, \{(p_E, n_M, p_T)\}, \{(n_E, n_M, p_T)\}, \\ \{(p_E, p_M, n_T), (n_E, p_M, n_T), (p_E, n_M, n_T), (n_E, n_M, n_T)\} \end{array} \right\},$
- $s(\omega) = \omega$ for every $\omega \in Y$.

9.6 An Aumann model of incomplete information is given by

- $N = \{I, (J)uliet\}$.
- $Y = \{\omega_1, \omega_2, \omega_3, \omega_4, \omega_5\}$. where both ω_1 and ω_2 are associated with the situation in which I love Juliet and Juliet loves me; ω_3 is associated with the situation in which I love Juliet but Juliet does not love me; ω_4 is associated with the situation in which I do not love Juliet but Juliet loves me; ω_5 is associated with the situation in which I do not love Juliet and Juliet does not love me.

- $\mathcal{F}_I = \{\{\omega_1, \omega_2\}, \{\omega_3\}, \{\omega_4, \omega_5\}\}$,
  $\mathcal{F}_J = \{\{\omega_1\}, \{\omega_2, \omega_4\}, \{\omega_3, \omega_5\}\}$.
- The true state of the world is ω_2.

9.7 In all parts there are two states of nature: s_1 and s_2; in s_1 Mary gave birth to a baby, in s_2 she did not. The players are (M)ary and (H)erod.

(a) $Y = \{\omega_1\}$, $s(\omega_1) = s_1$, $\mathcal{F}_1$ and $\mathcal{F}_2$ are the trivial partitions.

(b) $Y = \{\omega_1, \omega_2\}$, $s(\omega_1) = s_1$, $s(\omega_2) = s_2$, $\mathcal{F}_H = \{\{\omega_1, \omega_2\}\}$ and $\mathcal{F}_M = \{\{\omega_1\}, \{\omega_2\}\}$. The true state of the world is ω_1.

(c) Like in Part (a).

(d) $Y = \{\omega_1, \omega_2, \omega_3\}$, $s(\omega_1) = s_1$, $s(\omega_2) = s_1$, $s(\omega_3) = s_2$, $\mathcal{F}_H = \{\{\omega_1\}, \{\omega_2, \omega_3\}\}$ and $\mathcal{F}_M = \{\{\omega_1, \omega_2\}, \{\omega_3\}\}$. The true state of the world is ω_1.

(e) Like in Part (d), the true state of the world is ω_2.

9.8 (a) In this situation we consider 4 states of the world regarding the knowledge possessed by Romeo, Tybalt, and Juliet regarding the content of the letter: in ω_1 only Romeo knows the content of the letter; in ω_2 only Romeo and Tybalt know the content of the letter; in ω_3 only Romeo and Juliet know the content of the letter; and in ω_4 Romeo, Tybalt and Juliet know the content the letter. Hence, an Aumann model of incomplete information is given by $Y = \{\omega_1, \omega_2, \omega_3, \omega_4\}$. The player set is $N = \{$(R)omeo, (T)ybalt, (J)uliet$\}$. Finally,

$$\mathcal{F}_R = \{\{\omega_1, \omega_2, \omega_3, \omega_4\}\},$$
$$\mathcal{F}_T = \{\{\omega_1\}, \{\omega_2\}, \{\omega_3\}, \{\omega_4\}\},$$
$$\mathcal{F}_J = \{\{\omega_1, \omega_2\}, \{\omega_3, \omega_4\}\}.$$

The true state of the world is ω_4.

(b) In the true state of the world, Romeo does not know that Juliet has read the letter ($K_R(\{\omega_3, \omega_4\}) = \emptyset$) since it was not done in front of him, furthermore, he cannot be sure that Juliet got the letter.

(c) By Part (b) it follows that in the true state of the world Tybalt does not know that Romeo know that Juliet has read the letter ($K_T K_R(\{\omega_3, \omega_4\}) = \emptyset$). Furthermore, Tybalt knows that Romeo does not know that Juliet has read the letter ($K_T((K_R(\{\omega_3, \omega_4\})^c) = Y$).

(d) In this new situation we consider an additional state of the world ω_4^* in which Romeo, Tybalt and Juliet know the content of the letter and Juliet knows that Tybalt read the letter. An Aumann model of incomplete information is given by $Y = \{\omega_1, \omega_2, \omega_3, \omega_4, \omega_4^*\}$. The player set is $N = \{$(R)omeo, (T)ybalt, (J)uliet$\}$. Finally,

$$\mathcal{F}_R = \left\{\{\omega_1, \omega_2, \omega_3, \omega_4, \omega_4^*\}\right\},$$
$$\mathcal{F}_T = \left\{\{\omega_1\}, \{\omega_2\}, \{\omega_3\}, \{\omega_4, \omega_4^*\}\right\},$$
$$\mathcal{F}_J = \left\{\{\omega_1, \omega_2\}, \{\omega_3, \omega_4\}, \{\omega_4^*\}\right\}.$$

The true state of the world is ω_4.

9.9 (a) Denote by $\omega_{c_1c_2c_3}$ a state of the world where c_1, c_2 and c_3 are the hat colors of George, John, and Thomas, respectively, in the set of colors $\{$(R)ed,(W)hite$\}$.

An Aumann model of incomplete information is given by $Y = \{\omega_{RRR}, \omega_{RRW}, \omega_{RWR}, \omega_{WRR}, \omega_{RWW}, \omega_{WRW}, \omega_{WWR}\}$. The player set is $N = \{$(G)eorge, (J)ohn, (T)homas$\}$.

(b) The partitions of George, John, and Thomas after James's announcement are

$$\mathcal{F}_G = \left\{\{\omega_{RRR}, \omega_{RRW}, \omega_{RWR}, \omega_{RWW}, \omega_{WRR}, \omega_{WRW}, \omega_{WWR}\}\right\},$$
$$\mathcal{F}_J = \left\{\{\omega_{RRR}, \omega_{RRW}, \omega_{RWR}, \omega_{RWW}\}, \{\omega_{WRR}, \omega_{WRW}, \omega_{WWR}\}\right\}.$$
$$\mathcal{F}_T = \left\{\{\omega_{RRR}, \omega_{RRW}\}, \{\omega_{RWR}, \omega_{RWW}\}, \{\omega_{WRR}, \omega_{WRW}\}, \{\omega_{WWR}\}\right\}.$$

(c) The partitions of George and John after Thomas's response and before John responded to James's question are

$$\mathcal{F}_G = \left\{\{\omega_{RRR}, \omega_{RRW}, \omega_{RWR}, \omega_{RWW}, \omega_{WRR}, \omega_{WRW}\}, \{\omega_{WWR}\}\right\},$$
$$\mathcal{F}_J = \left\{\{\omega_{RRR}, \omega_{RRW}, \omega_{RWR}, \omega_{RWW}\}, \{\omega_{WRR}, \omega_{WRW}\}, \{\omega_{WWR}\}\right\}.$$

(d) George's partition after hearing John's response is

$$\mathcal{F}_G = \left\{\{\omega_{RRR}, \omega_{RRW}, \omega_{RWR}, \omega_{RWW}\}, \{\omega_{WRR}, \omega_{WRW}\}, \{\omega_{WWR}\}\right\}.$$

(e) Since both Thomas and John claimed that they do not know their hat color, then the true state of the world is necessarily $\{\omega_{RRR}, \omega_{RRW}, \omega_{RWR}, \omega_{RWW}\}$ so George knows that his hat color is Red, also he does not know the true state of the world.

9.10 **The proof Corollary 9.16:** let $(N, Y, (\mathcal{F}_i)_{i\in N}, s, \omega_*)$ be a situation of incomplete information over a set of states of nature S. We show that it uniquely determines a knowledge hierarchy among the players over the set of states of nature S in state of the world ω_*.
Let $C \subseteq S$ be a subset of states of nature. Denote

$$s^{-1}(C) = \{\omega \in Y : s(\omega) \in C\} \subseteq Y.$$

In state of the world ω_* player i_1 knows that player i_2 knows that player i_3 knows ...that player i_l knows event C in S if and only if $\omega_* \in K_{i_1}K_{i_2}...K_{i_l}C$.
However, by Theorem 9.15, every situation of incomplete information $(N, Y, (\mathcal{F}_i)_{i\in N}, s, \omega_*)$ uniquely determines a knowledge hierarchy over the set of states of the world Y in state of the world ω_*. Hence, for every $A \subseteq Y$, in particular for $A = s^{-1}(C)$, and any finite sequence $i_1, i_2, ..., i_l$ of players in N, either $\omega_* \in K_{i_1}K_{i_2}...K_{i_l}A$ or $\omega_* \notin K_{i_1}K_{i_2}...K_{i_l}A$, as needed.

9.11 The connected components in the graph corresponding to the given Aumann model are $\{1\}$, $\{2,3,4,5,7\}$ and $\{6,8,9\}$.
Every event $A \subset Y$ such that $1 \in A$ is common knowledge in state of the world $\omega = 1$.
Every event $A \subset Y$ such that $\{2,3,4,5,7\} \subseteq A$ is common knowledge in state of the world $\omega = 5$.
Every event $A \subset Y$ such that $\{6,8,9\} \subseteq A$ is common knowledge in state of the world $\omega = 9$.

9.12 **Example 9.12:** We first show that every two states of the world must be in the same connected component in the graph corresponding to the given Aumann model. Let $\omega_{c_1c_2c_3}, \omega_{c'_1c'_2c'_3} \in Y$ be two different states of the world where c_1, c_2, c_3 (similarly c'_1, c'_2, c'_3) are the hat colors of Anthony, Betty, and Carol, respectively. Indeed, $\omega_{c_1c_2c_3} \in F_1(\omega_{c'_1c_2c_3})$, $\omega_{c'_1c_2c_3} \in F_2(\omega_{c'_1c'_2c_3})$, and $\omega_{c'_1c'_2c_3} \in F_3(\omega_{c'_1c'_2c'_3})$. In conclusion, $\omega_{c_1c_2c_3}$ and $\omega_{c'_1c'_2c'_3}$ must be in the same connected component.
Therefore, there is only one connected component, Y, and hence the only event that is common knowledge is Y.
Example 9.13: We show that for every state of the world $\omega \in Y$, $\omega \in C(\omega_{b,\emptyset})$. Hence, there is only one connected component, Y from which it follows that the only event that is common knowledge is Y. Indeed,

$$\omega_{b,H}, \omega_{b,T}, \omega_{b,TH} \in F_A(\omega_{b,\emptyset}) \Rightarrow \omega_{b,H}, \omega_{b,T}, \omega_{b,TH} \in C(\omega_{b,\emptyset}),$$
$$\omega_{y,H}, \omega_{y,\emptyset} \in F_T(\omega_{b,\emptyset}) \Rightarrow \omega_{y,H}, \omega_{y,\emptyset} \in C(\omega_{b,\emptyset}),$$
$$\omega_{y,T} \in F_H(\omega_{b,\emptyset}) \Rightarrow \omega_{y,T} \in C(\omega_{b,\emptyset}).$$

Finally, since $\omega_{y,TH} \in F_T(\omega_{b,T})$ and $\omega_{b,T} \in F_A(\omega_{b,\emptyset})$ it follows that $\omega_{y,TH} \in C(\omega_{b,\emptyset})$, as claimed.

9.13 (a) The connected components are $\{1, 2, 3, 4, 5, 6\}$ and $\{7, 8, 9\}$.

(b) The events that are common knowledge at ω are all the events that contain the connected component that includes ω.

(c) At $\omega = 1$ it is true that $K_1 A$ and $K_2 A$, but it is not true that $K_2 K_1 A$ (player 2 cannot rule out the possibility that $\omega = 5$, in which case player 1 does not know A). Similarly, at $\omega = 1$ it is not true that $K_1 K_2 A$ (player 1 cannot rule out the possibility that $\omega = 2$, in which case player 2 does not know A).

9.14 (a) William knows the correct time at $d_1 d_2 : d_3 d_4$ where either (1) $d_1 \in \{0, 1\}$, $d_2 \in \{0, 1, 4, 7\}$, $d_3 \in \{0, 1, 4, 5\}$ and $d_4 \in \{0, 1, 4, 7\}$; or (2) $d_1 = 2$, $d_2 \in \{0, 1\}$, $d_3 \in \{0, 1, 4, 5\}$ and $d_4 \in \{0, 1, 4, 7\}$.

(b) Dan knows the correct time at $d_1 d_2 : d_3 d_4$ where either (1) $d_1 \in \{0, 1\}$, $d_2 \in \{0, 2, 4\}$, $d_3 \in \{0, 1, 2, 4\}$ and $d_4 \in \{0, 2, 4\}$; or (2) $d_1 = 2$, $d_2 \in \{0, 1, 2, 3\}$, $d_3 \in \{0, 1, 2, 4\}$ and $d_4 \in \{0, 2, 4\}$.

(c) William knows that Dan knows the correct time at $d_1 d_2 : d_3 d_4$ where either (1) $d_1 \in \{0, 1\}$, $d_2 \in \{0, 4\}$, $d_3 \in \{0, 1, 4\}$ and $d_4 \in \{0, 4\}$; or (2) $d_1 = 2$, $d_2 \in \{0, 1, 2, 3\}$, $d_3 \in \{0, 1, 4\}$ and $d_4 \in \{0, 4\}$.

(d) Dan knows that William knows the correct time at $d_1 d_2 : d_3 d_4$ where either (1) $d_1 \in \{0, 1\}$, $d_2 \in \{0, 1, 4, 7\}$, $d_3 \in \{0, 1, 4\}$ and $d_4 \in \{0, 1, 4, 7\}$; or (2) $d_1 = 2$, $d_2 \in \{0, 1\}$, $d_3 \in \{0, 1, 4\}$ and $d_4 \in \{0, 1, 4, 7\}$.

(e) The correct time is common knowledge among William and Dan at $d_1 d_2 : d_3 d_4$ where either (1) $d_1 \in \{0, 1\}$, $d_2 \in \{0, 4\}$, $d_3 \in \{0, 1, 4\}$ and $d_4 \in \{0, 4\}$; or (2) $d_1 = 2$, $d_2 \in \{0, 1\}$, $d_3 \in \{0, 1, 4\}$ and $d_4 \in \{0, 4\}$.

(f) Let $H = \{0, 1, ..., 23\}$ be the set of hours and let $D_3 = \{0, 1, ...5\}$ and $D_4 = \{0, 1, ..., 9\}$. Then $h : d_3 d_4$ is an hour in the digital clock, for every $h \in H, d_3 \in D_3$ and $d_4 \in D_4$.

Aumann model of incomplete information describing the given situation: The set of states of the world is $Y = H \times D_3 \times D_4$ which contains 1440 states. The players are $N = \{$(W)illiam, (D)an$\}$. To define the partition of William we first define partitions over H,

D_3 and D_4. Let

$$\mathcal{H}_{\mathrm{W}} = \left\{ \begin{array}{c} \{0\},\{1\},\{2,3\},\{4\},\{5,6\},\{7\},\{8,9\}, \\ \{10\},\{11\},\{12,13\},\{14\},\{15,16\},\{17\},\{18,19\}, \\ \{20\},\{21\},\{22,23\} \end{array} \right\}$$

$$\mathcal{D}_{3,\mathrm{W}} = \Big\{\{0\},\{1\},\{2,3\},\{4\},\{5\}\Big\}$$

$$\mathcal{D}_{4,\mathrm{W}} = \Big\{\{0\},\{1\},\{2,3\},\{4\},\{5,6\},\{7\},\{8,9\}\Big\}$$

$$\Rightarrow \mathcal{F}_{\mathrm{W}} = \left\{A \times B \times C : A \in \mathcal{H}_{\mathrm{W}},\ B \in \mathcal{D}_{3,\mathrm{W}},\ C \in \mathcal{D}_{4,\mathrm{W}}\right\}.$$

Similarly,

$$\mathcal{H}_{\mathrm{D}} = \left\{ \begin{array}{c} \{10\},\{11,17\},\{12\},\{13,15,19\}\{14\},\{16,18\}, \\ \{0\},\{1,7\},\{2\},\{3,5,9\}\{4\},\{6,8\}, \\ \{20\},\{21\},\{22\},\{23\} \end{array} \right\}$$

$$\mathcal{D}_{3,\mathrm{D}} = \Big\{\{0\},\{1\},\{2\},\{3,5\},\{4\}\Big\}$$

$$\mathcal{D}_{4,\mathrm{D}} = \Big\{\{0\},\{1,7\},\{2\},\{3,5,9\},\{4\},\{6,8\},\Big\}$$

$$\Rightarrow \mathcal{F}_{\mathrm{D}} = \left\{A \times B \times C : A \in \mathcal{H}_{\mathrm{D}},\ B \in \mathcal{D}_{3,\mathrm{D}},\ C \in \mathcal{D}_{4,\mathrm{D}}\right\}.$$

9.15 Suppose that both A and B are common knowledge at ω. Then they both contain the connected component $C(\omega)$ that includes ω, but then their intersection $A \cap B$ contains this connected component as well. In particular, $A \cap B$ is common knowledge at ω as well.

9.16 Given an Aumann model of incomplete information, we prove that event A is common knowledge in every state of the world in A if and only if $K_1K_2...K_nA = A$, where $N = \{1, 2, ..., n\}$ is the set of players.
Suppose first that event A is common knowledge in every state of the world in A. Thus, for every $\omega \in A$ and every finite sequence of players $i_1, i_2, ..., i_l$, $\omega \in K_{i_1}K_{i_2}...K_{i_{l-1}}K_{i_l}A$. Hence, for every finite sequence of players $i_1, i_2, ..., i_l$, $A \subseteq K_{i_1}K_{i_2}...K_{i_{l-1}}K_{i_l}A$. In particular, $A \subseteq K_1K_2...K_{n-1}K_nA$. On the other hand, applying Theorem 9.9 inductively leads to $K_1K_2...K_{n-1}K_nA \subseteq A$. Thus $K_1K_2...K_{n-1}K_nA = A$.
We next prove the other direction. Assume that $K_1K_2...K_{n-1}K_nA = A$. Therefore,

$$A = K_1K_2...K_{n-1}K_nA \subseteq K_2...K_{n-1}K_nA \subseteq ... \subseteq K_nA \subseteq A.$$

Therefore, all weak inclusions are equalities and thus $K_nA = A$, $K_{n-1}A = K_{n-1}K_nA = A$,..., $K_1A = A$. In conclusion, for every $\omega \in A$ and for every finite sequence of players $i_1, i_2, ..., i_l$, $\omega \in A = K_{i_1} K_{i_2} ... K_{i_{l-1}} K_{i_l} A$ so A is common knowledge in every state of the world in A.

9.17 Consider an Aumann model of incomplete information with n players and let $A \subseteq Y$ be an event that is common knowledge among the players in state of the world ω. That is, for every finite sequence of players $i_1, i_2, ..., i_l \in N$, $\omega \in K_{i_1} K_{i_2} ... K_{i_{l-1}} K_{i_l} A$. In particular, for every subset of players $M \subseteq N$, and every finite sequence of players $i_1, i_2, ..., i_l \in M \subseteq N$, $\omega \in K_{i_1} K_{i_2} ... K_{i_{l-1}} K_{i_l} A$. Therefore, A is also common knowledge among M.

9.18 Consider an Aumann model of incomplete information where $Y = \{\omega_1, \omega_2, \omega_3\}$, $N = \{1, 2, 3\}$, and the partitions are

$$\begin{aligned}\mathcal{F}_1 &= \{\{\omega_1, \omega_2\}, \{\omega_3\}\},\\ \mathcal{F}_2 &= \{\{\omega_1\}, \{\omega_2, \omega_3\}\},\\ \mathcal{F}_3 &= \{\{\omega_1\}, \{\omega_2\}, \{\omega_3\}\}.\end{aligned}$$

The event $A = \{\omega_1\}$ is not common knowledge among all the players N, but is common knowledge among players $2, 3$.

9.19 (a) In state of the world ω, Andrew knows that Sally knows the state of nature. This does not imply that Andrew knows the state of nature in ω. Furthermore, the fact that Sally knows the state of nature is not necessarily common knowledge among Andrew and Sally in ω.
Suppose that

$$\begin{aligned}Y &= \{\omega_1, \omega_2, \omega_3, \omega_4, \omega_5\},\\ S &= \{s_1, s_2, s_3, s_4\},\\ s(\omega) &= \begin{cases} s_1 & \omega = \omega_1 \\ s_2 & \omega = \omega_2, \omega_3 \\ s_3 & \omega = \omega_4 \\ s_4 & \omega = \omega_5 \end{cases},\\ \mathcal{F}_{\text{Andrew}} &= \left\{\{\omega_1, \omega_2\}, \{\omega_3, \omega_4, \omega_5\}\right\},\\ \mathcal{F}_{\text{Sally}} &= \left\{\{\omega_1\}, \{\omega_2, \omega_3\}, \{\omega_4, \omega_5\}\right\},\end{aligned}$$

Then in ω_2, Andrew knows that Sally knows the state of nature but Andrew does not know the state of nature and the fact that Sally knows the state of nature is not common knowledge among Andrew and Sally.

(b) In every state of the world, Andrew knows that Sally knows the state of nature. This does not imply that Andrew knows the state of nature in every state of the world (a counter example can be obtain by eliminating ω_5 in the situation presented in Part (a)).

However, the fact that Sally knows the state of nature is common knowledge among Andrew and Sally in every state of the world. Indeed, since in every state of the world Andrew knows that Sally knows the state of nature, it follows that in every state of the world Sally knows the state of nature, hence the set of states of the world where Sally knows the state of nature is Y, and the set Y is always common knowledge among the players.

(c) In state of the world ω, Andrew knows that Sally knows the state of the world. This does not imply that Andrew knows the state of the world in ω (for instance $\mathcal{F}_{\text{Sally}} = \{\{\omega\} : \omega \in Y\}$ and $\mathcal{F}_{\text{Andrew}} = \{Y\}$). However, Andrew knows that Sally knows the state of the world in ω hence for every $\omega' \in F_{\text{Andrew}}(\omega)$ one has $F_{\text{Sally}}(\omega) = \{\omega\}$. Thus, the fact that Sally knows the state of the world is common knowledge among Andrew and Sally in ω.

9.20 Let $(N, Y, (\mathcal{F}_i)_{i \in N}, s, P)$ be an Aumann model of incomplete information with beliefs, and let $W \subseteq Y$ be a nonempty event. We prove that $(N, W, (F_i \cap W)_{i \in N}, P(\cdot|W))$ is also an Aumann model of incomplete information with beliefs, where for each player $i \in N$, $\mathcal{F}_i \cap W = \{F \cap W : F \in \mathcal{F}_i\}$ is the partition $\mathcal{F}_i$ restricted to W, and $P(\cdot|W)$ is the conditional distribution of P over W.
Indeed,

- N is a finite set of players;
- W as a nonempty subset of Y is a finite set of states of the world.
- For each $i \in N$, $\mathcal{F}_i$ is a partition of Y, hence $(F_i \cap W)_{i \in N}$ is a partition of W as well:

$$(F \cap W) \cap (F' \cap W) = (F \cap F') \cap W = \emptyset \;\; \forall F \neq F' \in \mathcal{F}_i$$

$$\bigcup_{F \in \mathcal{F}_i} (F \cap W) = \left(\bigcup_{F \in \mathcal{F}_i} F \right) \cap W = Y \cap W = W.$$

- $s : W \to S$ is a function associating a state of nature to every state of the world.
- $P(\cdot|W)$ is a probability distribution over W such that $P(\omega) > 0$ for each $\omega \in W$:

$$P(\omega|W) = \frac{P(\omega)}{\sum_{\omega' \in W} P(\omega')} > 0$$

$$\sum_{\omega \in W} P(\omega|W) = \sum_{\omega \in W} \frac{P(\omega)}{\sum_{\omega' \in W} P(\omega')} = 1.$$

9.21 Without the assumption that $P(\omega) > 0$ for all $\omega \in Y$, Theorem 9.29 does not obtain. Suppose that $N = \{1,2\}$, $Y = \{1,2\}$, $\mathcal{F}_1 = \{Y\}$, $P(1) = 1$, $P(2) = 0$ and $A = \{1\}$. Then $P(A|F_1(1)) = 1$ but $F_1(1) \not\subseteq A$.

9.22 Given $(N, Y, (\mathcal{F}_i)_{i\in N}, s, P)$ an Aumann model of incomplete information with beliefs and with n players. Suppose that for each $i \in N$, the fact that player i ascribes probability q_i to an event A is common knowledge among the players.
For every two players $i, j \in N$, consider an Aumann model of incomplete information with beliefs $(\{i,j\}, Y, (\mathcal{F}_i, \mathcal{F}_j), s, P)$. By Exercise 9.17, the fact that player i (similarly player j) ascribes probability q_i (similarly q_j) to an event A is common knowledge among i, j.
Therefore, using Theorem 9.32, $q_i = q_j$.
However, since $q_i = q_j$ for every $i, j \in N$, it follows that

$$q_1 = q_2 = ... = q_n.$$

9.23 (a) The situation as a Harsanyi model of incomplete information is given by: a set of players $N = \{1,2,3\}$, for every player $i \in N$ the set of types is $T_i = \{R, W\}$ where R is for red hat and W for white hat. Denote by $T = T_1 \times T_2 \times T_3$ the set of type vectors. The vector of types corresponding to the true situation is (R, R, R).

(b) The situation as an Aumann model of incomplete information is given by: a set of players $N = \{1,2,3\}$, a set of world states $Y = \{(c_1, c_2, c_3) : c_1, c_2, c_3 \in \{R, W\}\}$. The partitions of the players:

$$\mathcal{F}_1 = \left\{ \begin{array}{c} \{(R,R,R),(W,R,R)\},\{(R,W,R),(W,W,R)\}, \\ \{(R,R,W),(W,R,W)\},\{(R,W,W),(W,W,W)\} \end{array} \right\},$$

$$\mathcal{F}_2 = \left\{ \begin{array}{c} \{(R,R,R),(R,W,R)\},\{(W,R,R),(W,W,R)\}, \\ \{(R,R,W),(R,W,W)\},\{(W,W,W),(W,R,W)\} \end{array} \right\},$$

$$\mathcal{F}_3 = \left\{ \begin{array}{c} \{(R,R,R),(R,R,W)\},\{(R,W,R),(R,W,W)\}, \\ \{(W,R,R),(W,R,W)\},\{(W,W,R),(W,W,W)\} \end{array} \right\}.$$

The true state of the world is (R, R, R).

(c) No one leaves the room after few rings since the knowledge of each player i remains $F_i(R, R, R)$ as was before the rings.

(d) Before the the first announcement: the knowledge of player 1 is $\{(R,R,R),(W,R,R)\}$; the knowledge of player 2 is $\{(R,R,R),(R,W,R)\}$;

and the knowledge of player 3 is $\{(R,R,R),(R,R,W)\}$. Therefore, the common knowledge is Y.

After the first announcement: it is common knowledge that the state of world is not (W,W,W). However, no one knows for sure the color of his hat.

After the second announcement: since no player left after the first announcement, it becomes a common knowledge that no player was sure that he is wearing a red hat thus, it becomes common knowledge that the state of world is also not (W,W,R), (W,R,W) and (R,W,W). Equivalently,

$$\Big\{(R,R,R),(R,R,W),(R,W,R),(W,R,R)\Big\}.$$

is common knowledge.

Finally, since no player left the room after the second announcement, it become common knowledge that each player see two red hats and thus each player knows that the state of the world is (R,R,R) and hence they all leave the room.

(e) The information that at least one person in the room was wearing a red hat becomes common knowledge by the announcement.

(f) In general: n individuals are seated in a room. Each one of them is wearing a hat, which may be either red or white. Each of them sees the hats worn by the others, but cannot see his own hat (and in particular does not know its color). The true situation is that every person in the room is wearing a red hat. A stranger enters the room, holding a bell. Once a minute, he rings the bell while saying "At least one of you is wearing a red hat." He continues to ring the bell once a minute and requesting that those who know their hat to be red to leave. Then after the n-th ring, all n hat-wearers will leave the room. The proof is similar to the proof for three players.

9.24 Prove that in an Aumann model of incomplete information with a common prior P, if in a state of the world ω Player 1 knows that Player 2 knows A, then $P(A|F_1(\omega)) = 1$. That is, $\omega \in K_1K_2A$. Therefore, $F_1(\omega) \subseteq K_2A \subseteq A$. However, $P(F_1(\omega)) > 0$ and so $P(A|F_1(\omega)) = 1$.

9.25 (a) Since $\omega_* = 9$, initially Player I learns that the true state is in $\{7,8,9\}$, and Player II learns that the true state is in $\{9\}$. Hence, the conditional probability that Player I (given his information) ascribes to the event A is $\mathbf{P}(A \mid F_1(9)) = \frac{1}{3}$.

(b) In $\omega_* = 9$ Player II knows the true state of the world, so that the conditional probability he ascribes to A is 1.

(c) The announcement of Player I will not affect the probability that Player II ascribes to the event A since Player II already knows the true state of the world.

(d) Only if $\omega_* = 9$ the conditional probability of player 2 is 1 (in other states of the world it is $\frac{1}{4}$). Thus, the announcement reveals to Player I that the true state is 9, and therefore the conditional probability of Player I becomes 1.

(e) Now both players know the true state of the world, and their conditional probabilities remain 1.

(f) Suppose $\omega_* = 8$. Initially Player I learns that the true state is in $\{7, 8, 9\}$, and Player II learns that the true state is in $\{5, 6, 7, 8\}$. The first announcement of Player I is $\frac{1}{3}$. That announcement does not effect Player II that announces the conditional probability $\frac{1}{4}$. This tells Player I that the true state is not 9 (since if 9 was the true state, Player II's announcement would have been 1). Therefore, Player I knows that the true state is in $\{7, 8\}$, and the conditional probability that he assigns to A is 0. Given this announcement Player II realizes that the true state is in $\{7, 8\}$: if it was either 5 or 6 Player I's announcement would have been $\frac{1}{3}$ and not 0. Thus, his conditional probability is 0, and from now on no player will learn anything new.

(g) Suppose $\omega_* = 6$. Initially Player I learns that the true state is in $\{4, 5, 6\}$, and Player II learns that the true state is in $\{5, 6, 7, 8\}$. The first announcement of Player II is $\frac{1}{4}$. This tells Player I that the true state is not 9, however its conditional probability that he assigns to A remains $\frac{1}{3}$ which he reveals. Therefore Player II knows that the true state is neither 7 nor 8 and thus the true state is in $\{5, 6\}$. In particular, the conditional probability that he assigns to A is $\frac{1}{2}$. Given this announcement Player I realizes that the true state is in $\{5, 6\}$ thus his conditional probability is $\frac{1}{2}$ as well, and from now on no player will learn anything new.

(h) Suppose $\omega_* = 4$. Initially Player I learns that the true state is in $\{4, 5, 6\}$, and Player II learns that the true state is in $\{1, 2, 3, 4\}$. The first announcement of Player II is $\frac{1}{4}$. This tells Player I that the true state is not 9 but its conditional probability that he assigns to A remains $\frac{1}{3}$. Player II knows that Player I knows that true state is neither 7 nor 8 and announces the same probability, $\frac{1}{4}$. Therefore, Player I knows that the true state is 4, so the conditional probability that he assigns to A is 1. Given this announcement Player II realizes that the true state is in 4 thus his conditional probability is 1, and from now on no player will learn anything new.

9.26 Suppose that $\omega_* = 3$. Initially Player I learns that the true state is in $\{3,4\}$, and he announces the conditional probability he ascribes to the event A which is $\frac{1}{2}$.
Player II learns that the true state is in $\{1,3,5\}$. Furthermore, by the announcement of Player I he knows that the true state is not 5 so he reveals his conditional probability which is $\frac{1}{2}$.
The announcement of Player II tells Player I that the true state is 3 hence the conditional probability he assign to A becomes 0.
Now both players know the true state, and their conditional probabilities remain 0.

9.27 Assume that the two players have different priors over Y :

$$P_I(\omega) = \frac{\omega}{45}, \quad P_{II}(\omega) = \frac{10-\omega}{45} \quad \forall \omega \in Y.$$

Suppose that $\omega = 9$**.** At the beginning

$$F_I(\omega) = \{7,8,9\} \Rightarrow \qquad P_I(A|F_I(\omega)) = \frac{\frac{9}{45}}{\frac{7+8+9}{45}} = \frac{3}{8}$$

$$F_{II}(\omega) = \{9\} \Rightarrow \qquad P_{II}(A|F_{II}(\omega)) = \frac{\frac{1}{45}}{\frac{1}{45}} = 1.$$

Since Player II knows the true state of the world, no announcement affects him.
On the other hand, as soon as Player II announce that he assign probability 1 to A, Player I learns that the true state of the world is 9, both of them announce 1.
Suppose that $\omega = 8$**.** At the beginning

$$F_I(\omega) = \{7,8,9\} \Rightarrow \qquad P_I(A|F_I(\omega)) = \frac{\frac{9}{45}}{\frac{7+8+9}{45}} = \frac{3}{8}$$

$$F_{II}(\omega) = \{5,6,7,8\} \Rightarrow \qquad P_{II}(A|F_{II}(\omega)) = \frac{\frac{5}{45}}{\frac{5+4+3+2}{45}} = \frac{5}{14}.$$

Player I announces the above probability, $\frac{3}{8}$, from which Player II deduces that the true state must also be in $\{7,8,9\}$ (whereas $P_I(A|\{4,5,6\}) = \frac{1}{3}$). Hence he knows that the true state is in $\{7,8\}$ and announces that the probability which he ascribes to the event A is 0. From now on it is a common knowledge that the true state is in $\{7,8\}$ and that the probability that they both assign to A is 0.
Suppose that $\omega = 6$**.** At the beginning

$$F_I(\omega) = \{4,5,6\} \Rightarrow \qquad P_I(A|F_I(\omega)) = \frac{\frac{5}{45}}{\frac{4+5+6}{45}} = \frac{1}{3}$$

$$F_{II}(\omega) = \{5,6,7,8\} \Rightarrow \qquad P_{II}(A|F_{II}(\omega)) = \frac{\frac{5}{45}}{\frac{5+4+3+2}{45}} = \frac{5}{14}.$$

Player I announces the above probability, $\frac{1}{3}$, from which Player II deduces that the true state must also be in $\{4,5,6\}$ (whereas $P_I(A|\{7,8,9\}) = \frac{3}{8}$). Hence he knows that the true state is in $\{5,6\}$ and announces that the probability which he ascribes to the event A is $\frac{5}{9}$. Hence, Player I knows that the true state cannot be 4. From now on, it is common knowledge that the true state is in $\{5,6\}$. Furthermore, from now on, the probabilities that Player I and Player II assign to A are $\frac{5}{11}$ and $\frac{5}{9}$, respectively.

9.28 Let $(N, Y, \mathcal{F}_I, \mathcal{F}_{II}, s, P)$ be an Aumann model of incomplete information with beliefs in which $N = \{I, II\}$ and let $A \subseteq Y$ be an event. Consider the following process:

- Player I informs Player II of the conditional probability $P(A|F_I(\omega))$.
- Player II informs Player I of the conditional probability that he ascribes to event A given the partition element $F_{II}(\omega)$ and Player I's announcement.
- Player I informs Player II of the conditional probability that he ascribes to event A given the partition element $F_I(\omega)$ and all the announcements so far.
- Repeat indefinitely.

(a) We prove that the sequence of conditional probabilities that Player I announces converges; that the sequence of conditional probabilities that Player II announces also converges; and that both sequences converge to the same limit.

Denote by Y^n the minimal set of states of the world that after the n-th announcement is common knowledge among the players that the true state is in that set. Denote by q_I^{2n-1} (q_{II}^{2n}) the conditional probability that Player I (Player II) assigns to A at stage $2n-1$ ($2n$).

At stage 1: Player I announces $q_I^1 = P(A|F_1(\omega))$ from which it become common knowledge that the true state is in

$$Y^1 = \{\omega' \in Y : P(A|F_1(\omega')) = q_I^1\}.$$

At stage 2: Player II announces $q_{II}^2 = P(A|F_2(\omega) \cap Y^1)$ from which it become common knowledge that the true state is in

$$Y^2 = \{\omega' \in Y^1 : P(A|F_2(\omega') \cap Y^1) = q_{II}^2\}.$$

In general, at stage $2n-1$ for $n > 1$: Player I announces $q_I^{2n-1} = P(A|F_1(\omega) \cap Y^{2n-2})$ so it is common knowledge that the true state is in

$$Y^{2n-1} = \{\omega' \in Y^{2n-2} : P(A|F_1(\omega') \cap Y^{2n-2}) = q_I^{2n-1}\}.$$

At stage $2n$: Player II announces $q_{II}^{2n} = P(A|F_2(\omega) \cap Y^{2n-1})$ and

$$Y^{2n} = \{\omega' \in Y^{2n} : P(A|F_2(\omega') \cap Y^{2n-1}) = q_{II}^{2n}\}.$$

Note that (Y^n) is weakly decreasing (with respect to set inclusion) and $Y^1 \subseteq Y$ is finite. Hence, there is $n^* \in \mathbf{N}$ such that $Y^m = Y^{n^*}$ for every $m > n^*$ (n^* is the minimal n for which there are three stages at which $Y^n = Y^{n+1} = Y^{n+2}$). In addition, since $\omega \in Y^n$ for every n, $Y^{n^*} \neq \emptyset$. In particular, from stage n^* onwards q_I^{2n-1} and q_{II}^{2n} converge to q_I and q_{II} respectively. We show that $q_I = q_{II}$.
As proved in Exercise 9.20 $\mathcal{F}_I \cap Y^{n^*} = \{F \cap Y : F \in \mathcal{F}_I\}$ is a partition of Y^{n^*}. For every $\omega' \in Y^{n^*}$, $P(A|F_1(\omega') \cap Y^{n^*}) = q_I$. Therefore,

$$\begin{aligned}
P(A|Y^{n^*}) &= \sum_{\substack{F\in\mathcal{F}_I \\ F\cap Y^{n^*}\neq\emptyset}} P(F \cap Y^{n^*}|Y^{n^*})P(A|F \cap Y^{n^*}) \\
&= \sum_{\substack{F\in\mathcal{F}_I \\ F\cap Y^{n^*}\neq\emptyset}} P(F \cap Y^{n^*}|Y^{n^*})q_I \\
&= q_I \sum_{\substack{F\in\mathcal{F}_I \\ F\cap Y^{n^*}\neq\emptyset}} P(F \cap Y^{n^*}|Y^{n^*}) = q_I.
\end{aligned}$$

The proof that $P(A|Y^{n^*}) = q_{II}$ is identical and thus it was omitted. In conclusion $q_I P(A|Y^{n^*}) = q_{II}$, as claimed.

(b) As mentioned before (in Part(a)), if along two sequential stages the set Y^N remains unchanged, the sequence of announcements made by the players becomes constant. Hence, the process continue as long as at least one state of the world is eliminated at every two stages at most. However, since $\omega \in Y^{n^*}$, after at most $2|Y|$ announcements the announcements made by the players becomes constant.

9.29 Let C be the connected component that contains ω_*. Then $C = \bigcup_{k=1}^K F_1^k$ is a union of disjoint atoms in $\mathcal{F}_1$. Then

$$\mathbf{E}[f \mid C] = \sum_{k=1}^{K} \mathbf{P}(F_1^k \mid C)\mathbf{E}[f \mid F_1^k] \geq 0. \tag{71}$$

Similarly, $C = \bigcup_{l=1}^L F_2^l$ is a union of disjoint atoms in $\mathcal{F}_2$, and

$$\mathbf{E}[f \mid C] = \sum_{l=1}^{L} \mathbf{P}(F_2^l \mid C)\mathbf{E}[f \mid F_2^l] \leq 0. \tag{72}$$

This implies that $\mathbf{E}[f \mid C] = 0$. Since the summation in (71) is of non-negative numbers this implies that each summand is 0, and since the summation in (72) is of non-negative numbers this implies that each summand is 0 as well. In particular, the event

$$D := \{\omega \in Y : E[f \mid \mathcal{F}_I](\omega) = E[f \mid \mathcal{F}_{II}](\omega) = 0\}$$

is common knowledge in the state of the world ω^*.

9.30 Let $(N, Y, \mathcal{F}_I, \mathcal{F}_{II}, s, P)$ be an Aumann model of incomplete information with beliefs where $N = \{I, II\}$, let $f : Y \mapsto \mathbf{R}$ be a function, and let $\omega \in Y$ be a state of the world.
Denote by Y^n the minimal set of states of the world in which it is common knowledge among the players after the n-th announcement that the true state is in that set.
At stage 1: Player I announces whether or not $E[f|\mathcal{F}_I](\omega) > 0$. If he says "no", the process ends here. Otherwise, it becomes common knowledge that the true state is in

$$Y^1 = \{\omega' \in Y : E[f|\mathcal{F}_I](\omega') > 0\}.$$

Since $E[f|\mathcal{F}_I](\omega) > 0$, $\omega \in Y^1$ and the set Y^1 is nonempty.
At stage 2: Player II announces whether or not $E[f|\mathcal{F}_{II} \cap Y^1](\omega) < 0$ (note that by Exercise 9.20 $\mathcal{F}_{II} \cap Y^1$ is a partition of $Y^1 \neq \emptyset$ hence the conditional expectation is well defined). If he says "no", the process ends here. Otherwise, it becomes common knowledge that the true state is in

$$Y^2 = \{\omega' \in Y^1 : E[f|\mathcal{F}_{II} \cap Y^1](\omega') < 0\}.$$

$\omega \in Y^2$ and so Y^2 is necessarily nonempty.
In general, at stage $2n-1$ for $n > 1$, if the process has not yet ended: Player I announces whether or not $E[f|\mathcal{F}_I \cap Y^{2n-2}](\omega) > 0$. If he says "no", the process ends here. Otherwise, it becomes common knowledge that the true state is in

$$Y^{2n-1} = \{\omega' \in Y^{2n-2} : E[f|\mathcal{F}_I \cap Y^{2n-2}](\omega') > 0\}.$$

At stage $2n$ if the process has not yet ended: Player II announces whether or not $E[f|\mathcal{F}_{II} \cap Y^{2n-1}](\omega) > 0$. If he says "no", the process ends here. Otherwise, it becomes common knowledge that the true state is in

$$Y^{2n} = \{\omega' \in Y^{2n-1} : E[f|\mathcal{F}_{II} \cap Y^{2n-1}](\omega') < 0\},$$

and so on. As long as the process continues, $\omega \in Y^n$ and thus the conditional expectation of f given Y^n and $\mathcal{F}_i$ at ω is well defined.

Furthermore, (Y^n) is weakly decreasing (with respect to set inclusion) where Y^1 is finite. Hence, if by contradiction the process never ends, then there must be some stage n such that $Y^{2n} = Y^{2n+1} = Y^{2n+2}$ where for every $\omega' \in Y^{2n} = Y^{2n+1} = Y^{2n+2}$

$$E[f|\mathcal{F}_I \cap Y^{2n}](\omega') > 0$$
$$E[f|\mathcal{F}_{II} \cap Y^{2n}](\omega') = E[f|\mathcal{F}_{II} \cap Y^{2n+1}](\omega') < 0.$$

Therefore,

$$E[f|Y^{2n}] = \sum_{\omega' \in Y^{2n}} P(\omega' \mid Y^{2n}) E[f|\mathcal{F}_I \cap Y^{2n}](\omega') > 0 \text{ and}$$
$$E[f|Y^{2n}] = \sum_{\omega' \in Y^{2n}} P(\omega' \mid Y^{2n}) E[f|\mathcal{F}_{II} \cap Y^{2n}](\omega') < 0,$$

a contradiction. In conclusion, after a finite number of stages, the process ends.
In fact, as proved above, as long as the process continues, at least one state of the world must be eliminated every two sequential stages. However, if for instance we eliminate w' at some stage $2n-1$ (similarly $2n$) then $E[f|\mathcal{F}_I \cap Y^{2n-2}](\omega') \leq 0$ but hence $E[f|\mathcal{F}_I \cap Y^{2n-2}](\omega'') = E[f|\mathcal{F}_I \cap Y^{2n-2}](\omega') \leq 0$ for every $\omega'' \in F_I(\omega')$ so all states in $F_I(\omega')$ must be eliminated as well. Therefore, the number of stages prior to the end of the process is at most $\max\{2|\mathcal{F}_I| - 1, 2|\mathcal{F}_{II}| - 1\}$.

9.31 (a) in the model there are 12 states of the world (and 12 states of nature, one for each state of the world). Each state of the world is a pair (x, y), where x is the amount in Mark's envelope and y is amount in Luke's envelope. The model is therefore as follows

$$N = \{M(ark), L(uke)\}$$
$$Y = \begin{Bmatrix} (10,100), (100,1000), ..., (100000,1000000) \\ (100,10), (1000,100), ..., (1000000,100000) \end{Bmatrix}$$
$$\mathcal{F}_M = \begin{Bmatrix} \{(10,100)\}, \{(100,10), (100,1000)\}, ..., \\ \{(100000,10000), (100000,1000000)\}, \{(1000000,100000)\} \end{Bmatrix}$$
$$\mathcal{F}_L = \begin{Bmatrix} \{(100,10)\}, \{(10,100), (1000,100)\}, ..., \\ \{(10000,100000), (1000000,100000)\}, \{(100000,1000000)\} \end{Bmatrix}$$
$$p(\omega) = \tfrac{1}{12} \quad \forall \omega \in Y.$$

(b) The true state of the world in the model is $(1000, 10000)$.

(c) According to Mark's information the expected amount in Luke's envelope is $\frac{1}{2} \times 100 + \frac{1}{2} \times 10000 = 5050$.

(d) According to Luke's information the expected amount in Luke's envelope is $\frac{1}{2} \times 1000 + \frac{1}{2} \times 100000 = 50500$.

(e) Note: Each participant would like to switch envelopes unless the amount he finds in his envelope is the maximal possible amount: 1,000,000. Therefore, if, say, Mark hears that Luke wants to switch envelopes he realizes that the amount the other has in his envelope is not 1,000,000. Therefore, if Mark's envelope contains 100,000 and he hears that Luke does not want to switch he realizes that Luke's envelope contains 10,000, in which case he will refuse to switch envelopes. Hence, if Luke's envelope contains 10,000, he agreed in the past to switch (confirming that he does not have 1,000,000) and then Mark agreed to switch (Mark confirmed that he does not have 100,000), then Luke realizes that Mark has 1,000, and he will refuse to switch.

9.32 In the current situation, both, Mark and Luke, give Peter sealed envelopes with "no", because they will realize that they cannot profit: If someone found 1,000,000 in his envelope he will surely refuse to switch. Therefore, if someone found 100,000 in his envelope he will refuse to switch, since he cannot profit. And this goes on.

9.33 Suppose that Peter (Exercise 9.31) chooses the integer k randomly according to a geometric distribution with parameter $\frac{1}{2}$. Obviously, each participant would like to switch envelopes in case the amount he finds in his envelope is the minimal possible amount: 10. For every $n > 1$, if a participant finds the amount of 10^n in his envelope, then he knows that with probability $\frac{P(k=n)}{P(k=n)+P(k=n-1)} = \frac{\frac{1}{2^n}}{\frac{1}{2^n}+\frac{1}{2^{n-1}}} = \frac{1}{3}$ Petter chooses $k = n$ so the other has 10^{n+1} and with probability $\frac{2}{3}$ Petter chooses $k = n - 1$ so the other has 10^{n-1}. Hence the expected amount in the other envelope is $\frac{1}{3} \times 10^{n+1} + \frac{2}{3} \times 10^{n-1} = 34 \times 10^{n-1} > 10^n$. In particular, each participant wants to switch envelopes regardless the amount in his envelope. Therefore, no participant learns new information by the desire of the other to switch envelopes so the process described in Exercise 9.31 is endless.

9.34 In order to be absolutely certain that both commanders are attacking simultaneously, they must have common knowledge about the attack. Hence, the two commanders cannot coordinate the attack in a finite time. Indeed, whenever the courier is on the way from Division A to Division B and back, the commander of Division A does not attack since he is not certain that the courier informed Division B, and thus even if the courier reaches Division B, the commander of Division B does not attack since he knows that Division A does not attack. Similarly, whenever the courier is on the way from Division B to Division A and back, the commander of Division B does not attack since he is not certain that the courier informed Division A, and thus even if the courier reaches Division A, the commander of Division A does not attack since he knows that Division B does not attack.

9.35 **Proof of Theorem 9.42:** We show that every (Harsanyi) game with incomplete information can be described as an extensive-form game (with moves of chance and information sets).
Let $(N, (T_i)_{i\in N}, p, S, (s_t)_{t\in \times_{i\in N} T_i})$ be a Harasanyi game with incomplete information.
$s_t = \big(N, (A_i(t_i))_{i\in N}, (u_i(t))_{i\in N}\big) \in S$ is the state game for the type vector t, for every $t \in T$. s_t is a strategic-form game so it has a representation as an extensive for game (with imperfect information)

$$\Gamma^t = \Big(N, V^t, E^t, x^{0,t}, (V_i^t)_{i\in N}, (U_i^t)_{i\in N}, \mathbf{R}^N, (u_i(t))_{i\in N}\Big),$$

where $(V^t, E^t, x^{0,t})$ is a game tree; V_i^t is the vertices controlled by player i for every $i \in N$; Each player $i \in N$ has only one information set $U_i^t = V_i^t$; and $(u_i(t))_{i\in N}$ is a function mapping each leaf of the game tree to a possible outcome in $\mathbf{R}^N$.
To define the extensive-form game that describes the Harsanyi game, let v^0 be a vertex controlled by the "Nature". v^0 is the root of the game tree. For every $t \in T$ let e^t be an edge connecting v^0 and $x^{0,t}$ representing a chance move under which Nature chooses a type vector $t \in T$. Let $E^0 = \{e^t : t \in T\}$.
The following extensive-form game describes the Harsanyi game:

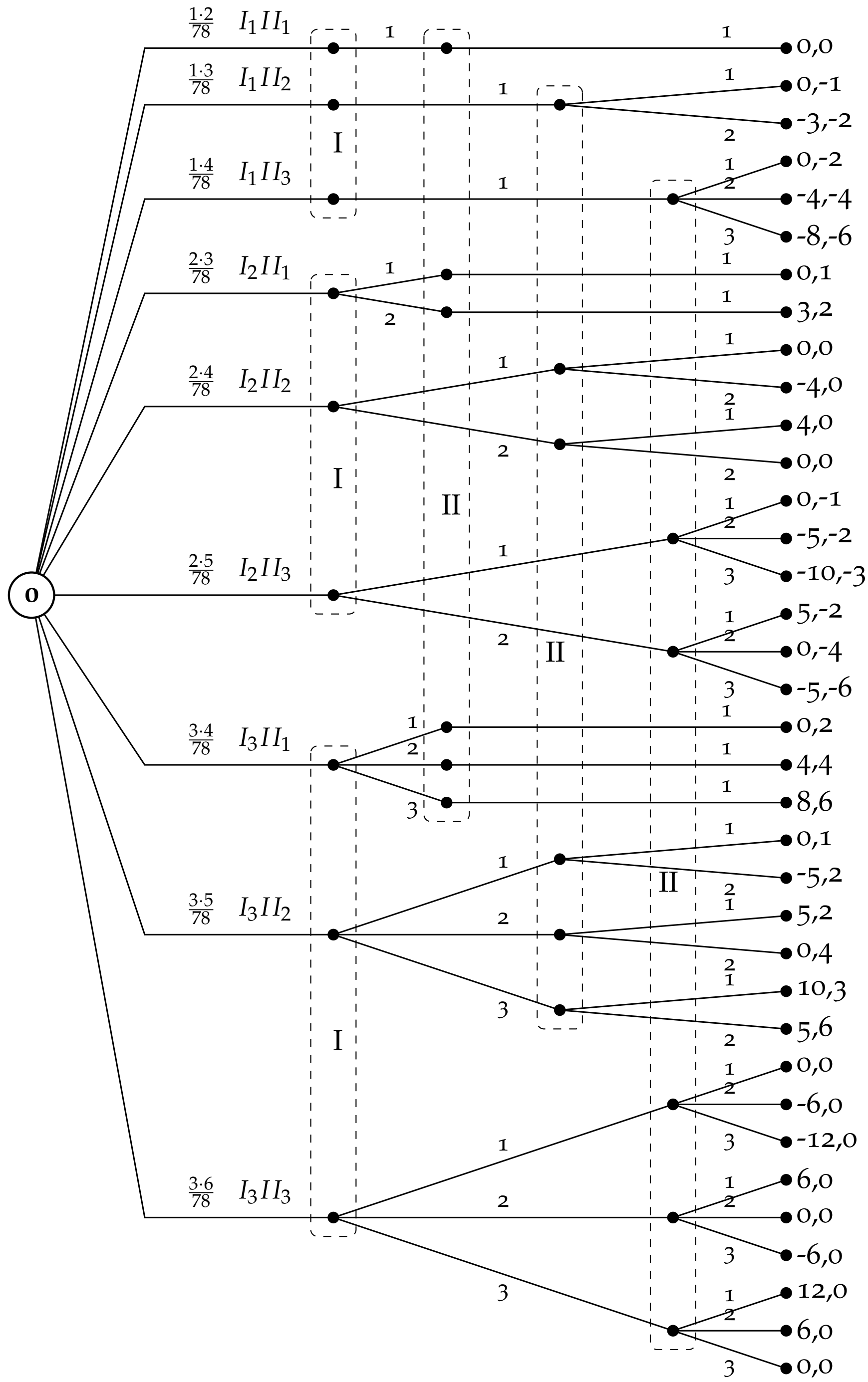
$$\Gamma = \Big(N, V, E, v^0, (V_i)_{i\in N}, p, (U_i^{t_i})_{i\in N}^{t_i=1,2,\ldots,|T_i|}, \mathbf{R}^N, u\Big),$$

where

- The set of vertices V is a union of the vertices corresponding the state games and the root v^0, i.e., $V = \{v^0\} \bigcup \big(\bigcup_{t\in T} V^t\big)$.
- The set of edges E is a union of the edges connecting the root v^0 to all state games and the edges corresponding these state games, i.e., $E = E^0 \bigcup \big(\bigcup_{t\in T} E^t\big)$.
- For every player $i \in N$, the vertices $V_i = \bigcup_{t\in T} V_i^t$ are controlled by player i.
- p is a distribution over the edges in E^0 such that an edge e^t is chosen with probability $p(t)$ for every $t \in T$.

- For every player $i \in N$ and every type t_i, let $U_i^{t_i} = \bigcup_{t_{-i} \in T_{-i}} V^{t_i, t_{-i}}$ and $A(U_i^{t_i}) = A_i(t_i)$, so $(U_i^{t_i}, A(U_i^{t_i}))$ is an information set of player i in which he only knows his type t_i but he does not know the types of the other players.
- u is a function mapping each leaf of the game tree to a possible outcome in $\mathbf{R}^N$ according to $(u_i(t))_{i \in N}$ where $t \in T$ is the type vector that is chosen by the Nature.

9.36



9.37 We find a Bayesian equilibrium in the game described in Example 9.38. Harry knows the state game that is played. Thus in case that G_2

is played he may eliminated the weakly dominated strategies C and B. It easy to show that the obtaining game (after eliminating C and B) has no Bayesian equilibrium in which William plays the pure action b. If William play the pure action t, then Harry plays T in both G_1 and G_2. Hence, this is a Bayesian equilibrium if and only if $1 - p \geq 2p \iff p \leq \frac{1}{3}$.
Otherwise, if $p > \frac{1}{3}$ then there is no equilibrium in which one of them plays pure action.
Denote by $[y(t), (1-y)(b)]$ the equilibrium strategy of William, and by $[x(T), (1-x)(B)]$ the equilibrium strategy of Harry in G_1.
Since Harry's strategy in G_1 is completely mixed, he is necessarily indifferent between T and B.

$$y = 1 - y \iff y = \tfrac{1}{2}.$$

Similarly, since William's strategy is completely mixed, he is necessarily indifferent between t and b.

$$3p(1-x) + (1-p) = 2px \iff x = \tfrac{1+2p}{5p}.$$

For $p > \frac{1}{3}$, $x \in (0,1)$ so it defines a completely mixed strategy, as needed.

9.38 Denote by $[y(L), (1-y)(R)]$ the equilibrium strategy of Player II, by $[x(T), (1-x)(B)]$ the equilibrium strategy of Player I of type I_1, and by $[z(T), (1-z)(B)]$ the equilibrium strategy of Player I of type I_2.
If $y = 1$ then Player I of type I_1 plays T ($x = 1$) and Player I of type I_2 plays B ($z = 0$). However, the best reply of Player II is R, i.e., $y = 0$, so this is not an equilibrium.
If $y = 0$ then Player I of type I_1 plays B ($x = 0$) and Player I of type I_2 plays T ($z = 1$). However, the best reply of Player II is L, i.e., $y = 1$, so this is not an equilibrium as well.
In particular, the equilibrium strategy of Player II is completely mixed, i.e., $y \in (0,1)$ so he is indifferent between L and R.

$$\tfrac{1}{3} \cdot 4(1-x) + \tfrac{2}{3} \cdot 3z = \tfrac{1}{3} \cdot 3x + \tfrac{2}{3} \Rightarrow x = \tfrac{2+6Z}{7}. \tag{73}$$

if $y < \frac{3}{5}$ then the best response of Player I of type I_2 is T so $Z = 1$ which case Equality (73) cannot be satisfied and thus it is not an equilibrium.
If $y > \frac{3}{5}$ then the best response of type I_1 is T ($x = 1$) and the best response of type I_2 is B ($Z = 0$) and again Equality (73) does not hold and thus it is not an equilibrium.
Finally, if $y = \frac{3}{5}$ then the best response of I_1 ($x = 1$) is T whereas I_2 is indifferent between T and B ($0 \leq z \leq 1$). Thus, $x = 1$, $z = \frac{5}{6}$ and

$y = \frac{3}{5}$, is the only equilibrium.

9.39 (a) The set of pure strategies of Player I contains 4 strategies: (T, T) in which he always plays T, (B, B) in which he always plays B, (T, B) in which he plays T only if his type is I_1 otherwise he plays B, and (B, T) in which he plays T only if his type is I_2 otherwise he plays B. The set of pure strategies of Player II contains 4 strategies: (L, L) in which he always plays L, (R, R) in which he always plays R, (L, R) in which he plays L only if his type is II_1 otherwise he plays R, and (R, L) in which he plays L only if his type is II_2 otherwise he plays R.

(b) The game in strategic form is given by

		Player II			
		(L,L)	(L,R)	(R,L)	(R,R)
Player I	(T,T)	7.6	8.8	6.2	7.4
	(T,B)	7	9.1	1	3.1
	(B,T)	8.8	13.6	14.6	19.4
	(B,B)	8.2	13.9	9.4	15.1

(c) We eliminate dominated strategies: first (T, T) and (T, B) of Player I, then (L, R), (R, L) and (R, R) of Player I and last (B, B) of Player I. Hence, the value of the game is 8.8, and the optimal strategies for Player I and Player II are (B, B) and (L, L), respectively.

9.40 (a) -(b) The situation as Harsanyi game consists of two players, E (Employer) and YP (Young person).

- E has one time I_1.
- YP has two types, II_1 (talented) and II_2 (not talented). Prior probability is $\mathbf{P}(I_1, II_1) = \frac{1}{4}$ and $\mathbf{P}(I_1, II_2) = \frac{3}{4}$.
- YP has two actions: Travel and Study.
- E has four actions:
  - AHire: Always hire.
  - ANHire: Always do not hire.
  - HifS: Hire only if studied.
  - HifT: Hire only if not studied.
- The state games are as follows:

Employer (Youth: Travel / Study)	Travel	Study
AHire	0,6	0,2
HifS	3,3	0,2
HifT	0,6	3,−3
ANHire	3,3	3,−3

State game
if youth is untalented.

Employer (Youth: Travel / Study)	Travel	Study
AHire	8,6	8,4
HifS	3,3	8,4
HifT	8,6	3,1
ANHire	3,3	3,1

State game
if youth is talented.

(c) There are two equilibria in pure strategies:

i. The employer does not hire anyone, both talented and untalented young people travel.

ii. The employer hires only young people who studied, talented young people study, untalented young people travel.

9.41 The lemon market game consist of two player $N = \{I, II\}$ where Player I is a buyer and Player II is a seller. Player I has only one type $T_1 = \{I_1\}$ and Player II has two types $T_2 = \{II_1, II_2\}$ where types II_1 indicates that the seller's car is in good condition, and types II_2 indicates that the seller's car is in bad condition. There is a value $q \in [0,1]$ such that $\mathbf{P}(I_1, II_1) = q$ and $\mathbf{P}(I_1, II_2) = 1 - q$. The game form that corresponds to each state game is given in the question.
We now turn to the calculation of Bayesian equilibria in this game.
For every $p \geq 0$ and $q \in [0,1]$ the strategy in which the buyer does not buy and the seller of each type does not sell is an equilibrium.
We next turn to the calculation of Bayesian equilibria in this game according to the value of p. Denote by $[x(\text{Buy}), (1-x)(\text{Don't buy})]$ the equilibrium strategy of buyer, by $[y(\text{Sell}), (1-y)(\text{Don't Sell})]$ the equilibrium strategy of the seller with a car in good condition, and by $[z(\text{Sell}), (1-z)(\text{Don't sell})]$ the equilibrium strategy of the seller with a car in bad condition.
For $0 < p < 4$: If the buyer buys with probability $x > 0$ then the seller with a car in bad condition sells with probability $z = 1$ and the seller with a car in good condition does not sell ($y = 0$) which case the buyer buys with probability $x = 1$. Hence, there is an equilibrium in which $x = 1$, $y = 0$ and $z = 1$.
For $p = 4$: If the buyer buys with probability $x > 0$ then the seller with a car in bad condition sells with probability $z = 1$ and the seller with a car in good condition does not sell ($y = 0$) which case the buyer is indifferent between buying and not buying. On the other hand, if the

seller with a car in good condition sells with positive probability then the buyer buys with probability $x = 1$. Hence, there are equilibria in which $x \in [0, 1]$, $y = 0$ and $z \in [0, 1]$.

For $4 < p < 5$: If the buyer buys with probability $x > 0$ then the seller with a car in bad condition sells with probability $z = 1$ and the seller with a car in good condition does not sell ($y = 0$) which case the buyer loses so it cannot be an equilibrium. Hence in an equilibrium the buyer does not buy and the strategies of the seller of both types must satisfy that the buyer will not deviates, i.e.,

$$qy(6-p) + (1-q)z(4-p) \leq 0 \Rightarrow y \leq \tfrac{(1-q)z(p-4)}{q(6-p)}. \qquad (74)$$

For $p = 5$: If the buyer buys with probability $x > 0$ then the seller with a car in bad condition sells with probability $z = 1$ and the seller with a car in good condition is indifferent between selling and not selling. Hence there are three kinds of equilibria: (1)$x = 0$, and z and y satisfy Inequality (74); (2) $x \in (0, 1)$, $z = 1$ and y satisfies Inequality (74) with equality; and (3) $x = 1$, $z = 1$ and y does not satisfy Inequality (74).

For $5 < p < 6$: If the buyer buys with probability $x > 0$ then the seller of both types sells with probability 1. Hence there are equilibria in which $x = 0$, and z and y satisfy Inequality (74). If in addition, $q(6-p) > (1-q)(p-4)$ then $x = 1$, $z = 1$ and $y = 1$ is also an equilibrium. Finally, if $q(6-p) = (1-q)(p-4)$ then $x \in [0, 1]$, $z = 1$ and $y = 1$ is an equilibrium as well.

For $p \geq 6$: If the buyer buys with probability $x > 0$ then the seller of both types sells with probability 1 which case the buyer lose (unless $p = 6$ and $q = 0$ and he is indifferent between buying and not buying). Hence, $x = 0$, and $z, y \in [0, 1]$ is an equilibrium.

9.42 In all of the situations presented below, there are two players Nicolas (who own the company) and Marc (who wants to buy it).

(a) There are 3 states of nature, each corresponds to one possible value of the company. The partitions of the two players contain one atom (no-one knows the true value).

If Marc offers a price that is lower than 10, Nicolas rejects, and if he offers a price that is higher than 12 Nicolas accepts so Marc loses, hence the action set of Marc contains all possible offers $\{c : c \in [10, 12]\}$.

An action of Nicolas determines whether he accepts or rejects any possible offer. We only consider threshold actions: Nicolas determines a value under which he rejects the offer otherwise he accepts (if the threshold is higher than 12 then Nicolas rejects any offer). The set of actions of Nicolas is $\{t : t \in [10, 12] \bigcup \{13\}\}$.

Since both players ascribe probability $\frac{1}{3}$ to each possible value, they both believe that expected value of the company is 11, hence Nicolas rejects any price which is less than 11 and Marc offer 11 at most. Therefore, there are equilibria in which Marc offers $c \leq 11$, Nicolas accepts an offer of at $t \geq 11$ and rejects all other offers. Note that only if $c = t = 11$ there is a trade.

(b) In all of the situations presented below, there are 3 states of nature, each corresponds to one possible value of the company. Marc's partition contains one atom, Nicolas' partition contains 3 atoms (he knows the true value). The set of actions of Nicolas if the true value of the company is $v \in \{10, 11, 12\}$ is $\{t : t \in [10, 12] \bigcup \{13\}\}$.

Now Nicolas rejects any offer that is below the true value of the company. Hence, Marc knows that if Nicolas accepts his offer then the true value of the company is at most the value he offers. Thus, unless he offers 10 he loses with positive probability. Hence there is an equilibrium in which Marc offers $c = 10$ and Nicolas rejects any offer that is below $t_v \geq v$ where v is the true value of the company. A trade may occurs only in case the true value is 10.

(c) There are 6 states of nature. Each state is a triple (v, k) where $v \in \{10, 11, 12\}$ is the true value of the company and k indicates whether Nicolas knows the true value which case $k = 1$ or he does not know which case $k = 0$.

Marc's partition contains one atom (he knows nothing), and Nicolas' partition contains 4 atoms:

$$\mathcal{F}_{\text{Nicolas}} = \{\{(10, 1)\}, \{(11, 1)\}, \{(12, 1)\}\{(10, 0), (11, 0), (12, 0)\}\}.$$

The prior is $\mathbf{P}(v, 1) = \frac{p}{3}$ and $\mathbf{P}(v, 0) = \frac{1-p}{3}$ for every $v \in \{10, 11, 12\}$. We assume that $p \in (0, 1)$ ($p = 0$ and $p = 1$ are presented in Parts (a) and (b) respectively).

As claimed in previous parts, if Nicolas does not know the true value he rejects any offer less than 11, and if he knows the true value then he necessarily rejects any offer less than his true value. Therefore, if Marc offers $c \geq 10$ then Nicolas accepts the offer if and only if it is either at least the true value (if he knows it) or at least the expected value (if he does not know) so he loses with positive probability. In particular, in an equilibrium Marc offers $c = 10$, and Nicolas rejects any offer that is less than $t_v \geq v$ if he knows the true value v and he rejects any offer that is less than $t \geq 11$ if he does not know the true value.

9.43 We prove that in each game with incomplete information with a finite set of players, where the set of types of each player is a countable set, and the set of possible actions of each type is finite, there exists a Bayesian equilibrium (in behavior strategies).
Suppose that the set of types of player i, T_i , is the set of natural numbers $\mathbf{N}$. Denote $T_i^k := \{1, 2, ..., k\}$ and $T^k = \times_{i \in N} T_i^k$. Note that since $\sum_{t \in \times_{i \in N} T_i} \mathbf{P}(t) = 1$ is a convergent series of non negative numbers then there is a sufficiently large $K \in \mathbf{N}$ such that for every $k \geq K$ the partial sum $\mathbf{P}(T^k) = \sum_{t \in \times_{i \in N} T_i^k} \mathbf{P}(t) > 0$.
Let p^k be the probability distribution p conditioned on the set T^k for $k \geq K$:

$$p^k(t) = \begin{cases} \frac{p(t)}{p(T^k)} & t \in T^k, \\ 0 & t \notin T^k. \end{cases}$$

Since the denominator $p(T^k)$ is positive the probability distribution p^k is well defined.
For each $k \geq K$, the game in which the set of types of player i is T_i^k and the probability distribution over the types is a game p^k is a game with incomplete information in which the set of types is finite and the set of actions of each type is finite hence, by Theorem 9.52, it has a Bayesian equilibrium (in behavior strategies) denoted by σ^k.
Denote by σ^* the accumulation point of such equilibria, as k goes to infinity. Then, since U_i is a continuous function, the following holds for every $i \in N$, every $t_i \in N$, and every $a_i \in A(t_i)$

$$\begin{aligned} U_i(\sigma^* | t_i) &= U_i \left(\lim_{\substack{k \to \infty \\ k > t_i}} \sigma^k | t_i \right) = \lim_{\substack{k \to \infty \\ k > t_i}} U_i \left(\sigma^k | t_i \right) \\ &\geq \lim_{\substack{k \to \infty \\ k > t_i}} U_i \left(a_i, \sigma^k_{-i} | t_i \right) = U_i \left(a_i, \lim_{\substack{k \to \infty \\ k > t_i}} \sigma^k_{-i} | t_i \right) = U_i(a_i, \sigma^*_{-i} | t_i), \end{aligned}$$

so σ^* is an equilibrium of the original game.

9.44 Proof of Theorem 9.51: Let Γ be a game with incomplete information and let $\widehat{\Gamma}$ be the agent-form game corresponding to Γ.
Every behavior strategy of player i in the game Γ naturally defines a mixed strategy for the players in T_i in the agent-form game $\widehat{\Gamma}$ and vice versa. For simplicity, we denote by σ the strategy vector in Γ as well as the corresponding strategy vector in $\widehat{\Gamma}$.

We next show that for every strategy vector σ the payoff of player t_i under σ in $\widehat{\Gamma}$ and the conditional expected payoff to player i of type t_i in Γ are equal.

$$
\begin{aligned}
\widehat{U}_{t_i}(\sigma) &= \sum_{a\in A} \sigma(a)\widehat{u}_{t_i}(a) \\
&= \sum_{a\in A} \sigma(a) \sum_{t_{-i}\in T_{-i}} p(t_{-i}|t_i) u_i(t_i, t_{-i}; a(t_i, t_{-i})) \\
&= \sum_{t_{-i}\in T_{-i}} p(t_{-i}|t_i) \sum_{a\in A} \sigma(a) u_i(t_i, t_{-i}; a(t_i, t_{-i})) \\
&= \sum_{t_{-i}\in T_{-i}} p(t_{-i}|t_i) \sum_{a\in A} \left(\prod_{j=1}^{n} \prod_{t_j\in T_j} \sigma_j(t_j; a(t_j)) \right) u_i(t_i, t_{-i}; a(t_i, t_{-i})) \\
&= \sum_{t_{-i}\in T_{-i}} p(t_{-i}|t_i) \sum_{a(t_i,t_{-i})\in A(t_i,t_{-i})} \sigma(a(t_i, t_{-i})) u_i(t_i, t_{-i}; a(t_i, t_{-i})) \qquad (75) \\
&= \sum_{t_{-i}\in T_{-i}} p(t_{-i}|t_i) u_i(t_i, t_{-i}; \sigma) \\
&= U_i(\sigma|t_i),
\end{aligned}
$$

where Equality (75) holds since $\sigma_j(t_j)$ is a distribution over $A_j(t_j)$ so for every type that is not in (t_i, t_{-i}) it does not effect the payoff $u_i(t_i, t_{-i}; a(t_i, t_{-i}))$ and thus it defines an expectation over a constant with regards to it.
In addition, for every strategy vector σ in $\widehat{\Gamma}$, for every $i \in N$, for every player $t_i \in T_i$, and for every pure strategy $a_i(t_i) \in A_i(t_i)$ denote by $(\sigma_{-t_i}, a_i(t_i))$ the strategy vector in $\widehat{\Gamma}$ in which all players are following σ except for player t_i. However, since the payoff of player i of type t_i depends only on the strategy that is taken by t_i and does not depends on the other types of i then $U_i(\sigma_{-t_i}, a_i(t_i)|t_i) = u_i(\sigma_{-i}, a_i(t_i))$.
In conclusion, a strategy vector $(\sigma_i^*(t_i))_{i\in N, t_i\in T_i}$ is a Nash equilibrium in the agent-form game $\widehat{\Gamma}$ if and only if for every $i \in N$, for every player $t_i \in T_i$ in the agent game, and for every possible action $a_i(t_i) \in A_i(t_i)$ one has

$$
\begin{gathered}
\widehat{U}_{t_i}(\sigma^*) \geq \widehat{U}_i(\sigma^*_{-t_i}, a_i(t_i)) \iff \\
U_i(\sigma^*|t_i) \geq U_i(\sigma^*_{-t_i}, a_i(t_i)|t_i) = U_i(\sigma^*_{-i}, a_i(t_i)|t_i) \qquad (76)
\end{gathered}
$$

if and only if σ^* is is a Bayesian equilibrium in Γ.

9.45 Let $\Gamma = (N, (T_i)_{i\in N}, p, S, (s_t)_{t\in \times_{i\in N} T_i})$, where $s_t = \big(N, (A_i(t_i), u_i(t))_{i\in N}\big)$ for each $t \in \times_{i\in N} T_i$, be a game with incomplete information in which there exists a player j and a type t_j^* of player j such that $|A_j(t_j^*)| = 1$.

Let $\widehat{\Gamma}$ be a game with incomplete information that is identical to Γ, except that the payoff function $\widehat{u}_j(t_j^*)$ of player j of type t_j^* may be different from $u_j(t_j^*)$.

We show that the two games Γ and $\widehat{\Gamma}$ have the same set of Bayesian equilibria.

The sets of strategy vectors in both Γ and $\widehat{\Gamma}$ are identical since the sets of players are identical as well as the sets of types for each player and the sets of actions for each type.

Assume that σ is a Bayesian equilibrium in Γ. Then for every player $i \in N$, every type $t_i \in T_i$ and every pure strategy $a_i(t_i) \in A_i(t_i)$ one has

$$U_i(\sigma|t_i) \geq U_i(\sigma_{-i}, a_i(t_i)|t_i).$$

However, player j of type t_j^* cannot deviate since he has only one pure/mixed action in Γ as well as in $\widehat{\Gamma}$. Furthermore, for every player i of type t_i such that either $i \neq j$ or $t_i \neq t_j^*$ then the payoff functions U_i and $\widehat{U}_i$ are identical so

$$\widehat{U}_i(\sigma|t_i) = U_i(\sigma|t_i) \geq U_i(\sigma_{-i}, a_i(t_i)|t_i) = \widehat{U}_i(\sigma_{-i}, a_i(t_i)|t_i),$$

so σ is a Bayesian equilibrium in $\widehat{\Gamma}$. By exchanging the roles of Γ and $\widehat{\Gamma}$, we deduce that the two games Γ and $\widehat{\Gamma}$ have the same set of Bayesian equilibria.

9.46 **Electronic Mail game**

(a) The game with incomplete information is given by: (1) A set of players $N = \{I, II\}$; (2) The set of types of Player I $T_1 = \{t_1^{-1}, t_1^0, t_1^1, t_1^2, ...\}$ where the type of Player I is t_1^{-1} if and only if the true state is $s = 1$, otherwise, the true state is $s = 2$ and the index of the type is the number of e-mail messages Player I received; (3) The set of types of Player II $T_2 = \{t_2^0, t_2^1, t_2^2, ...\}$ where the type is indexed by the number of e-mail messages he has received; (4) A distribution p over $T_1 \times T_2$ given by

$$p(t_1^n, t_2^m) = \begin{cases} 1 - p & n = -1, m = 0 \\ p\epsilon(1-\epsilon)^{n+m} & 0 \leq n \leq m \leq n+1 \\ 0 & \text{otherwise.} \end{cases}$$

(5) A set of action $A_i(t_i) = \{A, B\}$ for every player $i \in N$ and every type $t_i \in T_i$. The payoff functions of types t_1^{-1} and t_1^n for $n \geq 0$ is given by the left and the right tables, respectively, in the question.

(b) We show by induction over the index that in a Bayesian equilibrium each type t_i^n of each player i plays A.

First, Player I of type t_1^{-1} knows that the true state is $s = 1$ so he plays A in a Bayesian equilibrium, since A is strictly dominated over B in $s = 1$.

Player II of type t_2^0 ascribes probability $\frac{1-p}{1-p+p\epsilon}$ to the type t_1^{-1} of Player I (in which he plays A) and probability $\frac{p\epsilon}{1-p+p\epsilon}$ to the type t_1^0 of Player I. In particular, Player II expected payoff if he plays A is at least $\frac{1-p}{1-p+p\epsilon}M > 0$ whereas his expected payoff if he plays B is at most $\frac{1-p}{1-p+p\epsilon}(-L) + \frac{p\epsilon}{1-p+p\epsilon}M < 0$, hence he is better off playing A.

Assume that in a Bayesian equilibrium, Player I of type t_1^n and Player II of type t_2^{n+1} play A, for $n \geq 0$. We show that Player I of type t_1^{n+1} and Player II of type t_2^{n+2} play A as well.

Indeed, Player I of type t_1^{n+1} ascribes probability $\frac{1}{1+1-\epsilon}$ to the type t_2^{n+1} of Player II (in which he plays A) and probability $\frac{1-\epsilon}{2-\epsilon}$ to the type t_2^{n+2} of Player II. In particular, Player II expected payoff if he plays A is 0 whereas his expected payoff if he plays B is at most $\frac{1}{2-\epsilon}(-L) + \frac{1-\epsilon}{2-\epsilon}M < 0$, hence he is better off playing A.

Likewise, Player II of type t_2^{n+2} ascribes probability $\frac{1}{2-\epsilon}$ to the type t_1^{n+1} of Player I (in which he plays A) and probability $\frac{1-\epsilon}{2-\epsilon}$ to the type t_2^{n+2} of Player II. In particular, Player II expected payoff if he plays A is at least 0 whereas his expected payoff if he plays B is at most $\frac{1}{2-\epsilon}(-L) + \frac{1-\epsilon}{2-\epsilon}M < 0$, hence he is better off playing A.

9.47 Consider the example described in Section 9.5, where $\epsilon \in [0,1]$, α has uniform distribution over the interval $[\frac{1}{4}, \frac{1}{2}]$ and β has uniform distribution over the interval $[-\frac{1}{3}, \frac{2}{3}]$. We find Bayesian equilibria in threshold strategies:

$$s_1^{\alpha_0} = \begin{cases} T & \alpha > \alpha_0 \\ B & \alpha \leq \alpha_0, \end{cases}$$

$$s_2^{\beta_0} = \begin{cases} L & \beta > \beta_0 \\ R & \beta \leq \beta_0. \end{cases}$$

Since $\mathbf{P}(\beta > \beta_0) = \frac{2}{3} - \beta_0$ and $\mathbf{P}(\beta \leq \beta_0) = \beta_0 + \frac{1}{3}$, the expected payoff of Player I of type α facing strategy $s_2^{\beta_0}$ of Player II is

$$U_1(T, s_2^{\beta_0}) = \epsilon\alpha$$

if he plays T ; and it is

$$U_1(B, s_2^{\beta_0}) = (\tfrac{2}{3} - \beta_0) - (\beta_0 + \tfrac{1}{3}) = \tfrac{1}{3} - 2\beta_0$$

if he plays B. In order for $s_1^{\alpha_0}$ to be a best reply to $s_2^{\beta_0}$ the following conditions must hold:

$$\epsilon\alpha_0 = \tfrac{1}{3} - 2\beta_0. \tag{77}$$

Similarly, since $\mathbf{P}(\alpha > \alpha_0) = 2 - 4\alpha_0$ and $\mathbf{P}(\alpha \leq \alpha_0) = 4\alpha_0 - 1$, the expected payoff of Player II of type β facing strategy $s_1^{\alpha_0}$ of Player I is

$$U_2(L, s_1^{\alpha_0}) = \epsilon\beta$$

if he plays L; and it is

$$U_2(R, s_1^{\alpha_0}) = -(2 - 4\alpha_0) + 3(4\alpha_0 - 1) = 16\alpha_0 - 5$$

if he plays R. In order for $s_2^{\beta_0}$ to be a best reply to $s_1^{\alpha_0}$ the following conditions must hold:

$$\epsilon\beta_0 = 16\alpha_0 - 5. \tag{78}$$

By Equalities (77) and (78),

$$\begin{aligned} \alpha_0 &= \tfrac{30+\epsilon}{96+3\epsilon^2} \\ \beta_0 &= \tfrac{16-15\epsilon}{96+3\epsilon^2}. \end{aligned}$$

Note that under this strategy vector the probability that Player I plays T is

$$\mathbf{P}(\alpha > \alpha_0) = 2 - 4 \cdot \tfrac{30+\epsilon}{96+3\epsilon^2} \xrightarrow[\epsilon \to 0]{} \tfrac{3}{4}.$$

Similarly, the probability that Player II plays L is

$$\mathbf{P}(\beta > \beta_0) = \tfrac{2}{3} - \tfrac{16-15\epsilon}{96+3\epsilon^2} \xrightarrow[\epsilon \to 0]{} \tfrac{1}{2}.$$

9.48 **Game A:** The game has a unique equilibrium $\left(\left(\tfrac{1}{3}(T), \tfrac{2}{3}(B)\right); \left(\tfrac{4}{5}(L), \tfrac{1}{5}(R)\right)\right)$.
Consider the following games with incomplete information where $\epsilon \in [0,1]$, and α and β have uniform distribution over the interval $[0,1]$:

	L	R
T	$\epsilon\alpha + 1, \epsilon\beta + 5$	$\epsilon\alpha + 4, 1$
B	$2, \epsilon\beta + 1$	$0, 3$

We find Bayesian equilibria in threshold strategies:

$$s_1^{\alpha_0} = \begin{cases} T & \alpha > \alpha_0 \\ B & \alpha \le \alpha_0, \end{cases}$$

$$s_2^{\beta_0} = \begin{cases} L & \beta > \beta_0 \\ R & \beta \le \beta_0. \end{cases}$$

Since $\mathbf{P}(\beta > \beta_0) = 1 - \beta_0$ and $\mathbf{P}(\beta \le \beta_0) = \beta_0$, the expected payoff of Player I of type α facing strategy $s_2^{\beta_0}$ of Player II is

$$U_1(T, s_2^{\beta_0}) = (1-\beta_0)(\epsilon\alpha + 1) + \beta_0(\epsilon\alpha + 4) = \epsilon\alpha + 1 + 3\beta_0$$

if he plays T ; and it is

$$U_1(B, s_2^{\beta_0}) = (1-\beta_0)2 = 2 - 2\beta_0$$

if he plays B. In order for $s_1^{\alpha_0}$ to be a best reply to $s_2^{\beta_0}$ the following conditions must hold:

$$\epsilon\alpha + 1 + 3\beta_0 = 2 - 2\beta_0. \tag{79}$$

Similarly, since $\mathbf{P}(\alpha > \alpha_0) = 1 - \alpha_0$ and $\mathbf{P}(\alpha \le \alpha_0) = \alpha_0$, the expected payoff of Player II of type β facing strategy $s_1^{\alpha_0}$ of Player I is

$$U_2(L, s_1^{\alpha_0}) = \epsilon\beta + 5 - 4\alpha$$

if he plays L; and it is

$$U_2(R, s_1^{\alpha_0}) = 1 + 2\alpha$$

if he plays R. In order for $s_2^{\beta_0}$ to be a best reply to $s_1^{\alpha_0}$ the following conditions must hold:

$$\epsilon\beta + 5 - 4\alpha = 1 + 2\alpha. \tag{80}$$

By Equalities (79) and (80),

$$\alpha_0 = \tfrac{20+\epsilon}{30+\epsilon^2}$$
$$\beta_0 = \tfrac{6-4\epsilon}{30+\epsilon^2}.$$

Note that under this strategy vector the probability that Player I plays T is

$$\mathbf{P}(\alpha > \alpha_0) = 1 - \tfrac{20+\epsilon}{30+\epsilon^2} \xrightarrow[\epsilon \to 0]{} \tfrac{2}{3}.$$

Similarly, the probability that Player II plays L is

$$\mathbf{P}(\beta > \beta_0) = \tfrac{6-4\epsilon}{30+\epsilon^2} \xrightarrow[\epsilon \to 0]{} \tfrac{1}{5}.$$

Game B: The game has two equilibria (T, L) and (B, R).
Consider the following games with incomplete information where $\epsilon \in [0,1]$, and α and β have uniform distribution over the interval $[0,1]$:

	L	R
T	$\epsilon\alpha+3, \epsilon\beta+4$	$\epsilon\alpha+2, 2$
B	$1, \epsilon\beta+1$	$2, 1$

Since T strictly dominates B and L strictly dominates R, the only Bayesian equilibrium is (T, L) which clearly converges (T, L) as ϵ goes to zero.
Consider next the games with incomplete information where $\epsilon \in [0, 1]$, and α and β have uniform distribution over the interval $[0, 1]$:

	L	R
T	$3, 4$	$2, \epsilon\beta+2$
B	$\epsilon\alpha+1, 1$	$\epsilon\alpha+2, \epsilon\beta+1$

(B, R) is a Bayesian equilibrium of the game which clearly converges (B, L) as ϵ goes to zero.

9.49 The beliefs of the players in the given game are consistent. They can be derived from the common prior beliefs in the following table:

	II_1	II_2
I_1	3/24	9/24
I_2	8/24	4/24

By eliminating weakly dominated strategies one can deduce that the game has a Bayesian equilibrium under which Player I of type I_1 plays T, Player I of type I_2 plays B, Player II of type I_1 plays R, Player I of type I_2 plays L.

9.50 The the beliefs of the players are not consistent. Assume by contradiction that there is a common prior beliefs

	II_1	II_2
I_1	$x_{1,1}$	$x_{1,2}$
I_2	$x_{2,1}$	$x_{2,2}$

Then,

$$\begin{aligned}
\mathbf{P}(II_1 | I_1) &= \tfrac{x_{1,1}}{x_{1,1}+x_{1,2}} = \tfrac{1}{3} \quad \Rightarrow \quad 2x_{1,1} = x_{1,2} \\
\mathbf{P}(II_1 | I_2) &= \tfrac{x_{2,1}}{x_{2,1}+x_{2,2}} = \tfrac{3}{4} \quad \Rightarrow \quad x_{2,1} = 3x_{2,2} \\
\mathbf{P}(I_1 | II_1) &= \tfrac{x_{1,1}}{x_{1,1}+x_{2,1}} = \tfrac{3}{5} \quad \Rightarrow \quad 2x_{1,1} = 3x_{2,1}.
\end{aligned}$$

Therefore,

$$\mathbf{P}(I_1 | II_2) = \tfrac{x_{1,2}}{x_{1,2}+x_{2,2}} = \tfrac{3x_{2,1}}{3x_{2,1}+\frac{1}{3}x_{2,1}} = \tfrac{9}{10} \neq \tfrac{1}{6},$$

a contradiction.

9.51 (a) There are two states of nature in this game, s_1 in which Player III is not active and s_2 in which Player III is an active player.

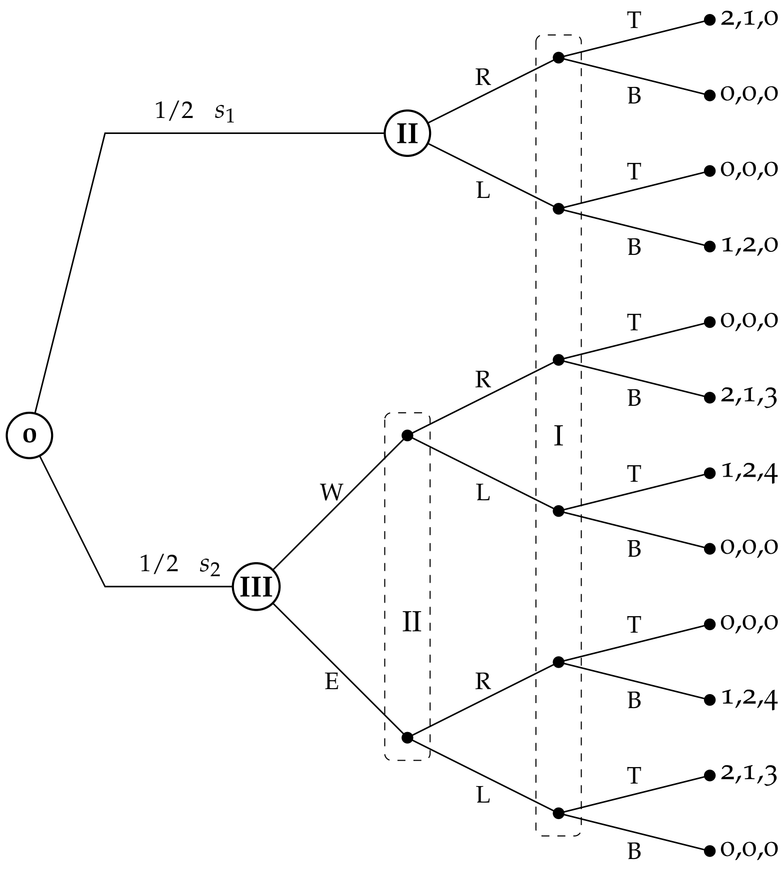
(b) Player I, who does not know the true state, has only two pure strategies, T and B. Player II has 4 pure strategies $\{(a_1, a_2) : a_1, a_2 \in \{L, R\}\}$ where a_k is the pure strategy in state s_k. Player III has only two pure strategies, W and E, since he is not active in state s_1.

(c) The game with incomplete information: $N = \{I, II, III\}$ is the set of players. The sets of types of each player are $T_1 = \{I_1\}$, $T_2 = \{II_1, II_2\}$ and $T_3 = \{III_1, III_2\}$ where types II_k and III_k corresponding to state s_k. For every $t \in T = T_1 \times T_2 \times T_3$,

$$p(t) = \begin{cases} \frac{1}{2} & t = (I_1, II_1, III_1) \text{ or } t = (I_1, II_2, III_2) \\ 0 & \text{otherwise.} \end{cases}$$

The action sets are $A_1(I_1) = \{T, B\}$, $A_2(II_1) = A_2(II_2) = \{L, R\}$, $A_3(III_1) = \{W\}$ (one can define any action), and $A_3(III_2) = \{W, E\}$. The payoff functions are given in the question.

(d)



(e) The game has two Bayesian equilibria in pure strategies: (1) Player I plays T, Player II plays R in s_1 and L in s_2, and Player III plays W; (2) Player I plays B, Player II plays L in s_1 and R in s_2, and Player III plays E.

(f) An additional Bayesian equilibrium in which all players of all types are indifferent between their two possible actions is the strategy vector under which Player I plays $(\frac{1}{2}(T), \frac{1}{2}(B))$, Player II plays $(\frac{1}{2}(L), \frac{1}{2}(R))$ in both states, and Player III plays $(\frac{1}{2}(W), \frac{1}{2}(E))$.

9.52 This exercise generalizes Theorems 9.47 and 9.53 to the case where the prior distributions of the players differ.

(a) Let $(N, (T_i)_{i\in N}, (p_i)_{i\in N}, S, (s_t)_{t\times i\in N} T_i)$ be a game with incomplete information where each player has a different prior distribution: for each $i \in N$, player i's prior distribution is p_i. We show that it can be described as a game in strategic form, hence by Nash Theorem (Theorem 5.10) it has an equilibrium in mixed strategies. Let $G = (N, (A_i)_{i\in N}, (v_i)_{i\in N})$, in which: N is a finite set of players. $A_i = \times_{t_i\in T_i} A_i(t_i)$ is the set of strategies of player i, for every player $i \in N$. We denote the set of all vectors of strategies by $A = \times_{i\in N} A_i$. Finally, the payoff function of player i is

$$v_i(a) = \sum_{t\in T} p_i(t) u_i(t; a).$$

The sets of mixed strategies in the original game and in G are equivalent.

By Nash Theorem, G has an equilibrium σ^* in mixed strategies. Thus, for every player i and every strategy σ_i of player i one has

$$V_i(\sigma^*) \geq V_i(\sigma_i, \sigma^*_{-i}).$$

Hence,

$$\begin{aligned}
U_i(\sigma^*) &= \sum_{t\in T} p_i(t) U_i(t; \sigma^*) = \sum_{t\in T} p_i(t) \sum_{a\in A} \sigma^*(a) u_i(t; a) \\
&= \sum_{a\in A} \sigma^*(a) \sum_{t\in T} p_i(t) u_i(t; a) = \sum_{a\in A} \sigma^*(a) v_i(a) \\
&= V_i(\sigma^*) \geq V_i(\sigma_i, \sigma^*_{-i}) = \sum_{a\in A} (\sigma_i, \sigma^*_{-i})(a) v_i(a) \\
&= \sum_{a\in A} (\sigma_i, \sigma^*_{-i})(a) \sum_{t\in T} p_i(t) u_i(t; a) = U_i(\sigma_i, \sigma^*_{-i}).
\end{aligned}$$

Therefore, σ^* is Nash equilibrium in the original game.

(b) We prove that if each player assigns positive probability to every type of every player, i.e., if $p_i(t_j) := \sum_{t_{-j}\in T_{-j}} p_i(t_j, t_{-j}) > 0$ for every $i, j \in N$ and every $t_j \in T_j$, then every Nash equilibrium is a Bayesian equilibrium, and every Bayesian equilibrium is a Nash equilibrium.

Under the assumption $p_i(t_i) > 0$ for every i and thus following holds for every player i and every σ

$$
\begin{aligned}
U_i(\sigma) &= \sum_{t\in T} p_i(t) U_i(t;\sigma) = \sum_{t_i\in T_i} p_i(t_i) \sum_{t_{-i}\in T_{-i}} \tfrac{p_i(t_i,t_{-i})}{p_i(t_i)} U_i(t_i, t_{-i}; \sigma) \\
&= \sum_{t_i\in T_i} p_i(t_i) U_i(\sigma | t_i). \qquad (81)
\end{aligned}
$$

In particular, if σ^* is a Bayesian equilibrium then for every player i, every type $t_i \in T_i$, and every strategy σ_i of player i

$$U_i(\sigma^* | t_i) \geq U_i(\sigma_i, \sigma^*_{-i} | t_i)$$

from which it follows that for every player i and every strategy σ_i of player i

$$U_i(\sigma^*) = \sum_{t_i\in T_i} p_i(t_i) U_i(\sigma^* | t_i) \geq \sum_{t_i\in T_i} p_i(t_i) U_i(\sigma_i, \sigma^*_{-i} | t_i) = U_i(\sigma_i, \sigma^*_{-i})$$

so σ^* is also Nash equilibrium.

To prove the other direction, assume by contradiction that σ^* is Nash equilibrium but it is not Bayesian equilibrium. That is, there is a player i with a type t_i who has a strategy σ_i in which $U_i(\sigma^* | t_i) < U_i(\sigma_i, \sigma^*_{-i})$. However, by Equality (81), player i is better off deviating to σ_i if his type is t_i, a contradiction.

9.53 (a) Let $\Gamma = (N, (T_i)_{i\in N}, p, S, (s_t)_{t\in\times_{i\in N} Ti})$ be a game with incomplete information, where the set of states of nature S contains only one state, which is a game in strategic form $G = (N, (A_i)_{i\in N}, (u_i)_{i\in N})$; that is, $s_t = G$ for every $t \in \times_{i\in N} T_i$. Denote the set of action vectors by $A = \times_{i\in N} A_i$. Every strategy vector σ in Γ naturally induces a distribution μ_σ over the vectors in A:

$$\mu_\sigma(a) = \sum_{t\in T} p(t) \times \sigma_1(t_1; a_1) \times \sigma_2(t_2; a_2) \times \cdots \times \sigma_n(t_n; a_n).$$

Let σ^* be a a Bayesian equilibrium of Γ. We prove that the distribution μ_{σ^*} is a correlated equilibrium in G.

By Theorem 9.53, every Bayesian equilibrium in Γ is also a Nash equilibrium. Therefore, for every player i and every strategy σ_i of player i,

$$U_i(\sigma^*) \geq U_i(\sigma_i, \sigma^*_{-i}). \tag{82}$$

In particular, Inequality (82) holds for every strategy σ'_i under which whenever $\widehat{a}_i$ is chosen player i plays a'_i instead.

However, since $s_t = G$ for every $t \in T$, $A(t) = A$ and $u_i(t; a) = u_i(a)$ so

$$\begin{aligned} U_i(\sigma^*) &= \sum_{t \in T} p(t) U_i(t; \sigma) = \sum_{t \in T} p(t) \sum_{a \in A} \sigma^*(t; a) u_i(a) \\ &= \sum_{a \in A} u_i(a) \sum_{t \in T} p(t) \sigma^*(t; a) = \sum_{a \in A} u_i(a) \mu_{\sigma^*}(a) \\ &= \sum_{a_i \in A_i} \sum_{a_{-i} \in A_{-i}} u_i(a_i, a_{-i}) \mu_{\sigma^*}(a_i, a_{-i}). \end{aligned} \tag{83}$$

In addition,

$$\begin{aligned} U_i(\sigma'_i, \sigma^*_{-i}) = &\sum_{a_i \neq \widehat{a}_i} \sum_{a_{-i} \in A_{-i}} u_i(a_i, a_{-i}) \mu_{\sigma^*}(a_i, a_{-i}) \\ &+ \sum_{a_{-i} \in A_{-i}} u_i(a'_i, a_{-i}) \mu_{\sigma^*}(\widehat{a}_i, a_{-i}). \end{aligned} \tag{84}$$

Using Inequality (82) and Equalities (83) and (84) one can deduce that

$$\begin{aligned} &\sum_{a_i \in A_i} \sum_{a_{-i} \in A_{-i}} u_i(a_i, a_{-i}) \mu_{\sigma^*}(a_i, a_{-i}) \geq \\ &\sum_{a_i \neq \widehat{a}_i} \sum_{a_{-i} \in A_{-i}} u_i(a_i, a_{-i}) \mu_{\sigma^*}(a_i, a_{-i}) + \sum_{a_{-i} \in A_{-i}} u_i(a'_i, a_{-i}) \mu_{\sigma^*}(\widehat{a}_i, a_{-i}) \\ &\Rightarrow \sum_{a_{-i} \in A_{-i}} u_i(\widehat{a}_i, a_{-i}) \mu_{\sigma^*}(\widehat{a}_i, a_{-i}) \geq \sum_{a_{-i} \in A_{-i}} u_i(a'_i, a_{-i}) \mu_{\sigma^*}(\widehat{a}_i, a_{-i}), \end{aligned}$$

from which it follows that that the distribution μ_{σ^*} is a correlated equilibrium in G.

(b) Let $G = (N, (A_i)_{i \in N}, (u_i)_{i \in N})$ be a strategic-form game, and let μ be a correlated equilibrium in this game. We define a game with incomplete information $\Gamma = (N, (T_i)_{i \in N}, p, S, (s_t)_{t \in \times_{i \in N} T_i})$ where $T_i = A_i$ for every $i \in N$, $p = \mu$ and the set of states of nature S contains only one state, i.e., $s_t = G$ for every $t \in \times_{i \in N} T_i$.

We show that the strategy vector σ^* under which each type $a_i \in T_i$ of player i plays the pure strategy a_i is a Bayesian equilibrium.

Indeed, for each player $i \in N$, each type $t_i \in T_i$, and each possible action $a_i \in A_i$

$$U_i(\sigma^*|t_i) = \sum_{t_{-i}\in T_{-i}} p(t_{-i}|t_i)U_i(t_i, t_{-i}; \sigma^*) = \sum_{a_{-i}\in A_{-i}} \mu(a_{-i}|a_i)U_i(a_i, a_{-i})$$
$$\geq \sum_{a_{-i}\in A_{-i}} \mu(a_{-i}|a_i)U_i(a'_i, a_{-i}) = U_i(a'_i, \sigma^*_{-i}|t_i),$$

so σ^* is a Bayesian equilibrium in the game Γ.
Further, since $\sigma^*_i(t_i; a_i) = 1$ if $t_i = a_i$, otherwise $\sigma^*_i(t_i; a_i) = 0$, then for every $a \in A$

$$\sum_{t\in T} p(t) \times \sigma^*_1(t_1; a_1) \times \sigma^*_2(t_2; a_2) \times \cdots \times \sigma^*_n(t_n; a_n) = p(a) = \mu(a).$$

9.54 (a) Denote Carolyn, Maurice by players I and II, respectively. In the game with incomplete information $N = \{I, II\}$ is the set of players. The set of types of Player I (Carolyn) is $T_I = \{t^1_I\}$ and the set of types of Player II (Maurice) is $T_{II} = \{t^1_{II}, t^2_{II}, t^3_{II}\}$ where t^1_{II}, t^2_{II} and t^3_{II} corresponding the situations in which Jim, John, and Arthur won the Wimbledon tennis tournament, respectively. $p(t^1_I, t^k_{II}) = \frac{1}{3}$ for $k = 1, 2, 3$. The set of states contains only one state- the game "Chicken". By Exercise 9.53, every Bayesian equilibrium in such a game with incomplete information corresponding a correlated equilibrium in the game "Chicken". Hence, there are three Bayesian equilibria (corresponding the equilibria in G): (1) Player I plays T and Player II of each type plays R; (2) Player I plays B and Player II of each type plays L; and (3) Player I plays $(\frac{2}{3}(T), \frac{1}{3}(B))$ and Player II of each type plays $(\frac{2}{3}(R), \frac{1}{3}(L))$.
Note that there is no strategy vector in the game of incomplete information corresponding the correlated equilibrium under which the three action vectors, (T, L), (T, R), and (B, L), are chosen with equal probability.

(b) In the current situation, the set of types of Player I (Carolyn) is $T_I = \{t^1_I, t^2_I\}$ and the set of types of Player II (Maurice) is $T_{II} = \{t^1_{II}, t^2_{II}\}$ where t^1_I and t^1_{II} corresponding the situation in which Jim won the Wimbledon tennis tournament, and t^2_I and t^2_{II} corresponding the situation in which Jim did not win the Wimbledon tennis tournament. $p(t^1_I, t^1_{II}) = \frac{1}{3}$ and $p(t^2_I, t^2_{II}) = \frac{2}{3}$. There are 9 Bayesian equilibria according to which the players of type vector (t^1_I, t^1_{II}) play one of the following (1) (T, R); (2) (B, L); and (3) Player I plays $(\frac{2}{3}(T), \frac{1}{3}(B))$ and Player II plays $(\frac{2}{3}(R), \frac{1}{3}(L))$. As well as, the players of type vector (t^2_I, t^2_{II}) play one of the strategy vectors (1)-(3).

(c) In the current situation, the set of types of Player I (Carolyn) is $T_I = \{t_I^1, t_I^2\}$ where t_I^1 and t_I^2 corresponding the situation in which John won and did not win the Wimbledon tennis tournament, respectively; and the set of types of Player II (Maurice) is $T_{II} = \{t_{II}^1, t_{II}^2\}$ where t_{II}^1 and t_{II}^2 corresponding the situation in which Jim won and did not win the Wimbledon tennis tournament, respectively. $p(t_I^1, t_{II}^2) = \frac{1}{3}$, $p(t_I^2, t_{II}^1) = \frac{1}{3}$ and $p(t_I^2, t_{II}^2) = \frac{1}{3}$. There are 27 Bayesian equilibria according to which the players of each type vector play one of the following strategy vector: (1) (T, R); (2) (B, L); and (3) Player I plays $(\frac{2}{3}(T), \frac{1}{3}(B))$ and Player II plays $(\frac{2}{3}(R), \frac{1}{3}(L))$.

In addition there are six equilibria according to which the pure strategy vectors (T, L), (T, R), and (B, L) are played by the three type vectors (which occurs with positive probability). These equilibria corresponding the correlated equilibrium under which the three action vectors, (T, L), (T, R), and (B, L), are chosen with equal probability.

Chapter 10: Games with incomplete information: the general model

10.1 (a) $(Y, \mathcal{Y}, s, (\pi_i)_{i \in N})$ is a belief space of N over S.

The function $(\pi_i)_{i \in N}$ satisfies the coherency condition:

For Player I,

$$\{\omega' \in Y : \pi_I(\omega') = \pi_I(\omega_1)\} = \{\omega_1\} \in \mathcal{Y} \text{ and } \pi_I(\{\omega_1\}|\omega_1) = 1,$$
$$\{\omega' \in Y : \pi_I(\omega') = \pi_I(\omega_2)\} = \{\omega_2, \omega_3\} \in \mathcal{Y} \text{ and } \pi_I(\{\omega_2, \omega_3\}|\omega_2) = 1.$$

For Player II,

$$\{\omega' \in Y : \pi_{II}(\omega') = \pi_{II}(\omega_1)\} = \{\omega_1, \omega_2, \omega_3\} = Y \in \mathcal{Y} \text{ and } \pi_{II}(\{Y\}|\omega_1) = 1.$$

The function $(\pi_i)_{i \in N}$ satisfies the measurability condition: let $E \in \mathcal{Y}$. Let $i \in N$. Define $f : Y \to [0, 1]$ by $f(\omega) = \pi_i(E|\omega)$ for every $\omega \in Y$. Since $\pi_i(\omega_2) = \pi_i(\omega_3)$, then $f(\omega_2) = f(\omega_3)$. Hence, for every measurable subset C of $[0, 1]$, $\omega_2 \in f^{-1}(C)$ if and only if $\omega_3 \in f^{-1}(C)$, and thus $f^{-1}(C) \in \mathcal{Y}$ so f is a measurable function.

(b) In this part, $(Y, \mathcal{Y}, s, (\pi_i)_{i \in N})$ is a belief space of N over S. The proof is similar to that in Part (a) and thus it was omitted.

(c) $(Y, \mathcal{Y}, s, (\pi_i)_{i\in N})$ is not a belief space of N over S. The function $(\pi_i)_{i\in N}$ does not satisfy the coherency condition:

$$\{\omega' \in Y : \pi_I(\omega') = \pi_I(\omega_1)\} = \{\omega_1\} \in \mathcal{Y}$$
$$\text{but } \pi_I(\{\omega_1\}|\omega_1) = \tfrac{1}{3} < 1.$$

10.2 Let $\Pi = (Y, \mathcal{Y}, s, (\pi_i)_{i\in N})$ and $\widehat{\Pi} = (\widehat{Y}, \widehat{\mathcal{Y}}, \widehat{s}, (\widehat{\pi}_i)_{i\in N})$ be two belief spaces satisfying $Y \cap \widehat{Y} = \emptyset$.
Let $\widetilde{\Pi} = (\widetilde{Y}, \widetilde{\mathcal{Y}}, \widetilde{s}, (\widetilde{\pi}_i)_{i\in N})$ where

- $\widetilde{Y} := Y \bigcup \widehat{Y}$.
- $\widetilde{\mathcal{Y}} := \{F \bigcup \widehat{F} : F \in \mathcal{Y}, \widehat{F} \in \widehat{\mathcal{Y}}\}$.
- $\widetilde{s}(\omega) = s(\omega)$ for every $\omega \in Y$ and $\widetilde{s}(\widehat{\omega}) = \widehat{s}(\widehat{\omega})$ for every $\widehat{\omega} \in \widehat{Y}$.
- $\widetilde{\pi}_i(\omega) = \pi_i(\omega)$ for every $\omega \in Y$ and $\widetilde{\pi}_i(\widehat{\omega}) = \widehat{\pi}_i(\widehat{\pi})$ for every $\widehat{\omega} \in \widehat{Y}$.

We prove that $\widetilde{\Pi}$ is a belief space.

- $\widetilde{\mathcal{Y}}$ is a σ-algebra:
  - The empty set is contained in $\mathcal{Y}$ and $\widehat{\mathcal{Y}}$, hence $\emptyset = \emptyset \bigcup \emptyset \in \widetilde{\mathcal{Y}}$.
  - $\widetilde{\mathcal{Y}}$ is closed under complementation: let $F \bigcup \widehat{F} \in \widetilde{\mathcal{Y}}$ where $F \in \mathcal{Y}$ and $\widehat{F} \in \widehat{\mathcal{Y}}$. Then, $Y \setminus F \in \mathcal{Y}$ and $\widehat{Y} \setminus \widehat{F} \in \widehat{\mathcal{Y}}$. Thus

    $$(F \bigcup \widehat{F})^c = F^c \cap \widehat{F}^c = (Y \setminus F \bigcup \widehat{Y}) \cap (\widehat{Y} \setminus \widehat{F} \bigcup Y) = (Y \setminus F) \bigcup (\widehat{Y} \setminus \widehat{F}) \in \widetilde{\mathcal{Y}}.$$

  - $\widetilde{\mathcal{Y}}$ is closed under countable intersections: let $\{F_k \cup \widehat{F}_k\}_{k=1}^{\infty} \in \widetilde{\mathcal{Y}}$ where $\{F_k\}_{k=1}^{\infty} \in \mathcal{Y}$ and $\{\widehat{F}_k\}_{k=1}^{\infty} \in \widehat{\mathcal{Y}}$. Then, $\bigcap_{k=1}^{\infty} F_k \in \mathcal{Y}$ and $\bigcap_{k=1}^{\infty} \widehat{F}_k \in \widehat{\mathcal{Y}}$. Thus

    $$\bigcap_{k=1}^{\infty} (F_k \bigcup \widehat{F}_k) = \left(\bigcap_{k=1}^{\infty} F_k\right) \bigcup \left(\bigcap_{k=1}^{\infty} \widehat{F}_k\right) \in \widetilde{\mathcal{Y}}.$$

- $\widetilde{s} : \widetilde{Y} \to S$ is a measurable function, mapping each state of the world to a state of nature:
  Let $S' \in \mathcal{S}$. Since s and $\widehat{s}$ are measurable, $s^{-1}(S') \in \mathcal{Y}$ and $\widehat{s}^{-1}(S') \in \widehat{\mathcal{Y}}$ hence
  $$\widetilde{s}^{-1}(S') = s^{-1}(S') \bigcup \widehat{s}^{-1}(S') \in \widetilde{\mathcal{Y}}.$$
  so s is measurable.
- The function $(\widetilde{\pi}_i)_{i\in N}$ satisfies the coherency condition:
  Let $i \in N$ and let $\omega \in \widetilde{Y}$. Hence, either $\omega \in Y \setminus \widehat{Y}$ or $\omega \in \widehat{Y} \setminus Y$. Assume without loss of generality that $\omega \in Y \setminus \widehat{Y}$. Since $Y \cap \widehat{Y} = \emptyset$,

for every $\omega' \in \widehat{Y}$, $\widetilde{\pi}_i(\widehat{Y}|\omega) = \pi_i(\widehat{Y}|\omega) = 0$, whereas $\widetilde{\pi}_i(\widehat{Y}|\omega') = \widehat{\pi}_i(\widehat{Y}|\omega') = 1$. Therefore, and by the coherency of $(\pi_i)_{i\in N}$ and $(\widehat{\pi}_i)_{i\in N}$,

$$\begin{aligned}\{\omega' \in \widetilde{Y} : \widetilde{\pi}_i(\omega') = \widetilde{\pi}_i(\omega)\} &= \{\omega' \in Y : \widetilde{\pi}_i(\omega') = \widetilde{\pi}_i(\omega)\} \\ &= \{\omega' \in Y : \pi_i(\omega') = \pi_i(\omega)\} \in \mathcal{Y} \subseteq \widetilde{\mathcal{Y}}.\end{aligned}$$

Furthermore,

$$\widetilde{\pi}_i(\{\omega' \in \widetilde{Y} : \widetilde{\pi}_i(\omega') = \widetilde{\pi}_i(\omega)\}|\omega) = \pi_i(\{\omega' \in Y : \pi_i(\omega') = \pi_i(\omega)\}|\omega) = 1,$$

as needed.

- The function $(\widetilde{\pi}_i)_{i\in N}$ satisfies the measurability condition:
  Let $i \in N$ and let $E \in \widetilde{\mathcal{Y}}$. Then $E = F \bigcup \widehat{F}$ where $F \in \mathcal{Y}$ and $\widehat{F} \in \widehat{\mathcal{Y}}$. Define $f : \widetilde{Y} \to [0,1]$ by $f(\omega) = \widetilde{\pi}_i(E|\omega)$ for every $\omega \in \widetilde{Y}$. For every measurable subset $C \subset [0,1]$,

  $$\begin{aligned}f^{-1}(C) &= \{\omega \in \widetilde{Y} : \widetilde{\pi}_i(E|\omega) \in C\} \\ &= \{\omega \in Y : \pi_i(E|\omega) \in C\} \bigcup \{\omega \in \widehat{Y} : \widehat{\pi}_i(E|\omega) \in C\} \in \widetilde{\mathcal{Y}}\end{aligned}$$

  where the last phase following the measurability condition of $(\pi_i)_{i\in N}$ and $(\widehat{\pi}_i)_{i\in N}$. In particular, $(\widetilde{\pi}_i)_{i\in N}$ satisfies the measurability condition as well.

10.3 Let $\Pi = (Y, \mathcal{Y}, s, (\pi_i)_{i\in N})$ be a belief space. Let $\mathcal{F}_i$ be a partition of Y based on his beliefs, i.e., $F_i(\omega) = \{\omega'' \in Y : \pi_i(\omega') = \pi_i(\omega)\}$.
For every player $i \in N$ and every $\omega \in Y$

$$\mathbf{P}(A|F_i(\omega)) = \tfrac{\mathbf{P}(A\cap F_i(\omega))}{\mathbf{P}(F_i(\omega))}$$

However,

$$\begin{aligned}\mathbf{P}(A \cap F_i(\omega)) &= \sum_{\omega'\in Y} \tfrac{1}{|Y|}\pi_i(A \cap F_i(\omega)|\omega') && (85)\\ &= \sum_{\omega'\in F_i(\omega)} \tfrac{1}{|Y|}\pi_i(A \cap F_i(\omega)|\omega') + \sum_{\omega'\in F_i(\omega)^c} \tfrac{1}{|Y|}\pi_i(A \cap F_i(\omega)|\omega') \\ &= \sum_{\omega'\in F_i(\omega)} \tfrac{1}{|Y|}\pi_i(A|\omega') + \sum_{\omega'\in F_i(\omega)^c} \tfrac{1}{|Y|} \cdot 0 && (86)\\ &= \tfrac{|F_i(\omega)|}{|Y|}\pi_i(A|\omega)\end{aligned}$$

where Equality (85) follows by Equality (10.9), and Equality (86) holds by the coherency condition. Similarly,

$$\begin{aligned}
\mathbf{P}(F_i(\omega)) &= \sum_{\omega' \in Y} \tfrac{1}{|Y|}\pi_i(F_i(\omega)|\omega') \\
&= \sum_{\omega' \in F_i(\omega)} \tfrac{1}{|Y|}\pi_i(F_i(\omega)|\omega') + \sum_{\omega' \in F_i(\omega)^c} \tfrac{1}{|Y|}\pi_i(F_i(\omega)|\omega') \\
&= \sum_{\omega' \in F_i(\omega)} \tfrac{1}{|Y|} \cdot 1 + \sum_{\omega' \in F_i(\omega)^c} \tfrac{1}{|Y|} \cdot 0 \\
&= \tfrac{|F_i(\omega)|}{|Y|}.
\end{aligned}$$

Therefore,

$$\mathbf{P}(A|F_i(\omega)) = \frac{\frac{|F_i(\omega)|}{|Y|}\pi_i(A|\omega)}{\frac{|F_i(\omega)|}{|Y|}} = \pi_i(A|\omega).$$

10.4 The set of players is $N = \{\text{Minerva, Hercules}\}$ and let the set of states of nature be $S = \{s_1, s_2\}$ where s_1 is the state of nature in which Hercules is able to lift a massive rock, and s_2 is the state of nature in which Hercules is not able to lift a massive rock. Consider a belief space $Y = \{\omega_1, \omega_2\}$; $\mathcal{Y} = 2^Y$; $s(\omega_1) = s_1$ and $s(\omega_2) = s_2$; $\pi_{\text{Minerva}}(\omega_1) = \pi_{\text{Minerva}}(\omega_2) = [0.7(\omega_1), 0.3(\omega_2)]$; and $\pi_{\text{Hercules}}(\omega_1) = \pi_{\text{Hercules}}(\omega_2) = [1(\omega_1)]$.

10.5 The set of players is $N = \{\text{Minerva, Hercules}\}$ and let the set of states of nature be $S = \{s_1, s_2\}$ where s_1 is the state of nature in which Hercules is able to lift a massive rock, and s_2 is the state of nature in which Hercules is not able to lift a massive rock. Consider a belief space $Y = \{\omega_1, \omega_2, \omega_3\}$; $\mathcal{Y} = 2^Y$; $s(\omega_1) = s_1$ and $s(\omega_2) = s(\omega_3) = s_2$; $\pi_{\text{Minerva}}(\omega_1) = \pi_{\text{Minerva}}(\omega_2) = \pi_{\text{Minerva}}(\omega_3) = [0.7(\omega_1), 0.3(\omega_2)]$; and $\pi_{\text{Hercules}}(\omega_1) = \pi_{\text{Hercules}}(\omega_2) = [1(\omega_1)]$ and $\pi_{\text{Hercules}}(\omega_3) = [1(\omega_3)]$. The described situation occurs at state of the world ω_3.

10.6 The set of players is $N = \{\text{Eric, Jack}\}$ and let the set of states of nature be $S = \{s_1, s_2\}$ where s_1 is the state of nature in which the New York Mets won the baseball World Series in 1969, and s_2 is the state of nature in which it did not win the baseball World Series in 1969. Consider a belief space $Y = \{\omega_1, \omega_2, \omega_3\}$; $\mathcal{Y} = 2^Y$; $s(\omega_1) = s(\omega_3) = s_1$ and $s(\omega_2) = s_2$; $\pi_{\text{Eric}}(\omega_1) = \pi_{\text{Eric}}(\omega_2) = [1(\omega_1)]$, $\pi_{\text{Eric}}(\omega_3) = [1(\omega_3)]$; and $\pi_{\text{Jack}}(\omega_1) = [1(\omega_1)]$, and $\pi_{\text{Jack}}(\omega_2) = \pi_{\text{Jack}}(\omega_3) = [0.5(\omega_2), 0.5(\omega_3)]$ The described situation occurs at state ω_2. Eric believes that the true state of the world is ω_1 at which $\{\omega_1\}$ is a common belief among him and Jack (by Theorem 10.10). Jack ascribes probability 0.5 to ω_2 (under which the New York Mets won the baseball World Series in 1969 and Jack believes that this is a common belief among them) and probability 0.5 to ω_3 (under which the New York Mets did not win the baseball World Series in 1969 and this is not a common belief among them).

10.7 In all situation $N = \{\text{Roger,Jimmy}\}$, $S = \{s_1, s_2\}$ where s_1 is the state of nature in which the Philadelphia Phillies winning the World Series, and s_2 is the state of nature in which the Philadelphia Phillies not winning the World Series.

(a) Let $Y = \{\omega_1, \omega_2\}$, where $s(\omega_1) = s_1$ and $s(\omega_2) = s_2$. The belief of Roger is $\pi_{\text{Roger}}(\omega_1) = \pi_{\text{Roger}}(\omega_2) = [0.4(\omega_1), 0.6(\omega_2)]$.

(b) Let $Y = \{\omega_1, \omega_2, \omega_3, \omega_4\}$, where $s(\omega_{2k-1}) = s_1$ and $s(\omega_{2k}) = s_2$ for $k = 1, 2$. The belief of Roger is $\pi_{\text{Roger}}(\omega_k) = [0.4(\omega_1), 0.6(\omega_2)]$. The belief of Jimmy is $\pi_{\text{Jimmy}}(\omega_k) = [0.3(\omega_3), 0.7(\omega_4)]$. Either one of the states may be the true state.

(c) Let $Y = \{\omega_1, \omega_2, ..., \omega_{14}\}$ and let $s(\omega_{2k-1}) = s_1$ and $s(\omega_{2k}) = s_2$. The belief function of Roger is

$$\pi_{\text{Roger}}(\omega) = \begin{cases} [0.3(\omega_1), 0.7(\omega_2)] & \omega = \omega_1, \omega_2 \\ [0.4(\omega_3), 0.6(\omega_4)] & \omega = \omega_3, \omega_4 \\ [0.2(\omega_5), 0.8(\omega_6)] & \omega = \omega_5, \omega_6 \\ [0.5(\omega_7), 0.5(\omega_8)] & \omega = \omega_7, \omega_8 \\ [0.4(\omega_9), 0.6(\omega_{10})] & \omega = \omega_9, \omega_{10} \\ [0.1(\omega_{11}), 0.9(\omega_{12})] & \omega = \omega_{11}, \omega_{12} \\ [0.1(\omega_{13}), 0.9(\omega_{14})] & \omega = \omega_{13}, \omega_{14}. \end{cases}$$

The belief function of Jimmy is

$$\pi_{\text{Jimmy}}(\omega) = \begin{cases} [0.3(\omega_3), 0.7(\omega_4)] & \omega = \omega_1, \omega_3, \omega_4 \\ [0.7(\omega_7), 0.3(\omega_8)] & \omega = \omega_5, \omega_6, \omega_7, \omega_8 \\ [0.2(\omega_{11}), 0.8(\omega_{12})] & \omega = \omega_{10}, \omega_{11}, \omega_{12} \\ [0.6(\omega_9), 0.4(\omega_2)] & \omega = \omega_2, \omega_9, \omega_{13}, \omega_{14}. \end{cases}$$

The first situation is at state of the world ω_1, the second is at state of the world ω_5 (or ω_6), the third is at ω_{10} and the last is at ω_{13} (or ω_{14}).

10.8 Proof of Theorem 10.7:

(a) By the coherency, $\pi_i(\{\omega' \in Y : \pi_i(\omega') = \pi_i(\omega)\}|\omega) = 1$ for every $\omega \in Y$. However, by the monotonicity of $\pi(\cdot|\omega)$, $\pi_i(Y|\omega) = 1$. Therefore,

$$B_i Y := \{\omega \in Y : \pi_i(Y|\omega) = 1\} = Y.$$

(b) For every $\omega \in Y$ and every two events $A, C \subseteq Y$ the following holds: If $P(A \cap C|\omega) = 1$ then, by the monotonicity of $\pi(\cdot|\omega)$, $\pi(A|\omega) = 1$ as well as $\pi(C|\omega) = 1$. In addition, if $\pi(A|\omega) = 1$ and $\pi(C|\omega) = 1$ then, by the Inclusion-Exclusion Principle, $\pi(A \cap C|\omega) = \pi(A|\omega) + \pi(C|\omega) - \pi(A \bigcup C|\omega) = 1$. Hence,

$$\begin{aligned} B_i A \cap B_i C &= \{\omega \in Y : \pi_i(A|\omega) = 1\} \cap \{\omega \in Y : \pi_i(C|\omega) = 1\} \\ &= \{\omega \in Y : \pi_i(A \cap C|\omega) = 1\} = B_i(A \cap C). \end{aligned}$$

(c) Let $\omega \in B_i A$, i.e., $\pi_i(A|\omega) = 1$. Hence, $\pi_i(A|\omega') = \pi_i(A|\omega) = 1$ for every $\omega' \in F_i(\omega)$. In particular, $F_i(\omega) \subseteq B_i A$. However, by the coherency, $\pi_i(F_i(\omega)|\omega) = 1$ and thus (by the monotonicity of π_i) $\pi_i(B_i A|\omega) = 1$ so $\omega \in B_i B_i A$. In conclusion, $B_i A \subseteq B_i B_i A$. Furthermore, for every $\omega \in B_i A$ then $\pi_i(A|\omega) = 1$ and (as proved) $\pi_i(B_i A|\omega) = 1$ from which it follows that $\pi_i(A \cap B_i A|\omega) = 1$, i.e., $\omega \in B_i(A \cap B_i A)$. Thus,

$$B_i A \subseteq B_i(A \cap B_i A) = B_i A \cap B_i B_i A \Rightarrow B_i A \subseteq B_i B_i A.$$

Therefore, $B_i A = B_i B_i A$.

(d) We first prove that $(B_i A)^c \subset B_i((B_i A)^c)$. Let $\omega \in (B_i A)^c$ hence $F_i(\omega) \subseteq (B_i A)^c$. However, by the coherency $\pi_i(F_i(\omega)|\omega) = 1$ and thus by the monotonicity $\pi_i((B_i A)^c|\omega) = 1$ so $\omega \in B_i((B_i A)^c)$. We next prove that $B_i((B_i A)^c) \subseteq (B_i A)^c$. Let $\omega \in B_i((B_i A)^c)$. That is $\pi_i((B_i A)^c|\omega) = 1$. Hence, $\pi_i(B_i A|\omega) = 0$ so (by Part (c)) $\pi_i(A|\omega) < 1$. Therefore, $\omega \in (B_i A)^c$.

10.9 Let Π be a belief space equivalent to an Aumann model of incomplete information and let B_i be player i's belief operator in Π. We prove that the knowledge operator K_i that is defined by the partition via Equation (10.9) is the same operator as the belief operator B_i.

$$\omega \in K_i A \Rightarrow F_i(\omega) \subseteq A \Rightarrow 1 \leq \pi_i(F_i(\omega)|\omega) \leq \pi_i(A|\omega) \leq 1 \Rightarrow \omega \in B_i A.$$

Therefore, $K_i A \subseteq B_i A$.

For the other direction, let $\omega \in B_i A$ so $\pi_i(A|\omega) = 1$. However, since $\pi_i(F_i(\omega)|\omega) = 1$ as well, it follows that $\pi_i(A \cap F_i(\omega)|\omega) = 1$. Therefore,

$$\mathbf{P}(F_i(\omega) \setminus A|F_i(\omega)) = \mathbf{P}(F_i(\omega)|F_i(\omega)) - \mathbf{P}(F_i(\omega) \cap A|F_i(\omega)) = 0. \tag{87}$$

However, by Definition 9.27, $\mathbf{P}(\omega') > 0$ for every $\omega' \in Y$. Therefore, Equation (87) , yields $F_i(\omega) \setminus A = \emptyset$ and thus $F_i(\omega) \subseteq A$ so $\omega \in K_i A$. In conclusion, $B_i A \subseteq K_i A$.

10.10 We show that the converse of the statement of Theorem 10.10 does not hold. Consider a belief space $Y = \{\omega_1, \omega_2, \omega_3, \omega_4\}$, where:

State of the world	$\pi_I(\cdot)$	$\pi_{II}(\cdot)$
ω_1	$[1(\omega_2)]$	$[0.5(\omega_1), 0.5(\omega_4)]$
ω_2	$[1(\omega_2)]$	$[1(\omega_2)]$
ω_3	$[0.5(\omega_3), 0.5(\omega_4)]$	$[1(\omega_2)]$
ω_4	$[0.5(\omega_3), 0.5(\omega_4)]$	$[0.5(\omega_1), 0.5(\omega_4)]$

The event $A = \{\omega_2, \omega_3\}$ is common belief among the players at the state of the world ω_2, but $\pi_I(A|\omega_3) < 1$.

10.11 Consider a an Aumann model of incomplete information in which $N = \{I\}$, $Y = \{\omega_1, \omega_2\}$, $\mathcal{F}_I = \{\{\omega_1, \omega_2\}\}$, $\mathbf{P}(\omega_1) = 1$ and $\mathbf{P}(\omega_2) = 0$. Then, $\pi_I(\omega_1) = \pi_I(\omega_2) = ([1(\omega_1)])$. Therefore, the knowledge operator in the Aumann model satisfies $K_I(\{\omega_1\}) = \{\omega_1, \omega_2\}$ whereas the belief operator $B_I(\{\omega_1\}) = \{\omega_1\}$.

10.12 The beliefs of the players in Example 10.4 are as follows:
At both states of the world ω_1 and ω_2 Player I believes that the state of nature is s_1. On the other hand, at state of the world ω_3 he believes that the state of nature is s_2. In addition, at every state of the world Player I believes that Player II believes that the state of nature is either s_1 or s_2 with equal probability and that Player I believes that the state of nature is s_1.
At every state of the world, Player II believes that the state of nature is either s_1 or s_2 with equal probability. Moreover, he believes that Player I believes that the state of nature is s_1.

10.13 Proof of Theorem 10.8: Let $i \in N$ be a player and let $A, C \subseteq Y$ be a pair of events such that $A \subseteq C$.
By a monotonicity of a probability function, $\pi_i(A|\omega) \leq \pi_i(C|\omega)$ for every $\omega \in Y$. Therefore, for every $\omega \in B_i A$

$$\pi_i(A|\omega) = 1 \Rightarrow 1 = \pi(A|\omega) \leq \pi(C|\omega) \leq 1 \Rightarrow \omega \in B_i C.$$

That is, $B_iA \subseteq B_iC$, as needed.

10.14 Let $\Pi = (Y, Y, s, (\pi_i)_{i \in N})$ be a belief space equivalent to an Aumann model of incomplete information $\Pi_A = (N, Y, (\mathcal{F}_i)_{i \in N}, s, \mathbf{P})$. That is, for every player $i \in N$ and every state of the world $\omega \in Y$, $\pi_i(\omega) = \mathbf{P}(\cdot|F_i(\omega))$ where $F_i(\omega) \in \mathcal{F}_i$ is the partition element containing the state of the world ω.
We prove that the partition defined by Equation (10.8): $\widehat{F}_i(\omega) = \{\omega' \in Y | \pi_i(\omega) = \pi_i(\omega')\}$ for the belief space Π is the same partition as the partition $\mathcal{F}_i$ in the equivalent Aumann model, for every player i. That is, $F_i(\omega) = \widehat{F}_i(\omega)$ for every player i and every $\omega \in Y$.
Let $\omega' \in F_i(\omega)$ so $F_i(\omega) = F_i(\omega')$. Hence,

$$\pi_i(\omega) = \mathbf{P}(\cdot|F_i(\omega)) = \mathbf{P}(\cdot|F_i(\omega')) = \pi_i(\omega'),$$

which yields to $\omega \in \widehat{F}_i(\omega)$. Hence, $F_i(\omega) \subseteq \widehat{F}_i(\omega)$.
Assume next that $\omega' \in \widehat{F}_i(\omega)$ so $\pi_i(\omega) = \pi_i(\omega')$. Therefore, $\mathbf{P}(\cdot|F_i(\omega)) = \mathbf{P}(\cdot|F_i(\omega'))$ and thus

$$1 = \mathbf{P}(F_i(\omega')|F_i(\omega')) = \mathbf{P}(F_i(\omega')|F_i(\omega)).$$

Hence, by Theorem 9.29, $\omega' \in F_i(\omega') \subseteq F_i(\omega)$. In conclusion, $\widehat{F}_i(\omega) \subseteq F_i(\omega)$, which completes the proof.

10.15 For every player $i \in N$ and every real number $p \in [0,1]$, define B_i^p to be the operator mapping each set $E \in \mathcal{Y}$ to the set of states of the world at which player i ascribes to E probability equal to or greater than p,

$$B_i^p(E) := \{\omega \in Y : \pi_i(E|\omega) \geq p\}.$$

(a) The property $B_i^p(Y) = Y$ holds for every $p \in [0,1]$: on the one hand, $B_i^p(Y) \subseteq Y$ by definition. On the other hand, for every $\omega \in Y$, $\pi_i(Y|\omega) = 1 \geq p$ so $y \in B_i^p(Y)$.

(b) The property $B_i^p(A) \subseteq A$ does not hold (for every $p \in [0,1]$): Example 10.5 is a counter example. $B_i^p(\{\omega_1\}) = \{\omega_1, \omega_2\} \not\subseteq \{\omega_1\}$ for every $p \in [0,1]$.

(c) The following property holds for every $p \in [0,1]$: if $A \subseteq C$ then $B_i^p(A) \subseteq B_i^p(C)$. Indeed, if $\omega \in B_i^p(A)$ then $\pi_i(A|\omega) \geq p$. However, by the monotonicity of a probability function, $\pi_i(C|\omega) \geq \pi_i(A|\omega) \geq p$. Thus $\omega \in B_i^p(C)$.

(d) The property $B_i^p(A) = B_i^p(B_i^p(A))$ holds for every $p \in [0,1]$. For $p = 0$, $B_i^0(A) = Y$ for every $A \in \mathcal{Y}$ and thus $B_i^p(B_i^p(A)) = B_i^p(Y) = Y = B_i^p(A)$.
Let $p > 0$. If $\omega \in B_i^p(A)$ then for every $\omega' \in F_i(\omega)$ one has

$$\pi_i(A|\omega') = \pi_i(A|\omega) \geq p \Rightarrow \omega' \in B_i^p(A).$$

Therefore, $F_i(\omega) \subseteq B_i^p(A)$. In particular, by the monotonicity of a probability,

$$p \leq 1 = \pi_i(F_i(\omega)|\omega) \leq \pi_i(B_i^p(A)|\omega) \Rightarrow \omega \in B_i^p(B_i^p(A)).$$

Hence, $B_i^p(A) \subseteq B_i^p(B_i^p(A))$. If on the contrary, $\omega \notin B_i^p(A)$ then $\pi_i(A|\omega) < p$. Therefore, for every $\omega' \in F_i(\omega)$ one has $\pi_i(A|\omega') = \pi_i(A|\omega) < p$ so $F_i(\omega) \cap B_i^p(A) = \emptyset$. Thus, $\pi_i(B_i^p(A)|\omega) = 0 < p$ from which it follows that $\omega \notin B_i^p(B_i^p(A))$.

(e) The property $B_i^p((B_i^p(A)^c) = (B_i^p(A))^c$ holds for every $p > 0$: if $\omega \in (B_i^p(A))^c$ then $\pi_i(A|\omega) < p$ from which it follows that $F_i(\omega) \subseteq (B_i^p(A))^c$. Hence,

$$\pi_i((B_i^p(A))^c|\omega) \geq \pi_i(F_i(\omega)|\omega) = 1 \geq p \Rightarrow w \in B_i^p((B_i^p)^c).$$

If on the contrary $\omega \notin (B_i^p(A))^c$ then $\pi_i(A|\omega) \geq p$ from which it follows that $F_i(\omega) \cap (B_i^p(A))^c = \emptyset$. Hence,

$$\pi_i((B_i^p(A))^c|\omega)0 < p \Rightarrow w \notin B_i^p((B_i^p)^c).$$

Note that for $p = 0$, for every $B_i^p(A) = Y$ for every event A and thus

$$B_i^p((B_i^p(A))^c) = Y \neq \emptyset = Y^c = (B_i^p(A))^c,$$

so the property does not hold.

10.16 Let $\widetilde{Y}$ be a belief subspace of a belief space $\Pi = (Y, \mathcal{Y}, s, (\pi_i)_{i\in N})$, and let $\omega \in \widetilde{Y}$.
Then, by Definition 10.21, for every player $i \in N$, $\pi_i(\widetilde{Y}|\omega) = 1$. Furthermore, for every $\omega' \in \widetilde{Y}$ and every player $i \in N$, $\pi_i(\widetilde{Y}|\omega') = 1$. Therefore, by Theorem 10.10, the event $\widetilde{Y}$ is common belief among the players at every state of the world $\omega \in \widetilde{Y}$.

10.17 Let $\Pi = (Y, \mathcal{Y}, s, (\pi_i)_{i\in N})$ be a belief space equivalent to an Aumann model of incomplete information. We prove that for each state of the world ω, the common knowledge component $C(\omega)$ among the players at ω is a belief subspace.
For every player $i \in N$ and every $\omega \in C(\omega)$ one has $F_i(\omega) \subseteq C(\omega)$. Therefore,

$$1 = \pi_i(F_i(\omega)|\omega) \leq \pi_i(C(\omega)|\omega) \leq 1$$

and thus $\pi_i(C(\omega)|\omega) = 1$, as needed.

10.18 Consider Example 10.19, and suppose that the beliefs of the types in that example are as given in the exercise, where $x, y, z, w \in [0, 1]$.
Assume first that $x, y, z, w \in (0, 1)$. Then, the belief system of the players consistent if there is a common prior $(p_{11}, p_{12}, p_{21}, p_{22})$ over $T_I \times T_{II}$ satisfying $p_{ij} > 0$ for every $i, j = 1, 2$ and

$$\begin{aligned} x &= \tfrac{p_{11}}{p_{11}+p_{12}} \\ y &= \tfrac{p_{21}}{p_{21}+p_{22}} \\ z &= \tfrac{p_{11}}{p_{11}+p_{21}} \\ w &= \tfrac{p_{12}}{p_{12}+p_{22}}. \end{aligned}$$

If and only if

$$\left.\begin{aligned} \tfrac{x}{1-x} &= \tfrac{p_{11}}{p_{12}} \\ \tfrac{y}{1-y} &= \tfrac{p_{21}}{p_{22}} \\ \tfrac{z}{1-z} &= \tfrac{p_{11}}{p_{21}} \\ \tfrac{w}{1-w} &= \tfrac{p_{12}}{p_{22}}. \end{aligned}\right\} \Rightarrow$$

$$\tfrac{x}{1-x} \cdot \tfrac{w}{1-w} = \tfrac{y}{1-y} \cdot \tfrac{z}{1-z}. \tag{88}$$

In conclusion, if the values x, y, z, and w are in $(0, 1)$ satisfying Equation (88), then one can define

$$\begin{aligned} p_{11} &= 1/(1 + \tfrac{1-x}{x} + \tfrac{1-x}{x} \cdot \tfrac{1-w}{w} + \tfrac{1-z}{z}) \\ p_{12} &= \tfrac{1-x}{x} p_{11} \\ p_{22} &= \tfrac{1-w}{w} \tfrac{1-x}{x} p_{11} \\ p_{21} &= \tfrac{1-z}{z} p_{11}, \end{aligned}$$

and thus the belief system of the players is consistent.
In addition, x, y, z and w satisfy one of the following conditions then the the belief system of the players is consistent: $x = y = 0$; $x = y = 1$; $z = w = 0$; $z = w = 1$; $x = 1 - z = 0$; $x = z = 1$; $x = 1 - w = 1$; $x = 1 - w = 0$; $y = 1 - z = 0$ and $y = z = 0$.

10.19 We show that the beliefs of the players in Example 10.20 are inconsistent.
Assume by contradiction that there is a a common prior over $[0, 1]^2$, in particular the probability $\mathbf{P}(|X - Y| \leq 0.1) \in [0, 1]$ is well defined, where X and Y indicate the the first and the second coordinates respectively.
Let Z be a random variable defined as follows:

$$Z = \begin{cases} 1 & \text{if } |X - Y| \leq 0.1 \\ 0 & \text{otherwise.} \end{cases}$$

Then, $E[Z] = \mathbf{P}(Z=1) = \mathbf{P}(|X-Y| \le 0.1)$. The law of total expectation, and the beliefs of Player I yield to

$$\mathbf{P}(|X-Y| \le 0.1) = E[Z] = E[E[Z|X]] = E[1] = 1.$$

However, the beliefs of Player II yield to

$$\mathbf{P}(|X-Y| \le 0.1) = E[Z] = E[E[Z|Y]] = E[0] = 0,$$

a contradiction.

10.20 (a) The set of types of Player I are $T_I = \{\mathrm{I}_1, \mathrm{I}_2\} = \{[\frac{2}{5}(\omega_1), \frac{3}{5}(\omega_2)], [1(\omega_1)]\}$. The set of types of Player II are $T_{\mathrm{II}} = \{\mathrm{II}_1, \mathrm{II}_2\} = \{[1(\omega_1)], [\frac{3}{4}(\omega_1), \frac{1}{4}(\omega_2)]\}$. The types of the two players at each state of the world are given in the next table.

State of the world	Type of Player I	Type of Player II
ω_1	I_1	II_1
ω_2	I_1	II_2
ω_3	I_2	II_2

(b) The beliefs of the players can be derived from the common prior $\mathbf{P}(\omega_1) = \frac{1}{3}$, $\mathbf{P}(\omega_2) = \frac{1}{2}$ and $\mathbf{P}(\omega_3) = \frac{1}{6}$.

10.21 (a) Consider the belief space Π of the players $N = \{Boris, Alex\}$. The set of natures is $S = \{s_1, s_2\}$ where s_1 corresponds to "Bruce Jenner won a gold medal at the Montreal Olympics" and s_2 corresponds to "Bruce Jenner won a silver medal at the Montreal Olympics". The set of world state is $Y = \{\omega_1, \omega_2, \omega_3\}$. The beliefs of the players are as follows:

	$\mathbf{s}(\cdot)$	π_{Boris}	π_{Alex}
ω_1	s_1	$[1(\omega_1)]$	$[1(\omega_1)]$
ω_2	s_2	$[1(\omega_1)]$	$[1(\omega_2)]$
ω_3	s_2	$[1(\omega_2)]$	$[1(\omega_2)]$

(b) Assume that Π is a belief space in which the set of states of the world is a finite set and contains a state ω describing the situation in this exercise. We prove that $\omega \notin \mathrm{supp}(\pi_i(\omega'))$ for any state of the world ω', for $i \in \{\text{Boris, Alex}\}$.

Denote by A the event that "Bruce Jenner won a gold medal at the Montreal Olympics". Then, $\widetilde{Y}_{\text{Boris}}(\omega) \subseteq A$ and $\widetilde{Y}_{\text{Alex}}(\omega) \subseteq A^c$ and thus $\widetilde{Y}_{\text{Boris}}(\omega) \cap \widetilde{Y}_{\text{Alex}}(\omega) = \emptyset$. Therefore, by Theorem 10.31, $w \notin \widetilde{Y}_i(\omega)$ and thus $\pi_i(\{\omega\}|\omega) = 0$ for every $i \in \{Boris, Alex\}$. Next, assume by contradiction that there is a state of the world ω' and a player i such that $\pi_i(\{\omega\}|\omega') > 0$. Then, by the coherency property of a belief space, $\pi_i(\omega) = \pi_i(\omega')$ so $\pi_i(\{\omega\}|\omega) = \pi_i(\{\omega\}|\omega') > 0$, a contradiction.

10.22 After Priam's declaration, the fact that "Laocöon ascribes probability 0.6 to the Greeks attacking from within a wooden horse" is not necessarily a common belief among the players. Assume for instance that the belief space is equivalent to an Aumann model of incomplete information where $N = \{\text{Laocöon,Priam}\}$; $S = \{s_1, s_2\}$ where s_1 corresponds to the Greeks attacking from within a wooden horse, and s_2 is its complement; $Y = \{\omega_1, \omega_2, \omega_3, \omega_4\}$ where $s(\omega_1) = s(\omega_3) = s_1$ and $s(\omega_2) = s(\omega_4) = s_2$; $\mathcal{F}_L = \{\{\omega_1, \omega_2, \omega_3, \omega_4\}\}$; $\mathcal{F}_P = \{\{\omega_1, \omega_2\}, \{\omega_3, \omega_4\}\}$; and the common prior $\mathbf{P}$ is $\mathbf{P}(\omega_1) = \frac{7}{20}$, $\mathbf{P}(\omega_2) = \frac{3}{20}$, $\mathbf{P}(\omega_3) = \frac{5}{20}$, and $\mathbf{P}(\omega_4) = \frac{5}{20}$. Then after Priam's declaration, the fact that "Laocöon ascribes probability 0.7 to the Greeks attacking from within a wooden horse" is a common belief among the players.

10.23 The corresponding belief space is:

- $N = \{\text{I,II}\}$.
- $S = \{s_1, s_2\}$.
- $Y = \{\omega_1, \omega_2, \omega_3, \omega_4, \omega_5\}$, where
  - ω_1 corresponding to s_1 is chosen and the message is R;
  - ω_2 corresponding to s_1 is chosen and the message is L;
  - ω_3 corresponding to s_2 is chosen, the message is M and Player I is informed;
  - ω_4 corresponding to s_2 is chosen, the message is M and Player I is not informed;
  - ω_5 corresponding to s_2 is chosen and the message is L.
- The beliefs of the players are as follows

	s$(\cdot)$	π_{I}	π_{II}
ω_1	s_1	$[0.6(\omega_1), 0.4(\omega_2)]$	$[1(\omega_1)]$
ω_2	s_1	$[0.6(\omega_1), 0.4(\omega_2)]$	$[0.47(\omega_2), 0.53(\omega_5)]$
ω_3	s_2	$[1(\omega_3)]$	$[0.2(\omega_3), 0.8(\omega_4)]$
ω_4	s_2	$[0.65(\omega_4), 0.35(\omega_5)]$	$[0.2(\omega_3), 0.8(\omega_4)]$
ω_5	s_2	$[0.65(\omega_4), 0.35(\omega_5)]$	$[0.47(\omega_2), 0.53(\omega_5)]$

10.24 The beliefs of the player in the current situation is as follows:

	s(·)	π_{I}	π_{II}
ω_1	s_1	$[0.6(\omega_1), 0.4(\omega_2)]$	$[1(\omega_1)]$
ω_2	s_1	$[0.6(\omega_1), 0.4(\omega_2)]$	$[0.57(\omega_2), 0.43(\omega_5)]$
ω_3	s_2	$[1(\omega_3)]$	$[0.2(\omega_3), 0.8(\omega_4)]$
ω_4	s_2	$[0.65(\omega_4), 0.35(\omega_5)]$	$[0.2(\omega_3), 0.8(\omega_4)]$
ω_5	s_2	$[0.65(\omega_4), 0.35(\omega_5)]$	$[0.57(\omega_2), 0.43(\omega_5)]$

10.25 The corresponding belief space is:

- $N = \{$J,B,T$\}$ (corresponding John, Bob and Ted).
- $S = \{s_1, s_2, s_3, s_4, s_5, s_6, s_7, s_8\}$ where $s_1, s_2, s_3, s_4, s_5, s_6, s_7$ and s_8 corresponding to the sets of poets JBT, JB, JT, BT, J, B, T and $\emptyset$ respectively.
- $Y = \{\omega_1, \omega_2, \omega_3, \omega_4, \omega_5, \omega_6, \omega_7, \omega_8\}$ and $s(\omega_k) = s_k$ for every $k = 1, 2, ..., 8$.
- The beliefs of the players are as follows

State of the world	π_{J}
ω_1	$[1(\omega_1)]$
ω_2	$[1(\omega_2)]$
ω_3	$[1(\omega_3)]$
ω_4	$[\frac{9}{16}(\omega_4), \frac{3}{16}(\omega_6), \frac{3}{16}(\omega_7), \frac{1}{16}(\omega_8)]$
ω_5	$[1(\omega_5)]$
ω_6	$[\frac{9}{16}(\omega_4), \frac{3}{16}(\omega_6), \frac{3}{16}(\omega_7), \frac{1}{16}(\omega_8)]$
ω_7	$[\frac{9}{16}(\omega_4), \frac{3}{16}(\omega_6), \frac{3}{16}(\omega_7), \frac{1}{16}(\omega_8)]$
ω_8	$[\frac{9}{16}(\omega_4), \frac{3}{16}(\omega_6), \frac{3}{16}(\omega_7), \frac{1}{16}(\omega_8)]$
State of the world	π_{B}
ω_1	$[1(\omega_1)]$
ω_2	$[1(\omega_2)]$
ω_3	$[\frac{9}{16}(\omega_3), \frac{3}{16}(\omega_5), \frac{3}{16}(\omega_7), \frac{1}{16}(\omega_8)]$
ω_4	$[1(\omega_4)]$
ω_5	$[\frac{9}{16}(\omega_3), \frac{3}{16}(\omega_5), \frac{3}{16}(\omega_7), \frac{1}{16}(\omega_8)]$
ω_6	$[1(\omega_6)]$
ω_7	$[\frac{9}{16}(\omega_3), \frac{3}{16}(\omega_5), \frac{3}{16}(\omega_7), \frac{1}{16}(\omega_8)]$
ω_8	$[\frac{9}{16}(\omega_3), \frac{3}{16}(\omega_5), \frac{3}{16}(\omega_7), \frac{1}{16}(\omega_8)]$

State of the world	π_T
ω_1	$[1(\omega_1)]$
ω_2	$[\frac{9}{16}(\omega_2), \frac{3}{16}(\omega_5), \frac{3}{16}(\omega_6), \frac{1}{16}(\omega_8)]$
ω_3	$[1(\omega_3)]$
ω_4	$[1(\omega_4)]$
ω_5	$[\frac{9}{16}(\omega_2), \frac{3}{16}(\omega_5), \frac{3}{16}(\omega_6), \frac{1}{16}(\omega_8)]$
ω_6	$[\frac{9}{16}(\omega_2), \frac{3}{16}(\omega_5), \frac{3}{16}(\omega_6), \frac{1}{16}(\omega_8)]$
ω_7	$[1(\omega_7)]$
ω_8	$[\frac{9}{16}(\omega_2), \frac{3}{16}(\omega_5), \frac{3}{16}(\omega_6), \frac{1}{16}(\omega_8)]$

10.26 (a) The table below presents the belief space. The state ω_5 represents the state of the world in which Walter's type is t_1, Karl's type is t_2, and Ferdinand's type is t_3.

State of the world	s(·)	π_{Walter}	π_{Karl}	$\pi_{\text{Ferdinand}}$
ω_1	s_1	$[1(\omega_1)]$	$[1(\omega_1)]$	$[1(\omega_1)]$
ω_2	s_2	$[1(\omega_2)]$	$[1(\omega_2)]$	$[1(\omega_2)]$
ω_3	s_1	$[1(\omega_1)]$	$[1(\omega_1)]$	$[0.5(\omega_3), 0.5(\omega_4)]$
ω_4	s_2	$[1(\omega_2)]$	$[1(\omega_2)]$	$[0.5(\omega_3), 0.5(\omega_4)]$
ω_5	s_1	$[1(\omega_1)]$	$[1(\omega_2)]$	$[0.5(\omega_3), 0.5(\omega_4)]$

(b) The minimal belief subspace of Walter at ω_5 is $\{\omega_1\}$. The minimal belief subspace of Karl at ω_5 is $\{\omega_2\}$. The minimal belief subspace of Ferdinand at ω_5 is $\{\omega_1, \omega_2, \omega_3, \omega_4\}$.

10.27 (a) The table below presents the belief space. The state ω_5 represents the state of the world in which

State	s(·)	π_{Walter}	π_{Karl}	$\pi_{\text{Ferdinand}}$
ω_1	s_1	$[1(\omega_1)]$	$[1(\omega_1)]$	$[1(\omega_1)]$
ω_2	s_2	$[1(\omega_2)]$	$[1(\omega_2)]$	$[1(\omega_2)]$
ω_3	s_1	$[0.5(\omega_3), 0.5(\omega_4)]$	$[1(\omega_1)]$	$[1(\omega_1)]$
ω_4	s_2	$[0.5(\omega_3), 0.5(\omega_4)]$	$[1(\omega_2)]$	$[1(\omega_2)]$
ω_5	s_1	$[1(\omega_1)]$	$[0.5(\omega_5), 0.5(\omega_6)]$	$[1(\omega_1)]$
ω_6	s_2	$[1(\omega_2)]$	$[0.5(\omega_5), 0.5(\omega_6)]$	$[1(\omega_2)]$
ω_7	s_1	$[1(\omega_1)]$	$[1(\omega_1)]$	$[0.5(\omega_7), 0.5(\omega_8)]$
ω_8	s_2	$[1(\omega_2)]$	$[1(\omega_2)]$	$[0.5(\omega_7), 0.5(\omega_8)]$
ω_9	s_1	$[0.5(\omega_3), 0.5(\omega_4)]$	$[0.5(\omega_5), 0.5(\omega_6)]$	$[0.5(\omega_7), 0.5(\omega_8)]$

(b) The minimal belief subspace of Walter at ω_5 is $\{\omega_1, \omega_2, \omega_3, \omega_4\}$. The minimal belief subspace of Karl at ω_5 is $\{\omega_1, \omega_2, \omega_5, \omega_6\}$. The minimal belief subspace of Ferdinand at ω_5 is $\{\omega_1, \omega_2, \omega_7, \omega_8\}$.

10.28 (a) Let $p = \sum_{\omega' \in Y} x_{\omega'} \pi_i(\cdot|\omega') \in P_i$ and let $\omega \in Y$. Since $\pi_i(F_i(\omega)|\omega') = 1$ for every $\omega' \in F_i(\omega)$ whereas $\pi_i(F_i(\omega)|\omega') = 0$ for every $\omega' \notin F_i(\omega)$, one has

$$p(A|F_i(\omega)) = \frac{p(A \cap F_i(\omega))}{p(F_i(\omega))} = \frac{\sum_{\omega' \in Y} x_{\omega'} \pi_i(A \cap F_i(\omega)|\omega')}{\sum_{\omega' \in Y} x_{\omega'} \pi_i(F_i(\omega)|\omega')}$$

$$= \frac{\pi_i(A \cap F_i(\omega)|\omega) \sum_{\omega' \in F_i(\omega)} x_{\omega'}}{\pi_i(F_i(\omega)|\omega) \sum_{\omega' \in F_i(\omega)} x_{\omega'}} = \frac{\pi_i(A|\omega)}{1} = \pi_i(A|\omega).$$

That is, if p is a common prior, the beliefs of player i would be given by π_i.

(b) Let $p = \sum_{\omega \in Y} x_\omega \pi_i(\cdot|\omega) \in P_i$ and $q = \sum_{\omega \in Y} y_\omega \pi_i(\cdot|\omega) \in P_i$ where $\sum_{\omega \in Y} x_\omega = \sum_{\omega \in Y} y_\omega = 1$ and $x_\omega, y_\omega > 0$ for every $\omega \in Y$. Let $\alpha \in (0,1)$. Then,

$$\sum_{\omega \in Y} (\alpha x_\omega + (1-\alpha) y_\omega) = \alpha \sum_{\omega \in Y} x_\omega + (1-\alpha) \sum_{\omega \in Y} y_\omega = \alpha + (1-\alpha) = 1,$$

and

$$\alpha x_\omega + (1-\alpha) y_\omega > 0 \quad \forall \omega \in Y.$$

Therefore,

$$\alpha p + (1-\alpha) q = \sum_{\omega \in Y} (\alpha x_\omega + (1-\alpha) y_\omega) \pi_i(\cdot|\omega) \in P_i.$$

Hence, the set P_i is a convex set in $\Delta(Y)$, for every $i \in N$.

(c) If there is $p \in P_I \cap P_{II}$ then, by (a), p is a prior for both Player I and Player II. In conclusion, if there is no common prior, then P_I and P_{II} are disjoint sets.

(d) Using Exercise 24.45, since P_I and P_{II} are convex (Part (b)), there exist $\alpha \in \mathbf{R}^{|Y|}$ and $\beta \in \mathbf{R}$ such that

$$\langle \alpha, p_I \rangle > \beta > \langle \alpha, p_{II} \rangle \quad \forall p_I \in P_I, p_{II} \in P_{II}.$$

(e) The beliefs of Players I and II about the state of the world are given by the probability distributions $(\pi_i)_{i \in N}$. The state of the world, while unknown to them today, will become known to them tomorrow. They decide that after the state of the world ω will be revealed to them, Player II will pay Player I the sum

$\alpha(\omega) - \beta$. If this quantity is negative, the payment will be from Player I to Player II. Under these conditions, for every prior of Player I $p_I = \sum_{\omega \in Y} x_\omega \pi_I(\cdot|\omega) \in P_I$, the expected payoff of Player I is

$$\sum_{\omega \in Y} (\alpha(\omega) - \beta) x_\omega = \langle \alpha, p_I \rangle > 0,$$

where the left inequality holds by Part (d). Similarly, for every prior of Player II $q_I = \sum_{\omega \in Y} y_\omega \pi_{II}(\cdot|\omega) \in P_{II}$, the expected payoff of Player II is

$$\sum_{\omega \in Y} (\alpha(\omega) - \beta) y_\omega = \langle \alpha, p_{II} \rangle < 0,$$

where the left inequality holds by Part (d). That is, given his subjective beliefs, the expected payoff of each player under this procedure is positive.

10.29 **Proof of Theorem 10.23:** Let $\widetilde{\Pi}_1 = (\widetilde{Y}_1, \mathcal{Y}_{|\widetilde{Y}_1}, s, (\pi_i)_{i \in N})$ and $\widetilde{\Pi}_2 = (\widetilde{Y}_2, \mathcal{Y}_{|\widetilde{Y}_2}, s, (\pi_i)_{i \in N})$ be two subspaces of a belief space $\Pi = (Y, \mathcal{Y}, s, (\pi_i)_{i \in N})$ satisfying $\widetilde{Y}_1 \cap \widetilde{Y}_2 = \emptyset$. That is,

$$\pi_i(\widetilde{Y}_1|\omega) = 1 \quad \forall i \in N, \ \forall \omega \in \widetilde{Y}_1,$$
$$\pi_i(\widetilde{Y}_2|\omega) = 1 \quad \forall i \in N, \ \forall \omega \in \widetilde{Y}_2.$$

Therefore, for every $i \in N$ and every $\omega \in \widetilde{Y}_1 \cap \widetilde{Y}_2$ one has

$$\pi_i(\widetilde{Y}_1 \cap \widetilde{Y}_2|\omega) = \pi_i(\widetilde{Y}_1|\omega) + \pi_i(\widetilde{Y}_2|\omega) - \pi_i(\widetilde{Y}_1 \bigcup \widetilde{Y}_2|\omega) = 1 + 1 - 1 = 1,$$

which yields to $(\widetilde{Y}_1 \cap \widetilde{Y}_2, \mathcal{Y}_{|\widetilde{Y}_1 \cap \widetilde{Y}_2}, s, (\pi_i)_{i \in N})$ is also a belief subspace of Π.

10.30 We prove that if there exists a minimal belief subspace, then it is unique. Assume by contradiction there are two minimal subspaces $\widetilde{\Pi}_1 = (\widetilde{Y}_1, \mathcal{Y}_{|\widetilde{Y}_1}, s, (\pi_i)_{i \in N})$ and $\widetilde{\Pi}_2 = (\widetilde{Y}_2, \mathcal{Y}_{|\widetilde{Y}_2}, s, (\pi_i)_{i \in N})$ at $\omega \in Y$. Hence, for every player i,

$$\pi_i(\widetilde{Y}_1|\omega) = 1 = \pi_i(\widetilde{Y}_2|\omega),$$

from which it follows $\pi_i(\widetilde{Y}_1 \cap \widetilde{Y}_2|\omega) = 1$ and so $\widetilde{Y}_1 \cap \widetilde{Y}_2 \neq \emptyset$. Thus, by Exercise 10.29, $(\widetilde{Y}_1 \cap \widetilde{Y}_2, \mathcal{Y}_{|\widetilde{Y}_1 \cap \widetilde{Y}_2}, s, (\pi_i)_{i \in N})$ is also a belief subspace of Π at ω, which contradicts the minimality of $\widetilde{\Pi}_1$ and $\widetilde{\Pi}_2$.

10.31 Let $\Pi = (Y, \mathcal{Y}, s, (\pi_i)_{i\in N})$ be a belief space in which the set of states of the world Y is countably infinite. We prove that there exists a minimal belief subspace at each state of the world $\omega \in Y$.
We construct the minimal subspace by induction. Set

$$\begin{aligned} Y_1 &= \{\omega\} \\ Y_{n+1} &= Y_n \bigcup \left(\bigcup_{\omega' \in Y_n} \bigcup_{i\in N} \operatorname{supp}\left(\pi_i(\omega')\right) \right) \quad \forall n \in \mathbf{N}. \\ \widetilde{Y} &= \bigcup_{n=1}^{\infty} Y_n. \end{aligned}$$

We first show that $\widetilde{\Pi} = (\widetilde{Y}, \mathcal{Y}_{|\widetilde{Y}}, s, (\pi_i)_{i\in N})$ is a belief subspace of Π at ω. Indeed, $\omega \in Y_1 \subseteq \widetilde{Y}$. In addition, for every $\omega' \in \widetilde{Y}$ and every $i \in N$, there is $n \in \mathbf{N}$ such that $\omega' \in Y_n$. Therefore, $\operatorname{supp}(\pi_i(\omega')) \subseteq Y_{n+1}$ and thus

$$\pi_i(\widetilde{Y}|\omega') \geq \pi_i(Y_{n+1}|\omega') \geq \pi_i(\operatorname{supp}(\pi_i(\omega'))|\omega') = 1,$$

from which it follows that $\widetilde{\Pi}$ is a belief subspace of Π at ω.
We finally prove that $\widetilde{\Pi}$ is minimal. Let $\widehat{\Pi} = (\widehat{Y}, \mathcal{Y}_{|\widehat{Y}}, s, (\pi_i)_{i\in N})$ be another belief subspace of Π at ω. We show by induction that $Y_n \subseteq \widehat{Y}$ for every $n \in \mathbf{N}$ which yields to $\widetilde{Y} \subseteq \widehat{Y}$. $\omega \in \widehat{Y}$ so $Y_1 = \{\omega\} \subseteq \widehat{Y}$. Assume that $Y_n \subseteq \widehat{Y}$. We prove that $Y_{n+1} \subseteq \widehat{Y}$. Let $\omega' \in Y_n$. By the induction hypothesis, $\omega' \in Y_n \subseteq \widehat{Y}$. Thus, since $\widehat{Y}$ is a belief subspace, for every $i \in N$ one has $\pi_i(\widehat{Y}|\omega') = 1$. Hence, $\operatorname{supp}(\pi_i(\omega)) \subseteq \widehat{Y}$ and thus $Y_{n+1} \subseteq \widehat{Y}$. In conclusion, $\widetilde{Y} = \bigcup_{n=1}^{\infty} Y_n \subseteq \widehat{Y}$, which completes the proof.

10.32 **Proof of Theorem 10.33:** let $\Pi = (Y, \mathcal{Y}, s, (\pi_i)_{i\in N})$ be a belief space, where Y is a finite set. We show that for each state of the world $\omega \in Y$,

$$\widetilde{Y}(\omega) = \{\omega\} \bigcup \left(\bigcup_{i\in N} \widetilde{Y}_i(\omega) \right).$$

The minimal belief subspace $\widetilde{Y}(\omega)$ of Π at ω is also a belief subspace of player i at ω, for every $i \in N$. Indeed, $\widetilde{Y}(\omega)$ is a belief subspace and since $\omega \in \widetilde{Y}(\omega)$, $\pi_i(\widetilde{Y}(\omega)|\omega) = 1$. Since $\widetilde{Y}_i(\omega)$ is the minimal a belief subspace of player i at ω, we have $\widetilde{Y}_i(\omega) \subseteq \widetilde{Y}(\omega)$ for every $i \in N$. In particular,

$$\{\omega\} \bigcup \left(\bigcup_{i\in N} \widetilde{Y}_i(\omega) \right) \subseteq \widetilde{Y}(\omega).$$

To prove the other direction, we show that $\{\omega\} \bigcup \left(\bigcup_{i \in N} \widetilde{Y}_i(\omega) \right)$ is a belief subspace of Π at ω. Let $\omega' \in \{\omega\} \bigcup \left(\bigcup_{i \in N} \widetilde{Y}_i(\omega) \right)$, then either (1) $\omega' = \omega$; or (2) there is $j \in N$ such that $\omega' \in \widetilde{Y}_j(\omega)$. If (1) holds, then for every $i \in N$,

$$\pi_i \left(\{\omega\} \bigcup \left(\bigcup_{i \in N} \widetilde{Y}_i(\omega) \right) \mid \omega \right) \geq \pi_i(\widetilde{Y}_i(\omega) \mid \omega) = 1.$$

If (2) holds, since $\widetilde{Y}_j(\omega)$ is a belief subspace of player j at ω,

$$\pi_i \left(\{\omega\} \bigcup \left(\bigcup_{i \in N} \widetilde{Y}_i(\omega) \right) \mid \omega' \right) \geq \pi_i(\widetilde{Y}_j(\omega) \mid \omega') = 1.$$

In conclusion, $\{\omega\} \bigcup \left(\bigcup_{i \in N} \widetilde{Y}_i(\omega) \right)$ is a belief subspace of Π at ω. By the minimality of $\widetilde{Y}(\omega)$,

$$\widetilde{Y}(\omega) \subseteq \{\omega\} \bigcup \left(\bigcup_{i \in N} \widetilde{Y}_i(\omega) \right).$$

10.33 **Proof of Theorem 10.35:** let $\Pi = (Y, \mathcal{Y}, s, (\pi_i)_{i \in N})$ be a belief space, where Y is a finite set. Then the subset $\widetilde{Y}$ of Y is a belief subspace if and only if $\widetilde{Y}$ is a closed component in the graph G defined by Π.
Let $\widetilde{Y} \subseteq Y$ be a closed component in the graph G defined by Π. Then, for every $\omega \in \widetilde{Y}$ and every $i \in N$,

$$\pi_i(\omega' | \omega) = 0, \quad \forall \omega' \in (\widetilde{Y})^c, \tag{89}$$

$$\pi_i((\widetilde{Y})^c | \omega) = \sum_{\omega' \in (\widetilde{Y})^c} \pi_i(\omega' | \omega) = 0, \tag{90}$$

$$\pi_i(\widetilde{Y} | \omega) = 1 - \pi_i((\widetilde{Y})^c | \omega) = 1, \tag{91}$$

where Equation (89) holds since $\widetilde{Y}$ is closed and Equation (90) follows the finiteness of Y. Therefore, by Equation (91), $\widetilde{Y}$ is a belief subspace.
Assume next that $\widetilde{Y}$ is a belief subspace. Let $\omega \in \widetilde{Y}$ and let $\omega' \in Y$ such that there is an edge from ω to ω'. That is, there is $i \in N$ such that $\pi_i(\{\omega'\} | \omega) > 0$. Assume by contradiction that $\omega' \notin \widetilde{Y}$. Then,

$\pi_i(\widetilde{Y}|\omega) \leq \pi_i((\{\omega'\})^c|\omega) = 1 - \pi_i(\{\omega'\}|\omega) < 1$, which contradict the assumption that $\widetilde{Y}$ is a belief subspace.

10.34 **Proof of Theorem 10.36:** let $\Pi = (Y, \mathcal{Y}, s, (\pi_i)_{i\in N})$ be a belief space, where Y is a finite set, let $\omega \in Y$, and let $i \in N$. For each state of the world ω', let $C(\omega')$ be the minimal closed component containing ω' in the graph corresponding to π. We prove that

$$\widetilde{Y}_i(\omega) = \bigcup_{\{\omega' : \pi(\{\omega\}|\omega)>0\}} C(\omega').$$

$\widetilde{Y}_i(\omega)$ satisfies $\pi(\widetilde{Y}_i(\omega)|\omega) = 1$. Therefore, for every $\omega' \in Y$ such that $\pi_i(\{\omega'\}|\omega) > 0$ then $\omega' \in \widetilde{Y}_i(\omega)$. In particular, $\widetilde{Y}_i(\omega)$ is also a belief subspace at ω' and thus the minimal belief subspace $\widetilde{Y}(\omega')$ at ω' is contained in $\widetilde{Y}_i(\omega)$ ($\widetilde{Y}(\omega') \subseteq \widetilde{Y}_i(\omega)$). However, following Exercise 10.33, $\widetilde{Y}(\omega') = C(\omega')$. In conclusion,

$$\bigcup_{\{\omega' : \pi(\{\omega\}|\omega)>0\}} C(\omega') \subseteq \widetilde{Y}_i(\omega).$$

On the other hand, by Exercise 10.33, $C(\omega')$ is a belief subspace containing ω'. Therefore, $\bigcup_{\{\omega' : \pi(\{\omega\}|\omega)>0\}} C(\omega')$ is a belief subspace containing the support of $\pi_i(\omega)$. Hence,

$$\pi_i\left(\bigcup_{\{\omega' : \pi(\{\omega\}|\omega)>0\}} C(\omega') \mid \omega\right) \geq \pi_i(\{\omega' : \pi(\{\omega\}|\omega) > 0\} \mid \omega) = 1,$$

from which it follows that $\bigcup_{\{\omega' : \pi(\{\omega\}|\omega)>0\}} C(\omega')$ is a belief subspace of player i at ω. Using the minimality of $\widetilde{Y}_i(\omega)$,

$$\widetilde{Y}_i(\omega) \subseteq \bigcup_{\{\omega' : \pi(\{\omega\}|\omega)>0\}} C(\omega').$$

10.35 Consider the following belief space:

State of the world	$s(\cdot)$	π_1	π_2	π_3
ω_1	s_1	$[1(\omega_1)]$	$[1(\omega_1)]$	$[1(\omega_1)]$
ω_2	s_2	$[1(\omega_1)]$	$[1(\omega_2)]$	$[1(\omega_2)]$
ω_3	s_3	$[1(\omega_1)]$	$[1(\omega_2)]$	$[1(\omega_3)]$

The minimal belief subspaces of Player 1, Player 2 and Player 3 at ω_3 are as follows: $\widetilde{Y}_1 = \{\omega_1\}$, $\widetilde{Y}_2 = \{\omega_1, \omega_2\}$ and $\widetilde{Y}_3 = \{\omega_1, \omega_2, \omega_3\}$ respectively.

10.36 We prove the claim. consider the belief space given by the next table:

State of the world	$s(\cdot)$	π_1	π_3
ω_1	s_1	$[1(\omega_2)]$	$[1(\omega_3)]$
ω_2	s_2	$[1(\omega_2)]$	$[1(\omega_4)]$
ω_3	s_3	$[1(\omega_4)]$	$[1(\omega_3)]$
ω_4	s_4	$[1(\omega_4)]$	$[1(\omega_4)]$

The belief space of Player 1 at ω_1 is $\widetilde{Y}_1(\omega_1) = \{\omega_2, \omega_4\}$. The belief space of Player 2 at ω_1 is $\widetilde{Y}_2(\omega_1) = \{\omega_3, \omega_4\}$. In particular,

$$\widetilde{Y}_1(\omega_1) \cap \widetilde{Y}_2(\omega_1) = \{\omega_4\} \subset \widetilde{Y}_1(\omega_1), \text{ and}$$
$$\widetilde{Y}_1(\omega_1) \cap \widetilde{Y}_2(\omega_1) = \{\omega_4\} \subset \widetilde{Y}_2(\omega_1).$$

10.37 (a) The corresponding belief space is as follows:

State of the world	$s(\cdot)$	π_1	π_2
ω_{11}	s_{11}	$[1(\omega_{11})]$	$[\frac{3}{5}(\omega_{11}), \frac{2}{5}(\omega_{21})]$
ω_{12}	s_{12}	$[1(\omega_{11})]$	$[1(\omega_{22})]$
ω_{21}	s_{21}	$[\frac{2}{3}(\omega_{21}), \frac{1}{3}(\omega_{22})]$	$[\frac{3}{5}(\omega_{11}), \frac{2}{5}(\omega_{21})]$
ω_{22}	s_{22}	$[\frac{2}{3}(\omega_{21}), \frac{1}{3}(\omega_{22})]$	$[1(\omega_{22})]$

(b)+(c) $\widetilde{Y}_1(\omega_{12}) = \{\omega_{11}, \omega_{21}, \omega_{22}\} = \widetilde{Y}_2(\omega_{12})$.

(d)+(e) Let p be a prior over S where $p(s_{11}) = \frac{1}{2}$, $p(s_{12}) = 0$, $p(s_{21}) = \frac{1}{3}$ and $p(s_{22}) = \frac{1}{6}$. Then, the players agree that the state of the world has been chosen according to p. Note that, the world s_{12} is ascribed a zero probability by p.

10.38 The corresponding belief space is as follows:

State of the world	$s(\cdot)$	π_1	π_2
ω_{11}	s_{11}	$[1(\omega_{11})]$	$[\frac{3}{5}(\omega_{11}), \frac{1}{5}(\omega_{21}), \frac{1}{5}(\omega_{31})]$
ω_{12}	s_{12}	$[1(\omega_{11})]$	$[1(\omega_{22})]$
ω_{13}	s_{13}	$[1(\omega_{11})]$	$[\frac{1}{3}(\omega_{13}), \frac{2}{3}(\omega_{23})]$
ω_{21}	s_{21}	$[\frac{2}{3}(\omega_{21}), \frac{1}{3}(\omega_{22})]$	$[\frac{3}{5}(\omega_{11}), \frac{1}{5}(\omega_{21}), \frac{1}{5}(\omega_{31})]$
ω_{22}	s_{22}	$[\frac{2}{3}(\omega_{21}), \frac{1}{3}(\omega_{22})]$	$[1(\omega_{22})]$
ω_{23}	s_{23}	$[\frac{2}{3}(\omega_{21}), \frac{1}{3}(\omega_{22})]$	$[\frac{1}{3}(\omega_{13}), \frac{2}{3}(\omega_{23})]$
ω_{31}	s_{31}	$[1(\omega_{33})]$	$[\frac{3}{5}(\omega_{11}), \frac{1}{5}(\omega_{21}), \frac{1}{5}(\omega_{31})]$
ω_{32}	s_{32}	$[1(\omega_{33})]$	$[1(\omega_{22})]$
ω_{33}	s_{33}	$[1(\omega_{33})]$	$[\frac{1}{3}(\omega_{13}), \frac{2}{3}(\omega_{23})]$

The minimal belief subspaces of the players at ω_{12} are $\widetilde{Y}_1(\omega_{12}) = \{\omega_{11}, \omega_{13}, \omega_{21}, \omega_{22}, \omega_{23}, \omega_{31}, \omega_{33}\} = \widetilde{Y}_2(\omega_{12})$.
There is no common prior p over S such that the players agree that the state of the world has been chosen according to p.

10.39 The corresponding belief space is as follows:

State of the world	$s(\cdot)$	π_1	π_2
ω_{11}	s_{11}	$[1(\omega_{11})]$	$[1(\omega_{11})]$
ω_{12}	s_{12}	$[1(\omega_{11})]$	$[1(\omega_{22})]$
ω_{21}	s_{21}	$[1(\omega_{22})]$	$[1(\omega_{11})]$
ω_{22}	s_{22}	$[1(\omega_{22})]$	$[1(\omega_{22})]$

The minimal belief subspaces of the players at ω_{12} are $\widetilde{Y}_1(\omega_{12}) = \{\omega_{11}\} \neq \{\omega_{22}\} = \widetilde{Y}_2(\omega_{12})$. Note that, no player can calculate the minimal belief subspace of the other player at ω_{12}. For instance, Player I believes that the state of the world is ω_{11} and thus he mistakenly believes that the minimal belief subspace of Player II is $\widetilde{Y}_2(\omega_{11}) = \{\omega_{11}\}$ instead of $\widetilde{Y}_2(\omega_{12}) = \{\omega_{22}\}$. Similarly, no player can calculate the minimal belief subspace of the other player at ω_{21}.
Let p be a prior over S where $p(s_{11}) = p = 1 - p(s_{22})$ for some $p \in (0,1)$, and $p(s_{12}) = p(s_{21}) = 0$. Then, the players agree that the state of the world has been chosen according to p. Note that, the world s_{12} is ascribed a zero probability by p.

10.40 The belief space is given by $N = \{I, II\}$; $S = \{s_{ij} : \ i, j \in \{1,2,3,4,5\}\}$; $Y = \{\omega_{ij} : \ i, j \in \{1,2,3,4,5\}\}$; $s(\omega_{ij}) = s_{ij}$ for every $i, j \in \{1,2,3,4,5\}$.
The beliefs of Player I are

$$\begin{aligned}
\pi_1(\omega_{1j}) &= [1(\omega_{11})] && \forall j = 1,2,3,4,5,\\
\pi_1(\omega_{2j}) &= [\tfrac{1}{3}(\omega_{21}), \tfrac{2}{3}(\omega_{22})] && \forall j = 1,2,3,4,5,\\
\pi_1(\omega_{3j}) &= [\tfrac{1}{4}(\omega_{34}), \tfrac{3}{4}(\omega_{35})] && \forall j = 1,2,3,4,5,\\
\pi_1(\omega_{4j}) &= [\tfrac{1}{4}(\omega_{44}), \tfrac{3}{4}(\omega_{45})] && \forall j = 1,2,3,4,5.
\end{aligned}$$

The beliefs of Player II are

$$\begin{aligned}
\pi_2(\omega_{i1}) &= [1(\omega_{21})] && \forall i = 1,2,3,4,\\
\pi_2(\omega_{i2}) &= [\tfrac{3}{5}(\omega_{12}), \tfrac{2}{5}(\omega_{22})] && \forall j = 1,2,3,4,\\
\pi_2(\omega_{i3}) &= [\tfrac{1}{2}(\omega_{33}), \tfrac{1}{2}(\omega_{43})] && \forall j = 1,2,3,4,\\
\pi_2(\omega_{i4}) &= [\tfrac{1}{2}(\omega_{34}), \tfrac{1}{2}(\omega_{44})] && \forall j = 1,2,3,4,\\
\pi_2(\omega_{i5}) &= [\tfrac{1}{3}(\omega_{35}), \tfrac{2}{3}(\omega_{45})] && \forall j = 1,2,3,4.
\end{aligned}$$

The minimal belief subspaces of the Player I and Player II at ω_{13} are $\widetilde{Y}_1(\omega_{13}) = \{\omega_{11}, \omega_{12}, \omega_{21}, \omega_{22}, \omega_{33}, \omega_{34}, \omega_{35}, \omega_{43}, \omega_{44}, \omega_{45}\}$ and $\widetilde{Y}_2(\omega_{13}) = \{\omega_{33}, \omega_{34}, \omega_{35}, \omega_{43}, \omega_{44}, \omega_{45}\}$ respectively.

There is no common prior p over S such that the players agree that the state of the world has been chosen according to p.

10.41 (a)

State of the world	$s(\cdot)$	π_1	π_2
ω_{11}	s_{11}	$[\frac{1}{4}(\omega_{11}), \frac{3}{4}(\omega_{12})]$	$[\frac{1}{5}(\omega_{11}), \frac{4}{5}(\omega_{21})]$
ω_{12}	s_{12}	$[\frac{1}{4}(\omega_{11}), \frac{3}{4}(\omega_{12})]$	$[\frac{3}{8}(\omega_{12}), \frac{5}{8}(\omega_{22})]$
ω_{21}	s_{21}	$[\frac{4}{9}(\omega_{21}), \frac{5}{9}(\omega_{22})]$	$[\frac{1}{5}(\omega_{11}), \frac{4}{5}(\omega_{21})]$
ω_{22}	s_{22}	$[\frac{4}{9}(\omega_{21}), \frac{5}{9}(\omega_{22})]$	$[\frac{3}{8}(\omega_{12}), \frac{5}{8}(\omega_{22})]$

(b) This belief space is consistent. It can be described as the following Aumann model of incomplete information: $N = \{I, II\}$; $Y = \{\omega_{11}, \omega_{12}, \omega_{21}, \omega_{22}\}$; $\mathcal{F}_I = \{\{\omega_{11}, \omega_{12}\}, \{\omega_{21}, \omega_{22}\}\}$, $\mathcal{F}_{II} = \{\{\omega_{11}, \omega_{21}\}, \{\omega_{12}, \omega_{22}\}\}$; $\mathbf{P}(\omega_{11}) = \frac{1}{13}$, $\mathbf{P}(\omega_{12}) = \frac{3}{13}$, $\mathbf{P}(\omega_{21}) = \frac{4}{13}$ and $\mathbf{P}(\omega_{22}) = \frac{5}{13}$.

10.42 (a)

State of the world	$s(\cdot)$	π_1	π_2
ω_{11}	s_{11}	$[\frac{1}{4}(\omega_{11}), \frac{3}{4}(\omega_{12})]$	$[\frac{2}{3}(\omega_{11}), \frac{1}{3}(\omega_{21})]$
ω_{12}	s_{12}	$[\frac{1}{4}(\omega_{11}), \frac{3}{4}(\omega_{12})]$	$[\frac{1}{2}(\omega_{12}), \frac{1}{2}(\omega_{22})]$
ω_{21}	s_{21}	$[\frac{1}{5}(\omega_{21}), \frac{4}{5}(\omega_{22})]$	$[\frac{2}{3}(\omega_{11}), \frac{1}{3}(\omega_{21})]$
ω_{22}	s_{22}	$[\frac{1}{5}(\omega_{21}), \frac{4}{5}(\omega_{22})]$	$[\frac{1}{2}(\omega_{12}), \frac{1}{2}(\omega_{22})]$

(b) This belief space is not consistent. Indeed, if by contradiction the belief space is consistent then $p_{11} = 2p_{21}$ (Player II of type II_1) hence $p_{12} = 6p_{21}$ (Player I of type I_1) from which it follows that $p_{12} = 6p_{21}$ (Player II of type II_2) but this contradicts the belief of Player I given that his type is I_1.

10.43 Let $\omega = (x, y)$ be the state of the world.
The minimal belief subspace of Player I at ω given $x \leq \frac{4}{9}$ is

$$\begin{aligned}\widetilde{Y}_I(\omega) = & \{(x, y') : y' \in [0.9x, 0.9x + 0.1]\} \\ & \bigcup \{(x', y') : y' \in [0.9x, 0.9x + 0.1], x' \in [0.7, 1]\} \\ & \bigcup \{(x', y') : y' \in [0.9x', 0.9x' + 0.1], x' \in [0.7, 1]\} \\ & \bigcup \{(x', y') : y' \in [0.63, 1], x' \in [0, 0.3]\} \\ & \bigcup \{(x', y') : y' \in [0.9x', 0.9x' + 0.1], x' \in [0, 0.3]\} \\ & \bigcup \{(x', y') : y' \in [0, 0.37], x' \in [0.7, 1]\}.\end{aligned}$$

If $x > \frac{5}{9}$ then the minimal belief subspace of Player I is

$$\begin{aligned}
\widetilde{Y}_I(\omega) =& \{(x, y') : y' \in [0.9x, 0.9x + 0.1]\} \\
& \bigcup \{(x', y') : y' \in [0.9x, 0.9x + 0.1], x' \in [0, 0.3]\} \\
& \bigcup \{(x', y') : y' \in [0.9x', 0.9x' + 0.1], x' \in [0, 0.3]\} \\
& \bigcup \{(x', y') : y' \in [0, 0.37], x' \in [0.7, 1]\} \\
& \bigcup \{(x', y') : y' \in [0.9x', 0.9x' + 0.1], x' \in [0.7, 1]\} \\
& \bigcup \{(x', y') : y' \in [0.63, 1], x' \in [0, 0.3]\}.
\end{aligned}$$

If $\frac{4}{9} < x \leq \frac{5}{9}$ then the minimal belief subspace of Player I is

$$\begin{aligned}
\widetilde{Y}_I(\omega) =& \{(x, y') : y' \in [0.9x, 0.9x + 0.1]\} \\
& \bigcup \{(x', y') : y' \in [0.9x, 0.5], x' \in [0.7, 1]\} \\
& \bigcup \{(x', y') : y' \in (0.5, 0.9x + 0.1], x' \in [0, 0.3]\} \\
& \bigcup \{(x', y') : y' \in [0.9x', 0.9x' + 0.1], x' \in [0, 0.3] \bigcup [0.7, 1]\} \\
& \bigcup \{(x', y') : y' \in [0, 0.37], x' \in [0.7, 1]\} \\
& \bigcup \{(x', y') : y' \in [0.63, 1], x' \in [0, 0.3]\}.
\end{aligned}$$

The minimal belief subspace of Player II given $y \leq \frac{1}{2}$ is

$$\begin{aligned}
\widetilde{Y}_{II}(\omega) =& \{(x', y) : x' \in [0.7, 1]\} \\
& \bigcup \{(x', y') : y' \in [0.9x', 0.9x' + 0.1], x' \in [0.7, 1]\} \\
& \bigcup \{(x', y') : y' \in [0.63, 1], x' \in [0, 0.3]\} \\
& \bigcup \{(x', y') : y' \in [0.9x', 0.9x' + 0.1], x' \in [0, 0.3]\} \\
& \bigcup \{(x', y') : y' \in [0, 0.37], x' \in [0.7, 1]\}.
\end{aligned}$$

The minimal belief subspace of Player II given $y > \frac{1}{2}$ is

$$\begin{aligned}
\widetilde{Y}_{II}(\omega) =& \{(x', y) : x' \in [0, 0.3]\} \\
& \bigcup \{(x', y') : y' \in [0.9x', 0.9x' + 0.1], x' \in [0, 0.3]\} \\
& \bigcup \{(x', y') : y' \in [0, 0.37], x' \in [0.7, 1]\} \\
& \bigcup \{(x', y') : y' \in [0.9x', 0.9x' + 0.1], x' \in [0.7, 1]\} \\
& \bigcup \{(x', y') : y' \in [0.63, 1], x' \in [0, 0.3]\}.
\end{aligned}$$

10.44 There exists a belief space in which the set of states of the world contains K states, and there are $2^K - 1$ different belief subspaces. Consider a belief space corresponds to a situation of perfect information, i.e., $\pi_i(\{\omega\}|\omega) = 1$ for every $\omega \in Y$ and every $i \in N$. Then, for every $\omega \in Y$, $\widetilde{Y}(\omega) = \{\omega\}$ and thus every non empty subset of states is also a subspace as a union of subspaces.

10.45 There does not exist a belief space comprised of three states of the world and six different belief subspaces. If the belief space equivalent to perfect information, i.e. $\pi_i(\{\omega\}|\omega) = 1$ for every $i \in N$ and every $\omega \in Y$, then there is $2^3 - 1 = 7$ subspaces (Exercise 10.44). Otherwise, there are a player i and at least two states of the world such that $\omega' \in F_i(\omega)$. If both $\omega, \omega' \in \mathrm{supp}(\pi_i(\omega))$ then every subspace containing ω also contains ω' and vice versa and thus there are at most 3 subspaces. If one of these states is not in $\mathrm{supp}(\pi_i(\omega))$, say $\omega' \notin \mathrm{supp}(\pi_i(\omega))$, then every subspace that contains ω' must also contains ω which case there are at most 5 subspaces.

10.46 There exists a belief space with $N = \{I, II\}$ and a finite set of states of the world containing a state of the world ω such that $\widetilde{Y}_I(\omega) \not\subseteq \widetilde{Y}_{II}(\omega)$ and $\widetilde{Y}_{II}(\omega) \not\subseteq \widetilde{Y}_I(\omega)$. For instance,

State of the world	$s(\cdot)$	π_1	π_3
ω_1	s_1	$[1(\omega_2)]$	$[1(\omega_3)]$
ω_2	s_2	$[1(\omega_2)]$	$[1(\omega_4)]$
ω_3	s_3	$[1(\omega_4)]$	$[1(\omega_3)]$
ω_4	s_4	$[1(\omega_4)]$	$[1(\omega_4)]$

The belief space of Player 1 at ω_1 is $\widetilde{Y}_1(\omega_1) = \{\omega_2, \omega_4\}$ whereas the belief space of Player 2 at ω_1 is $\widetilde{Y}_2(\omega_1) = \{\omega_3, \omega_4\}$.

10.47 The claim is wrong. There may be $\omega \in Y$ and a player $i \in N$ such that the set $\widetilde{Y}_i(\omega) \bigcup \{\omega\}$ is not a belief subspace. Indeed, consider the belief space presented in Exercise 10.46, then $\widetilde{Y}_1(\omega_1) \bigcup \{\omega_1\}$ is not a belief subspace since $\pi_2(\widetilde{Y}_1(\omega_1) \bigcup \{\omega_1\}|\omega_1) = 0$.

10.48 Let $G_H = (N, (T_i)_{i\in N}, p, S, (s_t)_{t \in \times_{i\in N} T_i})$ be a Harsanyi game with incomplete information. Then G_H is a game with incomplete information $G = (N, S, (A_i)_{i\in N}, \Pi)$ where $\Pi = (Y, Y, s, (\pi_i)_{i\in N})$ where $Y = \times_{i\in N} T_i$; for every $\omega = t = (t_i)_{i\in N} \in Y$ and every player $i \in N$ set $\pi_i(\omega) = p(\cdot|t_i)$; and $s(\omega) = s_t$ for every $\omega = t = (t_i)_{i\in N} \in Y$.

10.49 **Proof of Theorem 10.40:** We prove that the strategy vector $\sigma^* = (\sigma_i^*)_{i\in N}$ is a Bayesian equilibrium if and only if for each player $i \in N$, each state of the world $\omega \in Y$, and each action $a_{i,\omega} \in A_i(\omega)$,

$$\gamma_i(\sigma^*|\omega) \geq \gamma_i((\sigma^*; a_{i,\omega})|\omega) \tag{92}$$

.

The first direction that the later is is derived from the former, holds trivially by the definition of a Bayesian equilibrium. We prove the other direction. Assume that Equation (92) holds for each player $i \in N$, each state of the world $\omega \in Y$, and each action $a_{i,\omega} \in A_i(\omega)$. Then, for every $i \in N$, every $\omega \in Y$ and every $\sigma_i(\omega) \in (A_i(s(\omega)))$, one has

$$\begin{aligned}
\gamma_i(\sigma_i(\omega), \sigma^*_{-i}|\omega) &= \int_Y U_i(s(\omega'); \sigma_i(\omega), \sigma^*_{-i}(\omega'))d\pi_i(\omega'|\omega) \\
&= \int_Y \int_{A(s(\omega))} U_i(s(\omega'); a_{i,\omega}, \sigma^*_{-i}(\omega'))d\sigma_i(\omega)d\pi_i(\omega'|\omega) \\
&= \int_{A(s(\omega))} \int_Y U_i(s(\omega'); a_{i,\omega}, \sigma^*_{-i}(\omega'))d\pi_i(\omega'|\omega)d\sigma_i(\omega) \\
&\leq \int_{A(s(\omega))} \int_Y U_i(s(\omega'); \sigma^*(\omega'))d\pi_i(\omega'|\omega)d\sigma_i(\omega) \\
&= \int_Y U_i(s(\omega'); \sigma^*(\omega'))d\pi_i(\omega'|\omega) \\
&= \gamma_i(\sigma^*|\omega),
\end{aligned}$$

from which it follows that σ^* is a Bayesian equilibrium.

10.50 Let $G = (N, S, (A_i)_{i \in N}, \Pi)$ be a game with incomplete information such that the payoff functions $(u_i)_{i \in N}$ are uniformly bounded from below by $M \in \mathbf{R}$. Let $\widehat{G} = (N, S, (\widehat{A_i})_{i \in N}, \Pi)$ be the game with incomplete information defined where $\widehat{A}_i(s) = A_i$ for every player $i \in N$ and every state of nature $s \in S$; and for each player $i \in N$, the payoff function is given by

$$\widehat{u}_i(s; a) = \begin{cases} u_i(s; a) & a \in A(s) \\ M & a_i \in A_i(s), a \notin A(s) \\ M - 1 & a_i \notin A_i(s). \end{cases}$$

We prove that the set of Bayesian equilibria of the game G coincides with the set of Bayesian equilibria of the game $\widehat{G}$.
Let σ^* be an equilibrium of the game G. By the definition of $\widehat{G}$ and, in particular, its payoff function, σ^* is a strategy vector in $\widehat{G}$ satisfying $\widehat{\gamma}_i(\sigma^*|\omega) = \gamma(\sigma^*|\omega)$ for every $i \in N$ and every $\omega \in Y$. In addition, for every $i \in N$, every $\omega \in Y$ and every $a_{i,\omega} \in A_i$, either $a_{i,\omega} \in A_i(s(\omega))$ and thus, by Exercise 10.49,

$$\widehat{\gamma}_i(a_{i,\omega}, \sigma^*_{-i}|\omega) = \gamma_i(a_{i,\omega}, \sigma^*_{-i}|\omega) \leq \gamma_i(\sigma^*|\omega) = \widehat{\gamma}_i(\sigma^*|\omega); \tag{93}$$

or $a_{i,\omega} \notin A_i(s(\omega))$ and thus

$$\widehat{\gamma}_i(a_{i,\omega}, \sigma^*_{-i}|\omega) = M - 1 < M \leq \gamma_i(\sigma^*|\omega) = \widehat{\gamma}_i(\sigma^*|\omega). \tag{94}$$

Therefore, by Equation (93) and (94) and by Exercise 10.49, σ^* is also an equilibrium of $\widehat{G}$.
Assume next that σ^* is an equilibrium of $\widehat{G}$. Then, each action $a_{i,\omega} \in A_i \setminus A_i(s(\omega))$ is played with probability zero according to $\sigma^*_i(\omega)$, for every player $i \in N$ and every $\omega \in Y$. Indeed, if, by contradiction, there are $i \in N$, $\omega \in Y$ and $a_{i,\omega} \in A_i \setminus A_i(s(\omega))$ such that $\sigma^*_i(a_{i,\omega}|\omega) > 0$ then player i is better of deviating to any action in $A_i(s(\omega))$ whenever he is supposed to play $a_{i,\omega}$ and gain at least $\sigma^*_i(a_{i,\omega}|\omega) \cdot 1 > 0$ which contradicts the assumption that σ^* is an equilibrium of $\widehat{G}$. In conclusion, σ^* is also a strategy vector in G satisfying that for every $i \in N$, every $\omega \in Y$, and every $a_{i,\omega} \in A_i(s(\omega)) \subseteq A_i$,

$$\gamma_i(a_{i,\omega}, \sigma^*_{-i}|\omega) = \widehat{\gamma}_i(a_{i,\omega}, \sigma^*_{-i}|\omega) \leq \widehat{\gamma}_i(\sigma^*|\omega) = \gamma_i(\sigma^*|\omega),$$

which yields, using Exercise 10.49, that σ^* is an equilibrium of G.

10.51 See Exercise 9.43.

10.52 Suppose that for every player i in a game with incomplete information there exists a strategy σ^*_i that weakly dominates all his other strategies. Therefore, for every strategy vector σ_{-i} of the other players

$$U_i(s(\omega); \sigma_{-i}(\omega), \sigma^*(\omega)) \geq U_i(s(\omega); \sigma_{-i}(\omega), a_{i,\omega}), \quad \forall i \in N,\ \forall \omega \in Y,\ \forall a_{i,\omega} \in A_i(s(\omega)). \tag{95}$$

In particular, by Equation (95) and the monotonicity of the integral, the strategy vector $\sigma^* = (\sigma^*_i)_{i \in N}$ satisfies for each $i \in N$, each $\omega \in Y$ and each $a_{i,\omega} \in A_i(s(\omega))$,

$$\begin{aligned}
\gamma_i(\sigma^*|\omega) &= \int_Y U_i(s(\omega'); \sigma^*(\omega)) d\pi_i(\omega'|\omega) \\
&\leq \int_Y U_i(s(\omega'); \sigma^*_{-i}(\omega), a_{i,\omega}) d\pi_i(\omega'|\omega) \\
&= \gamma_i(\sigma^*_{-i}, a_{i,\omega}|\omega).
\end{aligned}$$

Using Exercise 10.49 it follows that σ^* is a Bayesian equilibrium.

10.53 Let $G = (N, S, (A_i)_{i \in N})$ be a game with incomplete information where the set of actions $(A_i)_{i \in N}$ is a finite set, and $\Pi = (Y, \mathcal{Y}, s, (\pi_i)_{i \in N})$ is a belief space with a finite set of states of the world Y. Define a strategic-form game Γ, where the set of players is N, the set of player

i's pure strategies is the set of functions σ_i that map each type of player i to an available action for that type, and player i's payoff function w_i is given by $w_i(\sigma) = \sum_{\omega \in Y} \gamma_i(\sigma|\omega)$.

(a) The strategic-form game Γ has a finite set of players and each player has a finite set of pure strategies where the number of pure strategies is bounded by $A_i^{|Y|}$ for every player i. Hence by Nash's theorem (Theorem 5.10), Γ has a Nash equilibrium in mixed strategies.

(b) Since all the players in the game have perfect recall, by Kuhn's Theorem (Theorem 6.15) and Theorem 6.11, for each player i and each mixed strategy of i in Γ there exists an equivalent behavior strategy and vice versa. Furthermore, Theorem 6.11 implies that the expected payoffs under every mixed strategy vector and its equivalent behavior strategy are equal. Hence, by item (a) one can deduce that the game Γ has a Nash equilibrium in behavior strategies.

(c) We prove that the set of Nash equilibria in behavior strategies of the game Γ coincides with the set of Bayesian equilibria of the game G.
Let σ^* be a Bayesian equilibrium of G. That is, for every player i and every behavior strategy σ_i of player i one has

$$\gamma_i(\sigma^*|\omega) \geq \gamma_i(\sigma^*_{-i}, \sigma_i|\omega) \ \ \forall \omega \in Y.$$

Therefore,

$$W_i(\sigma^*) = \sum_{\omega \in Y} \gamma_i(\sigma^*|\omega) \geq \sum_{\omega \in Y} \gamma_i(\sigma^*_{-i}, \sigma_i|\omega) = W_i(\sigma^*_{-i}, \sigma_i),$$

from which it follows that σ^* is an equilibrium of Γ in behavior strategies.
Let σ^* be an equilibrium of Γ in behavior strategies, i.e., $W_i(\sigma^*) \geq W_i(\sigma^*_{-i}, \sigma_i)$ for every player $i \in N$ and every behavior strategy σ_i of player i. For every player i, every $\omega \in Y$ and every pure action $a_{i,\omega} \in A_i(s(\omega))$, let σ_i be a behavior strategy of player i such that $\sigma_i(\omega) = [1(a_{i,\omega})]$and $\sigma_i(\omega') = \sigma_i^*(\omega')$ for every $\omega' \neq \omega$. In particular,

$$\begin{aligned}
\gamma_i(\sigma^*_{-i}, a_{i,\omega}|\omega) &= W_i(\sigma^*_{-i}, \sigma_i) - \sum_{\omega' \neq \omega} \gamma_i(\sigma^*_{-i}, \sigma_i'|\omega) \\
&= W_i(\sigma^*_{-i}, \sigma_i) - \sum_{\omega' \neq \omega} \gamma_i(\sigma^*|\omega') \\
&\leq W_i(\sigma^*) - \sum_{\omega' \neq \omega} \gamma_i(\sigma^*|\omega') \\
&= \gamma_i(\sigma^*|\omega).
\end{aligned}$$

Hence, σ^* is a Bayesian equilibrium in G.

10.54 Denote by $\sigma_I(x)$ the probability that Player I of type x, who has received the information $x \in (0,1)$, will choose action T. Denote by U_y the uniform distribution over $(0,y)$ which its density in the interval is $\frac{1}{y}$. The payoff to Player II of type $y \in (0,1)$ if he plays the action L is then

$$\gamma_{II}(L, \sigma_I|y) = \int_0^y (1-\sigma_I(x))dU_y = \int_0^y (1-\sigma_I(x))\tfrac{1}{y}dy = 1 - \tfrac{1}{y}\int_0^y \sigma_I(x)dy.$$

The payoff to Player II of type $y \in (0,1)$ if he plays the action R is then

$$\gamma_{II}(R, \sigma_I|y) = \int_0^y y\sigma_I(x)dU_y = \int_0^y y\sigma_I(x)\tfrac{1}{y}dy = \int_0^y \sigma_I(x)dy.$$

Player II of type y is indifferent between L and R if these two quantities are equal to each other, i.e., if

$$\tfrac{y}{y+1} = \int_0^y \sigma_I(x)dy.$$

Differentiating by y yields to

$$\sigma_I(y) = \frac{1}{(y+1)^2}.$$

By replacing the variable y by x, which is the information that Player I receives, we deduce that $\sigma_I(x) = \frac{1}{(x+1)^2}$.

10.55 At ω_1: Player II believes that the state of the world is ω_1 and thus the state of nature is s_1. Hence, he necessarily plays R that strictly dominates L.
At ω_2 and ω_3: Player II believes that the state of the world is either ω_2 or ω_3 with equal probabilities. Hence, he plays L that strictly dominates R (according to his expected patyoffs).
At ω_4: Player II believes that the state of the world is ω_4 and thus the state of nature is s_2. Hence, he necessarily plays L that strictly dominates R.
Player I, in both ω_1 and ω_2 believes that the state of the world is ω_1 and so the state of nature is s_1 which case Player II plays R. Therefore, he plays B which strictly dominates T (after the elimination of L by Player II).
Similarly, in both states of the world ω_3 and ω_4, Player I believes that the state of the world is ω_4 and so the state of nature is s_2 which case

Player II plays L. Therefore, he plays T which strictly dominates B (after the elimination of R by Player II).

Note that, since this equilibria was obtained by a process of eliminating strictly dominated strategies, there is no additional Bayesian equilibrium.

10.56 The following strategy vector is a Bayesian equilibrium in the given game:

- Player I of type I_1 plays B.
- Player I of type I_2 plays T.
- Player II of both types plays L.

Indeed, Player I of type I_1 ascribes probabilities of $\frac{0.4}{0.5} = 0.8$ and $\frac{0.1}{0.5} = 0.2$ to the events that Player II is of type II_1 and II_2 respectively. Hence he believes that his expected payoff is $2 \cdot 0.8 - 24 \cdot 0.2 = -3.2$ if he plays T; and at least $-1 \cdot 0.8 = -0.8$ if he plays B. Therefore he is better of Playing B.

Similarly, Player I of type I_2 ascribes probabilities of 0.4 and 0.6 to the events that Player II is of type II_1 and II_2 respectively. Hence he believes that his expected payoff is $28 \cdot 0.4 + 12 \cdot 0.6 = 18.4$ if he plays T; and $40 \cdot 0.4 + 2 \cdot 0.6 = 17.2$ if he plays B. Therefore he is better of Playing T.

Player II of type II_1 prefers L to R since he believes that his expected payment under L is higher than under R $(-1 \cdot 0.5 + 28 \cdot 0.5 < 20 \cdot 0.5 + 15 \cdot 0.5))$.

Finally, Player II of Type II_2 plays L since it strictly dominates R given Player I's strategy.

10.57 (a) The set of states of nature is $S = \{s_{head}, s_{tail}\}$, the set of players is $N = \{I, II\}$ where Player I is Ronald nd Player II is Jimmy, and the belief space is given by:

State of the world	$s(\cdot)$	π_1	π_2
ω_1	s_{head}	$[\frac{1}{3}(\omega_1), \frac{2}{3}(\omega_2)]$	$[\frac{3}{4}(\omega_1), \frac{1}{4}(\omega_2)]$
ω_2	s_{tail}	$[\frac{1}{3}(\omega_1), \frac{2}{3}(\omega_2)]$	$[\frac{3}{4}(\omega_1), \frac{1}{4}(\omega_2)]$

The state games are as follows:

s_{head}:

	H	T
H	0	1
T	-1	0

s_{tail}:

	H	T
H	0	-1
T	1	0

(b) The beliefs of Ronald and Jimmy are not consistent since each one of them has only one type. If by contradiction there is a common prior $p \in \Delta(Y)$ then for each $\omega' \in Y$ and each $i \in N$ one has

$$p(\omega') = \sum_{\omega \in Y} \pi_i(\{\omega'\}|\omega)p(\omega) = \pi_i(\{\omega'\}|\omega') \sum_{\omega \in Y} p(\omega) = \pi_i(\{\omega'\}|\omega').$$

Hence, $\pi_1(\{\omega_1\}|\omega') = \pi_2(\{\omega_1\}|\omega')$, a contradiction.

(d) a Bayesian equilibrium in this game is Ronald (Player I) selects *Tail* whereas Jimmy (Player II) selects *Head*.

10.58 Using the result of Exercise 10.52 and eliminating weakly dominated strategies it follows that the following strategy vector is a Bayesian equilibrium: Player I of type I_1 plays T, Player I of type I_2 plays B, Player II of type II_1 plays L, and Player II of type II_2 plays L.

10.59 (a) The situation in the exercise cannot be described as a Harsanyi game with incomplete information (with a common prior) since Marc ascribes probability zero to the event that Nicolas knows that the true value of the company is 11 whereas Nicolas ascribes probability 1 to this event.

(b) In this situation Marc thinks that the probability of every possible value is $\frac{1}{3}$, and he believes that this is common belief among him and Nicolas. Hence, he offers at most (or less than) 11 (the expected value of the company) and he believes that Nicolas accepts any offer which is at least (or higher than) 11 (the expected value of the company). Nicolas, who knows that the true value is 11, accepts any offer which is at least/higher 11 (the expected value of the company).

10.60 At the state ω_3, it is common belief among the players that they play the game form s_3. In this game there are three equilibria: (B, L), (T, R), and $([\frac{2}{5}(T), \frac{3}{5}(B)], [\frac{4}{11}(L), \frac{7}{11}(R)])$.
At states ω_1 and ω_2, Player 2 believes that they play the game form s_1, Player 1 believes that the play the game forms s_1 and s_2 with equal probability, and these beliefs are common beliefs among the two players. Hence the players face the following game form, where the payoff of Player 2 is her payoff in s_1 and the payoff of Player 1 is her average payoff in the two game forms:

		Player 2	
		L	R
Player 1	T	$2,0$	$1,1$
	B	$1,2$	$\frac{3}{2},0$

This game form has a unique equilibrium, $([\frac{2}{3}(T),\frac{1}{3}(B)],[\frac{1}{3}(L),\frac{2}{3}(R)])$. In total, the game has three equilibria:

- Types $[\frac{1}{2}(\omega_1),\frac{1}{2}(\omega_2)]$ (of Player 1) and $[1(\omega_1)]$ (of Player 2) play $([\frac{2}{3}(T),\frac{1}{3}(B)],[\frac{1}{3}(L),\frac{2}{3}(R)])$.
- Types $[1(\omega_3)]$ (of both players) play either (B,L), (T,R), or $([\frac{2}{5}(T),\frac{3}{5}(B)],[\frac{4}{11}(L),\frac{7}{11}(R)])$.

10.61 (a) The game with incomplete information is given by $N=\{I,II\}$; $S=\{s_{ll},s_{lh},s_{hl},s_{hh}\}$ where the left index represents the type of Player I and the right index represents the type of Player II; $A_i=[0,\infty)$ for every player $i\in N$; $Y=\{\omega_{ll},\omega_{lh},\omega_{hl},\omega_{hh}\}$; $s(\omega_{t_1,t_2})=s_{t_1,t_2}$ for every $t_1,t_2\in\{l,h\}$; and the beliefs of the players:

$$\begin{aligned}\pi_I(\omega_{ll})=\pi_I(\omega_{lh})&=[\tfrac{1}{3}(\pi_I(\omega_{ll})),\tfrac{2}{3}(\pi_I(\omega_{lh}))],\\ \pi_I(\omega_{hl})=\pi_I(\omega_{hh})&=[\tfrac{1}{3}(\pi_I(\omega_{hl})),\tfrac{2}{3}(\pi_I(\omega_{hh}))],\\ \pi_{II}(\omega_{ll})=\pi_{II}(\omega_{hl})&=[\tfrac{3}{4}(\pi_I(\omega_{ll})),\tfrac{1}{4}(\pi_I(\omega_{hl}))],\\ \pi_{II}(\omega_{lh})=\pi_{II}(\omega_{hh})&=[\tfrac{1}{4}(\pi_I(\omega_{lh})),\tfrac{3}{4}(\pi_I(\omega_{hh}))].\end{aligned}$$

(b) Assume by contradiction that there is a common prior $p_{i,j}=\mathbf{P}(\omega_{i,j})$ such that $\sum_{i,j}p_{i,j}=1$ and $p_{i,j}>0$ for all i,j. Then, by the beliefs of Player I, $2p_{11}=p_{12}$ and $2p_{21}=p_{22}$, and by the beliefs of Player II, $3p_{21}=p_{11}$ and $3p_{12}=p_{22}$. Therefore,

$$\begin{aligned}p_{22}&=2p_{21}=\tfrac{2}{3}3p_{21}=\tfrac{2}{3}p_{11}\\ p_{22}&=3p_{12}=6p_{11},\end{aligned}$$

a contradiction. Therefore, the beliefs of the manufacturers are inconsistent.

(c) Denote by $x_i^{t_i}$ the quantity of a product manufacturer $i\in N$ of type $t_i\in\{l,h\}$ produces.
Manufacturer I of type l expects to gain

$$x_I^l(\tfrac{5}{4}-x_I^l-(\tfrac{1}{3}x_{II}^l+\tfrac{2}{3}x_{II}^h)).$$

Hence, his best response is:

$$BR_I^l(x_{II}^l, x_{II}^h) = \begin{cases} \frac{1}{2}(\frac{5}{4} - \frac{1}{3}x_{II}^l - \frac{2}{3}x_{II}^h) & \frac{1}{3}x_{II}^l + \frac{2}{3}x_{II}^h < \frac{5}{4}, \\ 0 & \frac{1}{3}x_{II}^l + \frac{2}{3}x_{II}^h \geq \frac{5}{4}. \end{cases}$$

Manufacturer I of type h expects to gain

$$x_I^h(\tfrac{3}{4} - x_I^h - (\tfrac{1}{3}x_{II}^l + \tfrac{2}{3}x_{II}^h)).$$

Hence, his best response is:

$$BR_I^h(x_{II}^l, x_{II}^h) = \begin{cases} \frac{1}{2}(\frac{3}{4} - \frac{1}{3}x_{II}^l - \frac{2}{3}x_{II}^h) & \frac{1}{3}x_{II}^l + \frac{2}{3}x_{II}^h < \frac{3}{4}, \\ 0 & \frac{1}{3}x_{II}^l + \frac{2}{3}x_{II}^h \geq \frac{3}{4}. \end{cases}$$

Manufacturer II of type l expects to gain

$$x_{II}^l(\tfrac{5}{4} - x_{II}^l - (\tfrac{3}{4}x_I^l + \tfrac{1}{4}x_I^h)).$$

Hence, his best response is:

$$BR_{II}^l(x_I^l, x_I^h) = \begin{cases} \frac{1}{2}(\frac{5}{4} - \frac{3}{4}x_I^l - \frac{1}{4}x_I^h) & \frac{3}{4}x_I^l + \frac{1}{4}x_I^h < \frac{5}{4}, \\ 0 & \frac{3}{4}x_I^l + \frac{1}{4}x_I^h \geq \frac{5}{4}. \end{cases}$$

Manufacturer II of type h expects to gain

$$x_{II}^h(\tfrac{3}{4} - x_{II}^h - (\tfrac{1}{4}x_I^l + \tfrac{3}{4}x_I^h)).$$

Hence, his best response is:

$$BR_{II}^h(x_I^l, x_I^h) = \begin{cases} \frac{1}{2}(\frac{3}{4} - \frac{1}{4}x_I^l - \frac{3}{4}x_I^h) & \frac{1}{4}x_I^l + \frac{3}{4}x_I^h < \frac{3}{4}, \\ 0 & \frac{1}{4}x_I^l + \frac{3}{4}x_I^h \geq \frac{3}{4}. \end{cases}$$

Hence, a pure equilibrium of the game is $x_I^l = \frac{23}{48}$, $x_I^h = \frac{11}{48}$, $x_{II}^l = \frac{20}{48}$, $x_{II}^h = \frac{11}{48}$.

Are there additional equilibria? We argue that the answer is negative. If there is another equilibrium, then necessarily one of the players of one of the types is not indifferent among her actions: the action this type takes is not on interior point of the interval $[0, \infty)$. The only point that is not an interior point is 0.

Suppose that some type of Player i produces 0. The gain of Player i of high type cannot be higher than the gain of Player i of low type, hence if some type of Player i produces 0, it is necessarily type I_i^h (it is possible that both types of Player i will produce 0).

One can now check whether there are equilibria where some high type or both high types produce 0, and find out that there are no solutions to the corresponding system of equations.

10.62 Let G and $\widehat{G}$ be as presented in the exercise. We prove that the set of Bayesian equilibria of the game G coincides with the set of Bayesian equilibria of the game G.
First note that the set of behavior strategy vectors of G and G' coincides since in both games the set of states of the worlds are identical, as well as the sets of actions $A_i(s(\omega))$ for every $\omega \in Y$ and every $i \in N$. Furthermore, for every behavior strategy vector σ, every player $i \in N$ and every $\omega \in Y$ the following holds

$$\begin{aligned}
\widehat{\gamma}_i(\sigma|\omega) &= \sum_{\omega' \in F_i(\omega)} \frac{\mathbf{P}(\omega')}{\mathbf{P}(F_i(\omega))} \widehat{U}_i(\widehat{s}(\omega'); \sigma(\omega')) \\
&= \sum_{\omega' \in F_i(\omega)} \frac{\mathbf{P}_i(F_i(\omega))}{\mathbf{P}(F_i(\omega))} \cdot \frac{\mathbf{P}_i(\omega')}{\mathbf{P}_i(F_i(\omega))} U_i(s(\omega'); \sigma(\omega')) \\
&= \frac{\mathbf{P}_i(F_i(\omega))}{\mathbf{P}(F_i(\omega))} \sum_{\omega' \in F_i(\omega)} \frac{\mathbf{P}_i(\omega')}{\mathbf{P}_i(F_i(\omega))} U_i(s(\omega'); \sigma(\omega')) \\
&= \frac{\mathbf{P}_i(F_i(\omega))}{\mathbf{P}(F_i(\omega))} \gamma_i(\sigma|\omega).
\end{aligned}$$

Therefore, σ^* is an equilibrium in G if and only if for every $i \in N$, every $\omega \in Y$ and every $a_{i,\omega} \in A_i(\omega)$,

$$\begin{aligned}
&\gamma_i(\sigma^*|\omega) \geq \gamma_i(\sigma^*_{-i}, a_{i,\omega}|\omega) \\
\iff &\frac{\mathbf{P}_i(F_i(\omega))}{\mathbf{P}(F_i(\omega))} \gamma_i(\sigma^*|\omega) \geq \frac{\mathbf{P}_i(F_i(\omega))}{\mathbf{P}(F_i(\omega))} \gamma_i(\sigma^*_{-i}, a_{i,\omega}|\omega) \\
\iff &\widehat{\gamma}_i(\sigma^*|\omega) \geq \widehat{\gamma}_i(\sigma^*_{-i}, a_{i,\omega}|\omega),
\end{aligned}$$

if and only if σ^* is an equilibrium in $\widehat{G}$.

10.63 (a) $\widetilde{\Pi}$ is a belief space as a belief subspace of Π. Furthermore, since G is a game with incomplete information, for every $\omega, \omega' \in \widetilde{Y} \subseteq Y$, if $\pi_i(\omega) = \pi_i(\omega')$, then $A_i(s(\omega)) = A_i(s(\omega'))$. Therefore, $\widetilde{G} = (N, S, (A_i)_{i \in N}, \widetilde{\Pi})$ is a game with incomplete information.

(b) Let σ^* be a Bayesian equilibrium of G. Denote by $\sigma^*_{|\widetilde{Y}}$ the strategy vector σ^* restricted to the states of the world in $\widetilde{Y}$.

For every $i \in N$, every $\omega \in \widetilde{Y}$ and every $a_{i,\omega} \in A_i(s(\omega))$

$$\begin{aligned}
\widetilde{\gamma}_i(\sigma^*_{|\widetilde{Y}}|\omega) &= \int_{\widetilde{Y}} U_i(s(\omega'); \sigma^*_{|\widetilde{Y}}(\omega'))d\pi_i(\omega'|\omega) \\
&= \int_{\widetilde{Y}} U_i(s(\omega'); \sigma^*(\omega'))d\pi_i(\omega'|\omega) \\
&= \int_{Y} U_i(s(\omega'); \sigma^*(\omega'))d\pi_i(\omega'|\omega) \\
&= \gamma_i(\sigma^*|\omega) \\
&\geq \gamma_i(\sigma^*_{-i}, a_{i,\omega}|\omega) \\
&= \int_{Y} U_i(s(\omega'); \sigma^*_{-i}(\omega'), a_{i,\omega})d\pi_i(\omega'|\omega) \\
&= \int_{\widetilde{Y}} U_i(s(\omega'); \sigma^*_{-i}(\omega'), a_{i,\omega})d\pi_i(\omega'|\omega) \\
&= \int_{\widetilde{Y}} U_i(s(\omega'); \sigma^*_{-i|\widetilde{Y}}(\omega'), a_{i,\omega})d\pi_i(\omega'|\omega) \\
&= \widetilde{\gamma}_i(\sigma^*_{|\widetilde{Y}}|\omega).
\end{aligned}$$

Therefore, $\sigma^*_{|\widetilde{Y}}$ is a Bayesian equilibrium of the game $\widetilde{G}$.

(c) Let $\widetilde{\sigma}$ be a Bayesian equilibrium of $\widetilde{G}$. We prove that there exists a Bayesian equilibrium σ^* of G satisfying $\widetilde{\sigma}_i(\omega) = \sigma^*_i$ for each player $i \in N$ and each state of the world $\omega \in \widetilde{Y}$.

For every $\omega \in \widetilde{Y}$ let $s^\omega = (N, (\{a_i^\omega\})_{i \in N}, (\widehat{u}_i)_{i \in N})$ be a state game in which each player has only one pure action denoted by a_i^ω and the payoff is $\widehat{u}_i(a^\omega) = u_i(s(\omega); \widetilde{\sigma}(\omega))$. That is, the state game s^ω is obtained by the state game $s(\omega)$ when the players are restricted to play according to $\widetilde{\sigma}(\omega)$. Define a game with incomplete information $\widehat{G} = (N, \widehat{S}, (\widehat{A}_i)_{i \in N}, \widehat{\Pi})$ as follows: $\widehat{A}_i = A_i \bigcup \{a_i^\omega : \omega \in \widetilde{Y}\}$; $\widehat{S} = S \bigcup \{s^\omega : \omega \in \widetilde{Y}\}$; and $\widehat{\Pi} = (Y, \widehat{s}, (\pi_i)_{i \in N})$ where $\widehat{s}(\omega) = s(\omega)$ for every $\omega \notin \widetilde{Y}$ and $\widehat{s}(\omega) = s^\omega$ for every $\omega \in \widetilde{Y}$. From Theorem 10.42, since Y and $\widehat{A}_i$ are finite sets for every $i \in N$, $\widehat{G}$ has a Bayesian equilibrium $\widehat{\sigma}$ in behavior strategies. Define a profile σ^* in G as follows, $\sigma^*(\omega) = \widehat{\sigma}(\omega)$ for every $\omega \notin \widetilde{Y}$ and $\sigma^*(\omega) = \widetilde{\sigma}(\omega)$ for every $\omega \in \widetilde{Y}$. Therefore, for every $i \in N$, every $\omega \in \widetilde{Y}$ and every $a_i \in A_i(s(\omega))$,

$$\gamma_i(\sigma^*|\omega) = \widetilde{\gamma}_i(\sigma^*|\omega) \geq \widetilde{\gamma}_i(\sigma^*_{-i}, a_i|\omega) = \gamma_i(\sigma^*_{-i}, a_i|\omega).$$

In addition, for every $i \in N$, every $\omega \in Y \setminus \widetilde{Y}$ and every $a_i \in A_i(s(\omega))$,

$$\gamma_i(\sigma^*|\omega) = \widehat{\gamma}_i(\sigma^*|\omega) \geq \widehat{\gamma}_i(\sigma^*_{-i}, a_i|\omega) = \gamma_i(\sigma^*_{-i}, a_i|\omega).$$

In particular σ^* is a Bayesian equilibrium of G satisfying $\widetilde{\sigma}_i(\omega) = \sigma^*_i(\omega)$ for each player $i \in N$ and each state of the world $\omega \in \widetilde{Y}$.

10.64 Let p be the probability distribution defined by Equation (10.58). Let $A \subseteq Y$ be an event, i be a player, and $\omega \in \text{supp}(p)$. Since $\text{supp}(p) = \text{supp}(\widetilde{p}) \subseteq \widetilde{Y}$, $p(P_i(\omega)) > 0$ and

$$\pi(A|\omega) = \pi(A \cap P_i(\omega)|\omega) = p(A \cap P_i(\omega)|Pi(\omega)) = p(A|Pi(\omega)),$$

where the last equality holds since $p(A \setminus P_i(\omega)|Pi(\omega)) = 0$. In conclusion, p is consistent over the belief space described in Remark 10.47.

10.65 Let p be a probability distribution over Y whose support is a finite set. Then, for every event $A \subseteq \widetilde{Y}$, for each player i, and for each $\omega \in \text{supp}(p)$ such that $p(P_i(\omega)) > 0$, the following holds

$$\begin{aligned}
p(A|P_i(\omega)) = \frac{p(A \cap P_i(\omega))}{p(P_i(\omega))} &= \frac{\sum\limits_{\omega' \in A \cap P_i(\omega)} p(\omega')}{\sum\limits_{\omega' \in P_i(\omega)} p(\omega')} \\
&= \frac{\sum\limits_{\omega' \in A \cap F_i(\omega)} p(\omega')}{\sum\limits_{\omega' \in F_i(\omega)} p(\omega')} = p(A|F_i(\omega)).
\end{aligned}$$

Therefore, $\pi_i(A|\omega) = p(A|P_i(\omega))$ if and only if $\pi_i(A|\omega) = p(A|F_i(\omega))$. Hence, in Equation (10.57) one may condition on the set $\{\omega' \in Y : \pi_i(\omega') = \pi_i(\omega)\}$ instead of $P_i(\omega)$.

10.66 We show that, in Example 10.18, Equation (10.57) is satisfied for each event $A \subseteq Y$, for each player i, and for each $\omega \in Y$:

$$p(A|P_I(\omega_{11})) = p(A|P_I(\omega_{12})) = \begin{cases} 0 & \omega_{11}, \omega_{12} \notin A \\ \frac{0.3}{0.3+0.4} & \omega_{11} \in A, \omega_{12} \notin A \\ \frac{0.4}{0.3+0.4} & \omega_{11} \notin A, \omega_{12} \in A \\ 1 & \omega_{11}, \omega_{12} \in A \end{cases}$$
$$= \pi_I(A|\omega_{11}) = \pi_I(A|\omega_{12}).$$

$$p(A|P_I(\omega_{21})) = p(A|P_I(\omega_{22})) = \begin{cases} 0 & \omega_{21}, \omega_{22} \notin A \\ \frac{0.2}{0.2+0.1} & \omega_{21} \in A, \omega_{22} \notin A \\ \frac{0.1}{0.2+0.1} & \omega_{21} \notin A, \omega_{22} \in A \\ 1 & \omega_{21}, \omega_{22} \in A \end{cases}$$
$$= \pi_I(A|\omega_{21}) = \pi_I(A|\omega_{22}).$$

$$p(A|P_{II}(\omega_{11})) = p(A|P_{II}(\omega_{21})) = \begin{cases} 0 & \omega_{11}, \omega_{21} \notin A \\ \frac{0.3}{0.3+0.2} & \omega_{11} \in A, \omega_{21} \notin A \\ \frac{0.2}{0.3+0.2} & \omega_{11} \notin A, \omega_{21} \in A \\ 1 & \omega_{11}, \omega_{21} \in A \end{cases}$$

$$= \pi_{II}(A|\omega_{11}) = \pi_{II}(A|\omega_{21}).$$

$$p(A|P_{II}(\omega_{12})) = p(A|P_{II}(\omega_{22})) = \begin{cases} 0 & \omega_{12}, \omega_{22} \notin A \\ \frac{0.4}{0.4+0.1} & \omega_{12} \in A, \omega_{22} \notin A \\ \frac{0.1}{0.4+0.1} & \omega_{12} \notin A, \omega_{22} \in A \\ 1 & \omega_{12}, \omega_{22} \in A \end{cases}$$

$$= \pi_{II}(A|\omega_{12}) = \pi_{II}(A|\omega_{22}).$$

10.67 Assume that p is a consistent distribution according to Definition 10.46. Then, for every $i \in N$ and every $\omega' \in Y$,

$$\begin{aligned}
\sum_{\omega \in Y} \pi_i(\{\omega'\}|\omega)p(\omega) &= \sum_{\omega \in P_i(\omega')} \pi_i(\{\omega'\}|\omega)p(\omega) && (96) \\
&= \sum_{\omega \in P_i(\omega')} \pi_i(\{\omega'\}|\omega')p(\omega) && (97) \\
&= \pi_i(\{\omega'\}|\omega') \sum_{\omega \in P_i(\omega')} p(\omega) \\
&= p(\{\omega'\}|P_i(\omega'))p(P_i(\omega')) && (98) \\
&= p(\omega'),
\end{aligned}$$

where Equations (96) and (97) follows the coherency requirement of π_i and Equation (98) follows Definition 10.46. In conclusion, p is a consistent distribution according to Definition 10.49.
Assume next that p is a consistent distribution according to Definition 10.49. Let $i \in N$, let $\omega \in \text{supp}(p)$ and let $\omega' \in Y$.
Then, for every $\omega' \in F_i(\omega)$

$$\begin{aligned}
p(\omega') = \sum_{\omega'' \in Y} \pi_i(\{\omega'\}|\omega'')p(\omega'') = \pi_i(\{\omega'\}|\omega) \sum_{\omega'' \in P_i(\omega)} p(\omega'') \\
= \pi_i(\{\omega'\}|\omega)p(P_i(\omega)).
\end{aligned}$$

In particular, since $p(\omega) > 0$ then $p(P_i(\omega)) > 0$. In addition, for every $A \subseteq Y$.

$$
\begin{aligned}
p(A|P_i(\omega) &= \frac{p(A\cap P_i(\omega))}{p(P_i(\omega))} = \frac{\sum\limits_{\omega'\in A\cap P_i(\omega)} p(\omega')}{p(P_i(\omega))} \\
&= \frac{\sum\limits_{\omega'\in A\cap P_i(\omega)} \pi_i(\{\omega'\}|\omega)p(P_i(\omega))}{p(P_i(\omega))} \\
&= \sum_{\omega'\in A\cap P_i(\omega)} \pi_i(\{\omega'\}|\omega) = \pi(A|\omega),
\end{aligned}
$$

which yields to the fact that p is a consistent distribution according to Definition 10.46.

10.68 We show that Equation (10.65) is satisfied for Player II in Example 10.18:

$$
\begin{aligned}
p(\omega_{11}) &= \tfrac{3}{5}p(\omega_{11}) + \tfrac{3}{5}p(\omega_{21}) = \tfrac{3}{5}\cdot 0.3 + \tfrac{3}{5}\cdot 0.2 = 0.3,\\
p(\omega_{12}) &= \tfrac{4}{5}p(\omega_{12}) + \tfrac{4}{5}p(\omega_{22}) = \tfrac{4}{5}\cdot 0.4 + \tfrac{4}{5}\cdot 0.1 = 0.4,\\
p(\omega_{21}) &= \tfrac{2}{5}p(\omega_{11}) + \tfrac{2}{5}p(\omega_{21}) = \tfrac{2}{5}\cdot 0.3 + \tfrac{2}{5}\cdot 0.2 = 0.2,\\
p(\omega_{22}) &= \tfrac{1}{5}p(\omega_{12}) + \tfrac{1}{5}p(\omega_{22}) = \tfrac{1}{5}\cdot 0.4 + \tfrac{1}{5}\cdot 0.1 = 0.1.
\end{aligned}
$$

10.69 Example 10.17, Player I:

$$
\begin{aligned}
p(\omega_{11}) &= \tfrac{1}{2}p(\omega_{11}) + \tfrac{1}{2}p(\omega_{12}) = \tfrac{1}{2}\cdot\tfrac{1}{6} + \tfrac{1}{2}\cdot\tfrac{1}{6} = \tfrac{1}{6},\\
p(\omega_{12}) &= \tfrac{1}{2}p(\omega_{11}) + \tfrac{1}{2}p(\omega_{12}) = \tfrac{1}{2}\cdot\tfrac{1}{6} + \tfrac{1}{2}\cdot\tfrac{1}{6} = \tfrac{1}{6},\\
p(\omega_{21}) &= \tfrac{1}{2}p(\omega_{21}) + \tfrac{1}{2}p(\omega_{22}) = \tfrac{1}{2}\cdot\tfrac{1}{3} + \tfrac{1}{2}\cdot\tfrac{1}{3} = \tfrac{1}{3},\\
p(\omega_{22}) &= \tfrac{1}{2}p(\omega_{21}) + \tfrac{1}{2}p(\omega_{22}) = \tfrac{1}{2}\cdot\tfrac{1}{3} + \tfrac{1}{2}\cdot\tfrac{1}{3} = \tfrac{1}{3}.
\end{aligned}
$$

Example 10.17, Player II:

$$
\begin{aligned}
p(\omega_{11}) &= \tfrac{1}{3}p(\omega_{11}) + \tfrac{1}{3}p(\omega_{21}) = \tfrac{1}{3}\cdot\tfrac{1}{6} + \tfrac{1}{3}\cdot\tfrac{1}{3} = \tfrac{1}{6},\\
p(\omega_{12}) &= \tfrac{1}{3}p(\omega_{12}) + \tfrac{1}{3}p(\omega_{22}) = \tfrac{1}{3}\cdot\tfrac{1}{6} + \tfrac{1}{3}\cdot\tfrac{1}{3} = \tfrac{1}{6},\\
p(\omega_{21}) &= \tfrac{2}{3}p(\omega_{11}) + \tfrac{2}{3}p(\omega_{21}) = \tfrac{2}{3}\cdot\tfrac{1}{6} + \tfrac{2}{3}\cdot\tfrac{1}{3} = \tfrac{1}{3},\\
p(\omega_{22}) &= \tfrac{2}{3}p(\omega_{12}) + \tfrac{2}{3}p(\omega_{22}) = \tfrac{2}{3}\cdot\tfrac{1}{6} + \tfrac{2}{3}\cdot\tfrac{1}{3} = \tfrac{1}{3}.
\end{aligned}
$$

10.70 **Proof of Theorem 10.50:** Let $\Pi = (Y, \mathcal{Y}, s, (\pi_i)_{i\in N})$ be a belief space in which Y is a finite set. Let p be a consistent distribution over Y. Then, by Definition 10.46, for every $i \in N$ and every $\omega \in \mathrm{supp}(p)$,

$$\pi_i(\mathrm{supp}(p)|\omega) = p(\mathrm{supp}(p)|P_i(\omega)) = 1.$$

In particular, by Definition 10.21, $\text{supp}(p)$ is a belief subspace. Furthermore, the conditions of Definition 10.46 holds for every $A \subseteq Y$ and thus they also holds for every $A \subseteq \text{supp}(p) \subseteq Y$. Therefore, $\text{supp}(p)$ is a consistent belief subspace.

10.71 Let $\widetilde{Y}_1$ and $\widetilde{Y}_2$ be two consistent belief subspaces of the same belief space Π, and let p_1 and p_2 be consistent distributions over these two subspaces, respectively. Since $\widetilde{Y}_1$ and $\widetilde{Y}_2$ are subspaces then for every $i \in N$ and every $\omega \in \widetilde{Y}_1 \bigcup \widetilde{Y}_2$,

$$\pi_i(\widetilde{Y}_1 \bigcup \widetilde{Y}_2|\omega) \geq \max\{\pi_i(\widetilde{Y}_1|\omega), \pi_i(\widetilde{Y}_2|\omega)\} = 1.$$

Hence, $\widetilde{Y}_1 \bigcup \widetilde{Y}_2$ is a belief subspace. For each $k \in \{1,2\}$ we expand p_k to a probability distribution over $\widetilde{Y}_1 \bigcup \widetilde{Y}_2$ by setting $p_k(\omega) = 0$ for every $\omega \notin \widetilde{Y}_k$. Then, p_k is a consistent probability over $\widetilde{Y}_1 \bigcup \widetilde{Y}_2$ (the proof is identical to that in Exercise 10.64 and thus it was omitted).
In addition, for every $\lambda \in [0,1]$, $p = \lambda p_1 + (1-\lambda)p_2$ is a consistent probability distribution over $\widetilde{Y}_1 \bigcup \widetilde{Y}_2$. Indeed, for every $i \in N$ and every $\omega' \in \widetilde{Y}_1 \bigcup \widetilde{Y}_2$, Equation (10.66) of Definition 10.49 holds:

$$\begin{aligned}
&\sum_{\omega \in \widetilde{Y}_1 \bigcup \widetilde{Y}_2} \pi_i(\{\omega'\}|\omega)p(\omega) = \sum_{\omega \in \widetilde{Y}_1 \bigcup \widetilde{Y}_2} \pi_i(\{\omega'\}|\omega)(\lambda p_1(\omega) + (1-\lambda)p_2(\omega)) \\
&= \lambda \sum_{\omega \in \widetilde{Y}_1 \bigcup \widetilde{Y}_2} \pi_i(\{\omega'\}|\omega)p_1(\omega) + (1-\lambda) \sum_{\omega \in \widetilde{Y}_1 \bigcup \widetilde{Y}_2} \pi_i(\{\omega'\}|\omega)p_2(\omega) \\
&= \lambda p_1(\omega') + (1-\lambda)p_2(\omega') = p(\omega').
\end{aligned}$$

Using Exercise 10.64 it follows that p is also a consistent probability over Y.

10.72 Let Π be a consistent belief space, and let p be a consistent distribution. Let $\omega \in Y$ be a state of the world satisfying $p(\widetilde{Y}(\omega)) > 0$. We prove that the probability distribution p conditioned on the set $\widetilde{Y}(\omega)$, $p(\cdot|\widetilde{Y}(\omega))$, is consistent. For every $i \in N$ and every $\omega' \in \widetilde{Y}(\omega)$,

$$\begin{aligned}
p(\{\omega'\}|\widetilde{Y}(\omega)) &= \frac{p(\{\omega'\})}{p(\widetilde{Y}(\omega))} = \frac{\sum_{\omega'' \in \widetilde{Y}(\omega)} \pi_i(\{\omega'\})p(\omega'' \cap \widetilde{Y}(\omega))}{p(\widetilde{Y}(\omega))} \\
&= \sum_{\omega'' \in \widetilde{Y}(\omega)} \pi_i(\{\omega'\})p(\omega''|\widetilde{Y}(\omega)).
\end{aligned}$$

so by Definition 10.49, $p(\cdot|\widetilde{Y}(\omega))$, is a consistent probability over $\widetilde{Y}(\omega)$ which is therefore a consistent belief subspace.

10.73 The claim is false. There may be a finite belief space that has no consistent belief subspace. The belief space presented in Example 10.19

has no belief subspace except for itself, however it is inconsistent belief space.

10.74 The claim is false. In the next belief space, $\widetilde{Y}_1(\omega_2) = \{\omega_3, \omega_4\}$ which is not consistent, whereas $\widetilde{Y}(\omega_2) = \{\omega_1, \omega_2, \omega_3, \omega_4\}$ is a consistent belief space (with the consistent probability p where $p(\omega_1) = 1$ and $p(\omega) = 0$ for every $\omega \neq \omega_1$).

State of the world	s	π_1	π_2
ω_1	s_1	$[1(\omega_1)]$	$[1(\omega_1)]$
ω_2	s_2	$[0.5(\omega_3), 0.5(\omega_4)]$	$[1(\omega_1)]$
ω_3	s_3	$[0.5(\omega_3), 0.5(\omega_4)]$	$[1(\omega_4)]$
ω_4	s_4	$[0.5(\omega_3), 0.5(\omega_4)]$	$[1(\omega_4)]$

10.75 The claim is true. Suppose that $\widetilde{Y}_i(\omega)$ is inconsistent for every player $i \in N$. Assume by contradiction that there is a consistent probability p over $\widetilde{Y}(\omega)$. Then, by Exercise 10.72 and since $\widetilde{Y}_i(\omega)$ is inconsistent for every player $i \in N$, $p(\widetilde{Y}_i(\omega)) = 0$. However, by Theorem 10.33, $\widetilde{Y}(\omega) = \{\omega\} \cup \left(\bigcup_{i \in N} \widetilde{Y}_i(\omega)\right)$, and thus $p(\omega) = 1$. In particular, $\pi_i(\{\omega\}|\omega) > 0$ for every $i \in N$ and thus $\omega \in \widetilde{Y}_i(\omega)$ so $p(\omega) = 0$, a contradiction.

10.76 The claim is true. If there is a player $i \in N$ for whom $\widetilde{Y}_i(\omega)$ is consistent, then $\widetilde{Y}(\omega)$ is consistent as well (by Exercise 10.64, since $\widetilde{Y}_i(\omega) \subseteq \widetilde{Y}(\omega)$). Hence, if $\widetilde{Y}(\omega)$ is inconsistent then $\widetilde{Y}_i(\omega)$ is inconsistent for every player $i \in N$.

10.77 The following belief space Π with three players, contains a state of the world ω_1, such that the minimal belief subspaces of the players at ω_1 are inconsistent, and differ from each other: $\widetilde{Y}_1(\omega_1) = \{\omega_2, \omega_3\}$, $\widetilde{Y}_2(\omega_1) = \{\omega_4, \omega_5\}$, and $\widetilde{Y}_3(\omega_1) = \{\omega_6, \omega_7\}$.

State of the world	s	π_1	π_2	π_3
ω_1	s_1	$[0.5(\omega_2), 0.5(\omega_3)]$	$[1(\omega_4)]$	$[1(\omega_6)]$
ω_2	s_2	$[0.5(\omega_2), 0.5(\omega_3)]$	$[1(\omega_3)]$	$[1(\omega_3)]$
ω_3	s_3	$[0.5(\omega_2), 0.5(\omega_3)]$	$[1(\omega_3)]$	$[1(\omega_3)]$
ω_4	s_4	$[0.5(\omega_4), 0.5(\omega_5)]$	$[1(\omega_4)]$	$[1(\omega_4)]$
ω_5	s_5	$[0.5(\omega_4), 0.5(\omega_5)]$	$[1(\omega_4)]$	$[1(\omega_4)]$
ω_6	s_6	$[0.5(\omega_6), 0.5(\omega_7)]$	$[1(\omega_6)]$	$[1(\omega_6)]$
ω_7	s_7	$[0.5(\omega_6), 0.5(\omega_7)]$	$[1(\omega_6)]$	$[1(\omega_6)]$

10.78 The following belief space Π with three players, contains a state of the world ω_1, such that the minimal belief subspaces of two of the players at ω_1 are consistent and differ from each other, but the minimal belief space of the third player is inconsistent: $\widetilde{Y}_1(\omega_1) = \{\omega_4\}$ and $\widetilde{Y}_2(\omega_1) = \{\omega_5\}$ are consistent, whereas $\widetilde{Y}_3(\omega_1) = \{\omega_2, \omega_2\}$ is inconsistent.

State of the world	s	π_1	π_2	π_3
ω_1	s_1	$[1(\omega_4)]$	$[1(\omega_5)]$	$[0.5(\omega_2), 0.5(\omega_3)]$
ω_2	s_2	$[1(\omega_3)]$	$[1(\omega_3)]$	$[0.5(\omega_2), 0.5(\omega_3)]$
ω_3	s_3	$[1(\omega_3)]$	$[1(\omega_3)]$	$[0.5(\omega_2), 0.5(\omega_3)]$
ω_4	s_4	$[1(\omega_4)]$	$[1(\omega_4)]$	$[1(\omega_4)]$
ω_5	s_5	$[1(\omega_5)]$	$[1(\omega_5)]$	$[1(\omega_5)]$

10.79 Let $\widetilde{Y}$ be a minimal consistent belief subspace, and let $\widetilde{Y}'$ be an inconsistent belief subspace. Assume by contradiction that $\widetilde{Y} \cap \widetilde{Y}' \neq \emptyset$, i.e., there is $\omega \in \widetilde{Y} \cap \widetilde{Y}'$.
Denote by p the consistent probability over $\widetilde{Y}$. By Theorems 10.50 and 10.52, $\mathrm{supp}(p) = \widetilde{Y}$, in particular, $p(\omega) > 0$. Therefore, $\widetilde{Y} \cap \widetilde{Y}'$ is a belief subspace such that $p(\widetilde{Y} \cap \widetilde{Y}') > 0$. Hence, following Exercise 10.72, $\widetilde{Y} \cap \widetilde{Y}'$ is a consistent subspace, and thus $\widetilde{Y}'$ is consistent as well(Exercise 10.64), a contradiction.
Note that the claim does not necessarily without the condition that $\widetilde{Y}$ is minimal, since the support of p may strictly contains in $\widetilde{Y}$.

10.80 In Example 10.27, the event A that the state of the world is consistent contains two consistent states, ω_1 and ω_3.
By the players beliefs the following holds:

$$\begin{aligned}
\pi_1(A|\omega_2) &= \pi_1(\{\omega_1\}|\omega_2) = 1, \\
\pi_2(A|\omega_2) &= \pi_2(\{\omega_3\}|\omega_2) = 1, \\
\pi_i(A|\omega_1) &= \pi_1(\{\omega_1\}|\omega_1) = 1 \;\; \forall i \in \{1,2\}, \\
\pi_i(A|\omega_3) &= \pi_1(\{\omega_3\}|\omega_3) = 1 \;\; \forall i \in \{1,2\},
\end{aligned}$$

and thus, by Theorem 10.10, at the state of the world ω_2, the fact that the state of the world is consistent is common belief among the players.

10.81 (a) Let p be a consistent probability over Y. Then, by Definition 10.49,

$$p(\omega_2) = \sum_{\omega \in Y} \pi_{II}(\omega_2|\omega)p(\omega) = 0$$

$$p(\omega_1) = \sum_{\omega \in Y} \pi_I(\omega_1|\omega)p(\omega) = \frac{1}{2}p(\omega_1) + \frac{1}{2}p(\omega_2) \Rightarrow p(\omega_1) = p(\omega_2) = 0,$$

Hence, ω_1 and ω_2 are inconsistent states of the world. Note that, the probability p such that $p(\omega_3) = 1$ is consistent probability over Y. Hence, ω_3 is the only consistent state of the world in Y.

(b) Denote by A the event that the state of the world is consistent. By item (a), $A = \{\omega_3\}$. By the beliefs of Player II,

$$\pi_{II}(A|\omega_2) = \pi_{II}(\{\omega_3\}|\omega_2) = 1,$$

which yields to the fact that at the state of the world ω_2, Player II believes that the state of the world is consistent.

(c) At the state of the world ω_1, both players believe that the state of the world is inconsistent. Indeed, for every player $i \in \{I, II\}$

$$\pi_i(A^c|\omega_1) = \pi_i(\{\omega_1, \omega_2\}|\omega_1) = 1.$$

(d) At the state of the world ω_1, Player I believes that with probability $\frac{1}{2}$ the state of the world is ω_2 which case (item (b)) Player II believes that the state of the world is consistent. Therefore, it is not common belief among the players that the state of the world is inconsistent.

10.82 In the following belief space, where the set of players is $N = \{I, II\}$, at the state of the world ω_1, Player I believes that the state of the world is consistent, while Player II believes the state of the world is inconsistent.

State of the world	s	π_1	π_2
ω_1	s_1	$[1(\omega_4)$	$[0.5(\omega_2), 0.5(\omega_3)]$
ω_2	s_2	$[1(\omega_3)$	$[0.5(\omega_2), 0.5(\omega_3)]$
ω_3	s_3	$[1(\omega_3)$	$[0.5(\omega_2), 0.5(\omega_3)]$
ω_4	s_4	$[1(\omega_4)$	$[1(\omega_4)$

10.83 At the state of the world ω, Player I believes that the state of the world is consistent, while Player II believes the state of the world is inconsistent.

If by contradiction ω is a consistent state of the world then, by Theorem 10.54, it is common belief among the players at ω that ω is consistent. In particular, every player at ω believes that ω is consistent, which contradicts the assumption that Player II believes the state of the world is inconsistent.

10.84 The claim is false. In Example 10.27, it is common belief among the players at the state of the world ω_2 that the state of the world is consistent (See Exercise 10.80). However, $\widetilde{Y}_1 = \{\omega_1\}$ whereas $\widetilde{Y}_2 = \{\omega_3\}$.

10.85 Consider the following belief space where the set of players consists of two players:

State of the world	s	π_1	π_2
ω_1	s_1	$[0.5(\omega_2), 0.5(\omega_3)]$	$[1(\omega_3)]$
ω_2	s_1	$[0.5(\omega_2), 0.5(\omega_3)]$	$[1(\omega_3)]$
ω_3	s_2	$[0.5(\omega_2), 0.5(\omega_3)]$	$[1(\omega_3)]$

Then, ω_1 is an inconsistent state of the world at which $\pi_i(\omega_1|\omega_1) = 0$, and each player i believes that the state of the world is inconsistent.

10.86 Denote by A the set of all consistent states. Denote by B the set of all states of the world ω in which at ω it is a common belief among the players that the state is consistent. By Theorem 10.54, $A \subseteq B$.
In particular, if player i believes at the state of the world ω that the state of the world is consistent, then $\pi_i(A|\omega = 1)$ from which it follows that $\pi_i(B|\omega) = 1$. That is, player i believes that it is common belief that the state of the world is consistent. In particular, he believes that every player believes that the state of the world is consistent.

10.87 (a) The beliefs of the buyers are consistent. Indeed, there is a consistent distribution over $Y = [0,1]^2$ under which with probability p ($0 \leq p \leq 1$) a pair of private values (v_1, v_2) is chosen according to a uniform distribution over $[0, 0.5]^2$ and with probability $(1-p)$ a pair of private values (v_1, v_2) is chosen according to a uniform distribution over $(0.5, 1]^2$.

In a symmetric Bayesian equilibrium, a player whose private value is $x \in [0, \frac{1}{2}]$ submits $\frac{x}{2}$ whereas a player whose private value is $x \in (\frac{1}{2}, 1]$ submits $\frac{x}{2} + \frac{1}{4}$. For instance, if player i has a value $x \in (\frac{1}{2}, 1]$ and he submits $c \leq \frac{1}{2}$ then he believes that he will not win the item. If he submits $c \geq 1$ then he believes that he win the item but he pays at least the amount that equals his private value. Therefore, his optimal bid, according to his beliefs over his opponent private value y, must be $c \in (\frac{1}{2}, 1)$ which yields to the revenue of

$$\mathbf{P}(\frac{y}{2} + \tfrac{1}{4} \leq c)(x - c) = \mathbf{P}(y \leq 2c - \tfrac{1}{2})(x - c) = (c - \tfrac{1}{2})(v - c).$$

Differentiating the above function yields to the fact that $\frac{x}{2} + \frac{1}{4}$ is indeed the best response, as claimed.

(b) The beliefs of the buyers are consistent. There is a consistent distribution over $Y = [0,1]^2$ under which with probability p ($0 \leq p \leq 1$) a pair of private values (v_1, v_2) is chosen according to a uniform distribution over $[0, 0.5] \times [0.5, 1]$ and with probability

$(1-p)$ a pair of private values (v_1, v_2) is chosen according to a uniform distribution over $(0.5, 1] \times [0, 0.5]$.

In a symmetric Bayesian equilibrium, a player whose private value is $x \in [0, \frac{1}{2}]$ submits $\frac{x}{2} + \frac{1}{6}$ but he believes whereas a player whose private value is $x \in (\frac{1}{2}, 1]$ submits $\frac{x}{2} + \frac{1}{12}$.

Chapter 11: The universal belief space

11.1 (a) There are two states of nature corresponding to the given description: s_1 in which the wall will come tumbling down and s_2 in which the wall will not come tumbling down.

(b) The king's first-order belief:

$$[0.8(s_1), 0.2(s_2)].$$

The king's second-order belief

$$([0.8(s_1), 0.2(s_2)], [0.8(s_1, [0.6(s_1), 0.4(s_2)]), 0.2(s_2, [0.5(s_1), 0.5(s_2))]).$$

(c) Joshua's first-order belief cannot be ascertained from the given description since it only describes the kin's beliefs.

11.2 The next belief space of the set of players $N = \{Don, Phil\}$ describing the situation in Example 11.9. Note that at ω_1, Phil's belief hierarchy of order 3 is the hierarchy described in the example.

State of the world	s	π_{Phil}	π_{Don}
ω_1	s_r	$[\frac{1}{3}(\omega_1), \frac{2}{3}(\omega_6)]$	$[\frac{1}{2}(\omega_2), \frac{1}{2}(\omega_4)]$
ω_2	s_r	$[1(\omega_5)]$	$[\frac{1}{2}(\omega_2), \frac{1}{2}(\omega_4)]$
ω_3	s_r	$[1(\omega_3)]$	$[\frac{1}{2}(\omega_2), \frac{1}{2}(\omega_4)]$
ω_4	s_b	$[1(\omega_3)]$	$[\frac{1}{2}(\omega_2), \frac{1}{2}(\omega_4)]$
ω_5	s_p	$[1(\omega_5)]$	$[1(\omega_5)]$
ω_6	s_g	$[\frac{1}{3}(\omega_1), \frac{2}{3}(\omega_6)]$	$[1(\omega_7)]$
ω_7	s_g	$[1(\omega_8)]$	$[1(\omega_7)]$
ω_8	s_w	$[1(\omega_8)]$	$[1(\omega_8)]$

11.3 We prove by induction over k that $Z_k \subseteq X_k$ for each $k \geq 1$. For $k = 1$, $Z_1 = X_1$ so the claim holds trivially. Assume that $Z_{k'} \subseteq X_{k'}$ for every $k' \leq k$. We prove that it is also holds for $k + 1$. By the induction hypothesis, $Z_k \subseteq X_k$ and thus $S \times (Z_k)^{n-1} \subseteq S \times (X_k)^{n-1}$. However, if $X \subseteq Y$, then every probability distribution over X can be expanded to a probability over Y by $\mathbf{P}(\{y\}) = 0$ for every $y \in Y \setminus X$. In conclusion, $\Delta(S \times (Z_k)^{n-1}) \subseteq \Delta(S \times (X_k)^{n-1})$ which yields to

$$Z_{k+1} = Z_k \times \Delta(S \times (Z_k)^{n-1}) \subseteq X_k \times \Delta(S \times (X_k)^{n-1}) = X_{k+1},$$

as claimed.

11.4 The belief hierarchies of order 1, 2 and 3 of Player I, at ω_1 and at ω_2 are μ_1, $\mu_2 = (\mu_1, \nu_1)$ and $\mu_3 = (\mu_2, \nu_2)$ respectively where

$$\mu_1 = [\tfrac{1}{2}(s_1), \tfrac{1}{2}(s_2)],$$
$$\nu_2 = [\tfrac{1}{2}(s_1, [1(s_1)]), \tfrac{1}{2}(s_2, [\tfrac{3}{4}(s_2), \tfrac{1}{4}(s_3)])], \text{ and}$$
$$\nu_3 = [\tfrac{1}{2}(s_1, [1(s_1, [\tfrac{1}{2}(s_1), \tfrac{1}{2}(s_2)])]), \tfrac{1}{2}(s_2, [\tfrac{3}{4}(s_2, [\tfrac{1}{2}(s_1), \tfrac{1}{2}(s_2)]), \tfrac{1}{4}(s_3, [1(s_3)])])].$$

The belief hierarchies of order 1, 2 and 3 of Player I, at ω_3 are μ'_1, $\mu'_2 = (\mu'_1, \nu'_1)$ and $\mu'_3 = (\mu'_2, \nu'_2)$ respectively where

$$\mu'_1 = [1(s_1)],$$
$$\nu'_2 = [1(s_1, [\tfrac{3}{4}(s_2), \tfrac{1}{4}(s_1)])], \text{ and}$$
$$\nu'_3 = [1(s_1, [\tfrac{3}{4}(s_2, [\tfrac{1}{2}(s_1), \tfrac{1}{2}(s_2)]), \tfrac{1}{4}(s_1, [1(s_1)])])].$$

11.5 (a) The set of states of nature is $S = \{s_w, s_l\}$ where s_w corresponds to the situation in which the Philadelphia Phillies will win the World Series next year, and s_l corresponds to the situation in which the Philadelphia Phillies will not win the World Series next year.

(b) Roger's first-order belief is $\mu_1 = [1(s_w)]$. Roger's second-order belief is $\mu_2 = (\mu_1, \nu_1)$ where

$$\nu_1 = [0.5(s_w, [0.4(s_w), 0.6(s_l)]), 0.4(s_w, [0.2(s_w), 0.8(s_l)]),$$
$$0.1(s_w, [0.6(s_w), 0.4(s_l)])].$$

Roger's third-order belief is $\mu_3 = (\mu_2, \nu_2)$ where

$$\nu_2 = [0.3(s_w, [0.4(s_w, [0.4(s_w), 0.6(s_l)]), 0.6(s_l, [0.2(s_w), 0.8(s_l)])]),$$
$$0.2(s_w, [0.4(s_w, [0.5(s_w), 0.5(s_l)]), 0.6(s_l, [0.8(s_w), 0.2(s_l)])]),$$
$$0.4(s_w, [0.2(s_w, [0.1(s_w), 0.9(s_l)]), 0.8(s_l, [0.3(s_w), 0.7(s_l)])]),$$
$$0.1(s_w, [0.6(s_w, [0.4(s_w), 0.6(s_l)]), 0.4(s_l, [0.7(s_w), 0.3(s_l)])])].$$

11.6 **Proof of Theorem 11.7:** let $\mu_k = (\mu_1; \nu_1, \nu_2, ..., \nu_{k-1}) \in Z_k$ be a coherent belief hierarchy of order k and let l_1, l_2 be two integers such that $1 \leq l_1 \leq l_2 \leq k$.

(a) We prove that the marginal distribution of ν_{l_1} over S equals μ_1 by induction over l_1. For $l_1 = 1 \leq k$ the claim holds since μ_k is a coherent belief hierarchy (Equation 11.6). Assume that the claim holds for $l_1 < k$,. We prove that it also holds for $l_1 + 1$. Since μ_k is a coherent belief hierarchy, the marginal distribution of ν_{l_1+1} over $S \times (Z_{l_1})^{n-1}$ is ν_{l_1} (Equation 11.7). In addition, by the induction hypothesis, the marginal distribution of ν_{l_1} over S equals μ_1. Therefore, the marginal distribution of ν_{l_1+1} over S equals μ_1.

(b) We prove that the marginal distribution of ν_{l_2} over $S \times (Z_{l_1})^{n-1}$ is ν_{l_1} by induction over l_2. For $l_2 = l_1 + 1 \leq k$, the claim holds since since μ_k is a coherent belief hierarchy (Equation 11.7). Assume that it holds for $l_2 < k$, we prove that it also holds for $l_2 + 1$. Since μ_k is a coherent belief hierarchy, the marginal distribution of ν_{l_2+1} over $S \times (Z_{l_2})^{n-1}$ is ν_{l_2}(Equation 11.7). In addition, by the induction hypothesis, the marginal distribution of ν_{l_2} over $S \times (Z_{l_1})^{n-1}$ is ν_{l_1}. Therefore, the marginal distribution of ν_{l_2+1} over $S \times (Z_{l_1})^{n-1}$ is ν_{l_1}.

11.7 Let $s_0 \in S$ be a state of nature. Consider the sentence defines a belief hierarchy of order k: "I ascribe probability 1 to the state of nature being s_0, I ascribe probability 1 to all the other players ascribing probability 1 to the state of nature being s_0, I ascribe probability 1 to each of the other players ascribing probability 1 to every other player ascribing probability 1 to the state of nature being s_0, and so on, to level k."

The above hierarchy is defined as follows:

$$\begin{aligned} \mu_1 &= [1(s_0)] \\ \mu_{k+1} &= (\mu_k, \nu_k) \text{ such that } \nu_k = [1(s_0, \underbrace{\mu_k, ..., \mu_k}_{n-1 \text{ times}})] \quad \forall k \geq 1 \end{aligned}$$

We prove that μ_k is coherent by induction over k.

For $k = 1$ then $\mu(1) = [1(s_0)] \in \Delta(S) = Z_1$. Assume that μ_k is coherent, i.e., $\mu_k \in Z_k$. Then, by induction hypothesis $\nu_k \in \Delta(S \times (Z_k)^{n-1})$. If $k = 1$ then the marginal distribution of $\nu_1 = [1(s_0, \mu_1, ..., \mu_1)]$ over S is $[1(s_0)] = \mu_1$. If $k > 1$,

$$\nu_k = [1(s_0, \mu_k, ..., \mu_k)] = [1(s_0, (\mu_{k-1}, \nu_{k-1}), ..., (\mu_{k-1}, \nu_{k-1}))].$$

Hence, the marginal distribution of ν_k over $S \times (X_{k-1})^{n-1}$ equals $[1(s_0, \mu_{k-1}, ..., \mu_{k-1})] = \nu_{k-1}$. Therefore, by Theorem 11.8, μ_{k+1} is coherent, as needed.

11.8 (a) The belief hierarchy of Player I is of order 1 which is necessarily a **coherent belief hierarchy**.

(b) The belief hierarchy of Player I is **not a coherent belief hierarchy** since the probability Player I ascribes to the state of nature being s_1 according to the his second-order belief is $\frac{2}{3} \cdot 1 + \frac{1}{3} \cdot 1 = 1 \neq \mu_1(s_1)$.

(c) The belief hierarchy of Player I is a **coherent belief hierarchy**. The belief of the first-order and the the second-order are as follows:

$$\mu_1 = [\tfrac{1}{2}(s_1), \tfrac{1}{2}(s_2)],$$
$$\nu_1 = [\tfrac{1}{3}(s_1, [\tfrac{1}{4}(s_1), \tfrac{3}{4}(s_2)]), \tfrac{1}{6}(s_1, [\tfrac{1}{2}(s_1), \tfrac{1}{2}(s_2)]), \tfrac{1}{2}(s_2, [\tfrac{2}{3}(s_1), \tfrac{1}{3}(s_2)])]$$
$$\mu_1 = [\tfrac{1}{2}(s_1), \tfrac{1}{2}(s_2)].$$

The marginal distribution of ν_1 over S is μ_1.

(d) The belief hierarchy of Player I is **not a coherent belief hierarchy**. According to the third-order hierarchy, Player I ascribes probability $\frac{1}{6}$ to the state of nature being s_1 whereas he ascribes 0.5 following the first-order and second-order beliefs.

(e) The belief hierarchy of Player I is **not a coherent belief hierarchy**. According to the second-order belief, Player I ascribes probability 1 to the state of nature being s_1 whereas he ascribes 0.5 following the first-order belief.

(f) The belief hierarchy of Player I is a **coherent belief hierarchy**. The belief of the first-order, the the second-order and third-order are as follows:

$$\mu_1 = [\tfrac{1}{2}(s_1), \tfrac{1}{2}(s_2)],$$
$$\nu_1 = [\tfrac{1}{3}(s_1, [\tfrac{1}{4}(s_1), \tfrac{3}{4}(s_2)]), \tfrac{1}{6}(s_1, [\tfrac{1}{2}(s_1), \tfrac{1}{2}(s_2)]), \tfrac{1}{2}(s_2, [\tfrac{2}{3}(s_1), \tfrac{1}{3}(s_2)])]$$
$$\mu_1 = [\tfrac{1}{2}(s_1), \tfrac{1}{2}(s_2)],$$
$$\begin{aligned}\nu_2 = [&\tfrac{1}{3}(s_1, [\tfrac{1}{4}(s_1, [\tfrac{2}{5}(s_1), \tfrac{3}{5}(s_2)]), \tfrac{3}{4}(s_2, [\tfrac{4}{5}(s_1), \tfrac{1}{5}(s_2)])]),\\ &\tfrac{1}{6}(s_1, [\tfrac{4}{5}(s_1, [\tfrac{1}{5}(s_1), \tfrac{1}{2}(s_2)]), \tfrac{1}{2}(s_2, [\tfrac{3}{5}(s_1), \tfrac{2}{5}(s_2)])]),\\ &\tfrac{1}{2}(s_2, [\tfrac{2}{3}(s_1, [\tfrac{2}{5}(s_1), \tfrac{3}{5}(s_2)]), \tfrac{1}{3}(s_2, [\tfrac{1}{5}(s_1), \tfrac{4}{5}(s_2)])])].\end{aligned}$$

The marginal distribution of ν_1 over S is μ_1. The projection of ν_2 on $S \times Z_1$ is ν_1.

11.9 There is a single player $N = \{I\}$ and the set of states of nature is a finite set S. By Theorem 11.16, the universal type space in this case is $T = \Delta(S)$. The universal belief space $\Omega(N, S) = S \times \Delta(S)$.

11.10 Let $\widehat{Y}$ be the set defined in Equation (11.35) in the proof of Theorem 11.20. We prove that $\widehat{Y}$ is a belief subspace of Π^*. That is, $t_i(\widehat{Y}|\varphi(\omega)) = 1$ for every $i \in N$ and every $\varphi(\omega) \in \widehat{Y}$.
For every $\omega \in Y$ and every player $i \in N$, the belief hierarchy of the first order is $\pi_i(\omega)$ which defines uniquely the infinite belief hierarchy $t_i(\omega)$ that describes the beliefs of player i. Therefore, by the definition of Π, for every player i and every $\omega, \omega' \in Y$,

$$\pi_i(\omega) = \pi_i(\omega') \iff t_i(\omega) = t_i(\omega').$$

Hence, for every $\varphi(\omega) \in \widehat{Y}$ and every $i \in N$ one has

$$\begin{aligned}
t_i\left(\widehat{Y}|\varphi(\omega)\right) &= t_i\left(\left\{\varphi(\omega') : \omega' \in Y\right\}|\varphi(\omega)\right) \\
&= t_i\left(\left\{\varphi(\omega') : \omega' \in Y, t_i(\omega) = t_i(\omega')\right\}|\varphi(\omega)\right) \\
&= t_i\left(\left\{\varphi(\omega') : \omega' \in Y, \pi_i(\omega) = \pi_i(\omega')\right\}|\varphi(\omega)\right) \\
&= \pi_i\left(\left\{\omega' \in Y : \pi_i(\omega) = \pi_i(\omega')\right\}|\omega\right) \\
&= 1.
\end{aligned}$$

11.11 We prove that the universal type space T is a Hausdorff space.
We first present an auxiliary claim.

Claim 1 *(a) Every metric X space is a Hausdorff space.*

(b) Let X be a Hausdorff space and let $Z \subset X$. Then Z (regarded as a topological space via the subspace topology) is Hausdorff.

(c) If both X and Y are Hausdorff then $X \times Y$ is Hausdorff.

Proof.

(a) Let X be a metric space and let $x, y \in X$ with $x \neq y$. Set $r = d(x, y)$. Then, the open balls $U = B(x; \frac{r}{3})$ and $V = B(y; \frac{r}{3})$ contain x and y respectively. Furthermore, by the triangle inequality $U \cap V = \emptyset$ and X is Hausdorff.

(b) Let $x, y \in Z$ with $x \neq y$. Since X is Hausdorff there exist open sets $U, V \subset X$ such that $x \in U$, $y \in V$ and $U \cap V = \emptyset$. Then $U^* = U \cap Z$ and $V^* = V \cap Z$ are open in (the subspace topology on) Z, moreover $x \in U^*$, $y \in V^*$ and $U^* \cap V^* = \emptyset$. Hence Z is Hausdorff.

(c) Suppose that X and Y are Hausdorff. Let $w = (x,y)$, $w' = (x',y') \in X \times Y$ with $w \neq w'$. Assume without loss of generality that $x \neq x'$. Then there exist open sets $U, U' \subset X$ such that $x \in U$, $x' \in U'$ and $u \cap U' \neq \emptyset$. Set $W = U \times Y$, $W' = U' \times Y$. In particular, W and W' are open sets in $X \times Y$, $w \in W$, $w' \in W'$ and $W \cap W' \neq \emptyset$. In conclusion, $X \times Y$ is Hausdorff.

By the claim it follows that for every $k \in \mathbf{N}$, X_k is Hausdorff as a metric space (defined by the weak-* topology). Thus, by part (b), since $Z_k \subseteq X_k$ is Hausdorff (regarded as a topological space via the subspace topology). Using part (c) it follows that $\times_{k=1}^{\infty} Z_k$ is Hausdorff. In conclusion, by part (b), $T \subseteq \times_{k=1}^{\infty} Z_k$ is a Hausdorff space, as claimed.

■

Chapter 12: Auctions

12.1 (a) Andrew's expected revenue as a function of the price x that he sets is

$$\pi(x) = (1 - F_V(x, x, ..., x))x = (1 - x^n)x.$$

(b) π is a differentiable function. We find the maximum by differentiation.

$$\pi'(x) = 1 - (n+1)x^n = 0 \Rightarrow x^* = \sqrt[n]{\tfrac{1}{n+1}}.$$

Note that $\pi'(x) > 0$ for every $x < x^*$ and $\pi'(x) < 0$ for every $x > x^*$, and Andrew's expected revenue is maximized at x^*. expected revenue.

(c) The maximal expected revenue that Andrew can obtain, as a function of n is $\pi(x^*) = \frac{n}{(n+1)\sqrt[n]{(n+1)}}$.

(d) If $n \geq 3$ Andrew's maximal revenue is less than the revenue he would gain if he sells the car by way of a sealed-bid first-price auction.

12.2 (a) In an open-bid decreasing auction when the current announced price is x, a buyer knows that the values of all other buyers are at most x, an information he did not have prior to the auction.

(b) In an open-bid increasing auction when the current announced price is x, a buyer knows that the values of the buyers that raise their hand is at least x whereas the values of all other players is less than x (he may even know the private value of each player that lowered his hand).

12.3 We prove that in a symmetric sealed-bid second-price auction with independent private values the only monotonically increasing, symmetric equilibrium is the equilibrium β in which every buyer submits a bid equal to his private value.
Assume by contradiction that there is another monotonically increasing, symmetric equilibrium $\bar{\beta}$. That is $\bar{\beta}(\widehat{v}) \neq \widehat{v}$ for some $\widehat{v} \in V_i$.
Assume first that $\bar{\beta}(\widehat{v}) < \widehat{v}$ for some $\widehat{v} \in V_i$. Let $v' \in (\bar{\beta}(\widehat{v}), \widehat{v})$. By the monotonicity of $\bar{\beta}$, $\bar{\beta}(v') < \bar{\beta}(\widehat{v})$. Hence, if some buyer, say i has a private value v', then he is better off submitting a bid equals his private value. Indeed, if he wins the item submitting $\bar{\beta}(v')$ then he obviously wins the item submitting v' and his price remains unchanged. Furthermore, by Assumption (A5), $\max_{j \neq i} v_j \in [\bar{\beta}(v'), \bar{\beta}(\widehat{v})]$ with a positive probability for which case according to $\bar{\beta}$, buyer i does not win the item whereas under β he wins with positive revenue. In particular, the above contradicts the assumption that $\bar{\beta}$ is an equilibrium.
Assume next that $\bar{\beta}(\widehat{v}) > \widehat{v}$ for some $\widehat{v} \in V_i$. Since $\bar{\beta}$ is an equilibrium, there cannot be any other bid in $(\widehat{v}, \bar{\beta}(\widehat{v}))$ or else a buyer with a private value $\widehat{v}$ loses (so he better off deviate). Furthermore, by the montonicity, $\bar{\beta}(\widehat{v}) < \bar{\beta}(v')$ for every $v' \in (\widehat{v}, \bar{\beta}(\widehat{v}))$. In particular, the private values in $(\widehat{v}, \bar{\beta}(\widehat{v}))$ are mapped by $\bar{\beta}$ to disjoint segments of type $(v, \bar{\beta}(v))$ in which no bids are taken. However, there are only countable disjoint segments over R which contradicts the cardinality of the segment $(\widehat{v}, \bar{\beta}(\widehat{v}))$.

12.4 Suppose that $V = [0, \bar{v}]$ is a bounded interval. Consider a symmetric sealed bid second-price auction with independent private values. Let β be a strategy vector under which buyer 1 bids $\bar{v}$ and all the other buyers bid 0. We show that β is an (asymmetric) equilibrium.
Under β, buyer 1 wins the item and pays nothing, no matter what are the private values of his and the other buyers. As long as his bid is higher than 0, this result remains the same. However, if he submits 0 then his probability of wining the item may decreases, so he cannot profit by deviation.
Similarly, if some other buyer different from 1 deviates, he may win the item only in case he submits a bid of at least $\bar{v}$ which case the price of the item is $\bar{v}$. Thus again, he also cannot profit by deviation.
It is not an equilibrium in a sealed-bid first-price auction, since with probability 1, buyer 1 has a private value less than $\bar{v}$ which case he loses by paying his bid of $\bar{v}$.

An equilibrium in an open-bid ascending auction that is analogous to this equilibrium is as follows: buyer 1 declare that he buys the item regardless the price and the other buyers do not try to buy it, hence buyer 1's payment is low.

12.5 Consider a sealed-bid second-price auction where the private values of the buyers are independent and identically distributed with the uniform distribution over $[0,1]$. Then the strategy under which a buyer bids his private value does not strictly dominate all his other strategies. For instance, it does not strictly dominates the strategy according to which he submits 1, since given all of the other buyers submits 0, the revenue under both strategies are identical.

12.6 Brent and Stuart are the only buyers in a sealed-bid first-price auction of medical devices. Brent knows that Stuart's private value is uniformly distributed over $[0,2]$, and that Stuart's strategy is $\beta(v_s) = \frac{v_s^2}{3} + \frac{v_s}{3}$.

(a) Assume that the private value of Brent's is v_b and he submits x, then his expected profit is

$$\begin{aligned} u_b(x,\beta;v_b) &= \mathbf{P}\left(x > \tfrac{V_s^2}{3} + \tfrac{V_s}{3}\right)(v_b - x) \\ &= \begin{cases} (v_b - x)\frac{\sqrt{1+12x}-1}{4} & 0 \leq x \leq 2 \\ v_b - x & 2 \leq x < v_b \\ 0 & x \geq v_b \end{cases} \end{aligned}$$

Note that,

$$u_b(x,\beta;v_b) < u_b(2,\beta;v_b) \quad \forall x > 2.$$

In addition, if $v_b \geq \frac{16}{3}$ then

$$(v_b - x)\tfrac{\sqrt{1+12x}-1}{4} \leq v_b - 2 \quad \forall x < 2.$$

Therefore, the optimal bid is $x^* = 2$.
Otherwise, if $v_b < \frac{16}{3}$, then the optimal bid is in $(0,2)$ which can be found using the derivative of the function $x \mapsto u_b(x,\beta;v_b)$ for $x \in (0,2)$:

$$\begin{aligned} -\tfrac{\sqrt{1+12x}-1}{4} + (v_b - x)\tfrac{12}{2\cdot 4\sqrt{1+12x}} = 0 \iff \\ x^* = \tfrac{9v_b - 1 + \sqrt{9v_b+1}}{27}. \end{aligned}$$

Note that second order derivative is negative and the function is continuous and thus the calculated x^* is the optimal solution.

In conclusion, Brent's optimal strategy is

$$x^*(v_b) = \begin{cases} 2 & v_b \geq \frac{16}{3} \\ \frac{9v_b - 1 + \sqrt{9v_b+1}}{27} & v_b < \frac{16}{3}. \end{cases}$$

(b) Brent's expected payment if he implements this optimal strategy (as a function of his own private value) is

$$u_b(x^*, \beta; v_b) = \begin{cases} v_b - 2 & v_b \geq \frac{16}{3} \\ (v_b - \frac{9v_b-1+\sqrt{9v_b+1}}{27})\frac{\sqrt{1+12\cdot\frac{9v_b-1+\sqrt{9v_b+1}}{27}}-1}{4} & v_b < \frac{16}{3}. \end{cases}$$

12.7 A single indivisible object is sold in a first-price auction between two buyers, 1 and 2. The value of the object v is common (the same) to both buyers but it is uncertain; It has a uniform distribution in $[0, 1]$. Buyer 1 knows the (realized) value v while buyer 2 only knows that it is uniformly distributed in $[0, 1]$ (and that buyer 1 knows the realized value).

(a) We show that the following pair of strategies is Nash equilibrium: Buyer 1 bids $\frac{v}{2}$ and buyer 2 plays a mixed strategy choosing the bid by the uniform distribution on $[0; \frac{1}{2}]$.
Assume first that the value of the object is v and Buyer 1 bids x. Since Buyer 2 bids less than $\frac{1}{2}$ with probability 1 then Buyer 1 is better of bidding $x \leq \frac{1}{2}$. Hence, with probability $2x$ (as a result of Buyer 2's strategy) he wins the item. Therefore, the expected revenue of Buyer 1 bidding $x \leq \frac{1}{2}$ is

$$u_1(x, \sigma_2 | v) = 2x(v - x). \tag{99}$$

Differentiating Equation (99) and equating its directional derivatives to 0, we deduce that the optimal bid of Buyer 1 is $\frac{v}{2}$, as needed.
Assume next that buyer 2 bids y. Buyer 2 wins the item if $y > \frac{v}{2}$, therefore his expected revenue is

$$u_2(\sigma_1, y) = \int_0^{2y} (v - y) dy = 0. \tag{100}$$

In particular, Buyer 2 is indifferent between his pure strategies and so the mixed strategy choosing the bid by the uniform distribution on $[0; \frac{1}{2}]$ is a best reply as well.

(b) The expected profit of Buyer 1 is $\int_0^1 \frac{v^2}{2} dv = \frac{1}{6}$. The expected profit of Buyer 2 is 0.

12.8 (a) Suppose that the private values of two buyers in a sealed-bid first-price auction are independent and uniformly distributed over the set $\{0, 1, 2\}$. In other words, each buyer has three possible private values. The bids in the auction must be nonnegative integers. We find all the equilibria:

If a buyer whose private value is 0 submits a bid of 2, then with some positive probability he wins the item but his expected profit is negative. Hence, in an equilibrium such a buyer does not submits a bid of 2.

Similarly, if a buyer whose private value is 0 submits a bid of 1, then with some positive probability (e.g., if the other buyer's private value is zero as well) he wins the item but his expected profit is negative.

In conclusion, in an equilibrium a buyer whose private value is 0 submits a bid of 0.

A buyer whose private value is 1 does not submit 2 (for the same reasoning as presented above). Furthermore, if he submits 1 his expected profit is 0 whereas by submitting 0 he has a positive probability to gain the item with positive profit (for instance, if the other buyer's private value is zero). In particular, a buyer whose private value is 1 submits a bid of 0 in an equilibrium.

Finally, a buyer whose private value is 2 does not submit 2. If he submits 1 is profit is higher than $\frac{2}{3} \cdot 1$ whereas if he submits 0 then his private value is at most $\frac{1}{3} \cdot 2$.

Hence, in an equilibrium, a buyer whose private value is 2 submits 1 otherwise he submits 0. The expected revenue of the seller is $\frac{1}{9}$.

(b) There are several equilibria, under the same assumptions, when the auction is a sealed-bid second-price auction:

- Every buyer submits a bid equal to his private value.
- One buyer always submits 0 and the other always submits 2 which case the revenue of the seller is $\frac{5}{9}$.
- One buyer always submits 1 and the other submits 2 if his private value is 2 , otherwise he submits 0. The expected revenue of the seller is $\frac{1}{3}$.
- One buyer always submits 1. The other buyer submits 0 if his private value is 0, otherwise he submits 2. The expected revenue of the seller is $\frac{2}{3}$.
- One buyer submits 2 if his private value is 2, otherwise he submits 1. The other buyer submits 2 if his private value is 2,

otherwise he submits 0. The expected revenue of the seller is $\frac{4}{9}$.

- One buyer submits 0 if his private value is 0, otherwise he submits 1. The other buyer submits 0 if his private value is 0, otherwise he submits 2. The expected revenue of the seller is $\frac{4}{9}$.

(c) In most equilibria, the seller's expected revenue under the sealed-bid second-price is higher than the expected revenue under the sealed-bid first-price auction.

12.9 Suppose that there are n buyers, whose private values are independent and uniformly distributed over $[0,1]$. Denote by E^I the seller's revenue in a sealed-bid first-price auction, and by E^{II} the seller's revenue in a sealed-bid second-price auction.
We calculate the variance of E^I:

$$\begin{aligned}
E^I &= \max\{\beta(V_1),...,\beta(V_n)\} = \max\{\tfrac{n-1}{n}V_1,...,\tfrac{n-1}{n}V_n\}.\\
F_{E^I}(r) &= \mathbf{P}\left(\max\{\tfrac{n-1}{n}V_1,...,\tfrac{n-1}{n}V_n\} \le r\right)\\
&= \mathbf{P}\left(\tfrac{n-1}{n}V_1 \le r,...,\tfrac{n-1}{n}V_n \le r\right)\\
&= \left(\tfrac{n}{n-1}\right)^n r^n \qquad \forall 0 \le r \le \tfrac{n-1}{n}.\\
f_{E^I}(r) &= n\left(\tfrac{n}{n-1}\right)^n r^{n-1} \qquad \forall 0 \le r \le \tfrac{n-1}{n}\\
E[E^I] &= \int_0^{\frac{n-1}{n}} rn\left(\tfrac{n}{n-1}\right)^n r^{n-1}dr = \tfrac{n-1}{n+1}\\
E[(E^I)^2] &= \int_0^{\frac{n-1}{n}} r^2n\left(\tfrac{n}{n-1}\right)^n r^{n-1}dr = \tfrac{(n-1)^2}{(n+2)n}\\
Var(E^I) &= \tfrac{(n-1)^2}{(n+2)n} - \left(\tfrac{n-1}{n+1}\right)^2 = \tfrac{(n-1)^2}{n(n+2)(n+1)^2}.
\end{aligned}$$

To calculate the variance of E^{II} note that E^{II} is the second-highest bid. Denote by M the highest bid. Then,

$$\begin{aligned}
F_{E^{II}}(r) &= \mathbf{P}\left(E^{II} \leq r\right) \\
&= \mathbf{P}\left(E^{II} \leq r, M \leq r\right) + \mathbf{P}\left(E^{II} \leq r, M > r\right) \\
&= \mathbf{P}\left(V_1 \leq r, ..., V_n \leq r\right) + n\mathbf{P}\left(V_1 > r, V_2 \leq r, ..., V_n \leq r\right) \\
&= r^n + nr^{n-1}(1-r) \qquad \forall 0 \leq r \leq 1. \\
f_{E^{II}}(r) &= n(n-1)r^{n-2}(1-r) \qquad \forall 0 \leq r \leq 1 \\
E[E^{II}] &= \int_0^1 rn(n-1)r^{n-2}(1-r)dr = \tfrac{n-1}{n+1} \\
E[(E^{II})^2] &= \int_0^1 r^2 n(n-1)r^{n-2}(1-r)dr = \tfrac{n(n-1)}{(n+2)(n+1)} \\
Var(E^{II}) &= \tfrac{n(n-1)}{(n+2)(n+1)} - \left(\tfrac{n-1}{n+1}\right)^2 = \tfrac{2(n-1)}{(n+2)(n+1)^2}.
\end{aligned}$$

The expected revenues of both auctions are identical but the variance in the sealed-bid first-price auction is lower.

12.10 Consider a sealed-bid second-price auction with n buyers whose private values are independent and uniformly distributed over $[0, 1]$. Define the a pair of strategies

$$\beta_1^*(v) = \begin{cases} v & v \geq \frac{1}{2}, \\ \frac{1}{2} & v < \frac{1}{2}, \end{cases} \qquad \beta_2^*(v) = \begin{cases} v & v \geq \frac{1}{2}, \\ 0 & v < \frac{1}{2}. \end{cases}$$

We prove that the strategy profile according to which Buyer 1 follows β_1^* and each other buyer follows β_2^* is an equilibrium.
Each buyer whose private value is $v > \frac{1}{2}$ bids v which weakly dominates all other strategies so he has no incentive to deviate.
Suppose that the private value of Buyer 1 is less than $\frac{1}{2}$. If he bids $0 < x < \frac{1}{2}$ then his winning probability as well as his revenue remain unchanged. If he bids $x = 0$, his winning probability decreases whereas his revenue remains unchanged. Finally, if he bids $x > \frac{1}{2}$, his winning probability increases but just for cases in which he pays more than the value of the item. In conclusion, Buyer 1 cannot gain by deviation.
Suppose that the private value of a buyer $i \neq 1$ is less than $\frac{1}{2}$. If he bids $0 < x < \frac{1}{2}$ then his winning probability as well as his revenue remain unchanged. If he bids $x \geq \frac{1}{2}$, his winning probability increases but just for cases in which he pays more than the value of the item. Therefore, every buyer $i \neq 1$ cannot gain by deviation.

12.11 Consider a sealed-bid first-price auction with three buyers, where the private values of the buyers are independent and uniformly distributed over $[0, 2]$.

(a) Using Theorem 12.9, since Assumptions (A1)-(A6) are satisfied, a symmetric equilibrium of the game is

$$\beta(0) = 0 \text{ and } \beta(v) = E[Y|Y \leq V] \ \ \forall v \in V \setminus \{0\}$$

where Y is the second-highest private value. However,

$$F_Y(y) = \begin{cases} 1 & y \geq 2 \\ \frac{y^2}{4} & 0 < y < 2 \\ 0 & y \leq 0 \end{cases} \quad \text{and} \quad f_Y(y) = \begin{cases} \frac{y}{2} & 0 < y < 2 \\ 0 & \text{otherwise.} \end{cases}$$

Therefore,

$$\beta(v) = E[Y|Y \leq v] = \frac{\int_0^v y \frac{y}{2} dy}{\frac{v^2}{4}} = \tfrac{2}{3}v.$$

(b) The seller's expected revenue is

$$E = 3e_i = 3\int_0^2 F_y(v)E[Y|Y \leq v]f_i(v)dv = 3\int_0^2 \tfrac{v^2}{4} \cdot \tfrac{2}{3}v \cdot \tfrac{1}{2}dv = 1.$$

12.12 Consider a symmetric sealed-bid auction with independent private values the random variable. Denote $Y = \max\{V_2, V_3, ..., V_n\}$. We show the Y is a continuous random variable.
By Assumption (A5), for each i, the random variable V_i is continuous, and its density function, which we denote by f , is continuous and positive. Hence, for every $y \in \mathbf{R}$

$$\begin{aligned} F_Y(y) &= \mathbf{P}(Y \leq y) = \mathbf{P}(V_2 \leq y, ..., V_n \leq y) \\ &= \mathbf{P}(V_2 \leq y)\mathbf{P}(V_3 \leq y) \cdot ... \cdot \mathbf{P}(V_n \leq y) && (101) \\ &= F_{V_2}(y)F_{V_3}(y) \cdot ... \cdot F_{V_n}(y) \\ &= \left(F(y)\right)^{n-1}, && (102) \end{aligned}$$

where Equality (101) holds since $V_2, ..., V_n$ are independent and Equality (102) holds since they are identically distributed.
Set

$$f_Y(y) = (n-1)f(y)\left(F(y)\right)^{n-2} \quad \forall y \in \mathbf{R}.$$

Then f_Y is a positive and continuous (almost everywhere) function for every $v \in V$ since f and F are positive and continuous (almost everywhere). Furthermore, $\int_{-\infty}^{v} f(y)dy = F_Y(v)$ for every $v \in \mathbf{R}^N$, from which it follows that Y is continuous and f_Y is its density function.

12.13 The following auction methods satisfy the conditions of the Revenue Equivalence Theorem : sealed-bid first-price auctions and sealed-bid second-price auctions.
Sealed-bid first-price and second-price auctions with a reserve price and sealed-bid second-price auctions with entry fees do not satisfy the efficiency property: if both buyers have private values lower than the reserve price/entry fee, the auctioned object is not sold, despite the fact that there is a buyer willing to pay for it.

12.14 Consider a sealed-bid second-price auction with a reserve price ρ, in which there are two buyers whose private values are independent and uniformly distributed over $[0,1]$.
In an equilibriun, a buyer with a private value lower than or equal to ρ cannot profit no matter what bid he makes, therefore he does not participate. A buyer with a private value greater than ρ submits a bid equals to his private value (which is weakly dominates all other strategies). Therefore, the seller's expected revenue under this equilibrium is

$$\begin{aligned}
E &= 2\int_{\rho}^{1} \left(\mathbf{P}(V_2 \leq \rho)\rho + \mathbf{P}(\rho < V_2 \leq v)E[V_2|\rho < V_2 \leq v]\right) f(v)dv \\
&= 2\int_{\rho}^{1} \left(\rho^2 + \int_{\rho}^{v} y dy\right) dv = 2\int_{\rho}^{1} \left(\rho^2 + \tfrac{v^2}{2} - \tfrac{\rho^2}{2} dy\right) dv \\
&= \int_{\rho}^{1} v^2 + \rho^2 dv = \tfrac{1}{3} + \rho^2 - \tfrac{4\rho^3}{3}.
\end{aligned}$$

The revenue of the seller is identical to the revenue of the seller in a sealed-bid first-price auctions with a reserve price ρ.

12.15 (a) Consider a sealed-bid first-price auction with a reserve price ρ, with n buyers whose private values are independent and uniformly distributed over $[0,1]$. Assuming that there exists a monotonically increasing and differentiable symmetric equilibrium strategy β^*, we can compute it using the Envelope Theorem.
Denote $Y = \max\{V_2, ..., V_n\}$. Since $(V_j)_{j=2}^{n}$ are independent and uniformly distributed over $[0,1]$ and β is monotonically increas-

ing, for every $v > \rho$

$$u_1(b_1, \beta; v) = (v - b_1)P(Y \le \beta^{-1}(b_1)) = (v - b_1)(\beta^{-1}(b_1))^{n-1}$$
$$\Rightarrow \tfrac{\partial u_1}{\partial v}(\beta(b_1), \beta; v) = (\beta^{-1}(b_1))^{n-1}$$
$$\Rightarrow \tfrac{\partial u_1^*}{\partial v}(\beta; v) = \tfrac{\partial u_1}{\partial v}(b_1, \beta; v)_{|b_1=\beta(v)} = v^{n-1}.$$

For $v \le \rho$, buyer 1 cannot profit no matter what bid he makes, therefore he does not participate and $u_1^*(\beta, v) = 0$.
By integrating, we get

$$u_1^*(\beta, v) = \int_\rho^v \tfrac{\partial u_1^*}{\partial v}(\beta; t_1)dt_1 = \int_\rho^v t_1^{n-1}dt_1 = \tfrac{v^n - \rho^n}{n}. \qquad (103)$$

On the other hand,

$$u_1^*(\beta, v) = u_1(\beta(v), \beta; v) = (v - \beta(v))v^{n-1}. \qquad (104)$$

By Equations (103) and (104),

$$(v - \beta(v))v^{n-1} = \tfrac{v^n - \rho^n}{n} \Rightarrow \beta(v) = \tfrac{n-1}{n}v + \tfrac{\rho^n}{nv^{n-1}}.$$

Note that $\beta'(v) = \frac{n-1}{n} - \frac{(n-1)\rho^n}{nv^n} > 0$ for every $v > \rho$ and thus β satisfies the Envelope theorem's condition.
The seller's expected revenue is

$$\begin{aligned} E &= n\int_\rho^1 \beta(v)F_Y(v)f(v)dv \\ &= n\int_\rho^1 \left(\tfrac{n-1}{n}v + \tfrac{\rho^n}{nv^{n-1}}\right)v^{n-1}dv \\ &= \tfrac{n-1}{n+1} + \rho^n - \tfrac{2n}{n+1}\rho^{n+1}. \end{aligned}$$

(b) The reserve price $\rho^* = \frac{1}{2}$ maximizes the seller's expected revenue.

(c) In a symmetry equilibrium of a sealed-bid second-price auction with a reserve price ρ, a buyer with a private value lower than or equal to ρ cannot profit no matter what bid he makes, therefore he does not participate. A buyer with a private value greater than ρ submits a bid equals to his private value (which is weakly dominates all other strategies). Therefore, the seller's expected revenue under this equilibrium is

$$\begin{aligned} E &= n\int_\rho^1 \left(\mathbf{P}(Y \le \rho)\rho + \mathbf{P}(\rho < Y \le v)E[Y|\rho < Y \le v]\right)f(v)dv \\ &= n\int_\rho^1 \left(\rho^n + \int_\rho^v (n-1)y^{n-1}dy\right)dv = \int_\rho^1 (n-1)v^n + \rho^n dv \\ &= \tfrac{n-1}{n+1} + \rho^n - \tfrac{2n}{n+1}\rho^{n+1}. \end{aligned}$$

The revenue of the seller is identical to the revenue of the seller in a sealed-bid first-price auctions with a reserve price ρ.

(d) Both auction methods presented in previous items satisfy the condition of the general Revenue Equivalence Theorem (Corollary 12.50), hence the revenue of the seller in both auction methods are identical.

(e) The expected revenue in the auctions presented in previous items are higher than the expected revenue of a seller who is selling the object by setting the monopoly price.

(f) The optimal reserve price in items (b) and (c) above converge to $\rho = \frac{1}{2}$ when the number of buyers increases to infinity.

12.16 Consider a symmetric sealed-bid first-price auction with independent private values with n buyers whose cumulative distribution function F is given by $F(v) = v^2$ for every $v \in [0,1]$.
By theorem Theorem 12.19,

$$\beta(v) = E[Y|Y \leq v] = \frac{\int_0^v w f_Y(w)dy}{F_Y(v)} = \frac{\int_0^v w(n-1)(F(w))^{n-2}f(w)dy}{(F(v))^{n-1}}$$
$$= \frac{\int_0^v 2(n-1)w^{2n-2}dy}{v^{2n-2}} = \frac{2n-2}{2n-1}v.$$

The expected payment of each buyer i whose private value is v is

$$e_i(v) = F_Y(v)\beta(v) = \frac{2n-2}{2n-1}v^{2n-1}.$$

Hence, the expected payment of each buyer i is

$$e_i = \int_0^1 \frac{2n-2}{2n-1}v^{2n-1}2vdv = 4\frac{n-1}{2n-1}\int_0^1 v^{2n}dv = 4\frac{n-1}{(2n-1)(2n+1)}.$$

In conclusion, the seller's expected revenue is

$$\pi = ne_i = \frac{4n(n-1)}{(2n-1)(2n+1)}.$$

12.17 Consider a symmetric sealed-bid first-price auction with independent private values with n buyers whose cumulative distribution function F is given by $F(v) = v^3$ for every $v \in [0,1]$.
By theorem Theorem 12.19,

$$\beta(v) = E[Y|Y \leq v] = \frac{\int_0^v w f_Y(w)dy}{F_Y(v)} = \frac{\int_0^v w(n-1)(F(w))^{n-2}f(w)dy}{(F(v))^{n-1}}$$
$$= \frac{\int_0^v 3(n-1)w^{3n-3}dy}{v^{3n-3}} = \frac{3n-3}{3n-2}v.$$

The expected payment of each buyer i whose private value is v is

$$e_i(v) = F_Y(v)\beta(v) = \tfrac{3n-3}{3n-2}v^{3n-2}.$$

Hence, the expected payment of each buyer i is

$$e_i = \int_0^1 \tfrac{3n-3}{3n-2}v^{3n-2}3v^2dv = 9\tfrac{n-1}{3n-2}\int_0^1 v^{3n}dv = 9\tfrac{n-1}{(3n-2)(3n+1)}.$$

In conclusion, the seller's expected revenue is

$$\pi = ne_i = \tfrac{9n(n-1)}{(3n-2)(3n+1)}.$$

12.18 Consider a sealed-bid first-price auction with entry fee λ and n buyers, whose private values are independent and uniformly distributed over [0, 1].

(a) Let β be a monotonically increasing strategy corresponding a symmetric equilibrium. Denote $Y = \max\{V_2, ..., V_n\}$. Let v_0 be a private value where a buyer is indifferent between participating and not participating the auction.

Let $v > v_0$ and assume a player submits b then the expected revenue of a buyer is

$$u_1(b, \beta; v) = F_Y(\beta^-1(b))(v - b) - \rho.$$

Note that, in an auction without a fee the revenue is different only by the constant ρ, hence the equilibrium strategy in both auction method (with and without a fee) are identical. Therefore,

$$\beta(v) = \begin{cases} 0 & v \leq v_0 \\ \frac{n-1}{n}v & v > v_0, \end{cases}$$

where v_0 satisfies

$$\begin{aligned} F_Y(v_0)(v_0 - \tfrac{n-1}{n}v_0) - \rho = 0 \\ v_0^{n-1}(v_0 - \tfrac{n-1}{n}v_0) - \rho = 0 \\ v_0 = \sqrt[n]{n\rho}. \end{aligned}$$

(b) The seller's expected revenue is

$$\begin{aligned}
E &= n\int_{v_0}^{1}\left(\rho + F_Y(v)\beta(v)\right)f(v)dv \\
&= n\int_{v_0}^{1}\left(\rho + v^{n-1}\tfrac{n-1}{n}v\right)dv \\
&= n(1-v_0)\rho + \tfrac{n-1}{n+1}(1-v_0^{n+1}) \\
&= n\rho - n\rho\sqrt[n]{n\rho} + \tfrac{n-1}{n+1} - \tfrac{n(n-1)}{n+1}\rho\sqrt[n]{n\rho} \\
&= \tfrac{n-1}{n+1} + n\rho - \tfrac{2n^{(2n+1)/n}}{n+1}\rho^{(n+1)/n}.
\end{aligned}$$

(c) The function that maps $\rho \mapsto E(\rho)$ is differentiable and attains its maximum at $(0,1)$. We find the maximum using the derivative

$$E'(\rho) = n - \tfrac{n+1}{n}\cdot\tfrac{2n^{(2n+1)/n}}{n+1}\rho^{1/n} = 0 \Leftrightarrow \rho^* = \tfrac{1}{2^n}.$$

Note that $E''(\rho^*) < 0$ and E' is continuous so ρ^* is indeed the optimal entry fee maximizes the seller's expected revenue.

(d) The optimal entry fee approach 0 as the number of buyers increases to ∞.

12.19 Consider a sealed-bid third-price auction, with n buyers whose private values are independent; each V_i has uniform distribution over $[0,1]$. Denote the highest private value from among $V_2, V_3, \ldots, V_n$ by Y, and the second-highest private value from among $V_2, V_3, \ldots, V_n$ by W. Denote by F the cumulative distribution function of V_i, and by f its density function.

(a) Let $v_1 \in (0,1]$. The conditional cumulative distribution function of W at $w \le v_1$, given $Y \le v_1$, is

$$\begin{aligned}
F_{(W|Y\le v_1)}(w) &= \frac{\mathbf{P}(W\le w, Y\le v_1)}{\mathbf{P}(Y\le v_1)} \\
&= \frac{\mathbf{P}(W\le w, Y\le w)+\mathbf{P}(W\le w, w<Y\le v_1)}{\mathbf{P}(Y\le v_1)} \\
&= \frac{\left(F(w)\right)^{n-1}+(n-1)\left(F(w)\right)^{n-2}\left(F(v_1)-F(w)\right)}{\left(F(v_1)\right)^{n-1}} \\
&= \frac{\left(F(w)\right)^{n-2}\left((n-1)F(v_1)-(n-2)F(w)\right)}{\left(F(v_1)\right)^{n-1}}.
\end{aligned}$$

(b) The conditional density function is

$$f_{(W|Y\le v_1)}(w) = \frac{(n-1)(n-2)\left(F(w)\right)^{n-3}f(w)\left(F(v_1)-F(w)\right)}{\left(F(v_1)\right)^{n-1}}.$$

(c) Denote

$$h(y) = (n-2)\left(F(y)\right)^{n-3} f(y), \quad \forall y \in [0,1].$$

In particular,

$$f_{(W|Y \le v_1)}(w) = \frac{(n-1)h(w)\left(F(v_1)-F(w)\right)}{\left(F(v_1)\right)^{n-1}}.$$

The expected payment of buyer 1 with private value $v_1 \in (0,1]$ is given by $F_Y(v_1)\mathbf{E}[\beta(W) \mid Y \le v_1]$. Using the Revenue Equivalence Theorem it follows that

$$\begin{aligned} F_Y(v_1)\mathbf{E}[\beta(W) \mid Y \le v_1] &= F_Y(v_1)\mathbf{E}[Y \mid Y \le v_1] && \Rightarrow \\ F_Y(v_1)\int_0^{v_1} \beta(y) f_{W|Y\le v_1}(y)dy &= \int_0^{v_1} y f_Y(y)dy && \Rightarrow \\ \int_0^{v_1} \beta(y)(n-1)h(y)\left(F(v_1)-F(y)\right)dy &= \int_0^{v_1} y f_Y(y)dy. \end{aligned}$$

(d) We differentiate the last equation in Part (c) by v_1 using the Leibniz integral rule,

$$\begin{aligned} \int_0^{v_1} \beta(y)(n-1)h(y)f(v_1)dy + \beta(v_1)(n-1)h(v_1)\left(F(v_1)-F(v_1)\right) \\ = v_1 f_Y(v_1). \end{aligned}$$

Substituting $f_Y(v_1)$ in the last equation and deviating by the positive values $n-1$ and $f(v_1)$ leads to

$$\int_0^{v_1} \beta(y)h(y)dy = v_1\left(F(v_1)\right)^{n-2}$$

(e) Differentiating the last equation in Part (d) by v_1 using the Leibniz integral rule leads to

$$\beta(v_1)h(v_1) = \left(F(v_1)\right)^{n-2} + (n-2)v_1 f(v_1)\left(F(v_1)\right)^{n-3}.$$

We finally, deviate by $h(v_1)$ and substitute it so

$$\begin{aligned} \beta(v_1) &= \frac{\left(F(v_1)\right)^{n-2}+(n-2)v_1 f(v_1)\left(F(v_1)\right)^{n-3}}{h(v_1)} \\ &= \frac{\left(F(v_1)\right)^{n-2}+(n-2)v_1 f(v_1)\left(F(v_1)\right)^{n-3}}{(n-2)\left(F(v_1)\right)^{n-3} f(v_1)} \\ &= v_1 + \frac{F(v_1)}{(n-2)f(v_1)} \end{aligned}$$

(f) The density function f should be positive and differentiable function to ensure that the strategy β satisfies the conditions of Theorem 12.23

(g) Since buyer are symmetric as well as the equilibrium, all the computations done so far show that the strategy β is a symmetric equilibrium.

12.20 Consider a sealed-bid first-price auction with two buyers whose private values are independent; the private value of buyer 1 has uniform distribution over the interval $[0,3]$, and the private value of buyer 2 has uniform distribution over the interval $[3,4]$.

(a) We prove that the following pair of strategies form an equilibrium

$$\beta_1(v_1) = 1 + \tfrac{v_1}{2},$$
$$\beta_2(v_2) = \tfrac{1}{2} + \tfrac{v_2}{2}.$$

Indeed,

$$\begin{aligned} u_1(b,\beta;v_1) &= \mathbf{P}(\beta_2(V_2) \le b)v_1 = \mathbf{P}(\tfrac{1}{2} + \tfrac{V_2}{2} \le b)(v_1 - b) \\ &= \mathbf{P}(V_2 \le 2b-1)(v_1 - b) \end{aligned}$$

$$u_1(b,\beta;v_1) = \begin{cases} v_1 - b & b > 5/2 \\ (2b-4)(v_1-b) & 2 < b \le 5/2 \\ 0 & b \le 2. \end{cases}$$

This function is differentiable function at $(2, \frac{5}{2})$ hence we find its maximum using its derivative.

$$\tfrac{\partial u_1(b,\beta;v_1)}{\partial b} = 2v_1 - 2b - 2b + 4 = 0 \iff b = 1 + \tfrac{v_1}{2},$$

and the second order derivative is negative (-4) hence this is indeed the best response of the first buyer.
In addition,

$$\begin{aligned} u_2(b,\beta;v_2) &= \mathbf{P}(\beta_1(V_1) \le b)v_2 = \mathbf{P}(1 + \tfrac{V_1}{2} \le b)(v_2 - b) \\ &= \mathbf{P}(V_1 \le 2b-2)(v_2 - b) \end{aligned}$$

$$u_2(b,\beta;v_2) = \begin{cases} v_2 - b & b > 5/2 \\ \frac{2b-2}{3}(v_2-b) & 1 < b \le 5/2 \\ 0 & b \le 1. \end{cases}$$

This function is differentiable function at $(1, \frac{5}{2})$ hence we find its maximum using its derivative.

$$\tfrac{\partial u_2(b,\beta;v_2)}{\partial b} = \tfrac{2}{3}v_2 - \tfrac{2}{3}b - \tfrac{2}{3}b + \tfrac{2}{3} = 0 \iff b = \tfrac{1}{2} + \tfrac{v_2}{2},$$

and the second order derivative is negative hence this is indeed the best response of the second buyer.

(b) The probability that buyer 2 wins the auction equals

$$\begin{aligned}\mathbf{P}(1+\tfrac{V_1}{2} \le \tfrac{1}{2}+\tfrac{V_2}{2}) =& \mathbf{P}(1+V_1 \le V_2)\\ &\le 1-\mathbf{P}(V_1 > 2.5)\mathbf{P}(V_2 < 3.5)\\ &= 1-\tfrac{1}{6}\cdot\tfrac{1}{2} < 1.\end{aligned}$$

(c) The expected payment of buyer 1 is

$$\begin{aligned}e_1 &= \int_0^3 \mathbf{P}(\tfrac{1}{2}+\tfrac{V_2}{2} \le 1+\tfrac{v}{2})\left(1+\tfrac{v_1}{2}\right) f_{V_1}(v)dv = \int_2^3 \tfrac{v-2}{3}(1+\tfrac{v}{2})\tfrac{1}{3}dv\\ &= \tfrac{1}{9}\left(\tfrac{3^3}{6}-2\cdot 3-\tfrac{2^3}{6}+2\cdot 2\right) = \tfrac{7}{54}.\end{aligned}$$

The expected payment of buyer 2 is

$$\begin{aligned}e_2 &= \int_3^4 \mathbf{P}(1+\tfrac{V_1}{2} \le \tfrac{1}{2}+\tfrac{v}{2})\left(\tfrac{1}{2}+\tfrac{v_1}{2}\right) f_{V_2}(v)dv = \int_3^4 \tfrac{v-1}{3}(\tfrac{1}{2}+\tfrac{v}{2})dv\\ &= \tfrac{1}{6}\left(\tfrac{4^3}{3}-4-\tfrac{3^3}{3}+3\right) = \tfrac{19}{18}.\end{aligned}$$

Therefore, the seller's expected revenue if the buyers implement the strategies (β_1, β_2) is $E = \frac{7}{54}+\frac{19}{18} = \frac{32}{27}$.

(d) Suppose that the object is sold by a sealed-bid second-price auction (instead of a sealed-bid first price auction) and every bidder bids his private value. Hence,

$$\begin{aligned}e_1 &= 0,\\ e_2 &= \mathbf{E}[V_1] = 1.5,\end{aligned}$$

and thus the seller's ex- pected revenue is $E = e_1 + e_2 = 1.5$.

12.21 Consider a sealed-bid first-price auction with two buyers whose private values are independent; the private value of buyer 1 has uniform distribution over the interval $[0, m+z]$, and the private value of buyer 2 has uniform distribution over the interval $[\frac{3m}{2}, \frac{3m}{2}+z]$, where $m, z > 0$. We show that the following pair of strategies form an equilibrium.

$$\begin{aligned}\beta_1(v_1) &= \tfrac{v_1}{2}+\tfrac{m}{2},\\ \beta_2(v_2) &= \tfrac{v_2}{2}+\tfrac{m}{4}.\end{aligned}$$

Note that these equilibrium strategies are independent of z.

The expected revenue of buyer 1 is

$$\begin{aligned}u_1(b,\beta;v_1) &= \mathbf{P}(\beta_2(V_2) \le b)(v_1 - b) = \mathbf{P}(\tfrac{V_2}{2} + \tfrac{m}{4} \le b)(v_1 - b)\\ &= \mathbf{P}(V_2 \le 2b - \tfrac{m}{2})(v_1 - b)\end{aligned}$$

$$u_1(b,\beta;v_1) = \begin{cases} v_1 - b & b > \frac{m}{2} + \frac{z}{2} \\ \frac{2b-2m}{z}(v_1 - b) & m < b \le \frac{m}{2} + \frac{z}{2} \\ 0 & b \le m. \end{cases}$$

This function is differentiable function at $(\frac{m}{2}, \frac{m}{2} + \frac{z}{2})$ hence we find its maximum using its derivative.

$$\tfrac{\partial u_1(b,\beta;v_1)}{\partial b} = \tfrac{1}{z}\left(2v_1 - 2b - 2b + 2m\right) = 0 \iff b = \tfrac{v_1}{2} + \tfrac{m}{2},$$

and the second order derivative is negative hence this is indeed the best response of the first buyer.
In addition,

$$\begin{aligned}u_2(b,\beta;v_2) &= \mathbf{P}(\beta_1(V_1) \le b)(v_2 - b) = \mathbf{P}(\tfrac{V_1}{2} + \tfrac{m}{2} \le b)(v_2 - b)\\ &= \mathbf{P}(V_1 \le 2b - m)(v_2 - b)\end{aligned}$$

$$u_2(b,\beta;v_2) = \begin{cases} v_2 - b & b > m + \frac{z}{2} \\ \frac{2b-m}{m+z}(v_2 - b) & \frac{m}{2} < b \le m + \frac{z}{2} \\ 0 & b \le \frac{m}{2}. \end{cases}$$

This function is differentiable function at $(\frac{3m}{2}, \frac{3m}{2} + z)$ hence we find its maximum using its derivative.

$$\tfrac{\partial u_2(b,\beta;v_2)}{\partial b} = \tfrac{1}{m+z} \cdot (2v_2 - 2b - 2b + m) = 0 \iff b = \tfrac{v_2}{2} + \tfrac{m}{4},$$

and the second order derivative is negative hence this is indeed the best response of the second buyer, which completes the proof.

12.22 **Proof of Theorem 12.29:** in a sealed-bid second-price auction with a reserve price, for each buyer strategy β the following strategy $\widehat{\beta}$ weakly dominates β, if $\widehat{\beta} \neq \beta$,

$$\widehat{\beta}(v) = \begin{cases} {}''no'' & \beta(v) = {}''\ no'', \\ v & \beta(v) = x. \end{cases}$$

For every v for which $\widehat{\beta}(v) = \beta(v)$, e.g., if $\beta(v) = {}''\ no''$, then the revenues under both strategies are identical.
For every v for which $\beta(v) \neq \widehat{\beta}(v)$, $\beta(v) = x \neq v = \widehat{\beta}(v)$. Hence, the buyer participates the auctions and pays the fee under both strategies and thus it can be ignored.

Let y be the maximal-bid among all of the other buyers.
If $y \leq x < v$ then the buyer wins the item under $\beta(v) = x$ as well as under $\widehat{\beta}(v) = v$ and pays the same amount of money.
If $x < v < y$ then the buyer loses the item under both $\beta(v) = x$ and $\widehat{\beta}(v) = v$ and pays the fee only.
If $x < y \leq v$ then the buyer loses the item under $\beta(v) = x$ but wins the item under $\widehat{\beta}(v) = v$ and pays a payment that is lower than his private value and thus he gains.
If $y \leq v < x$ then the buyer wins the item under $\beta(v) = x$ as well as under $\widehat{\beta}(v) = v$ and pays the same amount of money.
If $v < x < y$ then the buyer loses the item under both $\beta(v) = x$ and $\widehat{\beta}(v) = v$ and pays the fee only.
If $v < y < x$ then the buyer loses the item under $\widehat{\beta}(v) = v$ but wins the item under $\beta(v) = x$ but pays a payment that is higher than his private value and thus he loses.

12.23 Consider a sealed-bid second-price auction with two buyers, whose private values are independent; buyer 1's private value is uniformly distributed over $[0,1]$, and buyer 2's private value is uniformly distributed over $[0,2]$.

(a) For buyer 1 the the strategy according to which he submits a bid equal to his private value. For buyer 2, each strategy according to which he submits a bid equal to his private value v whenever $v \leq 1$ and submits any bid higher than 1 whenever his private value is his higher than 1. Note that since the bid of buyer 1 is at most 1 then buyer 2 wins the item if he submits any bid higher than 1.

(b) Consider the equilibrium in which every buyer bids his private value. The probability that buyer 1 wins the auction, under this equilibrium is

$$\begin{aligned}
\mathbf{P}(V_2 \leq V_1) &= \mathbf{P}(V_2 \geq 1)\mathbf{P}(V_2 \leq V_1 \mid V_2 \geq 1) \\
&\quad + \mathbf{P}(V_2 \leq 1)\mathbf{P}(V_2 \leq V_1 \mid V_2 \leq 1) \\
&= \tfrac{1}{2} \cdot 0 + \tfrac{1}{2} \cdot \tfrac{1}{2} = \tfrac{1}{4}.
\end{aligned}$$

The expected payment of buyer 1 is

$$\begin{aligned}
e_1 &= \int_0^1 F_{V_2}(v) E[V_2 \mid V_2 \leq v] f_{V_1(v)} dv = \int_0^1 \int_0^v \tfrac{y}{2} dy dv \\
&= \int_0^1 \tfrac{v^2}{4} dv = \tfrac{1}{12}.
\end{aligned}$$

The expected payment of buyer 2 is

$$e_2 = \int_0^1 F_{V_1}(v)E[V_1 \mid V_1 \le v] f_{V_2}(v)dv + \int_1^2 E[V_1] f_{V_2}(v)dv$$
$$= \int_0^1 \int_0^v y dy \tfrac{1}{2} dv + \tfrac{1}{4} = \int_0^1 \tfrac{v^2}{4} dv + \tfrac{1}{4} = \tfrac{1}{3}.$$

Therefore, the seller's expected revenue in this case is $E = \frac{1}{12} + \frac{1}{3} = \frac{5}{12}$.

(c) We prove that at any equilibrium $\beta = (\beta_1, \beta_2)$ satisfying $\beta_1(v) = \beta_2(v)$ for all $v \in [0,1]$, one has $\beta_1(v) = \beta_2(v) = v$ for all $v \in [0,1]$.

Let $\beta = (\beta_1, \beta_2)$ be an equilibrium. We first prove that there is no segment $(a, b) \subseteq [0,1]$ such that

$$\mathbf{P}\left(\beta_i(V_i) \in (a,b)\right) = 0.$$

If by contradiction there is such a segment (a, b) then either

$$\mathbf{P}\left(V_i \in (a,b) \text{ and } \beta(V_i) \ge b\right) > 0 \tag{105}$$

or

$$\mathbf{P}\left(V_i \in (a,b) \text{ and } \beta(V_i) \le a\right) > 0. \tag{106}$$

If Equation (105) holds then there is $v' \in (a,b)$ such that

$$\mathbf{P}\left(V_i \in (a,b) \text{ and } b \le \beta(V_i) \le \beta(v')\right) > 0.$$

That is, a buyer with a private value of v' may wins the item with positive probability, but his payment exceeds his private value. Hence, he is better off submitting v', a contradiction.

If Equation (106) holds then there is $v' \in (a,b)$ such that

$$\mathbf{P}\left(V_i \in (a,b) \text{ and } \beta(v') \le \beta(V_i) \le a\right) > 0.$$

That is, a buyer with a private value of v' can increase his winning probability by submitting a bid equals his private value, a contradiction.

Finally, assume by contradiction that β is an equilibrium satisfying that $\beta_1(v) = \beta_2(v)$ for every $v \in [0,1]$ but there is $w \in [0,1]$ such that $\beta(w) \ne w$. Then with probability 1 there is no bid in the open segment between w and $\beta(w)$, which contradicts the previous claim, or else a buyer whose private value is w is better off submitting w rather than $\beta(w)$ (if $w < \beta(w)$ his payment exceeds his private value with positive probability; if $w > \beta(w)$ then by submitting w he increases his probability of earning).

(d) There are equilibria at which a buyer, whose private value v_i is less than 1, does not submit the bid $\beta_i(v_i) = v_i$. For instance, the first buyer always submits 1 and the second buyer submits 0 if his private value is at most 1, otherwise he submits 2.

12.24 (a) Let F be a cumulative distribution function over $[0,1]$, and let Y be the maximum of $n-1$ independent random variables with cumulative distribution function F, then

$$\begin{aligned}\mathbf{E}[Y \mid Y \leq v] &= \frac{\int_0^v x f_Y(x)dx}{F_Y(v)} = \frac{\int_0^v xF'_Y(x)dy}{F_Y(v)} \\ &= \frac{xF_Y(x)|_{x=0}^{v} - \int_0^v F_Y(x)dx}{F_Y(v)} \qquad (107) \\ &= \frac{vF_Y(v) - \int_0^v F_Y(x)dx}{F_Y(v)} \\ &= v - \frac{\int_0^v F_Y(x)dx}{F_Y(v)} \\ &= v - \frac{\int_0^v (F(x))^{n-1}dx}{(F(v))^{n-1}} \qquad (108)\end{aligned}$$

where Equality (107) follows integration by parts and Equality (108) follows substituting $F_Y(x) = (F(x))^{n-1}$.

(b)
- For $F(x) = x$ then

$$\beta^*(v) = \mathbf{E}[Y \mid Y \leq v] = v - \frac{\int_0^v x^{n-1}dx}{v^{n-1}} = \frac{n-1}{n}v.$$

- For $F(x) = x^2$ then

$$\beta^*(v) = \mathbf{E}[Y \mid Y \leq v] = v - \frac{\int_0^v x^{2(n-1)}dx}{v^{2(n-1)}} = \frac{2n-2}{2n-1}v.$$

12.25 In a sealed-bid second-price auction with risk-neutral buyer, the strategy under which a buyer submits a bid equal to his private value weakly dominates all his other strategies (Theorem 12.7). Hence, in order to prove that the same holds when the buyer is risk-averse or risk-seeking, it is enough to prove that if a lottery $X = (\alpha_1(x_1), ..., \alpha_n(x_m))$ weakly dominates $Y = (\alpha_1(y_1), ..., \alpha_n(y_m))$ then $U(X) \geq U(Y)$. Indeed, since X weakly dominates Y then $x_k \geq y_k$ for every $k = 1, .., m$. Hence, since U is monotonic increasing, $u(x_k) \geq u(y_k)$ for every $k = 1, ..., m$. Therefore,

$$U(X) = \sum_{k=1}^{m} \alpha_k u(x_k) \geq \sum_{k=1}^{m} \alpha_k u(y_k) = U(Y).$$

12.26 Consider a symmetric sealed-bid auction with independent private values in which all the buyers are risk-seeking, and have the same strictly convex, differentiable, and monotonically increasing utility function.

We prove that the seller's expected revenue is lower in a sealed-bid first-price auction than in a sealed-bid second-price auction.

Note that in a sealed-bid second-price auction, the strategy $\beta(v) = v$ weakly dominates all other strategies, even if the buyers are risk-seeking (see Exercise 12.25). Therefore, the seller's expected revenue in a second-price auction with risk-seeking buyers is identical to his revenue in a second-price auction with risk-neutral buyers, which, by the Revenue Equivalence Theorem, is identical to the seller's expected revenues in a first-price auction with risk-neutral.

We turn to a symmetric sealed-bid first-price auction with risk-seeking buyers. Let γ be a monotonically increasing, differentiable, and symmetric equilibrium strategy in an auction with risk-seeking buyers satisfying $\gamma(0) = 0$. Let β be a monotonically increasing, differentiable, and symmetric equilibrium strategy in an auction with risk-neutral buyers satisfying $\beta(0) = 0$.

Note that the proof of Theorem 12.35 does not rely on the fact that U is strictly concave (and thus it also holds for risk-seeking and risk neutral utility function), hence γ and β satisfy all of the necessary conditions of Theorem 12.35 (excluding concavity) so

$$\gamma'(x) = (n-1)\tfrac{f(x)}{F(x)} \times \tfrac{U(x-\gamma(x))}{U'(x-\gamma(x))} \text{ and}$$
$$\beta'(x) = (n-1)\tfrac{f(x)}{F(x)} \times \tfrac{x-\beta(x)}{1}.$$

Since U is a strictly convex function, and $U(0) = 0$, it follows that $U'(v) > \frac{U(v)}{v}$ or equivalently, $v > \frac{U(v)}{U'(v)}$. Therefore,

$$\gamma'(x) = (n-1)\tfrac{f(x)}{F(x)} \times \tfrac{U(x-\gamma(x))}{U'(x-\gamma(x))} < (n-1)\tfrac{f(x)}{F(x)} \times (x-\beta(x)) = \beta'(x).$$

Hence, since $\gamma(0) = 0 = \beta(0)$, $\gamma(x) < \beta(x)$ for every $x > 0$ (as shown at the end of the proof of Theorem 12.36).

In conclusion, the bids of risk seeking buyers in such a symmetric sealed-bid first-price auction are lower than the bids of risk-neutral buyers, so the expected revenue of the seller is lower as well. On the other hand, as claimed above, the expected revenue of the seller in both first-price auction with risk-neutral buyer and second-price auction with risk-seeking buyers are identical. Hence, the seller's expected revenue is lower in a sealed-bid first-price auction than in a sealed-bid second-price auction.

12.27 Suppose that in a sealed-bid first-price auction there are n buyers whose private values are independent. Suppose further that the utility function of all buyers is $U(x) = x^c$.

(a) The utility function must be monotonic increasing function. That is, for every $x > 0$

$$U'(x) = cx^{c-1} > 0 \Rightarrow c > 0.$$

For a risk-averse buyer:

$$U''(x) = c(c-1)x^{c-2} < 0 \Rightarrow 0 < c < 1.$$

For a risk-neutral buyer:

$$U''(x) = c(c-1)x^{c-2} = 0 \Rightarrow c = 1.$$

For a risk-seeking buyer:

$$U''(x) = c(c-1)x^{c-2} > 0 \Rightarrow c > 1.$$

(b) By Theorem 12.35, since U is monotonically increasing and differentiable, then a monotonically increasing, differentiable, and symmetric equilibrium γ must satisfy the following differential equation:

$$\begin{aligned}\gamma'(x) &= (n-1)\tfrac{f(x)}{F(x)} \times \tfrac{U(x-\gamma(x))}{U'(x-\gamma(x))}\\ &= (n-1)\tfrac{f(x)}{F(x)} \times \tfrac{(x-\gamma(x))^c}{c(x-\gamma(x))^{c-1}}\\ &= \tfrac{n-1}{c}\tfrac{f(x)}{F(x)}(x-\gamma(x)) \end{aligned} \tag{109}$$

.

(c) Let

$$\gamma(v) = v - \frac{\int_0^v F^{\frac{n-1}{c}}(x)dx}{F^{\frac{n-1}{c}}(v)}.$$

be a strategy. Then, a strategy vector according to which all buyers follow γ is symmetric. We show that γ is also differentiable and monotonic increasing.

$$\begin{aligned}
\gamma'(v) &= 1 - \frac{F^{2\frac{n-1}{c}}(v) - \frac{n-1}{c} F^{\frac{n-1}{c}-1}(v) f(v) \int_0^v F^{\frac{n-1}{c}}(x)dx}{F^{2\frac{n-1}{c}}(v)} \\
&= \frac{n-1}{c}\frac{f(v)}{F(v)} \frac{\int_0^v F^{\frac{n-1}{c}}(x)dx}{F^{\frac{n-1}{c}}(v)} \\
&= \frac{n-1}{c}\frac{f(v)}{F(v)} \left(v - \left(v - \frac{\int_0^v F^{\frac{n-1}{c}}(x)dx}{F^{\frac{n-1}{c}}(v)} \right) \right) \\
&= \frac{n-1}{c}\frac{f(x)}{F(x)}(x - \gamma(x)).
\end{aligned}$$

In particular, γ is indeed differentiable and monotonic increasing which is the solution for Equation (109) and it is therefore a a symmetric equilibrium (Theorem 12.35).

(d) Suppose that F is the uniform distribution over $[0,1]$. Then

$$\int_0^v F^{\frac{n-1}{c}}(x)dx = \int_0^v x^{\frac{n-1}{c}}(x)dx = \frac{v^{\frac{n-1}{c}+1}}{\frac{n-1}{c}+1}$$

Hence,

$$\gamma(v) = v - \frac{v^{\frac{n-1}{c}+1}}{(\frac{n-1}{c}+1)v^{\frac{n-1}{c}}} = \frac{n-1}{n-1+c}v.$$

(e) The symmetric equilibrium γ presented in previous item as a function of c, is monotonic decreasing function. Hence, (as claimed in Theorem 12.36 and Exercise 12.26) under this symmetric equilibrium, risk-averse buyers submit higher bids than risk-neutral buyers, and risk-seeking buyers submit lower bids than risk-neutral buyers.

(f) The seller's expected revenue as a function of c is

$$\mathbf{E} = n\int_0^1 F_Y(x)\gamma(x)dx = n\int_0^1 x^{n-1}\frac{n-1}{n-1+c}xdx = \frac{n(n-1)}{(n+1)(n-1+c)}.$$

12.28 **Proof of Theorem 12.42** Let $(N, (V_i, F_i)_{i\in N}, p, C)$ be a sealed-bid auction with risk-neutral buyers, then it can be presented as a selling mechanism $(\Theta_i, \widehat{q}_i, \widehat{\mu}_i)_{i\in N}$ for the selling problem $(N, (V_i, F_i))_{i\in N}$ as follows:

- $\Theta_i = [0, \bar{b}]$- is the space of messages (bids) that buyer i can send to the seller.

- $\widehat{q}_i = p_i$ is a function mapping each vector of messages to the probability that buyer i wins the object.
- $\widehat{\mu}_i = C_i$ is a function mapping every vector of messages to the payment that buyer i makes to the seller (whether or not he wins the object).

12.29 There can be more than one equilibrium for an incentive-compatible direct selling mechanism.
For instance, consider the incentive-compatible direct selling mechanism corresponds to a sealed-bid second-price auction with two buyers, whose private values are independent; buyer 1's private value is uniformly distributed over $[0,1]$, and buyer 2's private value is uniformly distributed over $[0,2]$. Then, as shown in Exercise 12.23, the mechanism is incentive-compatible and direct but it has infinitely many equilibria according to which every buyer whose private value is at most 1, reports his private value, whereas if the private value of buyer 2 is higher than 1, he may reports any value higher than 1.

12.30 **Proof of the revelation principle (Theorem 12.47):** let $(\Theta_i, \widehat{q}_i, \widehat{\mu}_i)_{i\in N}$ be a selling mechanism, and $\widehat{\beta}$ be an equilibrium of this mechanism. Let β^* be strategy vector in mechanism (q,μ) (defined by Equations (12.132)–(12.133)), when all buyers reveal their true values, .
Then, for each player $i \in N$, every $v \in V$, every private value $v_i' \in V_i$

$$\begin{aligned}
&u_i\left(v; \beta^*(v_1), ...\beta^*(v_{i-1}), v_i', \beta^*(v_{i+1}), ..., \beta^*(v_n)\right)\\
&= u_i\left(v; v_1, ...v_{i-1}, v_i', v_{i+1}, ..., v_n\right)\\
&= q_i\left(v_1, ..., v_{i-1}, v_i', v_{i+1}, ..., v_n\right) v_i - \mu_i\left(v_1, ..., v_{i-1}, v_i', v_{i+1}, ..., v_n\right)\\
&= \widehat{q}_i\left(\widehat{\beta}(v_1), ..., \widehat{\beta}(v_{i-1}), \widehat{\beta}(v_i'), \widehat{\beta}(v_{i+1}), ...\widehat{\beta}(v_n)\right) v_i\\
&\qquad - \widehat{\mu}_i\left(\widehat{\beta}(v_1), ..., \widehat{\beta}(v_{i-1}), \widehat{\beta}(v_i'), \widehat{\beta}(v_{i+1}), ...\widehat{\beta}(v_n)\right)\\
&= \widehat{u}_i\left(v; \widehat{\beta}(v_1), ..., \widehat{\beta}(v_{i-1}), \widehat{\beta}(v_i'), \widehat{\beta}(v_{i+1}), ...\widehat{\beta}(v_n)\right).
\end{aligned}$$

Hence, for each buyer $i \in N$, and pair of private values $v_i, v_i' \in V_i$, $\widehat{u}_i(\widehat{\beta}_i(v_i'), \widehat{\beta}_{-i}; v_i) = u_i(v_i', \beta^*_{-i}; v_i)$. That is, the outcome of the given mechanism under $\widehat{\beta}$ is identical to the outcome of (q,μ) under β^*. Furthermore, since $\widehat{\beta}$ is an equilibrium in $(\Theta_i, \widehat{q}_i, \widehat{\mu}_i)_{i\in N}$ then β^* is an equilibrium in (q,μ).

12.31 Let (p, C) and $(\widetilde{p}, \widetilde{C})$ be two symmetric sealed-bid auctions with independent private values defined over the same set N of risk-neutral buyers. Suppose that in both auctions the winner of the auction is the

buyer who submits the highest bid and a buyer who submits a bid of 0 pays nothing. Let β and $\widetilde{\beta}$ be symmetric and monotonically increasing equilibrium strategies in (p, C) and in $(\widetilde{p}, \widetilde{C})$, respectively (for the same distributions of private values). We prove that the seller's expected revenue is the same under both equilibria, and the buyer's expected profit given his private value is also identical in both auctions.
By Theorem 12.42 (Exercise 12.28) and the revelation principle (Exercise 12.30), both, (p, C) and $(\widetilde{p}, \widetilde{C})$, can be presented as incentive-compatible direct selling mechanisms (q, μ) and $(\widetilde{q}, \widetilde{\mu})$, respectively. Since both, (p, C) and $(\widetilde{p}, \widetilde{C})$, defined over the same set N of risk-neutral buyers for the same distributions of private values, (q, μ) and $(\widetilde{q}, \widetilde{\mu})$ are defined over the same selling problem $(N, (V_i, f_i)_{i \in N})$.
Furthermore, the rule determining the winner in both auctions is identical (the buyer who submits the highest bid) and thus $q = \widetilde{q}$. In addition, following the assumption mentioned above, a buyer who reports that his private value is 0 pays the same sum in both auctions and hence he pays the same sum in both mechanisms.
In conclusion, by Corollary 12.50, at the truth-telling equilibrium β^* the expected profit of each buyer is the same in both mechanisms. Therefore, by the revelation principle (Theorem 12.47) the buyer's expected profit given his private value is also identical in both auctions, and thus the seller's expected revenue is the same under both equilibria.

12.32 Consider the following cumulative distribution function

$$F_i(v_i) = (v_i)^k, \quad 0 \leq v \leq 1,$$

where $k \in \mathbf{N}$. Then

$$c_i(v_i) = v_i - \tfrac{1-F_i(v_i)}{f_i(v_i)} = v_i - \tfrac{1-(v_i)^k}{k(v_i)^{k-1}} = \tfrac{k+1}{k} v_i - \tfrac{1}{k(v_i)^{k-1}}$$

Differentiate c_i:

$$c_i'(v_i) = \tfrac{k+1}{k} + \tfrac{k-1}{k(v_i)^k}$$

hence c_i is monotonic increasing for every $v_i > 0$ if and only if $k \geq 1$.

12.33 Suppose that V_i is distributed according to the exponential distribution with parameter $\lambda > 0$.
Then

$$c_i(v_i) = v_i - \tfrac{1-F_i(v_i)}{f_i(v_i)} = v_i - \tfrac{1-(1-e^{-\lambda v_i})}{\lambda e^{-\lambda v_i}} = v_i - \tfrac{1}{\lambda} \quad \forall v_i > 0,$$

so c_i is monotonic increasing for every $\lambda > 0$.

12.34 For each of the following auctions and their respective equilibria β, we construct an incentive-compatible direct selling mechanism whose truth-telling equilibrium β^* is equivalent to the equilibrium β.

(a) Sealed-bid second-price auction and an equilibrium $\beta = (\beta_i)_{i\in N}$, where $\beta_i(v_i) = v_i$ for each buyer i and each $v_i \in V_i$:

$$\Theta_i = V_i$$
$$q_i(\theta) = \begin{cases} \frac{1}{|\{k:\theta_k=\max_{j\in N}\theta_j\}|} & \theta_i = \max_{j\in N}\theta_j \\ 0 & \theta_i < \max_{j\in N}\theta_j \end{cases}$$
$$\mu_i(\theta) = \begin{cases} \frac{\max_{j\neq k}\{\min\{\theta_j,\theta_k\}\}}{|\{k:\theta_k=\max_{j\in N}\theta_j\}|} & \theta_i = \max_{j\in N}\theta_j \\ 0 & \theta_i < \max_{j\in N}\theta_j \end{cases}$$

(b) Sealed-bid second-price auction in which $V_i = [0,1]$ for every buyer i with an equilibrium $\beta = (\beta_i)_{\in N}$ where $\beta_1(v_1) = 1$ for all $v_1 \in [0,1]$ and $\beta_i(v_i) = 0$ for each buyer $i \neq 1$ and all $v_1 \in [0,1]$:

$$\Theta_i = [0,1]$$
$$q_i(\theta) = \begin{cases} 1 & i = 1 \\ 0 & i \neq 1 \end{cases}$$
$$\mu_i(\theta) = 0.$$

(c) Sealed-bid first-price auction with $n = 2$, and the private values of the buyers are independent and uniformly distributed over $[0,1]$ with an equilibrium $\beta = (\beta_i)_{i=1,2}$, given by $\beta_i(v_i) = \frac{v_i}{2}$:

$$\Theta_i = [0,1]$$
$$q_i(\theta) = \begin{cases} \frac{1}{|\{k:\theta_k=\max_{j\in N}\theta_j\}|} & \theta_i = \max_{j\in N}\theta_j \\ 0 & \theta_i < \max_{j\in N}\theta_j \end{cases}$$
$$\mu_i(\theta) = \begin{cases} \frac{\max_{j\in N}\theta_j}{2|\{k:\theta_k=\max_{j\in N}\theta_j\}|} & \theta_i = \max_{j\in N}\theta_j \\ 0 & \theta_i < \max_{j\in N}\theta_j \end{cases}$$

(d) Sealed-bid first-price auction with a reserve price ρ and two buyers whose private values are independent and uniformly distributed over the interval $[0,1]$ with an equilibrium given in Example 12.34:

$$\Theta_i = [0,1]$$
$$q_i(\theta) = \begin{cases} \frac{1}{|\{k:\theta_k = \max_{j\in N}\theta_j\}|} & \theta_i = \max_{j\in N}\theta_j \geq \rho \\ 0 & \text{otherwise} \end{cases}$$
$$\mu_i(\theta) = \begin{cases} \frac{\frac{\max_{j\in N}\theta_j}{2} + \frac{\rho^2}{2\max_{j\in N}\theta_j}}{|\{k:\theta_k = \max_{j\in N}\theta_j\}|} & \theta_i = \max_{j\in N}\theta_j \geq \rho \\ 0 & \theta_i < \max_{j\in N}\theta_j \end{cases}$$

12.35 Suppose that there are n buyers whose private values are independent and uniformly distributed over $[0,1]$.

(a) By Exercise 12.32, $c_i(v_i) = 2v_i - 1$ for every $i \in N$. c_i is monotonic increasing, hence, using Theorem 12.59, the following mechanism (q^*, μ^*) defined below maximizes the seller's expected revenue within all incentive-compatible, individually rational direct selling mechanisms:

$$q_i^*(v) = \begin{cases} 0 & c_i(v_i) \leq 0, \\ 0 & c_i(v_i) < \max_{j\in N} c_j(v_j), \\ \frac{1}{|\{l:c_l(v_l) = \max_{j\in N} c_j(v_j)\}|} & c_i(v_i) = \max_{j\in N} c_j(v_j) > 0. \end{cases}$$
$$= \begin{cases} 0 & v_i \leq \frac{1}{2}, \\ 0 & v_i < \max_{j\in N} v_j, \\ \frac{1}{|\{l:v_l = \max_{j\in N} v_j\}|} & v_i = \max_{j\in N} v_j > 0. \end{cases}$$
$$\mu_i^*(v) = v_i q_i^*(v) - \int_0^{v_i} q_i^*(t_i, v_{-i}) d_{t_i}$$
$$= \begin{cases} v_i - (v_i - \max\{\max_{j\neq i} v_j, \frac{1}{2}\}) & v_i > \max\{\frac{1}{2}, \max_{j\neq i} v_j\} \\ 0 & \text{otherwise.} \end{cases}$$

That is, the mechanism that corresponding a sealed-bid second-price auction with a reserve price $\rho = \frac{1}{2}$.

(b) The probability that some buyer wins the object, assuming that each buyer reports his true private value, is $1 - (\frac{1}{2})^n$. Hence, each buyer's probability of winning the object is $\frac{1-(\frac{1}{2})^n}{n}$.

(c) The seller's expected revenue in this case:

$$\begin{aligned}
\pi &= n\left(\int_{0.5}^{1}\left(\tfrac{1}{2}\mathbf{P}(Y \le \tfrac{1}{2}) + E[Y|0.5 < Y \le v]\mathbf{P}(\tfrac{1}{2} < Y \le v)\right) f(v)dv\right) \\
&= n\left(\int_{0.5}^{1}\left(\tfrac{1}{2}\cdot(\tfrac{1}{2})^{n-1} + \int_{0.5}^{v} y\cdot(n-1)y^{n-2}dy\right) dv\right) \\
&= n\left(\int_{0.5}^{1}\left((\tfrac{1}{2})^{n} + \tfrac{n-1}{n}(v^n - (\tfrac{1}{2})^n)\right) dv\right) \\
&= n\left(\int_{0.5}^{1}\left(\tfrac{n-1}{n}v^n + \tfrac{1}{n}(\tfrac{1}{2})^n\right) dv\right) \\
&= \tfrac{n-1}{n+1}(1-(\tfrac{1}{2})^{n+1}) + (\tfrac{1}{2})^{n+1} \\
&= \tfrac{n-1}{n+1} + \tfrac{1}{n+1}(\tfrac{1}{2})^n.
\end{aligned}$$

12.36 Suppose that there are two buyers, and their private values V_1 and V_2 are independent; V_1 is uniformly distributed over $[0,2]$ and V_2 is uniformly distributed over $[0,3]$.

(a) $c_1(v_1) = v_1 - \frac{1-\frac{v_1}{2}}{\frac{1}{2}} = 2v_1 - 2$ and $c_2(v_2) = 2v_2 - 3$. c_i is monotonic increasing, hence, using Theorem 12.59, the following mechanism (q^*, μ^*) defined below maximizes the seller's expected revenue within all incentive-compatible, individually rational direct selling mechanisms:

$$q_1^*(v) = \begin{cases} 0 & v_1 \le 1 \text{ or } v_1 + \frac{1}{2} < v_2, \\ \frac{1}{2} & v_1 > 1 \text{ and } v_1 + \frac{1}{2} = v_2 \\ 1 & v_1 > 1 \text{ and } v_1 + \frac{1}{2} > v_2. \end{cases}$$

$$q_2^*(v) = \begin{cases} 0 & v_2 \le \frac{3}{2} \text{ or } v_1 + \frac{1}{2} > v_2, \\ \frac{1}{2} & v_2 > \frac{3}{2} \text{ and } v_1 + \frac{1}{2} = v_2 \\ 1 & v_2 > \frac{3}{3} \text{ and } v_1 + \frac{1}{2} < v_2. \end{cases}$$

$$\mu_1^*(v) = \begin{cases} 0 & v_1 \le 1 \text{ or } v_1 + \frac{1}{2} < v_2 \\ 1 & v_1 > 1 \text{ and } v_2 \le \frac{3}{2} \\ v_2 - \frac{1}{2} & v_1 > 1 \text{ and } v_1 + \frac{1}{2} > v_2. \end{cases}$$

$$\mu_2^*(v) = \begin{cases} 0 & v_2 \le \frac{3}{2} \text{ or } v_1 + \frac{1}{2} > v_2 \\ \frac{3}{2} & v_2 > \frac{3}{2} \text{ and } v_1 \le 1 \\ v_1 + \frac{1}{2} & v_2 > \frac{3}{2} \text{ and } v_1 + \frac{1}{2} < v_2. \end{cases}$$

(b) Assume that each buyer reports his true private value, then

$$\begin{aligned}\mathbf{P}(\text{buyer 1 wins}) &= \int_1^2 \mathbf{P}(V_2 < v_1 + \tfrac{1}{2}) f_{V_1}(v_1) dv_1 \\ &= \int_1^2 \tfrac{v_1 + \frac{1}{2}}{3} \cdot \tfrac{1}{2} dv_1 = \tfrac{1}{3}.\end{aligned}$$

$$\begin{aligned}\mathbf{P}(\text{buyer 2 wins}) &= \int_{1.5}^{2.5} \mathbf{P}(V_1 < v_2 - \tfrac{1}{2}) f_{V_2}(v_2) dv_2 + \int_{2.5}^{3} 1 \cdot f_{V_2}(v_2) dv_2 \\ &= \int_{1.5}^{2.5} \tfrac{v_2 - \frac{1}{2}}{2} \cdot \tfrac{1}{3} dv_2 + \int_{2.5}^{3} 1 \cdot \tfrac{1}{3} dv_2 = \tfrac{5}{12}.\end{aligned}$$

(c) The expected payments of the buyers:

$$\begin{aligned}\mu_1 &= \int_1^2 \int_0^3 \mu_1^*(v) f_V(v) dv_2 dv_1 \\ &= \tfrac{1}{6} \int_1^2 \left(\int_0^{1.5} 1 dv_2 + \int_{1.5}^{v_1 + 0.5} v_2 - \tfrac{1}{2} dv_2 \right) dv_1 \\ &= \tfrac{1}{12} \int_1^2 v_1^2 + 2 dv_1 = \tfrac{13}{36}. \\ \mu_2 &= \int_{1.5}^3 \int_0^2 \mu_2^*(v) f_V(v) dv_1 dv_2 \\ &= \tfrac{1}{6} \left(\int_{1.5}^{2.5} \left(\int_0^1 1.5 dv_2 + \int_1^{v_2 - 0.5} v_1 + \tfrac{1}{2} dv_2 \right) dv_2 \right. \\ &\quad \left. + \int_{2.5}^3 \left(\int_0^1 1.5 dv_2 + \int_1^2 v_1 + \tfrac{1}{2} dv_2 \right) dv_2 \right) \\ &= \tfrac{1}{12} \left(\int_{1.5}^{2.5} v_1^2 + \tfrac{3}{4} dv_1 + \tfrac{1}{12} \int_{1.5}^{2.5} 7 dv_1 \right) = \tfrac{25}{36}.\end{aligned}$$

The seller's expected revenue in this case $\pi = \frac{13}{36} + \frac{25}{36} = \frac{38}{36}$.

12.37 Consider a symmetric selling problem with independent private values and a symmetric and monotonic increasing equilibrium. Then, using Corollary 12.50, the given auction methods can be partitioned into three groups, each of which contains methods whose symmetric equilibrium yields the same expected revenue for the seller.
The first group includes all methods in which the winning buyer is the one who submits the highest bid and a buyer pays 0 if he

reports that his private value is 0. The group includes sealed-bid first-price auction, sealed-bid second-price auction, sealed-bid third-price auction and sealed-bid all-pay auction.
The second group includes all methods in which if there is a buyer that submits a bid of at least the reverse price then the winning buyer is the one who submits the highest bid (and a buyer pays 0 if he reports that his private value is 0), this group includes sealed-bid first-price auction with a reserve price ρ, sealed-bid second-price auction with a reserve price ρ and sealed-bid all-pay auction with a reserve price ρ.
The third group includes sealed-bid first-price auction with an entry fee λ and sealed-bid second-price auction with an entry fee λ.

12.38 There exist methods in which a buyer's submitted bid in an equilibrium may be greater than his private value, e.g., sealed-bid third-price auction (see Exercise 12.19).

12.39 Suppose that there are n buyers participating in an auction, where the private values $V_1, V_2, ..., V_n$ are independent and identically distributed, with the cumulative distribution function $F_i(v_i) = (v_i)^2$ for $v_i \in [0,1]$.
The function $c_i(v_i) = v_i - \frac{1-F_i(v_i)}{f_i(v_i)} = \frac{3v_i}{2} - \frac{1}{2v_i}$ is monotonically non-decreasing. Therefore, by Theorem 12.59, the mechanism (q^*, μ^*) defined below maximizes the seller's expected revenue.

$$\begin{aligned}
q_i^*(v) &= \begin{cases} 0 & v_i \leq \frac{1}{\sqrt{3}} \text{ or } v_i < \max_j v_j \\ \frac{1}{|\{l: v_l = \max_j v_j\}|} & v_i > \frac{1}{\sqrt{3}} \text{ and } v_i = \max_j v_j. \end{cases} \\
\mu_i^*(v) &= v_i q_i^*(v) - \int_0^{v_i} q_i^*(t_i, v_{-i}) dt_i \\
&= \begin{cases} 0 & v_i \leq \frac{1}{\sqrt{3}} \text{ or } v_i < \max_j v_j \\ \max\{\frac{1}{\sqrt{3}}, \max_{j\neq i} v_j\} & v_i > \frac{1}{\sqrt{3}} \text{ and } v_i = \max_j v_j. \end{cases}
\end{aligned}$$

In other words, the auction that maximizes the seller's revenue is a sealed-bid second-price auction with a reserve price $\rho^* = \frac{1}{\sqrt{3}}$.
The seller's expected revenue in that auction is a sealed-bid second-price auction with a reserve price $\rho = \frac{1}{\sqrt{3}}$.
The expected payment of buyer i with $v_i \leq \frac{1}{\sqrt{3}}$ is 0. To calculate the expected payment of buyer i with $v_i > \frac{1}{\sqrt{3}}$, denote $Y = \max_{j\neq i} V_j$. As shown before, for every $y \in [0,1]$, $F_Y(y) = (y^2)^{n-1} = y^{2n-2}$ and

$f_Y(y) = (2n-2)y^{2n-3}$. Therefore,

$$\begin{aligned}\mu_i^*(v_i) &= \tfrac{1}{\sqrt{3}}F_Y(\tfrac{1}{\sqrt{3}}) + \mathbf{P}(\tfrac{1}{\sqrt{3}} < Y \le v_i)E[Y|\tfrac{1}{\sqrt{3}} < Y \le v_i] \\ &= \tfrac{1}{\sqrt{3}} \cdot (\tfrac{1}{3})^{n-1} + \int_{\frac{1}{\sqrt{3}}}^{v_i} (2n-2)y^{2n-2}dy \\ &= \tfrac{1}{\sqrt{3}} \cdot (\tfrac{1}{3})^{n-1} + \tfrac{2n-2}{2n-1}v_i^{2n-1} - \tfrac{2n-2}{2n-1}(\tfrac{1}{\sqrt{3}})^{2n-1} \\ &= \tfrac{2n-2}{2n-1}v_i^{2n-1} + \tfrac{1}{2n-1}(\tfrac{1}{\sqrt{3}})^{2n-1}.\end{aligned}$$

Hence, the expected revenue of the seller is

$$\begin{aligned}\pi &= n\int_{\frac{1}{\sqrt{3}}}^{1} \mu_i^*(v_i)dv_i = n\int_{\frac{1}{\sqrt{3}}}^{1} \tfrac{2n-2}{2n-1}v_i^{2n-1} + \tfrac{1}{2n-1}(\tfrac{1}{\sqrt{3}})^{2n-1}dv_i \\ &= \tfrac{n-1}{2n-1} - \tfrac{n-1}{2n-1}(\tfrac{1}{3})^n + \tfrac{n}{2n-1}(\tfrac{1}{\sqrt{3}})^{2n-1} - \tfrac{n}{2n-1}(\tfrac{1}{3})^n \\ &= \tfrac{n-1}{2n-1} - (\tfrac{1}{3})^n + \tfrac{n}{2n-1}(\tfrac{1}{\sqrt{3}})^{2n-1}.\end{aligned}$$

12.40 Suppose there are n buyers participating in an auction, where the private values $V_1, V_2, ..., V_n$ are independent and identically distributed, with the cumulative distribution function $F_i(v_i) = \frac{1}{2}v_i + \frac{1}{2}(v_i)^2$, $v_i \in [0,1]$.

(a) The function

$$c_i(v_i) = v_i - \frac{1-\frac{1}{2}v_i-\frac{1}{2}(v_i)^2}{\frac{1}{2}+v_i} = \frac{3v_i^2+2v_i-2}{2v_i+1}$$

is monotonic nondecreasing. Hence, and since the private values of the buyers are independent and identically distributed, Corollary 12.61 can be implemented for this case, so the incentive compatible direct selling mechanism that maximizes the seller's expected revenue is a sealed-bid second-price auction with a reserve price $\rho = \frac{\sqrt{7}-2}{3}$ (that is obtain by solving the equation $c_i(v_i) = 0$).

(b) Let $Y = \max_{j \neq i} V_i$. The seller's expected revenue in this auction is $\pi = n\int_{\frac{\sqrt{7}-2}{3}}^{1} \mu_i^*(v_i)dv_i$ where

$$\mu_i^*(v_i) = \tfrac{\sqrt{7}-2}{3}F_Y(\tfrac{\sqrt{7}-2}{3}) + \mathbf{P}(\tfrac{\sqrt{7}-2}{3} < Y \le v_i)E[Y|\tfrac{\sqrt{7}-2}{3} < Y \le v_i]$$

(c) The seller's expected revenue monotonically increasing as the number of buyers in the auction increases. Indeed, as the number of buyers in the auction increases then the maximal value among all of the buyers except for buyer i increases, and thus F_Y, $\mathbf{P}(\frac{\sqrt{7}-2}{3} < Y \le v_i)$ and $E[Y|\frac{\sqrt{7}-2}{3} < Y \le v_i]$ increase as well.

12.41 Suppose there are n buyers participating in an auction, where the private values $V_1, V_2, ..., V_n$ are independent and for each $i \in \{1, 2, ..., n\}$, v_i is uniformly distributed over $[0, \bar{v}_i]$. Suppose further that $\bar{v}_1 < \bar{v}_2 < ... < \bar{v}_n$.

(a) The function

$$c_i(v_i) = v_i - \frac{1 - \frac{v_i}{\bar{v}_i}}{\frac{1}{\bar{v}_i}} = 2v_i - \bar{v}_i$$

is monotonic nondecreasing. Hence, by Theorem 12.59, the mechanism (q^*, μ^*) defined below maximizes the seller's expected revenue.

$$q_i^*(v) = \begin{cases} 0 & v_i \le \frac{\bar{v}_i}{2} \text{ or } 2v_i - \bar{v}_i < \max_j 2v_j - \bar{v}_j \\ \frac{1}{|\{l: 2v_l - \bar{v}_l = \max_j 2v_j - \bar{v}_j|} & v_i > \frac{\bar{v}_i}{2} \text{ and } 2v_i - \bar{v}_i = \max_{j \ne i} 2v_j - \bar{v}_j. \end{cases}$$

$$\begin{aligned} \mu_i^*(v) &= v_i q_i^*(v) - \int_0^{v_i} q_i^*(t_i, v_{-i}) dt_i \\ &= \begin{cases} 0 & v_i \le \frac{\bar{v}_i}{2} \text{ or } 2v_i - \bar{v}_i < \max_j 2v_j - \bar{v}_j \\ \frac{\bar{v}_i}{2} + \max\{0, \max_{j \ne i} v_j - \frac{\bar{v}_j}{2}\} & v_i > \frac{\bar{v}_i}{2} \text{ and } 2v_i - \bar{v}_i = \max_j 2v_j - \bar{v}_j. \end{cases} \end{aligned}$$

(b) Let $Y = \max_{j \ne i} 2V_i - \bar{v}_i$.

$$F_Y(y) = \prod_{j \ne i} \mathbf{P}(2V_j - \bar{v}_j \le y) = \prod_{\{j \ne i: \bar{v}_j > y\}} \left(\frac{y}{2\bar{v}_j} + \frac{1}{2} \right).$$

The payment of buyer i with $v_i > \frac{\bar{v}_i}{2}$ is

$$\begin{aligned} \mu_i^*(v_i) &= \frac{\bar{v}_i}{2} F_Y(0) + \int_0^{2v_i - \bar{v}_i} y f_Y(y) dy \\ &= (\tfrac{1}{2})^n \bar{v}_i + \int_0^{2v_i - \bar{v}_i} y f_Y(y) dy. \end{aligned}$$

The seller's expected revenue in this auction is $\pi = n \int_{\frac{\bar{v}_i}{2}}^{\bar{v}_i} \mu_i^*(v_i) dv_i$.

(c) The probability that buyer n wins the object under this mechanism is

$$\begin{aligned} &\int_{\frac{\bar{v}_n}{2}}^{\bar{v}_n} \mathbf{P}(Y \le 2v_n - \bar{v}_n) f_{V_n}(v_n) dv_n \\ &= \int_{\frac{\bar{v}_n}{2}}^{\bar{v}_n} \prod_{\{j \ne i: \bar{v}_j > 2v_n - \bar{v}_n\}} \left(\frac{2v_n - \bar{v}_n}{2\bar{v}_j} + \frac{1}{2} \right) \frac{1}{\bar{v}_n} dv_n \\ &= \frac{1}{\bar{v}_n} \sum_{j=1}^{n} \left(\int_{\frac{\bar{v}_n + \bar{v}_{j-1}}{2}}^{\frac{\bar{v}_n + \bar{v}_j}{2}} \prod_{k=j}^{n} \left(\frac{2v_n - \bar{v}_n}{2\bar{v}_j} + \frac{1}{2} \right) dv_n \right) \end{aligned}$$

where $\bar{v}_0 = 0$.

12.42 Suppose that there are N potential buyers whose private values $V_1, V_2, ..., V_N$ are independent and have identical continuous distribution over $[0,1]$. Denote by F the common cumulative distribution function. The number of buyers participating in this auction is unknown to any of the participating buyers. Each buyer ascribes probability p_n to the event that there are n participating buyers, in addition to himself, where $\sum_{n=0}^{N-1} p_n = 1$. Note that each buyer has the same belief about the distribution of the number of buyers in the auction.

(a) Let β be a strategy vector according to which each buyer submits a bid equals his private value. Then β is a symmetric equilibrium of this situation when the selling takes place by way of a sealed-bid second-price auction. Indeed, for every number of players the suggested strategy weakly dominates all other strategies.

(b) Denote $G^{(n)}(z) = (F(z))^n$. Let $Y_1^{(n)}$ be the maximum of n independent random variables sharing the same cumulative distribution function F. Let $X(v)$ be the payment of a participating buyer with a private value v. The expected payment to the seller of a participating buyer (say buyer 1) with a private value v is

$$\begin{aligned}
e_1(v) = E[X(v)] = E[E[X(v)|N]] &= \sum_{n=0}^{N-1} p_n E[X(v)|n] \\
&= \sum_{n=0}^{N-1} p_n P(Y_1^{(n)} < v) E[Y_1^{(n)} | Y_1^{(n)} < v] \\
&= \sum_{n=0}^{N-1} p_n (F(v))^n E[Y_1^{(n)} | Y_1^{(n)} < v] \\
&= \sum_{n=0}^{N-1} p_n G^{(n)}(v) E[Y_1^{(n)} | Y_1^{(n)} < v].
\end{aligned}$$

(c) We prove the Revenue Equivalence Theorem in this case. Consider a symmetric sealed-bid auction with independent private values, and let β be a monotonically increasing symmetric equilibrium satisfying the assumptions that (a) the winner is the buyer submitting the highest bid, and (b) the expected payment of a buyer whose private value is 0, is 0.

Let $Y = \max_{j \in N \setminus \{1\}} V_i$. In particular, by the law of total probability,

$$F_Y(y) = \sum_{n=0}^{N-1} p_n \mathbf{P}(Y \leq y | N = n) = \sum_{n=0}^{N-1} p_n (F(y))^n = \sum_{n=0}^{n} p_n G^{(n)}(y).$$

The expected revenue of each participating buyer with a private value v_1 who plays as if his private value is z_1 is

$$u_1(\beta(z_1), \beta_{-1}; v_1) = v_1 F_Y(z_1) - e_1(z_1) \tag{110}$$

Note that Equation (110) is identical to Equation (12.49) in the proof of Theorem 12.23. Furthermore, β, e_1 and Y satisfying the same conditions as in Theorem 12.23 and thus e_1 is a differentiable function and its derivative is $e_1'(v_1) = v_1 f_Y(v_1)$ for every $v_1 \in V_1$. Since $e_1(0) = 0$, by integration, for every $v_1 \in V_1$ (including the extreme points) we get that the the expected payment of a buyer whose private value is v is given by

$$\begin{aligned}
e_1(v1) = e_1(0) + \int_0^{v_1} e_1'(y)dy &= \int_0^{v_1} y f_Y(y)dy \\
&= \int_0^{v_1} y \sum_{n=0}^{N-1} p_n n(F(v))^{n-1} f(y)dy \\
&= \sum_{n=0}^{N-1} p_n \int_0^{v_1} yn(F(v))^{n-1} f(y)dy \\
&= \sum_{n=0}^{N-1} p_n \mathbf{P}(Y_1^{(n)} < v_1) E[Y_1^{(n)} | Y_1^{(n)} < v_1] \\
&= \sum_{n=0}^{N-1} p_n G^{(n)}(v_1) E[Y_1^{(n)} | Y_1^{(n)} < v_1].
\end{aligned}$$

(d) Let β be a symmetric equilibrium strategy in a sealed-bid first-price auction. Then the expected payment of a buyer whose private value is v is given by $F_Y(v)\beta(v)$. Hence, using the Revenue Equivalence Theorem (that was proved in Part (c))

$$F_Y(v)\beta(v) = \sum_{n=0}^{N-1} p_n G^{(n)}(v) E[Y_1^{(n)} | Y_1^{(n)} < v]$$

In particular,

$$\begin{aligned}
\beta(v) &= \frac{\sum_{n=0}^{N-1} p_n G^{(n)}(v_1) E[Y_1^{(n)} | Y_1^{(n)} < v]}{F_Y(v)} \\
&= \frac{\sum_{n=0}^{N-1} p_n G^{(n)}(v) E[Y_1^{(n)} | Y_1^{(n)} < v]}{\sum_{n=0}^{N-1} p_n G^{(n)}(v)} \\
&= E[Y | Y < v].
\end{aligned}$$

(e) By Theorem 12.59, for every selling problem which includes n buyers, the optimal selling mechanism is a sealed-bid second-price auction with a reserve price. However, the reserve price depends on F and v only and not on the number of buyers, hence it is optimal for every number of identical buyers, in particular it is optimal for the current case.

Chapter 13: Repeated games

13.1 Let T -stage game with n players, where the number of actions of each player i in the base game is $|S_i| = k_i$. Denote $k = \prod_{i=1}^{n} k_i$. The number of pure strategies player i has in this game is

$$\prod_{t=1}^{T} k_i^{k^{t-1}} = k_i^{\sum_{t=1}^{T} k^{t-1}} = k_i^{\frac{k^T-1}{k-1}}.$$

13.2 (a) The base game for this situation is given by the next table where Artemis is the row player and Diana is the column player.

	N	L	S
N (Not go hunting)	0,0	0,0	0,0
L (Lead hunter)	0,0	0,0	2,1
S (Second hunter)	0,0	1,2	0,0

(b) The equilibria of the one-stage game are (N, N), (L, S), (S, L) and $([\frac{2}{3}(L), \frac{1}{3}(S)], [\frac{2}{3}(L), \frac{1}{3}(S)])$.

(c) There are continuum of equilibria in the 2 stage game.

- At the first stage, play the one-stage pure equilibrium (L, S) (or (S, L)). At the second stage, if no player deviates, play either (L, S) or (S, L). After each other history, play an arbitrary pure/mixed action vector.
  The expected payoff corresponding this strategy vector is at least 2 to each player, whereas if some player deviates then he gain 0 at the first stage and at most 2 at the second stage. Hence, no player has an incentive to deviate.
- At the first stage, play (L, S). At the second stage, if no player deviates, play (N, N); if only Diana deviates, play any pure/mixed action vector in which Artemis plays S with probability $\frac{1}{2}$ at most; and after each other history, play an arbitrary pure/mixed action vector.
- At the first stage, play (S, L). At the second stage, if no player deviates, play (N, N); if only Artemis deviates, play any pure/mixed action vector in which Diana plays S with

probability $\frac{1}{2}$ at most; and after each other history, play an arbitrary pure/mixed action vector.

- At the first stage, play (L, S). At the second stage, if no player deviates, play $([\frac{2}{3}(L), \frac{1}{3}(S)], [\frac{2}{3}(L), \frac{1}{3}(S)])$; if only Diana deviates, play any pure/mixed action vector in which Artemis plays S with probability $\frac{5}{6}$ at most; and after each other history, play an arbitrary pure/mixed action vector.
- At the first stage, play (S, L). At the second stage, if no player deviates, play $([\frac{2}{3}(L), \frac{1}{3}(S)], [\frac{2}{3}(L), \frac{1}{3}(S)])$; if only Artemis deviates, play any pure/mixed action vector in which Diana plays S with probability $\frac{5}{6}$ at most; and after each other history, play an arbitrary pure/mixed action vector.
- At the first stage, play (N, N). At the second stage, if at most one player deviates at the first stage, play (N, N); and after each other history, play an arbitrary pure/mixed action vector.
- At the first stage, play (N, N). At the second stage: if no player deviates at the first stage, play (S, L); if only Artemis deviates then play any pure/mixed action vector in which Diana plays S with probability at most 0.5; and after each other history, play an arbitrary pure/mixed action vector.
- At the first stage, play (N, N). At the second stage, if no player deviates at the first stage, play (L, S); if only Diana deviates then play any pure/mixed action vector in which Artemis plays S with probability at most 0.5; and after each other history, play an arbitrary pure/mixed action vector.
- At the first stage, play (N, N). At the second stage, if no player deviates at the first stage, play $([\frac{2}{3}(L), \frac{1}{3}(S)], [\frac{2}{3}(L), \frac{1}{3}(S)])$.; if only one of the players deviates then play either (N, N)or the one-stage mixed equilibrium; and after each other history, play an arbitrary pure/mixed action vector.
- At the first stage, play the mixed equilibrium in the one-stage game, $([\frac{2}{3}(L), \frac{1}{3}(S)], [\frac{2}{3}(L), \frac{1}{3}(S)])$. At the second stage, if no player plays N at the first stage, play (S, L); if only Artemis plays N then play any pure/mixed action vector in which Diana plays S with probability at most $\frac{5}{6}$; and after each other history, play an arbitrary pure/mixed action vector.
- At the first stage, play the mixed equilibrium in the one-stage game, $([\frac{2}{3}(L), \frac{1}{3}(S)], [\frac{2}{3}(L), \frac{1}{3}(S)])$. At the second stage, if no player plays N at the first stage, play (L, S); if only Diana plays N then play any pure/mixed action vector in which Artemis plays S with probability at most $\frac{5}{6}$; and after each other history, play an arbitrary pure/mixed action vector.
- At the first stage, play the mixed equilibrium in the one-stage

game, $([\frac{2}{3}(L), \frac{1}{3}(S)], [\frac{2}{3}(L), \frac{1}{3}(S)])$. At the second stage, if no player plays N at the first stage, play (N, N); if only one of the players plays N then play either (N, N) or the one stage mixed equilibrium; and after (N, N), play an arbitrary pure/mixed action vector.

- At both stages, play the mixed equilibrium in the one-stage game, $([\frac{2}{3}(L), \frac{1}{3}(S)], [\frac{2}{3}(L), \frac{1}{3}(S)])$. However, if at the first stage one of the players deviates by playing N then his opponent plays S with probability $\frac{4}{6}$ at most. If both player plays N at the first stage, play an arbitrary pure/mixed action vector.

13.3 (a) The base game for this situation is given by the next table where Mark is the row player and Jim is the column player.

	T	B
T (train)	4,3	-1,-1
B (bus)	-1,-2	4,4

(b) The equilibria of the one-stage game are (T, T), (B, B), and $([\frac{3}{5}(T), \frac{2}{5}(B)], [\frac{1}{2}(T), \frac{1}{2}(B)])$.

(c) There are a continuum of equilibria of the two-stage game.

- At the first stage, play either (T, T) or (B, B). At the second stage, if no player deviates, play one of the equilibria in the one-stage game. After each other history, play an arbitrary pure/mixed action vector.
  Note that, if one player deviates he loses 5 at the first stage and gain 2 at most at the second stage. In particular, no player has an incentive to deviate.
- At both stages (after each history), play $([\frac{3}{5}(T), \frac{2}{5}(B)], [\frac{1}{2}(T), \frac{1}{2}(B)])$.
- At the first stage, play $([\frac{3}{5}(T), \frac{2}{5}(B)], [\frac{1}{2}(T), \frac{1}{2}(B)])$. At the second stage, after each action that is taken by Mark at the first stage (no matter the action that is taken by Jim at the first stage), play one of the one-stage pure equilibria, (T, T) or (B, L) (this description define 4 equilibria of the two-stage game).

13.4 (a) The base game for this situation is given by the next tables where Andrew is the row player, Mike is the column player and Ron chooses a table. The action of each player is F for first and S for second.

	F	S
F	1,1,1	2,0,2
S	0.2.2	2,2,0

F

	F	S
F	2,2,0	0,2,2
S	2,0,2	1,1,1

S

(b) The equilibria of the one-stage game are (F, F, F), (S, S, S), and $([\frac{1}{2}(F), \frac{1}{2}(S)], [\frac{1}{2}(F), \frac{1}{2}(S)], [\frac{1}{2}(F), \frac{1}{2}(S)])$.

(c) There are a continuum of equilibria of the two-stage game.

- At the first stage, play either (F, F, F) or (S, S, S). At the second stage, if no player deviates, play one of the equilibria in the one-stage game. After each other history, play an arbitrary pure/mixed action vector.
  The equilibrium expected payoff for each player is at least 2. If one player deviates he loses at least 1 at the first stage and gain 2 at most at the second stage so his expected payoff is at most 1. In particular, no player has an incentive to deviate.
- At both stages (after each history), play the one-stage mixed equilibrium.
- At the first stage, play the one-stage mixed equilibrium. At the second stage, after each history play one of the on-stage pure equilibrium.

13.5 The claim is not correct. Consider the following two-player one-stage game.

	L	R
T	2,1	0,0
B	0,0	1,2

(T, L), (B, R) and $([\frac{2}{3}(T), \frac{1}{3}(B)], [\frac{1}{3}(L), \frac{2}{3}(R)])$ are the equilibria in this one stage game. However, the strategy τ in the two-stage game under which at the first stage the player follow $([\frac{2}{3}(T), \frac{1}{3}(B)], [\frac{1}{3}(L), \frac{2}{3}(R)])$, then after the histories (T, L) and (T, R) they play the pure equilibrium (T, L) whereas after the histories (B, L) and (B, R) they play the pure equilibrium (B, R) is not an equilibrium. Indeed, Player I is better off playing T at the first stage under which his expected payoff in the first stage is unchanged but his second-stage expected payoff increases.

13.6 We prove that at every equilibrium of the T -stage Prisoner's Dilemma, both players play D in every stage.
Assume by contradiction that there is an equilibrium σ^* at which, with positive probability, the players do not play (D, D) at some

stage. Let $t \in \{1, 2, ..., T\}$ be the last stage in which there is a history $h \in H(t-1)$ that is occurred with some positive probability such that either $\sigma_I^*(h) \neq D$ or $\sigma_{II}^*(h) \neq D$. Suppose without loss of generality that after this history, Player I does not play the pure action$D$ in stage t.

Let σ_I be a strategy of Player I under which he follows σ_I^* unless h is occurred which case he plays D from h onwards.

The stage payoffs under $(\sigma_I, \sigma_{II}^*)$ and under σ^* are identical unless h is occurred. Thus it is left to show that by deviating to σ_I, Player I increases his payoff in the subgame starting at h. At stage t the payoff is higher since $\sigma_I^*(h) \neq D$ is strictly dominated by $\sigma_I(h) = D$. Furthermore, since the players was supposed to play (D, D) (stage t is the last stage in which they may not play (D, D)), Player I's stage payoff was supposed to be 1. On the other hand, by playing D in these stages, Player I's stage payoff is either 1 or 4 (depending on whether Player II plays D or C). In either case, Player I cannot lose from stage t onwards. However, since h is occurred with positive probability, $U_i(\sigma^*) < U_i(\sigma_I, \sigma_{II}^*)$ which contradict to σ^* being an equilibrium.

13.7 Let $\Gamma = (N, (S_i)_{i \in N}, (u_i)_{i \in N}$ be a game in strategic form that has a unique equilibrium σ, and let Γ^T be the T-stage repeated game corresponding to Γ. We prove that the strategy vector under which at each stage the players play σ is a unique subgame perfect equilibrium of Γ^T by induction over T.

For $T = 1$, the game has a unique subgame, Γ itself, for which σ is a unique equilibrium and the claim holds.

Assume that the claim holds for every Γ^t where $t < T$. We prove that it also holds for Γ^T.

First note that every subgame perfect equilibrium of Γ^T induces a subgame perfect equilibrium for every subgame Γ^{T-1} that starts at the second stage onwards (after each history in $H(1)$). However, by the induction hypothesis, Γ^{T-1} has a unique subgame according to which at each stage the players play σ. Therefore, in every subgame perfect equilibrium of Γ^T, the player necessarily play σ after each history, at each stage from stage 2 onwards. In particular, the equilibrium payoff of each player i from the second stage onwards is some constant c_i independent of the actions that are taken at the first stage. It is left to prove that in the first stage of a subgame perfect equilibrium of Γ^T the players also play σ. Consider a strategic form game $\widehat{\Gamma} = (N, (S_i)_{i \in N}, (\widehat{u}_i)_{i \in N})$ where $\widehat{u}_i(s) = u_i(s) + c_i$ for every $i \in N$ and every $s \in S$. Γ and $\widehat{\Gamma}$ are strategically equivalent strategic-form games. Therefore, by Theorem 5.35, σ is also a unique equilibrium of $\widehat{\Gamma}$. In conclusion, by Exercise 7.11, in the first stage of a subgame perfect equilibrium of Γ^T the players also play σ, which completes

the proof.
It is possible for Γ^T to have an additional Nash equilibrium, e.g., Example 13.2.

13.8 We prove that the minmax value of each player i in the T-stage repeated game is equal to his minmax value

$$\bar{v}_i = \min_{\sigma'_{-i} \in \Sigma_{-i}} \max_{\sigma_i \in \Sigma_i} U_i(\sigma_i, \sigma'_{-i})$$

in the base game.
Let σ_{-i} be the punishment strategy against player i, that is, $\bar{v}_i = \max_{\sigma_i \in \Sigma_i} U_i(\sigma_i, \sigma'_{-i})$. Then by playing σ_{-i} at each stage, the players can guarantee that the stage payoffs of player i are at most $\bar{v}_i$ and so his average expected payoff in Γ^T is at most $\bar{v}_i$ as well. In conclusion, the minmax value of player i in Γ^T is at most $\bar{v}_i$.
On the other hand, for every history h let $\sigma'_{-i}(h)$ be the action that is played by the other players. Consider the strategy of player i at which he plays $\sigma_i(h)$ after a history h such that $U_i(\sigma_i(h), \sigma'_{-i}(h)) = \max_{\sigma_i \in \Sigma_i} U_i(\sigma_i, \sigma'_{-i}(h))$ then his stage payoff is at least $\bar{v}_i$ and thus his average expected payoff in Γ^T is at least $\bar{v}_i$. In particular, the minmax value of player i in Γ^T is at least $\bar{v}_i$ from which it follows that it equals $\bar{v}_i$.

13.9 The value of the repeated game in A is 2.5 where the optimal strategy of Player I is to always play $[\frac{1}{2}(T), \frac{1}{2}(B)]$, and the optimal strategy of Player II is to always play $[\frac{5}{8}(L), \frac{3}{8}(R)]$.
The value of the repeated game in B is 0.5 where the optimal strategy of Player I is to always play $[\frac{1}{2}(T), \frac{1}{2}(B)]$, and the optimal strategy of Player II is to always play $[\frac{1}{2}(L), \frac{1}{2}(R)]$.
Therefore, the game may be reduced into the following one-stage game

	L	R
T	1+2.5T	4+0.5T
B	2+0.5T	2.5T

from which one can deduce that the value of the repeated game is $\frac{6T^2-0.5T-8}{4T^2-5T}$. The optimal strategy of player I is to play $[\frac{2T-2}{4T-5}(T), \frac{2T-3}{4T-5}(B)]$ at the first stage and then to play $[\frac{1}{2}(T), \frac{1}{2}(B)]$. The optimal strategy of Player II is to play $[\frac{2T-4}{4T-5}(L), \frac{2T-1}{4T-5}(R)]$ at the first strategy then to play the optimal strategy in the repeated game. Note that as T goes to infinity, the value goes to 1.5.

13.10 (a) Let $\tau_i^1, ..., \tau_i^L$ be all the pure strategies of player i in the T-stage repeated game. We prove that for every behavior strategy τ_i of player i in the repeated game, there exist nonnegative numbers $\alpha_1, \dots, \alpha_L$, whose sum is 1, such that for each strategy vector τ_{-i} of the other players,

$$\gamma_i^T(\tau_i, \tau_{-i}) = \sum_{l=1}^{L} \alpha_l \gamma_i^T(\tau_i^l, \tau_{-i}).$$

Indeed, by Theorem 6.11, the behavior strategy τ_i is equivalent to a mixed strategy σ_i satisfying that for every $l \in \{1, 2, \dots, L\}$ the probability of playing τ_i^l is

$$\sigma(\tau_i^l) = \prod_{h \in \bigcup_{t=0}^{T-1} H(t)} \tau_i(\tau_i^l(h))$$

where $\tau_i(\tau_i^l(h))$ is the probability under τ_i that an action $\tau_i^l(h)$ is taking at h. Therefore,

$$\begin{aligned}
\gamma_i^T(\tau_i, \tau_{-i}) &= E_{\tau_i, \tau_{-i}}\left[\frac{1}{T}\sum_{t=1}^{T} u_i^T\right] = \frac{1}{T}\sum_{t=1}^{T} E_{\tau_i, \tau_{-i}}[u_i^t] \\
&= \frac{1}{T}\sum_{t=1}^{T} \sum_{(s^1, \dots, s^t) \in H(t)} \mathbf{P}_{\tau_i, \tau_{-i}}(s^1, \dots, s^t) u_i(s^t) \\
&= \frac{1}{T}\sum_{t=1}^{T} \sum_{(s^1, \dots, s^t) \in H(t)} \left(\sum_{l=1}^{L} \sigma(\tau_i^l) \mathbf{P}_{\tau_i^l, \tau_{-i}}(s^1, \dots, s^t)\right) u_i(s^t) \\
&= \sum_{l=1}^{L} \sigma(\tau_i^l) \left(\frac{1}{T}\sum_{t=1}^{T} \sum_{(s^1, \dots, s^t) \in H(t)} \mathbf{P}_{\tau_i^l, \tau_{-i}}(s^1, \dots, s^t) u_i(s^t)\right) \\
&= \sum_{l=1}^{L} \sigma(\tau_i^l) \gamma_i^T(\tau_i^l, \tau_{-i}).
\end{aligned}$$

(b) $\alpha_l = \sigma(\tau_i^l) = \prod_{h \in \bigcup_{t=0}^{T-1} H(t)} \tau_i(\tau_i^l(h))$ for every $l = 1, 2, \dots, L$.

13.11 (a) We use the cyclicality of the strategies, hence the limit of the average payoffs in the infinitely repeated game are

$$\frac{2}{6}(0,2) + \frac{2}{6}(4,0) + \frac{1}{6}(-1,3) + \frac{1}{6}(1,-1) = (1\tfrac{1}{3}, 1).$$

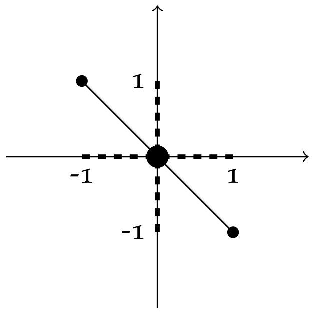
(b) The stage payoffs at the first stage do not effect the limit of the average payoffs which are

$$\frac{1}{2}\left(\frac{1}{4}(-1,3) + \frac{3}{4}(4,0)\right) + \frac{1}{2}(0,2) = (\tfrac{11}{8}, \tfrac{11}{8}).$$

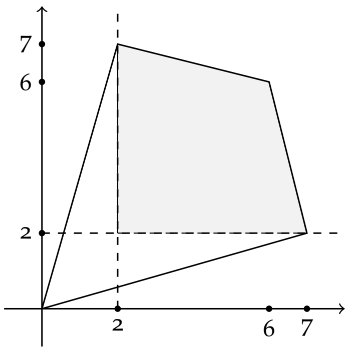
(c) At each stage, the probability that in the previous stage Player I played T is $\frac{2}{3}$ and the probability that in the previous stage he played B is $\frac{1}{3}$. Hence the limit of the average payoffs in the infinitely repeated game are

$$\frac{2}{3}\left(\frac{2}{3}(4,0)+\frac{1}{3}(0,2)\right)+\frac{1}{3}\left(\frac{2}{3}\cdot\frac{1}{4}(-1,3)+\frac{2}{3}\cdot\frac{3}{4}(4,0)+\frac{1}{3}\cdot\frac{1}{4}(1,-1)+\frac{1}{3}\cdot\frac{3}{4}(0,2)\right)$$
$$=\left(\frac{87}{36},\frac{2}{3}\right).$$

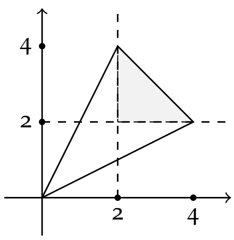
13.12 **Game A:** $F \cap V = \{(0,0)\}$:



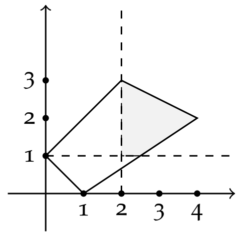
Game B: $F \cap V = \text{conv}\{(2,2),(7,2),(2,7),(6,6)\}$:



Game C: $F \cap V = \text{conv}\{(2,2),(4,2),(2,4)\}$:



Game D: $F \cap V = \text{conv}\{(1,2),(2,3),(4,2),(2.5,1)\}$:



13.13 The equilibrium corresponds to the payoff $(3, 3, \frac{1}{2})$ is

- Player I repeatedly cycles through the actions T, T, T, B, B, B, B, B.
- Player II repeatedly cycles through the actions L, R, R, L, L, R, R, R.
- Player III repeatedly cycles through the actions W, W, W, W, W, W, W, W.
- If Player I deviates, and fails to play the action he is supposed to play under this plan, the other player chooses (W, R) in every subsequent stage of the game, forever; if Player II deviates, and fails to play the action he is supposed to play under this plan, the other player chooses (B, R) in every subsequent stage of the game, forever; and if Player III deviates, and fails to play the action he is supposed to play under this plan, the other player chooses $([\frac{1}{2}(T), \frac{1}{2}(B)], [\frac{1}{2}(L), \frac{1}{2}(R)])$ in every subsequent stage of the game, forever.

13.14 Proof of Theorem 13.11:
Let $K \in N$ and let $x \in F$. We prove that there are nonnegative integers $(k_s)_{s\in S}$ summing to K satisfying

$$\left\| \sum_{s\in S} \tfrac{k_s}{K} u(s) - x \right\|_\infty \leq \tfrac{M\times|S|}{K}.$$

F is the convex hull of the one-stage payoff vectors. Since $x \in F$. There are nonnegative real numbers $(\lambda_s)_{s\in S}$ summing to 1 satisfying $\sum_{s\in S} \lambda_s u(s) = x$.
For every $s \in S$, let $\widehat{k}_s$ the highest integer such that

$$K\lambda_s - 1 \leq \widehat{k}_s \leq K\lambda_s.$$

Hence $\sum_{s\in S} \widehat{k}_s$ is an integer which satisfies

$$K - |S| = \sum_{s\in S}(K\lambda_s - 1) \leq \sum_{s\in S} \widehat{k}_s \leq \sum_{s\in S} K\lambda_s = K.$$

Let $(k_s)_{s\in S}$ be integers that are obtained by increasing $K - \sum_{s\in S} \widehat{k}_s$ arbitrary numbers of $(k_s)_{s\in S}$ by 1. Hence, $\sum_{s\in S} k_s = K$ and for every $s \in S$

$$\|\lambda_s - \tfrac{k_s}{K}\|_\infty \leq 1.$$

In particular, by the triangle inequality,

$$\begin{aligned}\left\|\sum_{s\in S} \tfrac{k_s}{K} u(s) - x\right\|_\infty &= \left\|\sum_{s\in S} \tfrac{k_s}{K} u(s) - \sum_{s\in S} \lambda_s u(s)\right\|_\infty \\ &\leq \sum_{s\in S} \left\|\tfrac{k_s}{K} - \lambda_s\right\|_\infty \|u(s)\|_\infty \\ &\leq \tfrac{|S|\times|M|}{K}.\end{aligned}$$

13.15 Suppose that each player i in a two-player zero-sum base game Γ has a unique optimal mixed strategy x_i. We prove that in the T-stage repeated game Γ^T, at each equilibrium in behavior strategies, at each stage each player implements the mixed strategy x_i.
A strategy vector in which both players following their optimal strategies is an equilibrium in the zero-sum base game. Hence, by Theorem 13.6, a behavior strategies at which each player i implements x_i at each stage is an equilibrium.
To prove the uniqueness, we assume by contradiction there is a last stage t and a history $h \in H(t-1)$ that is occurred with positive probability, such that at least one of the players, say Player I, does not implement x_I at h. Since x_I is a unique optimal strategy in the base game, Player II's best response leads to a stage payoff that is less than the value in the one stage game and the stage payoffs from stage $t+1$ onwards equal the value in the one stage game. Therefore, if Player II plays his best response at h then Player I is better off deviating to x_I at h and then continue implementing x_I from stage $t+1$ onwards. If Player II does not play his best response at h then Player II is better off deviating to his best response to Player I action at h and then continue implementing x_I from stage $t+1$ onwards, a contradiction.

13.16 The payoff $(5,6)$:
Note that

$$(5,6) = \frac{23}{30}(6,6) + \frac{1}{30}(0,0) + \frac{6}{30}(2,7).$$

An equilibrium would consists of 33,331 blocks of 30 stages each, plus a tail of 70 stages ($33,331 \times 30 + 70 = 1,000,000$):

- In each block of length 30, unless someone deviated before, Player I plays 23 times T, 1 times B, and then 6 times T; and Player II plays 23 times L, 7 times R.

- If no-one deviates, then after 33,331 blocks, Player I plays alternately T and B, and Player II plays alternately R and L.
- If someone deviates, both players switch to a punishing strategy: Player I plays B and Player II plays R for the rest of the game.

The payoff $(4,3)$:
Note that

$$(4,3) = \frac{22}{45}(7,2) + \frac{13}{45}(2,7) + \frac{10}{45}(0,0).$$

An equilibrium would consists of 22,219 blocks of 45 stages each, plus a tail of 145 stages ($22219 \times 45 + 145 = 1,000,000$):

- In each block of length 45, unless someone deviated before, Player I plays 22 times B, 13 times T, and then 10 times B; and Player II plays 22 times L, 23 times R.
- If no-one deviates, then after 22,219 blocks, Player I plays alternately T and B, and Player II plays alternately R and L.
- If someone deviates, both players switch to a punishing strategy: Player I plays B and Player II plays R for the rest of the game.

13.17 (a) Player II plays L in every stage: the limit of the average payoffs is $\frac{1}{2}(4,6) + \frac{1}{2}(7,3) = (\frac{11}{2}, \frac{9}{2}) = (5\frac{1}{2}, 4\frac{1}{2})$.

(b) Player II plays R in every stage: the limit of the average payoffs is $\frac{3}{4}(2,8) + \frac{1}{4}(1,0) = (\frac{7}{4}, 6) = (1\frac{3}{4}, 6)$.

(c) Player II plays L in odd stages, and R in even stages: the limit of the average payoffs is

$$\tfrac{1}{2}(\tfrac{1}{2}(2,8) + \tfrac{1}{2}(1,0)) + \tfrac{1}{2}(\tfrac{3}{4}(4,6) + \tfrac{1}{4}(7,3)) = (\tfrac{25}{8}, \tfrac{37}{8}) = (3\tfrac{1}{8}, 4\tfrac{5}{8}).$$

(d) Player II plays R in odd stages and L in even stages: the limit of the average payoffs is $(3\frac{1}{8}, 4\frac{5}{8})$.

13.18 Note that

$$(4\frac{1}{3}, 3) = \frac{37}{90}(6,6) + \frac{24}{90}(7,2) + \frac{29}{90}(0,0).$$

An equilibrium strategy plays in blocks of length 90: in each block Player I plays 37 times T and then 53 times B; Player II plays 61 times L and then 29 times R. Once someone deviates, both players switch to a punishment strategy forever: Player I plays B and Player II plays R.

13.19 **Game A:** $(2,1)$ is not an equilibrium payoff of the corresponding infinitely repeated game since it is not in the convex hull of the one-stage game payoffs.

Game B: $(2,1)$ is an equilibrium payoff of the corresponding infinitely repeated game. An equilibrium strategy plays in blocks of length 9: in each block Player I plays 6 times T and then 3 times B; Player II plays 4 times R, 2 times L and then 3 times R. Once someone deviates, both players switch to a punishment strategy forever: Player I plays $[0.75(T), 0.25(B)]$ and Player II plays $[0.8(L), 0.2(R)]$.

Game C: $(2,1)$ is not an equilibrium payoff of the corresponding infinitely repeated game since Player I's payoff is lower than his maxmin value ($2 < 2\frac{1}{5}$).

Game B: $(2,1)$ is an equilibrium payoff of the corresponding infinitely repeated game. An equilibrium strategy plays in blocks of length 3: in each block Player I plays B, B, T; Player II always plays L. Once someone deviates, both players switch to a punishment strategy forever: Player I plays T and Player II plays M.

13.20 Note that

$$(2, 3\tfrac{2}{3}) = \frac{1}{3}(1,4) + \frac{1}{3}(2,5) + \frac{1}{3}(3,2).$$

An equilibrium strategy plays in blocks of length 3: in each block Player I plays T, T, B; and Player II plays L, R, R. Once someone deviates, both players switch to a punishment strategy forever: Player I plays B and Player II plays L.

13.21 (a) The minmax value of Player I is $\frac{5}{3}$ which corresponding the punishment strategy in which Player III plays $[\frac{1}{3}(W), \frac{2}{3}(E)]$. The minmax value of Player II is 1 which corresponding the punishment strategy in which Player I plays T. The minmax value of Player III is 1 which corresponding the punishment strategy in which Player I plays T and Player II plays $[\frac{1}{2}(L), \frac{1}{2}(R)]$.

(b) Player II plays L only if Player I plays B and Player III plays E, since R weakly dominates L. However, if Player III plays E then Player I is better off plays T, hence there is no equilibrium in which Player I plays B and Player III plays E and thus in every equilibrium Player II plays R. Therefore, Player III is better of playing E which case Player I's best response is T. In conclusion, the game has a unique equilibrium (T, R, E) corresponding the payoff vector $(2, 1, 2)$. In particular, the game does not satisfy the conditions of Theorem 13.9.

(c) Note that

$$(2,1,2\frac{1}{2}) = \frac{1}{2}(2,1,2) + \frac{1}{4}(3,0,3) + \frac{1}{4}(1,2,3).$$

An equilibrium strategy whose payoff is within 0.1 of $(2,1,2\frac{1}{2})$ plays in 249 blocks of length 4: Player I plays B, B, T, T; Player II plays L, R, R, R; and Player III plays W, E, E, E. Then during the last 4 stages the players play the equilibrium (T, R, E). Once someone deviates, the players switch to a punishment strategies against the deviator as presented in Part (a) in every subsequent stage of the game.

No player can gain by deviating at the last 4 stages of the game in which the players follow the one-stage equilibrium. Player I may gain at most 1 by deviating at the second stage of a block but then during at least 6 stages he loses at least $6 \cdot \frac{1}{3} = 2$ so he has no incentive to deviate. Similarly, Player II may gain at most 2 by deviating at the first stage of a block but then at at the subsequent stage he loses 2 so he has no incentive to deviate.

(d) An equilibrium strategy in the infinitely repeated game whose payoff is $(2,1,2\frac{1}{2})$ plays in blocks of length 4: Player I plays T, T, B, B; Player II plays R, R, L, R; and Player III plays E, E, W, E. Once someone deviates the players switch to a punishment strategy against the deviator as presented in Part (a).

13.22 The maxmin value of Player I is $\frac{x}{1+x}$. The maxmin value of Player II is $\frac{3}{2}$. Hence, the set of equilibrium payoffs in the infinitely repeated game based on this game is $F \cap V = \{(\frac{x}{1+x}, \frac{3}{2})\}$ for every $x \in [\frac{1}{2}, 1]$, and $F \cap V = \text{conv}\{(\frac{x}{1+x}, \frac{3}{2}), (\frac{x}{1+x}, 2 - \frac{x}{1+x}), (0.5, 1.5)\}$ for every $x \in [0, \frac{1}{2}]$.

13.23 Consider a two-player repeated game based on this one-stage game

	L	R
T	-1,-1	0,0
B	1,1	-1,-1

Let τ be the strategy vector in the infinitely repeated game according to which at first stage they play the base game equilibrium (B, L), in the following 2^1 stages they play the base game equilibrium (T, R), then in the next 2^2 stages they play (B, L) again, then they play (T, R) for 2^3 stages and so on. By Theorem 13.6, this is indeed an equilibrium in the finite repeated game. However, $\lim \gamma^T(\tau)$ does not exist.

13.24 Proof of Theorem 13.17:

Let $x \in F \cap V$. By Exercise 13.14, for each $K \in N$, there are nonnegative integers $(k_s^K)_{s \in S}$ summing to K satisfying

$$\left\| \sum_{s \in S} \frac{k_s^K}{K} u(s) - x \right\|_\infty \leq \frac{M \times |S|}{K}.$$

We construct a strategy vector τ in which the players play in blocks, such that the length of the K-th block is K stages, and in the K-th block the players play each action vector $s \in S$ exactly k_s^K times, then the average payoff over the course of the K-th block is $\sum_{s \in S} \frac{k_s^K}{K} u(s)$, and the distance between this average payoff and x is at most $\frac{M \times |S|}{K}$. If at a certain stage, a player deviates from the action he is supposed to play at that stage, he is punished from the next stage onwards by a punishment strategy.

We next show that $\lim_{t \to \infty} \frac{1}{T} \sum_{t=1}^T E_\tau[u_i^t] = x$. Let $T \in \mathbf{N}$. Let $R \in \mathbf{N}$ be the maximal integer such that $T \geq \sum_{K=1}^R K = \frac{(1+R)R}{2}$. Denote $L = \frac{(1+R)R}{2}$. Note that $\frac{R}{T}$ and $\frac{T-L}{T}$ goes to zero as T goes to infinity.

$$\begin{aligned}
&\left\| \frac{1}{T} \sum_{t=1}^T E_\tau[u_i^\tau] - x \right\|_\infty = \left\| \frac{1}{T} \sum_{K=1}^R \sum_{s \in S} k_s^K u_i(s) + \frac{1}{T} \sum_{t=L+1}^T E_\tau[u_i^\tau] - x \right\|_\infty \\
&= \left\| \frac{1}{T} \sum_{K=1}^R K \left(\sum_{s \in S} \frac{k_s^K}{K} u_i(s) - x \right) + \frac{1}{T} \sum_{t=L+1}^T E_\tau[u_i^\tau] - \frac{T-L}{T} x \right\|_\infty \\
&\leq \frac{1}{T} \sum_{K=1}^R K \left\| \sum_{s \in S} \frac{k_s^K}{K} u_i(s) - x \right\|_\infty + \frac{1}{T} \sum_{t=L+1}^T \left\| E_\tau[u_i^\tau] \right\|_\infty + \frac{T-L}{T} \|x\|_\infty \\
&\leq \frac{R}{T} \times M \times |S| + 2 \frac{L-T}{T} \times M \xrightarrow[T \to \infty]{} 0.
\end{aligned}$$

In conclusion, the payoff in the infinitely repeated game under this strategy vector is x, whereas if a player deviates then his average payoff is reduced to his maxmin value. Therefore, x is an equilibrium payoff in the infinitely repeated game.

13.25 We prove that the strategy vector τ^* defined in Example 13.1 is an ϵ-equilibrium of the T-stage game, for every T sufficiently large.

Let $\epsilon > 0$. By Equation (13.43) the average payoff at this equilibrium converges to $(1, 2)$. Let T be an integer satisfying that $T > \frac{4}{\epsilon}$. If a player deviates at some stage and plays D, he receives 4 in that stage, but from that stage on, the other player plays D, and then the most that the deviating player receives in every stage is 1. Thus, his total payoff may increase by at most 4 and so his average payoff may increase by at most $\frac{4}{T} < \epsilon$.

13.26 Proof of Theorem 13.19:
Let $\epsilon > 0$ and let $x \in F \cap V$. Let $R_0 > \frac{12M|S|}{\epsilon}$, and set $T_0 = \frac{R_0(R_0+1)}{2}$. For each $K \in N$, let $(k_s^K)_{s\in S}$ be nonnegative integers summing to K satisfying

$$\left\|\sum_{s\in S} \frac{k_s^K}{K} u(s) - x\right\|_\infty \leq \frac{M\times|S|}{K}.$$

Let τ be the strategy vector that is played in blocks, such that the length of the K-th block is K stages, and in the K-th block the players play each action vector $s \in S$ exactly k_s^K times. If at a certain stage, a player deviates from the action he is supposed to play at that stage, he is punished from the next stage onwards by a punishment strategy. By Exercise 13.24, τ is an equilibrium in the infinitely repeated game with corresponding equilibrium payoff x.
Furthermore, for every $T > T_0$. Let $R \in \mathbf{N}$ be the maximal integer such that $T \geq \sum_{K=1}^{R} K = \frac{(1+R)R}{2}$. In particular, $R \geq R_0 > \frac{12M|S|}{\epsilon}$. Denote $L = \frac{(R+1)R}{2}$. Then (by Exercise 13.24)

$$\begin{aligned}
\left\|\frac{1}{T}\sum_{t=1}^{T} E_\tau[u_i^\tau] - x\right\|_\infty &\leq \frac{R}{T}M|S| + 2\frac{L-T}{T}M \\
&\leq \frac{R}{T}M|S| + 2\frac{R+1}{T}M \\
&< 3\frac{R+1}{\frac{R(R+1)}{2}}M|S| \\
&= \frac{6M|S|}{R} \leq \frac{\epsilon}{2}.
\end{aligned}$$

Therefore, the average payoff for every player i is higher than $x_i - \frac{\epsilon}{2}$. Whereas, if a player deviates at some stage, he gain at most $2M$ in that stage, but from that stage on, the other players play the punishment strategy so his average payoff is at most x_i. Hence he can gain at most $2M$ in the stages in the block in which he deviates as well at the last block. Thus, his average payoff is at most $x_i + \frac{4(R+1)M}{T} = x_i + \frac{2M}{R} < x_i + \frac{\epsilon}{2}$. In conclusion, τ is an ϵ-equilibrium in the T-stage game.

13.27 Consider Example 13.20. Suppose that Player II uses the strategy τ_{II}^* defined in the example, i.e., Player II plays L up to stage $(t_I)^2$ and plays R thereafter, where t_I is the first stage in which Player I plays B.

Then, the strategy of Player I under which Player I plays T for $\lceil\sqrt{T}\rceil$ stages and plays B thereafter yielding the payoff $\frac{T-\lceil\sqrt{T}\rceil}{T}$. Indeed, since $(\lceil\sqrt{T}\rceil+1)^2 > T$, Player II always play L and Player I plays T for $\lceil\sqrt{T}\rceil$ stages and B for $T-\lceil\sqrt{T}\rceil$ stages, and thus Player I's average payoff is

$$\frac{0\cdot\lceil\sqrt{T}\rceil+1\cdot(T-\lceil\sqrt{T}\rceil)}{T} = \frac{T-\lceil\sqrt{T}\rceil}{T}.$$

13.28 Let N be a set of players, and let $(S_i)_{i\in N}$ be finite sets of actions of the players. Let $u : S \to \mathbf{R}^N$ and $u' : S \to \mathbf{R}^N$ be two payoff functions. Consider a variation of the repeated game, in which in odd stages the payoff function is u, and in even stages the payoff function is u'. For every player i, let $\bar{v}_i$ and $\bar{v}'_i$ be the maxmin values of player i in the base game that is played in odd and even stages, respectively. Then the average maxmin value of player i in two successive stage games is $m_i = \frac{\bar{v}_i+\bar{v}'_i}{2}$. Denote by $V = \{x \in \mathbf{R}^N : x_i \geq m_i \ \ \forall i \in N\}$ the set of individually rational payoffs.
Let $F = \mathrm{conv}\{u(s) : s \in S\}$ and $F' = \mathrm{conv}\{u'(s) : s \in S\}$. Then $\bar{F} = \{\frac{1}{2}x + \frac{1}{2}x' : x \in F,\ x' \in F'\}$ is the set of feasible payoffs.

(a) Suppose that for every player $i \in N$ there exists either an equilibrium $\beta(i)$ in the base game $\Gamma = (N, (S_i)_{i\in N}, (u_i)_{i\in N})$ satisfying $u_i(\beta(i)) > \bar{v}_i$ or an equilibrium $\beta'(i)$ in the base game $\Gamma = (N, (S_i)_{i\in N}, (u'_i)_{i\in N})$ satisfying $u'_i(\beta'(i)) > \bar{v}'_i$. Then for every $\epsilon > 0$ there exists $T_0 \in N$ such that for every $T \geq T_0$, and every feasible and individually rational payoff vector $x \in \bar{F} \cap V$, there exists an equilibrium τ^* of the T-stage game Γ^T whose corresponding payoff is ϵ-close to x (in the maximum norm):

$$\|\gamma^T(\tau^*) - x\|_\infty < \epsilon.$$

(b) The set of equilibrium payoffs of Γ_∞ is the set $\bar{F} \cap V$

13.29 In this variation, the maxmin value of each player in each base game is the same as in Exercise 13.28. Let F, F' and V as defined in Exercise 13.28. The set of feasible payoffs is $\bar{F} = \mathrm{conv}(F \bigcup F')$. Hence, the set of equilibrium payoff in the infinitely repeated game (or in the finite game under the condition presented in Exercise 13.28 Part (a)) is $\bar{F} \cap V$.

13.30 The set of equilibrium payoff in this game is the set of equilibrium payoff in an auxiliary (standard) repeated game in which the payoff matrix is the average of the two payoff matrices of the original game.

13.31 In this variation, the set of equilibrium payoff in this game is the set of equilibrium payoff in an auxiliary repeated game in which the base game is $(N, (\widehat{S}_i)_{i\in N}, (\widehat{u}_i)i \in N)$ where the set of actions of each player $i \neq 1$ is $\widehat{S}_i = S_i$, whereas the set of actions of Player 1 is $\widehat{S}_i = S_1 \times S_1$; and the payoff function of each player i is

$$\widehat{u}_i(s_1^1, s_1^2, s_2, ..., s_n) = \tfrac{1}{2}u_i(s_1^1, s_1^2, s_2, ..., s_n) + \tfrac{1}{2}u_i'(s_1^1, s_1^2, s_2, ..., s_n).$$

13.32 We prove that for every discount factor $\lambda \in [0, 1)$, the minmax value of each player i in the λ-discounted game Γ^λ is equal to his minmax value in the base game Γ

$$\bar{v}_i = \min_{\sigma'_{-i}\in\Sigma_{-i}} \max_{\sigma_i\in\Sigma_i} U_i(\sigma_i, \sigma'_{-i}).$$

Let σ_{-i} be the punishment strategy against player i, that is, $\bar{v}_i = \max_{\sigma_i\in\Sigma_i} U_i(\sigma_i, \sigma'_{-i})$. Then by playing σ_{-i} at each stage, the players can guarantee that the stage payoffs of player i are at most $\bar{v}_i$ and so his λ-discounted payoff in Γ^λ is at most $(1-\lambda)\sum_{t=1}^{\infty} \lambda^{t-1}\bar{v}_i = \bar{v}_i$ as well. In conclusion, the minmax value of player i in Γ^λ is at most $\bar{v}_i$. On the other hand, for every history h let $\sigma'_{-i}(h)$ be the action that is played by the other players. Consider the strategy of player i at which he plays $\sigma_i(h)$ after a history h such that $U_i(\sigma_i(h), \sigma'_{-i}(h)) = \max_{\sigma_i\in\Sigma_i} U_i(\sigma_i, \sigma'_{-i}(h))$ then his λ-discounted payoff in Γ^λ is at least $(1-\lambda)\sum_{t=1}^{\infty} \lambda^{t-1}\bar{v}_i = \bar{v}_i$. In particular, the minmax value of player i in Γ^λ is at least $\bar{v}_i$ from which it follows that it is equal to $\bar{v}_i$.

13.33 Denote by (x_1, x_2) the λ-discounted payoff in each of the three cases.

(a) We use the cyclicality of the strategies,

$$\begin{aligned}
x_1 &= (1-\lambda)(4\lambda + \lambda^2 + 4\lambda^3 - \lambda^5 + \lambda^6 x_1)\\
&\Rightarrow x_1 = \tfrac{1-\lambda}{1-\lambda^6}(4\lambda + \lambda^2 + 4\lambda^3 - \lambda^5)\\
x_2 &= (1-\lambda)(2 - \lambda^2 + 2\lambda^4 + 3\lambda^5 + \lambda^6 x_2)\\
&\Rightarrow x_2 = \tfrac{1-\lambda}{1-\lambda^6}(2 - \lambda^2 + 2\lambda^4 + 3\lambda^5)
\end{aligned}$$

(b) Denote by $y_i = \sum_{t=2}^{\infty} \lambda^{t-1} u_i(s_t)$ for every player i.

$$\begin{aligned}
y_1 &= (2.75\lambda + \lambda^2 + \lambda^2 y_1) \Rightarrow y_1 = \tfrac{2.75\lambda+\lambda^2}{1-\lambda^2}\\
y_2 &= (0.75\lambda + 2\lambda^2 + \lambda^2 y_2) \Rightarrow y_2 = \tfrac{0.75\lambda+2\lambda^2}{1-\lambda^2}\\
x_1 &= (1-\lambda)(1+y_1) = 1 - \lambda + \tfrac{2.75\lambda+\lambda^2}{1+\lambda}\\
x_2 &= (1-\lambda)(-1+y_2) = -1 + \lambda + \tfrac{0.75\lambda+2\lambda^2}{1+\lambda}.
\end{aligned}$$

(c) Denote by $y = (y_1, y_2)$ and $z = (z_1, z_2)$ the λ-discounted payoffs in case the game starts with Player I playing T and B respectively.

$$y = \tfrac{2}{3}\left((4,0) + \lambda y\right) + \tfrac{1}{3}\left((0,2) + \lambda z\right)$$
$$z = \tfrac{1}{2}\left((4,0) + \lambda y\right) + \tfrac{1}{4}\left((0,2) + \lambda z\right) + \tfrac{1}{6}\left((-1,3) + \lambda y\right) + \tfrac{1}{12}\left((1,-1) + \lambda z\right)$$
$$\Rightarrow y = \left(\tfrac{3\lambda+14}{12(1-\lambda)}, \tfrac{\lambda+8}{12(1-\lambda)}\right), \quad z = \left(\tfrac{23-6\lambda}{12(1-\lambda)}, \tfrac{11-2\lambda}{12(1-\lambda)}\right)$$
$$(x_1, x_2) = \left(\tfrac{3\lambda+14}{12}, \tfrac{\lambda+8}{12}\right).$$

13.34 (a) The cartel game is given by $N = \{1, 2, ..., n\}$, $S_i = [0, \infty)$ for every $i \in N$, and $u_i(s) = (100,000 - \sum_{j=1}^{n} s_j)s_i$ for every $s \in \prod_{i=1}^{n} S_i$ and every $i \in N$.

(b) To find best response of player $i \in N$ to s_{-i} we find the maximum of $u_i(s_i, s_{-i})$ with respect to s_i. The function $s_i \mapsto u_i(s_i, s_{-i})$ is a quadratic function that attains its maximum at the point where the derivative of the function is zero:

$$\frac{\partial u_i}{\partial s_i}(s_i, s_{-i}) = 100,000 - \sum_{j=1}^{n} s_j - s_i = 0, \Rightarrow s_i = \tfrac{1}{2}(100,000 - \sum_{j \neq i} s_j).$$

Therefore, a unique symmetric equilibrium $(x, x, ..., x)$ must satisfies that

$$x = \tfrac{1}{2}(100,000 - \sum_{j \neq i} x) \Rightarrow x = \tfrac{100,000}{n+1}.$$

(c) Let C be the amount of cars the manufacturers will manufacture together. Then the total profit of all of them is $u(C) = (100,000 - C)C$ which attains its maximum at the point where the derivative of the function is zero $u'(C) = 100,000 - C - C = 0$. That is, they manufacture together $C = 50,000$ cars so each manufacturer manufactures $\frac{50,000}{n}$. Note that if $n > 1$ the profit of each manufacturer under the cartel is $\frac{50,000^2}{n}$ which is higher than the profit without the cartel $\frac{100,000^2}{(n+1)^2}$, despite the lower manufacturing numbers.

(d) No, the strategy vector at which each manufacturer manufactures $\frac{50,000}{n}$ cars in each stage is not an equilibrium of Γ^λ, for every $\lambda \in [0, 1)$. For instance, for λ that is close enough to zero, the first-stage payoff is the most important one, hence each player is better off deviating to the best response to this strategy.

(e) Let $n > 1$. The stage payoff of each player under τ is $\frac{50{,}000^2}{n}$. If player i deviates at stage t then his payoff at this stage is at most $\frac{25{,}000^2(n+1)^2}{n^2}$ and from the next stage on ward his stage-payoff is at most $\frac{100{,}000^2}{(n+1)^2}$. Hence, τ is an equilibrium in Γ^λ if it satisfies

$$\frac{25{,}000^2(n+1)^2}{n^2} + \frac{\lambda}{1-\lambda} \cdot \frac{100{,}000^2}{(n+1)^2} \leq \frac{50{,}000^2}{n} + \frac{\lambda}{1-\lambda} \cdot \frac{50{,}000^2}{n}$$
$$\Rightarrow 1 - \frac{4n}{(n+1)^2+4n} \leq \lambda.$$

In particular, one can deduce that a cartel is not stable as the number of manufacturers growths.

(f) Yes, there are similarities between the repeated Prisoner's Dilemma and the cartel game of this exercise. The equilibrium of the repeated Prisoner's Dilemma corresponds to the equilibrium described in items (b) is the strategy vector under which the players always play (D, D). The equilibrium of the repeated Prisoner's Dilemma corresponds to the equilibria described in items (e) is the strategy vector under which the players play (C, C) unless of deviation which leads to higher payoffs for all of the players but it is not stable.

13.35 Let Γ be the alternating offers game. Γ is given by $N = \{I, II\}$ where I is the first to offer a split and II is the second. Each player either offers $x \in [0, 100]$- the amount of money to Player I corresponding to a split $(x, 1-x)$ or accepts/rejects the last offer of his opponent. Therefore, the set of actions at each stage is given as follows

$$S_I^n = \begin{cases} [0,100] & n \text{ is odd} \\ \{\text{Accept, Reject}\} & n \text{ is even} \end{cases}$$
$$S_{II}^n = \begin{cases} [0,100] & n \text{ is even} \\ \{\text{Accept, Reject}\} & n \text{ is odd} \end{cases}.$$

For every $n \in \mathbf{N}$ such that at the first $n-1$ stages all offers were rejected and at stage n player i offers x and his offer was accepted then the payoff vector is $(\lambda^{n-1}x, \lambda^{n-1}(100-x)$ where $\lambda \in (0,1)$ is the discount factor in the game, in all other stages the payoff vector is $(0,0)$.

We show that there is a unique subgame perfect equilibrium according to which (1) every player offers a split according to which he gets $\frac{100}{1+\lambda}$ and his opponent gets $\frac{100\lambda}{1+\lambda}$; and (2) every player accepts offers in which he gets at least $\frac{100\lambda}{1+\lambda}$ and rejects all other offers.

We first show that this is indeed a subgame perfect equilibrium. Consider a subgame of Γ that starts at stage n (assume without a loss of generality that n is odd). If an offer was already accepted at the first $n-1$ stages then all payoff are zero for both players and thus no deviation is worthwhile. Otherwise, the equilibrium payoff vector in this subgame is $(\frac{\lambda^{n-1}100}{1+\lambda}, \frac{\lambda^{n}100}{1+\lambda})$. Player I has no incentive to offer $(x, 1-x)$ satisfying $x < \frac{100}{1+\lambda}$ since Player II accepts it and he loses. If Player I offers $x > \frac{100}{1+\lambda}$ at the first stage then Player II rejects this offer which case the payoff of Player I is either zero (in case no offer is accepted) or at most $\frac{100\lambda^n}{1+\lambda}$ (if he accepts the offer of Player II at the next stage or later as well as if he suggests $x \leq \frac{100}{1+\lambda}$ in later stages). Hence, Player I has no incentive to deviate. Similarly, Player II has no worthwhile deviation, since the lowest offer Player I accepts at any even stage is $\frac{100\lambda}{1+\lambda}$ hence by rejecting Player I's offer he can gets at most $\lambda^n \frac{100}{1+\lambda}$ which equals his equilibrium payoff.
We next prove that this subgame perfect is unique.
Note that every subgame of Γ is equivalent to Γ up to changing the role of the players. Therefore, the set of equilibria of Γ and the set of equilibria of its subgames are identical (Theorem 5.35).
Consider the set A of every $x \in [0, 100]$ for which there is a subgame equilibrium in which at the first stage Player I offers x (the split $(x, 1-x)$) and Player II accepts. The set A is not empty since $\frac{100}{1+\lambda} \in A$. Let $x^* = \max_{x \in A} x$. In particular, $x^* \geq \frac{100}{1+\lambda}$.
Assume that in a subgame perfect equilibrium Player I offers $x < x^*$. If player II rejects the offer then he has some counter offer in the next stage that can be expected. However, the best he can get as a first player is λx^* so

$$\lambda x^* \geq 100 - x > 100 - x^* \Rightarrow x^* < \tfrac{100}{1+\lambda},$$

a contradiction. Therefore, in every subgame perfect equilibrium Player II necessarily excepts any offer $x < x^*$. In particular, if by contradiction $x^* > \frac{100}{1+\lambda}$ then Player I can offer $x \in (\frac{100}{1+\lambda}, x^*)$ and Player II accepts, which contradicts that there is a subgame perfect equilibrium in which Player I offers $x = \frac{100}{1+\lambda}$. In every subgame equilibrium in which a player offers to opponent $1-x$ and his opponent accepts, $x = \frac{100}{1+\lambda}$. In particular, Player I has no incentive to suggest higher offer since his offer is rejected and he loses. Similarly, Player II cannot gain by accepting higher offers, and thus the above subgame perfect equilibrium is unique.

13.36 The value of the repeated game in A is 2.5 where the optimal strategy of Player I is to always play $[\frac{1}{2}(T), \frac{1}{2}(B)]$, and the optimal strategy of Player II is to always play $[\frac{5}{8}(L), \frac{3}{8}(R)]$.

The value of the repeated game in B is 0.5 where the optimal strategy of Player I is to always play $[\frac{1}{2}(T), \frac{1}{2}(B)]$, and the optimal strategy of Player II is to always play $[\frac{1}{2}(L), \frac{1}{2}(R)]$.
Therefore, the game may be reduced into the following one-stage game

	L	R
T	$1 + 2.5\frac{\lambda}{1-\lambda}$	$4 + 0.5\frac{\lambda}{1-\lambda}$
B	$2 + 0.5\frac{\lambda}{1-\lambda}$	$2.5\frac{\lambda}{1-\lambda}$

from which one can deduce that the value of the repeated game is $\frac{-1.5\lambda^2+15.5\lambda-8}{9\lambda-5}$. The optimal strategy of player I is to play $[\frac{4\lambda-2}{9\lambda-5}(T), \frac{5\lambda-3}{9\lambda-5}(B)]$ at the first stage and then to play $[\frac{1}{2}(T), \frac{1}{2}(B)]$. The optimal strategy of Player II is to play $[\frac{6\lambda-4}{9\lambda-5}(L), \frac{3\lambda-1}{9\lambda-5}(R)]$ at the first strategy then to play the optimal strategy in the continuous game. Note that as T goes to infinity, the value goes to 1.5.

13.37 Consider a λ-discounted two-player game based on this one-stage game

	L	R
T	1,1	0,0
B	0,0	2,2

Let τ be the strategy vector in the infinitely repeated game according to which at first stage they play the base game equilibrium (B, L), in the following 2^1 stages they play the base game equilibrium (T, R), then in the next 2^2 stages they play (B, L) again, then they play (T, R) for 2^3 stages and so on. Since the players play an equilibria of the base game at each stage, this is indeed an equilibrium in the counted game. However, $\lim_{\lambda\to 1} \gamma^\lambda(\tau)$ does not exist.

13.38 Suppose two players are playing the repeated Prisoner's Dilemma. Let τ be the strategy vector at which the players implement the grim-trigger strategy, i.e., every player plays C as long as the other player plays C, and otherwise plays D. We prove that if the discount factor λ is sufficiently close to 1, then τ is a λ-discounted equilibrium.
Let T be the first stage in which one of the players, deviates. At stage T the deviator get at most 4 (instead of 3), but from the next stage on he gets 1. Hence the discounted payoff of a deviator is at most

$$\begin{aligned}&\lim_{\lambda\to 1}(1-\lambda)\left(\sum_{t=1}^{T-1} 3\lambda^{t-1} + 4\lambda^{T-1} + \sum_{t=T+1}^{\infty} \lambda^{t-1}\right)\\ &= \lim_{\lambda\to 1} 3 - 2\lambda^{T-1}(3\lambda - 2) = 1,\end{aligned}$$

whereas by following τ his discounted payoff is 3.

13.39 (a) Let $|S_i| = m_i$ for every player $i \in N$. Set $m = \prod_{j=1}^{n} m_j$ the number of pure action vectors in the base game. The number of pure strategies of recall k player i has at stage $t \leq k+1$ is $(m_i)^{m^{t-1}}$, and the number of pure strategies at stage $t > k+1$ is $(m_i)^{m^k}$. In particular, the number of pure strategies of recall k in the T stage game is $\prod_{t=1}^{T-1}(m_i)^{m^{t-1}} = m_i^{\frac{m^{T-1}-1}{m-1}}$ for $T \leq k+1$ and $\left(\prod_{t=1}^{k+1}(m_i)^{m^{t-1}}\right)(m_i)^{m^k} = m_i^{\frac{(2m-1)m^k-1}{m-1}}$ for $T > k+1$.

(b) The grim-trigger strategy cannot be implemented by a pure strategy with recall k, since the grim trigger is an unforgiven strategy whereas a strategy with recall k is forgiven, i.e., if a player deviates at some stage but then plays C for k stages then the other player do not remember the deviation.

(c) In the T-stage repeated Prisoner's Dilemma, when the players are limited to playing only strategies with recall k (where $1 < k+1 < T$), (3, 3) is an equilibrium payoff. Let us look at following strategy: in the first stage play C. In every subsequent stage, if in the k stages you and your opponent chose C, choose C in the current stage; otherwise choose D. If one player deviates from that strategy by playing D at some stage t then from the next stage onwards the other player plays D since he remembers that either the other player played D in the previous stage (on stage $t+1$) or he himself played D in the previous stage (on stage $t' > t+1$). Note that since $T > k+1$ and the players use strategies with recall k, if a player deviates at stage T then he must have played that strategy also at stage $T-1$ so there is at least one punishing stage.

(d) Assume that in the T-stage repeated Prisoner's Dilemma, Player I is limited to strategies with recall k, and Player II is limited to strategies with recall l. Then if $1 < \min\{k+1, l+1\} < \max\{k+1, l+1\} < T$ then $(3,3)$ is an equilibrium payoff corresponding the strategy vector presented in the previous part.

13.40 Suppose two players are playing the repeated Prisoner's Dilemma with an unknown number of stages; after each stage, a lottery is conducted, such that with probability $1-\beta$ the game ends with no further stages conducted, and with probability β the game continues to another stage, where $\beta \in [0,1)$ is a given real number. Each player's goal is to maximize the sum total of payoffs received over all the stages of the game. We prove that if β is sufficiently close to 1, the strategy vector in which at the first stage every player plays C, and in each subsequent stage each player plays C if the other player played C in the previous stage, and he plays D otherwise, is an equilibrium. Under this strategy vector each player expects to gain 3 at each stage as long as the game goes on. Assume that a player deviates. At each stage that he plays D and the other player also plays D then his stage payoff is 1 instead of 3 so he does not gain. We show that he cannot gain also in each stage t in which he plays D whereas his opponent plays C. Although, in stage t he receives 4 instead of 3, but with probability β the game continuous to stage $t+1$ in which he receives at most 1 instead of 3 so as β goes to 1 $4+\beta < 3+3\beta$ and he loses by the deviation.

13.41 (a) The value in mixed strategies of the base game is $v = 0$. The optimal strategy of Player I is $[\frac{1}{2}(T), \frac{1}{2}(B)]$. The optimal strategy of Player II is $[\frac{1}{2}(L), \frac{1}{2}(R)]$.

(b) The discounted payoff of Player I under the given pair of strategies is

$$\begin{aligned}
&(1-\lambda)\left(-\sum_{t=1}^{t_0}\lambda^{t-1} + \sum_{t=t_0+1}^{\infty}\lambda^{t-1}\right) \\
&= (1-\lambda)\left(-\tfrac{1-\lambda^{t_0}}{1-\lambda} + \tfrac{\lambda^{t_0}}{1-\lambda}\right) \\
&= 2\lambda^{t_0} - 1.
\end{aligned}$$

Similarly, the discounted payoff of Player II under the given pair of strategies is $1 - 2\lambda^{2t_0}$.

(c) For $t_0 = -\frac{\ln 2}{\ln\lambda}$ the sum of the payoffs of the two players is maximized.

The sum of the payoffs for this t_0 is $2\left(\lambda^{-\frac{\ln 2}{\ln\lambda}} - \lambda^{-2\frac{\ln 2}{\ln\lambda}}\right) = \frac{1}{2}$.

(d) The mixed strategies $[\frac{1}{2}(T), \frac{1}{2}(B)]$ and $[\frac{1}{2}(L), \frac{1}{2}(R)]$ are equilibrium in the base game. The pair of strategies in which Player I plays the mixed action $[\frac{1}{2}(T), \frac{1}{2}(B)]$ at each stage, and Player II plays the mixed action $[\frac{1}{2}(L), \frac{1}{2}(R)]$ at each stage, is an equilibrium, because neither player can profit by deviating. No player can profit in a stage in which he deviates from equilibrium, because by definition in such a stage the players implement an equilibrium of the base game. In addition, his deviation in any stage cannot influence the future actions of the other players, because they are playing according to a strategy that does not depend on the history.

(e) Let $\tau^* = (\tau_I^*, \tau_{II}^*)$ be any equilibrium of this discounted game. For $t_0 \in \mathbf{N}$, denote by $A(\lambda, t_0)$ the λ-discounted payoff under strategy vector τ^* starting from stage t_0. We prove that for every $t_0 \in \mathbf{N}$, $A(\lambda, t_0) \geq v$. Indeed, consider the strategy τ_I of Player I under which he plays according to τ_I^* at the first $t_0 - 1$ stages. From stage t_0 onwards he plays his optimal strategy (Part (a)). Then, since τ^* is an equilibrium $\gamma_I^\lambda(\tau^*) \geq \gamma_I^\lambda(\tau_I, \tau_{II}^*)$, therefore

$$\begin{aligned}
\gamma_I^\lambda(\tau^*) &= (1-\lambda)\sum_{t=1}^{\infty}\lambda^{t-1}E_{\tau^*}[u^t] \qquad (111)\\
&= (1-\lambda)\sum_{t=1}^{t_0-1}\lambda^{t-1}E_{\tau^*}[u^t] + \lambda^{t_0-1}A(\lambda, t_0)\\
&\geq (1-\lambda)\sum_{t=1}^{\infty}\lambda^{t-1}E_{\tau_I,\tau_{II}^*}[u^t]\\
&\geq (1-\lambda)\sum_{t=1}^{t_0-1}\lambda^{t-1}E_{\tau^*}[u^t] + \lambda^{t_0-1}v.
\end{aligned}$$

Hence, $A(\lambda, t_0) \geq v$. Similarly, since the game is a zero-sum game, by switching the roles of the players it follows that $A(\lambda^2, t_0) \leq v$.

(f) Let $\lambda, \mu \in [0, 1)$.

$$\begin{aligned}
\sum_{k=0}^{\infty}\lambda^k A(\mu, t_0+k) &= \sum_{k=0}^{\infty}\lambda^k(1-\mu)\sum_{t=t_0+k}^{\infty}\mu^{t-t_0-k}E_{\tau^*}[u^t]\\
&= (1-\mu)\sum_{t=t_0}^{\infty}\mu^{t-t_0}E_{\tau^*}[u^t]\sum_{k=0}^{t-t_0}\left(\tfrac{\lambda}{\mu}\right)^k\\
&= (1-\mu)\sum_{t=t_0}^{\infty}\mu^{t-t_0}E_{\tau^*}[u^t]\frac{1-\left(\frac{\lambda}{\mu}\right)^{t-t_0+1}}{1-\frac{\lambda}{\mu}}\\
&= \tfrac{\mu}{\mu-\lambda}(1-\mu)\sum_{t=t_0}^{\infty}\mu^{t-t_0}E_{\tau^*}[u^t]\left(1-\left(\tfrac{\lambda}{\mu}\right)^{t-t_0+1}\right)\\
&= \tfrac{1}{\mu-\lambda}\left(\mu A(\mu, t_0) - \tfrac{\lambda(1-\mu)}{(1-\lambda)}A(\lambda, t_0)\right)
\end{aligned}$$

Therefore,

$$A(\lambda, t_0) = \tfrac{(1-\lambda)}{\lambda(1-\mu)}\left(\mu A(\mu, t_0) + (\lambda-\mu)\sum_{k=0}^{\infty}\lambda^k A(\mu, t_0+k)\right) \qquad (112)$$

Hence, for $\mu = \lambda^2$ it follows that

$$A(\lambda, t_0) = \tfrac{\lambda}{1+\lambda} A(\lambda^2, t_0) + \tfrac{1-\lambda}{1+\lambda} \sum_{k=0}^{\infty} \lambda^k A(\lambda^2, t_0 + k)$$

(g) By Part (e), $A(\lambda, t_0) \geq v = 0$. By Part (f) and (e),

$$\begin{aligned} A(\lambda, t_0) &= \tfrac{\lambda}{1+\lambda} A(\lambda^2, t_0) + \tfrac{1-\lambda}{1+\lambda} A(\lambda^2, t_0) \sum_{k=0}^{\infty} \lambda^k A(\lambda^2, t_0 + k) \\ &\leq \tfrac{\lambda}{1+\lambda} v + \tfrac{1-\lambda}{1+\lambda} A(\lambda^2, t_0) \sum_{k=0}^{\infty} \lambda^k v = v = 0. \end{aligned}$$

Therefore, $A(\lambda, t_0) = v = 0$ for every $t_0 \in \mathbf{N}$. Hence,

$$E_{\tau^*}[u^t] = \tfrac{A(\lambda,t)}{1-\lambda} - \tfrac{\lambda A(\lambda,t+1)}{1-\lambda} = v = 0$$

for every $t \in \mathbf{N}$.

(h) By Part (g), for equilibrium τ^*, $E_{\tau^*}[u^t] = 0$ for every $t \in \mathbf{N}$ so by Equality (111), the discounted payoff of each player is $v = 0$.

(i) The result of item (c) does not contradict the result of item (h) since the strategy vector presented in (c) is not an equilibrium.

(j) We next prove that if $\tau^* = (\tau_I^*, \tau_{II}^*)$ is an equilibrium of a two-player zero-sum game in which the discount factor of Player I is λ_I and the discount factor of Player II is λ_{II}, then $A(\lambda_i, t_0) = v$ for $i \in \{I, II\}$, for every $t_0 \in \mathbf{N}$, where v is the value in mixed strategies of the base game. In particular, at any equilibrium, the discounted payoff of each player (at his discount factor) is the value of the base game.

Assume without loss of generality that the discount factor of Player I is greater than the discount of Player II.

Then by Part (e) $A(\lambda_I, t_0) \geq v$ and $A(\lambda_{II}, t_0) \leq v$ for every $t_0 \in \mathbf{N}$. Furthermore, by Equality (112),

$$\begin{aligned} v \leq A(\lambda_I, t_0) &= \tfrac{(1-\lambda_I)}{\lambda_I(1-\lambda_{II})} \left(\lambda_{II} A(\lambda_{II}, t_0) + (\lambda_I - \lambda_{II}) \sum_{k=0}^{\infty} \lambda_I^k A(\lambda_{II}, t_0 + k) \right) \\ &\leq \tfrac{(1-\lambda_I)}{\lambda_I(1-\lambda_{II})} \left(\lambda_{II} v + (\lambda_I - \lambda_{II}) \sum_{k=0}^{\infty} \lambda_I^k v \right) \\ &\leq \tfrac{v(1-\lambda_I)}{\lambda_I(1-\lambda_{II})} \left(\lambda_{II} + (\lambda_I - \lambda_{II}) \tfrac{1}{1-\lambda_I} \right) = v. \end{aligned}$$

Therefore, all weak inequalities hold as equalities so $A(\lambda_I, t_0) = A(\lambda_{II}, t_0) = v$ for every $t_0 \in \mathbf{N}$. Hence,

$$E_{\tau^*}[u^t] = \tfrac{A(\lambda,t)}{1-\lambda} - \tfrac{\lambda A(\lambda,t+1)}{1-\lambda} = v$$

for every $t \in \mathbf{N}$.

13.42 Suppose that for every player i there exists an equilibrium $\beta(i)$ in the base game for which $U_i(\beta(i)) > \bar{v}_i$. We prove that there exists a vector $\widehat{x} \in F \cap V$ satisfying $\widehat{x}_i > \bar{v}_i$ for every $i \in \mathbf{N}$.
Denote $\hat{x} = \sum_{j \in N} \frac{1}{|N|} U(\beta(j))$. Since $U(\beta(i)) \in F$ and F is a convex set then $\hat{x} \in F$ as a linear combination of elements of F. In addition, By Theorem 5.42, $U_i(\beta(j)) \geq \bar{v}_i$ for every two players $i, j \in N$. Hence, and since $U_i(\beta(i)) > \bar{v}_i$, $\hat{x}_i > \sum_{j \in N} \frac{1}{|N|} \bar{v}_i = \bar{v}_i$, for every $i \in N$. In particular, $\widehat{x} \in V$.

13.43 Proof of the Folk Theorem for discounted games:
Let $\Gamma = (N, (S_i)_{i \in N}, (u_i)_{i \in N})$ be a base game in which there exists a vector $\widehat{x} \in F \cap V$ that satisfies $\widehat{x}_i > v_i$ for every player $i \in N$. Denote $M = \max_{i \in N} \max_{s \in S} \|u_i(s)\|$ and $|N| = n$.
Let $\epsilon > 0$ such that $\epsilon < 1$ and $\widehat{x}_i > v_i + 2M\epsilon$ for every $i \in N$. Let $\lambda_0 > 1 - \frac{\epsilon^3}{64M^2|S|}$. We show that for every vector $x \in F \cap V$, there exists an strategy vector τ^* in a repeated game such that for every $\lambda \in [\lambda_0, 1)$ τ^* is an equilibrium in Γ^λ satisfying

$$\|\gamma^\lambda(\tau^*) - x\|_\infty < \epsilon.$$

Let $x \in F \cap V$. Set $x' = (1 - \frac{\epsilon}{4M})x + \frac{\epsilon}{4M}\widehat{x}$. In particular $x' \in F \cap V$, $x'_i > \bar{v}_i + \frac{\epsilon^2}{2}$ for every $i \in N$ and $\|x' - x\|_\infty \leq \frac{\epsilon}{2}$. Let $K = \left\lceil \frac{4M|S|}{\epsilon} \right\rceil$ be the cycle length. By Exercise 13.14, there are nonnegative integers $(k_s)_{s \in S}$ summing to K satisfying

$$\left\| \sum_{s \in S} \tfrac{k_s}{K} u(s) - x' \right\|_\infty \leq \tfrac{M \times |S|}{K}.$$

Let τ^* be a strategy vector that is played in cycles. In each cycle, the players play every action vector $s \in S$ exactly k_s times. If some player deviates at some stage t, then from the next stage on, the other players punish the deviator: at every subsequent stage they implement a punishment strategy vector against the deviator.
Let $\lambda \in [\lambda_0, 1)$. We first show that $\|\gamma^\lambda(\tau^*) - x'\|_\infty < \frac{\epsilon}{2}$. Note that by Equality (13.72), $\gamma(\tau^*) = \sum_{T=1}^\infty \alpha_T(\lambda) S_T$ where $S_T = \frac{1}{T} \sum_{t=1}^T u_i(s^t)$.

For every $T \in \mathbf{N}$ let $R = \left\lceil \frac{T}{K} \right\rceil$. Then

$$\begin{aligned}
\|S_T - x'\|_\infty &= \|\tfrac{1}{T}\sum_{t=1}^{T}(u(s^t) - x')\|_\infty \\
&\le \tfrac{1}{T}\sum_{r=1}^{R}\|\sum_{k=1}^{K}(u(s^{(r-1)K+k}) - x')\|_\infty + \tfrac{1}{T}\sum_{t=RK+1}^{T}\|u(s^t) - x'\|_\infty \\
&\le \tfrac{M\times|S|}{K^2} + \tfrac{2MK}{T}.
\end{aligned}$$

Hence,

$$\begin{aligned}
&\|\gamma^\lambda(\tau^*) - x'\|_\infty = \|\sum_{T=1}^{\infty}\alpha_T(\lambda)S_T - x'\|_\infty = \|\sum_{T=1}^{\infty}\alpha_T(\lambda)(S_T - x')\|_\infty \\
&\le \sum_{T=1}^{\infty}\alpha_T(\lambda)\|S_T - x'\|_\infty \le \sum_{T=1}^{\infty}(\tfrac{M\times|S|}{K^2} + \tfrac{2MK}{T})\alpha_T(\lambda) \\
&\le \sum_{T=1}^{\infty}\tfrac{M\times|S|}{K^2}\alpha_T(\lambda) + \sum_{T=1}^{\infty}\tfrac{2MK}{T}\alpha_T(\lambda) = \tfrac{M\times|S|}{K^2} + 2MK(1-\lambda) \\
&\le \tfrac{\epsilon^2}{16} + \tfrac{\epsilon^2}{8} < \tfrac{\epsilon^2}{4}.
\end{aligned}$$

Thus,

$$\|\gamma^\lambda(\tau^*) - x\|_\infty \le \|\gamma^\lambda(\tau^*) - x'\|_\infty + \|x - x'\|_\infty < \epsilon.$$

Furthermore,

$$\gamma_i^\lambda(\tau^*) > x' - \tfrac{\epsilon^2}{4} > \bar{v}_i + \tfrac{\epsilon^2}{2} - \tfrac{\epsilon^2}{4} = \bar{v}_i + \tfrac{\epsilon^2}{4}.$$

However, if player i deviates at stage t, then his payoff is at most $2M$ at stage t and from the next stage his stage payoff is $\bar{v}_i$ therefore his payoff is at most $\bar{v}_i + (1-\lambda)2M < \bar{v}_i + \frac{\epsilon^2}{32M}$ so he loses.

13.44 **Proof of Theorem 13.25:** We show that the set of subgame perfect equilibrium payoffs of Γ_∞ is the set $F \cap V$. The construction of the subgame perfect equilibrium of Γ_∞ is similar to the construction of the equilibrium in Exercise 13.24. However, instead of having the punishment last forever, the punishment that follows the n-th deviation of player i is of finite length l_i^n, and, once the punishment phase ends, the players resume the equilibrium play in blocks as presented in Exercise 13.24 as if no deviation had occurred.

The length of the n-th punishment of player i for whom $x_i > \bar{v}_i$ is $l_i^n = (K+1)\lceil \frac{M}{x_i - \bar{v}_i} \rceil$. The length of the n-th punishment of player i for whom $x_i = \bar{v}_i$ is $l_i^n = 2^n KM$.

In every subgame, given that no player deviates at the subsequence stages, after finitely many stages at most, the players play the equilibrium play in blocks in the infinitely repeated game which leads to the payoff of x. The same holds in case that a player deviates only finitely many times.

If a player i for whom $x_i < \bar{v}_i$ deviates at some stage then he can gain at most KM in the block in which he deviates but then in the following $K\lceil \frac{M}{x_i - \bar{v}_i} \rceil$ he gets $\bar{v}_i$ instead of an average of x_i so he loses at least KM and thus he cannot gain by infinitely many deviations.

If a player i for whom $x_i = \bar{v}_i$ deviates then he can gain at most KM from each deviation. On the other hand, each deviation is detected by the other players and leads to a punishment phase of length l_i^n. In particular, for every $T = \sum_{N=1}^{n} l_i^N = KM \sum_{N=1}^{n} 2^n$ there can be at most n deviations of player i which case the stage payoff can be increased by at most $\frac{nKM}{KM \sum_{N=1}^{n} 2^n} = \frac{n}{\sum_{N=1}^{n} 2^n} \xrightarrow[n\to\infty]{} 0$ and thus he cannot gain by infinitely many deviations.

13.45 **Proof of the Folk Theorem for discounted games** (Theorem 13.24).
Let Γ be a base game in which there exists a vector $\widehat{x} \in F \cap V$ that satisfies $\widehat{x}_i > v_i$ for every player $i \in N$.

We prove that for every $\epsilon > 0$ there is $\lambda_0 \in [0,1)$ such that for every $\lambda \in [\lambda_0, 1)$ and every vector $x \in F \cap V$, there exists an equilibrium τ^* of Γ_λ satisfying

$$\|\gamma^\lambda(\tau^*) - x\|_\infty < \epsilon.$$

Let $\epsilon > 0$. Select $\alpha \in (1 - \frac{\epsilon}{6M}, 1)$. Set

$$\lambda_1 = \max_{i \in N} \frac{M - (\alpha v_i + (1-\alpha)\widehat{x}_i)}{M - v_i}.$$

Note that $v_i < \alpha v_i + (1-\alpha)\widehat{x}_i \le M$ for every $i \in N$ so $\lambda_1 \in [0,1)$. Let

$$0 < \epsilon_1 < \min\left\{\epsilon, \min_{i\in N}\Big(\big(\alpha v_i + (1-\alpha)\widehat{x}_i\big) - \big(\lambda_1 v_i + (1-\lambda_1)M\big)\Big)\right\}$$

(ϵ_1 is well defined by the definition of λ_1). Choose $K \ge \frac{M \times |S|}{\epsilon_1/3}$. Finally, let $\lambda_0 \in [\lambda_1, 1)$ satisfying

$$M \times K \times \max\left\{\frac{1}{k} - \frac{\lambda_0^{K-1}}{\sum_{k=1}^{K} \lambda_0^{k-1}}, \frac{1}{\sum_{k=1}^{K} \lambda_0^{k-1}} \frac{1}{k}\right\} \le \frac{\epsilon_1}{3}.$$

Let $x \in F \cap V$ and let $\lambda \in [\lambda_0, 1)$. Set $y = \alpha x + (1-\alpha)\widehat{x}$. Then

$$\|x - y\|_\infty = (1-\alpha)\|x - \widehat{x}\|_\infty \le \frac{\epsilon}{6M} 2M = \frac{\epsilon}{3}. \tag{113}$$

By Exercise 13.14, there are nonnegative integers $(k_s)_{s\in S}$ summing to

K satisfying

$$\left\|\sum_{s\in S}\tfrac{k_s}{K}u(s)-y\right\|_\infty \leq \tfrac{M\times|S|}{K} \leq \frac{\epsilon_1}{3} \tag{114}$$

We construct a strategy vector τ^* in which the players play in blocks. In each block the players play each action vector $s \in S$ exactly k_s times. Denote by $s^1, s^2, ..., s^K$ the action that are played over the course of the each block. If at a certain stage, a player deviates from the action he is supposed to play at that stage, he is punished from the next stage onwards by a punishment strategy.

We show that the distance between the λ-discounted payoff under τ^* and $\sum_{s\in S}\frac{k_s}{K}u(s)$ is at most $\frac{\epsilon_1}{2}$.

$$\begin{aligned}
\left\|\gamma^\lambda(\tau^*)-\sum_{s\in S}\tfrac{k_s}{K}u(s)\right\|_\infty &= \left\|(1-\lambda)\sum_{t=1}^{\infty}\lambda^{t-1}E_\tau[u^\tau]-\sum_{k=1}^{K}\tfrac{1}{K}u(s^k)\right\|_\infty\\
&= \left\|(1-\lambda)\sum_{R=1}^{\infty}\lambda^{R-1}\sum_{k=1}^{K}\lambda^{k-1}u(s^k)-\sum_{k=1}^{K}\tfrac{1}{K}u(s^k)\right\|_\infty\\
&= \left\|\frac{1-\lambda}{1-\lambda^K}\sum_{k=1}^{K}\lambda^{k-1}u(s^k)-\sum_{k=1}^{K}\tfrac{1}{K}u(s^k)\right\|_\infty\\
&= \left\|\frac{1}{\sum_{k=1}^{K}\lambda^{k-1}}\sum_{k=1}^{K}\lambda^{k-1}u(s^k)-\sum_{k=1}^{K}\tfrac{1}{K}u(s^k)\right\|_\infty\\
&\leq \sum_{k=1}^{K}\left|\frac{\lambda^{k-1}}{\sum_{k=1}^{K}\lambda^{k-1}}-\tfrac{1}{K}\right|\left\|u(s^k)\right\|_\infty\\
&\leq M\times K\times\max\left\{\frac{1}{k}-\frac{\lambda_0^{K-1}}{\sum_{k=1}^{K}\lambda_0^{k-1}},\frac{1}{\sum_{k=1}^{K}\lambda_0^{k-1}}\frac{1}{k}\right\}\leq\frac{\epsilon_1}{3}.
\end{aligned} \tag{115}$$

From Equations (113)-(115) and the triangle inequality it can be deduced that $\left\|\gamma^\lambda(\tau^*)-x\right\|_\infty \leq \frac{\epsilon}{3}+\frac{\epsilon_1}{3}+\frac{\epsilon_1}{3}\leq\epsilon$.

It is left to prove that τ^* is indeed an equilibrium.

If a player i deviates at stage t then he may get M at stage t but from stage $t+1$ and forth his stage payoff is v_i that is, his discounted payoff from stage t is at most

$$(1-\lambda)M+\lambda v_i.$$

On the other hand, by following τ^* his discounted payoff is at least

$$\begin{aligned}
&y_i-\epsilon_1=\alpha x_i+(1-\alpha)\widehat{x}_i-\epsilon_1\\
&>\alpha x_i+(1-\alpha)\widehat{x}_i-\Big(\big(\alpha v_i+(1-\alpha)\widehat{x}_i\big)-\big(\lambda_1 v_i+(1-\lambda_1)M\big)\Big)\\
&\geq(1-\lambda)M+\lambda v_i
\end{aligned}$$

and thus player i cannot gain by deviation.

13.46 **Exercise 13.20:** The strategy vector is played in blocks of length 3: in each block Player I plays T, T, B; and Player II plays L, R, R. Once someone deviates, both players switch to a punishment strategy forever: Player I plays B and Player II plays L. The discounted payoff is

$$\lim_{\lambda \to 1}(1-\lambda)((1,4)+\lambda(2,5)+\lambda^2(3,2)+\lambda^3(1,4)+\lambda^4(2,5)+\lambda^5(3,2)+...)$$
$$= \lim_{\lambda \to 1} \tfrac{1-\lambda}{1-\lambda^3}(1,4)+\tfrac{\lambda-\lambda^2}{1-\lambda^3}(2,5)+\tfrac{\lambda^2-\lambda^3}{1-\lambda^3}(3,2))$$
$$= \lim_{\lambda \to 1} \tfrac{-1}{-3\lambda^2}(1,4)+\tfrac{1-2\lambda}{-3\lambda^2}(2,5)+\tfrac{2\lambda-3\lambda^2}{-3\lambda^2}(3,2))$$
$$\tfrac{1}{3}(1,4)+\tfrac{1}{3}(2,5)+\tfrac{1}{3}(3,2))=(2,3\tfrac{2}{3}).$$

If Player I deviates at stage t then his discounted payoff from stage t onwards is bounded by :

$$(1-\lambda)(\lambda^{t-1}3+\sum_{l=t+1}^{\infty}\lambda^{l-1}\cdot 1)=\lambda^{t-1}(3-2\lambda).$$

Since for $\lambda = 1$ the discounted payoff from stage t in case of deviation is 1 which is less than the discounted payoff from stage t without deviation (which equals 2), then the same hold for $\lambda < 1$ sufficiently large.
If Player II deviates at stage t then his discounted payoff from stage t onwards is bounded by :

$$(1-\lambda)(\lambda^{t-1}5+\sum_{l=t+1}^{\infty}\lambda^{l-1}\cdot 2)=\lambda^{t-1}(5-3\lambda).$$

Since for $\lambda = 1$ the discounted payoff from stage t in case of deviation is 2 which is less than the discounted payoff from stage t without deviation (which equals $3\frac{2}{3}$), then the same hold for $\lambda < 1$ sufficiently large. Therefore it is indeed a uniform equilibria for discounted game.

Exercise 13.21: The strategy vector is played in blocks of length 4: Player I plays T, T, B, B; Player II plays R, R, L, R; and Player III plays E, E, W, E. Once someone deviates the players switch to a punishment strategy:

- If Player I deviates, Players II plays $[\frac{1}{3}(W), \frac{2}{3}(E)]$ forever (and Player I's payoff is $\frac{5}{3}$).
- If Player II deviates, Player I plays T forever (and Player II's payoff is at most 1).

- If Player III deviates, Player I plays T and Player II plays $[\frac{1}{2}(L), \frac{1}{2}(R)]$ forever (and Player III's payoff is at most 1).

The discounted payoff is

$$\lim_{\lambda\to 1}(1-\lambda)((3,0,3)+\lambda(1,2,3)+\lambda^2(2,1,2)+\lambda^3(2,1,2)+\lambda^4(3,0,3)+\lambda^5(1,2,3)+...)$$
$$= \lim_{\lambda\to 1} \tfrac{1-\lambda}{1-\lambda^4}(3,0,3)+\tfrac{\lambda-\lambda^2}{1-\lambda^4}(1,2,3)+\tfrac{\lambda^2-\lambda^4}{1-\lambda^4}(2,1,2))$$
$$= \lim_{\lambda\to 1} \tfrac{-1}{-4\lambda^3}(3,0,3)+\tfrac{1-2\lambda}{-4\lambda^2}(2,1,2)+\tfrac{2\lambda-4\lambda^3}{-4\lambda^3}(2,1,2))$$
$$\tfrac{1}{4}(3,0,3)+\tfrac{1}{4}(1,2,3)+\tfrac{1}{2}(2,1,2)) = (2,1,2\tfrac{1}{2}).$$

If Player I deviates at stage t then his discounted payoff from stage t onwards is bounded by :

$$(1-\lambda)(\lambda^{t-1}2+\sum_{l=t+1}^{\infty}\lambda^{l-1}\cdot\tfrac{5}{3}) = \lambda^{t-1}(2-\tfrac{\lambda}{3}).$$

Since for $\lambda = 1$ the discounted payoff from stage t in case of deviation is $\frac{5}{3}$ which is less than the discounted payoff from stage t without deviation (which equals 2), then the same hold for $\lambda < 1$ sufficiently large.
If Player II deviates at stage t then his discounted payoff from stage t onwards is bounded by :

$$(1-\lambda)(\lambda^{t-1}2+\sum_{l=t+1}^{\infty}\lambda^{l-1}\cdot 1) = \lambda^{t-1}(1-\lambda).$$

Since for $\lambda = 1$ the discounted payoff from stage t in case of deviation is 0 which is less than the discounted payoff from stage t without deviation (which equals 1), then the same hold for $\lambda < 1$ sufficiently large.
Player III cannot gain by deviating at any stage. Therefore it is indeed a uniform equilibria for discounted game.

13.47 Proof of the Folk Theorem for uniform equilibrium in discounted games:
Let $\Gamma = (N, (S_i)_{i\in N}, (u_i)_{i\in N})$ be a base game in which there exists a vector $\widehat{x} \in F\cap V$ that satisfies $\widehat{x}_i > v_i$ for every player $i \in N$. Let $\epsilon > 0$ and $x \in F \cap V$. Consider the strategy vector τ^* presented in Exercise 13.43. By Exercise 13.43, τ^* is an equilibrium in every discounted

game Γ^λ for every $\lambda \in [\lambda_0, 1)$ where $\lambda_0 > 1 - \frac{\epsilon^3}{64M^2|S|}$. Furthermore,

$$\begin{aligned}
\lim_{\lambda\to 1} \gamma^\lambda(\tau^*) &= \lim_{\lambda\to 1}(1-\lambda) \sum_{R=1}^{\infty} \sum_{k=1}^{K} \lambda^{(R-1)K+k-1} u(s^k) \\
&= \lim_{\lambda\to 1}(1-\lambda) \sum_{k=1}^{K} \lambda^{k-1} u(s^k) \sum_{R=1}^{\infty} \lambda^{(R-1)K} \\
&= \lim_{\lambda\to 1} \sum_{k=1}^{K} u(s^k) \tfrac{\lambda^{k-1}-\lambda^k}{1-\lambda^K} \\
&= \lim_{\lambda\to 1} \sum_{k=1}^{K} u(s^k) \tfrac{(k-1)\lambda^{k-2}-k\lambda^{k-1}}{-K\lambda^{K-1}} \\
&= \sum_{k=1}^{K} \tfrac{u(s^k)}{K} = \sum_{s\in S} \tfrac{k_s}{K} u(s).
\end{aligned}$$

In particular,

$$\left\| \lim_{\lambda\to 1} \gamma^\lambda(\tau^*) - x' \right\|_\infty = \left\| \sum_{s\in S} \tfrac{k_s}{K} u(s) - x' \right\|_\infty \leq \tfrac{M\times|S|}{K} < \epsilon.$$

which completes the proof.

13.48 Let Γ be a two-player zero-sum base game. The real number v is called the uniform value if for each $\epsilon > 0$ there exist strategies τ_I^* of Player I and τ_{II}^* of Player II in Γ^∞, and an integer T_0, such that the following condition is satisfied: for each $T \geq T_0$, and each pair of strategies (τ_I, τ_{II}) in Γ^T , $\gamma^T(\tau_I, \tau_{II}^*) \leq v + \epsilon$, and $\gamma^T(\tau_I^*, \tau_{II}) \geq v - \epsilon$. We prove that the uniform value for finite games equals the value of the base game $\widehat{v}$. Indeed, let $\widehat{\tau}_I$ and $\widehat{\tau}_{II}$ be the strategies of Player I and Player II respectively according to which at each stage they played their optimal strategies in the base game. Then for every $\epsilon > 0$, let τ_I^*, τ_{II}^* the strategies in Γ^∞ and an integer T_0 as presented above. Then,

$$v - \epsilon \leq \gamma^T(\tau_I^*, \widehat{\tau}_{II}) \leq \widehat{v} \leq \gamma^T(\widehat{\tau}_I, \tau_{II}^*) \leq v + \epsilon.$$

Let ϵ goes to zero, hence by the sandwich theorem $v = \widehat{v}$.

13.49 **Exercise 13.20:** Let $\epsilon > 0$. Consider the strategy vector that is played in blocks of length 3: in each block Player I plays T, T, B; and Player II plays L, R, R. Once someone deviates, both players switch to a punishment strategy forever: Player I plays B and Player II plays L. The limit of the payoffs in the finite games is

$$\tfrac{1}{3}(1,4) + \tfrac{1}{3}(2,5) + \tfrac{1}{3}(3,2) = (2, 3\tfrac{2}{3}).$$

If Player I deviates at stage t then at stage t he can gain at most 1 but in the next stages he gains nothing. If Player II deviates at stage t then at stage t can gain at most 1 but in the next stages he gains nothing. Let $T_0 = \lceil \frac{1}{\epsilon} \rceil$. Hence for every $T > T_0$,the above strategy is indeed an ϵ-equilibrium.

Exercise 13.21: Let $\epsilon > 0$. Consider the strategy vector that is played in blocks of length 4: Player I plays T, T, B, B; Player II plays R, R, L, R; and Player III plays E, E, W, E. Once someone deviates the players switch to a punishment strategy:

- If Player I deviates, Players II plays $[\frac{1}{3}(W), \frac{2}{3}(E)]$ forever (and Player I's payoff is $\frac{5}{3}$).
- If Player II deviates, Player I plays T forever (and Player II's payoff is at most 1).
- If Player III deviates, Player I plays T and Player II plays $[\frac{1}{2}(L), \frac{1}{2}(R)]$ forever (and Player III's payoff is at most 1).

The limit of the payoffs in the finite games is

$$\tfrac{1}{4}(3,0,3) + \tfrac{1}{4}(1,2,3) + \tfrac{1}{2}(2,1,2) = (2, 3\tfrac{2}{3}).$$

If Player I deviates at stage t then at stage t can gain at most 1 but in the next stages he gains nothing. If Player II deviates at stage t then at stage t can gain at most 2 but in the next stages he gains nothing. Player III cannot gain by any deviation. Let $T_0 = \lceil \frac{2}{\epsilon} \rceil$. Hence for every $T > T_0$,the above strategy is indeed an ϵ-equilibrium.

13.50 Let E_T be the set of equilibrium payoffs of a T-stage repeated game Γ^T.

(a) Let $T, k \in \mathbf{N}$ and let $x \in E_T$. Hence there is an equilibrium τ^T of Γ^T corresponding to x. Consider the strategy vector in Γ^{kT} under which the players plays k cycles of τ^T. This strategy is an equilibrium in Γ^{kT}, by Theorem 13.6 where Γ^T is the base game for Γ^{kT}. Furthermore, the average payoff at each repetition of τ^T is x and thus the average payoff of Γ^{kT} is x as well. In conclusion, $x \in E_{kT}$ and $E_T \subseteq E_{kT}$.

(b) We next prove that $\frac{T}{T+1}E_T + \frac{1}{T+1}E_1 \subseteq E_{T+1}$ for every $T \in \mathbf{N}$.
Let $x \in E_1$ and $y \in E_T$. Denote by τ^1 and τ^T the equilibria of Γ^1 and Γ^T corresponding to x and y, respectively. Hence, the strategy vector in Γ^{T+1} under which at the first T stages the players follow τ^T and then they follow τ^1 is an equilibrium in Γ^{T+1} which corresponding the average payoff

$$\tfrac{T \cdot y + 1 \cdot x}{T+1} = \tfrac{T}{T+1}y + \tfrac{1}{T+1}x \in E^{T+1},$$

as needed.

Note that, by induction, it can be deduced that for every $T, K \in \mathbf{N}$,

$$\tfrac{T}{T+K} E_T + \tfrac{K}{T+K} E_1 \subseteq E_{T+K}.$$

(c) For every set $S \subseteq \mathbf{R}^k$, the set $\bar{S}$ denotes the closure of S. Let

$$E_\infty := \limsup_{T' \to \infty} E_{T'} = \bigcap_{T' \in \mathbf{N}} \bigcup_{K \geq T'} \bar{E}_K.$$

We first prove that $E_T \subseteq E_\infty$ for every $T \in \mathbf{N}$. Let $T \in \mathbf{N}$. For every $T' \in \mathbf{N}$ there is $k \in \mathbf{N}$ such that $kT \geq T'$. Then, by Part (a), $E_T \subseteq E_{kT} \subseteq \bar{E}_{kT} \subseteq \bigcup_{K \geq T'} \bar{E}_K$. Hence, $E_T \subseteq \bigcap_{T' \in \mathbf{N}} \bigcup_{K \geq T'} \bar{E}_K = E_\infty$.

We next show that E_∞ is not empty. E_1 is the set of equilibrium payoffs in the one stage game which, by Nash Theorem, is not empty. Hence and since $E_1 \subseteq E_\infty$ so $E_\infty \neq \emptyset$ as well.

We finally prove that for every $x \in E_\infty$ and every $\epsilon > 0$, there exists $T_0 \in \mathbf{N}$ such that for every $T \geq T_0$ there exists $y \in E_T$ satisfying $\|x - y\|_\infty \leq \epsilon$.

Let $\epsilon > 0$ and let $x \in E_\infty$. Hence, there is $T' \in \mathbf{N}$ such that $x \in \bar{E}_{T'}$. Therefore, there is $z \in E_{T'}$ such that $\|x - z\|_\infty \leq \frac{\epsilon}{2}$. Set $T_0 = T' \left\lceil \frac{4M}{\epsilon} \right\rceil$ (where M is a bound over the base game payoffs). For every $T \geq T_0$, let $K = \left\lceil \frac{T}{T'} \right\rceil$ and $L = T - T'K$. Choose an arbitrary $w \in E_1$ and set $y = \frac{T'K}{T} z + \frac{L}{T} w$. Then, by Part (a), $z \in E_{T'K}$ and so, Part (b), $y \in \frac{T'K}{T} E_T + \frac{L}{T} E_1 \subseteq E_T$. In addition,

$$\|z - y\|_\infty = \|z - (\tfrac{T'K}{T} z + \tfrac{L}{T} w)\|_\infty \leq \tfrac{L}{T} \|z - w\|_\infty \leq \tfrac{T'}{T'K} 2M \leq \tfrac{\epsilon}{2}$$

so $\|x - y\|_\infty \leq \|x - z\|_\infty + \|z - y\|_\infty \leq \epsilon$, as claimed.

13.51 Let τ^* be a uniform 0-equilibrium for finite games. That is, τ^* is a strategy vector in the infinitely repeated game satisfying (1)$\lim_{T \to \infty} \gamma^T(\tau^*)$ exists; and (2) there is $T_0 \in \mathbf{N}$ such that for every $T \geq T_0$, τ^* is an equilibrium in Γ^T.

Therefore, if by contradiction, according to τ^* there is a stage $T \geq T_0$ such that the players does not play an equilibrium of the base game then in the last stage of Γ^T there is a player that is better off deviating from τ_i^*, which contradict τ^* is an equilibrium in Γ^T. In conclusion, from stage T_0 onwards the players play an equilibrium of the base game at each stage.

13.52 Proof of the Folk Theorem for uniform equilibrium in finite games: Let $\epsilon > 0$ and let $x \in F \cap V$. We show that there exists a strategy vector τ^* such that: (1) τ^* is a uniform ϵ-equilibrium for finite games; and (2) $\| \lim_{T\to\infty} \gamma^T(\tau^*) - x\|_\infty < \epsilon$.

Let K be an integer such that $K > \frac{M\times|S|}{\epsilon}$ where S is the set of action vector in the base game and M is a bound over the base game payoffs. By Exercise 13.14, there are nonnegative integers $(k_s)_{s\in S}$ summing to K satisfying

$$\left\| \sum_{s\in S} \frac{k_s}{K} u(s) - x \right\|_\infty \leq \frac{M\times|S|}{K}.$$

Let τ^* be a strategy vector in the infinitely repeated game that is played in cycles. In each cycle, the players play every action vector $s \in S$ exactly k_s times. If some player deviates at some stage t, then from the next stage onwards the other players punish the deviator: at every subsequent stage they implement a punishment strategy vector against the deviator.

For every $T \in \mathbf{N}$ denote by $R_T = \left\lceil \frac{T}{K} \right\rceil$ the number of repetition of the cycle in Γ^T.

The average payoff in Γ^T corresponding τ^* is

$$\begin{aligned}\gamma^T(\tau^*) &= \frac{1}{T}\sum_{t=1}^{T} u(s^T) = \frac{1}{T}\sum_{r=1}^{R_T}\sum_{s\in S}\frac{k_s}{K}u(s) + \frac{1}{T}\sum_{t=R_TK+1}^{T} u(s^T)\\ &= \frac{R_TK\sum_{s\in S}\frac{k_s}{K}u(s)}{T} + \frac{\sum_{t=R_TK+1}^{T} u(s^T)}{T}.\end{aligned}$$

Therefore, since $\lim_{T\to\infty}\frac{R_TK}{T} = 1$ and $\|\sum_{t=R_TK+1}^{T} u(s^T)\| \leq 2K2M$ then $\lim_{T\to\infty}\gamma^T(\tau^*) = \sum_{s\in S}\frac{k_s}{K}u(s)$.

Furthermore, let $T_0 > \frac{4MK}{\epsilon}$. Then, for every $T > T_0$, τ^* is an ϵ-equilibrium of Γ^T. Indeed, if some player deviates then he can gain at most $2M$ at every stage in the cycle in which he deviates and in the last $T - R_TK$ stages, that is he can gain at most $\frac{2K\times 2M}{T} \leq \frac{2K\times 2M}{T_0} < \epsilon$ (note that in all other cycles his average payoff is at least $\bar{v}_i$ his average punishing payoff). In conclusion, τ^* is a uniform ϵ-equilibrium for finite games.

Finally,

$$\| \lim_{T\to\infty}\gamma^T(\tau^*) - x\|_\infty = \|\sum_{s\in S}\frac{k_s}{K}u(s) - x\|_\infty \leq \frac{M\times|S|}{K} \leq \epsilon.$$

13.53 Let $(q_t)_{t\in N}$ be a sequence of natural numbers. Define a sequence $(p_t)_{t\in\mathbf{N}}$ as follows: $p_1 := 0$, $p_t := q_1 + q_2 + ... + q_{t-1}$. Define a sequence $(x_t)_{t\in\mathbf{N}}$ as follows:

$$x_t = \begin{cases} 1 & \text{when there exists } k \text{ such that } 2p_k < t \leq 2p_k + q_k, \\ 0 & \text{otherwise.} \end{cases}$$

That is, the first q_1 elements of the sequence $(x_t)_{t\in\mathbf{N}}$ equal 1, the next q_1 elements of the sequence equal 0, the next q_2 elements of the sequence equal 1, the next q_2 elements of the sequence equal 0, and so on.

(a) For every T, denote by $R_T = \max\{t \in \mathbf{N} : 2p_{t+1} \leq T\}$. Then

$$\liminf_{T\to\infty} \frac{\sum_{t=1}^T x_k}{T} = \liminf_{T\to\infty} \left(\frac{T/2}{T} + \sum_{t=p_{R_T}+1}^{T} \frac{x_t}{T} \right) = \frac{1}{2}$$

(b) Denote $A(\lambda) = (1-\lambda)\sum_{t=1}^\infty \lambda^{t-1}x_t$. We prove that $A(\lambda) = \sum_{k=1}^\infty \lambda^{2p_k}(1-\lambda^{q_k})$.

$$\begin{aligned} A(\lambda) &= (1-\lambda)\sum_{t=1}^\infty \lambda^{t-1}x_t \\ &= (1-\lambda)\sum_{k=1}^\infty \sum_{t=2p_k+1}^{2p_k+q_k} \lambda^{t-1}1 \\ &= (1-\lambda)\sum_{k=1}^\infty \frac{\lambda^{2p_k} - \lambda^{2p_k+q_k}}{1-\lambda} \\ &= \sum_{k=1}^\infty \lambda^{2p_k}(1-\lambda^{q_k}). \end{aligned}$$

(c) Denote $\alpha_k = \lambda^{p_k} - \lambda^{p_{k+1}}$ for every $k \in \mathbf{N}$. Note that,

$$\alpha_k = \lambda^{p_k} - \lambda^{p_{k+1}} = \lambda^{p_k} - \lambda^{p_k+q_k} = \lambda^{p_k}(1-\lambda^{q_k}) \text{ and}$$
$$\sum_{n=k}^\infty \alpha_n = \lambda^{p_k}.$$

Hence and by Part (b),

$$\begin{aligned} A(\lambda) &= \sum_{k=1}^\infty \lambda^{2p_k}(1-\lambda^{q_k}) = \sum_{k=1}^\infty \lambda^{p_k}\alpha_k \\ &= \sum_{k=1}^\infty (\sum_{n=k}^\infty \alpha_n)\alpha_k = \frac{1}{2}\left((\sum_{k=1}^\infty \alpha_k)^2 + \sum_{k=1}^\infty (\alpha_k)^2 \right) \\ &= \frac{1}{2}\left((\lambda^{p_1})^2 + \sum_{k=1}^\infty (\alpha_k)^2 \right) = \frac{1}{2}\left(\sum_{k=1}^\infty \alpha_k^2 + 1 \right). \end{aligned}$$

(d) Let $\epsilon(0, \frac{1}{4})$, and define $c := \frac{\ln(\epsilon)}{\ln(1-\sqrt{\epsilon})}$. The function $\epsilon \mapsto \frac{\ln(\epsilon)}{\ln(1-\sqrt{\epsilon})}$ is monotonic decreasing for every $\epsilon \in (0, \frac{1}{4})$ hence it attains its minimum at $\epsilon = \frac{1}{4}$. In particular, $c := \frac{\ln(\epsilon)}{\ln(1-\sqrt{\epsilon})} > \frac{\ln(\frac{1}{4})}{\ln(1-\sqrt{\frac{1}{4}})} = 2$.

(e) Suppose that the sequence $(q_t)_{t\in N}$ satisfies $q_k > \frac{2p_k}{c-2}$ for every $k \in \mathbf{N}$. Define $a_k := \frac{|\ln(1-\sqrt{\epsilon})|}{q_k}$, $b_k := \frac{|\ln(\epsilon)|}{2p_k}$. Then, since $(p_k)_{k\in\mathbf{N}}$ is strictly increasing to infinity, then $\lim_{k\to\infty} b_k = 0$ and for every $k \in \mathbf{N}$, $b_{k+1} < b_k$.

Moreover, for every $k \in \mathbf{N}$, since $q_k > \frac{2p_k}{c-2}$, then $cq_k > 2p_k + 2q_k$. Therefore,

$$b_{k+1} = \frac{|\ln(\epsilon)|}{2p_{k+1}} = \frac{|\ln(\epsilon)|}{2p_k+2q_k} > \frac{|\ln(\epsilon)|}{cq_k} = \frac{|\ln(1-\sqrt{\epsilon})|}{\ln\epsilon}\frac{|ln(\epsilon)|}{q_k} = a_k.$$

(f) By Part (e), $a_k < b_{k+1} < b_k$ so $(a_k, b_k) \cap (a_{k+1}, b_{k+1}) \neq \emptyset$. Hence, $\bigcup_{k=1}^{\infty}(a_k, b_k) = (a, b)$. In addition, since $p_1 = 0$, $b = b_1 = \infty$, and $a = \lim_{k\to\infty} a_k$ but $0 \leq \lim_{k\to\infty} a_k \leq \lim_{k\to\infty} b_k = 0$ so $a = 0$.

In particular, for every $\lambda \in [0, 1)$, $|\ln \lambda| \in (0, \infty)$ and hence there exists $k(\lambda) \in \mathbf{N}$ satisfying $a_{k(\lambda)} \leq |\ln \lambda| < b_{k(\lambda)}$.

(g) By Part (f), for every $\lambda \in [0, 1)$ there is $k(\lambda)$ for which

$$\frac{|\ln(1-\sqrt{\epsilon})|}{q_{k(\lambda)}} = a_{k(\lambda)} \leq |\ln \lambda| < b_{k(\lambda)} = \frac{|\ln(\epsilon)|}{2p_{k(\lambda)}}.$$

Therefore, $1 - \sqrt{\epsilon} \geq \lambda^{q_{k(\lambda)}}$ and $\epsilon < \lambda^{2p_{k(\lambda)}}$.

In conclusion,

$$(\alpha_{k(\lambda)})^2 = \lambda^{2p_{k(\lambda)}}(1 - \lambda^{q_{k(\lambda)}})^2 > \epsilon(1 - (1 - \sqrt{\epsilon}))^2 = \epsilon^2.$$

(h) By Part (c) and (g),

$$\liminf_{\lambda\to 1} A(\lambda) = \liminf_{\lambda\to 1} \frac{1}{2}\left(\sum_{k=1}^{\infty} \alpha_k^2 + 1\right) \geq \liminf_{\lambda\to 1} \frac{1}{2}\left((\alpha_{k(\lambda)})^2 + 1\right) \geq \frac{\epsilon^2+1}{2}.$$

(i) By Parts (a) and (h)

$$\liminf_{T\to\infty} \frac{\sum_{t=1}^{T} x_t}{T} = \frac{1}{2} < \frac{\epsilon^2+1}{2} \leq \liminf_{\lambda\to 1} A(\lambda).$$

(j) Let $y_t = 1 - x_t$ for every $t \in \mathbf{N}$, then

$$\liminf_{T\to\infty} \frac{\sum_{t=1}^{T} y_t}{T} = 1 - \liminf_{T\to\infty} \frac{\sum_{t=1}^{T} x_t}{T}$$
$$> 1 - \liminf_{\lambda\to 1}(1-\lambda)\sum_{t=1}^{\infty} \lambda^{t-1}x_t = \liminf_{\lambda\to 1}(1-\lambda)\sum_{t=1}^{\infty} \lambda^{t-1}y_t.$$

13.54 For the given sequence,

$$\liminf_{T\to\infty} \frac{\sum_{t=1}^T y_t}{T} = 0, \limsup_{T\to\infty} \frac{\sum_{t=1}^T y_t}{T} = \frac{1}{1}.$$

In addition,

$$\lim_{\lambda\to 1}(1-\lambda)\sum_{t=1}^{\infty}\lambda^{t-1}x_t = \lim_{T\to\infty}(1-\lambda)\left(\sum_{t=1}^{\infty}\lambda^{2t-2}t - \sum_{t=1}^{\infty}\lambda^{2t-1}t\right) = \frac{1}{4}.$$

13.55 The strategy vector $(\sigma_I^A, \sigma_{II})$ defined in Example 13.35 is not an equilibrium of the infinitely repeated game, since

$$\lim_{T\to\infty}\gamma_I^T((\sigma_I^A,\sigma_{II}) = \lim_{T\to\infty} c = c < \limsup_{T\to\infty}\frac{1}{T}\sum_{t=1}^{\infty}x^t = \limsup_{T\to\infty}\gamma_I^T((\sigma_I^B,\sigma_{II}).$$

13.56 (a) For $T = 1$: the players play the one-stage zero-sum game given by

		Tom: Red	Tom: Yellow
David	Red	-1	1
	Yellow	1	-1

In this game, David can guarantee that the payoff is at least 0 by playing $[\frac{1}{2}(\text{Red}), \frac{1}{2}(\text{yellow})]$, and Tom can guarantee that the payoff is at most 0 by playing $[\frac{1}{2}(\text{Red}), \frac{1}{2}(\text{yellow})]$. In particular, the only equilibrium payoff is $(0, 0)$.

(b) For $T \geq 2$: at each stage t in the given T-stage game, the players play the following one-stage game denoted by G^{T-t}

		Tom: Red	Tom: Yellow
David	Red	-1	$1 \times (T-t+1)$
	Yellow	1	$-1 \times (T-t+1)$

This is a zero-sum game. David has a strategy that guarantees that the payoff is at least 0 by playing $[\frac{1}{2}(\text{Red}), \frac{1}{2}(\text{yellow})]$ at each stage t, which is an optimal strategy in each game G^{T-t}. Tom has a strategy that guarantees that the payoff is at least 0 by playing $[\frac{2+2(T-t)}{4+2(T-t)}(\text{Red}), \frac{2}{4+2(T-t}(\text{yellow})]$ at each stage, which is an optimal strategy in each game G^{T-t}. Therefore, the only equilibrium payoff in the T-stage game for every T is (0, 0).

13.57 Let x and y be two numbers in the interval $[0,1]$. Suppose that in each stage David chooses "red" with probability x and "yellow" with probability $1-x$, and in each stage Tom guesses "red" with probability y and "yellow" with probability $1-y$.

(a) Let $\lambda \in [0,1)$. The expected λ-discounted payoff in this infinite game is

$$\begin{aligned}
&(1-\lambda)\sum_{t=1}^{\infty}\lambda^{t-1}\left(-xy+(1-x)y+(x(1-y)-(1-x)(1-y))\sum_{r=1}^{\infty}\lambda^{r-1}\right)\\
&= -xy+y(1-x)+\frac{x(1-y)-(1-x)(1-y)}{1-\lambda}\\
&= y-2xy+\frac{2x+y-2xy-1}{1-\lambda}\\
&= (1-2x)\left(y-\tfrac{1-y}{1-\lambda}\right).
\end{aligned}$$

(b) If the players are restricted to these i.i.d. strategies, then Player I can guarantee that the discounted payoff is at least 0 by playing $[\frac{1}{2}(\text{Red}), \frac{1}{2}(\text{Yellow})]$ and Player II can guarantee that the discounted payoff is at most 0 by playing $[\frac{1}{2-\lambda}(\text{Red}), \frac{1-\lambda}{2-\lambda}(\text{Yellow})]$, hence $(0,0)$ is a λ-discounted equilibrium payoff, for each $\lambda \in [0,1)$.

Chapter 14: Repeated games with vector payoffs

14.1 Let (X,d) be a metric space and let C be a subset of X. Let $x,y \in X$ then by the triangle inequality,

$$\begin{aligned}
d(x,C) = \inf_{z\in C} d(x,z) &\le \inf_{z\in C}\big(d(x,y)+d(y,z)\big) = d(x,y)+\inf_{z\in C} d(y,z)\\
&= d(x,y)+d(y,C).
\end{aligned}$$

14.2 Proof of Markov's inequality:
Let X be a nonnegative random variable, and let $c>0$. Define a

random variable Y as follows:

$$Y = \begin{cases} c & \text{if } X \geq c \\ 0 & \text{otherwis.} \end{cases}$$

Note that since X is nonnegative, $X \geq Y$ and thus, by the monotonicity property of the expectation

$$E[X] \geq E[Y] = c \cdot \mathbf{P}(Y = c) + 0 \cdot \mathbf{P}(Y = 0) = c \cdot \mathbf{P}(X \geq C) \Rightarrow$$
$$\mathbf{P}(X \geq c) \leq \tfrac{E[X]}{c}.$$

14.3 The repeated game with vector payoffs is given by the following table where the left and right values in each payoff vector represent the payoffs of M. Goriot to Anastasia and Delphine respectively.

		The daughters	
		Ignore	Visit
M. Goriot	Generous	(10,12)	(18;16)
	Stingy	(3,2)	(5,8)

14.4 (a) Suppose that a strategy σ_k approaches a set C for player k. That is, for every $\epsilon > 0$ there exists $T \in \mathbf{N}$ such that for every strategy σ_{-k} of the other player $\mathbf{P}_{\sigma_k,\sigma_{-k}}(d(\bar{g}_t, C) < \epsilon,\ \forall t \geq T) > 1 - \epsilon$.
Denote by $\bar{C}$ the closure of C. Then, since $C \subseteq \bar{C}$, $d(\bar{g}_t, C) \geq d(\bar{g}_t, \bar{C})$ and hence for every $\epsilon > 0$ there exists $T \in \mathbf{N}$ such that for every strategy σ_{-k} of the other player

$$\mathbf{P}_{\sigma_k,\sigma_{-k}}(d(\bar{g}_t, \bar{C}) < \epsilon,\ \forall t \geq T) > 1 - \epsilon.$$

In particular, σ_k approaches the closure of C for that player.

(b) Assume that a strategy σ_k excludes a set C for player k, i.e., there exists $\delta > 0$ such that the set $\{x \in \mathbf{R}^m : d(x, C) \geq \delta\}$ is approachable by player k.
We first show that $d(x, C) = d(x, \bar{C})$ for every $x \in \mathbf{R}^m$. Let $x \in \mathbf{R}^m$. Since $C \subseteq \bar{C}$, $d(x, C) \geq d(x, \bar{C})$. On the other hand, $d(x, \bar{C}) = \inf\{d(x, \bar{y}) : \bar{y} \in \bar{C})$ so for every $\epsilon > 0$ there is $\bar{y} \in \bar{C}$ such that $d(x, \bar{y}) < d(x, \bar{C}) + \frac{\epsilon}{2}$. In addition, since $\bar{C}$ is the closure of C, for every $\bar{y} \in \bar{C}$ there is $y \in C$ such that $d(\bar{y}, y) < \frac{\epsilon}{2}$. In conclusion, $d(x, y) \leq d(x, \bar{y}) + d(\bar{y}, y) < d(x, \bar{C}) + \epsilon$ and thus $d(x, C) \leq d(x, \bar{C})$. Hence, $d(x, C) = d(x, \bar{C})$ from which it follows that

$$\{x \in \mathbf{R}^m : d(x, C) \geq \delta\} = \{x \in \mathbf{R}^m : d(x, \bar{C}) \geq \delta\}.$$

Thus $\{x \in \mathbf{R}^m : d(x, \bar{C}) \geq \delta\}$ is approachable by player k and so σ_k excludes the closure of C for that player.

14.5 Suppose that $C \subseteq \mathbf{R}^m$ is a closed set. Let M be the maximal payoff of the game, in absolute value. Denote $C_M = \{y \in C : \|y\| \leq M\}$.
Before proving the claims presented in the exercise, we prove the following auxiliary claim:

Claim 2 *Let $\delta, \epsilon > 0$. Denote $S^\delta = \{y \in \mathbf{R}^m : \|y\| \leq M,\ \ d(y, C) > \delta\}$. Similarly, let $S_M^\epsilon = \{y \in \mathbf{R}^m : \|y\| \leq M,\ \ d(y, C_M) > \epsilon\}$. Then, for every $\epsilon > 0$ there is $\delta > 0$ such that $S_M^\epsilon \subseteq S^\delta$.*

Proof. We prove that there is $\delta > 0$ such that for every $y \in S_M^\epsilon$, $d(y, C) > \delta$. Assume by contradiction that the claim does not hold. That is, for every $n \in \mathbf{N}$, there are $y_n \in S_M^\epsilon$ and $x_n \in C$ such that $d(y_n, x_n) < \frac{1}{n}$. Since $\|y_n\| \leq M$ and $\|x_n\| \leq M + 1$, there is a subsequence $\big((x_{n_l}, y_{n_l})\big)_l$ that converges to (x, y). Since C is closed $x \in C$. In addition $\|y\| \leq M$. Moreover,

$$0 \leq d(x, y) \leq d(x, x_{n_l}) + d(x_{n_l}, y_{n_l}) + d(y_{n_l}, y) \to 0,$$

from which it follows that $d(x, y) = 0$. Hence, $x = y$ so $\|x\| \leq M$. In particular, $x \in C_M$, and $0 \leq d(y_{n_l}, x) \leq d(x, x_{n_l}) + d(x_{n_l}, y_{n_l}) \to 0$, which contradict $d(y_{n_l}, x) \geq d(y_{n_l}, C_M) > \epsilon$ for every l. ■

(a) We prove that C is approachable by a player if and only if the set C_M is approachable by that player.
Since $C_M \subseteq C$, if C_M is approachable by a player then C is also approachable by that player (see Exercise 14.6).
On the other hand, assume that C is approachable by a player k. Then, there exists a strategy σ_k of player k such that for every $\epsilon > 0$ there exists $T_\epsilon \in \mathbf{N}$ such that for every strategy σ_{-k} of the other player

$$\mathbf{P}_{\sigma_k, \sigma_{-k}}\left(d(\bar{g}^t, C) < \epsilon, \forall t \geq T_\epsilon\right) > 1 - \epsilon. \tag{116}$$

Assume by contradiction that σ_k does not approach C_M for player k. Then, there is an $\epsilon > 0$ such that for every $T \in \mathbf{N}$ there is a strategy σ_{-k} of the other player such that

$$\begin{aligned} 1 - \epsilon &> \mathbf{P}_{\sigma_k, \sigma_{-k}}\left(d(\bar{g}^t, C_M) < \epsilon, \forall t \geq T\right) \\ &= 1 - \mathbf{P}_{\sigma_k, \sigma_{-k}}\left(\exists t \geq T;\ \ d(\bar{g}^t, C_M) \geq \epsilon\right) \\ \iff \mathbf{P}_{\sigma_k, \sigma_{-k}}&\left(\exists t \geq T;\ \ d(\bar{g}^t, C_M) \geq \epsilon\right) > \epsilon. \end{aligned}$$

However, by Claim 2, there must be $\delta > 0$ such that whenever $d(\bar{g}^t, C_M) \geq \epsilon$ then $d(\bar{g}^t, C) \geq \delta$ so σ_k does not approach C for player k as well, a contradiction.

(b) We next prove that C is excludable by a player if and only if the set C_M is excludable by that player.

If C is excludable by a player then the C_M is excludable by that player since $C_M \subseteq C$ (see Exercise 14.6).

Assume that C_M is excludable by a player k then there exists $\delta > 0$ such that the set $\{y \in \mathbf{R}^m : d(x, C_M) \geq \delta\}$ is approachable by player k. Hence, by Part (a), $S_M^\delta = \{y \in \mathbf{R}^m : \|y\| \leq M, \ \ d(x, C_M) \geq \delta\}$ is approachable by player k. However, by Claim 2 there is $\epsilon > 0$ such that $S_M^\delta \subseteq S^\epsilon$ so, by Exercise 14.6, S^ϵ is approachable by player k. Using Part (a) again, one can obtain that $\{y \in \mathbf{R}^m : d(x, C) \geq \epsilon\}$ is approachable by player k so C is excludable by him.

(c) If C is not closed then item (a) above does not necessarily hold. For instance, consider a game where each player has two possible actions, and the payoff function u is given by the following matrix

	L	R
T	(1,1)	(1,1)
B	(1,0)	(0,1)

The set $C = \{(x_1, x_2) : x_1 > 1, x_2 > 1\}$ is approachable by Player I (using T) however, $C_M = \{x \in C : \|x\| \leq 1\} = \emptyset$ and thus it is not approachable.

14.6 (a) Let C' be a superset of C. Thus, for every $x \in \mathbf{R}^m$,

$$d(x, C) = \inf_{y \in C} d(x, y) \geq \inf_{y \in C'} d(x, y) = d(x, C'),$$

In particular, if a strategy σ_k approaches a set C for player k then for every $\epsilon > 0$ there exists $T \in \mathbf{N}$ such that for every strategy σ_{-k} of the other player

$$1 - \epsilon < \mathbf{P}_{\sigma_k, \sigma_{-k}} \left(d(\bar{g}^t, C) < \epsilon, \forall t \geq T \right) \leq \mathbf{P}_{\sigma_k, \sigma_{-k}} \left(d(\bar{g}^t, C') < \epsilon, \forall t \geq T \right),$$

from which it follows that σ_k approaches C' as well.

(b) If strategy σ_k excludes a set C for player k, then there exists $\delta > 0$ such that σ_k approaches the set $\{x \in \mathbf{R}^m : d(x, C) \geq \delta\}$. However, for every $C' \subseteq C$, the set $\{x \in \mathbf{R}^m : d(x, C') \geq \delta\}$ is a superset of $\{x \in \mathbf{R}^m : d(x, C) \geq \delta\}$, therefore, by Part (a), σ_k

approaches the set $\{x \in \mathbf{R}^m : d(x, C') \geq \delta\}$ as well, i.e., σ_k also excludes the set C' for player k.

14.7 Proof of Theorem 14.8:
We prove that a set cannot be both approachable by one player and excludable by the other player.
Assume that σ_k approaches a set C for player k. That is, for every $\epsilon > 0$ there exists $T \in \mathbf{N}$ such that for every strategy σ_{-k} of the other player

$$1 - \epsilon < \mathbf{P}_{\sigma_k, \sigma_{-k}} \left(d(\bar{g}^t, C) < \epsilon, \forall t \geq T \right).$$

We show that, there is no strategy σ_{-k} of the opponent that approaches the set $C^\delta = \{x \in \mathbf{R}^m : d(x, C) \geq \delta\}$ for every $\delta > 0$.
Let $\epsilon < \frac{\delta}{2}$, and let $T' > T$, then

$$\begin{aligned}
\mathbf{P}_{\sigma_k, \sigma_{-k}} \left(d(\bar{g}^t, C^\delta) < \epsilon, \forall t \geq T' \right) &\leq \mathbf{P}_{\sigma_k, \sigma_{-k}} \left(d(\bar{g}^t, C) \geq \delta - \epsilon, \forall t \geq T' \right) \\
&= 1 - \mathbf{P}_{\sigma_k, \sigma_{-k}} \left(d(\bar{g}^t, C) < \delta - \epsilon, \exists t \geq T' \right) \\
&= 1 - \mathbf{P}_{\sigma_k, \sigma_{-k}} \left(d(\bar{g}^t, C) < \delta - \epsilon, \forall t \geq T' \right) \\
&< \epsilon.
\end{aligned}$$

In particular C^δ is not approachable by $-k$ and thus C is not excludable by him.

14.8 In this exercise we will prove that the set C_3 defined in Example 14.4 is approachable by Player 1, using the strategy described there.

(a) Suppose that $\bar{g}^{t-1}$ is above the diagonal $x_1 + x_2 = 1$.
Note that for every $x = (x_1, x_2) \in \mathbf{R}^2$ such that $x_1 + x_2 \geq 1$ one has $d(x, \overrightarrow{0}) = \sqrt{x_1^2 + x_2^2} \geq \sqrt{\left(\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^2} > \frac{1}{2}$.
Since $\bar{g}^{t-1}$ is above the diagonal $x_1 + x_2 = 1$, Player I plays T in stage t so $\bar{g}^t = \frac{t-1}{t} \bar{g}^{t-1}$. Hence,

$$\begin{aligned}
&d(\bar{g}^t, \overrightarrow{0}) = d(\tfrac{t-1}{t} \bar{g}^{t-1}, \overrightarrow{0}) = \tfrac{t-1}{t} d(\bar{g}^{t-1}, \overrightarrow{0}) \\
&= d(\bar{g}^{t-1}, \overrightarrow{0}) - \tfrac{1}{t} d(\bar{g}^{t-1}, \overrightarrow{0}) \leq d(\bar{g}^{t-1}, \overrightarrow{0}) - \tfrac{1}{2t}.
\end{aligned}$$

(b) Assume that $\bar{g}^{t-1}$ is under the diagonal $x_1 + x_2 = 1$, so Player I plays B in stage t and thus the payoff in stage t is $(1, y)$ for $y \in [0, 1]$ and $\bar{g}^t = \frac{t-1}{t} \bar{g}^{t-1} + \frac{1}{t}(1, y)$.

$$\begin{aligned}
d(\bar{g}^t, C_2) &= d(\tfrac{t-1}{t} \bar{g}^{t-1} + \tfrac{1}{t}(1, y), C_2) = (1 - \tfrac{1}{t}) d(\bar{g}^{t-1}, C_2) \\
&\leq (1 - \tfrac{1}{2t}) d(\bar{g}^{t-1}, C_2).
\end{aligned}$$

(c) By Part (a), as long as the average payoff $\bar{g}^t$ is above this diagonal, the distance from it to $\overrightarrow{0}$ reduces (by at least $\frac{1}{2t}$) and thus it become closer to C_3. However, since $\sum_{t=1}^{\infty} \frac{1}{t}$ diverges, after finitely many consecutive stages $\bar{g}^t$ will be under the diagonal $x_1 + x_2 = 1$ (or else $d(\bar{g}^t, \overrightarrow{0}) < 0$, a contradiction).
Similarly, by Part (b), as long as the average payoff $\bar{g}^t$ is under the diagonal $x_1 + x_2 = 1$, the distance from it to C_2 reduces and thus it become closer to C_3. However, since $\sum_{t=1}^{\infty} \frac{1}{t}$ diverges, after finitely many consecutive stages $\bar{g}^t = \frac{t-1}{t}\bar{g}^{t-1} + \frac{1}{t}(1, y)$ will be above the diagonal $x_1 + x_2 = 1$.
Since the difference is by $\frac{1}{2t}$ and using the fact that the sequence $(\frac{1}{2t})_{t \in \mathbf{N}}$ converges to 0, one can deduce that $\lim_{t\to\infty} d(\bar{g}^t, C_3) = 0$.

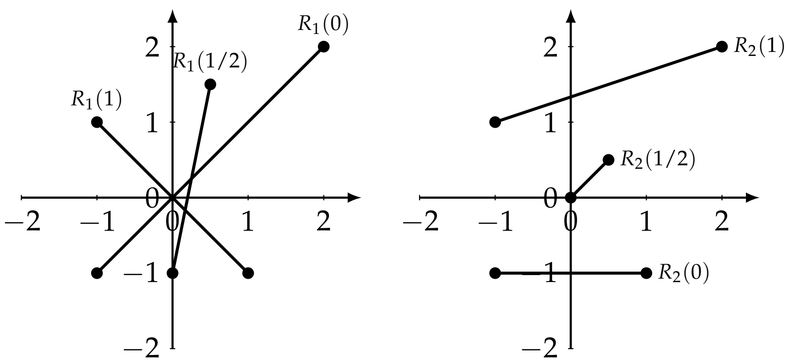
14.9 Consider Example 14.4.

(a) As shown in Exercise 14.8 the set C_3 is approachable by Player 1. In addition, $T = \{(x, x) : x \in [0, 1]\}$ is approachable by Player 1 (by playing T at every stage). However, the set $C_3 \cap T = \{(\frac{1}{2}, \frac{1}{2})\}$ is excludable by Player 2 (by playing R at every stage).

(b) The set C_2 is approachable by Player 1. The set $S = \{(1, x) : x \in [0, \frac{1}{2}]\}$ is excludable by Player 2 (by playing L at every stage), and the set $C_2 \setminus S = \{(1, x) : x \in (\frac{1}{2}, 1]\}$ is excludable by Player 2 (by playing R at every stage).

(c) The sets S and $C_2 \setminus S$ (as defined in Part (b)) are excludable by Player 2, but the set $S \bigcup (C_2 \setminus S) = C_2$ is approachable by Player 1.

14.10 (a)



(b) The set $C_1 = [(1,-1),(-1,1)] = R_1(1)$ and hence it is approachable by Player 1.

The set $C_2 = [(0,0),(2,2)] \bigcup [(0,0),(1,-1)]$ is a B-set for Player 1 and thus it is approachable by him. Indeed, let $x \in F \setminus C_2$. If the point $y \in C_2$ that is closest to x is in $((0,0),(2,2)]$ then $R_1(0) \subseteq H$ so the mixed action $p = 0$ (B) satisfies the conditions of Definition 14.12. If the point $y \in C_2$ that is closest to x is in $((0,0),(1,-1)]$ then $R_1(1) \subseteq H$ so the mixed action $p = 1$ (T) satisfies the conditions of Definition 14.12. Otherwise, $(0,0)$ is the closest to x so the separating hyperplane is the y-axis so the mixed action $p = 1/2$ satisfies the conditions of Definition 14.12.

(c) The set $C3 = [(-1,1),(2,2)]$ is not approachable by Player 1 since it is excludable by Player 2 by using the action R at every stage.

The set $C_4 = [(0,0),(-1,1)] \bigcup [(0,0),(-1,-1)]$ is not approachable by Player 1 since for every strategy σ_1 of Player 1 there is a strategy of Player 2 such that the average payoffs are in $\{(x_1,x_2) : x_1 > 0.1\}$. Indeed, whenever Player 1 plays T with probability 0.6 at most, play L otherwise play R.

The set $C_5 = \{(x,\frac{1}{2}) : -\infty < x < \infty\}$ is not approachable by Player 1 since it is excludable by Player 2 by using the action R at every stage.

14.11 Let C_1 be a closed and convex set and let C_2 be a closed and convex set containing C_1.
Suppose that C_1 is a B-set for player k. By Theorem 14.14 (Blackwell's Theorem), C_1 is approachable by player k. Hence, by Theorem 14.7, C_2 is also approachable by him, since $C_1 \subseteq C_2$. Finally, by Theorem 14.24, since C_2 is a closed and convex set which is approachable by player k, the set C_2 is a B-set for player k.

14.12 Consider the game presented in Exercise 14.10.

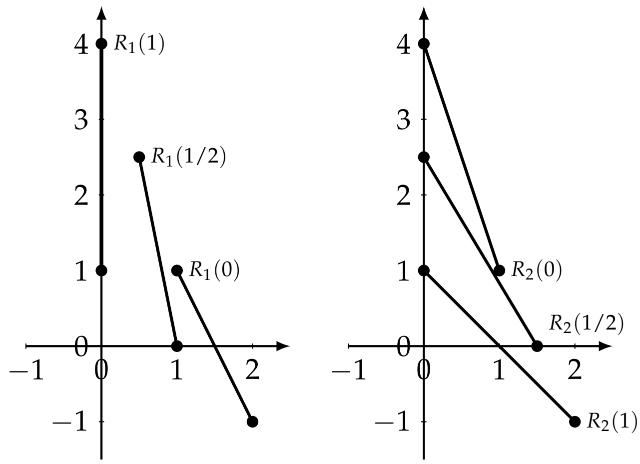
(a) The set C_1 is a B-set as shown in Exercise 14.10 Part (b).

(b) Let $\epsilon \in (0, 0.1)$. Let C_ϵ be the union of C_1 with the triangle whose vertices are $(0,0), (\epsilon,\epsilon)$, and $(\epsilon,-\epsilon)$. We show that C_ϵ is not a B-set for Player 1.

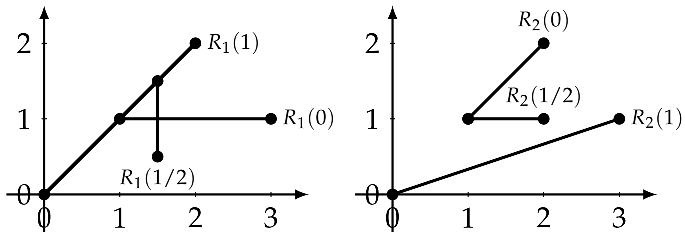
Select a point $x = (x_1, 0) \in F$ for $x_1 \in (\epsilon, 2\epsilon)$. In particular, $x \notin C_\epsilon$. Then, $y = (\epsilon, 0) \in C_\epsilon$ is the closest to x. However, the hyperplane $H(y-x, \langle y-x, y\rangle)$ is $\{z \in \mathbf{R}^2 : z_1 = \epsilon\}$ which does not separates x from $R_1(p)$ for every $p \in [0,1]$. Indeed, $R_1(p) = [(2-3p, 2-p),(2p-1,-1)]$. For every $p < 0.6$ then

$2-3p > 2-3\times 0.6 = 0.2 > \epsilon$ thus both x and $(2-3p, 2-p)$ on the right half-space so the hyperplane does not separate x and $R_1(p)$. In addition, for every $p > 0.55$, $2p-1 > 2\times 0.55 - 1 = 0.1 > \epsilon$ thus both x and $(2p-1, -1)$ on the right half-space so the hyperplane does not separate x and $R_1(p)$. Since $[0, 0.6) \bigcup (0.55, 1] = [0, 1]$, the set C_2 is not a B-set.

14.13 **Game A:**

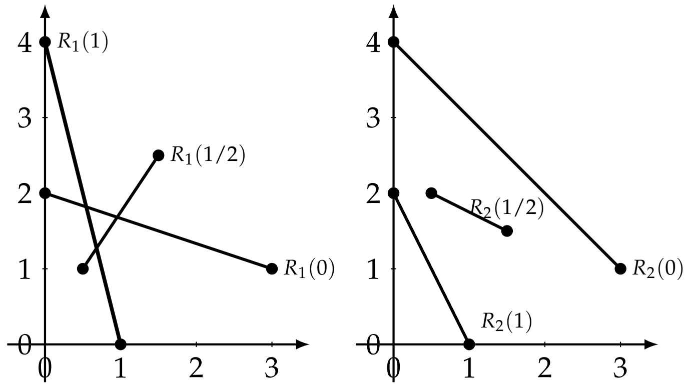


B-sets of Player 1: $S_1 = [(0,1),(0,4)]$, $S_2 = [(1,0),(0.5,2.5)]$, $S_3 = [(2,-1),(1,1)]$ and $S_4 = [(0.5,0.5),(0.25,3.25)]$.
B-sets of Player 2: $S_1 = [(0,4),(1,1)]$, $S_2 = [(0,1),(2,-1)]$, $S_3 = [(0,2.5),(1.5,0)]$ and $S_4 = [(0,3.25),(1.75,-0.5)]$.
Game B:



B-sets of Player 1: $S_1 = [(1,1),(3,1)]$, $S_2 = [(0,0),(2,2)]$, $S_3 = [(1.5,0.5),(1.5,1.5)]$ and $S_4 = [(0.75,0.25),(1.75,1.75)]$.
B-sets of Player 2: $S_1 = [(1,1),(2,2)]$, $S_2 = [(0,0),(3,1)]$, $S_3 = [(1,1),(2,1)]$ and $S_4 = [(0.5,0.5),(2.5,1)]$.

14.14 (a)



(b) The set $C_1 = [(1,0),(0,4)]$ is a B-set of Player 1 since $C_1 = R_1(1)$. The set $C_2 = [(0,4),(\frac{6}{11},\frac{20}{11})] \bigcup [(0,2),(\frac{6}{11},\frac{20}{11})]$ is neither a B-set for Player 1 nor a B-set for Player 2: let $x = (\frac{7}{11},\frac{20}{11})$. Then $y = (\frac{6}{11},\frac{20}{11})$ is the closest point to x. However, the hyperplane $H(y-x,\langle y-x,y\rangle)$ is $\{z \in \mathbf{R}^2 : z_1 = \frac{6}{11}\}$ which does not separates x from $R_1(p)$ and from $R_2(q)$ for every $p,q \in [0,1]$. Indeed, $R_1(p) = [(p,2-2p),(3-3p,1+3p)]$. For every $p > \frac{6}{11}$, both x and $(p,2-2p)$ on the right half-space so the hyperplane does not separate x and $R_1(p)$. In addition, for every $p \leq \frac{6}{11}$, $3-3p > 3-3\times\frac{6}{11} = \frac{12}{11} > \frac{6}{11}$ thus both x and $(3-3p,1+3p)$ on the right half-space so the hyperplane does not separate x and $R_1(p)$. Similarly, $R_2(q) = [(q,4-4q),(3-3q,1+q)]$. For every $q > \frac{6}{11}$, both x and $(q,4-4q)$ on the right half-space so the hyperplane does not separate x and $R_2(q)$. In addition, for every $q \leq \frac{6}{11}$, $3-3q > \frac{6}{11}$ thus both x and $(3-3q,1+q)$ on the right half-space so the hyperplane does not separate x and $R_2(q)$.

The set $C_3 = [(3,1),(\frac{6}{11},\frac{20}{11})] \bigcup [(1,1),(\frac{6}{11},\frac{20}{11})]$ is not a B-set for Player 1 since it is excludable by Player 2 (by playing R at every stage). However, it is a B-set for Player 2. Indeed, let $x \in F \setminus C_3$ and let $y \in C_3$ be the closest point to x. If $y \in [(\frac{6}{11},\frac{20}{11}),(3,1)]$ then for the mixed action $q = \frac{6}{11}$ satisfies the condition of Definition 14.12, since $R_2(6/11) \subseteq [(\frac{6}{11},\frac{20}{11}),(3,1)]$. If $y \in [(\frac{6}{11},\frac{20}{11}),(1,1)]$ then either the mixed action $q = 1$ or the mixed action $q = \frac{6}{11}$ or the mixed action $q = \frac{2}{3}$ satisfies the condition of Definition 14.12.

The set $C_4 = \{(\frac{6}{11},\frac{20}{11})\}$ is neither a B-set for Player 1 nor a B-set for Player 2 (the proof is similar to that of the set C_2).

14.15 (a) The set C, which is the union of two intervals, $[(0,0),(\frac{1}{2},0)]$ and

$[(\frac{1}{2},0),(1,\frac{1}{2})]$, is not a B-set for Player 1. Indeed, let $x = (\frac{1}{2},\frac{1}{2}) \in F \setminus C$ and let $y = (\frac{3}{4},\frac{1}{4}) \in C$. Then y is the closest point to x in C. However, the hyperplane $\{z \in \mathbf{R}^2 : z_2 = z_1 - \frac{1}{2}\}$ is orthogonal to $y - x$ but it does not separates x and $R_1(p) = [(1-p,0),(1-p,1-p)]$ for every $p \in [0,1]$ $(x_1 - \frac{1}{2} = 0 < x^2$ and $(1-p) - \frac{1}{2} = \frac{1}{2} - p < 1 - p)$.

(b) The set $\{(0,0)\}$ is a B-set of Player 1 since $R_1(1) = \{(0,0)\}$.

14.16 We prove that a closed and convex set C is approachable by Player 1 if and only if $C \cap R_2(q) \neq \emptyset$ for every $q \in \Delta(\mathcal{J})$.
Let C be a closed and convex set. Assume first that C is approachable by Player 1. By Theorem 14.8, C is not excludable by Player 2. Hence, by Corollary 14.20, $C \cap R_2(q) \neq \emptyset$ for every $q \in \Delta(\mathcal{J})$.
Assume next that C is not approachable by Player 1. Thus, by Theorem 14.24, there is a half-space H^+ containing C such that for every $p \in \Delta(\mathcal{I})$ one has $R_1(p) \not\subseteq H^+$. Therefore, by Theorem 14.21, there exists $q \in \Delta(\mathcal{J})$ such that $R_2(q) \cap H^+ = \emptyset$.

14.17 (a) For every $q = (q_1, 1-q_1) \in \Delta(\mathcal{J})$, $R_2(q) = [-2(1-q_1), 2q_1]$. Therefore, if $q_1 \leq \frac{1}{2}$ then $\emptyset \neq [-2(1-q_1), -1] \subseteq C \cap R_2(q)$. If $q_1 \geq \frac{1}{2}$ then $\emptyset \neq [1, 2q_1] \subseteq C \cap R_2(q)$.

(b)-(c) We show that C is excludable by Player 2 and thus it is not acceptable by Player 1 (Theorem 14.8). The following strategy of Player 2 guarantees that the average payoff approaches the set $[-0.9, 0.9]$:

- If $\bar{g}^{t-1} > 0.9$ play R.
- If $-0.9 \leq \bar{g}^{t-1} \leq 0.9$ play $[0.5(L), 0.5(R)]$.
- If $\bar{g}^{t-1} < -0.9$ play L.

Following the above strategy, if $\bar{g}^{t-1} > 0.9$ then

$$\bar{g}^t \leq \tfrac{t-1}{t}\bar{g}^{t-1} \leq \bar{g}^{t-1} - \tfrac{0.9}{t}. \tag{117}$$

If $\bar{g}^{t-1} < -0.9$ then

$$\bar{g}^t \geq \tfrac{t-1}{t}\bar{g}^{t-1} \geq \bar{g}^{t-1} + \tfrac{0.9}{t}. \tag{118}$$

By Equation (117), as long as the average payoff $\bar{g}^t$ is higher than 0.9, it reduces by at least $\frac{0.9}{t}$. However, since $\sum_{t=1}^{\infty} \frac{0.9}{t}$ diverges, after finitely many consecutive stages $\bar{g}^t$ will be less than 0.9 (or else $\bar{g}^t < -2$, a contradiction).
Similarly, by Equation (118), as long as the average payoff $\bar{g}^t$ is less than -0.9, it increases. However, since $\sum_{t=1}^{\infty} \frac{0.9}{t}$ diverges, after finitely many consecutive stages $\bar{g}^t$ will be higher than -0.9.

Since the difference is by $\frac{0.9}{t}$ and using the fact that the sequence $(\frac{0.9}{t})_{t\in\mathbf{N}}$ converges to 0, one can deduce that $\lim_{t\to\infty}\bar{g}^t \notin C$.

14.18 Let $\Gamma = (N, S_1, S_2, u)$ be a two-player zero-sum game. Denote by

$$\underline{v}_1 = \max_{\sigma_1\in\Delta(S_1)} \min_{\sigma_2\in\Delta(S_2)} U_1(\sigma_1,\sigma_2)$$

(respectively $\overline{v}_1 = \min_{\sigma_2\in\Delta(S_2)} \max_{\sigma_1\in\Delta(S_1)} U_1(\sigma_1,\sigma_2)$) the maxmin (respectively minmax) value in mixed strategies. Consider the game to be a game with one-dimensional vector payoffs, and define the sets $C = [\overline{v}_1, \infty)$ and $D = (\infty, \underline{v}_1]$.

(a) Assume by contradiction that C is not approachable by Player 1. By Theorem 14.25, since C is a closed and convex set, it is necessarily excludable by Player 2. That is, there is $\delta > 0$ such that $(-\infty, \overline{v}_1 - \delta]$ is approachable by Player 2. Furthermore, by Corollary 14.23, there is a stationary strategy $\sigma_2 \in \Delta(S_2)$ which approaches $(-\infty, \overline{v}_1 - \delta]$ for Player 2. Hence, using Theorem 14.22, it follows that $R_2(\sigma_2) \subseteq (-\infty, \overline{v}_1 - \delta]$. However, by the definition of $\overline{v}_1$, there is a strategy $\sigma_1 \in \Delta(S_1)$ such that $\overline{v}_1 \leq U_1(\sigma_1, \sigma_2)$ so $R_2(\sigma_2) \nsubseteq (-\infty, \overline{v}_1 - \delta]$, a contradiction. In conclusion, C is approachable by Player 1.

(b) The proof that D is approachable by Player 2 is similar to that of Part (a) and thus it was omitted.

(c) By Part (a) and Corollary 14.23, there is a stationary strategy $\sigma_1^* \in \Delta(S_1)$ which approaches C for Player 1. Hence, by Theorem 14.22, $R_1(\sigma_1^*) \subseteq C$ so $U_1(\sigma_1^*, \sigma_2) \geq \overline{v}_1$ for every $\sigma_2 \in \Delta(S_2)$.
Similarly, by Part (b) and Corollary 14.23, there is a stationary strategy $\sigma_2^* \in \Delta(S_2)$ which approaches D for Player 2. Hence, by Theorem 14.22, $R_2(\sigma_2^*) \subseteq D$ so $U_1(\sigma_1, \sigma_2^*) \leq \underline{v}_1$ for every $\sigma_1 \in \Delta(S_1)$.
In conclusion, $\overline{v}_1 \leq U_1(\sigma_1^*, \sigma_2^*) \leq \underline{v}_1$. Using the fact that $\overline{v}_1 \geq \underline{v}_1$ (Exercise 4.34), one can deduce that $\overline{v}_1 = \underline{v}_1$.

14.19 Let $(Y_i)_{i=1}^{\infty}$ be a sequence of random variables, $Y_i \in [0,1]$ for every $i \in N$. Let $(X_i)_{i=1}^{\infty}$ be a sequence of independent random variables, with Bernoulli distribution with parameter p, where $p \in (0,1)$. In other words, $\mathbf{P}(X_i = 1) = p$, and $\mathbf{P}(X_i = 0) = 1 - p$.
We prove that

$$\mathbf{P}\left(\lim_{n\to\infty}\left(\frac{1}{n}\sum_{i=1}^{n} Y_i - \frac{\sum_{\{i:X_i=1\}} Y_i}{|\{i : X_i = 1\}|}\right) = 0\right) = 1.$$

Consider the following game with vector payoffs in $\mathbf{R}^4$:

	1	0
1	(1,0,0,0)	(0,1,0,0)
0	(0,0,1,0)	(0,0,0,1)

Interpret Y_n as the mixed action of Player 2 in stage n, and X_n as the pure action of Player 1 in stage n.
Consider the stationary strategy σ_1 of Player 1, under which he plays action 1 with probability p, and action 0 with probability $1-p$ (independently of previous choices) in each stage. Then by Theorem 14.19, σ_1 approaches the set $R_1(p)$ where

$$\begin{aligned} R_1(p) &= \left\{(yp,(1-y)p,y(1-p),(1-y)(1-p)) : 0 \le y \le 1\right\} \\ &= \left\{y(p,0,1-p,0)+(1-y)(0,p,0,1-p) : 0 \le y \le 1\right\}. \end{aligned}$$

In particular, for every strategy σ_2 of Player 2

$$\mathbf{P}_{\sigma_1,\sigma_2}(\lim_{n\to\infty} d(\bar{g}^n, R_1(p)) = 0) = 1,$$

from which it can be deduced that

$$\mathbf{P}_{\sigma_1,\sigma_2}\left(\lim_{n\to\infty}\left(\tfrac{\bar{g}_3^n}{1-p} - \tfrac{\bar{g}_1^n}{p}\right) = 0\right) = 1. \tag{119}$$

Let X_n be the pure action chosen by Player 1 at stage n for every stage n according to the stationary strategy σ_1 of Player 1, under which he plays action 1 with probability p, and action 0 with probability $1-p$. Let Y_n be the mixed action taken by Player 2 at stage n (it may depends on $X_1, X_2, ..., X_{i-1}$).
Then

$$\bar{g}^n = \left(\tfrac{1}{n}\sum_{i=1}^n X_iY_i, \tfrac{1}{n}\sum_{i=1}^n X_i(1-Y_i), \tfrac{1}{n}\sum_{i=1}^n (1-X_i)Y_i, \tfrac{1}{n}\sum_{i=1}^n (1-X_i)(1-Y_i)\right).$$

Hence, by Equation (119)

$$\begin{aligned} 1 &= \mathbf{P}\left(\lim_{n\to\infty}\left(\tfrac{1}{(1-p)n}\sum_{i=1}^n (1-X_i)Y_i - \tfrac{1}{pn}\sum_{i=1}^n X_iY_i\right) = 0\right) \\ &= \mathbf{P}\left(\lim_{n\to\infty}\left(\tfrac{p}{n}\sum_{i=1}^n Y_i - \tfrac{1}{n}\sum_{i=1}^n X_iY_i\right) = 0\right) \\ &= \mathbf{P}\left(\lim_{n\to\infty}\left(\tfrac{1}{n}\sum_{i=1}^n Y_i - \tfrac{1}{pn}\sum_{i=1}^n X_iY_i\right) = 0\right) \\ &= \mathbf{P}\left(\lim_{n\to\infty}\left(\frac{1}{n}\sum_{i=1}^n Y_i - \frac{\sum_{\{i\le n : X_i=1\}} Y_i}{|\{j \le n : X_j = 1\}|}\right) = 0\right) \end{aligned}$$

where the last equality holds since $\lim_{n\to\infty} \frac{|\{j\leq n: X_j=1\}|}{n} = p$ (by the strong law of large number).

14.20 We prove that Blackwell's Theorem obtains even when for every $i \in \mathcal{I}$ and $j \in \mathcal{J}$, the payoff is a random variable $X(i,j)$ taking values in $\mathbf{R}^m$ with expectation $u(i,j)$, and Player 1 observes, after every stage t, the value of $X_t(i_t, j_t)$ and the action j_t of Player 2 (where $X_1, X_2, \ldots$ are independent random variables that are distributed like X).
Suppose that a set C is a B-set for player k in the repeated game where the payoff is a random variable $X(i,j)$ for every $(i,j) \in \mathcal{I} \times \mathcal{J}$. Denote by $\bar{g}^T$ the average payoff up to stage T in this game. Note that $\bar{g}^T = \frac{1}{T}\sum_{t=1}^T X_t(i_t, j_t)$ is a random variable even if the players play pure strategies.
Consider an auxiliary repeated game in which the payoff is $u(i,j) = E[X(i,j)]$ for every $(i,j) \in \mathcal{I} \times \mathcal{J}$. Denote by $\bar{h}^T$ the average payoff up to stage T in the auxiliary game.
Since C is a B-set for player k in the original game it is also a B-set for player k in the auxiliary game. By Theorem 14.14, there is a strategy σ_k for player k in the auxiliary game that approaches C. That is, $g(\bar{h}^T, C)$ converges to zero almost surely.
We show that σ_k approaches C for player k in the original game as well. Denote by $n^T_{i,j}$ the number of stages in which the players play the action vector (i,j) up to stage T. Note that

$$\bar{g}^T = \frac{1}{T}\sum_{t=1}^T X(i_t, j_t) = \sum_{(i,j)\in\mathcal{I}\times\mathcal{J}} \frac{n^T_{i,j}}{T} \times \frac{\sum_{\{t\geq T: (i_t,j_t)=(i,j)\}} X_t(i,j)}{n^T_{i,j}},$$

whereas

$$\bar{h}^T = \frac{1}{T}\sum_{t=1}^T u(i_t, j_t) = \sum_{(i,j)\in\mathcal{I}\times\mathcal{J}} \frac{n^T_{i,j}}{T} \times u(i,j).$$

By the strong law of large numbers, if $\lim_{T\to\infty} n^T_{i,j} = \infty$ then $\frac{\sum_{\{t\geq T: (i_t,j_t)=(i,j)\}} X_t(i,j)}{n^T_{i,j}}$ converges to $u(i,j)$ almost surely. Hence, $d(\bar{g}^T, \bar{h}^T)$ converges to zero almost surely.
Finally, by the the triangle inequality,

$$0 \leq d(\bar{g}^t, C) \leq d(\bar{g}^T, \bar{h}^T) + d(\bar{h}^T, C)$$

so $d(\bar{g}^T, C)$ converges to zero almost surely by σ_k. In particular, σ_k approaches C for player k in the original game.

14.21 A compact and convex set C is approachable by Player 2 if and only if

$$\mathrm{val}(G^T_\alpha) \geq \min_{x\in C} \langle \alpha, x\rangle, \quad \forall \alpha \in \mathbf{R}^m,$$

where G_α^T is the two-player zero-sum game (with real values payoffs) defined by the transpose of the game matrix G_α.
This is obtained by switching the role of the players.

14.22 Let $C \subseteq \mathbf{R}^m$ be a closed and convex set. We prove that C is approachable by player 1 if and only if $C \cap R_2(q) \neq \emptyset$ for every $q \in \Delta(\mathcal{J})$.
Suppose that C is approachable by player 1 and assume by contradiction that there is $q \in \Delta(\mathcal{J})$ such that $C \cap R_2(q) = \emptyset$. Hence, by Corollary 14.20, C is excludable by Player 2, which contradicts Theorem 14.8 according to which a set cannot be both approachable by one player and excludable by the other player.
Assume next that $C \cap R_2(q) \neq \emptyset$ for every $q \in \Delta(\mathcal{J})$. Then, for every half space H^+ containing C, $H^+ \cap R_2(q) \neq \emptyset$ for every $q \in \Delta(\mathcal{J})$. Therefore, by Theorem 14.21, there is $p \in \Delta(\mathcal{I})$ satisfying $R_1(p) \subseteq H^+$. Using Theorem 14.24 it follows that C is approachable by Player 1.

14.23 We prove Corollary 14.26 for closed and convex sets that are not compact: a closed and convex set C is approachable by Player 1 if and only if

$$\mathrm{val}(G_\alpha) \geq \inf_{x \in C} \langle \alpha, x \rangle, \quad \forall \alpha \in \mathbf{R}^m.$$

If C is approachable by Player 1 then $\{x \in C : \|x\| \leq M\}$ is also approachable by him (Exercise 14.5). Hence, by Corollary 14.26,

$$\mathrm{val}(G_\alpha) \geq \min_{x \in \{x \in C : \|x\| \leq M\}} \langle \alpha, x \rangle \geq \inf_{x \in C} \langle \alpha, x \rangle, \quad \forall \alpha \in \mathbf{R}^m.$$

On the other direction, assume that

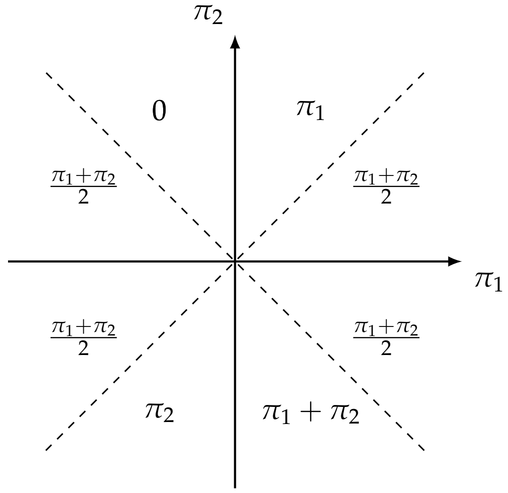
$$\mathrm{val}(G_\alpha) \geq \inf_{x \in C} \langle \alpha, x \rangle, \quad \forall \alpha \in \mathbf{R}^m.$$

Therefore, for every half space $H^+(\alpha, \beta)$ such that $C \subseteq H^+(\alpha, \beta)$ one has $\beta \leq \inf_{x \in C} \langle \alpha, x \rangle$. Thus, by the first part of the proof of Corollary 14.26, $R_1(p) \subseteq H^+(\alpha, \beta)$. In conclusion, by Theorem 14.24, C is approachable by Player 1.

14.24 (a) The game G_π for every $\pi = (\pi_1, \pi_2) \in \mathbf{R}^2$ is given by the following matrix.

	L	R
T	π_2	π_1
B	$\pi_1 + \pi_2$	0

(b)



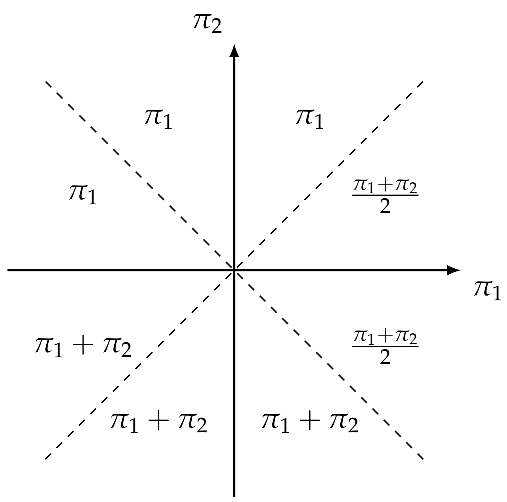
(c) All of the sets in the current part are compact and convex, hence it is enough to check Inequality (14.69) to determine whether the set is approachable or not.

i. $C_1 = \{(\frac{1}{2}, \frac{1}{2})\}$: $\min_{x\in S}\langle \pi, x\rangle = \frac{\pi_1+\pi_2}{2}$.
For $\pi_2 > \pi_1 > 0$, $\min_{x\in S}\langle \pi, x\rangle = \frac{\pi_1+\pi_2}{2} > \pi_1 = \mathrm{val}(G_\pi)$. In particular, C is not approachable by Player 1.

ii. $C_2 = [(0,1),(1,0)]$: $\min_{x\in S}\langle \pi, x\rangle = \min\{\pi_1, \pi_2\}$.
For every π, $\min_{x\in S}\langle \pi, x\rangle = \min\{\pi_1, \pi_2\} \leq \mathrm{val}(G_\pi)$. In particular, C is approachable by Player 1.

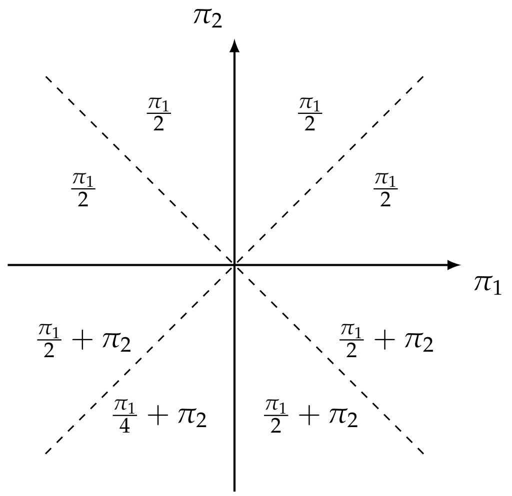
iii. $C_3 = [(0,1),(0,0)]$: $\min_{x\in S}\langle \pi, x\rangle = \min\{0, \pi_2\}$.
However, for $\pi_1 < \pi_2 < 0$, $\min_{x\in S}\langle \pi, x\rangle = \pi_2 > \frac{\pi_1+\pi_2}{2} = \mathrm{val}(G_\pi)$ so C is not approachable by Player 1.

iv. C_4 is the triangle whose vertices are $(1,1),(1,0)$, and $(\frac{1}{2}, \frac{1}{2})$: we present $\min_{x\in S}\langle \pi, x\rangle = \min\{\pi_1, \pi_2\}$ in the following graph.



For every π, $\min_{x\in S}\langle \pi, x\rangle \leq \mathrm{val}(G_\pi)$ and thus C is approachable by Player 1.

v. C_5 is the parallelogram whose vertices are $(\frac{1}{2},1)$, $(\frac{1}{4},1)$, $(\frac{1}{2},0)$, and $(\frac{3}{4},0)$:$\min_{x\in S}\langle \pi, x\rangle = \min\{\frac{\pi_1}{2}+\pi_2, \frac{\pi_1}{4}+\pi_2, \frac{\pi_1}{2}, \frac{3\pi_1}{4}\}$. In particular, an upper bounds over $\min_{x\in S}\langle \pi, x\rangle$ is presented in the following graph.



from which one can deduce that for every π, $\min_{x\in S}\langle \pi, x\rangle \leq \mathrm{val}(G_\pi)$ and thus C is approachable by Player 1.

14.25 Let $C \subset \mathbf{R}^m$ be a compact and convex set, and let $(x^t)_{t\in\mathbf{N}}$ be a bounded sequence of vectors in $\mathbf{R}^m$. Denote by $\bar{x}^t = \frac{1}{t}\sum_{j=1}^{t} x^j$ the average of the first t elements in the sequence $(x^t)_{t\in\mathbf{N}}$ and denote by $y(\bar{x}^t, C)$ the point in C that is closest to $\bar{x}^t$. Assume that for each t , the following inequality holds:

$$\langle \bar{x}^t - y(\bar{x}^t, C), x^{t+1} - y(\bar{x}^t, C)\rangle \leq 0, \quad \forall t \in \mathbf{N}. \tag{120}$$

(a) Suppose that $\bar{x}^t \notin C$. Hence, by the Separating Hyperplane Theorem (Theorem 23.39), the hyperplane

$$H(\bar{x}^t - y(\bar{x}^t, C), \langle \bar{x}^t - y(\bar{x}^t, C), y(\bar{x}^t, C)\rangle)$$

which tangent to C at $y(\bar{x}^t, C)$ separates $\bar{x}^t$ from C. In particular, $\langle \bar{x}^t - y(\bar{x}^t, C), \bar{x}^t)\rangle \leq \langle \bar{x}^t - y(\bar{x}^t, C), y(\bar{x}^t, C)\rangle$.
Therefore, $H(\bar{x}^t - y(\bar{x}^t, C), \langle \bar{x}^t - y(\bar{x}^t, C), y(\bar{x}^t, C)\rangle)$ separates x^{t+1} from $\bar{x}^t$ if and only if

$$\langle \bar{x}^t - y(\bar{x}^t, C), x^{t+1}\rangle \leq \langle \bar{x}^t - y(\bar{x}^t, C), y(\bar{x}^t, C)\rangle,$$

if and only if Inequality (120) holds, as needed.

(b) We next prove that $\lim_{t\to\infty} d(\bar{x}^t, y(\bar{x}^t, C)) = 0$.
Let $M = 2\max\{d(\overrightarrow{0}, C), \sup_{t\in\mathbf{N}} x_t\}$. Then,

$$\begin{aligned}
0 &\le d^2(\bar{x}^{t+1}, y(\bar{x}^{t+1}, C)) \le d^2(\bar{x}^{t+1}, y(\bar{x}^t, C)) \\
&= \langle \bar{x}^{t+1} - y(\bar{x}^t, C), \bar{x}^{t+1} - y(\bar{x}^t, C)\rangle \\
&= \tfrac{t}{t+1}\langle \bar{x}^t - y(\bar{x}^t, C), \bar{x}^{t+1} - y(\bar{x}^t, C)\rangle + \tfrac{1}{t+1}\langle x^{t+1} - y(\bar{x}^t, C), \bar{x}^{t+1} - y(\bar{x}^t, C)\rangle \\
&\le \tfrac{1}{t+1}\langle x^{t+1} - y(\bar{x}^t, C), \bar{x}^{t+1} - y(\bar{x}^t, C)\rangle \qquad (121)\\
&\le (\|x^{t+1}\| + \|y(\bar{x}^t, C)\|)(\|\bar{x}^{t+1}\| + \|y(\bar{x}^t, C)\|) \qquad (122)\\
&\le \tfrac{4M^2}{t+1} \xrightarrow[t\to\infty]{} 0.
\end{aligned}$$

where Inequality (121) follows Inequality (120) and Inequality (122) follows the triangle inequality. In conclusion, $\lim_{t\to\infty} d(\bar{x}^t, y(\bar{x}^t, C)) = 0$.

(c) In this exercise, the vectors (x^t) are arbitrary vectors, whereas in Blackwell's Theorem the vectors contains in the finite set $\{u(i,j) : (i,j) \in \mathcal{I} \times \mathcal{J}\}$.

14.26 Let C be a compact and convex set in $\mathbf{R}^m$. Let $(x^t)_{t=1}^{\infty}$ be a sequence of points in C satisfying

$$\langle \bar{x}^t, x^{t+1}\rangle \le \langle \bar{x}^t, z\rangle, \quad \forall z \in C.$$

We prove that $\lim_{t\to\infty} d(\bar{x}^t, \arg\min_{z\in C}\langle z, z\rangle) = 0$.
The function $z \mapsto \langle z, z\rangle$ for every $z \in C$ is a continuous function over a compact and convex set, so it attains its minimum in C so $\arg\min_{z\in C}\langle z, z\rangle \ne \emptyset$. Assume by contradiction that there are $z_1, z_2 \in \arg\min_{z\in C}\langle z, z\rangle$ such that $z_1 \ne z_2$. Then, by the convexity $\frac{z_1+z_2}{2} \in C$ and

$$\begin{aligned}
\langle \tfrac{z_1+z_2}{2}, \tfrac{z_1+z_2}{2}\rangle &= \tfrac{1}{4}\left(\langle z_1, z_1\rangle + \langle z_2, z_2\rangle + 2\langle z_1, z_2\rangle\right) \\
&< \left(\langle z_1, z_1\rangle + \langle z_2, z_2\rangle + 2\|z_1\|\|z_2\|\right) \\
&= \langle z_1, z_1\rangle,
\end{aligned}$$

which contradict the minimality of z_1. In conclusion, $\arg\min_{z\in C}\langle z, z\rangle$ contains a single point and hence it is compact and convex. Furthermore, by Inequality (14.129) it follows that Inequality (14.128) holds so by Exercise 14.25, $\lim_{t\to\infty} d(\bar{x}^t, \arg\min_{z\in C}\langle z, z\rangle) = 0$.

14.27 (a) Let $C \subseteq \mathbf{R}^m$ be a convex and closed set. Suppose that for every $k \in \mathbf{N}$ the set C is a B-set for Player 1 in $G(k)$. We prove that C is approachable by Player 1 in the game $\widehat{G}$.

We will first define a strategy σ_1^* for Player 1, and then prove that it guarantees that the average payoff approaches the set C. In the first stage, the strategy σ_1^* chooses any action. For each $t \geq 1$ the strategy σ_1^* instructs Player 1 to play as follows in stage $t+1$: If the average payoff $\bar{x}^t$ up to stage t is in C, the definition of σ_1^* is immaterial (play any action). If $\bar{x}^t \notin C$, the strategy σ_1^* instructs the player to choose the mixed action $p(\bar{x}^t, C)$ (as defined in Definition 14.12).

For every $t \in \mathbf{N}$, if $\bar{x}^t \notin C$ then, since C is a B-set of for Player 1 in $G(t)$,

$$x^{t+1} \in R(p(\bar{x}^t, C)) \subseteq H^+(y(\bar{x}^t, C) - \bar{x}^t, \langle y(\bar{x}^t, C) - \bar{x}^t, y(\bar{x}^t, C)\rangle).$$

That is,

$$\langle \bar{x}^t - y(\bar{x}^t, C), x^{t+1}\rangle \leq \langle \bar{x}^t - y(\bar{x}^t, C), y(\bar{x}^t, C)\rangle.$$

so x^{t+1} satisfies Equation (14.128) and thus, by Exercise 14.25, $\lim_{t\to\infty} d(\bar{x}^t, C) = 0$.

(b) We show that the result in the previous item does not necessarily hold if C is not convex. Let $m = 1$. Suppose that $u(2k-1) = 0$ is a constant function (for every strategy of the players) and $u(2k) = 1$ is a constant function as well, for every $k \in \mathbf{N}$. Let $C = \{0, 1\}$. Then, C is a B-set in $G(k)$ for every k since $F_k \subseteq C$ for every k, but C is not approachable by any player since $\bar{x}^t \to \frac{1}{2}$.

14.28 Consider the game described in Section 14.7.

(a) In the one-stage game, Player 1 can guarantee that the payoff is at least $\frac{1}{2}$ by Playing T in $s = 1$ and B in $s = 2$. Under this strategy, given Player 2 plays $[y(L), (1-y)(R)]$, the payoffs are $\frac{1}{2} \cdot y + \frac{1}{2} \cdot (1-y) = \frac{1}{2}$. On the other hand, Player 2 can guarantee that the payoff is at most $\frac{1}{2}$ by playing $[(\frac{1}{2}(L), \frac{1}{2}(R)]$.

(b) We prove the value of the two-stage game is $\frac{3}{8}$.

The optimal strategy of Player 1 is: at the first stage, play $[\frac{1}{2}(T), \frac{1}{2}(B)]$ (which lead to a stage payoff of $\frac{1}{4}$). At the second stage, if $s = 1$ play T and if $s = 2$ play B (which leads to a stage payoff of $\frac{1}{2}$) thus the payoff is $\frac{1}{2}(\frac{1}{4} + \frac{1}{2}) = \frac{3}{8}$.

The optimal strategy of Player 2 is : at the first stage, play $[\frac{1}{2}(L), \frac{1}{2}(R)]$ (which lead to a stage payoff of $\frac{1}{2}$). At the second stage, play R if at the first stage (T, L) was played, play L if at the first stage (B, R) was played, otherwise, play $[\frac{1}{2}(L), \frac{1}{2}(R)]$. Suppose that if $s = 1$ then at the first stage Player 1 plays T with probability x and if $s = 2$ he plays T with probability y. Then the average payoff is at most

$$\mathbf{P}(s=1)\left(\tfrac{1}{2}(x\cdot\tfrac{1+0}{2}+(1-x)\cdot\tfrac{1}{2}\cdot\tfrac{0+1}{2})+\tfrac{1}{2}(x\cdot\tfrac{1}{2}\cdot\tfrac{0+1}{2}+(1-x)\cdot\tfrac{0+1}{2})\right)$$
$$+\mathbf{P}(s=2)\left(\tfrac{1}{2}(y\cdot\tfrac{0+1}{2}+(1-y)\cdot\tfrac{1}{2}\cdot\tfrac{0+1}{2})+\tfrac{1}{2}(y\cdot\tfrac{1}{2}+(1-y)\cdot\tfrac{1+0}{2})\right)=\tfrac{3}{8}.$$

(c) Denote by v_T the value of the T -stage repeated game. By Proposition 14.37, the uniform value of the game is $\frac{1}{4}$. Hence, by Theorem 14.33, $\lim_{T\to\infty} v^T = \frac{1}{4}$.

14.29 (a) For $p = 0$ and $p = 1$, both players knows which game is played. Hence, Player 2 can guarantee that the payoff is at most 0 by playing R if $s = 1$ and L if $s = 2$. Hence, since 0 is the minimal payoff in the game, the value is 0. That is, $v(0) = v(1) = 1$.

(b) Player 1, knowing s, can guarantee $p(1-p)$ by not using his information, and playing the same mixed action $[(1-p)(T), p(B)]$ in every stage. The payoff at stage $t \in \mathbf{N}$ given Player 2 plays L with probability y^t is $p\cdot(1-p)\cdot y^t + (1-p)\cdot p\cdot(1-y^t) = p(1-p)$ so the average payoff is $p(1-p)$.

(c) Define

$$C := \operatorname{conv}\{(0,0),(\tfrac{1-p}{2},0),(0,\tfrac{p}{2}),(\tfrac{1-p}{2},\tfrac{p}{2})\}.$$

Suppose that C is approachable by Player 2 in the auxiliary repeated game with vector payoff. That is, Player 2 has a strategy σ^2 guaranteeing that in the T stage repeated game the action vector (T, L) will be played at most $\frac{(1-p)T}{2}+1$ stages and the action (B, R) will be played at most $\frac{pT}{2}+1$ stages, for every strategy of Player 1, given T is sufficiently large. Hence, if Player 2 follows σ in the original game, then the action vector (T, L) will be played at most $\frac{(1-p)T}{2}+1$ stages and the action vector (B, R) will be played at most $\frac{pT}{2}+1$ stages and thus the average payoff is at most

$$\mathbf{P}(s=1)\cdot\tfrac{1}{T}\cdot(\tfrac{(1-p)T}{2}+1)+\mathbf{P}(s=2)\cdot\tfrac{1}{T}\cdot(\tfrac{pT}{2}+1)\xrightarrow[T\to\infty]{} p(1-p),$$

so $v(p) \le p(1-p)$.

(d) We prove that C is a B-set for Player 2 from which it follows that C is approachable by Player 2 in the auxiliary repeated game with vector payoff.
Let $x = (x_1, x_2) \in F \setminus C$.

- If $x_1 \le \frac{1-p}{2}$ and $x_2 \ge \frac{p}{2}$ then the closest point in C to x is $y = (x_1, \frac{p}{2})$ and the hyperplane separating x and C that orthogonal to $y - x$ is $x_2' = \frac{p}{2}$ and it separates x and $R_2(1)$ (corresponding to the action L).
- If $x_1 \ge \frac{1-p}{2}$ and $x_2 \le \frac{p}{2}$ then the closest point in C to x is $y = (\frac{1-p}{2}, x_2)$ and the hyperplane separating x and C that orthogonal to $y - x$ is $x_1' = \frac{1-p}{2}$ and it separates x and $R_2(1)$ (corresponding to the action L).
- Otherwise, $x_1 > \frac{1-p}{2}$ and $x_2 > \frac{p}{2}$ so the closest point in C to x is $y = (\frac{1-p}{2}, \frac{p}{2})$. Hence, the slope of the hyperplane separating x and C that orthogonal to $y - x$ is $-\frac{x_1 - \frac{1-p}{2}}{x_2 - \frac{p}{2}}$ and it intersects the y-axis at $\frac{p}{2} + \frac{1-p}{2} \cdot \frac{x_1 - \frac{1-p}{2}}{x_2 - \frac{p}{2}}$. However, for $q = \frac{x_2 - \frac{p}{2}}{x_1 - \frac{1-p}{2} + x_2 - \frac{p}{2}} \in (0,1)$, the slope of $R_2(q)$ equals the slope of the separating hyperplane. Furthermore, assume without loss of generality that $p \le \frac{1}{2}$, then by

$$0 \le \quad \frac{p}{2}\left((x_1 - \tfrac{1-p}{2}) - (x_2 - \tfrac{p}{2})\right)^2 + \left(\tfrac{1-p}{2} - \tfrac{p}{2}\right)(x_1 - \tfrac{1-p}{2})^2.$$

it follows that $R_2(q)$ intersects the y-axis at $1 - q = \frac{x_1 - \frac{1-p}{2}}{x_1 - \frac{1-p}{2} + x_2 - \frac{p}{2}}$ which is lower than the intersection point of the hyperplane. Thus, the hyperplane separating x and C also separates x and $R_2(q)$.

(e) By Part (d), the set C is approachable by Player 2, hence, by Part (c), $v(p) \le p(1-p)$. Hence and by Part (b) and (a), it can be deduced that $v(p) = p(1-p)$ for every $p \in [0,1]$.

(f) For every $\epsilon > 0$, the following strategy for Player 2 guarantees $p(1-p) + \epsilon$: at the first stage play $[(1-p)(L), p(R)]$ (or any other action). For every $t \in \mathbf{N}$, let x^t and y^t be the proportions of stages in which (T, L) and (B, R) respectively are played in the first t stages. Then at stage $t+1$, if $x^t \ge \frac{1-p}{2}$ and $y^t \le \frac{p}{2}$ then play R; if $x^t \le \frac{1-p}{2}$ and $y^t \ge \frac{p}{2}$ then play L; otherwise, play $[q(L), (1-q)(R)]$ where $q = \frac{x_2 - \frac{p}{2}}{x_1 - \frac{1-p}{2} + x_2 - \frac{p}{2}}$.

14.30 (a) Denote by $w(p)$ the value of the game in which Player 1 is restricted to play the strategies in $\mathcal{B}_1^*$; in other words, he does not make use of his information regarding the state of nature. Then, the players are actually playing the following zero-sum game.

	L	R
T	$-p$	0
B	0	$-(1-p)$

in which Player 1 can guarantee a payoff of $-p(1-p)$ by playing $[(1-p)(T), p(B)]$ and Player 2 can guarantee it by playing $[(1-p)(L), p(R)]$.

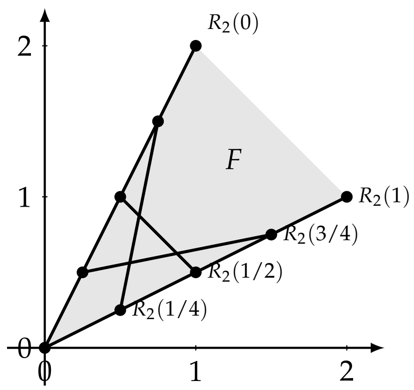
(b) Player 1 can guarantee 0 by playing B if $s = 1$ and T if $s = 2$. Since the maximal payoff in the game is 0, every strategy of Player 2 guarantees him 0. Therefore, the uniform value of the game is $v(p) = 0$, for every $p \in [0,1]$. Note that, since the function that assign to each state the optimal pure action of Player 1 is a bijection, then Player 1 reveals the true state by playing it: by playing B in $s = 1$ Player 1 reveals that $s = 1$ whereas by playing T in $s = 2$ he reveals that the state is $s = 2$.

14.31 (a) Player 1 can guarantee $\frac{(2-p)(1+p)}{3}$ by playing $[\frac{2-p}{3}(T), \frac{1+p}{3}(B)]$ and Player 2 can guarantee it by playing $[\frac{2-p}{3}(L), \frac{1+p}{3}(R)]$.

(b) The corresponding repeated game with vector payoffs G_V is given in the next matrix.

	L	R
T	$(2,1)$	$(0,0)$
B	$(0,0)$	$(1,2)$

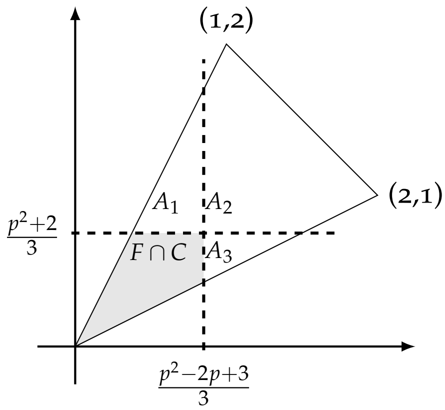
(c) Note that $R_2(q) = [(2q,q),(1-q,2(1-q)]$.



(d) Let $p \in [0,1]$. Set

$$C := \text{conv}\left\{(0,0), \left(0, \tfrac{p^2+2}{3}\right), \left(\tfrac{p^2-2p+3}{3}, 0\right), \left(\tfrac{p^2-2p+3}{3}, \tfrac{p^2+2}{3}\right)\right\}.$$

We prove that C is a B-set for Player 2.



Let $x \in F \setminus C$.

- If $x \in A_1$ then the closest point in C to x is $y = (x_1, \frac{p^2+2}{3})$. The hyperplane separating x and C that orthogonal to $y - x$ is $x_2' = \frac{p^2+2}{3}$ and it separates x and $R_2(\frac{p^2+2}{3})$ (since $q = \frac{p^2+2}{3} \geq \frac{2}{3}$ and thus $2(1-q) \leq q \leq \frac{p^2+2}{3}$).
- If $x \in A_3$ then the closest point in C to x is $y = (\frac{p^2-2p+3}{3}, x_2)$. The hyperplane separating x and C that orthogonal to $y - x$ is $x_1' = \frac{p^2-2p+3}{3}$ and it separates x and $R_2(\frac{p^2-2p+3}{6})$ (since $2q = \frac{p^2-2p+3}{3} \geq \frac{1}{3}$ and thus $1 - q \leq 2q \leq \frac{p^2-2p+3}{3}$).
- Otherwise, $x \in A_2$ so the closest point in C to x is $y = (\frac{p^2-2p+3}{3}, \frac{p^2+2}{3})$. Hence, the slope of the hyperplane separating x and C that orthogonal to $y - x$ is $a = -\frac{x_1 - \frac{p^2-2p+3}{3}}{x_2 - \frac{p^2+2}{3}} < 0$ and it intersects the y-axis at $b = \frac{p^2+2}{3} - \frac{p^2-2p+3}{3} \cdot a$. However, for $q = \frac{2-a}{3(1-a)} \in [0,1]$, the slope of $R_2(q)$ equals the slope of the separating hyperplane. Furthermore, the hyperplane containing $R_2(q)$ is $w_2 = a \cdot w_1 + q(1-2a)$ which intersects the y-axis at $q(1-2a) < b$. In particular, the hyperplane separating x and C also separates x and $R_2(q)$.

(e) By Part (d), Player 2 can guarantee $p \cdot \frac{p^2-2p+3}{3} + (1-p) \cdot \frac{p^2+2}{3} = \frac{(2-p)(1+p)}{3}$. Hence and by Part (a), $v(p) = \frac{(2-p)(1+p)}{3}$

14.32 (a) When Player 1 does not use the information in his possession, then the players play the game

	L	M	R
T	2	2	0
B	2	2	0

In this game Player 2 can guarantee 0 by playing R. In addition, since 0 is the minimal stage payoff, every strategy of player 1 guarantee at least 0.

(b) If Player 2 also knows the state of nature, then Player 2 can play M if $s = 1$ and L if $s = 2$ so the payoff is 0. Again, since 0 is the minimal payoff, Player 1 guarantees that the payoff is at least 0, so 0 is the value of this game.

(c) Returning to the game in which only Player 1 knows the state of nature, define the strategy $\widehat{\sigma}_1$ of Player 1 as follows: In the first stage, if $s = 1$, play the mixed action $[\frac{3}{4}(T), \frac{1}{4}(B)]$. If $s = 2$, play the mixed action $[\frac{1}{4}(T), \frac{3}{4}(B)]$. In each stage $t > 1$, repeat the action that was played in the first stage; i.e., play either T in every stage, or B in every stage, according to the result of the lottery in the first stage.

Then

$$\mathbf{P}(s = 1 | i^1 = T) = \frac{\mathbf{P}(i^1{=}T|s{=}1)\mathbf{P}(s{=}1)}{\mathbf{P}(i^1{=}T|s{=}1)\mathbf{P}(s{=}1)+\mathbf{P}(i^1{=}T|s{=}2)\mathbf{P}(s{=}2)} = \frac{\frac{3}{4}\cdot\frac{1}{2}}{\frac{3}{4}\cdot\frac{1}{2}+\frac{1}{4}\cdot\frac{1}{2}} = \frac{3}{4},$$

$$\mathbf{P}(s = 1 | i^1 = B) = \frac{\mathbf{P}(i^1{=}B|s{=}1)\mathbf{P}(s{=}1)}{\mathbf{P}(i^1{=}B|s{=}1)\mathbf{P}(s{=}1)+\mathbf{P}(i^1{=}B|s{=}2)\mathbf{P}(s{=}2)} = \frac{\frac{1}{4}\cdot\frac{1}{2}}{\frac{1}{4}\cdot\frac{1}{2}+\frac{3}{4}\cdot\frac{1}{2}} = \frac{1}{4}.$$

(d) We show that for every strategy of Player 2, the conditional expectation of the average payoff $\bar{g}^t$, given the action of Player 1 in the first stage, is at least 1.

Assume first that the action of Player 1 in the first stage is T. Let y_1^t, y_2^t and y_3^t be the probabilities that at stage t Player 2 plays L, M and R, respectively. Hence, by Part (c), at each stage the conditional expected stage payoff of Player 2 is:

$$g^t = \tfrac{3}{4}(4y_1^t + 2y_3^t) + \tfrac{1}{4}(4y_2^t - 2y_3^t) = y_1^t + y_2^t + y_3^t = 1.$$

Similarly, suppose that the action of Player 1 in the first stage is B, and let z_1^t, z_2^t and z_3^t be the probabilities that at stage t Player 2 plays L, M and R, respectively. Hence, by Part (c), at each stage the conditional expected stage payoff of Player 2 is:

$$g^t = \tfrac{1}{4}(4y_1^t - 2y_3^t) + \tfrac{3}{4}(4y_2^t + 2y_3^t) = y_1^t + y_2^t + y_3^t = 1.$$

Therefore, the conditional expectation of the average payoff $\bar{g}^t$, given the action of Player 1 in the first stage is at least 1.

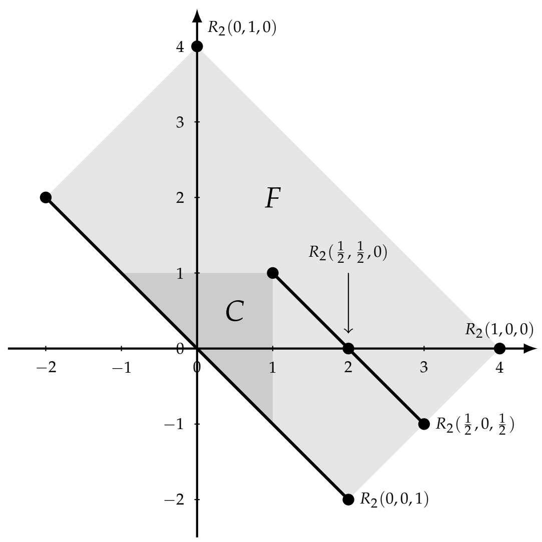
In conclusion, Player 1 can guarantee at least 1 in the infinitely repeated game.

(e) Consider the repeated game with vector payffs given by the matrix:

	L	M	R
T	(4,0)	(0,4)	(2,-2)
B	(4,0)	(0,4)	(-2,2)

Suppose that the set $C = [0,1]^2$ is approachable by Player 2. Then Player 2 has a strategy guaranteeing that the payoff of both component are at most 1. Therefore, by following this strategy in the original game, the average payoff is at most $p \cdot 1 + (1 - p) \cdot 0 = 1$. In conclusion, the uniform value of the game is at most 1.

(f)



(g) Let $x = (x_1, x_2) \in F \setminus C$ and let y be closest point in C to x. If $x_1 \le x_2$ (that is, x is a point on (or above) the diagonal $x_1 = x_2$) then the hyperplane separating x and C that is orthogonal to $y - x$ also separates x and $R_2(\frac{1}{2}, 0, \frac{1}{2}) = [(3,-1),(1,1)]$. If $x_1 > x_2$ (that is, x is a point below the diagonal $x_1 = x_2$) then the hyperplane separating x and C that is orthogonal to $y - x$ also separates x and $R_2(0, \frac{1}{2}, \frac{1}{2}) = [(-1,3),(1,1)]$. Hence, C is approachable by Player 2.

(h) Player 1 reveals information on the state of nature at the first stage only, but he does not entirely reveal the state of nature.

14.33 (a) Player 1 can guarantee $w(p)$ by ignoring his knowledge and playing the optimal strategy of $D(p)$ at each stage in $G(p)$ no matter what is the true state.

(b) Let $p^1, p^2, \ldots, p^L \in \Delta(S)$, and let $\alpha = (\alpha^l)_{l=1}^L$ be a distribution over $\{1,2,\ldots,L\}$ satisfying $p = \sum_{l=1}^L \alpha^l p^l$. Consider the following strategy σ_1^* of Player 1: if u_{k_0} is the payoff function that has been selected in the chance move at the beginning of the game, randomly choose $l_0 \in \{1,2,\ldots,L\}$ according to the following distribution $\lambda_{k_0} = (\lambda_{k_0}^1, \lambda_{k_0}^2, \ldots, \lambda_{k_0}^L)$:

$$\lambda_{k_0}^l = \frac{\alpha^l p_{k_0}^l}{\sum_{l'=1}^L \alpha^{l'} p_{k_0}^{l'}} = \frac{\alpha^l p_{k_0}^l}{p_{k_0}}.$$

In each stage of the game, play the optimal strategy in the game $D(p^{l_0})$. We will prove that this strategy guarantees Player 1 the payoff $\sum_{l=1}^L \alpha^l w(p^l)$ in $G(p)$. To accomplish this, consider the game $G'(p)$, a variation of $G(p)$, where Player 2 is informed of the index l_0 that Player 1 chose. After Player 2 knows l_0, the conditional probability that the chosen payoff matrix is u_k, given l_0, equals

$$\frac{\lambda_k^{l_0} p_k}{\sum_{k'=1}^K \lambda_{k'}^{l_0} p_{k'}} = \frac{\frac{\alpha^{l_0} p_k^{l_0}}{p_k} p_k}{\sum_{k'=1}^K \frac{\alpha^{l_0} p_{k'}^{l_0}}{p_{k'}} p_{k'}} = \frac{p_k^{l_0}}{\sum_{k'=1}^K p_{k'}^{l_0}} = p_k^{l_0}.$$

Hence, by Part (a), in $G'(p)$, if l_0 is chosen then the player are actually playing the game $G(p^{l_0})$ in which Player 1 can guarantee a payoff $w(p^{l_0})$ by following the strategy σ_1^*. In conclusion. The strategy σ_1^* guarantees Player 1 the payoff $\sum_{l=1}^L \alpha^l w(p^l)$.

(c) In the game $G(p)$, Player 2 has fewer strategies than he has in $G'(p)$ (because he has more information in $G'(p)$) whereas Player 1 can implement each strategy of $G'(p)$ in $G(p)$. Thus, by Exercise 4.27, in $G(p)$, the strategy σ_1^* guarantees Player 1 the payoff $\sum_{l=1}^L \alpha^l w(p^l)$.

(d) The function cav w is the smallest concave function that is pointwise greater than or equal to w. That is, for every $(p, (\text{cav } w)(p))$

there are L, $p^1, ..., p^L \in \Delta(S)$ and $(\alpha_1, ..., \alpha_l) \in \Delta(\{1, 2, ...L\})$ such that $p = \sum_{l=1}^{L} \alpha_l p^l$ and $(\text{cav } w)(p) = \sum_{l=1}^{L} \alpha_l w(p^l)$. Hence, by Part (c), Player 1 can guarantee $(\text{cav } w)(p)$ in $G(p)$.

(e) We next prove that Player 2 can also guarantee $(\text{cav } w)(p)$ in $G(p)$.

Set $Z := \{z \in \mathbf{R}^K : \langle q, z \rangle \geq w(q) \quad \forall q \in \Delta(S)\}$. We first prove that $(\text{cav } w)(p) = \min_{z \in Z} \langle p, z \rangle$.

For every $(p, (\text{cav } w)(p))$ there are L, $p^1, ..., p^L \in \Delta(S)$ and $(\alpha_1, ..., \alpha_l) \in \Delta(\{1, 2, ..., L\})$ such that $p = \sum_{l=1}^{L} \alpha_l p^l$ and $(\text{cav } w)(p) = \sum_{l=1}^{L} \alpha_l w(p^l)$. Hence, for every $z \in Z$.

$$(\text{cav } w)(p) = \sum_{l=1}^{L} \alpha_l w(p^l) \leq \sum_{l=1}^{L} \alpha_l \langle z, p^l \rangle = \langle z, \sum_{l=1}^{L} \alpha_l p^l \rangle = \langle z, p^l \rangle.$$

Furthermore, since $\Delta(S)$ is a convex set and the function cav w is concave function, there is a supporting hyperplane in $(p, (\text{cav } w)(p))$. That is, there is $z' \in Z$ such that $(\text{cav } w)(p) = \langle z', p \rangle$ and $(\text{cav } w)(p) \leq \langle z, p \rangle$ for every $z \in Z$. In conclusion,

$$(\text{cav } w)(p) = \langle z', p \rangle = \min_{z \in Z} \langle z, p \rangle.$$

(f) For every $z \in Z$, define $M_z := z - \mathbf{R}^K_+$. The set M_z is closed and convex so, by reversing the roles of the players in Corollary 14.26, it is approachable by Player 2 if and only if

$$\text{val}(G_\pi) = \max_{y \in \Delta(\mathcal{J})} \min_{x \in \Delta(\mathcal{I})} U_\pi(x, y) \geq \min_{z' \in M_z} \langle \pi, z' \rangle \quad \forall \pi \in \mathbf{R}^K.$$

(g) Equation (14.136) obtains for $\pi = \overrightarrow{0}$ since both sides of the equation equal 0.

(h) Let $\pi \neq \overrightarrow{0}$ satisfy the property that there exists $k \in S$ such that $\pi_k > 0$. For every integer $r < \min\{0, z_k\}$ there is $z^r \in M_z$ such that $z_k^R = r$ and $z_{k'}^r = 0$ for every $k' \neq k$. Therefore,

$$\langle \pi, z^r \rangle = r \cdot \pi_k \xrightarrow[r \to -\infty]{} -\infty.$$

In particular, $\min_{z' \in M_z} \langle \pi, z' \rangle = -\infty$. On the other hand, following Nash-Theorem $\text{val}(G_\pi)$ is finite and thus Equation (14.136) holds.

(i) Let $\pi \neq \overrightarrow{0}$ satisfying $\pi_k \leq 0$ for all $k \in S$. Notes that from the last two parts, M_z is approachable if and only if Equation (14.136) holds for every $\pi \in \mathbf{R}^k$ satisfying the above property.

Define $\widehat{q}_k = \frac{\pi_k}{\sum_{k'=1}^{K} \pi_{k'}}$. By the property of π, $\pi_k \leq 0$ for every $k \in S$ and there is $k \in S$ such that $\pi_k < 0$ so $-\sum_{k'=1}^{K} \pi_{k'} > 0$

and $-\sum_{k'=1}^{K}\pi_{k'} \geq -\pi_k$. Hence, $0 \leq \frac{\pi_k}{\sum_{k'=1}^{K}\pi_{k'}} = \widehat{q}_k$, and $\sum_{k=1}^{K}\widehat{q}_k = \sum_{k=1}^{K}\frac{\pi_k}{\sum_{k'=1}^{K}\pi_{k'}} = 1$ which leads to $\widehat{q}_k \in \Delta(S)$.

(j) Equation (14.136) obtains for every π satisfying the property of Part (i), if and only if

$$\frac{1}{\sum_{k'=1}^{K}\pi_{k'}} \cdot \max_{y\in\Delta(\mathcal{J})}\min_{x\in\Delta(\mathcal{I})} U_\pi(x,y) \leq \frac{1}{\sum_{k'=1}^{K}\pi_{k'}} \cdot \min_{z'\in M_z}\langle \pi, z'\rangle \Leftrightarrow$$
$$\min_{y\in\Delta(\mathcal{J})}\max_{x\in\Delta(\mathcal{I})} U_{\widehat{q}}(x,y) \leq \max_{z'\in M_z}\langle \widehat{q}, z'\rangle$$

(k) Since M_z is an infinite rectangle bounded from above, the function that assign $z' \mapsto \langle \widehat{q}, z'\rangle$ for each $z' \in M_z$ attains its maximum at an extreme point of M_z. However, z is a unique extreme point of M_z so $\max_{z'\in M_z}\langle \widehat{q}, z'\rangle = \langle \widehat{q}, z\rangle$.

(l) The left-hand side of Equation (14.137) represents the value of the game $G_{\widehat{q}}$ which is a two-player zero-sum game (with real values payoffs) in which Player 1's set of pure strategies is $\mathcal{I}$, Player 2's set of pure strategies is$\mathcal{J}$, and the payoff function is $U_{\widehat{q}}(i,j) = \langle \widehat{q}, u(i,j)\rangle = \sum_{k=1}^{K}\widehat{q}_k u_k(i,j)$. That is, $G_{\widehat{q}} = D(\widehat{q})$. Therefore, the left-hand side of Equation (14.137) equals $w(\widehat{q})$.

(m) M_z is approachable by Player 2 if and only if (Part (f))

$$\mathrm{val}(G_\pi) = \max_{y\in\Delta(\mathcal{J})}\min_{x\in\Delta(\mathcal{I})} U_\pi(x,y) \geq \min_{z'\in M_z}\langle \pi, z'\rangle \quad \forall \pi \in \mathbf{R}^K,$$

if and only if (Parts (g)-(j))

$$\min_{y\in\Delta(\mathcal{J})}\max_{x\in\Delta(\mathcal{I})} U_{\widehat{q}}(x,y) \leq \max_{z'\in M_z}\langle \widehat{q}, z'\rangle$$

if and only if (Parts (k)and (l)) $w(\widehat{q}) \leq \langle \widehat{q}, z\rangle$ for every $\widehat{q} \in \Delta(\mathcal{J})$, if and only if $z \in Z$.

(n) By Part (m), M_z is approachable for every $z \in Z$, in particular, M_{z_0} is approachable where $z_0 = \arg\min_{z\in Z}\langle z, p\rangle$ (by Part (e), there is such z_0).

(o) By Part (n), there is a strategy of Player 2 that approaches M_{z_0} in G_V. Furthermore, using Part (e) it follows that $(\mathrm{cav}\ w)(p) = \langle z_0, p\rangle$, this strategy of Player 2 guarantees him $(\mathrm{cav}\ w)(p)$ in $G(p)$.

14.34 We prove Theorem 14.44 for any set E of experts.
Define auxiliary games with vector payoffs $(G^t)_{t=1}^{\infty}$. In each auxiliary game, Player 1 represents the decision maker whose set of actions is E, and Player 2 represents nature, whose set of actions is S. The

payoffs are $|E|$-dimensional and the payoff function w^t is defined as follows: the vector payoff $w^t(e,s) = (w^t_{e'}(e,s))_{e' \in E}$ that obtains when Player 1 implements action e and Player 2 implements s is given by $w^t_{e'}(e,s) := u(a^t_e, s) - u(a^t_{e'}, s)$.

Let $\widehat{G}$ be the game in which the payoffs change from one stage to another: the payoffs in stage t are given by the function u^t.

A strategy for Player 1 in this auxiliary game with vector payoffs is a function $\widehat{\sigma} : \bigcup_{t=1}^{\infty}(E \times S)^{t-1} \to \Delta(E)$. Every strategy in this auxiliary game defines a strategy in the decision problem with experts.

Denote by $C = \{x \in \mathbf{R}^E : x_i \geq 0 \ \ \forall i \in E\}$ the nonnegative orthant in $\mathbf{R}^E$.

By Theorem 14.45, C is approachable by Player 1 in each repeated game with vector payoff G^t. Hence, using Exercise 14.27 since C is closed and convex, C is aproachable by Player 1 in $\widehat{G}$ as well.

Let σ be the strategy of Player 1 that approaches C in $\widehat{G}$. Then, for every $\epsilon > 0$ there is $T \in \mathbf{N}$ such that for strategy τ of Player 2,

$$\mathbf{P}_{\sigma,\tau}\left(\lim_{T\to\infty} d\left(\tfrac{1}{T}\sum_{t=1}^{T} w^t(e^t,s^t), C\right) = 0\right) = 1.$$

Since $w^t_{e'}(e^t,s^t) = u(a^t_e,s) - u(a^t_{e'},s)$, and since C is the nonnegative orthant, we deduce that for every $e' \in E$,

$$\begin{aligned} 1 &= \mathbf{P}_{\sigma,\tau}\left(\liminf_{T\to\infty} \tfrac{1}{T}\sum_{t=1}^{T} w^t_{e'}(e^t,s^t) \geq 0\right) \\ &= \mathbf{P}_{\sigma,\tau}\left(\liminf_{T\to\infty} \tfrac{1}{T}\sum_{t=1}^{T} u(a^t_e,s) \geq \liminf_{T\to\infty} \tfrac{1}{T}\sum_{t=1}^{T} u(a^t_{e'},s)\right). \end{aligned}$$

This equation holds for every strategy τ of Player 2, and in particular for every sequence of states $(s^1,s^2,...)$ (constituting a pure strategy) in which Player 2 plays the action s^t in stage t. This states precisely that σ is a no-regret strategy.

14.35 Consider a decision problem without experts given by a set A of actions, A finite set S of states and a payoff function $u : A \times S \to \mathbf{R}$. A strategy σ of a decision maker in such a decision problem is a no-regret strategy if for every action $\widehat{a} \in A$, and every sequence $(s^1,s^2,...)$ of states of nature,

$$\mathbf{P}_{\sigma}\left(\liminf_{T\to\infty} \tfrac{1}{T}\sum_{t=1}^{T} u(a^t,s^t) - \tfrac{1}{T}\sum_{t=1}^{T} u(\widehat{a}^t,s^t) \geq 0\right) = 1.$$

We show that a game without an experts has a no-regret strategy. Consider a decision problem with experts given by A, S and u as defined above and a set of $|A|$ experts where each expert is identified with different action.

Following Theorem 14.44, the decision maker has a no-regret strategy $\widehat{\sigma}$ in the decision problem with experts. Hence, the strategy in the decision problem without experts corresponding σ according to which at stage t the decision maker follows the action that is identified the expert under σ is a no-regret strategy.

14.36 Consider a decision problem with experts, where $E = A$, and for every action $a \in A$ there exists an expert $e \in E$ such that at $e = a$ for every $t \in \mathbf{N}$. Define a game with vector payoffs, where the set of actions of Player 1 is A, the set of actions of Player 2 is S, and the payoffs are $(|S|+1)$-dimensional vectors $(w(s,a))_{a\in A}^{s\in S}$, where $w(s,a) = ((w_{s'}(s,a))_{s'\in S}, U(s,a))$, defined as follows: $w_{s'}(s,a) = 1$ if $s' = s$, and $w_{s'}(s,a) = 0$ if $s' \neq s$. Define a set $C \subset \mathbf{R}^{|S|+1}$ as follows:

$$C := \{(q,x) \in \Delta(S) \times \mathbf{R} : x \geq \max_{p\in\Delta(A)} U(q,p)\}.$$

(a) We prove that C is a convex set. Let $(q_1,x_1),(q_2,x_2) \in C \subseteq \Delta(S) \times \mathbf{R}$ and let $\alpha \in (0,1)$.
$\Delta(S)$ and $\mathbf{R}$ are convex so $\Delta(S) \times \mathbf{R}$ is also convex. In particular,

$$\alpha(q_1,x_1) + (1-\alpha)(q_2,x_2) = (\alpha q_1 + (1-\alpha)q_2, \alpha x_1 + (1-\alpha)x_2) \in \Delta(S) \times \mathbf{R}.$$

In addition,

$$\max_{p\in\Delta(A)} U(p,\alpha q_1 + (1-\alpha)q_2) = \max_{p\in\Delta(A)} (\alpha U(p,q_1) + (1-\alpha)U(p,q_2))$$
$$\leq \alpha \max_{p\in\Delta(A)} U(p,q_1) + (1-\alpha) \max_{p\in\Delta(A)} U(p,q_2)) \leq \alpha x_1 + (1-\alpha)x_2.$$

In conclusion, $\alpha(q_1,x_1) + (1-\alpha)(q_2,x_2) \in C$.

(b) For every $q \in \Delta(S)$ there is $p' \in \Delta(A)$ such that $U(p',q) = \max_{p\in\Delta(A)} U(p,q)$. Hence, $(q,U(p',q)) \in C$. Therefore, $C \cap R_2(q) \neq \emptyset$. In particular, by Exercise 14.22, C is approachable by Player 1 and thus (Theorem 14.8) it is not excludable by Player 2.

(c) Since C is approachable by Player 1, for every strategy σ of Player 1 that approaches C and every strategy τ of Player 2,

$$\mathbf{P}_{\sigma,\tau}\left(\liminf_{T\to\infty} d(\tfrac{1}{T}\sum_{t=1}^{T} w(s^t,a^t),C) = 0\right) = 1$$

However, $\left(\frac{1}{T}\sum_{t=1}^{T} w_{s'}(s^t,a^t)\right)_{s'\in S} \in \Delta(S)$ so

$$\liminf_{T\to\infty} d(\tfrac{1}{T}\sum_{t=1}^{T} w(s^t,a^t),C) = 0$$

if and only if

$$\liminf_{T\to\infty} \tfrac{1}{T}\sum_{t=1}^{T} u(a^t, s^t) \geq \liminf_{T\to\infty} \max_{p\in\Delta(A)} U\left(p, \left(\tfrac{1}{T}\sum_{t=1}^{T} w_{s'}(s^t, a^t)\right)_{s'\in S}\right)$$

$$\geq \liminf_{T\to\infty} \max_{a\in A} U\left(a, \left(\tfrac{1}{T}\sum_{t=1}^{T} w_{s'}(s^t, a^t)\right)_{s'\in S}\right).$$

Note that $\left(\frac{1}{T}\sum_{t=1}^{T} w_{s'}(s^t, a^t)\right)_{s'\in S}$ represents the average strategy of Player 2. Therefore and by the linearity of U

$$\mathbf{P}_{\sigma,\tau}\left(\liminf_{T\to\infty}\left(U(\bar{x}^T, \bar{y}^T) - \max_{p\in\Delta(A)} U(p, \bar{y}^T)\right) \geq 0\right) = 1 \Rightarrow$$

$$\mathbf{P}_{\sigma,\tau}\left(\liminf_{T\to\infty}\left(U(\bar{x}^T, \bar{y}^T) - \max_{a\in\Delta(A)} U(a, \bar{y}^T)\right) \geq 0\right) = 1.$$

The last equation holds for every strategy τ of Player 2, and in particular for every sequence of states $(s^1, s^2, \dots)$ (constituting a pure strategy) in which Player 2 plays the action s^t in stage t. This states precisely that σ is a no-regret strategy.

14.37 Consider a decision problem with experts, where $E = A$, and for every action $a \in A$ there exists an expert $e \in E$ such that at $e = a$ for every $t \in N$. Define a game with vector payoffs, where the set of actions of Player 1 is A, the set of actions of Player 2 is S, and the payoffs are $(|S| \times |A|)$-dimensional vectors $(w(s,a))_{a\in A}^{s\in S}$, where $w(s,a) = (w_{s',a'}(s,a))_{a'\in A}^{s'\in S}$, defined as follows: $w_{s',a'}(s,a) = 1$ if $(s',a') = (s,a)$, and $w_{s',a'}(s,a) = 0$ if $(s',a') \neq (s,a)$. Define a set $C \subset \mathbf{R}^{|S|\times|A|}$ as follows:

$$C := \{q \in \Delta(S\times A) : U(1_a, q_{|a}) \geq \max_{p\in\Delta(A)} U(p, q_{|a}),\ \ \forall a \in A\},$$

where U is the multilinear extension of u, and for every action $a \in A$, $q_{|a}$ is the conditional distribution over S given a, with this conditional distribution defined by

$$q_{|a}(s) := \begin{cases} \frac{q(s,a)}{\sum_{s'\in S} q(s',a)} & \text{if } \sum_{s'\in S} q(s',a) > 0, \\ 0 & \text{if } \sum_{s'\in S} q(s',a) = 0. \end{cases}$$

(a) We prove that C is a convex set. Let $q^1, q^2 \in C \subseteq \Delta(S \times A)$ and let $\alpha \in (0,1)$.
By the convexity of $\Delta(S \times A)$,

$$\alpha \cdot q^1 + (1-\alpha)q^2 \in \Delta(S \times A).$$

For every $a \in A$, denote $q^1(a) = \sum_{s' \in S} q^1(s', a)$ and $q^2(a) = \sum_{s' \in S} q^2(s', a)$.

$$\sum_{s' \in S} \Big(\alpha \cdot q^1(s', a) + (1-\alpha) \cdot q^2(s', a)\Big) = \alpha q^1(a) + (1-\alpha) q^2(a).$$

Therefore, if $\alpha \cdot q^1(a) + (1-\alpha) \cdot q^2(a) > 0$ then

$$(\alpha \cdot q^1 + (1-\alpha) \cdot q^2)_{|a}(s) = \tfrac{\alpha \cdot q^1(a)}{\alpha \cdot q^1(a) + (1-\alpha) \cdot q^2(a)} q^1_{|a}(s) + \tfrac{(1-\alpha) \cdot q^2(a)}{\alpha \cdot q^1(a) + (1-\alpha) \cdot q^2(a)} q^2_{|a}(s)$$

otherwise it equals 0.
Hence, for every $a \in A$. If $\alpha \cdot q^1(a) + (1-\alpha) \cdot q^2(a) = 0$ then $(\alpha \cdot q^1 + (1-\alpha)q^2)_{|a} = 0$ and $U(1_a, 0) = 0 = \max_{p \in \Delta(A)} U(p, 0)$.
If $\alpha \cdot q^1(a) + (1-\alpha) \cdot q^2(a) > 0$, then

$$\begin{aligned}
&U(1_a, (\alpha \cdot q^1 + (1-\alpha)q^2)_{|a}) \\
&= \tfrac{\alpha \cdot q^1(a)}{\alpha \cdot q^1(a) + (1-\alpha) \cdot q^2(a)} U(1_a, q^1_{|a}) + \tfrac{(1-\alpha) \cdot q^2(a)}{\alpha \cdot q^1(a) + (1-\alpha) \cdot q^2(a)} U(1_a, q^2_{|a}) \\
&\geq \tfrac{\alpha \cdot q^1(a)}{\alpha \cdot q^1(a) + (1-\alpha) \cdot q^2(a)} \max_{p \in \Delta(A)} U(p, q^1_{|a}) + \tfrac{(1-\alpha) \cdot q^2(a)}{\alpha \cdot q^1(a) + (1-\alpha) \cdot q^2(a)} \max_{p \in \Delta(A)} U(p, q^2_{|a}) \\
&\geq \max_{p \in \Delta(A)} \left(\tfrac{\alpha \cdot q^1(a)}{\alpha \cdot q^1(a) + (1-\alpha) \cdot q^2(a)} U(p, q^1_{|a}) + \tfrac{(1-\alpha) \cdot q^2(a)}{\alpha \cdot q^1(a) + (1-\alpha) \cdot q^2(a)} U(p, q^2_{|a}) \right) \\
&= \max_{p \in \Delta(A)} U(p, (\alpha q^1 + (1-\alpha) q^2)_{|a}).
\end{aligned}$$

In conclusion, $\alpha \cdot q_1 + (1-\alpha) \cdot q^2 \in C$.

(b) For every $q_2 \in \Delta(S)$ there is $a \in A$ such that $U(1_a, q_2) = \max_{p \in \Delta} U(p, q_2)$ (a continuous function attains its maximum at extreme point). Hence, $(q_2, 1_a) \in C$. Therefore, $C \cap R_2(q_2) \neq \emptyset$. In particular, by Exercise 14.22, C is approachable by Player 1 and thus (Theorem 14.8) it is not excludable by Player 2.

(c) Since C is approachable by Player 1, for every strategy σ of Player 1 that approaches C and every strategy τ of Player 2,

$$\mathbf{P}_{\sigma,\tau}\left(\lim_{T \to \infty} d\Big(\tfrac{1}{T} \sum_{t=1}^{T} w(s^t, a^t), C\Big) = 0 \right) = 1$$

However, $\left(\frac{1}{T}\sum_{t=1}^{T} w_{s',a'}(s^t,a^t)\right)_{a'\in A}^{s'\in S} \in \Delta(S\times A)$ so

$$\liminf_{T\to\infty} d(\tfrac{1}{T}\sum_{t=1}^{T} w(a^t,s^t), C) = 0$$

if and only if for every $a \in A$

$$\liminf_{T\to\infty} u(1_a, \bar{y}^T_{|a}) \geq \liminf_{T\to\infty} \max_{p\in\Delta(A)} U(p, \bar{y}^T_{|a}),$$

where

$$\bar{y}^T_{|a}(s) = \begin{cases} \frac{|\{t : t\leq T \text{ and } (s^t,a^t)=(s,a)\}|}{|\{t:t\leq T \text{ and } a^t=a\}|} & \text{if } |\{t : t \leq T \text{ and } a^t = a\}| > 0 \\ 0 & \text{if } |\{t : t \leq T \text{ and } a^t = a\}| = 0. \end{cases}$$

Therefore,

$$\mathbf{P}_{\sigma,\tau}\left(\liminf_{T\to\infty}\left(u(1_a,\bar{y}^T_{|a}) - \max_{p\in\Delta(A)} U(p,\bar{y}^T_{|a})\right) \geq 0 \ \ \forall a \in A\right) = 1 \Rightarrow$$

$$\mathbf{P}_{\sigma,\tau}\left(\liminf_{T\to\infty}\sum_{a\in A}\bar{x}^T(a)\left(u(1_a,\bar{y}^T_{|a}) - \max_{p\in\Delta(A)} U(p,\bar{y}^T_{|a})\right) \geq 0\right) = 1 \Rightarrow$$

$$\mathbf{P}_{\sigma,\tau}\left(\liminf_{T\to\infty}\left(U(\bar{x}^T,\bar{y}^T) - \max_{a\in\Delta(A)} U(a,\bar{y}^T)\right) \geq 0\right) = 1.$$

The last equation holds for every strategy τ of Player 2, and in particular for every sequence of states $(s^1, s^2, ...)$ (constituting a pure strategy) in which Player 2 plays the action s^t in stage t. This states precisely that σ is a no-regret strategy.

14.38 Let S be a finite set of states of nature, and let A be a finite set of actions available to a decision maker. Let $F : \Delta(S) \to 2^A$ be a set-valued function represents the best actions given a distribution over the states satisfying the property that for every $a \in A$, the set $F^{-1}(a)$ is closed and convex. Define a game with $(|A|\times|S|)$-dimensional vector payoffs as follows: the payoff $u(a,s)$ that is attained when the state of nature is s, and the action chosen is a, is the unit vector whose value is 0 at every coordinate except the coordinate (a,s), where its value is 1. Denote this vector by 1_{as}. Define a set $C \subseteq \mathbf{R}^{|S|\times|A|}$ as follows:

$$C := \text{conv}\left\{\sum_{s\in S} y_s 1_{as} : a \in A, y \in F^{-1}(a)\right\}.$$

(a) C as a convex hull of a set is necessarily a convex set.

(b) For every $y \in \Delta(S)$ an action of Player 2, there is an action $a \in F(y)$ of Player 1. Hence, $\sum_{s\in S} y_s 1_{a,s} \in R_2(y) \cap C$ so $R_2(y) \cap C \neq \emptyset$. Furthermore, C is convex (Part (a)) and closed (since $F^{-1}(a')$ is closed for every $a' \in A$ and $\sum_{s\in S} y_s 1_{a,s}$ is continuous). In conclusion, by Exercise 14.22, C is approachable by Player 1.

(c) Let n_a^t a be the number of stages, up to t, in which the action a is chosen, and let y_a^t be the empirical distribution of the states of nature in all stages up to stage t in which the action a is chosen. Suppose that the decision maker plays a strategy σ that approaches C. Assume also that $\liminf_{t\to\infty} \frac{n_a^t}{t} > 0$. We prove that $\lim_{t\to\infty} d(y_a^t, F^{-1}(a)) = 0$.

We first prove the following auxiliary claim:

Claim 3 *Let $v = (v_{a,s})_{a\in A}^{s\in S} \in C$. Then, there are $x = (x_a)_{a\in A} \in \Delta(A)$ and $y^a \in F^{-1}(a)$ for every $a \in A$ such that $x_a > 0$ satisfying*

$$v = \sum_{\{a\in A: x_a>0\}} x_a \sum_{s\in S} y_s^a 1_{a,s}.$$

Proof. Let $v \in C$. Hence, for every $a \in A$ and $y \in F^{-1}(a)$ there is $\alpha_{a,y} \geq 0$ such that $\sum_{a\in A} \sum_{y\in F^{-1}(a)} \alpha_{a,y} = 1$ and

$$v = \sum_{a\in A} \sum_{y\in F^{-1}(a)} \alpha_{a,y} \sum_{s\in S} y_s 1_{a,s}.$$

Denote $x_a = \sum_{y'\in F^{-1}(a)} \alpha_{a,y'}$. In particular, $(x_a)_{a\in A} \in \Delta(A)$. In addition,

$$v = \sum_{\{a\in A: x_a>0\}} x_a \sum_{s\in S} 1_{a,s} \left(\sum_{y\in F^{-1}(a)} \tfrac{\alpha_{a,y}}{x_s} y_s \right).$$

However, since $F^{-1}(a)$ is convex, $y^a := \left(\sum_{y\in F^{-1}(a)} \frac{\alpha_{a,y}}{x_s} y_s \right)_{s\in S} \in F^{-1}(a)$, as claimed. ■

Now, since σ approaches C for the decision maker (Player 1), for every strategy τ of Player 2 (the nature),

$$\begin{aligned}
1 &= \mathbf{P}_{\sigma,\tau}\left(\lim_{T\to\infty} d\big(\tfrac{1}{T} \sum_{t=1}^{T} u(s^t, a^t), C\big) = 0 \right) \\
&= \mathbf{P}_{\sigma,\tau}\left(\lim_{T\to\infty} d\big(\tfrac{1}{T} \sum_{t=1}^{T} 1_{a^t s^t}, C\big) = 0 \right) \\
&= \mathbf{P}_{\sigma,\tau}\left(\lim_{T\to\infty} d\big(\sum_{a\in A} \tfrac{n_a^t}{T} \sum_{s\in S} y_a^t 1_{a,s}, C\big) = 0 \right).
\end{aligned}$$

However, by the auxiliary claim it follows that with probability 1, if $\liminf_{t\to\infty} \frac{n_a^t}{t} > 0$ then $\lim_{t\to\infty} d(y_a^t, F^{-1}(a)) = 0$.

14.39 Morris is not correct. The proof relies on Corollary 14.26, in which the set C must be compact and convex set. However, the minimal compact and convex set containing the vectors $\overrightarrow{0}$ and $(x^k)_{k\in N}$ is the nonnegative quadrant of R.

14.40 Let $G = (N, (S_i)_{i\in N}, (u_i)_{i\in N})$ be a two-player game in strategic form, and denote by $E_c \subseteq \Delta(S_1 \times S_2)$ the set of correlated equilibria of this game.

(a) For every $s_1 \in S_1$ define

$$\Sigma_2(s_1) := \{\sigma_2 \in \Sigma_2 \colon u_1(s_1, \sigma_2) = \max_{t_1 \in S_1} u_1(t_1, \sigma_2)\}. \tag{123}$$

The set $\Sigma_2(s_1)$ is a subset of the bounded set Σ_2 and thus it is also bounded.

We next prove that $\Sigma_2(s_1)$ is a closed set. Let $(\sigma_2(n))_{n\in\mathbf{N}}$ be a sequence of mixed actions of Player 2 for which s_1 is a best response. Assume that $(\sigma_2(n))_{n\in\mathbf{N}}$ convergences to a limit σ_2. Since u and the maximum are continuous functions

$$\begin{aligned} u_1(s_1, \sigma_2) &= u_1(s_1, \lim_{n\to\infty} \sigma_2(n)) = \lim_{n\to\infty} u_1(s_1, \sigma_2(n)) \\ &= \lim_{n\to\infty} \max_{t_1\in S_1} u_1(t_1, \sigma_2(n)) = \max_{t_1\in S_1} u_1(t_1, \lim_{n\to\infty} \sigma_2(n)) \\ &= \max_{t_1\in S_1} u_1(t_1, \sigma_2) \end{aligned}$$

which leads to $\sigma_2 \in \Sigma_2(s_1)$ so $\Sigma_2(s_1)$ is closed.

We finally prove that $\Sigma_2(s_1)$ is convex. Let $\sigma_2, \widehat{\sigma}_2 \in \Sigma_2$, and let $\alpha \in [0,1]$. Then, by Exercise 5.1, $\alpha\sigma_2 + (1-\alpha)\widehat{\alpha}_2 \in \Sigma_2$. Furthermore,

$$\begin{aligned} \max_{t_1\in S_1} u_1(t_1, \alpha\sigma_2 + (1-\alpha)\widehat{\sigma}_2) &\geq u_1(s_1, \alpha\sigma_2 + (1-\alpha)\widehat{\sigma}_2) \\ &= \alpha u_1(s_1, \sigma_2) + (1-\alpha) u_1(s_1, \widehat{\sigma}_2) \\ &= \alpha \max_{t_1\in S_1} u_1(t_1, \sigma_2) + (1-\alpha) \max_{t_1\in S_1} u_1(t_1, \widehat{\sigma}_2) \\ &\geq \max_{t_1\in S_1} u_1(t_1, \alpha\sigma_2 + (1-\alpha)\widehat{\sigma}_2) \end{aligned}$$

from which it follows that

$$u_1(s_1, \alpha\sigma_2 + (1-\alpha)\widehat{\sigma}_2) = \max_{t_1\in S_1} u_1(t_1, \alpha\sigma_2 + (1-\alpha)\widehat{\sigma}_2).$$

That is, $\Sigma_2(s_1)$ is a convex set.

(b) Consider the following repeated game with vector payoffs Γ:

- The set of players is $N = \{1, 2\}$.
- The sets of actions are $\mathcal{I} = S_1$ and $\mathcal{J} = S_2$,
- The payoffs are $|S_1| \cdot |S_2|$ dimensional. The payoff vector at the entry (s_1, s_2) is a unit vector whose coordinate (s_1, s_2) is equal to 1, and all other coordinates are 0.

Note that the strategy sets of the players are the same in G and in Γ. Define

$$C_1(s_1) := \{p_{s_1,\sigma_2} \colon \sigma_2 \in \Sigma_2(s_1)\} \subseteq \Delta(S_1 \times S_2), \tag{124}$$

and let $C_1 := \text{conv}\ (\cup_{s_1 \in S_1} C_1(s_1))$. We prove that C_1 is approachable by Player 1.

We first claim that C_1 is a convex and compact set as a convex hull of a finite union of convex and compact sets. We prove the claim for a union of two sets. By induction, the claim also holds for any finite union.

Claim 4 *Let A and B be convex and compact sets. Then* conv $(A \cup B)$ *is convex and compact.*

Proof. We prove that conv $(A \cup B)$ is closed. The other properties hold trivially.

Let $(x^n)_{n \in \mathbf{N}} \subseteq \text{conv}\ (A \cup B)$ be a sequence converging to a limit x^*.

For every $n \in \mathbf{N}$ there are $a^n \in A$,$b^n \in B$ and $\alpha^n \in [0, 1]$ satisfying $x^n = \alpha^n a^n + (1 - \alpha^n) b^n$. Since A, B and $[0, 1]$ are compact there are a subsequences $(a^{n_k})_k$, $(b^{n_k})_k$ and $(\alpha^{n_k})_k$ converging to $a \in A$, $b \in B$ and $\alpha \in [0, 1]$ respectively. In particular,

$$\begin{aligned} x = \lim_{k \to \infty} x^{n_k} &= \lim_{k \to \infty} \alpha^{n_k} a^{n_k} + (1 - \alpha^{n_k}) b^{n_k} \\ &= \alpha a + (1 - \alpha) b \in \text{conv}\ (A \cup B), \end{aligned}$$

which completes the proof. ■

We next prove that the the other condition of Exercise 14.22 holds. For every $\sigma_2 \in \Delta(S_2)$ a mixed action of Player 2, there is an action $s_1 \in S_1$ of Player 1 such that $u(s_1, \sigma_2) = \max_{t_1 \in S_1} u_1(t_1, \sigma_2)$. Hence, $\emptyset \neq R_2(\sigma_2) \cap C_1(s_1) \subseteq R_2(\sigma_2) \cap C_1$. Therefore, by Exercise 14.22, C_1 is approachable by Player 1.

(c) Let $p \in C_1$. There is a unique linear combination $p = \sum_{s_1 \in S_1} \alpha_{s_1} p_{s_1}$ where $p_{s_1} \in C_1(s_1)$ and $\alpha_{s_1} \geq 0$ for every $s_1 \in S_1$ and $\sum_{s_1 \in S_1} \alpha_{s_1} = 1$. In particular, $\alpha_{p,s_1} = \sum_{s_2 \in S_2} p(s_1, s_2)$ and

$$p_{s_1}(s_1', s_2') = \begin{cases} \frac{p(s_1', s_2')}{\alpha_{p,s_1}} & \text{if } s_1' = s_1 \\ 0 & \text{otherwise.} \end{cases}$$

Therefore, for every $s_1, s_1' \in S_1$

$$\begin{aligned}
&\sum_{s_2 \in S_2} p_{s_1}(s_1, s_2) u(s_1, s_2) \geq \sum_{s_2 \in S_2} p_{s_1}(s_1', s_2) u(s_1', s_2) \Rightarrow \\
&\sum_{s_2 \in S_2} \frac{p(s_1', s_2')}{\alpha_{p,s_1}} u(s_1, s_2) \geq \sum_{s_2 \in S_2} \frac{p(s_1', s_2')}{\alpha_{p,s_1}} u(s_1', s_2) \Rightarrow \\
&\sum_{s_2 \in S_2} p(s_1, s_2) u_1(s_1, s_2) \geq \sum_{s_2 \in S_2} p(s_1, s_2) u_1(s_1', s_2)
\end{aligned}$$

(d) Define the analog sets for Player 2: For every $s_2 \in S_2$ let

$$\Sigma_1(s_2) := \{\sigma_1 \in \Sigma_1 \colon u_2(\sigma_1, s_2) = \max_{t_2 \in S_2} u_2(\sigma_1, t_2)\}, \quad (125)$$

$$C_2(s_2) := \{p_{\sigma_1, s_2} \colon \sigma_1 \in \Sigma_1(s_2)\} \subseteq \Delta(S_1 \times S_2), \quad (126)$$

$$C_2 := \text{conv}\,(\cup_{s_2 \in S_2} C_2(s_2)). \quad (127)$$

By replacing the roles of the players in previous items one can deduce that C_2 is approachable by Player 2.

(e) Let σ_1^* be a strategy of Player 1 that approaches C_1 in Γ and let σ_2^* be a strategy of Player 2 that approaches C_2 in Γ. Denote by $p_N \in \Delta(S_1 \times S_2)$ the empirical play up to stage N:

$$p_N(s_1, s_2) := \frac{1}{N} \#\{n \leq N \colon (a_1^n, a_2^n) = (s_1, s_2)\}. \quad (128)$$

Note that p_N is also the average payoff in the repeated game with vector payoffs.

Next, since C_1 is approachable by Player 1 (Item (b)),

$$1 = \mathbf{P}_{\sigma_1^*, \sigma_2^*} \left(\lim d(p_N, C_1) = 0 \right).$$

Since C_2 is approachable by Player 2 (Item (d)),

$$1 = \mathbf{P}_{\sigma_1^*, \sigma_2^*} \left(\lim d(p_N, C_2) = 0 \right).$$

Hence,

$$1 = \mathbf{P}_{\sigma_1^*, \sigma_2^*} \left(\lim d(p_N, C_1 \cap C_2) = 0 \right).$$

However, by Item (c), $C_1 \cap C_2 = E_c$ from which it follows that

$$P_{\sigma_1^*, \sigma_2^*} \left(\lim_{N \to \infty} d(p_N, E_c) = 0 \right) = 1. \quad (129)$$

14.41 Suppose that a set C is weakly approachable by Player 1. We show that C cannot be weakly excludable by Player 2.
Let $\delta > 0$. Denote $C^\delta = \{x \in \mathbf{R}^m : d(x, C) \geq \delta\}$. We show that C^δ is not weakly approachable by Player 2. That is, there is $\epsilon > 0$ such that for every $T \in \mathbf{N}$ there is $t \geq T$ such that for every strategy σ_2 of Player 2 (that depends on t) there is a strategy σ_1 of Player 1 satisfying

$$E_{\sigma_1,\sigma_2}(d(\bar{g}^t, C^\delta)) > \epsilon.$$

Let $\epsilon = \frac{\delta}{2}$. Since C is weakly approachable by Player 1, there is $T \in \mathbf{N}$ such that for every $t \geq T$ there is a strategy σ_1 of Player 1 satisfying the property that for every strategy σ_2 of the Player 2 $E_{\sigma_1,\sigma_2}(d(\bar{g}^t, C)) \leq \epsilon$.
Let $t > T$ and let σ_2 be any strategy of Player 2. Define a random variable Y^t as follows: $Y^t = 1$ if and only if $d(\bar{g}^t, C) < 2\epsilon$ otherwise $Y^t = 0$. Then, by Markov Inequality,

$$\mathbf{P}_{\sigma_1,\sigma_2}(Y^t = 0) = \mathbf{P}_{\sigma_1,\sigma_2}(d(\bar{g}^t, C) \geq 2\epsilon) \leq \frac{E_{\sigma_1,\sigma_2}(d(\bar{g}^t,C))}{2\epsilon} \leq \frac{\epsilon}{2\epsilon} = \frac{1}{2}.$$

In particular,

$$\begin{aligned}
E_{\sigma_1,\sigma_2}(d(\bar{g}^t, C^\delta)) &= \mathbf{P}(Y^t = 1)E_{\sigma_1,\sigma_2}(d(\bar{g}^t, C^\delta)|Y^t = 1) \\
&\quad + \mathbf{P}(Y^t = 0)E_{\sigma_1,\sigma_2}(d(\bar{g}^t, C^\delta)|Y^t = 0) \\
&\geq \tfrac{1}{2}E_{\sigma_1,\sigma_2}(d(\bar{g}^t, C^\delta)|Y^t = 1) \\
&\geq \tfrac{1}{2}E_{\sigma_1,\sigma_2}(d(C, C^\delta) - d(C, \bar{g}^t)|Y^t = 1) \qquad (130) \\
&\geq \tfrac{1}{2}(\delta - \epsilon) = \tfrac{\delta}{4},
\end{aligned}$$

where Inequality (130) follows the triangle inequality.
In conclusion, C is not weakly excludable by Player 2.

Chapter 15: Stochastic Games

15.1 The situation can be described as a stochastic game in the following way:

- The players are $N = \{\text{The United States, Russia}\}$.
- The set of states is $Z = \{z_{us}, z_r, z_e\}$ where z_{us} corresponds the case that the United States leads in military power, z_r corresponds the case that Russia leads in military power, and z_e corresponds to equal military power.
- At each state z, each player i has two actions $A_i(z) = \{a_l; a_h\}$ where a_l corresponds to a low budget of 4 and a_h correspond to a high budget of 10.

- The transition probabilities and the payoff functions are as follows

Z_{us} (Row: United State; Column: Russia)

	a_l	a_h
a_l	$6,-14$ $[1(z_{us})]$	$6,-20$ $[0.4(z_{us}),0.6(z_e)]$
a_h	$0,-14$ $[1(z_{us})]$	$0,-20$ $[1(z_{us})]$

Z_r (Row: United State; Column: Russia)

	a_l	a_h
a_l	$-14,6$ $[1(z_r)]$	$-14,0$ $[1(z_r)]$
a_h	$-20,6$ $[0.4(z_r),0.6(z_e)]$	$-20,0$ $[1(z_r)]$

Z_e (Row: United State; Column: Russia)

	a_l	a_h
a_l	$-4,-4$ $[1(z_e)]$	$-4,-10$ $[0.4(z_e),0.6(z_r)]$
a_h	$-10,-4$ $[0.4(z_e),0.6(z_{us})]$	$-10,-10$ $[1(z_e)]$

15.2 (The Big Match game)

(a) The stochastic game is given by
- $N = \{\text{David, Tom}\}$ the set of players.
- The set of states is $Z = \{z_0, z_1, z_2\}$.
- For each player i, $A_i(z_0) = \{\text{Red,Yellow}\}$ and $A_i(z_1) = A_i(z_2) = \{a\}$ are the sets of actions.
- The transition probabilities and the payoff functions are as follows:

Z_0 (Row: David; Column: Tom)

	Red	Yellow
Red	-1 $[1(z_0)]$	1 $[1(z_1)]$
Yellow	1 $[1(z_0)]$	-1 $[1(z_2)]$

Z_1 (Row: David; Column: Tom)

	a
a	1 $[1(z_1)]$

Z_2 (Row: David; Column: Tom)

	a
a	-1 $[1(z_2)]$

Present the game as a stochastic game and specify its initial state.

(b) The shortened description of the game is

		Tom	
		Red	Yellow
David	Red	-1 $[1(z_0)]$	1 $*$
	Yellow	1 $[1(z_0)]$	-1 $*$
		Z_0	

(c) For $T = 1$ the game at the initial state is actually the matching pennies game (Example 3.20) for which 0 is its unique equilibrium payoff.

(d) Following previous item, the equilibrium payoff in the second stage is $0, 1$ and -1 at state z_0,z_1 and z_2, respectively. In particular, at the first state (of two) the players face the following game

		Tom	
		Red	Yellow
David	Red	$\frac{-1+0}{2}$	$\frac{1+1}{2}$
	Yellow	$\frac{1+0}{2}$	$\frac{-1-1}{2}$
		Z_0	

Therefore, David can guarantee that the payoff is at least 0 by playing at z_0 the stationary strategy $[\frac{1}{2}(Red), \frac{1}{2}(Yellow)]$. Tom can guarantee that the payoff is at most 0 by playing at the first stage at the initial state z_0, $[\frac{2}{3}(Red), \frac{1}{3}(Yellow)]$ and then in the second stage, if the game returns to z_0, $[\frac{1}{2}(Red), \frac{1}{2}(Yellow)]$. Therefore, the only equilibrium payoff at the initial state when $T = 2$ is 0.

(e) We prove that for every $T \in \mathbf{N}$, David and Tom has optimal strategies guaranteeing a value of 0 and thus the only equilibrium payoff at the initial state is 0 using an induction over T.

For $T = 1$ the claim is proved in item (a). Assume that the claim holds for every $T' \leq T$. Consider a $T + 1$-stage game. By the induction hypothesis if the second state is z_0 then the induction hypothesis holds for the subgame that starts at the second stage. Thus, at the initial state z_0, David can guarantee that the payoff is at least zero by playing at the first stage $[\frac{1}{2}(Red), \frac{1}{2}(Yellow)]$ and if the next state is also z_0, then from stage 2 onwards he follows his optimal strategy guaranteeing him a zero, hence his expected payoff is at least :

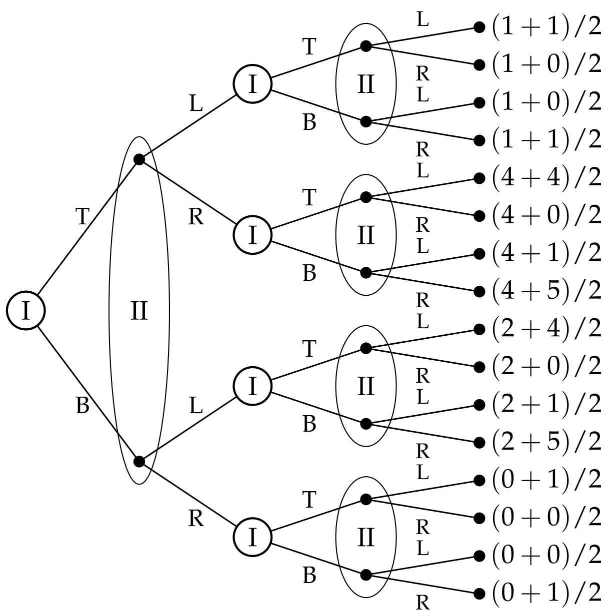
$$\min\left\{\frac{(0.5(-1)+0.5\cdot 1)+(T-1)\cdot 0}{T}, 0.5\frac{1+(T-1)\cdot 1}{T} + 0.5\frac{-1+(T-1)(-1)}{T}\right\} = 0.$$

Similarly, Tom can guarantee that the payoff is at most zero by playing at the first stage $[\frac{T}{T+1}(Red), \frac{1}{T+1}(Yellow)]$ and if the next state is also z_0, then from stage 2 onwards he follows his optimal strategy guaranteeing him a zero, hence his expected payoff is at least :

$$\min\left\{\frac{\frac{T}{T+1}((-1)+(T-1)\cdot 0)+\frac{1}{T+1}(1+(T-1)\cdot 1)}{T},\right.$$
$$\left.\frac{\frac{T}{T+1}((1)+(T-1)\cdot 0)+\frac{1}{T+1}(-1+(T-1)\cdot(-1))}{T}\right\} = 0.$$

Therefore, the only equilibrium payoff at the initial state for $T+1$ is 0.

15.3



15.4 Given a stochastic game, for every initial state $z^1 \in Z$ let τ^{z^1} be an equilibrium of the T-stage game with this initial state. Define the vector of strategies $\tau^* = (\tau_i^*)_{i\in N}$ by

$$\tau_i^*(h) := \tau_i^{z^1}(h), \quad \forall h = (z^1, a^1, \cdots, z^t) \in H^T. \tag{131}$$

Hence, for every initial state $z^1 \in Z$ and every $\tau_i \in \mathcal{B}_i^T$

$$\gamma_i^T(z^1, \tau^*) = \gamma_i^T(z^1, \tau^{z^1}) \geq \gamma_i^T(z^1, \tau_1, \tau_{-i}^{z^1}) = \gamma_i^T(z^1, \tau_1, \tau_{-i}^*).$$

Therefore, for every initial state $z^1 \in Z$, the strategy vector τ^* is an equilibrium of the T-stage stochastic game with initial state z^1 and thus it is an equilibrium of the T-stage game.

15.5 Proof of Theorem 15.16: We prove that every T-stage stochastic game has an equilibrium in Markovian strategies by induction over T.
For $T = 1$: Every T-stage stochastic game is equivalent to an extensive-form game with perfect recall and thus it has a subgame perfect equilibrium in behavior strategies (Theorem 6.16). However, every behavior strategy in a 1-stage stochastic game is a Markovian strategy and thus the claim holds for $T = 1$.
Suppose that the claim holds for every T'-stage stochastic game where $T' \leq T$. Consider a $T+1$-stage stochastic game. By the induction hypothesis, the T-stage game has an equilibrium τ^T in Markovian strategies. For every state $z \in Z$, let $\widehat{\Gamma}^z = (N, (A_i(z))_{i \in N}, (\widehat{u}_i^z)_{i \in N}$ be a one-stage game where

$$\widehat{u}_i^z(a) = \tfrac{1}{T+1}\left(u_i(a) + T\sum_{z' \in Z} q(z'|z,a)\gamma_i^T(z',\tau^T)\right) \quad \forall a \in A(z).$$

The game $\widehat{\Gamma}^z$ has an equilibrium in mixed strategies τ^z (Theorem 6.16). Define a strategy vector τ^* in the $T+1$-stage stochastic game as follows. For every $h = (z^1)$, $\tau^*(h) = \tau^{z^1}$. For every $t > 1$ and every $h = (z^1, a^1, z^2, a^2, ..., a^{t-1}, z^t)$ let $h' = (z^2, a^2, ..., a^{t-1}, z^t)$. Then $\tau^*(h) = \tau^T(h')$.
The strategy vector τ^* is Markovian. Furthermore, τ^* is an equilibrium. Indeed, in the $T+1$-stage stochastic game with an initial state z^1 for every player i and every behavior strategies $\tau_i \in \mathcal{B}_i^{T+1}$ and τ_i'

which is the the strategy induced by τ_i from stage 2 on,

$$\begin{aligned}
&\gamma_i^{T+1}(z^1, \tau_i, \tau^*_{-i}) \\
&= \tfrac{1}{T+1} \sum_{a \in A(z^1)} \tau_i(a_i)\tau^{z^1}_{-i}(a_{-i}) \left(u_i(a) + T \sum_{z' \in Z} q(z'|z^1, a)\gamma_i^T(z', \tau_i', \tau_{-i}^T) \right) \\
&= \tfrac{1}{T+1} \sum_{a \in A(z^1)} \tau_i(a_i)\tau^{z^1}_{-i}(a_{-i}) u_i(a) \\
&\quad \tfrac{T}{T+1} \sum_{a \in A(z^1)} \tau_i(a_i)\tau^{z^1}_{-i}(a_{-i}) \sum_{z' \in Z} q(z'|z^1, a)\gamma_i^T(z', \tau_i', \tau_{-i}^T) \\
&\leq \tfrac{1}{T+1} \sum_{a \in A(z^1)} \tau_i(a_i)\tau^{z^1}_{-i}(a_{-i}) u_i(a) \\
&\quad \tfrac{T}{T+1} \sum_{a \in A(z^1)} \tau_i(a_i)\tau^{z^1}_{-i}(a_{-i}) \sum_{z' \in Z} q(z'|z^1, a)\gamma_i^T(z', \tau^T) \\
&= \tfrac{1}{T+1} \sum_{a \in A(z^1)} \tau_i(a_i)\tau^{z^1}_{-i}(a_{-i}) \left(u_i(a) + T \sum_{z' \in Z} q(z'|z^1, a)\gamma_i^T(z', \tau^T) \right) \\
&\leq \tfrac{1}{T+1} \sum_{a \in A(z^1)} \tau^{z^1}(a) \left(u_i(a) + T \sum_{z' \in Z} q(z'|z^1, a)\gamma_i^T(z', \tau^T) \right) \\
&= \gamma_i^{T+1}(z^1, \tau^*)
\end{aligned}$$

where the first inequality holds by the induction hypothesis and the second inequality holds since τ^{z^1} is an equilibrium in $\widehat{\Gamma}^{z^1}$.

15.6 Consider the following two-stage two-player stochastic game when the initial state is z_0.

	L	R
T	$0,0^{z_0}$	$3,2^{z_1}$
B	$2,1^{z_1}$	$0,0^{z_0}$

state z_0

	L	R
T	$0,0^{z_0}$	$1,2^{z_0}$
B	$3,1^{z_1}$	$0,0^{z_1}$

state z_1

We first find all subgame-perfect Markovian equilibria using the technique presented in Theorem 15.18.

There are three equilibria in a 1-stage stochastic game starting with z_0: (T, R) corresponding to the payoff $(3, 2)$, (B, L) corresponding to the payoff $(2, 1)$ and $([\frac{1}{3}(T), \frac{2}{3}(B)], [\frac{3}{5}(L), \frac{2}{5}(R)])$ corresponding to the payoff $(\frac{6}{5}, \frac{2}{3})$.

There are three equilibria in a 1-stage stochastic game starting with z_1: (T, R) corresponding to the payoff $(1, 2)$, (B, L) corresponding to the payoff $(3, 1)$ and $([\frac{1}{3}(T), \frac{2}{3}(B)], [\frac{1}{4}(L), \frac{3}{4}(R)])$ corresponding to the payoff $(\frac{3}{4}, \frac{2}{3})$.

We next continue to a 2-stage stochastic game assuming that at the

second stage at each state the equilibria in a 1-stage stochastic game is played.
Assume first that at the second stage in both states the players play (T,R) then by backward induction at the first stage they face the following game

	L	R
T	$\frac{1}{2}((0,0)+(3,2))$	$\frac{1}{2}((3,2)+(1,2))$
B	$\frac{1}{2}((2,1)+(1,2))$	$\frac{1}{2}((0,0)+(3,2))$

which has two equilibria: (T,R), (B,L) (note that if R is played with positive probability then player 1 is better off playing T so these are the only equilibria). In conclusion, the above define two equilibria in the 2-stage stochastic game when the initial state is z_0: Play (T,R) (or (B,L)) at the first stage then play (T,R) in both states.
Similarly we find the following equilibria:

- At the first stage play $(T,R)/(B,L)/([\frac{1}{3}(T),\frac{2}{3}(B)],[\frac{4}{7}(L),\frac{3}{7}(R)])$ then at the second stage in both states play (B,L).
- At the first stage play $(T,R)/(B,L)$ then at the second stage play (T,R) at state z_0 and (B,L) at state z_1.
- At the first stage play $(T,R)/(B,L)/([\frac{2}{5}(T),\frac{3}{5}(B)],[\frac{2}{3}(L),\frac{1}{3}(R)])$ then at the second stage play (B,L) at state z_0 and (T,R) at state z_1.
- At the first stage play (T,R) then at the second stage play (T,R) at state z_0 and $([\frac{1}{3}(T),\frac{2}{3}(B)],[\frac{1}{4}(L),\frac{3}{4}(R)])$ at state z_1.
- At the first stage play $(T,R)/(B,L)/([\frac{7}{17}(T),\frac{10}{17}(B)],[\frac{14}{23}(L),\frac{9}{23}(R)])$ then at the second stage play $([\frac{1}{3}(T),\frac{2}{3}(B)],[\frac{3}{5}(L),\frac{2}{5}(R)])$ at state z_0 and (T,R) at state z_1.
- At the first stage play $(T,R)/(B,L)/([\frac{4}{11}(T),\frac{7}{11}(B)],[\frac{24}{43}(L),\frac{19}{43}(R)])$ then at the second stage play $([\frac{1}{3}(T),\frac{2}{3}(B)],[\frac{3}{5}(L),\frac{2}{5}(R)])$ at state z_0 and (B,L) at state z_1.
- At the first stage play $(T,R)/(B,L)/([\frac{1}{3}(T),\frac{2}{3}(B)],[\frac{51}{82}(L),\frac{31}{82}(R)])$ then at the second stage play $([\frac{1}{3}(T),\frac{2}{3}(B)],[\frac{3}{5}(L),\frac{2}{5}(R)])$ at state z_0 and (T,R) at state z_1.

In addition to these equilibria, there are also equilibria under which at the first stage they play (T,R), at the second stage at state z_1 they play one of the three equilibria in the 1-stage stochastic game when the initial state is z_1 (presented above), and at the second stage at state z_0 they may play any pure/mixed action vector.
There are also equilibria under which at the first stage they play (B,L), at the second stage at state z_1 they play one of the pure equilibria in the 1-stage stochastic game when the initial state is z_1 (presented above) and at the second stage at state z_0 they may play any pure/mixed action vector.

Finally, there are equilibria under which at the first stage they play (B, L), at the second stage at state z_1 they play the mixed equilibrium in the 1-stage stochastic game when the initial state is z_1 (presented above) and at the second stage at state z_0 they may play any pure/mixed action vector as long as player 1 plays T with probability $\frac{5}{6}$ at most.

15.7 We prove that the set of equilibria of a T-stage stochastic game is closed.
A T-stage stochastic game has a finite set of player, a finite set of states and a finite set of actions for each player at each state, from which it follows that it has a finite set of histories. Therefore, it can be represented as a finite extensive-form game. Every path in the corresponding extensive form game emanating from the initial state z_1 passes through each information set at most once, then by Theorem 6.11, every behavior strategy has an equivalent mixed strategy. On the other hand, in a T-stage stochastic game, every player has a perfect recall so, by Kuhn's Theorem (Theorem 6.15), it follows that every mixed strategy is equivalent to a behavior strategy. Therefore, in the T-stage stochastic game, the set of mixed equilibria and the set of behavior equilibria are equivalent. However, using Exercise 5.39, the set of mixed equilibria of a T-stage stochastic game is compact hence the set of behavior equilibria is compact as well.

15.8 **Proof of Theorem 15.18:** Let τ^* be a vector of Markovian strategies in the T-stage stochastic game satisfying the following properties: for every history $h = (z^1, a^1, \cdots, z^t) \in H^T$, the vector of mixed actions $(\tau_i^*(h))_{i\in N}$ is an equilibrium in the game $G(z^t; w)$, where $w(z) := (T-t)\gamma^{T-t}(z; \tau')$ for every $z \in Z$ and τ' is the vector of strategies induced by τ^* from stage $(t+1)$ on.
We prove by backward induction over t, that for every $t \in \{1, 2, ..., T\}$ and every initial state $z^t \in Z$, the strategy vector τ^t induced by τ^* from stage t on is an equilibrium in the $T-t+1$-stage game when the initial state is z^t.
For $t = T$, then $(T-t)\gamma^{T-t}(z; \tau') = 0$ for every $z \in Z$, thus $G(z^t; w)$ is actually a 1-stage stochastic game. Hence, by the properties of τ^*, the strategy vector τ^T induced by τ^* from stage T on is an equilibrium in the 1-stage game when the initial state is z^T.
Assume that the assumption holds for every $t' \geq t+1$ we prove that it also holds for t and every $z^t \in Z$. For every player i, every strategy τ_i of player i in the $T-t+1$-stage stochastic game and the strategy

τ_i' that is induced by τ_i from stage $(t+1)$ on the following holds

$$\gamma_i^{T-t+1}(z^t;\tau_i,\tau_{-i}^t)$$
$$= \tfrac{1}{T-t+1} \sum_{a\in A(Z^t)} \tau_i(a_i;z^t)\tau_{-i}^t(a_{-i};z^t)\left(u_i(a) + (T-t)\sum_{z\in Z} q(z|z^t,a)\gamma_i^{T-t}(z,\tau_i',\tau_{-i}^{t+1})\right)$$

$$\le \tfrac{1}{T-t+1} \sum_{a\in A(Z^t)} \tau_i(a_i;z^t)\tau_{-i}^t(a_{-i};z^t)\left(u_i(a) + (T-t)\sum_{z\in Z} q(z|z^t,a)\gamma_i^{T-t}(z,\tau^{t+1})\right)$$

$$\le \tfrac{1}{T-t+1} \sum_{a\in A(Z^t)} \tau^t(a;z^t)\left(u_i(a) + (T-t)\sum_{z\in Z} q(z|z^t,a)\gamma_i^{T-t}(z,\tau^{t+1})\right)$$
$$= \gamma_i^{T-t+1}(z^t;\tau^t)$$

where the first inequality holds by the induction hypothesis and the second inequality holds by the properties of τ^*. In particular, the strategy vector τ^t induced by τ^* from stage t on is an equilibrium in the $T-t+1$-stage game when the initial state is z^t, as claimed.

15.9 $T = 1$**:** There are three equilibria, (T,L) and (B,R) which correspond to the equilibrium payoff of $(0,0)$ and $([\frac{1}{2}(T),\frac{1}{2}(B)],[\frac{1}{2}(L),\frac{1}{2}(R)]$ which corresponds to the equilibrium payoff of $(-\frac{1}{2},-\frac{1}{2})$.

$T = 2$**:** We first consider all equilibria in which at the second stage the players follow one of the 1-stage equilibria in case that the transition at the first stage is to z_0. Denote by (a,a) the expected payoff at the second stage given that at the first stage they play (T,L). Hence, the two stage game corresponds to the following one-stage game:

	L	R
T	$\frac{1}{2}((0,0)+(a,a))$	$(-1,-1)$
B	$(-1,-1)$	$(0,0)$

from which the next equilibria can be obtained.

- $a = 0$: At the first stage play one of the above three equilibria for $T = 1$, then after (T,L) play one of the above pure equilibria for $T = 1$ (either (T,L) or (B,R)). These set of equilibria correspond to the equilibrium payoff of $(0,0)$ and $(-0.5,-0.5)$.
- $a = -0.5$: At the first stage play either (T,L) or (B,R) or $([\frac{4}{7}(T),\frac{3}{7}(B)],[\frac{4}{7}(L),\frac{3}{7}(R)])$, then after (T,L) play the above

mixed equilibria for $T = 1$. These set of equilibria correspond to the equilibrium payoff of $(0,0)$, $(-0.25,-0.25)$ and $(-\frac{4}{7},-\frac{4}{7})$.

Note that, if according to an equilibrium the players play a pure action then after every history that is not supposed to be played the payers may play a vector that is not an equilibrium in the 1-stage stochastic game as long as the payoff after deviation is lower than the equilibrium payoff. For instance, at the first stage play (B,R) and at the second stage after (T,L) play an arbitrary action vector. This is an equilibrium, since 0 is the maximal payoff in the game for both players so no player can gain by deviation.

$T = 3$**:** The equilibrium payoffs in the T-stage game for $T = 3$ are the equilibrium payoffs in the the following one stage game

	L	R
T	$\frac{1}{3}((0,0)+2(a,a))$	$(-1,-1)$
B	$(-1,-1)$	$(0,0)$

where (a,a) is an equilibrium payoff in the 2-stage game starting at z_0. Therefore, the equilibrium payoffs are $(0,0)$, $(-0.5,-0.5)$ (for $a = 0$), $(-\frac{1}{3},-\frac{1}{3})$, $(-0.6,-0.6)$ (for $a = -0.5$), $(-\frac{1}{6},-\frac{1}{6})$, $(-\frac{6}{11},-\frac{6}{11})$ (for $a = -0.25$), $(-\frac{8}{21},-\frac{8}{21})$ and $(-\frac{21}{34},-\frac{21}{34})$ (for $a = -\frac{4}{7}$).

15.10 $T = 1$**:** There is no pure equilibrium. There is a completely mixed equilibrium $([x(T),(1-x)(B)],(y(L),(1-y)(R)])$, which can be calculated using the Indifference Principle

$$\left.\begin{aligned} 4x &= x+3(1-x) \\ 2y+5(1-y) &= 8y+2(1-y) \end{aligned}\right\} \Rightarrow x = \tfrac{1}{2}, y = \tfrac{1}{3}.$$

Hence, there is a inique equilibrium payoff, $(4,2)$.

$T = 2$**:** Consider the next auxiliary one-stage game:

	L	R
T	$\frac{1}{2}((2,4)+(4,2))$	$\frac{1}{2}((5,1)+(3,8))$
B	$\frac{1}{2}((8,0)+(0,8))$	$\frac{1}{2}((2,3)+(4,2))$

There are three types of equilibria

- At the first stage play (T,R) or (B,L). If the transition at the first stage is to z_0, play $([x(T),(1-x)(B)],(y(L),(1-y)(R)])$ where $x \in [0,1]$ and $y \leq \frac{2}{3}$.

- Using the Indifference Principle in the auxiliary game there is also an equilibrium under which at the first stage they play

$$([\tfrac{1}{2}(T), \tfrac{1}{2}(B)], (\tfrac{1}{2}(L), \tfrac{1}{2}(R)]),$$

then if the transition at the first stage is to z_0, play the mixed equilibrium of the 1-stage stochastic game (for $T = 1$).

15.11 Let Γ^T be a two-player zero-sum symmetric stochastic T-stage game. We prove by induction on T that the value of Γ^T is 0 for every initial state.
For $T = 1$, the claim holds by Exercise 5.12. Assume that the claim holds for every two-player zero-sum symmetric stochastic T-stage game. We prove that it also holds for two-player zero-sum symmetric stochastic $(T+1)$-stage game.
We first show that Player I has a strategy guaranteeing a payoff of at least 0 at the $(T+1)$-stage stochastic game starting with an initial state z^1. Let τ_1^* be a strategy according to which at the first stage at state z_1, Player I play an optimal strategy at a 1-stage stochastic game with an initial state z_1 which guaranteeing him at least zero (Exercise 5.12). Then, if the transition at the first stage is to $z^2 \in Z$, Player I play the optimal strategy at the T-stage stochastic game with an initial state z^2 which guaranteeing him a zero (by the induction hypothesis). Therefore, for every strategy τ_2

$$\begin{aligned}\gamma_1^{T+1}(z^1, \tau_1^*, \tau_2) = {} & \underbrace{\tfrac{1}{T+1} \sum_{a \in A(z^1)} \tau_1^*(a_1)\tau_2(a_2)u_1(z_1, a)}_{\geq 0 \text{ (Exercise 5.12)}} \\ & + \tfrac{T}{T+1} \sum_{a \in A(z^1)} \tau_1^*(a_1)\tau_2(a_2) \sum_{z \in Z} q(z|z^1, a) \underbrace{\gamma_1^T(z, \tau_1^*, \tau_2|z^1, a)}_{\geq 0 \text{ (Induction hpothesis)}} \geq 0.\end{aligned}$$

By exchanging the roles of Player I and Player II, we deduce that Player II has an optimal strategy guaranteeing him at most zero and thus the value of the $(T+1)$ stochastic game is zero, as we claimed.

15.12 **Proof of Theorem 15.34:** Let z be an absorbing state in a two-player zero-sum stochastic game and let $\lambda \in [0, 1)$ be discount factor. We show that the quantity $v^\lambda(z)$ is equal to $\mathrm{val}(\Gamma^1(z))$, the value of the one-stage game in state z.
Let τ_1^* be the strategy of Player I according to which at stage z Player I plays the optimal strategy of the game $\Gamma^1(z)$ guaranteeing a discounted payoff of at least $\mathrm{val}(\Gamma^1(z))$. Then, for every strategy τ_2 of

Player II, the λ-discounted payoff is

$$\gamma^\lambda(z, \tau_1^*, \tau_2) = (1-\lambda)\sum_{t=1}^{\infty} \lambda^{t-1} E_{z,\tau_1^*,\tau_2}(u_i^t). \tag{132}$$

However, since z is an absorbing state,

$$\begin{aligned} E_{z,\tau_1^*,\tau_2}(u_i^t) &= \sum_{h\in H(t)} \mathbf{P}_{z,\tau_1^*}(h) U(z, \tau_1^*(h), \tau_2(h)) \\ &\geq \sum_{h\in H(t)} \mathbf{P}_{z,\tau_1^*}(h) \mathrm{val}(\Gamma^1(z)) = \mathrm{val}(\Gamma^1(z)). \end{aligned}$$

Therefore, by setting the last inequality in Equation (132), it follows that

$$\gamma^\lambda(z, \tau_1^*, \tau_2) \geq (1-\lambda)\sum_{t=1}^{\infty} \lambda^{t-1} \mathrm{val}(\Gamma^1(z)) = \mathrm{val}(\Gamma^1(z)).$$

Similarly, Player II has an optimal strategy guaranteeing that the λ-discounted payoff is at most $\mathrm{val}(\Gamma^1(z))$ according to which at stage z he plays his optimal strategy of the game $\Gamma^1(z)$ (the proof is identical and thus it was omitted).

15.13 Given a two-player zero-sum stochastic game, denote the minimal payoff in the game by $m := \min_{(z,a)\in\Lambda} u(z,a)$ and the maximal payoff in the game by $M := \max_{(z,a)\in\Lambda} u(z,a)$. We prove that for every state $z \in Z$ and for every discount factor $\lambda \in [0,1)$, the value $v^\lambda(z)$ of the discounted stochastic game with initial state z satisfies $m \leq v^\lambda(z) \leq M$.
By Theorem 15.32, $v^\lambda = (v^\lambda(z))_{z\in Z}$ is the unique solution of

$$v^\lambda(z) = \mathrm{val}(G^\lambda(z; v^\lambda)) \quad \forall z \in Z,$$

where $G^\lambda(z; v^\lambda)$ is the game in strategic form in which the set of players is N, the set of pure strategies of every player $i \in N$ is $A_i(z)$, and the payoff function of every player i is

$$u_i^\lambda(a) := (1-\lambda)u_i(z;a) + \lambda \sum_{z'\in Z} q(z' \mid z; a) v^\lambda(z') \ \ \forall a \in A(z).$$

Let $\underline{z} \in \arg\min_{z'\in Z} v^\lambda(z')$. Assume by contradiction that $m > v^\lambda(\underline{z})$.

Then, for every $a \in A(\underline{z})$,

$$\begin{aligned}
u_i^\lambda(a) &= (1-\lambda)u_i(z;a) + \lambda \sum_{z' \in Z} q(z' \mid z;a) v^\lambda(z') \\
&\geq (1-\lambda)u_i(z;a) + \lambda \sum_{z' \in Z} q(z' \mid z;a) v^\lambda(\underline{z}) \\
&\geq (1-\lambda)m + \lambda v^\lambda(\underline{z}) \sum_{z' \in Z} q(z' \mid z;a) \\
&> (1-\lambda)v^\lambda(\underline{z}) + \lambda v^\lambda(\underline{z}) \sum_{z' \in Z} q(z' \mid z;a) \\
&= v^\lambda(\underline{z}).
\end{aligned}$$

Therefore,

$$\operatorname{val}(G^\lambda(\underline{z}, v^\lambda)) \geq \min u_i^\lambda(a) > v^\lambda(\underline{z}),$$

which contradicts Theorem 15.32.
To prove the second part of the claim, let $\bar{z} \in \arg\max_{z' \in Z} v^\lambda(z')$. Assume by contradiction that $M < v^\lambda(\bar{z})$. Then, for every $a \in A(\bar{z})$,

$$\begin{aligned}
u_i^\lambda(a) &= (1-\lambda)u_i(z;a) + \lambda \sum_{z' \in Z} q(z' \mid z;a) v^\lambda(z') \\
&\leq (1-\lambda)u_i(z;a) + \lambda \sum_{z' \in Z} q(z' \mid z;a) v^\lambda(\bar{z}) \\
&\leq (1-\lambda)M + \lambda v^\lambda(\bar{z}) \sum_{z' \in Z} q(z' \mid z;a) \\
&< (1-\lambda)v^\lambda(\bar{z}) + \lambda v^\lambda(\bar{z}) \sum_{z' \in Z} q(z' \mid z;a) \\
&= v^\lambda(\bar{z}).
\end{aligned}$$

Therefore,

$$\operatorname{val}(G^\lambda(\bar{z}, v^\lambda)) \leq \max u_i^\lambda(a) < v^\lambda(\bar{z}),$$

which contradicts Theorem 15.32.

15.14 As shown in Example 15.33, $v^\lambda(z_0) = \frac{1-\sqrt{1-\lambda}}{\lambda}$, $v^\lambda(z_1) = 1$ and $v^\lambda(z_2) = 0$. Hence, the game $G^\lambda(z_0, v^\lambda)$ is

	L	R
T	$1-\sqrt{1-\lambda}$	1
B	1	0

Using the Indifference Principle, the optimal strategy of Player I in this game is $[x(T), (1-x)(B)]$ which satisfies

$$x(1-\sqrt{1-\lambda}) + (1-x) = x$$

so $x = \frac{1}{1+\sqrt{1-\lambda}}$. Similarly, the optimal strategy of Player II is

$$[\tfrac{1}{1+\sqrt{1-\lambda}}(L), \tfrac{\sqrt{1-\lambda}}{1+\sqrt{1-\lambda}}(R)].$$

In conclusion, the stationary strategy of each player according to which at each state $z \in Z$ he follows his optimal strategy in $G^\lambda(z, v^\lambda)$ is a λ-discounted optimal strategy for the two players game in Example 15.33.

15.15 (a) For every discount factor λ, the *maxmin* value of the λ-discounted game with initial state z_0 is positive. Indeed, Player I guarantees a positive payoff by playing the stationary strategy $[\frac{1}{2}(T), \frac{1}{2}(B)]$ which leads to a payoff that is higher than $\frac{1-\lambda}{2}$.

(b) If Player I always play T (B) at z_0 then Player II's best response is the stationary strategy L (R, respectively) so the discounted payoff is 0 which contradicts the previous item. Hence Player I has no stationary optimal strategy that chooses a pure action in state z_0.

(c) Let $\lambda \in [0, 1)$ be a discount factor. By Theorem 15.34, $v^\lambda(z_1) = 1$, $v^\lambda(z_2) = 0$ and $v^\lambda(z_3) = 3$. Denote $v = v^\lambda(z_0)$. Therefore, $G^\lambda(z_0)$ is

	L	R
T	λv	$(1-\lambda) + 3\lambda$
B	$2(1-\lambda) + \lambda$	0

By item (b) the optimal strategy is mixed. Denote by $[x(T), y(B)]$ and $[y(L), (1-y)(R)]$ the λ-discounted optimal strategies of Player I and Player II respectively. By the Indifference Principle and Theorem 15.32,

$$\begin{aligned}
v &= x\lambda v + (1-x)(2-\lambda)\\
v &= x(1+2\lambda)\\
v &= y\lambda v + (1-y)(1+2\lambda)\\
v &= y(2-\lambda).
\end{aligned}$$

Therefore,

$$\begin{aligned}
v &= \frac{3+\lambda-\sqrt{8\lambda^3-11\lambda^2-2\lambda+9}}{2\lambda}\\
x &= \frac{3+\lambda-\sqrt{8\lambda^3-11\lambda^2-2\lambda+9}}{2\lambda(1+2\lambda)}\\
y &= \frac{3+\lambda-\sqrt{8\lambda^3-11\lambda^2-2\lambda+9}}{2\lambda(2-\lambda)}.
\end{aligned}$$

15.16 (a) Let λ be a discounted value. By playing the mixed action $[\frac{1}{2}(T), \frac{1}{2}(B)]$ in both states z_0 and z_1 Player I guarantees a positive payoff in each initial state (of at least $\frac{1}{4}\lambda$ if the initial state is z_0 and of at least $\frac{1}{2}$ if the initial state is z_1). Hence, $v^\lambda(z_0)$ and $v^\lambda(z_1)$ are positive for every $\lambda \in (0,1)$. By playing the mixed action $[\frac{1}{2}(L), \frac{1}{2}(R)]$ in both states z_0 and z_1 Player II guarantees a payoff less than 1 in each initial state, hence $v^\lambda(z_0)$ and $v^\lambda(z_1)$ are less than 1 for every $\lambda \in (0,1)$.

(b) If Player I always play the pure action T (B) at z_0 then Player II's best response is the stationary strategy R (L, respectively) so the discounted payoff is 0 which contradicts the previous item. Similarly, if Player I always B at z_1 then Player II's best response is the stationary strategy B so the discounted payoff is 0 which contradicts the previous item. Finally, if Player I always T at z_1 then by playing L in z_1 and $[\frac{1}{2}(L), \frac{1}{2}(R)]$ the discounted payoff with initial state z_1 is at most $\lambda\frac{1}{2}$ which contradicts item (a). In conclusion, Player I has no stationary optimal strategy that chooses a pure action in state z_0 as well as in state z_1.

(c) Let $\lambda \in [0,1)$ be a discount factor. By Theorem 15.34, $v^\lambda(z_2) = 0$, and $v^\lambda(z_3) = 1$. Denote $v = v^\lambda(z_0)$ and $w = v^\lambda(z_1)$. Therefore, the games $G^\lambda(z_0)$ and $G^\lambda(z_1)$ are as follows

	L	R
T	λw	0
B	0	1

state z_0

	L	R
T	λv	1
B	1	0

state z_1

In the game $G^\lambda(z_0, (v, w))$ the diagonal entries are over the off-diagonal entries, hence the optimal mixed action of both players is completely mixed. In the game $G^\lambda(z_1, (v, w))$ the off-diagonal entries are over the diagonal entries, hence the optimal mixed actions of both players are completely mixed. Denote by $[x(T), (1-x)(B)], [y(T), (1-y)(B)]$ the λ-discounted optimal stationary strategy of Player I ($[x(T), (1-x)(B)]$ is the optimal mixed action in z_0). By the Indifference Principle,

$$
\begin{aligned}
v &= \lambda w x & \qquad w &= \lambda v y + (1-y) \\
v &= 1 - x & \qquad w &= y.
\end{aligned}
$$

The solution of this system of equations is

$$v = \frac{\lambda + 2 - \sqrt{4\lambda + 4 - 3\lambda^2}}{2\lambda},$$
$$w = \frac{-1 + \sqrt{\lambda - 2 - \frac{4\lambda}{\lambda - 2}}}{2\lambda}$$
$$x = \frac{\lambda - 2 + \sqrt{4\lambda + 4 - 3\lambda^2}}{2\lambda},$$
$$y = \frac{-1 + \sqrt{\lambda - 2 - \frac{4\lambda}{\lambda - 2}}}{2\lambda}.$$

Similarly, the λ-discounted optimal stationary strategy of Player II is $[x(L),(1-x)(R)],[y(L),(1-y)(R)]$ (where $[x(L),(1-x)(R)]$ is the optimal mixed action in z_0).

15.17 (a) In the stochastic game described in the question, the λ-discounted game with the initial state z_0 is equivalent to the λ-discounted game with the initial state z_1 in which the roles of Player I and Player II is exchanged (so Player II become the maximizer). Hence $v^\lambda(z_0) = -v^\lambda(z_1)$.

(b) We prove that $v^\lambda(z_0) > 0$. Assume by contradiction that $v^\lambda(z_0) \leq 0$. Then, by item (a), $v^\lambda(z_1) \geq 0$. Hence, Player I necessarily has a strategy guaranteeing a positive payoff in $\Gamma^\lambda(z_0)$: play $[\frac{1}{2}(T), \frac{1}{2}(B)]$ as long as the play stays at z_0, then as soon as it moves to z_1, follow the optimal strategy in $\Gamma^\lambda(z_1)$ guaranteeing $v^\lambda(z_1)$, a contradiction.

(c) By Theorem 15.34, $v^\lambda(z_2) = 1$ and $v^\lambda(z_3) = -1$. Denote $v = v^\lambda(z_0)$. Therefore, by item (a), the game $G^\lambda(z_0, v^\lambda)$ is

	L	R
T	$1-\lambda+\lambda v$	$1-\lambda-\lambda v$
B	$1-\lambda-\lambda v$	1

state z_0

Since, by item (b), $v > 0$, the diagonal entries are over the off-diagonal entries, hence the optimal mixed actions of both players are completely mixed. That is, the stationary optimal strategies of both players involve completely mixed actions in z_0 as well as in z_1. In particular, by Theorem 15.32

$$v = (1-\lambda+\lambda v)x + (1-\lambda-\lambda v)(1-x)$$
$$v = (1-\lambda-\lambda v)x + (1-x).$$

from which it follows that

$$v = \frac{1-\lambda+2\sqrt{1-\lambda}}{3+\lambda}$$
$$x = \frac{1-\sqrt{1-\lambda}}{\lambda}.$$

Similarly, the optimal action of Player II in $G^\lambda(z_0, v^\lambda)$ is $[x(L), (1-x)(R)]$. Hence, by item (a), the optimal actions of the players in $G^\lambda(z_0, v^\lambda)$ as well as in $G^\lambda(z_1, v^\lambda)$ are identical.

15.18 Let τ_{I} be an optimal strategy of Player I in a T-stage two-player zero-sum stochastic game. Then, it is not necessarily true that for every history $h \in H^T$ the mixed action $\tau_1(h)$ is an optimal strategy in the one-stage game in the current state. Consider the following T-stage two-player zero-sum stochastic game

	L	R
T	1 z_0	1 z_0
B	0 z_1	0 z_1

state z_0

	L	R
T	1 z_1	1 z_1
B	0 z_0	0 z_0

state z_1

In this game Player I has an optimal strategy according to which he plays T as long as the game stays in the initial state. However, as soon as the game transfers to the other state he plays B which is obviously not an optimal action in the one stage game.

15.19 Let τ_{I} be a stationary strategy of Player I in a two-player zero-sum stochastic game satisfying the following condition: for every state $z \in Z$ the mixed action $\tau_{\mathrm{I}}(z)$ is not optimal in the game $G^\lambda(z; v^\lambda)$. We prove that for every initial state, the strategy τ_{I} is not an optimal strategy in the stochastic game.
For every $z \in Z$, let $\tau_2(z)$ be the best response of Player II to $\tau_1(z)$ in the game $G^\lambda(z; v^\lambda)$. Since $\tau_1(z)$ is not an optimal strategy of Player I in the game and by Theorem 15.32,

$$U^\lambda(z; v^\lambda; \tau_1(z), \tau_2(z)) < \mathrm{val}(G^\lambda(z; v^\lambda)) = v^\lambda(z).$$

Let τ_2 be the stationary strategy of Player II under which at each $z \in Z$, he follows $\tau_2(z)$. Hence and by Lemma 15.30,

$$v^\lambda(z^1) > \gamma^\lambda(z^1, \tau_1, \tau_2).$$

That is, τ_1 is not an optimal strategy of Player I.

15.20 Every discounted repeated two-player zero-sum game is a discounted two-player zero-sum stochastic game with only one state which is therefore an absorbing state. Hence, by Theorem 15.34 (Exercise 15.12), for every discount factor, the value of the discounted game is equal to the value of the one-stage game.

15.21 (a) The game on a graph is a stochastic game where $N = \{I, II\}$ is the set of players; V is the set of states; for every state $v \in V$, the set of actions for player $i(v)$ is $A_{i(v)}(v) = \{(v, v') \in E\}$ and the set of actions of player $-i(v)$ contains only one action, $A_{-i(v)}(v) = \{N\}$, so it does not really affect; and for every two states $v, v' \in V$ and an action $a \in A(v)$ the transition probability

$$q(v'|v,a) = \begin{cases} 1 & a_{i(v)} = (v, v') \\ 0 & \text{otherwise.} \end{cases}$$

(b) Let $\lambda \in [0, 1)$ be a discount factor.

We first show that each player has a pure optimal strategy in $G^\lambda(v; v^\lambda)$ for every state $v \in V$. For every $v \in V$, player $-i(v)$ has only one action and thus his optimal strategy is trivially pure. On the other hand, since player $-i(v)$ has only one action at v, there is necessarily a pure action of player $i(v)$ that dominates all other actions and thus he has a pure optimal strategy in $G^\lambda(v; v^\lambda)$.

Therefore, by Theorem 15.32, Lemma 15.30 and Lemma 15.31, the stationary strategy of each player i according to which at each state $v \in V$, player i plays the pure optimal strategy in $G^\lambda(v; v^\lambda)$ is an optimal strategy for the λ-discounted stochastic game.

In conclusion, the game has a λ-discounted optimal strategy that is pure and stationary.

(c) Assume that the two players follow optimal pure stationary strategies. Then, it is not necessarily true that there exist a vertex v^* and a stage $t_0 \in \mathbf{N}$ such that $v^t = v^*$ for all $t \geq t_0$. For instance, let $V = \{v_1, v_2\}$, $E = V \times V$, $i(v_1) = I$, $i(v_2) = II$, $u(v_1) = -1$ and $u(v_2) = 1$ then the stationary strategy vector where at v_1 Player I chooses (v_1, v_2) and at v_2 Player II chooses (v_2, v_1) is optimal but there are no such v^* and t_0 satisfying the above properties.

15.22 In this exercise we generalize one part of Shapley's Theorem (Theorem 15.32) to games with any number of players.

Let $(N, Z, (A_i)_{i\in N}, (u_i)_{i\in N}, q)$ be a stochastic game. The *λ-discounted maxmin value* of player i in state z is

$$\underline{v}_i^\lambda(z) = \sup_{\tau_i\in\mathcal{B}_i^\infty} \inf_{\tau_{-i}\in\mathcal{B}_{-i}^\infty} \gamma_i^\lambda(z, \tau_i, \tau_{-i}). \tag{133}$$

We prove that every player i has a stationary strategy τ_i^* satisfying

$$\gamma_i^\lambda(z, \tau_i^*, \tau_{-i}) \geq \underline{v}_i^\lambda(z), \quad \forall z \in Z, \forall \tau_{-i} \in \mathcal{B}_i^\infty. \tag{134}$$

The proof is similar to the proof of Shapley's Theorem.
We first define a contraction mapping $T : \mathbf{R}^{|Z|} \to \mathbf{R}^{|Z|}$ as follows:

$$T_z(w) := \underline{v}_i(G^\lambda(z; w)), \quad \forall w \in \mathbf{R}^{|Z|}.$$

Note that, for every w and $\widehat{w}$ two vectors in $\mathbf{R}^{|Z|}$

$$\begin{aligned} |T_z(w) - T_z(\widehat{w})| = & \quad \left|\underline{v}_i(G^\lambda(z; w)) - \underline{v}_i(G^\lambda(z; \widehat{w}))\right| \\ \leq & \quad \|U^\lambda(z; w; \cdot) - U^\lambda(z; \widehat{w}; \cdot)\|_\infty. \end{aligned}$$

Hence, by Equations (15.87)-(15.95), T is indeed a contraction mapping. By Theorem 15.39, the function T has a unique fixed point w^*. We next prove that $\underline{v}_i^\lambda(z) \leq w^*(z)$ for every $z \in Z$.
Let $\tau_i \in \mathcal{B}_i^\infty$ be any strategy of player i. Let $\tau_{-i}^* \in \mathcal{B}_{-i}^\infty$ be a strategy vector according to which after each history $h = (z^1, a^1, ..., z^t) \in H^\infty$ all of the other players try to lower player i's payoff in $G^\lambda(z^t; w^*)$ against $\tau_i(h)$. That is

$$U^\lambda(z^t; w^*; \tau_i(h), \tau_{-i}^*(h)) \leq \underline{v}_i(G^\lambda(z^t; w^*)) = w^*(z^t).$$

Therfore, by Lemma 15.30, $\gamma_i^\lambda(z^1, \tau_i, \tau_{-i}^*) \leq w^*(z^1)$ for every $z^1 \in Z$. In conclusion,

$$\underline{v}_i^\lambda(z) \leq \sup_{\tau_i\in\mathcal{B}_i^\infty} \gamma_i^\lambda(z^1, \tau_i, \tau_{-i}^*) \leq w^*(z^1). \tag{135}$$

On the other hand, let τ_i^* be a stationary strategy of player i, according to which at each state z he plays his maxmin strategy in the game $G^\lambda(z; w^*)$. Then, at each state z, for every mixed action $\sigma_{-i,z}$ of the other players,

$$U^\lambda(z; w^*; \tau_i(z), \sigma_{-i,z}) \geq \underline{v}_i(G^\lambda(z^t; w^*)) = w^*(z^t).$$

Hence, by Lemma 15.29 and Equation (135), for every $z \in Z$ and every $\tau_{-i} \in \mathcal{B}_{-i}^\infty$ one has

$$\gamma_i^\lambda(z, \tau_i^*, \tau_{-i}) \geq w^*(z) \geq \underline{v}_i^\lambda(z),$$

as claimed.

15.23 In this exercise we generalize another part of Shapley's Theorem (Theorem 15.32) to games with any number of players.

Let $(N, Z, (A_i)_{i\in N}, (u_i)_{i\in N}, q)$ be a stochastic game. The *λ-discounted minmax value* of player i in state z is

$$\overline{v}_i^\lambda(z) = \inf_{\tau_{-i}\in\mathcal{B}_{-i}^\infty} \sup_{\tau_i\in\mathcal{B}_i^\infty} \gamma_i^\lambda(z, \tau_i, \tau_{-i}). \tag{136}$$

We prove that there exists a stationary strategy vector τ_{-i}^* satisfying

$$\gamma_i^\lambda(z, \tau_i, \tau_{-i}^*) \le \overline{v}_i^\lambda, \quad \forall z \in Z, \forall \tau_i \in \mathcal{B}_i^\infty. \tag{137}$$

We first define a contraction mapping $T : \mathbf{R}^{|Z|} \to \mathbf{R}^{|Z|}$ as follows:

$$T_z(w) := \overline{v}_i(G^\lambda(z; w)), \quad \forall w \in \mathbf{R}^{|Z|}.$$

Note that, for every w and $\widehat{w}$ two vectors in $\mathbf{R}^{|Z|}$

$$\begin{aligned} |T_z(w) - T_z(\widehat{w})| = &\quad \left|\overline{v}_i(G^\lambda(z; w)) - \overline{v}_i(G^\lambda(z; \widehat{w}))\right| \\ \le &\quad \|U^\lambda(z; w; \cdot) - U^\lambda(z; \widehat{w}; \cdot)\|_\infty. \end{aligned}$$

Hence, by Equations (15.87)-(15.95), T is indeed a contraction mapping. By Theorem 15.39, the function T has a unique fixed point w^*. We next prove that $\overline{v}_i^\lambda(z) \ge w^*(z)$ for every $z \in Z$.

Let $\tau_{-i} \in \mathcal{B}_{-i}^\infty$ be any vector of strategies. Let $\tau_i^* \in \mathcal{B}_i^\infty$ be a strategy vector according to which after each history $h = (z^1, a^1, ..., z^t) \in H^\infty$ player i plays his best response against $\tau_{-i}(h)$ in $G^\lambda(z^t; w^*)$. That is

$$U^\lambda(z^t; w^*; \tau_i^*(h), \tau_{-i}(h)) \ge \overline{v}_i(G^\lambda(z^t; w^*)) = w^*(z^t).$$

Therfore, by Lemma 15.29, $\gamma_i^\lambda(z^1, \tau_i^*, \tau_{-i}) \ge w^*(z^1)$ for every $z^1 \in Z$. In conclusion,

$$\overline{v}_i^\lambda(z) \ge \inf_{\tau_{-i}\in\mathcal{B}_{-i}^\infty} \gamma_i^\lambda(z^1, \tau_i^*, \tau_{-i}) \ge w^*(z^1). \tag{138}$$

On the other hand, let τ_{-i}^* be a stationary strategy of $-i$, according to which at each state z all of the players except for player i try to lower player i's payoff in $G^\lambda(z; w^*)$ by playing the minmax strategy vector. Then, at each state z, for every mixed action $\sigma_{i,z}$ of player i

$$U^\lambda(z; w^*; \sigma_{i,z}, \tau_{-i}^*(z)) \le \overline{v}_i(G^\lambda(z^t; w^*)) = w^*(z^t).$$

Hence, by Lemma 15.30 and Equation (138),for every $z \in Z$ and every $\tau_i \in \mathcal{B}_i^\infty$ one has

$$\gamma_i^\lambda(z, \tau_i, \tau_{-i}^*) \le w^*(z) \le \overline{v}_i^\lambda(z),$$

as claimed.

15.24 Let $\lambda \in [0,1)$ be a discount factor. We find a number $p_\lambda \in [0,1]$ such that the vector of stationary strategies

$$([p_\lambda(T),(1-p_\lambda)(B)],[p_\lambda(L),(1-p_\lambda)(R)],[p_\lambda(W),(1-p_\lambda)(E)])$$

is a λ-discounted equilibrium.

Denote by $w^* = (w_1^*, w_2^*, w_3^*)$ the λ-discounted equilibrium payoff corresponds to the symmetric equilibrium. Hence, the game $G^\lambda(z_0, w^*)$ is

$$\begin{array}{c|c|c|}
 & p_\lambda & 1-p_\lambda \\ \hline
p_\lambda & \lambda w_1^*, \lambda w_2^*, \lambda w_3^* & 0,1,3 \\ \hline
1-p_\lambda & 1,3,0 & 1,0,1 \\ \hline
\multicolumn{3}{c}{p_\lambda}
\end{array}
\qquad \text{state } z_0 \qquad
\begin{array}{c|c|c|}
 & p_\lambda & 1-p_\lambda \\ \hline
p_\lambda & 3,0,1 & 1,1,0 \\ \hline
1-p_\lambda & 0,1,1 & 0,0,0 \\ \hline
\multicolumn{3}{c}{1-p_\lambda}
\end{array}$$

Note that there is no symmetric equilibrium in which the players play pure: if $p_\lambda = 0$ then Player I is better of playing $p_\lambda = 1$ and if $p_\lambda = 1$ then Player I is better of playing $p_\lambda = 0$. Hence, by the indifference principle, for every player i one has

$$\left.\begin{aligned} w_i^* &= \lambda w_i^* p_\lambda^2 + 3p_\lambda(1-p_\lambda) + (1-p_\lambda)^2 \\ w_i^* &= p_\lambda \end{aligned}\right\} \Rightarrow \lambda p_\lambda^3 - 2p_\lambda^2 + 1 = 0.$$

That is, the solution to the cubic equation $\lambda p_\lambda^3 - 2p_\lambda^2 + 1 = 0$ is the needed value of p_λ. Note that such a solution exists, since the function that assign to each $p_\lambda \in (0,1)$ the value of $\lambda p_\lambda^3 - 2p_\lambda^2 + 1$ is a continuous function that assigns a positive value to $p_\lambda = 0$ and a negative value to $p_\lambda = 1$.

15.25 Fix a discount factor $\lambda \in [0,1)$.

(a) We prove that the set $\mathcal{B}_i^\infty$ is a compact set in the metric d_i defined in Equation (15.19).

For every $\tau_i, \widehat{\tau}_i \in \mathcal{B}_i^\infty$,

$$\begin{aligned} d(\tau_i, \widehat{\tau}_i) &= \sum_{t=1}^{\infty} \tfrac{1}{2^t |H(t)|} \sum_{h \in H(t)} \|\tau_i(h) - \widehat{\tau}_i(h)\|_1 \\ &\le \sum_{t=1}^{\infty} \tfrac{1}{2^t |H(t)|} \sum_{h \in H(t)} \max_{z \in Z} |A_i(z)| = \max_{z \in Z} |A_i(z)|, \end{aligned}$$

hence the set $\mathcal{B}_i^\infty$ is bounded.

Let $\tau_i^1, \tau_i^2, \ldots \in \mathcal{B}_i^\infty$ such that $\lim_{k\to\infty} \tau_i^k = \tau_i$. We show that for every $h = (z^1, a^1, \ldots, z^t)$, $\tau_i(h) \in \Delta(A(z^t))$ and thus $\tau_i \in \mathcal{B}_i^\infty$ and $\mathcal{B}_i^\infty$ is closed.

Let $t \in \mathbf{N}$ and let $h = (z^1, a^1, ..., z^t) \in H(t)$. Since, $\lim_{k\to\infty} \tau_i^k = \tau_i$, for every $\epsilon > 0$ there is $K \in \mathbf{N}$ such that for every $k > K$

$$\frac{\epsilon}{2^t|H(t)|} > d(\tau_i^k, \tau_i) = \sum_{t'=1}^{\infty} \frac{1}{2^{t'}|H(t')|} \sum_{h' \in H(t')} \|\tau_i^k(h') - \tau_i(h')\|_1$$
$$\Rightarrow \|\tau_i^k(h) - \tau_i(h)\|_1 < \epsilon.$$

from which it follows that $\tau_i(h) = \lim_{k\to\infty} \tau_i^k(h)$. However, since $\Delta(A(z))$ is compact for every $z \in Z$ then for every $t \in \mathbf{N}$ and every $h = (z^1, a^1, ..., z^t) \in H(t)$, $\tau_i(h) \in \Delta(A(z^t))$ so $\tau_i \in \mathcal{B}_i^\infty$, as claimed.

Note that, the set $\mathcal{B}^\infty = \times_{i\in N}\mathcal{B}_i^\infty$ is compact as well (by Exercise 5.2).

(b) We prove that for every state $z^1 \in Z$, the function $\tau \mapsto \gamma_i^\lambda(z^1, \tau)$ is a continuous function in the metric d defined in Equation (15.20). That is, for every $\epsilon > 0$ there is $\delta > 0$ such that for every $\tau, \widehat{\tau} \in \mathcal{B}^\infty$ satisfying $d(\tau, \widehat{\tau}) < \delta$ one has

$$|\gamma_i^\gamma(z^1, \tau) - \gamma_i^\gamma(z^1, \widehat{\tau})| < \epsilon.$$

Let $\epsilon > 0$. Denote by M a bound over the absolute values of the stage payoffs. Let $K \in \mathbf{N}$ sufficiently large where

$$|(1-\lambda)\sum_{t=K+1}^{\infty} \lambda^{t-1}(E_{z^1,\tau}[u_i^t] - E_{z^1,\widehat{\tau}}[u_i^t])| \le (1-\lambda)\sum_{t=K+1}^{\infty} \lambda^{t-1}2M < \tfrac{\epsilon}{2}.$$

Choose $\delta < \frac{\epsilon}{MK\cdot 2^{K+1}|H(K)|}$. Then,

$$\begin{aligned}
&|\gamma_i^\lambda(z^1, \tau) - \gamma_i^\lambda(z^1, \widehat{\tau})| \\
&= \left|(1-\lambda)\sum_{t=1}^{\infty} \lambda^{t-1}\left(E_{z^1,\tau}[u_i^t] - E_{z^1,\widehat{\tau}}[u_i^t]\right)\right| \\
&\le \left|(1-\lambda)\sum_{t=1}^{K} \lambda^{t-1}\left(E_{z^1,\tau}[u_i^t] - E_{z^1,\widehat{\tau}}[u_i^t]\right)\right| + \tfrac{\epsilon}{2} \\
&\le \sum_{t=1}^{K}\sum_{h\in H(t)}\sum_{a\in A(z^t)} u_i(z^t, a^t)\left|\mathbf{P}(h|z^1,\tau)\tau(a|h) - \mathbf{P}(h|z^1,\widehat{\tau})\widehat{\tau}(a|h)\right| + \tfrac{\epsilon}{2} \\
&\le M\sum_{t=1}^{K}\sum_{h\in H(t)}\sum_{i\in N}(K-t+1)\|\tau_i(h) - \widehat{\tau}_i(h)\|_1 + \tfrac{\epsilon}{2} \\
&\le MK2^K|H(K)|\sum_{i\in N}\sum_{t=1}^{K}\frac{1}{2^t|H(t)|}\sum_{h\in H(t)}\|\tau_i(h) - \widehat{\tau}_i(h)\|_1 + \tfrac{\epsilon}{2} \\
&\le MK2^k|H(K)|d(\tau, \widehat{\tau}) + \tfrac{\epsilon}{2} \le MK2^k|H(K)|\delta + \tfrac{\epsilon}{2} < \epsilon.
\end{aligned}$$

(c) For every initial state $z^1 \in Z$ and every $k \in \mathbf{N}$, let $\widehat{\Gamma}^{k,\lambda}(z^1)$ be the game in which the player set is N, the strategy set of each player i is $\mathcal{B}_i^\infty$, and the payoff function is the k'th stage payoff defined in Equation (15.21). The game $\widehat{\Gamma}^{k,\lambda}(z^1)$ is a finite game so every player has a finite number of pure strategies. Therefore, by Nash's Theorem (Theorem 5.10) the game with each initial state z^1 has an equilibrium $\tau^k(z^1)$.

(d) Let $(\tau^k(z^1))_{k\in\mathbf{N}}$ be a sequence of equilibria in the games $(\widehat{\Gamma}^{\lambda,k}(z^1))_{k\in\mathbf{N}}$ respectively. For every $k \in \mathbf{N}$ denote by $\widehat{\tau}^k(z_1)$ the strategy vector in $\Gamma^\lambda(z^1)$ according to which at the first k stages the players follow the strategy vector $\tau^k(z^1)$ and from stage $k+1$ onwards they choose arbitrary actions. Each strategy vector $\widehat{\tau}^k(z^1)$ is in $\mathcal{B}^\infty$ which is compact, by Part (a). Hence, the sequence $(\widehat{\tau}^k(z^1))_{k\in\mathbf{N}}$ has a converging subsequence. Denote by $\tau^*(z^1) \in \mathcal{B}^\infty$ the limit point of such a subsequence. That is $\lim_{l\to\infty} \widehat{\tau}^{k_l}(z^1) = \tau^*(z^1)$.

(e) For every player $i \in N$ and every $\sigma_i \in \mathcal{B}_i^\infty$ denote by σ_i^k the reduction of σ_i to the first k stages.

Let $\epsilon > 0$ and let K be sufficiently large such that $(1-\lambda)\sum_{t=K+1}^\infty \lambda^t M < \frac{\epsilon}{2}$ hence the following holds for every $k > K$,

$$\gamma_i^\lambda(z^1,\widehat{\tau}^k) \geq \gamma_i^{\lambda,k}(z^1,\widehat{\tau}^k) - \tfrac{\epsilon}{2} \;\; \forall k > K,$$
$$\gamma_i^\lambda(z^1,\sigma_i,\widehat{\tau}_{-i}^k) \leq \gamma^{\lambda,k}(z^1,\sigma_i^k,\widehat{\tau}_{-i}^k) + \tfrac{\epsilon}{2} \;\; \forall k > K.$$

In addition, since $\tau^k(z^1)$ is an equilibrium in $\Gamma^{\lambda,k}(z^1)$,

$$\gamma^{\lambda,k}(z^1,\sigma_i^k,\widehat{\tau}_{-i}^k) \leq \gamma_i^{\lambda,k}(z^1,\widehat{\tau}^k) \;\; \forall k > K.$$

In conclusion, for every $k \geq K$,

$$\gamma_i^\lambda(z^1,\widehat{\tau}^k) \geq \gamma_i^\lambda(z^1,\sigma_i,\widehat{\tau}_{-i}^k) - \epsilon$$

Therefore, by the continuity of the function $\tau \mapsto \gamma_i^\lambda(z^1,\tau)$ and letting ϵ converge to 0 yields

$$\gamma_i^\lambda(z^1,\tau^*) \geq \gamma_i^\lambda(z^1,\sigma_i,\tau_{-i}^*).$$

That is, the vector of strategies $\tau^*(z^1)$ is a λ-discounted equilibrium of the stochastic game with initial state z^1.

15.26 (a) By playing $[\frac{1}{3}(T),\frac{1}{3}(M),\frac{1}{3}(B)]$ at z_0 Player I guarantees that his payoff is positive. Therefore, the λ-discounted maxmin value is positive.

(b) If at z_0, Player I never plays one of his pure actions, then Player II has a counter strategy yielding a zero payoff. For instance, if Player I never plays T then Player II's best response is R and the payoff is 0, which contradicts Part (a). Hence, the optimal stationary strategy of Player I (in the nonabsorbing state z_0) is completely mixed.

(c) The game $G^\lambda(z_0; v)$ is the following game:

	L	I	R
T	λv	λv	1
M	λv	1	0
B	1	0	0

state z_0

We have to find $v \in \mathbf{R}$ for which there exists a value of $G^\lambda(z_0; v)$ with payoff of v. By Part (b), the optimal stationary strategy of Player I is completely mixed, denoted by $[x_1(T), x_2(M), x_3(B)]$ where $x_1, x_2, x_3 > 0$ and $x_1 + x_2 + x_3 = 1$. We use the indifference principle to obtain the following equations:

$$\begin{aligned} v &= x_1 \lambda v + x_2 \lambda v + x_3 \\ v &= x_1 \lambda v + x_2 \\ v &= x_1. \end{aligned}$$

The solution of this system is

$$\begin{aligned} v &= \tfrac{1-\sqrt[3]{1-\lambda}}{\lambda}, \\ x_1 &= v, \\ x_2 &= v - \lambda v^2, \\ x_3 &= 1 - 2v + \lambda v^2. \end{aligned}$$

By similar arguments, the optimal stationary strategy of Player II is $[x_1(L), x_2(I), x_3(R)]$.

15.27 Let $\lambda \in [0, 1)$ be a discount factor.
The game $G^\lambda(z_0; v)$ is the following game:

	L	R
T	$1 - \lambda + 0.6\lambda v$	$1 - \lambda + 0.3\lambda v$
B	$0.4\lambda v$	$2(1 - \lambda) + 0.5\lambda v$

state z_0

We have to find $v \in \mathbf{R}$ for which there exists a value of $G^\lambda(z_0; v)$ with payoff of v. In the game $G^\lambda(z_0, v)$ the off-diagonal entries are below the diagonal entries, hence the optimal actions of both players are completely mixed. Denote by $[x(T), (1-x)(B)]$ the optimal stationary strategy of Player I. We use the indifference principle to obtain the following equations:

$$\begin{aligned} v &= x(1-\lambda+0.6\lambda v) + (1-x)0.4\lambda v, \\ v &= x(1-\lambda+0.3\lambda v) + (1-x)(2-2\lambda+0.5\lambda v). \end{aligned}$$

The solution of this system is

$$\begin{aligned} v &= \tfrac{20(1-\lambda)}{20-9\lambda}, \\ x &= \tfrac{4}{5} \cdot \tfrac{2\lambda-5}{\lambda-4}. \end{aligned}$$

By similar arguments we get that the optimal stationary of Player II is $[\frac{1}{2}(L), \frac{1}{2}(R)]$.

15.28 Let $\lambda \in [0,1)$ be a discount factor.
The game $G^\lambda(z_0; v)$ is the following game:

	L	R
T	$3(1-\lambda)+0.6\lambda v$	$1-\lambda+0.3\lambda v$
B	$0.4\lambda v$	$2(1-\lambda)+0.5\lambda v$

state z_0

We have to find $v \in \mathbf{R}$ for which there exists a value of $G^\lambda(z_0; v)$ with payoff of v. In the game $G^\lambda(z_0, v)$ the off-diagonal entries are below the diagonal entries, hence the optimal actions of both players are completely mixed. Denote by $[x(T), (1-x)(B)]$ the optimal stationary strategy of Player I. We use the indifference principle to obtain the following equations:

$$\begin{aligned} v &= x(3-3\lambda+0.6\lambda v) + (1-x)0.4\lambda v, \\ v &= x(1-\lambda+0.3\lambda v) + (1-x)(2-2\lambda+0.5\lambda v). \end{aligned}$$

The solution of this system is

$$\begin{aligned} v &= \frac{5\left(23\lambda^2-63\lambda+40+(\lambda-1)\sqrt{97\lambda^2-880\lambda+1600}\right)}{2\lambda(9\lambda-20)}, \\ x &= \frac{15\lambda-40+\sqrt{97\lambda^2-880\lambda+1600}}{8\lambda}. \end{aligned}$$

By similar arguments we get that the optimal stationary of Player II is $[y(L), (1-y)(R)]$ where

$$y = \frac{40-9\lambda+\sqrt{97\lambda^2-880\lambda+1600}}{8\lambda}.$$

15.29 Let Γ^λ be a two-player zero-sum symmetric discounted stochastic game. That is, (1) $A_I(z) = A_{II}(z) = A(z)$ for every $z \in Z$; (2) $u(z, a_I, a_{II}) = -u(z, a_{II}, a_I)$ for every $z \in Z$ and every $a_I, a_{II} \in A(z)$; and (3) $q(z, a_I, a_{II}) = q(z, a_{II}, a_I)$ for every $z \in Z$ and $a_I, a_{II} \in A(z)$. We prove that the value of $\Gamma^\lambda(z^1)$ is 0 for every initial state z^1. For every history $h = (z^1, a^1_I, a^1_{II}, z^2, a^2_I, a^2_{II}, ..., z^t)$ let $\widetilde{h} = (z^1, a^1_{II}, a^1_I, z^2, a^2_{II}, a^2_I, ..., z^t)$ be the history in which the actions of the players at each stage were switched one by the other. For every strategy τ of Player I let $\widehat{\tau}$ be a strategy of player II such that $\widehat{\tau}(\widehat{h}) = \tau(h)$. Similarly, for every strategy σ of Player II let $\widehat{\sigma}$ be a strategy of player II such that $\widehat{\sigma}(\widehat{h}) = \sigma(h)$. We first prove by induction over t that for every $t \in \mathbf{N}$ and every $h \in H(t)$, $\mathbf{P}(h|z^1, \tau, \sigma) = \mathbf{P}(\widetilde{h}|z^1, \widehat{\sigma}, \widehat{\tau})$ where τ and σ are strategies of Player I and Player II respectively.

For $t = 1$, the claim holds trivially since $h = \widetilde{h} = \{z\}$ for every $h \in H(1)$. Assume that the claim holds for every $t' \leq t$. We prove that it also holds for $t+1$. Let $h = (z^1, a^1_I, a^1_{II}, z^2, a^2_I, a^2_{II}, ..., z^{t+1})$. Denote by $h_{|t} = (z^1, a^1_I, a^1_{II}, z^2, a^2_I, a^2_{II}, ..., z^t) \in H(t)$ the prefix of h. Then

$$\begin{aligned}
&\mathbf{P}(h|z^1, \tau, \sigma) = \mathbf{P}(h_{|t}|z^1, \tau, \sigma)\tau(a^t_I|h_{|t})\sigma(a^t_{II}|h_{|t})q(z^{t+1}|z^t, a^t_I, a^t_{II}) \\
&= \mathbf{P}(\widehat{h}_{|t}|z^1, \widehat{\sigma}, \widehat{\tau})\tau(a^t_I|h_{|t})\sigma(a^t_{II}|h_{|t})q(z^{t+1}|z^t, a^t_I, a^t_{II}) && (139)\\
&= \mathbf{P}(\widehat{h}_{|t}|z^1, \widehat{\sigma}, \widehat{\tau})\widehat{\tau}(a^t_I|\widehat{h}_{|t})\widehat{\sigma}(a^t_{II}|\widehat{h}_{|t})q(z^{t+1}|z^t, a^t_I, a^t_{II}) && (140)\\
&\mathbf{P}(\widehat{h}_{|t}|z^1, \widehat{\sigma}, \widehat{\tau})\widehat{\tau}(a^t_I|\widehat{h}_{|t})\widehat{\sigma}(a^t_{II}|\widehat{h}_{|t})q(z^{t+1}|z^t, a^t_{II}, a^t_I) && (141)\\
&= \mathbf{P}(\widetilde{h}|z^1, \widehat{\sigma}, \widehat{\tau}), && (142)
\end{aligned}$$

where Equality (139) holds by the induction hypothesis, Equality (140) holds by the definition of $\widehat{h}_{|t}$, and Equality (141) follows the symmetry of the game (condition (3)).

The above claim and condition (2) yield the following for every initial state $z^1 \in Z$ and every τ and σ strategies of Player I and Player II respectively

$$\begin{aligned}
E_{z^1,\tau,\sigma}(u^t_i) &= \sum_{h \in H(t)} q(h|z^1, \tau, \sigma)U_i(z^t, \tau(h), \sigma(h)) \\
&= -\sum_{h \in H(t)} q(\widehat{h}|z^1, \widehat{\sigma}, \widehat{\tau})U_i(z^t, \widehat{\sigma}(\widehat{h}), \widehat{\tau}(\widehat{h})) \\
&= -E_{z^1,\widehat{\sigma},\widehat{\tau}}(u^t_i).
\end{aligned}$$

and thus

$$\gamma^\lambda(z^1, \tau, \sigma) = (1-\lambda)\sum_{t=1}^{\infty} \lambda^{t-1} E_{z^1,\tau,\sigma}(u^t_i) \tag{143}$$

$$= -(1-\lambda)\sum_{t=1}^{\infty} \lambda^{t-1} E_{z^1,\widehat{\sigma},\widehat{\tau}}(u^t_i) = -\gamma^\lambda(z^1, \widehat{\sigma}, \widehat{\tau}) \tag{144}$$

We next show that Equation (144) yields to $v^\lambda(z^1) = 0$ for every initial state $z^1 \in Z$. Assume by contradiction that there is an initial state $z^1 \in Z$ such that $v^\lambda(z^1) > 0$. Let τ^* be an optimal strategy of Player I. Hence, for every strategy τ of Player II,

$$\gamma^\lambda(z^1, \sigma^*, \tau) \geq v^\lambda(z^1).$$

However, by playing $\widehat{\tau}^*$ Player II can guarantee that the payoff is at most $-v$

$$\gamma^\lambda(\tau, \widehat{\tau}^*) = -\gamma^\lambda(\tau^*, \widehat{\tau}) \leq -v^\lambda(z^1) < 0 \;\; \forall \tau \in \mathcal{B}_I^\infty,$$

a contradiction. The case in which $v^\lambda(z^1) < 0$ can be proved by switching the roles of the players.

15.30 (a) Every discounted Markov decision process can be represented as a two-player zero-sum discounted stochastic game by adding a player to the game (Player II) who has only one action at each state.

Hence, by Shapley Theorem (Theorem 15.32), the player (Player I in the two-player game) has an optimal stationary strategy according to which he plays at each state $z \in Z$ the optimal strategy in the one shot game $G^\lambda(z, v^\lambda)$. However, since Player II has only one strategy at each state, at each state Player I has a pure optimal strategy that dominates all other strategies . In conclusion, for every discount factor $\lambda \in [0, 1)$, the player has an optimal strategy that is stationary and pure.

(b) Let σ be a stationary strategy. For every discount factor $\lambda \in [0, 1)$, the discounted payoffs $(\gamma^\lambda(z, \sigma))_{z \in Z}$ satisfy the following system of equations

$$\gamma^\lambda(z, \sigma) = (1 - \lambda) U(z, \sigma(z)) + \lambda \sum_{z'} \mathbf{P}_{z,\sigma}(z^2 = z') \gamma^\lambda(z', \sigma) \;\; \forall z \in Z.$$

We next prove that for every discount factor λ, the solution of the above system of equations defined a rational function $\lambda \mapsto \gamma^\lambda(z^1, \sigma)$ for every initial state z^1 the payoff function is a rational function.

We prove a more general claim: if there is a solution to a rational system of equations

$$X_i = f_{i,0}(\lambda) + \sum_{j=1}^{n} f_{i,j}(\lambda) X_j \;\; \forall i = 1, 2, ..., n \tag{145}$$

where $f_{i,j}(\lambda)$ is a rational function for every $i = 0, 1, ..., n$ and $j = 1, 2, ..., n$, then the solution $X_i(\lambda)$ is a rational function. We prove the claim by induction over n.

For $n = 1$,

$$X_1 = \frac{f_{1,0}(\lambda)}{1 - f_{1,1}(\lambda)},$$

which is a rational function, as a division of two rational function. Assume that the claim holds for n, we prove that it also holds for $n+1$. By Equation (145),

$$X_{n+1} = \frac{f_{n+1,0}(\lambda)+\sum_{j=1}^{n} f_{n+1,j}(\lambda)X_j}{1-f_{n+1,n+1}(\lambda)}.$$

Hence, for every $i = 1, 2, ..., n$

$$X_i = f_{i,0}(\lambda) + \sum_{j=1}^{n} f_{i,j}(\lambda)X_j + f_{i,n+1}(\lambda)\frac{f_{n+1,0}(\lambda)+\sum_{j=1}^{n} f_{n+1,j}(\lambda)X_j}{1-f_{n+1,n+1}(\lambda)},$$

which is a rational system of equations with n variable and a solution. Therefore, by the induction hypothesis, the solution $X_i(\lambda)$ is a rational function for every $i = 1, 2, ..., n$ and thus $X_{n+1}(\lambda)$ is a rational function as well, as claimed.

(c) By Part (a), for every $\lambda \in [0,1)$, the optimal strategy is stationary and pure. That is, if a pure stationary strategy τ^* satisfies $\gamma^\lambda(z^1, \tau^*) \geq \gamma^\lambda(z^1, \tau)$ for every pure stationary strategy τ and every initial state $z^1 \in Z$, then τ^* is an optimal strategy in the discounted Markov decision process. However, by Part (b), the function $\lambda \mapsto \gamma^\lambda(z^1, \tau') - \gamma^\lambda(z^1, \tau)$ is a rational function for every two stationary strategies τ and τ', so it is a continuous function with finitely many roots. That is, there is a segment $(\lambda', 1) \subset [0,1)$ for which the function has the same sign, so one of the strategies is preferred to the other. In conclusion, since there is only finitely many pure stationary strategies, there is a segment $(\lambda', 1) \subset [0,1)$ for which one pure stationary strategy is preferred to all other pure stationary strategies. Thus, there is a pure stationary strategy σ that is λ-discounted optimal for every discount factor λ close enough to 1.

15.31 We generalize Shapley's Theorem (Theorem 15.32) to a stochastic game $\Gamma = (N, Z, (A_i)_{i\in N}, (u_i)_{i\in N}, q)$ where the state space is a countable set $Z = \mathbf{N}$, each player has a finite number of actions in each state and the absolute values of the payoff functions are bounded by M.

Let $\lambda \in [0,1)$. For every $n \in \mathbf{N}$, let $\Gamma^n = (N, Z^n, (A_i)_{i\in N}, (u_i)_{i\in N}, q^n)$ be the stochastic game where $Z^n = \{1, 2, ..., n\}$ and for every $z, z' \in Z^n$

$$q^n(z'|z,a) = \begin{cases} q(z'|z,a) & z' < n \\ \sum_{\hat{z}=n}^{\infty} q(\hat{z}|z,a) & z' = n. \end{cases}$$

The game Γ^n is the game that is obtained by eliminating all states higher than n.
For every $n \in \mathbf{N}$ denote by $\Gamma^{\lambda,n}$ the λ-discounted infinite stochastic game which corresponds to Γ^n. By Theorem 15.32, for every $n \in \mathbf{N}$, the value $v^{\lambda,n}$ of $\Gamma^{\lambda,n}$ exists. Furthermore, each player has a stationary optimal strategy, and the value $v^{\lambda,n} = (v^{\lambda,n}(z))_{z\in Z^n}$ is the unique solution of the following system of equations:

$$v^{\lambda,n}(z) = \mathrm{val}(G^{\lambda,n}(z; v^{\lambda,n})). \tag{146}$$

Let $z \in Z$. We next show that $(v^{\lambda,n}(z))_{n\in\mathbf{N}}$ converses to $\overline{v}^{\lambda}(z)$ as well as to $\underline{v}(z)$ from which it follows that $v^{\lambda}(z)$ exist.
Let $\epsilon > 0$. Let $K \in \mathbf{N}$ be sufficiently large such that $\frac{\lambda^K 2M}{1-\lambda} < \frac{\epsilon}{2}$. Let $N > K$ be sufficiently large such that the probability that during the first K stages the game $\Gamma(z)$ reaches a state $z' \notin Z^N$ under every strategy profile is less than $\frac{\epsilon}{4M}$.
Denote by $\tilde{\tau}^n_{\mathrm{I}}$ ($\tilde{\tau}^n_{\mathrm{II}}$) the strategy of Player I (Player II) in Γ^{λ} according to which he follows his stationary optimal strategy in $\Gamma^{\lambda,n}$ as long as the game visits states in Z^n, however, as soon as the game visits a state $z > n$ he plays arbitrary actions.
For every $n > N$,

$$\begin{aligned} v^{\lambda,n}(z) - \epsilon &< v^{\lambda,n}(z) - \tfrac{\lambda^K 2M}{1-\lambda} - \tfrac{\epsilon}{4M} \cdot 2M \\ &\leq \min_{\tau_{\mathrm{II}} \in \mathcal{B}^{\infty}_{\mathrm{II}}} \gamma^{\lambda}(z; \tilde{\tau}^n_{\mathrm{I}}, \tau_{\mathrm{II}}) \\ &\leq \underline{v}^{\lambda}(z), \end{aligned} \tag{147}$$

where Inequality (147) holds since the stage payoffs at the first n stages in both in Γ^{λ} and $\Gamma^{\lambda,n}$ under $\tilde{\tau}^n_I$ and τ^n_I respectively are identical unless the game $\Gamma^{\lambda,n}$ reaches a state $z \notin Z^n$ which occurs with probability less than $\frac{\epsilon}{4M}$.
Similarly, for every $n > N$,

$$\begin{aligned} v^{\lambda,n}(z) + \epsilon &> v^{\lambda,n}(z) + \tfrac{\lambda^K 2M}{1-\lambda} + \tfrac{\epsilon}{4M} \cdot 2M \\ &\geq \max_{\tau_{\mathrm{I}} \in \mathcal{B}^{\infty}_{\mathrm{I}}} \gamma^{\lambda}(z; \tau_{\mathrm{I}}, \tilde{\tau}^n_{\mathrm{II}}) \\ &\geq \overline{v}^{\lambda}(z). \end{aligned}$$

Since $\underline{v}^{\lambda}(z) \leq \overline{v}^{\lambda}(z)$,

$$v^{\lambda,n}(z) - \epsilon < \underline{v}^{\lambda}(z) \leq \overline{v}^{\lambda}(z) < v^{\lambda,n}(z) + \epsilon.$$

In conclusion, for every $n, m > N$

$$|v^{\lambda,n}(z) - v^{\lambda,n}(z)| < |v^{\lambda,n}(z) - \underline{v}^{\lambda}(z)| + |\underline{v}^{\lambda}(z) - v^{\lambda,n}(z)| < 2\epsilon.$$

Therefore, $(v^{\lambda,n}(z))_{n\in\mathbf{N}}$ is a Cauchy sequence. Consequently, this sequence has a limit and thus

$$\lim_{n\to\infty} v^{\lambda,n}(z) - \epsilon < \underline{v}^{\lambda}(z) \leq \overline{v}^{\lambda}(z) < \lim_{n\to\infty} v^{\lambda,n}(z) + \epsilon. \tag{148}$$

Since Equation (148) holds for every $\epsilon > 0$,

$$\lim_{n\to\infty} v^{\lambda,n}(z) = \underline{v}^{\lambda}(z) = \overline{v}^{\lambda}(z) = v^{\lambda}(z).$$

Furthermore, by Equation (146) and since the value function is continuous (Exercise 5.40)

$$\lim_{n\to\infty} v^{\lambda,n}(z) = \mathrm{val}(\lim_{n\to\infty} G^{\lambda,n}(z;v^{\lambda,n}))$$
$$v^{\lambda}(z) = \mathrm{val}(G^{\lambda}(z;v^{\lambda})).$$

That is, each player has an optimal strategy in Γ^{λ}. By Exercise 15.32, the strategy in Γ^{λ} according to which at each state z a player follows his optimal strategy in $G^{\lambda}(z;v^{\lambda})$ is a stationary optimal strategy in Γ^{λ}.

15.32 Proof of Theorem 15.40: let Γ^{λ} be a two-player zero-sum stochastic game. Let τ_{I} be a strategy of Player I such that for every finite history $h = (z^1, a^1, z^2, a^2, \cdots, z^{t-1}, a^{t-1}, z^t) \in H^{\infty}$, the mixed action $\tau_{\mathrm{I}}(h)$ is an optimal strategy in the game $G^{\lambda}(z^t; v^{\lambda})$. We show that τ_{I} is an optimal strategy of Player I in the stochastic game.
By Theorem 15.32, there is an optimal stationary strategy τ_I^* of Player I. For every $t \in \mathbf{N}$, denote by τ_I^t the strategy according to which Player I plays according to τ_I for the first t stages, and from stage $t+1$ onwards he follows τ_I^*. The expected payoff in $\Gamma^{\lambda}(z^1)$ under τ_I^t from stage $t+1$ onwards against every strategy of Player II is at least the value of the game $v^{\lambda}(z^{t+1})$. Thus, using backward induction and by the property of τ_I, the payoff of player I in $\Gamma^{\lambda}(z^1)$ is at least $v^{\lambda}(z^1)$. However, $\lim_{t\to\infty} \tau_I^* = \tau_I$ so for every strategy τ_{II} of Player II,

$$\gamma^{\lambda}(z^1, \tau_I, \tau_{II}) \geq \gamma^{\lambda}(z^1, \tau_I^t, \tau_{II}) - \lambda^t(2M) > v^{\lambda}(z^1) - \lambda^t(2M) \quad \forall t \in \mathbf{N}.$$

Converging t to infinity, yields

$$\gamma^{\lambda}(z^1, \tau_I, \tau_{II}) \geq v^{\lambda}(z^1) \quad \forall \tau_{II}.$$

Note that the converse of the above result is not necessarily true. Consider the following discounted two-player zero-sum stochastic game

	L	R
T	0 $*$	1 z_1
B	1 z_1	2 z_1

z_1

Define for a strategy τ_I as follows

$$\tau_I(h) = \begin{cases} B & h = z_1 \\ B & h = (z_1, (B, a_2^1), z_1, (B, a_2^2), ..., (B, a_2^{t-1}), z_1) \\ T & \text{otherwise.} \end{cases}$$

Then, τ_I is an optimal strategy of Player I corresponding $v^\lambda(z^1) = 1$ and $v^\lambda(z_2) = 0$. But for $h = (z^1, (B, a_2^1), z^1, (T, a_2^2), z^1)$ the action $\tau_{\mathrm{I}}(h) = T$ is not an optimal strategy in the game $G^\lambda(z^t; v^\lambda)$ since T is strictly dominated by B.

15.33 We prove that in a two-player zero-sum stochastic game, a pair of strategies $(\tau_{\mathrm{I}}, \tau_{\mathrm{II}})$ is a subgame-perfect equilibrium if and only if the next property holds: for every finite history

$$h = (z^1, a^1, z^2, a^2, \cdots, z^{t-1}, a^{t-1}, z^t) \in H^\infty,$$

the mixed action $\tau_{\mathrm{I}}(h)$ is an optimal strategy of Player I in the game $G^\lambda(z^t; v^\lambda)$ and the mixed action $\tau_{\mathrm{II}}(h)$ is an optimal strategy of Player II in this game.

Assume first that the property holds. Then by Exercise 15.32, for every history $h = (z^1, a^1, ..., z^t) \in H^\infty$ both strategies, τ_I and τ_{II}, are optimal strategies for Player I and Player II respectively in the subgame starting after h occurs. Therefore, by Theorem 4.45, the pair of strategies (τ_I, τ_{II}) defines an equilibrium in the subgame starting after h and so (τ_I, τ_{II}) is a subgame perfect equilibrium.

Assume next that the property does not hold. Then there are an history $h = (z^1, a^1, ..., z^t) \in H^\infty$ and a player, say Player I, such that $\tau_1(h)$ is not optimal in the game $G^\lambda(z^t; v^\lambda)$. Therefore, one can derive that τ_I is not optimal in the subgame starting after h (Exercise 15.19) and thus, by Theorem 4.44, the pair of strategies (τ_I, τ_{II}) is not an equilibrium in the subgame starting after h and it is not a subgame perfect equilibrium.

15.34 An infinite stochastic game has *perfect information* if each one of the $|Z|$ extensive-form games assigned to it, one for every initial state, has perfect information.

(a) Let Γ^λ be a λ-discounted infinite stochastic game.

Assume first that in every state $z \in Z$, the action sets of all players but one (denote by i_z) is a singleton. Hence, at every state z, each player $i \neq i_z$ has only one action, so he is not really an active player. Player i_z is the only active player at state z who has perfect recall and he knows the actions that are taken by all other players (since each has only one action). Thus, he has perfect information. In conclusion the game has perfect information as well.

On the other hand, if there is at least one state such that there are at least two players with more than one action, then each one of them does not know the action that is chosen by the other and thus the game has no perfect information.

(b) We prove that in every two-player zero-sum λ-discounted stochastic game with perfect information, each player has a pure optimal strategy.

By Shapley's Theorem (Theorem 15.32) in every two-player zero-sum λ-discounted stochastic game, each player has a stationary optimal strategy according to which he plays at each state $z \in Z$ an optimal strategy of the one shot game $G^\lambda(z, v^\lambda)$. However, there are pure optimal strategies for both players in every game $G^\lambda(z, v^\lambda)$: player $-i_z$ has only one pure action which is necessarily a unique optimal action of $G^\lambda(z, v^\lambda)$; and thus player i_z has an action that dominates all other actions so he has pure optimal strategy. In conclusion, in every two-player zero-sum λ-discounted stochastic game with perfect information, each player has a pure optimal strategy.

15.35 Consider a two-player zero-sum stochastic game with perfect information.

(a) Let $\tau = (\tau_I, \tau_{II})$ be a stationary strategy vector. For every discount factor $\lambda \in [0, 1)$, the discounted payoffs $(\gamma^\lambda(z, \tau))_{z \in Z}$ satisfy the following system of equations

$$\gamma^\lambda(z, \tau) = (1 - \lambda) U(z, \tau(z)) + \lambda \sum_{z'} \mathbf{P}_{z,\tau}(z^2 = z') \gamma^\lambda(z', \tau) \quad \forall z \in Z.$$

However, by Exercise 15.30 (Part (b)), for every discount factor λ, the solution of the above system of equations defined a rational function $\lambda \mapsto \gamma^\lambda(z^1, \tau)$ for every initial state z^1.

(b) Let τ_{II} be a stationary strategy of Player II and let λ be a discount factor. The best reply of Player I in the two-player zero-sum stochastic game with perfect information, can be equivalently presented as an optimal strategy in a Markov decision problem in which the payoff function of Player I is $\widehat{u}(z, a) = u(z, (a, \tau_{\mathrm{II}}))$ for every state $z \in Z$ and every action $a \in A_I(z)$. Hence, following Exercise 15.30 (Part (a)), Player I has a best reply that is a stationary pure strategy. Furthermore, by Exercise 15.30 (Part (c)), he has a stationary pure best reply for every discount factor that is close enough to 1.

Note that, by exchanging the roles of the players one can deduce a similar results for Player II.

(c) Since the game has perfect information, by Exercise 15.34 (Part (b)), for every $\lambda \in [0,1)$, each player has a stationary pure optimal strategy.

By Part (b), for every statinary pure strategy τ_I of Player I there is a segment $(\lambda_{\tau_I}, 1) \subseteq [0,1)$ and a stationary pure strategy $\tau_{\text{II}}^{\tau_I}$ of Player II, such that for every $\lambda \in (\lambda_{\tau_I}, 1)$, the strategy $\tau_{\text{II}}^{\tau_I}$ is a best reply of Player II against τ_I in $\Gamma^\lambda(z)$ for every initial state $z \in Z$. However, there is only finitely many stationary pure strategies of Player I and thus $\lambda_0 = \max_{\tau_I \in \times_{z\in Z} A_I(z)} \lambda_{\tau_I}$ is well defined. Therefore, there is a stationary pure strategy τ_I^* which is an optimal strategy of Player I for every $\lambda \in (\lambda_0, 1)$, which satisfy

$$\gamma^\lambda(z, \tau_I^*, \tau_{\text{II}}^{\tau_I^*}) = \max_{\tau_I \in \times_{z\in Z} A_I(z)} \gamma^\lambda(z, \tau_I, \tau_{\text{II}}^{\tau_I}).$$

15.36 In this exercise we show that the result in Exercise 15.35 does not hold for non-zero-sum games; that is, a two-player non-zero-sum stochastic game with perfect information may not have a stationary equilibrium in pure strategies.

Consider the following two-player stochastic game with perfect information where $|A_2(z_1)| = |A_1(z_2)| = 1$.

State z_1 (Player I chooses):

Player I	
T	$2,1$ $\quad z_2$
B	$2,3$ $\quad z_1$

state z_1

State z_2 (Player II chooses):

Player II: L	R
$3,2$ $\quad z_1$	$1,2$ $\quad z_2$

state z_2

We show that for every discount factor $\lambda \in [0,1)$ there is a unique λ-discounted equilibrium in stationary strategies, and this equilibrium is not pure.

Let w be a vector of equilibrium payoffs corresponds to a stationary equilibrium in the given stochastic game. Denote by w_i^j the equilibrium payoff of player i if the initial state is z_j. Note that $2 \le w_1^1 < 3$ and $2 \le w_2^2 < 3$.

By Theorem 15.41, for every state z_j, w^j is the equilibrium payoffs of both players in the game $G^\lambda(z_j, w)$ given by

	state z_1
T	$2(1-\lambda)+\lambda w_1^2\ ,\ 1-\lambda+\lambda w_2^2$
B	$2(1-\lambda)+\lambda w_1^1\ ,\ 3(1-\lambda)+\lambda w_2^1$

L	R
$3(1-\lambda)+\lambda w_1^1\ ,\ 2(1-\lambda)+\lambda w_2^1$	$1-\lambda+\lambda w_1^2\ ,\ 2(1-\lambda)+\lambda w_2^2$

state z_2

Assume by contradiction that there is a stationary equilibrium in which Player I plays the pure action B in state z_1. Hence,

$$2 = w_1^1 \geq w_1^2 \text{ and } w_2^1 = 3,$$

from which it follows that at z_2 Player II plays L so $w_1^2 > w_1^1$, a contradiction.
Next, assume by contradiction that there is a stationary equilibrium in which Player I plays the pure action T in state z_1. Hence,

$$2 \leq w_1^1 \leq w_1^2 \text{ and } w_2^1 < w_2^2,$$

from which it follows that at z_2 Player II plays R so $w_1^2 = 1 < 2 \leq w_1^1$, a contradiction.
By the symmetry of the players, there is no stationary equilibrium in which Player II plays a pure action at z_2
In conclusion, at an equilibrium both players play completely mixed actions $([x(T),(1-x)(B)],[(y(L),(1-y)(R)])$ which satisfies the following system of equation

$$\begin{aligned}
w_1^1 &= 2(1-\lambda)+\lambda w_1^2 \\
w_1^1 &= 2(1-\lambda)+\lambda w_1^1 \\
w_2^1 &= x(1-\lambda+\lambda w_2^2)+(1-x)(3(1-\lambda)+\lambda w_2^1) \\
w_1^2 &= y(3(1-\lambda)+\lambda w_1^1)+(1-y)(1-\lambda+\lambda w_1^2) \\
w_2^2 &= 2(1-\lambda)+\lambda w_2^1 \\
w_2^2 &= 2(1-\lambda)+\lambda w_2^2.
\end{aligned}$$

In conclusion the unique equilibrium is $([0.5(T),0.5(B)],[(0.5(L),0.5(R)])$ corresponding the payoff $w_i^j = 2$ for every i,j.

15.37 We prove that in a stochastic game with any number of players and with perfect information, the λ-discounted maxmin value of player i in state z is equal to player i's λ-discounted minmax value in that state.
Let i be a player. Define an auxiliary two-player zero-sum stochastic game with perfect information where Player I indicates player i,

Player II indicates all of the other players. For every state $z \in Z$, the set of actions of Player I is $A_i(z)$ and the set of actions of Player II is $\times_{j \neq i} A_j(z)$. Finally, the payoff function is $u_i(z, a)$ for every $z \in Z$ and every $a \in A(z)$.
The original game has perfect information so $\Delta(\times_{j \neq i} A_j(z)) = \times_{j \neq i} \Delta(A_j(z))$, and thus the set of behavior strategies of Player II in the auxiliary game equals the set $\mathcal{B}^{\infty}_{-i}$ in the original game. In conclusion, the λ-discounted maxmin(minmax) value of player i in state z in the original game equals the λ-discounted maxmin(minmax) value of player i in state z in the auxiliary game. However, by Shapley's Theorem (Theorem 15.32), the auxiliary game has a value and thus in the original game the λ-discounted maxmin value of player i in state z and player i's λ-discounted minmax value in that state is equals v.

15.38 Consider a stochastic game with perfect information and fix a discount factor $\lambda \in [0, 1)$.

(a) By Exercise 15.22, each player i has a stationary strategy τ_i^* that guarantees him the maxmin value in any state, that is,

$$\gamma_i^{\lambda}(z, \tau_i^*, \tau_{-i}) \geq \underline{v}_i^{\lambda}, \quad \forall z \in Z, \forall \tau_{-i} \in \mathcal{B}^{\infty}_{-i}. \tag{149}$$

Furthermore, if $\tau_i^*(z)$ is a maxmin action of player i in $G^{\lambda}(z, \underline{v}_i^{\lambda})$ for every state $z \in Z$ then τ_i^* satisfies Equation (149). However, since the game has perfect information, there is a maxmin pure action of player i in $G^{\lambda}(z, \underline{v}_i^{\lambda})$ for every state $z \in Z$ and thus player i has a pure stationary strategy τ_i^* that guarantees him the maxmin value in any state.

(b) By Exercise 15.23, for every player i there is a vector of stationary strategies of the other players $\widetilde{\tau}^i_{-i} = (\widetilde{\tau}^i_j)_{j \neq i}$, that guarantees that player i's payoff will be no more than his minmax value:

$$\gamma_i^{\lambda}(z, \tau_i, \widetilde{\tau}^i_{-i}) \leq \overline{v}_i^{\lambda}, \quad \forall z \in Z, \forall \tau_i \in \mathcal{B}_i^{\infty}. \tag{150}$$

Furthermore, if $\tau^*_{-i}(z)$ is a minmax action against player i in $G^{\lambda}(z, \underline{v}_i^{\lambda})$ for every state $z \in Z$ then τ^*_{-i} satisfies Equation (150). However, since the game has perfect information, there is a minmax pure action against player i in $G^{\lambda}(z, \underline{v}_i^{\lambda})$ for every state $z \in Z$ and thus for every player i there is a vector of pure stationary strategies of the other players $\widetilde{\tau}^i_{-i} = (\widetilde{\tau}^i_j)_{j \neq i}$, that guarantees that player i's payoff will be no more than his minmax value.

(c) Consider the following vector of strategies:

- Each player i plays the strategy τ_i^*, as long as no other player $j \in N$ has deviated from his strategy τ_j^*.

- As soon as some player deviates from this plan, say at stage t player i_0 deviates from the strategy $\tau^*_{i_0}$, all the other players play $\widetilde{\tau}^{i_0}_j$ from stage $(t+1)$ on.

Assume that at stage t in state z^t player i deviates from τ^*_i for the first time. Hence, if at stage t the game transfer to state z^{t+1} then from stage $t+1$ onwards the other players play $\widetilde{\tau}^i_{-i}$ so player i's payoff is at most $\overline{v}^\lambda_i(z^{t+1})$. Therefore, his payoff from stage t onwards is at most i's minmax value in $G^\lambda(z^t, \overline{v}^\lambda_i)$ which equals to $\overline{v}^\lambda_i(z^t)$ (by Exercise 15.23). But then player i cannot gain by deviating since under τ^*_i his payoff from stage t onwards is at least $\underline{v}^\lambda_i(z^t) = \overline{v}^\lambda_i(z^t)$ (by Exercise 15.37). In conclusion, the given profile is indeed a λ-discounted equilibrium.

15.39 A state $\widehat{z}$ is *recurrent* if for every strategy vector τ and every state $z \in Z$, the probability of eventually reaching $\widehat{z}$ under τ when the initial state is z is 1:

$$\mathbf{P}_{z,\tau}\left(\bigcup_{t=1}^{\infty}\{z^t = \widehat{z}\}\right) = 1, \quad \forall z \in Z, \forall \tau \in \mathcal{B}^\infty. \tag{151}$$

A stochastic game is called a *recurrent game* if it has a recurrent state.

(a) In a recurrent game not necessarily every state is recurrent. For instance, in the following recurrent stochastic game z_1 is recurrent but z_0 is not.

	L	R
T	$1,1$ $\quad z_1$	$2,2$ $\quad [0.5(z_0), 0.5(z_1)]$
B	$0,0$ $\quad [0.9(z_0), 0.1(z_1)]$	$2,2$ $\quad z_1$

state z_0

	L	R
T	$0,0$ $\quad z_1$	$0,1$ $\quad z_1$
B	$1,0$ $\quad z_1$	$0,0$ $\quad z_1$

state z_1

(b) Let $\widehat{z}$ be a recurrent state in a stochastic game. We prove that for every $\varepsilon > 0$ there exists $T_0 \in \mathbf{N}$ that satisfies

$$\mathbf{P}_{z,\tau}(z^t = \widehat{z} \text{ for some } t \in \{1, 2, \ldots, T_0\}) \geq 1 - \varepsilon, \quad \forall z \in Z, \forall \tau \in \mathcal{B}^\infty \tag{152}$$

For every $t \in \mathbf{N}$, denote by A_t the event according to which $z^t = \widehat{z}$ for the first time (i.e., for every $t' < t$, $z^{t'} \neq \widehat{z}$). Then $(A_t)_{t \in \mathbf{N}}$ are disjoint events satisfying

$$\bigcup_{t \in \mathbf{N}} A_t = \bigcup_{t \in \mathbf{N}} \{z^t = z\} \quad \text{and}$$

$$\bigcup_{t=1}^{T_0} A_t = \left\{z^t = \widehat{z} \text{ for some } t \in \{1, 2, \ldots, T_0\}\right\} \quad \forall T_0 \in \mathbf{N}.$$

Hence, for every $z \in Z$ and every $\tau \in \mathcal{B}^\infty$ one has

$$1 = \mathbf{P}_{z,\tau}\left(\bigcup_{t=1}^{\infty}\{z^t = \widehat{z}\}\right) = \mathbf{P}_{z,\tau}\left(\bigcup_{t=1}^{\infty} A_t\right) = \sum_{t=1}^{\infty}\mathbf{P}_{z,\tau}\left(A_t\right),$$

where the right equality holds by the sigma-additivity of the probability (over disjoint events). In particular, for every $\epsilon > 0$, there is $T_0 \in \mathbf{N}$ such that

$$\begin{aligned}1 - \varepsilon < \sum_{t=1}^{T_0}\mathbf{P}_{z,\tau}\left(A_t\right) &= \mathbf{P}_{z,\tau}\left(\bigcup_{t=1}^{T_0} A_t\right)\\ &= \mathbf{P}_{z,\tau}\Big(z^t = \widehat{z} \text{ for some } t \in \{1,2,\ldots,T_0\}\Big).\end{aligned}$$

(c) We prove that in a recurrent two-player zero-sum stochastic game, the limit $\lim_{\lambda\to 1} v^\lambda(z)$ is independent of the initial state z.
Let $\widehat{z}$ be recurrent state. Let $\delta > 0$ and choose $0 < \varepsilon < \frac{\delta}{2M}$ where $M = \max_{z\in Z, a\in A}|u(z,a)|$. By Part (b), there is $T_0 \in \mathbf{N}$ satisfying Equation (152). Then for every initial state $z \in Z$ and every discount factor $\lambda > 1 - \frac{\delta}{2T_0 M}$ Player I can guarantee a payoff of at least $(1-\varepsilon)\lambda^{T_0} v^\lambda(\widehat{z}) + \delta$ by playing his stationary optimal strategy. Indeed, for every strategy of Player II, with probability of least $1-\varepsilon$ the game reaches $\widehat{z}$ from which the payoff is at least $v^\lambda(\widehat{z})$ and thus the payoff is bound from below by

$$\begin{aligned}&\varepsilon(-M) + (1-\varepsilon)((1-\lambda)T_0(-M) + \lambda^{T_0}v^\lambda(\widehat{z}))\\ &\geq \varepsilon(-M) + (1-\lambda)T_0(-M) + (1-\varepsilon)\lambda^{T_0}v^\lambda(\widehat{z}))\\ &> \delta + (1-\tfrac{\delta}{2M})\lambda^{T_0}v^\lambda(\widehat{z}))\end{aligned}$$

Therefore,

$$\begin{aligned}v^\lambda(z) &> \delta + (1-\tfrac{\delta}{2M})\lambda^{T_0}v^\lambda(\widehat{z}) \quad \forall \lambda > 1 - \tfrac{\delta}{2T_0M}\\ &\Rightarrow \lim_{\lambda\to 1} v^\lambda(z) \geq \delta + (1-\tfrac{\delta}{2M})\lim_{\lambda\to 1} v^\lambda(\widehat{z}).\end{aligned} \tag{153}$$

Since Equation (153) holds for every $\delta > 0$ then

$$\lim_{\lambda\to 1} v^\lambda(z) \geq \lim_{\lambda\to 1} v^\lambda(\widehat{z}).$$

By exchanging the roles of the players one can deduced that $\lim_{\lambda\to 1} v^\lambda(z) \leq \lim_{\lambda\to 1} v^\lambda(\widehat{z})$ and thus $\lim_{\lambda\to 1} v^\lambda(z) = \lim_{\lambda\to 1} v^\lambda(\widehat{z})$ for every $z \in Z$, as claimed.

15.40 Let $\lambda \in [0,1)$ be a discount factor. Denote by (v,w) the vector of equilibrium payoffs of both players corresponds to a stationary equilibrium in the given stochastic game. Note that in this stochastic game, each player can guarantee a positive payoff by playing each action with probability 0.5 so $v,w > 0$.
By Theorem 15.41, (v,w) is the equilibrium payoff vector in the game $G^\lambda(z_0,(v,w))$ given by

	L	R
T	$1-\lambda+\lambda v, \lambda w$	$\lambda v, 1-\lambda+\lambda w$
B	$0,2$	$1,0$

$G^\lambda(z_0,(v,w))$

Since $0 < v \leq 1$, $0 < w \leq 2$ and $\lambda < 1$, the game $G^\lambda(z_0,(v,w))$ has no equilibrium in which at least one of the players plays pure. In particular, it has a unique equilibrium $([x(T),(1-x)(B)],[y(L),(1-y)(R)])$ which is completely mixed. By the indifference principle,

$$\begin{aligned} v &= (1-\lambda+\lambda v)y + \lambda v(1-y) \\ v &= 1-y \\ w &= \lambda w x + 2(1-x) \\ w &= (1-\lambda+\lambda w)x \end{aligned}$$

Therefore,

$$\begin{aligned} x &= \tfrac{2}{3-\lambda}, \\ y &= \tfrac{1}{2}, \\ v &= \tfrac{1}{2}, \\ w &= \tfrac{2}{3}. \end{aligned}$$

15.41 Let w be a vector of equilibrium payoffs corresponds to a stationary equilibrium in the given stochastic game. Denote by w_i^j the equilibrium payoff of player i if the initial state is z_j.
By Theorem 15.41, for every state z_j, w^j is the equilibrium payoffs of both players in the game $G^\lambda(z_j,w)$ given by

	L	R
T	$\lambda w_1^2, \lambda w_2^2$	$1,3$
B	$1,3$	$3,1$

$G^\lambda(z_1,w)$

	L	R
T	$\lambda w_1^1, \lambda w_2^1$	$0,0$
B	$0,0$	$3,1$

$G^\lambda(z_2,w)$

(a) For every discount factor $\lambda \in [0,1)$, we find a λ-discounted equilibrium in which in state z_2 the players play (T,L).

By the assumption

$$w_i^2 = \lambda w_i^1 \quad \forall i = 1, 2.$$

Hence, $G^\lambda(z_1, w)$ is

	L	R
T	$\lambda^2 w_1^1, \lambda^2 w_2^1$	$1, 3$
B	$1, 3$	$3, 1$

$G^\lambda(z_1, w)$

Recall that (w_1^1, w_2^1) should be an equilibrium payoff in $G^\lambda(z_1, w)$. Assume by contradiction that there is an equilibrium in which Player 1 plays T with positive probability. Thus, T is not strictly dominated by B, so $\lambda^2 w_1^1 \geq 1$. Furthermore, since w_1^1 is a weighted arithmetic mean of 1 and λw_1^1, it can be derives that $w_1^1 \leq \lambda^2 w_1^1$. Hence, $\lambda \geq 1$, a contradiction.

In conclusion, there is a unique stationary equilibrium satisfying the assumption: at z_1 play (B, L) and at z_2 play (T, L). The equilibria payoffs are $w^1 = (1, 3)$ and $w^2 = (\lambda, 3\lambda)$.

(b) Assume next that in a λ-discounted stationary equilibrium the players play (B, R) in state z_2.

Then $w^2 = (3, 1)$ and thus $G^\lambda(z_1, w)$ is

	L	R
T	$3\lambda, \lambda$	$1, 3$
B	$1, 3$	$3, 1$

$G^\lambda(z_1, w)$

We find an equilibrium $([x(T), (1-x)(B)], (y(L), (1-y)(R)]$ in the one-shot game.

If $\lambda < \frac{1}{3}$ $(\backslash \lambda = \frac{1}{3})$ then by eliminating strictly ($\backslash$ weakly) dominated strategies, the only equilibrium is (B, L). Hence it define a stationary equilibrium in the stochastic game in which at z_1 the players play (B, L) and at z_2 the players play (B, R).

If $\lambda > \frac{1}{3}$ then there is no equilibrium in which one of the players play pure. We use the indifference principle to find a completely mixed equilibrium.

$$\begin{aligned}
w_1^1 &= 3\lambda y + (1-y) \\
w_1^1 &= \lambda + 3(1-y) \\
w_2^1 &= \lambda x + 3(1-x) \\
w_2^1 &= 3x + (1-x).
\end{aligned}$$

Hence,

$$\begin{aligned} x &= \tfrac{2}{5-\lambda} \\ y &= \tfrac{2+\lambda}{2+3\lambda} \\ w_1^1 &= \lambda\tfrac{8+3\lambda}{2+3\lambda} \\ w_2^1 &= \tfrac{9-\lambda}{5-\lambda}. \end{aligned}$$

15.42 Let us generalize the model of stochastic games by allowing different discount factors for different players. That is, for every player $i \in N$ there is a discount factor $\lambda_i \in [0,1)$ and the payoff to player i given the strategy vector τ and the initial state z^1 is $\gamma_i^{\lambda_i}(z^1, \tau)$.
Let $\lambda := (\lambda_i)_{i \in N}$ be the vector of discount factors. Let $z \in Z$ be a state, and let $w = (w_i(z))_{i \in N, z \in Z} \in \mathbf{R}^{N \times Z}$ be a vector of $|N| \cdot |Z|$ real numbers. For every vector of discount factor, we define a variant of Definition 15.28: the game $\widehat{G}^{\lambda}(z; w)$ is the game in strategic form in which the set of players is N, the set of pure strategies of every player $i \in N$ is $A_i(z)$, and the payoff function of every player i is

$$u_i^{\lambda}(a) := (1-\lambda_i) u_i(z, a) + \lambda_i \sum_{z' \in Z} q(z' \mid z, a) w_i(z'), \quad \forall a \in A(z).$$

The proof that in the above generalized model there exists a discounted equilibrium in stationary strategies for every vector of discount factors $(\lambda_i)_{i \in N}$ is obtained by using the variant of the one-shot game $\widehat{G}^{\lambda}(z; w)$ in the proof of Theorem 15.41. Note that, since the proof of Theorem 15.41 does not rely on the fact that the discount factor is identical to all players, the proof of the current generalization is identical one by one and thus it was omitted.

Chapter 16: Bargaining games

16.1 The solution concept defined in Example 16.23 satisfies the following the properties:
Symmetry: If (S, d) is a symmetric bargaining problem then $\varphi(S, d) = d$ is symmetric.
Covariance under positive affine transformations: For every $a, b \in \mathbf{R}^2$ such that $a \gg 0$ one has $\varphi(aS + b, ad + b) = ad + b = a\varphi(S, d) + b$.
Independence of irrelevant alternatives: Let (T, d) be a bargaining problem, and let $S \subseteq T$ then $\varphi(S, d) = d = \varphi(T, d)$, in particular, it holds in case $\varphi(T, d) \in S$.
However, the solution does not satisfy **efficiency** since, by Definition

16.3, for every bargaining problem (S, d) there exists an alternative $x \in S$ satisfying $x \gg d$.

16.2 The solution concept defined in Example 16.24 satisfies the following the properties:
Efficiency: The solution selects a point among efficient points so it is necessarily efficient.
Covariance under positive affine transformations: A positive linear transformation preserves the set of efficient points as well the order of these points with regards to player I. Thus,

$$\underset{\{x \in PO(aS+b): ax+b \geq ad+b\}}{\arg\max} (a_1 x_1 + b_1) = a_1 \underset{\{x \in PO(S): x \geq d\}}{\arg\max} x_1 + b_1.$$

Independence of irrelevant alternatives: Let (T, d) be a bargaining problem, and let $S \subseteq T$ such that $\varphi(T, d) \in S$. Then, $\varphi(T, d) \in PO(S)$. Furthermore, for every $x \in PO(S) \setminus \{\varphi(T, d)\}$, there is $y \in PO(T) \setminus \{\varphi(T, d)\}$ such that $x \leq y$, hence $x_1 \leq y_1 < \varphi_1(T, d)$ so $\varphi(S, d) = \varphi(T, d)$.
However, the solution does not satisfy **symmetry**. Let S be a triangle whose vertices are $(0, 0), (1, 0)$ and $(0, 1)$ and $d = (0, 0)$. Then the solution of the symmetric bargaining problem (S, d) is the non-symmetric solution $\varphi(S, d) = (1, 0)$.

16.3 The set $PO(S)$ is closed. In addition, if $x, y \in PO(S)$ then every point in the boundary of S between x, y is also efficient. Therefore, if $x_1 - d_1 > x_2 - d_2$ for every $x = (x_1, x_2) \in PO(S)$ then the solution concept is $x^* = \arg\max_{x \in PO(S)} x_2$. Similarly, if $x_1 - d_1 < x_2 - d_2$ for every $x = (x_1, x_2) \in PO(S)$ then the solution concept is $x^* = \arg\max_{x \in PO(S)} x_1$. Otherwise, the solution concept is $x^* = \lambda_{45^\circ}(S, d)$. In particular, the given solution concept is well defined. This solution concept satisfies the following the properties:
Symmetry: Let (S, d) be a symmetric bargaining game. Then, the set of efficient points is symmetric and thus the current solution concept is $\lambda_{45^\circ}(S, d)$. However, by the symmetry assumption, the line emanating from d at angle 45° relative to the line $x_2 = d_2$ is the line $x_2 = x_1$, so $\lambda_{45^\circ}(S, d)$, the highest point in S on this line, is symmetric, as needed.
Efficiency: is satisfied by the definition.
Independence of irrelevant alternatives: Let (T, d) be a bargaining game, and let $S \subseteq T$ such that $\varphi(T, d) \in S$. Then, $\varphi(T, d)$ is efficient in both T and S. Furthermore, if by contradiction there is an efficient point in S that is closer to $\lambda_{45^\circ}(S, d)$ than $\varphi(T, d)$ then, since $S \subseteq T$ and by the convexity of T, there must be an efficient point in T

that is closer to $\lambda_{45^\circ}(S,d)$ than $\varphi(T,d)$, a contradiction. In particular, $\varphi(S,d) = \varphi(T,d)$.

However, the solution does not satisfy **Covariance under positive affine transformations**. Indeed, let S be a triangle whose vertices are $(0,0),(1,0)$ and $(0,1)$ and $d = (0,0)$. Let $a = (1,2)$ and $b = (0,0)$ so $aS + b$ is the triangle whose vertices are $(0,0),(1,0)$ and $(0,2)$ and $ad + b = (0,0)$. Therefore, $\varphi(S) = (\frac{1}{2},\frac{1}{2})$ whereas $\varphi(aS+b, ad+b) = (\frac{2}{3},\frac{2}{3}) \neq a\varphi(S,d) + b$.

16.4 The solution concept defined in Example 16.26 satisfies following the properties:

Symmetry: For every symmetric bargaining problem (S,d) the solution is $\varphi(S,d) = \mathcal{N}(S,d)$ which is symmetric.

Efficiency: Both solutions, Nash solution and the solution in Example 5.24, are efficient hence the suggested solution is also efficient.

Covariance under positive affine transformations: For every problem (S,d) that satisfies the condition according to which it has a positive affine transformation L satisfying the property that $(L(S), L(d))$ is a symmetric bargaining game, then the same condition holds for every problem $(aS+b, ad+b)$ where $a, b \in \mathbf{R}^2$ such that $a \gg 0$. Therefore, the solution of both (S,d) and $(aS+b, ad+b)$ is Nash solution, which satisfies the property of covariance under positive affine transformations. Similarly, if (S,d) does not satisfy the condition, then for every $a,b \in \mathbf{R}^2$ such that $a \gg 0$, the problem $(aS+b, ad+b)$ does not satisfy the condition as well, so for these problems the solution is as in Exercise 16.2 which satisfies the property.

However, the solution does not satisfy **independence of irrelevant alternatives:** Let $d = (0,0)$. Let S be a triangle whose vertices are $(0,0),(1,1)$ and $(2,0)$ and let T be a triangle whose vertices are $(0,0),(2,0)$ and $(0,2)$. In particular, $S \subset T$ and $\varphi(T,d) = (1,1) \in S$ but $\varphi(S,d) = (2,0)$.

16.5 Let $a, b \in \mathbf{R}^2$ such that $a \gg 0$ then

$$
\begin{aligned}
& a \cdot \arg\max_{x\in S}(x_1 - d_1)(x_2 - d_2) + b \\
&= a\left\{x \in S : (x_1 - d_1)(x_2 - d_2) \geq (z_1 - d_1)(z_2 - d_2) \quad \forall z \in S\right\} + b \\
&= a\left\{x \in S : \begin{array}{l} (a_1x_1 + b_1 - a_1d_1 - b_1)(a_2x_2 + b_2 - a_2d_2 - b_2) \\ \geq (a_1z_1 + b_1 - a_1d_1 - b_1)(a_2z_2 + b_2 - a_2d_2 - b_2) \quad \forall z \in S \end{array}\right\} + b \\
&= \arg\max_{y\in T}(y_1 - d_1)(y_2 - d_2)
\end{aligned}
$$

16.6 Let $S \subset \mathbf{R}^2$, let $x \in S$ be an alternative on the boundary of S, and let l be a supporting line of S at x. Namely, $x \in l$ and for every $y, z \in S \setminus l$ and every $\alpha \in (0,1)$ one has $\alpha y + (1-\alpha)z \notin l$.
Let $a, b \in \mathbf{R}^2$ such that $a \gg 0$. We prove that the line $al + b$ is a supporting line of the set $aS + b$ at the point $ax + b$. Assume by contradiction that there are $y, z \in S$ and $\alpha \in (0,1)$ such that $ay + b, az + b \in (aS + b) \setminus (al + b)$ but $\alpha(ay + b) + (1-\alpha)(az + b) \in (al + b)$. However, by the counter assumption it follows that $y, z \in S \setminus l$. Therefore, $a(\alpha y + (1-\alpha)z) + b = \alpha(ay + b) + (1-\alpha)(az + b) \in (al + b)$ so $\alpha y + (1-\alpha)z \in l$, which contradicts l being a supporting line of S.

16.7 Let $S \subset \mathbf{R}^2$ be a convex set, and let $a, b \in \mathbf{R}^2$. We prove that the set $aS + b$ is convex. Let $v, w \in aS + b$ and let $\alpha \in (0,1)$. Then, there are $x, y \in S$ for which $v = ax + b$ and $w = ay + b$. Hence, by the convexity of S, $\alpha v + (1-\alpha)w = \alpha(ax + b) + (1-\alpha)(ay + b) = a(\alpha x + (1-\alpha)y) + b \in aS + b$, as needed.

16.8 Let φ be a solution concept (for $\mathcal{F}$) satisfying symmetry and efficiency. Let (S, d) be a symmetric bargaining game. By the symmetry property, $\varphi_1(S, d) = \varphi_2(S, d)$ so the the outcome $\varphi(S, d)$ is a point on the line $x_1 = x_2$ that is also in S. By the efficiency, there is no other point (y, y) in S such that $(y, y) \gg \varphi(S, d)$. Therefore, $\varphi(S, d)$ is the point in S and on the line $x_1 = x_2$ for which x_1 is maximal.

16.9 Let φ be a solution concept (for $\mathcal{F}$) satisfying symmetry, efficiency, and independence of the units of measurement. Let $a, b > 0$ be positive numbers, and let S be the triangle whose vertices are $(0,0)$, $(a,0)$, and $(0,b)$. We prove that $\varphi(S,(0,0)) = (\frac{a}{2}, \frac{b}{2})$.
Let T be a triangle whose vertices are $(0,0)$, $(1,0)$, and $(0,1)$. By the properties of symmetry and efficiency, $\varphi(T,(0,0)) = (\frac{1}{2}, \frac{1}{2})$. Hence, by the property of independence of the units of measurement, $\varphi(S,(0,0)) = \varphi((a,b)T, (a,b)(0,0)) = (a,b)\varphi(T,(0,0)) = (a,b)(\frac{1}{2}, \frac{1}{2}) = (\frac{a}{2}, \frac{b}{2})$, as needed.

16.10 Let φ be a solution concept (for $\mathcal{F}$) satisfying symmetry, efficiency, and independence of the units of measurement. Let $a, b > 0$ be positive numbers, and let $x = (x1, x2)$ be a point on the ray emanating from $(0,0)$ and passing through (a,b), satisfying $x_1 > \frac{a}{2}$. Let S be the quadrangle whose vertices are $(0,0)$, $(a,0)$, $(0,b)$, and x. We prove that $\varphi(S,(0,0)) = x$.
Consider the transformation $L(y) = (\frac{1}{a}, \frac{1}{b})y$ for every $y \in \mathbf{R}^2$. Then, $L(S)$ is a quadrangle whose vertices are $L(0,0) = (0,0)$, $L(a,0) = (1,0)$, $L(0,b) = (0,1)$, and $L(x) = (\frac{x_1}{a}, \frac{x_2}{b})$. Note that $L(x)$ is a point on the ray emanating from $L(0,0) = (0,0)$ and passing through

$L(a,b) = (1,1)$ so $L(x)$ is symmetric and thus $L(S)$ is a symmetric quadrangle. Furthermore, since $L(x) > \frac{1}{2}$, the point $L(x)$ is efficient, so by the properties of symmetry and efficiency, $\varphi(L(S),(0,0)) = L(x)$. Finally, by the property of independence of the units of measurement, $\varphi(S,(0,0)) = \varphi((a,b)L(S),(a,b)(0,0)) = (a,b)\varphi(L(S),(0,0)) = (a,b)L(x) = x$.

16.11 Let L be a positive affine transformation of the plane and let T be an equilateral triangle whose base is the x_1-axis, i.e., there are $a,b,c \in \mathbf{R}$ where $a < b$ such that T's vertices are $(a,0)$, $(b,0)$ and $(\frac{a+b}{2},c)$. Then, $L(T)$ is a triangle whose vertices are $L(a,0) = (L(a),L(0))$, $L(b,L(0)) = (L(b),L(0))$ and $L(\frac{a+b}{2},c) = (L(\frac{a+b}{2}),L(c)) = (\frac{L(a)+L(b)}{2},L(c))$. Therefore, $L(T)$ is an equilateral triangle.

16.12 A set solution concept for a family of bargaining games $\widetilde{F}$ is a function φ associating every bargaining game (S,d) in $\widetilde{F}$ with a subset of S (which may contain more than a single point). Let $f : \mathbf{R}^4 \to \mathbf{R}$ be a function. Define a set solution concept φ as follows: $\varphi(S,d) = \arg\max_{x\in S} f(d,x)$.

(a) Let

$$f(d,x) = \begin{cases} x_1 + x_2 - d_1 - d_2 & \text{if } x \geq d \\ 0 & \text{otherwise} \end{cases}$$

Then for $S = \{(x_1,x_2) : x_1 + x_2 = 1\}$ and $d = (0,0)$ one has $\varphi(S,d) = S$ which is not a single point.

(b) Let $S \subseteq T$ and $x \in \varphi(T,d) \cap S$. Then

$$f(x,d) \leq \max_{x'\in S} f(x',d) \leq \max_{x'\in T} f(x',d) = f(x,d)$$

where the left inequality holds since $x \in S$, the second inequality holds since $S \subseteq T$, the equality holds by the assumption that $x \in \varphi(T,d)$. Therefore, all inequalities hold as equalities so $x \in \varphi(S,d)$.

16.13 Let φ_1 and φ_2 be two solution concepts for the family of bargaining games $\mathcal{F}$. Define another solution concept φ for the family of bargaining games $\mathcal{F}$ as follows: $\varphi(S,d) = \frac{1}{2}\varphi_1(S,d) + \frac{1}{2}\varphi_2(S,d)$.
For each of the following properties, we prove or disprove the following claim: if φ_1 and φ_2 satisfy the property, then the solution concept φ also satisfies the same property.

(a) If φ_1 and φ_2 satisfy the property of symmetry, then the solution concept φ also satisfies the same property. Indeed, let (S,d) be a symmetric game. Then $\varphi_1(s,d) = (x,x)$ and $\varphi_2(S,d) = (y,y)$ are symmetric points, hence $\varphi(S,d) = (\frac{x+y}{2}, \frac{x+y}{2})$ is also symmetric.

(b) Even if both φ_1 and φ_2 satisfy the property of efficiency, the solution concept φ does not necessarily satisfy this property. Consider, for example, $\varphi_1(S,d) = \arg\max_{x \in PO(S)} x_1$ and $\varphi_2(S,d) = \arg\max_{x \in PO(S)} x_2$ are efficient. However, for a quadrangle S whose vertices are $(0,0)$, $(1,0)$, $(0,1)$ and $(\frac{3}{4}, \frac{3}{4})$ one has $\varphi(S,(0,0)) = \frac{1}{2}\varphi_1(S,(0,0)) + \frac{1}{2}\varphi_2(S,(0,0)) = \frac{1}{2}(1,0) + \frac{1}{2}(0,1) = (\frac{1}{2}, \frac{1}{2})$, which is not efficient.

(c) Even if both φ_1 and φ_2 satisfy the property of independence of irrelevant alternatives, the solution concept φ does not necessarily satisfy this property. Let φ_1, φ_2 and S be as defined in Part (b) and let T be a triangle whose vertices are $(0,0)$, $(1.5,0)$, $(0,1.5)$. Then, even though φ_1 and φ_2 satisfy the property of independence of irrelevant alternatives (Exercise 16.2) $S \subseteq T$, $\varphi(T,(0,0)) = (\frac{3}{4}, \frac{3}{4}) \in S$, however $\varphi(S,(0,0)) = (\frac{1}{2}, \frac{1}{2}) \neq \varphi(T,(0,0)$.

(d) Assume that φ_1 and φ_2 satisfy the property of covariance under positive affine transformations. Then, for every bargaining game (S,d) and every positive transformation L, one has

$$\begin{aligned}
\varphi(L(S), L(d)) &= \tfrac{1}{2}\varphi_1(L(S), L(d)) + \tfrac{1}{2}\varphi_2(L(S), L(d)) \\
&= \tfrac{1}{2}L(\varphi_1(S,d)) + \tfrac{1}{2}L(\varphi_2(S,d)) \\
&= L(\tfrac{1}{2}\varphi_1(S,d) + \tfrac{1}{2}\varphi_2(S,d)) \\
&= L(\varphi(S,d)).
\end{aligned}$$

In conclusion, φ satisfies the property of covariance under positive affine transformations.

16.14 We find Nash solution in the current exercise using the properties of symmetry, efficiency and covariant under positive affine transformations. Let $T = \{(x,y) : x + y = 2000\}$ and $d = (0,0)$. Then by the symmetry and efficiency properties of Nash solution $\mathcal{N}(T,d) = (1000, 1000)$.

(a) $S = \{(x, 0.6y) : x + y = 2000\}$, $d = (0,0)$ and $\mathcal{N}(S,d) = (1, 0.6)\mathcal{N}(T,d) = (1000, 600)$.

(b) $S = \{(x, 0.4y) : x + y = 2000\}$, $d = (0,0)$ and $\mathcal{N}(S,d) = (1, 0.4)\mathcal{N}(T,d) = (1000, 400)$.

(c) $S = \{(0.8x, 0.7y) \; : \; x + y = 2000\}$, $d = (0,0)$ and $\mathcal{N}(S,d) = (0.8, 0.7)\mathcal{N}(T,d) = (800, 700)$.

16.15 (a) $S = \{(x_1, \sqrt{x_2}) \; : \; x_1 + x_2 \in [0, 2000]\}$, $d = (0,0)$ and

$$\begin{aligned}\mathcal{N}(S,d) &= \operatorname*{arg\,max}_{\{x \in S : c_1 + x_2 = 2000\}} (x_1 - 0)(\sqrt{x_2} - 0) \\ &= \operatorname*{arg\,max}_{\{x \in S : c_1 + x_2 = 2000\}} x_1 \sqrt{2000 - x_1}.\end{aligned}$$

To compute that, differentiate the obtained function, and set the derivative to 0, which leads to $\mathcal{N}(S,d) = (1333.33, 25.82)$.

(b) $S = \{(x_1, \sqrt{x_2}) \; : \; x_1 + x_2 \in [0, 2000]\}$, $d = (16, 7)$ and

$$\mathcal{N}(S,d) = \operatorname*{arg\,max}_{\{x \in S : c_1 + x_2 = 2000\}} (x_1 - 16)(\sqrt{x_2} - 7) = (1207.27, 28.15).$$

16.16 Let $S = \left\{ x \in \mathbf{R}^2 \; : \; \frac{x_1^2}{16^2} + \frac{x_2^2}{20^2} \leq 1 \right\}$. We will find Nash solution of (S,d).

(a) Assume $d = (0,0)$. Consider the bargaining game $(T, (0,0))$ where $T = \{x \in \mathbf{R}^2 \; : \; x_1^2 + x_2^2 \leq 1\}$. By the symmetry and efficiency properties of Nash solution, $\mathcal{N}(T, (0,0)) = (\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}})$. Hence, by the property of covariance under positive affine transformations $\mathcal{N}(S,d) = (16, 20)\mathcal{N}(T, (0,0)) = (\frac{16}{\sqrt{2}}, \frac{20}{\sqrt{2}})$.

(b) Assume $d = (10, 0)$. Then

$$\begin{aligned}\mathcal{N}(S,d) &= \operatorname*{arg\,max}_{x \in S} (x_1 - 10)(x_2 - 0) \\ &= \operatorname*{arg\,max}_{\{x \in \mathbf{R}^2 \; : \; x_1 \in [10,16]\}} (x_1 - 10) 20 \sqrt{1 - \frac{x_1^2}{16^2}}.\end{aligned}$$

To compute that, differentiate the obtained function, and set the derivative to 0, which leads to $\mathcal{N}(S,d) = (14.09, 9.48)$.

16.17 **Proof of Theorem 16.22:** Let (S,d) be a bargaining game such that for each efficient point satisfying $y \gg d$ there exists a unique tangent to S at y. We first present two observations:

- an alternative $x = (x_1, x_2) \in S$ and a positive number c satisfies the egalitarianism property if and only if c is the slope of the line connecting d with x.

- an alternative $x = (x_1, x_2) \in S$ and a positive number c satisfies the utilitarianism property if and only if

$$(y_2 - d_2) + c(y_1 - d_1) \le (x_2 - d_2) + c(x_1 - d_1) \Leftrightarrow$$
$$y_2 \le -cy_1 + (x_2 + cx_1),$$

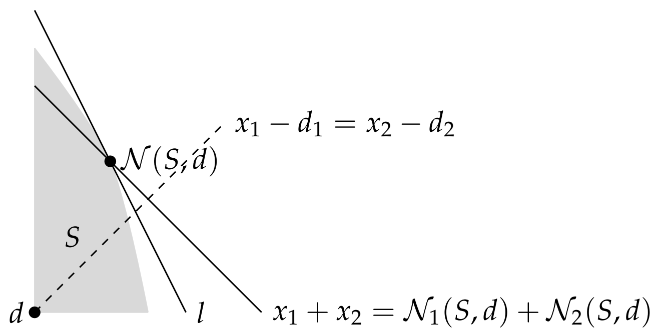
  if and only if $l(y_1) = -cy_1 + (x_2 + cx_1)$ is a supporting line of S at x which is unique by the above assumption.

In conclusion, an alternative x has a positive number c satisfying the egalitarianism and the utilitarianism properties, if and only if the slope of the supporting line of S at x is minus the slope c of the line connecting d with x, if and only if the angle between the line connecting d with x and the x_1-axis equals the angle between the supporting line l and the x_1-axis, if and only if, by Theorem 16.20, $x = \mathcal{N}(S, d)$.

16.18 Let $(S, d) \in \mathcal{F}$ be a bargaining game.

(a) Note that $f : (x_1, x_2) \mapsto (x_1 - d_1) - (x_2 - d_2)$ as well as $|f| : (x_1, x_2) \mapsto |(x_1 - d_1) - (x_2 - d_2)|$ are continuous functions defined over a compact set $PO(S)$ hence a minimum is attained.
Assume by contradiction that there are two efficient points x and y minimizing the function $|f|$. In particular, $|f(x)| = |f(y)|$.
If $f(x) = f(y)$ then $x_1 - x_2 = y_1 - y_2$ which contradicts the efficiency assumption. Therefore, $f(x) = -f(y) \neq 0$. However, by the continuity of f over the compact set $PO(S)$ there is $z \in PO(S)$ such that $f(z) = 0 = |f(z)|$ which contradict the minimization of x and y.

(b) Assume first that $\mathcal{N}(S, d)$ is above the line $x_1 - d_1 = x_2 - d_2$.



Then (1) the slope of the supporting line l of S in $\mathcal{N}(S, d)$ is less than -1; and (2) $x_1^* \ge \mathcal{N}_1(S, d)$. Hence, for every $y \in S$ such that $y_1 > \mathcal{N}_1(S, d)$ one has $y_1 + y_2 < \mathcal{N}_1(S, d) + \mathcal{N}_2(S, d)$ so $\mathcal{N}(S, d)$ is on the efficient boundary between x^* and the point in Y that is closest to x^*.
Similarly, If $\mathcal{N}(S, d)$ is under the line $x_1 - d_1 = x_2 - d_2$ then (1)

the slope of the supporting line of S in $\mathcal{N}(S,d)$ is higher than -1; and (2) $x_1^* \leq \mathcal{N}_1(S,d)$. Hence, for every $y \in S$ such that $y_1 < \mathcal{N}_1(S,d)$ one has $y_1 + y_2 < \mathcal{N}_1(S,d) + \mathcal{N}_2(S,d)$ so $\mathcal{N}(S,d)$ is on the efficient boundary between x^* and the point in Y that is closest to x^*.
Finally, if $\mathcal{N}(S,d)$ is on the line $x_1 - d_1 = x_2 - d_2$ then $x^* = \mathcal{N}(S,d)$ so the line $y_1 + y_2 = \mathcal{N}_1(S,d) + \mathcal{N}_2(S,d)$ is the supporting line of S so $\mathcal{N}(S,d) \in Y$.

16.19 Let $(S,d) \in \mathcal{F}$ be a bargaining game. Denote by $x_2 = g(x_1)$ the equation defining the north-east boundary of S. Assume that g is strictly concave and twice differentiable. Then, by Exercise 16.17, the point $x^* = \mathcal{N}(S,d)$ is the only efficient point x in S satisfying that the slope of the supporting line of S at x, $g'(x_1)$, is minus the slope $\frac{x_2-d_2}{x_1-d_1}$ of the line connecting d with x, i.e.,

$$g'(x_1) = -\tfrac{x_2-d_2}{x_1-d_1} \Rightarrow -g'(x_1)(x_1-d_1) = x_2 - d_2.$$

16.20 Suppose two players have utility functions for money given by $u_1(x) = x^{\alpha_1}$ and $u_2(x) = x^{\alpha_2}$, where $0 < \alpha_1 < \alpha_2 < 1$.

(a) For every rational player i, $u_i'(x) = \alpha_i x^{\alpha_i - 1}$ must be positive, so $u_i(x)$ is defined for $x > 0$. In addition, $u_i''(x) = \alpha_i(\alpha_i - 1)x^{\alpha_i - 2} < 0$ so player i is risk-averse.

(b) The Arrow–Pratt risk aversion index of player i is $r_{u_i}(x) = -\frac{\alpha_i x^{\alpha_i-1}}{\alpha_i(\alpha_i-1)x^{\alpha_i-2}} = \frac{1-\alpha_i}{x}$. Hence, since $\alpha_1 < \alpha_2$, one has $r_{u_1}(x) = \frac{1-\alpha_1}{x} > \frac{1-\alpha_2}{x} = r_{u_2}(x)$ so player 1 is more risk-averse than player 2.

(c) In this case $d = (0,0)$ and

$$S = \left\{(x,y) \colon x \in [0, A^{\alpha_1}], y = (A - x^{1/\alpha_1})^{\alpha_2}\right\}.$$

Nash solution in utility units is

$$\mathcal{N}(S,d) = \operatorname*{arg\,max}_{x \in [0, A^{\alpha_1}]} x(A - x^{1/\alpha_1})^{\alpha_2}.$$

To compute that, differentiate the obtained function, and set the derivative to 0, which leads to

$$\mathcal{N}(S,d) = \left(\left(\frac{\alpha_1}{\alpha_1+\alpha_2}A\right)^{\alpha_1}, \left(\frac{\alpha_2}{\alpha_1+\alpha_2}A\right)^{\alpha_2}\right),$$

so that player 1 receives $\frac{\alpha_1}{\alpha_1+\alpha_2}A$ and player 2 receives the greater payoff of $\frac{\alpha_2}{\alpha_1+\alpha_2}A$.

(d) In this example, if the risk aversion of a player increases with respect to his opponent, then his payoff obtained by the Nash outcome decreases.

16.21 Let

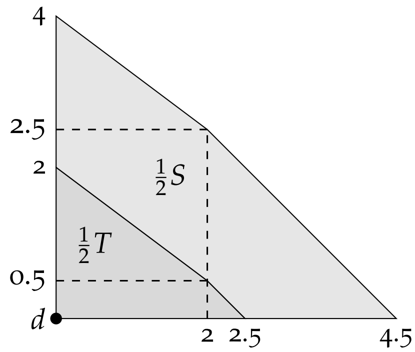
$$S = \{x \in \mathbf{R}^2_+ \ : \ 2x_1 + x_2 \leq 100\}$$
$$T = \{x \in \mathbf{R}^2_+ \ : \ x_1 + 2x_2 \leq 100\}.$$

By Exercise 16.9, $\mathcal{N}(S,(0,0)) = (\frac{100}{2}, \frac{50}{2}) = (50,25)$, $\mathcal{N}(T,d) = (25,50)$ and $\frac{1}{2}\mathcal{N}(S,d) + \frac{1}{2}\mathcal{N}(T,d) = (37.5, 37.5)$. On the other hand, the bargaining game $(\frac{1}{2}S + \frac{1}{2}T, (0,0))$ is symmetric so its Nash solution is $\mathcal{N}(\frac{1}{2}S + \frac{1}{2}T, (0,0)) = (50,50)$, a better solution for both players.

16.22 Let

$$S = \{x \in \mathbf{R}^2_+ \ : \ x_1 + x_2 \leq 4\}$$
$$T = \{x \in \mathbf{R}^2_+ \ : \ x_1 + x_2 \leq 5, \tfrac{3}{4}x_1 + x_2 \leq 4\}.$$

The bargaining game (S,d) is symmetric hence $\mathcal{N}(S,(0,0)) = (2,2)$. Let $T' = \{x \in \mathbf{R}^2_+ \ : \ \frac{3}{4}x_1 + x_2 \leq 4\}$. By Exercise 16.9, $\mathcal{N}(T',(0,0)) = (\frac{8}{3}, 2) \in T$ thus $\mathcal{N}(T,d) = (\frac{8}{3}, 2)$. In conclusion, $\frac{1}{2}\mathcal{N}(S,d) + \frac{1}{2}\mathcal{N}(T,d) = (\frac{7}{3}, 2)$. We next consider the bargaining game $(\frac{1}{2}S + \frac{1}{2}T, (0,0))$.



Denote by W the symmetric triangle whose vertices are $(0,0)$, $(4.5,0)$ and $(0,4.5)$. Then $\frac{1}{2}S + \frac{1}{2}T \subseteq W$, $\mathcal{N}(W,(0,0)) = (2.25, 2.25) \in \frac{1}{2}S + \frac{1}{2}T$ therefore, $\mathcal{N}(\frac{1}{2}S + \frac{1}{2}T, (0,0)) = (2.25, 2.25)$.

16.23 Let $0° < \alpha < 90°$. The solution λ_α satisfies the following properties. **Weak efficiency:** Assume by contradiction that there is $x \in S$ such that $x \gg \lambda_\alpha(S)$. By the comprehensiveness, the rectangle $[0,x_1] \times [0,x_2] \subseteq S$. In particular, λ_α is in the interior of this rectangle, so the line emanating from the origin at angle α relative to the x-axis intersects the rectangle, and thus there is a higher point in S on the line, a contradiction.

Homogeneity: Let S be a bargaining game, and let $c > 0$. The solution $(x_1^*, x_2^*) = \lambda_\alpha(S)$ is on the line emanating from the origin at angle α relative to the x-axis. However, since $\frac{x_2^*}{x_1^*} = \frac{cx_2^*}{cx_1^*}$ and since $\lambda_\alpha(S) \gg y \iff c\lambda_\alpha(S) \gg cy$ one has that $c\lambda_\alpha(S)$ is the highest point in cS on the line emanating from the origin at angle α relative to the x-axis.

Strict individual rationality: There is a point $x \gg (0,0)$. By the comprehensiveness, the rectangle $[0, x_1] \times [0, x_2] \subseteq S$. In particular, since $0^\circ < \alpha < 90^\circ$, the line emanating from the origin at angle α relative to the x-axis intersects the rectangle, and thus the a highest point in S on the line $\lambda_\alpha(S) \gg (0,0)$.

Full monotonicity: Let $S \subseteq T$. Then, every point in S on the line emanating from the origin at angle α relative to the x-axis is also in T, hence the highest point in T on the line is at least the highest point in T on it, i.e., $\lambda_\alpha(T) \geq \lambda_\alpha(S)$.

In addition, if $\alpha = 45^\circ$, the line emanating from the origin at angle 45° relative to the x-axis, satisfies $x_1 = x_2$ for every point x on the line, hence the solution λ_α satisfies symmetry.

16.24 We next prove the minimality of the set of properties characterizing the solution concept λ_α. In the following examples, for each property in turn, we will exhibit a solution concept for the family $\mathcal{F}_0$ with respect to which that property fails to hold, but the other three properties do hold.

Leaving out weak efficiency: Consider the solution $\varphi(S) = \frac{1}{2}\lambda_\alpha(S)$. Then, since λ_α satisfies homogeneity, strict individual rationality, and full monotonicity, φ satisfies these properties as well. However, $\varphi(S) \ll \lambda_\alpha(S)$, so φ does not satisfy weak efficiency.

Leaving out strict individual rationality: The solution concept presented in Exercise 16.2 satisfies all properties but strict individual rationality.

Leaving out homogeneity: Let $\varphi(S)$ be the highest point in S on the parabola $x_2 = x_1^2$. The proof that this solution satisfies weak efficiency, strict individual rationality and full monotonicity is similar to the proof in Exercise 16.23. However, $\varphi(cS) = c^2\varphi(S)$ therefore the solution is not homogeneous.

Leaving out full monotonicity: By Theorem 16.15, Nash solution satisfies efficiency, strict individual rationality and homogeneity. However, as presented in Figure 16.16, it does not satisfy full monotonicity.

16.25 **The Kalai– Smorodinsky solution of the game in Exercise 16.14:** Here $m_1 = 2000$ and $m_2 = \sqrt{2000}$. The intersection of the line that connects $(0,0)$ and (m_1, m_2) with the pareto frontier of S is $x_1 \approx 1,236$, $x_2 \approx 27.64$.

The Kalai– Smorodinsky solution of the game in Exercise 16.15: Here we should delete from S all points which are not above the threat point d for both players. We then get $m_1 = 2000 - 49 = 1951$ and $m_2 = \sqrt{2000 - 16} \approx 44.54$. The intersection of the line that connects $(16,7)$ and (m_1, m_2) with the pareto frontier of S is $x_1 \approx 1,154.24$, $x_2 \approx 29.08$.

16.26 **The Kalai– Smorodinsky solution of the game in Exercise 16.21:** For the bargaining game S, $m_1 = 50$ and $m_2 = 100$. The intersection of the line $x_2 = 2x_1$ that connects $(0,0)$ and (m_1, m_2) with the pareto frontier $2x_1 + x_2 = 100$ of S is $\mathcal{KS}(S) = (25,50)$. Similarly, $\mathcal{KS}$ solution of T is $\mathcal{KS}(T) = (50,25)$. Hence $\frac{1}{2}\mathcal{KS}(S) + \frac{1}{2}\mathcal{KS}(T) = (37.5, 37.5)$. Whereas, by symmetry, $\mathcal{KS}(\frac{1}{2}S + \frac{1}{2}T) = (50,50)$.

The Kalai– Smorodinsky solution of the game in Exercise 16.22: By symmetry, $\mathcal{KS}(S) = (2,2)$. For the bargaining game T, $m_1 = 5$ and $m_2 = 4$. The intersection of the line $x_2 = \frac{4}{5}x_1$ that connects $(0,0)$ and (m_1, m_2) with the pareto frontier of T is $\mathcal{KS}(S) = (\frac{80}{31}, \frac{64}{31})$. Hence $\frac{1}{2}\mathcal{KS}(S) + \frac{1}{2}\mathcal{KS}(T) \approx (2.29, 2.03)$. Whereas, $\mathcal{KS}(\frac{1}{2}S + \frac{1}{2}T) = (2.38, 2.11)$.

16.27 We prove the minimality of the set of properties defining the Kalai-Smorodinsky solution. In the following examples, for each property in turn, we will exhibit a solution concept for the family $\mathcal{F}_0$ with respect to which that property fails to hold, but the other three properties do hold.

Leaving out symmetry: Consider the solution presented in Example 16.24, $\varphi(S) = \arg\max_{\{x \in PO(S)\}} x_1$. By Exercise 16.2, this solution satisfies the properties of efficiency and independence of the units of measurement, and violates the property of symmetry. Furthermore, if $S \subseteq T$ then for every $x \in PO(S)$ there is $y \in PO(T)$ such that $x \geq y$, hence the solution also satisfies the property of monotonicity.

Leaving out efficiency: The constant solution $\varphi(S) = (0,0)$ for every S satisfies all properties but efficiency.

Leaving out independence of the units of measurement: Let $\varphi(S)$ be the closet point in $PO(S)$ to the line $x_2 = x_1$. Obvious, the solution is efficient. In addition, if S is symmetric then there is an efficient point on the line $x_1 = x_2$ hence the solution is symmetric. We show that the solution satisfies limited monotonicity. Let $S \subseteq T$ such that

$m_1(S) = m_1(T)$ and $m_2(S) = m_2(T)$. If $\varphi(S)$ is a point above the line $x_1 = x_2$ then $\varphi(S)$ must be the the leftmost point in $PO(S)$ which satisfies $\varphi_1(S) = m_1(S)$. Hence, since $m_1(S) = m_1(T)$ and $S \subseteq T$, the leftmost point in $PO(T)$ is (y_1, y_2) where $y_1 = \varphi_1(S)$ and $y_2 \geq \varphi_2$ which must be the closet point in $PO(T)$ to the line $x_1 = x_2$. Thus $\varphi(T) = y$ and $\varphi(T) \geq \varphi(S)$. Similarly, if $\varphi(S)$ is a point below the line $x_1 = x_2$ then $\varphi(T) \geq \varphi(S)$. If $\varphi(S)$ is a point on the line $x_1 = x_2$ then there must be an efficient point $y \in T$ such that $y \geq \varphi(S)$. In particular, the solution $\varphi(T)$ is in $PO(T)$ between y and the line $x_1 = x_2$, for $x_1 \geq \varphi_1(S)$, so $\varphi(T) \geq \varphi(S)$. Finally, this solution does not satisfy the property of independence of the units of measurement. For instance, let S be a triangle whose vertices are $(0,0)$, $(1,0)$ and $(0,1)$. Then $\varphi(S) = (\frac{1}{2}, \frac{1}{2})$. However, $\varphi((2,1)S) = (\frac{2}{3}, \frac{2}{3}) \neq (2,1)\varphi(S)$.
Leaving out limited monotonicity: By Theorem 16.15, Nash solution satisfies symmetry, efficiency and independence of the units of measurement. However, as presented in Figure 16.16, it does not satisfy limited monotonicity.

16.28 **Proof of Theorem 16.35:**
Step I: For every bargaining game (N, S, d) there is a unique point in $\arg\max_{\{x \in S, x \geq d\}} \prod_{i \in N}(x_i - d_i)$. Let (N, S, d) be a bargaining game. Let $D = \{x \in S, x \geq d\}$. Let $f_d(x) = \prod_{i \in N}(x_i - d_i)$, for every $x \in \mathbf{R}^N$. Since D is compact and f is continuous, there is at least one point in $\arg\max_{x \in D} f(x)$. Assume by contradiction that there are at least two points $y, z \in \arg\max_{x \in D} f_d(x)$. Then, by the convexity of D, $\frac{y+z}{2} \in D$, furthermore, by the inequality of arithmetic and geometric means, it follows that

$$\begin{aligned}
f_d(\tfrac{y+z}{2}) &= \prod_{i \in N} \left(\tfrac{y_i + z_i}{2} - d_i\right) = \prod_{i \in N} \tfrac{(y_i - d_i) + (z_i - d_i)}{2} \\
&> \prod_{i \in N} \sqrt{(y_i - d_i)(z_i - d_i)} = \prod_{i \in N} \sqrt{y_i - d_i} \prod_{i \in N} \sqrt{z_i - d_i} \\
&= \sqrt{f_d(y)}\sqrt{f_d(z)} = f_d(y),
\end{aligned}$$

a contradiction.
By Step I it follows that $\mathcal{N}^*(N, S, d) = \arg\max_{x \in D} f(x)$ is indeed a solution to (N, S, d). We next prove that this solution satisfies the properties presented in the theorem.
Step II: $\mathcal{N}^*$ satisfies the properties of symmetry, efficiency, covariance under positive affine transformations, and independence of irrelevant alternatives. The property of efficiency holds by the increasing monotonicity of f_d.
The property of independence of irrelevant alternatives holds since the solution defines by the argmax function.

We next prove that a symmetry holds. Assume that (N,S,d) is a symmetric bargaining game. Let $y = \mathcal{N}^*(N,S,d)$, and let $\pi(y)$ be a permutation of y. Since (N,S,d) is symmetric and by the definition of f_d, $f_d(\pi(y)) = f_d(y)$, however, by Step I, one has $\pi(y) = y$ from which it can be derived that y is symmetric.
Finally, we show that the property of covariance under positive affine transformations holds. Let (N,S,d) be a bargaining game and let $a,b \in \mathbf{R}^N$ where $a \gg 0$. Then

$$\begin{aligned}\mathcal{N}^*(N,aS+b,ad+b) &= \underset{\{ax+b\in aS+b, ax+b\geq ad+b\}}{\arg\max} \prod_{i\in N}(a_i x_i + b_i - a_i d_i - b_i)\\ &= a\left(\underset{\{x\in S, x\geq d\}}{\arg\max} \prod_{i\in N}^{n}(x_i - d_i)\right) + b = a\mathcal{N}^*(N,S,d) + b.\end{aligned}$$

Step III: If a solution concept φ for the family of bargaining games $\mathcal{F}^*$ satisfies the properties of symmetry, efficiency, covariance under positive affine transformations, and independence of irrelevant alternatives then $\varphi = \mathcal{N}^*$. Let (N,S,d) be a bargaining game where $N = \{1,2,...,n\}$. Since there is a point $x \in S$ such that $x \gg d$ one has $y := \mathcal{N}^*(N,S,d) \gg d$. Let L be the following positive affine transformations,

$$L(x_1,...,x_n) = \left(\tfrac{x_1-d_1}{y_1-d_1},...,\tfrac{x_n-d_n}{y_n-d_n}\right).$$

Note that $L(d) = 0$, $L(y) = (1,...,1)$ and by covariance under positive affine transformations $\mathcal{N}^*(N,L(S),L(d)) = \mathcal{N}^*(N,L(S),0) = (1,...,1)$. Let $x \in L(S)$ such that $x \gg 0$. By the convexity of $L(S)$, for every $0 < \epsilon < 1$ one has $z^\epsilon = \epsilon x + (1-\epsilon)(1,...,1) \in L(S)$ and $z^\epsilon \gg 0$. Hence, for a sufficiently small $\epsilon > 0$ one has

$$\begin{aligned}1 = f_0(1,...,1) &\geq f_0(z^\epsilon) = \prod_{i\in N}(\epsilon x_i + (1-\epsilon)1) = \prod_{i\in N}(1+\epsilon(x_i - 1))\\ &\geq 1 + \epsilon(\sum_{i\in N} x_i - n) - \epsilon^2 \sum_{E\subset N}\prod_{i\in E}(x_i - 1)\\ &\Rightarrow n \geq \sum_{i\in N} x_i - \epsilon \sum_{E\subset N}\prod_{i\in E}(x_i - 1)\\ &\Rightarrow n \geq \sum_{i\in N} x_i.\end{aligned}$$

Hence, let T be a symmetric cube relative to the diagonal $x_1 = x_2 = ... = x_n$ that contains $L(S)$, with one face along the hyperplane $n = \sum_{i\in N} x_i$. Since $L(S)$ is compact (and thus bounded), such a cube exists. By the symmetry and efficiency of φ, one has $\varphi(N,T,0) = (1,...,1)$. Since the solution concept φ satisfies independence of irrelevant alternatives, and since $L(S)$ is a subset of T containing $(1,...,1)$, it follows that $\varphi(N, L(S), 0) = (1, ..., 1)$. Since the solution concept φ satisfies covariance under positive affine transformations, one can implement the inverse transformation L^{-1} to deduce that $\varphi(N, S, d) = y = \mathcal{N}^*(N, S, d)$, as required.

Chapter 17: Coalitional games with transferable utility

17.1 The United Nations Security Council game can be presented as a weighted majority game. Let $w_i = 0.2$ for every permanent member i; let $w_i = 0.01$ for every nonpermanent member i; and set $q = 1.04$. Note that if a coalition S contains all permanent member and at least 4 nonpermanent member then $\sum_{i \in S} w_i \geq 1.04$. On the other hand, if S contains either at most 4 permanent member or at most 3 nonpermanent members then $\sum_{i \in S} w_i < 1.04$.

17.2 (a) Denote by p the President of the United States, by N_1 the set of Senators of the Senate, and by N_2 the set of members of the House of Representatives. Let $N = \{p\} \bigcup N_1 \bigcup N_2$. The coalitional function of the corresponding game is defined as follows: for every $S \subseteq N$

$$v(S) = \begin{cases} 1 & \text{if } p \in S \text{ and } |N_1 \cap S| \geq 51 \text{ and } |N_2 \cap S| \geq 218 \\ 1 & \text{if } |N_1 \cap S| \geq 67 \text{ and } |N_2 \cap S| \geq 290 \\ 0 & \text{otherwise.} \end{cases}$$

(b) This is not a weighted majority game. Assume by contradiction that this is a weighted majority game. Hence the weights of all Senators are identical, as well as the weights of all members of the House of Representatives are identical.
Let S be a coalition such that $p \in S$, $|S \cap N_1| = 50$ and $|S \cap N_2| = 435$ and let S' be a coalition such that $p \in S$, $|S \cap N_1| = 51$ and $|S \cap N_2| = 218$. Since $v(S) = 0$ and $v(S') = 1$, the weight of a Senior is at least the total weights of 217 members of the House of Representatives.
Similarly, let S be a coalition such that $p \in S$, $|S \cap N_1| = 100$ and $|S \cap N_2| = 217$ and let S' be a coalition such that $p \in S$, $|S \cap N_1| = 51$ and $|S \cap N_2| = 218$. Since $v(S) = 0$ and $v(S') = 1$, the weight of a member of the House of Representatives is at least the total weights of 49 Senators, a contradiction.

17.3 The coalitional function of the corresponding sequencing game: $N = \{\text{Henry}, \text{Ethan}, \text{Tom}\}$.

$$
\begin{aligned}
&v(\text{Henry}) = 0\\
&v(\text{Ethan}) = 0\\
&v(\text{Tom}) = 0\\
&v(\text{Henry}, \text{Ethan}) = (5 \cdot 200 + 8 \cdot 550) - (3 \cdot 550 + 8 \cdot 200) = 2150\\
&v(\text{Henry}, \text{Tom}) = (5 \cdot 200 + 12 \cdot 400) - (4 \cdot 400 + 12 \cdot 200) = 1800\\
&v(\text{Ethan}, \text{Tom}) = 0\\
&v(\text{Henry}, \text{Ethan}, \text{Tom}) = (5 \cdot 200 + 8 \cdot 550 + 12 \cdot 400)\\
&\qquad - (3 \cdot 550 + 7 \cdot 400 + 12 \cdot 200) = 3350.
\end{aligned}
$$

17.4 In Parts (a)-(c) denote the players by 1 to 4 from left to right.

(a)

$$v(S) = \begin{cases} 1 & \text{if } 1 \in S \text{ and } |S| \geq 2 \\ 1 & \text{if } |S| \geq 3 \\ 0 & \text{otherwise} \end{cases}$$

(b)

$$v(S) = \begin{cases} 1 & \text{if } \{1, 2\} \subseteq S \\ 0 & \text{otherwise} \end{cases}$$

(c)

$$v(S) = \begin{cases} 1 & \text{if } 1 \in S \text{ and } |S| \geq 2 \\ 0 & \text{otherwise} \end{cases}$$

(d) Denote the players by 1 to 15 from left to right.

$$v(S) = \begin{cases} 1 & \text{if } \{1, 2, 3, 4, 5\} \subset S \text{ and } |S| \geq 9 \\ 0 & \text{otherwise} \end{cases}$$

17.5 Denote the board of directors by $N = \{1, 2, 3, 4\}$ where 1 represents the chairman of the board.

$$v(S) = \begin{cases} 1 & \text{if } 1 \in S \text{ and } |S| \geq 3 \\ 0 & \text{otherwise} \end{cases}$$

17.6 Let $N = \{\text{Bethesda}, \text{Silver Spring}, \text{McLean}\}$. For every $S \in N$,

$$v(S) = \begin{cases} 2 & \text{if } |S| = 1 \\ 3 & \text{if } |S| = 2 \\ 5 & \text{if } |S| = 3 \end{cases}$$

17.7 **First network:** Let $N = \{\text{Richmond}, \text{Baltimore}, \text{Wilmington}\}$. The coalitional function is given as follows.

$v(\text{Richmond}) = 30, v(\text{Baltimore}) = 10, v(\text{Wilmington}) = 10.$
$v(\text{Richmond}, \text{Baltimore}) = 40, v(\text{Richmond}, \text{Wilmington}) = 30,$
$v(\text{Baltimore}, \text{Wilmington}) = 15,$
$v(N) = 35.$

Second network: Let $N = \{\text{Silver Spring}, \text{Rockville}, \text{Alexandria}\}$. The coalitional function is given as follows.

$v(\text{Silver Spring}) = 4, v(\text{Rockville}) = 4, v(\text{Alexandria}) = 5.$
$v(\text{Silver Spring}, \text{Rockville}) = 8, v(\text{Silver Spring}, \text{Alexandria}) = 8,$
$v(\text{Rockville}, \text{Alexandria}) = 6,$
$v(N) = 10.$

Third network: Let $N = \{\text{Albany}, \text{Binghamton}, \text{Buffalo}, \text{Newburgh}, \text{Montreal}\}$.

The coalitional function is given as follows.

$v(\text{Albany}) = 12$
$v(\text{Binghamton}) = v(\text{Albany, Binghamton}) = 20$
$v(\text{Buffalo}) = v(\text{Albany, Buffalo}) = 21$
$v(\text{Newburgh}) = 7$
$v(\text{Montreal}) = v(\text{Albany, Montreal}) = 19$
$v(\text{Albany, Newburgh}) = 14$
$v(\text{Binghamton, Buffalo}) = v(\text{Albany, Binghamton, Buffalo}) = 29$
$v(\text{Binghamton, Montreal}) = v(\text{Albany, Binghamton, Montreal}) = 27$
$v(\text{Binghamton, Newburgh}) = v(\text{Albany, Binghamton, Newburgh}) = 22$
$v(\text{Buffalo, Montreal}) = v(\text{Albany, Buffalo, Montreal}) = 25$
$v(\text{Buffalo, Newburgh}) = v(\text{Albany, Buffalo, Newburgh}) = 23$
$v(\text{Newburgh, Montreal}) = v(\text{Albany, Newburgh, Montreal}) = 21$
$v(\text{Binghamton, Buffalo, Montreal}) = 33$
$v(\text{Albany, Binghamton, Buffalo, Montreal}) = 33$
$v(\text{Binghamton, Buffalo, Newburgh}) = 31$
$v(\text{Albany, Binghamton, Buffalo, Newburgh}) = 31$
$v(\text{Binghamton, Newburgh, Montreal}) = 29$
$v(\text{Albany, Binghamton, Newburgh, Montreal}) = 29$
$v(\text{Buffalo, Newburgh, Montreal}) = 27$
$v(\text{Albany, Buffalo, Newburgh, Montreal}) = 27$
$v(\text{Binghamton, Buffalo, Newburgh, Montreal}) = v(N) = 35.$

17.8 The coalitional function correspond all three games is :

$$v(S) = \begin{cases} 1 & \text{if } |S| \geq 2 \\ 0 & \text{otherwise.} \end{cases}$$

17.9 (a) The weighted majority game $[10; 9, 1, 2, 3, 4]$ has a homogeneous representation $[4; 3, 1, 1, 1, 1]$.

(b) The weighted majority game $[8; 5, 4, 2]$ has a homogeneous representation $[2; 1, 1, 0]$.

(c) The weighted majority game $[9; 7, 5, 3, 1]$ has a homogeneous representation $[5; 3, 2, 2, 1]$.

(d) The weighted majority game $[10; 7, 5, 3, 1]$ has a homogeneous representation $[6; 4, 2, 2, 1]$.

Note that not necessarily every weighted majority game has a homogeneous representation. For instance, the weighted majority game $[9; 6, 5, 3, 2, 2]$ does not have a homogeneous representation. Assume by contradiction that there is a homogeneous representation $[q; w_1, w_2, w_3, w_4, w_5]$. Then

$$q = w_1 + w_2 = w_1 + w_3 \Rightarrow w_2 = w_3 \Rightarrow q = w_2 + w_4 + w_5 = w_3 + w_4 + w_5,$$

which contradicts the fact that $\{3, 4, 5\}$ is a loosing coalition.

17.10 Assume by contradiction that a projective game with seven players can be represented as a weighted majority game $[q; w_1, w_2, w_3, w_4, w_5, w_6, w_7]$. Then the following inequalities holds

$$\begin{aligned}
w_1 + w_2 + w_4 &\geq q \\
w_2 + w_3 + w_5 &\geq q \\
w_1 + w_3 + w_6 &\geq q \\
w_3 + w_4 + w_7 &\geq q \\
w_1 + w_5 + w_7 &\geq q \\
w_2 + w_6 + w_7 &\geq q \\
w_4 + w_5 + w_6 &\geq q
\end{aligned}$$

Summing up the above inequalities leads to

$$3w_1 + 3w_2 + 3w_3 + 3w_4 + 3w_5 + 3w_6 + 3w_7 \geq 3q.$$

In addition,

$$\begin{aligned}
w_1 + w_2 + w_3 &< q \\
w_2 + w_3 + w_4 &< q \\
w_3 + w_4 + w_5 &< q \\
w_4 + w_5 + w_6 &< q \\
w_5 + w_6 + w_7 &< q \\
w_6 + w_7 + w_1 &< q \\
w_7 + w_1 + w_2 &< q
\end{aligned}$$

Summing up the above inequalities leads to

$$3w_1 + 3w_2 + 3w_3 + 3w_4 + 3w_5 + 3w_6 + 3w_7 < 3q,$$

a contradiction. Therefore, this game cannot be represented as a weighted majority game.

17.11 Let $(N; v)$ be the simple game where $N = \{1, 2, 3, 4\}$ and the winning coalitions are $\{1, 2\}$, $\{3, 4\}$, and every coalition that contains one of these two coalitions.
Assume by contradiction that $(N; v)$ is a weighted majority game. That is, there are $q, w_1, w_2, w_3, w_4 \in \mathbf{R}$ such that for every $S \subseteq N$,

$$v(S) = \begin{cases} 1 & \sum_{i \in N} w_i \geq q \\ 0 & \text{otherwise.} \end{cases}$$

Since $v(1, 2) = 1$ then $w_1 + w_2 \geq q$ and thus there is $i \in \{1, 2\}$ satisfying $w_i \geq \frac{q}{2}$. Similarly, since $v(3, 4) = 1$ then $w_3 + w_4 \geq q$ and thus, there is $j \in \{3, 4\}$ satisfying $w_j \geq \frac{q}{2}$. However, $w_i + w_j \geq q$ whereas $v(i, j) = 0$, a contradiction.

17.12 (a) Consider a simple game satisfying the property that $v(S) + v(N \setminus S) = 1$ for every coalition $S \subseteq N$. Assume, the game has a veto player, say player i. Then, for every player $j \neq i$, one has $i \notin \{j\}$ so $v(j) = 0$. Thus $v(N \setminus \{j\}) = 1 - v(j) = 1$, i.e., player j is not a veto player. In conclusion, there exists at most one veto player.
Furthermore, for every coalition $S \subseteq N$ such that $i \in S$, $v(N \setminus S) = 0$ so $v(S) = 1 - v(N \setminus S) = 1$ so player i, the veto player, is a dictator.

(b) A simple three-player game where the coalitional function is

$$v(S) = \begin{cases} 1 & \text{if } |S| \geq 2 \\ 0 & \text{otherwise.} \end{cases}$$

satisfies $v(S) + v(N \setminus S) = 1$ for every coalition $S \subseteq N$ but has no veto player.

17.13 Let $N = \{1, 2, 3, 4, 5\}$, $N_1 = \{1, 2\}$ and $N_2 = \{3, 4, 5\}$. The coalitional function is given as follows.

$$v(S) = \begin{cases} 0 & \text{if } |S \cap N_1| = 0 \text{ or } |S \cap N_2| = 0 \\ \frac{1}{2} & \text{if } |S \cap N_1| = 1 \text{ and } |S \cap N_2| \geq 1 \\ 1 & \text{if } |S \cap N_1| = 2 \text{ and } |S \cap N_2| \geq 1. \end{cases}$$

17.14 Let $N = \{1, 2, 3, 4, 5\}$, $N_1 = \{1, 2\}$ and $N_2 = \{3, 4, 5\}$. The coalitional function is given as follows.

$$v(S) = \begin{cases} 0 & \text{if } |S \cap N_1| = 0 \text{ or } |S \cap N_2| = 0 \\ \frac{2}{3} & \text{if } |S \cap N_1| = 1 \text{ and } |S \cap N_2| \geq 1 \\ 1 & \text{if } |S \cap N_1| = 2 \text{ and } |S \cap N_2| = 1 \\ \frac{4}{3} & \text{if } |S \cap N_1| = 2 \text{ and } |S \cap N_2| \geq 2. \end{cases}$$

17.15 We show that the game are not strategically equivalent. Assume by contradiction that the games are strategically equivalent. Then there are $a > 0$ and $b_1, b_2, b_3 \in \mathbf{R}$ such that $w(S) = av(S) + \sum_{i \in S} b_i$. Therefore,

$$\begin{aligned} w(1) &= av(1) + b_1 & &\Rightarrow 13 = 6a + b_1 \\ w(2) &= av(2) + b_2 & &\Rightarrow 10 = 5a + b_2 \\ w(1,2) &= av(1,2) + b_1 + b_2 & &\Rightarrow 23 = 10a + b_1 + b_2 \end{aligned}$$

However, the solution of the above system of equations is $a = -2$, a contradiction.

17.16 To find a $0-1$ normalized coalitional game $(N; w)$ that is strategically equivalent to $(N; v)$ the following system of equations has to be solved.

$$\left.\begin{aligned} 0 &= w(1) = av(1) + b_1 \\ 0 &= w(2) = av(2) + b_2 \\ 0 &= w(3) = av(3) + b_3 \\ 1 &= w(1,2,3) = av(1,2,3) + \textstyle\sum_{i=1}^3 b_i \end{aligned}\right\} \Rightarrow \left\{\begin{aligned} 0 &= 3a + b_1 \\ 0 &= 6a + b_2 \\ 0 &= 8a + b_3 \\ 1 &= 80a + \textstyle\sum_{i=1}^3 b_i \end{aligned}\right.$$

Thus,$a = \frac{1}{63}$, $b_1 = -\frac{3}{63}$, $b_2 = -\frac{6}{63}$ and $b_1 = -\frac{8}{63}$. In conclusion, the coalitional function of the $0-1$ normalized coalitional game is

$$\begin{aligned} &w(1) = w(2) = w(3) = 0, \\ &w(1,2) = \tfrac{3}{63}, w(1,3) = \tfrac{4}{63}, w(2,3) = \tfrac{4}{63}, \\ &w(1,2,3) = 1. \end{aligned}$$

17.17 The coalitional function of the game derived from $(\{1,2,3\}, v)$ if each player is given an initial sum of \$1,000 is

$$\begin{aligned} &w(1) = 1020, w(2) = 1030, w(3) = 1050, \\ &w(1,2) = 2010, w(1,3) = 2015, w(2,3) = 2040, \\ &w(1,2,3) = 3005. \end{aligned}$$

17.18 **Proof of Theorem 17.7:** Let $(N;v)$ be a coalitional game and let $(N;w))$ be a a coalitional game where $w(i) = 0$ for every $i \in N$. Then $(N;v)$ and $(N;w)$ are equivalent if and only if there are $a > 0$ and $b_i \in \mathbf{R}$ for every $i \in N$ such that

$$\begin{aligned} 0 &= a \cdot v(i) + b_i, \quad \forall i \in N \text{ and} \\ w(N) &= a \cdot v(N) + \sum_{i \in N} b_i. \end{aligned}$$

if and only if, by setting $b_i = -av(i)$ for every $i \in N$, there is $a > 0$ such that

$$w(N) = a\left(v(N) - \sum_{i \in N} v(i)\right). \tag{154}$$

Therefore, $(N;v)$ is strategically equivalent to a $0-1$ normalized game if and only if $(N;v)$ is strategically equivalent to $(N;w)$ where $w(i) = 0$ for every $i \in N$ and $w(N) = 1$, if and only if, by Equality (154), $v(N) > \sum_{i \in N} v(i)$ which case $a = \frac{1}{v(N) - \sum_{i \in N} v(i)} > 0$.

Similarly, $(N;v)$ is strategically equivalent to a $0-0$ normalized game if and only if $(N;v)$ is strategically equivalent to $(N;w)$ where $w(i) = 0$ for every $i \in N$ and $w(N) = 0$, if and only if, by Equality (154), $v(N) = \sum_{i \in N} v(i)$ which case a can be chosen arbitrarily.

Finally, $(N;v)$ is strategically equivalent to a $0-(-1)$ normalized game if and only if $(N;v)$ is strategically equivalent to $(N;w)$ where $w(i) = 0$ for every $i \in N$ and $w(N) = 1$, if and only if, by Equality (154), $v(N) < \sum_{i \in N} v(i)$ which case $a = -\frac{1}{v(N) - \sum_{i \in N} v(i)} > 0$.

17.19 The family of all superadditive games in which the set of players is $N = \{1,2\}$ is $\{(N;v) : v(1) + v(2) \leq v(1,2)\}$. That is, the set of all games with $N = \{1,2\}$ that are strategically equivalent to 0-1 or 0-0 normalized games.

17.20 (a) $(N;v)$ is not a superadditive game since $v(1,2) + v(3) = 4 + 2 = 6 > 5 = v(1,2,3)$.

(b) The set of imputations of this game is the set of all vectors $x \in \mathbf{R}^3$ which are (1) efficient, i.e., $x_1 + x_2 + x_3 = 5$; and (2) individually rational, i.e., $x_1, x_2 \geq 1$ and $x_3 \geq 2$. Therefore, it is the triangle whose vertices are $(1,1,3)$, $(1,2,2)$ and $(2,1,2)$.

17.21 We prove that the convex combination of superadditive games is also superadditive.
Let $(N;v)$ and $(N;w)$ be superadditive games, and let $0 \leq \lambda \leq 1$. Let $(N, \lambda v + (1-\lambda)w)$ be a game defined by $(\lambda v + (1-\lambda)w)(S) := \lambda v(S) + (1-\lambda)w(S)$. We show that $(N, \lambda v + (1-\lambda)w)$ is also superadditive. Indeed, any pair of disjoint coalitions S and T one has

$$\begin{aligned}
&(\lambda v + (1-\lambda)w)(S) + (\lambda v + (1-\lambda)w)(T) \\
&= \big(\lambda v(S) + (1-\lambda)w(S)\big) + \big(\lambda v(S) + (1-\lambda)w(T)\big) \\
&= \lambda \left(v(S) + v(T)\right) + (1-\lambda) \left(w(S) + w(T)\right) \\
&\leq \lambda v(S \bigcup T) + (1-\lambda)w(S \bigcup T) \\
&(\lambda v + (1-\lambda)w)(S \bigcup T)
\end{aligned}$$

where the inequality holds by the superadditivity of v and w.

17.22 **A monotonic game that is not superadditive:** let $(N;v)$ be a game where $N = \{1,2\}$, $v(1) = v(2) = 2$ and $v(1,2) = 3$. Then, $(N;v)$ is monotonic but it is not superadditive since $v(1) + v(2) > v(1,2)$.
A superadditive game that is not monotonic: let $(N;v)$ be a game where $N = \{1,2\}$, $v(1) = -1, = v(2) = 2$ and $v(1,2) = 1$. Then, $(N;v)$ is not monotonic since $v(1,2) < v(2)$ but it is superadditive since $v(1) + v(2) = v(1,2)$.

17.23 Let $(N;v)$ be a game that is strategically equivalent to a superadditive game $(N;w)$. By Theorem 17.5, $(N;w)$ is also strategically equivalent to $(N;v)$ so there are $a > 0$ and $b_i \in \mathbf{R}$ for every $i \in N$ such that $v(S) = a \cdot w(S) + \sum_{i \in S} b_i$.
Let S and T be a pair of disjoint coalitions then

$$\begin{aligned}
v(S) + v(T) &= \left(a \cdot w(S) + \sum_{i \in S} b_i\right) + \left(a \cdot w(T) + \sum_{i \in T} b_i\right) \\
&= a \cdot (w(S) + w(T)) + \sum_{i \in S \bigcup T} b_i \\
&\leq a \cdot w(S \bigcup T) + \sum_{i \in S \bigcup T} b_i \\
&= v(S \bigcup T)
\end{aligned}$$

so $(N;v)$ is itself superadditive.

17.24 We prove that a convex combination of monotonic games is also monotonic. Let $(N;v)$ and $(N;w)$ be monotonic games, and let $0 \leq$

$\lambda \leq 1$. Consider the game $(N; \lambda v + (1-\lambda)w)$ defined by

$$(\lambda v + (1-\lambda)w)(S) := \lambda v(S) + (1-\lambda)w(S).$$

By the monotonicity, for every pair of coalitions $S \subseteq T$ one has

$$v(S) \leq v(T) \text{ and } w(S) \leq w(T).$$

Therefore,

$$\begin{aligned}
\big(\lambda v + (1-\lambda)w\big)(S) &= \lambda v(S) + (1-\lambda)w(S) \\
&\leq \lambda v(T) + (1-\lambda)w(T) \\
&= \big(\lambda v + (1-\lambda)w\big)(T),
\end{aligned}$$

so $(N, \lambda v + (1-\lambda)w)$ is also monotonic.

17.25 (a) The given three-player game $(N; v)$ is not monotonic since

$$v(1,3) = 15 > v(1,2,3) = 10.$$

(b) Let $b_i = 20$ for every $i \in N$. Then the game $(N; w)$ where $w(S) = v(S) + \sum_{i \in N} b_i$ is strategically equivalent to $(N; v)$ and monotonic. Indeed,

$$\begin{aligned}
&w(1) = 23, w(2) = 33, w(3) = 24, \\
&w(1,2) = 52, w(1,3) = 55, w(2,3) = 41, \\
&w(1,2,3) = 70.
\end{aligned}$$

(c) Let $(N; v)$ be a game. Let $M > 0$ such that $M > v(S)$ for every $S \subseteq N$. Set $b_i = M$ for every $i \in S$. Then $(N; v)$ is strategically equivalent to $(N; w)$ where $w(S) = v(S) + \sum_{i \in N} b_i = v(S) + M|S|$ for every $S \subseteq N$. Furthermore, for every $S \subset T$ one has

$$w(S) = v(S) + M|S| < M + M|S| = M(1 + |S|) \leq M|T|.$$

17.26 Let $(N; v)$ be a nonnegative coalitional game, i.e., $v(S) \geq 0$ for every coalition $S \subseteq N$.

(a) Assume that $(N; v)$ is superadditive. Then, for every $S \subseteq T$, the coalitions S and $T \setminus S$ are disjoint sets hence

$$v(S) \leq v(S) + v(T \setminus S) \leq v(S \bigcup (T \setminus S)) = v(T).$$

where the left inequality holds by the non-negativity of the game, and the second inequality holds by the superadditivity. Hence, $(N; v)$ is monotonic.

(b) By the counter example presented in Exercise 5.22, it is possible for $(N;v)$ to be nonnegative and monotonic but not superadditive.

17.27 (a) i. The o-strategically equivalent game (N,w) to (N,v) is obtains by $a>0$ and $b_1,b_2,b_3 \in \mathbf{R}$ such that

$$
\begin{aligned}
&0 = w(1) = 5a + b_1 \Rightarrow b_1 = -5a \\
&0 = w(2) = 8a + b_2 \Rightarrow b_2 = -8a \\
&\Rightarrow w(1,2) = 10a + b_1 + b_2 = 10a - 5a - 8a = -3a < 0 = w(1).
\end{aligned}
$$

so the game (N,v) is not a o-monotonic.

ii. The o-normalization of a coalitional game $(N;v)$ is a game $(N;w)$ for which

$$
\begin{aligned}
&w(1) = w(2) = w(3) = 0 \\
&w(1,2) = 6a, w(1,3) = 18a, w(2,3) = 12a \\
&w(1,2,3) = 20a.
\end{aligned}
$$

As one can observe, for every $S \subseteq T$, $w(S) \leq w(T)$ so the game $(N;v)$ is a o-monotonic.

(b) The game $(N;v)$ where $N=\{1,2\}$, $v(1)=0$, $v(2)=-2$ and $v(1,2)=-1$ is not monotonic. However, its o-normalization, the game $(N;w)$ where $w(1)=w(2)=0$ and $w(1,2)=0$, is a monotonic game.

17.28 We prove that a coalitional game $(N;v)$ is 0-monotonic if and only if

$$v(S \bigcup \{i\}) \geq v(S) + v(i) \quad \forall S \subset N,\ \forall i \in N \setminus S. \tag{155}$$

Let $(N;v)$ be a coalitional game. (N,v) is 0-monotonic if and only its 0-normalized strategically equivalent game (N,w) is monotonic. Namely, if and only if there are $a>0$ and $b_i \in \mathbf{R}$ for every $i \in N$ such that (1) $w(i)=0$ for every $i \in N$; (2) $w(S) = a \cdot v(S) + \sum_{i \in S} b_i$ for every $S \subseteq N$; and (3) $w(S) \leq w(T)$ for every $S \subseteq T$. However, there are $a>0$ and $b_i \in \mathbf{R}$ for every $i \in N$ satisfying (1), (2) and(3) if and on if

$$0 = a \cdot v(i) + b(i) \quad \forall i \in N \text{ and} \tag{156}$$

$$a \cdot v(S) + \sum_{i \in S} b_i \leq a \cdot v(T) + \sum_{i \in T} b_i \quad \forall S \subseteq T. \tag{157}$$

By setting $b_i = -a \cdot v(i)$ and since $a > 0$, one can conclude that Equalities (156) and (157) hold if and only if

$$\sum_{i \in T \setminus S} v(i) \leq v(T) - v(S) \quad \forall S \subseteq T. \tag{158}$$

It is left to prove that Equation (155) holds if and only if Equation (158) holds.
Equation (155) is a special case of Equation (158) where $T = S \bigcup \{i\}$. On the other direction, assume that Equation (155) holds. We prove that Equation (158) holds by induction over $|T \setminus S|$. If $|T \setminus S| = 1$, then Equation (158) holds by Equation (155). Assume that Equation (158) holds for every S and T where $S \subseteq T$ and $|T \setminus S| \leq k$ for $k \geq 1$. Then, for every S and T where $S \subseteq T$ and $|T \setminus S| = k+1$, there is at least one player $i \in T \setminus S$ hence, by the induction hypothesis

$$v(T) \geq v(T \setminus \{i\}) + v(i) \geq v(S) + \sum_{i \in T \setminus S} v(i),$$

as needed.

17.29 The superadditive cover of the given game is (N, w) where

$w(1) = 3, w(2) = 5, w(3) = 7,$
$w(1,2) = \max\{v(1,2), w(1) + w(2)\} = 8,$
$w(1,3) = \max\{v(1,3), w(1) + w(3)\} = 10,$
$w(2,3) = \max\{v(2,3), w(2) + w(3)\} = 12,$
$w(1,2,3) = \max\{v(1,2,3), w(1) + w(2,3), w(2) + w(1,3), w(3) + w(1,2)\} = 15.$

17.30 Let $(N; v)$ be a coalitional game. Let $(N; w)$ be a coalitional game satisfying $w(S) = \max_{T \subseteq S} v(T)$. We show that $(N; w)$ is the monotonic cover of $(N; v)$.
First, $(N; w)$ is a monotonic game, since $w(S) = \max_{T \subseteq S} v(T) \leq \max_{T' \subseteq S'} v(T') = w(S')$ for every $S \subseteq S'$ (in $w(S)$ the maximization is taken over a smaller set). By similar arguments, for every $S \subseteq N$, $w(S) = \max_{T \subseteq S} v(T) \geq v(S)$. Finally, for every monotonic game $(N; u)$ satisfying $u(S) \geq v(S)$ for every $S \subseteq N$ one has

$$u(S) \geq u(T) \geq v(T) \quad \forall T \subseteq S$$

In particular, $u(S) \geq \max_{T \subseteq S} v(T) = w(S)$. In conclusion, $(N; w)$ is the monotonic cover of $(N; v)$.

17.31 (a) There are 5 different coalitional structures in a three-player game:

$$\big\{\{1,2,3\}\big\}, \big\{\{1\},\{2,3\}\big\}, \big\{\{1,3\},\{2\}\big\}, \big\{\{1,2\},\{3\}\big\}, \big\{\{1\},\{2\},\{3\}\big\}.$$

(b) There are 15 different coalitional structures in a four-player game:

$$\big\{\{1,2,3,4\}\big\}, \big\{\{1,2,3\},\{4\}\big\}, \{1,2,4\},\{3\}\big\}, \big\{\{1,3,4\},\{2\}\big\},$$
$$\big\{\{2,3,4\},\{1\}\big\} \big\{\{1,2\},\{3,4\}\big\}, \big\{\{1,3\},\{2,4\}\big\}, \big\{\{1,4\},\{2,3\}\big\},$$
$$\big\{\{1,2\},\{3\},\{4\}\big\}, \big\{\{1,3\},\{2\},\{4\}\big\}, \big\{\{1,4\},\{2\},\{3\}\big\},$$
$$\big\{\{2,3\},\{1\},\{4\}\big\}, \big\{\{2,4\},\{1\},\{3\}\big\}, \big\{\{3,4\},\{1\},\{2\}\big\}$$
$$\big\{\{1\},\{2\},\{3\},\{4\}\big\}.$$

17.32 Let $(N;v)$ be a coalitional game and let $\mathcal{B}$ be a coalitional structure. For every $i \in N$ set

$$X_i = \{x \in \mathbf{R}^{|N|} : x_i \geq v(i)\}.$$

In addition, for every coalition $S \in \mathcal{B}$ set

$$\begin{aligned} X_S &= \{x \in \mathbf{R}^{|N|} : x(S) = v(S)\} \\ &= \{x \in \mathbf{R}^{|N|} : x(S) \geq v(S)\} \cap \{x \in \mathbf{R}^{|N|} : x(S) \leq v(S)\}. \end{aligned}$$

Thus

$$X(\mathcal{B},v) = \left(\bigcap_{i\in N} X_i\right) \bigcap \left(\bigcap_{S\in\mathcal{B}} X_S\right)$$

is an intersection of a finite number of closed half-spaces, hence it is convex and compact.

17.33 Let $(N;v)$ coalitional game where $N = \{1,2,...,n\}$. Then $\mathcal{B} = \big\{\{i\} \ : \ i \in N\big\}$ is a coalitional structure for which the set of imputations $X(\mathcal{B};v) = \Big\{\big(v(1),...,v(n)\big)\Big\}$ is nonempty.

17.34 (a)

$$\begin{aligned} X\left(\big\{\{1\},\{2\},\{3\}\big\},v\right) &= \{(3,5,7)\} \\ X\left(\big\{\{1\},\{2,3\}\big\},v\right) &= \{(3,x_2,15-x_2) \ : \ 5 \leq x_2 \leq 8\} \\ X\left(\big\{\{1,3\},\{2\}\big\},v\right) &= \{(x_1,5,12-x_1) \ : \ 3 \leq x_1 \leq 5\} \\ X\left(\big\{\{1,2,3\}\big\},v\right) &= X\left(\big\{\{1,2\},\{3\}\big\},v\right) = \emptyset. \end{aligned}$$

(b) The the set of imputations of the three-player game given in this part is the same as the set of imputations of the game in Part (a).

(c) All the imputation sets are the same as in Part (a) but the following:

$$X\left(\{\{1,2,3\}\},v\right)$$
$$=\left\{(x_1,x_2,34-x_1-x_2)\ :\ x_1\geq 3, x_2\geq 5, x_1+x_2\leq 27\right\}.$$

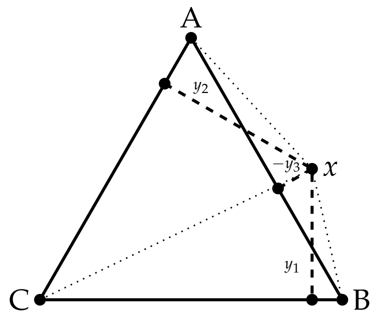
17.35 Let $(N;v)$ and $(N;w)$ be two coalitional games with the same set of players. Let $x\in X(\mathcal{B};v)$ and $y\in X(\mathcal{B};w)$.
Then $x+y\in X(\mathcal{B};v+w)$ necessarily holds. Indeed, since $x_i\geq v(i)$ and $y_i\geq w(i)$ one has $x_i+y_i\geq v(i)+w(i)$. In addition, $x(B)=v(B)$ and $y(B)=w(B)$ for every coalition $B\in\mathcal{B}$ therefore, $(x+y)(B)=x(B)+y(B)=v(B)+w(B)=(v+w)(B)$.
On the other hand, $x-y\in X(\mathcal{B};v-w)$ does not necessarily hold. A counter example: $N=\{1,2\}$, $v(1)=v(2)=w(1)=w(2)=0$, $v(1,2)=w(1,2)=1$ and $\mathcal{B}=\{\mathcal{N}\}$. Then $x=(1,0)\in X(\mathcal{B};v)$ and $y=(0,1)\in X(\mathcal{B};w)$ but $x-y=(1,-1)\notin X(\mathcal{B};v-w)$ since $-1<v(2)-w(2)$.

17.36 (a) Let a be the length of the triangle side. Denote by $S_{\triangle ABC}$ the area of a triangle whose vertices are A,B and C. Then,

$$S_{\triangle 12x}+S_{\triangle 13x}+S_{\triangle 23x}=S_{\triangle 123}$$
$$\Rightarrow \tfrac{x_2a}{2}+\tfrac{x_3a}{2}+\tfrac{x_1a}{2}=\tfrac{k\cdot a}{2}.$$

which leads to $x_1+x_2+x_3=k$.

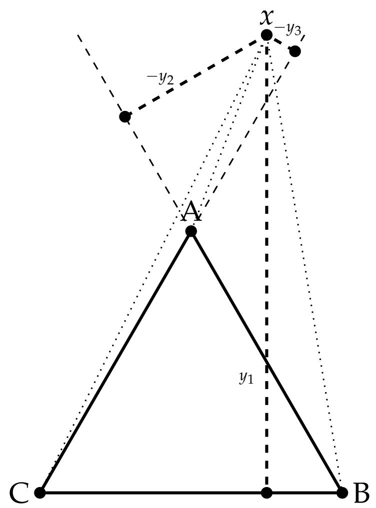
(b) Consider an equilateral triangle with x being a point located in the plane of the triangle, but not necessarily in the triangle. Denote by y_1,y_2,y_3 the distance of the point x from each side of the triangle, where the distance from the point to the side of the triangle is negative if the line on which the side lies separates the triangle from the point. We first consider the case where there is only one negative distance, say y_3.



Then,

$$S_{\triangle ACx} + S_{\triangle BCx} - S_{\triangle ABx} = S_{\triangle ABC}$$
$$\Rightarrow \frac{y_2 a}{2} + \frac{y_1 a}{2} - \frac{-y_3 a}{2} = \frac{k \cdot a}{2}.$$

which leads to $y_1 + y_2 + y_3 = k$.
We next consider the case where there are two negative distance, say y_2 and y_3.



Then,

$$S_{\triangle BCx} - S_{\triangle ABx} - S_{\triangle ACx} = S_{\triangle ABC}$$
$$\Rightarrow \frac{y_1 a}{2} - \frac{-y_2 a}{2} - \frac{-y_3 a}{2} = \frac{k \cdot a}{2}.$$

which leads to $y_1 + y_2 + y_3 = k$.

(c) Let AB be a one-dimensional line segment whose length is k. Let x be a point located in the line of the segment. Denote by x_1 and x_2 the distance of the point x from A and B, respectively, where the distance from the point to a vertex is negative if the vertex separates the segment from the point. Then $x_1 + x_2 = k$.

Chapter 18: The core

18.1 We prove that the core is a convex set. Let x, y two imputations in the core of a coalitional game $(N; v)$, and $\alpha \in [0,1]$. Then $x(N) = y(N) = v(N)$, so

$$(\alpha x + (1-\alpha)y)(N) = \alpha x(N) + (1-\alpha)y(N) = \alpha v(N) + (1-\alpha)v(N) = v(N).$$

In addition, for every $S \subseteq N$ one has $x(S) \geq v(S)$ and $y(S) \geq v(S)$ hence

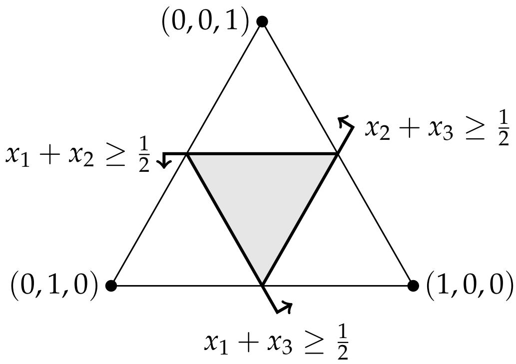
$$(\alpha x + (1-\alpha)y)(S) = \alpha x(S) + (1-\alpha)y(S) \geq \alpha v(S) + (1-\alpha)v(S) = v(S).$$

Therefore, the imputation $\alpha x + (1-\alpha)y$ is also in the core of the game $(N; v)$.

18.2 (a) Let $(N; v)$ be a three-player coalitional game where

$$v(S) = \begin{cases} 0 & |S| \leq 1 \\ \frac{1}{2} & |S| = 2 \\ 1 & |S| = 3. \end{cases}$$

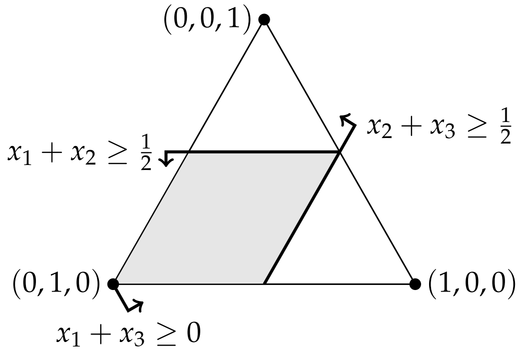
Then the core is a triangle.



(b) Let $(N; v)$ be a three-player coalitional game where

$$v(S) = \begin{cases} 0 & |S| \leq 1 \text{ or } S = \{1,3\} \\ \frac{1}{2} & S = \{1,2\} \text{ or } S = \{2,3\} \\ 1 & |S| = 3. \end{cases}$$

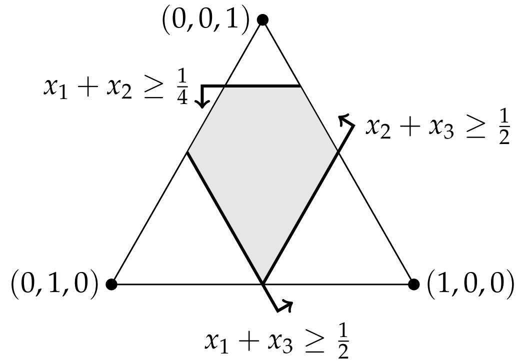
Then the core is a parallelogram..



(c) Let $(N; v)$ be a three-player coalitional game where

$$v(S) = \begin{cases} 0 & |S| \leq 1 \\ \frac{1}{4} & S = \{1,2\} \\ \frac{1}{2} & S = \{1,3\} \text{ or } S = \{2,3\} \\ 1 & |S| = 3. \end{cases}$$

Then the core is a pentagon.



18.3 Let (N, v) be a coalitional game where $N = \{1,2\}$, $v(1) = 5$, $v(2) = 3$ and $v(1,2) = 6$. Then $(N; v)$ is a monotonic game but its core is empty, since there is no solution that satisfies $x_1 \geq 5$, $x_2 \geq 3$ and $x_1 + x_2 = 6$.

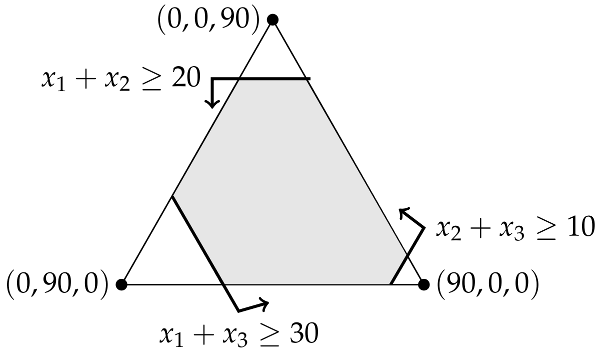
18.4 Let (N, v) be a coalitional game where $N = \{1,2,3\}$ and

$$v(S) = \begin{cases} 0 & |S| \leq 1 \\ 1 & |S| \geq 2. \end{cases}$$

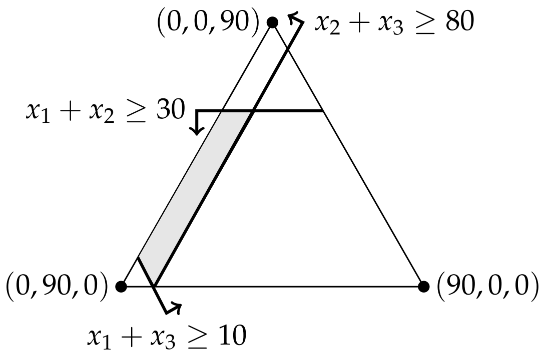
Then $(N; v)$ is a superadditive game but its core is empty. Assume by contradiction that there is $x \in \mathcal{C}(N; v)$. Hence, $x_1 = 0$ since

$0 = v(1) \leq x_1 = x(1,2,3) - x(2,3) \leq v(1,2,3) - v(2,3) = 0$. Similarly, $x_2 = x_3 = 0$ which leads to $x_1 + x_2 + x_3 = 0 < 1 = v(1,2,3)$, a contradiction.

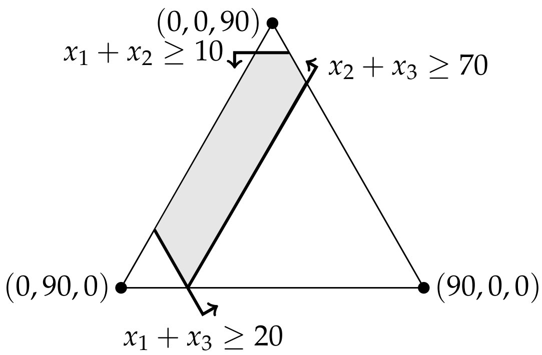
18.5 (a)



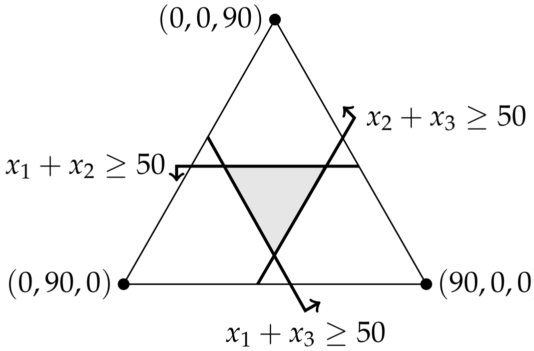
(b)



(c)

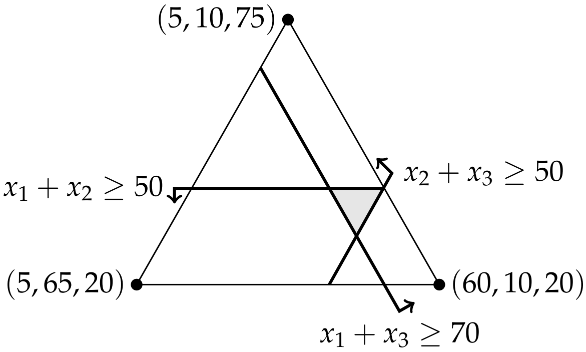


(d)



(e) The core is empty.

18.6



18.7 (a) We prove that the symmetry relation between two players is transitive. Assume that i and j are symmetric players, and j and k are symmetric players. Let S be a coalition that does not include i and k. Then, if $j \notin S$, then $v(S \bigcup \{i\}) = v(S \bigcup \{j\}) = v(S \bigcup \{k\})$. If $j \in S$, then $v(S \bigcup k) = v(S \setminus j \bigcup \{i, k\}) = v(S \bigcup \{i\})$. Thus, i and k are symmetric players.

(b) Assume that the core is nonempty. Let $x \in \mathcal{C}(N; v)$. Denote by $\Pi^S(N)$ the set of all permutations π of the set of players N, such that for every player $i \in N$ either $\pi(i) = i$ or players i and $\pi(i)$ are symmetric. Denote by $\pi(S) = \{\pi(i) : i \in S\}$ for every coalition $S \subseteq N$.
We first prove that, $v(S) = v(\pi(S))$ for every coalition $S \subseteq N$. The proof proceeds by induction on $n_S = |S \setminus \pi(S)|$. For $n_S = 0$ the claim holds trivially. Assume that it holds for every S such that $n_S \leq n$. Let S be a coalition such that $n_S = n + 1$. Thus there must be a player $i \in S \setminus \pi(S)$. Since the symmetric relation is

transitive and since a player in S is only replaced by a symmetric player, the number of players in S that are symmetric to i must be equals to the number of players in $\pi(S)$ that are symmetric to $\pi(i)$. Thus, there must be a player $j \in \pi(S) \setminus S$ that is symmetric to i. Hence,

$$v(S) = v(S \setminus \{i\} \bigcup \{j\}) = v(\pi(S))$$

where the last equality holds by the induction hypothesis.
We next prove that $\pi(x) = (x_{\pi(i)})_{i \in N} \in \mathcal{C}(N; v)$. Indeed, (1) $v(N) = \sum_{i \in N} x_i = \sum_{i \in N} x_{\pi(i)}$; and (2) By previous claim, for every coalition S, $\sum_{i \in S} x_{\pi(i)} \geq v(\pi(S)) = v(S)$.
Finally, by Exercise 18.1, $\frac{\sum_{\pi \in \Pi^S(N)} \pi(x)}{|\Pi^S(N)|} \in \mathcal{C}(N; v)$ which grants every pair of symmetric players the same payoff.

18.8 Let $(a_i)_{i \in N}$ be nonnegative real numbers.
For $k = n$**:** $v(S) = 0$ for every $S \subseteq N$, thus $\mathcal{C}(N; v) = \{(0, 0, ..., 0)\}$.
For $k = n - 1$**:** $v(S) = 0$ for every $S \subset N$ and $v(N) = \sum_{i \in N} a_i$. Hence, $\mathcal{C}(N; v) = \left\{(x_i)_{i \in N} \ : \ \forall i \in N, x_i \geq 0, \text{ and } \sum_{i \in N} x_i = \sum_{i \in N} a_i\right\}$.
For $0 < k < n - 1$**:** let $x \in \mathcal{C}(N; v)$. Therefore, for every $i \in N$,

$$0 = v(i) \leq x_i = \sum_{j \in N} x_j - \sum_{j \neq i} x_j \leq v(N) - v(N \setminus \{i\}) = a_i.$$

If, by contradiction, there is a player i such that $x_i < a_i$, then $\sum_{j \in N} x_j < \sum_{j \in N} a_j = v(N)$, a contradiction. In conclusion, $\mathcal{C}(N; v) = \left\{(a_i)_{i \in N}\right\}$.
For $k = 0$**:** $v(S) = \sum_{i \in S} a_i$ for every $S \subseteq N$, thus $\mathcal{C}(N; v) = \left\{(a_i)_{i \in N}\right\}$.

18.9 Let $(N; v)$ be a three-player 0-normalized game whose core is nonempty, and satisfying $v(S) \geq 0$ for every coalition S. We prove that $(N; v)$ is monotonic, that is, $v(S) \leq v(T)$ for every $S \subset T$. For $|S| = 1$ one has $v(S) = 0$ (since the game is 0-normalized) so by the nonnegativity $v(T) \geq 0 = v(S)$ for every T such that $S \subset T$. Assume else that $|S| = 2$ and $T = N$. By Bondareva–Shapley condition, since the core is not empty, $v(N) \geq v(S) + v(N \setminus S) = v(S)$, so the game is indeed monotonic.
Note that the claim does not hold true without the condition that $v(S) \geq 0$ for every coalition, for instance, a three-player 0-normalized game $(N; v)$ where $v(N) = 1$ and $v(i, j) = -1$ for every two players $i, j \in N$ is not monotonic, but $(1, 0, 0) \in \mathcal{C}(N; v)$.
In addition, the claim is not necessarily true for games with more than three players. Consider, a four-player 0-normalized game $(N; v)$

where $v(N) = 4$, $v(S) = 0$ for every S where $|S| = 3$ and $v(S) = 1$ for every S where $|S| = 2$. Then, the game is not monotonic, however, $(1,1,1,1) \in \mathcal{C}(N;v)$.

18.10 Let $(N;v)$ be a coalitional game whose core is not empty. Assume that player i is a null player. Then, for every $x \in \mathcal{C}(N;v)$, $x_i \geq v(i) = v(\emptyset) = 0$. On the other hand, $x_i = \sum_{j\in N} x_j - \sum_{j\neq i} x_j \leq v(N) - v(N \setminus \{i\}) = 0$. Hence, $x_i = 0$.

18.11 (a) If $\frac{v(S)}{|S|} \leq \frac{v(N)}{|N|}$ for every S then the vector $x = (\frac{v(N)}{|N|}, \frac{v(N)}{|N|}, \cdots, \frac{v(N)}{|N|})$ is in the core.

Assume now that there is S such that $\frac{v(S)}{|S|} > \frac{v(N)}{|N|}$, and let x be any imputation. Assume without loss of generality that $x_1 \leq x_2 \leq \cdots \leq x_{|N|}$. Let $S' = \{1, 2, \ldots, |S|\}$ be the coalition that consists of the $|S|$ players who receive the least according to x. By the strong symmetry property, $v(S') = v(S)$. Then $\sum_{i\in S'} x_i \leq |S|\frac{v(N)}{|N|}$. Indeed, if $\sum_{i\in S'} x_i > |S|\frac{v(N)}{|N|}$ then in particular $x_{|S|} > \frac{v(N)}{|N|}$, and then $x_{|N|} \geq x_{|N|-1} \geq \cdots \geq x_{|S|} > \frac{v(N)}{|N|}$, so that

$$\sum_{i\in N} x_i = \sum_{i\in S'} x_i + \sum_{i=|S|+1}^{|N|} x_i > |S|\frac{v(N)}{|N|} + (|N| - |S|)\frac{v(N)}{|N|} = v(N),$$

a contradiction. We conclude that

$$\sum_{i\in S'} x_i \leq |S|\frac{v(N)}{|N|} < v(S) = v(S'),$$

so that x is not in the core. Since this is true for every imputation x, the core is empty.

(b) Assume that the core of the game is nonempty, and there exists a coalition $\emptyset \neq S \subset N$ satisfying $\frac{v(S)}{|S|} = \frac{v(N)}{|N|}$. Let $x \in \mathcal{C}(N;v)$. Denote by $n = |N|$ and $k = |S|$. Then, since there are $\binom{n}{k}$ coalitions of size k and there are $\binom{n-1}{k-1}$ coalitions of size k that includes player i, the following holds.

$$\binom{n-1}{k-1} v(N) = \binom{n-1}{k-1} \sum_{i\in N} x_i = \sum_{S':|S'|=k} \sum_{i\in S'} x_i \geq \sum_{S':|S'|=k} v(S')$$

$$= \sum_{S':|S'|=k} v(S) = \binom{n}{k} v(S) = \binom{n}{k} \frac{k}{n} v(N) = \binom{n-1}{k-1} v(N).$$

Therefore, one can deduce that $\sum_{i \in S'} x_i = v(S') = \frac{|S|}{|N|} v(N)$ for every coalition S' of size k. In particular, if $k = 1$ then $x_i = v(i) = \frac{1}{|N|} v(N)$, for every player i, as needed. Otherwise, for every two players $i, j \in N$, let $S' \subset N$ such that $\{i, j\} \cap S' = \emptyset$ and $|S'| = k - 1$ one has, $x_i - x_j = x(S' \bigcup \{i\}) - x(S' \bigcup \{j\}) = \frac{k}{|N|} v(N) - \frac{k}{|N|} v(N) = 0$ which leads to $x_i = \frac{1}{n} x(N) = \frac{1}{n} v(N)$ for every player i, and we are done.

18.12 (a) We first show that every vector x for which $x_i = 0$ for every non-veto player, $x_i \geq 0$ for every player i and $\sum_{i \in N} x_i = 1$ is in the core. Let x be such a vector. Denote by I the subset of veto players. All veto players must be in every coalition S that satisfies $v(S) = 1$. Therefore,

$$\sum_{i \in I} x_i = v(N) - \sum_{i \notin I} x_i = 1 - 0 = 1.$$

In particular, for every coalition S, if $v(S) = 0$ then $\sum_{i \in S} x_i \geq 0 = v(S)$, while if $v(S) = 1$ then S contains all veto players so that

$$\sum_{i \in S} x_i = \sum_{i \in I} x_i + \sum_{i \in S \setminus I} x_i = 1 \geq v(S).$$

We now show that every vector x in the core satisfies the desired conditions. Every vector in the core satisfies $\sum_{i \in N} x_i = v(N) = 1$ and $x_i \geq v(i) = 0$. We now show that $x_i = 0$ for every non-veto player. Indeed, let x be in the core, and let i_0 be a non-veto player. Then there is a coalition S, not containing i_0, such that $v(S) = 1$. But then $\sum_{i \in S} x_i \geq v(S) = 1$. Since $x_i \geq 0$ for every $i \notin S$, it follows that $x_i = 0$ for every $i \notin S$, and in particular $x_{i_0} = 0$.

(b) The gloves game is a simple game with only one veto player- Player 3. By Part (a), its core contains every imputation x satisfying $x_1 = x_2 = 0$. Since $x_1 + x_2 + x_3 = v(1, 2, 3) = 1$, $x_3 = 1$ so the core contains only the imputation $(0, 0, 1)$.

(c) For $n = 2$, both players of a a simple majority game are veto players, thus $\mathcal{C}(N; v) = \{(x_1, 1 - x_1) : x_1 \in [0, 1]\}$.

For $n > 2$ the core of a simple majority game is empty. Assume by contradiction that there is an imputation x in the core. Since $n > 2$, all players of a simple majority game are not veto players, so, by Part (a), $x_i = 0$ for every player $i \in N$. However, $\sum_{i \in N} x_i = 0 \neq 1 = v(N)$,which contradicts the efficiency condition.

(d) By Part (a), the core of a simple coalitional game $(N;v)$ without veto players where $v(N) = 0$ is $\mathcal{C}(N;v) = \{(0,0,\cdots,0)\}$, otherwise, in case that $v(N) \neq 0$, the core is empty.

18.13 Let (N,v) be a buyer–seller game where $N = B \bigcup S$ for B a set of buyers and S a set of sellers. For every imputation $x \in \mathcal{C}(N;v)$ one has $x_i \geq v(i) = 0$.
Assume first that $|B| > |S|$. Then, for every $i \in B$ one has $x_i = 0$. Indeed,

$$x_i = \sum_{i \in N} x_i - \sum_{i \in N \setminus \{i\}} \leq v(N) - v(N \setminus \{i\}) = |S| - |S| = 0.$$

For every $i \in B$ and every $j \in S$, $1 = v(i,j) \leq x_i + x_j = x_j$. Therefore, $|S| \leq \sum_{j \in S} x_j = \sum_{i \in N} x_i = v(N) = |S|$, from which $x_j = 1$ can be derived. In conclusion, there is only one imputation x in the core where $x_i = 0$ for every $i \in B$ and $x_i = 1$ for every $i \in S$.
Similarly, if $|S| > |B|$, there is only one imputation x in the core where $x_i = 0$ for every $i \in S$ and $x_i = 1$ for every $i \in B$.
Finally, assume that $|S| = |B|$. Then, for every $i \in B$ and every $j \in S$, $1 = v(i,j) \leq x_i + x_j$. Therefore, $|S| \leq \sum_{i \in N} x_i = v(N) = |S|$ which leads to $x_i + x_j = 1$, for every $i \in B$ and every $j \in S$. Hence, the core of the game is

$$\mathcal{C}(N;v) = \left\{ (x_i)_{i \in N} \ : \ \begin{array}{rl} x_i = a & \forall i \in B \\ x_j = 1 - a & \forall i \in S \end{array} \ a \in [0,1] \right\}.$$

Note that the gloves game can be represented as a buyer-seller game where $B = \{1,2\}$ and $S = \{3\}$ hence, since $|B| > |S|$, the core contains only one imputation $(0,0,1)$ as shown in Exercise 18.12b.

18.14 The core of $(N;c)$ contains the imputation $(1,1,1,1)$ which satisfies all the conditions. We next show that this is the only imputation in the core. Let $x \in \mathcal{C}(N;c)$ then $x_i + x_j \leq c(\{i,j\}) = 2$, for every two players $i,j \in N$. On the other hand, $x_i + x_j = \sum_{k \in N} x_k - \sum_{k \in N \setminus \{i,j\}} x_k \geq c(N) - c(N \setminus \{i,j\}) = 4 - 2 = 2$. Therefore, $x_i + x_j = 2$ for every two players $i,j \in N$. Furthermore, $x_i + x_k = x_j + x_k$ for every three players $i,j,k \in N$, so $x_i = x_j = 1$ for every $i,j \in N$, as claimed before.

18.15 Even if the core of a coalitional game $(N;v)$ is nonempty, the core of its dual $(N;v^*)$ may be empty.
Consider a coalitional game (N,v) where $N = 1,2$, $v(1) = v(2) = 0$ and $v(1,2) = 1$. The core of $(N;v)$ is not empty. For instance, $(0,1) \in \mathcal{C}(N;v)$. However, by the Bondareva–Shapley condition, the core of

the dual game (N, v^*) is empty since $v^*(N) = 0 < v^*(1) + v^*(2) = 2$.

18.16 Let $x \in \mathcal{C}(N; v)$. We prove that x is a reasonable solution. Let i be a player in N then the following holds.

$$x_i = \sum_{j \in N} x_j - \sum_{j \in N \setminus \{i\}} x_j \leq v(N) - v(N \setminus \{i\}) \leq \max_{S \subseteq N \setminus \{i\}} \left\{ v(S \bigcup \{i\}) - v(S) \right\}.$$

18.17 (a) Let S be an inessential coalition. We prove that there exists a partition $(S_j)_{j=1}^r$ of S into essential coalitions such that $v(S) \leq \sum_{j=1}^r v(S_j)$. The proof will be accomplished by induction on the size of S. Every coalition S where $|S| = 1$ is essential, so the claim holds for inessential coalition S of size $|S| = 2$. Assume that the claim holds for every inessential coalition of size $|S| < n$. Let S be an inessential coalition where $|S| = n$. Then, there exists a partition $(S_j)_{j=1}^r$ of S into coalitions such that $r \geq 2$, $|S_j| < n$ for every j and $v(S) \leq \sum_{j=1}^r v(S_j)$. However, by the induction hypothesis, for every j, either S_j is essential or there exists a partition $(S_i^j)_{i=1}^{r_j}$ of S_j into essential coalitions such that $v(S_j) \leq \sum_{i=1}^{r_j} v(S_i^j)$. For simplicity set $r_j = 1$ and $S_k^j = S_j$ in case that S_j is essential. Therefore, $v(S) \leq \sum_{j=1}^r v(S_j) \leq \sum_{j=1}^r \sum_{k=1}^{r_j} v(S_k^j)$, as needed.

(b) Condition (a) is the efficiency condition. Condition (b) is a weaker condition of the coalitionally rational condition. Therefore, it is enough to prove that if an imputation satisfies (b) then it is also coalitionally rational. Let x be an imputation satisfying (a) and (b). Let S be a coalition. If S is essential, then, by (b), $\sum_{i \in S} x_i \geq v(S)$. Otherwise, S is inessential, so by Part (a), there exists a partition $(S_j)_{j=1}^r$ of S into essential coalitions such that $v(S) \leq \sum_{j=1}^r v(S_j)$. Therefore, $\sum_{i \in S} x_i = \sum_{j=1}^r \sum_{i \in S_j} x_i \geq \sum_{j=1}^r v(S_j) \geq v(S)$, which completes the proof.

(c) Assume by contradiction that coalition S is essential in the game $(N; v)$ but it is not essential in the game $(N; u)$. Then, by Part (a), there exists a partition $(S_j)_{j=1}^r$ of S into essential coalitions in $(N; u)$ such that $u(S) \leq \sum_{j=1}^r u(S_j)$ hence $v(S) = u(S) \leq \sum_{j=1}^r u(S_j) = \sum_{j=1}^r v(S_j)$ so S is inessential in $(N; v)$, a contradiction.

(d) If $v(N) = u(N)$, then an imputation $x \in \mathcal{C}(N; v)$ if and only if, by Part (b), $x(N) = v(N)$ and $v(S) \leq x(S)$ for every essential coalition S in (N, v), if and only if, by Part (c), $x(N) = v(N) =$

$u(N)$ and $u(S) = v(S) \leq x(S)$ for every essential coalition S in (N, u), if and only if $x \in \mathcal{C}(N; u)$.

(e) Assume that the cores of the games $(N; v)$ and $(N; u)$ are nonempty. If N is essential then $v(N) = u(N)$. Otherwise, there exists a partition $(S_j)_{j=1}^r$ of N into essential coalitions in $(N; v)$ such that $v(N) \leq \sum_{j=1}^r v(S_j)$. Let $x \in \mathcal{C}(N; v)$. Then

$$v(N) = x(N) = \sum_{j=1}^{r} x(S_j) \geq \sum_{j=1}^{r} v(S_j) \geq v(N) \Rightarrow v(N) = \sum_{j=1}^{r} v(S_j).$$

Similarly, $u(N) = \sum_{j=1}^r u(S_j)$. Finally, whereas $u(S_j) = v(S_j)$ for every essential coalition S_j it follows that $v(N) = u(N)$ and therefore by Part (d), $\mathcal{C}(N; v) = \mathcal{C}(N; u)$.

(f) Let (N, v) be a game where $N = \{1, 2\}$, $v(1) = v(2) = 1$ and $v(1, 2) = 2$. The core of the game contains only one imputation $\mathcal{C}(N; v) = \{(1, 1)\}$. However, the core of the coalitional game $(N; u)$ where $u(1) = u(2) = 1$ and $u(1, 2) = 0$ is empty, although $v(S) = u(S)$ for every essential coalition.

18.18 Let $(N; v)$ be a coalitional game with a nonempty core $\mathcal{C}(N; v)$. For every coalition $S \subseteq N$, define $v^E(S) := \min_{x \in \mathcal{C}(N;v)} x(S)$.

(a) Let $S \subseteq N$. Since the core is compact and convex and since the function $x(S)$ is continuous, the minimum of $x(S)$ over the core is attained. Let $x^S \in \mathcal{C}(N; v)$ such that $x^S(S) = \min_{x \in \mathcal{C}(N;v)} x(S)$. Hence, by the coalitionally rational, $v^E(S) = x^S(S) \geq v(S)$.

(b) By Part (a) and the efficiency of every imputation in the core, $v^E(N) = x^N(N) = v(N)$.

(c) We first prove that $\mathcal{C}(N; v) \subseteq \mathcal{C}(N; v^E)$. Let $x \in \mathcal{C}(N, v)$. Then $x(N) = v(N)$ and $v(S) \leq x(S)$ for every $S \subseteq N$. Therefore, by Part (b) and the definition of v^E, $x(N) = v(N) = v^E(N)$ and $v^E(S) = \min_{x' \in \mathcal{C}(N;v)} x'(S) \leq x(S)$ for every $S \subseteq N$ so $x \in \mathcal{C}(N; v^E)$.

We next prove that $\mathcal{C}(N; v^E) \subseteq \mathcal{C}(N; v)$. Let $x \in \mathcal{C}(N, v^E)$. Then $x(N) = v^E(N)$ and $v^E(S) \leq x(S)$ for every $S \subseteq N$. Therefore, by Parts (b) and (a), $x(N) = v^E(N) = v(N)$ and $v(S) \leq v^E(S) \leq x(S)$ for every $S \subseteq N$ so $x \in \mathcal{C}(N; v)$. In conclusion $\mathcal{C}(N; v^E) = \mathcal{C}(N; v)$. In particular, every $S \subseteq N$ satisfies $v^E(S) = \min_{x \in \mathcal{C}(N;v)} x(S) = \min_{x \in \mathcal{C}(N;v^E)} x(S)$ and thus $(N; v^E)$ is exact.

18.19 Assume Equations (18.23)–(18.27) hold for a coalitional game $(N; v)$, where $N = \{1,2,3\}$, then the game has a nonempty core.
We will consider three cases:

- If $v(1,2) + v(1,3) \geq v(N) + v(1)$, then the imputation

$$x = (v(1,2) + v(1,3) - v(N), v(N) - v(1,3), v(N) - v(1,2))$$

is in the core. Indeed, the imputation is coalitionally rational:

$$\begin{aligned}
&x_1 = v(1,2) + v(1,3) - v(N) \geq v(1) \text{ (by the assumtion)} \\
&x_2 = v(N) - v(1,3) \geq v(2) \text{ (by Inequality (18.25))} \\
&x_3 = v(N) - v(1,2) \geq v(3) \text{ (by Inequality (18.24))} \\
&x(1,2) = v(1,2) \\
&x(1,3) = v(1,3) \\
&x(2,3) = v(N) - v(1,3) + v(N) - v(1,2) \\
&\quad \geq v(N) - v(1) \text{ (by the assumtion)} \\
&\quad \geq v(2,3) \text{ (by Inequality (18.26)).}
\end{aligned}$$

In addition, the imputation is efficient so it is indeed in the core.

- If $v(1,2) + v(1,3) < v(N) + v(1)$ and $v(1,3) \geq v(1) + v(3)$, then the imputation

$$x = (v(1), v(N) - v(1,3), v(1,3) - v(1))$$

is in the core. Indeed, the imputation is coalitionally rational:

$$\begin{aligned}
&x_1 = v(1) \\
&x_2 = v(N) - v(1,3) \geq v(2) \text{ (by Inequality (18.25))} \\
&x_3 = v(1,3) - v(1) \geq v(3) \text{ (by the assumption)} \\
&x(1,2) = v(N) + v(1) - v(1,3) > v(1,2) \text{ (by the assumption)} \\
&x(1,3) = v(1,3) \\
&x(2,3) = v(N) - v(1) \geq v(2,3) \text{ (by Inequality (18.26)).}
\end{aligned}$$

In addition, the imputation is efficient so it is indeed in the core.

- If $v(1,2) + v(1,3) < v(N) + v(1)$ and $v(1,3) < v(1) + v(3)$, then the imputation

$$x = (v(1), v(N) - v(1) - v(3), v(3))$$

is in the core. Indeed, the imputation is coalitionally rational:

$$\begin{aligned}
&x_1 = v(1)\\
&x_2 = v(N) - v(1) - v(3) \geq v(2) \text{ (by Inequality (18.23))}\\
&x_3 = v(3)\\
&x(1,2) = v(N)) - v(3) \geq v(1,2) \text{ (by Inequality (18.24))}\\
&x(1,3) = v(1) + v(3) > v(1,3) \text{ (by the assumption)}\\
&x(2,3) = v(N) - v(1) \geq v(2,3) \text{ (by Inequality (18.26)).}
\end{aligned}$$

In addition, the imputation is efficient so it is indeed in the core.

18.20 Let $\mathcal{D}_1$ and $\mathcal{D}_2$ be two balanced collections. That is, for every $k \in \{1,2\}$ there exists a vector of positive numbers $(\delta_S^k)_{S\in\mathcal{D}_k}$ such that

$$\sum_{\{S\in\mathcal{D}_k : i\in S\}} \delta_S^k = 1, \quad \forall i \in N.$$

In particular, for every $\lambda \in (0,1)$ and every $S \in \mathcal{D}_1 \bigcup \mathcal{D}_2$ let

$$\delta_S = \begin{cases} \lambda \cdot \delta_S^1 & S \in \mathcal{D}_1 \setminus \mathcal{D}_2 \\ (1-\lambda)\cdot \delta_S^2 & S \in \mathcal{D}_2 \setminus \mathcal{D}_1 \\ \lambda \cdot \delta_S^1 + (1-\lambda)\cdot\delta_S^2 & S \in \mathcal{D}_1 \cap \mathcal{D}_2. \end{cases}$$

$(\delta_S)_{S\in\mathcal{D}_1\bigcup\mathcal{D}_2}$ is a vector of positive number. Furthermore, since $\mathcal{D}_1 = \big(\mathcal{D}_1 \setminus \mathcal{D}_2\big) \bigcup (\mathcal{D}_1 \cap \mathcal{D}_2)$ and $\big(\mathcal{D}_1 \setminus \mathcal{D}_2\big) \cap (\mathcal{D}_1 \cap \mathcal{D}_2) = \emptyset$ the following holds for every $i \in N$

$$\sum_{\{S\in\mathcal{D}_1\bigcup\mathcal{D}_2 : i\in S\}} \delta_S = \lambda \sum_{\{S\in\mathcal{D}_1 : i\in S\}} \delta_S^1 + (1-\lambda) \sum_{\{S\in\mathcal{D}_2 : i\in S\}} \delta_S^2 = \lambda + (1-\lambda) = 1,$$

so $\mathcal{D}_1 \bigcup \mathcal{D}_2$ is a balanced collection.

18.21 Let $\mathcal{D}$ be a balanced collection of coalitions. Suppose that there is a player i contained in every coalition in $\mathcal{D}$. Hence, there exists a vector of positive numbers $(\delta_S)_{S\in\mathcal{D}}$ such that

$$1 = \sum_{\{S\in\mathcal{D} : i\in S\}} \delta_S = \sum_{S\in\mathcal{D}} \delta_S.$$

Assume by contradiction that there is a coalition $S' \in \mathcal{D}$ such that $\emptyset \neq S' \subset N$. In particular, there is a player $j \in N \setminus S'$. By the assumption $j \neq i$, and

$$\sum_{\{S\in\mathcal{D} : j\in S\}} \delta_S \leq \sum_{S\in\mathcal{D}\setminus\{S'\}} \delta_S < \sum_{S\in\mathcal{D}} \delta_S = 1,$$

a contradiction. In conclusion, $\mathcal{D}$ contains a single coalition, $\mathcal{D} = \{N\}$.

18.22 Let $\mathcal{D}$ be a balanced collection of coalitions, and let $S' \in \mathcal{D}$. We prove that there is a minimal balanced collection $\mathcal{T} \subseteq \mathcal{D}$ containing S'.
If $\mathcal{D}$ is a minimal balanced collection then the claim holds trivially. Otherwise, there is a minimal balanced collection $\mathcal{T} \subset \mathcal{D}$.
If $S' \in \mathcal{T}$ we are done. On the other hand, if $S' \notin \mathcal{T}$, then as will be shown below, there is a balanced collection $\mathcal{T}' \subset \mathcal{D}$ containing S'. Note that, if $\mathcal{T}'$ is not minimal one may repeat this procedure finitely many times (by the finiteness of $\mathcal{D}$) until an appropriate minimal balanced collection is obtained.
Since $\mathcal{D}$ is a balanced collection, there is a vector of positive numbers $(d_S)_{S\in\mathcal{D}}$ such that

$$\sum_{\{S\in\mathcal{D}: i\in S\}} d_S = 1 \quad \forall i \in N.$$

Similarly, since $\mathcal{T}$ is a balanced collection, there is a vector of positive numbers $(t_S)_{S\in\mathcal{T}}$ such that

$$\sum_{\{S\in\mathcal{T}: i\in S\}} t_S = 1 \quad \forall i \in N.$$

There must be at least one coalition $S \in \mathcal{T}$ for which $t_S > d_S$. Assume by contradiction that $d_S \geq t_S$ then for $i \in S'$ one has

$$\sum_{\{S\in\mathcal{T}: i\in S\}} t_S \leq \sum_{\{S\in\mathcal{T}: i\in S\}} d_S < \sum_{\{S\in\mathcal{D}: i\in S\}} d_S = 1,$$

a contradiction. Set $\alpha = \min\left\{\frac{d_S}{t_S - d_S} \;:\; S \in \mathcal{T} \text{ and } t_S > d_S\right\}$, so $\alpha > 0$. Define

$$\delta_S = \begin{cases} (1+\alpha)d_S - \alpha t_S & S \in \mathcal{T} \\ (1+\alpha)d_S & S \in \mathcal{D} \setminus \mathcal{T}. \end{cases}$$

We first show that $\delta_S \geq 0$ for every $S \in \mathcal{D}$. For every $S \in \mathcal{D} \setminus \mathcal{T}$, $\delta_s = (1+\alpha)d_S > 0$ as a product of positive numbers. In addition, for every $S \in \mathcal{T}$ where $d_S \geq t_S$ one has $\delta_S = d_S + \alpha(d_S - t_S) \geq d_S > 0$. Finally, for every $S \in \mathcal{T}$ where $d_S < t_S$ one has $\delta_S = d_S + \alpha(d_S - t_S) \geq d_S + \frac{d_S}{t_S - d_S}(d_S - t_S) = \alpha d_S > 0$.
We next show that $\sum_{\{S\in\mathcal{D}: i\in S\}} \delta_S = 1$. Indeed,

$$\sum_{\{S\in\mathcal{D}: i\in S\}} \delta_S = (1+\alpha)\left(\sum_{\{S\in\mathcal{D}: i\in S\}} d_S\right) - \alpha\left(\sum_{\{S\in\mathcal{T}: i\in S\}} t_S\right) = (1+\alpha) - \alpha = 1.$$

Therefore, the collection $\mathcal{T}' = \{S \in \mathcal{D} : \delta_S > 0\}$ is balanced. Moreover, since $S' \notin \mathcal{T}$, $\delta_{S'} = (1+\alpha)d_{S'} > 0$ hence $S' \in \mathcal{T}'$. Finally, let $S \in \mathcal{T}$ such that $\alpha = \frac{d_S}{t_S - d_S}$ then $\delta_S = d_S - \alpha(d_S - t_S) = d_S - \frac{d_S}{t_S - d_S}(d_S - t_S) = 0$ so $S \notin \mathcal{T}'$ and $S' \in \mathcal{T}' \subset \mathcal{D}$ is a balanced collection, as needed.
In conclusion, $\mathcal{D}$ is the union of all the minimal balanced collections contained in $\mathcal{D}$.

18.23 There is a balanced collection $\mathcal{D}$ that is not minimal, and a coalition $S \in \mathcal{D}$, such that every minimal balanced collection $\mathcal{T} \subseteq \mathcal{D}$ contains S. Consider the balanced collection $\mathcal{D} = \{\{1\}, \{2\}, \{3,4\}, \{2,3,4\}\}$, then every minimal balanced collection $\mathcal{T} \subseteq \mathcal{D}$ must contains $\{1\}$.

18.24 Let $\mathcal{D}$ be a minimal balanced collection of coalitions. Hence, there is a vector of positive numbers $(\delta_S)_{S \in \mathcal{D}}$ such that

$$\sum_{S \in \mathcal{D}} \delta_S \mathcal{X}^S = \mathcal{X}^N.$$

Assume by contradiction that the vectors $\{\mathcal{X}^S, S \in \mathcal{D}\}$ are not linearly independent. Thus, there are real numbers $(\lambda_S)_{S \in \mathcal{D}}$ not all zeros satisfying $\sum_{S \in \mathcal{D}} \lambda_S \mathcal{X}^S = 0$. In particular, since $\mathcal{X}^S$ is non-negative for every S, there is at least one coalition $S \in \mathcal{D}$ for which $\lambda_S < 0$. Set $\alpha = \min\left\{-\frac{\delta_S}{\lambda_S} : \lambda_S < 0\right\} > 0$. Define $\mu_S = \delta_S + \alpha\lambda_S$ for every $S \in \mathcal{D}$. Then, $\sum_{S \in \mathcal{D}} \mu_S \mathcal{X}^S = \sum_{S \in \mathcal{D}} (\delta_S + \alpha\lambda_S)\mathcal{X}^S = 1$. In addition, for every $S \in \mathcal{D}$ such that $\lambda_S \geq 0$ one has $\mu_S > 0$. Moreover, for every $S \in \mathcal{D}$ such that $\lambda_S < 0$ one has $\mu_S = \delta_S + \alpha\lambda_S \geq \delta_S - \frac{\delta_S}{\lambda_S}\lambda_S = 0$. Let $S \in \mathcal{D}$ such that $-\frac{\delta_S}{\lambda_S} = \alpha$ then $\mu_S = 0$. In conclusion, the collection $\mathcal{T} = \{S \in \mathcal{D} : \mu_S > 0\}$ is a balanced collection such that $\mathcal{T} \subset \mathcal{D}$, which contradicts the minimality of $\mathcal{D}$, a contradiction.

18.25 (a) Assume by contradiction that $\{\{1\}, \{2\}, \{3\}, \{3,4\}, \{1,3,4\}\}$ is a balanced collection.then there are positive numbers $\delta_1, ..., \delta_5$ satisfying

$$\begin{aligned} &\delta_1 + \delta_5 = 1 \\ &\delta_2 = 1 \\ &\delta_3 + \delta_4 + \delta_5 = 1 \qquad (159) \\ &\delta_4 + \delta_5 = 1. \qquad (160) \end{aligned}$$

By Equalities (159) and (160) $\delta_3 = 0$, a contradiction.

(b) The collection $\{\{1,2\}, \{1,3\}, \{1,4\}, \{3\}, \{4\}, \{2,3,4\}\}$ is a balanced collection with the balancing weights $(\frac{1}{2}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{2})$. However, the collection is not minimal since the sub-collection $\{\{1,2\},\{3\},\{4\}\}$ is also balanced.

18.26 (a) Let $\mathcal{D}$ be a balanced collection of coalitions that is not minimal. By Exercise 18.22, $\mathcal{D}$ is the union of all the minimal balanced collections contained in $\mathcal{D}$. As shown in Exercise 18.20 a union of at least two balanced collection has an infinite set of balancing weights.

(b) A weakly balanced collection $\mathcal{D}$ of coalitions that is not minimal, does not necessarily have an infinite set of balancing weights. For instance, $\{\{1\},\{1,2\}\}$ is a weakly balanced collection but it has a unique balancing weights $\delta_{\{1\}} = 0, \delta_{\{1,2\}} = 1$.

18.27 Let $N = \{1,2,3\}$. By Exercise 18.24, if $\mathcal{D}$ is a minimal balanced collection of coalitions, then the vectors $\{\mathcal{X}^S, S \in \mathcal{D}\}$ in $\mathbf{R}^3$ are linearly independent. Hence, a minimal balanced collection of coalitions contains at most three coalitions.

There is a unique balanced collection with only one coalition which is $\{1,2,3\}$.

By Exercise 18.21, a minimal balanced collection with only two coalitions S_1 and S_2 must satisfies $S_1 \cap S_2 = \emptyset$ so the only minimal balanced collections with two coalitions are $\{\{1,2\},\{3\}\}$, $\{\{1,3\},\{2\}\}$ and $\{\{2,3\},\{1\}\}$ with the balanced weights $(1,1)$.

Finally, consider a minimal balanced collection with three coalitions. By the minimality of the collection, N cannot be contained in this collection. If the collection contains a two player coalition, say $\{1,2\}$, then, by the minimality of the collection, there must be a two player coalition containing player 3, say $\{1,3\}$. However the only balanced collection containing these coalition is $\{\{1,2\},\{1,3\},\{2,3\}\}$. If, on the other hand the collection only contains coalitions of size 1, the only balanced collection is $\{\{1\},\{2\},\{3\}\}$.

18.28 We prove that the two formulations of the Bondareva–Shapley are equivalent.

A collection containing all of the coalitions of N with the weakly balancing weights $(\delta_S)_{S\in\mathcal{P}(N)}$ define a balanced collection of coalitions $\mathcal{D} = \{S \in P(N) \ : \ \delta_S > 0\}$ with positive balancing weights $(\delta_S)_{S\in\mathcal{D}}$. In the opposite direction, every balanced collection of coalitions $\mathcal{D}$ with positive balancing weights $(\delta_S)_{S\in\mathcal{D}}$ define a weakly balancing weights $(\delta_S)_{S\in\mathcal{P}(N)}$ over the collection containing all of the coalitions of N, by assigning $\delta_S = 0$ for every $S \in \mathcal{P}(N) \setminus \mathcal{D}$.

In conclution, Equation (18.36) holds for every balanced collection of coalitions $\mathcal{D}$ with balancing weights $(\delta_S)_{S\in\mathcal{D}}$ if and only if Equation (18.62) holds for every weakly balancing weights collection containing all of the coalitions of N.

18.29 Let $\mathcal{D} = \{\{1,2\},\{3,4\}\}$. $\mathcal{D}$ is a balanced collection with balancing weights $(1,1)$. Then, $v(1,2) + v(3,4) = 30 + 30 > 60$ which contradicts Equation 18.36., that is, it does not satisfy the Bondareva–Shapley condition. In conclusion, the core is empty.

18.30 By the Bondareva–Shapley condition and Exercise 18.27, the game (N, v) where $N = \{1,2,3\}$ has a nonempty core if $v(N)$ satisfies the following:

(a) $v(N) \geq \max\{12+10+20, 12+70, 10+50, 20+20, 0.5(20+50+ +70)\} = 82.$

(b) $v(N) \geq \max\{30+40+70, 30+5, 40+20, 70+10, 0.5(10+20+ +5)\} = 140.$

18.31 Note that for every player i, $w(i) = v(i)$, and since the core of $(N; v)$ is not empty $w(N) = v(N)$.

(a)

$$\begin{aligned}
&w(1) = v(1) = 12\\
&w(2) = v(2) = 10\\
&w(3) = v(3) = 20\\
&w(1,2) = \max\{v(1,2), w(1)+w(2)\} = \max\{20, 22\} = 22\\
&w(1,3) = \max\{v(1,3), w(1)+w(3)\} = \max\{50, 32\} = 50\\
&w(2,3) = \max\{v(2,3), w(2)+w(3)\} = \max\{70, 30\} = 70\\
&w(N) = v(N) = 82.
\end{aligned}$$

(b)

$$\begin{aligned}
&w(1) = v(1) = 30\\
&w(2) = v(2) = 40\\
&w(3) = v(3) = 70\\
&w(1,2) = \max\{v(1,2), w(1)+w(2)\} = \max\{10, 70\} = 70\\
&w(1,3) = \max\{v(1,3), w(1)+w(3)\} = \max\{20, 100\} = 100\\
&w(2,3) = \max\{v(2,3), w(2)+w(3)\} = \max\{5, 110\} = 110\\
&w(N) = v(N) = 140.
\end{aligned}$$

18.32 The claim follows directly by Exercise 18.24. A balanced collection of coalitions $\mathcal{D}$ is a minimal balanced collection of coalitions, then the vectors $\{\mathcal{X}_S, S \in \mathcal{D}\}$ (which are vectors in $R^{|N|}$) are linearly independent. Hence $\mathcal{D}$ contains at most $|N|$ coalitions.

18.33 (a) Let $\epsilon \in \mathbf{R}$. We show that $\mathcal{C}_\epsilon(N;v)$ is a polytope. For each coalition S the set $\{x \in \mathbf{R}^n : x(S) \geq v(S) - \epsilon\}$ is a closed half-space. Therefore, $\mathcal{C}_\epsilon(N;v)$ is the intersection of $2^n - 1$ half-spaces $\{x \in \mathbf{R}^N : x(S) \geq v(S) - \epsilon\}$, for all $\emptyset \neq S \subset N$, and the half-spaces $\{x \in \mathbf{R}^N : x(N) \leq v(N)\}$ and $\{x \in \mathbf{R}^N : x(N) \geq v(N)\}$. $\mathcal{C}_\epsilon(N;v)$ is therefore a polytop.

(b) We prove that the least core $\mathcal{C}_{\epsilon_0}(N;v)$ is non-empty. First note that the set $E := \{\epsilon \in \mathbf{R} \;:\; \mathcal{C}_\epsilon(N;v) \neq \emptyset\}$ is not empty and bounded from below since v is bounded. In particular, ϵ_0 is well defined.

Let $(\epsilon_n)_{n\in\mathbf{N}} \subset E$ be a non-increasing sequence converges to ϵ_0. Then $\cap_{n\in\mathbf{N}}\mathcal{C}_{\epsilon_n}(N;v)$ is an intersection of a countable decreasing non-empty compact nested sets which is therefore non-empty (Cantor's intersection Theorem). Let $x \in \cap_{n\in\mathbf{N}}\mathcal{C}_{\epsilon_n}(N;v)$. Then, $x(N) = v(N)$. In addition, for every coalition $\emptyset \neq S \subset N$, $x(S) \geq v(S) - \epsilon_n$ for every $n \in \mathbf{N}$ which leads by convergences to $x(S) \geq v(S) - \epsilon_0$, so $x \in \mathcal{C}_{\epsilon_0}(N;v)$, as needed.

(c) i.

$$\left.\begin{aligned}
&x_1 + x_2 + x_3 = v(1,2,3) = 12\\
&\forall i \in N,\ x_i \geq v(i) - \epsilon = -\epsilon\\
&x_1 + x_2 \geq v(1,2) - \epsilon = 2 - \epsilon \Rightarrow x_3 \leq 10 + \epsilon\\
&x_1 + x_3 \geq v(1,3) - \epsilon = 8 - \epsilon \Rightarrow x_2 \leq 4 + \epsilon\\
&x_2 + x_3 \geq v(2,3) - \epsilon = 3 - \epsilon \Rightarrow x_1 \leq 9 + \epsilon
\end{aligned}\right\} \Rightarrow \epsilon_0 = -2,$$

$$\mathcal{C}_{\epsilon_0}(N;v) = \{\alpha(2,2,8) + (1-\alpha)(7,2,3) \;:\; \alpha \in [0,1]\}.$$

ii.

$$\left.\begin{array}{l}\left.\begin{array}{l}x_1+x_2+x_3=v(1,2,3)=12\\ x_1\geq v(1)-\epsilon=6-\epsilon\\ x_2\geq v(2)-\epsilon=5-\epsilon\end{array}\right\}\Rightarrow 12=x_1+x_2+x_3\geq 15-3\epsilon\\ x_3\geq v(3)-\epsilon=4-\epsilon\\ x_1+x_2\geq v(1,2)-\epsilon=2-\epsilon\Rightarrow x_3\leq 10+\epsilon\\ x_1+x_3\geq v(1,3)-\epsilon=4-\epsilon\Rightarrow x_2\leq 8+\epsilon\\ x_2+x_3\geq v(2,3)-\epsilon=3-\epsilon\Rightarrow x_1\leq 9+\epsilon\end{array}\right\}\Rightarrow \epsilon_0=-1,$$

$$\mathcal{C}_{\epsilon_0}(N;v)=\{(5,4,3)\}.$$

iii.

$$\left.\begin{array}{l}x_1+x_2+x_3=v(1,2,3)=12\\ \forall i\in N,\ x_i\geq v(i)-\epsilon=-\epsilon\\ \left.\begin{array}{l}x_1+x_2\geq v(1,2)-\epsilon=8-\epsilon\Rightarrow x_3\leq 4+\epsilon\\ x_1+x_3\geq v(1,3)-\epsilon=8-\epsilon\Rightarrow x_2\leq 4+\epsilon\\ x_2+x_3\geq v(2,3)-\epsilon=8-\epsilon\Rightarrow x_1\leq 4+\epsilon\end{array}\right\}\Rightarrow 12\leq 12+3\epsilon\end{array}\right\}\Rightarrow \epsilon_0=0,$$

$$\mathcal{C}_{\epsilon_0}(N;v)=\{(4,4,4)\}.$$

iv.

$$\left.\begin{array}{l}x_1+x_2+x_3=v(1,2,3)=12\\ \forall i\in N,\ x_i\geq v(i)-\epsilon=-\epsilon\\ x_1+x_2\geq v(1,2)-\epsilon=3-\epsilon\Rightarrow x_3\leq 9+\epsilon\\ x_1+x_3\geq v(1,3)-\epsilon=6-\epsilon\Rightarrow x_2\leq 6+\epsilon\\ x_2+x_3\geq v(2,3)-\epsilon=2-\epsilon\Rightarrow x_1\leq 10+\epsilon\end{array}\right\}\Rightarrow \epsilon_0=-3,$$

$$\mathcal{C}_{\epsilon_0}(N;v)=\{\alpha(6,3,3)+(1-\alpha)(3,3,6)\ \ \alpha\in[0,1]\}.$$

v.

$$\left.\begin{array}{l}x_1+x_2+x_3=v(1,2,3)=12\\ \forall i\in N,\ x_i\geq v(i)-\epsilon=-\epsilon\\ x_1+x_2\geq v(1,2)-\epsilon=12-\epsilon\Rightarrow x_3\leq \epsilon\\ x_1+x_3\geq v(1,3)-\epsilon=15-\epsilon\Rightarrow x_2\leq -3+\epsilon\\ x_2+x_3\geq v(2,3)-\epsilon=12-\epsilon\Rightarrow x_1\leq \epsilon\end{array}\right\}\Rightarrow \epsilon_0=5,$$

$$\mathcal{C}_{\epsilon_0}(N;v)=\{(5,2,5)\}.$$

18.34 (a) If $\mathcal{C}(N;v) \neq \emptyset$ then $\mathcal{C}(N;w) \neq \emptyset$. Indeed, let $x \in \mathcal{C}(N;v)$ then $x(N) = v(N)$ and $x(S) \geq v(S)$ for every $\emptyset \neq S \subset N$. Therefore, $x(N) = v(N) = w(N)$ and $x(S) \geq v(S) \geq w(S)$ for every $\emptyset \neq S \subset N$ so $x \in \mathcal{C}(N;w)$.

(b) If $\mathcal{C}(N;w) \neq \emptyset$ then not necessarily $\mathcal{C}(N;v) \neq \emptyset$. A counter example: let $N = \{1,2\}$, then the core of $(N;v)$ where $v(1) = 1$ and $v(2) = v(1,2) = 2$ is empty, whereas the core of $(N;w)$ where $w(1) = w(2) = 1$ and $v(1,2) = 2$ contains the imputation $(1,1)$.

18.35 Let $(N;v)$ be a coalitional game and let $(N;u)$ be a coalitional game that is strategically equivalent to $(N;v)$. We prove that if $(N;v)$ is a market game then $(N;u)$ is also a market game.
By Theorem 18.40, $(N;v)$ is totally balanced. Hence, the core of a sub game (S,v) is non-empty, for every $\emptyset \neq S \subset N$. Following Corollary 18.8, the core of a sub game (S,u) is non-empty, for every $\emptyset \neq S \subset N$ so $(N;u)$ is totally balanced and is therefore a market game (Theorem 18.40).

18.36 We complete the computation of the market game derived from the market in Example 18.24:
Consider first $S = \{1,2\}$:
Since $a(1) + a(2) = (1,1)$ then

$$v(1,2) = \max\{(x_{1,1} + x_{1,2}) + (x_{2,1} + x_{2,2}) \ : \ x_{1,1} + x_{2,1} = 1, x_{1,2} + x_{2,2} = 1\} = 3.$$

Consider next $S = \{1,3\}$:
Since $a(1) + a(3) = (3,2)$ then

$$\begin{aligned} v(1,3) &= \max\{(x_{1,1} + x_{1,2}) + (\sqrt{x_{3,1}} + \sqrt{x_{3,2}}) \ : \ x_{1,1} + x_{3,1} = 3, x_{1,2} + x_{3,2} = 2\} \\ &= \max\{(x_{1,1} + x_{1,2}) + (\sqrt{3 - x_{1,1}} + \sqrt{2 - x_{1,2}}) \ : \ 0 \leq x_{1,1} \leq 3, 0 \leq x_{1,2} \leq 2\} \\ &= \max\{x_{1,1} + \sqrt{3 - x_{1,1}} \ : \ 0 \leq x_{1,1} \leq 3\} + \max\{x_{1,2} + \sqrt{2 - x_{1,2}} \ : \ 0 \leq x_{1,2} \leq 2\} \end{aligned}$$

Using the derivatives of $x_{1,1} + \sqrt{3 - x_{1,1}}$ one can derive that the maximum of this function is obtained at $x_{1,1} = 2.75$. Similarly, using the derivatives of $x_{1,2} + \sqrt{2 - x_{1,2}}$ one can derive that the maximum of this function is obtained at $x_{1,1} = 1.75$. Therefore, the optimal endowments of the producers are $x_1 = (2.75, 1.75)$ and $x_3 = (0.25, 0.25)$ so $v(1,3) = 5.5$.
Finally, consider $S = \{2,3\}$:

Since $a(2) + a(3) = (2, 3)$ then

$$\begin{aligned} v(2,3) &= \max\{(x_{2,1} + 2x_{2,2}) + (\sqrt{x_{3,1}} + \sqrt{x_{3,2}}) \ : \ x_{2,1} + x_{3,1} = 3, x_{2,2} + x_{3,2} = 2\} \\ &= \max\{(x_{2,1} + 2x_{2,2}) + (\sqrt{2 - x_{2,1}} + \sqrt{3 - x_{2,2}}) \ : \ 0 \leq x_{2,1} \leq 3, 0 \leq x_{2,2} \leq 2\} \\ &= \max\{x_{2,1} + \sqrt{2 - x_{2,1}} \ : \ 0 \leq x_{2,1} \leq 2\} + \max\{x_{2,2} + \sqrt{3 - x_{22}} \ : \ 0 \leq x_{2,2} \leq 3\} \end{aligned}$$

Using the derivatives of $x_{1,1} + \sqrt{3 - x_{1,1}}$ one can derive that the maximum of this function is obtained at $x_{1,1} = 2.75$. Similarly, using the derivatives of $x_{1,2} + \sqrt{2 - x_{1,2}}$ one can derive that the maximum of this function is obtained at $x_{1,1} = 1.75$. Therefore, the optimal endowments of the producers are $x_1 = (2.75, 1.75)$ and $(0.25, 0.25)$

18.37 (a) The core of (N, v) is not empty, e.g., $(1.5, 1.5) \in \mathcal{C}(N; v)$, as well as the core of each of its sub game is non-empty. The game $(N; v)$ is therefore totally balanced and hence it is a market game.

(b) Consider the following market: the market contains two producers, i.e., $N = \{1, 2\}$. There is only one commodity. The initial endowments of the producers are $a_1 = 1, a_2 = 0.5$. The production functions of the producers are $u_1(x_1) = x_1, u_2(x_2) = 2x_2$. Then $(N; v)$ is derived from this market.

18.38 Let $(N; v)$ be a coalitional game, and let $(N; w)$ be a coalitional game strategically equivalent to it.

(a) We prove that if the game $(N; v)$ is a market game then the game $(N; w)$ is also a market game. Note that since $(N; v)$ is strategically equivalent to $(N; w)$ the opposite direction holds as well.

Assume that $(N; v)$ is a market game, then there are a set of producer N, a set of commodities L, initial endowment a_i of each producers $i \in N$ and production functions of the producer $(u_i)_{i \in N}$ such that $v(S) = \max \left\{ \sum_{i \in S} u_i(x_i) : \ (x_i)_{i \in N} \in \mathcal{X}^S \right\}$. Then the market $(N, L, (a_i)_{i \in N}, (\widehat{u}_i)_{i \in N})$ where $\widehat{u}_i(x_i) = a \cdot u_i(x_i) + b_i$ is a market from which the game $(N; w)$ is derived. Therefore, $(N; w)$ is a market game.

(b) This part was presented in Exercise 18.35.

18.39 The core of the game $(N; v)$ is non-empty as well as the core of its sub games: $(\frac{2}{3}, \frac{2}{3}, \frac{2}{3}) \in \mathcal{C}(N; v)$; $(0.5, 0.5) \in \mathcal{C}(\{i, j\}, v)$ for every $i, j \in N$;

and $(0) \in \mathcal{C}((\{i\};v)$ for every $i \in N$. Therefore, the game $(N;v)$ is totally balanced.
The directed market from which this game derived is a three producers market with three commodities. The initial endowments of the producers are $a_1 = (1,0,0)$, $a_2 = (0,1,0)$ and $a_3 = (0,0,1)$. The production function of each of the producers is

$$\begin{aligned} u(x_1,x_2,x_3) &= \max \left\{ \sum_{\emptyset \neq T \subseteq N} \delta_S v(S) : \begin{array}{c} \delta_T \geq 0, \forall \emptyset \neq T \subseteq N \\ \sum_{\emptyset \neq T \subseteq N} \delta_S \mathcal{X}^S = x \end{array} \right\} \\ &= \max \left\{ \delta_{1,2} + \delta_{1,3} + \delta_{2,3} + 2\delta_N \;:\; \begin{array}{c} \delta_1 + \delta_{1,2} + \delta_{1,3} + \delta_N = x_1 \\ \delta_2 + \delta_{1,2} + \delta_{2,3} + \delta_N = x_2 \\ \delta_3 + \delta_{1,3} + \delta_{2,3} + \delta_N = x_3 \\ \delta_T \geq 0, \forall \emptyset \neq T \subseteq N \end{array} \right\} \end{aligned}$$

Note that if $x_i = 0$ then $\delta_T = 0$ for every coalition T containing i. Denote by $N_x = \{i \in N : x_i > 0\}$. Then

$$u(x_1,x_2,x_3) = \begin{cases} 0 & |N_x| = 1 \\ \min_{i \in N_x} x_i & |N_x| = 2 \\ \sum_{i \in N} x_i - \max_{i \in N} x_i & |N_x| = 3 \end{cases}$$

18.40 Let $\epsilon > 0$, let $x \in \mathbf{R}^N$, and let $(\delta_S)_{\{S \subseteq N, S \neq \emptyset\}}$ be nonnegative weights satisfying $\sum_{\{S \subseteq N, S \neq \emptyset\}} \delta_S \mathcal{X}^S = x$. Let $y \in \mathbf{R}^N$ be a nonnegative vector satisfying $|x_i - y_i| < \epsilon$ for all $i \in N$. We prove that there exists a collection of nonnegative weights $(\mu_S)_{\{S \subseteq N, S \neq \emptyset\}}$ satisfying (a) $\sum_{\{S \subseteq N, S \neq \emptyset\}} \mu_S \mathcal{X}^S = y$ and (b) $|\delta_S - \mu_S| < 2^{|N|}\epsilon$.
We first, define $\widehat{\mu}_S := \max\{\delta_S - \epsilon, 0\}$ for every nonempty coalition S. Then for every $i \in N$ the following holds

$$|\sum_{\substack{S \subseteq N, \\ i \in S}} \widehat{\mu}_S - x_i| = |\sum_{\substack{S \subseteq N, \\ i \in S}} \widehat{\mu}_S - \sum_{\substack{S \subseteq N, \\ i \in S}} \delta_S| = |-\sum_{\substack{S \subseteq N, \\ i \in S \\ \delta_S \geq \epsilon}} \epsilon - \sum_{\substack{S \subseteq N, \\ i \in S \\ \delta_S \leq \epsilon}} \delta_S| \leq 2^{|N|-1}\epsilon,$$

where the right inequality holds since there are at most $2^{|N|-1}$ coalitions containing i. Hence, $\sum_{\{S \subseteq N, S \neq \emptyset\}} \widehat{\mu}_S \mathcal{X}^S$ is approximately x.
Note that for every player i, $\sum_{\substack{S \subseteq N, \\ i \in S}} \widehat{\mu}_S \leq y_i$. Indeed, if $\delta_S < \epsilon$ for every coalition S containing player i, then $\sum_{\substack{S \subseteq N, \\ i \in S}} \widehat{\mu}_S = 0 \leq y_i$. Otherwise, there is at least one coalition for which $\delta_S \geq \epsilon$ so $\widehat{\mu}_S = \delta_S - \epsilon$, hence

$$\sum_{\substack{S \subseteq N, \\ i \in S}} \widehat{\mu}_S = \sum_{\substack{S \subseteq N, \\ i \in S \\ \delta_S \geq \epsilon}} (\delta_S - \epsilon) \leq \sum_{\substack{S \subseteq N, \\ i \in S}} \delta_S - \epsilon = x_i - \epsilon \leq y_i.$$

For every coalition $\emptyset \neq S \subseteq N$ set

$$\mu_S = \begin{cases} \widehat{\mu}_S & |S| > 1 \\ \widehat{\mu}_i + y_i - \sum_{\substack{S \subseteq N, \\ i \in S}} \widehat{\mu}_S & S = \{i\}. \end{cases}$$

Then the vector $(\mu_S)_{\{S \subseteq N, S \neq \emptyset\}}$ satisfies condition (a): for every player $i \in N$ one has $\sum_{\substack{S \subseteq N, \\ i \in S}} \mu_S = \sum_{\substack{S \subseteq N, \\ i \in S \\ |S|>1}} \widehat{\mu}_S + \widehat{\mu}_i + y_i - \sum_{\substack{S \subseteq N, \\ i \in S}} \widehat{\mu}_S = y_i$. In addition, it satisfies condition (b): for every coalition S where $|S| > 1$, $|\delta_S - \mu_S| = |\delta_S - \widehat{\mu}_S| \leq \epsilon$ and for coalition containing only one player, say player i one has

$$|\delta_i - \mu_i| = |\delta_i - \widehat{\mu}_i - y_i + \sum_{\substack{S \subseteq N, \\ i \in S}} \widehat{\mu}_S| \leq |\delta_i - \widehat{\mu}_i| + |y_i - x_i| + |x_i - \sum_{\substack{S \subseteq N, \\ i \in S}} \widehat{\mu}_S|$$
$$\leq 2^{|N|}\epsilon.$$

It is left to show that the function u defined in Equation (18.88) is a continuous function. Denote by $M = \max\{|v(S)| : \emptyset \neq S \subseteq N\}$. Let $x \in \mathbf{R}^N_+$ and let $(\delta_S)_{\{S \subseteq N, S \neq \emptyset\}}$ be a nonnegative vector at which the maximum in Equation (18.88) is attained, i.e., $\sum_{S \subseteq N} \delta_S \mathcal{X}^S = x$ and $u(x) = \sum_{S \subseteq N} \delta_S v(S)$. Then, for every vector $y \in \mathbf{R}^N_+$ satisfying $|x_i - y_i| < \epsilon$ for every $i \in N$, there is a collection of nonnegative weights $(\mu_S)_{\{S \subseteq N, S \neq \emptyset\}}$ satisfying (a) and (b) hence the following holds:

$$|\sum_{\substack{S \subseteq N \\ S \neq \emptyset}} \delta_S v(S) - \sum_{\substack{S \subseteq N \\ S \neq \emptyset}} \mu_S v(S)| \leq \sum_{\substack{S \subseteq N \\ S \neq \emptyset}} |\delta_S - \mu_S||v(S)| \leq 2^{2|N|}\epsilon M.$$

Therefore,

$$u(x) = \sum_{\substack{S \subseteq N \\ S \neq \emptyset}} \delta_S v(S) \leq \sum_{\substack{S \subseteq N \\ S \neq \emptyset}} \mu_S v(S) + 2^{2|N|}\epsilon M \leq u(y) + 2^{2|N|}\epsilon M.$$

Similarly, by exchanging the roles of x and y the following holds

$$u(y) \leq u(x) + 2^{2|N|}\epsilon M,$$

so

$$|u(y) - u(x)| \leq 2^{2|N|}\epsilon M,$$

and the function u is continuous.

18.41 (a) Assume that the collection $(Y_S)_{\{S \subseteq N, S \neq \emptyset\}}$ is totally balanced. We show that $(N; v)$ is a totally balanced game. Let $T \subseteq N$ be a nonempty coalition and let $\mathcal{D}$ be a balanced collection of coalitions of T with balancing weights $(\delta_S)_{S \in \mathcal{D}}$. Since Y_S is compact and nonempty and f is a concave function, a maximum of f over Y_S is attained so there is $y_S \in Y_S$ such that $v(S) = f(y_S)$, for every $S \in \mathcal{D}$. In addition, since $(Y_S)_{\{S\subseteq N, S\neq\emptyset\}}$ is totally balanced $\sum_{S\in\mathcal{D}} \delta_S y_S \in Y_T$. Therefore,

$$\sum_{S\in\mathcal{D}} \delta_S v(S) = \sum_{S\in\mathcal{D}} \delta_S f(y_S) \leq f(\sum_{S\in\mathcal{D}} \delta_S y_S) \leq \max\{f(y_T) : y_T \in Y_T\}$$
$$= v(T),$$

where the left inequality follows the concavity of f. In particular, by the Bondareva–Shapley condition, $(N; v)$ is a totally balanced game.

(b) Let $(N; v)$ be a market game then there are a set of commodities L, initial endowment a_i of each producers $i \in N$ and production functions of the producer $(u_i)_{i\in N}$ such that

$$v(S) = \max\left\{\sum_{i\in S} u_i(x_i) : \ (x_i)_{i\in N} \in X^S\right\}.$$

Set $d = |N| \cdot |L|$ and define a partition $\{B_1, B_2, ..., B_n\}$ of $\{1, 2, ..., d\}$ by $B_i = \{i, 2i, ..., |L|i\}$. For every $y \in \mathbf{R}^d$ denote by $y^i \in \mathbf{R}^{|L|}$ the vector in which $y^i_l = y_{i\cdot l}$. Define $f(y) = \sum_{\{i\in N, y^i\neq 0\}} u_i(y^i)$. Note that f is concave as a summation of concave functions. Finally, for every nonempty coalition $S \subseteq N$ define

$$Y_S = \left\{ y \in \mathbf{R}^d : \begin{array}{c} \forall j \notin S, y_j = 0, \\ \sum_{i\in S} y_{i\cdot l} = \sum_{i\in S} a^l_i \end{array} \right\}.$$

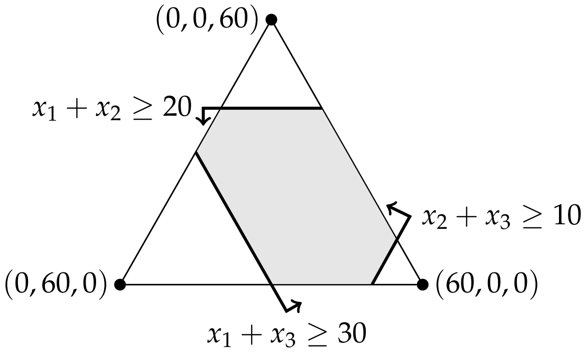
The collection $(Y_S)_{\{S\subseteq N, S\neq\emptyset\}}$ is a collection of compact, nonempty subsets of $\mathbf{R}^d$. Furthermore, it is totally balanced. Indeed, let $T \subseteq N$ be a nonempty coalition and let $\mathcal{D}$ be a balanced collection of coalitions of T with balancing weights $(\delta_S)_{S\in\mathcal{D}}$, then for every list of points $(y_S)_{\{S\subseteq T, S\neq\emptyset\}}$ such that $y_S \in Y_S$ for every $S \in \mathcal{D}$, one has (a) For every $j \notin T$ one has $j \notin S \in \mathcal{D}$ hence $\sum_{S\in\mathcal{D}} \delta_S y_{S,j} = 0$; and (b) For every $l \in L$ one has

$$\sum_{i\in T}\sum_{S\in\mathcal{D}} \delta_S y_{S,i\cdot l} = \sum_{S\in\mathcal{D}} \delta_S \sum_{i\in T} y_{S,i\cdot l} = \sum_{S\in\mathcal{D}} \delta_S \sum_{i\in T} a^l_i = \sum_{i\in T} a^l_i$$

hence $\sum_{S\in\mathcal{D}} \delta_S y_S \in Y_T$ and $(Y_S)_{\{S\subseteq N, S\neq\emptyset\}}$ is totally balanced. Finally,

$$v(S) = \max\left\{\sum_{i\in S} u_i(x_i) : \ (x_i)_{i\in N} \in X^S\right\} = \max\left\{f(y_S) : y_S \in Y_S\right\}.$$

18.42 (a)



(b) By the Bondareva–Shapley condition the core is empty since $v(N) = 60 < 80 = v(1) + v(2,3)$.

(c) By the Bondareva–Shapley condition the core is empty since $v(N) = 60 < 70 = v(1) + v(2,3)$.

(d) By the Bondareva–Shapley condition the core is empty since $v(N) = 60 < 75 = \frac{1}{2}v(1,2) + \frac{1}{2}v(1,3) + \frac{1}{2}v(2,3)$.

(e) By the Bondareva–Shapley condition the core is empty since $v(N) = 60 < 80 = v(2) + v(1,3)$.

18.43 Let $(N; v)$ be a coalitional game with a coalitional structure $\mathcal{B}$. Let k and l be two players who are members of different coalitions B_k and B_l, respectively, in $\mathcal{B}$. Assume that k and l are symmetric players, i.e., $v(S \bigcup \{k\}) = v(S \bigcup \{l\})$ for every coalition S that does not contain either of them, and let $x \in \mathcal{C}(N; v; \mathcal{B})$. We show that $x_k = x_l$. Since x in the core of the game with coalitional structure $\mathcal{B}$ and by the symmetry of the players the following holds

$$x_k + x(B_k \setminus \{k\}) = x(B_k) = v(B_k) = v(B_k \setminus \{k\} \bigcup \{l\}) \leq v(B_k \setminus \{k\}) + x_l \Rightarrow x_k \leq x_l.$$

Similarly,

$$x_l + x(B_l \setminus \{l\}) = x(B_l) = v(B_l) = v(B_l \setminus \{l\} \bigcup \{k\}) \leq v(B_l \setminus \{l\}) + x_k \Rightarrow x_l \leq x_k.$$

Therefore, $x_k = x_l$.

18.44 Let $(N; v)$ be an additive game. We prove that $(N; v)$ is convex. $(N; v)$ is additive so $v(S) = \sum_{i \in S} v(i)$ for every nonempty coalition $S \subseteq N$.

Then, for every two nonempty coalitions $S, T \subseteq N$ one has

$$\begin{aligned} v(S \bigcup T) + v(S \cap T) &= \sum_{i \in S \bigcup T} v(i) + \sum_{i \in S \cap T} v(i) = \sum_{i \in S} v(i) + \sum_{i \in T} v(i) \\ &= v(S) + v(T), \end{aligned}$$

which leads to the convexity of $(N; v)$.

18.45 Let (N, v) be a coalitional game where $N = \{1, 2, 3\}$ and

$$v(S) = \begin{cases} 0 & |S| \leq 1 \\ 1 & |S| \geq 2. \end{cases}$$

Then, $(N; v)$ is superadditive but it is not convex, e.g.,

$$v(1, 2) + v(2, 3) = 2 > 1 = v(1, 2, 3) + v(2).$$

18.46 Let $(N; v)$ be a convex game, i.e., for every two nonempty coalitions $S_1, S_2 \subseteq N$ one has

$$v(S_1 \bigcup S_2) + v(S_1 \cap S_2) \geq v(S_1) + v(S_2). \tag{161}$$

In particular, Equation (161) holds for every two nonempty coalitions $S_1, S_2 \subseteq T \subseteq N$. Thus, the subgame $(T; v)$ is also a convex game.
By Theorem 18.58, since the game $(N; v)$ is convex as well as all of its subgames, the core of $(N; v)$ and the cores of its subgames are nonempty, so $(N; v)$ is totally balanced.

18.47 Let $(N; v)$ be a convex game whose core contains exactly one imputation. Since $(N; v)$ is convex, by Theorem 18.55, there is an imputation $x^i \in \mathcal{C}(N; v)$ such that $x^i_i = v(i)$ for every player $i \in N$. However, the core contains exactly one imputation, and therefore $x^i = x^j$ for every two players $i, j \in N$, so $\mathcal{C}(N; v) = \{(v(i))_{i \in N}\}$.
We next prove that $v(S) = \sum_{i \in S} v(i)$ for every $S \subseteq N$ by induction over the size of S.
For $|S| = 1$, the claim holds trivially. Assume that it holds for every coalition S of size less than k where $k > 1$. For every S of size k, let $i \in S$, then

$$\sum_{j \in S \setminus \{i\}} v(j) + v(i) = v(S \setminus \{i\}) + v(i) \leq v(S) \leq x(S) = \sum_{j \in S} v(j)$$

where the first equality holds by the induction hypothesis and the second inequality holds by the convexity of $(N; v)$. Therefore, all the above weak inequalities obtain as equalities so $v(S) = \sum_{i \in S} v(i)$. That is, $(N; v)$ is an additive game.

18.48 (a) For every $i \in N$, one has

$$v(i) = \begin{cases} p_0(i) & \{i\} \in \mathcal{B} \\ 0 & \text{Otherwise.} \end{cases} \geq 0.$$

In addition,

$$v(N) = \sum_{\{B \in \mathcal{B}, B \subseteq N\}} \sum_{i \in B} p_0(i) = \sum_{B \in \mathcal{B}} \sum_{i \in B} p_0(i) = \sum_{i \in N} p_0(i) = 1.$$

Moreover, $v(B) = \sum_{i \in B} p_0(i)$ for every $B \in \mathcal{B}$. Hence, for every imputation $x \in X(\mathcal{B}; v)$ the following holds: (i) $x_i \geq v(i) \geq 0$ for every $i \in N$; and (ii) $x(B) = v(B) = \sum_{i \in B} p_0(i)$ for every $B \in \mathcal{B}$, which leads to

$$x(N) = \sum_{B \in \mathcal{B}} x(B) = \sum_{B \in \mathcal{B}} v(B) = \sum_{i \in N} p_0(i) = 1.$$

The imputation x is therefore a distribution over N. That is $X(\mathcal{B}; v) \subseteq \Delta(N)$. This inclusion holds as an equality for $\mathcal{B} = \{N\}$.

(b) Let $T, S \subseteq N$ be two nonempty coalitions.

$$\begin{aligned} v(S) + v(T) &= \sum_{\substack{B \in \mathcal{B} \\ B \subseteq S}} \sum_{i \in B} p_0(i) + \sum_{\substack{B \in \mathcal{B} \\ B \subseteq T}} \sum_{i \in B} p_0(i) \\ &= \sum_{\substack{B \in \mathcal{B} \\ B \subseteq S \cap T}} \sum_{i \in B} p_0(i) + \sum_{\substack{B \in \mathcal{B} \\ B \subseteq S \\ B \not\subseteq T}} \sum_{i \in B} p_0(i) + \sum_{\substack{B \in \mathcal{B} \\ B \subseteq S \cap T}} \sum_{i \in B} p_0(i) + \sum_{\substack{B \in \mathcal{B} \\ B \subseteq T \\ B \not\subseteq S}} \sum_{i \in B} p_0(i) \\ &\leq \sum_{\substack{B \in \mathcal{B} \\ B \subseteq S \cap T}} \sum_{i \in B} p_0(i) + \sum_{\substack{B \in \mathcal{B} \\ B \subseteq S \bigcup T}} \sum_{i \in B} p_0(i) \\ &= v(S \cap T) + v(S \bigcup T), \end{aligned}$$

where the last inequality holds since the collection $\{B \in \mathcal{B} : B \subseteq S \cap T\}$, $\{B \in \mathcal{B} : B \subseteq S \text{ and } B \not\subseteq T\}$ and $\{B \in \mathcal{B} : B \subseteq T \text{ and } B \not\subseteq S\}$ are disjoint subsets of $\{B \in \mathcal{B} : B \subseteq S \bigcup T\}$. (N, v) is therefore a convex game.

(c) We first show that $\mathcal{C}(N; v) \subseteq C(p_0)$. Let $p \in \mathcal{C}(N; v)$. Then, by Part (a), (1) $p_i \geq v(i) \geq 0$ for every $i \in N$; (2) $p(N) = v(N) = 1$; and (3) $p(B) \geq v(B) = \sum_{i \in B} p_0(i)$ for every $B \in \mathcal{B}$. Hence, by (1) and (2), $p \in \Delta(N)$. Furthermore, by (1) and (3),

$$1 = p(N) = \sum_{B \in \mathcal{B}} p(B) \geq \sum_{B \in \mathcal{B}} \sum_{i \in B} p_0(i) = \sum_{i \in N} p_0(i) = 1,$$

so the weak inequalities obtain as equalities and $p(B) = \sum_{i \in B} p_0(i)$. In particular, $p \in C(p_0)$.

We next prove that $C(p_0) \subseteq \mathcal{C}(N;v)$, which conclude the proof. Let $p \in C(p_0)$. Then, $v(N) = 1 = p(N)$. In addition, for every $B \in \mathcal{B}$ one has $v(B) = \sum_{i \in B} p_0(i) = \sum_{i \in B} p_i = p(B)$. Finally, let $S \subseteq N$ be a nonempty coalition. Then,

$$v(S) = \sum_{\substack{B \in \mathcal{B} \\ B \subseteq S}} \sum_{i \in B} p_0(i) = \sum_{\substack{B \in \mathcal{B} \\ B \subseteq S}} \sum_{i \in B} p_i \leq \sum_{i \in S} p_i = p(S),$$

where the last inequality holds since $p_i \geq 0$ for every $i \in N$ and $\bigcup_{\substack{B \in \mathcal{B} \\ B \subseteq S}} B \subseteq S$ is a union of disjoint sets. Therefore, one can conclude that $C(p_0) \subseteq \mathcal{C}(N;v)$ and therefore $C(p_0) = \mathcal{C}(N;v)$, as needed.

18.49 Let $(N;v)$ be a convex coalitional game which is strategically equivalent to the coalitional game $(N;w)$. We show that $(N;w)$ is also a convex game.
There are $a > 0$ and $b \in \mathbf{R}^{|N|}$ such that $w(S) = a \cdot v(S) + b(S)$ for every $S \subseteq N$. Note that for every two coalitions $S, T \subseteq N$ the following holds

$$b(S) + b(T) = b(S \setminus T) + b(S \cap T) + b(T \setminus S) + b(S \cap T) = b(S \cap T) + b(S \bigcup T). \tag{162}$$

Let $S, T \subseteq N$ be two nonempty coalitions. Then

$$\begin{aligned} w(S) + w(T) &= a \cdot v(S) + b(S) + a \cdot v(T) + b(T) \\ &= a(v(S) + v(T)) + b(S \cap T) + b(S \bigcup T) \qquad (163) \\ &\leq a(v(S \cap T) + v(S \bigcup T)) + b(S \cap T) + b(S \bigcup T) \qquad (164) \\ &= a \cdot v(S \cap T) + b(S \cap T) + a \cdot v(S \bigcup T) + b(S \bigcup T) \\ &= w(S \cap T) + w(S \bigcup T), \end{aligned}$$

where Equality (163) follows from Equality (162) and Inequality (164) follows by the convexity of $(N;v)$. In conclusion, $(N;w)$ is also a convex game.

18.50 Let N be a set of players, and let $f : N \to \mathbf{R}$ be a function. Define a coalitional game $(N;v)$ by $v(S) := f(|S|)$. We prove that $(N;v)$ is a convex game if and only if f is a convex function.
Assume first that f is convex, and let $S, T \subseteq N$ be two different nonempty coalitions. We show that

$$v(S) + v(T) \leq v(S \cap T) + v(S \bigcup T). \tag{165}$$

Case I: $S \subseteq T$ then Inequality (165) holds as equality since $S \cap T = S$ and $S \bigcup T = T$. Similarly, if $T \subseteq S$ then the condition holds.

Otherwise, $S \setminus T \neq \emptyset$ and $T \setminus S \neq \emptyset$. Denote by $s = |S|$, $t = |T|$, $k = |S \cap T|$ and hence $s + t - k = |S \bigcup T|$. In particular, $k < s < s + t - k$ and $k < t < s + t - k$.
Case II: $s = t$. Then, by the convexity of f, since $0 \leq k < s < 2s - k$ one has

$$(2s - 2k)f(s) \leq (s - k)f(k) + (s - k)f(2s - k) \Rightarrow 2f(s) \leq f(k) + f(2s - k), \tag{166}$$

from which one can derive Inequality (165), using the equalities $v(T) = v(S) = f(s)$, $v(S \cap T) = f(k)$ and $v(S \bigcup T) = f(2s - k)$.
Case III: $s < t$ (similarly for case $t < s$). Then, by the convexity of f, since $0 \leq k < s < t$ one has

$$(t - k)f(s) \leq (t - s)f(k) + (s - k)f(t). \tag{167}$$

In addition, since $0 < s < t < s + t - k$ one has

$$(t - k)f(t) \leq (s - k)f(s) + (t - s)f(s + t - k). \tag{168}$$

Summing Inequalities (167) and (168) leads to the following inequality

$$(t - s)\left(f(s) + f(t)\right) \leq (t - s)\left(f(k) + f(s + t - k)\right)$$

and therefore, $v(S) + v(T) \leq v(S \cap T) + v(S \bigcup T)$ and the game is convex.
We next prove the opposite direction. Assume that $(N; v)$ is a convex game. Let Let $k \leq m \leq l \leq |N|$ and $k < l$. We show that

$$f(m) \leq \tfrac{l-m}{l-k} f(k) + \tfrac{m-k}{l-k} f(l). \tag{169}$$

If either $k = m$ or $l = m$ then Inequality (169) holds trivially. Assume next that $k < m < l$.
By Theorem 18.54, for every $S \subseteq T \subset N$ and every $i \in N \setminus T$ one has $v(S \bigcup \{i\}) - v(S) \leq v(T \bigcup \{i\}) - v(T)$. Therefore,

$$f(s + 1) - f(s) \leq f(t + 1) - f(t) \tag{170}$$

for every $0 \leq s \leq t < n$.

Then, by Inequality (170),

$$\left.\begin{array}{ll} f(k+1)-f(k) & \leq f(m+1)-f(m) \\ f(k+2)-f(k+1) & \leq f(m+1)-f(m) \\ \vdots & \\ f(m)-f(m-1) & \leq f(m+1)-f(m) \end{array}\right\}$$

$$\Rightarrow \frac{f(m)-f(k)}{m-k} \leq f(m+1)-f(m).$$

Similarly,

$$\left.\begin{array}{ll} f(m+1)-f(m) & \leq f(m+1)-f(m) \\ f(m+1)-f(m) & \leq f(m+2)-f(m+1) \\ \vdots & \\ f(m+1)-f(m) & \leq f(l)-f(l-1) \end{array}\right\}$$

$$\Rightarrow f(m+1)-f(m) \leq \frac{f(l)-f(m)}{l-m}.$$

Therefore,

$$\frac{f(m)-f(k)}{m-k} \leq \frac{f(l)-f(m)}{l-m}$$

$$\Rightarrow f(m) \leq \tfrac{l-m}{l-k} f(k) + \tfrac{m-k}{l-k} f(l),$$

and f is therefore convex.

18.51 Let $(N;v)$ be a convex game and let $(N;\widetilde{v})$ be the monotonic cover of $(N;v)$ defined by $\widetilde{v}(S) := \max_{R\subseteq S} v(R)$.
Then, for every two coalition $S,T \subseteq N$, there are $S' \subseteq S$ and $T' \subseteq T$ satisfying $\widetilde{v}(S) = v(S')$ and $\widetilde{v}(T) = v(T')$. Hence,

$$\widetilde{v}(S)+\widetilde{v}(T) = v(S')+v(T') \leq v(S'\bigcup T')+v(S'\cap T') \leq \widetilde{v}(S'\bigcup T')+\widetilde{v}(S'\cap T'),$$

where the left inequality obtains by the convexity of $(N;v)$ and the right inequality obtains by the definition of $\widetilde{v}$. $(N;\widetilde{v})$ is therefore a convex game.

18.52 Consider the following coalitional game: $N = \{1,2,3\}$. $v(1) = 0$, $v(2) = v(3) = 1$, $v(1,2) = 2$, $v(1,3) = v(2,3) = 3$ and $v(N) = 4$. Then $(N;v)$ is not convex, e.g., $v(1,3)+v(2,3) = 6 > 5 = v(3)+v(N)$. The core of this game is not empty, since $(1,1,2) \in \mathcal{C}(N;v)$. However, the core does not contain the Weber set, since

$$(v(1), v(1,2)-v(1), v(N)-v(1,2)) = (0,2,2) \notin \mathcal{C}(N;v).$$

18.53 The game $(N; v)$ where $v(S) = |S|$ satisfies the conditions required in the exercise.

18.54 The coalitional function of the spanning tree game corresponding to Example 18.62 where

$$N = \{D(Detroit), L(Lansing), G(GrandRapids), A(AnnArbor)\}$$

is as follows

$$\begin{aligned}
&c(A) = 1, c(D) = 1.5, c(G) = 2, c(L) = 2.5,\\
&c(AD) = 2.5, c(AG) = 2, c(AL) = 3,\\
&c(DG) = 3.5, c(DL) = 2.5, c(GL) = 3,\\
&c(ADG) = 3.5, c(ADL) = 3.5, c(AGL) = 3,\\
&c(DGL) = 3.5, c(ADGL) = 4.
\end{aligned}$$

The above game is a convex game.

18.55 We proof that a connected graph over a set V with $n+1$ vertices is a tree if and only if it contains n edges.
We first prove that a tree with $n+1$ vertices has n edges by induction on n. The result is obviously true for $n = 1, 2$ and 3. Assume the result holds for every tree with at most n vertices. Let T be a tree with $n+1$ vertices and let e be an edge connecting $u \in V$ and $v \in V$. Hence, e is the only path between u and v. Therefore deletion of e from T separates T into two connected components, say T_1 and T_2, and as there were no cycles to begin with, each component is a tree. Let n_1 and n_2 be the number of vertices in T_1 and T_2 respectively, so that $n_1 + n_2 = n + 1$. However, by induction hypothesis since $n_1 \leq n$ and $n_2 \leq n$, the number of edges in T_1 and T_2 are respectively $n_1 - 1$ and $n_2 - 1$. Hence, the number of edges in $T = n_1 - 1 + n_2 - 1 + 1 = n_1 + n_2 - 1 = n + 1 - 1 = n$, as needed.
We next prove that every connected graph G with $n+1$ vertices and n edges is a tree, that is, G contains no cycles. Assume to the contrary that G contains cycles. Remove an edge from a cycle so that the resulting graph is again connected. Continue this process of removing one edge from one cycle at a time till the resulting graph H is a connecting graph with no cycle. H is a tree with $n+1$ vertices (since the set of vertices in H and G are identical) so the number of edges in H is n. By the preceding process it follows that the number of edges in G is greater than the number of edges in H. So $n > n$, a contradiction. Hence, G has no cycles and therefore it is a tree.

18.56 A spanning tree system (N, V, E, v_0, a), where (V, E) is a tree, is called a tree system. Let $(N; c)$ be the spanning tree game corresponding to such a system. We prove that $(N; c)$ is a convex game.
For every coalition $S \subseteq N$, denote by (V^S, E^S) the minimal-cost spanning tree of coalition S. In particular, $c(S) = \sum_{e \in E^S} a(e)$.
Note that, since (V, E) is a tree there is a unique acyclic path connecting a vertex v to v_0 for every $v \in V$. Hence, for every coalition S, the minimal-cost spanning tree of coalition S is a union of all the acyclic paths connecting every $v \in S$ to v_0. Therefore, (1) if $S \subseteq T \subseteq N$ then $V^S \subseteq V^T$ and $E^S \subseteq E^T$; (2) $V^{S \cap T} \subseteq V^S \cap V^T$ and $E^{S \cap T} \subseteq E^S \cap E^T$ for every two coalitions $S, T \subseteq N$; and (3) $V^{S \cup T} = V^S \bigcup V^T$ and $E^{S \cup T} = E^S \bigcup E^T$ for every two coalitions $S, T \subseteq N$.
Let $S, T \subseteq N$ be two coalitions. Then, by (1)-(3) and since a is a positive function,

$$\begin{aligned} c(S) + c(T) &= \sum_{e \in E^S} a(e) + \sum_{e \in E^T} a(e) = \sum_{e \in E^S \bigcup E^T} a(e) + \sum_{e \in E^S \cap E^T} a(e) \\ &\geq \sum_{e \in E^{S \cup T}} a(e) + \sum_{e \in E^{S \cap T}} a(e) = c(S \bigcup T) + c(S \cap T), \end{aligned}$$

and the game is therefore convex.

18.57 (a) The spanning tree game corresponding to the given spanning tree system is $(N; c)$ where $N = \{1, 2, 3\}$ and

$$c(S) = \begin{cases} 0 & |S| = 0 \\ 2 & |S| = 1 \\ 3 & |S| = 2 \\ 5 & |S| = 3. \end{cases}$$

(b) By the Bondareva–Shapley condition, the core is empty since

$$\tfrac{1}{2}c(1,2) + \tfrac{1}{2}c(1,3) + \tfrac{1}{2}c(2,3) < c(1,2,3).$$

18.58 The game $(N; v)$ satisfies the following properties: (1) $v(S) \geq 0$ for every $S \subseteq N$; and (2) $v(S) \leq v(T)$ for every $S \subseteq T \subseteq N$, since $E - \sum_{i \notin S} d_i \leq E - \sum_{i \notin T} d_i$.
Let $S, T \subseteq N$. Assume first that $v(S) = 0$, then

$$v(S) + v(T) = v(T) \leq v(S \bigcup T) \leq v(S \bigcup T) + v(S \cap T),$$

where the first inequality holds by (2) and the second inequality holds by (1). The proof in case $v(T) = 0$ is similar.

Assume next that $v(S) > 0$ and $v(T) > 0$, i.e., $v(S) = E - \sum_{i \notin S} d_i > 0$ and $v(T) = E - \sum_{i \notin T} d_i > 0$. Therefore,

$$\begin{aligned}
v(S) + v(T) &= (E - \sum_{i \notin S} d_i) + (E - \sum_{i \notin T} d_i) \\
&= (E - \sum_{\substack{i \notin S \\ i \notin T}} d_i) + (E - \sum_{\substack{i \notin S \\ i \notin T}} d_i - \sum_{\substack{i \in S \\ i \notin T}} d_i - \sum_{\substack{i \notin S \\ i \in T}} d_i) \\
&= (E - \sum_{i \notin S \bigcup T} d_i) + (E - \sum_{i \notin S \cap T} d_i) \\
&\leq v(s \bigcup T) + v(S \cap T),
\end{aligned}$$

so the game is convex.

18.59 We show that $x = \left(\frac{d_i}{\sum_{j=1}^n d_j} E \right)_{i=1}^n$ is in the core of $(N; v)$.
The efficiency condition:

$$x(N) = \sum_{i=1}^n x_i = \sum_{i=1}^n \frac{d_i}{\sum_{j=1}^n d_j} E = E = v(N).$$

The coalitional rationality condition: for every nonempty coalition $S \subseteq N$ such that $v(S) = 0$ then $x(S) \geq 0 = v(S)$. Otherwise,

$$\begin{aligned}
x(S) &= \sum_{i \in S} \frac{d_i}{\sum_{j=1}^n d_j} E = \sum_{i=1}^n \frac{d_i}{\sum_{j=1}^n d_j} E - \sum_{i \notin S} \frac{d_i}{\sum_{j=1}^n d_j} E = E - \frac{E}{\sum_{j=1}^n d_j} \sum_{i \notin S} d_i \\
&\geq E - \sum_{i \notin S} d_i = v(S),
\end{aligned}$$

where the inequality holds since $E < \sum_{i=1}^n d_i$, and we are done.

18.60 Let $x = (x_1, x_2, ..., x_n)$ be an imputation in the core of $(N; v)$, and let S be a nonempty coalition. Denote by $(S; w_S^x)$ the reduced game of $(N; v)$ to coalition S relative to x where

$$w_S^x(R) = \begin{cases} \max_{Q \subseteq N \setminus S} \{v(R \bigcup Q) - x(Q)\} & \emptyset \neq R \subset S \\ x(S) & R = S \\ 0 & R = \emptyset. \end{cases}$$

We prove that $(S; w_S^x)$ is the game corresponding to the bankruptcy problem $\Gamma = [x(S); (d_i)_{i \in S}]$. That is $w_S^x = w$ where

$$w(R) = \max\{x(S) - \sum_{i \in S \setminus R} d_i, 0\}.$$

First note that $w(S) = x(S) = w_S^x(S)$ and $w(\emptyset) = 0 = w_S^x(\emptyset)$. Let R be a coalition such that $\emptyset \neq R \subset S$. We show that $w_S^x(R) \geq w(R)$ as well as $w(R) \geq w_S^x(R)$ and hence $w_S^x(R) = w(R)$.
If $w(R) = 0$ then

$$w_S^x \geq v(R \bigcup Q) - x(Q) \geq x(R \bigcup Q) - x(Q) = x(R) \geq 0 = w(R),$$

where $Q \subseteq N \setminus S$. Otherwise, $w(R) = x(S) - \sum_{i \in S \setminus R} d_i$ and therefore

$$\begin{aligned} w_S^x(R) &\geq v(R \bigcup (N \setminus S)) - x(N \setminus S) && (171) \\ &= v(R \bigcup (N \setminus S)) - \big(x(N) - x(S)\big) \\ &= v(R \bigcup (N \setminus S)) - E + x(S) && (172) \\ &\geq (E - \sum_{i \notin R \bigcup (N \setminus S)} d_i) - E + x(S) && (173) \\ &= x(S) - \sum_{i \in S \setminus R} = w(R), \end{aligned}$$

where Inequality (171) follows by the definition of w_S^x, Equality (172) holds since $x \in \mathcal{C}(N; v)$, and Inequality (173) follows by the definition of v. Thus, $w_S^x(R) \geq w(R)$ for every coalition $R \subseteq S$.
We show the other direction. If $w_S^x(R) = 0$ then $w_S^x(R) = 0 \leq w(R)$. Otherwise, there is $Q \subseteq N \setminus S$ such that $w_S^x(R) = v(R \bigcup Q) - x(Q) > 0$. In particular $v(R \bigcup Q) > x(Q) \geq 0$ so $v(R \bigcup Q) = E - \sum_{i \notin R \bigcup Q} d_i$. Therefore,

$$\begin{aligned} w_S^x(R) &= v(R \bigcup Q) - x(Q) \\ &= E - \sum_{i \notin R \bigcup Q} d_i - x(Q) \\ &= E - \sum_{i \notin R \bigcup Q} d_i - (x(N) - x(S) - x(N \setminus (S \bigcup Q)) \\ &= E - \sum_{i \in S \setminus R} d_i - \sum_{i \in N \setminus (S \bigcup Q)} d_i - E + x(S) + x(N \setminus (S \bigcup Q)) \\ &= x(S) - \sum_{i \in S \setminus R} d_i + x(N \setminus (S \bigcup Q)) - \sum_{i \in N \setminus (S \bigcup Q)} d_i \\ &\leq x(S) - \sum_{i \in S \setminus R} d_i = w(R), \end{aligned}$$

where the inequality holds since $x(N \setminus T) \leq \sum_{i \in N \setminus T} d_i$, for every $\emptyset \neq T \subseteq N$. Indeed,

$$E - \sum_{i \in N \setminus T} d_i \leq v(T) \leq x(T) = x(N) - x(N \setminus T) = E - x(N \setminus T).$$

Thus, $w_S^x(R) \leq w(R)$ for every $R \subseteq S$ which completes the proof.

18.61 By Exercise 18.59, the suggested solution concept is in the core of $(N; v)$. By theorem 18.48, the core satisfies the Davis–Maschler reduced game property. Hence, the suggested solution concept satisfies the Davis–Maschler reduced game property.

18.62 **Proof of Theorem 18.68:** Let $F = (V, E, v_0, v_1, c, N, I)$ be a flow problem, let $(N; v)$ be the corresponding flow game, and let S be a coalition. Denote by $F_{|S} = (V, E_{|S}, v_0, v_1, c, S, I)$ the flow problem where the set of players is S and the set of edges $E_{|S}$ consists of the edges in E controlled by the members of S, that is, $E_{|S} = I^{-1}(S)$. Let $(S; v_{|S})$ be the flow game corresponding to $F_{|S}$. We show that $v_{|S}(R) = v(R)$ for every coalition $R \subseteq S$.
For every $R \subseteq S$,

$$\begin{aligned}(E_{|S})_{|R} &= \{e \in E_{|S} : I(e) \in R\} = \{e \in E : I(e) \in R \text{ and } I(e) \in S\} \\ &= \{e \in E : I(e) \in R\} = E_{|R}.\end{aligned}$$

Therefore, $(F_{|S})_{|R} = F_{|R}$. Hence, for every $R \subseteq S$, $v_{|S}(R)$, the magnitude of the maximal flow of the flow problem $(F_{|S})_{|R}$, equals $v(R)$, the magnitude of the maximal flow of the flow problem $F_{|R}$, as needed.

18.63 We prove that the function $\hat{f}$ that is defined in Equation (18.160) is a flow in the graph G, and its magnitude is $M(f^*) + \epsilon$.

Let $(u, v) \in E$ be an edge. We prove that $0 \leq \hat{f}(u, v) \leq c(u, v)$.
Since f^* is a flow and $\epsilon > 0$, the following inequalities hold:(1) $0 \leq f^*(u, v) \leq c(u, v)$; (2) $0 \leq f^*(u, v) \leq f^*(u, v) + \epsilon$; and (3) $f^*(u, v) - \epsilon \leq f^*(u, v) \leq c(u, v)$. Therefore, by (1), if (u, v) is not in the path σ then $0 \leq \hat{f}(u, v) = f^*(u, v) \leq c(u, v)$. If (u, v) is in the path σ and $f^*(u, v) < c(u, v)$, then $(u, v) \in \hat{E}$ and therefore $\epsilon \leq f^*(u, v) - c(u, v)$ from which one can derive that $0 \leq \hat{f}(u, v) = f^*(u, v) + \epsilon \leq c(u, v)$ (where the left inequality holds by (2)). If (u, v) is in the path σ and $f^*(u, v) > 0$, then $(v, u) \in \hat{E}$ and therefore $\epsilon \leq f^*(u, v)$ from which one can derive that $0 \leq \hat{f}(u, v) = f^*(u, v) - \epsilon \leq c(u, v)$ (where the right inequality holds by (3)).

We next prove that

$$\sum_{\{u \in V \setminus \{v\} : (u,v) \in E\}} \hat{f}(u, v) = \sum_{\{u \in V \setminus \{v\} : (v,u) \in E\}} \hat{f}(v, u) \tag{174}$$

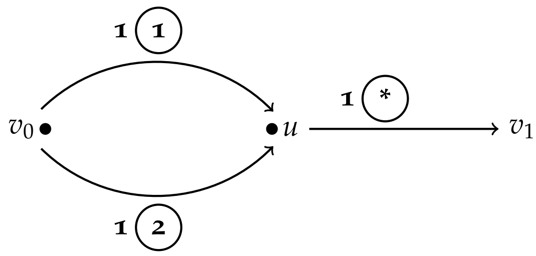
for every vertex $v \in V \setminus \{v_0, v_1\}$.
If there is no edge e in the path σ which is contained in $\{(u, v) \in E : u \in V \setminus \{v\}\} \bigcup \{(v, u) \in E : u \in V \setminus \{v\}\}$ then Equality (174) holds since f^* is a flow.

Otherwise, v is in the path σ. However, since Equality (174) holds for the flow f^* (instead of $\hat{f}$), it is left to deal with edges e in the above summations for which $\hat{f}(e) = f^*(e) \pm \epsilon$.
For every appearance of v in the path σ, there are two successive edges in the path, $(u,v),(v,w) \in \hat{E}$. If $(u,v) \in E$ and $(v,w) \in E$ then $\hat{f}(u,v) = f^*(u,v) + \epsilon$ and $\hat{f}(v,w) = f^*(v,w) + \epsilon$, so one ϵ offsets the other ϵ. If $(u,v) \notin E$ and $(v,w) \notin E$ then $(v,u),(w,v) \in E$ thus $\hat{f}(u,v) = f^*(u,v) - \epsilon$ and $\hat{f}(v,w) = f^*(v,w) - \epsilon$. Hence, again, one ϵ offsets the other ϵ. If $(u,v) \in E$ and $(v,w) \notin E$ then $(w,v) \in E$ thus $\hat{f}(u,v) = f^*(u,v) + \epsilon$ and $\hat{f}(w,v) = f^*(v,w) - \epsilon$ so again $\epsilon - \epsilon = 0$. The case where $(u,v) \notin E$ and $(v,w) \in E$ is similar. In conclusion, Equality (174) holds, $\hat{f}$ is a flow.

Finally, $M(\hat{f}) := \sum_{\{u \in V\setminus\{v_1\} \,:\, (u,v_1)\in E\}} \hat{f}(u,v_1)$. However, there must be exactly one edge $(u_1,v_1) \in E \cap \hat{E}$ in the path σ so $\hat{f}(u_1,v_1) = f^*(u_1,v_1) + \epsilon$ whereas all other edges $(u,v_1) \in E$ are not in the path. Therefore,

$$\begin{aligned} M(\hat{f}) &:= \sum_{\substack{u\in V\setminus\{u_1,v1\}\\ (u,v_1)\in E}} \hat{f}(u,v_1) + \hat{f}(u_1,v_1) \\ &= \sum_{\substack{u\in V\setminus\{u_1,v1\}\\ (u,v_1)\in E}} f^*(u,v_1) + f^*(u_1,v_1) + \epsilon \\ &= M(f^*) + \epsilon. \end{aligned}$$

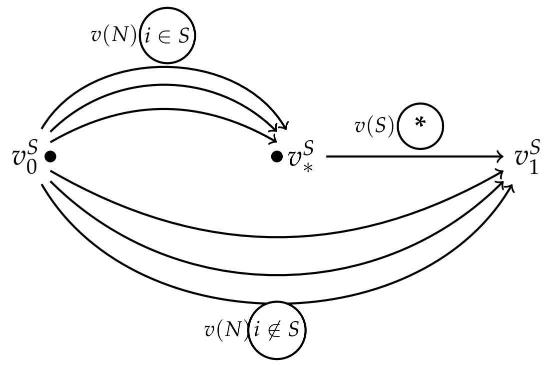
18.64 (a) Consider the following problem where $N = \{1,2\}$.



Then $v(1) = v(2) = v(1,2) = 1$ so $v(1) + v(2) > v(1,2)$. Therefore, by the Bondareva–Shapley condition, the core is empty.

(b) Every flow game (N,v) corresponding to a flow problem F with public edges is a monotonic game. Indeed, let $S \subseteq T \subseteq N$ be two coalitions. Then, the maximal flow in the flow problem $F_{|S}$ is a feasible flow in $F_{|T}$ where no flow passes through edges controlled by players in $T \setminus S$. Hence, the magnitude $v(S)$ of

the maximal flow of $F_{|S}$ is at least the magnitude $v(T)$ of the maximal flow in $F_{|T}$, so $(N;v)$ is a monotonic game.
We next show that for every monotonic game $(N;v)$ there is a flow problem in which there may be public edges such that $(N;v)$ is the flow game corresponding to this flow problem. Let $S \subseteq N$ be a nonempty coalition. Consider the next flow problem F^S



That is, $F^S = (V^S, E^S, v_0^S, v_1^S, c^S, N, I^S)$ where:

- The set of vertices V^S is $V = \{v_0^S, v_1^S, v_*^S\}$ where v_0^S is the source and v_1^S is the sink.
- There are $|N|$ edges $(e_i^S)_{i \in N}$ connecting the source to v_*^S and v_1^S. In addition, there is a unique edge e_*^S connects v_*^S to v_1^S. Therefore, the set of edges E satisfies
$$E = \{e_1^S, ..., e_{|N|}^S, e_*^S\}.$$
- The capacity function is
$$c(e) = \begin{cases} v(N) & e = e_i^S,\ i \in N \\ v(S) & e = e_*^S. \end{cases}$$
- $I(e_i^S) = i$ for every $i \in N$, and $I(e_*^S) = *$ (i.e., e_*^S is a public edge).

Then the flow game $(N;v^S)$ corresponding the flow problem F^S satisfies, $v^S(S) = v(S)$ and $v^S(T) \geq v(T)$ for every nonempty coalition $T \subseteq N$. Indeed, if $T \subseteq S$ then by the monotonicity $v^S(T) = v(S) \geq v(T)$; otherwise, there is at least $i \in T \setminus S$ so, by the monotonicity, $v^S(T) \geq v(N) \geq v(T)$. In conclusion, (N, v^S) is a flow game, for every nonempty coalition $S \subseteq N$. Furthermore, $v(T) = \min_{\substack{S \subseteq N \\ S \neq \emptyset}} v^S(T)$. Hence, by Theorem 18.71, (N, v) is a flow game.

18.65 The games in Exercise 18.5, are all 0-normalized and positive. Hence, $v^*(i) = v(i) = 0$ and $v^*(i,j) = v(i,j)$, for every two players $i \neq j$. Finally, since the games are 0-normalized and monotonic, $v^*(N) = v(N) = 90$.

18.66 The core of the game in Exercise 18.6, with respect to each possible coalitional structure is as follows:

- The core with respect to $\mathcal{B} = \{N\}$ is presented in Exercise 18.6.
- The core with respect to $\mathcal{B} = \{\{1\},\{2\},\{3\}\}$ is a single point $(5,10,20)$.
- The core with respect to $\mathcal{B} = \{\{1,2\},\{3\}\}$ is a the one-dimensional line segment between the points $(5,45,20)$ and $(40,10,20)$.
- The core with respect to $\mathcal{B} = \{\{1,3\},\{2\}\}$ is a the one-dimensional line segment between the points $(5,10,65)$ and $(50,10,20)$.
- The core with respect to $\mathcal{B} = \{\{1\},\{2,3\}\}$ is a the one-dimensional line segment between the points $(5,10,40)$ and $(5,30,20)$.

18.67 **The proof of Theorem 18.79:** Let $(N;v)$ be a coalitional game and let $v^*(S) = \max_{\mathcal{T}} \sum_{T\in\mathcal{T}} v(T)$ where $\mathcal{T}$ is a partition of S.

(a) We show that $(N;v^*)$ is superadditive. Let $S,T \subseteq N$ be two disjoint coalitions. Let $\mathcal{S}$ be a partition of S such that $v^*(S) = \sum_{R\in\mathcal{S}} v(R)$ and let $\mathcal{T}$ be a partition of T such that $v^*(T) = \sum_{R\in\mathcal{T}} v(R)$. Hence, since S and T are disjoint sets, $\mathcal{S}\bigcup\mathcal{T}$ is a partition of $S\bigcup T$ so

$$v^*(S\bigcup T) \geq \sum_{R\in\mathcal{S}\bigcup\mathcal{T}} v(R) = v^*(S) + v^*(T),$$

so (N,v^*) is superadditive.

(b) Let $(N;w)$ be a superadditive game satisfying $w(S) \geq v(S)$ for every coalition S. We prove that for every n disjoint coalitions $S_1, S_2, ..., S_n$ one has $w(\bigcup_{i=1}^n S_i) \geq \sum_{i=1}^n w(S_i)$ by induction over n. For $n = 2$ it holds by the superadditivity. Assume that it holds for n disjoint coalitions. Let $S_1, S_2, ..., S_{n+1}$ be $n+1$ disjoint coalitions. Then, by the induction hypothesis,

$$w(\bigcup_{i=1}^{n+1} S_i) \geq w(\bigcup_{i=1}^{n} S_i) + w(n+1) \geq \sum_{i=1}^{n+1} w(S_i).$$

Thus, for every partition $\mathcal{S}$ of S one has

$$w(S) \geq \sum_{R\in\mathcal{S}} w(R) \geq \sum_{R\in\mathcal{S}} v(R)$$

from which one can derive that

$$w(S) \geq \max_{\mathcal{S}} \sum_{R \in \mathcal{S}} v(R) = v^*(S).$$

In conclusion, the game $(N; v^*)$ is the smallest superadditive game that is larger than $(N; v)$.

(c) The game $(N; v)$ is superadditive if and only if $v^* = v$. Indeed, by Part (a), if $v^* = v$ then $(N; v)$ is superadditive. On the other hand, if $(N; v)$ is superadditive, then it is the smallest super additive game that is at least $(N; v)$, however, by Part (b), $v^* = v$.

18.68 Let $x \in \mathcal{C}(N; v; \mathcal{B})$ and let $y \in \mathcal{C}(N; w; \mathcal{B})$. Then, for every $S \in \mathcal{B}$,

$$x(S) + y(S) = v(S) + w(S) = (v + w)(S).$$

In addition, for every $S \subseteq N$,

$$x(S) + y(S) \geq v(S) + w(S) = (v + w)(S).$$

Therefore, $x + y \in \mathcal{C}(N; v + w; \mathcal{B})$.
On the other hand, consider the 0-normalized game $(N; v)$ where $N = \{1, 2\}$ and $v(1, 2) = 3$. Then $(1, 2), (2, 1) \in \mathcal{C}(N; v)$ but

$$(1, 2) - (2, 1) = (-1, 1) \notin \mathcal{C}(N, v - v).$$

18.69 Consider the game $(N; v)$ presented in Exercise 18.4. Then, there is no coalitional structure $\mathcal{B}$ such that $\mathcal{C}(N; v; \mathcal{B}) \neq \emptyset$. For $\mathcal{B} = \{N\}$, the core is empty as shown in Exercise 18.4. For $\mathcal{B} = \{\{1\}, \{2\}, \{3\}\}$, the point $x = (0, 0, 0)$ is the only point in $X(\mathcal{B}; v)$, but $x(i, j) = 0 < v(i, j)$, thus the core is empty. Finally, for $\mathcal{B} = \{\{1, 2\}, \{3\}\}$ (similarly, the cases where either $\mathcal{B} = \{\{1, 3\}, \{2\}\}$ or $\mathcal{B} = \{\{2, 3\}, \{1\}\}$), every point $x \in X(\mathcal{B}; v)$ satisfies $x_3 = 0$ and either $x_1 < 1$ or $x_2 < 1$. Hence, either $x(1, 3) < 1 = v(1, 3)$ or $x(2, 3) < 1 = v(2, 3)$, thus the core is empty.

18.70 Let $(N; v)$ be a coalitional game, let (N, v^*) be its superadditive cover, and let $(N; \widetilde{v})$ be its monotonic cover. Clearly, $v(N) \leq v^*(N)$ and $v(N) \leq \widetilde{v}(N)$. However, none of the quantities $v^*(N)$, and $\widetilde{v}(N)$ always greater than or equal the other quantity. For instance, for $(N; v)$ where $N = \{1, 2\}$ and $v(1) = v(2) = v(1, 2) = 1$ one has $v(N) = \widetilde{v}(N) = 1$ but $v^*(N) = 2$. On the other hand, for $(N; v)$ where $N = \{1, 2\}$ and $v(1) = -1, v(2) = 2, v(1, 2) = 1$ one has $v(N) = v^*(N) = 1$ but $\widetilde{v}(N) = 2$.

Chapter 19: The Shapley value

19.1 Consider the solution concept defined in Example 19.9, $\psi_i(N;v) = v(i)$. We show that it satisfies the following properties.
Additivity: $\psi_i(N;v) + \psi_i(N;w) = v(i) + w(i) = \psi_i(N;v+w)$.
Symmetry: let $i,j \in N$ be two symmetric players of $(N;v)$, hence $v(i) = v(j)$ which leads to $\psi_i(N;v) = \psi_j(N;v)$.
The null player property: let i be a null player of $(N;v)$, hence $\psi_i(N,v) = v(i) = v(i \bigcup \emptyset) = v(\emptyset) = 0$.
Covariance under strategic equivalence: for every $a > 0$ and every $b \in \mathbf{R}^n$, $\psi_i(N; av+b) = (av+b)(i) = av(i) + b_i = a\psi_i(N;v) + b_i$.
However, this solution concept **does not satisfy efficiency**. For instance, for $N = \{1,2\}$, $v(1) = v(2) = 0$ and $v(1,2) = 1$ then $\psi_1(N;v) = \psi_2(N;v) = 0$ but $\psi_1(N;v) + \psi_2(N;v) = 0 \neq 1 = v(N)$.

19.2 Consider the solution concept defined in Example 19.10. We show that it satisfies the following properties.
Efficiency: denote by D the set of dummy players. Then

$$\begin{aligned}\sum_{i\in N} \psi_i(N;v) &= \sum_{i\in N} v(i) + \sum_{i \in N\setminus D} \tfrac{v(N)-\sum_{j\in N} v(j)}{n-d(v)} \\ &= \sum_{i\in N} v(i) + (n-d(v))\tfrac{v(N)-\sum_{j\in N} v(j)}{n-d(v)}) = v(N).\end{aligned}$$

Symmetry: let $i,j \in N$ be two symmetric players of $(N;v)$. We first prove that either both of the players are dummy players or none of them is a dummy player. Assume that player i is a dummy player. Let $S \subseteq N \setminus \{j\}$. If $i \notin S$ then

$$v(S \bigcup \{j\}) = v(S \bigcup \{i\}) = v(S) + v(i) = v(S) + v(j)$$

where the left equality and the right equality hold since i and j are symmetric, and the second equality holds since i is a dummy player. If $i \in S$ then

$$v(S \bigcup \{j\}) = v\left((S \setminus \{i\} \bigcup \{j\}) \bigcup \{i\}\right) = v(S \setminus \{i\} \bigcup \{j\}) + v(i) = v(S) + v(j)$$

where the left equality and the right equality hold since i and j are symmetric, and the second equality holds since i is a dummy player. Hence j is also a dummy player. Thus, since $v(i) = v(j)$ either both i and j are dummy player so $\psi_i(N;v) = v(i) = v(j) = \psi_j(N;v)$, or both i and j are non-dummy so $\psi_i(N;v) = v(i) + \frac{v(N)-\sum_{j\in N} v(j)}{n-d(v)} = v(j) + \frac{v(N)-\sum_{j\in N} v(j)}{n-d(v)} = \psi_j(N;v)$.
The null player property: let i be a null player of $(N;v)$, hence $v(i) = v(i \bigcup \emptyset) = v(\emptyset) = 0$ and for every $S \subseteq N \setminus \{i\}$, one has $v(S \bigcup \{i\}) =$

$v(S) = v(S) + v(i)$ so i is a dummy player which leads to $\psi_i(N, v) = v(i) = 0$.

Covariance under strategic equivalence: Let $a > 0$ and let $b \in \mathbf{R}^n$. We first prove that i is a dummy player of $(N; v)$ if and only if he is a dummy player of $(N; av + b)$. Let i be a dummy player of $(N; v)$. Let $S \subseteq N \setminus \{i\}$. Then

$$\begin{aligned}(av+b)(S \bigcup \{i\}) &= av(S \bigcup \{i\}) + b(S \bigcup \{i\}) = a(v(S) + v(i)) + b(S) + b_i \\ &= (av+b)(S) + (av+b)(i),\end{aligned}$$

so i is a dummy player of $(N; av + b)$. In particular, if i is a dummy player of $(N; av + b)$ then, by the above proof, he is a dummy player of $(N; \frac{1}{a}(av + b) - \frac{b}{a}) = (N; v)$.

Therefore, for every dummy player i, the following holds

$$\psi_i(N; av+b) = (av+b)(i) = av(i) + b_i = a\psi_i(N; v) + b_i.$$

If i is not a dummy player then

$$\begin{aligned}\psi_i(N; av+b) &= (av+b)(i) + \tfrac{(av+b)(N) - \sum_{j\in N}(av(j)+b_j)}{n-d(av+b)} \\ &= (av+b)(i) + \tfrac{(av+b)(N) - \sum_{j\in N}(av(j)+b_j)}{n-d(v)} \\ &= a\left(v(i) + \tfrac{v(N) - \sum_{j\in N} v(j)}{n-d(v)}\right) + b_i = a\psi_i(N; v) + b_i.\end{aligned}$$

19.3 Consider the solution concept defined in Example 19.11. We show that it satisfies the following properties.

Symmetry: let $i, j \in N$ be two symmetric players of $(N; v)$. For every $S \subseteq N \setminus \{i, j\}$, $v(S \bigcup i) - v(S) = v(S \bigcup \{j\}) - v(S)$ and $v(S \bigcup \{i, j\}) - v(S \bigcup \{i\}) = v(S \bigcup \{i, j\} - v(S \bigcup \{j\})$. Hence,

$$\psi_i(N; v) = \max_{S\subseteq N\setminus\{i\}} (v(S \bigcup \{i\}) - v(S)) = \max_{S\subseteq N\setminus\{j\}} (v(S \bigcup \{j\}) - v(S)) = \psi_j(N; v).$$

The null player property: let i be a null player of $(N; v)$, hence for every $S \subseteq N \setminus \{i\}$ one has $v(S \bigcup \{i\}) - v(S) = 0$ so $\psi_i(N, v) = \max_{S\subseteq N\setminus\{i\}}(v(S \bigcup \{i\}) - v(S)) = 0$.

Covariance under strategic equivalence: Let $a > 0$ and let $b \in \mathbf{R}^n$. Then, for every $S \subseteq N \setminus \{i\}$,

$$(av+b)(S \bigcup \{i\}) - (av+b)(S) = a(v(S \bigcup \{i\}) - v(S)) + b_i.$$

Therefore,

$$\begin{aligned}\psi_i(N; av+b) &= \max_{S\subseteq N\setminus\{i\}} ((av+b)(S \bigcup \{j\}) - (av+b)(S)) \\ &= a \max_{S\subseteq N\setminus\{i\}} (v(S \bigcup \{i\}) - v(S)) + b_i = a\psi_i(N; v) + b_i.\end{aligned}$$

19.4 Consider the solution concept defined in Example 19.12, $\psi_i(N;v) = v(1,2,...,i) - v(1,2,...,i-1)$. We show that it satisfies the following properties.
Efficiency: $\sum_{i=1}^n \psi_i(N;v) = \sum_{i=1}^n (v(1,2,...,i) - v(1,2,...,i-1)) = v(N)$.
Additivity:

$$\begin{aligned}
&\psi_i(N;v) + \psi_i(N;w) \\
&= \big((v(1,2,...,i) - v(1,2,...,i-1)\big) + \big(w(1,2,...,i) - w(1,2,...,i-1)\big) \\
&= (v(1,2,...,i) + (w(1,2,...,i)) - (v(1,2,...,i-1) + (w(1,2,...,i-1)) \\
&= \psi_i(N;v+w).
\end{aligned}$$

The null player property: let i be a null player of $(N;v)$, hence $\psi_i(N,v) = v(1,2,...,i) - v(1,2,...,i-1) = 0$.
Covariance under strategic equivalence: for every $a > 0$ and every $b \in \mathbf{R}^n$,

$$\begin{aligned}
\psi_i(N;av+b) &= (av+b)(1,2,...,i) - (av+b)(1,2,...,i-1) \\
&= a(v(1,2,...,i) - v(1,2,...,i-1) + b_i = a\psi_i(N;v) + b_i.
\end{aligned}$$

19.5 Let $(\{1,2\};v)$ be a two player game and let φ be a single-valued solution concept satisfying efficiency, covariance under strategic equivalence, and symmetry.
Set $a = 1$ and $b = (-v(1), -v(2))$. The game $(\{1,2\};av+b)$ is a 0-normalized game where $(av+b)(1,2) = v(1,2) - v(1) - v(2)$. In this game, both players are symmetric, so by the symmetry,

$$\varphi_1(\{1,2\};av+b) = \varphi_2(\{1,2\};av+b).$$

Furthermore, by the efficiency,

$$\varphi_1(\{1,2\};av+b) = \varphi_2(\{1,2\};av+b) = \tfrac{v(1,2)-v(1)-v(2)}{2}.$$

Finally, by the covariance under strategic equivalence,

$$\varphi_i(\{1,2\};av+b) = a\varphi_i(\{1,2\};v) + b_i$$

which leads to

$$\varphi_i(\{1,2\};v) = \varphi_i(\{1,2\};av_b) - b_i = \tfrac{v(1,2)-v(1)-v(2)}{2} + v(i).$$

That is,

$$\begin{aligned}
\varphi_1(\{1,2\};v) &= \tfrac{v(1,2)+v(1)-v(2)}{2} \\
\varphi_2(\{1,2\};v) &= \tfrac{v(1,2)+v(2)-v(1)}{2}.
\end{aligned}$$

19.6 Let ψ^1 and ψ^2 be two solution concepts both satisfying symmetry, efficiency, the null player property, and covariance under strategic equivalence, and let $\lambda \in [0,1]$. Define a solution concept ψ by $\psi := \lambda\psi^1 + (1-\lambda)\psi^2$. We prove that ψ also satisfies symmetry, efficiency, the null player property, and covariance under strategic equivalence.
Symmetry: let $i, j \in N$ be two symmetric players of $(N; v)$. Then, by the symmetry of ψ^1 and ψ^2,

$$\begin{aligned}\psi_i(N;v) &= \lambda\psi_i^1(N;v) + (1-\lambda)\psi_i^2(N;v)\\ &= \lambda\psi_j^1(N;v) + (1-\lambda)\psi_j^2(N;v)\\ &= \psi_j(N;v).\end{aligned}$$

Efficiency: since ψ^1 and ψ^2 satisfy efficiency,

$$\begin{aligned}\sum_{i=1}^n \psi_i(N;v) &= \sum_{i=1}^n \Big(\lambda\psi_i^1(N;v) + (1-\lambda)\psi_i^2(N;v)\Big)\\ &= \lambda\sum_{i=1}^n \psi_i^1(N;v) + (1-\lambda)\sum_{i=1}^n \psi_i^2(N;v)\\ &= \lambda v(N) + (1-\lambda)v(N) = v(N).\end{aligned}$$

The null player property: let $i \in N$ be a null player. Then

$$\psi_i(N;v) = \lambda\psi_i^1(N;v) + (1-\lambda)\psi_i^2(N;v) = \lambda\cdot 0 + (1-\lambda)\cdot 0 = 0.$$

Covariance under strategic equivalence: for every $a > 0$ and every $b \in \mathbf{R}^n$,

$$\begin{aligned}\psi_i(N;av+b) &= \lambda\psi_i^1(N;av+b) + (1-\lambda)\psi_i^2(N;av+b)\\ &= \lambda(a\psi_i^1(N;v) + b_i) + (1-\lambda)(a\psi_i^2(N;v) + b_i)\\ &= a(\lambda\psi_i^1(N;v) + (1-\lambda)\psi_i^2(N;v)) + b_i\\ &= a\psi_i(N;v) + b_i.\end{aligned}$$

19.7 We decompose the given game $(N; v)$ as a linear combination of carrier games

$$u_T(S) = \begin{cases} 1 & T \subseteq S \\ 0 & \textit{otherwise}. \end{cases}$$

for every nonempty coalition $T \subset N$.

That is, we find $\alpha_T \in \mathbf{R}$ for every nonempty coalition $T \subseteq N$, such that $v = \sum_{T \subseteq N} \alpha_T u_T$.
For every player $i \in N$,

$$u_T(i) = \begin{cases} 1 & T = \{i\} \\ 0 & otherwise \end{cases}$$

so $v(i) = \sum_{T \subseteq N} \alpha_T u_T(i) = \alpha_{\{i\}}$.

$$\Rightarrow \alpha_{\{1\}} = 6, \alpha_{\{2\}} = 12, \alpha_{\{3\}} = 0, \alpha_{\{4\}} = 18.$$

For every two players $i, j \in N$,

$$u_T(i,j) = \begin{cases} 1 & T \in \{\{i\}, \{j\}, \{i,j\}\} \\ 0 & otherwise \end{cases}$$

so $v(i,j) = \sum_{T \subseteq N} \alpha_T u_T(i,j) = \alpha_{\{i\}} + \alpha_{\{j\}} + \alpha_{\{i,j\}} = v(i) + v(j) + \alpha_{\{i,j\}}$.
Therefore, $\alpha_{\{i,j\}} = v(i,j) - v(i) - v(j)$.

$$\Rightarrow \alpha_{\{1,2\}} = 6, \alpha_{\{1,3\}} = 42, \alpha_{\{1,4\}} = 36, \alpha_{\{2,3\}} = 0, \alpha_{\{2,4\}} = 2, \alpha_{\{3,4\}} = 20.$$

For every three players $i, j, k \in N$,

$$u_T(i,j,k) = \begin{cases} 1 & T \in \{\{i\}, \{j\}, \{k\}, \{i,j\}, \{i,k\}, \{j,k\}, \{i,j,k\}\} \\ 0 & otherwise \end{cases}$$

so

$$\begin{aligned} v(i,j,k) &= \alpha_{\{i\}} + \alpha_{\{j\}} + \alpha_{\{k\}} + \alpha_{\{i,j\}} + \alpha_{\{i,k\}} + \alpha_{\{j,k\}} + \alpha_{\{i,j,k\}} \\ &= v(i,j) + v(i,k) + v(j,k) - v(i) - v(j) - v(k) + \alpha_{\{i,j,k\}} \\ \Rightarrow \alpha_{\{i,j,k\}} &= v(i,j,k) - v(i,j) - v(i,k) - v(j,k) + v(i) + v(j) + v(k). \end{aligned}$$

$$\Rightarrow \alpha_{\{1,2,3\}} = 54, \alpha_{\{1,2,4\}} = 9, \alpha_{\{1,3,4\}} = 28, \alpha_{\{2,3,4\}} = 127.$$

Finally,

$$v(N) = \sum_{T \subseteq N} \alpha_T u_T(N) = \sum_{T \subseteq N} \alpha_T \Rightarrow \alpha_N = v(N) - \sum_{T \subset N} \alpha_T = -120.$$

19.8 We prove that the Shapley value is a linear solution concept: for every list of K games $\left((N; v_k)\right)_{k=1}^K$, and every list of K real numbers $(\alpha_k)_{k=1}^K$

$$\mathrm{Sh}\left(N; \sum_{k=1}^{k} \alpha_k v_k\right) = \sum_{k=1}^{K} \alpha_k \mathrm{Sh}(N; v_k). \tag{175}$$

We prove that Equality (175) holds for every K, by induction over K. For $K = 2$, Equality (175) holds by the additivity and the covariance under strategic equivalence of the Shapley value (Claim 19.18):

$$\mathrm{Sh}\left(N; \sum_{k=1}^{2} \alpha_k v_k\right) = \sum_{k=1}^{2} \mathrm{Sh}(N; \alpha_k v_k) = \sum_{k=1}^{2} \alpha_k \mathrm{Sh}(N; v_k).$$

Assume that Equality (175) holds for every list of at most K games. Assume that we have a list of $K+1$ games

$$\mathrm{Sh}\left(N; \sum_{k=1}^{k+1} \alpha_k v_k\right) = \mathrm{Sh}\left(N; \sum_{k=1}^{k} \alpha_k v_k\right) + \mathrm{Sh}\left(N; \alpha_{k+1} v_{k+1}\right) \tag{176}$$

$$= \mathrm{Sh}\left(N; \sum_{k=1}^{k} \alpha_k v_k\right) + \alpha_{k+1}\mathrm{Sh}\left(N; v_{k+1}\right) \tag{177}$$

$$= \sum_{k=1}^{K+1} \alpha_k \mathrm{Sh}(N; v_k). \tag{178}$$

where Equality (176) holds by the additivity of the Shapley value, Equality (177) holds by the covariance under strategic equivalence of the Shapley value, and Equality (178) holds by induction hypothesis.

19.9 In Theorem 19.15 no property can be replaced by covariance under strategic equivalence, while maintaining the conclusion of the theorem.
The solution concept defined in Example 19.9 satisfies additivity, symmetry, the null player property, and covariance under strategic equivalence but it does not satisfy efficiency and thus it does not define the Shapley value.
The solution concept defined in Example 19.10 satisfies efficiency, symmetry, covariance under strategic equivalence, and the null player property but it does not satisfy additivity and thus it does not define the Shapley value.
The solution concept defined in Example 19.12 satisfies efficiency, additivity, the null player property, and covariance under strategic equivalence but it does not satisfy symmetry and thus it does not define the Shapley value.
Finally, the solution concept $\psi(N, v) = \frac{v(N)}{n}$ satisfies efficiency, symmetry, additivity and covariance under strategic equivalence but it does not satisfy the null player property and thus it does not define the Shapley value.

19.10 A set of coalitional functions over the set of players N that form a basis for the space of games is the set $\{w_T, \emptyset \subset T \subseteq N\}$, where

$w_T(S) = 1$ if and only if $T = S$. However, as opposed to the $2^n - 1$ carrier games, in the suggested games, no player is a null player and thus this basis was not used in the proof of Theorem 19.15.

19.11 We show that the solution concept φ satisfies anonymity.
Let $(N; v)$ be a coalitional game, let π be a permutation of the set N, and let $i \in N$ be a player. Then,

$$\begin{aligned}
&Sh_{\pi(i)}(N; \pi(v)) \\
&= \sum_{S \subseteq N \setminus \pi(i)} \frac{|S|! \times (n - |S| - 1)!}{n} \left(\pi(v)(S \bigcup \{\pi(i)\}) - \pi(v)(S) \right) \\
&= \sum_{S \subseteq N \setminus \pi(i)} \frac{|S|! \times (n - |S| - 1)!}{n} \left(v(\pi^{-1}(S)) \bigcup \pi^{-1}(\pi(\{i\})) - v(\pi^{-1}(S)) \right) \\
&= \sum_{S \subseteq N \setminus i} \frac{|S|! \times (n - |S| - 1)!}{n} \left(v(S \bigcup \{i\}) - v(S) \right) \\
&= Sh_i(N; v).
\end{aligned}$$

19.12 Consider a family of coalitional games $\mathcal{F}$ such that $(N, v) \in \mathcal{F}$ if and only if $v(N) = \sum_{i=1}^{n} v(i)$. $\mathcal{F}$ is additively closed. Furthermore, it is also closed under strategic equivalence. Indeed, for every $a > 0$ and every $b \in \mathbf{R}^n$, if $v(N) = \sum_{i=1}^{n} v(i)$ then $(av + b)(N) = av(N) + b(N) = a(\sum_{i=1}^{n} v(i)) + b(N) = \sum_{i=1}^{n}(av + b)(i)$.
Let $\psi_i(N; v) = v(i)$ be a single-valued solution concept defined over $\mathcal{F}$. By Exercise 19.1, ψ satisfies additivity, symmetry, the null player property and covariance under strategic equivalence (note that the properties of additivity and covariance under strategic equivalence are well defined since $\mathcal{F}$ is closed under additivity and under strategic equivalence, respectively). Moreover, ψ also satisfies efficiency since $\sum_{i=1}^{n} \psi_i(N; v) = \sum_{i=1}^{n} v(i) = v(N)$.
However, this solution concept is not the Shapley value. For instance, for $N = \{1, 2, 3\}$, $v(1) = v(2) = 0$ and $v(S) = 1$ for every $S \neq \{1\}, \{2\}$ then $\psi_i(N; v) = (0, 0, 1)$ but $\mathrm{Sh}(N; v) = (\frac{1}{6}, \frac{1}{6}, \frac{2}{3})$. This exercise does not contradict Theorem 19.15, since Theorem 19.5 claims that the value is the only solution concept satisfying the four properties proposed by Shapley for the class of all coalitional games whereas the proposed solution concept satisfies the four properties for only subclass of coalitional games.

19.13 In an additive game $(N; v)$, the marginal contribution $v(S \bigcup \{i\}) - v(S) = v(i)$ of every player i and every coalition $S \in N \setminus \{i\}$. Therefore, the Shapley value of the game is $\mathrm{Sh}_i(N; v) = v(i)$, for every

$i \in N$.

19.14 Let $a \in \mathbf{R}^N$ be a vector. Let $(N;v)$ be a coalitional game defined by $v(S) = \left(\sum_{i \in S} a_i\right)^2$ for every $S \subseteq N$. The marginal conribution of player $i \in N$ to coalition $S \subseteq N \setminus \{i\}$ is

$$\begin{aligned} v(S \bigcup \{i\}) - v(S) &= \left(\sum_{j \in S} a_j + a_i\right)^2 - \left(\sum_{j \in S} a_j\right)^2 \\ &= \left(\sum_{j \in S} a_j\right)^2 + 2a_i \sum_{j \in S} a_j + a_i^2 - \left(\sum_{j \in S} a_j\right)^2 \\ &= a_i^2 + 2a_i \sum_{j \in S} a_j \end{aligned}$$

Note that, in half of the permutations player j precedes player i so $\sum_{\substack{S \subseteq N \setminus \{i\} \\ j \in S}} \frac{|S|!(n-|S|-1)!}{n!} = \frac{1}{2}$ Therefore,

$$\begin{aligned} \mathrm{Sh}(N;v) &= \sum_{S \subseteq N \setminus \{i\}} \tfrac{|S|!(n-|S|-1)!}{n!} (v(S \bigcup \{i\}) - v(S)) \\ &= \sum_{S \subseteq N \setminus \{i\}} \tfrac{|S|!(n-|S|-1)!}{n!} \left(a_i^2 + 2a_i \sum_{j \in S} a_j\right) \\ &= a_i^2 \sum_{S \subseteq N \setminus \{i\}} \tfrac{|S|!(n-|S|-1)!}{n!} + 2a_i \sum_{j \in N \setminus \{i\}} a_j \sum_{\substack{S \subseteq N \setminus \{i\} \\ j \in S}} \tfrac{|S|!(n-|S|-1)!}{n!} \\ &= a_i^2 + a_i \sum_{j \in N \setminus \{i\}} a_j \tfrac{1}{2} = a_i \sum_{j \in N} a_j = a_i \sqrt{v(N)}. \end{aligned}$$

19.15 Let $(N;v)$ be a coalitional game. For every stage k of the algorithm (including Parts (a) and (b)), denote by T_k the nonempty coalition that is chosen at stage k, and by v_k the new coalitional function after stage k is completed. For simplicity, set $v_0 = v$. Denote by K the last stage of the algorithm, if the algorithm terminates then $v_K = 0$, otherwise $K = \infty$.

(a) We first prove that the algorithm always terminates. Assume by contradiction that the algorithm never terminates. Since there are only finitely many coalitions, there must be at least one nonempty coalition that is chosen infinitely many times. Let T be a minimal such a coalition. That is, there are infinitely many stages $k_1, k_2, ...$ such that $T_{k_l} = T$ for every $l \in \mathbf{N}$, and every coalition $S \subset T$ is chosen at most finitely many times. Hence, for every $l \in \mathbf{N}$ one has $v_{k_l}(T) = 0$ and $v_{k_{l+1}-1}(T) \neq 0$ which leads to an existence of a coalition $S \subset T$ such that $T_k = S$ for some $k \in \{k_l + 1, ..., k_{l+1} - 1\}$. However, by the finitely number of coalitions at least one S is cosen infinitely many times, which contradicts the minimality of T.

(b) For every $k \in \{1, 2, ..., K\}$ the new coalitional function satisfies $v_k = v_{k-1} - v_{k-1}(T_k)u_{T_k}$ where u_{T_k} is the carrier game over T_k defined by

$$u_{T_k}(S) = \begin{cases} 1 & T_k \subseteq S \\ 0 & otherwise. \end{cases}$$

By induction, since $v_K = 0$ and $v = v_0$, one can deduce that v is a linear combination of carrier games,

$$v = \sum_{k=1}^{K} v_{k-1}(T_k)u_{T_k}.$$

However, by Theorem 19.21, $\mathrm{Sh}(N; v_{k-1}(T_k)u_{T_k}) = \frac{v_{k-1}(T_k)}{|T_k|}$ which is the dividend in Part (a). Finally, by the additivity the Shapley value one has

$$\mathrm{Sh}(N; v) = \mathrm{Sh}(N; \sum_{k=1}^{K} v_{k-1}(T_k)u_{T_k}) = \sum_{k=1}^{K} \mathrm{Sh}(N; v_{k-1}(T_k)u_{T_k})$$

which is the summation of all of the dividends.

19.16

1	**2**	**3**	**4**	**1,2**	**1,3**	**1,4**	**2,3**	**2,4**	**3,4**	**1,2,3**	**1,2,4**	**1,3,4**	**2,3,4**	**N**	**Coalition**	**1**	**2**	**3**	**4**
6	12	0	18	24	48	60	12	32	38	120	89	150	179	240	1	6	0	0	0
0	12	0	18	18	42	54	12	32	38	114	83	144	179	234	2	0	12	0	0
0	0	0	18	6	42	54	0	20	38	102	71	144	167	222	4	0	0	0	18
0	0	0	0	6	42	36	0	2	20	102	53	126	149	204	1,2	3	3	0	0
0	0	0	0	0	42	36	0	2	20	96	47	126	149	198	1,3	21	0	21	0
0	0	0	0	0	0	36	0	2	20	54	47	84	149	156	1,4	18	0	0	18
0	0	0	0	0	0	0	0	2	20	54	11	48	149	120	2,4	0	1	0	1
0	0	0	0	0	0	0	0	0	20	54	9	48	147	118	3,4	0	0	10	10
0	0	0	0	0	0	0	0	0	0	54	9	28	127	98	1,2,3	18	18	18	0
0	0	0	0	0	0	0	0	0	0	0	9	28	127	44	1,2,4	3	3	0	3
0	0	0	0	0	0	0	0	0	0	0	0	28	127	35	1,3,4	$9\frac{1}{3}$	0	$9\frac{1}{3}$	$9\frac{1}{3}$
0	0	0	0	0	0	0	0	0	0	0	0	0	127	7	2,3,4	0	$42\frac{1}{3}$	$42\frac{1}{3}$	$42\frac{1}{3}$
0	0	0	0	0	0	0	0	0	0	0	0	0	0	-120	N	-30	-30	-30	-30
																$48\frac{1}{3}$	$49\frac{1}{3}$	$70\frac{2}{3}$	$71\frac{2}{3}$

Therefore, $\mathrm{Sh}(N;v) = (48\frac{1}{3}, 49\frac{1}{3}, 70\frac{2}{3}, 71\frac{2}{3})$.

19.17 For every game $(N;v)$, define the dual game $(N;v^*)$ as follows: $v^*(S) = v(N) - v(N \setminus S)$.

(a) Assume that $(N;v^*)$ is the dual game to $(N;v)$. Then

$$\begin{aligned} v^*(N) - v^*(N \setminus S) &= (v(N) - v(N \setminus N)) - (v(N) - v(N \setminus (N \setminus S))) \\ &= (v(N) - v(\emptyset)) - (v(N) - v(S)) = v(S) \end{aligned}$$

so $(N;v)$ is the dual game to $(N;v^*)$.

(b) We show that $\mathrm{Sh}(N;v^*) = \mathrm{Sh}(N;v)$. For every player $i \in N$,

$$\begin{aligned} \mathrm{Sh}_i(N;v^*) &= \sum_{S \subseteq N \setminus \{i\}} \tfrac{|S|!(n-|S|-1)!}{n!} (v^*(S \bigcup \{i\}) - v^*(S)) \\ &= \sum_{S \subseteq N \setminus \{i\}} \tfrac{|S|!(n-|S|-1)!}{n!} (v(N \setminus S) - v(N \setminus (S \bigcup \{i\}))). \end{aligned}$$

Set $T = N \setminus (S \bigcup \{i\})$. Then $|T| = n - |S| - 1$ and $n - |T| - 1 = |S|$. Furthermore, there is bijection according to which each $S \subset N \setminus \{i\}$ correspond to $T = N \setminus (S \bigcup \{i\}) \subseteq N \setminus \{i\}$. In conclusion,

$$\begin{aligned} \mathrm{Sh}_i(N;v^*) &= \sum_{S \subseteq N \setminus \{i\}} \tfrac{|S|!(n-|S|-1)!}{n!} (v(N \setminus S) - v(N \setminus (S \bigcup \{i\}))) \\ &= \sum_{T \subseteq N \setminus \{i\}} \tfrac{|T|!(n-|T|-1)!}{n!} (v(T \bigcup \{i\}) - v(T)) = \mathrm{Sh}_i(N;v). \end{aligned}$$

19.18 Define a cost game $(N;c)$, where N is the set of all planes landing at an airport, and $c(S)$, for each coalition S, is the maintenance cost of the shortest runway that can accommodate all the planes in the coalition.

Divide the runway into K disjoint segments $L = \{l_1, l_2, ..., l_K\}$. Each plane requires the entire length of the runway up to (and including) some segment in L. For every segment l_k, denote by S_k the set of planes using that segment.

For every $k \in \{1, 2, ..., K\}$ define a cost game $(N;c_k)$ where $c_k(S)$ is the cost of the runway segment l_k if $S \cap S_k \neq \emptyset$, otherwise $c_k(S) = 0$. Every plane $i \in N \setminus S_k$ is a zero player in $(N;c_k)$ so $\mathrm{Sh}_i(N;c_k) = 0$. All of the planes in S_k are symmetric players so $\mathrm{Sh}_i(N;c_k) = \frac{c_k(S_k)}{|S_k|}$ for every $i \in S_k$.

The cost $c(S)$ which is the maintenance cost of the shortest runway that can accommodate all the planes in the coalition, satisfies

$c(S) = \sum_{k=1}^{K} c_k(S)$. Hence, by the additivity of the Shpley value, $\mathrm{Sh}_i(N;c) = \sum_{k=1}^{K} \mathrm{Sh}_i(N;c_k)$, according to which the maintenance cost of each runway segment is borne equally by the planes using that segment.

19.19 Consider maintenance costs associated with a road network connecting villages to a central township. The network is depicted as a tree, with the central township at the root of the tree. Each village is associated with a node of the tree, and there are additional nodes of the tree that represent road intersections. The villages vary in their numbers of inhabitants.

A cost game $(N;c)$ is derived from the network, where N is the set of residents in all the villages, and for each coalition $S \subseteq N$, $c(S)$ is the maintenance cost of the minimal subtree required to maintain the network of roads connecting all the members in S to the central township.

Let $E = \{e_1, e_2, ..., e_L\}$ be the segments of road connecting two intersections of that network. For every $e_l \in E$, let c_l be maintenance cost of e_l, and let $S_l \subseteq N$ be the set of residents using that segment.

For every segment of road $e_l \in E$, define a cost game $(N;c_l)$ as follows

$$c_l(S) = \begin{cases} c_l & S \cap S_l \neq \emptyset \\ 0 & \textit{otherwise.} \end{cases}$$

Every $i \in N \setminus S_l$ is a zero player and all of the players in S_l are symmetric, hence

$$\mathrm{Sh}_i(N, c_l) = \begin{cases} \frac{c_l}{|S_l|} & i \in S_l \\ 0 & \textit{otherwise.} \end{cases}$$

However, since $(N;c) = \sum_{e_l \in E}(N;c_l)$ it follows, by the additivity of the Shapley value that the maintenance cost of each road segment is borne equally by all the people using that segment.

19.20 Let $(N;v)$ and $(N;w)$ are two weighted majority games.

The game $(N; v \vee w)$ is not necessarily a weighted majority game. Consider the two majority games $(N;v) = [2.5;1,1,1,0,0,0]$ and $(N;w) = [2.5;0,0,0,1,1,1]$. Assume by contradiction that $(N; v \vee w)$ is a weighted majority $[q; w_1, ..., w_6]$. Since $(v \vee w)(1,2,3) = 1$, $w_1 + w_2 + w_3 \geq q$ which leads to the existence of two players $i, j \in \{1,2,3\}$ such that $w_i + w_j \geq \frac{2}{3}q$. Since $(v \vee w)(4,5,6) = 1$, $w_4 + w_5 + w_6 \geq q$ which leads to the existence of a player $k \in \{4,5,6\}$ such that $w_k \geq \frac{1}{3}q$. In particular, $w_i + w_j + w_k \geq q$ but $(v \vee w)(i,j,k) = 0$, a contradiction.

The game $(N; v \wedge w)$ is not a necessarily weighted majority game. Consider the two majority games $(N; v) = [10; 3, 3, 3, 4, 4, 4, 4]$ and $(N; w) = [10; 4, 4, 4, 4, 3, 3, 3]$. Assume by contradiction that $(N; v \wedge w)$ is a weighted majority $[q; w_1, ..., w_7]$. Since $(v \vee w)(1, 2, 3) = 0$, $w_1 + w_2 + w_3 < q$ which leads to the existence of two players $i, j \in \{1, 2, 3\}$ such that $w_i + w_j < \frac{2}{3}q$. Since $(v \wedge w)(5, 6, 7) = 0$, $w_5 + w_6 + w_7 < q$ which leads to the existence of a player $k \in \{5, 6, 7\}$ such that $w_k < \frac{1}{3}q$. In particular, $w_i + w_j + w_k < q$ but $(v \wedge w)(i, j, k) = 1$, a contradiction.

19.21 Let $(N; v)$ and $(N; w)$ be two coalitional games over the same set of players. Then $\text{Sh}_i(N; v \vee w) \geq \text{Sh}_i(N; v)$ does not necessarily hold for every player $i \in N$.
For instance, let $(\{1, 2\}, v) = [1; 1, 0]$ and $(\{1, 2\}, v) = [1; 0, 1]$ be two majority games. By the null player property and the efficiency of the Shapley value, $\text{Sh}_1(N; v) = 1$. However, $(\{1, 2\}, v \vee w) = [1; 1, 1]$ and by the symmetry and the efficiency of the Shapley, value $\text{Sh}_1(N; v \vee w) = \frac{1}{2} < \text{Sh}_1(N; v)$.

19.22 Let Σ_N be the set of simple monotonic games over a set of players N.

(a) Let $(N; v), (N; w) \in \Sigma_N$. Since, $(N; v)$ and $(N; w)$ are simple games, $v(S), w(S) \in \{0, 1\}$ for every coalition $S \subseteq N$. Thus,

$$(v \vee w)(S) = \max\{v(S), w(S)\} \in \{0, 1\} \text{ and}$$
$$(v \wedge w)(S) = \min\{v(S), w(S)\} \in \{0, 1\}$$

for every coalition $S \subseteq N$. Therefore, $(N; v \vee w)$ and $(N; v \wedge w)$ are also simple games.
By the monotonicity of $(N; v)$ and $(N; w)$, for every two coalitions $S \subseteq T$ one has

$$(v \vee w)(S) = \max\{v(S), w(S)\} \leq \max\{v(T), w(T)\} = (v \vee w)(T)$$
$$(v \wedge w)(S) = \min\{v(S), w(S)\} \leq \min\{v(T), w(T)\} = (v \wedge w)(T),$$

so $(N; v \vee w)$ and $(N; v \wedge w)$ are also monotonic games. In particular, $(N; v \vee w)$ and $(N; v \wedge w)$ are games in Σ_N.

(b) We prove that the Shapley value is the unique solution concept over Σ_N satisfying efficiency, symmetry, the null player property, and the valuation property.
By Claim 19.18, the Shapley value satisfies efficiency, symmetry, the null player property. We prove that it also satisfies the valuation property. For every two real numbers $a, b \in \mathbf{R}$ one has $a + b = \min\{a, b\} + \max\{a, b\}$ so $v + w = v \vee w + v \wedge w$, thus,

by the additivity of the Shapley value

$$\begin{aligned}\mathrm{Sh}(N,v)+\mathrm{Sh}(N,w) &= \mathrm{Sh}(N,v+w)=\mathrm{Sh}(N,v\vee w+v\wedge w)\\ &= \mathrm{Sh}(N,v\vee w)+\mathrm{Sh}(N,v\wedge w).\end{aligned}$$

We next prove the uniqueness of the Shapley value. Let ψ be a solution concept over Σ_N which satisfies the four properties. We will show that for every game $(N;v)\in\Sigma_N$ one has $\mathrm{Sh}(N;v)=\psi(N;v)$ by induction over the number of minimal winning coalition of $(N;v)$. If $(N;v)$ has only one minimal winning coalition S, then every player $i\notin S$ is a zero player, and all of the players in S are symmetric. Therefore, by properties of efficiency, symmetry and the null player property,

$$\mathrm{Sh}_i(N;v)=\psi_i(N;v)=\begin{cases}\frac{v(N)}{|S|} & i\in S\\ 0 & \textit{otherwise.}\end{cases}$$

Assume that for every game $(N;v)\in\Sigma_N$ with at most m minimal winning coalitions one has $\mathrm{Sh}_i(N;v)=\psi_i(N;v)$. Let $(N;v)\in\Sigma_N$ with $m+1$ minimal winning coalitions $\{S_1,S_2,...,S_{m+1}\}$. Denote by $(N;v^{m+1})$ the simple monotonic game define as follows

$$v^{m+1}(S)=\begin{cases}1 & S_{m+1}\subseteq S\\ 0 & \textit{otherwise.}\end{cases}$$

Denote by $(N;v^*)$ the simple monotonic game define as follows

$$v^*(S)=\begin{cases}1 & \exists k\in\{1,2,...,k\};S_k\subseteq S\\ 0 & \textit{otherwise.}\end{cases}$$

Then, $(N;v^{m+1})$ and $(N;v^*)$ and $(N;v^{m+1}\wedge v^*)$ are simple monotonic games with at most m minimal coalition, hence, by the induction hypothesis, $\mathrm{Sh}(N;v^{m+1})=\psi(N;v^{m+1})$, $\mathrm{Sh}(N;v^*)=\psi(N;v^*)$ and $\mathrm{Sh}(N;v^{m+1}\wedge v^*)=\psi(N;v^{m+1}\wedge v^*)$. Finally, since $v^{m+1}\vee v^*=v$ and by the valuation property, it follows that

$$\begin{aligned}\mathrm{Sh}(N;v) &= \mathrm{Sh}(N;v^{m+1}\vee v^*)\\ &= \mathrm{Sh}(N;v^{m+1})+\mathrm{Sh}(N;v^*)-\mathrm{Sh}(N;v^{m+1}\wedge v^*)\\ &= \psi(N;v^{m+1})+\psi(N;v^*)-\psi(N;v^{m+1}\wedge v^*)\\ &= \psi(N;v^{m+1}\vee v^*)=\psi(N;v),\end{aligned}$$

as needed.

19.23 Let $(a_i)_{i\in N}$ be real numbers. Let v be the following coalitional function:

$$v(S) = \begin{cases} 0 & |S| \leq k \\ \sum_{i\in S} a_i & |S| > k. \end{cases}$$

If $k = n$ then $(N; v)$ is a zero game, where all of the player are zero player so $\mathrm{Sh}(N; v) = 0$.
Assume, $k \in \{, 1, ..., n-1\}$. Let $i \in N$ and let $S \subseteq N \setminus \{i\}$. Then

$$v(S \bigcup \{i\}) - v(S) = \begin{cases} 0 & |S| < k \\ \sum_{j\in S} a_j + a_i & |S| = k \\ a_i & |S| > k. \end{cases}$$

In particular, the value a_i corresponds to every permutation where i is one of the last $n-k$ players (there are $(n-k)(n-1)!$ such permutations). Similarly, the value a_j of player $j \neq i$ corresponds to every permutation where i is the $k+1$-th player and player j is one of the first k players (there are $k(n-2)!$ such permutations). Therefore,

$$\begin{aligned} \mathrm{Sh}_i(N, v) &= \tfrac{(n-k)(n-1)!}{n!} a_i + \sum_{j\neq i} \tfrac{k(n-2)!}{n!} a_j \\ &= \tfrac{n-k}{n} a_i + \tfrac{k}{n(n-1)} \sum_{j\neq i} a_j. \end{aligned}$$

19.24 Consider a weighted majority game with four players, with quota $q = \frac{1}{2}$, and weights $(0.1, 0.2, 0.3, 0.4)$.

(a) The coalitional game (N, v) corresponding to this weighted majority game is defined as follows

$$v(S) = \begin{cases} 0 & |S| = 1 \text{ or } S \in \{\{1,2\}, \{1,3\}\} \\ 1 & |S| \geq 3 \text{ or } S \in \{\{1,4\}, \{2,3\}, \{2,4\}, \{3,4\}\}. \end{cases}$$

(b) The marginal contribution of Player 1 is 1 in case he joins to coalition $\{4\}$. Hence $\mathrm{Sh}_1(N; v) = \frac{2!}{4!} \cdot 1 = \frac{1}{12}$.
The marginal contribution of Player 2 is 1 in case he joins to coalitions $\{3\}, \{4\}$ and $\{1,3\}$. Hence $\mathrm{Sh}_2(N; v) = 3 \cdot \frac{2!}{4!} \cdot 1 = \frac{1}{4}$.
The marginal contribution of Player 3 is 1 in case he joins to coalition $\{2\}, \{4\}$ and $\{1,2\}$. Hence $\mathrm{Sh}_3(N; v) = 3 \cdot \frac{2!}{4!} \cdot 1 = \frac{1}{4}$.
The marginal contribution of Player 4 is 1 in case he joins to coalition $\{1\}, \{2\}, \{3\}, \{1,2\}$ and $\{1,3\}$. Hence $\mathrm{Sh}_4(N; v) = 5 \cdot \frac{2!}{4!} \cdot 1 = \frac{5}{12}$.

(c) The core of the game is empty whereas this is a simple game with no veto player.

19.25 Consider a weighted majority game $\left[\frac{N}{2}; \frac{N}{3}, 1, 1, 1, ..., 1\right]$ with $N+1$ players. Suppose that N is divisible by 6. The player with weight $\frac{N}{3}$ turns a losing coalition into a winning coalition if he joins a coalition of k players where $\frac{N}{6} \le k < \frac{N}{2}$. Thus the Shapley value of this player is

$$\frac{N!\frac{N}{3}}{(N+1)!} = \frac{1}{3} \cdot \frac{N}{N+1}$$

which converges to $\frac{1}{3}$ as N goes to infinity. All other players are symmetric. By the efficiency and symmetric properties, the Shapley value of each other player is

$$\frac{1}{N}\left(1 - \frac{1}{3} \cdot \frac{N}{N+1}\right) = \frac{2N+3}{3N(N+1)}.$$

19.26 Consider a weighted majority game $\left[\frac{5N}{6}; \frac{N}{3}, \frac{N}{3}, 1, 1, 1, ..., 1\right]$ with $N+2$ players. Suppose that N is divisible by 6. Each player with weight 1 turns a losing coalition into a winning coalition if one of the following holds: (1) He is the $\frac{5N}{6}$-th player joins the coalition where all previous players are with weight 1; or (2) He is the $\left(\frac{N}{2}+1\right)$-th player joins the coalition where only one player with weight $\frac{N}{3}$ joined previously; (3) He is the $\left(\frac{N}{6}+2\right)$-th player joins the coalition where both players with weight $\frac{N}{3}$ joined previously.
Thus the Shapley value of this player is

$$\frac{(N-1)!\left(\left(N+2-\frac{5N}{6}\right)\left(N+2-\frac{5N}{6}-1\right)+2\frac{N}{2}\left(N+2-\frac{N}{2}-1\right)+\frac{N}{6}\left(\frac{N}{6}+1\right)\right)}{(N+2)!}$$
$$= \frac{5N^2+15N+18}{9(N+2)(N+1)N}$$

The other players are symmetric. By the efficiency and symmetric properties, the Shapley value of each other player is

$$\frac{1}{2}\left(1 - N \cdot \frac{5N^2+15N+18}{9(N+2)(N+1)N}\right) = \frac{2N(N+3)}{9(N+2)(N+1)}.$$

which converges to $\frac{2}{9}$ as N goes to infinity.

19.27 Consider a weighted majority game $\left[N; \frac{N}{3}, \frac{N}{3}, 1, 1, 1, ..., 1\right]$ with $N+2$ players. Suppose that N is divisible by 6. Each player with weight 1 turns a losing coalition into a winning coalition if one of the following holds: (1) He is the N-th player joins the coalition where all previous players are with weight 1; or (2) He is the $\left(\frac{2N}{3}+1\right)$-th player joins the coalition where only one player with weight $\frac{N}{3}$ joined previously; (3) He is the $\left(\frac{N}{3}+2\right)$-th player joins the coalition where both players with weight $\frac{N}{3}$ joined previously.
Thus the Shapley value of this player is

$$\frac{(N-1)!\left((N+2-N)(N+2-N-1)+2\frac{2N}{3}\left(N+2-\frac{2N}{3}-1\right)+\frac{N}{3}\left(\frac{N}{3}+1\right)\right)}{(N+2)!}$$
$$= \frac{5N^2+15N+18}{9(N+2)(N+1)N}$$

The other players are symmetric. By the efficiency and symmetric properties, the Shapley value of each other player is

$$\frac{1}{2}\left(1 - N \cdot \frac{5N^2+15N+18}{9(N+2)(N+1)N}\right) = \frac{2N(N+3)}{9(N+2)(N+1)}.$$

which converges to $\frac{2}{9}$ as N goes to infinity.

19.28 (a) We prove that for every nonempty set of players U, there is a unique potential function $P : \Gamma_U^* \to \mathbf{R}$.
Every potential function P satisfies

$$\begin{aligned} v(N) &= \sum_{i\in N} D_i P(N;v) \\ &= \sum_{i\in N} \left(P(N;v) - P(N\setminus\{i\};v)\right) \\ &= |N|P(N;v) - \sum_{i\in N} P(N\setminus\{i\};v), \end{aligned}$$

for every $|N| \geq 2$. Hence, every potential function P can be constructed recursively for every $|N| \geq 2$ as follows

$$P(N;v) = \frac{1}{|N|}\left(v(N) + \sum_{i\in N} P(N\setminus\{i\};v)\right).$$

However, for $N = \{i\}$ a potential function must satisfies

$$v(i) = D_i P(N;v) = P(N;v)$$

which leads to a unique potential function defined recursively

$$P(N;v) = \begin{cases} v(i) & N = \{i\} \\ \frac{1}{|N|}\left(v(N) + \sum_{i\in N} P(N\setminus\{i\};v)\right) & |N| \geq 2. \end{cases}$$

(b) Let P be a potential function. We prove that $D_iP(N;v) = Sh_i(N;v)$ for every $N \subseteq U$ and every $i \in N$.

By the definition of a potential function, $D_iP(N;v)$ satisfies the efficiency property and therefore the claim holds for $|N| = 1$.

We next prove by induction over $|N|$ that for every $|N| \geq 2$, $D_iP(N;v)$ also satisfies the null player property, symmetry and additivity which leads, by Theorem 19.15, $D_iP(N;v) = Sh_i(N;v)$. Assume that $D_iP(N;v)$ satisfies all four properties whenever $1 \leq |N| < n$. Let $|N| = n$.

The null player property: $D_iP(N;v)$ satisfies the null player property if and only if $P(N;v) = P(N \setminus \{i\};v)$, for every null player i of $(N;v)$. Let i be a null player of $(N;v)$, so $v(N) = v(N \setminus \{i\})$ and i is a null player in every game $(N \setminus \{j\};v)$ where $j \neq i$. By the induction hypothesis

$$\begin{aligned} P(N;v) &= \tfrac{1}{|N|}\left(v(N) + \sum_{j\neq i} P(N \setminus \{j\};v) + P(N \setminus \{i\};v)\right) \\ &= \tfrac{1}{|N|}\left(v(N \setminus \{i\}) + \sum_{j\neq i} P(N \setminus \{i,j\};v) + P(N \setminus \{i\};v)\right) \\ &= \tfrac{|N|-1}{|N|}P(N \setminus \{i\};v) + \tfrac{1}{|N|}P(N \setminus \{i\};v) = P(N \setminus \{i\};v). \end{aligned}$$

where the last equality holds by Part (a) either in case that $|N| = 2$ or in case that $|N| > 2$.

Symmetry: $D_iP(N;v)$ satisfies symmetry if and only if $P(N \setminus \{i\};v) = P(N \setminus \{j\};v)$, for two symmetric players i,j of $(N;v)$. Let i,j be symmetric players of $(N;v)$, so $v(N \setminus \{i\}) = v(N \setminus \{j\})$. If $|N| = 2$ then $N = \{i,j\}$ and thus

$$P(N \setminus \{i\};v) = v(j) = v(i) = P(N \setminus \{i\};v).$$

If $|N| > 2$ then

$$\begin{aligned} P(N \setminus \{i\};v) &= \tfrac{1}{|N|}\left(v(N \setminus \{i\}) + \sum_{k\neq i} P(N \setminus \{i,k\};v)\right) \\ &= \tfrac{1}{|N|}\left(v(N \setminus \{j\}) + \sum_{k\neq j} P(N \setminus \{j,k\};v)\right) \\ &= P(N \setminus \{j\};v). \end{aligned}$$

Additivity: $D_i P(N; v)$ satisfies additivity if and only if $P(N; v + w) = P(N; v) + P(N; w)$. By the induction hypothesis

$$\begin{aligned}
P(N; v + w) &= \tfrac{1}{|N|}\left((v + w)(N) + \sum_{i \in N} P(N \setminus \{i\}; v + w)\right) \\
&= \tfrac{1}{|N|}\left(v(N) + \sum_{i \in N} P(N \setminus \{i\}; v)\right) \\
&+ \tfrac{1}{|N|}\left(w(N) + \sum_{i \in N} P(N \setminus \{i\}; w)\right) \\
&= P(N; v) + P(N; w).
\end{aligned}$$

19.29 Let $(N; v)$ be a simple monotonic game satisfying $v(N) = 1$. For each player i, define $B_i(N; v)$ to be the number of coalitions S satisfying $v(S) = 0$ and $v(S \bigcup \{i\}) = 1$. The Banzhaf value of player i is defined to be $BZ_i(N; v) := \frac{B_i(N;v)}{\sum_{j \in N} B_j(N;v)}$.

(a) The Banzhaf value satisfies the following properties:

Efficiency: $\sum_{i \in N} BZ_i(N; v) = \sum_{i \in N} \frac{B_i(N;v)}{\sum_{j \in N} B_j(N;v)} = 1 = v(N)$.

The null player property: every null player turns no losing coalition into a winning coalition, so its Banzhaf value is zero.

Symmetry: let i, j be two symmetric players. Then the number of losing coalitions which player i turns into winning coalitions is identical to the number of losing coalitions which player j turns to winning coalitions. Hence, $B_i(N; v) = B_j(N; v)$ so $BZ_i(N; v) = BZ_j(N; v)$.

Additivity: Banzhaf value does not satisfies additivity since a sum of two simple game is not simple which case the Banzhaf value is not defined.

Marginality: every two simple games which satisfies Equality (19.55) are necessarily identical hence have their Banzhaf values are identical as well.

(b) Consider the game in Exercise 19.24. Player 1 turns the losing coalition $\{4\}$ into a winning coalition.

Player 2 turns the losing coalitions $\{3\}, \{4\}$ and $\{1, 3\}$ into winning coalitions.

Player 3 turns the losing coalitions $\{2\}, \{4\}$ and $\{1, 2\}$ into winning coalitions.

Player 4 turns the losing coalitions $\{1\}, \{2\}, \{3\}, \{1,2\}$ and $\{1,3\}$ into winning coalitions.

Therefore, the Banzhaf values are $BZ_1(N;v) = \frac{1}{12}$,$BZ_2(N;v) = \frac{1}{4}$, $BZ_3(N;v) = \frac{1}{4}$ and $BZ_4(N;v) = \frac{5}{12}$.

(c) The game in Exercises 19.25:

A player with weight 1 turns a losing coalition into a winning coalition if he join a coalition with either $\frac{N}{2} - 1$ players with weight 1, or a coalition including the player with weight $\frac{N}{3}$ and $\frac{N}{6} - 1$ players with weight 1. Thus, for every player $i \in \{2, 3, ..., N+1\}$

$$B_i = \binom{N-1}{\frac{N}{2}-1} + \binom{N-1}{\frac{N}{6}-1}.$$

The player with weight $\frac{N}{3}$ turns a losing coalition into a winning coalition if he joins a coalition of k players where $\frac{N}{6} \leq k < \frac{N}{2}$. Hence

$$B_1(N;v) = \sum_{k=\frac{N}{6}}^{\frac{N}{2}-1} \binom{N}{k}.$$

The Banzhaf value is $BZ_i(N;v) = \frac{B_i(N;v)}{B_1(N;v)+N \cdot B_2(N;v)}$.

The game in Exercise 19.26:

Each player with weight 1 turns a losing coalition into a winning coalition if one of the following holds: (1) He is the $\frac{5N}{6}$-th player joins the coalition where all previous players are with weight 1; or (2) He is the $\left(\frac{N}{2}+1\right)$-th player joins the coalition where only one player with weight $\frac{N}{3}$ joined previously; (3) He is the $\left(\frac{N}{6}+2\right)$-th player joins the coalition where both players with weight $\frac{N}{3}$ joined previously. Thus, for $i = 3, ..., N+2$

$$B_i = \binom{N-1}{\frac{5N}{6}-1} + 2\binom{N-1}{\frac{N}{2}-1} + \binom{N-1}{\frac{N}{6}-1}.$$

A player with weight $\frac{N}{3}$ turns a losing coalition into a winning coalition if one of the following holds (1) he joins a coalition of k players where $\frac{N}{2} \leq k < \frac{5N}{6}$; or (2) he joins a coalition including the other player with weight $\frac{N}{3}$ and k players with weight 1 where $\frac{N}{6} \leq k < \frac{N}{2}$ Hence, for $i = 1, 2$

$$B_i(N;v) = \sum_{k=\frac{N}{2}}^{\frac{5N}{6}-1} \binom{N}{k} + \sum_{k=\frac{N}{6}}^{\frac{N}{2}-1} \binom{N}{k}$$

The Banzhaf value is $BZ_i(N;v) = \frac{B_i(N;v)}{2B_1(N;v)+N\cdot B_3(N;v)}$.

(d) The Banzhaf value of the members of the United Nations Security in its pre-1965 structure:

Denote by P the set of permanent members of the council, and by NP the set of nonpermanent members. Every nonpermanent member $i_{np} \in NP$ turns every losing coalition containing the five permanent members along with one nonpermanent member other than i into a winning coalition. Hence, $B_{i_{np}} = 5$.

Every permanent member $i_p \in P$ turns every losing coalition containing the four permanent members other than i along with at least two nonpermanent member into a winning coalition. Hence, $B_{i_p} = \sum_{k=2}^{6} \binom{6}{k} = 57$.

Therefore, the Banzhaf value of a nonpermanent member is $BZ_{np} = \frac{5}{6\cdot 5+5\cdot 57} = 0.0158$ and the Banzhaf value of a permanent member is $BZ_p = \frac{57}{6\cdot 5+5\cdot 57} = 0.1809$.

The Banzhaf value of the members of the United Nations Security in its post-1965 structure :

Every nonpermanent member $i_{np} \in NP$ turns every losing coalition containing the five permanent members along with three nonpermanent members other than i into a winning coalition. Hence, $B_{i_{np}} = \binom{9}{3} = 84$.

Every permanent member $i_p \in P$ turns every losing coalition containing the four permanent members other than i along with at least four nonpermanent member into a winning coalition. Hence, $B_{i_p} = \sum_{k=4}^{1} 0\binom{6}{k} = 848$.

Therefore, the Banzhaf value of a nonpermanent member is $BZ_{np} = \frac{84}{10\cdot 84+5\cdot 848} = 0.0165$ and the Banzhaf value of a permanent member is $BZ_p = \frac{848}{10\cdot 84+5\cdot 848} = 0.1669$.

19.30 A cost game $(N;c)$ is convex if and only if

$$c(S) + c(T) \geq c(S \bigcup T) + c(S \cap T), \quad \forall S, T \subseteq N,$$

if and only if the the coalitional game $(N;-c)$ is convex. However, by Theorem 18.54, the above holds if and only if

$$-c(S \bigcup \{i\}) + c(S) \leq -c(T \bigcup \{i\}) + c(T), \quad \forall S \subseteq T \subseteq N, i \in N \setminus T$$

if and only if

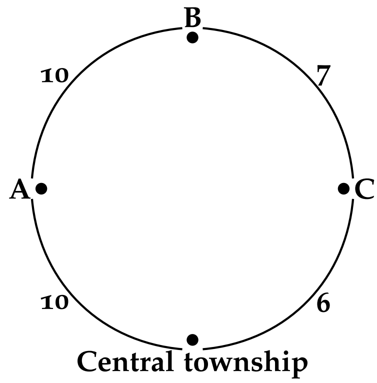
$$c(S \bigcup \{i\}) - c(S) \geq c(T \bigcup \{i\}) - c(T), \quad \forall S \subseteq T \subseteq N, i \in N \setminus T.$$

Consider the airport game in Exercise 19.18. The marginal cost obtained by a plane i joins a coalition S is the maintenance cost of any runway segment that is used by i and not by the members of S. Therefore, if $S \subseteq T$ then every segment that is not used by the members of T is not used by members of S from which it follows that the marginal cost obtained by a plane i joins a coalition S is at least marginal cost obtained by a plane i joins a coalition T. Thus, Inequality 19.148 holds so $(N; c)$ is convex.

19.31 The cost game corresponding the following road network game, is not convex. Let $S = \{A, B\}$ and $T = \{B, C\}$. Then, $c(S) = 20$, $c(T) = 13$, $c(S \bigcup T) = 23$ and $c(S \cap T) = 13$. Hence,

$$c(S) + c(T) < c(S \bigcup T) + c(S \cap T)$$

in contradiction to Inequality 19.147.



19.32 Let i be a null player in a coalitional game $(N; v)$. The Hart–Mas-Colell reduced game over $N \setminus \{i\}$ relative to the Shapley value is $(N \setminus \{i\}; v)$. Indeed, by the null player property of the Shapley value, for every nonempty coalition $R \subseteq N \setminus \{i\}$,

$$\widetilde{v}(R) = v(R \bigcup \{i\}) - \mathrm{Sh}_i(R \bigcup \{i\}; v) = v(R) - 0 = v(R).$$

19.33 Let $(N; v)$ be an additive game and let S be a nonempty coalition. The Hart–Mas-Colell reduced game over S relative to the Shapley value is $(S; v)$. Indeed, by Exercise 19.13, for every nonempty coalition $R \subseteq S$,

$$\widetilde{v}(R) = v(R \bigcup S^c) - \sum_{i \in S^c} \mathrm{Sh}_i(R \bigcup S^c; v) = \sum_{i \in R \bigcup S^c} v(i) - \sum_{i \in S^c} v(i)$$
$$= \sum_{i \in R} v(i) = v(R).$$

19.34 Let $(N; v)$ be a coalitional game, and let $(N; v^*)$ be its dual game. For a nonempty coalition S, the coalition function of the Hart–Mas-Colell reduced game of $(N; v^*)$ over S relative to the Shapley value is

$$\begin{aligned}
\widetilde{v}^*_{S,\mathrm{Sh}}(R) &= v^*(R \bigcup S^c) - \sum_{i \in S^c} \mathrm{Sh}_i(R \bigcup S^c; v^*) \\
&= v(N) - v((R \bigcup S^c)^c) - \sum_{i \in S^c} \mathrm{Sh}_i(R \bigcup S^c; v) \\
&= v(N) - v((R \bigcup S^c)^c) - v(R \bigcup S^c) + v(R \bigcup S^c) - \sum_{i \in S^c} \mathrm{Sh}_i(R \bigcup S^c; v) \\
&= v(N) - v((R \bigcup S^c)^c) - v(R \bigcup S^c) + \widetilde{v}_{S,\mathrm{Sh}}(R)
\end{aligned}$$

where the second equality holds by Exercise 19.17.

19.35 Let φ be a linear solution concept and let $S \subseteq N$ be a nonempty coalition. We prove that the function that assigns to each coalition game its Hart–Mas-Colell reduced game over S relative to φ is a linear function. That is, for every pair of coalitional games $(N; u)$ and $(N; v)$ and for every pair of real numbers α and β,

$$\widetilde{w}_{S,\varphi} = \alpha \widetilde{u}_{S,\varphi} + \beta \widetilde{v}_{S,\varphi},$$

where $w = \alpha u + \beta v$. For every $R \subseteq S$,

$$\begin{aligned}
\widetilde{w}_{S,\varphi}(R) &= w(R \bigcup S^c) - \sum_{i \in S^c} \varphi_i(R \bigcup S^c, w) \\
&= \alpha u(R \bigcup S^c) + \beta v(R \bigcup S^c) - \sum_{i \in S^c} \varphi_i(R \bigcup S^c, \alpha u + \beta v) \\
&= \alpha u(R \bigcup S^c) + \beta v(R \bigcup S^c) - \alpha \sum_{i \in S^c} \varphi_i(R \bigcup S^c, u) - \beta \sum_{i \in S^c} \varphi_i(R \bigcup S^c, v) \\
&= \alpha \left(u(R \bigcup S^c) - \sum_{i \in S^c} \varphi_i(R \bigcup S^c, u) \right) + \beta \left(v(R \bigcup S^c) - \sum_{i \in S^c} \varphi_i(R \bigcup S^c, v) \right) \\
&= \alpha \widetilde{u}_{S,\varphi}(R) + \beta \widetilde{v}_{S,\varphi}(R).
\end{aligned}$$

Chapter 20: The bargaining set

20.1 We prove that the core of the coalitional game $(N; v)$ for the coalitional structure $\mathcal{B} = \{N\}$ is the set of all imputations at which no player has an objection against another player.

Let $x \in \mathcal{C}(N;v)$. Let $C \subset N$ be a nonempty coalition, and let $y \in \mathbf{R}^C$ such that $y(C) = v(C)$. Then, $y(C) = v(C) \leq x(C)$ so there is at least one player $i \in C$ such that $y_i \leq x_i$. Hence, at x no player has an objection against any other player.
Let $x \in X(\{N\};v) \setminus \mathcal{C}(N;v)$. Then, $x(N) = v(N)$, $x_i \leq v(i)$ for every $i \in N$, and there is a nonempty coalition $C \subset N$ such that $x(C) < v(C)$. In particular, set $y_i = x_i + \frac{v(C)-x(C)}{|C|}$, then (C,y) is an objection of every player $k \in C$ against every player $l \in N \setminus C$ at x. That completes the proof.

20.2 We prove that for every coalitional structure $\mathcal{B}$, when the core relative to this coalitional structure is not empty, the core is the set of all imputations in $X(\mathcal{B};v)$ at which no player has an objection against any other player.
The proof that at $x \in \mathcal{C}(N;v;\mathcal{B})$ no player has an objection against any other player is similar to that of Exercise 20.1.
Let $x \in X(\mathcal{B};v) \setminus \mathcal{C}(N;v;\mathcal{B})$. We show that there is a player that has an objection against some other player.
Since $x \in X(\mathcal{B};v) \setminus \mathcal{C}(N;v;\mathcal{B})$, $x(B) = v(B)$ for every $B \in \mathcal{B}$, $x_i \leq v(i)$ for every $i \in N$, and there is a nonempty coalition $C \subseteq N$ such that $x(C) < v(C)$.
If, by contradiction, $C = \bigcup_{k=1}^{K} B_k$ where $\{B_1, ..., B_K\} \subseteq \mathcal{B}$ then

$$v\left(\bigcup_{k=1}^{K} B_k\right) > X\left(\bigcup_{k=1}^{K} B_k\right) = \sum_{k=1}^{K} v(B_k) \Rightarrow$$

$$v^*(N) \geq v\left(\bigcup_{k=1}^{K} B_k\right) + \sum_{B \in \mathcal{B} \setminus \{B_1,...,B_K\}} v(B) > \sum_{B \in \mathcal{B}} v(B).$$

Hence, by Theorem 18.81, $\mathcal{C}(N;v;\mathcal{B}) = \emptyset$, a contradiction. In conclusion, there are two players $k,l \in B \in \mathcal{B}$ such that $k \in C$ and $l \notin C$ so (C,y) where $y_i = x_i + \frac{v(C)-x(C)}{|C|}$, this an objection of every player $k \in C$ against every player $l \in N \setminus C$ at x.

20.3 We show that Theorem 20.6 does not hold without the condition that the core is nonempty. Let $(N;v)$ be a two-player coalitional game with payoff function $v(1) = v(2) = 0, v(1,2) = 1$, and let $\mathcal{B} = \{\{1\},\{2\}\}$.

(a) The core relative to the coalitional structure $\mathcal{B}$ is empty since $X(\mathcal{B};v) = \{x\}$ where $x = (0,0)$ but $x(1,2) = 0 < 1 = v(1,2)$.

(b) The imputation $x = (0,0)$ is not in the core however since there is no objections in a one-player coalition, no player has an objection against any other player which leads $x \in \mathcal{M}(N;v;\mathcal{B})$.

20.4 Assume that player k has an objection (C, y) against player l at x , and player l has an objection (D, z) against player k at x. We show that at least one of these objections is not justified.

If $C \cap D = \emptyset$ then (D, z) is a counterobjection of l against player k at x and thus (C, y) is not justified.

Otherwise, $C \cap D \neq \emptyset$. Assume without loss of generality that $z(C \cap D) \geq y(C \cap D)$. Set

$$\widehat{z}(i) = \begin{cases} y_i & i \in C \cap D \\ z_i & C \setminus (D \bigcup \{l\}) \\ z_i + z(C \cap D) - y(C \cap D) & i = l. \end{cases}$$

In particular,

$$\begin{aligned} \widehat{z}(D) &= \widehat{z}(C \cap D) + \widehat{z}(C \setminus (D \bigcup \{l\})) + \widehat{z}(l) \\ &= y(C \cap D) + z(C \setminus (D \bigcup \{l\})) + z_l + z(C \cap D) - y(C \cap D) \\ &= z(D) = v(D). \end{aligned}$$

In addition,

$$\begin{aligned} \widehat{z}_i &\geq z_i > x_i && \forall i \in D \setminus C, \\ \widehat{z}_i &= y_i && \forall i \in C \cap D. \end{aligned}$$

Therefore, $(D, \widehat{z})$ is a counterobjection of l against player k at x and thus (C, y) is not justified.

20.5 Let $(N; v)$ be a three-player coalitional game. Assume that for an imputation x none of the three equations in (20.16) holds. That is,

$$x_1 + x_3 < v(1,3) \tag{179}$$

$$x_2 > v(2) \tag{180}$$

$$v(2,3) - x_2 < v(1,3) - x_1. \tag{181}$$

Hence, $\max\{x_3, v(2,3) - x_2\} < v(1,3) - x_1$. Select

$$y_3 \in \big(\max\{x_3, v(2,3) - x_2\}, v(1,3) - x_1\big)$$

and set $y_1 = v(1,3) - y_3$. By the definition of y, $y_3 > x_3$ and $y(\{1,3\}) = v(1,3)$, moreover, by Inequality (179), $y_1 > x_1$, so $(\{1,3\}, y)$ is an objection of Player 1 against Player 2. In addition, by Inequality (180) and Inequality (181), Player 2 has no counterobjection neither with a coalition $\{2\}$ nor with a coalition $\{2,3\}$, respectively. In particular the objection $(\{1,3\}, y)$ is justified.

20.6 We prove that the bargaining set is covariant under strategic equivalence. In other words, if $(N;v)$ and $(N;w)$ are two coalitional games with the same set of players, and if there exist $a > 0$ and $b \in \mathbf{R}^N$ such that

$$w(S) = av(S) + b(S), \quad \forall S \subseteq N,$$

then for every coalitional structure $\mathcal{B}$,

$$x \in \mathcal{M}(N;v;\mathcal{B}) \Leftrightarrow ax + b \in \mathcal{M}(N;w;\mathcal{B}).$$

Note that since the strategic equivalence relation is symmetric, it is enough to show that

$$x \in \mathcal{M}(N;v;\mathcal{B}) \Rightarrow ax + b \in \mathcal{M}(N;w;\mathcal{B}).$$

We first prove that

$$x \in X(\mathcal{B};v) \Rightarrow ax + b \in X(\mathcal{B};w).$$

Let $x \in X(\mathcal{B};v)$ then $x(B) = v(B)$ for every $B \in \mathcal{B}$ and $x_i \geq v(i)$ for every $i \in N$. Hence, $ax(B) + b(B) = av(B) + b(B)$ for every $B \in \mathcal{B}$, and $ax_i + b_i \geq av(i) + b_i$ for every $i \in N$ so $ax + b \in X(\mathcal{B};w)$.
Assume next that $x \in \mathcal{M}(N;v;\mathcal{B})$. Then, $x \in X(\mathcal{B};v)$ and if there is a player $k \in B \in \mathcal{B}$ who has an objection (C,y) against another player $l \in B$ at x then player l has a counterobjection (D,z) against k. We show that there is no player in $(N;w)$ who has a justified objection at $ax + b$ so $ax + b \in \mathcal{M}(N;w;\mathcal{B})$.
Assume next that there is a player $k \in B \in \mathcal{B}$ in $(N;w)$ who has has an objection $(C, ay + b)$ against player $l \in B$ at $ax + b$. That is, $k \in C$, $l \notin C$, $ay(C) + b(C) = av(C) + b(C)$ and $ax_i + b_i < ay_i + b_i$ for every $i \in C$. Thus, (C,y) is an objection of k in $(N;v)$ against l at x. Therefore, player l has counterobjection (D,z) against k: $l \in D$, $k \notin D$, $v(D) = z(D)$, $x_i \leq z_i$ for every $i \in D \setminus C$, and $y_i \leq z_i$ for every $i \in D \cap C$. In particular, $l \in D$, $k \notin D$, $av(D) + b(D) = az(D) + b(D)$, $ax_i + b_i \leq az_i + b_i$ for every $i \in D \setminus C$, and $ay_i + b_i \leq az_i + b_i$ for every $i \in D \cap C$. In conclusion, player l has a counterobjection against k in $(N;w)$ so no player has a justified objection against any other player and $ax + b \in \mathcal{M}(N;w;\mathcal{B})$, as needed.

20.7 Let $(N;v)$ be the game in Example 20.1 where $N = \{Anna, Ben, Carl\}$, $v(Anna, Ben) = v(Anna, Carl) = 1{,}000$, $v(Ben, Carl) = 500$, and $v(S) = 0$ for every other coalition.
For every coalitional structure the bargaining set contains a single payoff vector as follows.

$\mathcal{B} = \{\{Anna, Ben\}, \{Carl\}\}$**:** By Therem 20.15, $x \in \mathcal{M}(N; v; \mathcal{B})$ if and only if $x \in X(\mathcal{B}; v)$, i.e.,

$$x_1 + x_2 = 1000, \quad x_1, x_2 \geq 0, \text{ and } \quad x_3 = 0, \tag{182}$$

and at least one expression in each line of the table (20.17) holds. However, if the left expression or the middle expression in the first line holds no expression in the second line holds, as well as if the left expression or the middle expression in the second line holds no expression in the first line holds so x must satisfies

$$\left.\begin{array}{ll} v(Anna, Carl) - x_1 & \leq v(Ben, Carl) - x_2 \\ v(Ben, Carl) - x_2 & \leq v(Anna, Carl) - x_1 \end{array}\right\} \Rightarrow 1000 - x_1 = 500 - x_2 \tag{183}$$

The solution of Equations (182) and (183) is $x = (750, 250, 0)$.

$\mathcal{B} = \{\{Anna, Carl\}, \{Ben\}\}$**:** This case is similar to the previous case so the only imputation in the bargaining set is $(750, 0, 250)$.

$\mathcal{B} = \{\{Anna\}, \{Ben, Carl\}$**:** By Therem 20.15, $x \in \mathcal{M}(N; v; \mathcal{B})$ if and only if $x \in X(\mathcal{B}; v)$, i.e.,

$$x_2 + x_3 = 500, \quad x_2, x_3 \geq 0, \text{ and } \quad x_3 = 0, \tag{184}$$

and at least one expression in each line of the table (20.17), relative to players 2 and 3 holds. However, niether $x_1 + x_2 \geq v(Anna, Ben)$ nor $x_1 + x_3 \geq v(Anna, Carl)$ holds. Furthermore, if either $x_2 = 0$ or $x_3 = 0$ then one of the players has a justified objection. Thus, x must satisfies

$$\left.\begin{array}{ll} v(Anna, Ben) - x_2 & \leq v(Anna, Carl) - x_3 \\ v(Anna, Carl) - x_3 & \leq v(Anna, Ben) - x_2 \end{array}\right\} \Rightarrow 1000 - x_1 = 1000 - x_2 \tag{185}$$

The solution of Equations (184) and (185) is $x = (250, 250, 0)$.

$\mathcal{B} = \{\{Anna\}, \{Ben\}, \{Carl\}\}$**:** By Theorem 20.11, the bargaining set contains a single payoff vector $(v(Anna), v(Ben), v(Carl)) = (0, 0, 0)$.

$\mathcal{B} = \{Anna, Ben, Carl\}$ **if** $v(N) = 0$**:** $x \in X(\mathcal{B}; v)$ if and only if $x_1 + x_2 + x_3 = 0$ and $x_i \geq 0$ for every player i. Hence the only imputation in $X(\mathcal{B}; v)$ is $(0, 0, 0)$ and thus, by Theorem 20.19, it is the only point in the bargaining set.

$\mathcal{B} = \{Anna, Ben, Carl\}$ **if** $v(N) = 1000$**:** Let $x \in X(\mathcal{B}; v)$ be an imputation so $x_1 + x_2 + x_3 = 1000$ and $x_i \geq 0$ for every player i. This imputation $x \in \mathcal{M}(N;v;\mathcal{B})$ if $x \in X(\mathcal{B};v)$ if and only if it satisfies at least one expression in each line in the system presented in (20.18).

If, by contradiction, $x_2 = 0$ then, by $2 \to 3$, $x_1 = 1000$ which case Carl has a justified objection against Anna, so $x_2 > 0$. Similarly, $x_3 > 0$. If, by contradiction, $x_1 = 0$ then, by $1 \to 2$, $x_3 = 0$ which case Anna ha a justified objection against Carl. In conclusion, $x_1, x_2, x_3 > 0$.

Therefore, the left and middle expressions in both lines 5 and 5 cannot hold so by the right expression it follows that $x_2 = x_3$. Furthermore, by the rightexpression of $1 \to 2$, $500 + x_2 \leq x_1$. In particular, only the right expressions in lines 2 and 4 holds, so $x_2 = x_3 = 166\frac{2}{3}$ and $x_1 = 666\frac{2}{3}$.

20.8 For $N = \{1,2,3\}$, we compute the bargaining set of the coalitional game $(N;v)$ relative to the coalitional structure $\{\{1,2\},\{3\}\}$, for each of the following coalitional functions. By Theorem 20.15, the bargaining set relative to the coalitional structure $\mathcal{B} = \{\{1,2\},\{3\}\}$ is the set of vector payoffs in $X(B;v)$ satisfying at least one expression in each line of the following table.

	L	M	R
1	$x_1 + x_3 \geq v(1,3)$	$x_2 = v(2)$	$v(1,3) - x_1 \leq v(2,3) - x_2$
2	$x_2 + x_3 \geq v(2,3)$	$x_1 = v(1)$	$v(2,3) - x_2 \leq v(1,3) - x_1$

(a) $v(1) = v(2) = v(3) = 0$, $v(1,2) = 40$, $v(1,3) = 50$, $v(2,3) = 60$, $v(1,2,3) = 100$.

Let $x \in \mathcal{M}(N;v;\mathcal{B})$. Then, $x_1 + x_2 = 40$, $x_1, x_2 \geq 0$ and $x_3 = 0$. Thus, $x_1 + x_3, x_2 + x_3 \leq 40$ from which it is obtained that expressions $L1$ and $L3$ cannot hold. If $M1$ holds then $x_1 = 40$ and $v(2,3) - 0 > v(1,3) - 40$, so no expression in the second line holds, hence $M2$ cannot hold. Similarly, $M2$ cannot hold as well. Thus expressions $R1$ and $R2$ holds from which it follows that the bargaining set includes only one point $(15,25,0)$.

(b) $v(1) = v(2) = v(3) = 0$, $v(1,2) = 80$, $v(1,3) = 20$, $v(2,3) = 30$, $v(1,2,3) = 100$.

Let $x \in \mathcal{M}(N;v;\mathcal{B})$. Then, $x_1 + x_2 = 80$, $x_1, x_2 \geq 0$ and $x_3 = 0$. If $M1$ holds then $x_2 + x_3 = 0 < v(2,3)$, $x_1 = 80$ and $v(2,3) - 0 > v(1,3) - 80$, so no expression in the second line holds, hence $M1$ cannot hold. Similarly, $M2$ cannot hold as well.

In particular, expressions $L1$ and $L2$ are satisfied for

$$\{(x_1, 80 - x_1, 0) : 20 \leq x_1 \leq 50\}).$$

Expressions $L1$ and $R2$ are satisfied for

$$\{(x_1, 80 - x_1, 0) : 20 \leq x_1 \leq 35\}).$$

Expressions $L2$ and $R1$ are satisfied for

$$\{(x_1, 80 - x_1, 0) : 35 \leq x_1 \leq 50\}).$$

Expressions $R1$ and $R2$ are satisfied for $\{(35, 45, 0)$.
In conclusion, the bargaining set is

$$\{(x_1, 80 - x_1, 0) : 20 \leq x_1 \leq 50\}).$$

20.9 The games in Exercise 20.8 are 0-monotonic, three player games with nonempty core. Following Theorem 20.16, the bargaining set for the coalitional structure N coincides with the core.
The bargaining set of the game in Part (a) is

$$\left\{(x_1, x_2, 100 - x_1 - x_2) : 0 \leq x_1 \leq 40,\ 40 - x_1 \leq x_2 \leq 50\right\}.$$

The bargaining set of the game in Part (b) is

$$\left\{(x_1, x_2, 100 - x_1 - x_2) : 0 \leq x_1 \leq 70,\ 80 - x_1 \leq x_2 \leq \min\{80, 100 - x_1\}\right\}.$$

20.10 Let $(N; v)$ be "My Aunt and I" coalitional game where the Players 1 and 2 are my aunt and I, respectively. In particular, $N = \{1, 2, 3, 4, 5\}$, $v(S) = 1000$ for every coalition $S \in \big\{\{1, 2\}, \{1, 3\}, \{1, 4\}, \{1, 5\}, \{2, 3, 4, 5\}\big\}$ and $v(S) = 0$ for every other coalition S.
Consider the coalitional structure $\mathcal{B} = \{\{1, 2\}, \{3\}, \{4\}, \{5\}\}$.
Let $x \in X(\mathcal{B}; v)$, that is $x_1 + x_2 = 1000$, $x_1, x_2 \geq 0$, and $x_3 = x_4 = x_5 = 0$.
This imputation is in the bargaining set of this game for the coalitional structure $\mathcal{B}$ if and only if, Player 1 has no justified objection against Player 2 and vice versa.
Player 1 has no justified objection against Player 2 if either $x_2 = v(2) = 0$ or $v(1, k) - x_1 - x_k \leq v(2, 3, 4, 5) - \sum_{m=2}^{5} x_m$; if and only if either $x_2 = 0$ or $x_2 \leq x_1$.
Similarly, Player 2 has no justified objection against Player 1 if either $x_1 = 0$ or $x_1 \leq x_2$.
However, if, by contradiction, $x_2 = 0$ then $x_1 = 1000 > 0$ and $x_1 > x_2$ which case Player 2 has a justified objection against 1, a contradiction. In particular, $x_2 > 0$ and similarly $x_1 > 0$ from which it can be conclude that $x2 \leq x_1$ and $x_1 \leq x_2$ so $x_1 = x_2$.

In conclusion, $(500, 500, 0, 0, 0)$ is the unique point in the bargaining set.

20.11 **The proof of Theorem 20.16:** Let $(N; v)$ be a 0-monotonic three-player game, and consider the coalitional structure $\{N\}$. We show that the bargaining set coincides with the core if the core is nonempty, and is a single point if the core is empty.

By Theorem 20.5 (which is proved in Exercise 20.1), the core of the coalitional game $(N; v)$ for the coalitional structure $\mathcal{B} = \{N\}$ is the set of all imputations at which no player has an objection against another player. Therefore, $\mathcal{C}(N; v) \subseteq \mathcal{M}(N; v; \mathcal{B})$. It is left to prove that if there is $x \in \mathcal{M}(N; v; \mathcal{B}) \setminus \mathcal{C}(N; v)$ then $\mathcal{C}(N; v) = \emptyset$ and $\mathcal{M}(N; v; \mathcal{B}) = \{x\}$.

Let $x \in \mathcal{M}(N; v; \mathcal{B}) \setminus \mathcal{C}(N; v)$. Then $x_1 + x_2 + x_n = v(N)$, $x_i \geq v(i)$ for every $i \in N$ and there is a two player coalition, say $\{1, 2\} \subset N$ such that

$$x_1 + x_2 < v(1, 2). \tag{186}$$

Since $(N; v)$ is a 0-monotonic game, $x_3 > v(3)$, otherwise, by Inequality (186),

$$v(1, 2) + v(3) > x_1 + x_2 + x_3 = v(N) \tag{187}$$

which contradict Exercise 17.28. Now, since Player 1 has no justified objection against 3,

$$v(1, 2) - x_1 \leq v(2, 3) - x_3. \tag{188}$$

Similarly, since Player 2 has no justified objection against 3,

$$v(1, 2) - x_2 \leq v(1, 3) - x_3. \tag{189}$$

Summing Inequalities (187) and (188) as well as Inequalities (187) and (189), since $v(3) < x_3$, one can deduce that

$$x_2 + x_3 < v(2, 3) \text{ and } x_1 + x_3 < v(1, 3). \tag{190}$$

Using Inequalities (187) and (190) the following is obtained

$$v(N) = x_1 + x_2 + x_3 < \tfrac{1}{2}v(1, 2) + \tfrac{1}{2}v(1, 3) + \tfrac{1}{2}v(2, 3),$$

so by the Bondareva–Shapley conditions the core is empty.

Consider the system of Inequalities (20.18). All of the left hand inequalities of each line do not hold, by Inequalities (186) and (190). Therefore and by Exercise 17.28, all of the inequalities in the middle of each line do not hold. Therefore x satisfies all of the right hand inequalities from which it follows that

$$\begin{aligned} x_1 &= \tfrac{v(N)+v(1,2)+v(1,3)-2v(2,3)}{3} \\ x_2 &= \tfrac{v(N)+v(1,2)+v(2,3)-2v(1,3)}{3} \\ x_3 &= \tfrac{v(N)+v(1,3)+v(2,3)-2v(1,2)}{3} \end{aligned}$$

so $\mathcal{M}(N;v;\mathcal{B}) = \{x\}$, as claimed.

20.12 We prove that in Definition 20.7, the inequality in Condition 4 can be replaced by a strict inequality.
Assume that player k has a justified objection $(C;y)$ against player l at x according to Definition 20.7 where the inequalities in Condition 4 are replaced by strict inequalities. That is, for every coalition D such that $l \in D$ and $k \notin D$ and every $z \in \mathbf{R}^D$ satisfying $z(D) = v(D)$ and $z_j \leq x_j$ for every $j \in D \setminus C$, there is a player $i \in D \cap C$ such that $z_i \leq y_i$.
Since (C,y) is an objection of k against l at x, $y_k > x_k$. Select $\epsilon \in (0, y_k - x_k)$. Set $\widetilde{y}_i = y_i + \frac{\epsilon}{|C|}$. Then $(C,\widetilde{y})$ is a justified objection of player k against l according to the original Definition 20.7.

20.13 Consider a three player game where the coalitional function is as follows

$$v(1) = v(2) = v(3) = 2, v(1,2) = v(1,3) = 4, v(2,3) = 5, v(1,2,3) = 6.$$

The only point in the bargaining game is $(2,2,2)$ since every player l has a counter objection $(\{l\},2)$ against any other player k. However, if strict inequalities are required in Condition 3 of Definition 20.7, then Player 2 has a justified objection $(\{2,3\},(2.5,2.5))$ against Player 1 whom has no counter objection since $v(1) = x_1$ and $v(1,3) = x_1 + x_3$. Hence the bargaining set in this case is empty.

20.14 Let $N = \{1,2,3,4\}$. Player 1 has no justified objection against Player 2 at x if and only if x satisfies at least one expression in each part in the following conditions:

Player 1 has no justified objection with the coalition $\{1,3\}$:

$$x_1 + x_3 \geq v(1,3) \text{ or} \tag{191}$$

$$x_2 = v(2) \text{ or } x_2 + x_4 \leq v(2,4) \text{ or} \tag{192}$$

$$v(1,3) - x_1 \leq v(2,3) - x_2 \text{ or } v(1,3) - x_1 \leq v(2,3,4) - x_2 - x_4. \tag{193}$$

If Inequality (191) holds then Player 1 has no objection consists of the coalition $\{1,3\}$. If Inequality (192) holds then Player 2 has a counter objection consists of a coalition containing players that are not in $\{1,3\}$. If Inequality (193) holds then Player 2 has a counter objection consists of a coalition containing players that are in $\{1,3\}$.

Player 1 has no justified objection with the coalition $\{1,4\}$**:**

$$\begin{gathered} x_1 + x_4 \geq v(1,4) \text{ or} \\ x_2 = v(2) \text{ or } x_2 + x_3 \leq v(2,3) \text{ or} \\ v(1,4) - x_1 \leq v(2,4) - x_2 \text{ or } v(1,4) - x_1 \leq v(2,3,4) - x_2 - x_3. \end{gathered}$$

Player 1 has no justified objection with the coalition $\{1,3,4\}$**:**

$$\begin{gathered} x_1 + x_3 + x_4 \geq v(1,3,4) \text{ or} \\ x_2 = v(2)\text{or} \\ v(1,3,4) - x_1 - x_4 \leq v(2,3) - x_2 \text{ or} \\ v(1,3,4) - x_1 - x_3 \leq v(2,4) - x_2 \text{ or} \\ v(1,3,4) - x_1 \leq v(2,3,4) - x_2. \end{gathered}$$

20.15 (a) Denote by $N_1 = \{1,2\}$ the set of sellers with left-hand gloves, and by $N_2 = \{3,4,5\}$ the set of sellers with right-hand gloves. Set $N = N_1 \bigcup N_2$. The coalitional function of the gloves game is $v(S) = 10 \cdot \min\{|S \cap N_1|, |S \cap N_2|\}$ for every $S \subseteq N$.

(b) We prove that

$$\mathcal{M}(N;v;\mathcal{B}) = \big\{(10,10,0,0,0)\big\}.$$

Clearly, $\big\{(10,10,0,0,0)\big\} \subseteq X(\mathcal{B};v)$. Furthermore, since Players 1 and 2 are symmetric they have no justified objection against each other (else the other player can use the symmetric objection as a counterobjection). Similarly, since Players 3,4, and 5 are symmetric they have no justified objection against each other. Finally, for every $i \in \{1,2\}$ and $j \in \{3,4,5\}$ player i has no objection at all against j since $v(D) \leq \sum_{i \in D} x_i$ for every $i \in D \subseteq N \setminus \{j\}$, and vice versa. Thus, $(10,10,0,0,0) \in \mathcal{M}(N;v;\mathcal{B})$.

We next prove the other direction. Let $x \in \mathcal{M}(N;v;\mathcal{B})$. In particular, $\sum_{i=1}^{5} x_i = 20$, $x_i \geq 0$ for every $i \in N$ and no player has a justified objection against any other player at x.

From now on, assume without loss of generality that $x_1 \leq x_2$ and $x_3 \leq x_4 \leq x_5$.

We first claim that $x_i + x_j \leq 10$ for every $i \in \{1,2\}$ and $j \in \{2,3,4\}$.

Assume by contradiction that $x_2 + x_5 > 10$, so $x_1 + x_3 < 10$. Therefore, if $x_5 = 0$, then Player 1 has a justified objection against Player 2 with a coalition $\{1,3\}$ which leads to a contradiction. Note that Player 2 has no counterobjection since his payoff is greater than the worth of any coalition D such that $2 \in D$ and $1 \notin D$. If on the other hand $x_5 > 0$ then Player 1 has a justified objection against Player 5 with a coalition $\{1,3\}$ whereas Player 5 has no counterobjection since $v(D) = 0 < x_5 \leq \sum_{i \in D} x_i$ for every $D \subseteq N_1$ and $v(D) = 10 < x_2 + x_5 \leq \sum_{i \in D} x_i$ for every $D \subseteq N_1 \bigcup \{2\}$, a contradiction.

We now claim that $x_1 = x_2$.

Assume by contradiction that $x_1 < x_2$. Then, if $x_5 = 0$, then $x_3 = x_4 = x_5 = 0$ so Player 1 has a justified objection against Player 2 with the coalition of $\{1,3\}$ whereas Player 2 has no counter objection. Assume, on the other hand, that $x_5 > 0$. Therefore, if $x_2 + x_5 = 10$ then Player 1 has a justified objection against Player 5 with a coalition $\{1,3\}$, since $x_1 + x_3 < x_2 + x_5 = 10$ and Player 5 has no counter objection. If on the other hand $x_2 + x_5 < 10$ then Player 1 has a justified objection against Player 5 with a coalition $\{1,2,3,4\}$. Player 5 has no counter objection since

$$\begin{aligned} v(1,2,3,4) - x_1 - x_3 - x_4 &= v(1,3) - x_1 - x_3 + v(2,4) - x_4 \\ &> v(2,4) - x_4 \geq v(2,5) - x_5. \end{aligned}$$

In conclusion, in all three cases there is a player who has a justified objection against some other player, a contradiction.

We next claim that $x_3 = x_4 = x_5$.

Assume by contradiction that $x_3 < x_5$. Then Player 1 has a justified objection against Player 5 with the coalition $\{1,2,3,4\}$ since $v(1,2,3,4,5) = v(N) > x_1 + x_2 + x_3 + x_4$ and Player 5 has no counter objection since

$$\begin{aligned} v(1,2,3,4) - x_1 - x_3 - x_4 &= v(1,3) - x_1 - x_3 + v(2,4) - x_4 \\ &> v(2,4) - x_4 \geq v(2,5) - x_5, \end{aligned}$$

a contradiction.

We finally claim that $x_1 = x_2 = 10$ and $x_3 = x_4 = x_5 = 0$.

Assume by contradiction that $x_1 = x_2 = a < 10$ hence $x_3 = x_4 = x_5 = b > 0$ such that $2a + 3b = 20$. In particular, $a + b < 10$. Therefore, Player 1 has a justified objection against Player 5 with the coalition $\{1,2,3,4\}$ since $v(1,2,3,4,5) = v(N) = 2a + 3b >$

$x_1 + x_2 + x_3 + x_4$. However, Player 5 has no counter objection since $v(5) = 0 < x_5$ and

$$\begin{aligned} v(2,5) - x_5 &= 10 - b < 10 - b + 10 - b - a \\ &= v(2,4) - x_4 + v(1,3) - x_1 - x_3 \\ &= v(1,2,3,4) - x_1 - x_3 - x_4, \end{aligned}$$

a contradiction.

All the above claims lead to $x = (10, 10, 0, 0, 0)$, and we are done.

20.16 Consider the market game in which the initial endowment of each member of $N_1 = \{1,2\}$ is $(1,0)$, the initial endowment of each member of $N_2 = \{3,4,5\}$ is $(0, \frac{1}{2})$ and for each coalition S,

$$v(S) = \min\left\{|S \cap N_1|, \tfrac{1}{2}|S \cap N2|\right\}$$

(a) We find the core of this game. Let $x \in \mathcal{C}(N;v)$. By the efficiency property,$x_1 + x_2 + x_3 + x_4 + x_5 = 1.5$. In addition,

$$\begin{aligned} x_2 + x_5 &\geq v(2,5) = \tfrac{1}{2} \\ x_1 + x_3 + x_4 &\geq v(1,3,4) = 1 \end{aligned}$$

From which it follows that

$$\begin{aligned} &1 \leq x_1 + x_3 + x_4 = 1.5 - x_2 - x_5 \leq 1.5 - 0.5 = 1 \\ &\Rightarrow x_1 + x_3 + x_4 = 1 \text{ and } x_2 + x_5 = \tfrac{1}{2}. \end{aligned}$$

Similarly, $x_i + x_j + x_k = 1$ and $x_i + x_j = \frac{1}{2}$ for every $i \in \{1,2\}$ and $j \neq k \in \{3,4,5\}$, so $x_k = \frac{1}{2}$ for every $k \in \{3,4,5\}$ and $x_i = 0$ for every $i \in \{1,2\}$.

In conclusion, $(0, 0, \frac{1}{2}, \frac{1}{2}, \frac{1}{2})$ is the only vector in the core of this game relative to the coalitional structure $\mathcal{B} = \{N\}$.

(b) We prove that the bargaining set relative to the coalitional structure $\{N\}$ is

$$\mathcal{M}(N;v;\mathcal{B}) = \left\{(\alpha, \alpha, \beta, \beta, \beta) : 0 \leq \alpha \leq \tfrac{3}{4}, \beta = \tfrac{1}{2} - \tfrac{2}{3}\alpha\right\}.$$

We start by proving that

$$\left\{(\alpha, \alpha, \beta, \beta, \beta) : 0 \leq \alpha \leq \tfrac{3}{4}, \beta = \tfrac{1}{2} - \tfrac{2}{3}\alpha\right\} \subseteq \mathcal{M}(N;v;\mathcal{B}).$$

Clearly,

$$\left\{(\alpha, \alpha, \beta, \beta, \beta) : 0 \leq \alpha \leq \tfrac{3}{4}, \beta = \tfrac{1}{2} - \tfrac{2}{3}\alpha\right\} \subseteq X(\mathcal{B};v).$$

Furthermore, since Players 1 and 2 are symmetric they have no justified objection against each other (else the other player can use the symmetric objection as a counter objection). Similarly, since Players 3,4, and 5 are symmetric they have no justified objection against each other. Finally, $\alpha + \beta = \frac{1}{2} + \frac{1}{3}\alpha \geq \frac{1}{2}$ so no player has a justified objection against any other player from its opposite group.

We prove the other direction. Let $x \in \mathcal{M}(N; v; \mathcal{B})$. In particular, $\sum_{i=1}^{5} x_i = 1.5$, $x_i \geq 0$ for every $i \in N$ and no player has a justified objection against any other player at x.

From now on, assume without loss of generality that $x_1 \leq x_2$ and $x_3 \leq x_4 \leq x_5$.

We first claim that $x_i \leq \frac{1}{2}$ for every $i \in \{3, 4, 5\}$.

Assume by contradiction that $x_5 > \frac{1}{2}$, so $x_1 + x_2 + x_3 + x_4 < 1 = v(1, 2, 3, 4)$. Then Player 1 has a justified objection against Player 5 with the coalition $\{1, 2, 3, 4\}$, a contradiction. Note that Player 5 has no counter objection by at most two player coalition since his payoff is greater than the worth of any such coalition. In addition, Player 5 has no counter objection by at least 3 player coalition since Player 5 can offer less than Player 1, for instance

$$v(1, 2, 3, 4) - x_1 - x_4 > v(2, 3, 5) - x_2 - x_5.$$

We next claim that $x_1 = x_2$.

Assume by contradiction that $x_1 < x_2$. If in addition $x_5 = 0 \geq x_4 \geq x_3 \geq 0$, then $x_1 + x_2 = 1.5$, $x_1 < \frac{3}{4}$ and $x_2 > \frac{3}{4}$. Therefore, Player 1 has a justified objection against Player 2 with coalition $\{1, 3, 4\}$, a contradiction. Indeed, Player 2 has no counter objection by a two-player coalition since his payoff is greater than the worth of any such coalition. Furthermore, Player 2 has no counter objection by at least 3 players coalition since Player 2 can offer to either Player 3 and\or Player 4 less than $\frac{1}{4}$ whereas Player 1 can offer more than $\frac{1}{4}$.

If, $x_5 > 0$ then

$$\begin{aligned} 1.5 = x_1 + x_2 + x_3 + x_4 + x_5 \leq 2x_2 + 3x_5 < 3(x_2 + x_5) \\ \Rightarrow x_2 + x_5 > \tfrac{1}{2}. \end{aligned}$$

Hence, Player 1 has a justified objection against Player 5 with the coalition $\{1, 3, 4\}$, a contradiction. Player 5 has no counter objection by at most two player coalition since the payoff according to x is greater than than worth of any such coalition. Player 5 has no counter objection by at least 3 players coalition since

$$\begin{aligned}
v(1,3,4) - x_1 - x_4 &> v(2,3,5) - x_2 - x_5 \\
v(1,3,4) - x_1 - x_3 &> v(2,4,5) - x_2 - x_5 \\
v(1,3,4) - x_1 &> v(2,3,4,5) - x_2 - x_5,
\end{aligned}$$

a contradiction.

We finally claim that $x_3 = x_4 = x_5$**.**

Assume by contradiction that $x_3 < x_5$. Then Player 1 has a justified objection against Player 5 with the coalition $\{1,2,3,4\}$ since $v(1,2,3,4) = v(N) > x_1 + x_2 + x_3 + x_4$ so either $x_1 + x_3 < v(1,3)$ or $x_1 + x_4 < v(1,4)$. Assume without loss of generality that $x_1 + x_3 < v(1,3)$ then Player 5 has no counter objection since

$$\begin{aligned}
v(1,2,3,4) - x_1 - x_3 - x_4 &= v(1,3) - x_1 - x_3 + v(2,4) - x_4 \\
&> v(2,4) - x_4 \geq v(2,5) - x_5,
\end{aligned}$$

a contradiction.

All the above claims leads to

$$\mathcal{M}(N;v;\mathcal{B}) \subseteq \left\{(\alpha,\alpha,\beta,\beta,\beta) : 0 \leq \alpha \leq \tfrac{3}{4}, b = \tfrac{1}{2}\tfrac{2}{3}\alpha\right\},$$

and we are done.

(c) Note that Players 1 and 2 has no incentive to join Players 3-5 if their payoffs are zero, therefore the points in the bargaining set that are not in the core may be more reasonable than points in the core.

20.17 Consider the market game in which the initial endowment of each member of $N_1 = \{1,2\}$ is $(1,0)$, the initial endowment of each member of $N_2 = \{3,4,5\}$ is $(0,1)$ and for each coalition S,

$$v(S) = \min\left\{|S \cap N_1|, |S \cap N2|\right\}$$

(a) We find the core of this game. Let $x \in \mathcal{C}(N;v)$. By the efficiency property, $x_1 + x_2 + x_3 + x_4 + x_5 = 2$. In addition,

$$\begin{aligned}
x_2 + x_5 &\geq v(2,5) = 1 \\
x_1 + x_3 + x_4 &\geq v(1,3,4) = 1
\end{aligned}$$

From which it follows that

$$\begin{aligned}
1 \leq x_1 + x_3 + x_4 &= 2 - x_2 - x_5 \leq 2 - 1 = 1 \\
&\Rightarrow x_1 + x_3 + x_4 = 1 \text{ and } x_2 + x_5 = 1.
\end{aligned}$$

Similarly, $x_i + x_j + x_k = 1$ and $x_i + x_j = 1$ for every $i \in \{1,2\}$ and $j \neq k \in \{3,4,5\}$, so $x_k = 0$ for every $k \in \{3,4,5\}$. Finally, $x_i = x_i + x_5 \geq v(i,5) = 1$ for every $i \in \{1,2\}$ so $x_i = 1$.

In conclusion, $(1,1,0,0,0)$ is the only vector in the core of this game relative to the coalitional structure $\mathcal{B} = \{N\}$.

(b) We prove that the bargaining set relative to the coalitional structure $\{N\}$ is

$$\mathcal{M}(N;v;\mathcal{B}) = \left\{(\alpha,\alpha,\beta,\beta,\beta) : 0 \leq \alpha \leq 1, \beta = \tfrac{2}{3} - \tfrac{2}{3}\alpha\right\}.$$

We start by proving that

$$\left\{(\alpha,\alpha,\beta,\beta,\beta) : 0 \leq \alpha \leq 1, \beta = \tfrac{2}{3} - \tfrac{2}{3}\alpha\right\} \subseteq \mathcal{M}(N;v;\mathcal{B}).$$

Clearly,

$$\left\{(\alpha,\alpha,\beta,\beta,\beta) : 0 \leq \alpha \leq 1, \beta = \tfrac{2}{3} - \tfrac{2}{3}\alpha\right\} \subseteq X(\mathcal{B};v).$$

Furthermore, since Players 1 and 2 are symmetric they have no justified objection against each other (else the other player can use the symmetric objection as a counter objection). Similarly, since Players 3,4, and 5 are symmetric they have no justified objection against each other. Finally, $\alpha + \beta = \frac{2}{3} + \frac{1}{3}\alpha \geq \frac{2}{3}$ so no player has a justified objection against any other player from its opposite group.

We prove the other direction. Let $x \in \mathcal{M}(N;v;\mathcal{B})$. In particular, $\sum_{i=1}^{5} x_i = 2$, $x_i \geq 0$ for every $i \in N$ and no player has a justified objection against any other player at x.

From now on, assume without loss of generality that $x_1 \leq x_2$ and $x_3 \leq x_4 \leq x_5$.

We first claim that $x_i \leq 1$ for every $i \in \{1,2\}$.

Assume by contradiction that $x_2 > 1$, so $x_1 + x_3 < 1 = v(1,3)$. Then Player 1 has a justified objection against Player 2 with the coalition $\{1,3\}$, a contradiction. Note that Player 2 has no counter objection since his payoff is greater than the worth of any coalition not including Player 1.

We next claim that $x_1 = x_2$.

Assume by a contradiction that $x_1 < x_2$. Then

$$2 = x_1 + x_2 + x_3 + x_4 + x_5 < 2x_2 + 3x_5 \leq 2 + 3x_5$$
$$\Rightarrow x_5 > 0.$$

Hence, Player 1 has a justified objection against Player 5 with the coalition $\{1,2,3,4\}$, a contradiction. Player 5 has no counter objection by himself since $x_5 > 0 = v(5)$. In addition, Player 5 has no counter objection by at least 2 players coalition since if $x_2 + x_5 < v(2,5)$ then $x_1 + x_3 < v(1,3)$ so

$$\begin{aligned} v(1,2,3,4) - x_1 - x_3 - x_4 &= v(1,3) - x_1 - x_3 + v(2,4) - x_4 \\ &> v(2,5) - x_5, \end{aligned}$$

a contradiction.

We finally claim that $x_3 = x_4 = x_5$.

Assume by contradiction that $x_3 < x_5$. Then Player 1 has a justified objection against Player 5 with the coalition $\{1,2,3,4\}$ since $v(1,2,3,4) = v(N) > x_1 + x_2 + x_3 + x_4$ so either $x_1 + x_3 < 1$ or $x_1 + x_4 < 1$, assume without loss of generality that $x_1 + x_3 < 1$. Hence, Player 5 has no counter objection since

$$\begin{aligned} v(1,2,3,4) - x_1 - x_3 - x_4 &= v(1,3) - x_1 - x_3 + v(2,4) - x_4 \\ &> v(2,4) - x_4 \geq v(2,5) - x_5. \end{aligned}$$

All the above claims leads to

$$\mathcal{M}(N;v;\mathcal{B}) \subseteq \left\{(\alpha,\alpha,\beta,\beta,\beta) : 0 \leq \alpha \leq 1, b = \tfrac{2}{3}\tfrac{2}{3}\alpha\right\},$$

and we are done.

20.18 Let $N = \{1,2,3,4,5\}$. Consider the coalitional game $(N;v)$ where $v(1,2,3) = 30$, $v(2,4,5) = 40$, $v(1,4) = 40$ and for every other coalition S, $v(S) = 0$. Consider the coalitional structure $\mathcal{B} = \{\{1,2,3\},\{4\},\{5\}\}$ and an imputation $x = (10,10,10,0,0)$. Then Player 1 has a justified objection $(\{1,4\},(20,20))$ against 3 who has no counter objection since $v(D) = 0 < 10 = x_3$ for every $3 \in D \subseteq N \setminus \{1\}$. In particular, $1 \succ_x 3$. On the other hand, $1 \sim_x 2$ and $2 \sim_x 3$ since Players 1,2 and 3 are symmetric players in $\{1,2,3\} \in \mathcal{B}$ thus every objection of one of them has a symmetric counter objection of the other.

20.19 In the weighted majority game $[3;1,1,1,1,1,0]$ with six players, Player 6 is a null player. We prove that despite this, the vector $x = (\frac{1}{7},\frac{1}{7},\frac{1}{7},\frac{1}{7},\frac{1}{7},\frac{2}{7})$ is in the bargaining set relative to the coalitional structure $\{N\}$.
Clearly, x is efficient and individually rational. We next show that no player has a justified objection.
No Player $i \in \{1,2,3,4,5\}$ has a justified objection against Player $j \in \{1,2,3,4,5\} \setminus \{i\}$, since i and j are symmetric and $x_i = x_j$ (Player j has a counter objection using the symmetric objection of player i).
Player 6 has no justified objection against any player $j \in \{1,2,3,4,5\}$. Indeed, for every coalition C containing Player 6 such that $|C| \leq 3$,

$v(C) = 0 < x_6$. For every objection (C, y) of Player 6 against j such that $6 \in C$, $j \notin C$ and $|C| > 3$, let $D = C \setminus \{6\} \bigcup \{j\}$ then $v(C) - x_6 = v(D) - x_6 < v(D) - x_j$. Therefore,$(C, y)$ is not a justified objection.
There is no player $i \in \{1, 2, 3, 4, 5\}$ that has a justified objection against Player 6. For every coalition $C \subset \{1, 2, 3, 4, 5\}$ satisfying $i \in C$, $j \notin C$ then one of the following holds
(1) $|C| < 3$ which case $v(C) = 0 < x_i \leq x(C)$ so player i has no objection $(C; y)$.
(2) $|C| = 3$. However, if (C, y) is an objection of i against Player 6, then there is a player $j \in C \setminus i$ whose payoff $y_j < \frac{3}{7}$. Hence, Player 6 has a counter objection $(N \setminus C \bigcup \{j\}, z)$ where $z_j = \frac{3}{7}$ and $z_k = x_k$ for every $k \in D \setminus \{j\}$.
(3) $|C| \geq 4$. However, for every player $j \in C \setminus \{i\}$, the coalition $D = N \setminus C \bigcup \{j\}$ satisfies $v(C) - \sum_{k \in C \setminus \{j\}} x_k \leq v(D) - \sum_{k \in C \setminus \{j\}} x_k$ so Player 6 has a counter objection. In conclusion, x is indeed a vector in the bargaining set relative to the coalitional structure $\{N\}$.
The intuitive explanation for the fact that a null player can obtain a positive payoff in the bargaining set is that he can be the one who motivating the other players to join into a coalition.

20.20 A simple coalitional game $(N; v)$ is called a veto control game if there is a player i such that $v(S) = 0$ for every coalition that does not contain i. Let $(N; v)$ be a monotonic, veto control game where player i is the veto player. We show that the core relative to the coalitional structure $\{N\}$ coincides with the bargaining set relative to the same coalitional structure.
In case $v = 0$, then $x = (0, 0, ..., 0) \in X(B; v) = \mathcal{C}(N; v) = \mathcal{M}(N; v; \{N\})$.
Assume next that $v(N) = 1$.
By Exercise 18.12, $\mathcal{C}(N; v) = \{x\}$ such that $x_i = v(N)$ and $x_k = 0$ for every $k \in N \setminus \{i\}$. By Exercise 20.1, $\mathcal{C}(N; v) \subseteq \mathcal{M}(N; v; \{N\})$.
We prove the opposite. Let $x \in \mathcal{M}(N; v; \{N\})$. Then, $\sum_{j \in N} x_j = v(N)$ and $x_j \geq v(j)$ for every $j \in N$. Assume by contradiction that $x \notin \mathcal{C}(N; v)$. That is, there is a nonempty coalition $C \subset N$ satisfying $v(C) > \sum_{k \in S} x_k$. In particular, $v(C) > 0$ so the veto player $i \in S$. Furthermore, by the monotonicity, $v(N) \geq v(C)$ so there is a player $j \notin C$ whose payoff is $x_j > 0$. Set $y_k = x_k + \frac{v(C) - \sum_{l \in C} x_l}{|C|}$ for every $k \in C$. Then, player i has a justified objection (C, y) against player j, since x_j is greater than the worth of any coalition D not including player i. Thus $x \notin \mathcal{M}(N; v; \{N\})$, a contradiction.

20.21 **The proof of Theorem 20.17** Let $(N; v)$ be a coalitional game and let $\mathcal{B}$ be a coalitional structure. We prove that the bargaining set is a finite union of polytopes.

The bargaining set is an intersection of $N(N-1)+1$ sets:

$$\mathcal{M}(N;v;\mathcal{B}) = X(\mathcal{B};v) \bigcap \left(\bigcap_{i\in N} \bigcap_{j\neq i} A_{i,j} \right)$$

where $A_{i,j}$ is the set of all vector such that player i has no justified objection against j for every two players $i \neq j \in N$.
The set $X(\mathcal{B};v)$ is a finite intersection of half spaces

$$X(\mathcal{B};v) = \left(\bigcap_{i\in N} \{x \in \mathbf{R}^N : x_i \geq v(i)\} \right) \bigcap \{x \in \mathbf{R}^N : x(N) \geq v(N)\} \\ \bigcap \{x \in \mathbf{R}^N : x(N) \leq v(N)\}.$$

For every player $i \in N$ and player $j \in N \setminus \{i\}$, denote by $\mathcal{S}_{i,j}$ the set of all coalitions of $N \setminus \{j\}$ containing player i. Denote by $\mathcal{S}^{no}_{i,j}$ the set of coalition in which player i has no objection against j, thus the set $\mathcal{S}_{i,j} \setminus \mathcal{S}^{no}_{i,j}$ is the set of all coalition in which player i has an objection against i.
The set $A_{i,j}$ of vectors according to which player i has no justified objection against j is as follows:

$$A_{i,j} = \bigcup_{\mathcal{S}^{no}_{i,j} \subseteq \mathcal{S}_{i,j}} \left\{ \left(\bigcap_{C \in \mathcal{S}^{no}_{i,j}} \{x \in \mathbf{R}^N : x(C) \geq v(C)\} \right) \right. \\ \bigcap \left(\bigcap_{C \in \mathcal{S}_{i,j} \setminus \mathcal{S}^{no}_{i,j}} \Big(\{x \in \mathbf{R}^N : x(C) < v(C)\} \right. \\ \left. \left. \bigcup_{\{D \subset N \setminus \{i\} : j \in D\}} \{x \in \mathbf{R}^N : v(C) - x(C) \leq v(D) - x(D)\} \Big) \right) \right\}.$$

In particular, the bargaining set is an intersection and union of a finite number of half spaces and thus it is a a finite union of polytopes.

20.22 The statement of Theorem 20.28 is formulated for the coalitional structure $\mathcal{B} = \{N\}$. This was used in the assumption that

$$e(N) = v(N) - x(N) = 0$$

in the proof of the theorem, for other coalitional structure $e(N) = 0$ does not necessarily hold.

Chapter 21: The nucleolus

21.1 The function $x \mapsto \sum_{S\subseteq N} e(S,x)$ is constant on the set of preimputations $X^0(N;v)$.
Indeed,

$$\sum_{S\subseteq N} e(S,x) = \sum_{S\subseteq N} v(S) - \sum_{S\subseteq N} x(S) = \sum_{S\subseteq N} v(S) - \sum_{i\in N} x_i \cdot |\{S\subseteq N : i \in S\}|$$
$$= \sum_{S\subseteq N} v(S) - \sum_{i\in N} x_i 2^{n-1} = \sum_{S\subseteq N} v(S) - v(N)2^{n-1},$$

where the last equality holds since $x \in X^0(N;v)$.

21.2 We prove that the lexicographic relation is transitive.
Let $x,y,z \in \mathbf{R}^n$ such that $x \prec_L y$ and $y \prec_L z$. Then, there is $k \in \{1,2,...,n\}$ for which $x_i = y_i$ for every $i < k$ and $x_k < y_k$. Similarly, there is $m \in \{1,2,...,n\}$ for which $y_i = z_i$ for every $i < m$ and $y_m < z_m$. If $k \leq m$, then $x_i = y_i = z_i$ for every $i < k$, and $x_k < y_k \leq z_k$, so $x \prec_L z$. If $m < k$, then $x_i = y_i = z_i$ for every $i < m$, and $x_m = y_m < z_m$, so $x \prec_L z$, as needed.

21.3 Let $K \subseteq \mathbf{R}^n$, and let $f : \mathbf{R}^n \to \mathbf{R}$ be a function. Denote the set of points in K at which the minimum of f is attained by $\arg\min_{x\in K} f(x) = \{y \in K : f(y) = \min_{z\in K} f(z)\}$. We prove that if K is a compact set and f is a continuous function, then the set $\arg\min_{x\in K} f(x)$ is compact and nonempty.
We first prove that the function f over K is bounded from below. Assume by contradiction that for every $n \in \mathbf{N}$ there is $x_n \in K$ satisfying $f(x_n) < -n$. However, the sequence $(x_n)_{n\in\mathbf{N}}$ is contained in the compact set K and therefore contains a convergent subsequence $(x_{n_k})_{k\in\mathbf{N}}$ such that $\lim_{k\to\infty} x_{n_k} = x \in K$. Furthermore, by the continuity of f,

$$-\infty = \lim_{k\to\infty} f(x_{n_k}) = f(\lim_{k\to\infty} x_{n_k}) = f(x),$$

a contradiction. Therefore, $\inf_{x\in X} f(x)$ is finite.
We next proof that the set $\arg\min_{x\in K} f(x)$ is not empty. For every $n \in \mathbf{N}$, there is $x_n \in K$ such that $f(x_n) - \frac{1}{n} < \inf_{x\in X} f(x)$. The sequence $(x_n)_{n\in\mathbf{N}}$ is contained in the compact set K and therefore contains a convergent subsequence $(x_{n_k})_{k\in\mathbf{N}}$ such that $\lim_{k\to\infty} x_{n_k} = x^* \in K$. Thus,

$$f(x^*) \geq \inf_{x\in X} f(x) \geq \lim_{k\to\infty} \left(f(x_{n_k}) - \tfrac{1}{n_k} \right) = f(\lim_{k\to\infty} x_k) = f(x^*),$$

so $f(x^*) = \inf_{x\in X} f(x) = \min_{x\in X} f(x)$ an therefore $\arg\min_{x\in K} f(x)$ is not empty.

Next, since $\arg\min_{x\in K} f(x)$ is a nonempty subset of a bounded set, it is also bounded.
We finally proved that $\arg\min_{x\in K} f(x)$ is closed. Let $(y_n)_{n\in\mathbf{N}}$ be a sequence in $\arg\min_{x\in K} f(x)$. By the compactness of K, $\lim y_n \in K$. In addition, $(f(y_n)_{n\in\mathbf{N}}$ is a constant sequence satisfying $f(y_n) = \min_{z\in K} f(z)$ so it is also the limit of the sequence so $f(\lim y_n) = \min_{z\in K} f(z)$, i.e., $\arg\min_{x\in K} f(x)$ is closed.

21.4 Consider the sequence of vectors $\left((\frac{1}{n},0)\right)_{n\in\mathbf{N}}$ in $\mathbf{R}^2$ converging to $(0,0)$, and a vector $(0,1) \in \mathbf{R}^2$. Then, $(\frac{1}{n},0) \succ_L (0,1)$ for all $n \in \mathbf{N}$, but $(0,0) \prec_L (0,1)$.

21.5 Consider the three-player coalitional game with the following coalitional function:

$$v(1) = 0, v(2) = 1, v(3) = 4, v(1,2) = 2, v(1,3) = 6,$$
$$v(2,3) = -1, v(1,2,3) = 5.$$

(a)

S	$e\left(S,(1,1,3)\right)$	$e\left(S,(1,3,1)\right)$	$e\left(S,(3,1,1)\right)$
$\emptyset$	0	0	0
$\{1\}$	-1	-1	-3
$\{2\}$	0	-2	0
$\{3\}$	1	3	3
$\{1,2\}$	0	-2	-2
$\{1,3\}$	2	4	2
$\{2,3\}$	-5	-5	-3
$\{1,2,3\}$	0	0	0

Hence

$$\theta((1,1,3)) = (2,1,0,0,0,0,-1,-5),$$
$$\theta((1,3,1)) = (4,3,0,0,-1,-2,-2,-5),$$
$$\theta((3,1,1)) = (3,2,0,0,-2,-3,-3).$$

(b) $\theta(1,3,1) \succ_L \theta(3,1,1) \succ_L \theta(1,1,3)$.

21.6 The statement is not necessarily true.
Let $(N;v)$ be a two-player coalitional game where $v(1) = v(2) = 0$

and $v(1,2) = 1$. Let $x = (1,0)$ and $y = (0.1, 0.9)$. Then,

$$\begin{aligned}
\theta(x) &= (0,0,0,-1),\\
\theta(y) &= (0,0,-0.1,-0.9),\\
\theta(\tfrac{1}{2}x + \tfrac{1}{2}y) &= (0,0,-0.45,-0.55),\\
\Rightarrow \theta(x) &\succ_L \theta(y) \succ_L \theta(\tfrac{1}{2}x + \tfrac{1}{2}y).
\end{aligned}$$

21.7 (a) Let $y = (0,0,1)$ be an imputation in the gloves game. We prove that $\theta(y) \prec_L \theta(u)$ for every $u \in X(N;v)$, $u \neq y$.

Since $\theta_1(y) = 0$, it is enough to show that $\theta_1(u) > 0$ for every $u \in X(N;v) \setminus \{y\}$.

Let $u \in X(N;v)$ where $u \neq y$. That is, $u_1 + u_2 + u_3 = 1$, $0 \leq u_1$, $0 \leq u_2$ and $0 \leq u_3 < 1$. Therefore, either $u_1 + u_3 < 1$ or $u_2 + u_3 < 1$ (or both). Hence, at least one of the following holds $e(\{1,3\}, u) > 0$ or $e(\{2,3\}, u) > 0$ from which it follows that $\theta_1(u) > 0$.

In particular, $\theta(y) \prec_L \theta(u)$.

(b) We prove that this imputation also satisfies $\theta(y) \prec_L \theta(u)$ for every $u \in X^0(N;v)$, $u \neq y$.

For every $u \in X(N;v) \subseteq X^0(N;v)$ such that $u \neq y$, it follow by Part (a). Assume next that $u \in X^0(N;v) \setminus X(N;v)$. That is $u_1 + u_2 + u_3 = 1$ and there is $i \in N$ such that $u_i < 0$ so $e(\{i\}, u) = 0 - u_i > 0$. In particular $\theta_1(u) \geq e(\{i\}, u) > 0$ from which it follows that $\theta(y) \prec_L \theta(u)$, as claimed.

21.8 (a) An example of a two-player coalitional game $(N;v)$, and a bounded set K that is not closed, such that the nucleolus $\mathcal{N}(N;v;K)$ is the empty set: let $(N;v)$ be a coalitional game where N is a finite set of players. Let $M \in \mathbf{N}$ such that $M > \max_{S \subseteq N} v(S)$. Set $K = \{x \in \mathbf{R}^n : x_i < M \ \ \forall i \in N\}$.

We show that for every $x \in K$ there is $y \in K$ such that $\theta(y) \prec_L \theta(x)$ so the nucleolus $\mathcal{N}(N;v;K)$ is the empty set.

Let $x \in K$,then there is an $\epsilon > 0$ such that $y_i = x_i + \epsilon < M$ for every $i \in N$. Hence, $y = (y_1, ..., y_n) \in K$ and

$$e(S,y) = v(S) - y(S) = v(S) - x(S) - |S|\epsilon < e(S,x)$$

for every nonempty coalition $S \subseteq N$. In particular $\theta(y) \prec_L \theta(x)$, as claimed.

(b) An example of a two-player coalitional game $(N; v)$, and a closed and unbounded set K, such that the nucleolus $\mathcal{N}(N; v; K)$ is the empty set: let $(N; v)$ be a coalitional game where N is a finite set of players. Set $K = \mathbf{R}^n$.

We show that for every $x \in K$ there is $y \in K$ such that $\theta(y) \prec_L \theta(x)$ so the nucleolus $\mathcal{N}(N; v; K)$ is the empty set.

Let $x \in K$. Set $y_i = x_i + 1$ for every $i \in N$. Hence, $y = (y_1, ..., y_n) \in K$ and

$$e(S, y) = v(S) - y(S) = v(S) - x(S) - |S| < e(S, x)$$

for every nonempty coalition $S \subseteq N$. In particular $\theta(y) \prec_L \theta(x)$, as claimed.

21.9 Let $(N; v)$ be a coalitional game satisfying the following property: there is an imputation $x \in X(N; v)$ such that all the excesses at x are nonnegative.

(a) We prove that x is the only imputation in the game, and in particular it is the nucleolus.

By the individually rational property,

$$x_i \geq v(i) \quad \text{for every } i \in N. \tag{194}$$

On the other hand, since all the excesses are nonnegative,

$$0 \leq e(\{i\}, x) = v(i) - x_i \quad \text{for every } i \in N. \tag{195}$$

By Inequalities (194) an(195), $x_i = v(i)$ for every $i \in N$.

By the efficiency, $\sum_{i \in N} v(i) = \sum_{i \in N} x_i = v(N)$.

Let $y \in X(N; v)$. By the individually rational property, $y_i \geq v(i)$ for every $i \in N$. By the efficiency, for every $i \in N$,

$$\begin{aligned} y_i &= v(N) - \sum_{j \neq i} y_j = \sum_{j \in N} v(j) - \sum j \neq i y_j \leq \sum_{j \in N} v(j) - \sum_{j \neq i} v(j) \\ &= v(i) \end{aligned}$$

In conclusion, $y_i = v(i) = x_i$ for all $i \in N$ so x is the only imputation in the game, and in particular it is the nucleolus.

(b) x is not necessarily also the prenucleolus. For instance, let $(N; v)$ be a three player coalitional game with the following coalitional function

$$v(1) = v(2) = 1, v(3) = \tfrac{1}{2}, v(1,2) = v(1,3) = v(2,3) = 2,$$
$$v(1,2,3) = 2\tfrac{1}{2}.$$

The excesses at $x = (1, 1, \frac{1}{2})$ are nonnegative, whereas $y = (\frac{3}{4}, \frac{3}{4}, 1) \in X^0(N; v)$ satisfies

$$\theta(y) = (\tfrac{1}{2}, \tfrac{1}{4}, \tfrac{1}{4}, \tfrac{1}{4}, \tfrac{1}{4}, 0) \prec_L (\tfrac{1}{2}, \tfrac{1}{2}, 0, 0, 0, 0) = \theta(x),$$

so $x \notin \mathcal{N}(N; v; X^0(N; v))$.

21.10 Player i in a coalitional game $(N; v)$ is a dummy player if

$$v(S \bigcup \{i\}) = v(S) + v(i), \ \ \forall S \subseteq N \setminus \{i\}.$$

We prove that if player i is a dummy player in a game $(N; v)$ then under both the nucleolus and the prenucleolus, player i's payoff is $v(i)$, that is, $\mathcal{N}_i(N; v) = v(i)$ and $\mathcal{PN}_i(N; v) = v(i)$.
Let $b \in \mathbf{R}^n$ such that $b_i = -v(i)$ and $b_j = 0$ for every $j \in N \setminus \{i\}$. Set $w(S) = v(S) + b(S)$ for every coalition $S \subseteq N$. Player i is a null player in $(N; w)$. Hence, by Theorem 21.19, $\mathcal{N}_i(N; w) = \mathcal{PN}_i(N; w) = 0$. By the covariant under strategic equivalence property of the nucleolus and the prenucleolus, $\mathcal{N}_i(N; v) = \mathcal{N}_i(N; w) - b_i = v(i)$ and $\mathcal{PN}_i(N; v) = \mathcal{PN}_i(N; w) - b_i = v(i)$.

21.11 The proof of Theorem 21.7: We prove that for every k, $2 \leq k \leq 2^n$,

$$\theta_k(x) = \max_{\text{different } S_1, \dots, S_k} \min \left\{ e(S_1, x), \dots, e(S_k, x) \right\}.$$

Denote by $\widetilde{S}_1, \dots \widetilde{S}_{2^n}$ all the coalitions ordered in decreasing excess order. That is, $e(\widetilde{S}_1, x) \geq e(\widetilde{S}_2, x) \geq \dots \geq e(\widetilde{S}_{2^n}, x)$ so $\theta_k(x) = e(\widetilde{S}_k, x)$ for every $k \in \{1, 2, \dots, 2^n\}$.
For every set of k different coalition $\{S_1, \dots S_k\}$, either (1) $\{S_1, \dots, S_k\} = \{\widetilde{S}_1, \dots, \widetilde{S}_k\}$ and therefore

$$\min\{e(S_1, x), \dots, e(S_k, x)\} = \min\{e(\widetilde{S}_1, x), \dots, e(\widetilde{S}_k, x)\} = \theta_k;$$

or (2) there is $m > k$ such that $\widetilde{S}_m \in \{S_1, \dots, S_k\}$ thus

$$\min\{e(S_1, x), \dots, e(S_k, x)\} \leq e(\widetilde{S}_m, x) = \theta_m(x) \leq \theta_k.$$

In conclusion,

$$\theta_k(x) = \max_{\text{different } S_1, \dots, S_k} \min \left\{ e(S_1, x), \dots, e(S_k, x) \right\}.$$

21.12 Consider the three-player coalitional game with the following coalitional function:

$$v(1) = v(2) = v(3) = v(2,3) = 0, v(1,2) = v(1,3) = v(1,2,3) = 1.$$

We compute the nucleolus $y = (y_1, y_2, y_3)$ of the game. Player 2 and Player 3 are symmetric, so by Theorem 21.17, $y_2 = y_3$. By the efficiency, $y_1 = 1 - 2y_2$. By the individually rational property, $0 \leq y_2 \leq \frac{1}{2}$. For every $S \in \{\{1\}, \{2\}, \{3\}, \{2,3\}, \{1,2,3\}\}$, $e(S, y) \leq 0$, whereas $e(\{1,2\}, y) = e(\{1,3\}, y) = y_2 \geq 0$. Hence, $\theta_1(y) = y_2$ which attains its minimum at 0. In conclusion, $y = (1, 0, 0)$ is the nucleolus.

21.13 Consider the three-player coalitional game with the following coalitional function:

$$v(1) = v(2) = v(3) = v(2,3) = 0, v(1,2) = v(1,3) = v(1,2,3) = -1.$$

We compute the prenucleolus $y = (y_1, y_2, y_3)$ of this game.
Player 2 and Player 3 are symmetric, so by Theorem 21.17, $y_2 = y_3$.
By the efficiency, $y_1 = 1 - 2y_2$.
The excesses of the coalitions is as follows: $e(\{1\}, y) = 2y_2 - 1$, $e(\{2\}, y) = e(\{3\}, y) = -y_2$, $e(\{1,2\}, y) = e(\{1,3\}, y) = y_2 - 2$, $e(\{2,3\}, y) = -2y_2$. For $y_2 = \frac{1}{3}$, $\theta_1(y) = -\frac{1}{3}$. For every $y_2 < \frac{1}{3}$, $\theta_1(y) \geq e(\{2\}, y) = -y_2 > -\frac{1}{3}$. For every $y_2 > \frac{1}{3}$, $\theta_1(y) \geq e(\{1\}, y) = 2y_2 - 1 > -\frac{1}{3}$. In conclusion, $y = (\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$ is the prenucleolus.

21.14 Let $(N; v)$ be the three-player coalitional game with the following coalitional function:

$$v(S) = 0 \Longleftrightarrow |S| \leq 1,$$
$$v(S) = 1 \Longleftrightarrow |S| \geq 2.$$

Let K be the triangle in $\mathbf{R}^3$ whose vertices are $(1,0,0)$, $(0,1,0)$, and $(0,0,1)$, and let K_0 be its boundary. Then the nucleolus of the game $(N; v)$ relative to K is $(\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$ and relative to K_0 is $\{(\frac{1}{2}, \frac{1}{2}, 0), (\frac{1}{2}, 0, \frac{1}{2}), (0, \frac{1}{2}, \frac{1}{2})\}$.

21.15 Let $(N; v)$ be a three-player coalitional game in which $v(1,2) = 1$ and $v(S) = 0$ for every other coalition S. Player 3 in this game is a null player so by Theorem 21.19, $\mathcal{N}_3(N; v) = \mathcal{PN}_3(N; v) = 0$. In addition, Players 1 and 2 are symmetric. Therefore, by Theorem 21.17, $\mathcal{N}_1(N; v) = \mathcal{N}_2(N; v)$ and $\mathcal{PN}_1(N; v) = \mathcal{PN}_2(N; v)$. However, by the efficiency,

$$\mathcal{N}_1(N; v) = \mathcal{N}_2(N; v) = \mathcal{PN}_1(N; v) = \mathcal{PN}_2(N; v) = 0.$$

In particular, the nucleolus coincide with the prenucleolus.

21.16 Let $(N;v)$ be a five-player coalitional game with the following coalitional function:

$$v(S) = \begin{cases} 1 & S = \{1,2\} \text{ or } S = \{1,2,5\} \\ 2 & S = \{3,4\} \text{ or } S = \{3,4,5\} \\ 0 & \text{otherwise.} \end{cases}$$

(a) Player 5 is a null player since $v(S \bigcup \{5\}) = v(S)$ for every $S \subseteq N \setminus \{5\}$.

(b) We prove that the nucleolus and the prenucleolus for the coalitional structure $\mathcal{B} = \{\{1,2,5\},\{3\},\{4\}\}$ are

$$\mathcal{N}(N;v;\mathcal{B}) = \mathcal{PN}(N;v;\mathcal{B}) = (0,0,0,0,1).$$

Since $x = (0,0,0,0,1) \in X(N;v;\mathcal{B}) \subseteq X^0(N;v;\mathcal{B})$, it is enough to prove that $x \in \mathcal{PN}(N;v;\mathcal{B})$.

Note that, $\theta_1(x) = e(\{3,4\},x) = 2$, $\theta_2(x) = e(\{1,2\},x) = 1$, $\theta_3(x) = e(\{3,4,5\},x) = 1$. and $\theta_k(x) \leq 0$, for every $k \geq 4$.

Let $y \in X^0(N;v;\mathcal{B})$ then $y_3 = y_4 = 0$ and $y_1 + y_2 + y_5 = 1$. In addition, one of the following holds:

i. $y_5 > 1$. Thus $e(\{3,4\},y) = 2$ and $e(\{1,2\},y) = 1 - y_1 - y_2 = y_5 > 1$. Hence, $\theta(y) \succ_L \theta(x)$.

ii. $y_5 < 1$ from which it follows that $e(\{3,4\},y) = 2$ ans $e(\{3,4,5\},y) = 2 - y_5 > 1$. Hence, $\theta(y) \succ_L \theta(x)$.

iii. $y_5 = 1$ and $y_1 < 0$ (the case where $y_2 < 0$ is similar and thus it was omitted). Thus, $e(\{3,4\},y) = 2$, $e(\{1,2\},y) = e(\{3,4,5\},y) = 1$ and $e(\{1,3\},y) = -y_1 > 0$. Hence, $\theta(y) \succ_L \theta(x)$.

In conclusion, $x \in \mathcal{PN}(N;v;\mathcal{B})$ and thus $x \in \mathcal{N}(N;v;\mathcal{B})$, as needed.

21.17 We prove that the nucleolus is covariant under strategic equivalence. Let $(N;v)$ be a coalitional game, and let $K \subseteq \mathbf{R}^N$. We show that for every $a > 0$, and every $b \in \mathbf{R}^N$,

$$\mathcal{N}(N;av+b;aK+b) = a\mathcal{N}(N;v;K) + b.$$

Denote by $(N;w)$ the coalitional game with the coalitional function $w = av + b$. Denote by $e_v(S,x)$ the excess of coalition S at x in the

game $(N;v)$ and by $e_w(S,x)$ the excess of coalition S at x in the game $(N;w)$.
For every S,

$$\begin{aligned} e_w(S, ax+b) &= w(S) - (ax(S) - b(S)) = av(S) - b(S) - (ax(S) - b(S)) \\ &= ae_v(S,x). \end{aligned}$$

Hence,

$$\begin{aligned} e_v(S,x) = e_v(S,y) &\iff e_w(S,ax+b) = e_w(S,ay+b) \text{ and} \\ e_v(S,x) < e_v(S,y) &\iff e_w(S,x) < e_w(S,y). \end{aligned}$$

From which it follows

$$ax + y \in \mathcal{N}(N; av+b; aK+b) \iff x \in \mathcal{N}(N;v;K),$$

which completes the proof.

21.18 Let $(N;v)$ be a two-player coalitional game where $v(1)=1$, $v(2)=2$ and $v(1,2)=3$. Let K be a V-shaped set that is the union of two line segments $[(0,0),(1,1)]$ and $[(0,0),(0,2)]$ sharing an edge point $(0,0)$. Then $\theta((1,1)) = (1,1,0) = \theta((0,2))$. Furthermore, for every $y = (y_1,y_1) \in [(0,0),(1,1)]$, $\theta(y) = (3-2y_1, 2-y_1, 1-y_1) \succ_L \theta((1,1))$. In addition, for every $y=(0,y_2) \in [(0,0),(0,2)]$, $\theta_1(y) = 3-y_2$ so $\theta(y) \succ_L \theta((0,2))$.
In conclusion, the nucleolus of $(N;v)$ relative to K is the two edge points of K.

21.19 The nucleolus of the weighted majority game $[q;2,2,3,3]$ for every quota $q>0$ presented below.
Note that in all games Players 1 and 2 are symmetric, as well as Player 3 and 4. Hence, by the efficiency and individually rational property, the nucleolus whenever $q \le 10$ is of the form $(a, a, \frac{1-2a}{2}, \frac{1-2a}{2})$ where $0 \le a \le \frac{1}{2}$. Whereas, in case $q > 10$ then the coalitional function of the majority game is $v=0$ which case $X(N;v) = \{(0,0,0,0)\}$ so $(0,0,0,0)$ is the nucleolus.

q	$\mathcal{W}^m$	$\mathcal{N}(N;v)$
$0 < q \le 2$	$\{1\},\{2\},\{3\},\{4\}$	$(\frac{1}{4},\frac{1}{4},\frac{1}{4},\frac{1}{4})$
$2 < q \le 3$	$\{3\},\{4\},\{1,2\}$	$(\frac{1}{6},\frac{1}{6},\frac{1}{3},\frac{1}{3})$
$3 < q \le 4$	$\{1,2\},\{1,3\},\{1,4\},\{2,3\},\{2,4\},\{3,4\}$	$(\frac{1}{4},\frac{1}{4},\frac{1}{4},\frac{1}{4})$
$4 < q \le 5$	$\{1,3\},\{1,4\},\{2,3\},\{2,4\},\{3,4\}$	$(\frac{1}{6},\frac{1}{6},\frac{1}{3},\frac{1}{3})$
$5 < q \le 6$	$\{3,4\},\{1,2,3\},\{1,2,4\}$	$(\frac{1}{6},\frac{1}{6},\frac{1}{3},\frac{1}{3})$
$6 < q \le 7$	$\{1,2,3\},\{1,2,4\},\{1,3,4\},\{2,3,4\}$	$(\frac{1}{4},\frac{1}{4},\frac{1}{4},\frac{1}{4})$
$7 < q \le 8$	$\{1,3,4\},\{2,3,4\}$	$(0,0,\frac{1}{2},\frac{1}{2})$
$8 < q \le 10$	$\{1,2,3,4\}$	$(\frac{1}{4},\frac{1}{4},\frac{1}{4},\frac{1}{4})$
$q > 10$	$\emptyset$	$(0,0,0,0)$

21.20 Let $(N; v)$ be of the coalitional game where $N = \{1, 2, 3, 4\}$ and the coalitional function v is given by

$$v(S) = \begin{cases} i & S = \{i\}, \\ 0 & \text{otherwise.} \end{cases}$$

The set

$$X(N; v) = \left\{ x \in \mathbf{R}^4 : \begin{array}{c} x_i \geq i \ \ \forall i \in N \text{ and} \\ x_1 + x_2 + x_3 + x_4 = 0 \end{array} \right\} = \emptyset,$$

so the nucleolus is empty as well.

21.21 **My Aunt and I**

(a) Let $(N; v)$ be "My Aunt and I" coalitional game where the Players 1 and 2 are my aunt and I, respectively. In particular, $N = \{1, 2, 3, 4, 5\}$, $v(S) = 1000$ for every coalition

$$S \in \big\{\{1, 2\}, \{1, 3\}, \{1, 4\}, \{1, 5\}, \{2, 3, 4, 5\}\big\}$$

and $v(S) = 0$ for every other coalition S.

(b) By the efficiency and individually rational property

$$X(N, v) = \{(0, 0, 0, 0, 0)\} = \mathcal{N}(N, v).$$

(c) Consider the coalitional structure

$$\begin{aligned} \mathcal{B} &= \big\{\{\text{Auntie Betty, Me}\}, \{\text{Brother A}\}, \{\text{Brother B}\}, \{\text{Brother C}\}\big\} \\ &= \big\{\{1, 2\}, \{3\}, \{4\}, \{5\}\big\}. \end{aligned}$$

Then $X(\mathcal{B}; v) = \{(a, 1000 - a, 0, 0, 0) : 0 \leq a \leq 1000\}$. Let $x \in X(\mathcal{B}; v)$ then

$$e(S, x) = \begin{cases} 0 & S = \{1, 2\} \text{ or } \{1, 2\} \cap S = \emptyset \\ -a & 1 \in S \text{ and } 2 \notin S \\ a - 1000 & 1 \notin S \text{ and } 2 \in S \\ -1000 & 1, 2 \in S \text{ and } |S| \geq 3. \end{cases}$$

In particular, the nucleolus of the game for the coalitional structure $\mathcal{B}$ is $(500, 500, 0, 0, 0)$.

21.22 Let $(N;c)$ be a cost game and let $\mathcal{B}$ be a coalitional structure. Then, the set of imputation is

$$X(\mathcal{B};c) = \left\{ x \in \mathbf{R}^N : \begin{array}{c} x(B) = c(B) \ \ \forall B \in \mathcal{B} \\ x_i \leq c(i) \ \ \forall i \in N \end{array} \right\}.$$

For every coalition S, denote by $e(S,x) = x(S) - c(S)$ the excess of coalition S. If $e(S,x) > 0$ then the members of the coalition pay more than their obligations, whereas, if $e(S,x) < 0$, they pay less than their obligation.
Let $\theta(x)$ be the excess vector of all the coalitions written in decreasing order from left to right.
The nucleolus of $(N;c)$ is

$$\mathcal{N}(N;v;\mathcal{B}) = \left\{ x \in X(\mathcal{B};c) : \theta(x) \precsim_L \theta(y) \quad \forall y \in X(\mathcal{B};v) \right\}.$$

21.23 The coalitional game corresponding the spanning tree game is (N,c) where $N = \{1,2,3\}$ and

$$c(S) = \begin{cases} 2 & |S| = 1 \\ 3 & |S| = 2 \\ 5 & |S| = 3. \end{cases}$$

Let $x \in X(N;c)$. Since x is individually rational $x_i \leq 2$ for every $i \in N$. By the efficiency , $\sum_{i \in N} x_i = 5$.
Therefore, $x \in \mathcal{C}(N;c)$ if and only if $x \in X(N;c)$ and it coalitionally rational. Hence $x_1 \leq 2$ and

$$x_1 = 5 - \sum_{j=2}^{3} x_j \geq 5 - c(2,3) = 2.$$

Thus $x_1 = 2$. Similarly, $x_2 = x_3 = 2$ so $x_1 + x_2 + x_3 = 6 > 5$, which contradict the efficiency. In particular, the core is empty.
On the other hand, since all of the players are symmetric and by the efficiency the nucleolus is $(1\frac{2}{3}, 1\frac{2}{3}, 1\frac{2}{3})$.

21.24 Let $K \subseteq \mathbf{R}^N$ be a closed set satisfying the following property: there exists a real number c such that $\sum_{i \in N} x_i \leq c$ for all $x \in K$. We prove that the nucleolus relative to K is nonempty.
Choose an arbitrary point $y \in K$. Define,

$$\widetilde{K} = \left\{ x \in K : e(S,x) \leq \theta_1(y) \ \ \forall S \subseteq N \right\}.$$

Since $y \in \widetilde{K}$, the set $\widetilde{K}$ is not empty. K is closed and $\widetilde{K} \subseteq K$ and defined by weak inequalities hence it is closed as well.
We show that $\widetilde{K}$ is bounded. For every $x \in \widetilde{K}$,

$$x_i = v(i) - e(\{i\}, x) \geq v(i) - \theta_1(y)$$

In addition,

$$x_i = \sum_{j=1}^{n} c_j - \sum_{j \neq i} x_j \leq c - \sum_{j \neq i} v(j) + |N-1|\theta_1(y).$$

Therefore, $\widetilde{K}$ is compact and nonempty, so, by Theorem 21.9, the nucleolus of the game $(N; v)$ relative to $\widetilde{K}$ is a nonempty compact set. Let $z \in \mathcal{N}(N; v; \widetilde{K})$ then clearly, $\theta(z) \precsim \theta(x)$ for every $x \in \widetilde{K}$. In addition, for every $x \in K \setminus \widetilde{K}$, $\theta(z) \precsim \theta(y) \prec \theta(x)$. Hence, $z \in \mathcal{N}(N; v; K)$ which is necessarily nonempty.

21.25 Let $(N; v)$ be a coalitional game, let $(c_i)_{i \in N}$ be a collection of positive real numbers, let $c \in \mathbf{R}$, and let $K := \left\{x \in \mathbf{R}^N \colon \sum_{i \in N} c_i x_i \leq c\right\} \subseteq \mathbf{R}^N$. We prove that the set $\mathcal{N}(N; v; K)$ contains exactly one point.
Let $M = \max_{S \subseteq N} |v(S)|$. Define

$$K^* = \left\{ x \in K \ : \ x_i \geq \frac{-2M - |c||N|}{\min\{\sum_{j \in N} c_j, 1\}} \quad \forall i \in N \right\}. \tag{196}$$

We first show that for every $x \in K \setminus K^*$ there is $x^* \in K^*$ such that $\theta(x) \succ_L \theta(x^*)$ from which it follows that $\mathcal{N}(N; v; K) = \mathcal{N}(N; v; K^*)$. Then we prove that $\mathcal{N}(N; v; K^*)$ contains exactly one point.
Let $x \in K \setminus K^*$. That is, there is $i \in N$ such that $x_i < \frac{-2M - |c||N|}{\min\{\sum_{j \in N} c_j, 1\}}$. Define $x^* \in K^*$ by $x^*_j = \frac{c}{\min\{\sum_j c_j, 1\}}$ for every $j \in N$. For every $S \subseteq N$,

$$\begin{aligned}
\theta_1(x) \geq e(\{i\}, x) = v(i) - x_i &> v(i) - \frac{-2M - |c||N|}{\min\{\sum_{j \in N} c_j, 1\}} \\
&= v(i) + \frac{2M + |c||N|}{\min\{\sum_{j \in N} c_j, 1\}} \geq v(S) - \frac{c|S|}{\min\{\sum_{j \in N} c_j, 1\}} \\
&= v(S) - x^*(S) = e(S, x^*).
\end{aligned}$$

Therefore, $\theta(x) \succ \theta(x^*)$, as claimed.
The set K^* is convex: for every $x, y \in K^*$ and every $\alpha \in (0, 1)$,

$$\sum_{i \in N} c_i(\alpha x_i + (1-\alpha) y_i) = \alpha \sum_{i \in N} c_i x_i + (1-\alpha) \sum_{i \in N} c_i y_i \leq \alpha c + (1-\alpha) c = c,$$

and for every $i \in N$

$$\begin{aligned}
\alpha x_i + (1-\alpha) y_i &\geq \alpha \cdot \frac{-2M - |c||N|}{\min\{\sum_{j\in N} c_j, 1\}} + (1-\alpha) \cdot \frac{-2M - |c||N|}{\min\{\sum_{j\in N} c_j, 1\}} \\
&= \frac{-2M - |c||N|}{\min\{\sum_{j\in N} c_j, 1\}}.
\end{aligned}$$

Hence, $\alpha x + (1-\alpha) y \in K^*$. Therefore, using Theorem 21.13, $\mathcal{N}(N; v; K^*)$ contains at most one point.
In addition, The set K^* is bounded from below by Equation (196) and thus, by the definition of K, it also bounded from above.
Furthermore,K^* is also closed: let $(x^n)_{n=1}^{\infty} \subseteq K^*$ such that $\lim_{n\to\infty} x^n = x$. Then

$$\begin{aligned}
&\sum_{i\in N} c_i x_i = \sum_{i\in N} c_i \lim_{n\to\infty} x_i^n = \lim_{n\to\infty} \sum_{i\in N} c_i x_i^n \leq c \text{ and} \\
&x_i = \lim_{n\to\infty} x_i^n \geq \frac{-2M - |c||N|}{\min\{\sum_{j\in N} c_j, 1\}} \quad \forall i \in N,
\end{aligned}$$

from which it follows that $x \in K^*$. Therefore, K^* is a compact set and thus, by Theorem 21.9, $\mathcal{N}(N; v; K^*)$ is a nonempty compact set.
In conclusion, $\mathcal{N}(N; v; K^*) = \mathcal{N}(N; v; K)$ contains exactly one point, as needed.

21.26 We prove that the vector $x \in X(\mathcal{B}; v)$ is in the core $\mathcal{C}(N; v; \mathcal{B})$ if and only if $\theta_1(x) \leq 0$.
A vector $x \in X(\mathcal{B}; v)$ is in the core if and only if it is coalitionally rational, if and only if

$$\begin{aligned}
&x(S) \geq v(S) && \forall S \subseteq N \\
\iff &0 \geq e(S, x) = v(S) - x(S) && \forall S \subseteq N
\end{aligned}$$

if and only if $\theta_1(x) = \max_{S\subseteq N} e(S, x) \leq 0$.

21.27 We prove that if the ϵ-core $\mathcal{C}_\epsilon(N; v)$ is nonempty, then $\mathcal{N}(N; v) \subseteq \mathcal{C}_\epsilon(N; v)$.
By Exercise 18.33, the ϵ-core $\mathcal{C}_\epsilon(N; v)$ is a polytope. Since it is nonempty, let $x \in \mathcal{C}_\epsilon(N; v)$. Therefore, $e(S, x) \leq \epsilon$ for every coalition $S \subseteq N$. For ever $y \in \mathcal{N}(N; v)$,

$$e(S, y) \leq \theta_1(y) \leq \theta_1(x) \leq \epsilon.$$

Thus, $y \in \mathcal{C}_\epsilon(N; v)$, as needed.
In conclusion, the nucleolus $y \in \mathcal{C}_\epsilon(N; v)$ for every $\epsilon \geq 0$ such that $\mathcal{C}_\epsilon(N; v) \neq \emptyset$. In particular, for every nonempty coalition $S \subset N$, its

excess at y, $e(S,y) \leq \epsilon$. Hence, one can deduce that the nucleolus is contained in the minimal core, and that if the core is nonempty, then the nucleolus is contained in the core.

21.28 (a) We prove that the process presented in the exercise indeed finds the nucleolus. By Exercise 21.27, the nucleolus is contained in the minimal core. In a three-player game, the minimal core is either a point or a segment in a barycentric coordinates. For every coalition S, the line H_S that cannot be displaced without causing the ϵ-core to become empty, corresponding a constrain holds as equality, $e(S,x) = \epsilon_0$ for every x in the minimal core $\mathcal{C}_\epsilon(N;v)$. However, since the nucleolus y is contained in the minimal core, $e(S,y) = \epsilon_0$. If the minimal core is not a single point, then there must be two lines that can be moved to the middle of the segment where the maximal excess is minimal.

(b) i.

$$\begin{aligned} \mathcal{C}_{\epsilon_0}(N;v) &= \{\alpha(2,2,8) + (1-\alpha)(7,2,3) \ : \ \alpha \in [0,1]\} \\ &\Rightarrow \mathcal{N}(N;v) = \{(4\tfrac{1}{2}, 2, 5\tfrac{1}{2})\}. \end{aligned}$$

ii.

$$\begin{aligned} \mathcal{C}_{\epsilon_0}(N;v) &= \{(5,4,3)\} \\ &\Rightarrow \mathcal{N}(N;v) = \{(5,4,3)\}. \end{aligned}$$

iii.

$$\begin{aligned} \mathcal{C}_{\epsilon_0}(N;v) &= \{(4,4,4)\} \\ &\Rightarrow \mathcal{N}(N;v) = \{(4,4,4)\}. \end{aligned}$$

iv.

$$\begin{aligned} \mathcal{C}_{\epsilon_0}(N;v) &= \{\alpha(6,3,3) + (1-\alpha)(3,3,6) \ \ \alpha \in [0,1]\} \\ &\Rightarrow \mathcal{N}(N;v) = \{(4\tfrac{1}{2}, 3, 4\tfrac{1}{2})\}. \end{aligned}$$

v.

$$\begin{aligned} \mathcal{C}_{\epsilon_0}(N;v) &= \{(5,2,5)\} \\ &\Rightarrow \mathcal{N}(N;v) = \{(5,2,5)\}. \end{aligned}$$

21.29 Suppose that $(N;v)$ is a coalitional game such that the set of imputations $X(N;v)$ is nonempty, and such that the nucleolus x^* differs

from the prenucleolus $\hat{x}$. We prove that the nucleolus is located on the boundary of the set $X(N;v)$.
Assume by contradiction that the nucleolus is not on the boundary of $X(N;v)$. Therefore, since, $X(N;v)$ is nonempty, compact and convex, there is a $\alpha \in (0,1)$ such that the point $z = \alpha x^* + (1-\alpha)\hat{x} \in X(N;v)$. For every nonempty coalition $S \subseteq N$,

$$e(S,z) = \alpha e(S,x^*) + (1-\alpha)e(S,\hat{x}).$$

Denote

$$\begin{aligned}
\theta(x^*) &= \big(e(S_1,x^*),...,e(S_{2^n},x^*)\big)\\
\theta(\hat{x}) &= \big(e(R_1,\hat{x}),...,e(R_{2^n},\hat{x})\big)\\
\theta(z) &= \big(e(T_1,z),...,e(T_{2^n},z)\big)
\end{aligned}$$

Since x^* is a nucleolus,

$$\begin{aligned}
\alpha e(T_1,x^*) + (1-\alpha)e(T_1,\hat{x}) = e(T_1,z) \geq e(S_1,x^*) \geq e(T_1,x^*)\\
\Rightarrow e(T_1,\hat{x}) \geq e(T_1,x^*).
\end{aligned}$$

Similarly, Since $\hat{x}$ is a prenucleolus,

$$\begin{aligned}
\alpha e(T_1,x^*) + (1-\alpha)e(T_1,\hat{x}) = e(T_1,z) \geq e(R_1,\hat{x}) \geq e(T_1,\hat{x})\\
\Rightarrow e(T_1,x^*) \geq e(T_1,\hat{x}).
\end{aligned}$$

Hence,

$$e(T_1,z) = e(T_1,x^*) = e(T_1,\hat{x}) = e(S_1,x^*) = e(R_1,\hat{x}).$$

Therefore, T_1 maximizes the excess at x^* and at $\hat{x}$; i.e, by changing the order of the coalitions, one can assume that $T_1 = S_1 = R_1$.
Continuing by induction, for every k such that $1 \leq k \leq 2^n$, we show that $e(T_k,z) = e(S_k,x^*) = e(R_k,\hat{x})$ from which one can assume that $T_k = S_k = R_k$. In other words,

$$e(S,x^*) = e(S,\hat{x}), \quad \forall S \subseteq N.$$

In particular, setting $S = \{i\}$, we deduce that for every player $i \in N$,

$$v(i) - x_i^* = e(\{i\},x^*) = e(\{i\},\hat{x}) = v(i) - \hat{x}_i,$$

and therefore $x_i^* = \hat{x}_i$. This means that $x^* = \hat{x}$, a contradiction.

21.30 We prove that the nucleolus is independent of the names given to the players.
Let $\Pi : N \to N$ be a permutation of N, and let $(N;v)$ and $(N;w)$ be coalitional games satisfying $w(S) = v(\Pi(S))$ for every $S \subseteq N$.

We prove that if x^v and x^w are the nucleoli of these games, then $x_i^w = x_{\pi(i)}^v$ for every player $i \in N$. By Corollary 21.14, the nucleolus of every coalitional game with nonempty set of imputations consists of a single imputation, hence it is enough to show that if x^v is the nucleolus of $(N;v)$, then the vector x^w defined by $x_i^w = x_{\pi(i)}^v$ for every player $i \in N$ is a nucleolus of $(N;w)$.
For every $y^v \in X(N;v)$, denote by y^w the vector in $\mathbf{R}^n$ such that $y_i^w = y_{\Pi(i)}^v$. We first prove that for every $y^v \in X(N;v)$, $y^w \in X(N;w)$.
Let $y^v \in X(N;v)$. y^v is individually rational, hence,

$$y_i^w = y_{\Pi(i)}^v \geq v(\Pi(i)) = w(i) \quad \forall i \in N.$$

In addition, by the efficiency of y^v,

$$\sum_{i \in N} y_i^w = \sum_{i \in N} y_{\Pi(i)}^v = \sum_{i \in N} y_i^v = v(N) = v(\Pi(N)) = w(N).$$

Thus $y^w \in X(N;w)$.
Furthermore, since a permutation Π is an injective mapping, the inverse function $\Pi^{-1} : N \to N$ is well defined permutation satisfying $v(S) = v(\Pi(\Pi^{-1}(S)) = w(\Pi^{-1}(S))$. Hence, for every $y^w \in X(N;w)$, denote by y^v the vector in $\mathbf{R}^n$ such that $y_i^v = y_{\Pi^{-1}(i)}^w$, then, $y^v \in X(N;v)$.
Let x^v be a nucleolus of $(N;v)$. Then $x^v \in X(N;v)$ so the corresponding vector defined above $x^w \in X(N;w)$. Moreover, for every $y^w \in X(N;w)$ and every $S \subset N$

$$e^w(S, y^w) = w(S) - y^w(S) = v(\Pi(S)) - y^v(\Pi(S)) = e^v(\Pi(S), y^v).$$

Therefore, $\theta^w(y^w) = \theta^v(y^v)$ from which it follows that, for every $y^w \in X(N;w)$,

$$\theta^w(x^w) = \theta^v(x^v) \precsim_L \theta^v(y^v) = \theta^w(y^w),$$

so x^w is the nucleolus of $(N;w)$, as needed.

21.31 We prove that the algorithm described in Section 21.4 is well defined. Assume that $X(N;v)$ is nonempty so the nucleolus contains a single point in $X(N,v)$ (otherwise the nucleolus is empty as well as the feasible region of the linear program defined in (21.67)). Set $X_0 = X(N;v)$.
We prove by induction over the step number k, that as long as $|\bigcup_{l=1}^{k-1} \Sigma_l| < 2^n$: (1) $\emptyset \neq X_k \subseteq X_{k-1}$; (2) $\theta(x) \prec_L \theta(y)$ for every $x \in X_k$ and $y \in X(n;v) \setminus X_k$; and (3) $\Sigma_k \neq \emptyset$.

For step $k = 1$: the set $X(N;v)$ is an intersection of finitely many half spaces. In addition, for every coalition $S \subseteq N$, the constraint

$e(S,x) \leq t$ is a half space. Hence, the set of constraints defined in the linear program (21.67), is a nonempty intersection of finitely many half spaces so it is nonempty, compact and convex. Thus the minimum is attained at some point (θ_1, x^1) such that $x^1 \in X(N; v)$, and

$$e(x^1, S) \leq \theta_1 \ \ \forall S \subseteq N. \tag{197}$$

In particular, $x^1 \in X_1 \subseteq X(N; v) = X_0$ so (1) holds. In addition, for every $x \in X_1$ one has $\theta_1(x) \leq \theta_1$, whereas for every $y \in X(N; v) \setminus X_1$ there is $S \subseteq N$ such that $\theta_1(y) \geq e(S, y) > \theta_1$ so $\theta(x) \prec \theta(y)$ and (2) holds. Assume by contradiction that (3) does not hold, i.e., $\Sigma_1 = \emptyset$. Therefore, for every $R \subseteq N$ there is $x^R \in X_1$ such that $e(R, x^R) < \theta_1$. Set $z = \frac{\sum_{R \subseteq N} x^R}{2^n}$. Then, $z \in X(N; v)$ and for every $S \subseteq N$ one has

$$e(S, z) = \frac{\sum_{R \subseteq N} e(S, x^R)}{2^n} < \frac{\sum_{R \subseteq N} \theta_1}{2^n} = \theta_1,$$

which contradicts the minimality of θ_1.

Assume that (1)-(3) hold for every step $k < K$ such that $|\bigcup_{l=1}^{k-1} \Sigma_l| < 2^n$.

Consider step K. The set of constraints in the K-th linear program is given by

$$\begin{aligned}
& e(S,x) = \theta_l && \forall l = 1, 2, ..., K-1, \forall S \in \Sigma_l && (198)\\
& e(S,x) \leq t && \forall S \notin \bigcup_{l=1}^{K-1} \Sigma_l && (199)\\
& x(N) = v(N) && && (200)\\
& x_i = v(i) && \forall i \in N. && (201)
\end{aligned}$$

This set of constrains is a nonempty intersection of finitely many half spaces so it is compact and convex. Furthermore, every point (θ_{K-1}, x) where $x \in X_{K-1}$ is a feasible point so the feasible space is not empty. Thus the minimum is attained at some point (θ_K, x^K) which necessarily satisfies $\theta_K \leq \theta_{K-1}$. By Constrains (198), (199) and (201), every vector x at which the minimum is attained satisfies $x \in X_{K-1}$ and thus $x^K \in X_K \subseteq X_{K-1}$ as claimed in (1).
(2) For every $y \in X_K \setminus X_{K-1}$ Equation (199) does not hold for some $S \notin \bigcup_{l=1}^{K-1} \Sigma_l$. Hence, $\theta(y) \succ_L \theta(x)$ for every $x \in X_K$ since

$$\begin{aligned}
& e(S,x) = \theta_l = e(S,y) && \forall S \in \Sigma_l, l \in \{1, 2, ..., k-1\}\\
& e(S,x) \leq \theta_K && \forall S \notin \bigcup_{l=1}^{K-1} \Sigma_l.
\end{aligned}$$

Furthermore, by the induction hypothesis (condition (2)) and the transitivity of the lexicographic preference, $\theta(y) \succ_L \theta(x)$ for every $y \in X(N;v) \setminus X_{K-1}$ and every $x \in X_K$.
(3) For every $x \in X_K$, $e(R,x) = \theta_l$ for every $R \in \Sigma_l$ for some $\{1,2,...,K-1\}$; and $e(R,x) \le \theta_l$ for every $R \notin \bigcup_{l=1}^{K-1} \Sigma_l$. Assume by contradiction that (3) does not hold, i.e., $\Sigma_K = \emptyset$. Therefore, for every $R \notin \bigcup_{l=1}^{K-1} \Sigma_l$ there is $x^R \in X_K$ such that $e(R,x^R) < \theta_K$. Set $z = \frac{\sum_{R \notin \bigcup_{l=1}^{K-1} \Sigma_l} x^R}{2^n - |\bigcup_{l=1}^{K-1} \Sigma_l|}$. Then, $z \in X(N;v)$ and for every $S \subseteq N$ one has

$$e(S,z) = \frac{\sum_{R \notin \bigcup_{l=1}^{K-1} \Sigma_l} e(S,x^R)}{2^n - |\bigcup_{l=1}^{K-1} \Sigma_l|} = \theta_l, \qquad \textbf{if } S \in \Sigma_l, l \in \{1,2,...,K-1\}$$

$$e(S,z) = \frac{\sum_{R \notin \bigcup_{l=1}^{K-1} \Sigma_l} e(S,x^R)}{2^n - |\bigcup_{l=1}^{K-1} \Sigma_l|} < \theta_K, \qquad \textbf{if } S \notin \bigcup_{l=1}^{K-1} \Sigma_l.$$

which contradicts the minimality of θ_K.

Since $(\Sigma_l)_l$ are nonempty disjoint subsets of $\{S : S \subseteq N\}$, after at most 2^n steps, $|\bigcup_{l=1}^{K-1} \Sigma_l| = 2^n$.

Finally, let K be such that $|\bigcup_{l=1}^{K-1} \Sigma_l| = 2^n$. Then, for every $S \subseteq N$ one has $S \in \Sigma_l$ for some $l \in \{1,2,...,K-1\}$. In particular, $e(S,x) = \theta_l$ for every $x \in X_{K-1}$. Therefore, $\theta(x) = \theta(x')$ for every $x, x' \in X_{K-1}$. However, by condition (2), $\theta(x) \prec_L \theta(y)$ for every $x \in X_{K-1}$ and $y \in X(N;v) \setminus X_{k-1}$. In conclusion, $X_{K-1} \subseteq \mathcal{N}(N;v)$. However, by Corollary 21.14, since $X(N;v)$ is nonempty, then the nucleolus consists of a single imputation so $X_{K-1} = \mathcal{N}(N;v)$, as needed.

21.32 Let $x_1, x_2, ..., x_L$ be imputations in the game $(N;v)$ that include the nucleolus. Assume without loss of generality that x_1 is the nucleolus. Suppose that $\theta_1(x_l) = \theta_1(x_{l'})$ for every l, l' satisfying $1 \le l, l' \le L$, and denote $\Sigma_1 := \{S \subseteq N : e(S,x_l) = \theta_1(x_l) \forall l \in \{1,2,...,L\}\}$. We prove that the collection of coalitions Σ_1 is nonempty.
Since $\theta_1(x_l) = \theta_1(x_{l'})$ for every l, l' satisfying $1 \le l, l' \le L$,

$$e(S,x_l) \le \theta_1(x_l) = \theta_1(x_1) \quad \forall S \subseteq N, \quad \forall 1 \le l \le L.$$

Assume by contradiction that $\Sigma_1 = \emptyset$. That is, for every coalition $S \subseteq N$ there is $1 \le l_S \le L$ such that $e(S, x_{l_S}) < \theta_1(x_{l_S})$. Set $x = \frac{1}{L}\sum_{l=1}^{L} x_l$. Hence,

$$e(S,x) = \tfrac{1}{L} \sum_{l=1}^{L} e(S,x_l) < \theta_1(x_1) \quad \forall S \subseteq N, \tag{202}$$

which contradicts the minimality of the nucleolus.

21.33 An example of a monotonic game that is not 0-monotonic: $(\{1,2\};v)$ where $0 < v(1) = v(2) = v(1,2)$.

21.34 We show that a simple, constant-sum game need not be 0-monotonic. Consider the following counter example.

$$v(1) = v(2) = 1, v(3) = 0, v(1,2) = 1, v(1,3) = v(2,3) = 0, v(N) = 1.$$

The game is a simple zero-sum game, but

$$v(1,2) = 1 < 2 = v(1) + v(2),$$

so it is not 0-monotonic.

21.35 We prove that every convex game is 0-monotonic.
Let $(N;v)$ be a convex game. That is, for every pair of coalitions $S,T \subseteq N$, $v(S) + v(T) \leq v(S \bigcup T) + v(S \cap T)$.
In particular, for every $i \in N$ and every $S \subseteq N \setminus \{i\}$,

$$v(S) + v(i) \leq v(S \bigcup \{i\}) + v(S \cap \{i\}) = v(S \bigcup \{i\}).$$

21.36 Every null player is a dummy player. Indeed, let $i \in N$ be a null player, so $v(i) = 0$ and for every $S \subseteq N \setminus \{i\}$, $v(S \bigcup \{i\}) = v(S)$. Therefore,

$$v(S) + v(i) = v(S) = v(S \bigcup \{i\}).$$

However, there are dummy players that are not null players, for instance, let $(\{1,2\}, v)$ be a game where $v(1) = v(2) = 1$ and $v(1,2) = 2$. In this game, both players are dummy players but none of them is a null player.

21.37 Using Theorem 21.29, we prove that the imputation $x = (\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$ is not the prenucleolus of the gloves game that appears in Example 21.3 and that the prenucleolus of this game is $y = (0,0,1)$.
First,

$$\begin{aligned}\mathcal{D}(\alpha, x) &= \{S \subset N : S \neq \emptyset, v(S) - x(S) \geq \alpha\} \\ &= \begin{cases} \{\{1,3\},\{2,3\}\} & -\frac{1}{3} < \alpha \leq \frac{1}{3} \\ \{\{1\},\{2\},\{3\},\{1,3\},\{2,3\}\} & -\frac{2}{3} < \alpha \leq -\frac{1}{3} \\ \{\{1\},\{2\},\{3\},\{1,2\},\{1,3\},\{2\}\} & -\frac{2}{3} \geq \alpha. \end{cases}\end{aligned}$$

However, $\mathcal{D}(0, x) = \{\{1,3\}, \{2,3\}\}$ is not a balanced collection and thus x is not the prenucleolus of the game.
Next,

$$\mathcal{D}(\alpha, y) = \{S \subset N : S \neq \emptyset, v(S) - y(S) \geq \alpha\}$$
$$= \begin{cases} \{\{1\}, \{2\}, \{1,2\}, \{1,3\}, \{2,3\}\} & -1 < \alpha \leq 0 \\ \{\{1\}, \{2\}, \{3\}, \{1,2\}, \{1,3\}, \{2\}\} & -1 \geq \alpha. \end{cases}$$

However, $\mathcal{D}(\alpha, x)$ is a balanced collection and thus y is the prenucleolus of the game.

21.38 Eilon is not correct. Denote the nucleolus by x^*, and by α_1 be the greatest value of the excesses at $\theta(x^*)$. Then for every imputation x, there is at least one coalition S such that

$$y(S) = (x^* - x)(S) = e(S, x) - e(S, x^*) \geq 0,$$

but not necessarily for every coalition S.

21.39 Let $[E; \frac{E}{2}, \frac{E}{2}]$ be a weighted majority game. This game is not constant sum since $v(1) + v(2) = 0 \neq 1 = v(1,2)$. However, it satisfies the property that that if S is a winning coalition, then S^c is a losing coalition. Indeed, the only winning coalition is $N = \{1,2\}$ and $N^c = \emptyset$ is a losing coalition.

21.40 Let $[E; \frac{E}{2}, \frac{E}{2}, \frac{E}{2}, \frac{E}{2}]$ be a weighted majority game. In this game a coalition is a winning coalition if and only if its size is at least 2. The game is not constant sum since $v(1,2) + v(3,4) = 2 \neq 1 = v(N)$. In addition, it satisfies the property that if S is a losing coalition, then $|S| \leq 1$ so $|S^C| \geq 2$ and thus S^c is a winning coalition.

21.41 We prove that the game $[5; 2, 2, 2, 1, 1, 1]$ does not have a homogeneous representation.
Assume by contradiction that the game has a homogeneous representation $[q; w_1, w_2, w_3, w_4, w_5, w_6]$ such that $\sum_{i \in S} w_i$ is the same for every minimal winning coalition $S \in \mathcal{W}^m$. Then

$$w_1 + w_2 + w_3 = w_1 + w_2 + w_4 \Rightarrow w_3 = w_4.$$

In a similar way, one can deduce that

$$w_1 = w_2 = w_3 = w_4 = w_5 = w_6,$$

from which it follows that $w_1 + w_2 + w_3 = w_4 + w_5 + w_6$ so $\{3, 4, 5\}$ is also a winning coalition, a contradiction.

21.42 Let $[q; w_1, w_2, ..., w_n]$ and $[\hat{q}; \hat{w}_1, \hat{w}_2, ..., \hat{w}_n]$ be two representations of the same simple game.
Therefore, S is a winning coalition if and only if $\sum_{i\in S} w_i \geq q$ if and only if $\sum_{i\in S} \hat{w}_i \geq \hat{q}$.
Hence, for every winning coalition $S \subseteq N$,

$$\sum_{i\in S}(w_i + \hat{w}_i) = \sum_{i\in S} w_i + \sum_{i\in S} \hat{w}_i \geq q + \hat{q}.$$

Likewise, for every losing coalition $S \subseteq N$,

$$\sum_{i\in S}(w_i + \hat{w}_i) = \sum_{i\in S} w_i + \sum_{i\in S} \hat{w}_i < q + \hat{q}.$$

In particular, $[[q + \hat{q}; w_1 + \hat{w}_1, w_2 + \hat{w}_2, ..., w_n + \hat{w}_n]$ is also a representation of that game.

21.43 Let $[q; w]$ be a representation of a weighted majority game $(N; v)$, then for every winning coalition S there is a minimal winning coalition $\hat{S}$ such that $\hat{S} \subseteq S$, hence

$$w(S) \geq w(\hat{S}) \geq \min_{S'\in\mathcal{W}^m} w(S') = q(w).$$

On the other hand, since $q(w) = \min_{S'\in\mathcal{W}^m} w(S') \geq q$, for every losing coalition $S \subseteq N$, $w(S) < q \leq q(w)$.
Therefore, $[q(w); w]$ is also a representation of that game.

21.44 Let $(N; v)$ be a simple strong game, and let $x \in X(N; v)$ be an imputation. The following example show that the game $[q(x); x]$ is not necessarily a simple strong game. Let $(N; v)$ be a three player simple strong game where S is a winning coalition if and only if $|S| \geq 2$ and let $x = [0, \frac{1}{2}, \frac{1}{2}]$ be an imputation. Then the game $[q(x); x]$ is a simple game where a coalition S is a winning coalition if and only if $x(S) \geq q(x) = \frac{1}{2}$. However, this game is not strong since both coalitions, $\{1, 2\}$ and $\{3\}$, are winning coalitions in the game $[q(x); x]$ from which it follows that the game $[q(x); x]$ in not a constant sum game.

21.45 Let $(N; v)$ be a coalitional game with $N = \{1, 2, 3\}$ and a coalitional function given by

$$v(S) = \begin{cases} 1 & |S| \geq 2, \\ 0 & |S| \leq 1. \end{cases}$$

(a) The game, as claimed in previous exercise is a simple strong game, since it is monotonic, $v(N) = 1$, and for every $S \subseteq N$, $v(S) = 1$ if and only if $|S| \geq 2$, if and only if $|S^c| \leq 1$, if and only if $v(S^c) = 0$.

(b) Let $x \in X(N;v)$ be an imputation, i.e., $x(N) = 1$ and $x_i \geq 0$ for every $i \in N$. Then

$$q(x) = \min\{x(1,2), x(1,3), x(2,3)\} = \min\{1 - x_3, 1 - x_2, 1 - x_1\}.$$

Let $(N;w)$ be the coalitional game corresponding to $[q(x);x]$. By the definition of $q(x)$,

$$w(1,2) = w(1,3) = w(2,3) = w(N) = 1.$$

Hence, the game $(N;w)$ is a simple strong game if and only if $w(1) = w(2) = w(3) = 0$ if and only if $\max_{i \in N} < \min_{i \in N}(1 - x_i)$ if and only if $x_i < 1 - x_i$ for every $i \in N$. In conclusion, the game $[q(x);x]$ is a simple strong game if and only if $x_i < \frac{1}{2}$ for every $i \in N$.

(c) Following previous part, if there is a player $i \in N$ such that $x_i \geq \frac{1}{2}$ then $q(x) \leq \frac{1}{2}$, hence, by Theorem 21.50, $[q(x);x]$ does not represent $(N;v)$.

21.46 The Rabbi Nathan solutions of the given bankruptcy problems are:

(a) There are four wives, who are owed 100, 200, 300, and 400, respectively, out of an estate of 350. Then the solution is 50, 100, 100, and 100, respectively.

(b) There are four wives, who are owed 100, 200, 300, and 400, respectively, out of an estate of 720. Then the solution is 50, $123\frac{1}{3}$, $223\frac{1}{3}$, and $323\frac{1}{3}$, respectively.

(c) There are five wives, who are owed 60, 120, 180, 240, and 300, respectively, out of an estate of 600. Then the solution is 30, 60, 110, 170, and 230, respectively.

21.47 The function f defined in Equation (21.166) is bilinear function in every range so it is continuous in the interior of each range. Furthermore, every appropriate boundaries are identical, for instance if $E = d_1 = d_2$ then the solution according all the ranges is $(\frac{E}{2}, \frac{E}{2})$. Hence, the function is continuous.
We next prove that for every fixed d_2, the first coordinate of f is monotonically nondecreasing in d_1.
Assume first that $E \leq d_2$, then the first coordinate of f is

$$f_1(E; d_1, d_2) = \begin{cases} \frac{E}{2} & E \leq d_1 \\ \frac{d_1}{2} & E > d_1. \end{cases}$$

However, for every $E \leq d_1$, $f_1(E; d_1, d_2)$ is constant with regards to d_1, and for every $E > d_1$, $f_1(E; d_1, d_2)$ is a monotonic increasing linear function. In conclusion, the first coordinate of f is monotonically nondecreasing in d_1.
Assume next that $E > d_2$, then the first coordinate of f is

$$f_1(E; d_1, d_2) = \begin{cases} E - \frac{d_2}{2} & E \leq d_1 \\ \frac{E + d_1 - d_2}{2} & E > d_1. \end{cases}$$

However, for every $E \leq d_1$, $f_1(E; d_1, d_2)$ is constant with regards to d_1, and for every $E > d_1$, $f_1(E; d_1, d_2)$ is a monotonic increasing linear function. In conclusion, the first coordinate of f is monotonically nondecreasing in d_1.
By the symmetry of d_1 and d_2, it follows that for every fixed d_1, the second coordinate of f is monotonically nondecreasing in d_2.

21.48 (a) The Rabbi Nathan solution to bankruptcy problems satisfies the property of symmetry (creditors with identical claims receive identical payments). Symmetric players have identical containers, hence, by the communicating vessels law, the height of the liquid on both sides of the system must be equal.

(b) The Rabbi Nathan solution to bankruptcy problems satisfies the property of null player (a creditor with a claim of 0 receives nothing). A null player has two containers of height zero, so no liquid flows into these containers.

(c) The Rabbi Nathan solution to bankruptcy problems does not satisfy the property of covariance under strategic equivalence. For instance, $\varphi(7; 1, 8) = (\frac{1}{2}, 6\frac{1}{2})$, whereas

$$\varphi(7 + (1+1); 1+1, 8+1) = \varphi(9; 2, 9) = (1, 8) \neq \varphi(7; 1, 8) + (1, 1).$$

(d) The Rabbi Nathan solution to bankruptcy problems does not satisfy the property of additivity. A counter example: $\varphi(3; 3, 1) = (2\frac{1}{2}, \frac{1}{2})$, $\varphi(3; 1, 8) = (\frac{1}{2}, 2\frac{1}{2})$ but

$$\varphi(3; 3, 1) + \varphi(3; 1, 8) = (3, 3) \neq (2, 4) = \varphi(6; 4, 9).$$

21.49 A solution concept to bankruptcy problems with n creditors that is consistent with **John's solution** for the bankruptcy problem with two creditors is the Rabbi Nathan solution.
A solution concept to bankruptcy problems with n creditors that is consistent with **Clarence's solution** for the bankruptcy problem with

two creditors is

$$\varphi_i(E;d_1,...,d_n) = d_i - \tfrac{\sum_{j=1}^n d_j - E}{n}.$$

A solution concept to bankruptcy problems with n creditors that is consistent with **Ruth's solution** for the bankruptcy problem with two creditors is

$$\varphi_i(E;d_1,...,d_n) = \tfrac{d_i}{\sum_{j=1}^n d_j}E.$$

21.50 (a) A system of containers that implements the proportional division solution includes n containers on the ground connected by a dimensionless pipe. The claim of creditor i is depicted by a container of unit height, where the claim of d_i is represented by the base area. To compute the solution, we let a volume of liquid equal to E flow into the system. By the communicating vessels law, the height of the liquid in all containers of the system must be equal. The volume of liquid in each container represents the payment given to creditor.

(b) The proportional division solution is a consistent solution concept. Indeed, for every $T \subseteq N$ set of creditors , let $d^t = (d_i)_{i\in T}$, and let

$$x^T = \sum_{i\in T} \varphi_i(E;d) = \tfrac{\sum_{i\in T} d_i}{\sum_{k=1}^n d_k}E.$$

Then, for every $i \in T$,

$$\varphi_i(x^T;d^T) = \tfrac{d_i}{\sum_{k\in T} d_k}x = \tfrac{d_i}{\sum_{k\in T} d_k} \cdot \tfrac{\sum_{k\in T} d_k}{\sum_{k=1}^n d_k}E = \tfrac{d_i}{\sum_{k=1}^n d_k}E = \varphi_i(E;d),$$

as needed.

(c) The properties that are satisfied by the proportional division solution are (1) Symmetry: if $d_i = d_j$ then $\frac{d_i}{\sum_{k=1}^n d_k}E = \frac{d_j}{\sum_{k=1}^n d_k}E$; and (2) The null player property: if $d_i = 0$ then $\frac{0}{\sum_{k=1}^n d_k}E = 0$.

However, it does not satisfy the covariance under strategic equivalence property: since $\varphi(2;2,2) = (1,1)$ but

$$\varphi(2+0+1;2+0,2+1) = (\tfrac{4}{5},\tfrac{6}{5}) \neq (1+0,1+1).$$

In addition it does not satisfy additivity since

$$\varphi(2;3,3) + \varphi(3;2,4) = (1,1) + (1,2) = (2,3) \neq (2\tfrac{1}{12}, 2\tfrac{11}{12}) = \varphi(5;5,7).$$

21.51 (a) A system of containers that implements the given solution includes n containers on the ground connected by a dimensionless pipe. The claims of the creditors are depicted by containers of unit base area, where the claim of d_i is represented by a container of height d_i. Assume that each container is closed from above. To compute the solution, we let a volume of liquid equal to E flow into the system. By the communicating vessels law, the height of the liquid in all containers of the system must be equal, unless the container is full. The volume of liquid in each container represents the payment given to creditor.

(b) The solution is a consistent solution by the communicating vessels law since the containers of each subset of creditors are the same as in the original bankruptcy problem as well as the volume of liquid that is flowed into them.

(c) The solution satisfies the properties of symmetry (symmetry creditors own identical containers) and the null player property (a null player owns a container of height zero) by the communicating vessels law.

21.52 Let φ be a solution concept to the bankruptcy problem. The dual solution of φ is the solution concept φ^* defined as follows:

$$\varphi_i^*(E; d) = d_i - \varphi_i(D - E, d), \quad \forall i \in N.$$

where $D := \sum_{i=1}^n d_i$ is the sum of the claims.

(a) φ^* is a solution concept. Indeed,

$$0 \le \varphi_i(D - E, d) \le d_i \Rightarrow 0 \le d_i - \varphi_i(D - E, d) \le d_i.$$

Furthermore,

$$\sum_{i=1}^n \varphi_i^*(E; d) = \sum_{i=1}^n d_i - \sum_{i=1}^n \varphi_i(D - E, d) = D - (E - D) = E.$$

(b) Let φ be the Rabbi Nathan solution. Since the claims of the creditors in both games are identical, the same system of containers implements both $\varphi(E; d)$ and $\varphi(D - E; d)$. Assume that we let a volume of liquid equal to E into the system, mark the height of the liquid and empty the system. Then we turn the system upside down and let a volume of liquid equal to $D - E$ into the system. Since $E + (D - E) = D$, the height of the liquid in the second phase must reach the marking of the liquid from first phase, so $\varphi_i(E; d) + \varphi_i(D - E; d) = d_i$ for every $i \in N$. In particular,$\varphi_i^*(E; d) = d_i - \varphi_i(D - E, d) = \varphi_i(E; d)$ for every $i \in N$ and thus the Rabbi Nathan solution is self-dual.

(c) Let $\varphi(E;d)$ be the proportional division solution. Then

$$\varphi_i^*(E;d) = d_i - \varphi(D-E;d) = d_i - \tfrac{d_i}{D}(D-E) = \tfrac{d_i}{D}E = \varphi(E;d).$$

(d) Let $\varphi(E;d)$ be the solution proposed by Clarence in Exercise 21.49. Then

$$\varphi_i^*(E;d) = d_i - \varphi(D-E;d) = d_i - (d_i - \tfrac{D-E}{n}) = \tfrac{D-E}{n}.$$

21.53 Let $[E;d_1,d_2,...,d_n]$ be a bankruptcy problem, where $d_1 \le d_2 \le ... \le d_n$. Denote $D := \sum_{i=1}^n d_i$, and for every $i \in N$, denote $D^i := \sum_{j=1}^i d_j$. Let φ be the Rabbi Nathan solution.
If $E \le n\frac{d_1}{2} = n\frac{D^1}{2}$, then the height of the bottom container corresponding each creditor i is at least d_1. Hence, as long as the liquid volume being streamed into the system is at most nd_1, the volume in all bottom containers is identical. Thus the solution is $(\frac{E}{n},...,\frac{E}{n})$.
If $\frac{D^i}{2} + (n-i)\frac{d_i}{2} < E \le \frac{D^{i+1}}{2} + (n-i-1)\frac{d_{i+1}}{2}$ for some $i \in N$, then by letting a volume of liquid equal to E flow into the system, (1) the height of the liquid of every container $j \le i$, is $\frac{d_j}{2}$ (that is, the bottom container is full); (2) the height of the liquid of every container $j > i$ is more than $\frac{d_i}{2}$ and it identical to all of the containers of creditors $i+1,...,n$. In particular, the solution is $\frac{d_j}{2}$ for every $j \le i$, and $\frac{E-D^i}{n-i}$ for every $j > i$.
Finally, if $\frac{D}{2} < E \le D$ then $0 \le D-E < \frac{D}{2}$ and therefore $\varphi(D-E;d)$ is as mentioned above. Hence, using Exercise 21.52, the Rabbi Nathan solution is $\varphi(E;d) = \varphi^*(E;d) = d - \varphi(D-E;d)$.

21.54 Let $[E;d_1,...,d_n;d_{n+1},...,d_{n+m}]$ be a corporation bankruptcy problem.

(a) We construct a system of containers implementing a solution to the corporate bankruptcy problem as follows. We first construct two systems of containers accordung to the Rabbi Nathan solution, one to each class of shares. We then put the system corresponding the class of preferred shareholders on the ground, and on the top of it, we put the other system corresponding the class of the regular shareholders. Finally, we connect the two systems by a dimensionless pipe. If $E < \sum_{i=1}^N d_i$ then the solutions to the preferred shareholders and the regular shareholders are identical to the Rabbi Nathan solutions of $(E;d_1,...,d_n)$ and $(0;d_{n+1},...,d_{n+m})$, respectively, which are consistent with the "contested garment" solution. If, on the other hand, $E \ge \sum_{i=1}^N d_i$ then the solutions to the preferred shareholders and the regular shareholders are identical to the Rabbi Nathan solutions of

$(\sum_{i=1}^{n} d_i; d_1, ..., d_n)$ and $(E - \sum_{i=1}^{n} d_i; d_{n+1}, ..., d_{n+m})$, respectively, which are consistent with the "contested garment" solution.

(b) As claimed before, if $E < \sum_{i=1}^{N} d_i$, then under each solution concept, every regular player receives zero, hence the problem can be reduced to $[E; d_1, ..., d_n]$ which has, by Theorem 21.59, a unique solution concept that is consistent with the "contested garment" solution-the Rabbi Nathan solution. Similarly, if $E \geq \sum_{i=1}^{N} d_i$, then under each solution concept, every preferred player receives the value of the share he holds, hence the problem can be reduced to $[E - \sum_{i=1}^{n} d_i; d_{n+1}, ..., d_{n+m}]$ which has, by Theorem 21.59, a unique solution concept that is consistent with the "contested garment" solution-the Rabbi Nathan solution.

21.55 Let $[E; d_1, d_2, ..., d_n]$ be a bankruptcy problem satisfying $\sum_{i=1}^{n} d_i > E$. Consider the game $(N; v)$ defined by

$$v(S) := \max\{E - \sum_{i \notin S} d_i, 0\}.$$

We prove that $(N; v)$ is convex using Theorem 18.54. That is, we prove that for every $i \in N$ and every $S \subseteq T \subseteq N \setminus \{i\}$ one has

$$v(S \bigcup \{i\}) - v(S) \leq v(T \bigcup \{i\}) - v(T). \tag{203}$$

First note that since $d_j \geq 0$ for every j, $(N; v)$ is monotonic.
In addition, $v(S \bigcup \{i\}) - v(S) \leq d_i$. Indeed, if $v(S) > 0$ then $v(S \bigcup \{i\}) > 0$ so

$$v(S \bigcup \{i\}) - v(S) = (E - \sum_{j \notin S \bigcup \{i\}} d_j) - (E - \sum_{j \notin S} d_j) = d_i.$$

If $v(S) = 0$, then $E \leq \sum_{j \notin S} d_j$ from which it follows that either $v(S \bigcup \{i\}) = 0$ which case

$$v(S \bigcup \{i\}) - v(S) = 0,$$

or $v(S \bigcup \{i\}) = E - \sum_{j \notin S \bigcup \{i\}} d_j$ which case

$$v(S \bigcup \{i\}) - v(S) = v(S \bigcup \{i\}) = E - \sum_{j \notin S \bigcup \{i\}} d_j = E - \sum_{j \notin S} d_j + d_i \leq 0 + d_i,$$

as claimed.
Therefore, if $v(T) > 0$ then $v(T \bigcup \{i\}) > 0$ so $v(T \bigcup \{i\}) - v(T) = d_i$, whereas $v(S \bigcup \{i\}) - v(S) \leq d_i$ and Inequality (203) holds. Otherwise, if $v(T) = 0$ then, by the monotonicity of $(N; v)$, $v(S) = 0$, hence again Inequality (203) holds by the monotonicity, and we are done.

21.56 We prove that if one associates a bankruptcy problem $[E; d_1, d_2, ..., d_n]$ with the coalitional game $(N; w)$ where $w(S) = E - \sum_{i \in S^c} d_i$ (unless $S = \emptyset$ which case $w(\emptyset) = 0$), and if $\widetilde{x}$ is the Shapley value of this game, then the Hart–Mas-Colell reduced game $(S; w_S^{\widetilde{x}})$ is the coalitional game corresponding to the bankruptcy problem $[\widetilde{x}(S); (di)_{i \in S}]$.
Denote $D = \sum_{j=1}^{n} d_j$ and $D^T = \sum_{j \in T} d_j$ for every coalition $T \subset N$.
We first calculate the Shapley value. For every $i \in N$,

$$\left.\begin{aligned} v(i) - v(\emptyset) &= E - D^{N \setminus \{i\}} \\ v(S \bigcup \{i\}) - v(S) &= E - D^{S^c \setminus \{i\}} - (E - D^{S^c}) = d_i \quad \forall S \subseteq N \setminus \{i\} \end{aligned}\right\}$$
$$\Rightarrow \widetilde{x}_i = \tfrac{1}{n}(E - D^{N \setminus \{i\}}) + \tfrac{n-1}{n} d_i = \tfrac{1}{n}(E - D) + d_i.$$

Hence, for every nonempty coalition $S \subseteq N$,

$$\widetilde{x}(S) = \tfrac{|S|}{n}(E - D) + D^S.$$

Denote by $(S; v^S)$ the coalitional game corresponding to the bankruptcy problem $[\widetilde{x}(S); (d_i)_{i \in S}]$. Then $v^S(\emptyset) = 0$ and for every coalition $R \subseteq S$,

$$v^S(R) = \widetilde{x}(S) - D^{S \setminus R} = \tfrac{|S|}{n}(E - D) + D^R.$$

We now show that $w_S^{\widetilde{x}} = v^S$. For $R = \emptyset$, $w_S^{\widetilde{x}}(\emptyset) = 0 = v^S(\emptyset)$. For $R = S$, $w_S^{\widetilde{x}}(S) = \widetilde{x}(S) = v^S(S)$.
Finally, let $R \subset S$ be a nonempty coalition. For every $Q \subseteq N \setminus S$,

$$v(R \bigcup Q) = E - D^{N \setminus (R \cup Q)} = E - D + D^{R \cup Q}$$
$$\Rightarrow v(R \bigcup Q) - \widetilde{x}(Q) = \tfrac{n - |Q|}{n}(E - D) + D^R.$$

However, since $E < D$ (by the definition of a bankruptcy problem), the maximum of $v(R \cup Q) - \widetilde{x}(Q)$ is obtain at $Q = N \setminus S$, so

$$w_S^{\widetilde{x}}(R) = \max_{Q \subseteq N \setminus S} (v(R \bigcup Q) - \widetilde{x}(Q)) = \tfrac{|S|}{n}(E - D) + D^R = v^S(R),$$

as needed.

21.57 We prove that the Rabbi Nathan solution is consistent according to Definition 21.67.
Let $(N; v)$ be the coalitional game corresponding to a bankruptcy problem $[E; (d_i)_{i \in N}]$. Let $x^* \in \mathbf{R}^n$ be the Rabbi Nathan solution to the bankruptcy problem.
Then by Theorem 21.66, x^* is the nucleolus of the game $(N; v)$.
Let $T \subseteq N$ be a nonempty coalition. Let $[x^*(T), (d_i)_{i \in T}]$ be the bankruptcy problem restricted to T with estate $x^*(T)$, and let $(T; w)$ be the corresponding coalitional game. By Theorem 21.63, since

$x^*(N) = E$ and $0 \leq x_i^* \leq d_i$ for every $i \in N$, $(T; w) = (T; w_T^{x^*})$, where $(T; w_T^{x^*})$ is the Davis–Maschler reduced game of $(N; v)$ to coalition T at x. Furthermore, by Theorem 21.64, $(T; w_T^{x^*})$ is 0-monotonic. Therefore, following Theorem 21.38, x_T^* is the nucleolus of the game $(T; w_T^{x^*})$ and thus it is the Rabbi Nathan solution to the bankruptcy problem restricted to T.

Chapter 22: Social choice

22.1 (a) A is the set of all subsets of some set S, and $a \succsim b$ if and only if b is a subset of a. The relation is reflexive (and hence it is not irreflexive), and transitive. However, it is not complete, for instance $A \not\subseteq A^c$ and $A^c \not\subseteq A$.

(b) A is the set of all natural numbers, and $a \succsim b$ if and only if b is a divisor of a. The relation is reflexive, transitive, but it is not complete ($3 \not\succsim 4$ and $4 \not\succsim 3$).

(c) A is the set of all 26 letters in the Latin alphabet, and $\alpha \succsim \beta$ if and only if $\alpha\beta$ is a standard word in English. The relation is not complete, reflexive, irreflexive, and transitive.

(d) A is the set of all natural numbers, and $a \succsim b$ if and only if $a \times b = 30$. The relations is irreflexive (there is no $n \in \mathbf{N}$ such that $n^2 = 30$) but it is neither complete nor transitive.

(e) A is the set of all human beings, past and present, and $b \succsim a$ if and only if a is a descendant of b. The relation is irreflexive and transitive but it is not complete (if a and b are brothers then neither $a \succsim b$ nor $b \succsim a$).

(f) A is the set of people living in a particular neighborhood, and $a \succsim b$ if and only if a likes b. The relation is not necessarily complete, reflexive, irreflexive and transitive.

None of the above relations is complete and thus none of them is a preference relation.

22.2 Let P^* be a strict preference relation over A, i.e., P^* is a complete, irreflexive, and transitive binary relation. Set

$$P := P^* \bigcup \{(a, a) : a \in A\}.$$

We show that P is reflexive, complete and transitive, and thus P is a preference relation.

- P is reflexive since $\{(a, a) : a \in A\} \subset P$.

- P is complete: let $a, b \in A$. If $a = b$ then $(a, b) = (a, a) \in P$. If $a \neq b$, then since P^* is complete, either $(a, b) \in P^* \subset P$ or $(b, a) \in P^* \subset P$.
- P is transitive: let $x, y, z \in A$ such that $(x, y) \in P$ and $(y, z) \in P$. If $x = y$(or $y = z$), it follows trivially that $(x, z) \in P$. Otherwise, $(x, y) \in P^*$ and $(y, z) \in P^*$. Hence, by the transitivity of P^*, $(x, z) \in P^* \subset P$.

In conclusion, the relation P is transitive, in addition to reflexive and complete. In particular, P is a preference relation.

22.3 Let V be a set of voters and C be a set of candidates.

(a) For every pair of candidates a and b, let $N_{a,b}$ be the number of voters ranking a ahead of b. For every voter $i \in V$ and every pair of candidates $a, b \in C$ such that $a \neq b$ set

$$N^i_{a,b} = \begin{cases} 1 & \text{if } i \text{ ranking } a \text{ ahead of } b \\ 0 & \text{otherwise.} \end{cases}$$

For every voter i and a candidate a, the Borda points from i to a is the number of candidates i are ranked below a. Hence the Borda points from i to a is $\sum_{b \neq a} N^i_{a,b}$. Therefore, the Borda ranking of candidate a equals

$$\sum_{i \in V} \sum_{b \neq a} N^i_{a,b} = \sum_{b \neq a} \sum_{i \in V} N^i_{a,b} = \sum_{b \neq a} N_{a,b}.$$

(b)

No. of committee members	**A**	**B**	**C**
23	2	1	0
2	1	2	0
17	0	2	1
10	1	0	2
8	0	1	2
Borda ranking	$23 \cdot 2 + 12 \cdot 1 =$ **58**	$19 \cdot 2 + 31 \cdot 1 =$ **69**	$18 \cdot 2 + 17 \cdot 1 =$ **53**

B is Borda winner.

22.4 (a) The strict preference relation amassing the greatest number of Condorcet points is not necessarily unique. For instance, Let

$\{1,2,3\}$ be a set of voter and let $\{A,B,C\}$ be a set of Candidates. Suppose that the preference relations are

$$\begin{aligned}
1 :& A \succ^1 B \succ^1 C \\
2 :& B \succ^2 C \succ^2 A \\
3 :& C \succ^3 A \succ^3 B.
\end{aligned}$$

Then the Condorcet points amassed by the strict preferences relations:

Preference relation	**Condorcet points**
$A \succ B \succ C$	$3+1+1=5$
$A \succ C \succ B$	$2+0+2=4$
$B \succ A \succ C$	$2+2+0=4$
$B \succ C \succ A$	$1+3+1=5$
$C \succ A \succ B$	$1+1+3=5$
$C \succ B \succ A$	$0+2+2=4$

where there are three strict preference relations amassing the greatest number of Condorcet points.

(b) We show that if there exists a Condorcet winner a, then every strict preference relation receiving the maximal number of Condorcet points ranks the Condorcet winner highest in its preference ordering.

Let P be a strict preference that does not rank a highest in its preference ordering. Consider the strict preference $Z(P,P;\{a\})$. The preferences over the set of alternatives $A \setminus \{a\}$ according to both P and $Z(P,P;\{a\})$ are identical and thus each pair of alternatives in $A \setminus \{a\}$ grants the same number of Condorcet points to both preference relations P and $Z(P,P;\{a\})$.

Furthermore, since a is the Condorcet winner, he defeats every other alternative by majority vote in a head to-head competition. Thus, for every $b \in A \setminus \{a\}$, there are more voters that grant more Condorcet points to a than to b. Therefore, the preference $Z(P,P;\{a\})$ that ranks a highest in its preference ordering, amasses more Condorcet points than P, which contradicts P beeing the strict preference relation receiving the maximal number of Condorcet points.

(c) Example 22.1:

The preference relation	A, B, C **(1)**	A, C, B **(7)**	B, C, A **(7)**	C, B, A **(6)**	Condorcet points
$A \succ B \succ C$	$3 \cdot 1$	$2 \cdot 7$	$1 \cdot 7$	$0 \cdot 6$	**24**
$A \succ C \succ B$	$2 \cdot 1$	$3 \cdot 7$	$0 \cdot 7$	$1 \cdot 6$	**29**
$B \succ A \succ C$	$2 \cdot 1$	$1 \cdot 7$	$2 \cdot 7$	$1 \cdot 6$	**29**
$B \succ C \succ A$	$1 \cdot 1$	$0 \cdot 7$	$3 \cdot 7$	$2 \cdot 6$	**34**
$C \succ A \succ B$	$1 \cdot 1$	$2 \cdot 7$	$1 \cdot 7$	$2 \cdot 6$	**34**
$C \succ B \succ A$	$0 \cdot 1$	$1 \cdot 7$	$2 \cdot 7$	$3 \cdot 6$	**39**

The preference relation that the Condorcet Method chooses: $C \succ B \succ A$.
Example 22.2:

The preference relation		Condorcet points
$A \succ B \succ C$	$23 \cdot 3 + 2 \cdot 2 + 17 \cdot 1 + 10 \cdot 1 + 8 \cdot 0 =$	**100**
$A \succ C \succ B$	$23 \cdot 2 + 2 \cdot 1 + 17 \cdot 0 + 10 \cdot 2 + 8 \cdot 1 =$	**76**
$B \succ A \succ C$	$23 \cdot 2 + 2 \cdot 3 + 17 \cdot 2 + 10 \cdot 0 + 8 \cdot 1 =$	**94**
$B \succ C \succ A$	$23 \cdot 1 + 2 \cdot 2 + 17 \cdot 3 + 10 \cdot 1 + 8 \cdot 2 =$	**104**
$C \succ A \succ B$	$23 \cdot 1 + 2 \cdot 0 + 17 \cdot 1 + 10 \cdot 3 + 8 \cdot 2 =$	**86**
$C \succ B \succ A$	$23 \cdot 0 + 2 \cdot 1 + 17 \cdot 2 + 10 \cdot 2 + 8 \cdot 3 =$	**80**

The preference relation that the Condorcet Method chooses: $B \succ C \succ A$.

22.5 (a) The Condorcet winner is A who defeats both B (by a vote of 41 to 40) and C (by a vote of 60 to 21).

(b) The Borda ranking of A, B and C are 101, 109, and 33 respectively. In particular, the Borda winner is B.

(c) Condorcet's critique of the Borda Method is that it is not reasonable to choose B as a winner whereas A is preferred to both B and C by most voters.

A counterclaim to Condorcet's critique:

(a) There are two kinds of Condorcet components,

No. of voters	First candidate	Second candidate	Third candidate
n	A	B	C
n	B	C	A
n	C	A	B

where $n = 1, 2, ..., 10$, and

No. of voters	First candidate	Second candidate	Third candidate
1	A	C	B
1	C	B	A
1	B	A	C

(b) Repeatedly removing all the individuals appearing in one of the Condorcet components leads to the following.

No. of voters	First candidate	Second candidate	Third candidate
20	A	B	C
28	**B**	A	C

The Condorcet winner is B.

(c) The counterclaim asserted that all the difficulties come from Condorcet components. That is, since all voters in Condorcet components "neutralize" each other, they should be removed which leads to a Condorcet winner that is identical to Borda winner.

22.6 We show that there is at most one dictator in every social welfare function F over A. Assume by contradiction that there are $i, j \in N$ where $i \neq j$ such that both i and j are dictators in F. Let $a, b \in A$. Let P^N be a profile of strict preferences satisfying $a \succ_{P_i} b$ and $b \succ_{P_j} a$. Hence, since i is a dictator, $a \succ_{F(P^N)} b$, whereas since j is a dictator, $b \succ_{F(P^N)} a$, a contradiction.

22.7 (a) The social welfare function described here is not dictatorial. No member i of the committee is a dictator. For instance, if blue is the preferred color of a member i, and red is the preferred color of all other members of the committee then red is determined to be the prettiest color, so i is not a dictator, as claimed.

(b) The social welfare function described here satisfies the unanimity property. If red is the preferred color of all of the committee' members, then it is preferred by at least five committee members hence red is determined to be the prettiest color. If blue is the preferred color of all of the committee' members, then red is not preferred by at least five committee members, whereas blue is preferred by at least five committee members hence blue is determined to be the prettiest color. Finally, if yellow is the preferred color of all of the committee' members, then neither red nor blue is preferred by at least five committee members, and yellow is determined to be the prettiest color.

(c) The social welfare function described here does not satisfy the independence of irrelevant alternatives property. For example, if the preference relation of the committee members are

No. of members	First color	Second color	Third color
4	Red	Blue	Yellow
10	Blue	Yellow	Red
1	Blue	Red	Yellow

then Yellow is preferred to Red (and Blue is determined as the prettiest color). However, if the committee member in the bottom line replaces his preference relation to Red $\succ$ Blue $\succ$ Yellow, then under the social welfare function Red is preferred to Yellow.

22.8 (a) The social welfare function F_k is not dictatorial, for every k. We show that no voter i is a dictator. Let $k < n$. If i prefers b over a and all other voters prefer a over b then a is preferred over b by F_k, so i is not a dictator. For $k = n$, assume that i prefers a over b and all other voters prefer b over a then b is preferred over a by F_n, so i is not a dictator.

(b) The social welfare function F_k satisfies the unanimity property for $k > 0$. Indeed, if all of the voters prefer a over b then at least k voter prefer a over b and a is chosen by F_k. If all of the voters prefer b over a then at most $k - 1$ voter prefer a over b and b is chosen by F_k. On the other hand, for $k = 0$, F_0 is not unanimity since a is chosen even if all of the voters prefers b over a.

(c) The social welfare function F_k satisfies the independence of irrelevant alternatives property since there are only two alternatives.

22.9 Let F be a social welfare function satisfying the independence of irrelevant alternatives property, and let a, b be two distinct alternatives. Let P^N and Q^N be two strict preference profiles satisfying

$$a \succ^{P_i} b \iff a \succ^{Q_i} b, \quad \forall i \in N.$$

Then by the independence of irrelevant alternatives property

$$a \approx_{F(P^N)} b \iff \begin{pmatrix} a \succsim_{F(P^N)} b \\ b \succsim_{F(P^N)} a \end{pmatrix} \iff \begin{pmatrix} a \succsim_{F(Q^N)} b \\ b \succsim_{F(Q^N)} a \end{pmatrix} \iff a \approx_{F(Q^N)} b.$$

22.10 Denote by $K = |A|$ the number of alternatives in A. For each alternative $a \in A$, denote by $j_a(P_i)$ the ranking of a in the strict preference relation P_i. Define a social welfare function as follows. For each alternative a, compute the sum $s_a = \sum_{i \in N} j_a(P_i)$. We say that alternative a is (weakly) preferred to alternative b if and only if $s_a \leq s_b$.

(a) Since the preferences are equivalent to the preference relation $\leq$ over a set of numbers which is complete, reflexive and transitive, the preferences in this exercise define a social welfare function.

(b) This is not a dictatorial function, since for every $i \in N$, if $j_a(P_i) = 1$ and $j_b(P_i) = 2$ whereas $j_a(P_l) = 2$ and $j_b(P_l) = 1$ for every $l \neq i$, then $s_b \leq s_a$ and hence b is (weakly) preferred to alternative a.

This function satisfies the unanimity property. If $a \succ_{P_i} b$ for every $i \in N$, then $j_a(P_i) < j_b(P_i)$ for every $i \in N$. Therefore,

$$s_a = \sum_{i \in N} j_a(P_i) < \sum_{i \in N} j_b(P_i) = s_b,$$

from which it follows that a is (weakly) preferred b.

The function does not necessarily satisfy the independence of irrelevant alternatives property. If, for instance there are three alternative and 3 players. Two players rank a by 2 and b by 3, and three players rank a by 3 and b by 2, then $s_a = 13 > 12 = s_b$ so b is preferred over a. However, if the two players change their ranking of a to 1, then $s_a = 11 < 12 = s_b$ so a is preferred over b.

(c) This voting system and the Borda system are equivalent.

22.11 Let F be a social welfare function satisfying the unanimity property. Let $a, b \in A$ be two different alternatives. For $S = N$, Condition 2 in Definition 22.15 always holds in a null manner. Therefore, if Conditions 1 holds, i.e.,

$$a \succ_{P_i} b \quad \forall i \in N,$$

then by the unanimity, $a \succ_F b$. Therefore, the coalition N is decisive for a over b.

Similarly, for $S = \emptyset$, Condition 1 in Definition 22.15 always holds in a null manner. Therefore, if Conditions 2 holds, i.e.,

$$b \succ_{P_i} a \quad \forall i \in N,$$

then by the unanimity, $b \succ_F a$. Therefore, the coalition $\emptyset$ is not decisive for a over b.

22.12 Denote by $\mathcal{D}$ the set of all decisive coalitions for "guilty" over "innocent", and by $\mathcal{D}^m$ the set of all minimal decisive coalitions for "guilty" over "innocent". Denote by N the set of jurors.

(a) $\mathcal{D} = \{N\} = \mathcal{D}^m$.

(b) $\mathcal{D} = \{S \subseteq N : |S| \geq 4\}$ and $\mathcal{D}^m = \{S \subseteq N : |S| = 4\}$.

(c) $\mathcal{D} = \{S \subseteq N : |S| \geq 4, \text{Debbie} \in S\}$ and $\mathcal{D}^m = \{S \subseteq N : |S| = 4, \text{Debbie} \in S\}$.

(d)

$$\mathcal{D} = \left\{ S \subseteq N : \begin{array}{c} |S| \geq 4, \\ \text{Debbie} \in S \end{array} \right\} \quad \bigcup \left\{ S \subseteq N : \begin{array}{c} |S| \geq 5, \\ \text{Bobby,Jack} \in S \end{array} \right\}$$

$$\mathcal{D}^m = \left\{ S \subseteq N : \begin{array}{c} |S| = 4, \\ \text{Debbie} \in S \end{array} \right\} \bigcup \left\{ S \subseteq N : \begin{array}{c} |S| = 5, \\ \text{Bobby,Jack} \in S \\ \text{Debbie} \notin S \end{array} \right\}.$$

(e)

$$\mathcal{D} = \{S \subseteq N : |S| \geq 4\} \quad \bigcup \{S \subseteq N : \text{Debbie,Bobby,Jack} \in S\}$$

$$\begin{aligned}\mathcal{D}^m = &\left\{ S \subseteq N : \begin{array}{c} |S| = 4 \\ \text{Debbie} \notin S \end{array} \right\} \bigcup \left\{ S \subseteq N : \begin{array}{c} |S| = 4 \\ \text{Bobby} \notin S \end{array} \right\} \\ &\bigcup \left\{ S \subseteq N : \begin{array}{c} |S| = 4 \\ \text{Jack} \notin S \end{array} \right\} \bigcup \{\{\text{Debbie,Bobby,Jack}\}\}.\end{aligned}$$

(f)

$$\mathcal{D} = \left\{ S \subseteq N : \begin{array}{c} |S| \geq 5, \\ \text{Debbie,Bobby} \in S \end{array} \right\} \bigcup \left\{ S \subseteq N : \begin{array}{c} |S| \geq 5, \\ \text{Debbie,Jack} \in S \end{array} \right\}.$$

$$\begin{aligned}\mathcal{D}^m = &\left\{ S \subseteq N : \begin{array}{c} |S| = 5, \\ \text{Debbie,Bobby, Jack} \in S \end{array} \right\} \bigcup \left\{ S \subseteq N : \begin{array}{c} |S| = 5, \\ \text{Debbie,Bobby} \in S \\ \text{Jack} \notin S \end{array} \right\} \\ &\bigcup \left\{ S \subseteq N : \begin{array}{c} |S| = 5, \\ \text{Debbie,Jack} \in S \\ \text{Bobby} \notin S \end{array} \right\}.\end{aligned}$$

22.13 Suppose that $|A| \geq 3$, and let F be a social welfare function satisfying the properties of unanimity and independence of irrelevant alternatives. Then, by Theorem 22.10, F is dictatorial. In particular, all of the coalitions containing the dictator are all the decisive coalitions.

22.14 We show that the assumptions of Theorem 22.18 are necessary for its conclusion:
Without the assumption $|A| \geq 3$: Let $A = \{a, b\}$. Let F be a social welfare function where a is preferred over b if and only if at least one voter prefers a over b. Then, F satisfies the unanimity and independence of irrelevant alternatives properties. Let V be a coalition containing at least one voter who prefers a over b and at least one voter who prefers b over a. Then, coalition V is decisive for a a over b but it is not decisive for b over a.
Without the assumption of unanimity: Let $A = \{a, b, c\}$. Let F be a social welfare function where $a \succ_F b \succ_F c$ regardless the preference relations of the voters. Then, F satisfies independence of irrelevant alternatives property but it does not satisfy the unanimity. Then every coalition V is decisive for a a over b but it is not decisive for b over a.
Without the assumption of independence of irrelevant alternatives: Let $A = \{a, b, c\}$. Let F be a social welfare function where a is preferred over b and c if and only if at least one voter prefers a over b, all other preferences are determined by the first voter. Then, F satisfies unanimity but it does not satisfy the independence of irrelevant alternatives property. Then every coalition V is decisive for a b over c but it is not decisive for b over a.

22.15 We prove Claim 22.20. Let P^N be a strict preference profile, and let $a, b \in A$ be two different alternatives such that $a \succ_{P_j} b$. We show that $a \succ_{F(P^N)} b$.
Since A contains at least three alternatives, there exists an alternative $c \in A \setminus \{a, b\}$.
Consider the following strict preference profile Q^N:

$$\begin{cases} a \succ_{Q_i} c \succ_{Q_i} b & i = j, \\ a \succ_{Q_i} b \succ_{Q_i} c & i \neq j, \quad a \succ_{P_i} b, \\ b \succ_{Q_i} a \succ_{Q_i} c & i \neq j, \quad b \succ_{P_i} a. \end{cases}$$

The other alternatives in A are ordered by each individual arbitrarily. Since $V = \{j\}$ is decisive for any two alternatives, it is in particular decisive for c over b, and therefore $c \succ_{F(Q^N)} b$. Since F satisfies the unanimity property, it follows that $a \succ_{F(Q^N)} c$. Since $F(Q^N)$ is a transitive ordering relation, we deduce that $a \succ_{F(Q^N)} b$.

Now, individual j prefers a over b, according to both P_j and Q_j, and every individual $i \neq j$ prefers a over b according to P_i if and only if he prefers a over b according to Q_i. Since F satisfies the independence of irrelevant alternatives property, and since $a \succ_{F(Q^N)} b$, we deduce that $a \succ_{F(P^N)} b$, as required.

22.16 The proof of Claim 22.20 fails when using the preference relation defined by Ben.
To determine what are the preferences of $F(Q^N)$ with regards to the pair b, c, by using the assumption that $V = \{j\}$ is decisive for any two alternatives, one must define opposite preferences over b and c for j and for all other players. Whereas, in Ben preferences, some of the players in $N \setminus \{j\}$ prefer a over b while other prefer b over a.

22.17 (a) **Example 22.7:** The social welfare function is not monotonic. If the preference relations are

No. of members	**First color**	**Second color**	**Third color**
5	Blue	Red	Yellow
10	Yellow	Red	Blue

then the function determined that Blue is prettiest color. On the other hand, if the preference relations are

No. of members	**First color**	**Second color**	**Third color**
5	Blue	Red	Yellow
10	Red	Yellow	Blue

then the function determined that Red is prettiest color, also the ranking of Blue relative to Red is not lowered.

Exercise 22.10: The social welfare function is not monotonic. Assume that $A = \{a, b, c, d\}$ and there are three voters. If the preference relation of the first two voters is $a \succ b \succ c \succ d$, and the preference relation of the last voter is $c \succ d \succ b \succ a$, then $s_a = 6 < s_b, s_c, s_d$ and a is the most preferred. However, if the last voter changes his preference relation to $b \succ c \succ d \succ a$, then $s_a = 6 > 5 = s_b$ and b is the most preferred.

(b) A monotonic social welfare function does not necessarily satisfy the unanimity property, e.g., a constant social welfare function that determine the same preference relation regardless the preference profile is monotonic, but does not satisfy the unanimity property.

(c) Not every monotonic social welfare function satisfy the independence of irrelevant alternatives property. Let $\{a, b, c\}$ be a set of alternatives. Consider the social welfare function F according to which the preferences over all of the alternatives is determined by the most preferred alternative of Voter 1. If Voter 1 most prefers a then the preference relation is $a \succ_F b \succ_F c$. If Voter I most prefers b then the preference relation is $b \succ_F c \succ_F a$. If Voter I most prefers c then the preference relation is $c \succ_F a \succ_F b$. This social welfare function is monotonic. Indeed, the least preferred alternative under F can not get lower ranking, the most preferred alternative get lower ranking under F if and only if it gets lower ranking by Voter I relative to the other alternative, and the middle alternative can change its position only if Voter I most prefers it. However, this function does not satisfy the independence of irrelevant alternatives property since for every preference profile such that

$$\text{Voter I: } a \succ c \succ b$$

one has $a \succ_F b \succ_F c$ so $b \succ_F c$, whereas for the preference profile in which only Voter I changes his preference relation

$$\text{Voter I } : c \succ a \succ b$$

one has $c \succ_F a \succ_F b$. In conclusion, also the preferences among b and c are the same in both preference profiles, the preference under the social welfare function is not the same.

(d) Not every social welfare function satisfying the independence of irrelevant alternatives property is monotonic. For instance, consider a set of two alternatives $\{a, b\}$, three voters and the social welfare function determines the preference according to the minority opinion. Since there are only two alternatives, the independence of irrelevant alternatives property is satisfied. However, the function is not monotonic, since

$$\left.\begin{array}{l} P_1 : a \succ b \\ P_2 : b \succ a \\ P_3 : b \succ a \end{array}\right\} \Rightarrow a \succ_F b$$

$$\left.\begin{array}{l} Q_1 : a \succ b \\ Q_2 : a \succ b \\ Q_3 : b \succ a \end{array}\right\} \Rightarrow b \succ_F a.$$

22.18 (a) The decision rule is a social welfare function. However, it is not monotonic. If the preference profile P is given by

Veronica:	Abigail $\succ$ Iris $\succ$ Irene $\succ$ Olga
Ron:	Irene $\succ$ Olga $\succ$ Iris $\succ$ Abigail

Abigail is the chosen name. However, if the preference relation of Ron is changed which leads to the preference profile Q

Veronica:	Abigail $\succ$ Iris $\succ$ Irene $\succ$ Olga
Ron:	Irene $\succ$ Iris $\succ$ Olga $\succ$ Abigail

Iris is the chosen name. Using the same preference profiles, it follows that the social welfare function is manipulable. If P_{Ron} is the true preferences of Ron, then by pretending that his preference relation is Q_{Ron} the chosen name is preferred by him.

(b) It is not a social preference relation since there are preference profiles that are map to two different names.

(c) The decision rule is a social welfare function. However, it is not monotonic. If the preference profile P is given by

Veronica:	Abigail $\succ$ Iris $\succ$ Irene $\succ$ Olga
Ron:	Olga $\succ$ Irene $\succ$ Iris $\succ$ Abigail

Abigail is the chosen name. However, if the preference relation of Ron is changed which leads to the preference profile Q

Veronica:	Abigail $\succ$ Iris $\succ$ Irene $\succ$ Olga
Ron:	Irene $\succ$ Olga $\succ$ Iris $\succ$ Abigail

Irene is the chosen name. Using the same preference profiles, it follows that the social welfare function is manipulable. If P_{Ron} is the true preferences of Ron, then by pretending that his preference relation is Q_{Ron} the chosen name is preferred by him.

(d) The decision rule is a social welfare function which satisfies the unanimity property. For the preference profile

Veronica:	Abigail $\succ$ Iris $\succ$ Irene $\succ$ Olga
Ron:	Irene $\succ$ Olga $\succ$ Iris $\succ$ Abigail

Olga is the chosen name. Therefore, this social welfare function is not dictatorial and it not necessarily chooses one of the couple's preferred name from among the names. Hence, by Theorems 22.39 and 22.35, this function is manipulable and it is not monotonic, respectively.

22.19 (a) The function is not dictatorial. Indeed, if some committee member i prefers Blue to Red while all other committee members prefer Red to Blue, Red is chosen, so i is not a dictator.

(b) The function is not monotonic. Consider the following preference profile,

No. of members	**First color**	**Second color**	**Third color**
14	Blue	Red	Yellow
1	Yellow	Red	Blue

then Blue is chosen. Whereas, under the following preference profile

No. of members	**First color**	**Second color**	**Third color**
14	Blue	Red	Yellow
1	Red	Yellow	Blue

Red is chosen.

(c) By Theorem 22.39, since the function is not dictatorial, the function is manipulable.

22.20 (a) We show that this electoral method is not monotonic. Let $A = \{a, b, c\}$ a set of candidates where the social security number of a is the smallest. Suppose there are only three residents whose preference profile is:

$$a \succ_{P_1} b \succ_{P_1} c$$
$$b \succ_{P_2} c \succ_{P_2} a$$
$$c \succ_{P_3} a \succ_{P_3} b$$

then a is chosen. On the other hand, under

$$a \succ_{Q_1} b \succ_{Q_1} c$$
$$c \succ_{Q_2} b \succ_{Q_2} a$$
$$c \succ_{Q_3} a \succ_{Q_3} b$$

c is chosen, which contradicts monotonicity.

(b) Consider the example in Part (a). If P^N is the true preference profile, and resident 2 pretends his preference relation is Q_2 then c is chosen whom resident 2 prefers to a.

22.21 (a) We show that this electoral method is not monotonic. Let $A = \{a, b, c\}$ a set of candidates where the social security number of a is the smallest. Suppose that there are only three residents whose preference profile is:

$$a \succ_{P_1} b \succ_{P_1} c$$
$$b \succ_{P_2} c \succ_{P_2} a$$
$$c \succ_{P_3} a \succ_{P_3} b$$

then a is chosen. On the other hand, under

$$a \succ_{Q_1} c \succ_{Q_1} b$$
$$b \succ_{Q_2} c \succ_{Q_2} a$$
$$c \succ_{Q_3} a \succ_{Q_3} b$$

c is chosen, which contradicts monotonicity.

(b) Consider the example in Part (a). If Q^N is the true preference profile, and resident 1 pretends his preference relation is P_1 then a is chosen whom resident 2 prefers to c.

22.22 (a) It is possible for the winner of the election to not be the most preferred candidate of any resident. Consider a set of 4 candidates a, b, c, d and three residents with the preference profile P given by

$$a \succ_{P_1} b \succ_{P_1} c \succ_{P_1} d$$
$$d \succ_{P_2} b \succ_{P_2} c \succ_{P_2} a$$
$$c \succ_{P_3} b \succ_{P_3} a \succ_{P_3} d.$$

Then, b is the winner who is obtained after eliminating d, a and then c.

(b) It is possible for the winner of the election to be ranked least preferred by at least half of the residents. Consider a set of 3 candidates a, b, c assume that the social security number of c is smallest and the social security number of c is smallest. There are four residents with the preference profile P given by

$$
\begin{aligned}
a &\succ_{P_1} b \succ_{P_1} c \\
a &\succ_{P_2} b \succ_{P_2} c \\
c &\succ_{P_3} b\_P3a \\
c &\succ_{P_4} b \succ_{P_4} a.
\end{aligned}
$$

Then, c is the winner who is obtained after eliminating a then b.

(c) By Part (a), it is possible for the winner of the election to not be the most preferred candidate of any resident, hence the social choice function described here is not dictatorial.

(d) By Part (a) it is possible for the winner of the election to not be the most preferred candidate of any resident. In addition the function satisfies the unanimity property so every candidate can be chosen. Hence, by Theorem 22.35, the function is not monotonic. Indeed, whereas b is the winner under the preference profile P in Part (a), in the preference profile Q given by

$$
\begin{aligned}
a &\succ_{Q_1} b \succ_{Q_1} c \succ_{Q_1} d \\
d &\succ_{Q_2} b \succ_{Q_2} a \succ_{Q_2} c \\
c &\succ_{Q_3} b \succ_{Q_3} a \succ_{Q_3} d.
\end{aligned}
$$

a is the winner who is obtained after eliminating d, c and then b.

(e) By Part (c) the fuction is not dictatorial so by Theorem 22.39 it follows that the function is manipulable. Consider the preference profiles in Parts (a) and (d). If the true preferences are as given in Part (d) and resident 2 pretends his preference relation is P_2 then b is chosen whom resident 2 prefers to a.

(f) Not necessarily the same candidate is chosen under both electoral methods. Consider for instance the example in Part (a). Whereas b is chosen under the electoral method used in Hobbiton, b is not chosen under the electoral method used in Whoville.

22.23 The given function is not dictatorial. Indeed, if all of the residents but one has the same preference relation the majority selection is the winner. Thus no resident is a dictator and every candidate can be the winner. Therefore, by Theorem 22.39, the function is manipulable. Furthermore, there is no player that the most preferred candidate necessarily chosen, hence, by Theorem 22.35, if there are at least 3 candidates, the function is not monotonic.

22.24 (a) A Condorcet winner does not necessarily exist for every strict preference profile P^N. Consider the following strict preference profile over $\{a, b, c\}$

$$\begin{aligned}&\text{Voter I:} \quad a \succ b \succ c\\&\text{Voter II:} \ \ b \succ c \succ a\\&\text{Voter III:} \ c \succ a \succ b.\end{aligned}$$

Then, this strict preference profile does not has a Condorcet winner.

(b) **Exercise 22.19:** the social choice function does not satisfy the Condorcet criterion. If for instance, one committee member prefers Red to Blue to Yellow and all other committee members prefer Blue to Red to Yellow, then Blue is the Condorcet winner whereas Red is chosen by the social choice function.

Exercise 22.20: the social choice function does not satisfy the Condorcet criterion. Indeed, for the next strict preference relations over 4 alternatives

$$\begin{aligned}P_1 &: a \succ b \succ c \succ d\\P_2 &: a \succ b \succ c \succ d\\P_3 &: b \succ c \succ d \succ a\\P_4 &: c \succ b \succ d \succ a\\P_5 &: d \succ b \succ c \succ a\end{aligned}$$

the Condorcet winner is b whereas a is chosen by the social choice function.

Exercise 22.21: the social choice function does not satisfy the Condorcet criterion. Indeed, for the next strict preference relations over 6 alternatives

$$\begin{aligned}P_1 &: a \succ b \succ c \succ d \succ e \succ f\\P_2 &: a \succ c \succ b \succ d \succ e \succ f\\P_3 &: b \succ a \succ c \succ d \succ e \succ f\\P_4 &: e \succ c \succ b \succ d \succ a \succ f\\P_5 &: f \succ d \succ b \succ c \succ a \succ e\end{aligned}$$

the Condorcet winner is b whereas a is chosen by the social choice function.

Exercise 22.22: the social choice function does not satisfy the Condorcet criterion. Indeed, for the next strict preference relations over 4 alternatives

$$\begin{aligned}
P_1 &: a \succ b \succ c \succ d \\
P_2 &: a \succ b \succ c \succ d \\
P_3 &: d \succ a \succ b \succ c \\
P_4 &: d \succ a \succ c \succ b \\
P_5 &: d \succ b \succ c \succ a
\end{aligned}$$

the Condorcet winner is d whereas d is the first to be removed during the process so it is not chosen under the social welfare function.

Exercise 22.23: the social choice function does not satisfy the Condorcet criterion. Indeed, for the next strict preference relations over 6 alternatives

$$\begin{aligned}
P_1 &: a \succ b \succ c \succ d \succ e \succ f \\
P_2 &: a \succ c \succ b \succ d \succ e \succ f \\
P_3 &: b \succ a \succ c \succ d \succ e \succ f \\
P_4 &: e \succ c \succ b \succ d \succ a \succ f \\
P_5 &: f \succ d \succ b \succ c \succ a \succ e
\end{aligned}$$

the Condorcet winner is b whereas a is chosen by the social choice function (for $k = 2$).

22.25 Let G be a monotonic social choice function, let P^N and Q^N be two strict preference profiles, and let $R \subseteq A$. We show that if $a \in R$ and $G(P^N) = a$, then $G(Z(P^N, Q^N; R)) = a$.
Let $G(P^N) = a$. Since $a \in R$, for every $i \in N$ and every $b \in A$ one has

$$\begin{aligned}
&a \succ_{Z(P_i,Q_i;R)} b \iff a \succ_{P_i} b && \text{if } b \in R, \\
&a \succ_{Z(P_i,Q_i;R)} b && \text{if } b \in A \setminus R.
\end{aligned}$$

Therefore, by the monotonicity of G, $G(Z(P^N, Q^N; R)) = a$, as claimed.

22.26 Let G be a monotonic social choice function, let $a \in \text{range}(G)$, and let P^N be a strict preference profile. Assume that

$$a \succ_{P_i} b, \quad \forall i \in N. \tag{204}$$

We show that $G(P^N) \neq b$. Assume by contradiction that $G(P^N) = b$.

Since $a \in \text{range}(G)$, there is a strict preference profile Q^N such that $G(Q^N) = a$. Let $W^N = Z(Q^N, P^N; \{a, b\})$.
Then, for every $i \in N$ and every $c \neq a$,

$$a \succ_{Q_i} c \Rightarrow a \succ_{W_i} c,$$

by the definition of W^N. Hence, by the monotonicity of G, since $G(Q^N) = a$, $G(W^N) = a$.
In addition, by the definition of W^N and by the assumption in Inequality (204), for every $i \in N$ and every $c \neq b$,

$$b \succ_{P_i} c \Rightarrow b \succ_{W_i} c.$$

Hence, by the monotonicity of G, since $G(P^N) = b$ hence $G(W^N) = b$, a contradiction.

22.27 **Proof of Theorem 22.35:**

(a) We prove that if G is a monotonic social choice function, then for any pair of strict preference relations P^N and Q^N satisfying

$$a \succ_{P_i} b \iff a \succ_{Q_i} b, \quad \forall a, b \in \text{range}(G), \ \forall i \in N, \tag{205}$$

$G(P^N) = G(Q^N)$.
Let $G(P^N) = c$. Set $W^N = Z(Q^N, P^N; \text{range}(G))$. Hence, by Inequality (205),

$$c \succ_{P_i} b \Rightarrow c \succ_{W_i} b \ \ \forall b \in \text{range}(G) \ \ \forall i \in N.$$

In addition, for every $b \notin \text{range}(G)$ and every $i \in I$

$$c \succ_{P_i} b \Rightarrow c \succ_{W_i} b,$$

since the right condition always holds.
Using the monotonicity, one can derive that since $G(P^N) = c$ then $G(W^N) = G(Z(Q^N, P^N; \text{range}(G)) = c$.
Finally, by Theorem 22.31, since $G(Q^N) \in \text{range}(G)$ and G is monotonic it follows that $G(Z(Q^N, P^N; \text{range}(G)) = G(Q^N) = c$.
The other direction is identical.

(b) Assume that G is a monotonic social choice function satisfying $|\text{range}(G)| \geq 3$. Following Exercise 22.26, since G is monotonic, it also satisfies the property of unanimity. In conclusion, by Theorem 22.27, G is dictatorial. In particular, there is an individual i such that for every strict preference profile P^N the alternative $G(P^N)$ is the most preferred alternative from among all the alternatives in range(G).

22.28 Define a choice function H^* as follows: an alternative a is a winning alternative if it maximizes the value of $|\{i \in N : a \in B(P_i)\}|$. If there are two or more alternatives receiving the greatest number of approval votes, the alternative whose serial number is lowest is chosen. We show that this choice function is monotonic. Let P^N and Q^N be different two-leveled preference profiles where $H(P^N) = a$, and every individual i satisfies $B(Q_i) = B(P_i)$ or $B(Q_i) = B(P_i) \bigcup \{a\}$. Then, since $P^N \neq Q^N$

$$|\{i \in N : a \in B(P_i)\}| < |\{i \in N : a \in B(Q_i)\}|$$

and for every $b \neq a$

$$|\{i \in N : b \in B(P_i)\}| = |\{i \in N : b \in B(Q_i)\}|.$$

Hence, since a is a winning alternative that is approved by the greatest number of individuals under P^N, then it is obviously approved by the unique greatest number of individuals under Q^N, from which it follows that $H(Q^N) = a$.

We show that this choice function is nonmanipulable. Let P^N be a two-leveled preference profile and let i be an individual. If $H(P^N) \in B(P_i)$ then by the definition of the relation preference of i, $H(P^N) \succsim_{P_i} H(Q_i, P_{-i})$. If, on the other hand, $H(P^N) \notin B(P_i)$ then by pretending his preference is Q_i, i can only weaken alternatives in $B(P_i)$ and strengthen alternatives that are not in $B(P_i)$, hence $H(P^N) \succ_{P_i} H(Q_i, P_{-i})$.

22.29 Let H^* be a function where $H^*(P^N)$ is the set of alternatives that are approved by the greatest number of individuals according to P^N. We show that the function is manipulable.

Let $N = \{1, 2, 3, 4\}$ and $A = \{a, b, c\}$. Assume that $B(P_1) = \{a, b\}$, $B(P_2) = \{a\}$, $B(P_3) = B(P_4) = \{b\}$. Then $a \notin B(H^*(P^N)) = \{b\}$ whereas, by pretending $B(Q_1) = \{a\}$ then $a \in B(H^*(Q_1, P_{-1})) = \{a, b\}$.

Chapter 23: Stable matching

23.1 Let $\succ$ be a complete, irreflexive, and transitive relation. Let $x \neq y$. Then, by the completeness, either $x \succ y$ or $y \succ x$. Assume by contradiction that both hold. Then, by the transitivity, $x \succ x$, which contradict the irreflexive property. In conclusion, the relation is asymmetric.

23.2 We prove that the Gale–Shapley men's courtship algorithm satisfies the following property: if Cleopatra asks Mark to stay in front of her house at stage k, and at a later stage l she asks Julius to stay in front of her house, then she prefers Julius to Mark.

Denote by $y_k, ..., y_l$ the men whom Cleopatra asks to wait at stages $k, ..., l$, respectively (y_k is Mark and y_l is Julius). We show by induction over t that for every $t \in \{k, ..., l\}$, either $y_t = y_k$ or $y_t \succ y_k$. For $t = k$ the claim holds. Assume that it holds for $t < l$ we show that it also holds for $t + 1$. If $y_{t+1} = y_k$ we are done. If $y_{t+1} = y_t$ then the claim holds for $t + 1$ by the induction hypothesis. Otherwise, $y_t \neq y_{t+1}$ so at the beginning of stage $t + 1$ there are at least two men in front of Cleopatra house: y_t whom was asked to wait at stge t and y_{t+1} whom is asked to wait at stage $t + 1$. Since Cleopatra asks y_{t+1} to wait, $y_{t+1} \succ y_t$. However, by the induction hypothesis $y_t \succeq y_k$ and thus, by the transitivity of the preference relation, $y_{t+1} \succ y_k$. In particular, the claim holds for l which also satisfies $y_k \neq y_l$ so $y_l \succ y_k$, i.e., Cleopatra prefers Julius to Mark.

23.3 The matching:

(Chris – Anne, Alfredo – Betty, Dean – Claire, Ben – Donna)

is not stable. This is because Donna and Dean have an objection to the matching: Donna prefers Dean (number 2 on her list) to Ben (number 4 on her list), and Dean prefers Donna (number 1 on his list) to Claire (number 2 on his list).

The matching:

(Chris – Anne, Alfredo – Betty, Dean – Claire, Ben – Donna)

is stable.

23.4 We prove that Definitions 23.4 and 23.5 are equivalent.

Assume that a matching A is stable according definition 23.4. Assume also that there is a man x who prefers another woman y to the woman to whom he is matched under A, then by definition 23.4, y prefers the man to whom she is matched to x, or else, x and y object to a matching which contradicts definition 23.4.

Assume next that a matching A is stable according definition 23.5. Then no man and a woman object to a matching, since, by Definition 23.5, if a man x prefers a woman y to the woman to whom he is matched under A, y prefers the man to whom she is matched to him. Therefore, A is stable according definition 23.4.

23.5 (a) The men's courtship algorithm:

	Anne	Barbara	Claire	Dismissed man
Stage 1(a)	Andre	Boris	Chris	

The women's courtship algorithm:

	Andre	Boris	Chris	Dismissed woman
Stage 1(a)		Anne	Barbara, Claire	
Stage 1(b)		Anne	Claire	Barbara
Stage 2(a)	Barbara	Anne	Claire	

(b) The men's courtship algorithm:

Stage	Ellen	Flora	Gail	Hillary	Dismissed man
1(a)	Alex, Bill	Colin	-	David	
1(b)	Alex	Colin	-	David	Bill
2(a)	Alex	Colin	-	David,Bill	
2(b)	Alex	Colin	-	Bill	David
3(a)	Alex	Colin, David	-	Bill	
3(b)	Alex	David	-	Bill	Colin
4(a)	Alex,Colin	David	-	Bill	
4(b)	Colin	David	-	Bill	Alex
5(a)	Colin	David,Alex	-	Bill	
5(b)	Colin	David	-	Bill	Alex
6(a)	Colin	David	Alex	Bill	

The women's courtship algorithm:

Stage	Alex	Bill	Colin	David	Dismissed woman
1(a)	-	Flora	Hillary	Ellen,Gail	
1(b)	-	Flora	Hillary	Gail	Ellen
2(a)	-	Flora	Hillary,Ellen	Gail	
2(b)	-	Flora	Ellen	Gail	Hillary
3(a)	-	Flora,Hillary	Ellen	Gail	
3(b)	-	Hillary	Ellen	Gail	Flora
4(a)	-	Hillary	Ellen	Gail,Flora	
4(b)	-	Hillary	Ellen	Flora	Gail
5(a)	Gail	Hillary	Ellen	Flora	

(c) The men's courtship algorithm:

Stage	Mary	Netty	Olivia	Patty	Dismissed man
1(a)	Kevin	-	Peter	Jacob,Larry	
1(b)	Kevin	-	Peter	Jacob	Larry
2(a)	Kevin	-	Peter,Larry	Jacob	
2(b)	Kevin	-	Peter	Jacob	Larry
3(a)	Kevin,Larry	-	Peter	Jacob	
3(b)	Kevin	-	Peter	Jacob	Larry
4(a)	Kevin	Larry	Peter	Jacob	

The women's courtship algorithm:

Stage	Peter	Jacob	Kevin	Larry	Dismissed woman
1(a)	Mary,Netty	Olivia	Patty	-	
1(b)	Mary	Olivia	Patty	-	Netty
2(a)	Mary	Olivia,Netty	Patty	-	
2(b)	Mary	Netty	Patty	-	Olivia
3(a)	Mary	Netty	Patty,Olivia	-	
3(b)	Mary	Netty	Olivia	-	Patty
4(a)	Mary,Patty	Netty	Olivia	-	
4(b)	Patty	Netty	Olivia	-	Mary
5(a)	Patty	Netty,Jacob	Olivia	-	
5(b)	Patty	Mary	Olivia	-	Netty
6(a)	Patty	Mary	Olivia,Netty	-	
6(b)	Patty	Mary	Netty	-	Olivia
7(a)	Patty,Olivia	Mary	Netty	-	
7(b)	Olivia	Mary	Netty	-	Patty
8(a)	Olivia	Mary,Patty	Netty	-	
8(b)	Olivia	Patty	Netty	-	Mary
9(a)	Olivia	Patty	Netty,Mary	-	
9(a)	Olivia	Patty	Mary	-	Netty
10(a)	Olivia	Patty	Mary	Netty	

(d) The stable matching that is obtained by both the men's courtship and the women's courtship algorithms is

Ernest-Felicia, Felix-Emma,George-Donna,Henry-Carol.

(e) The stable matching that is obtained by both the men's courtship and the women's courtship algorithms is

Peter-Natasha, Quentin-Octavia,Ron-Lisa,Sam-Melissa.

23.6 Suppose that the number of men equals the number of women.

(a) For every pair of matchings there exist preference relations for which these are two stable matchings. Indeed, let A and B be two matching. Set preference relations of the men such that number 1 of every man list is the woman to whom he is matched according to A. Then, A is a stable matching that is obtained by the men's courtship algorithm regardless the women's preference relations. Similarly, set preference relations of the women such that number 1 of every woman list is the man to whom she is matched according to B, then B is a stable matching that is obtained by the women's courtship algorithm regardless the men's preference relations.

(b) In this part we show that the claim in Part (a) cannot be generalized to more than two matchings; that is, it is not true that for every three matchings there are preference relations for which these three matchings are stable.

Suppose that the set of men is $M = \{X, Y, Z, W\}$ and the set of women is $\{x, y, z, w\}$. Consider the following three matching:

$$\begin{aligned} \text{A1} &: Y - w, Z - z, W - y, X - x, \\ \text{A2} &: Z - w, W - z, Y - y, X - x, \\ \text{A3} &: X - w, Z - z, Y - y, W - x. \end{aligned}$$

Assume by contradiction that there is a preference relations to the set of men and the set of women for which the above matchings are stable.

If $x : X \succ W$ then since A1-A3 are stable the following must holds:

$$\text{(A3) } X : w \succ x$$
$$\Downarrow$$
$$\begin{Bmatrix} \text{(A1) } w : Y \succ X \\ \text{(A3) } Y : y \succ w \\ \text{(A1) } y : W \succ Y \\ \\ \\ \end{Bmatrix} \text{ and } \begin{Bmatrix} \text{(A2) } w : Z \succ X \\ \text{(A3) } Z : z \succ w \\ \text{(A2) } z : W \succ Z \\ \text{(A1) } W : y \succ z \\ \text{(A2) } y : Y \succ W \end{Bmatrix}$$

which leads to a contradiction.

Similarly, if on the other hand, $x : W \succ X$ then

$$\left\{\begin{array}{l} (\text{A1})\ W : y \succ x \\ (\text{A3})\ y : Y \succ W \\ (\text{A1})\ Y : w \succ y \\ (\text{A2})\ w : Z \succ Y \\ \end{array}\right\} \text{ and } \left\{\begin{array}{l} (\text{A2})\ W : z \succ x \\ (\text{A3})\ z : Z \succ W \\ (\text{A2})\ Z : w \succ z \\ (\text{A1})\ w : Y \succ Z \\ \end{array}\right\}$$

which again leads to a contradiction.

23.7 Gary is at the top of Gail's preference list, and Gail is at the top of Gary's preference list. We show that under every stable matching Gary and Gail are matched to each other.
Since Gary is at the top of Gail's preference list, he is matched to her under a stable matching that is obtained by the women's courtship algorithm.
However, by Theorem 23.12, this is the worst stable matching from the perspective of all the men, in particular, from the perspective of Gary. In conclusion, under every stable matching he is matched to Gail at least, but since Gail is at the top of his preference list he is necessarily matched to her under every stable matching.

23.8 Even in case that Dan is at the bottom of Donna's preference list, and Donna is at the bottom of Dan's preference list, it is possible that there is a stable matching that matches Dan to Donna. A counter example appears in Exercise 23.5c (where Larry is replaced by Dan and Netty is replaced by Donna).

23.9 Suppose that Romeo and Juliet are matched to each other under both the men's courtship and the women's courtship algorithms.
Assume by contradiction that there is a stable matching in which Romeo is matched to $x \neq$ Juliet. By Theorem 23.12, since Romeo is matched to Juliet under the men's courtship, she is the most preferred by him from among the women to whom he is matched under any stable matching. In particular, Romeo: Juliet $\succ x$. Likewise, since Romeo is matched to Juliet under the women's courtship, she is the least preferred by him from among the women to whom he is matched under any stable matching. Hence, Romeo: $x \succ$ Juliet. Since the preference relation is transitive, Romeo: Juliet $\succ$ Juliet, a contradiction. Therefore, Romeo is matched to Juliet under every stable matching.

23.10 If the result of the men's courtship algorithm yields the same result as the women's courtship algorithm, then every man is matched to the same woman under both algorithms so, by Exercise 23.9, he is matched to her under every stable matching. In particular, this resulting matching is the unique stable matching.

23.11 Let A be a stable matching of n men and n women.

(a) It is not possible to find three pairs such that if the matching among them is changed, each man will be matched to a woman whom he prefers, and each woman will be matched to a man whom she prefers.

Suppose that under a stable matching A, the men X, Y and Z are matched to the women x, y and z, respectively. Let B be a matching such that the matching among the three pairs is changed. We show that necessarily at least one of the man/woman prefers the mate to whom he/she is matched under A. Assume that X is matched to y under B and X prefers y to x. Then, since A is stable, y prefers Y to X otherwise X and y object to a matching to A which contradicts the stability of A. Similarly, if X is matched to z under B and X prefers z to x. Then, since A is stable, z prefers Z to X, otherwise X and y object to a matching to A which contradicts the stability of A. Hence, under B at least one of each couple will be unsatisfied with the change.

(b) Let $4 \leq k \leq n$. There is no set of k pairs such that if the matching among them is changed, each man will be matched to a woman whom he prefers, and each woman will be matched to a man whom she prefers. The proof is identical to the proof in Part (a). Let $X - x$ and $Y - y$ two couple of pairs under A. If X is matched to y whom he prefers, under the new matching, then necessarily y prefers Y to X, otherwise X and y object to a matching to A which contradicts the stability of A.

23.12 In Julius's list of preferences, Agrippina appears first, Messalina appears second, and Cleopatra appears third. Suppose there is a stable matching under which Julius is matched to Agrippina, and that there is a stable matching under which Julius is matched to Cleopatra. There is not necessarily a stable matching under which Julius is matched to Messalina. For example if Messalina is in the top of Romeo list and Romeo is in the top of Messalina list, they will be matched to each other under every stable matching (Exercise 23.7) and hence there is no matching under which Julius is matched to Messalina.

23.13 In this exercise we consider a situation with n men and n women.

(a) Assume that in stage t of the men's courtship algorithm, a particular man, say Mark, is dismissed for the $(n-1)$-th time. According to the algorithm, if Mark has been dismissed $(n-1)$ times, then he visited $(n-1)$ women and at stage $t+1$ he will visit the n-th woman. In particular, at stage $t+1$ all the women has been visited. However, by the "once a woman is courted, she will always be courted" property, at the end of stage $t+1$, one man is standing in front of the house of each woman. Since the numbers of men and women are equal, no man is dismissed at this stage so the algorithm terminates.

(b) We prove that the men's courtship algorithm terminates after at most $(n-1)^2+1$ stages. Note that as long as the algorithm goes on, at list one man must be dismissed at each stage. By Part (a), if at some stage a man is dismissed for the $(n-1)$-th time, the algorithm terminates at the next stage. In conclusion the men's courtship algorithm terminates after at most $(n-1)(n-2)+n=(n-1)^2+1$ stages.

(c) Let $x_1, ..., x_n$ be a set of men, and let $y_1, ..., y_n$ be a set of women. Consider the following system of preferences.

$$\begin{aligned}
&x_1 : y_1 \succ y_2 \succ ... \succ y_{n-1} \succ y_n \\
&x_2 : y_2 \succ y_3 \succ ... \succ y_n \succ y_1 \\
&\vdots \\
&x_{n-1} : y_{n-1} \succ y_n \succ ... \succ y_{n-3} \succ y_{n-2} \\
&x_n : y_1 \succ y_2 \succ ... \succ y_{n-1} \succ y_n
\end{aligned}$$

$$\begin{aligned}
&y_1 : x_2 \succ x_3 \succ ... \succ x_n \succ x_1 \\
&y_2 : x_3 \succ x_4 \succ ... \succ x_2 \succ x_1 \\
&\vdots \\
&y_{n-1} : x_n \succ x_1 \succ ... \succ x_{n-2} \succ x_{n-1} \\
&y_n : x_1 \succ x_2 \succ ... \succ x_{n-1} \succ x_n
\end{aligned}$$

Since the men's preferences are circular but the preferences of x_n that are identical to x_1, at each stage there is only one women whom there are two men in front of her house (and x_n is not courted). By the women's preferences, every new man is preferred than the man that stood at the previous stage. Therefore, the order of the dismissed men is $x_1, x_2, ..., x_n, x_1, x_2 ...$ and so forth. In particular, the algorithm terminates after precisely $(n-1)^2+1$ stages (hence this is the lowest bound for the length of the algorithm).

23.14 Suppose that Fara is preferred by every man to all the other women. Then the man that Fara prefers at most is on her top list, and by the assumption she is in the top of his list. Thus, by Exercise 23.7, they are matched to each other under every stable matching.

23.15 Suppose that the number of men equals the number of women, and that Vera is last on the preference list of every man. Consider the man that is matched to Vera under the men's courtship algorithm. By theorem 23.12, Vera is necessarily the most preferred among all the women to whom he is matched under every stable match. However, since Vera is last on his preference list, it follows that under every stable list he is matched to Vera.

23.16 Assume that there are n men, $x_1, ..., x_n$ and n women, $y_1, ..., y_n$. Suppose every man has the same preference relation over the set of women $x_i : y_1 \succ y_2 \succ ... \succ y_n$. Then under every stable matching y_1 is matched to the man whom she prefers most, and every y_i is matched to the man she prefers most among the man that are not matched to $y_1, ..., y_{i-1}$. We prove it by induction over i. For $i = 1$ it follows by Exercise 23.14. Assume that under every stable matching every woman y_k is matched to the same man, for every $k < i$. Then, y_k is necessarily is not matched to the mates of $y_1, ..., y_{i-1}$. Furthermore, she is the most preferred among al the remaining women by the remaining men. Thus, using similar arguments as presented in Exercise 23.14, she must be matched to the man she prefers most among the man that are not matched to $y_1, ..., y_{i-1}$.

23.17 (a) We prove that this algorithm satisfies the three properties of the Gale–Shapley algorithm:

i. **The preferred women have been courted in the past:** If, in stage k, Henry stands in front of Anne, but prefers Catherine to Anne, then it must be the case that he previously courted Catherine by standing in front of her, and was dismissed. In other words, the men go down their preference list along the course of the algorithm.

ii. **The preferred men will come courting in the future:** The proof identical to that in Exercise 23.2.

iii. **Once a woman is courted, she will always be courted:** If, in stage k, Mark stands in front of Cleopatra, then from stage k onwards, there will always be at least one man courting Cleopatra by standing in front of her.

(b) We first prove that this algorithm terminates with a matching. A man who is dismissed by a women never returns to court her again. This means that each man courts at most n woman. Assume by contradiction that after n^2 stages there is at least one man outside the room. By the algorithm's construction, he must courts every woman. By the "once a women is courted, she will always be courted" property, after he has courted every woman, every woman has a man standing in front of her house. Since the number of women equals the number of men, and he is not standing in front of any woman, there must be a woman who has no man in front of her, a contradiction. Hence there cannot be a man who has been dismissed by every woman. It follows that the algorithm terminates by producing a matching.

The resulting matching is stable, by the same arguments presented in Theorem 23.7.

(c) Denote by O the matching obtained by the given algorithm. Assume by contradiction that the algorithm O does not always produce the men's courtship matching O^m. However, by Theorem 23.12, the men's courtship matching O^m is the best stable matching from the perspective of all the men. Therefore, every man who is matched to different women under O^m and O, prefers his mate under O^m. Furthermore, by the "The preferred women have been courted in the past" property, the woman to whom he is matched under O^m dismisses him during the algorithm O.

Let X be the first man that during the algorithm O, his mate x under O^m dismisses him and asks Y to to stay. That is, x prefers Y to X. Let z be the mate of Y under O^m. Then, it cannot be that Y prefers x to z or else, Y and x object the matching under O^m, a contradiction. That is, Y prefers z to x. However, by the "The preferred women have been courted in the past" property, z must dismiss Y before x dismisses X, which contradicts X being the first man that during the algorithm O, his mate x under O^m dismisses him.

23.18 We prove that the preference relation on matchings $\succsim^m$ over the set of stable matchings is transitive relation.

Let A, B, C be a stable matching. Assume that $A \succsim^m B$ and $B \succsim^m C$. That is, every man who is matched under A and B to different women prefers the woman to whom he is matched under A to the woman

to whom he is matched under B. In addition, every man who is matched under B and C to different women prefers the woman to whom he is matched under B to the woman to whom he is matched under C. Therefore, for every man who is matched under A and C to different women one of the following holds: (1) He is matched in A and B to the same woman hence, by $B \succsim_m C$, he prefers the woman to whom he is matched under A (and B) to the woman to whom he is matched under C. (2)He is matched in B and C to the same woman hence, by $A \succsim^m B$, he prefers the woman to whom he is matched under A to the woman to whom he is matched under C (and B). (3) He prefers the woman to whom he is matched under A to the woman to whom he is matched under B and he prefers the woman to whom he is matched under B to the woman to whom he is matched under C. Thus, by the transitivity of the preference relation of this man, he prefers the woman to whom he is matched under A to the woman to whom he is matched under C.
The proof that the preference relation on matchings $\succsim^w$ over the set of stable matchings is transitive relation is identical (by changing the role of men and women).

23.19 Theorem 23.11 does not hold if the matchings A and B are not stable. Let X, Y, Z be three men and x, y, z be three women with the following preferences relations:

$$\begin{array}{ll} x : Y \succ X \succ Z & X : x \succ y \succ z \\ y : Z \succ Y \succ X & Y : y \succ z \succ x \\ z : X \succ Y \succ Z & Z : z \succ x \succ y \end{array}$$

The stable matching A that is obtained by the men's courtship algorithm is $X - x, Y - y$ and $Z - z$. The matching B: $X - y$, $Y - x$, $Z - z$ is not a stable matching. However, $A \succsim^m B$ whereas $B \not\succsim^w B$.

23.20 We prove that when the partial ordering over the set of stable matchings is $\succsim^w$, then

$$\max\{A, B\} = A \vee^w B \text{ and } \min\{A, B\} = A \vee^m B.$$

By Theorem 23.14, if A and B are stable matchings, then $A \vee^w B$ is also a stable matching. Furthermore, by the definition of $A \vee^w B$ (Definition 23.13), $A \vee^w B \succsim^w A$ and $A \vee^w B \succsim^w B$. Assume that C is another stable matching satisfying $C \succsim^w A$ and $C \succsim^w B$. That is, every woman who is matched under C and $A(B)$ to different men prefers the man to whom she is matched under C to the man to whom she is matched under A (B). Therefore, every woman who is matched under C and $A \vee^w B$ to different men, prefers the man to

whom she is matched under C to the man to whom she is matched under $A \vee^w B$, since the man under $A \vee^w B$ is either the man under A or the man under B. In conclusion, $\max\{A, B\} = A \vee^w B$.
By reversing the roles of the men and the women, it follows that that when the partial ordering over the set of stable matchings is $\succsim^m$, then

$$\overset{m}{\max}\{A, B\} = A \vee^m B.$$

However, by Theorem 23.11 it follows that $\max^m\{A, B\} = \min^w\{A, B\}$. Hence, when the partial ordering over the set of stable matchings is $\succsim^w$, $\min\{A, B\} = A \vee^m B$.

23.21 We show that if A and B are two matchings (not necessarily stable), then $A \vee^w B$ is not necessarily a matching. For instance, suppose that there are two men X, Y and two women x, y. Assume that both women prefers X to Y. Let A be a matching under which X and x are matched and Y and y are matched. Let B be a matching under which X and y are matched and Y and x are matched (obviously, one of these matching is not stable). Then, under $A \vee^w B$, both women are matched to X which is not a matching.

23.22 See exercise 23.20.

23.23 Suppose that the number of women equals the number of men. Let $A_1, A_2, ..., A_K$ be stable matchings. Define a function B from the set of men to the set of women as follows. Under the function B, a man m is matched to the woman he most prefers from among the women to whom he is matched under the matchings $A_1, A_2, ..., A_K$.

(a) We prove that the function B is a matching. Assume by contradiction that it is not a matching, so there are two men m_1 and m_2 that are matched to the same woman w. In particular, there is a stable matching A_{l_1} under which m_1 and w are matched and there is a stable matching A_{l_2} under which m_2 and w are matched. The preference relation of w is asymmetric so she prefers one of these men to the other. Assume without loss of generality that w prefers m_1 to m_2. Then w and m_1 have an objection to a matching A_{l_2}, a contradiction.

(b) We prove that the function B is a stable matching. Assume by contradiction that there are a man m_1 who prefers a woman w_2 to the woman w_1 to whom he is matched, and assume that w_2 prefers m_1 to m_2 to whom she is matched. Since m_2 and w_2 are matched there is a stable matching A_k under which m_2 and w_2 are matched. Let w be the woman to whom m_1 is matched under A_k. However, since w_1 is the woman whom m_1 most prefers among those matched to him under $A_1, ..., A_k$, m_1 prefers w_1 to m and so, by the transitivity, he prefers w_2 to m. In particular m_1 and w_2 has an objection to A_k.

(c) Let C be a stable matching satisfying $C \succsim^m A_k$ for every $k \in \{1, 2, ..., K\}$. That is, every woman who is matched under C and A_k to different men prefers the man to whom she is matched under C to the man to whom she is matched under A_k, for every $k \in \{1, 2, ..., K\}$. Therefore, every woman who is matched under C and B to different men, prefers the man to whom she is matched under C to the man to whom she is matched under B, since the man under B is one of the men to whom she is matched under one of the matching A_k for $k \in \{1, 2, ..., K\}$. Therefore, matching C is the maximum of $A_1, A_2, ..., A_K$ according to the preference relations of the men.

23.24 Suppose that the matching problem is to match n_m men to n_w women, where $n_w < n_m$.

(a) Consider the generalization of the Gale-Shapley algorithm which is similar to the Gale-Shapley algorithm (the proof of Theorem 23.7) with the following minor change. In Stage k(a)(5): every man who was dismissed by a woman in the previous stage goes to stand in front of the house of the woman he most prefers from among the women who have not previously dismissed him, **unless all of the women dismissed him which case he stops trying and stays single from now on**.

We first prove that this algorithm terminates. All three properties, "the preferred women have been courted in the past", "the preferred men will come courting in the future" and "once a woman is courted, she will always be courted", are satisfied by the above version of Gale-Shapley algorithm.

A man who is dismissed by a women never returns to court her again. This means that each woman dismisses at most $n_m - 1$ men. It follows that after at most $n_w(n_m - 1)$ stages, there are no more rejections, and therefore after at most $n_w(n_m - 1)$ stages we arrive at a stage at which no woman dismisses any man.

After $n_w(n_m - 1)$ stages there is no rejection, hence at most one man stands in front of every woman. Hence the number of single men is at least $n_m - n_w \geq 1$. Assume Joe is single. According to the algorithm, Joe is single so he must have paid a visit to the house of every woman. By the "once a women is courted, she will always be courted" property, after Joe has courted every woman, every woman has a man standing in front of her house, so the algorithm terminates by producing a matching, and there are exactly $n_m - n_w$ single men.

We next show that this matching is stable. Assume by contradiction that there are a man, say Joe, and a woman, say Cleopatra, that have an objection. That is, either Joe is matched to Cornelia or Joe is a single, and Cleopatra is matched to Julius, but Joe and Cleopatra prefer each other. Since Joe prefers Cleopatra to Cornelia or being a single, by "the preferred women were courted in the past" property, Joe must have courted Cleopatra before but was dismissed by her. Since Cleopatra has been matched by the algorithm to Julius, by "the preferred men will come courting in the future" property, she prefers Julius to Joe. This contradicts the assumption that Joe and Cleopatra have an objection to the matching.

(b) The maximal number of stages in the men's courtship algorithm is $(n_w - 1)^2 + 1 + n_w$. Denote by M the set of men, and by S the set of singles at the end of the algorithm. Following the same arguments as those presented in Exercise 23.13c, the number of stages in which at least one man in $M \setminus S$ is dismissed is at most $(n_w - 1)^2 + 1$. It is left to show that the number of stages in which no man in $M \setminus S$ is dismissed is at most n_w. Assume that at some stage t no man in $M \setminus S$ is dismissed. Since $|M \setminus S| = n_w$, every woman asks some man in $M \setminus S$ to stay. In particular, every man in S is dismissed. However, every single man is dismissed at most n_w times, therefore the number of stages in which no man in $M \setminus S$ is dismissed is at most n_w.

(c) To show that this is the lowest bound for the length of the algorithm consider the following preference relations. Let $x_1, ..., x_{n_m}$ be a set of men, and let $y_1, ..., y_{n_w}$ be a set of women. Consider the following system of preferences.

$x_1 : y_1 \succ y_2 \succ ... \succ y_{n_w-1} \succ y_{n_w}$

$x_2 : y_2 \succ y_3 \succ ... \succ y_{n_w} \succ y_1$

$\vdots$

$x_{n_w-1} : y_{n_w-1} \succ y_{n_w} \succ ... \succ y_{n_w-3} \succ y_{n_w-2}$

$x_{n_w} : y_{n_w} \succ y_1 \succ y_2 \succ ... \succ y_{n_w-1}$

$x_i : y_1 \succ y_2 \succ ... \succ y_{n_w-1} \succ y_{n_w} \quad \forall i \in \{n_w + 1, ..., n_m\}$

$y_1 : x_2 \succ x_3 \succ ... \succ x_{n_w} \succ x_1 \succ x_{n_w+1} \succ x_{n_w+2} + ... + x_{n_m}$

$y_2 : x_3 \succ x_4 \succ ... \succ x_2 \succ x_1 \succ x_{n_w+1} \succ x_{n_w+2} + ... + x_{n_m}$

$\vdots$

$y_{n_w-1} : x_{n_w} \succ x_1 \succ ... \succ x_{n_w-2} \succ x_{n_w-1} \succ x_{n_w+1} \succ x_{n_w+2} + ... + x_{n_m}$

$y_{n_w} : x_1 \succ x_2 \succ ... \succ x_{n_w-1} \succ x_{n_w+1} \succ x_{n_w} \succ x_{n_w+2} + ... + x_{n_m}$

This preference relation is similar to that in Exercise 23.13c with regards to $x_1, ..., x_{n_w}, y_1, .., y_{n_w}$. Hence, it is enough to show that the process described in Exercise 23.13c starts after n_w stages. Indeed, during the first $n_w - 1$ stages, each man x_i stands in front of y_i for every $i = 1, ..., n_w$ whereas $x_{n_w+1}, ..., x_{n_m}$ visit each woman from y_1 to y_{n_w-1} and are dismissed. At stage n_w, $x_{n_w+1}, ..., x_{n_m}$ visit y_{n_w} whom asks x_{n_w+1} to stay and dismisses $x_{n_w}, x_{n_w+2}, x_{n_w+3}, ..., x_{n_m}$. Therefore, $x_{n_w+2}, x_{n_w+3}, ..., x_{n_m}$ become singles and the process described in Exercise 23.13c starts.

23.25 Assume that the number of men n_m is greater than the number of women n_w. We show that if some woman rejects a man during the men's courtship algorithm, there is no stable matching under which they are matched to each other.
Note that, by proving the above calim, one can deduce that every single man under the men's courtship matching must, have been rejected by all woman so he must be single in every stable matching. However, since being a single is the least preferred, every man that is matched to a woman under the men's courtship mathcing is not single under any stable matching. Hence, the set of singles are the same under every stable matching.
Assume by contradiction that the above claim is wrong. Let m be the first man in the men's courtship algorithm that was rejected by a woman w but he is matched to her under some stable matching

A. Let m^* be the man to whom w is matched under men's courtship matching. In particular, w prefers m^* to m. Since A is stable so m^* and w have no objection to A, it follows that m^* is not a single under A (since being a single is the least preferred) but he is matched to w^* who he prefers to w. However, by "the preferred women have been courted in the past" property, m^* was rejected before m by a woman to whom he is matched under some stable matching, a contradiction.

23.26 (a) The matching under both woman's and men's courtship algorithms is Ralph-Lilly, Quinn-Kara, Oscar-Mary, and Peter is a single.

(b) The matching under both woman's and men's courtship algorithms is Jack-Hanna, Isaac: Gloria, and Grant, Howard and Ida are singles.

23.27 Let A be a matching (not necessarily stable), and let M_{A,O^m} be the set of men who prefer the women to whom they are matched under A to the women to whom they are matched under the men's courtship matching O^m.

(a) If A is a stable matching then, by Theorem 23.12, $M_{A,O^m} = \emptyset$.

(b) Denote by W_1 the set of the women who are matched to men in M_{A,O^m} under the matching A, and by W_2 the set of the women matched to men in M_{A,O^m} under O^m. Assume that $W_1 \neq W_2$, and assume by contradiction that $W_1 \setminus W_2 = \emptyset$. Then $W_2 \setminus W_1 \neq \emptyset$, so

$$W_1 = W_1 \cap W_2 \subset W_2 \Rightarrow |W_1| < |W_2|,$$

which contradict that $|W_1| = |M_{A,O^m}| = |W_2|$.

(c) Assume that $W_1 \neq W_2$, and let $w \in W_1 \setminus W_2$. That is, under O^m, w is matched to a man $m_1 \notin M_{A,O^m}$ that prefers her to the woman he is matched under A. In addition, under A, w is matched to a man $m_2 \in M_{A,O^m}$ that prefers her to the woman he is matched under O^m. However, by the property "the preferred women have been courted in the past" and "the preferred men will come courting in the future" properties, w prefers m_1 to m_2. In conclusion, w and m_1 have an objection to A.

(d) From here to the end of the exercise, assume that $W_1 = W_2$. Let $w \in W_1$. Then, there are is a man $m \in M_{A;O^m}$ such that w is matched to m under A. Since, $m \in M_{A,O^m}$, he prefers w to the woman to whom he is matched under O^m. By the "the preferred women have been courted in the past" property, w necessarily dismisses m under the men's courtship algorithm. dismisses at least one man under the men's courtship algorithm.

(e) Consider the woman $w^* \in W_1$ who was the last woman approached by a man $m^* \in M_{A,O^m}$ under the men's courtship algorithm. If at the time t^*, w^* receives an offer from m^* and there was no man at her doorstep, then by Part (d), the man in M_{A,O^m} to whom she is matched under A was dismissed at some time $t > t^*$ from which it follows that the woman to whom he is matched under O^m was approached by a man in M_{A,O^m} under the men's courtship algorithm after w, which contradict w is the last.

(f) Denote by m' the man that that stands at w^* doorstep when she receives an offer from m^*. Using the fact that w^* is the last woman to get an offer from a man in M_{A,O^m} under O^m, and since m' offer another woman under O^m after dismissed by w^*, m' is not in M_{A,O^m}.

(g) Since $m' \notin M_{A,O^m}$, m' prefers the woman to whom he is matched under O^m then the one he is matched under A. Furthermore, by "the preferred women have been courted in the past" property, m' prefers w^* to the woman he is matched under O^m so by transitivity, he prefers w^* to woman he is matched under A. By Parts (d) and (e), the man to whom w^* is matched under A dismiss before m', hence, by "the preferred men will come courting in the future" property, she prefers m' to the man to whom she is matched under A. In conclusion, w^* and m' has an objection to A.

23.28 Let $\hat{G}$ be the matching problem derived from G by changing the preference relations of some of the men: let $\hat{M} \subseteq M$ be a set of men whom we term "dishonest," whose preference relations in matching problem $\hat{G}$ differ from their preference relations in matching problem G.

Denote by $M_{\hat{O}^m,O_m}$ the set of men who truly prefer the women to whom they are matched under $\hat{O}^m$ to the women to whom they are matched under the men's courtship matching O^m (in G).

We prove that there is no stable matching $\hat{A}$ for matching problem $\hat{G}$ satisfying the following properties: every man in $\hat{M}$ prefers (according to his true preference relation) the woman to whom he is matched under $\hat{A}$ to the woman to whom he is matched under O^m.

Assume by contradiction that there is a stable matching satisfying the above condition. Using Exercise 23.27, then since $\emptyset \neq \hat{M} \subseteq M_{\hat{A},O_m}$, there exists a pair (m, w) who object to the matching $\hat{A}$ in G, and $m \notin M_{\hat{O}^m, O_m}$. However, the preference relation of both m and w in G and $\hat{G}$ are identical, so (m, w) object to the matching $\hat{A}$ in $\hat{G}$, so $\hat{A}$ is not stable in $\hat{G}$, a contradiction.

23.29 Suppose that under the men's courtship matching O^m Messalina is matched to Claudius. Consider the matching A that results from the men's courtship algorithm when Messalina pretends that her preference relation is different from her true preference relation.

(a) We prove that the matching A is a stable matching under the true preferences of the men and women if and only if the man to whom Messalina is matched under A is the man she most prefers from among the men that came to her door throughout the algorithm; that is, she does not regret any rejection she made.

Since A is a result of the men's courtship algorithm where everyone except for Messalina are honest (which is stable matching as long as Messalina keeps lying), no woman except for Messalina has a objection.

Furthermore, by "the preferred women have been courted in the past" property of the men's courtship algorithm, every man who prefers Messalina over the woman to whom he is matched under A, came to her door throughout the algorithm and was dismissed. Hence, A is stable, if and only if the man to whom Messalina is matched under A is the man she most prefers from among the men that came to her door throughout the algorithm.

(b) Assume that messalina's mates under O^m and A are Jack and Mark, respectively. Suppose that Messalina prefers Mark to Jack. By "the preferred men will come courting in the future" property, Mark did not come to her door throughout the algorithm when she was honest. Hence, he prefers the woman to whom he is matched under O^m to Messalina. In particular, there is at least one man who is matched under A to a woman whom he prefers less than the woman to whom he is matched under O^m.

23.30 **The Gale–Shapley algorithm for the case in which the single life is not universally considered the worst possible outcome:** Consider the matching problem G presented in Section 23.5.3. Let M be a set of men and let W a set of women. For every woman $w \in W$, a preference relation over the set $M \bigcup \{(w - single)\}$. For every man, a preference relation over the set $W \bigcup \{(m - single)\}$.
We define a new matching problem G' as follows:

- The set of men is $M' := M \bigcup \{(w - single) : w \in W\}$. The set of women is $W' := W \bigcup \{(m - single) : m \in M\}$.
- Every woman w prefers the element $(w - single)$ to the element $(\hat{w} - single)$, for every woman $\hat{w} \neq w$. The precise ordering under which the elements $\{(\hat{w} - single), \hat{w} \in W \setminus \{w\}\}$ are ordered is immaterial.
- Every man m prefers the element $(m - single)$ to the element $(\hat{m} - single)$, for every man $\hat{m} \neq m$. The precise ordering under which the elements $\{(\hat{m} - single), \hat{m} \in M \setminus \{m\}\}$ are ordered is immaterial.
- The element $(w - single)$ prefers w to every other element in W'. The precise ordering under which the other elements are ordered in the preference relation of this element is immaterial.
- The element $(m - single)$ prefers m to every other element in M' The precise ordering under which the other elements are ordered in the preference relation of this element is immaterial.

We implement the men's courtship algorithm over G' as presented in Theorem 23.7 but the algorithm terminates when there can be no more dismissals. We next show that the algorithm terminates with stable matching in G'. We use this matching to define a matching in G by eliminating couples in which the man and the woman are not in M and W, respectively.
Note that, since we only changed the termination rule the algorithm over G' satisfies all three properties of Gale-Shapley algorithm.
In addition, a man who is dismissed by a women in W' never returns to court her again. Hence, every woman rejects at most $|W'| - 1$ men and the algorithm over G' necessarily terminates.
We show that that the algorithm terminates with a matching. First, $(w - single)$ most prefers w among all woman in W' so at first stage she courts w, moreover, by "once a woman is courted, she will always be courted" property, all women are matched at the end of the algorithm. Every $(m - single)$ most prefers m among all man in M', so if m is dismissed by all the woman who he prefers to $(m - single)$, he eventually comes to the door of $(m - single)$ he is asked to stay from then on. In conclusion the algorithm end with a matching.
We proof that this matching is stable in G'. Assume that there is a man $m \in M$ that prefers a woman w to his mate under this matching. By "the preferred women have been courted in the past" property, he came to w and was dismissed by her. Therefore, by "the preferred men will come courting in the future" property, w necessarily prefers her mate to m, so the matching is stable.
In particular, in this matching there is no woman w that has been matched to $(\hat{w} - single)$ where $w \neq \hat{w}$ or else w and $(w - single)$

have an objection since w prefers $(w - single)$ to $(\hat{w} - single)$ and $(w - single)$ most prefers w. Similarly, there is no man m that has been matched to $(\hat{m} - single)$ where $m \neq \hat{m}$. That is, in G' each man $m \in M$ is matched to either a woman $w \in W$ or to $(m - single)$, and each woman $w \in W$ is matched to either a man $m \in M$ or to $(w - single)$. Hence, by eliminating couples in which the man and the woman are not in M and W, respectively, we get a matching in G. Furthermore, since every objection in G is an objection in G', the matching is necessarily stable.

23.31 **The generalization of the Gale–Shapley algorithm (the students' courtship algorithm) in the case in which each university has a quota for the maximal number of students that it can accept:**
Let U be a finite set of universities and let S be a finite set of students. For each university $u \in U$ a quota $q_u \in \mathbf{N}$ represents the maximal number of students it will accept. For each student $s \in S$, there is a preference relation over the set U. For each university $u \in U$, a preference relation over the set S. The students' courtship algorithm is as follows:
Stage 1(a): Every student applies to the university he most prefers.
Stage 1(b): Every university $u \in U$ asks the q_u students that it most prefers from among the students gathered at the university's gate to remain (instead of asking only one) and rejecting all the rest; if less than q_u students have applied, it asks all of them to remain.
Stage k(a): Every rejected student goes to the university next down on his or her preference list.
Stage k(b): Every university $u \in U$ asks asks the q_u students that it most prefers from among the students gathered at the university's gate to remain (instead of asking only one) and rejecting all the rest; if less than q_u students have applied, it asks all of them to remain.
The algorithm terminates when there is no rejection.
We next prove that the algorithm terminates with stable matching. As long as the algorithm goes on, at least one student is rejected. Every students is rejected by at most $|U|$ universities. Hence, after finitely many stages there is no rejection and the game terminates. Since there is no rejection, in front of each university $u \in U$ there are at most q_u students, hence it terminate with a matching. We next prove that this matching is stable. Let $s \in S$ be a students that prefers university u to the university $\hat{u}$ to whom he is matched. Then s must applied to u but was rejected by her, hence by "the preferred students will come courting in the future" property, u prefers the students to whom it is matched. Similarly, if a students s is not matched by any university under this matching, then he applied to all university and was rejected, so again no university u prefers him to the students to

whom it is matched. In conclusion, the matching is stable.
The universities' courtship algorithm:
Stage 1(a): Every university $u \in U$ applies to q_u students it most prefers (instead of applying only one).
Stage 1(b): Every student asks the university that he most prefers from among the universities gathered at his door to remain and rejecting all the rest.
Stage k(a): Every university u that is rejected by r_u students ($r_u \leq q_u$) apply to r_u students next down on its preference list.
Stage k(b): Every student asks the university that he most prefers from among the universities gathered at his door to remain and rejecting all the rest.
The algorithm terminates when there is no rejection.
The last algorithm is preferred by the students (Exercise 23.32).

23.32 (a) We prove that in the students' courtship algorithm, every student is matched to the university that is most preferred by him from among all the universities to which he is matched under all possible stable matchings. Assume by contradiction that this is not true. Let s be the first student under the students' courtship algorithm that was rejected by a university u but there is another stable matching A under which s and u are matched. In particular, there must be at least one student $\hat{s}$ that is matched to u under the students' courtship but under A he is matched to $\hat{u} \neq u$. By the property, "the preferred student will come courting in the future" property, u prefers $\hat{s}$ to s. Hence, since A is stable, $\hat{u}$ prefers $\hat{u}$ to u. In particular, by "the preferred university have been courted in the past" property of the students' courtship algorithm, $\hat{s}$ was rejected by $\hat{u}$ before s was rejected by u, a contradiction.

(b) Let A be a stable matching, and let s be a student who is not matched by any university under the students' courtship algorithm. Then, s is rejected by all the universities, so by Part (a), he is not matched to any university under any stable matching.

(c) Consider the universities' courtship algorithm. A university that was rejected by a student will not be matched to him under any stable matching (the proof is similar to the proof of Part (a)). Therefore, a university u that is matched to only m_u students where $m_u \leq q_u$, is matched to at most m_u students under every stable matching. However, if by contradiction there is a university u that is matched to less that m_u students under A so it does not fill its student quota, then there is at least one student s who is not matched to any university. Thus u and s have an objection to A, which contradicts that stability of A. In conclusion, the number of students to whom a university is matched under any stable matching is identical. Hence, if under one stable matching there is a university that does not fill its student quota, then that university does not fill its student quota under any stable matching.

23.33 (a) Given a matching,

- A university $u \in U$ objects to the matching, if it is matched to a student $s \notin S_u$.
- A student $s \in S$ objects to the matching if he is matched to a university $u \in U_s$.
- A university u and a student $s \in S$ object to the matching if (a) $u \in U_s$ and $s \in S_u$; (b) the university u is matched to fewer than q_u students, or it is matched with q_u students but prefers s to one of the students that are matched to it; and (c) the student s prefers university u to the university to which he is matched, or he is not matched to any university under the matching.

A matching is stable if no university, no student, and no pair of a university and a student object to it.

(b) Consider a model of polygamous matchings denoted by G. Let U be the set of universities. For every $u \in U$, let $q_u \geq 1$ be the student quota, and let S_u be the the set of students that university u is willing to accept. Let S be a set of students. For every student $s \in S$, denote by U_s the set of universities that the student is willing to attend. We define a model of polygamous matchings as presented in Exercise 23.31, which we denote by $\hat{G}$.

The set of universities is given by

$$\hat{U} = U \bigcup \{(s - \text{not attend}) : s \in S\}.$$

For every university $u \in U$, the student quota is q_u, for every $(s - \text{not attend})$ the quota is 1.

The set of students are

$$\hat{S} = S \bigcup \{(u_k - \text{saving space}) : u \in U, k = 1, 2, ..., q_u\}.$$

The preference relations are as follows:

- On the top of the preference list of a university u is the set of S_u where the preference over S_u is as given in G. In the middle of the list is the set $\{(u_k - \text{saving space}) : k = 1, 2, ..., q_u\}$

where the precise ordering over this set is immaterial. On the bottom of the list is the set which includes the students in $S \setminus S_u$ and

$$\{(u'_k - \text{saving space}) : k = 1, 2, ..., q_{u'}, u' \in U \setminus \{u\}\}$$

where the ordering over this set is immaterial.

- On the top of the preference list of a student s is the set of U_s where the preference over U_s is as given in G. In the middle of the list is $(s - \text{not attend})$. On the bottom of the list is the set which includes the universities in $U \setminus U_s$ and

$$\{(s' - \text{not attend}) : s' \in S \setminus \{s\}\}$$

where the ordering over this set is immaterial.

- $(u_k - \text{saving space})$ most prefers u over every other university in U', for every $u \in U$ and $k = 1, 2, ..., q_u$. The ordering over $U' \setminus \{u\}$ is immaterial.
- $(s - \text{not attend})$ most prefers s over every other student in S', for every $s \in S$. The ordering over $S' \setminus \{s\}$ is immaterial.

Using Exercise 23.31, one can find a stable matching to G'. In this matching, there is no university u that is matched to either $s \in S \setminus S_u$ or $(u'_k - \text{saving space})$ where $u' \neq u$, else u and $(u_k - \text{saving space})$ object to the matching. Furthermore, every university fills its student quota by either students in S_u and/or one or more "saving spaces". Similarly, there is no student that is matched to either $u \in U \setminus U_s$ or $(s' - \text{not attend})$ where $s' \neq s$. In addition, every student is matched to either university in U_s or to $(s - \text{not attend})$.

In particular, we can use this matching to define a matching in G, by eliminating the auxiliary universities and students from the matching. If there will be an objection in G to the matching then this will be an objection to the matching in G'. Hence the matching in G is also stable.

(c) Every stable matching in G' define a stable matching in G and vice versa. Hence, using Exercise 23.32 Part (c) one can obtain that a student who is not accepted by any university under a particular stable matching will not be accepted into any university under any stable matching.

23.34 (a) A stable matching between the universities and the applicants that is preferred by all the applicants to any other stable matching is:

$$X - j, k, g, d$$
$$Y - h, a$$
$$Z - f, c, b$$
i and e do not attend.

(b) A stable matching between the universities and the applicants that is preferred by all the applicants to any other stable matching is:

$$X - a, b, c, d, e, f$$
$$Y - n, m, l, k, j, i$$
h and g do not attend.

23.35 (a) Given each man's weak preference relation, construct a strict preference relation by ordering the women to whom he is indifferent in an arbitrary way, while maintaining transitivity of preferences, and do the same with the preference relation of each woman. Denote by G^w the matching problem defined by weak preference relations, and by G^s the corresponding matching problem defined by strictly preference relations. Let A be a stable matching in G^s. We show that A is a stable matching in G^w. If in G^w there is a man m that strictly prefers a woman w to the woman to whom he is matched under A, then he strictly prefers this woman in G^s. Hence, since A is stable in G^s, w strictly prefers the man to whom she is matched under A to m, from which it follows that in G she necessarily does not strictly prefer m to her mate, so there is no objection. In conclusion, A is stable in the original matching problem with weak preferences G^w.

(b) Stable matchings for the given preference relations:

- Tina-Albert, Ursula-Boris, Victoria-Chester, and David is single (when Ursula prefers Chester to Albert, Victoria prefers Chester to David and Albert prefers Tina to Victoria).
- Tina-Albert, Ursula-Chester, Victoria-David, and Boris is single (when Ursula prefers Chester to Albert, Victoria prefers David to Chester and Albert prefers Tina to Victoria).

23.36 Let A be a stable matching to a matching problem G^w in which the preference relations of the men and women may include instances of indifference.
We use G^w to define a matching problem G^s where the weak preference relations (i.e., with instances of indifference) are replaced with strict preference relations. For every woman w, if w is indifferent between a man m to whom she is matched and some other man $\hat{m}$, then in the strict preference she prefers m to $\hat{m}$, all other indifference become strict preference in an arbitrary way, while maintaining transitivity of preferences. Similarly, for every man m, if m is indifferent between a woman w to whom he is matched and some other woman $\hat{w}$, then in the strict preference he prefers w to $\hat{w}$, all other indifference become strict preference in an arbitrary way, while maintaining transitivity of preferences.
We show that A is a stable matching in G^s.
If a man m is matched to w but prefers a woman $\hat{w}$, then in G he strictly prefers $\hat{w}$ to w. Therefore, since A is a stable matching, either $\hat{w}$ prefers her mate to m in G and thus in G^s as well, or $\hat{w}$ is indifferent between her mate and m in G, which case in G^s she strictly prefers her mate to m. In conclusion, A is a stable matching in G^s.

23.37 The example in Exercise 23.35 is a counter example which shows that Theorem 23.23 does not hold if there may be instances of indifference in the preference relations of the men and the women.

23.38 We show that Example 23.32 has no stable matching.
Suppose next, that Franklin is paired with Benjamin, which then means that Alex is paired with Chris: (Franklin – Benjamin , Alex – Chris). Then Alex and Benjamin have an objection: Alex prefers Benjamin (number 1 on his list) to Chris (number 2 on his list), and Benjamin prefers Alex (number 2 on his list) to Franklin (number 3 on his list).
Suppose finally, that Franklin is paired with Chris, which then means that Alex is paired with Benjamin: (Franklin – Chris , Alex – Benjamin). Then Chris and Benjamin have an objection: Chris prefers Benjamin (number 2 on his list) to Franklin (number 3 on his list), and Benjamin prefers Chris (number 1 on his list) to Alex (number 2 on his list), which completes the proof.

23.39 Suppose that a population of size 3n is partitioned into three subsets: n contractors, n carpenters, and n plumbers. Each person in this population has two preference relations: a preference relation over each one of the two subsets listed above to which he does not belong. We construct a stable matching in this model. We first use Gale-

Shapley contractors' courtship algorithm to find a stable matching A^1 in the matching problem G^1 under which contractors and carpenters are matched. We next use Gale-Shapley contractors' courtship algorithm to find a stable matching A^2 in the matching problem G^2 under which contractors and plumbers are matched. Then we define a matching A in the original matching problem under which a trio composed of a contractor x, a carpenter y, and a plumber z is in A if and only if $x - y$ is in A^1 and $x - z$ is in A^2. A is a stable matching. Indeed, if a contractor x prefers a carpenter y to the carpenter to whom he is matched, then by the stable matching A^1, y prefers the contractor to whom he is matched under A^1 and so under A. Similarly, if a contractor x prefers a plumber z to the plumber to whom he is matched, then by the stable matching A^2, z prefers the contractor to whom he is matched under A^2 and so under A.

23.40 We show by a counter example that the matching problem in Exercise 23.39 does not necessarily has a stable matching in case that each member of the population has a preference relation over pairs of potential co-workers. Let $\{x_1, x_2\}$ be a set of contractors, $\{y_1, y_2\}$ a set of carpenters, and $\{z_1, z_2\}$ a set of plumbers. Suppose that the preference relations of the members of the population are as follows.

$$\begin{aligned}
x_1 &: \{y_1, z_2\} \succ \{y_1, z_1\} \succ \{y_2, z_1\} \succ \{y_2, z_2\} \\
x_2 &: \{y_1, z_1\} \succ \{y_2, z_1\} \succ \{y_2, z_2\} \succ \{y_1, z_2\} \\
y_1 &: \{x_2, z_1\} \succ \{x_1, z_2\} \succ \{x_1, z_1\} \succ \{x_2, z_2\} \\
y_2 &: \{x_2, z_2\} \succ \{x_1, z_1\} \succ \{x_1, z_2\} \succ \{x_2, z_1\} \\
z_1 &: \{x_1, y_2\} \succ \{x_2, y_1\} \succ \{x_2, y_2\} \succ \{x_1, y_1\} \\
z_2 &: \{x_1, y_1\} \succ \{x_2, y_2\} \succ \{x_2, y_1\} \succ \{x_1, y_2\}.
\end{aligned}$$

In this matching problem there is 4 possible matching,none of them is stable.

- The matching $\{x_1, y_1, z_1\}, \{x_2, y_2, z_2\}$ is not stable because the trio $\{x_1, y_1, z_2\}$ has an objection.
- The matching $\{x_1, y_1, z_2\}, \{x_2, y_2, z_1\}$ is not stable because the trio $\{x_2, y_1, z_1\}$ has an objection.
- The matching $\{x_1, y_2, z_2\}, \{x_2, y_1, z_1\}$ is not stable because the trio $\{x_1, y_2, z_1\}$ has an objection.
- The matching $\{x_1, y_2, z_1\}, \{x_2, y_1, z_2\}$ is not stable because the trio $\{x_2, y_2, z_2\}$ has an objection.

Chapter 24: Appendices

24.1 Let $\{x^0, x^1, ..., x^K\}$ be a set of vectors in $\mathbf{R}^n$. Define

$$E = \{x \in \mathbf{R}^n : x \text{ is a convex combination of } x^0, x^1, ..., x^K\}.$$

We prove that $E = \operatorname{conv}(x^0, x^1, ..., x^K)$.

We start by proving that $\operatorname{conv}(x^0, x^1, ..., x^K) \subseteq E$. For every $k \in \{0, 1, ..., K\}$,

$$x^k = 0 \cdot x^0 + 0 \cdot x^{k-1} + 1 \cdot x^k + 0 \cdot x^K \in E.$$

In addition, let $x, y \in E$ and let $\lambda \in (0,1)$. Then, there are $(\alpha^k)_{k=0}^K, (\beta^k)_{k=0}^K \in \Delta(\{0,1,...,K\})$ such that $x = \sum_{k=0}^K \alpha^k x^k$ and $y = \sum_{k=0}^K \beta^k x^k$. Hence,

$$\lambda x + (1-\lambda)y = \lambda \sum_{k=0}^K \alpha^k x^k + (1-\lambda) \sum_{k=0}^K \beta^k x^k = \sum_{k=0}^K (\lambda\alpha^k + (1-\lambda)\beta^k) x^k.$$

However, since the simplex is convex, it can be deduced that $\lambda x + (1-\lambda)y \in E$. In particular, E is a convex set containing $x^0, ..., x^K$ and thus

$$\operatorname{conv}(x^0, x^1, ..., x^K) \subseteq E.$$

On the other hand, we show that $\operatorname{conv}(x^0, x^1, ..., x^K)$ contains every convex combination of $x^0, x^1, ..., x^K$ from which it follows that $E \subseteq \operatorname{conv}(x^0, x^1, ..., x^K)$.

Let $\alpha = (\alpha^k)_{k=0}^K \in \Delta(\{0,1,...,K\})$. Denote by $N_\alpha = |\{k \in \{0,1,...,K\} : \alpha^k > 0\}|$ the size of the support of α. We prove that

$$\sum_{k=0}^K \alpha^k x^k \in \operatorname{conv}(x^0, x^1, ..., x^K) \tag{206}$$

by induction over N_α.

For $N_\alpha = 1$ and $N_\alpha = 2$ Equation (206) follows the definition of the convex hull. Assume that Equation (206) holds for $N_\alpha \leq n$, we prove it also holds for $N_\alpha = n + 1 \leq K + 1$.

Assume without loss of generality that $\alpha^0 < 1$, then

$$\sum_{k=0}^K \alpha^k x^k = \alpha^0 x^0 + (\sum_{k'=1}^K \alpha^{k'}) \sum_{k=1}^K \frac{\alpha^k}{\sum_{k'=1}^K \alpha^{k'}} x^k.$$

However, by the induction hypothesis $\sum_{k=1}^K \frac{\alpha^k}{\sum_{k'=1}^K \alpha^{k'}} x^k \in \operatorname{conv}(x^0, x^1, ..., x^K)$ and thus by the convexity of the convex hull

$$\sum_{k=0}^K \alpha^k x^k \in \operatorname{conv}(x^0, x^1, ..., x^K),$$

as claimed.

24.2 Let $x^0, x^1, ..., x^k$ be vectors in $\mathbf{R}^n$, such that none of them is a convex combination of the other vectors.
We prove that the extreme points of $\text{conv}(x^0, x^1, ..., x^k)$ are the vectors $x^0, x^1, ..., x^k$.
We first show that x^i is an extreme point for every $i \in \{0, 1, ..., k\}$. Assume by contradiction that x^i is not an extreme point. That is, there are $x, y \in \text{conv}(x^0, x^1, ..., x^k)$ where $x \neq y$ and $\lambda \in (0, 1)$ satisfying $x^i = \lambda x + (1 - \lambda)y$. Let $(\alpha^{i'})_{i'=0}^K, (\beta^k)_{i'=0}^k \in \Delta(\{0, 1, ..., k\})$ such that $x = \sum_{i'=0}^k \alpha^{i'} x^{i'}$ and $y = \sum_{i'=0}^k \beta^{i'} x^{i'}$. Therefore,

$$x^i = \sum_{i'=0}^{k} (\lambda \alpha^{i'} + (1 - \lambda)\beta^{i'})x^{i'}.$$

Since x^i is not a convex combination of the other vectors, $\alpha^i = \beta^i = 1$ and $\alpha^{i'} = \beta^{i'} = 0$ for every $i' \neq i$. Hence, $x = y$, a contradiction.
To complete the proof, we show that for every $x \in \text{conv}(x^0, x^1, ..., x^k)$ such that $x \neq x^i$ for every $i \in \{0, 1, ..., k\}$, x is not an extreme point. Indeed, there are $(\alpha^i)_{i=0}^k \in \Delta(\{0, 1, ..., k\})$ in which at least two of them are positive such that $x = \sum_{i=0}^k \alpha^i x^i$. Assume without loss of generality that $\alpha^0 > 0$, then $1 - \alpha^0 = \sum_{i=1}^k \alpha > 0$ as well. Hence,

$$x = \alpha^0 \cdot x^0 + (1 - \alpha^0) \sum_{i=1}^{k} \tfrac{\alpha^i}{1-\alpha^0} x^i.$$

But x^0 and $\sum_{i=1}^k \frac{\alpha^i}{1-\alpha^0} x^i$ are in the convex hull (by Exercise 24.1). Moreover, since none of the vectors $x^0, x^2, ..., x^k$ is a convex combination of the other vectors, $x^0 \neq \sum_{i=1}^k \frac{\alpha^i}{1-\alpha^0} x^i$. In conclusion, x is a combination of two points in the convex hull so it is not an extreme point of it.

24.3 Let x^0, x^1, x^2 be three vectors in $\mathbf{R}^n$. We prove that these vectors are affine independent if and only if none of them is a convex combination of the other two.
We prove an equivalent argument: x^0, x^1, x^2 are not affine independent if and only if one of them is a convex combination of the other two.
Indeed, x^0, x^1, x^2 are not affine independent if and only if there are $(\alpha^l)_{l=0}^2$ in $\mathbf{R}$, not all zero, satisfying

$$\sum_{l=0}^{2} \alpha^l x^l = 0, \tag{207}$$

$$\sum_{l=0}^{2} \alpha^l = 0. \tag{208}$$

if and only there are $(\alpha^l)_{l=0}^2$ in $\mathbf{R}$ satisfying Equations (207) and (208), where only one of them is negative (say α_0) and at least one of the other two is positive. If and only if,

$$x_0 = \tfrac{\alpha_1}{-\alpha_0} x_1 + \tfrac{\alpha_2}{-\alpha_0} x_2 = \tfrac{\alpha_1}{\alpha_1+\alpha_2} x_1 + \tfrac{\alpha_2}{\alpha_1+\alpha_2} x_2,$$

i.e., x_0 is a is a convex combination of the other two.

24.4 (a) The vectors in the set are affine independent. Indeed, consider the following system of equations

$$\sum_{i=0}^{4} \alpha^i x^i = (\alpha^0 + \alpha^3 + \alpha^4, \alpha^0 + \alpha^1 + \alpha^4, \alpha^1 + \alpha^2 + \alpha^4, \alpha^2 - \alpha^3) = \overrightarrow{0},$$

and $\alpha^0 + \alpha^1 + \alpha^2 + \alpha^3 + \alpha^4 = 0$.

Hence,

$$0 = (\alpha^0 + \alpha^3 + \alpha^4) + (\alpha^1 + \alpha^2 + \alpha^4) \tag{209}$$

$$= (\alpha^0 + \alpha^1 + \alpha^2 + \alpha^3 + \alpha^4) + \alpha^4 \tag{210}$$

$$= 0 + \alpha^4 \Rightarrow \alpha^4 = 0. \tag{211}$$

That leaves the following system of equation

$$\sum_{i=0}^{3} \alpha^i x^i = (\alpha^0 + \alpha^3, \alpha^0 + \alpha^1, \alpha^1 + \alpha^2, \alpha^2 - \alpha^3) = \overrightarrow{0}, \tag{212}$$

$$\text{and } \alpha^0 + \alpha^1 + \alpha^2 + \alpha^3 = 0, \tag{213}$$

from which it follows that $\alpha^0 = \alpha^1 = \alpha^2 = \alpha^3 = 0$.

(b) The vectors in the set are affine independent. Indeed, as in the previous part

$$\sum_{i=0}^{4} \alpha^i x^i = (\alpha^0 + \alpha^3 + \alpha^4, \alpha^0 + \alpha^1 + 2\alpha^4, \alpha^1 + \alpha^2 + \alpha^4, \alpha^2 - \alpha^3) = \overrightarrow{0}$$

and $\alpha^0 + \alpha^1 + \alpha^2 + \alpha^3 + \alpha^4 = 0$,

leads to Equation (209) holds. Then, by placing $\alpha^4 = 0$, we obtain the same system of equations given in Equation (212) from which it follows that $\alpha^0 = \alpha^1 = \alpha^2 = \alpha^3 = 0$.

(c) The vectors in the set are not affine independent. There is a solution $\alpha^0 = \alpha^2 = \alpha^3 = 1$, $\alpha^1 = -1$ and $\alpha^4 = -2$ to the system

$$\sum_{i=0}^{4} \alpha^i x^i = (\alpha^0 + \alpha^3 + \alpha^4, \alpha^0 + \alpha^1, \alpha^1 + \alpha^2, \alpha^2 - \alpha^3) = \overrightarrow{0}$$

and $\alpha^0 + \alpha^1 + \alpha^2 + \alpha^3 + \alpha^4 = 0$.

24.5 Let $x^0, x^1, ..., x^k$ be affine-independent vectors in $\mathbf{R}^n$, and let y be a vector that is linearly independent of $\{x^0, x^1, ..., x^k\}$. We prove that $x^0, x^1, ..., x^k, y$ are affine independent vectors.
Assume by contradiction that $x^0, x^1, ..., x^k, y$ are not affine independent. Hence there are $(\alpha^i)_{i=0}^{k+1}$ in $\mathbf{R}$ not all zero such that

$$\sum_{i=1}^{k} \alpha^k x^k + \alpha^{k+1} y = \overrightarrow{0} \text{ and } \sum_{i=1}^{k+1} \alpha^i = 0. \tag{214}$$

If $\alpha^{k+1} = 0$ then Equation (214) contradicts the fact that $x^0, x^1, ..., x^k$ are affine-independent vectors in $\mathbf{R}^n$. If, on the other hand, $\alpha^{k+1} \neq 0$, then

$$y = \sum_{i=1}^{n} \tfrac{-\alpha^i}{\alpha^{k+1}} x^i,$$

which contradicts the assumption that y is linearly independent of $\{x^0, x^1, ..., x^k\}$.

24.6 Let $x^0, x^1, ..., x^k$ be affine-independent vectors in $\mathbf{R}^n$ and let $(\beta^l)_{l=0}^k$ be positive numbers summing to 1. Denote $y = \sum_{l=0}^k \beta^l x^l$. We prove that $x^1, ..., x^k, y$ are affine independent vectors. As will be shown in the proof it is enough to assume that $(\beta^l)_{l=0}^k$ are numbers summing to 1 where $\beta^0 > 0$.
Consider the system of equations with unknowns $(\alpha^l)_{l=1}^{k+1}$ in $\mathbf{R}$:

$$\sum_{l=1}^{k} \alpha^l x^l + \alpha^{k+1} y = \overrightarrow{0}, \tag{215}$$

$$\sum_{l=1}^{k+1} \alpha^l = 0. \tag{216}$$

$$\tag{217}$$

By Equation (215) and the definition of y,

$$\overrightarrow{0} = \sum_{l=1}^{k} \alpha^l x^l + \alpha^{k+1} y = \alpha^{k+1} \beta^0 x^0 + \sum_{l=1}^{k} (\alpha^l + \alpha^{k+1} \beta^l) x^l. \tag{218}$$

Similarly, by Equation (216) and since $(\beta^l)_{l=0}^k$ are numbers summing to 1,

$$0 = \sum_{l=1}^{k+1} \alpha^l = \alpha^{k+1} \beta^0 + \sum_{l=1}^{k} (\alpha^l + \alpha^{k+1} \beta^l). \tag{219}$$

However, since $x^0, x^1, ..., x^k$ are affine-independent vectors, the system of equations given by Equations (218) and (219) has a unique solution

$\alpha^l + \alpha^{k+1}\beta^l = 0$ for every $l \in \{1, 2, ..., k\}$ and $\alpha^{k+1}\beta^0 = 0$. Since $\beta^0 > 0$, it follows that $\alpha^{k+1} = \frac{0}{\beta^0} = 0$ and thus $\alpha^l = -\alpha^{k+1}\beta^l = 0$ for every $l \in \{1, ..., k\}$. In conclusion, $x^1, ..., x^k, y$ are affine independent vectors.

24.7 We prove that vectors $x^0, x^1, ..., x^k$ are affine independent in $\mathbf{R}^n$ if and only if every vector $y \in \mathrm{conv}\{x^0, x^1, ..., x^k\}$ can be represented in a unique way as a convex combination of $x^0, x^1, ..., x^k$.
Assume that $x^0, x^1, ..., x^k$ are affine independent in $\mathbf{R}^n$. Let $y \in \mathrm{conv}\{x^0, x^1, ..., x^k\}$. Then, there are $(\alpha^i)_{i=0}^k \in \Delta(\{0, 1, ..., k\})$ satisfying $y = \sum_{i=0}^k \alpha^i x^i$. For every $(\beta^i)_{i=0}^k \in \Delta(\{0, 1, ..., k\})$ satisfying $y = \sum_{i=0}^k \beta^i x^i$ the following holds

$$\overrightarrow{0} = y - y = \sum_{i=0}^k \alpha^i x^i - \sum_{i=0}^k \beta^i x^i = \sum_{i=0}^k (\alpha^i - \beta^i) x^i$$
$$\text{and } 0 = 1 - 1 = \sum_{i=0}^k \alpha^i - \sum_{i=0}^k \beta^i = \sum_{i=0}^k (\alpha^i - \beta^i).$$

However, since $x^0, x^1, ..., x^k$ are affine independent, the only solution to the above system of equations is $\alpha^i - \beta^i = 0$ for every $i \in \{0, 1, ..., k\}$. Hence, y can be represented in a unique way as a convex combination of $x^0, x^1, ..., x^k$.
Suppose next that $x^0, x^1, ..., x^k$ are not affine independent in $\mathbf{R}^n$. Hence there are $(\alpha^i)_{i=0}^k$ not all zero satisfying

$$\sum_{i=0}^k \alpha^i x^i = \overrightarrow{0} \text{ and } \sum_{i=0}^k \alpha^i = 0. \tag{220}$$

Note that there must be at least one $\alpha^i < 0$ and at least one $\alpha^j > 0$.
We show that for every $y = \sum_{i=0}^k \beta^i x^i \in \mathrm{conv}\{x^0, x^1, ..., x^k\}$ where $(\beta^i)_{i=0}^k \in \Delta(\{0, 1, ..., k\})$, there is another representation as a convex combination of $x^0, x^1, ..., x^k$.
Set $\lambda = \min\{-\frac{\beta^i}{\alpha^i} : \alpha^i < 0\}$. $\lambda > 0$ is well defined.
Then, $(\beta^i + \lambda\alpha^i)_{i=0}^k \in \Delta(\{0, 1, ..., k\})$. Indeed, for every i such that $\alpha^i \geq 0$, $\beta^i + \lambda\alpha^i \geq 0$. For every i such that $\alpha^i < 0$, $\beta^i + \lambda\alpha^i \geq \beta^i - \frac{\beta^i}{\alpha^i}\alpha^i = 0$. In addition,

$$\sum_{i=0}^k (\beta^i + \lambda\alpha^i) = \sum_{i=0}^k \beta^i + \lambda \sum_{i=0}^k \alpha^i = \sum_{i=0}^k \beta^i = 1.$$

Finally,

$$\sum_{i=0}^k (\beta^i + \lambda \cdot \alpha^i) x^i = \sum_{i=0}^k \beta^i x^i + \lambda \sum_{i=0}^k \alpha^i x^i = \sum_{i=0}^k \beta^i x^i = y.$$

In conclusion, $\sum_{i=0}^{k}(\beta^i + \lambda \cdot \alpha^i)x^i$ is a representation of y as a convex combination of $x^0, x^1, ..., x^k$.
Notice that we obtain different representation than $\sum_{i=0}^{k} \beta^i x^i$ since there is at least one i for which $\alpha^i > 0$ so $\beta^i + \lambda\alpha^i \neq \beta^i$.

24.8 Let $S = \langle\langle x^0, ..., x^k \rangle\rangle$ be a simplex in $\mathbf{R}^n$ and let $y \in \mathbf{R}^n$. We prove that $y \in S$ if and only if all the solutions of the following system of equations in the variables $(\alpha^l)_{l=0}^{k}$ satisfy $\alpha^l \geq 0$ for all $l \in \{1, 2, ..., k\}$:

$$\sum_{l=0}^{k} \alpha^l x^l = y \text{ and } \sum_{l=0}^{k} \alpha^l = 1. \tag{221}$$

First, by Definition 24.8, since S is a simplex, $x^0, ..., x^k$ are affine-independent vectors. Therefore, the system of equations in Equation (221) has at most one solution. If by contradiction there are two different solutions $(\alpha^l)_{l=0}^{k}$ and $(\beta^l)_{l=0}^{k}$ then

$$\begin{aligned} \overrightarrow{0} = y - y = \sum_{l=0}^{k} \alpha^l x^l - \sum_{l=0}^{k} \beta^l x^l = \sum_{l=1}^{k} (\alpha^l - \beta^l) x^l, \\ \text{and } 0 = 1 - 1 = \sum_{l=0}^{k} \alpha^l - \sum_{l=0}^{k} \beta^l = \sum_{l=1}^{k} (\alpha^l - \beta^l). \end{aligned}$$

Thus it can be derived that $\alpha^l = \beta^l$ for every $l \in \{0, 1, ..., k\}$, a contradiction.
Therefore, $y \in S$, if and only if there are non-negative numbers $(\alpha^l)_{l=0}^{k}$ summing to 1 such that $y = \sum_{l=0}^{k} \alpha^l x^l$ if and only if there is a unique solution to Equation (221) which is non-negative.

24.9 Let $x^0, x^1, ..., x^n$ be affine-independent vectors in $\mathbf{R}^n$. We prove that for every vector $y \in \mathbf{R}^n$ there exists a unique solution to the following system of equations in unknowns $(\alpha^l)_{l=0}^{n}$:

$$\sum_{l=0}^{n} \alpha^l x^l = y \text{ and } \sum_{l=0}^{n} \alpha^l = 1. \tag{222}$$

By Theorem 24.6, the vectors $x^0, x^1, ..., x^n, y$ are affine dependent vectors. In particular, there are $(\beta^l)_{l=0}^{n+1}$ in $\mathbf{R}$ not all zero satisfying

$$\begin{aligned} \sum_{l=0}^{n} \beta^l x^l + \beta^{n+1} y = \overrightarrow{0}, \\ \text{and } \sum_{l=0}^{n+1} \beta^l = 0. \end{aligned}$$

Since, $x^0, ..., x^n$ are affine independent, $\beta^{n+1} \neq 0$. Therefore,

$$y = \sum_{l=0}^{n} -\frac{\beta^l}{\beta^{n+1}} x^l,$$
$$\text{and } \sum_{l=0}^{n} -\frac{\beta^l}{\beta^{n+1}} = 1.$$

so $(\alpha^l)_{l=0}^n$ where $\alpha^l = -\frac{\beta^l}{\beta^{n+1}}$ is a solution of Equation (222).
The proof that that there cannot be more than one solution is presented in Exercise 24.8.

24.10 Let $S = \langle\langle x^0, x^1, ..., x^k \rangle\rangle$ be a simplex. The boundary of S is the union of all the $(k-1)$-dimensional subsimplices of S, i.e.,

$$\bigcup_{l=0}^{k} \langle\langle x^0, x^1, ..., x^{l-1}, x^{l+1}, ..., x^k \rangle\rangle \subseteq S.$$

We prove that the boundary of S is the set of all the points y in S whose barycentric coordinate representation has at least one zero coordinate.
Let $y \in S$. Then y is a point in the boundary of S if and only if there is $l \in \{0, 1, ..., k\}$ such that $y \in \langle\langle x^0, x^1, ..., x^{l-1}, x^{l+1}, ..., x^k \rangle\rangle$. However, by Exercise 24.8, $y \in \langle\langle x^0, x^1, ..., x^{l-1}, x^{l+1}, ..., x^k \rangle\rangle$ if and only if there is a unique solution of Equation (221) (relative to the vectors $x^0, x^1, ..., x^{l-1}, x^{l+1}, ..., x^k$) $(\alpha^{l'})_{l' \neq l}$ of non-negative numbers satisfying $\sum_{l' \neq l} \alpha^{l'} x^{l'} = y$ and $\sum_{l' \neq l} \alpha^{l'} = 0$. In particular, by defining $\widehat{\alpha}^l = 0$ and $\widehat{\alpha}^{l'} = \alpha^{l'}$ for every $l' \neq l$, $(\widehat{\alpha}^{l'})_{l'=0}^k$ is a unique solution of Equation (221) (relative to the vectors $x^0, x^1, ..., x^k$) of non-negative numbers where satisfying $\sum_{l'=0}^k \widehat{\alpha}^{l'} x^{l'} = y$ and $\sum_{l'=0}^k \alpha^{l'} = 0$ so $(\widehat{\alpha}^{l'})_{l'=0}^k$ is a barycentric coordinate representation of y (relative to the vectors $x^0, x^1, ..., x^k$) that has at least one zero coordinate.

24.11 Let $S^1 = \langle\langle x^0, x^1, ..., x^k \rangle\rangle$ and $S^2 = \langle\langle y^0, y^1, ..., y^k \rangle\rangle$ be two simplices of dimension k in $\mathbf{R}^n$.
Let

$$H^1 = \left\{ \sum_{l=0}^{k} \alpha^l x^l : \sum_{l=0}^{k} \alpha^l = 1 \right\} \text{ and}$$
$$H^2 = \left\{ \sum_{l=0}^{k} \alpha^l y^l : \sum_{l=0}^{k} \alpha^l = 1 \right\}$$

be two affine spaces of the same dimension k in $\mathbf{R}^n$. Denote by $M = |\{y^0, ..., y^k\} \setminus \{x^0, ..., x^k\}|$ the number of vectors in $\{y^0, ..., y^k\}$ that are not in $\{x^0, ..., x^k\}$.

Suppose that $H^1 \subseteq H^2$. We prove that $H^1 = H^2$ by induction over M. If $M = 0$ then $\{y^0, ..., y^k\} = \{x^0, ..., x^k\}$ and thus $H^1 = H^2$.
Assume that for $M \leq m$ one has $H^1 = H^2$. We prove that the same holds for $M = m + 1 \leq k + 1$.
Assume without loss of generality that $x^m \notin \{y^0, ..., y^n\}$. Since $x^m \in H^1 \subseteq H^2$, there are numbers summing to 1, not all zero, $(\alpha^l)_{l=1}^k$ such that $x^m = \sum_{l=1}^{m-1} \alpha^l y^l$. Therefore, there is $l \in \{0, 1, ..., k\}$ such that $\alpha^l > 0$. Assume without loss of generality that $\alpha^0 > 0$. Hence, by Exercise 24.6, $x^m, y^1, ..., y^k$ are affine independent vectors (also Exercise 24.6 assumes that α^l are all positive, the proof only relies on the fact that $\alpha^0 > 0$). Furthermore, the affine space H spanned by these vectors satisfies $H \subseteq H^2$. We next prove that $H^2 \subseteq H$. Indeed, for every $z \in H^2$, there are numbers $(\beta^l)_{l=0}^k$ summing to 1 such that $z = \sum_{l=0}^k \beta^l y^l$. Hence, by defining $\delta^0 = \frac{\beta^0}{\alpha^0}$ and $\delta^l = \beta^l - \delta^0 \alpha^l$ it follows that $z \in H$:

$$\delta^0 x^n + \sum_{l=1}^k \delta^l y^l = \delta^0 \alpha^0 y^0 + \sum_{l=1}^k (\delta^l + \delta^0 \alpha^l) y^l = \sum_{l=0}^k \beta^l y^l = z,$$
$$\delta^0 + \sum_{l=1}^k \delta^l = \delta^0 + \sum_{l=1}^k (\beta^l - \delta^0 \alpha^l) = \delta^0 (1 - \sum_{l=1}^k \alpha^l) + \sum_{l=1}^k \beta^l = \sum_{l=0}^k \beta^l = 1.$$

from which it follows that $H^2 \subseteq H$ so $H^2 = H$. Finally, applying the induction hypothesis on H and H^1 it follows that $H = H^1$. In conclusion $H^2 = H = H^1$ which completes the proof.

24.12 The barycentric coordinates relative to the three affine-independent vectors $x^0 = (0, 2), x^1 = (1, 0)$, and $x^2 = (2, 0)$:
The barycentric coordinates relative of $y^1 = (1, \frac{1}{2})$ are $(\frac{1}{4}, \frac{1}{2}, \frac{1}{4})$.
The barycentric coordinates relative of $y^2 = (3, 0)$ are $(0, -1, 2)$.
The barycentric coordinates relative of $y^3 = (2, 2)$ are $(1, -2, 2)$.
The barycentric coordinates relative of $y^4 = (\frac{3}{2}, \frac{1}{4})$ are $(\frac{2}{8}, \frac{5}{8}, \frac{1}{8})$.

24.13 **Partition A:** it is not a simplicial partition since the intersection of $\langle\langle x^1, x^3, x^4 \rangle\rangle$ and $\langle\langle x^2, x^5, x^6 \rangle\rangle$ is x^6 which is not a face of $\langle\langle x^1, x^3, x^4 \rangle\rangle$.
Partition B: it is a simplicial partition.
Partition C: it is not a simplicial partition since the convex hull of $\{x^3, x^4, x^5, x^6\}$ is not a simplex (4 points in $\mathbf{R}^2$ are necessarily affine dependent).

24.14 Proof of Theorem 24.14: let S be a k-dimensional simplex in $\mathbf{R}^n$, and let $\mathcal{T}$ be a simplicial partition of it. Denote by $\mathcal{T}^k$ the set of all the k-dimensional simplices in $\mathcal{T}$. We show that $S = \bigcup_{T \in \mathcal{T}^k} T$.
By property (1) in Definition 24.12, $S = \bigcup_{T \in \mathcal{T}} T$. Hence, since $\mathcal{T}^k \subseteq \mathcal{T}$ then $\bigcup_{T \in \mathcal{T}} T \subseteq S$.

On the other hand, by property (4) in Definition 24.12, for every l-dimensional simplex $T \in \mathcal{T}$ where $l < k$ there is $l+1$-dimensional simplex $T' \in \mathcal{T}$ satisfying $T \subseteq T'$. Hence, by induction, for every l-dimensional simplex $T \in \mathcal{T}$ where $l < k$ there is k-dimensional simplex $T^k \in \mathcal{T}$ satisfying $T \subseteq T^k$. Therefore and by property (4) in Definition 24.12, $S = \bigcup_{T \in \mathcal{T}} T \subseteq \bigcup_{T \in \mathcal{T}^k} T$. In particular, $S = \bigcup_{T \in \mathcal{T}^k} T$ as claimed.

24.15 Let $S = \langle\langle x^0, ..., x^k \rangle\rangle$ be a simplex, and let $\mathcal{T}$ be the collection of all the faces of S. We show that all four properties of Definition 24.12 hold so $\mathcal{T}$ is a simplicial partition of S.

(1) Every face T of S is contained in S so $\bigcup_{T \in \mathcal{T}} T \subseteq S$. In addition, S is a face of itself, i.e. $S \in \mathcal{T}$. Thus $S \subseteq \bigcup_{T \in \mathcal{T}} T$. In conclusion, $S = \bigcup_{T \in \mathcal{T}} T$.

(2) Let $S^1 = \langle\langle x^{l_0}, ..., x^{l_n} \rangle\rangle$ and $S^2 = \langle\langle x^{r_0}, ..., x^{r_m} \rangle\rangle$ be two faces of S. Assume that $S^1 \cap S^2 \neq \emptyset$. We show that $S^1 \cap S^2$ is also a face of S.
Denote by $V = \{x^{l_0}, ..., x^{l_n}\} \cap \{x^{r_0}, ..., x^{r_m}\}$ and $T = \mathrm{conv}(V)$. T is a face of S. Furthermore, $T \subseteq S^1$ and $T \subseteq S^2$ and thus $T \subseteq S^1 \cap S^2$. It only remains to prove that $S^1 \cap S^2 \subseteq T$. Let $x \in S^1 \cap S^2$. Then, x can be represented as a convex combination of $x^{l_0}, ..., x^{l_n}$ as well as a convex combination of $x^{r_0}, ..., x^{r_m}$. These two representation must be identical or else there are two way to represent x as a convex combination of $x^0, ..., x^k$ (Exercise 24.7). Therefore, it can be deduced that $x \in T$.

(3) Let T be a face of S. Then its face is also a face of S. Hence, all of T faces are also elements of the given collection.

(4) If T is an l-dimensional face of S, for $l < k$, then it is contained in an $l+1$- dimensional face of S.

24.16 The claim is wrong. Even if $T^1, T^2, .., T^L$ are simplices whose union is a simplex S, the collection containing all the faces of $T^1, T^2, .., T^L$ is not necessarily a simplicial partition of S. Partition B in Example 24.13 is a counter example.

24.17 Let $\mathcal{T}$ be a simplicial partition of a k-dimensional simplex $S = \langle\langle x^0, x^1, ..., x^k \rangle\rangle$. Define $\widehat{S} = \langle\langle x^0, x^1, ..., x^{k-1} \rangle\rangle$ and $\widehat{\mathcal{T}} := \{T \in \mathcal{T} : T \subseteq \widehat{S}\}$.
We show that all four properties of Definition 24.12 hold so $\widehat{\mathcal{T}}$ is a simplicial partition of $\widehat{S}$.

(1) By the definition of $\widehat{\mathcal{T}}$, $\bigcup_{T \in \widehat{\mathcal{T}}} T \subseteq \widehat{S}$. We next prove that $\widehat{S} \subseteq \bigcup_{T \in \widehat{\mathcal{T}}} T$.
Let $x \in \widehat{S}$. Therefore, there is a simplex $T = \langle\langle y^0, y^1, ..., y^n \rangle\rangle \in \mathcal{T}$ such that $x \in T$.

Since $T \subseteq S$, for every $i \in \{0, 1, ..., n\}$ there are non non-negative numbers $(\alpha_i^l)_{l=0}^k$ summing to 1 for which

$$y^i = \sum_{l=0}^{k} \alpha_i^l x^l \quad \forall i \in \{0, 1, ..., n\}.$$

In addition, since $x \in T$, there are non-negative numbers $(\beta^i)_{i=0}^n$ summing to 1 such that

$$x = \sum_{i=0}^{n} \beta^i y^i.$$

Therefore,

$$x = \sum_{i=0}^{n} \beta^i \left(\sum_{l=0}^{k} \alpha_i^l x^l \right) = \sum_{l=0}^{k} \left(\sum_{i=0}^{n} \beta^i \alpha_i^l \right) x^l. \tag{223}$$

where $\left(\sum_{i=0}^n \beta^i \alpha_i^l\right)_{l=0}^k$ are non-negative numbers summing to 1. In particular, Equation (223) is the unique representation of x as a convex combination of $x^0, ..., x^k$. However, $x \in \widehat{S}$ so $\sum_{i=0}^n \beta^i \alpha_i^{k+1} = 0$ from which the following holds: for every $i \in \{0, 1, ..., n\}$, if $\beta^i > 0$ then $\alpha_i^{k+1} = 0$ so $y^i \in \widehat{S}$. In conclusion, $x \in \operatorname{conv}\{y^i : y^i \in \widehat{S}\}$ which is a face of T that is contained in $\widehat{S}$ and thus it is an element in $\widehat{\mathcal{T}}$.

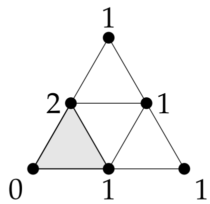
(2) For every two simplices $T_1, T_2 \in \widehat{\mathcal{T}}$ one has T_1, T_2 are elements in simplicial partition $\mathcal{T}$ and thus $T_1 \cap T_2$ is either the empty set or a face both of T_j and of T_m.

(3) If T is a simplex in the collection $\widehat{\mathcal{T}}$ then it is a simplex in the simplicial partition $\mathcal{T}$. Hence, all of its faces are also elements of $\mathcal{T}$. Furthermore, all of T's faces are subsets of $T \subseteq \widehat{S}$ and thus all of its faces are also elements of $\widehat{\mathcal{T}}$.

(4) If $T = \langle\!\langle y^0, ..., y^l \rangle\!\rangle$ is an l-dimensional simplex in $\widehat{\mathcal{T}}$, for $l < k-1$, then it is also an l-dimensional simplex in the simplicial partition $\mathcal{T}$. Therefore, it is contained in a k-dimensional simplex $T^k \in \mathcal{T}$. By property (2), T is the face of T^k. That is, $T^k = \langle\!\langle y^0, y^1, ..., y^l, y^{l+1}, ..., y^k \rangle\!\rangle$. Assume by contradiction that for every $i > l$, $y_i \notin \widehat{S}$. Then, every $x^j \in \{x_0, ..., x^{k-1}$ that is lineary independent of $y^0, ..., y^l$ is also linear independent of $y^0, ..., y^k$. Since the dimension of T is less than k there must be at least one such a vector, say x^j. But then, $y_0, ..., y^k, x^j$ are affine independent vectors satisfying $y_0, ..., y^k, x^j \subseteq S$ which contradict S is k-dimensional simplex. Finally, since there is $y^i \in \widehat{S}$ where $i > k$, then by property (3), T is contained in the $l+1$-dimensional simplex $\langle\!\langle y^0, y^1, ..., y^l, y^i \rangle\!\rangle \in \widehat{\mathcal{T}}$.

24.18 There are $2^6 \cdot 3 = 192$ proper colorings are there of the simplicial partition in Example 24.17.
Indeed, there are 3 vertices in which the support includes only one extreme point (so has only one optional color), 6 vertices have two extreme colors in their support and another vertex which his support is of size 3.

24.19 In following example there are a 2-dimensional simplex, a simplicial partition of this simplex, and a nonproper coloring of the simplicial partition, for which the number of perfectly colored 2-dimensional simplices is odd.



24.20 Let $\mathcal{T}$ be a simplicial partition of a k-dimensional simplex $S = \langle\langle x^0, x^1, ..., x^k \rangle\rangle$, and let $c : Y(\mathcal{T}) \to \{0, 1, ..., k\}$ be a proper coloring. Denote $\widehat{S} = \langle\langle x^0, x^1, ..., x^{k-1} \rangle\rangle$ and $\widehat{\mathcal{T}} := \{T \in \mathcal{T} : T \subseteq \widehat{S}\}$. We prove that c restricted to $Y(\widehat{\mathcal{T}})$ is a proper coloring of $\widehat{\mathcal{T}}$.
Following Exercise 24.17, $\widehat{\mathcal{T}}$ is a simplicial partition of $\widehat{S}$. Furthermore, since $\widehat{\mathcal{T}} \subseteq \mathcal{T}$ one has $Y(\widehat{\mathcal{T}}) \subseteq Y(\mathcal{T})$. Let $y \in Y(\widehat{\mathcal{T}})$. By Exercise 24.9, y has a unique representation as a convex combination over $x^0, x^1, ..., x^{k-1}$ which is identical to its representation as a convex combination over $x^0, x^1, ..., x^k$. Hence, since c is a proper coloring of $\mathcal{T}$, the color of y is is one of the indices in the support of y using its barycentric representation relative to S which is identical to its barycentric representation relative to $\widehat{S}$. In conclusion, c is a proper coloring of $\widehat{\mathcal{T}}$.

24.21

$$Y(\mathcal{T}) = \left\{ \frac{1}{m} \sum_{j=1}^{m} x^{i_j} : m \in \{1, 2, ..., k+1\}, \{x^{i_1}, ..., x^{i_m}\} \subseteq \{x^0, x^1, ..., x^k\} \right\}.$$

24.22 We prove that the diameter $\rho(S)$ of a simplex S equals the greatest (Euclidean) distance between two vectors in the simplex.
Let $x, y \in S$. By Exercise 24.9, thereare non-negative numbers $(\alpha^i)_{i=0}^k$ and $(\beta^i)_{i=0}^k$ satisfying similarly

$$x = \sum_{i=1}^k \alpha^i x^i \text{ and } \sum_{i=1}^k \alpha^i = 1, \text{ similarly}$$
$$y = \sum_{i=1}^k \beta^i x^i \text{ and } \sum_{i=1}^k \beta^i = 1.$$

Therefore,

$$\begin{aligned}\|x - y\| &= \left\| \sum_{i=0}^k \sum_{j=0}^k \alpha^i \beta^j (x^i - x^j) \right\| \leq \sum_{i=0}^k \sum_{j=0}^k \alpha^i \beta^j \left\| x^i - x^j \right\| \leq \sum_{i=0}^k \sum_{j=0}^k \alpha^i \beta^j \rho(S) \\ &= \rho(S),\end{aligned}$$

where the first inequality follows the triangle inequality, and the second inequality follows the definition of ρ. In particular, $\rho(S) \geq \max_{x,y \in S} \|x - y\|$.
On the other hand,

$$\rho(S) = \max_{0 \leq i < j \leq k} \|x^i - x^j\| \leq \max_{x,y \in S} \|x - y\|$$

and thus $\rho(S) = \max_{x,y \in S} \|x - y\|$.

24.23 **The completion of the proof of Theorem 24.23:** We show that the intersection of any two simplices in the simplicial partition $\mathcal{T}$ that is defined in the proof of the theorem is either empty or is contained in $\mathcal{T}$.
Let $T, T' \in \mathcal{T}$ be two simplices. Hence there are $l, l' \in \{0, 1, ..., k\}$ such that

$$T \subseteq \langle\!\langle x^0, ..., x^{l-1}, x^{l+1}, ..., x^k \rangle\!\rangle,$$
$$T' \subseteq \langle\!\langle x^0, ..., x^{l'-1}, x^{l'+1}, ..., x^k \rangle\!\rangle.$$

If $l = l'$ then, by Exercise 24.15, the collection of all the faces of $\langle\!\langle x^0, ..., x^{l-1}, x^{l+1}, ..., x^k \rangle\!\rangle$ is a simplicial partition of it, and thus $T \cap T'$ is either empty or is a face of both T and T' and so it is contained in $\mathcal{T}$.
Assume next that $l \neq l'$ and $T \cap T' \neq \emptyset$.
Since $Y(T) \cap Y(T') \subseteq Y(T)$ it follows that

$$\langle\!\langle Y(T) \cap Y(T') \rangle\!\rangle = \text{conv}(Y(T) \cap Y(T')) \subseteq \text{conv}(Y(T)) = T,$$
$$\langle\!\langle Y(T) \cap Y(T') \rangle\!\rangle = \text{conv}(Y(T) \cap Y(T')) \subseteq \text{conv}(Y(T')) = T'.$$

Therefore, if $\langle\!\langle Y(T) \cap Y(T') \rangle\!\rangle$ then it is contained in the intersection of T and T', and it is a face of both T and T'.
We next prove that $T \cap T' \subseteq \langle\!\langle Y(T) \cap Y(T') \rangle\!\rangle$ which completes the proof.
Let $x \in T \cap T'$. There are non-negative numbers $(\alpha^i)_{i \neq l}$, α^y, $(\beta^i)_{i \neq l'}$ and β^y satisfying

$$x = \sum_{i \neq l} \alpha^i x^i + \alpha^y \text{ and } \sum_{i \neq l} \alpha^i + \alpha^y = 1, \tag{224}$$

$$x = \sum_{i \neq l'} \beta^i x^i + \beta^y \text{ and } \sum_{i \neq l} \beta^i + \beta^y = 1. \tag{225}$$

However, by Equation (224),

$$x = \sum_{i \neq l} (\alpha^i + \tfrac{\alpha^y}{k+1}) x^i + \tfrac{\alpha^y}{k+1} x^l.$$

Similarly, by by Equation (225),

$$x = \sum_{i \neq l'} (\beta^i + \tfrac{\beta^y}{k+1}) x^i + \tfrac{\beta^y}{k+1} x^{l'}.$$

However, following Exercise (24.8), there is a unique representation of x as a convex combination of S. Therefore,

$$\beta^l = \tfrac{\alpha^y}{k+1} - \tfrac{\beta^y}{k+1} = -\alpha^{l'} \Rightarrow \alpha^{l'} = \beta^l = 0 \text{ and } \alpha^y = \beta^y,$$

from which it follows that $\alpha^i = \beta^i$.
Finally, by Exercise 24.8, Equation (224) is also the representation of x as a convex combination of $Y(T)$. Similarly, Equation (225) and thus Equation (224) are also the representation of x as a convex combination of $Y(T')$. Therefore, $x \in \langle\!\langle Y(T) \cap Y(T') \rangle\!\rangle$, as needed.

24.24 **The completion of the proof of Theorem 24.25:**
We show that the intersection of any two simplices in the simplicial partition $\mathcal{T}$ that is defined in the proof of the theorem is either empty or is contained in $\mathcal{T}$.
Note that since $\mathcal{T}$ is a simplicial partition of S, for all of the following three cases the intersection of the given two simplices of $\widehat{\mathcal{T}}$ is either empty or is contained in $\mathcal{T}$: (1) $T, T' \in \mathcal{T} \subset \widehat{\mathcal{T}}$; (2) $\langle\!\langle T, y \rangle\!\rangle, \langle\!\langle T', y \rangle\!\rangle \in \widehat{\mathcal{T}}$ where $T, T' \in \mathcal{T}$, since

$$\langle\!\langle T, y \rangle\!\rangle \cap \langle\!\langle T', y \rangle\!\rangle = \langle\!\langle T \cap T', y \rangle\!\rangle;$$

and (3) $T, \langle\!\langle T', y \rangle\!\rangle \in \widehat{\mathcal{T}}$ where $T, T' \in \mathcal{T}$, since

$$T \cap \langle\!\langle T', y \rangle\!\rangle = T \cap T'.$$

In addition, if $T \in \mathcal{T} \subset \widehat{\mathcal{T}}$ and $\langle\!\langle y \rangle\!\rangle \in \widehat{\mathcal{T}}$ then $T \cap \langle\!\langle y \rangle\!\rangle = \emptyset$ since y is affine independent of the vectors $x^0, x^1, ..., x^k$.
Finally, if $\langle\!\langle T, y \rangle\!\rangle, \langle\!\langle y \rangle\!\rangle \in \widehat{\mathcal{T}}$ then $\langle\!\langle T, y \rangle\!\rangle \cap \langle\!\langle y \rangle\!\rangle = \langle\!\langle y \rangle\!\rangle \in \widehat{\mathcal{T}}$.

24.25 The partition $\widehat{\mathcal{T}}$ constructed in the proof of Theorem 24.25 is

$$\widehat{\mathcal{T}} = \mathcal{T} \bigcup \langle\!\langle y \rangle\!\rangle \bigcup \{\langle\!\langle T, y \rangle\!\rangle : T \in \mathcal{T}\}.$$

By definition 24.20, for every $T \in \mathcal{T}$,

$$\begin{aligned}\rho(\langle\!\langle T, y \rangle\!\rangle) &= \max\left\{\max\{\|x^i - x^j\| : x^i, x^j \in Y(T), x^i \neq x^j\}, \max\{\|x^i - y\| : x^i \in Y(T)\}\right\} \\ &= \max\left\{\rho(T), \{\|x^i - y\| : x^i \in Y(T)\}\right\}.\end{aligned}$$

Moreover, $\rho(\langle\!\langle y \rangle\!\rangle) = 0$.
Therefore,

$$\begin{aligned}\rho(\widehat{\mathcal{T}}) &= \max_{T \in \widehat{\mathcal{T}}} \rho(T) \\ &= \max\left\{\max\{\rho(T) : T \in \mathcal{T}\}, \rho(\langle\!\langle y \rangle\!\rangle), \{\|x^i - y\| : x^i \in Y(T)\}\right\} \\ &= \max\left\{\rho(\mathcal{T}), \||x^0 - y\|, ..., \|x^k - y\|\right\},\end{aligned}$$

where the last equality holds since $\{x^i\} \in \mathcal{T}$ for every $i \in \{0, 1, ..., k\}$.

24.26 Suppose $A \subseteq \mathbf{R}^n$ is a convex set. Let $B(A, \epsilon)$ be the ϵ-neighborhood of A. We show that $B(A, \epsilon)$ is also a convex set.
Let $x^1, x^2 \in B(A, \epsilon)$. Then, there are $y^1, y^2 \in A$ such that $d(x^1, y^1), d(x^2, y^2) \leq \epsilon$. Since A is convex, for every $\lambda \in (0, 1)$, $\lambda y^1 + (1 - \lambda) y^2 \in A$. Therefore,

$$\begin{aligned}d(\lambda x^1 + (1 - \lambda) x^2, A) &\leq d(\lambda x^1 + (1 - \lambda) x^2, \lambda y^1 + (1 - \lambda) y^2) \\ &\leq \lambda d(x^1, y^1) + (1 - \lambda) d(x^2, y^2) \leq \epsilon,\end{aligned}$$

from which it follows that $\lambda x^1 + (1 - \lambda) x^2 \in B(A, \epsilon)$.

24.27 **Proof of Theorem 24.29:** let $X \subset \mathbf{R}^n$ be a closed and convex set. Define a function $g : \mathbf{R}^n \to X$ as follows: $g(x)$ is the closest point to x in X.
We show that $d(g(x), g(\widehat{x})) \leq d(x, \widehat{x})$ for all $x, \widehat{x} \in \mathbf{R}^n$.
By Theorem 24.39

$$\langle x - g(x), g(x) \rangle \geq \langle x - g(x), g(x') \rangle \Rightarrow \langle g(x') - g(x), g(x) - x \rangle \geq 0, \tag{226}$$

$$\langle x' - g(x'), g(x') \rangle \geq \langle x' - g(x'), g(x) \rangle \Rightarrow \langle g(x) - g(x'), g(x') - x' \rangle \geq 0. \tag{227}$$

Define

$$\begin{aligned}
D(t) &= \|t(x-x') + (1-t)(g(x)-g(x'))\|^2 \\
&= \|t(x-x'-g(x)+g(x')) + (g(x)-g(x'))\|^2 \\
&= t^2\|x-x'-g(x)+g(x')\|^2 + \|g(x)-g(x')\|^2 \\
&+ 2t(\langle g(x)-g(x'), g(x')-x'\rangle + \langle g(x')-g(x), g(x)-x\rangle).
\end{aligned}$$

However, since

$$\begin{aligned}
D'(t) =& 2t\|x-x'-g(x)+g(x')\|^2 + 2\langle g(x)-g(x'), g(x')-x'\rangle \\
&+ 2\langle g(x')-g(x), g(x)-x\rangle \geq 0 \quad \forall t \geq 0, \\
d^2(x,x') =& \|x-x'\|^2 = D(1) \geq D(0) = \|g(x)-g(x')\|^2 = d(g(x), g(x')).
\end{aligned}$$

24.28 (a) Let $X \subseteq \mathbf{R}^n$. Suppose that $f : X \to X$ is a continuous function. Let $F : X \to X$ be the correspondence defined by $F(x) = \{f(x)\}$ for every $x \in X$. We show that F is upper semi-continuous.
Let $\big((x^n, y^n)\big)_{n=1}^{\infty} \subseteq \big\{(x,y) \in X \times Y : y = f(x)\big\}$ be a sequence such that $\lim_{n\to\infty}(x^n, y^n) = (x^*, y^*)$. However, since f is continuous,

$$y^* = \lim_{n\to\infty} y_n = \lim_{n\to\infty} f(x_n) = f(\lim_{n\to\infty} x_n) = f(x^*).$$

(b) Let $X \subseteq \mathbf{R}^n$ be a convex, compact, and nonempty set and let $f : X \to X$ be a continuous function.
Using the result of the previous item, the correspondence F defined in previous item is upper semi-continuous. Therefore, since $F(x) = \{f(x)\}$ is nonempty and convex, by Kakutani's Fixed Point Theorem (Theorem 24.32) there exists a point $x^* \in X$ satisfying $x^* \in F(x^*) = \{f(x^*)\}$. In conclusion, there is $x^* \in X$ such that $x^* = f(x^*)$.

24.29 Let $X \subseteq \mathbf{R}^n$ be a compact, and nonempty set that are homeomorphic to a convex set Y. Let $f :\to X \to X$ be a continuous function.
Since X and Y are homeomorphic, there exists a continuous bijection $g : X \to Y$ satisfying the property that g^{-1} is also continuous. In particular, Y is closed and convex and the function $g \circ f \circ g^{-1} : Y \to Y$ is continuous (as a composition of continuous functions). Therefore, by Exercise 24.28, there is $y^* \in Y$ such that $g(f(g^{-1}(y^*))) = y^*$ and thus $f(g^{-1}(y^*)) = g^{-1}(g(f(g^{-1}(y^*))) = g^{-1}(y^*) \in X$. In particular, f has a fixed point.

24.30 Let N be a nonempty, finite set of players. For each player $i \in N$, let X_i be a convex and compact subset of $\mathbf{R}^{d_i}$, where d_i is a natural number. Denote by $X = \times_{i \in N} X_i$ the Cartesian product of the sets $(X_i)_{i \in N}$. For every player $i \in N$, let $u_i : X \to R$ be a function that is continuous and quasi-concave in x_i; that is, for every real number c and all $x_{-i} \in X_{-i}$, the set $\{x_i : u_i(x_{i,-i}) \geq c\}$ is convex.
Define a correspondence br from X to X as follows. For every $x \in X$ and for each $i \in N$,

$$\mathrm{br}_i(x) := \left\{ y_i \in X_i : u_i(y_i, x_{-i}) = \max_{z_i \in X_i} u_i(z_i, x_{-i}) \right\}, \quad \forall i \in N,$$

and

$$br(x) := \times_{i \in N} \mathrm{br}_i(x).$$

(a) Since X_i is convex and compact and u_i is continuous, the maximum over X_i is attained so $\mathrm{br}_i(x)$ is not empty. Furthermore, u_i is quasi-concave in x_i so the set

$$\mathrm{br}_i(x) = \{y_i : u_i(y_i, x_{-i}) \geq \max_{z_i \in X_i} u_i(z_i, x_{-i})\}$$

is convex. Therefore, br as a Cartesian product of upper semi-continuous correspondences with nonempty convex values, is an upper semi-continuous correspondence with nonempty convex values as well. Using Kakutani's Fixed Point Theorem, it can be deduced that br has a fixed point.

(b) Let $x^* \in X$ be a fixed point of br. Hence, for every player i x_i^* is in the set of best response to x_{-i}^* and it is thus a Nash equilibrium in the game $G = (N, (X_i)_{i \in N}, (u_i)_{i \in N})$.
Note that X_i is not necessarily a set of mixed strategies over a pure strategies and u_i is not necessarily a linear function (as assumed in Theorem 5.10).

24.31 In this exercise, we prove the KKM Theorem (Theorem 24.36) using Brouwer's Fixed Point Theorem (Theorem 24.30). Suppose by contradiction that the conditions of the KKM Theorem hold yet its conclusion does not hold, i.e., $\bigcap_{i=1}^{n} X_i = \emptyset$.

(a) For each $\epsilon > 0$ define $Y^{i,\epsilon} = \{x \in X(n) : d(x, X^i) \leq \epsilon\} \subseteq X(n)$. Then, for every $x \in X^i$, $d(x, X^i) = 0 < \epsilon$ so $X^i \subseteq Y^{i,\epsilon}$. Hence, for all $\epsilon > 0$

$$X(n) = \bigcup_{i=1}^{n} X^i \subseteq \bigcup_{i=1}^{n} Y^{i,\epsilon} \subseteq X(n).$$

Assume by contradiction that $\bigcap_{i=1}^n Y^{i,\epsilon} \neq \emptyset$ for every $\epsilon > 0$. Let $(\epsilon_m)_{m=1}^{\infty}$ be a decreasing sequence which converges to 0. By Cantor's intersection theorem,

$$\bigcap_{i=1}^n X^i = \bigcap_{i=1}^n \lim_{m\to\infty} Y^{i,\epsilon_n} = \bigcap_{i=1}^n (\bigcap_{m=1}^{\infty} Y^{i,\epsilon_n}) = \bigcap_{m=1}^{\infty} (\bigcap_{i=1}^n Y^{i,\epsilon_n}) \neq \emptyset,$$

a contradiction. In conclusion, there exists $\epsilon_0 > 0$ such that $\bigcap_{i=1}^n Y^{i,\epsilon_0} = \emptyset$.

(b) Assume next that $\epsilon < \epsilon_0$. Denote $Z^{i,\epsilon} = X(n) \setminus Y^{i,\epsilon}$ for every $i = 1, 2, ..., n$. We prove that $\sum_{i=1}^n d(x, Z^{i,\epsilon}) \geq \epsilon$ for every $x \in X(n)$. Let $x \in X(n)$. Then, there is $j \in \{1, 2, ..., n\}$ such that $x \in X_j$. Thus, for every $z \in Z^{j,\epsilon}$

$$d(x,z) \geq d(z, X^j) > \epsilon,$$

from which it follows that

$$\sum_{i=1}^n d(x, Z^{i,\epsilon}) \geq d(x, Z^{j,\epsilon}) \geq \min_{z\in Z^{j,\epsilon}} d(x,z) \geq \epsilon.$$

(c)

$$\begin{aligned}\sum_{i=1}^n x_i d(x, Y^{i,\epsilon}) &= \sum_{\{i: x\in Y^{i,\epsilon}\}} x_i d(x, Y^{i,\epsilon}) + \sum_{\{i: x\notin Y^{i,\epsilon}\}} x_i d(x, Y^{i,\epsilon}) \\ &= \sum_{\{i: x\in Y^{i,\epsilon}\}} x_i \cdot 0 + \sum_{\{i: x\notin Y^{i,\epsilon}\}} x_i d(x, Y^{i,\epsilon}) \\ &= \sum_{\{i: x\notin Y^{i,\epsilon}\}} x_i d(x, Y^{i,\epsilon}).\end{aligned}$$

(d) Define a function $f^\epsilon : X(n) \to X(n)$ as follows:

$$f_i^\epsilon(x) = \begin{cases} x_i - x_i d(x, Y^{i,\epsilon}) & \text{if } x \notin Y^{i,\epsilon}, \\ x_i + \left(\sum_{j=1}^n x_j d(x, Y^{j,\epsilon})\right) \frac{d(x, Z^{i,\epsilon})}{\sum_{j=1}^n d(x, Z^{j,\epsilon})} & \text{if } x \in Y^{i,\epsilon}. \end{cases}$$

By Part (c), f^ϵ is well defined function.

We next prove that $f^\epsilon(x) \in X(n)$. Let $x \in X(n)$. For every $i = \{1, 2, ..., n\}$, $f^\epsilon(x) \geq 0$ since $x_j \geq 0$ and $d(x, Y^{j,\epsilon}) \in [0, 1]$ for every $j \in \{0, 1, ..., n\}$. In addition,

$$\begin{aligned}
\sum_{i=1}^{n} f^{\epsilon}(x) &= \sum_{\{i: x \notin Y^{i,\epsilon}\}} (x_i - x_i d(x, Y^{i,\epsilon})) \\
&+ \sum_{\{i: x \in Y^{i,\epsilon}\}} \left(x_i + \left(\sum_{j=1}^{n} x_j d(x, Y^{j,\epsilon}) \right) \frac{d(x, Z^{i,\epsilon})}{\sum_{j=1}^{n} d(x, Z^{j,\epsilon})} \right) \\
&= \sum_{\{i: x \notin Y^{i,\epsilon}\}} (x_i - x_i d(x, Y^{i,\epsilon})) \\
&+ \sum_{\{i: x \in Y^{i,\epsilon}\}} \left(x_i + \left(\sum_{\{j: x \notin Y^{j,\epsilon}\}} x_j d(x, Y^{j,\epsilon}) \right) \frac{d(x, Z^{i,\epsilon})}{\sum_{\{j: x \in Y^{j,\epsilon}\}} d(x, Z^{j,\epsilon})} \right) \\
&= \sum_{i=1}^{n} x_i - \sum_{\{i: x \notin Y^{i,\epsilon}\}} x_i d(x, Y^{i,\epsilon}) + \sum_{\{j: x \notin Y^{j,\epsilon}\}} x_j d(x, Y^{j,\epsilon}) \\
&= \sum_{i=1}^{n} x_i = 1,
\end{aligned}$$

where the second equality holds by Part (c) and since $d(x, Z^{j,\epsilon}) = 0$ if $x \notin Y^{j,\epsilon}$.

We now prove that the function f^{ϵ} is continuous.

Let $(x^m)_{m=1}^{k} \subseteq X(n)$ be a sequence which converges to x. Let $i \in \{1, 2, ..., n\}$. If $x \notin Y^{i,\epsilon}$ and there is $M \in \mathbf{N}$ such that for every $m \geq M$, $x^m \notin Y^{i,\epsilon}$ then, by the continuity of the function $x_i - x_i d(x, Y^{i,\epsilon})$, $\lim_{m\to\infty} f_i^{\epsilon}(x^m) = f_i^{\epsilon}(x)$. The same holds in case that $x \in Y^{i,\epsilon}$ and there is $M \in \mathbf{N}$ such that for every $m \geq M$, $x^m \in Y^{i,\epsilon}$. Assume next that $x \notin Y^{i,\epsilon}$ but there is $M \in \mathbf{N}$ such that for every $m \geq M$, $x^m \in Y^{i,\epsilon}$. In this case,

$$\lim_{m\to\infty} d(x^m, Z^{i,\epsilon}) = \lim_{m\to\infty} 0 = 0$$

so

$$\lim_{m\to\infty} f_i^{\epsilon}(x) = x_i.$$

whereas

$$\begin{aligned}
f_i^{\epsilon}(x) = x_i - x_i d(x, Y^{i,\epsilon}) &= x_i - x_i d(\lim_{m\to\infty} x^m, Y^{i,\epsilon}) \\
&= x_i - x_i \lim_{m\to\infty} d(x^m, Y^{i,\epsilon}) = x_i - x_i \cdot 0 = x_i.
\end{aligned}$$

Therefore, $\lim_{m\to\infty} f_i^{\epsilon}(x^m) = f_i^{\epsilon}(x)$ which completes the proof that f^{ϵ} is continuous (indeed, $Y^{i,\epsilon}$ is compact so if there is $M \in \mathbf{N}$ such that for every $m \geq M$, $x^m \in Y^{i,\epsilon}$ then $x \in Y^{i,\epsilon}$).

In conclusion, by the Brouwer's Fixed Point Theorem , since $X(n)$ is a convex, compact, and nonempty set and f^ϵ is a continuous function, f^ϵ has a fixed point x^ϵ.

(e) Assume by contradiction that $x^\epsilon \notin \bigcap_{i=1}^n Y^{i,\epsilon}$. Then, there is $i \in \{1, 2, ..., n\}$ such that $x^\epsilon \notin Y^{i,\epsilon}$ from which it follows that $x_i^\epsilon = f_i^\epsilon(x^\epsilon) = x_i^\epsilon - x_i^\epsilon d(x^\epsilon, Y^{i,\epsilon})$. Hence either $x_i^\epsilon = 0$ so $x^\epsilon \in X^i \subseteq Y^{i,\epsilon}$ or $d(x^\epsilon, Y^{i,\epsilon}) = 0$ from which it follows that $x^\epsilon \in Y^{i,\epsilon}$ since $Y^{i,\epsilon}$ is compact, either way we get a contradiction to $x^\epsilon \notin Y^{i,\epsilon}$, which completes the proof.

(f) Let $(\epsilon_k)_{k\in\mathbf{N}}$ be a sequence converging to 0 such that $x^* := \lim_{k\to\infty} x^{\epsilon_k}$ exists. Then, for every $i = 1, 2, ..., n$,

$$0 \leq d(x^*, X^i) = d(\lim_{k\to\infty} x^{\epsilon_k}, X^i) = \lim_{k\to\infty} d(x^{\epsilon_k}, X^i) \leq \lim_{k\to\infty} \epsilon_k = 0.$$

However, since $X^1, ..., X^n$ are compacts $x^* \in \cap_{i=1}^n X^i$, contradicting the assumption that $\cap_{i=1}^n X^i = \emptyset$.

24.32 We prove Brouwer's Fixed Point Theorem (Theorem 24.30) using the KKM Theorem (Theorem 24.36).

Let $X \in \mathbf{R}^n$ be a convex, compact, and nonempty set. Let $f : X \to X$ be a continuous function.

Since X is compact, there exists a simplex S sufficiently large to contain X. Define a function $h : S \to S$ by $h(x) := f(g(x))$, where g is the function associating every point x with the point in X that is closest to it. Note that the function g is well defined by Theorem 24.28 and that it satisfies $g(x) = x$ for every $x \in X$. By Theorem 24.29, the function g is continuous; hence h, as the composition of two continuous functions, is also continuous.

Define

$$X^i = \{x \in S : h_i(x) \geq x_i\}.$$

Since $h_i(x) \in S$, then $h_i(x) \geq 0$ for every $x \in X(n)$. In particular, if $x_i = 0$ then $x_i \in X^i$. Therefore,

$$\{x \in S : x_i = 0\} \subseteq X^i, \quad \forall i = 1, 2, ..., n.$$

Note that, by the definition $\bigcup_{i=1}^n X^i \subseteq S$. To prove that $\bigcup_{i=1}^n X^i = S$, assume by contradiction that there is $x \in S \setminus \left(\bigcup_{i=1}^n X^i\right)$. Therefore, for every i, $x \notin X^i$ so $h_i(x) < x_i$. In particular

$$1 = \sum_{i=1}^n h_i(x) < \sum_{i=1}^n x_i = 1,$$

a contradiction. Hence, all of the conditions of KKM Theorem hold and thus there is $x^* \in \bigcap_{i=1}^n X^i \neq \emptyset$.
In conclusion, for every i, $1 = \sum_{i=1}^n h_i(x^*) \geq \sum_{i=1}^n x_i^* = 1$ and thus necessarily $h_i(x^*) = x_i^*$ for every i. From which it follows that $x^* \in X$, and thus $f_i(x^*) = f_i(g(x^*)) = h_i(x^*) = x_i^*$. That is x^* is a fixed point.

24.33 Let $\alpha \in \mathbf{R}^n$ and $\beta^{\mathbf{R}}$. Then,

$$\begin{aligned}
&x \in H^+(\alpha, \beta) && \iff \\
&\langle x, \alpha \rangle \geq \beta && \iff \\
&\sum_{i=1}^n x_i \alpha_i \geq \beta && \iff \\
&\sum_{i=1}^n x_i(-\alpha_i) \leq -\beta && \iff \\
&\langle x, -\alpha \rangle \leq -\beta && \iff \\
&x \in H^-(-\alpha, -\beta).
\end{aligned}$$

Hence, $H^+(\alpha, \beta) = H^-(-\alpha, -\beta)$.

24.34 Let $x, y \in \mathbf{R}^n$. Then,

$$\langle x - y, x \rangle - \langle x - y, y \rangle = \langle x - y, x - y \rangle = \|x - y\| \geq 0.$$

In conclusion, $\langle x, x - y \rangle \geq \langle x - y, y \rangle$ so $x \in H^+(x - y, \langle x - y, y \rangle)$.

24.35 Let $x, y \in \mathbf{R}^m$ be two different vectors.

(a) Since $\langle y, y - x \rangle = \langle y - x, y \rangle$ then $y \in H(y - x, \langle y - x, y \rangle)$.

(b) For every $z \in H(y - x, \langle y - x, y \rangle)$,

$$\langle y - x, y - z \rangle = \langle y - x, y \rangle - \langle y - x, z \rangle = \langle y - x, y \rangle - \langle y - x, y \rangle = 0.$$

Therefore, the hyperplane $H(y - x, \langle y - x, y \rangle)$ is perpendicular to $y - x$.

(c) For every $z \in H(y - x, \langle y - x, y \rangle, z \neq y, x$ the following holds

$$\begin{aligned}
&\langle z - x, z - x \rangle = \langle z - y, z - x \rangle + \langle y - x, z - x \rangle \\
&= \langle z - y, z - y \rangle + \langle z - y, y - x \rangle + \langle y - x, z - y \rangle + \langle y - x, y - x \rangle \\
&= \langle z - y, z - y \rangle + \langle y - x, y - x \rangle && (228) \\
&> \langle y - x, y - x \rangle, && (229)
\end{aligned}$$

where Equality (228) follows Part (b), and Inequality (229) holds since $z \neq y$. In particular, y is the point in $H(y - x, \langle (y - x, y \rangle)$ that is closest to x.

24.36 Let H be a hyperplane, let $x \notin H$ and let $y \in H$ be the point in H that is closest to x.
H is a hyperplane so it is a closed and convex set. Hence, by the Separating Hyperplane Theorem, since $x \notin H$ and y is the closest point in H to x, the hyperplane $H(x-y, \langle x-y, y\rangle)$ satisfies that $H \subseteq H^{-}(x-y, \langle x-y, y\rangle)$. Hence, either H is parallel to $H(x-y, \langle x-y, y\rangle)$ or it equals to it. However, since

$$x \in H \cap H(x-y, \langle x-y, y\rangle),$$

the intersection is not empty so $H = H(x-y, \langle x-y, y\rangle)$.

24.37 Let $H(\alpha, \beta)$ be a hyperplane and let $x \notin H(\alpha, \beta)$. Define

$$y := x + \tfrac{\beta - \langle \alpha, x\rangle}{\langle \alpha, \alpha\rangle}\alpha.$$

(a) $y \in H(\alpha, \beta)$. Indeed,

$$\begin{aligned}\langle y, \alpha\rangle &= \langle x + \tfrac{\beta-\langle\alpha,x\rangle}{\langle\alpha,\alpha\rangle}\alpha, \alpha\rangle \\ &= \langle x, \alpha\rangle + \tfrac{\beta-\langle\alpha,x\rangle}{\langle\alpha,\alpha\rangle}\langle \alpha, \alpha\rangle = \alpha.\end{aligned}$$

(b) We first show that for every $a \neq 0$, $H(\alpha, \beta) = H(a\alpha, a\beta)$:

$$x \in H(\alpha, \beta) \iff \langle x, \alpha\rangle = \beta \iff \langle x, a\alpha\rangle = a\beta \iff x \in H(a\alpha, a\beta).$$

Hence, since

$$\begin{aligned}&y - x = \tfrac{\beta-\langle\alpha,x\rangle}{\langle\alpha,\alpha\rangle}\alpha \text{ and} \\ &\langle y-x, y\rangle = \langle \tfrac{\beta-\langle\alpha,x\rangle}{\langle\alpha,\alpha\rangle}\alpha, y\rangle = \tfrac{\beta-\langle\alpha,x\rangle}{\langle\alpha,\alpha\rangle}\langle\alpha, y\rangle = \tfrac{\beta-\langle\alpha,x\rangle}{\langle\alpha,\alpha\rangle}\beta,\end{aligned}$$

from which it follows that

$$H(\alpha, \beta) = H(\tfrac{\beta-\langle\alpha,x\rangle}{\langle\alpha,\alpha\rangle}\alpha, \tfrac{\beta-\langle\alpha,x\rangle}{\langle\alpha,\alpha\rangle}\beta) = H(y-x, \langle y-x, y\rangle).$$

24.38 The claim is wrong. Consider the following counter example: the set S is a triangle whose vertices are $(1,-1)$, $(1,1)$ and $(-1,1)$; $x = (\frac{1}{2}, \frac{1}{2})$; the set $\widehat{S}$ is a triangle whose vertices are $(1,-1)$, $(-1,-1)$ and $(-1,1)$; $\widehat{x} = (-\frac{1}{2}, -\frac{1}{2})$. Then the set $S + \widehat{S}$ is a square whose vertices are $(1,-1)$, $(1,1)$, $(-1,1)$, and $(-1,-1)$ whereas $x + \widehat{x} = \overrightarrow{0} \in S + \widehat{S}$. Therefore, there is no hyperplane which separating $S + \widehat{S}$ and $x + \widehat{x}$.

24.39 Let S be a closed set (not necessarily convex) and let $x \notin S$. Then there does not necessarily exist a hyperplane separating x from S. For instance, $S = \{x \in R^n : 1 \leq \|x\| \leq 2\}$ and $x = \overrightarrow{0}$.

24.40 (a) There is no hyperplane separating x from S since $x \in S$.
(b) The hyperplane $H = \{x \in \mathbf{R}^2 : x_1 + x_2 = 1.5\}$ separates x from S.
(c) There is no hyperplane separating x from S since $x \in S$.
(d) There is no hyperplane separating x from S since x is in the interior of the convex hull of S.
(e) The hyperplane $H = \{x \in \mathbf{R}^2 : x_2 = 1\}$ separates x from S.
(f) The hyperplane $H = \{x \in \mathbf{R}^2 : x_2 = 1.5\}$ separates x from S.
(g) There is no hyperplane separating x from S since x is in the interior of the convex hull of S.
(h) The hyperplane $H = \{x \in \mathbf{R}^2 : x_1 + x_2 + x_3 = 1.9\}$ separates x from S.

24.41 (a) $H\left((2,2,2) - (1,1,1), \langle (2,2,2) - (1,1,1), (1,1,1) \rangle\right) = H\left((1,1,1), 3\right)$.
(b) $H\left((2,3,4) - (1,1,1), \langle (2,3,4) - (1,1,1), (1,1,1) \rangle\right) = H\left((1,2,3), 6\right)$.
(c)

$$\begin{aligned}
&H\left((2,2,2) - (\tfrac{1}{\sqrt{3}}, \tfrac{1}{\sqrt{3}}, \tfrac{1}{\sqrt{3}}), \langle (2,2,2) - (\tfrac{1}{\sqrt{3}}, \tfrac{1}{\sqrt{3}}, \tfrac{1}{\sqrt{3}}), (\tfrac{1}{\sqrt{3}}, \tfrac{1}{\sqrt{3}}, \tfrac{1}{\sqrt{3}}) \rangle\right) \\
&= H\left((\tfrac{2\sqrt{3}-1}{\sqrt{3}}, \tfrac{2\sqrt{3}-1}{\sqrt{3}}, \tfrac{2\sqrt{3}-1}{\sqrt{3}}), 2\sqrt{3} - 1\right) \\
&= H\left((\tfrac{1}{\sqrt{3}}, \tfrac{1}{\sqrt{3}}, \tfrac{1}{\sqrt{3}}), 1\right).
\end{aligned}$$

(d) $H\left((1,1) - (\frac{1}{2}, \frac{1}{2}), \langle (1,1) - (\frac{1}{2}, \frac{1}{2}), (\frac{1}{2}, \frac{1}{2}) \rangle\right) = H\left((\frac{1}{2}, \frac{1}{2}), \frac{1}{2}\right)$.
(e) $H\left((2,3) - (\frac{1}{2}, \frac{1}{2}), \langle (2,3) - (\frac{1}{2}, \frac{1}{2}), (\frac{1}{2}, \frac{1}{2}) \rangle\right) = H\left((\frac{3}{2}, \frac{5}{2}), 2\right)$.

24.42 Let $S \subseteq \mathbf{R}^n$ be a closed and convex set, and let $x \notin S$. Denote by $y \in S$ the closest point in S to x. Then the hyperplane

$$H\left(x - y, \tfrac{\langle x-y,y\rangle + \langle x-y,x\rangle}{2}\right)$$

strictly separates x from S.
Note that since $x \neq y$ then

$$0 < \langle x - y, x - y \rangle = \langle x - y, x \rangle - \langle x - y, y \rangle$$

so $\langle x-y, y\rangle < \langle x-y, x\rangle$. Therefore and by Theorem 24.39, for every $z \in S$,

$$\langle x-y, z\rangle \leq \langle x-y, y\rangle < \tfrac{\langle x-y,x\rangle + \langle x-y,y\rangle}{2},$$

Hence $S \subseteq H^{-}\left(x-y, \tfrac{\langle x-y,y\rangle+\langle x-y,x\rangle}{2}\right) \setminus H\left(x-y, \tfrac{\langle x-y,y\rangle+\langle x-y,x\rangle}{2}\right)$.
On the other hand,

$$\langle x-y, x\rangle = \tfrac{\langle x-y,x\rangle+\langle x-y,x\rangle}{2} > \tfrac{\langle x-y,y\rangle+\langle x-y,x\rangle}{2}$$

Thus $x \in H^{+}\left(x-y, \tfrac{\langle x-y,y\rangle+\langle x-y,x\rangle}{2}\right) \setminus H\left(x-y, \tfrac{\langle x-y,y\rangle+\langle x-y,x\rangle}{2}\right)$.

24.43 let $v \in \mathbf{R}^n$ be a vector, and let T be an $n \times m$ matrix. We prove that the following two claims are equivalent:

(a) $\langle u, v\rangle = uv^T \geq 0$ for every vector $u \in \mathbf{R}^n$ satisfying $uT \geq \overrightarrow{0}$.

(b) There exists a vector $w \in \mathbf{R}^m$, $w \geq \overrightarrow{0}$, satisfying $Tw^T = v^T$.

We first prove that a $\Rightarrow$ b. Suppose that claim (a) holds and assume by contradiction that claim (b) does not hold.
Define the set

$$A := \left\{(Tw^T)^T : w \in \mathbf{R}^m, w \geq \overrightarrow{0}\right\} \subseteq \mathbf{R}^n.$$

A is closed and convex. However, by the contradiction assumption, $v \notin A$. Hence, by the Separating Hyperplane Theorem, there exists a hyperplane $H(\alpha, \beta)$ separating A from v. Furthermore, following Exercise 24.42 one can assume that $H(\alpha, \beta)$ is a strictly separating hyperplane. That is,

$$\langle \alpha, (Tw^T)^T\rangle > \beta > \langle \alpha, v\rangle \quad \forall (Tw^T)^T \in A. \tag{230}$$

In particular, for $\overrightarrow{0} = (T\overrightarrow{0}^T)^T \in A$, Equation (230) holds so

$$\langle \alpha, \overrightarrow{0}\rangle = 0 > \beta > \langle \alpha, v\rangle.$$

However, since $H(\alpha, \beta) = H(\frac{1}{k}\alpha, \frac{1}{k}\beta)$ then one can assume without loss of generality that the there is a separating hyperplane (not necessarily strictly separating hyperplane) $H(\alpha, 0)$ that separates x from A.
Therefore, for every $w \geq \overrightarrow{0}$

$$\alpha v^T = \langle \alpha, v\rangle < 0 \leq \langle \alpha, (Tw^T)^T\rangle = (\alpha T)w^T,$$

from which it follows that $\alpha T \geq \overrightarrow{0}$ whereas $\alpha v^T < 0$ in contradiction to claim (a).

We next prove that b $\Rightarrow$ a. Let $w \in \mathbf{R}^m$ satisfying the condition in claim (b). Then, for every vector $u \in \mathbf{R}^n$ where $uT \geq \overrightarrow{0}$, the following holds

$$\langle u, v\rangle = uv^T = u(Tw^T) = (uT)w^T \geq 0,$$

so claim (a) also holds.

24.44 **Proof of Theorem 24.47:** Let X and Y be two closed and convex subsets of $\mathbf{R}^n$. Assume that X and Y are disjoint sets. We show that there exists a hyperplane $H(\alpha, \beta)$ strictly separating X from Y , i.e.,

$$\langle \alpha, x\rangle > \beta > \langle \alpha, y\rangle \quad \forall x \in X, y \in Y.$$

Denote by $d(X, Y)$ the distance between X and Y,

$$d(X, Y) := \min_{x \in X, y \in Y} d(x, y).$$

The minimum in the definition of $d(X, Y)$ is attained because X and Y are closed sets. Let $x^* \in X$ and $y^* \in Y$ be points that minimize the distance between the points in X and the points in Y, $d(X, Y) = d(x^*, y^*) > 0$.
Using Theorem 24.39, the hyperplane $H(x^* - y^*, \langle x^* - y^*, y^*\rangle)$ separating x^* from Y, i.e.,

$$\langle x^* - y^*, y\rangle \leq \langle x^* - y^*, y^*\rangle < \langle x^* - y^*, x^*\rangle \ \ \forall y \in Y. \tag{231}$$

Similarly, the hyperplane $H(y^* - x^*, \langle < y^* - x^*, x^*\rangle)$ separating y^* from X. Hence,

$$\begin{aligned} &\langle y^* - x^*, x\rangle \leq \langle y^* - x^*, x^*\rangle < \langle y^* - x^*, y^*\rangle \ \ \forall x \in X \Rightarrow \\ &\langle x^* - y^*, x\rangle \geq \langle x^* - y^*, x^*\rangle > \langle x^* - y^*, y^*\rangle \ \ \forall x \in X. \end{aligned} \tag{232}$$

Set $\alpha = x^* - y^*$ and $\beta = \langle x^* - y^*, \frac{x^*+y^*}{2}\rangle$. Then, by Equations (231) and (232) it follows that for every $x \in X$ and every $y \in Y$,

$$\langle \alpha, y\rangle \leq \langle \alpha, y^*\rangle < \beta < \langle \alpha, x^*\rangle \leq \langle \alpha, x\rangle.$$

24.45 Let X and Y be two convex subsets of $\mathbf{R}^n$. Assume that X and Y are disjoint sets.

(a) For every $\epsilon > 0$ denote by X_ϵ the set of all points whose distance from X^c, the complement of X, is at least ϵ:

$$X_\epsilon = \{x \in X : d(x, X^c) \geq \epsilon\}.$$

Let $x, y \in X_\epsilon$ and $\alpha \in (0,1)$. Denote by $z = \alpha x + (1-\alpha)y$. We show that for every $w \in \mathbf{R}^n$ such that $d(z,w) < \epsilon$ one has $w \in X$ from which it follows that $d(z, X^c) \geq \epsilon$ and $z \in X_\epsilon$.
Set $x' = x + (w-z)$ and $y' = y + (w-z)$. Then,

$$d^2(x', x) = \langle x' - x, x' - x \rangle = \langle w - z, w - z \rangle = d^2(z, w) < \epsilon^2.$$

In particular, since $x \in X_\epsilon$, $x' \in X$. Similarly, $y' \in X$. Therefore, by the convexity of X,

$$\begin{aligned} w &= z + w - z = \alpha(x + (w-z)) + (1-\alpha)(y + (w-z)) \\ &= \alpha x' + (1-\alpha)y' \in X. \end{aligned}$$

In conclusion, $z = \alpha x + (1-\alpha)y \in X_\epsilon$ and X_ϵ is convex.
Assume that $(x^t)_{t=1}^\infty \subseteq X_\epsilon$ where $\lim_{t\to\infty} x^t = x^*$. In particular, there is $t \in \mathbf{N}$ such that $d(x^t, x^*) < \frac{\epsilon}{2}$ whereas $d(x^t, X^c) \geq \epsilon$ so $x^* \in X$. In addition,

$$d(x^*, X^c) \geq d(x^t, X^c) - d(x^*, x^t) \geq \epsilon - d(x^*, x^t) \xrightarrow[t\to\infty]{} \epsilon,$$

so $x^* \in X_\epsilon$ and thus X_ϵ is closed.

(b) Note that $X_\epsilon \subseteq X$ and $Y_\epsilon \subseteq Y$. Hence,

$$X_\epsilon \cap Y_\epsilon \subseteq X \cap Y = \emptyset.$$

That is, the sets X_ϵ and Y_ϵ are disjoint for every $\epsilon > 0$.

(c) Since X_ϵ and Y_ϵ are closed and convex disjoint sets, by Exercise 24.44, for every $\epsilon > 0$ there exists a hyperplane $H(\alpha_\epsilon, \beta_\epsilon)$ satisfying

$$\langle \alpha_\epsilon, x \rangle > \beta_\epsilon > \langle \alpha_\epsilon, y \rangle, \quad \forall x \in X_\epsilon, y \in Y_\epsilon$$

(d) Suppose that the hyperplane $H(\alpha_\epsilon, \beta_\epsilon)$ is the hyperplane constructed in the proof of Theorem 24.39.
For every $0 < \widehat{\epsilon} < \epsilon$,

$$d(X, Y) \leq d(X_{\widehat{\epsilon}}, Y_{\widehat{\epsilon}}) \leq d(X_\epsilon, Y_\epsilon).$$

Therefore, the sequence $(\alpha_{1/n})_{n=1}^\infty = (x_{1/n} - y_{1/n})_{n=1}^\infty$ is bounded and thus it has a convergent subsequence $(\alpha_{1/n_k})_{k=1}^\infty$ that converges to α_*. Hence, there is a sufficiently large K, such that for every $k > K$

$$\langle \alpha_*, y_1 \rangle - 1 \leq \langle \alpha_{1/n_k}, y_1 \rangle < \beta_{1/n_k} < \langle \alpha_{1/n_k}, x_1 \rangle \leq \langle \alpha_*, x_1 \rangle + 1.$$

Then the sequence $(\alpha_{1/n_k}, \beta_{1/n_k})$ has a convergent subsequence $(\alpha_{1/n_{k_l}}, \beta_{1/n_{k_l}})$ which converges to the accumulation point (α_*, β_*).

(e) We prove that the hyperplane $H(\alpha_*, \beta_*)$ separates X from Y.
Let $x \in X$. We prove that $\langle \alpha_*, x \rangle \geq \beta_*$. First assume that there is $\epsilon > 0$ such that $x \in X_\epsilon$. Then, there is $L_x \in \mathbf{N}$ such that $x \in X_{1/n_{k_l}}$ for every $l \geq L_x$. In particular,

$$\begin{aligned}
\langle \alpha_*, x \rangle &= \langle \alpha_* - \alpha_{1/n_{k_l}} + \alpha_{1/n_{k_l}}, x \rangle \\
&= \langle \alpha_* - \alpha_{1/n_{k_l}}, x \rangle + \langle \alpha_{1/n_{k_l}}, x \rangle \\
&\geq -\|\alpha_* - \alpha_{1/n_{k_l}}\| \|x\| + \langle \alpha_{1/n_{k_l}}, x \rangle \\
&= -\|\alpha_* - \alpha_{1/n_{k_l}}\| \|x\| + \beta_{1/n_{k_l}}
\end{aligned}$$

As l goes to infinity, the following holds

$$\langle \alpha_*, x \rangle \geq \lim_{l \to \infty} \left(\|\alpha_* - \alpha_{1/n_{k_l}}\| \|x\| + \beta_{1/n_{k_l}} \right) = \beta_*.$$

Next, let $x \in X$, then for every $\epsilon > 0$ let x_ϵ be the closest point to x in X_ϵ. In particular, $\|x - x_\epsilon\| \leq \epsilon$. Therefore,

$$\langle \alpha_*, x \rangle = \langle \alpha_*, x_\epsilon \rangle + \langle \alpha_*, x - x_\epsilon \rangle \geq \beta_* - \|\alpha_*\| \|x - x_\epsilon\| \geq \beta_* - \epsilon \|\alpha_*\|.$$

However, since the last inequality holds for every $\epsilon > 0$ then $\langle \alpha_*, x \rangle \geq \beta_*$.
The proof that $\langle \alpha_*, y \rangle \leq \beta_*$ for every $y \in Y$ is similar, and thus it was omitted.

24.46 (a)

$$\begin{array}{ll}
\text{Compute:} & Z_D = \min\{x_1 + 4x_2 + 7x_3\} \\
\text{subject to:} & x_1 - x_2 - 6x_3 \geq 2 \\
& 2x_1 - 4x_2 + 5x_3 \geq -3 \\
& x_1, x_2, x_3 \geq 0.
\end{array}$$

(b)

$$\begin{array}{ll}
\text{Compute:} & Z_D = \min\{-3x_1 - 6x_2 + 17x_3\} \\
\text{subject to:} & -2x_1 + 5x_2 + 7x_3 \geq 6 \\
& 4x_1 + 2x_3 \geq 2 \\
& -x_2 - 4x_3 \geq -3 \\
& x_1, x_2, x_3 \geq 0.
\end{array}$$

24.47 let x be a feasible solution of Problem (24.100). Define $w = \max\{x, 0\}$ and $z = -\min\{x, 0\}$.

If $x \geq 0$ then $w = x \geq 0$ and $z = 0$, whereas if $x < 0$ then $w = 0$ and $z = -x > 0$. Either way, $x = w - z$ and $w, z \geq 0$. Furthermore,

$$(w - z)A = xA \geq c,$$

from which it follows that (w, z) is a feasible solution of Problem (24.99) satisfying $x = w - z$.

24.48 The dual program

$$\begin{array}{ll} \text{Compute:} & Z_D = \min xb^T \\ \text{subject to:} & xA \geq c \\ & x \geq 0 \end{array}$$

is equivalent to the following linear program

$$\begin{array}{ll} \text{Compute:} & -\max(-b)x^T \\ \text{subject to:} & (-A^T)x^T \leq -c^T \\ & x \geq 0. \end{array}$$

The dual problem is then

$$\begin{array}{ll} \text{Compute:} & -\min y(-c)^T \\ \text{subject to:} & y(-A^T) \geq -b \\ & x \geq 0. \end{array}$$

However, the last linear program is equivalent to

$$\begin{array}{ll} \text{Compute:} & \max cy^T \\ \text{subject to:} & Ay^T \leq b^T \\ & x \geq 0 \end{array}$$

which is the primal program.
In conclusion, the dual program of the dual program (Equation (24.93))is actually the primal program (Equation (24.92)).

24.49 (a) In the given problem there is no constraint of the form $y \geq 0$. Hence, we set $y = z - w$. The representation of the problem as a standard linear program is thus as follows:

$$\begin{array}{ll} \text{Compute:} & Z_P = \max cz^T - cw^T \\ \text{subject to:} & Az^T - Aw^T \leq b^T \\ & -Az^T + Aw^T \leq -b^T \\ & z, w \geq 0. \end{array}$$

(b) The dual program is

$$\begin{array}{ll} \text{Compute:} & Z_D = \min rb^T - vb^T \\ \text{subject to:} & rA - vA \geq c \\ & -rA + vA \geq -c \\ & r, v \geq 0. \end{array}$$

We set $x = r - v$, so the dual program equivalent to the linear program

$$\begin{array}{ll} \text{Compute:} & Z_D = \min xb^T \\ \text{subject to:} & xA = c. \end{array}$$

24.50 (a) In the given problem there is no constraint of the form $y \geq 0$. Hence, we set $y = z - w$. The representation of the problem as a standard linear program is thus as follows:

$$\begin{array}{ll} \text{Compute:} & Z_P = \max cz^T - cw^T \\ \text{subject to:} & Az^T - Aw^T \leq b^T \\ & z, w \geq 0. \end{array}$$

(b) The dual program is

$$\begin{array}{ll} \text{Compute:} & Z_D = \min xb^T \\ \text{subject to:} & xA \geq c \\ & -xA \geq -c \\ & x \geq 0. \end{array}$$

which is equivalent to

$$\begin{array}{ll} \text{Compute:} & Z_D = \min xb^T \\ \text{subject to:} & xA = c \\ & x \geq 0. \end{array}$$